
Double holographic projection of an aperiodic time crystal

**Construction of the continuum as an inverse limit of compatible cylinders,
coinductive generation of π, φ, e, α and exceptional
stabilizers of boundary incidence**

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Abstract

An arithmetic of oriented histories is constructed in which the unit is preserved under refinement while resolution, holonomic height, and memory increase. The finite support of all possible paths simultaneously maintains the additive and multiplicative evaluations, together with their residues, quotients, and carries; a tritic fiber distinguishes the elliptic, parabolic, and hyperbolic regimes; and a topological phase kernel selects, transports, and updates each state without confusing phase return with state return. The composition of these three structures determines a noncommutative associative algebra, graded by memory, and a system of compatible refinements whose history space is an inverse limit. Cylindrical observables extend covariantly and constitute the dual direct limit. This double variance provides the operator definition of holography used in the article.

The dynamics combines a finite nonadic phase, a mechanical coding of slope $\delta = \log_{10}(10/9)$, and a carry cocycle. The result is an aperiodic topological-temporal order of linear complexity and zero topological entropy, endowed with a holonomy that raises memory at each nonadic return. Its Weyl relations rigorously separate local periodicity from global aperiodicity. On this support, coinductive prolongation correlatively generates π , φ , e , and α : the first three coordinates realize, respectively, elliptic closure, hyperbolic autoscaling, and exponential transport; the fine-structure constant is the compatibility invariant that brings together, through two internal derivations, dodecaphasic closure and aperiodic prolongation.

The same structure produces a unit multiplicative-carry vacancy in an invariant additive fiber. Its local algebra is $Cl_{1,1} \cong M_2(\mathbb{R})$ and simultaneously contains elliptic and hyperbolic generators and null Witt directions. This section determines the electronic kernel and extends the genealogy toward action, the Planck and gravity scales, the Barbero–Immirzi parameter, the area spectrum, and boundary information. The second projection preserves incidence and leads, through the Paley and Witt configurations, the Golay code, the Leech lattice, and the vertex algebra $V^{\natural}$, to the exceptional stabilizers and the Monster group. Thus, the Archimedean realization and the exceptional realization do not constitute independent sources: they are two representations of a unique discrete structure of the continuum.

The uniform dodecaphasic reader also determines a sequence of first roots of critical return. Its separation ratio is proved to converge to the Feigenbaum constant, preserving the original selection, orientation and memory of the construction.

Keywords: triadic modular holography; inverse limit; nonadic holonomy; Sturmian word; aperiodic time crystal; path algebra; fine-structure constant; split Clifford; Barbero–Immirzi; Leech lattice; Monster group.

Abstract

We construct an arithmetic of oriented histories in which the unit is preserved under refinement while resolution, holonomic height and memory increase. The finite support of all possible paths retains the additive and multiplicative evaluations simultaneously, together with their remainders, quotients and carries; a tritic fibre distinguishes the elliptic, parabolic and hyperbolic regimes; and a topological phase kernel selects, transports and updates each state without identifying phase return with state return. Their composition determines a memory-graded, associative and noncommutative algebra and a compatible refinement system whose history space is an inverse limit. Cylindrical observables extend covariantly and form the dual direct limit. This variance duality is the operative notion of holography used throughout the article.

The dynamics combines a finite nonadic phase, a mechanical coding of slope $\delta = \log_{10}(10/9)$ and a carry cocycle. It yields an aperiodic topological–temporal order with linear complexity and zero topological entropy, endowed with a holonomy that raises memory at every nonadic return. On this support the coinductive prolongation generates π , φ , e and α as correlated outputs. The same structure produces a unit multiplicative-carry vacancy on an invariant additive fibre, whose local algebra is $\text{Cl}_{1,1} \cong M_2(\mathbb{R})$. The Archimedean realization and the exceptional boundary-incidence realization are thereby shown to be two representations of one discrete structure of the continuum.

The uniform dodecaphasic reader also determines a sequence of first roots of critical return. Its separation ratio is proved to converge to the Feigenbaum constant, preserving the original selection, orientation and memory of the construction.

Keywords: triadic modular holography; inverse limit; nonadic holonomy; Sturmian word; aperiodic time crystal; path algebra; fine-structure constant; split Clifford algebra; Barbero–Immirzi parameter; Leech lattice; Monster group.

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PART I

**Two-sheeted arithmetic, trigeometric orientation, and phase
dynamics**

The unit as a compatible section of an arithmetic of histories

The uniform dodecaphasic reader also determines a sequence of first roots of critical return. Its separation ratio is proved to converge to the Feigenbaum constant, preserving the original selection, orientation and memory of the construction.

1.1 The unit as a refinement invariant

The central question is not to obtain a collection of scalar values from a finite table. It is to determine the minimal structure that preserves a unit through unlimited refinements, distinguishes three classes of curvature, transports the memory of its paths, and generates, through typed representations, numbers, constants, sections, fields, and symmetries. The real line is not presupposed as the universal domain of this construction. It appears at the end of a compatible contraction of cylinders; it preserves the scalar value of the history, but not necessarily the entirety of its genealogy.

For each depth $n \geq 0$, let X_n be the finite set of admissible states. A state is not an isolated digit, but a structured fiber

$$x = (a, b; \Sigma, \Pi; R^+, q^+, R^\times, q^\times; \kappa; \tau, o; \theta_9, \theta_{12}; h; \partial; \mu; \rho; s), \quad (1.1)$$

where (a, b) locates the cell; Σ and Π are the additive and multiplicative sheets; (R^+, q^+) and $(R^\times, q^\times)$ preserve residue and quotient; κ is the carry; $\tau \in \{+1, 0, -1\}$ types the curvature; o fixes the orientation; θ_9 and θ_{12} are the nonadic and dodecaphasic phases; h measures the helical height; ∂ records boundary incidence; μ is memory; ρ preserves the path and s the survival condition. These coordinates do not form a free product: they satisfy the relations of path arithmetic, trigeometric typing, and the transport dynamics to be constructed in the following chapters.

The maps

$$r_{m,n} : X_m \longrightarrow X_n, \quad m > n, \quad (1.2)$$

forget the last layer of resolution, but preserve every relation visible at depth n . The space of infinitely refinable histories is

$$X_\infty = \varprojlim_n X_n. \quad (1.3)$$

If e_x denotes the characteristic idempotent of the cylinder determined by $x \in X_n$, the prolongation of observables satisfies

$$r_{m,n}^* e_x = \sum_{y: r_{m,n}(y)=x} e_y. \quad (1.4)$$

Consequently,

$$r_{m,n}^* \left(\sum_{x \in X_n} e_x \right) = \sum_{y \in X_m} e_y, \quad (1.5)$$

so that the unit is preserved even as the number of extensions grows. Evolution does not preserve a stationary configuration: it preserves the augmentation, the relations, and the possibility of reconstructing each antecedent from the complete history.

1.2 Double variance and double projection

The depth of states and the extension of observables advance in opposite categorical directions. The restrictions of (1.2) correspond to prolongation morphisms

$$r_{m,n}^* : \mathfrak{H}_n \longrightarrow \mathfrak{H}_m, \quad f \longmapsto f \circ r_{m,n}, \quad (1.6)$$

and the algebra of cylindrical observables is obtained as a direct limit:

$$\mathfrak{H}_{\text{cyl}} = \varinjlim_n (\mathfrak{H}_n, r_{n+1,n}^*). \quad (1.7)$$

The pair formed by (1.3) and (1.7) provides the fundamental holographic structure: the state gains resolution through contravariant compatibility and the language of observation gains scope through covariant prolongation. This is not a figurative similarity between scales, but the naturality of the evaluation

$$\langle r_{m,n}^* a, y \rangle_m = \langle a, r_{m,n} y \rangle_n. \quad (1.8)$$

The double projection in the title designates two representations of this same pair. The first contracts compatible cylinders toward an Archimedean realization; the second retains boundary incidence and extends it toward exceptional structures:

$$\rho_{\text{arq}} : \mathfrak{H}_{\text{cyl}} \longrightarrow \mathcal{A}_{\mathbb{R}}, \quad \rho_{\text{exc}} : \mathfrak{H}_{\text{cyl}} \longrightarrow \mathcal{A}_{\text{exc}}. \quad (1.9)$$

Both have the same source and preserve different invariants. The first retains the scalar coordinate of a compatible section; the second retains the incidence patterns that determine its stabilizers. The word *double* does not duplicate holography: it distinguishes two non-equivalent codomains of a single refinement and reading mechanism.

1.3 Central statement

Theorem 1.1 (Conservative evolution and production of symmetries). *There exists a graded algebra of oriented histories*

$$\mathfrak{H} = \left[\varinjlim_n \frac{\mathbb{R}\langle e_x, J_x, T_u \rangle}{\mathcal{R}_{\text{APP-TRIT-TPK}}} \right] \rtimes_{\Gamma_9} \mathbb{N}, \quad (1.10)$$

that acts on X_∞ and satisfies the following properties:

- (i) *the unit and augmentation are preserved under every compatible refinement;*
- (ii) *the nonadic return closes the phase and raises memory;*
- (iii) *the elliptic, parabolic, and hyperbolic regimes belong to a single trigeometric functional calculus;*
- (iv) *the symbolic factor has linear complexity, zero topological entropy, and long-range aperiodic order;*
- (v) *the discrete structure of the continuum arises jointly from the space of histories and the algebra of observables;*
- (vi) *π , φ , e , and α are correlated outputs of the same coinductive prolongation;*
- (vii) *a symmetry of a realization is the stabilizer of the relations and boundary incidence that this realization preserves.*

The proof will be distributed throughout the article. No known constant will be used to select its own genealogy, nor a symmetry group to retrospectively construct the object on which it acts. The logical order

will always be the same: two-sheeted arithmetic, trigeometric typing, holonomic transport, compatible refinement, continuum, constants, physical sections, and boundary stabilizers.

1.4 Irreducibility of the generating kernel

The structure has three functionally irreducible components. If the two arithmetic sheets are identified, the potential $\Delta_{\Sigma\Pi}$ and the commutator of their projectors disappear. If the three tritic regimes are identified, the spectrum disappears

$$J_+^2 = -I, \quad J_0^2 = 0, \quad J_-^2 = I. \quad (1.11)$$

If the transport cocycle is trivialized, the increment

$$[D_M, \Gamma_9] = \Gamma_9 \quad (1.12)$$

vanishes and the phase turn becomes indistinguishable from state return. Each forgetful quotient destroys an invariant of a different type that the other two factors do not reconstruct. This irreducibility explains why the structure sought cannot be reduced to an arithmetic table, an aperiodic word, a Clifford algebra, or an isolated matrix.

The conceptual consequence is precise. A number is a scalar coordinate of a representation; a constant is a stable return or autoscale invariant; a particle is a persistent section; a field transports sections; a symmetry stabilizes the relations of its boundary. These five classes come from the same algebra, but do not share a codomain and cannot be confused with one another.

Arithmetic support of all possible paths

2.1 APP: support, operations, and paths

The acronym APP derives from *All Possible Paths*. It designates a single path structure on a finite arithmetic support: the underlying graph has eighty-one vertices, whereas the set of finite paths, taken over all lengths, is countable. Its principal arithmetic matrix is the nonadic multiplication table. Addition is the internal law whose accumulation generates each row and each column of that table, and it also acts as a comparison observable on the same support. APP thus brings together addition, multiplication, and path composition before the ternary projection or the Archimedean evaluation is performed. The following definition exhaustively fixes the universe of paths under consideration.

2.1.1 Domain, generators and preformal relations of the APP support

The support of APP is a finite arithmetic torus: each of its two coordinates ranges over the nine residue classes and returns to its origin after nine unit translations. Its eighty-one points are therefore pairs of classes modulo nine. The associated Cayley graph preserves this cyclic structure: a vertex represents a position, an oriented edge represents a unit cardinal displacement, and a finite sequence of edges represents a path. Thus, *all possible paths* denotes the set of finite cardinal words with an initial point on the torus.

Each vertex simultaneously receives two arithmetic evaluations. The *additive sheet* publishes the sum of its two coordinates, and the *multiplicative sheet* publishes their product. The term *sheet* here designates an evaluation of the same point and the same path; APP preserves a single support and two functions upon it.

The operative difference between the two sheets is exact. In a row or column, addition acts by translation and repeatedly accumulates the same increment. Multiplication by a unit also permutes the classes; multiplication by a zero divisor folds several classes onto one and the same residue. To preserve the information that this folding conceals, APP lifts each evaluation to its residue–quotient coordinates:

$$i + j = R_{ij}^+ + 9q_{ij}^+, \quad ij = R_{ij}^\times + 9q_{ij}^\times.$$

The pairs (R^+, q^+) and $(R^\times, q^\times)$ recover the integer values prior to reduction. When blocks are concatenated, additive carry κ_9 records the discrepancy between adding integer representatives and first adding their residues; together with both quotients, it makes recovery compositional. Thus, additive accumulation constructs visible trajectories, whereas the residue–quotient lift unfolds the fibers created by multiplicative folding.

For a path $\gamma = (x_0, \dots, x_r)$, the genealogy visible is the sucesion of positions and pairs of evaluations

$$\mathcal{G}_{\text{vis}}(\gamma) = (\gamma; (\Sigma(x_k), \Pi(x_k))_{k=0}^r).$$

At each step, its enriched genealogy adjoins the quotients, carry, and transported orientation. The enriched–visible projection preserves the path and residues and forgets precisely those memory coordinates. This distinction will be the input to the complete state of TRIT and of the Topological Phase Kernel. The complete domain of (104 976) double-cursor initial conditions and the (468) emission fibers they produce are published in full in the chapter devoted to the APP seed domain and the canonical emission catalogue; there each seed is distinguished from the emission to which it belongs, and the multiplicity of the fiber is preserved.

2.1.2 Arithmetic support, translations and observables

Let

$$\text{dr}_9(n) = 1 + ((n - 1) \bmod 9) \in \{1, \dots, 9\}.$$

This map chooses the representative 9 for the null class. We define $X_9 = (\mathbb{Z}/9\mathbb{Z})^2$, on which acts the group of translations $(\mathbb{Z}/9\mathbb{Z})^2$, and fix the symmetric set of generators

$$\mathcal{D} = \{(1, 0), (0, 1), (-1, 0), (0, -1)\}.$$

The Cayley graph

$$\Gamma_{\text{APP},9} = \text{Cay}(X_9, \mathcal{D})$$

has 81 vertices and four oriented edges originating at each vertex. An oriented path of length r is a sequence $\gamma = (x_0, \dots, x_r)$ in X_9 such that $x_{k+1} - x_k \in \mathcal{D}$ for every $k < r$. We denote by $\text{Path}(\Gamma_{\text{APP},9})$ the set of all finite paths so defined.

Definition 2.1 (APP Structure). The arithmetic structure of possible paths is the tuple

$$\mathcal{A}_9 = (X_9, \Gamma_{\text{APP},9}, \Sigma, \Pi, \text{Path}(\Gamma_{\text{APP},9})),$$

where the two vertex observables are

$$\Sigma(i, j) = \text{dr}_9(i + j), \quad \Pi(i, j) = \text{dr}_9(ij), \quad 1 \leq i, j \leq 9.$$

Marked cardinal and path words

To make the orientation of the torus explicit, we use coordinates (row, column), with the row increasing southward and the column increasing eastward, and write

$$\delta(N) = (-1, 0), \quad \delta(E) = (0, 1), \quad \delta(S) = (1, 0), \quad \delta(O) = (0, -1).$$

Let $\mathcal{D}_{\text{card}} = \{E, N, O, S\}$, where O denotes west. Given an initial point $x_0 \in X_9$ and a word $w = d_1 \cdots d_r \in \mathcal{D}_{\text{card}}^r$, its trajectory is

$$\gamma(x_0; w)_k = x_0 + \sum_{j=1}^k \delta(d_j) \pmod{9}, \quad 0 \leq k \leq r.$$

The correspondence $(x_0, w) \mapsto \gamma(x_0; w)$ is bijective between $X_9 \times \mathcal{D}_{\text{card}}^r$ and the paths oriented of length r . In particular, existen exactly $81 \cdot 4^r$ paths of that length when it allowed returns and backtracks.

The word w closes in the torus precisely when

$$\#S(w) - \#N(w) \equiv 0 \pmod{9}, \quad \#E(w) - \#O(w) \equiv 0 \pmod{9}.$$

For a closed word, the integer displacement computed before reduction modulo nine defines the winding vector

$$\text{wind}(w) = \frac{1}{9} (\#S(w) - \#N(w), \#E(w) - \#O(w)) \in \mathbb{Z}^2.$$

Reversal of orientation reverses the order of the word, exchanges $E \leftrightarrow O$ and $N \leftrightarrow S$, and transforms $\text{wind}(w)$ into its opposite. This cardinal framework is subsequently transported in the phase kernel and forms part of the definition of each trajectory.

The expressions are written here in the chart of representatives $\{1, \dots, 9\}$. The formula $\tilde{\Sigma}(i, j) = \text{dr}_9(i + j - 1)$ is a translated chart. It represents the same torus only when the translation is also

applied to indices, initial conditions, positional trajectories, and labels. All the following proofs employ $\Sigma(i, j) = \text{dr}_9(i + j)$ and do not alternate both charts.

APP arithmetic matrix
 $\Pi(i, j) = \text{dr}_9(ij)$

	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9
2	2	4	6	8	1	3	5	7	9
3	3	6	9	3	6	9	3	6	9
4	4	8	3	7	2	6	1	5	9
5	5	1	6	2	7	3	8	4	9
6	6	3	9	6	3	9	6	3	9
7	7	5	3	1	8	6	4	2	9
8	8	7	6	5	4	3	2	1	9
9	9	9	9	9	9	9	9	9	9

subconjunto coordinado de la clase nula

Ley interna de acumulación

$$\Pi(i, j) = \text{dr}_9(\underbrace{i + \dots + i}_j) = \text{dr}_9(ij).$$

Addition generates the multiplicative rows and, in the transverse direction, the columns. The observable $\Sigma(i, j) = \text{dr}_9(i + j)$ preserves that elementary law in order to compare both readings at the same point.

$\mathcal{A}_9 = (X_9, \Gamma_{\text{APP},9}, \Pi; \Sigma, \text{Path}(\Gamma_{\text{APP},9}))$
a single APP, one principal matrix, one internal additive law, and four oriented directions N, E, S, O .

Figure 2.1: The unique APP matrix and its internal law. The nonadic multiplication table occupies the main body; each of its rows is generated by additive accumulation. The violet shading indicates the nonunitary rows or columns; the blue line identifies the subset determined by the representative 9 of the null class; the gold frame highlights the central block 2×2 .

Every point of X_9 simultaneously receives the product and sum values, but it does not thereby belong to two distinct APP matrices. The paths of $\text{Path}(\Gamma_{\text{APP},9})$ traverse a single support; Π publishes its main table and Σ conserves the internal operation that generates the accumulations and allows its incompatibility to be measured.

Proposition 2.2 (Structure of Rows and Columns). *The matrix of Σ is a Latin square of order nine. The matrix of Π is not; nevertheless, for each $u \in (\mathbb{Z}/9\mathbb{Z})^\times = \{1, 2, 4, 5, 7, 8\}$, the row and column corresponding to u are permutations of the nine residue classes.*

Proof. With i fixed, the map $j \mapsto i + j$ is a translation of $\mathbb{Z}/9\mathbb{Z}$; the same holds when j is fixed. Hence each representative occurs exactly once in every row and column of Σ . For Π , multiplication by u permutes the classes if and only if u is a unit. The rows corresponding to 3, 6, 9 contain repetitions, so the Latin property does not extend to the entire multiplicative matrix. $\square$

2.1.3 Multiplicative structure of the local ring

Each row of the multiplication table is a repeated sum. Row i accumulates i along one direction; column j accumulates j along the perpendicular direction. Cell (i, j) is the intersection of both accumulations and returns $ij = ji$. The table thus represents product and commutativity through two transverse accumulations on the same lattice.

The second row is

$$2, 4, 6, 8, 1, 3, 5, 7, 9.$$

Before the modular return the even ones appear and after the odd ones. The third row is

$$3, 6, 9, 3, 6, 9, 3, 6, 9,$$

and makes the radical visible. Let

$$R := \mathbb{Z}/9\mathbb{Z}, \quad \mathfrak{m} := (3) = \{\bar{0}, \bar{3}, \bar{6}\}.$$

Multiplication by three defines

$$M_3 : R \longrightarrow R, \quad x \longmapsto 3x, \quad \ker M_3 = \text{im } M_3 = \mathfrak{m}.$$

By the first isomorphism theorem induces a linear identification

$$\bar{M}_3 : R/\mathfrak{m} \xrightarrow{\sim} \mathfrak{m}, \quad \bar{x} \longmapsto 3x, \quad (2.1)$$

of vector spaces on $\mathbb{F}_3$. No is an isomorphism of rings: $\mathfrak{m}^2 = 0$, whereas the quotient is a field.

In positive representatives, the three phases of (2.1) are

$$369 \longmapsto 693 \longmapsto 936 \longmapsto 369, \quad (2.2)$$

produced by the phase shift $j \mapsto j+1$. The row corresponding to $6 \equiv -3 \pmod{9}$ reverses the orientation of the same cycle. This radical orbit must not be conflated with the integer 369 693 100, which will subsequently appear as the digit count of 9^{9^9} .

In direct reduction modulo three,

$$3, 6, 9 \longmapsto 0, 0, 0.$$

If we first extract the factor three and retain its internal coordinate,

$$\eta_{\mathfrak{m}}(3a) = a \bmod 3, \quad 3, 6, 9 \longmapsto 1, 2, 0.$$

The second map preserves the internal coordinate of the ideal generated by three. It must not be confused with the oriented ternary projection to be defined in the section devoted to the TRIT.

2.1.4 Coordinate subset of the null class

In the graph of the multiplicative observable,

$$\Pi(9, j) = \Pi(i, 9) = 9.$$

The last row and the last column form a union of seventeen positions, all with value nine. They do not constitute an intrinsic boundary of the arithmetic torus: they are the subset of the chart in which at least one coordinate represents the null class by means of 9. There are also four interior null representatives in

$$(3, 3), (3, 6), (6, 3), (6, 6).$$

The figure thus distinguishes this subset from the internal zeros of the nonunital ideal. The additional symbol used later in the cubic completion belongs to an enlarged space and is not identified with any of them.

2.1.5 Residue–quotient extension

Reduction modulo nine preserves the residue and eliminates the quotient of the Euclidean division. To study both components, we define, at each point,

$$i + j = 9q_{ij}^+ + R_{ij}^+, \quad ij = 9q_{ij}^\times + R_{ij}^\times,$$

where $R^+ = \Sigma$, $R^\times = \Pi$ and the q preserve the quotient. The residue–quotient extension of APP it is denoted by

$$\mathcal{A}_9^\# = (R^+, q^+; R^\times, q^\times).$$

Let $s_9 : \mathbb{Z}/9\mathbb{Z} \rightarrow \{1, \dots, 9\}$ be the section of representatives fixed by dr_9 . The additive carry associated with this section is

$$\kappa_9(a, b) = \frac{s_9(a) + s_9(b) - s_9(a + b)}{9} \in \mathbb{Z}.$$

Lemma 2.3 (Cocycle Identity of the Carry). *For arbitrary $a, b, c \in \mathbb{Z}/9\mathbb{Z}$,*

$$\kappa_9(a, b) + \kappa_9(a + b, c) = \kappa_9(b, c) + \kappa_9(a, b + c).$$

Proof. Both sides are equal to

$$\frac{s_9(a) + s_9(b) + s_9(c) - s_9(a + b + c)}{9}.$$

□

This cocycle records the discrepancy between summing integer representatives and first summing in the quotient. It is the compositional form of the quotient coordinate that will reappear in base 1000 concatenation.

The gross totals are

$$\sum_{i,j} (i + j) = 9 \sum_i i + 9 \sum_j j = 2 \cdot 9 \cdot 45 = 810,$$

and

$$\sum_{i,j} ij = \left(\sum_i i \right) \left(\sum_j j \right) = 45^2 = 2025.$$

The visible sums are obtained from the tables:

$$\sum R^+ = 405, \quad \sum R^\times = 459.$$

Therefore, the total sums of the quotient coordinates are

$$\sum q^+ = \frac{810 - 405}{9} = 45, \quad \sum q^\times = \frac{2025 - 459}{9} = 174.$$

Decomposition exacta of the difference integrada

$$2025 - 810 = 1215 = (459 - 405) + 9(174 - 45) = 54 + 9 \cdot 129.$$

The difference between the sums of residues is 54, whereas the difference between the sums of quotients is 129. The equality recomposes the total difference between the integer observables and proves that the quotient component contains arithmetic information not determined by the residual class.

2.2 Arithmetic Operators and Local Geometry of the Path Space

2.2.1 Localization of the excess in the nilpotent radical

The group of units of $\mathbb{Z}/9\mathbb{Z}$ is

$$(\mathbb{Z}/9\mathbb{Z})^\times = \{1, 2, 4, 5, 7, 8\},$$

and the maximal ideal is

$$\mathfrak{m} = (3) = \{0, 3, 6\} = \{9, 3, 6\}_{\text{visual}}, \quad \mathfrak{m}^2 = 0.$$

Each unit row of Π is a permutation of $1, \dots, 9$ and sums to 45. Rows 3 and 6 sum to 54; row 9 sums to 81. Hence

$$459 = 6 \cdot 45 + 2 \cdot 54 + 81$$

and

$$459 - 405 = 2(54 - 45) + (81 - 45) = 54.$$

All the excess is located in the nonunitary sector.

The union of rows and columns 3, 6, 9 contains 45 cells and sums to 297. Its central square 3×3 sums to 81 = $3 \cdot 27$; the outer arms, to $216 = 3 \cdot 72$. Thus,

$$297 = 216 + 81 = 3(72 + 27) = 3 \cdot 99.$$

The interpretation of 72 and 27 is obtained later from the uniquely selected central block.

2.2.2 Conditional expectation with respect to the projection $\mathbb{Z}/9\mathbb{Z} \rightarrow \mathbb{Z}/3\mathbb{Z}$

The canonical reduction $\mathbb{Z}/9\mathbb{Z} \rightarrow \mathbb{Z}/3\mathbb{Z}$ determines the partition

$$C_1 = \{1, 4, 7\}, \quad C_2 = \{2, 5, 8\}, \quad C_0 = \{3, 6, 9\}.$$

The uniform conditional expectation averages each observable on the nine points of each block $C_a \times C_b$. The aggregate matrices are

$$M_\Pi = \begin{pmatrix} 4 & 5 & 6 \\ 5 & 4 & 6 \\ 6 & 6 & 9 \end{pmatrix}, \quad M_\Sigma = \begin{pmatrix} 5 & 6 & 4 \\ 6 & 4 & 5 \\ 4 & 5 & 6 \end{pmatrix}.$$

Since each block contains nine positions,

$$9 \sum_{a,b} (M_\Pi)_{ab} = 459, \quad 9 \sum_{a,b} (M_\Sigma)_{ab} = 405.$$

Thus, the averaging preserves the totals of both observables and relocates their difference:

$$\sum M_\Pi - \sum M_\Sigma = 51 - 45 = 6, \quad 9 \cdot 6 = 54.$$

Theorem 2.4 (Spectrum of the aggregated multiplicative operator). *The matrix operator M_Π satisfies*

$$\det M_\Pi = -9$$

and its real spectrum is

$$\text{Spec}(M_\Pi) = \{-1, 9 - 6\sqrt{2}, 9 + 6\sqrt{2}\}.$$

In particular, the associated quadratic form has signature (2, 1).

Proof. The expansion of the determinant gives -9 . The characteristic polynomial factors as

$$(\lambda + 1)((\lambda - 9)^2 - 72).$$

Its roots are those indicated and $9 - 6\sqrt{2} > 0$; therefore, there exist two positive directions and one negative direction. $\square$

The spectral decomposition admits an integral form over an invariant sublattice. For

$$v_- = (1, -1, 0), \quad v_+ = (1, 1, 0), \quad v_0 = (0, 0, 1),$$

one has $M_{\Pi}v_- = -v_-$, whereas the restriction of M_{Π} to the submodule generated by v_+, v_0 is represented by

$$\begin{pmatrix} 9 & 6 \\ 12 & 9 \end{pmatrix} = 3H, \quad H = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix} \in SL(2, \mathbb{Z}).$$

The eigenvalues of H are

$$3 \pm 2\sqrt{2} = (1 \pm \sqrt{2})^2.$$

The three vectors v_-, v_+, v_0 generate a sublattice of index two in $\mathbb{Z}^3$, since the determinant of the matrix having them as columns equals -2 . Therefore, an integral direct sum of all of $\mathbb{Z}^3$ is not asserted here: on that invariant sublattice, the aggregate operator separates an eigendirection of eigenvalue -1 and a hyperbolic submodule governed by the quadratic unit $3 + 2\sqrt{2}$; over $\mathbb{R}^3$, the spectral decomposition is complete.

2.2.3 Commutator of the arithmetic projectors and separation of their images

The difference between Σ and Π is not reducible to their global sums. For $d \geq 2$, let $\mathcal{A}_d = \{1, \dots, 9\}^d$, and define

$$\Sigma_d(x) = \text{dr}_9(x_1 + \dots + x_d), \quad \Pi_d(x) = \text{dr}_9(x_1 \cdots x_d).$$

In the Hilbert space $\ell^2(\mathcal{A}_d)$, endowed with the uniform measure, let $P_{\Sigma,d}$ and $P_{\Pi,d}$ denote the orthogonal conditional expectations onto the functions measurable with respect to Σ_d and Π_d , respectively. The normalized joint table is

$$C_{ab}^{(d)} = \frac{N_{ab}^{(d)}}{\sqrt{n_a m_b}}, \quad N_{ab}^{(d)} = \#\{x : \Sigma_d(x) = a, \Pi_d(x) = b\},$$

where $n_a = \sum_b N_{ab}^{(d)}$ and $m_b = \sum_a N_{ab}^{(d)}$.

On the original two-dimensional support, the exact joint correlation is

$$N^{(2)} = \begin{pmatrix} 0 & 0 & 2 & 0 & 0 & 2 & 3 & 0 & 2 \\ 3 & 0 & 2 & 0 & 0 & 2 & 0 & 0 & 2 \\ 0 & 2 & 0 & 0 & 2 & 0 & 0 & 2 & 3 \\ 0 & 0 & 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 0 & 0 & 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 0 & 2 & 0 & 0 & 2 & 0 & 0 & 2 & 3 \\ 3 & 0 & 2 & 0 & 0 & 2 & 0 & 0 & 2 \\ 0 & 0 & 2 & 0 & 0 & 2 & 3 & 0 & 2 \\ 0 & 2 & 0 & 0 & 2 & 0 & 0 & 2 & 3 \end{pmatrix}.$$

The rows, ordered by the value of Σ , sum to nine; the columns, ordered by the value of Π , sum to $(6, 6, 12, 6, 6, 12, 6, 6, 21)$. Complete enumeration thus determines the joint correlation of the two observables before any subsequent evaluation map is introduced.

Proposition 2.5 (Switches in two and three dimensions). *The two pairs of projectors satisfy*

$$\| [P_{\Sigma,2}, P_{\Pi,2}] \| = \frac{\sqrt{2}}{3}, \quad \| [P_{\Sigma,3}, P_{\Pi,3}] \| = \frac{\sqrt{3}}{4}.$$

In particular, the additive and multiplicative observables do not induce simultaneously compatible decompositions.

Proof. The singular values s of $C^{(d)}$ are the cosines of the principal angles between the two subspaces. In the corresponding two-dimensional block, the norm of the commutator is $s\sqrt{1-s^2}$. The exact enumeration of the 9^d words produces, for $d = 2$, the nontrivial eigenvalues

$$s^2 \in \left\{ \frac{5}{7}, \frac{1}{3}, \frac{1}{3} \right\}.$$

The maximum of $s^2(1 - s^2)$ is $2/9$, whence $\sqrt{2}/3$. For $d = 3$, the same construction gives the maximum $s^2(1 - s^2) = 3/16$, attained at $s^2 = 1/4$, and therefore the norm is $\sqrt{3}/4$. In both cases, the commutator is nonzero. $\square$

Proposition 2.6 (Intersection of the observable subspaces). *The exact enumeration of the words of $\mathcal{A}_d$, for $2 \leq d \leq 15$, verifies*

$$\text{Ran } P_{\Sigma,d} \cap \text{Ran } P_{\Pi,d} = \mathbb{C}\mathbf{1}.$$

In that interval of dimensions, the constant functions constitute the only subspace common to the two arithmetic decompositions.

Proof. For two orthogonal projections, the dimension of the intersection of their ranges coincides with the multiplicity of the singular value 1 in the overlap matrix $C^{(d)}$. The integer computation of the tables $N^{(d)}$, followed by the exact computation of that multiplicity, gives the value one for each $d = 2, \dots, 15$. The constant vector belongs to both ranges; it therefore generates the entire intersection. $\square$

The sequence of norms is not extrapolated from the first two cases. In particular, for $d = 4$ the same calculation gives

$$\| [P_{\Sigma,4}, P_{\Pi,4}] \| = \sqrt{\frac{2618939}{22240656}} \simeq 0.3431538652,$$

which does not coincide with the elementary prolongation $2/(d+1)$. This case constitutes a counterexample to any formula inferred solely from $d = 2, 3$.

2.2.4 Central block and spectral decomposition

The characterization of the central block is obtained by means of a congruence condition. For two consecutive representatives $k, k + 1$, consider

$$B_k = \begin{pmatrix} k^2 & k(k+1) \\ k(k+1) & (k+1)^2 \end{pmatrix} \pmod{9}.$$

Theorem 2.7 (Uniqueness of the Adjacent Diagonal Block). *There is a unique block B_k whose diagonal entries coincide. It is determined by $k = 4$ and is*

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}.$$

Proof. The equality of the diagonals is equivalent to

$$k^2 \equiv (k+1)^2 \pmod{9},$$

that is, $2k + 1 \equiv 0 \pmod{9}$. Since 2 is a unit of $\mathbb{Z}/9\mathbb{Z}$, the congruence has the unique solution $k \equiv 4 \pmod{9}$. $\square$

Therefore, the positions 4, 5 of the multiplicative observable form

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 7I + 2R, \quad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad R^2 = I.$$

With $P_{\pm} = (I \pm R)/2$,

$$B_c = 9P_+ + 5P_-.$$

Theorem 2.8 (Typed hierarchy of the sum-product separation). *The central block has a local spectral gap ($9-5=4$). Its off-diagonal coefficient is (2). Compression by the three residue classes modulo three produces the aggregate difference ($51-45=6$), and unfolding across the nine positions produces the global defect*

$$9 \cdot 6 = 459 - 405 = 54.$$

The integers (4), (2), (6), and (54) belong, respectively, to the spectral jump, the local coupling, the modular compression, and the two-dimensional integration.

Proof. The decomposition $B_c = 7I + 2R = 9P_+ + 5P_-$ gives the jump and the local coefficient. The aggregated matrices M_Π, M_Σ constructed above satisfy $\sum M_\Pi = 51$ and $\sum M_\Sigma = 45$. Each aggregated entry represents nine positions, so unfolding multiplies the difference by nine and recovers the totals (459) and (405). $\square$

Their ordered concatenations produce the blocks 72 and 27:

$$72 + 27 = 99, \quad 72 - 27 = 45 = \det B_c.$$

The concatenation of the diagonal and antidiagonal entries produces, respectively, 77 and 22. Therefore,

$$72 + 27 = 77 + 22 = 99, \quad 77 - 22 = 55 = 54 + 1.$$

These equalities express two exact partitions of the same block; they do not by themselves identify an external constant.

Proposition 2.9 (Local–global relationship of the radical sector). *The partition by rows of the central block reproduces, with scale factor three, the partition of the radical sector:*

$$3 \cdot 72 = 216, \quad 3 \cdot 27 = 81, \quad 3 \cdot 99 = 297.$$

Proof. The block $C_0 \times C_0$ contains nine null representatives and sums to 81. The remaining four arms of the rows and columns of C_0 sum to 216. The identities follow from $72 + 27 = 99$. $\square$

Moreover,

$$M_c := \frac{B_c}{\sqrt{45}} = \exp\left(\frac{1}{2} \log \frac{9}{5} R\right) \in SO^+(1, 1).$$

If $\eta = \text{diag}(1, -1)$, the normalized matrix M_c satisfies $M_c^\top \eta M_c = \eta$, $\det M_c = 1$, and belongs to $SO^+(1, 1)$. This finite realization is confined to the constructed block 2×2 ; its space-time interpretation requires the dimensional bridge developed in the treatise on Dimensional Mechanics.

Proposition 2.10 (Spectral family APP–TRIT and reading Markov–Lorentz local). *For $\tau \in \{-1, 0, +1\}$, let*

$$B_\tau = (4 + 3\tau)I + (5 - 3\tau)R.$$

Then

$$\text{Spec}(B_\tau) = \{9, 6\tau - 1\}, \quad B_\tau = 9P_+ + (6\tau - 1)P_-,$$

and the differential mode recovers the orientation through

$$\tau = \frac{\lambda_- + 1}{6}.$$

In particular, $B_{+1} = B_c$. Its stochastic normalization

$$P_B := \frac{1}{9}B_c = P_+ + \frac{5}{9}P_-$$

satisfies, for all $n \geq 0$,

$$P_B^n = P_+ + \left(\frac{5}{9}\right)^n P_-.$$

Therefore, the contraction of the antisymmetric mode is $\kappa_B = 5/9$ and the spectral gap of this local chain is $4/9$. If

$$\eta_c := \frac{1}{2} \log \frac{9}{5},$$

the lorentzian normalization of the same block satisfies

$$\frac{B_c}{\sqrt{45}} = \exp(\eta_c R), \quad \boxed{\frac{5}{9} = e^{-2\eta_c}}.$$

Proof. Since R acts with eigenvalues $+1$ and -1 on $\text{im } P_+$ and $\text{im } P_-$, respectively, the two eigenvalues of B_τ are

$$(4 + 3\tau) + (5 - 3\tau) = 9, \quad (4 + 3\tau) - (5 - 3\tau) = 6\tau - 1.$$

The formula for the powers follows from $P_\pm^2 = P_\pm$ and $P_+P_- = 0$. Finally, $e^{-2\eta_c} = e^{-\log(9/5)} = 5/9$. $\square$

This coincidence identifies two exact readings of the local block: the differential memory decreases by the same quotient that encodes the speed in the decomposed realization. It does not turn B_τ into the global Markov kernel of TPK, or $5/9$ into a physical constant. Nor does it by itself extend the realization $SO^+(1, 1)$ to a Minkowski space $3 + 1$: those steps require additional dimensional maps.

2.2.5 Discrete curvature induced by the interaction of the sum and product leaves and its commutation defect

On the torus we write, now in residues,

$$S(x, y) = x + y, \quad P(x, y) = xy \pmod{9}.$$

For every function $T : X_9 \rightarrow \mathbb{Z}/9\mathbb{Z}$, we fix the forward differences

$$\Delta_x T(x, y) = T(x + 1, y) - T(x, y), \quad \Delta_y T(x, y) = T(x, y + 1) - T(x, y),$$

where the coordinates are taken modulo nine. This convention also fixes the positive orientation of each plaquette: first the direction x is traversed, then the direction y , and one returns along the opposite edges. It is important to type these two objects correctly. S and P are vertex potentials; the link variables are their discrete differences

$$A_x^T = \Delta_x T, \quad A_y^T = \Delta_y T.$$

For an ordered mixture of potentials (T_x, T_y) , we define the plaquette curvature by

$$F(T_x, T_y) = \Delta_x A_y^{T_y} - \Delta_y A_x^{T_x} = \Delta_x \Delta_y T_y - \Delta_y \Delta_x T_x.$$

Since

$$\Delta_x S = \Delta_y S = 1, \quad \Delta_x P = y, \quad \Delta_y P = x,$$

is obtained in the 81 plaquettes:

$$F(S, S) = F(P, P) = 0, \quad F(S, P) = +1, \quad F(P, S) = -1 \pmod{9}.$$

The connections associated with a single observable are flat; the mixed composition of the two observables produces curvature of opposite sign according to the order. For a coordinate rectangle $R_{a,b}$ contained in

the chart, we define the additive boundary Wilson observable as the oriented sum of the link variables. Cancellation of the interior edges gives

$$W_{\partial R_{a,b}}(S, P) = ab, \quad W_{\partial R_{a,b}}(P, S) = -ab \pmod{9},$$

which is the finite Stokes identity for this convention.

The distinction between potential and linkage is algebraically necessary. If A_x, A_y were directly substituted by S, P , terms depending on x, y would appear and the preceding constant values would not be obtained. The typed formulation is what produces the mixed cocycle used later in the finite Heisenberg closure.

Remark 2.11. The integrated difference 54 and the mixed curvature ± 1 are distinct invariants. The former compares global sums; the latter depends on the order of the two potentials and is localized on each plaquette. The formulation in terms of self-adjoint operators and commutators will be developed after the corresponding state space has been constructed.

2.2.6 Path Transition Operator Expansion

The definition of APP includes the space $\text{Path}(\Gamma_{\text{APP},9})$. Once a transition operator U has been fixed on the vertices of $\Gamma_{\text{APP},9}$, we assume that $U_{yx} = 0$ whenever $x \rightarrow y$ is not an edge of the graph. Under this support hypothesis, its powers admit the expansion

Proposition 2.12 (Finite Path Expansion). *For each $R \geq 1$ and each pair of vertices i, f ,*

$$(U^R)_{fi} = \sum_{\substack{\gamma: i \rightarrow f \\ |\gamma|=R}} \prod_{k=0}^{R-1} U_{x_{k+1}, x_k}, \quad \gamma = (x_0 = i, x_1, \dots, x_R = f).$$

Proof. For $R = 1$ the identity is the definition of the matrix elements of U . If it holds for R , then

$$(U^{R+1})_{fi} = \sum_j U_{fj} (U^R)_{ji}.$$

Substituting the induction hypothesis concatenates to every path of length R from i to j the final edge $j \rightarrow f$; thus, every path of length $R + 1$ appears exactly once. $\square$

Consequently, APP brings together in a single structure the finite arithmetic torus, the sum and product observables, the residue–quotient decomposition, the nilpotent radical, and an oriented mixed curvature. The representative 9 of the null class fixes a visible chart of the quotient; replacing it with 0 without simultaneously transporting the chart alters the registers used by the subsequent dynamics.

2.3 Arithmetic Matrix of APP and Local Ring Structure

The section 2.4 constructed a single support X_9 . APP has a single principal arithmetic matrix: nonadic multiplication published by the positive digital root. Addition does not constitute a second APP table; it is the internal accumulation law that generates the rows and columns of the multiplicative matrix and provides an auxiliary observable for comparison. This distinction will be maintained throughout the work.

2.3.1 Nonadic multiplicative matrix and digital root

For $a, b \in \mathcal{D}_9 = \{1, \dots, 9\}$, the APP entry is obtained directly through

$$\Pi(a, b) = \text{dr}_9(ab) = 1 + ((ab - 1) \bmod 9). \quad (2.3)$$

The only APP matrix is, therefore,

Π	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9
2	2	4	6	8	1	3	5	7	9
3	3	6	9	3	6	9	3	6	9
4	4	8	3	7	2	6	1	5	9
5	5	1	6	2	7	3	8	4	9
6	6	3	9	6	3	9	6	3	9
7	7	5	3	1	8	6	4	2	9
8	8	7	6	5	4	3	2	1	9
9	9	9	9	9	9	9	9	9	9

Each row can also be reconstructed by reiterated addition of its index, but that recurrence describes the internal law of the matrix and not a second sheet. The rows and columns of indices 1, 2, 4, 5, 7, 8 traverse the nine classes; those of indices 3, 6, 9 are exactly those that contract the support.

In the residual chart, the set of units is

$$R_9^\times = (\mathbb{Z}/9\mathbb{Z})^\times = \{[1], [2], [4], [5], [7], [8]\}.$$

The maximal ideal is

$$\mathfrak{m} = (3) = \{[0], [3], [6]\}.$$

Theorem 2.13 (Classification of multiplicative rows). *For $a \in \mathcal{D}_9$, the map*

$$m_a : \mathcal{D}_9 \longrightarrow \mathcal{D}_9, \quad b \longmapsto \Pi(a, b)$$

is bijective if and only if $[a]_9 \in R_9^\times$. If $[a]_9 \in \mathfrak{m} \setminus \{0\}$, its image has three elements. If $[a]_9 = [0]_9$, its image has a single element.

Proof. The residual reduction of m_a is multiplication by $[a]_9$. In a finite ring, that multiplication is bijective exactly when $[a]_9$ is a unit. If $[a]_9 = [3]$ or $[6]$, its image is the ideal $\mathfrak{m}$, of cardinality three. If $[a]_9 = [0]$, every entry is sent to the null class, published as 9. $\square$

Proposition 2.14 (Internality of the additive law). *The auxiliary observable*

$$\Sigma(a, b) = \text{dr}_9(a + b)$$

acts by translations: for each $a, c \in \mathcal{D}_9$ there exists a unique $b \in \mathcal{D}_9$ such that $\Sigma(a, b) = c$, and analogously in the other coordinate.

Proof. In the residual chart, $[a + b]_9 = [c]_9$ has the unique solution $[b]_9 = [c - a]_9$. The positive section transports that class to a unique element of $\mathcal{D}_9$. $\square$

The difference of geometry of fibers remains thus typed without duplicar the matrix APP: the law additive internal only translates, whereas the matrix multiplicative permutes on the units and contracts on the radical.

Proposition 2.15 (Linear structure of the radical). *The map*

$$M_3 : R_9 \longrightarrow R_9, \quad x \longmapsto 3x$$

satisfies

$$\ker M_3 = \operatorname{im} M_3 = \mathfrak{m}.$$

The induced map

$$R_9/\mathfrak{m} \longrightarrow \mathfrak{m}, \quad [x] \longmapsto 3x$$

is an isomorphism of vector spaces over $\mathbb{F}_3$, but not a ring isomorphism.

Proof. The congruence $3x \equiv 0 \pmod{9}$ is equivalent to $x \equiv 0 \pmod{3}$, whence the kernel $\mathfrak{m}$. The image is $\{[0], [3], [6]\} = \mathfrak{m}$. The first isomorphism theorem gives the linear isomorphism. It is not a ring isomorphism because $\mathfrak{m}^2 = 0$ in R_9 , whereas the quotient $R_9/\mathfrak{m} \cong \mathbb{F}_3$ has a nonzero unit. $\square$

The positive section publishes the non-trivial radical cycle as

$$369 \longmapsto 693 \longmapsto 936 \longmapsto 369,$$

according to the position from which the triple is read. This notation preserves the order of the coordinates; it does not turn the ideal into three independent integers.

2.3.2 Residue-quotient extensions and nonadic carry cocycle

Let $s_0 : R_9 \rightarrow \{0, \dots, 8\}$ be the normalized residual section. The binary carry of the sum is

$$\kappa_0(a, b) = \frac{s_0(a) + s_0(b) - s_0(a + b)}{9}.$$

Lemma 2.16 (Cocyclic Identity). *For $a, b, c \in R_9$,*

$$\kappa_0(a, b) + \kappa_0(a + b, c) = \kappa_0(b, c) + \kappa_0(a, b + c).$$

Proof. Multiply by nine and expand either of the two sides produces

$$s_0(a) + s_0(b) + s_0(c) - s_0(a + b + c).$$

Therefore, both are equal. $\square$

Define $\chi_0(a) = 1$ if $a = [0]_9$ and 0 otherwise. Since

$$s_+(a) = s_0(a) + 9\chi_0(a),$$

the carryovers of the two sections are related by

$$\kappa_+(a, b) = \kappa_0(a, b) + \chi_0(a) + \chi_0(b) - \chi_0(a + b).$$

Changing the representative modifies the cocycle by a coboundary. This is the exact form for transporting memory when the null class is published as 9 instead of 0.

2.3.3 Integrated censuses of residues and quotients

Summing over all pairs $(i, j) \in \mathcal{D}_9^2$, one obtains

$$\sum_{i,j} (i + j) = 810, \quad \sum_{i,j} ij = 2025.$$

The positive residues satisfy

$$\sum_{i,j} R_{i,j}^+ = 405, \quad \sum_{i,j} R_{i,j}^\times = 459.$$

By the raised identities,

$$\sum_{i,j} Q_{i,j}^+ = \frac{810 - 405}{9} = 45, \quad \sum_{i,j} Q_{i,j}^\times = \frac{2025 - 459}{9} = 174.$$

Theorem 2.17 (Exact separation of the sum-product excess). *The integrated difference between product and sum decomposes uniquely into a published part and a quotient part:*

$$2025 - 810 = (459 - 405) + 9(174 - 45) = 54 + 9 \cdot 129.$$

Proof. We subtract, term to term, the identities $ij = R_{i,j}^\times + 9Q_{i,j}^\times$ and $i + j = R_{i,j}^+ + 9Q_{i,j}^+$, and sum on the eighty and a pairs. The censuses preceding dan the equality. The uniqueness proceeds of the division Euclidean in each pair. $\square$

The value 54 is not the complete difference between two supposed matrices. It is the visible defect between the multiplicative matrix and the additive accumulator in the positive chart. The term $9 \cdot 129$ is the contribution that remains in the quotients. Omitting it would confuse publication with genealogy.

2.3.4 Bivariate central block of APP and decomposition into 2 spectral modes

For $k \in \mathbb{Z}/9\mathbb{Z}$, consideramos the block symmetric

$$B_k = \begin{pmatrix} k^2 & k(k+1) \\ k(k+1) & (k+1)^2 \end{pmatrix} \pmod{9}.$$

The diagonals coincide exactly when

$$k^2 \equiv (k+1)^2 \pmod{9} \iff 2k+1 \equiv 0 \pmod{9}.$$

Since $2^{-1} = 5 \pmod{9}$, the unique solution is $k = 4$. Thus appears the block

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}.$$

Let

$$R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad P_\pm = \frac{1}{2}(I \pm R).$$

Then

$$B_c = 7I + 2R = 9P_+ + 5P_-.$$

Theorem 2.18 (Uniqueness and spectrum of the central block). *The block B_c is the unique adjacent block in the family B_k with equal diagonals. Its eigenvalues on the symmetric and antisymmetric subspaces are, respectively, 9 and 5.*

Proof. The congruence before proves the uniqueness. Since that R acts by $+1$ on $\text{im } P_+$ and by -1 on $\text{im } P_-$, the matrix $7I + 2R$ acts by 9 and 5, respectively. $\square$

The two ordered readings of its digits give

$$72 + 27 = 99, \quad 72 - 27 = 45 = \det B_c,$$

and the two diagonals give

$$77 + 22 = 99, \quad 77 - 22 = 55 = 54 + 1.$$

These identities are internal controls of the block; they do not by themselves define the global defect.

2.3.5 Discrete curvature of the interaction between additive and multiplicative leaves

On the residual chart we write

$$S(x, y) = x + y, \quad P(x, y) = xy.$$

For a function $T : X_9 \rightarrow R_9$, let

$$\Delta_x T(x, y) = T(x + 1, y) - T(x, y), \quad \Delta_y T(x, y) = T(x, y + 1) - T(x, y).$$

We define the ordered curvature of a pair of laws by

$$\mathcal{F}(T_x, T_y) := \Delta_x \Delta_y T_y - \Delta_y \Delta_x T_x.$$

Proposition 2.19 (Orientations of the mixed composition). *The following hold*

$$\mathcal{F}(S, S) = 0, \quad \mathcal{F}(P, P) = 0,$$

and, for mixed compositions,

$$\mathcal{F}(S, P) = +1, \quad \mathcal{F}(P, S) = -1 \quad \text{in } R_9.$$

Proof. The mixed second differences of the linear law S vanish. For $P(x, y) = xy$, both pure mixed differences equal one and cancel when compared in the same order. Combining a linear law and a bilinear one leaves a single unit whose sign changes when the order of the sheets is reversed. $\square$

The curvature not creates another matrix APP. Is recorded that the order $\Sigma \rightarrow \Pi$ and the order $\Pi \rightarrow \Sigma$ are orientations opposite, whereas the sectors pure remain flat. This triad $-1, 0, +1$ will be typed in the unit TRIT.

2.3.6 Obstructions of the arithmetic matrix of APP: translations, local radical, central block, mixed curvature, and conservation of carry

Test criterion 2.20 (Refutation criteria for the APP matrix). The unit is refuted if:

1. the internal additive law ceases to act by translations;
2. a nonunitary multiplicative row is declared bijective;
3. $R_9/\mathfrak{m}$ is identified with $\mathfrak{m}$ as a ring;
4. the exeso 1215 it replaces by 54 without conservar quotients;
5. the central block is chosen without resolving $2k + 1 = 0 \pmod{9}$;

6. reversal of the mixed order does not change the sign of the curvature;
7. the carry it eliminates to the cambiar between the sections s_0 and s_+ .

APP is now constructed as support, category of trajectories, nonadic multiplicative matrix, internal additive law, local radical, carry cocycle, and mixed orientation. The following unit will not summarize these structures: it will take the triad of orientations that has just emerged and construct the complete TRIT state with its three quadratic regimes.

2.4 Gnomonic completion and arithmetic support of APP

This unit constructs the APP support from the nonadic cell of the [section 2.5](#). It does not presuppose a continuous plane, an infinite lattice, or a numerical constant. The starting object is finite: two ordered copies of the positive chart of $\mathbb{Z}/9\mathbb{Z}$. Completion of its interior interstices produces exactly eighty-one states, and the residual translation law turns that square support into a discrete torus.

Throughout the unit, APP denotes a mathematical structure and not an image. Henceforth, the acronym APP names this structure; the properties that follow depend on its objects and morphisms, not on a verbal expansion of the acronym.

2.4.1 Conjugation between the positive and residual charts of the same class modulo 9

We define

$$\mathcal{D}_9 := \{1, 2, \dots, 9\}, \quad \mathcal{D}_8 := \{1, 2, \dots, 8\}.$$

The residual publication application is

$$\lambda_9 : \mathcal{D}_9 \longrightarrow \mathbb{Z}/9\mathbb{Z}, \quad \lambda_9(a) = [a]_9.$$

Therefore, $\lambda_9(9) = [0]_9$. Its inverse as a set map is the positive section

$$s_+([a]_9) = \begin{cases} a, & [a]_9 \neq [0]_9, \quad 1 \leq a \leq 8, \\ 9, & [a]_9 = [0]_9. \end{cases}$$

The two expressions 9 and $[0]_9$ are not two phases: they are two representatives of the same element in different charts.

For two coordinates we will write

$$\lambda_9^{\times 2} : \mathcal{D}_9^2 \longrightarrow X_9 := (\mathbb{Z}/9\mathbb{Z})^2, \quad (a, b) \longmapsto ([a]_9, [b]_9).$$

This map is a bijection of sets. The chart $\mathcal{D}_9^2$ makes the positive marks visible; the group X_9 makes the residual sum and the translations visible.

Lemma 2.21 (State-preserving chart change). *The maps $\lambda_9^{\times 2}$ and $s_+^{\times 2}$ are inverses. In particular, the positive representation and the residual representation contain exactly the same eighty-one states.*

Proof. In each coordinate, $\lambda_9 \circ s_+ = \text{id}_{\mathbb{Z}/9\mathbb{Z}}$ and $s_+ \circ \lambda_9 = \text{id}_{\mathcal{D}_9}$ hold. The Cartesian product of these identities gives the two required identities. $\square$

2.4.2 Decomposition of the APP orthogram into inner core and gnomonic complement

The inner core is

$$\mathcal{N}_8 := \mathcal{D}_8 \times \mathcal{D}_8, \quad |\mathcal{N}_8| = 8^2 = 64.$$

In the plane of indices, $\mathcal{N}_8$ is the square that not uses the mark of closure in no coordinate. For publish also the null class it must completar the ninth row and the ninth column. We define the complement gnomonic disjoint by

$$\mathcal{G}_9 := (\{9\} \times \mathcal{D}_9) \sqcup (\mathcal{D}_8 \times \{9\}).$$

The corner $(9, 9)$ belongs to the first piece; for that reason the second uses $\mathcal{D}_8$ and not $\mathcal{D}_9$. Consequently,

$$|\mathcal{G}_9| = 9 + 8 = 17, \quad \mathcal{D}_9^2 = \mathcal{N}_8 \sqcup \mathcal{G}_9, \quad 81 = 64 + 17.$$

Proposition 2.22 (Minimality of the Completion). *Let Y be a subset of $\mathcal{D}_9^2$ that contains $\mathcal{N}_8$ and whose projection onto each coordinate is all of $\mathcal{D}_9$. If, in addition, Y contains all pairs required to translate one coordinate while the other remains arbitrary, then $\mathcal{G}_9 \subseteq Y$. Therefore, the addition of seventeen states is the minimal completion compatible with the two translations.*

Proof. To transport the first coordinate through the closure class while the second remains at an arbitrary value $b \in \mathcal{D}_9$, all pairs $(9, b)$ are necessary. Nine states are obtained. To transport the second coordinate through closure while the first assumes an arbitrary value $a \in \mathcal{D}_8$, the remaining eight pairs $(a, 9)$ are necessary. The corner $(9, 9)$ was already counted in the first family. In total, $9 + 8 = 17$ states are necessary, exactly those of $\mathcal{G}_9$. $\square$

The hypothesis of compatibility with the two translations is essential. It would suffice to add a single mark per coordinate for both projections to be surjective, but that set would not contain the boundary states needed to move one coordinate while holding the other fixed. APP requires the complete product, not two disconnected axes.

2.4.3 Exact realization of the interstitial interval via APP cells and boundary coordinates

Let P_9 be the ordered path with nine marks and eight interior edges

$$E(P_9) = \{e_1, \dots, e_8\}.$$

Closing the endpoints incorporates an edge e_9 and produces the cycle C_9 :

$$E(C_9) = E(P_9) \sqcup \{e_9\}.$$

The pairs of interior interstices form $E(P_9)^2$. The pairs of interstices of the complete cycle form $E(C_9)^2$. The difference is

$$E(C_9)^2 \setminus E(P_9)^2 = (\{e_9\} \times E(C_9)) \sqcup (E(P_9) \times \{e_9\}). \quad (2.4)$$

The decomposition is disjoint because the corner (e_9, e_9) is reserved for the first piece.

Theorem 2.23 (Equivalence of the mark model and the interstice model). *The map*

$$\beta : E(C_9)^2 \longrightarrow \mathcal{D}_9^2, \quad \beta(e_i, e_j) = (i, j),$$

is bijective. Its restrictions satisfy

$$\beta(E(P_9)^2) = \mathcal{N}_8, \quad \beta(E(C_9)^2 \setminus E(P_9)^2) = \mathcal{G}_9.$$

Therefore, the identity $81 = 64 + 17$ simultaneously counts pairs of published marks and pairs of closed interstices.

Proof. Each edge of C_9 has a unique index in $\mathcal{D}_9$. The map $(i, j) \mapsto (e_i, e_j)$ is the explicit inverse of β . If $i, j \leq 8$, the pair belongs to $E(P_9)^2$ and its image belongs to $\mathcal{D}_8^2$. If at least one of the indices is nine, the pair belongs to the complement of equation (2.4); separating the case $i = 9$ from the case $i \leq 8, j = 9$ produces exactly the two pieces of $\mathcal{G}_9$. $\square$

2.4.4 Toroidal closure and translations

Let

$$\mathbf{e}_1 = ([1]_9, [0]_9), \quad \mathbf{e}_2 = ([0]_9, [1]_9).$$

The set of elementary directions is

$$\mathcal{D}_{\text{APP}} := \{+\mathbf{e}_1, -\mathbf{e}_1, +\mathbf{e}_2, -\mathbf{e}_2\}.$$

We define the oriented Cayley graph.

$$\Gamma_{\text{APP}} := \text{Cay}(X_9, \mathcal{D}_{\text{APP}}).$$

Its vertices are the eighty-one elements of X_9 . For every $x \in X_9$ and $d \in \mathcal{D}_{\text{APP}}$, there is an oriented edge $x \xrightarrow{d} x + d$. The inverse edge bears the label $-d$.

In the positive chart, unit translation is written by means of the published successor and its carry:

$$u_+(a) = \begin{cases} a + 1, & a < 9, \\ 1, & a = 9, \end{cases} \quad \kappa_+(a) = \begin{cases} 0, & a < 9, \\ 1, & a = 9. \end{cases}$$

The raised identity is

$$a + 1 = u_+(a) + 9\kappa_+(a). \quad (2.5)$$

Upon applying λ_9 , the carry term vanishes, leaving

$$\lambda_9(u_+(a)) = \lambda_9(a) + [1]_9.$$

Lemma 2.24 (Compatibility between closure, residue, and elevation). *The addition of e_9 , the identification $9 \sim [0]_9$, and the translation $x \mapsto x + \mathbf{e}_i$ describe three compatible operations in distinct domains. The residual quotient publishes the return; the carry of equation (2.5) preserves the elevation that the quotient omits.*

Proof. Topological closure incorporates the edge joining the endpoints of the path. The residual chart identifies the published endpoint with the null class. If $a < 9$, the successor has zero carry; if $a = 9$, the lifted equality is $10 = 1 + 9$. Both cases descend to the same residual sum, but the value of κ_+ distinguishes them. $\square$

Corollary 2.25 (Toroidal support). *Identifying opposite sides of the chart $\mathcal{D}_9^2$ yields a geometric realization of the group X_9 . Each elementary translation is defined at every vertex and has an inverse.*

2.4.5 APP Path Category: Objects, Prolongation Morphisms and Compatible Composition

Definition 2.26 (Oriented trajectory). A trajectory of length $n \geq 0$ is a pair

$$\gamma = (x_0; d_1, \dots, d_n), \quad x_0 \in X_9, \quad d_k \in \mathcal{D}_{\text{APP}}.$$

Its vertices are

$$x_k = x_0 + \sum_{r=1}^k d_r, \quad 0 \leq k \leq n.$$

The source is $s(\gamma) = x_0$ and the target is $t(\gamma) = x_n$. For $n = 0$ the identity trajectory in x_0 is obtained.

If $t(\gamma) = s(\delta)$, its concatenation is

$$\delta \circ \gamma = (x_0; d_1, \dots, d_n, e_1, \dots, e_m),$$

where $e_1, \dots, e_m$ are the directions of δ . Concatenation is associative, and paths of zero length are identities. We thus obtain the free category

$$\text{Path}(\Gamma_{\text{APP}}).$$

Proposition 2.27 (Finiteness of Support and Countability of Trajectories). *The APP state set is finite, of cardinality 81. For a fixed source, there are 4^n direction words of length n . The union of all finite trajectories is countable.*

Proof. The first assertion follows from $|X_9| = 9^2$. At each of the n steps there are four independent choices, whence 4^n . Finally, $\bigcup_{n \geq 0} X_9 \times \mathcal{D}_{\text{APP}}^n$ is a countable union of finite sets. $\square$

This proposition separates two levels that will not be identified: APP has a finite support, but admits trajectories of arbitrary length. The prolongation of a route does not create new vertices of the support; it creates new genealogy over them.

2.4.6 Multiplicative matrix and internal additive law on a single support

For $n \in \mathbb{Z}_{>0}$, the root digital positive is

$$\text{dr}_9(n) := 1 + ((n - 1) \bmod 9).$$

On the positive chart we define the principal matrix and the internal law.

$$\Sigma(a, b) := \text{dr}_9(a + b), \quad \Pi(a, b) := \text{dr}_9(ab).$$

Both maps have the same domain $\mathcal{D}_9^2$ and the same codomain $\mathcal{D}_9$, but they do not have the same status: Π publishes the unique APP matrix, and Σ generates its accumulations. Their residual versions are

$$\bar{\Sigma}([a]_9, [b]_9) = [a + b]_9, \quad \bar{\Pi}([a]_9, [b]_9) = [ab]_9.$$

The change of chart is compatible with both operations:

$$\begin{aligned} \lambda_9 \circ \Sigma &= \bar{\Sigma} \circ \lambda_9^{\times 2}, \\ \lambda_9 \circ \Pi &= \bar{\Pi} \circ \lambda_9^{\times 2}. \end{aligned}$$

To preserve the integer quotient, lifts are introduced.

$$a + b = R_{a,b}^+ + 9Q_{a,b}^+, \quad 1 \leq R_{a,b}^+ \leq 9, \quad (2.6)$$

$$ab = R_{a,b}^\times + 9Q_{a,b}^\times, \quad 1 \leq R_{a,b}^\times \leq 9. \quad (2.7)$$

Here

$$R_{a,b}^+ = \Sigma(a, b), \quad R_{a,b}^\times = \Pi(a, b),$$

and the quotients are uniquely determined by

$$Q_{a,b}^+ = \frac{a + b - R_{a,b}^+}{9}, \quad Q_{a,b}^\times = \frac{ab - R_{a,b}^\times}{9}.$$

Definition 2.28 (APP genealogical). The APP structure at this stage is the tuple

$$\text{APP} := (X_9, \Gamma_{\text{APP}}, \text{Path}(\Gamma_{\text{APP}}), \Sigma, \Pi, R^+, Q^+, R^\times, Q^\times).$$

The genealogy of a trajectory preserves, for each vertex visited, the position, incoming direction, multiplicative entry, additive accumulation, and their respective residues and quotients.

Theorem 2.29 (Causal construction of the APP support). *The chain*

$$\mathcal{D}_9 \longrightarrow \mathcal{D}_8^2 \longrightarrow \mathcal{D}_8^2 \sqcup \mathcal{G}_9 \longrightarrow \mathcal{D}_9^2 \xrightarrow{\lambda_9^{\times 2}} X_9 \longrightarrow \Gamma_{\text{APP}} \longrightarrow \text{Path}(\Gamma_{\text{APP}}) \longrightarrow (\Sigma, \Pi)$$

constructs, without loss of cardinality, the support, motion, paths, APP multiplicative matrix and its internal additive law.

Proof. The [proposition 2.22](#) constructs the complete chart from the kernel. The [theorem 2.23](#) identifies its interstitial model. The [lemma 2.21](#) transports the eighty-one states to X_9 . The Cayley graph incorporates the four reversible translations without changing the vertices. The free category incorporates all finite concatenations. Finally, Σ and Π are distinct total maps on the same domain; the equations the corresponding identity preserve the part that residual reduction does not publish. $\square$

2.4.7 Decomposition of the APP orthogram into kernel 8×8 , gnomon of 17 states, and chart 9×9 with opposite sides identified

The [Figure 2.1](#) must be read in three layers: the inner square of sixty-four states, the gnomon of seventeen states, and the identification of opposite sides. The nine-by-nine chart thereby completes the eight-by-eight core by means of the closure gnomon. The multiplicative matrix and the internal additive law are evaluated at every vertex of the same state, preserve distinct residue–quotient lifts, and do not constitute two APP squares.

2.4.8 Obstructions to the fidelity of the APP support: cardinals 81/17, reversible translations, and separation of the sum and product sheets

Test criterion 2.30 (Refutation criteria for the APP support). The construction is refuted if any of the following facts occurs:

1. the complete support has cardinality other than 81;
2. the gnomonic complement does not have cardinal 17;
3. a state of X_9 lacks any of the four translations;
4. it identify Σ and Π because coincide in a value;
5. Q^+ or $Q^\times$ is eliminated upon returning through a boundary;
6. trajectories of different lengths are presented as new support vertices;

7. it interpreta 9 and $[0]_9$ as phases independent.

The unit has constructed APP up to its complete genealogical evaluation. It has not yet oriented the difference between the matrix and its accumulator, has not introduced the TRIT, and has not defined the Topological Phase Kernel. The following unit will study the algebraic structure of rows and columns, the sector of units, and the carry cocycle that prepares that orientation.

2.5 Construction of the nonadic cell from marked segments and interstices

The construction begins before every constant and before every choice of physical unit. The inaugural datum is not an already constituted real number, but an ordered segment, a family of marks, and the adjacency relation between consecutive marks. This separation is necessary because a mark, an interval, the length of that interval, and the numeral that publishes that length are objects of different types.

2.5.1 Decimal residual projection of the finite positional chart of APP

Definition 2.31 (Marked segment of order ten). Let

$$M_{10} := \{m_0, m_1, \dots, m_{10}\}$$

a set totally ordered by $m_0 < m_1 < \dots < m_{10}$. Its adjacency graph is the path

$$P_{10} : m_0 \longrightarrow m_1 \longrightarrow \dots \longrightarrow m_{10}.$$

We shall denote by

$$l_k := (m_k, m_{k+1}), \quad 0 \leq k \leq 9,$$

the abstract interstice between two consecutive marks. The family of interstices is $E_{10} := \{l_0, \dots, l_9\}$.

The open notation (m_k, m_{k+1}) does not yet presuppose the real line. It indicates only the ordered pair of endpoints and the edge joining them. When the Archimedean chart is chosen

$$\chi_{10}(m_k) = k,$$

the abstract interstice l_k is published by the semiopen interval $[k, k + 1)$. Therefore, the index k , the edge l_k , and the interval $[k, k + 1)$ will not be identified in the text: the first is a label, the second is a combinatorial element, and the third is its realization in a chart.

Lemma 2.32 (Census of Marks and Gaps). *The segment P_{10} contains eleven marks and ten interstices. More generally, a path with $n + 1$ marks has exactly n edges.*

Proof. The map

$$\{0, \dots, n - 1\} \longrightarrow E(P_n), \quad k \longmapsto (m_k, m_{k+1}),$$

is bijective. The particular assertion is obtained with $n = 10$. □

The endpoint m_0 fixes the origin of the chart, and m_{10} fixes its threshold. Neither by itself constitutes an additional interstice. This elementary remark prevents three subsequent confusions: counting endpoints as cells, confusing closure with a new phase state, and treating zero as a tenth phase independent of the nonadic cycle.

2.5.2 Construction of the nonadic cell of 9 phases via residual projection

The decimal chart distinguishes the anchor $[0, 1)$ of the periodic cell. The nine positive marks

$$D_9 := \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

it utilizaran as representantes publicos of $\mathbb{Z}/9\mathbb{Z}$, haciendo corresponder the mark 9 with the class residual $\bar{0}$. We define

$$\lambda_9 : D_9 \longrightarrow \mathbb{Z}/9\mathbb{Z}, \quad \lambda_9(a) = \bar{a}.$$

Proposition 2.33 (Positive chart of the nonadic classes). *The map λ_9 is bijective. Its inverse is the positive section*

$$s_+(\bar{a}) = \begin{cases} a, & 1 \leq a \leq 8, \\ 9, & \bar{a} = \bar{0}. \end{cases}$$

In particular, the public representation 9 and the residual representation 0 represent the same class, but belong to different charts.

Proof. The nine classes of $\mathbb{Z}/9\mathbb{Z}$ have unique representatives in $\{0, 1, \dots, 8\}$. Replacing the representative 0 by 9 produces another complete transversal, so λ_9 is bijective and s_+ is its inverse. $\square$

The phase cell is not obtained by arbitrarily erasing the origin. The segment $[0, 1)$ fixes the decimal chart from which the cycle is observed; the nine phases are the nine classes published as $1, \dots, 9$. In the lifted chart, the step that closes the cell is represented by $[9, 10)$; in the residual chart, that same step returns from $\bar{0}$ to $\bar{1}$. Closure and change of representative are not the same operation.

2.5.3 Position and height coordinates induced by the Euclidean division

Definition 2.34 (Nonadic Coordinates). For each integer $n \geq 1$, we define

$$\rho_9(n) := 1 + ((n - 1) \bmod 9), \quad q_9(n) := \left\lfloor \frac{n - 1}{9} \right\rfloor.$$

The first coordinate is called phase position; the second, turn height or memory quotient.

Theorem 2.35 (Unique nonadic decomposition). *Every integer $n \geq 1$ admits a unique expression*

$$n = 9q + \rho, \quad q \in \mathbb{Z}_{\geq 0}, \quad \rho \in D_9.$$

In esa escritura $q = q_9(n)$ and $\rho = \rho_9(n)$.

Proof. We apply Euclidean division to $n - 1$: there exist unique $q \geq 0$ and $r \in \{0, \dots, 8\}$ such that $n - 1 = 9q + r$. Taking $\rho = r + 1$ gives $n = 9q + \rho$, with $1 \leq \rho \leq 9$. The uniqueness of q, r implies that of q, ρ . $\square$

Corollary 2.36 (Phase return with memory increment). *For every $n \geq 1$,*

$$\rho_9(n + 9) = \rho_9(n), \quad q_9(n + 9) = q_9(n) + 1.$$

Consequently, a full rotation preserves the visible phase and changes the raised state.

Proof. If $n = 9q + \rho$, then $n + 9 = 9(q + 1) + \rho$; uniqueness of theorem 2.35 gives the two identities. $\square$

The one-step evolution is written unambiguously as

$$(q, \rho) \mapsto \begin{cases} (q, \rho + 1), & 1 \leq \rho < 9, \\ (q + 1, 1), & \rho = 9. \end{cases}$$

The second line is not a restart. It recovers the initial phase label and increases the quotient that records how many turns have been performed.

2.5.4 Circular projection periodicity and injectivity of the nonadic helical lifting

Let

$$\vartheta_9(\rho) = \frac{2\pi(\rho - 1)}{9}.$$

The phase projection onto the unit circle is

$$\mathcal{R}_9(n) = (\cos \vartheta_9(\rho_9(n)), \sin \vartheta_9(\rho_9(n))).$$

To simultaneously preserve position and height we introduce the helical immersion.

$$\mathcal{H}_9(n) = \left(\cos \vartheta_9(\rho_9(n)), \sin \vartheta_9(\rho_9(n)), q_9(n) + \frac{\rho_9(n) - 1}{9} \right).$$

Proposition 2.37 (The circular projection does not determine the state). *The map $\mathcal{R}_9$ is periodic with period nine. The map $\mathcal{H}_9$ is injective. In particular,*

$$\mathcal{R}_9(n + 9) = \mathcal{R}_9(n), \quad \mathcal{H}_9(n + 9) \neq \mathcal{H}_9(n).$$

Proof. The first equality follows from [corollary 2.36](#). The third coordinate of $\mathcal{H}_9(n + 9)$ exceeds that of $\mathcal{H}_9(n)$ by one unit, hence both points are distinct. Finally, n is recovered from the third coordinate by multiplying by nine and adding one to the phase part; therefore, $\mathcal{H}_9$ is injective. $\square$

This proposition is the minimal model of the principle that will run through the entire work:

$\text{phase return} \neq \text{state return}.$

The nonadic connection of the Topological Phase Kernel will preserve more fields than the elementary helix —sheet, orientation, carry, boundary, terminal, and route memory—, but it will not alter this inaugural distinction.

2.5.5 Correspondence between Discrete Cardinality and Projective Measure

Definition 2.38 (Counting). The count of a finite set X is its cardinality $|X|$. It requires neither an orientation, a scale, nor an additional reader.

Definition 2.39 (Measured data). A discrete measurement datum is a triple

$$(x, \mathcal{L}, C),$$

where x is a state, $\mathcal{L}$ is a typed reader, and C is a publication chart compatible with the codomain of $\mathcal{L}$. Its visible output is

$$C(\mathcal{L}(x)).$$

The same set may have a single cardinality and admit multiple measures, because the local cochain, the orientation of integration, or the publication chart may vary. Conversely, two distinct states may publish the same coordinate. The work will therefore always preserve two levels:

state with genealogy $\longrightarrow$ published coordinate.

Test criterion 2.40 (Refutation criteria for the inaugural unit). The construction is refuted if any of the following facts occurs:

1. ten independent phases are counted in the residue cell;
2. the mark 9 is identified with the integer 0 without declaring the change of chart;
3. the passage $9 \rightarrow 1$ forces $q \mapsto 0$;
4. two distinct integers have the same pair (q_9, ρ_9) ;
5. a physical unit is attributed to the cell before defining the dimensional transducer.

2.5.6 Autonomous arithmetic domain of the nonadic cell and causal precedence with respect to APP–TRIT–TPK and the constants analíticas

The unit constructed so far contains only:

$$P_{10}, \quad D_9, \quad \lambda_9, \quad (q_9, \rho_9), \quad \mathcal{R}_9, \quad \mathcal{H}_9.$$

It does not yet contain APP, TRIT, the Topological Phase Kernel, or an analytic constant. The following unit will take two copies of the nonadic cell, construct its gnomonic completion, and prove how the bidimensional support appears over which the multiplicative matrix of APP and its internal additive law act.

2.6 Cyclotomic skeleton of APP and binomial completion of the unit

The defect between the APP matrix and its internal accumulator is not an isolated number. Its law in Cartesian length, its decomposition into orbits of order three, and the completion of the nonadic cell produce three different objects that must remain typed:

$$\text{scalar defect } D_d, \quad \text{cyclotomic module } W_{\text{APP}}, \quad \text{completion } \overline{C}_9.$$

The first counts; the second preserves orientation and rank; the third adds boundary strata. Equalities between their cardinals do not authorize identifying their elements.

2.6.1 Cyclotomic Defect Invariance at Fixed Modulus Under Length Variation

For each integer $d \geq 1$, let

$$B_d := \{1, \dots, 9\}^d.$$

The nonary digital root is represented by

$$\text{dr}_9(n) = \begin{cases} n \bmod 9, & n \not\equiv 0 \pmod{9}, \\ 9, & n \equiv 0 \pmod{9}. \end{cases}$$

We define the additive accumulation and the multiplicative matrix evaluation on the same domain:

$$A_d(x_1, \dots, x_d) = \text{dr}_9\left(\sum_{j=1}^d x_j\right),$$

$$M_d(x_1, \dots, x_d) = \text{dr}_9\left(\prod_{j=1}^d x_j\right).$$

Their visible masses and their defect are

$$S_{\Sigma}(d) := \sum_{x \in B_d} A_d(x), \quad S_{\Pi}(d) := \sum_{x \in B_d} M_d(x),$$

$$D_d := S_{\Pi}(d) - S_{\Sigma}(d).$$

Here d counts Cartesian positions. The module 9 does not vary, and no physical dimension has yet been introduced.

Lemma 2.41 (Mass of the additive accumulator). *For every $d \geq 1$,*

$$S_{\Sigma}(d) = 45 \cdot 9^{d-1}.$$

Proof. Once the first $d-1$ coordinates and a published residue $r \in \{1, \dots, 9\}$ have been fixed, there exists a unique last coordinate that makes the sum have residue r . Each visible value thus has 9^{d-1} preimages. Summing the weights $1 + \dots + 9 = 45$ gives the formula. $\square$

Lemma 2.42 (Census of the multiplicative matrix). *For each of the six unit residues there are 6^{d-1} preimages for each fixed product value. The residues 3 and 6 have $d \cdot 6^{d-1}$ preimages each. The zero residue, published as 9, has*

$$9^d - 6^d - 2d \cdot 6^{d-1}$$

preimages. Consequently,

$$S_{\Pi}(d) = 9^{d+1} - 9(d+3)6^{d-1}.$$

Proof. The unit group of $\mathbb{Z}/9\mathbb{Z}$ has six elements. Upon fixing $d-1$ units, the last is determined by the prescribed unit product, giving 6^{d-1} solutions for each of the six classes.

To obtain a product of ternary valuation exactly one, one chooses which of the d coordinates belongs to one of the two classes 3, 6, while the others are units. Once the product 3 or 6 is fixed, the last unit is determined; hence $d \cdot 6^{d-1}$ for each class. All remaining tuples have product zero modulo 9. Subtracting the two preceding counts from 9^d gives their multiplicity. The sum weighted by the visible representatives yields the final identity. $\square$

Theorem 2.43 (Internal dimensional law of APP). *For every $d \geq 1$,*

$$D_d = 4 \cdot 9^d - 9(d+3)6^{d-1}.$$

In particular,

$$D_1 = 0, \quad D_2 = 54,$$

and

$$\sum_{d \geq 1} D_d z^d = \frac{54z^2(1-3z)}{(1-9z)(1-6z)^2}.$$

Moreover,

$$\frac{D_d}{9^d} = 4 - (d+3) \left(\frac{2}{3}\right)^{d-1} \longrightarrow 4.$$

Proof. The formulas from the two lemmas are subtracted. For the generating function, one uses

$$\sum_{d \geq 1} 9^d z^d = \frac{9z}{1-9z}, \quad \sum_{d \geq 1} (d+3)6^{d-1} z^d = \frac{z(4-18z)}{(1-6z)^2},$$

and the common numerator is reduced. The normalized formula follows by dividing by 9^d ; the exponential term multiplied by $d + 3$ converges to zero. $\square$

The equality $D_2 = 54$ reproduce the census $459 - 405$, but the formula before and the law of modulo variable $\Delta_{3m}^{\text{pl}} = m3^{2m-1}$ are enunciados distintos: the first varia d manteniendo the modulo 9; the second varia the modulo.

2.6.2 Cyclotomic operator of APP and action on the 12 A_2 -type fibers

Let

$$G_{A_2} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}.$$

Proposition 2.44 (Coxeter of type A_2). *They are verified.*

$$C^3 = I_2, \quad C^T G_{A_2} C = G_{A_2}, \quad \det(XI_2 - C) = X^2 + X + 1.$$

Therefore C preserves the form A_2 , has order exactly three, and has no nonzero fixed vector.

Proof. The multiplication gives

$$C^2 = \begin{pmatrix} -1 & 1 \\ -1 & 0 \end{pmatrix}, \quad C^3 = I_2.$$

Direct substitution into $C^T G_{A_2} C$ yields G_{A_2} . Finally, 1 is not a root of $X^2 + X + 1$, hence $\ker(C - I_2) = 0$. $\square$

2.6.3 Decomposition of APP into 12 fibers of type A_2

Consider the action

$$a : \mathbb{Z}/9\mathbb{Z} \longrightarrow \mathbb{Z}/9\mathbb{Z}, \quad a(r) = 4r.$$

Since $4^3 \equiv 1 \pmod{9}$, its restriction to the units has two free orbits:

$$O_1 = (1, 4, 7), \quad O_2 = (2, 8, 5).$$

For an orbit $O = (r, 4r, 7r)$, we define the augmentation module

$$A(O) := \ker \left(\mathbb{Z}[O] \xrightarrow{\Sigma} \mathbb{Z} \right).$$

At the base

$$b_1 = e_r - e_{4r}, \quad b_2 = e_{4r} - e_{7r},$$

the restricted Euclidean form has Gram matrix G_{A_2} , and the cycle $r \mapsto 4r \mapsto 7r \mapsto r$ acts by C .

For $x = (x_1, x_2, x_3) \in (\mathbb{Z}/9\mathbb{Z})^3$, the three unordered pairs are

$$\{1, 2\}, \quad \{2, 3\}, \quad \{3, 1\}.$$

Over each pair, two typed laws are evaluated:

$$\ell_{ij}^{\Sigma}(x) = x_i + x_j, \quad \ell_{ij}^{\Pi}(x) = x_i x_j.$$

Since

$$\ell_{ij}^{\Sigma}(ax) = a \ell_{ij}^{\Sigma}(x), \quad \ell_{ij}^{\Pi}(ax) = a^{-1} \ell_{ij}^{\Pi}(x),$$

the sum carries the orientation C and the product the inverse orientation C^{-1} .

Definition 2.45 (Cyclotomic module of APP). It is defined

$$W_{\text{APP}} := \bigoplus_{\{i,j\}} \bigoplus_{\mu \in \{\Sigma, \Pi\}} \bigoplus_{O \in \{O_1, O_2\}} A(O).$$

There are $3 \cdot 2 \cdot 2 = 12$ summands, each of rank two; therefore

$$W_{\text{APP}} \cong A_2^{12}, \quad \text{rank } W_{\text{APP}} = 24.$$

The par-law-orbit labels are part of the object.

For $u \in \mathbb{Z}/9\mathbb{Z}$, we write

$$\beta_O(u) = \begin{cases} (1, 0), & u = r, \\ (0, 1), & u = 4r, \\ (-1, -1), & u = 7r, \\ (0, 0), & u \notin O, \end{cases}$$

in the basis (b_1, b_2) . The state map is

$$\Psi(x)_{ij, \mu, O} := \beta_O(\ell_{ij}^\mu(x)).$$

Theorem 2.46 (Equivariance and rank of the state skeleton). *Let $\rho(a)$ be the action using C on the six additive fibers and C^{-1} on the six multiplicative fibers. Then*

$$\Psi(ax) = \rho(a)\Psi(x)$$

for the 729 states of $(\mathbb{Z}/9\mathbb{Z})^3$. The rational submodule generated by the images has rank 24.

Proof. Under the additive law, multiplying all coordinates by 4 advances one position in each orbit; under the multiplicative law, it multiplies the value by $4^2 \equiv 7 \equiv 4^{-1} \pmod{9}$ and therefore reverses the cycle. This proves componentwise equivariance.

In order for the rank not to remain hidden in a computational statement, we fix the coordinate order.

$$(12, \Sigma, O_1), (12, \Sigma, O_2), (12, \Pi, O_1), (12, \Pi, O_2),$$

followed by the same order for the pairs 23 and 31, and, within each fiber, the two coordinates (b_1, b_2) . Consider the states

$$\begin{aligned} &001, 002, 004, 005, 010, 011, 012, 013, 014, 015, 016, 020, \\ &022, 023, 040, 050, 101, 102, 104, 105, 110, 120, 140, 150. \end{aligned}$$

The integer matrix M_Ψ whose rows are its twenty-four images has entries in $(-1, 0, 1)$. Bareiss's fraction-free elimination yields the following successive leading principal minors

$$1, 1, 1, -1, -1, -1, -1, 1, -1, 1, 1, 1, 1, -2, -1, 1, 1, 1, 2, 1, 4, -6, 12, 9.$$

In particular,

$$\det M_\Psi = 9 \neq 0.$$

These twenty-four images are linearly independent. Since the codomain already has rank 24, they generate $W_{\text{APP}} \otimes \mathbb{Q}$. $\square$

The proof exhibits a finite reproducible certificate; it does not deduce full rank from the mere surjectivity of each component projection.

2.6.4 Derivation of the sum-product defect $459-405=54$ in the central block of APP

Let

$$\mathcal{M} := \mathbb{Z}[\mathbb{Z}/9\mathbb{Z}]$$

and, for $\mu \in \{\Sigma, \Pi\}$,

$$\nu_\mu := \sum_{x,y \in \mathbb{Z}/9\mathbb{Z}} e_{\ell^\mu(x,y)}.$$

The census of the two tables gives

$$\delta_{\Pi\Sigma} := \nu_\Pi - \nu_\Sigma = 12e_0 + 3e_3 + 3e_6 - 3 \sum_{u \in (\mathbb{Z}/9\mathbb{Z})^\times} e_u.$$

Its coefficients are constant on the orbits of a ; therefore

$$(I - a)\delta_{\Pi\Sigma} = 0.$$

We define the increase projection

$$T : \mathcal{M} \longrightarrow A(\mathcal{O}_1) \oplus A(\mathcal{O}_2),$$

$$T(\nu) = ((I - a)\nu|_{\mathcal{O}_1}, ((I - a)\nu|_{\mathcal{O}_2}).$$

Then

$$T(\delta_{\Pi\Sigma}) = 0.$$

However, the visible reader

$$w(e_0) = 9, \quad w(e_r) = r \quad (1 \leq r \leq 8)$$

satisfies

$$w(\delta_{\Pi\Sigma}) = 54.$$

Corollary 2.47 (Isotopic Obstruction). *The scalar 54 belongs to the C_3 -invariant sector and cannot be recovered by means of a linear functional that factors exclusively through the augmentation modules. In particular,*

$$\sum_x \Psi(x) = 0$$

and, for every linear functional L on $W_{\text{APP}} \otimes \mathbb{Q}$,

$$L\left(\sum_x \Psi(x)\right) = 0 \neq 54.$$

This obstruction prevents presenting 54 as a retrospective linear sum of the twelve fibers. The subsequent APP–Fock coupling preserves the orientation and uses an additional object; it does not contradict the corollary.

2.6.5 Cubic completion of the nonad cell

Let E_9 be the underlying set of $\mathbb{Z}/9\mathbb{Z}$, and let $\star \notin E_9$ be a boundary state. We define

$$C_9 := E_9^3, \quad \overline{C}_9 := (E_9 \sqcup \{\star\})^3.$$

The symbol $\star$ is not the zero class: it indicates that a coordinate has reached the closure stratum.

Theorem 2.48 (Binomial stratification of the unit cube). *The stratum with exactly r coordinates equal to $\star$ contains*

$$\binom{3}{r} 9^{3-r}$$

elements. Therefore,

$$1000 = 729 + 243 + 27 + 1 = 729 + 271.$$

Upon removing only the apex $(\star, \star, \star)$, there remains

$$999 = 729 + 243 + 27 = 729 + 270.$$

Proof. The r boundary coordinates are chosen in $\binom{3}{r}$ ways; each remaining coordinate admits nine values. The four strata are disjoint and exhaust the Cartesian product. The sum is the expansion of $(9 + 1)^3$. $\square$

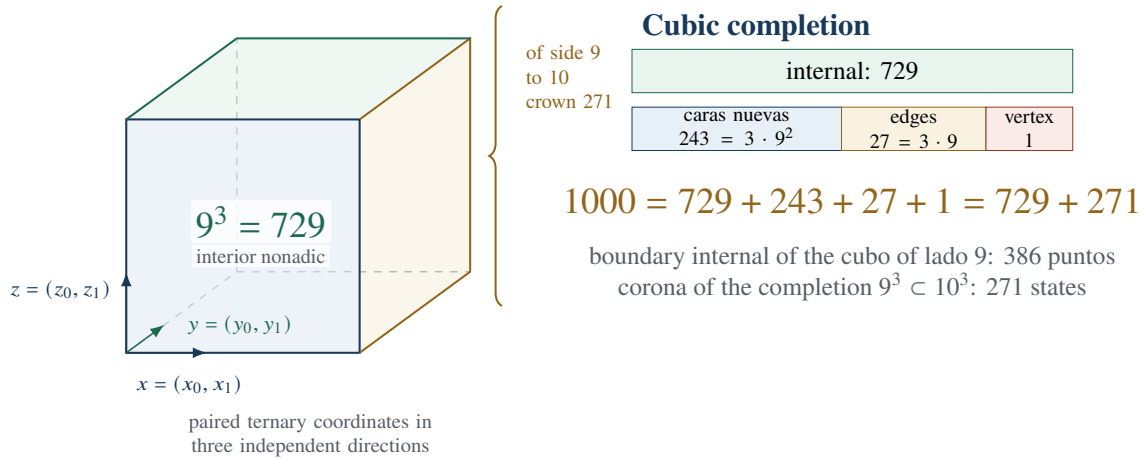


Figure 2.2: Cubic completion of the nonadic cell. The interior of cardinality 729 and the boundary strata of cardinalities 243, 27, and 1 are disjoint parts of a single object.

This stratification receives in HMT the name of extreme and mean holographic ratio. Mathematically it is the typed binomial completion of the nonadic cell; the name does not replace its definition.

For every $d \geq 1$, let

$$\overline{C}_9^{(d)} = (E_9 \sqcup \{\star\})^d.$$

Asignemos peso 1/10 to each symbol of a coordinate and tomemos the measure product μ_d .

Theorem 2.49 (Projective compatibility of the measure). *The projection that forgets the last coordinate satisfies*

$$(\pi_d)_* \mu_d = \mu_{d-1}.$$

Moreover,

$$10^d = \sum_{r=0}^d \binom{d}{r} 9^{d-r}, \quad \mu_d(\overline{C}_9^{(d)}) = 1.$$

The number of states increases with d , while the normalized total mass is conserved.

Proof. For every cylinder $A \times (E_9 \sqcup \{\star\})$,

$$\mu_d(A \times (E_9 \sqcup \{\star\})) = \mu_{d-1}(A) \sum_{u \in E_9 \sqcup \{\star\}} \frac{1}{10} = \mu_{d-1}(A).$$

The cilindros generate the algebra finite of subconjuntos, what that determines the measure image. The identity cardinal again becomes the binomio of Newton. $\square$

2.6.6 Morphism from the central block of APP to its Euclidean realization

The already constructed central block is

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}.$$

Row reading and diagonal reading give

$$72 + 27 = 77 + 22 = 99,$$

$$72 - 27 = 45 = \det B_c, \quad 77 - 22 = 55 = 54 + 1.$$

The relation with the completion is not obtained by concatenating symbols, but by the Euclidean division.

$$729 = 10 \cdot 72 + 9, \quad 271 = 10 \cdot 27 + 1.$$

Theorem 2.50 (Uniqueness of the Decimal Lift). *For $L_d(n) = 10n + d$, with $d \in \{0, \dots, 9\}$, there exists a single pair (d, d') such that*

$$L_d(72) = 729, \quad L_d(72) + L_{d'}(27) = 1000, \quad d + d' = 10.$$

It is the pair

$$(d, d') = (9, 1).$$

Proof. The first equality is $720 + d = 729$, hence $d = 9$. The second becomes $729 + 270 + d' = 1000$, whence $d' = 1$. The third is verified, and the uniqueness of quotient and remainder rules out any other solution. $\square$

Thus the causal diagram is obtained.

$$B_c \longrightarrow (72, 27) \longleftarrow ((72, 9), (27, 1)) \longleftarrow (729, 271).$$

The block and its reading belong to APP; the dividends and remainders belong to the completion. The diagram shares quotients; it does not invert causality.

2.6.7 Incidence of the cube and separation with respect to sextets and octets

Let M_d be the incidence matrix of the $2d$ codimension-one faces of a hypercube on its 2^d vertices. Each face contains 2^{d-1} vertices; two opposite faces are disjoint, and two transversal faces share 2^{d-2} vertices.

Theorem 2.51 (Hypercube incidence spectrum).

$$\text{Spec}(M_d M_d^T) = \{d 2^{d-1} \binom{1}{1}, 2^{d-1} \binom{d}{d}, 0^{(d-1)}\}.$$

Para $d = 3$,

$$\text{Spec}(M_3 M_3^T) = \{12, 4, 4, 4, 0, 0\}.$$

Proof. The row space decomposes into: the constant line; the d differences of each pair of opposite faces; and the $(d-1)$ combinations of pair sums whose total sum is zero. The intersections described above cause the Gram matrix to act by $d2^{d-1}$, 2^{d-1} , and 0, respectively. $\square$

The three-dimensional cube has six faces, eight vertices, and twelve edges. Those cardinals anticipate an incidence pattern (6/8), but a face is not a hexad and the set of vertices is not an octad. The exceptional identification requires a subsequent transporter that preserves supports and gauge.

2.6.8 Range 6 of the incidence operator and multiplicity of its resonant mode

Define the strata that preserve exactly k nonadic coordinates in rank d :

$$\mathcal{I}_k(d) := \binom{d}{k} 9^k.$$

The defects of APP are

$$D_{\text{vis}} = 54, \quad D_{\text{tot}} = 1215, \quad D_{\text{coc}} = 129, \quad 1215 = 54 + 9 \cdot 129.$$

Theorem 2.52 (Uniqueness of the Binomial Resonance). *The unique positive integer d that simultaneously satisfies*

$$\mathcal{I}_1(d) = 54, \quad \mathcal{I}_2(d) = 1215$$

is $d = 6$. In that range,

$$\frac{\mathcal{I}_2(6) - \mathcal{I}_1(6)}{9} = 129.$$

Proof. The first equality is $9d = 54$, which forces $d = 6$. Then

$$\mathcal{I}_2(6) = 81 \binom{6}{2} = 81 \cdot 15 = 1215,$$

and

$$\frac{1215 - 54}{9} = 129.$$

There remains no other possible rank because the first equality already determines d . $\square$

2.6.9 Obstructions of the cyclotomic skeleton of APP: fibers A_2^{12} , completion $729 + 271$, rank resonance 6, and cubic incidence

Test criterion 2.53 (Typed falsification of the unit). The construction is refuted if any of the following facts occurs:

1. the law of D_d not reproduce $D_1 = 0$ and $D_2 = 54$;
2. variation of length is fused with variation of modulus;
3. C ceases to preserve G_{A_2} or acquires a fixed vector;
4. the labels pair-sheet-orbit not produce twelve fibers;
5. the smallest certificate of Ψ has zero determinant;
6. a functional that factors through the augmentation module recovers 54;
7. the symbol of boundary $\star$ it identifica with the zero residue;
8. the four strata do not sum to 1000;

9. a cubic face is identified with a hexad without an incidence transporter.

The unit has constructed, without recognizing them retrospectively, the module A_2^{12} , the completion $729 + 271$, the projective measure, and the rank-six resonance. These structures will feed, in distinct ways, the generation of constants and the exceptional branches.

2.7 TRIT as a ternary oriented unit: internal geometries, temporal order, and dynamic conditions of time crystal

The mixed curvature of APP produced three oriented values: zero on the pure sheets and opposite signs when the sum–product order is exchanged. The TRIT types that triad and preserves, in addition to the visible residue, the quotient required to reconstruct the lifted integer. This unit explicitly separates four levels: ternary class, balanced representative, carry, and quadratic regime.

Definition 2.54 (Topological Information Unit TRIT). The *TRIT* is the elementary oriented unit whose state space is $\text{TRIT} = \{-1, 0, +1\}$. Its native logarithmic capacity is $\log 3$, and its expression in binary units is $\log_2 3$. Unlike a bit, its three states do not reduce to a cardinal encoding: they respectively type opening, threshold, and return, and preserve the carry quotient required to reconstruct the transition that produced the visible residue.

2.7.1 Oriented ternary system and balanced representation of TRIT states

We define the oriented set

$$\text{TRIT} := \{-1, 0, +1\}.$$

The balanced chart is the bijection of sets.

$$\text{bal} : \mathbb{F}_3 \longrightarrow \text{TRIT}, \quad [0]_3 \mapsto 0, \quad [1]_3 \mapsto +1, \quad [2]_3 \mapsto -1.$$

It is not of an equality of types. $\mathbb{F}_3$ carries sum and product of field; TRIT carries order and orientation. Every operation transported between both must indicate the chart used.

The orientation of the nine phases is

$$\varepsilon_9(r) = \begin{cases} +1, & r \in \{1, 4, 7\}, \\ -1, & r \in \{2, 5, 8\}, \\ 0, & r \in \{3, 6, 9\}. \end{cases}$$

His complete table is

r	1	2	3	4	5	6	7	8	9
$\varepsilon_9(r)$	+1	-1	0	+1	-1	0	+1	-1	0

Periodicity of three does not eliminate the nonadic phase: the phases 1, 4, 7, for example, have the same orientation and different positions in the circuit.

2.7.2 Balanced ternary division algorithm and uniqueness of the quotient–residue pair

Theorem 2.55 (Local ternary lifting). *For each $n \in \{-4, -3, \dots, 4\}$ there exists a single pair $(r, c) \in \text{TRIT}^2$ such that*

$$n = r + 3c.$$

Proof. The existence is verified in the table.

n	-4	-3	-2	-1	0	1	2	3	4
r	-1	0	+1	-1	0	+1	-1	0	+1
c	-1	-1	-1	0	0	0	+1	+1	+1

If $r + 3c = r' + 3c'$, then $r - r' = 3(c' - c)$. Since $r - r' \in \{-2, -1, 0, 1, 2\}$, the only possible multiple of three is zero; therefore $r = r'$ and $c = c'$. $\square$

The visible projection $(r, c) \mapsto r$ has three local states over the value zero:

$$0^- := (0, -1), \quad 0^0 := (0, 0), \quad 0^+ := (0, +1).$$

They publish the same residue, but reconstruct $-3, 0, +3$. This triple fiber is the minimal form of a distinction that will reappear in the integral holonomic state of the Topological Phase Kernel.

Corollary 2.56 (Balanced Iteration). *Every integer N admits a finite balanced ternary expansion*

$$N = r_0 + 3r_1 + \cdots + 3^m r_m, \quad r_j \in \text{TRIT}.$$

It is obtained by iterating the division $N_k = r_k + 3N_{k+1}$, where r_k is the balanced representative of $N_k \bmod 3$.

Proof. Euclidean division modulo three yields a residue in $\{0, 1, 2\}$; replacing 2 with -1 increases the quotient by one and produces $r_k \in \text{TRIT}$. The absolute value of the quotient decreases outside a finite set, so the process terminates. Uniqueness follows by comparing the first differing digit and reducing modulo three. $\square$

2.7.3 Carry cocycle of the oriented ternary composition

For $r, s \in \text{TRIT}$, we define $r \oplus s \in \text{TRIT}$ and $\kappa_3(r, s) \in \mathbb{Z}$ by the unique equality

$$r + s = (r \oplus s) + 3\kappa_3(r, s).$$

The table is

$$\begin{array}{c|ccc} \oplus & -1 & 0 & +1 \\ \hline -1 & +1 & -1 & 0 \\ 0 & -1 & 0 & +1 \\ +1 & 0 & +1 & -1 \end{array}, \quad \begin{array}{c|ccc} \kappa_3 & -1 & 0 & +1 \\ \hline -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ +1 & 0 & 0 & +1 \end{array}.$$

Proposition 2.57 (Ternary cocyclic identity). *For $r, s, t \in \text{TRIT}$,*

$$\kappa_3(r, s) + \kappa_3(r \oplus s, t) = \kappa_3(s, t) + \kappa_3(r, s \oplus t).$$

Proof. Both sides are the total quotient that appears when writing $r + s + t$ as a balanced representative plus a multiple of three. The uniqueness of the lift proves the equality. $\square$

If (r, c) and (s, d) are states levantados, its sum is

$$(r, c) \boxplus (s, d) := (r \oplus s, c + d + \kappa_3(r, s)).$$

The proposition 2.57 proves that $\boxplus$ is associative and that the reconstruction $\text{rec}_3(r, c) = r + 3c$ satisfies

$$\text{rec}_3(x \boxplus y) = \text{rec}_3(x) + \text{rec}_3(y).$$

2.7.4 Orientation and Temporal Order Involution

For to tritic word $\tau = (\tau_1, \dots, \tau_n)$, define

$$C_{\text{rev}}(\tau_1, \dots, \tau_n) := (-\tau_n, \dots, -\tau_1).$$

Lemma 2.58 (Oriented reversal). *The map C_{rev} is an involution. It interchanges the sectors +1 and -1 and fixes the sector 0.*

Proof. Applying it twice reverses the order twice and changes every sign twice. The initial word is thereby recovered. The remaining assertions are immediate in each component. $\square$

When a time-oriented parameter is fixed, the TPK convention is

+1 : return with memory, 0 : present threshold, -1 : prolongation opening.

This assignment does not add a fourth temporal state. The direction is preserved in the sign and the exact position is preserved in the nonadic phase.

2.7.5 Unified classification of the 3 quadratic algebras of the TRIT

For $\tau \in \text{TRIT}$, define

$$\mathbb{A}_\tau := \mathbb{R}[J_\tau] / (J_\tau^2 + \tau I).$$

The three cases are:

τ	J_τ^2	regime	$\exp(tJ_\tau)$
+1	$-I$	elliptic	$\cos t I + \sin t J_{+1}$
0	0	parabolic	$I + tJ_0$
-1	$+I$	hyperbolic	$\cosh t I + \sinh t J_{-1}$

Theorem 2.59 (Uniform quadratic exponential). *If $J_\tau^2 = -\tau I$, then*

$$\exp(tJ_\tau) = C_\tau(t)I + S_\tau(t)J_\tau,$$

where

τ	+1	0	-1
$C_\tau(t)$	$\cos t$	1	$\cosh t$
$S_\tau(t)$	$\sin t$	t	$\sinh t$

Proof. We separate the exponential series into even and odd powers. The relations $J_{+1}^{2m} = (-1)^m I$, $J_0^2 = 0$, and $J_{-1}^{2m} = I$ respectively produce the cosine and sine series, the linear truncation, and the hyperbolic series. $\square$

On $\mathbb{F}_3$ the analogues appear

$$\mathbb{F}_9 = \mathbb{F}_3[\iota]/(\iota^2 + 1), \quad \mathbb{D}_3 = \mathbb{F}_3[\epsilon]/(\epsilon^2), \quad \mathbb{S}_3 = \mathbb{F}_3[j]/(j^2 - 1).$$

The first is a field of nine elements; the second contains a nonzero nilpotent; the third decomposes through the idempotents $(1 \pm j)/2$ and is isomorphic to $\mathbb{F}_3 \times \mathbb{F}_3$. The coincidence of cardinality does not identify these three algebras.

The integral source of the elliptic regime is

$$\mathbb{Q}_Z := \mathbb{Z}[u]/(u^2 + 1).$$

Reduction modulo three sends u to ι and produces $\mathbb{F}_9$. The extension of scalars to $\mathbb{R}$ and a $u \mapsto J_{+1}$ representation produce the real elliptic realizer. There is therefore no untyped map $\mathbb{F}_9 \rightarrow \mathbb{R}$: both realizations follow from a common integral source.

2.7.6 Matrix representations of the TRIT selected by the sign of J_τ^2

Let

$$C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad C^3 = I,$$

and let us define

$$J_e := \frac{2C + I}{\sqrt{3}} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & -2 \\ 2 & -1 \end{pmatrix}.$$

A direct multiplication gives

$$(2C + I)^2 = -3I, \quad J_e^2 = -I.$$

The parabolic realizer is

$$J_p := N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad N^2 = 0.$$

For the hyperbolic regime we take

$$F_{av} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad A := F_{av}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix},$$

and

$$J_h := \frac{2A - 3I}{\sqrt{5}} = \frac{1}{\sqrt{5}} \begin{pmatrix} -1 & 2 \\ 2 & 1 \end{pmatrix}.$$

Then

$$(2A - 3I)^2 = 5I, \quad J_h^2 = I.$$

The bilinear form

$$G = \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}$$

satisfies

$$A^T G A = G.$$

Moreover,

$$\cosh(2 \log \varphi) = \frac{3}{2}, \quad \sinh(2 \log \varphi) = \frac{\sqrt{5}}{2},$$

for which

$$A = \exp(2 \log \varphi J_h) = \cosh(2 \log \varphi) I + \sinh(2 \log \varphi) J_h.$$

This identity will be recovered causally in the biography of φ ; here only the matrix representation is verified.

The three exponential controls are

$$\exp(\pi J_e) + I = 0, \quad \exp(J_p) = I + J_p, \quad \exp(2 \log \varphi J_h) = F_{av}^2.$$

The presence of π and φ in these verification identities does not authorize using them as constructors of those constants. Their constructors will appear later from the nonadic genealogy.

2.7.7 Local coupling APP–TRIT

With the exchange

$$R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

we define, for $\tau \in \text{TRIT}$,

$$B_\tau = (4 + 3\tau)I + (5 - 3\tau)R.$$

Since $P_\pm = (I \pm R)/2$,

$$B_\tau = 9P_+ + (6\tau - 1)P_-.$$

Theorem 2.60 (Spectral Recovery of Orientation). *The spectrum of B_τ is*

$$\text{Spec}(B_\tau) = \{9, 6\tau - 1\}.$$

The differential eigenvalue λ_- uniquely determines the TRIT:

$$\tau = \frac{\lambda_- + 1}{6}.$$

Moreover, the block central of APP is B_{+1} .

Proof. The projectors P_+ and P_- diagonalize R with eigenvalues $+1$ and -1 . The spectral formula is obtained by substitution. The second identity solves for τ . For $\tau = +1$, $B_{+1} = 7I + 2R = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}$, which is the unique block of [theorem 2.18](#). $\square$

Therefore, the TRIT not it anade as retrospective label: can be recovered of the differential mode of a family constructed on the internal exchange of APP.

2.7.8 Obstructions of TRIT: residue–representative–carry separation, involutivity, and spectral recovery of τ

Test criterion 2.61 (Refutation Criteria for the TRIT). The unit is refuted if:

1. it identify class residual, representative balanceado and carry;
2. it is asserted that 0^- , 0^0 , and 0^+ are the same state;
3. the composition $\boxplus$ ceases to reconstruct the integer sum;
4. C_{rev} is not involutive;
5. the three algebras are identified because they have the same cardinal;
6. some realizer fails to satisfy $J_\tau^2 = -\tau I$;
7. the value of τ is chosen after observing the outcome and is not recovered from the spectrum of B_τ .

APP provides the support, the multiplicative matrix, and the internal additive law; TRIT provides orientation, lift, and quadratic type. The next unit will construct the Topological Phase Kernel as the transport category of these data, without confusing the internal selector with the complete state.

2.8 TPK: observable base, integral holonomic state, and typed separation of its projections

APP has supplied a support, a multiplicative matrix, and an internal additive law; TRIT has supplied a lifted orientation. The Topological Phase Kernel, hereinafter TPK, specifies how those data are transported jointly, which coordinates are preserved when a route is closed, and which readers can be published without identifying the publication with the state. Its mathematical object is a category over a base of paths, endowed with an explicit family of fibers and prolongation relations.

2.8.1 Bivariant category of APP paths as the base of the integral holonomic TPK state

Let $\mathbf{P}_{\text{APP}}$ be the category libre constructed in the [section 2.4](#). We define

$$\mathbf{P}_{\text{APP}}^{(2)} := \mathbf{P}_{\text{APP}} \times \mathbf{P}_{\text{APP}}.$$

An object is a pair of positions $(p^\Sigma, p^\Pi) \in X_9^2$. A morphism is a pair of trajectories; one of them may be the identity. The first component is evaluated by Σ , the second by Π . They are not two lattices: they are two typed cursors over the same APP support.

The set of cardinal directions is

$$\mathcal{D}_{\text{card}} := \{N, E, S, O\},$$

with shifts

$$\delta(N) = (-1, 0), \quad \delta(E) = (0, 1), \quad \delta(S) = (1, 0), \quad \delta(O) = (0, -1) \quad \text{in } X_9.$$

The orientation involution is

$$\overline{N} = S, \quad \overline{S} = N, \quad \overline{E} = O, \quad \overline{O} = E.$$

2.8.2 Non-split extension $\mathbf{C}_{12} \rightarrow \mathbf{C}_{108} \rightarrow \mathbf{C}_9$ and dodecaphase record of the nonadic phase

The complete clock of this stage is

$$\Theta_{108} := \mathbb{Z}/108\mathbb{Z}.$$

For $\theta \in \Theta_{108}$, let $\tilde{\theta} \in \{0, \dots, 107\}$ be its normalized representative. We define three distinct coordinates:

$$\begin{aligned} g_0(\theta) &= [\tilde{\theta} + 1]_9 \in C_9, \\ r(\theta) &= 1 + (\tilde{\theta} \bmod 9) \in \mathcal{D}_9, \\ \beta(\theta) &= \left[\left[\frac{\tilde{\theta}}{9} \right] \right]_{12} \in C_{12}. \end{aligned}$$

The first is the phase class, the second its positive label, and the third the dodecaphasic sector. The map β is a block map; it is not the canonical reduction $\theta \bmod 12$.

The signature and the active sheet are

$$\tau(\theta) := \varepsilon_9(r(\theta)) \in \text{TRIT},$$

and

$$\mu(\theta) = \begin{cases} \Sigma, & \tau(\theta) = +1, \\ \Pi, & \tau(\theta) = -1, \\ \emptyset, & \tau(\theta) = 0. \end{cases}$$

The symbol $\emptyset$ denotes a threshold step in which neither cursor moves; it does not denote the absence of a state.

Definition 2.62 (Observable configuration). The observable base is

$$\mathcal{Q}_{\text{obs}} := X_9^2 \times \mathcal{D}_{\text{card}}^2 \times \Theta_{108}.$$

An element is written

$$q = (p^\Sigma, p^\Pi, d^\Sigma, d^\Pi, \theta).$$

It contains two positions, two cardinal orientations, and a calendar index.

2.8.3 Sheet selector $M_{\text{ph}} : \mathcal{D}_9 \rightarrow \text{TRIT}$ and joint evolution of position, orientation, and calendar

The function

$$M_{\text{ph}} : \mathcal{D}_9 \longrightarrow \text{TRIT}, \quad M_{\text{ph}}(r) = \varepsilon_9(r)$$

is the internal sheet selector. It selects which evaluation acts; it does not by itself displace any position.

We define the orientation inversion indicator by

$$\eta_{27}(\theta) = \begin{cases} -1, & \tilde{\theta} + 1 \in \{27, 54, 81, 108\}, \\ +1, & \text{otherwise.} \end{cases}$$

The new directions are

$$d^{\nu,+} = \begin{cases} \overline{d^\nu}, & \eta_{27}(\theta) = -1, \\ d^\nu, & \eta_{27}(\theta) = +1, \end{cases} \quad \nu \in \{\Sigma, \Pi\}.$$

The positions evolve via

$$(p^{\Sigma,+}, p^{\Pi,+}) = \begin{cases} (p^\Sigma + \delta(d^\Sigma), p^\Pi), & \tau(\theta) = +1, \\ (p^\Sigma, p^\Pi + \delta(d^\Pi)), & \tau(\theta) = -1, \\ (p^\Sigma, p^\Pi), & \tau(\theta) = 0. \end{cases}$$

Definition 2.63 (Observable evolution). The map

$$F_{\text{obs}} : \mathcal{Q}_{\text{obs}} \longrightarrow \mathcal{Q}_{\text{obs}}$$

is defined by

$$F_{\text{obs}}(q) = (p^{\Sigma,+}, p^{\Pi,+}, d^{\Sigma,+}, d^{\Pi,+}, [\theta + 1]_{108}).$$

The category libre of the graph $q \rightarrow F_{\text{obs}}(q)$ it is denoted by $\mathbf{P}_{\text{obs}}$.

Each arrow of $\mathbf{P}_{\text{obs}}$ induces a pair of trajectories in APP, so there exists a functor

$$B : \mathbf{P}_{\text{obs}} \longrightarrow \mathbf{P}_{\text{APP}}^{(2)}.$$

If $\tau = +1$, the second component of B is the identity; if $\tau = -1$, the first is the identity; if $\tau = 0$, both are the identity.

Proposition 2.64 (Separation between selection, motion, and remark). *The selector M_{ph} , the translations $T_d(p) = p + \delta(d)$, and any temporal sampling have distinct domains and codomains. They are not interchangeable operators.*

Proof. M_{ph} takes a phase label and returns a tritic orientation. T_d is an automorphism of X_9 . A sampling acts on a sequence or a category of arrows. Identifying two of them would violate typing. F_{obs} composes them in a declared order without erasing that distinction. $\square$

2.8.4 Decomposition of the integral holonomic state into 6 fibered families

Over each $q \in Q_{\text{obs}}$ we fix the total fiber

$$\mathcal{F}_q = \mathcal{O}_q \times \mathcal{H}_q \times \mathcal{C}_q \times \mathcal{S}_q \times \mathcal{B}_q \times \Lambda_q.$$

Each factor has its own type and function.

2.8.4.1 Joint determination of the orientation and sheet of the integral holonomic state

$$\mathcal{O}_q \subseteq \mathcal{D}_{\text{card}}^2 \times \{+1, -1\}.$$

The two directions coincide with those stored in q ; the last coordinate records the sheet orientation when the realization requires it. The sheet label does not replace the cardinal directions.

2.8.4.2 Euclidean decomposition of the state into compatible residue and quotient

The first lift fiber is

$$\mathcal{H}^{(1)} = \mathcal{D}_9 \times \mathbb{Z}, \quad \text{rec}_9(r, u) = r + 9u.$$

For each integer n there exist unique

$$r = \text{dr}_9(n), \quad u = \frac{n - r}{9}.$$

A greater depth consists of a declared tower of these fibers and their truncation maps. The term ‘‘Hensel’’ does not replace the definition of that tower.

2.8.4.3 Carry operator and joint update of residue, quotient, and memory

$\mathcal{C}_q$ is an abelian group. In the readers of decimal blocks one takes $\mathcal{C}_q = \mathbb{Z}$. The increment is not an independent coordinate, but a cochain over the arrows:

$$\kappa(r) \in \mathcal{C}_{q'} \quad \text{for } r : q \rightarrow q'.$$

2.8.4.4 Construction of the full signature from orientation, carry, and boundary

The signature is derived via a typed application.

$$\text{Sig}_q : \mathcal{H}_q \times \mathcal{C}_q \times \Lambda_q \longrightarrow \mathcal{S}_q.$$

In the nonadic boundary realization, individual conservation is maintained,

$$q_\partial, a_\partial, c_\partial \in \mathbb{F}_3^6, \quad \text{colw} \in \mathbb{Z}_{\geq 0}^6, \quad \rho_W, r_\partial \in \mathbb{F}_3^3.$$

These coordinates are margin, axial combination, cubic condition, column weights, Witt residual class, and ternary return. They do not condense into a single word if a subsequent proof uses any of them.

2.8.4.5 Cylindrical compatibility and boundary signature of the integral holonomic state

$$\mathcal{B}_q := \text{Cyl}_q \times \text{Fr}_q.$$

The cylinder localizes a family of prefixes; the boundary determines which extensions are compatible. Survival will be a relation between states endowed with both data, not a number added a posteriori.

2.8.4.6 Genealogical transport record

Λ_q contains the transport records whose endpoint is q . A prolongation $r : q \rightarrow q'$ acts by concatenation:

$$\lambda \longmapsto r \circ \lambda.$$

The genealogical register preserves the route; its observable endpoint does not determine it.

2.8.5 Cardinal operator T_d of transport between enriched fibers with conservation of path and memory

For each arrow $r : q \rightarrow q'$ maps are declared

$$H(r) : H_q \rightarrow H_{q'}, \quad C(r) : C_q \rightarrow C_{q'}, \quad \Lambda(r) : \Lambda_q \rightarrow \Lambda_{q'}.$$

They respect identities and composition, and $C(r)$ is a homomorphism. The carry satisfies

$$\kappa(r_2 \circ r_1) = \kappa(r_2) + C(r_2)(\kappa(r_1)). \quad (2.8)$$

Therefore, the transport of the determined coordinates is

$$\begin{aligned} h' &= H(r)(h), \\ c' &= C(r)(c) + \kappa(r), \\ \lambda' &= \Lambda(r)(\lambda), \\ \sigma' &= \text{Sig}_{q'}(h', c', \lambda'). \end{aligned}$$

Lemma 2.65 (Associativity of the transported carry). *The equation equation (2.8) guarantees that transporting a state by r_1 and then by r_2 produces the same carry as transporting it by $r_2 \circ r_1$.*

Proof. The successive transport gives

$$C(r_2)(C(r_1)c + \kappa(r_1)) + \kappa(r_2).$$

By functoriality of C , the first term is $C(r_2 r_1)c$; the sum of the other two is $\kappa(r_2 r_1)$ by equation (2.8). $\square$

2.8.6 Typed prolongation relation Ext_r between enriched fibers

Orientation and boundary are not imposed by means of a universal function. For each arrow $r : q \rightarrow q'$, a relation is declared

$$\text{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'}.$$

The two conditions are required.

$$\Delta_q \subseteq \text{Ext}_{\text{id}_q}, \quad \text{Ext}_{r_2} \circ \text{Ext}_{r_1} \subseteq \text{Ext}_{r_2 \circ r_1}.$$

The first preserves identities; the second preserves composition. A relation may admit several local extensions and leave only one after a larger horizon.

Definition 2.66 (The Topological Phase Kernel). An object of the TPK is a tuple

$$x = (q, o, h, c, \sigma, C, b, \lambda)$$

such that

$$\begin{aligned} q &\in \mathcal{Q}_{\text{obs}}, \quad o \in \mathcal{O}_q, \quad h \in H_q, \quad c \in C_q, \quad \lambda \in \Lambda_q, \\ \sigma &= \text{Sig}_q(h, c, \lambda), \quad (C, b) \in \text{Cyl}_q \times \text{Fr}_q. \end{aligned}$$

We associate to each object its fiber component.

$$\mathbf{f}(x) := (o, h, c, \sigma, (C, b), \lambda) \in \mathcal{F}_q.$$

A morphism $x \rightarrow x'$ is a triple (r, x, x') in which $r : q \rightarrow q'$ belongs to $\mathcal{P}_{\text{obs}}$, the transportable coordinates satisfy the preceding laws, and $(\mathbf{f}(x), \mathbf{f}(x')) \in \text{Ext}_r$. Composition is

$$(r_2, x', x'') \circ (r_1, x, x') = (r_2 r_1, x, x''),$$

and the identity of x is (id_q, x, x) .

Theorem 2.67 (Categorical structure of the TPK). *The definition 2.66 defines a category and the map*

$$p : \text{TPK} \longrightarrow \mathbf{P}_{\text{obs}}, \quad p(x) = q, \quad p(r, x, x') = r,$$

is a functor.

Proof. Route composition is associative. The lemma 2.65 proves compatibility of the carry coordinate; the declared functoriality of $\mathbf{H}$, $\mathbf{C}$, and $\mathbf{\Lambda}$ proves compatibility of the other determined coordinates. The stability of Ext guarantees that the composite remains admissible, and its diagonal condition provides the identities. The map p preserves identities and composition by construction. $\square$

2.8.7 Observable quotients of the integral holonomic state and preserved genealogical fields

The elementary projections are

$$\phi(x) = g_0(\theta), \quad \tau(x) = \varepsilon_9(r(\theta)), \quad \mu(x) = \mu(\theta).$$

The projection to the local TRIT state is

$$\pi_{\text{TRIT}}(x) = (\tau, \mu, o, h, c, \sigma, \lambda, \phi).$$

It forgets the cylinder, the boundary, and part of q ; it does not define another container. Two states may have the same image under π_{TRIT} and differ in the forgotten information.

2.8.8 Criterion of route survival via conservation of the projected boundary

Let $\mathcal{X}_{\text{TPK}}^{(n)}$ be the family of states at depth n . For $N \geq n$, we define

$$\text{Surv}_n^{(N)} = \left\{ x_n : \begin{array}{l} \text{there exist } x_{n+1}, \dots, x_N \text{ with} \\ (x_j, x_{j+1}) \in \text{Ext}_{r_j} \text{ for } n \leq j < N \end{array} \right\}.$$

Proposition 2.68 (Survival filtration). *For $N \geq n$,*

$$\text{Surv}_n^{(N+1)} \subseteq \text{Surv}_n^{(N)}.$$

If the scheme is defined to all depth, the states indefinitely prolongable are

$$\text{Surv}_n = \bigcap_{N \geq n} \text{Surv}_n^{(N)}.$$

Proof. A prolongation to $N + 1$ truncates to a prolongation to N , which proves the inclusion. The intersection expresses exactly prolongability for every finite horizon. $\square$

2.8.9 Insertion of the 6 fibered families of the integral holonomic state into the TPK operator hierarchy

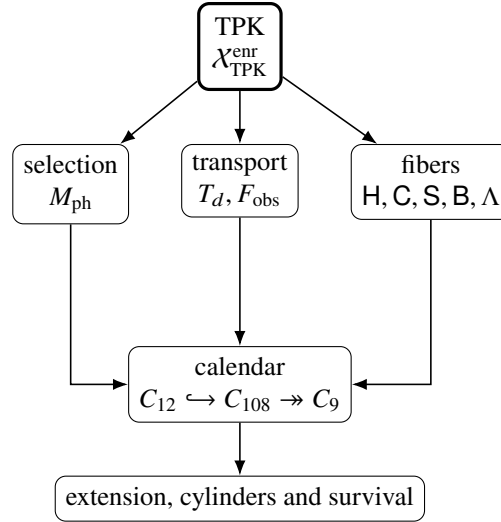


Figure 2.3: Operator containment of the Topological Phase Kernel. The selector, transport, readers, and fibers are not parallel containers: they are maps or coordinates of the same integral holonomic state.

2.8.10 Obstructions of the integral holonomic TPK state: selector typing, carry cochain, compatible extension, and genealogical preservation

Test criterion 2.69 (Refutation criteria for the integral holonomic state). The definition is refuted if:

1. the selector M_{ph} it usa as translation;
2. θ , $g_0(\theta)$, $r(\theta)$, and $\beta(\theta)$ are identified;
3. a signature is allowed to be chosen independently of (h, c, λ) ;
4. the composition of carries incumple [equation \(2.8\)](#);
5. Ext_r is replaced by an unproved functional choice;
6. a projection that has forgotten the cylinder, boundary, or genealogical record is called a complete state;
7. the phase return erases any of the coordinates of fiber.

This unit has defined the container. The next will enumerate its readers, operators, odometer, and calendars, and will prove why the returns of nine, twenty-seven, and one hundred eight steps are not restarts of the state.

2.9 Typed hierarchy of the 34 heterogeneous canonical constructions and causal dynamics of TPK updating

The Topological Phase Kernel is neither a reduction modulo three nor a clock with one hundred and eight positions. It is a category of integral holonomic states equipped with selection, transport, reading, memory, periodicity, prolongation, and publication operators. To prevent an abbreviation from replacing the object, each operator is recorded by its type and by the information it preserves.

2.9.1 Typed schema of the 34 canonical TPK constructions: domain, codomain, action, invariants, and memory

Definition 2.70 (Operator specification). A canonical realization of the TPK is a sextuple

$$O = (D_O, C_O, f_O, I_O, L_O, M_O), \quad (2.9)$$

where:

1. $f_O : D_O \rightarrow C_O$ is its rule;
2. I_O is the family of conserved relations;
3. L_O declares the information published or lost;
4. M_O enumerates the memory remaining in the integral holonomic state.

Two realizations represent the same operator only when there exists a typed equivalence that intertwines rules and invariants. Sharing a period, cardinality, or visible output is not sufficient.

2.9.2 Typed L_0 – L_7 hierarchy of the 34 heterogeneous canonical TPK constructions

The inventory is partitioned into:

- C1.** supports and states;
- C2.** internal selection;
- C3.** transporte;
- C4.** readers and quotients;
- C5.** memory and boundary;
- C6.** periodicity and return;
- C7.** prolongation;
- C8.** salidas.

The class is determined by domain, codomain, and function, not by the name of a number associated with it.

2.9.3 Functional grouping of the 34 canonical constructions in 14 families of composition APP–TRIT–TPK

Grouping	Operators or spaces	Function and invariants
State and route memory	$\mathcal{X}_{\text{TPK}}^{\text{enr}}, \mathcal{F}_q$	Preserve position, sheet, orientation, residue, quotient, carry, signature, boundary, cylinder, prefix, provenance and memory.
Internal selection	$M_{\text{ph}} : \{1, \dots, 9\} \rightarrow \{-1, 0, +1\}$	Activate the additive, multiplicative evaluation or the threshold step; does not move the state.
Transporte espacial	$T_N, T_E, T_S, T_O, F_{\text{loc}}, F_{\text{obs}}$	Perform cardinal translations, sheet feedback, and timeline of two cursors.
Lectores modulares	$\rho_9, q_9, \rho_3^{\text{or}}, c_3, q_{1000}, r_{1000}$	Publish phase, orientation and decimal block preserving the lift quotients.
Block and memory	$\mathcal{K}_{27} = F_{\text{obs}}^{27}, \mathcal{N}_{27}$	The former composes twenty-seven transports; the latter filters memory events. An equal index does not imply an equal operator.
Twelve-phase state	$\mathcal{R}_{12}, C_{12}, \Theta_{108}, \Gamma_{108}$	Distinguish enriched record, sector, cyclic law, and ordered support.
Periodicity and depths	$C_{12} \hookrightarrow C_{108} \twoheadrightarrow C_9, w_{108}, w_{1080}$	Preserves the non-splittable extent, words of different lengths and the vertical memory.
Channels 90/120	$\mathcal{F}_6, \mathcal{F}_8, \mathcal{I}_{6 \rightarrow 8}, S_{90}, S_{120}$	Separate finite incidences, analytic tails, and their normalizations.
Bidirectionality	$P_{\pm}, \text{rev}, \text{Ext}_{\pm}, N_{\pm}, \beta, C_M$	Perform exchange, route inversion, future and past prolongation, valence, balance, and conjugation.
Emission and coinduction	$L_0, L_1, R_{36}, G_9, \varprojlim \text{Cyl}_m$	Lift the seed, open the boundary, and organize finite and compatible extensions.
Numeric Readers	$\mathcal{C}_{\pi}, \mathcal{P}_e, F_{\text{av}}$	They construct rational forks, factorial propagation, and Fibonacci self-scaling.
Exceptional Projection	$\Gamma_{Aw}, \text{weight}, \text{projective support}$	Transport the same emission to ternary incidence without selecting Archimedean numbers.
Atlas and realization	route signature, atlas 81/56/13/324, C_M	Classify stories and characters before applying a dimensional chart.
Finite validation	censuses, structural necessity tests, integrity identities and independence controls	Checks coverage at the executed depth and separates construction, independent verification, and comparison data.

The eight classes describe mathematical nature; the fourteen groupings describe execution function. They are two gradings of the same inventory.

2.9.4 Mathematical classes, domains, codomains, and causal dependencies of the 34 canonical constructions

$$\mathfrak{D}_{\text{TPK}} := (\mathfrak{o}_1, \dots, \mathfrak{o}_{34}). \quad (2.10)$$

#	Symbol	Domain and codomain	Rule or function
1	$\mathcal{A}_9$	$(\mathbb{Z}/9)^2$, matrix and internal law	Additive-multiplicative arithmetic cell.
2	(ρ_9, q_9)	$\mathbb{Z}_{>0} \rightarrow \{1, \dots, 9\} \times \mathbb{Z}_{\geq 0}$	Nonadic division with positive representative and quotient.
3	ρ_3^{or}, c_3	$\mathbb{Z} \rightarrow \{-1, 0, +1\} \times \mathbb{Z}$	Lift ternary oriented $n = \rho_3^{\text{or}}(n) + 3c_3(n)$.
4	$\iota_{1000} := (q_{1000}, r_{1000})$	$\mathbb{Z} \rightarrow \mathbb{Z} \times \{0, \dots, 999\}$	Euclidean division $N = 1000q_{1000} + r_{1000}$.
5	T_d	$X_9 \rightarrow X_9, d \in \{N, E, S, O\}$	Cardinal oriented translation.
6	F_{loc}	$\mathcal{Q}_{\text{loc}} \times \mathcal{D}_{\text{card}} \rightarrow \mathcal{Q}_{\text{loc}}$	Local position feedback and sheet.
7	T_{min}	$\Omega_{\text{seed}} \rightarrow \mathcal{U}_6$	Minimum emitter of the two cursors, including the coordinate signature.
8	G	$\text{im } s_{\Sigma} \times \text{im } s_{\Pi} \rightarrow \{0, \dots, 999\}^6$	Coordinate-by-coordinate decimal coupling.
9	Ψ_6	$\mathcal{U}_6 \rightarrow \mathbb{F}_3^6$	Tertiary reduction of the catalog.
10	$\mathcal{R}_3$	$\mathcal{Q}_{\text{loc}} \times \mathcal{D}_{\text{card}}^3 \rightarrow \{-1, 0, +1\}$	Local three-step observable factor.
11	$\mathcal{K}_{27}$	$\mathcal{Q}_{\text{obs}} \rightarrow \mathcal{Q}_{\text{obs}}$	Composition F_{obs}^{27} .
12	$\mathcal{X}_{\text{TPK}}^{\text{enr}}$	category of integral holonomic states	Preserve all the typed coordinates of the nucleus.
13	$\mathcal{H}$	relation on integral holonomic states	Hensel Transition that retains residue and quotient.
14	$C_{3,1000}$	crossed states (N, L, D, K, A, B)	Exact update of ternary-decimal cylinders.
15	Σ_B	boundary memories $\rightarrow$ signature joint	Publish the block orientation preserving its genealogical record.
16	$\text{EF-}G_9$	extension relation	Filter compatible extensions and organize the nonadic genealogy.
17	Γ_9	enriched histories $\longrightarrow$ histories	Holonomy of a phase twist with boundary memory.
18	$X_{\mathcal{X}}$	$\lim_{\longleftarrow m} \text{Surv}_m^{(\mathcal{X})}$	Global section when non-emptiness, compatibility, and uniqueness are demonstrated in full depth.
19	$\mathcal{P}_{\text{APP}}^{(2)}$	bivariate path category	Transports simultaneously the multiplicative evaluation and the additive accumulation.
20	F_{obs}	$\mathcal{Q}_{\text{obs}} \rightarrow \mathcal{Q}_{\text{obs}}$	Deterministic observable chronology with two cursors.
21	$\mathcal{A}_{42}$	alphabet of forty-two triads	Extended decimal support of finite emission.
22	Ext_r	$\text{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'}$	Typed relation of fiber prolongation.
23	$\mathcal{K}_{\text{ph}}$	$\mathcal{U}_6 \times C_9$ about itself	Transfer of phase-compatible explicit continuation.
24	(P, H, Q)	$\mathcal{X}_{\times} \rightarrow \mathcal{X}_{\times}$	Advance, half-turn, and quarter-turn of seeds.
25	$(c_{\text{mode}}, c_{\text{or}})$	$\text{caminos} \rightarrow \mathbb{Z}^2$	Mode and orientation change cocycles.
26	E_{20}	finite states $\rightarrow$ twenty blocks	Finite triadic prolongation with sourcebook.
27	$\mathcal{R}_{12}$	history enriched $\rightarrow$ record of twelve sectors	Timed system with route, signature, carry, and genealogical record.
28	R_{sgn}	terminal state $\rightarrow U_{12}^{\text{sgn}} \subseteq \mathbb{Z}^{12}$	Reader of the signed harmonic observable.
29	$\mathcal{H}_{12}$	$\mathbb{Z}^{12} \rightarrow \mathbb{Z}^{12}$	Transformation by three orbits of step three; inverse $\mathcal{H}_{12}/4$ over the pertinent image.
30	$\mathcal{E}_{\text{dod}} = (D_3, D_4, Q)$	$\mathbb{Z}^{12} \rightarrow \mathbb{Z}^{25}$	Transversal extractor set of arities three and four.
31	C_{α}	blocks and carries $\rightarrow$ blocks	Dodecaphasic closure recurrence in base thousand.
32	$(L_{\text{tick}}, \chi_{\text{mass}})$	enriched history $\rightarrow$ temporal character	Posterior dimensional interface; does not participate in the numeric constructor of this work.
33	$\mathcal{W}_{\pm}$	bidirectional lattice of index twelve	Reading in both directions preserving orientation and origin.
34	$\mathcal{A}_{\text{species}}$	extension families $\rightarrow$ atlas	Posterior realization interface, not a primitive fourth.

2.9.5 Update of the integral holonomic state through $U_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t$

Definition 2.71 (Complete canonical state). Let

$$\mathcal{E} := \mathbb{Z}_{\text{quo}} \times \mathbb{Z}_{\text{car}} \times \text{Mem} \times \text{Term} \times \text{Cyl} \times \text{Fr} \times \text{Pref} \times \Lambda.$$

The canonical integral holonomic state at time t is

$$x_t = (p_t, m_t, \sigma_t, b_t, r_t; q_t, c_t, \eta_t, s_t, C_t, \partial_t, N_t, \lambda_t) \in X_9 \times \{\Sigma, \Pi\} \times \text{TRIT} \times C_{12} \times C_9 \times \mathcal{E}. \quad (2.11)$$

The state abbreviations used in other units are projections of this tuple. In particular, the quotient q_t , carry c_t , terminal s_t , cylinder C_t , boundary ∂_t , prefix N_t , and ledger λ_t are not omitted.

Let ρ_t be the observable arrow determined by x_t . The transport laws of U005 and the transition $C_{3,1000}$ determine a memory endomorphism

$$\begin{aligned} \mathbf{E}_t : \mathcal{E} &\longrightarrow \mathcal{E}, \\ (q, c, \eta, s, C, \partial, N, \lambda) &\longmapsto \begin{pmatrix} \mathbf{H}(\rho_t)q, \mathbf{C}(\rho_t)c + \kappa(\rho_t), \mathbf{M}(\rho_t)\eta, \\ \mathbf{T}(\rho_t)s, \mathbf{Cyl}(\rho_t)C, \mathbf{Fr}(\rho_t)\partial, \\ \mathbf{Pref}(\rho_t)N, \mathbf{\Lambda}(\rho_t)\lambda \end{pmatrix}. \end{aligned} \quad (2.12)$$

Each component published in (2.12) has the domain indicated by its fiber; $\kappa(\rho_t)$ is the carry cochain.

We define three endomorphisms of the canonical product. If $\sigma' = M_{\text{ph}}(g(t))$ and $m' = \mu(\sigma', m)$, then

$$\mathbf{Sel}_t(p, m, \sigma, b, r; e) := (p, m', \sigma', b, r; e), \quad (2.13)$$

$$\mathbf{Tra}_t(p, m, \sigma, b, r; e) := (T_{d(t)}p, m, \sigma, b, r; e), \quad (2.14)$$

$$\mathbf{Upd}_t(p, m, \sigma, b, r; e) := (p, m, \sigma, b', r'; \mathbf{E}_t(e)), \quad (2.15)$$

where (b', r') is the destination of $\rho_t : (b, r) \rightarrow (b', r')$. The causal update is a composition of maps, not the composition of a map with a tritic value:

$$\boxed{\mathcal{U}_t := \mathbf{Upd}_t \circ \mathbf{Tra}_t \circ \mathbf{Sel}_t.} \quad (2.16)$$

Theorem 2.72 (Closure of causal update). *The map $\mathcal{U}_t$ is an endomorphism of the complete canonical state. Readers, prolongations, and projections are applied after $\mathcal{U}_t$ only when their domain has been produced.*

Proof. $\mathbf{Sel}_t$ changes only the mode and the TRIT; $\mathbf{Tra}_t$ changes only the APP position; and $\mathbf{Upd}_t$ applies the typed product (2.12) and the phase arrow. All three codomains are the same product of definition 2.71; their composition is therefore defined and preserves all state coordinates. $\square$

2.9.6 Typed separation between operators, states, projections, and publications of the TPK

The inventory imposes

$$S_4 \neq D_4 \neq \text{sd}_4, \quad \mathcal{R}_{12} \neq C_{12} \neq \Theta_{108} \neq \Gamma_{108}, \quad F_{\text{obs}} \neq \mathcal{K}_{\text{ph}}, \quad (2.17)$$

and

$$U_{12}^{\text{cat}} \neq U_{12}^{\text{sgn}}, \quad \mathcal{E}_{\text{dod}} \neq D_{\text{raw}}^{90,120} \neq \mathcal{K}_{6 \rightarrow 8}. \quad (2.18)$$

They are type incompatibilities, not mere value inequalities.

Test criterion 2.73 (Verification of operator completeness). An execution is operator-complete only if:

1. declares the inputs of $\mathfrak{D}_{\text{TPK}}$ used;
2. is recorded the domain and the codomain of each composition;
3. preserves the causal order of (2.16);
4. distinguishes dynamics, transfer, reading, prolongation, and output;
5. permits reconstruction from λ_t of every block change and every carry;
6. does not promote a finite check to an infinite section without proving non-emptiness, compatibility, and uniqueness at every depth.

Any symbol used without a type, or any fusion prohibited by (2.17) and (2.18), constitutes a verifiable failure.

2.10 Tritic incidence 90/120 and typed separation of the temporal, nilpotent, and angular operators

The integers 90 and 120 appear in several strata of the construction. Their numerical coincidence does not establish an identity of objects. Likewise, the letter epsilon denotes three constructions with incompatible domains. The purpose of this unit is to fix their types before using them and to indicate precisely which realizations require structures that have not yet been constructed.

2.10.1 Temporal, nilpotent, and angular operators: domains and codomains

2.10.1.1 Action of the tritic temporal selector over $\{-1, 0, +1\}$: opening, threshold, and return with memory

The internal calendar selector is

$$\varepsilon_9 : \{1, \dots, 9\} \longrightarrow \{-1, 0, +1\}, \quad \varepsilon_9(r) = \begin{cases} +1, & r \in \{1, 4, 7\}, \\ -1, & r \in \{2, 5, 8\}, \\ 0, & r \in \{3, 6, 9\}. \end{cases} \quad (2.19)$$

Its codomain is the oriented TRIT. It determines which sheet acts and does not itself shift any position.

2.10.1.2 Nilpotent parabolic generator $J_0^2 = 0$ and evolution at the temporal threshold

The class

$$\varepsilon_{\text{nil}} \in \mathcal{Q}_n := \mathbb{F}_3[\varepsilon_{\text{nil}}]/(\varepsilon_{\text{nil}}^2) \quad (2.20)$$

satisfies

$$\varepsilon_{\text{nil}} \neq 0, \quad \varepsilon_{\text{nil}}^2 = 0. \quad (2.21)$$

It is an element of a degenerate quadratic algebra; it is not a temporal selector.

2.10.1.3 Angular residue induced after temporal selection and orientation preservation

The symbol

$$\varepsilon_{\text{res}}^\circ \in \mathbb{R} \quad (2.22)$$

designates, when its geometric parameters are introduced, the residue of an angular identity. At this stage only its type remains reserved. It is not obtained from ε_9 or from ε_{nil} , and it does not intervene in the construction of π , e , φ or α .

Proposition 2.74 (No identification of the epsilon symbols). *There is no well-typed equality between*

$$\varepsilon_9, \quad \varepsilon_{\text{nil}}, \quad \varepsilon_{\text{res}}^\circ.$$

Proof. The first object is a function between finite sets; the second, a nilpotent element of an algebra over $\mathbb{F}_3$; the third, a real scalar with angular unit. Their domains, codomains, and laws of composition are distinct. $\square$

2.10.2 Typed residences of the indices 90/120 in 6 non-equivalent operator objects

Definition 2.75 (Dodecaphasic Extractor). The integral extractor

$$\mathcal{E}_{\text{dod}} := (D_3, D_4, Q) : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{25} \quad (2.23)$$

reads differences of arities three and four and a sum coordinate. The subscripts are not angular lengths.

Definition 2.76 (Raw difference of sequences). For $|q| < 1$, define

$$S_m^{\text{raw}}(q) := \frac{q^m}{1 - q^{3m}}, \quad D_{\text{raw}}^{90,120}(q) := S_{90}^{\text{raw}}(q) - S_{120}^{\text{raw}}(q). \quad (2.24)$$

This object is an analytic function of q ; it is not a finite census.

Definition 2.77 (Marked families of incidence). Once a hexad H , its image $\ell(H)$ in a binary dodecad D , and the lifted octad O_H have been constructed, one will define

$$\mathcal{F}_6(H) := \ell(H) \times \binom{\ell(H)}{2}, \quad (2.25)$$

$$\mathcal{F}_8(H) := O_H \times \binom{\ell(H)}{2}. \quad (2.26)$$

By construction,

$$|\mathcal{F}_6(H)| = 6 \binom{6}{2} = 90, \quad |\mathcal{F}_8(H)| = 8 \binom{6}{2} = 120. \quad (2.27)$$

The inclusion $\ell(H) \hookrightarrow O_H$ inducira

$$\mathcal{I}_{6 \rightarrow 8} : \mathcal{F}_6(H) \hookrightarrow \mathcal{F}_8(H). \quad (2.28)$$

The the corresponding identity fix the type from now on. Their effective existence will be proved after constructing C_W , W_{12} , the binary datum G_{24} , D , ℓ , and W_{24} . That branch is not anticipated here.

Definition 2.78 (Normalized Queue Combination). For parameters $q_-, q_+ \in (0, 1)$, let

$$S_m(q) := \frac{q^m}{1 - q^{3m}}, \quad \Delta S_m := S_m(q_-) - S_m(q_+). \quad (2.29)$$

The functional

$$\mathcal{K}_{6 \rightarrow 8} := 12 \Delta S_{90} - \Delta S_{120} \quad (2.30)$$

is an analytic combination of tails. Its coefficient twelve belongs to its normalization and does not turn it into the record $\mathcal{R}_{12}$.

Definition 2.79 (Semigroup of Degrees and Moments). The additive semigroup generated by both degrees is

$$\mathfrak{S}_{90,120} := \langle 90, 120 \rangle = \{90a + 120b : a, b \in \mathbb{N}_0\}. \quad (2.31)$$

For $n \in \mathfrak{S}_{90,120}$, the oriented moment is

$$s_n := q_-^n - q_+^n. \quad (2.32)$$

The moments are differential monomials indexed by a semigroup; the series of (2.29) are infinite sums of those moments over specific progressions.

2.10.3 Normalized defect 1–90/120=1/4 of the tritic incidence

When the inclusion $\mathcal{I}_{6 \rightarrow 8}$ has been constructed, its difference of cardinalities will be

$$|\mathcal{F}_6(H)| - |\mathcal{F}_8(H)| = 90 - 120 = -30.$$

If normalization is declared

$$24 \binom{6}{2} = 360,$$

we obtain

$$\frac{90 - 120}{24 \binom{6}{2}} = \frac{6 - 8}{24} = -\frac{1}{12}. \quad (2.33)$$

The inclusion determines -30 ; the rational $-1/12$ depends moreover of the declared normalization. No it what must present as if the cardinality by if alone fixed the denominator.

2.10.4 Domain invariants, orientation, and memory of the 90/120 realizations

Theorem 2.80 (Separation of the six objects). *The objects*

$$\mathcal{E}_{\text{dod}}, \quad D_{\text{raw}}^{90,120}, \quad \mathcal{I}_{6 \rightarrow 8}, \quad \mathcal{K}_{6 \rightarrow 8}, \quad \mathfrak{S}_{90,120}, \quad \mathcal{E}_{\text{res}}^{\circ} \quad (2.34)$$

are mutually distinct by type: they are, respectively, an integer reader, an analytic function, a finite morphism, a combination of series, an index semigroup, and a real angular scalar.

Proof. The conclusion is obtained by comparing their domains and codomains:

$$\mathbb{Z}^{12} \rightarrow \mathbb{Z}^{25}, \quad \mathbb{D} \rightarrow \mathbb{C}, \quad \mathcal{F}_6(H) \rightarrow \mathcal{F}_8(H), \quad (0,1)^2 \rightarrow \mathbb{R}, \quad \mathbb{N}_0^2 \rightarrow \mathbb{N}_0, \quad \{*\} \rightarrow \mathbb{R}.$$

There are no two identical signatures. $\square$

Test criterion 2.81 (Refutation criteria). The classification is refuted if any of its domains or codomains is altered, if the census of (2.27) does not reproduce 90 and 120, if $-1/12$ is attributed to the inclusion without declaring the normalization, or if the angular residual is presented as an output of the constants generator of this monograph.

2.11 Local transport factor and ternary-decimal transition

This unit separates three operations involved in the transport of the Topological Phase Kernel that do not share a domain: the cardinal realization of a trit, the stroboscopic sampling of a sequence, and the exact update of a pair of ternary–decimal cylinders. The first is a finite dynamics; the second is a temporal reader; the third preserves the integers that make it possible to decide compatibility without floating-point arithmetic.

2.11.1 Oriented transition local operator and APP sheet update

Let

$$X_9 = (\mathbb{Z}/9\mathbb{Z})^2, \quad \mathcal{D}_{\text{card}} = \{N, E, S, O\},$$

with

$$\delta(N) = (-1, 0), \quad \delta(E) = (0, 1), \quad \delta(S) = (1, 0), \quad \delta(O) = (0, -1).$$

The visible evaluations of APP are

$$L_{\Sigma}(x, y) = \rho_9((x+1) + (y+1)), \quad L_{\Pi}(x, y) = \rho_9((x+1)(y+1)),$$

where $\rho_9(n) = 1 + ((n-1) \bmod 9)$. We further define

$$\rho_3^{\text{or}}([0]) = 0, \quad \rho_3^{\text{or}}([1]) = +1, \quad \rho_3^{\text{or}}([2]) = -1.$$

Definition 2.82 (Local factor of APP–TRIT). The local state space is

$$\mathcal{Q}_{\text{loc}} = X_9 \times \{\Sigma, \Pi\}, \quad |\mathcal{Q}_{\text{loc}}| = 162.$$

For $q = (p, m)$, $d \in \mathcal{D}_{\text{card}}$, we set

$$p_d = p + \delta(d), \quad a_d(q) = \rho_3^{\text{or}}([L_m(p_d)]_3),$$

and

$$\mu(a, m) = \begin{cases} \Sigma, & a = +1, \\ \Pi, & a = -1, \\ m, & a = 0. \end{cases}$$

The local transition is

$$F_d^{\text{loc}}(p, m) = (p_d, \mu(a_d(p, m), m)).$$

The position it is modified before evaluating the sheet. The null trit preserves the last active mode; not eliminates the sheet nor reinitializes the history.

For a cardinal word $u = d_1 d_2 d_3$, let

$$q_j = F_{d_j}^{\text{loc}}(q_{j-1}), \quad q_0 = q,$$

and let us denote by

$$R_3(q, u) \in \{-1, 0, +1\}$$

the trit produced in the third displacement.

Definition 2.83 (Remark on step three). On an oriented sequence $a = (a_t)_{t \geq 1}$ we define

$$\text{Obs}_3(a) = (a_3, a_6, a_9, \dots).$$

The map R_3 takes a state and a cardinal word of length three; Obs_3 takes a sequence and publishes a subsequence. They are operators of different types.

Theorem 2.84 (Exact surjectivity of the sampled factor). *For every $q \in \mathcal{Q}_{\text{loc}}$ and every $b \in \{-1, 0, +1\}$, the fiber*

$$\mathcal{R}_3(q, b) = \{u \in \mathcal{D}_{\text{card}}^3 : R_3(q, u) = b\}$$

is non-empty and satisfies

$$8 \leq |\mathcal{R}_3(q, b)| \leq 40.$$

In particular, the $162 \cdot 3 = 486$ state–output fibers are non-empty.

Proof. From each of the 162 states, the $4^3 = 64$ cardinal words are enumerated, and the recurrence of definition 2.82 is applied literally. The exhaustive census of the 486 pairs (q, b) has a minimum of eight, a maximum of forty, and no zero value. The statement is therefore a finite identity on the defined system, not a mixing hypothesis. $\square$

Corollary 2.85 (Realization of an arbitrary oriented sequence). *Remark. For every sequence $(b_k)_{k \geq 1} \in \{-1, 0, +1\}^{\mathbb{N}}$ there exists a cardinal history whose observation satisfies*

$$(a_3, a_6, a_9, \dots) = (b_1, b_2, b_3, \dots).$$

Proof. Fixing a total order on $\mathcal{D}_{\text{card}}^3$, at stage k the first element of $\mathcal{R}_3(q_{k-1}, b_k)$ is taken. The terminal state of the block again belongs to $\mathcal{Q}_{\text{loc}}$, so that the choice may be iterated indefinitely. $\square$

The corollary proves the realization capacity of the sampled factor. It does not select the sections of π, e, φ , does not fix the two intermediate trits, and does not replace the compatibility of the integral holonomic state.

2.11.2 Cylindrical systems of the 2 arithmetic readers and truncation compatibility

A ternary word of length L , interpreted as the integer N , determines

$$I_3(N, L) = \left[\frac{N}{3^L}, \frac{N+1}{3^L} \right).$$

A prefix of K decimal triads, interpreted as the integer D , determines

$$I_{1000}(D, K) = \left[\frac{D}{1000^K}, \frac{D+1}{1000^K} \right).$$

We introduce the cross residues

$$A = N1000^K - D3^L, \quad B = (N+1)1000^K - D3^L = A + 1000^K.$$

Lemma 2.86 (Integer compatibility criterion). *The cylinders $I_3(N, L)$ and $I_{1000}(D, K)$ intersect if and only if*

$$A < 3^L, \quad B > 0.$$

Equivalently,

$$N1000^K < (D+1)3^L, \quad (N+1)1000^K > D3^L.$$

Proof. Two half-open intervals $[a, b)$ and $[c, d)$ intersect exactly when $a < d$ and $c < b$. Multiplication by $3^L 1000^K > 0$ produces the two inequalities. Subtracting $D3^L$ and $N1000^K$, respectively, gives the form in A, B . $\square$

2.11.3 Completion operator $C_{3,1000}$: ternary extension and preservation of the unit in 10^3 positions

A new block of six trits is represented by

$$u \in \{0, \dots, 3^6 - 1\} = \{0, \dots, 728\}.$$

The ternary update is

$$N' = 729N + u, \quad L' = L + 6.$$

Decimal publication has two cases. If the refined cylinder does not yet determine a new triad, then $K' = K$, $D' = D$. If it determines $\delta \in \{0, \dots, 999\}$, then

$$K' = K + 1, \quad D' = 1000D + \delta.$$

Definition 2.87 (Crossed state and exact transition). The cylinder state is the sextuple

$$\mathbf{c} = (N, L, D, K, A, B)$$

subject to

$$A = N1000^K - D3^L, \quad B = A + 1000^K.$$

The transition $C_{3,1000}$ adjoins u and, when corresponds, δ , and updates

$$(N, L, D, K, A, B) \mapsto (N', L', D', K', A', B')$$

through means of

$$A' = \begin{cases} 729A + u1000^K, & K' = K, \\ 729000A + u1000^{K+1} - \delta 729 3^L, & K' = K + 1, \end{cases} \quad B' = A' + 1000^{K'}.$$

The prolongation is admissible exactly when

$$A' < 3^{L'}, \quad B' > 0.$$

Theorem 2.88 (Accuracy and Sufficiency of the Cross State). *The transition of the definition 2.87 exactly preserves the identities*

$$A' = N'1000^{K'} - D'3^{L'}, \quad B' = (N' + 1)1000^{K'} - D'3^{L'}.$$

Consequently, (A', B', L', K') suffices to decide the compatibility of the new pair of cylinders, while (N', D') preserves its reconstructive provenance.

Proof. If a decimal triplet does not appear,

$$N'1000^K - D3^{L+6} = (729N + u)1000^K - 729D3^L = 729A + u1000^K.$$

Si aparece δ ,

$$\begin{aligned} N'1000^{K+1} - D'3^{L+6} &= (729N + u)1000^{K+1} - (1000D + \delta)7293^L \\ &= 729000A + u1000^{K+1} - \delta 729 3^L. \end{aligned}$$

In both cases, increasing N' by one unit adds $1000^{K'}$, which yields B' . The criterion of the lemma 2.86 concludes. $\square$

Proposition 2.89 (Compatibility with concatenation). *Successively applying $C_{3,1000}$ to two blocks produces the same pair (N'', D'') and the same cross-residues as appending at once the two ternary words and the decimal triads published in the same order.*

Proof. The recurrences $N \mapsto 729N + u$ and $D \mapsto 1000D + \delta$ are the laws of positional concatenation in bases 729 and 1000. The theorem 2.88 identifies A, B with functions of those four integers; hence its stagewise update coincides with the evaluation subsequent to concatenation. $\square$

2.11.4 Factorization of the TPK transition as $U_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t$

The correct sequence is

$$(q, u) \xrightarrow{R_3} b, \quad (a_t)_{t \geq 1} \xrightarrow{\text{Obs}_3} (a_{3k})_{k \geq 1}, \quad (\mathfrak{c}, u, \delta) \xrightarrow{C_{3,1000}} \mathfrak{c}'.$$

R_3 proves local realizability; Obs_3 publishes a subsequence; $C_{3,1000}$ transports compatibility between two bases. None of the three can replace the survival relation of the full TPK.

Test criterion 2.90 (Refutation criteria). The construction is refuted in its domain if any of the following facts occurs:

1. any of the 486 fibers $\mathcal{R}_3(q, b)$ is empty or remains outside of the interval $[8, 40]$;
2. the null trit changes the active mode;
3. R_3 is identified with Obs_3 ;
4. the recurrence of A' does not reproduce its direct definition;

5. a declared compatible prolongation violates $A' < 3^{L'}$ or $B' > 0$;
6. the decision depends on a numerical tolerance absent from the integer inequalities.

2.12 Record dodecafasico integral, extension ciclica and decomposition $1 + 5 + 6$

The observable period of one hundred eight positions contains twelve sectors of nine positions. This equality of cardinalities does not suffice to describe the temporal state: the ordered support, cyclic law, sector index, and enriched register of transports, orientations, carries, and incidences must be distinguished. This unit constructs the four objects and proves the integral reconstruction of the twelve memory coordinates.

2.12.1 Cyclic coordinate on the time-oriented 12-sector support

For $m \in \{1, \dots, 12\}$, define

$$E_m := \{9(m-1) + 1, \dots, 9m\}. \quad (2.35)$$

The ordered support and the cyclic group are

$$\Gamma_{108} := E_1 \sqcup \dots \sqcup E_{12}, \quad \Theta_{108} := \mathbb{Z}/108\mathbb{Z}. \quad (2.36)$$

Γ_{108} preserves position, order and pertenencia to block; Θ_{108} preserves the law of translation. No are the same object.

For $\theta \in \Theta_{108}$, let $\tilde{\theta} \in \{0, \dots, 107\}$ be its representative. We define

$$\beta(\theta) := \left[\left[\frac{\tilde{\theta}}{9} \right] \right]_{12}, \quad r(\theta) := 1 + (\tilde{\theta} \bmod 9). \quad (2.37)$$

The first coordinate locates the duodecaphase sector; the second publishes the positive position $1, \dots, 9$ within the sector.

2.12.2 nonsplit extension $0 \rightarrow C_{12} \rightarrow C_{108} \rightarrow C_9 \rightarrow 0$ and carry cocycle

Let $C_n = \mathbb{Z}/n\mathbb{Z}$. The applications

$$\iota : C_{12} \longrightarrow C_{108}, \quad \iota([b]_{12}) = [9b]_{108}, \quad p : C_{108} \longrightarrow C_9, \quad p([\theta]_{108}) = [\theta]_9 \quad (2.38)$$

form the sequence

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0. \quad (2.39)$$

For the set-theoretic section

$$s([r]_9) := [\tilde{r}]_{108}, \quad 0 \leq \tilde{r} < 9,$$

the defect of additivity is

$$c(r, r') := \left[\left[\frac{\tilde{r} + \tilde{r}'}{9} \right] \right]_{12}, \quad s(r) + s(r') = s(r + r') + \iota(c(r, r')). \quad (2.40)$$

Theorem 2.91 (Accuracy, non-division, and cohomological class). *The sucesion (2.39) is exacta and not it escinde. For the action trivial,*

$$H^2(C_9, C_{12}) \simeq C_{12}/9C_{12} \simeq C_3, \quad (2.41)$$

and the cocycle (2.40) represents its generating class.

Proof. The image of ι is the subgroup of multiples of nine, which coincides with $\ker p$; p is surjective. This proves exactness. If the extension were split, C_{108} would be isomorphic to $C_{12} \times C_9$, but

$$\exp(C_{108}) = 108, \quad \exp(C_{12} \times C_9) = \text{mcm}(12, 9) = 36.$$

The cocycle identity is the Euclidean division of $\tilde{r} + \tilde{r}'$ by nine. For a cyclic group with trivial action, $H^2(C_9, C_{12}) \simeq C_{12}/9C_{12}$; the unit carry projects to $[1]$, which generates the quotient of order three. $\square$

The non-splitting typifies the return:

$$g_0(\theta + 9) = g_0(\theta), \quad \beta(\theta + 9) = \beta(\theta) + 1 \pmod{12}.$$

Nine iterations close the visible phase and shift one sector; one hundred and eight close both clock coordinates. The integral holonomic state also preserves vertical memory, the cylinder, the boundary, and genealogy.

2.12.3 Enhanced temporal registration: joint encoding of phase, sheet, boundary, and memory

Definition 2.92 (Oriented dodecaphonic register). For an enriched history, let τ_m be the ordered subroute on E_m , σ_m its orientation, κ_m the carry crossing the block boundary, and Λ_m its local incidence ledger. The complete record is

$$\mathcal{R}_{12} := ((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m))_{m=1}^{12}. \quad (2.42)$$

Thus four levels are distinguished:

$$\boxed{\mathcal{R}_{12} \neq C_{12} \neq \Theta_{108} \neq \Gamma_{108}.} \quad (2.43)$$

The first is an enriched register; the second, a sector index; the third, a group of translations; the fourth, an ordered support. There exist projections between them, but no identifications.

Let $q_{\text{mem}} \in \mathbb{Z}_{\geq 0}$ be the unbounded number of nonadic turns. The duodecaphase coordinate is only its residue

$$\beta = [q_{\text{mem}}]_{12}.$$

After one hundred and eight steps,

$$\beta \mapsto \beta, \quad q_{\text{mem}} \mapsto q_{\text{mem}} + 12.$$

The calendar returns; the vertical memory is not reset.

2.12.4 Extraction of the signed dodecaphase state from the enriched time record

Let $(d_t)_{t=1}^{108}$ be the digital record on Γ_{108} . We define

$$\mathcal{D}_{\text{evt}} := \{2, 4, 6, 8\}, \quad \Sigma_{12} := (+, +, +, -, -, -, +, +, +, -, -, -). \quad (2.44)$$

For $m = 0, \dots, 11$, let

$$e_m := \Sigma_{12}(m+1) \# \{t \in E_{m+1} : d_t \in \mathcal{D}_{\text{evt}}\} \in \mathbb{Z}. \quad (2.45)$$

The application preserves the local census and the sectorial orientation as separate factors.

We pair the two semicrowns by means of

$$p_i := e_i + e_{i+6}, \quad a_i := e_i - e_{i+6}, \quad 0 \leq i \leq 5. \quad (2.46)$$

We define the total census, the five differences with respect to the sixth pair, and the six antisymmetries:

$$s_C := \sum_{i=0}^5 p_i, \quad M_i := p_i - p_5 \quad (0 \leq i \leq 4), \quad A_6 := (a_0, \dots, a_5). \quad (2.47)$$

Definition 2.93 (Integral reader 1 + 5 + 6). The linear reader is

$$\mathcal{L}_{12} : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{12}, \quad \mathcal{L}_{12}(e) := (s_C, M_0, \dots, M_4, a_0, \dots, a_5). \quad (2.48)$$

The coordinate distribution is

$$\underbrace{1}_{s_C} + \underbrace{5}_{M_0, \dots, M_4} + \underbrace{6}_{a_0, \dots, a_5} = 12. \quad (2.49)$$

Theorem 2.94 (Exact reconstruction and integral image). *The reader $\mathcal{L}_{12}$ is injective. On its image,*

$$p_5 = \frac{s_C - \sum_{i=0}^4 M_i}{6}, \quad p_i = p_5 + M_i \quad (0 \leq i \leq 4), \quad (2.50)$$

$$e_i = \frac{p_i + a_i}{2}, \quad e_{i+6} = \frac{p_i - a_i}{2} \quad (0 \leq i \leq 5). \quad (2.51)$$

An output tuple belongs to $\text{im } \mathcal{L}_{12}$ if and only if

$$s_C - \sum_{i=0}^4 M_i \equiv 0 \pmod{6} \quad (2.52)$$

and, for the reconstructed p_i ,

$$p_i \equiv a_i \pmod{2} \quad (0 \leq i \leq 5). \quad (2.53)$$

Proof. From $M_i = p_i - p_5$ is obtained

$$s_C = \sum_{i=0}^4 (p_5 + M_i) + p_5 = 6p_5 + \sum_{i=0}^4 M_i,$$

which gives the first equation of (2.50); the others are immediate. Adding and subtracting $p_i = e_i + e_{i+6}$ and $a_i = e_i - e_{i+6}$ produces (2.51). Divisibility by six and the six parity congruences are precisely the necessary and sufficient conditions for all reconstructed coordinates to be integers. $\square$

Division by six appears after the oriented census, not during it. The integrality conditions for this unit are exactly (2.52) and (2.53); no Smith normal form is attributed here to a matrix that has not previously been defined.

2.12.5 Genealogical distinction between 2 dodecaphasic words of equal period

The distinction must be preserved.

$$U_{12}^{\text{cat}} := U_6 \| U_6, \quad U_{12}^{\text{sgn}} := \mathcal{H}_{12}(K). \quad (2.54)$$

The first concatenates two catalog emissions of length six. The second is a signed coordinate reversibly linked to K by the operator $\mathcal{H}_{12}$. Their common length does not imply the same provenance, semantics, or transport law.

Test criterion 2.95 (Falsification criteria). The construction is refuted if the sequence (2.39) is not exact, if its cohomology class is trivial, if a tuple in the image violates (2.52) or (2.53), if the inverse formulas do not recover the twelve initial integers, or if the return of β is interpreted as a reset of q_{mem} .

2.13 Cyclotomic phase functor–length and modular elevation of paths

The odometric quotient $O_{9,12}$ simultaneously preserves the nonadic position, the dodecaphase register, and the integral length of each arrow. In this unit, its phase is realized over fibers A_2 , and the cardinal lift that replaces each edge with four oriented edges is proved. The cyclotomic representation and the absolute length will be retained as distinct coordinates.

2.13.1 Coxeter representation of order 3 on the dodecaphasic fibers of type A_2

Let

$$G_{A_2} := \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad C := \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}. \quad (2.55)$$

By the proposition 2.44,

$$C^3 = I_2, \quad C^\top G_{A_2} C = G_{A_2}, \quad \det(XI_2 - C) = X^2 + X + 1. \quad (2.56)$$

Thus, C realizes the character cyclotomic of C_3 on A_2 .

For each $b \in \mathbb{Z}/12\mathbb{Z}$, let $A_{2,b}$ be a copy of A_2 , let

$$\iota_b : A_{2,b} \xrightarrow{\sim} A_2$$

an integral identification, and let $P_b \in O(G_{A_2})$ be a frame that depends on b , but not on the residue r . We define

$$\rho_b := \iota_b^{-1} P_b C P_b^{-1} \iota_b, \quad \rho_b^3 = \text{id}_{A_{2,b}}. \quad (2.57)$$

The independencia respect of r liga the character to the fiber dodecaphasic; r preserves the position internal of the turn nonadic.

2.13.2 Odometer category of the TPK: cylinders, truncations, and sequence morphisms

Recall that $O_{9,12}$ has objects

$$(b, r) \in \mathbb{Z}/12\mathbb{Z} \times \{0, \dots, 8\}.$$

For $n \in \mathbb{N}$, the flecha unique of length n is

$$\gamma_{(b,r),n} : (b, r) \longrightarrow (b + \kappa_r(n), [r + n]_9), \quad \kappa_r(n) := \left\lfloor \frac{r + n}{9} \right\rfloor. \quad (2.58)$$

The composition relies on

$$\kappa_r(n + m) = \kappa_r(n) + \kappa_{[r+n]_9}(m). \quad (2.59)$$

The length $\ell(\gamma_{(b,r),n}) = n$ is additive.

Definition 2.96 (Cyclotomic Phase–Length Functor). Let $\mathbf{Rep}_{\mathbb{C}}(C_3)$ be the category of complex representations of C_3 , and let $\mathbf{BN}$ be the monoidal category with a single object whose endomorphisms are $(\mathbb{N}, +)$. We define

$$\mathfrak{F}_{\text{cyc}} : O_{9,12} \longrightarrow \mathbf{Rep}_{\mathbb{C}}(C_3) \times \mathbf{BN} \quad (2.60)$$

by

$$\mathfrak{F}_{\text{cyc}}(b, r) = (A_{2,b} \otimes_{\mathbb{Z}} \mathbb{C}, \rho_b)$$

and

$$\mathfrak{F}_{\text{cyc}}(\gamma) := \left(\iota_{b'}^{-1} P_{b'} C^{\ell(\gamma)} P_b^{-1} \iota_b, \ell(\gamma) \right) \quad (2.61)$$

for $\gamma : (b, r) \rightarrow (b', r')$.

Theorem 2.97 (Functoriality and separation of the records). *The assignment of the definition 2.96 is a functor. Its first component preserves only the length class modulo three; the second preserves the*

complete integer and, by reduction, determines that class. The cyclotomic component does not allow the absolute length to be reconstructed.

Proof. If $\gamma_1 : (b_0, r_0) \rightarrow (b_1, r_1)$ and $\gamma_2 : (b_1, r_1) \rightarrow (b_2, r_2)$ have lengths n_1, n_2 , the intermediate trivialization cancels:

$$\begin{aligned} & (\iota_{b_2}^{-1} P_{b_2} C^{n_2} P_{b_1}^{-1} \iota_{b_1}) (\iota_{b_1}^{-1} P_{b_1} C^{n_1} P_{b_0}^{-1} \iota_{b_0}) \\ &= \iota_{b_2}^{-1} P_{b_2} C^{n_1+n_2} P_{b_0}^{-1} \iota_{b_0}. \end{aligned} \quad (2.62)$$

The second component composes by addition. Since C^n depends only on $[n]_3$, whereas $\ell = n$ preserves the integer, the two coordinates are not identified. $\square$

If

$$Q : \text{TPK} \longrightarrow \mathcal{O}_{9,12}$$

is the odometer functor of U006, the typed realization is

$$\mathfrak{F}_{\text{cyc}} \circ Q : \text{TPK} \longrightarrow \mathbf{Rep}_{\mathbb{C}}(C_3) \times \mathbf{BN}. \quad (2.63)$$

2.13.3 Modular elevation of cardinal paths

Let $\mathcal{A}_{N,d}$ be the category libre of paths cardinal of $(\mathbb{Z}/N\mathbb{Z})^d$. Let

$$N \geq 2, \quad u \geq 1, \quad \gcd(u, N) = 1, \quad k := \text{ord}_N(u), \quad q := \frac{u^k - 1}{N}.$$

Definition 2.98 (Cardinal Substitution by u). The functor

$$\mathcal{S}_u : \mathcal{A}_{N,d} \longrightarrow \mathcal{A}_{N,d}$$

sends an object p to up and each oriented edge to the ordered concatenation of u copies with the same direction.

Theorem 2.99 (Modular lifting to endomorphism of paths). *For every object p , path γ , and*

$$\kappa_{N,r}(n) := \left\lfloor \frac{r+n}{N} \right\rfloor,$$

we have

$$\mathcal{S}_u^k(p) = p, \quad (2.64)$$

$$\ell(\mathcal{S}_u^k \gamma) = u^k \ell(\gamma), \quad (2.65)$$

$$\kappa_{N,r}(u^k n) - \kappa_{N,r}(n) = qn. \quad (2.66)$$

Proof. The category is free, so the images of objects and generators determine a unique functor. Each map multiplies the length by u . Since $u^k = 1 + qN$, the objects return modulo N , and

$$\left\lfloor \frac{r+n+qNn}{N} \right\rfloor = \left\lfloor \frac{r+n}{N} \right\rfloor + qn.$$

$\square$

In the nonadic case,

$$(N, u, k, q) = (9, 4, 3, 7), \quad 4^3 = 64 = 1 + 7 \cdot 9. \quad (2.67)$$

Therefore,

$$\boxed{\mathcal{S}_4^3(e) \text{ preserves the endpoints and contains seven nonadic turns oriented per initial edge.}} \quad (2.68)$$

The endpoints and residual position return in three applications; the integral holonomic state need not return. Without a proof of invertibility, $\mathcal{S}_4^3$ is a vertical path endomorphism, not a monodromy.

Remark 2.100 (Separation with respect to clock dilation). $\mathcal{S}_4$ replaces objects and edges in a path category. The map D_4 of U006C will multiply the coordinate Θ_{108} by four. They share an integer factor, but not a domain, codomain, or composition law.

Test criterion 2.101 (Criteria of refutation). The block remains refuted if fails any of the identities (2.56), (2.59), (2.62) or (2.67); also remains badly typed if it identify the phase C^n , the length n , the substitution $\mathcal{S}_4$ and the dilation D_4 .

2.14 Quaternary dilation of the register, observable obstruction, and refined semiconjugacy

The cardinal substitution $\mathcal{S}_4$ constructed in the preceding unit acts on paths. In this unit, a second operation, arithmetic in nature, is defined on the register $\mathbb{Z}/12\mathbb{Z} \times \mathbb{Z}/9\mathbb{Z}$, and a third, categorical operation subdivides each observable transition into four microtransitions. The aim is not to merge them, but to prove the exact relationship between their realizations.

2.14.1 Dilation on the product register

Let

$$\mathcal{Q}_{12,9} := \mathbb{Z}/12\mathbb{Z} \times \mathbb{Z}/9\mathbb{Z}.$$

For $r \in \mathbb{Z}/9\mathbb{Z}$, we denote its normalized representative by $\tilde{r} \in \{0, \dots, 8\}$.

Definition 2.102 (Quaternary dilation of the register). We define

$$D_4(b, r) := \left(\left[4b + \left\lfloor \frac{4\tilde{r}}{9} \right\rfloor \right]_{12}, [4r]_9 \right). \quad (2.69)$$

The total coordinate is

$$\Theta : \mathcal{Q}_{12,9} \xrightarrow{\sim} \mathbb{Z}/108\mathbb{Z}, \quad \Theta(b, r) := [9b + \tilde{r}]_{108}. \quad (2.70)$$

Proposition 2.103 (Realization by modular multiplication). *In the total coordinate,*

$$\boxed{\Theta \circ D_4 \circ \Theta^{-1}(\theta) = [4\theta]_{108}.} \quad (2.71)$$

In particular,

$$\text{im}(D_4) = 4\mathbb{Z}/108\mathbb{Z} \simeq \mathbb{Z}/27\mathbb{Z}, \quad (2.72)$$

and the induced restriction on the image has order nine.

Proof. For $\theta = 9b + \tilde{r}$,

$$4\theta = 9 \left(4b + \left\lfloor \frac{4\tilde{r}}{9} \right\rfloor \right) + (4\tilde{r} \bmod 9).$$

This is exactly the quotient–residue decomposition of (2.69). The image consists of the classes that are multiples of four. Dividing by four yields $\mathbb{Z}/27\mathbb{Z}$, and

$$\begin{aligned} 4^1 &\equiv 4, & 4^2 &\equiv 16, & 4^3 &\equiv 10, \\ 4^4 &\equiv 13, & 4^5 &\equiv 25, & 4^6 &\equiv 19, \\ 4^7 &\equiv 22, & 4^8 &\equiv 7, & 4^9 &\equiv 1 \pmod{27}. \end{aligned}$$

No preceding power is one; therefore $\text{ord}_{27}(4) = 9$. $\square$

2.14.2 Exact carry identity

For $r \in \{0, \dots, 8\}$ and $n \in \mathbb{N}$, define

$$\kappa_r(n) := \left\lfloor \frac{r+n}{9} \right\rfloor, \quad r' := [r+n]_9.$$

Theorem 2.104 (Quaternary carry cocycle). *We have*

$$\kappa_{[4r]_9}(4n) = 4\kappa_r(n) + \left\lfloor \frac{4\tilde{r}'}{9} \right\rfloor - \left\lfloor \frac{4\tilde{r}}{9} \right\rfloor. \quad (2.73)$$

The last difference is the coboundary produced by the choice of representatives $r \mapsto \tilde{r}$.

Proof. Write

$$r + n = 9\kappa_r(n) + \tilde{r}'.$$

When multiplied by four,

$$4r + 4n = 36\kappa_r(n) + 4\tilde{r}'.$$

Separating quotient and residue modulo nine and subtracting the quotient already contained in $4\tilde{r}$ yields (2.73). $\square$

Corollary 2.105 (Third iteration). *For every $(b, r) \in \mathcal{Q}_{12,9}$,*

$$D_4^3(b, r) = ([4b + 7\tilde{r}]_{12}, r). \quad (2.74)$$

Proof. By proposition 2.103,

$$\Theta(D_4^3(b, r)) = [4^3(9b + \tilde{r})]_{108} = [64(9b + \tilde{r})]_{108}.$$

Since $64 = 1 + 7 \cdot 9$, the residue modulo nine is again r , whereas the block coordinate becomes $[4b + 7\tilde{r}]_{12}$. $\square$

The return of r is not return of the complete register. The difference $7\tilde{r}$ preserves the memory of the internal representative traversed.

2.14.3 Obstruction in the observable quotient

In the observable configuration of definition 2.62, we write

$$q = (p^\Sigma, p^\Pi, d^\Sigma, d^\Pi, \theta) \in \mathcal{Q}_{\text{obs}}.$$

The positional projection is

$$\pi_{\text{pos}}(q) := (p^\Sigma, p^\Pi) \in X_9^2. \quad (2.75)$$

The temporal signature is

$$\tau(q) := \varepsilon_9(r(\theta)) \in \{-1, 0, +1\}.$$

We define the oriented observable shift

$$a(q) := \begin{cases} (\delta(d^\Sigma), 0), & \tau(q) = +1, \\ (0, \delta(d^\Pi)), & \tau(q) = -1, \\ (0, 0), & \tau(q) = 0. \end{cases} \quad (2.76)$$

Then

$$\pi_{\text{pos}}(F_{\text{obs}}q) = \pi_{\text{pos}}(q) + a(q),$$

with the address update and clock defined in U005.

Theorem 2.106 (Obstruction to the ordinary observable endofunctor). *There is no endofunctor of the ordinary observable space that simultaneously satisfies:*

1. *multiplying the geometric position by four;*
2. *replace each transition by four genuine microtransitions;*
3. *preserving sheet and orientation in each microtransition;*
4. *sending a complete duodecaphase fiber to a single duodecaphase fiber.*

The obstruction persists in a step of umbral with $a(q) = 0$.

Proof. Let

$$\mathcal{B}_b := \{(b, r) : 0 \leq r \leq 8\}.$$

Replace one elementary iteration with four force

$$h(\theta + 1) = h(\theta) + 4$$

and, by induction,

$$h(\theta) = 4\theta + c \pmod{108}.$$

Take $0 \leq \tilde{c} < 108$ and write $\tilde{c} = 9c_b + c_0$, with $0 \leq c_0 < 9$. If the whole fiber $\mathcal{B}_b$ were sent to a single fiber, the destination block index would have to be independent of r ; however, it contains

$$4b + c_b + \left\lfloor \frac{4r + c_0}{9} \right\rfloor \pmod{12}, \quad (2.77)$$

which is not constant in r . Moreover, the neutral transition operator satisfies $P_0^{\text{TPK}} = I$, distinct from the cyclotomic trivializations P_b ; the ordinary calendar $(+1, -1, 0)$ causes four elementary iterations to visit both arithmetic modes and the neutral mode. $\square$

2.14.4 Minimal temporal subdivision compatible with semiconjugation

Definition 2.107 (Subdivided observable state). Let

$$\tilde{\mathcal{Q}}_4 := \mathcal{Q}_{\text{obs}} \times \{0, 1, 2, 3\}, \quad \tilde{\pi}(q, j) := 4\pi_{\text{pos}}(q) + j a(q). \quad (2.78)$$

We define

$$\tilde{F}(q, j) := \begin{cases} (q, j+1), & 0 \leq j < 3, \\ (F_{\text{obs}}q, 0), & j = 3, \end{cases} \quad \iota_4(q) := (q, 0). \quad (2.79)$$

Theorem 2.108 (Four-step semiconjugacy). *The inclusion ι_4 satisfies*

$$\boxed{\tilde{F}^4 \circ \iota_4 = \iota_4 \circ F_{\text{obs}}}. \quad (2.80)$$

The intermediate positions are exactly

$$4\pi_{\text{pos}}(q) + j a(q), \quad j = 0, 1, 2, 3.$$

Proof. Starting from $(q, 0)$, the first three applications increment only the microinstant index. The fourth applies F_{obs} and returns the index to zero. This gives

$$(q, 0) \mapsto (q, 1) \mapsto (q, 2) \mapsto (q, 3) \mapsto (F_{\text{obs}}q, 0).$$

The formula for the positions is the definition of $\tilde{\pi}$. The refinement makes visible a determined subdivision; it does not add a dynamic decision. $\square$

2.14.5 Subdivided category and cyclotomic elevation

Let $\text{sd}_4 \mathbf{P}_{\text{obs}}$ be the category obtained by replacing each generator $\gamma : q \rightarrow F_{\text{obs}}q$ with the chain

$$(q, 0) \rightarrow (q, 1) \rightarrow (q, 2) \rightarrow (q, 3) \rightarrow (F_{\text{obs}}q, 0). \quad (2.81)$$

The functor

$$\text{sd}_4 : \mathbf{P}_{\text{obs}} \longrightarrow \text{sd}_4 \mathbf{P}_{\text{obs}}$$

sends each generator to that string. The collapse functor

$$\text{col}_4 : \text{sd}_4 \mathbf{P}_{\text{obs}} \longrightarrow \mathbf{P}_{\text{obs}}$$

recompose the four microarrows and satisfies

$$\text{col}_4 \circ \text{sd}_4 = \text{id}_{\mathbf{P}_{\text{obs}}}. \quad (2.82)$$

In a chain that remains in the fiber b , the first three microarrows act by ρ_b ; the fourth incorporates the interfiber transport if a dodecaphase crossing occurs. By $\rho_b^3 = I$, the ordered product of the four actions coincides with the cyclotomic morphism of the original arrow.

Proposition 2.109 (Cyclotomic Invariance under Subdivision). *There exists a typed lift*

$$\tilde{\mathfrak{F}}_{\text{cyc}} : \text{sd}_4 \mathbf{P}_{\text{obs}} \longrightarrow \mathbf{Rep}_{\mathbb{C}}(C_3) \times \mathbf{BN}$$

such that, for every arrow γ of $\mathbf{P}_{\text{obs}}$,

$$\boxed{\tilde{\mathfrak{F}}_{\text{cyc}}(\text{sd}_4 \gamma) = (\mathfrak{F}_{\text{cyc}} \circ Q)(\gamma)}. \quad (2.83)$$

Proof. The successive powers of ρ_b are assigned to the internal microarrows, and transport between the frames of the source and target fibers is assigned to the fourth. The composition of the first three factors is the identity because $\rho_b^3 = I$. The remaining factor is exactly the morphism that $\mathfrak{F}_{\text{cyc}} \circ Q$ assigns to the collapsed arrow. The subdivided category is free on those microarrows, so the assignment extends uniquely. $\square$

2.14.6 Commutators of 3 quaternary operations and separation of their actions

The architecture contains three distinct operations:

1. $\mathcal{S}_4$, cardinal substitution of objects and paths;
2. D_4 , multiplication by four in Θ_{108} ;
3. sd_4 , functorial subdivision of an observable transition.

Their compatibilities are given, respectively, by

$$4^3 = 1 + 7 \cdot 9, \quad \Theta D_4 = 4\Theta, \quad \tilde{F}^4 \iota_4 = \iota_4 F_{\text{obs}}.$$

Sharing the integer four converts none into another.

Test criterion 2.110 (Refutation criteria). The construction is refuted if any of the identities (2.71), (2.73), (2.74), (2.80), or (2.83) fails; likewise if an observable realization satisfies the four conditions of theorem 2.106 without preserving a register equivalent to the microinstant.

2.15 Modular readers, odometer, and dodecaphasic calendar

The integral holonomic TPK state is not identified with any of its readings. This unit defines the readers modulo nine, modulo three, and modulo one thousand, the phase–sector odometer, the predictive quotient of thirty-six states, and the cyclic extension of one hundred and eight positions. Each construction declares what it publishes and what memory it preserves.

2.15.1 Typed scheme of the modular readers of the TPK: domain, codomain, publication, and conserved memory

Definition 2.111 (Operator specification). A canonical TPK operator is recorded by means of a sextuple

$$O = (D_O, C_O, f_O, I_O, L_O, M_O),$$

where D_O is the domain, C_O the codomain, f_O the rule, I_O the preserved invariants, L_O the published or forgotten information, and M_O the memory remaining in the integral holonomic state.

Two symbols with equal cardinality or equal period do not represent the same operator unless there exists an equivalence intertwining their domains, rules, and invariants. This convention will prevent confusing the predictive quotient C_{36}^{pred} , the boundary R_{36} , and a cardinal defect of value thirty-six.

2.15.2 Nonadic reader of the integral holonomic state: phase, transition, and return with memory

For $n \geq 1$, define

$$\rho_9(n) = 1 + ((n - 1) \bmod 9), \quad q_9(n) = \left\lfloor \frac{n - 1}{9} \right\rfloor.$$

Then

$$n = 9q_9(n) + \rho_9(n).$$

The card of ρ_9 is

$$(\mathbb{Z}_{>0}, \mathcal{D}_9, \rho_9, n \equiv \rho_9(n) \pmod{9}, \text{positive phase}, q_9(n)).$$

The positive digital root numerically coincides with ρ_9 , but its semantics is distinct: one is presented as a section of the quotient; the other, as an iterated sum of digits.

2.15.3 Triticale reader with quotient and carry transport

The balanced division is written

$$n = r_3(n) + 3c_3(n), \quad r_3(n) \in \text{TRIT}, \quad c_3(n) \in \mathbb{Z}.$$

Its card is

$$(\mathbb{Z}, \text{TRIT} \times \mathbb{Z}, n \mapsto (r_3(n), c_3(n)), \text{rec}_3(r, c) = r + 3c, r_3, c_3).$$

The publication may display only r_3 , but the TPK preserves c_3 if a subsequent composition requires reconstruction.

2.15.4 Reader positional by blocks in base 10^3

The Euclidean division defines

$$N = 1000q_{1000}(N) + r_{1000}(N), \quad 0 \leq r_{1000}(N) < 1000.$$

Its card is

$$(\mathbb{Z}, \mathbb{Z} \times \{0, \dots, 999\}, N \mapsto (q_{1000}, r_{1000}), \text{rec}_{1000}(q, r) = 1000q + r, r_{1000}, q_{1000}).$$

Proposition 2.112 (Concatenation and residual reading). *If $u_1, \dots, u_m \in \{0, \dots, 999\}$ and*

$$D_m = 1000D_{m-1} + u_m, \quad D_0 = 0,$$

then

$$D_m = \sum_{j=1}^m u_j 1000^{m-j}.$$

Moreover,

$$D_m \equiv \sum_{j=1}^m u_j \pmod{9}$$

because $1000 \equiv 1 \pmod{9}$.

Proof. The first equality is obtained by induction on m . The second follows by reducing it modulo nine. $\square$

The congruence does not reconstruct the order of the blocks. The genealogical register preserves the sequence $(u_1, \dots, u_m)$; the residual reader publishes only its sum modulo nine.

2.15.5 Odometer category of the TPK: cylindrical objects and composition by truncation

Definition 2.113 (Nonadic odometer–dodecaphasic). The category $\mathcal{O}_{9,12}$ has objects

$$(b, r) \in C_{12} \times \{0, \dots, 8\}.$$

An arrow of length $n \in \mathbb{N}$ is

$$(b, r) \xrightarrow{n} \left(b + \left\lfloor \frac{r+n}{9} \right\rfloor, r+n \bmod 9 \right).$$

The composition sums lengths.

Let

$$\kappa_r(n) = \left\lfloor \frac{r+n}{9} \right\rfloor.$$

Lemma 2.114 (Cocycle of the odometer). *If $r_1 = r + n_1 \bmod 9$, then*

$$\kappa_r(n_1 + n_2) = \kappa_r(n_1) + \kappa_{r_1}(n_2).$$

Proof. We write $r + n_1 = 9\kappa_r(n_1) + r_1$, with $0 \leq r_1 < 9$. Adding n_2 and again dividing by nine yields the identity. $\square$

For a state x whose base contains θ , we define

$$Q_0(x) = (\beta(\theta), \tilde{\theta} \bmod 9).$$

A TPK arrow has length equal to the number of observable generators that compose it.

Theorem 2.115 (Phase-length functor). *The assignment sending a state x to $Q_0(x)$ and an arrow of length n to the odometric arrow of length n defines a functor*

$$Q : \text{TPK} \longrightarrow \mathcal{O}_{9,12}.$$

Proof. Each generator increases θ by one unit. After n steps, the phase residue increases by n modulo nine and the sector increases by $\lfloor (r+n)/9 \rfloor$. The identity has zero length and the lemma 2.114 proves compatibility with composition. $\square$

After nine steps,

$$(b, r) \mapsto (b+1, r).$$

After one hundred and eight,

$$(b, r) \mapsto (b, r).$$

These identities belong to the odometric quotient. They do not assert that the cylinder, boundary, genealogical register, or vertical memory have returned.

2.15.6 Memory quotient of order 1 and predictive sufficiency of the modal projection

Let

$$X_{9,12} = \{(a, m) : a \in \mathbb{Z}/9\mathbb{Z}, 0 \leq m < 12\}$$

with successor

$$S(a, m) = \begin{cases} (a, m+1), & m < 11, \\ (a+1 \bmod 9, 0), & m = 11. \end{cases}$$

We define the observables

$$\beta_{\text{pred}}(a, m) = 4a \pmod{12}, \quad d_{\text{pred}}(a, m) = 1 + (m + \beta_{\text{pred}}(a, m) \pmod{12}).$$

For an observable q , the inclusive future word of horizon h is

$$Q_h^q(x) = (q(x), q(Sx), \dots, q(S^h x)).$$

Theorem 2.116 (Minimum predictive quotient). *The observables β_{pred} , d_{pred} , and the pair $(\beta_{\text{pred}}, d_{\text{pred}})$ produce the same stable quotient*

$$\pi_{36}(a, m) = (a \bmod 3, m).$$

It has thirty-six states and fibers of cardinality three. The minimal horizons are 11, 8, and 0, respectively.

Proof. For a finite system (X, T) , let

$$E_h = \text{Eq}(Q_h^q).$$

Then

$$E_{h+1} = E_h \cap (T \times T)^{-1}(E_h).$$

The decreasing chain stabilizes and its limit is the largest T -compatible equivalence contained in $\text{Eq}(q)$. In the declared system, the exact enumeration produces the following number of classes:

h	0	1	2	3	4	5	6	7	8	9	10	11
$\#(X/E_h^\beta)$	3	6	9	12	15	18	21	24	27	30	33	36

and

h	0	1	2	3	4	5	6	7	8
$\#(X/E_h^d)$	12	15	18	21	24	27	30	33	36

The pair already distinguishes the thirty-six classes at horizon zero. In all cases, the stable classes preserve m and $a \bmod 3$; the three possibilities for a with the same residue modulo three form each fiber. $\square$

We denote this object by C_{36}^{pred} . No is the boundary R_{36} , that will contain three ordered channels of six trits, nor a census of thirty-six elements of another construction.

2.15.7 Central extension of the calendar

We define

$$\iota : C_{12} \longrightarrow C_{108}, \quad \iota([b]_{12}) = [9b]_{108},$$

and

$$p : C_{108} \longrightarrow C_9, \quad p([\theta]_{108}) = [\theta]_9.$$

The exact sequence is obtained.

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0. \quad (2.84)$$

A section of sets $s([r]_9) = [r]_{108}$, $0 \leq r < 9$, induces the cocycle

$$c(r, r') = \left[\left[\frac{r + r'}{9} \right] \right]_{12}.$$

Proposition 2.117 (The extension does not split). *The sucesion equation (2.84) not is isomorfa to the extension product $C_{12} \times C_9$.*

Proof. The exponent of C_{108} is 108. The exponent of $C_{12} \times C_9$ is $\text{mcm}(12, 9) = 36$. Two isomorphic finite groups have the same exponent; therefore, they are not isomorphic. $\square$

Theorem 2.118 (Necessity and exact period of one hundred eight positions). *The translation $\theta \mapsto \theta + 1$ on C_{108} has order exactly 108. Consequently, an iteration n returns the complete calendar if and only if $108 \mid n$.*

Proof. The n -th power sends $[\theta]$ to $[\theta + n]_{108}$. It is the identity for every θ exactly when $[n]_{108} = 0$, that is, when $108 \mid n$. $\square$

In particular, 1008 is not another return of the calendar:

$$1008 \equiv 36 \pmod{108}.$$

After 1008 steps, the phase modulo nine returns, but the coordinate of C_{108} has shifted by thirty-six positions. Naturally, 216, 324, $\dots$ also return the calendar, but they are repetitions of the fundamental circuit of length 108.

2.15.8 Unbounded Growth of Memory Depth Under Phase Returns

Let $q_{\text{mem}} \in \mathbb{Z}_{\geq 0}$ be the unbounded number of nonadic turns, and let

$$\beta = [q_{\text{mem}}]_{12}.$$

Then a nine-step turn satisfies

$$q_{\text{mem}} \mapsto q_{\text{mem}} + 1, \quad \beta \mapsto \beta + 1 \pmod{12}.$$

After one hundred eight steps,

$$\beta \mapsto \beta, \quad q_{\text{mem}} \mapsto q_{\text{mem}} + 12.$$

The calendar returns; the vertical memory does not.

Corollary 2.119 (Hierarchy of returns). *In the phase quotient, 9 steps constitute one turn. The composite morphisms*

$$\mathcal{K}_{27} = F_{\text{obs}}^{27}, \quad F_{\text{obs}}^{54}, \quad F_{\text{obs}}^{108}$$

contain, respectively, three, six, and twelve nonadic turns. None erases the genealogical record or the boundary by definition.

2.15.9 Minimal type system of the integral holonomic state and compatibilities between its readers

The following table answers explicitly which internal objects already exist in the TPK before introducing seeds:

Objeto u operador	Dominio y codominio	Función
X_9	$(\mathbb{Z}/9)^2$	soporte APP
T_N, T_E, T_S, T_O	$X_9 \rightarrow X_9$	traslaciones cardinales
M_{ph}	$\mathcal{D}_9 \rightarrow \text{TRIT}$	selección de hoja
F_{obs}	$\mathcal{Q}_{\text{obs}} \rightarrow \mathcal{Q}_{\text{obs}}$	sucesor observable
(ρ_9, q_9)	$\mathbb{Z}_{>0} \rightarrow \mathcal{D}_9 \times \mathbb{Z}_{\geq 0}$	fase y altura
(r_3, c_3)	$\mathbb{Z} \rightarrow \text{TRIT} \times \mathbb{Z}$	orientación y carry
(q_{1000}, r_{1000})	$\mathbb{Z} \rightarrow \mathbb{Z} \times \{0, \dots, 999\}$	bloque decimal y cociente
Q	$\text{TPK} \rightarrow \mathcal{O}_{9,12}$	fase y longitud
C_{36}^{pred}	cociente de $X_{9,12}$	memoria predictiva mínima
C_{108}	calendario cíclico	extensión no escindida
q_{mem}	$\mathbb{Z}_{\geq 0}$	memoria vertical no acotada
Ext_r	$\text{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'}$	prolongaciones compatibles

2.15.10 Obstructions of the readers and the TPK calendar: Euclidean quotients, C_{36}^{pred} , extension C_{108} , and vertical memory

Test criterion 2.120 (Reader Refutation Criteria and Calendar). The unit is refuted if:

1. a residual reader is presented as a complete transition;
2. the quotient of some subsequently used Euclidean division is omitted;
3. C_{36}^{pred} is identified with R_{36} ;
4. it is asserted that the extension C_{108} is the product $C_{12} \times C_9$;
5. complete return is declared for a number not divisible by 108;
6. the return of β fuerza $q_{\text{mem}} = 0$;
7. $\mathcal{K}_{27}$ is presented as a distinct clock and not as a power of the transport.

2.16 Complete operatorial structure of APP and arithmetic-geometric hierarchy

The operatorial development of APP fixes the hierarchy $4 \rightarrow 2 \rightarrow 6 \rightarrow 54$, the arithmetic projectors, the central block, and the local geometry of paths. These consequences remain subordinated to the already constructed formal kernel and do not introduce a second origin of APP.

The next unit will construct the quadruple subdivision, the cyclotomic functor, and the internal channels of ninety and one hundred twenty positions. Only afterward will the exhaustive space of seeds be introduced.

Operator architecture of selection, transport, and update

3.1 Categorical structure of TPK: integral holonomic states, transport morphisms, and operator classification

The TPK is the categorical container that transports over APP the oriented states of TRIT. Its definition precedes the inventory of realizations: first objects, morphisms, composition, and operatorial classes are fixed; only afterward are the operators that realize each class introduced.

Definition 3.1 (TPK as a category over the observable base with fiber family). Let $\mathbf{P}_{\text{obs}}$ be the category of observable paths over the Cayley graph of APP. For each configuration q the fiber is fixed

$$\mathcal{F}_q = \mathbf{O}_q \times \mathbf{H}_q \times \mathbf{C}_q \times \mathbf{S}_q \times \mathbf{B}_q \times \mathbf{\Lambda}_q,$$

whose factors retain, respectively, orientation and sheet; residue and quotient; carry; signature; cylinder and boundary; and transport register. A TPK object is a pair $x = (q, f)$, with $f \in \mathcal{F}_q$. A morphism $\Gamma : x \rightarrow y$ is a route $\gamma : q \rightarrow q'$ of $\mathbf{P}_{\text{obs}}$ together with the fiber transport that preserves the declared extension relations between f and f' . The identity is the empty route with identity transport, and composition is the concatenation of routes with composition of the fiber transports. The projection

$$p : \mathcal{X}_{\text{TPK}}^{\text{enr}} \longrightarrow \mathbf{P}_{\text{obs}}$$

sends each state to its observable configuration and each morphism to the path that transports. This definition fixes a category over the base and a family of fibers; the existence is not presupposed the existence of Cartesian lifts for every flecha of $\mathbf{P}_{\text{obs}}$.

Definition 3.2 (Ontological–operational classification of the TPK). The internal structure of the TPK is organized into eight classes distinguished by type of domain, codomain, and function:

1. **Support and state.** The base is $\mathbf{P}_{\text{obs}}$ and the carrier is $\mathcal{X}_{\text{TPK}}^{\text{enr}} \rightarrow \mathbf{P}_{\text{obs}}$. This class contains objects; the next seven contain maps between objects or readers of those objects.
2. **Internal selection.** The selector

$$M_{\text{ph}} : \{1, \dots, 9\} \longrightarrow \{+1, -1, 0\}$$

activates additive evaluation, multiplicative evaluation, or the threshold passage. It acts on the leaf coordinate and does not by itself modify the position in APP.

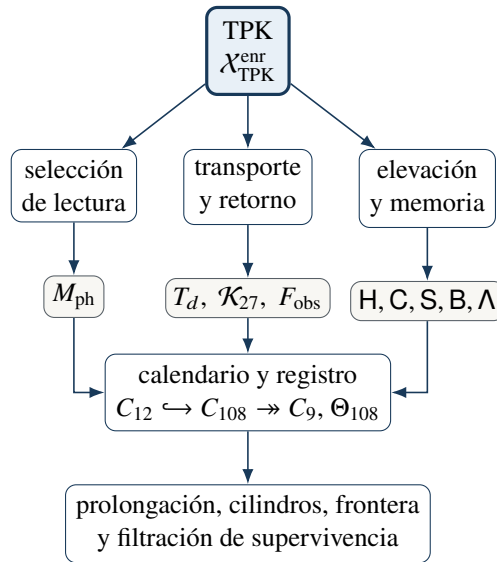
3. **Transport.** For each cardinal direction d , $T_d : X_9 \rightarrow X_9$, $p \mapsto p + \delta(d)$, it transports the position. The evolution $F_{\text{obs}} : \mathbf{Q}_{\text{obs}} \rightarrow \mathbf{Q}_{\text{obs}}$ composes selection, translation, orientation, and phase advance.
4. **Readers and quotients.** ρ_9 publishes the nonadic class; (r_3, c_3) conserves tritic residue and quotient; and (q_{1000}, r_{1000}) separates decimal block and carry. They are reading maps, not state transitions.
5. **Memory and boundary.** The transports of $\mathbf{H}_q, \mathbf{C}_q, \mathbf{S}_q, \mathbf{B}_q$ and $\mathbf{\Lambda}_q$ update quotient, carry, signature, cylinder, boundary, and register. Their visible projection may coincide even though the integral holonomic states differ.
6. **Periodicity and return.** The extension

$$0 \longrightarrow C_{12} \longrightarrow C_{108} \longrightarrow C_9 \longrightarrow 0$$

types sector, calendar, and visible phase. The nonadic return restores the phase; the periodicity of one hundred eight iterations restores the calendar; neither erases by definition the boundary memory. The loop $\mathcal{K}_{27} = F_{\text{obs}}^{27}$ is a composite morphism, not another clock.

7. **Prolongation.** The relations $\text{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'}$ select compatible continuations. The cylinders and their inverse system publish finite survival and, under the stated compatibility hypotheses, a global section.
8. **Outputs.** The evaluation functors contract the same integral holonomic state toward the Archimedean output, the exceptional incidence output, or the atlas classifiers. These outputs are reading codomains; they do not constitute new containers alongside TPK.

Each operator appearing below is determined by the sextuple (D, C, f, I, L, M) : domain D , codomain C , rule f , preserved invariants I , published or lost information L , and retained memory M . Equality of cardinalities, periods, or historical names does not identify two operators with distinct sextuples. Once this classification has been fixed, the following diagram summarizes five functional groupings of realizations that refine the eight operator classes of TPK:



The diagram establishes a containment, not a succession of theories. In particular, the internal selector, cardinal translations, stroboscopic observation, and modular readers preserve the types fixed in [definition 3.2](#). TPK coordinates their compositions and preserves their provenance in the same integral holonomic state.

Proposition 3.3 (Separation of selection, motion, and remark). *The selector M_{ph} , cardinal transport T_d , and sampling Obs_3 are noninterchangeable morphisms: the first chooses the active evaluation; the second changes position and winding; the third publishes a temporal subsequence. Their composition within the TPK determines an observable route, while the fiber $\mathcal{F}_q$ preserves the data required to distinguish prolongations with the same endpoint.*

Proof. The three morphisms have different codomains: M_{ph} takes values in the ternary activation set, T_d is an automorphism of the torus, and Obs_3 acts on sequences. An identification between any two of them would violate the domain or codomain typing. The extension by fibers further incorporates residue–quotient, carry, boundary, and register, so that projection to the endpoint of the path is an explicit forgetful map. $\square$

Canonical nomenclature

The public name of the container is *TPK*. When it is necessary to name the transported data, *integral holonomic TPK state* or $\mathcal{X}_{\text{TPK}}^{\text{enr}}$ will be used. The morphism M_{ph} designates exclusively internal phase selection.

3.2 Fundamental operators of the TPK: domains, actions, invariants, and memory

The definition 3.2 first fixed the eight ontological–operatorial classes: support and state; selection; transport; readers and quotients; memory and boundary; periodicity and return; prolongation; and outputs. This section defines their realizations. For each operator, the sextuple (D, C, f, I, L, M) of domain, codomain, rule, invariants, published or lost information, and preserved memory is declared. The subsequent nominal census groups symbols already defined by these six data and does not replace their domains or rules.

Definition 3.4 (Typed Record for a TPK Operator). A canonical operator is a sextuple

$$O = (D_O, C_O, f_O, I_O, L_O, M_O),$$

where $f_O : D_O \rightarrow C_O$ is the action rule, I_O is the family of relations it preserves, L_O declares the published coordinates and the equivalence relation induced by forgetting, and M_O enumerates the data remaining in the integral holonomic state after publication. Two realizations represent the same operator only when there is a typed equivalence intertwining their rules and invariants; sharing a period, cardinality, or visible output is not enough.

3.2.1 Modular readers built

The nonadic reduction is fixed by the published representative.

$$\rho_9(n) = 1 + ((n - 1) \bmod 9) = n - 9 \left\lfloor \frac{n - 1}{9} \right\rfloor, \quad (3.1)$$

so that the null class is written as 9. For $n \geq 1$, the digital root satisfies

$$\text{dr}_9(n) = \rho_9(n),$$

but the two expressions preserve distinct semantics: the first is a section of the quotient; the second publishes the iterated sum of digits. The tritic lift writes uniquely

$$n = r_3(n) + 3c_3(n), \quad r_3(n) \in \{-1, 0, +1\}, \quad c_3(n) \in \mathbb{Z}, \quad (3.2)$$

and simultaneously preserves oriented residue and carry quotient. Finally, the decimal-block reader is the Euclidean division

$$N = 1000q_{1000}(N) + r_{1000}(N), \quad 0 \leq r_{1000}(N) < 1000, \quad (3.3)$$

therefore basis 1000, residue modulo 1000 and carry q_{1000} are three data related and not a same object.

3.2.2 Classification of Operator Realizations by Domain and Function

The preceding definitions govern the reading of the table. Its fourteen functional groupings refine the eight operator classes and summarize the realizations that share a function; a class may therefore occupy more than one row. The table does not operate as a replacement definition. Symbols that share a cardinal remain separated by domain and codomain.

Functional grouping	Operators and Spaces	Domain, Action and Invariants
State and route memory	$\mathcal{X}_{\text{TPK}}^{\text{enr}}$, fiber $\mathcal{F}_q$, route memory	They preserve position, leaf, orientation, residue, quotient, carry, signature, boundary, cylinder, prefix, genealogy, and memory.
Sheet selection	$M_{\text{ph}} : \{1, \dots, 9\} \rightarrow \{-1, 0, +1\}$	It activates the additive evaluation, the multiplicative evaluation, or the threshold passage; it does not by itself move the state.
Transporte espacial	T_N, T_E, T_S, T_O , reversible two-cursor engine, F_{obs}	It realizes the translations of the Cayley graph, updates sheet and orientation and preserves each transition of the path.
Lectores modulares	$\rho_9, (r_3, c_3), (q_{1000}, r_{1000})$	Publish nonadic phase, tritic orientation and decimal block without erasing the lift quotients.
Block and memory	$\mathcal{K}_{27} = F_{\text{obs}}^{27}$, memory filter $\mathcal{N}_{27}$	The first composes twenty-seven transports; the second selects the memory events. Equal cardinality does not imply an equal operator.
Twelve-phase state	$\mathcal{R}_{12}$, centers $\theta_m = 30m - 10$ degrees, refinement $\mathcal{R}_{120}$	Organize the twelve phases, their opposition, and the angular refinement in increments of three degrees.
Periodicity and depths	$C_{12} \hookrightarrow C_{108} \twoheadrightarrow C_9, \Pi_{108}^{\text{cur}}, w_{108}, w_{1080}, \Gamma_0(1080)$	It distinguishes the total periodicity of order 108, the cursor projection, the words of 108 and 1080 trits, and modular level 1080.
Channels (90/120)	mask C_m , S_{90}, S_{120} , orbital grammar, supertrace, and invariants v_{120}, v_{270}	It separates the two genealogies, their geometric tails, their weights, and the coordinates that they extract from the state.
Bidirectionality	$P_{\pm}$, route inversion, $\text{Ext}_{\pm}, N_{\pm}, \beta, C_M = -\text{rev}$	It implements, respectively, spectral exchange, spatial outward passage and return, future/past prolongation, valence, balance, and word conjugation.
Emission and coinduction	L_0, L_1, R_{36}, G_9 , partition $3^6 = 729$, $\varprojlim \text{Cyl}_m$	It lifts the seed, opens the boundary, and exhaustively enumerates the finite prolongations. The numerical limit and the enriched lift belong to distinct types.
Numeric Readers	$\mathcal{C}_{\pi}, \mathcal{P}_e, F_{\text{av}}$	Through rational forks, factorial recurrence, and Fibonacci self-scaling, they construct the published sections of π, e, φ .
Exceptional Projection	Witt vector lifting, weight enumerator, and non-selective shadow	Transports the same state to ternary incidence without intervening in the Archimedean selection of digits.
Atlas and realization	route signature, atlas (81/56/13/324), involution C_M , electronic center, and fiber operators	Convert the route state into classifiers, characters, and operators before applying any dimensional chart.
Finite validation	exhaustive censuses, ablations, and causal independence	It checks coverage and uniqueness at the computed depth, and verifies that no target prefix intervenes in the generating map.

3.2.3 Functional Index of the Canonical Operators

The preceding classification organizes the realizations by function. The following table gathers the operators after the categorical definition and the constructed readers. Each row declares domain, codomain, rule, and invariants; the symbols make it possible to recognize a single definition throughout its different realizations without identifying operators of different types.

#	Canonical Operator	Symbol	Capa tipada
1	Additive–multiplicative arithmetic cell of order nine	$\mathcal{A}_9$	arithmetic support
2	Residual decomposition and nonadic quotient	(ρ_9, q_9)	reduction and elevation
3	Ternary oriented reduction	ρ_3^{or}	reduction and elevation
4	Ternary-decimal positional reader of base thousand	ι_{1000}	reduction and elevation
5	Cardinal Action Oriented on the Arithmetic Torus	$T_d, d \in \{N, E, S, O\}$	transporte cardinal
6	Additive–multiplicative local feedback	F_{loc}	transporte cardinal
7	Minimum emitter of two cursors	T_{min}	finite emission
8	Decimal Arithmetic Sign Coupler	G	finite emission
9	Ternary projection of the catalog	Ψ_6	finite emission
10	Local Surjectivity of Cardinal Transport	$\mathcal{R}_3$	transporte cardinal
11	Transportation loop of twenty-seven iterations	$\mathcal{K}_{27}$	transporte cardinal
12	Integral holonomic state of the topological phase kernel	$\chi_{\text{TPK}}^{\text{enr}}$	integral holonomic state
13	Hensel State Transition	$\mathcal{H}$	integral holonomic state
14	Exact Transition of Ternary-Decimal Cylinders	$C_{3,1000}$	integral holonomic state
15	Joint boundary signature	Σ_{B}	integral holonomic state
16	Relationship of extension and nonadic survival	$\text{EF-}G_9$	prolongation and monodromy
17	Nonadic monodromy with boundary holonomy	$\mathcal{M}_9$	prolongation and monodromy
18	Global Section of the Inverse Survivor System	$X_{\chi} = \varprojlim_m \text{Surv}_m^{(\chi)}$	prolongation and monodromy
19	Bivariant Category of Arithmetic Paths	$\mathcal{P}_{\text{APP}}^{(2)}$	transporte cardinal
20	Observable chronology of two cursors	F_{obs}	transporte cardinal
21	Decimal Alphabet of Forty-Two Trinities	$\mathcal{A}_{42}$	finite emission
22	Typed extension relation of fibers	Ext_r	integral holonomic state
23	Nonadic transfer of compatible fibers	K_{ph}	prolongation and monodromy
24	Refinement of Seeds by Advancement and Rotation	(P, H, Q)	transporte cardinal
25	Mode and Orientation Cocycles	$(c_{\text{mode}}, c_{\text{or}})$	integral holonomic state
26	Triadic prolongation of twenty blocks	E_{20}	prolongation and monodromy
27	Dodecaphonic oriented system	$\mathcal{R}_{12}$	dodecaphonic reading
28	Signed harmonic dodecaphonic observable	U_{12}^{sgn}	dodecaphonic reading
29	Harmonic transformation by three-step orbits	$\mathcal{H}_{12}$	dodecaphonic reading
30	Cyclic difference transversal extractor	$\mathcal{E}_{3,4}$	dodecaphonic reading
31	Dodecaphasic closure recurrence in base thousand	C_{α}	close in base thousand
32	Enhanced history transducer to angular character	$(L_{\text{tick}}, \chi_{\text{mass}})$	interfaz dimensional
33	Bidiirectional lattice of index twelve	$\mathcal{W}_{\pm}$	interfaz dimensional
34	Nonadic Atlas of Extension Families	$\mathcal{A}_{\text{species}}$	interfaz dimensional

The symbol L_{tick} denotes the iteration and return reader. Its codomain is the temporal memory of the state; it does not define an independent physical time unit.

The internal selector M_{ph} belongs to the TPK. Cardinal transport T_d occupies another factor of the transition $\mathcal{U}_t$ and effects the motion on APP. This separation preserves a single carrier, distinguishes phase selection from transport, and preserves the complete architecture by types.

Finite depth and enriched limit

The enumeration, censuses, and contraction of numerical cylinders are exact results in their domains. The existence and uniqueness of an enriched section jointly subject to all survival predicates follow when the lifting fibers are nonempty, singletons, and compatible at every depth. A calculation on a finite window verifies the fibers in that window, but does not by itself imply the quantifier over all depths.

Theorem 3.5 (Composition of the internal dynamics). *A transition observable of the TPK is the composition typed*

$$\mathcal{U}_t = \text{Rec}_t \circ \text{Mem}_t \circ T_{d(t)} \circ M_{\text{ph}}(g(t))$$

on $\mathcal{X}_{\text{TPK}}^{\text{enr}}$, followed, when appropriate, by a modular reader, a cylindrical extension, or one of the two projections. Each factor acts on a distinct component of the state; for that reason the selector, the movement, the clock, the memory, and the publication cannot collapse into a single modular reduction.

The preceding table enumerates the operator families. Equivalences among local realizations must intertwine their domains, rules, and invariants. Subsequent developments select compositions of these operations; they introduce no fourth primitive object alongside APP, TRIT, and TPK.

Transport, selection, and prolongation in the Topological Phase Kernel

The acronym TPK designates the *Topological Phase Kernel*, or the topological phase kernel.¹ It does not denote a single 108-step algorithm, a table of results, or the catalog of 468 emissions. It is the system of operators that enables the additive and multiplicative readings to coexist over the APP path category without identifying their genealogies.

The word *topological* has a precise content here. The support of APP is the Cayley graph of the quotient group

$$X_9 = (\mathbb{Z}/9\mathbb{Z})^2$$

with its cardinal edges. A closed route admits a winding class after being lifted to cover $\mathbb{Z}^2$. TPK transports that class, the route orientation, and phase holonomy. Two paths ending in the same cell may therefore represent distinct states. The phase kernel is precisely the information that remains invisible when publishing only the final cell or its residue.

The dynamical basis is the free category $\mathbf{P}_{\text{APP}}$ of the cardinal graph of APP. Since the two arithmetic evaluations do not induce the same partition, transport resides on the bivariate category

$$\mathbf{P}_{\text{APP}}^{(2)} = \mathbf{P}_{\text{APP}} \times \mathbf{P}_{\text{APP}}.$$

The first component is evaluated through the additive sheet Σ and the second through the multiplicative sheet Π . This separation is constitutive, but it does not duplicate APP: it records two ordered readings of one and the same matrix of paths. The multiplicative component is the principal table; the additive component preserves the law of accumulation that generates its rows and makes it possible to measure what information a visible projection eliminates.

4.1 Residual reductions and decimal representation

The architecture uses three reductions, each with a distinct purpose:

$$\begin{aligned} \mathbb{Z} &\xrightarrow{(\rho_9, q_9)} \{1, \dots, 9\} \times \mathbb{Z}, & \rho_3^{\text{or}} : \mathbb{Z}/3\mathbb{Z} &\longrightarrow \{-1, 0, +1\}, \\ (d_1, d_2, \dots) &\in \{0, \dots, 999\}^{\mathbb{N}} &\xrightarrow{t_{1000}} &\sum_{m \geq 1} d_m 1000^{-m}. \end{aligned}$$

The ternary orientation is

$$\rho_3^{\text{or}}([0]) = 0, \quad \rho_3^{\text{or}}([1]) = +1, \quad \rho_3^{\text{or}}([2]) = -1.$$

The first separates the visible residue from the nonadic quotient; the second assigns the oriented type of the transition; the third groups the Archimedean reading into decimal triads, whose register preserves the carry. None can replace another. In particular, reduction modulo three does not reconstruct the quotient modulo nine, and a decimal triad is not a six-trit word. Addition accumulates displacements; multiplication folds classes and contracts the noninvertible rows. The need for TPK arises because the two readings share a support but do not share a partition, period, or quotient.

¹The denomination likewise preserves the authorial dedication to Tan Peng Kian.

4.2 Correspondence between TPK symbols, domains, and compositions

Historical names are fixed here by object, not by numerical similarity. This table forms part of the public definition of TPK.

Name	Object	Function within the core
nonadic coordinate	$\rho_9(n)$ together with $q_9(n)$	Separate the representative visible modulo 9 of the quotient that preserves the memory of the cruce of boundary. No is a root digital.
ternary temporal selector	$M_{ph}(t) = \varepsilon(t) \in \{+1, -1, 0\}$	At 1, 4, 7 it activates additive accumulation; at 2, 5, 8 it activates multiplicative accumulation; at 3, 6, 9 it maintains both cursors and records the threshold. It does not denote the spatial directions.
transporte cardinal	T_N, T_E, T_S, T_O	It shifts the position on the APP torus and preserves the path word and its winding. It is independent of the temporal selector.
ternary reader	ρ_3^{or}	It converts the three residual classes into orientation 0, +1, -1. It does not by itself reconstruct the nonadic quotient.
reader in base thousand	ι_{1000}	It groups the Archimedean publication into decimal triads and transports their carries. It is a positional reader, not a congruence “modulo 1000”.
return of twenty-seven	$\mathcal{K}_{27} = F_{\text{obs}}^{27}$	It composes twenty-seven positional and oriented updates. Its fourth iterate restores the observable phase, while the complete register preserves the accumulated memory.
dodecaphonic register	$\Lambda \in \mathcal{R}_{12}$	It preserves six positions, five differences, and one total. It is the support of the reversible transport $1 + 5 + 6 = 12$.
bidirectional reading	q_+, q_- and its conjugate weights	It simultaneously preserves the outward and return orientations. In Dimensional Mechanics, a signature (n_A, s_C) first produces $W_{\pm} = 6n_A \pm s_C$; only afterward does $s_C/6$ appear in the angular character.

The temporal selector, cardinal transport, and ternary orientation are therefore three distinct maps. Their separation preserves the difference between *when* a reading is activated, *in which direction* the route advances, and *with which orientation* the step is recorded. TPK contains all three and preserves their relations.

4.3 Cardinal Action and Arithmetic Feedback

Let

$$\nu_N = (-1, 0), \quad \nu_E = (0, 1), \quad \nu_S = (1, 0), \quad \nu_O = (0, -1)$$

the four generators cardinal. For $d \in \{N, E, S, O\}$,

$$T_d(i, j) = (i, j) + \nu_d \pmod{9}.$$

The letter O means west. These directions are displacements over APP and not names of the channels π, e, φ .

A word $w = d_1 \cdots d_r$ determines

$$\gamma(x_0; w)_k = x_0 + \sum_{j=1}^k \nu_{d_j} \pmod{9}.$$

When it closes, its lift preserves

$$\text{wind}(w) = \frac{1}{9}(\#S - \#N, \#E - \#O) \in \mathbb{Z}^2.$$

The bidirectional reading preserves word, orientation, and sheet. Its formal reversal inverts the order and exchanges $N \leftrightarrow S, E \leftrightarrow O$. Its eventual reading as advance–delay is a later physical interface, not part of this route definition.

The locally observable state is

$$s_t = (i_t, j_t, m_t), \quad m_t \in \{\Sigma, \Pi\}.$$

The active sheet is evaluated,

$$a_t = \rho_3^{\text{or}}(m_t(i_t, j_t)),$$

and the mode is updated via

$$m_{t+1} = \begin{cases} \Sigma, & a_t = +1, \\ \Pi, & a_t = -1, \\ m_t, & a_t = 0. \end{cases}$$

Thus, each cardinal step modifies both the position and the future arithmetic reading. The path is not superimposed on a previously constructed word: it is the positional history that produces it.

Theorem 4.1 (Surjectivity of the sampled three-step factor). *For each of the $162 = 9^2 \cdot 2$ states (i, j, m) and each target oriented trit $b \in \{-1, 0, +1\}$, there exists a route $u \in \{N, E, S, O\}^3$ whose third step emits b . In the 486 state–target cases, the number of admissible routes is between 8 and 40. Consequently, for every target succession $(b_k)_{k \geq 1}$, there exists a cardinal history whose output satisfies*

$$(a_3, a_6, a_9, \dots) = (b_1, b_2, b_3, \dots).$$

The two intermediate trits a_{3k-2} and a_{3k-1} of each block are not prescribed by this assertion.

Proof. The $4^3 = 64$ three-step routes from each state are enumerated. The included exhaustive certificate records, for each of the $162 \cdot 3$ cases, the route executed, the final state, and the total number of witnesses; its indices 0, 1, 2 are transported by ρ_3^{or} to 0, +1, −1. All counts are positive; the minimum is 8 and the maximum is 40. After each block, the same result is applied to the state reached. The choice, for example, of the first witness in lexicographic order recursively produces a history for any target sequence. $\square$

Surjectivity concerns the factor observed every three steps, not the word emitted at unit time. This theorem establishes that sampled capacity of the route support. The selection of the sections π, e, φ belongs to the enriched EF–G9 relation defined later: capacity for realization and selection that is genealogical are two distinct operators.

4.4 Chronological evolution of the cursor pair and determination of the observable state

The calendar is formulated over

$$\Theta_{108} = \mathbb{Z}/108\mathbb{Z}, \quad \beta(\theta) = \left[\left[\frac{\tilde{\theta}}{9} \right] \right]_{12}, \quad r(\theta) = 1 + (\theta \bmod 9),$$

where $\tilde{\theta} \in \{0, \dots, 107\}$. The active signature is

$$\varepsilon(r) = \begin{cases} +1, & r \in \{1, 4, 7\}, \\ -1, & r \in \{2, 5, 8\}, \\ 0, & r \in \{3, 6, 9\}. \end{cases}$$

The observable space of two cursors is

$$\mathcal{Q}_{\text{obs}} = X_9^2 \times \{N, E, S, O\}^2 \times \Theta_{108}.$$

If $\varepsilon = +1$, the additive cursor advances; if $\varepsilon = -1$, the multiplicative cursor advances; if $\varepsilon = 0$, neither advances. The two orientations are reversed after steps 27, 54, 81, 108. This rule defines a permutation

$$F_{\text{obs}} : \mathcal{Q}_{\text{obs}} \longrightarrow \mathcal{Q}_{\text{obs}}.$$

Positions and orientations are restored after 54 steps and the complete phase after 108:

$$F_{\text{obs}}^{108} = I.$$

Therefore, the chronology of a single route is reversible and does not by itself select a distribution on the continuations; selection of all compatible prolongations belongs to the fiber transfer operator defined below.

4.5 Finite emitter associated with the cursor pair and generation of admissible states

A cursor seed is

$$\mathcal{S} = \mathbb{T}_9^2 \times \{N, E, S, O\}, \quad |\mathcal{S}| = 324.$$

Minimal TPK receives an additive cursor $P \in \mathcal{S}$ and a multiplicative cursor $M \in \mathcal{S}$. Their six-window signatures are $s(P)$ and $e(M)$. In each coordinate one applies

$$G(s, e) = 100s + 10((s + e) \bmod 10) + ((3s + 5e + 1) \bmod 10).$$

The figure of hundreds recovers s ; known s , the of tens recovers e . Therefore G is injective on the effective signatures. The emitter factorizes as

$$\mathcal{S}^2 \xrightarrow{s \times e} \text{im } s \times \text{im } e \xrightarrow{G} \mathcal{U}_6,$$

with

$$|\mathcal{S}^2| = 104\,976, \quad |\text{im } s| = 18, \quad |\text{im } e| = 26, \quad |\mathcal{U}_6| = 18 \cdot 26 = 468.$$

The factorization $468 = 18 \cdot 26$ is therefore a decomposition of the object: each emission conserves an additive and a multiplicative signature.

The projection

$$\Psi_6 : \mathcal{U}_6 \longrightarrow \mathbb{F}_3^6, \quad \Psi_6(U) = -U \pmod{3},$$

has an image of 243 words. The chain typed is

$$104\,976 \text{ seeds} \longrightarrow 468 \text{ decimal emissions } U_6 \longrightarrow 243 \text{ ternary words } w_6.$$

The ambient space $\mathbb{F}_3^6$ contains 729 words; it is not identified with the effective image of 243. Nor are U_6 and w_6 the same type.

The emitter minimo repite, in the chart dodecaphasic of catalogo,

$$U_{12}^{\text{cat}} = U_6 || U_6.$$

Its visible period six explains why it cannot replace the signed state U_{12}^{sgn} . This difference is not an insufficiency of TPK: it separates the catalogue projection from the enriched history.

Each emitted decimal block belongs to an exact alphabet of $7 \cdot 6 = 42$ triads. If $\overline{d_1 d_2 d_3}$ is one of them,

$$\boxed{d_3 \equiv 5d_2 - 2d_1 + 1 \pmod{10}.}$$

The sweep of the 104 976 seeds produces no triad outside this law. It is therefore useful to distinguish the complete chain of quotients:

$$\mathcal{S}^2 \longrightarrow \Omega_{\text{seed}} \longrightarrow \mathcal{U}_6 \xrightarrow{\Psi_6} \text{im } \Psi_6 \subset \mathbb{F}_3^6.$$

The microscopic presentations, the 468 decimal emissions, the 243 ternary words, and the environment of 729 words are four distinct objects.

Orbital Commutativity of Typed Morphisms. Let $W_{243} := \text{im } \Psi_6 \subset \mathbb{F}_3^6$. The complete funnel and its dihedral quotient are represented without turning inclusions into selections:

$$\underbrace{\mathcal{S}^2}_{104\,976 \text{ semillas}} \xrightarrow[\text{fiber census}]{\text{emission}} \underbrace{\mathcal{U}_6}_{468 \text{ } U_6} \xrightarrow[\text{visible projection}]{\Psi_6} \underbrace{W_{243}}_{243 \text{ } w_6} \hookrightarrow \underbrace{\mathbb{F}_3^6}_{729 \text{ palabras}}, \quad W_{243} \xrightarrow{/D_3^{\text{cat}}} \underbrace{O_{43}}_{43 \text{ orbits}}. \quad (\text{TPK-embudo})$$

The first arrow groups initial conditions by the emission they produce; the second forgets the decimal presentation; the third is an inclusion in the algebraic ambient space; the last is a quotient by the action D_3^{cat} on W_{243} . In particular, there is no implicit quotient $729 \rightarrow 43$. The exact census is

$$243 = 38 \cdot 6 + 5 \cdot 3,$$

that is, thirty-eight orbits of size six and five of size three. These 43 orbital classes are not the 43 complete nonadic turns per channel of the thousand-digit G9 witness either: the first 43 belongs to the finite quotient W_{243}/D_3^{cat} ; the second is a depth counter of a certified run.

4.6 Configuration of the integral holonomic TPK state: sheet, orientation, residue, quotient, carry, boundary, path, and memory

The complete dynamic unit is a history, not an isolated word. At depth m we write

$$v_m = (B_m, \delta_m, x_m, C_m, \Sigma_m, \omega_m, \ell_m, g(m), \Lambda_m),$$

where

$$B_m = (u_m^\pi, u_m^e, u_m^\varphi) \in (\mathbb{F}_3^6)^3.$$

Their components have distinct functions:

1. B_m is the joint visible block of the three channels;
2. δ_m retains the new decimal triads and the associated carry;
3. x_m is the deep Hensel state;
4. C_m contains the ternary and decimal cylinders and their integer residues;
5. Σ_m is the signature joint of boundary;
6. ω_m and ℓ_m conserve orientation and bidirectional sheet;
7. $g(m) \in \mathbb{Z}/9\mathbb{Z}$ is the nonadic phase;
8. Λ_m is the dodecaphasic record oriented.

Definition 4.2 (Operatorial presentation and TPK category). The operator inventory of the integral holonomic TPK state is

$$\mathfrak{T}_{\text{HMT}} = (\mathcal{A}_9, \widehat{\mathbb{T}}, T_{N,E,S,O}, T_{\min}, H, C_{3,1000}, \Sigma_B, EF, \mathcal{G}_9, \mathcal{R}_{12})$$

which transports histories $(v_0, \dots, v_m)$ and their truncations. Its formal carrier is the category TPK over the observable base: its objects are integral holonomic states and its morphisms are the prolongations Ext_r , stable under identity and concatenation, defined below. The inventory does not replace that category. The visible projection eliminates components of v_m ; it does not define an equivalence with the complete state.

This definition contains exactly what disappears when cursors are reset or only the residue is conserved: the path, the sheet, the quotient, the carry, the orientation, and the boundary. TPK is, for that reason, the point at which modular arithmetic ceases to be an operation without history.

4.7 Ternary-decimal cylinders and state of extension

Let N be the integer determined by a ternary prefix of length L , and let D be the integer determined by K decimal triads. We define

$$A = N1000^K - D3^L, \quad B = (N+1)1000^K - D3^L.$$

The cylinders intersect exactly when

$$\boxed{A < 3^L, \quad B > 0.}$$

When incorporating a ternary block $u \in \{0, \dots, 728\}$,

$$N' = 729N + u.$$

If no appears a new triad decimal,

$$A' = 729A + u1000^K.$$

Si aparece $\delta \in \{0, \dots, 999\}$,

$$D' = 1000D + \delta,$$

$$A' = 729000A + u1000^{K+1} - \delta 729 3^L.$$

In both cases,

$$B' = A' + 1000^{K+\Delta K}.$$

These integer recurrences transport the decimal datum without turning it into an external target of the update.

The Hensel state preserves the information that does not appear in the residue. The active matrix is

$$A_H = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \quad \det A_H = 1,$$

and the residue–quotient division is

$$y = xA_H, \quad u = y \bmod 3, \quad x^+ = \frac{y - u}{3}.$$

For the three channels, the joint cocycles are additionally formed

$$\begin{aligned} q &= (\pi + e + \varphi) \bmod 3, & c &= \frac{\pi + e + \varphi - q}{3}, \\ a &= (\pi + e - \varphi) \bmod 3, & h &= \frac{\pi + e - \varphi - a}{3}. \end{aligned}$$

The second-order update corrects the next state with c and h . Two branches with the same visible block are therefore not necessarily the same state: they may differ in their Hensel extension and in their cylinders. Since $\det A_H = 1$, the reduction of A_H belongs to $\text{GL}_6(\mathbb{F}_3)$. Hence, before cylinders, boundaries, or survival are imposed, the residual Hensel chart covers exactly $\mathbb{F}_3^6$. This raw coverage does not imply coverage of the filtered system.

Formally, over each observable configuration q a fiber is fixed

$$\mathcal{F}_q = \mathcal{O}_q \times \mathcal{H}_q \times \mathcal{C}_q \times \mathcal{S}_q \times \mathcal{B}_q \times \mathcal{A}_q,$$

corresponding, respectively, to orientation and sheet, residue and quotient, carry, signature, cylinder and boundary, and transport record. For a cardinal arrow $r : q \rightarrow q'$, the admissible prolongation does not reduce to a function on residues; it is the relation

$$\text{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'}, \quad \text{Ext}_{r_2} \circ \text{Ext}_{r_1} \subseteq \text{Ext}_{r_2 \circ r_1}.$$

The carry increment is a cocycle:

$$\kappa(r_2 \circ r_1) = \kappa(r_2) + \mathbf{C}(r_2)(\kappa(r_1)).$$

Thus the forgetful map is typed: projecting an extension to its observable route eliminates components of $\mathcal{F}_q$, but does not authorize identifying both objects.

4.8 Compound returns and transport calendars

The length of a trace does not determine its type. The academic presentation separates the composite return, calendar, trace, and atlas according to their respective preserved states.

The symbol $\mathcal{K}_{27}$ is reserved for the composition of twenty-seven updates of a positional and oriented transport loop. The integer 27 is its composition length; this does not turn it into a trace of 108 iterations or into an infinite emitter. Another construction, historical in origin, groups four loops whose joint periods reach 108; that joint rotor is not denoted by $\mathcal{K}_{27}$ in isolation.

Object	Estado conservado	Significado de 108
Contador simplificado	position and velocity	its real period is 18; it is a historical control.
Rotor set of four loops	four routes, reading, rotation, and motion	the individual periods are 54, 54, 36, 36; the joint state has period 108.
Raw clocked trace	four positions, orientations, and phase	the dynamic content repeats at 54.
Trace of $T_{\min}$	two positions and two directions	the state returns to 54 and the output is $U_6 \parallel U_6$.
Reinitialized pseudocode	cursors reset each window	collapses to $(222, 222, 222, 333, 333, 333)^2$; it does not represent the integral holonomic TPK state.
Rutas condicionadas Sectorial odometer	path and prescribed word $\mathbb{Z}/9\mathbb{Z} \times \{0, \dots, 11\}$	108 is horizon, not period. the complete state has period 108; the sectorial observable, 36.
Projection raw	quotient by forgetfulness of $X_{\text{TPK}}^{\text{enr}}$	does not possess by itself the signed reading.
Traza enriquecida	all the fields of v_m during 108 updates	is the temporal restriction of the integral holonomic TPK state.
Atlas of Routes	81 cells/56 signatures or 324 oriented seeds	is a taxonomy, not an orbit of 108 states.

The loop $\mathcal{K}_{27}$ is the finite transport operator of twenty-seven updates. The windows, joint rotor, and calendrical projection receive distinct symbols, so that the shared length does not replace the domain or composition of each object.

4.9 Fiber transfer and spin memory

The factorization $\mathcal{U}_6 \simeq \mathcal{S}_+ \times \mathcal{S}_\times$ allows one to define, with respect to the uniform count, the conditional expectations

$$(E_+ f)(s, e) = \frac{1}{18} \sum_{s' \in \mathcal{S}_+} f(s', e), \quad (E_\times f)(s, e) = \frac{1}{26} \sum_{e' \in \mathcal{S}_\times} f(s, e').$$

E_+ renews the additive fiber and preserves the multiplicative one; $E_\times$ realizes the dual operation. They are the only Markov projectors that completely marginalize the active coordinate, preserve the inactive one, and preserve the uniform count.

Let $\tau = (+1, -1, 0)^3$ be the phase word on C_9 , and $P_{+1} = E_+$, $P_{-1} = E_\times$, $P_0 = I$. The nonadic transfer is

$$\mathcal{K}_{\text{ph}}((u, r), (v, r + 1)) = P_{\tau(r)}(u, v).$$

It is doubly stochastic, irreducible, of period 9, and has a unique stationary distribution $1/(9 \cdot 468)$. Moreover,

$$\mathcal{K}_{\text{ph}}^9 = E_+ E_\times, \quad (E_+ E_\times)(u, v) = \frac{1}{468}.$$

This equality expresses the simultaneous renewal of all compatible continuations, not the deterministic successor of a single path.

Three geometric operations on the seeds preserve additional information: P , advance in the active orientation; H , a half-turn; and Q , a quarter-turn. The pair (P, H) refines the 26 product signatures into 28 classes. In particular, the signature 555555 splits into three sheets of eight representatives, and

$$18 \cdot 28 = 504 = 468 + 36.$$

The term 36 counts additional sheets of the refinement and is not the object R_{36} . The pair (P, Q) produces 162 classes of two elements, transitively permuted by

$$J(i, j, d) = (7 - i, 7 - j, \bar{d}) \pmod{9}.$$

In a trace of 108 steps, there appear 36 signs of each type, 72 effective changes between the two sheets, and four orientation reversals; the total signed cocycle is zero. These counts are invariants of the evolution record and link nonadic dynamics with the dodecaphase record.

4.10 Tertiary prolongation and joint boundary

Relative to the named seeds, boundaries, and readers of the published profile, the initial visible words are

$$w_6^\pi = 010211, \quad w_6^e = 201101, \quad w_6^\varphi = 121200.$$

After five blocks, the prefixes are

$$\begin{aligned} w_{30}^\pi &= 010211|012222|010211|002111|110221, \\ w_{30}^e &= 201101|121221|102011|012222|102011, \\ w_{30}^\varphi &= 121200|112202|121020|010210|010200. \end{aligned}$$

The first deep boundary is

$$R_{36} = \begin{pmatrix} 222220 \\ 021222 \\ 102011 \end{pmatrix}.$$

For each channel the prefix $w_{36}^\chi := w_{30}^\chi | u_5^\chi$ is defined; R_{36} is the ordered triple of the three sixth blocks. Neither is an emitter nor a terminal state.

For $B_k = (u_k^\pi, u_k^e, u_k^\varphi)$, the signatures

$$q_k = u_k^\pi + u_k^e + u_k^\varphi, \quad a_k = u_k^\pi + u_k^e - u_k^\varphi$$

determine

$$u_k^\varphi = 2(q_k - a_k), \quad u_k^\pi + u_k^e = q_k - u_k^\varphi \quad \text{in } \mathbb{F}_3^6.$$

Thus, φ is the fixed axis and the residual freedom is the internal distribution π/e . In the ninth phase,

$$B_9 = (100100|020112|010122)$$

admits three local outputs with the same axis φ and the same aggregate $\pi + e$; all three survive one horizon and only

$$B_{10} = (101222|112221|112002)$$

survives the second. This is the executable $3 \rightarrow 1$ descent of EF–G9. The published branch continues through twenty blocks per channel: w_{30} and R_{36} are prefix and boundary, not closure.

4.11 Relation of extension and nonadic monodromy

Let $\mathcal{H}_m$ be the set of admissible enriched histories of depth m , and let

$$p_{n,m} : \mathcal{H}_n \longrightarrow \mathcal{H}_m$$

the truncation. For $N \geq m$,

$$\text{Surv}_m^{(N)} = p_{N,m}(\mathcal{H}_N), \quad \text{Surv}_m = \bigcap_{N \geq m} \text{Surv}_m^{(N)}.$$

The EF–G9 relation conserves the extensions that belong to this stable core. The phase satisfies

$$g(m+9) = g(m),$$

but the transported state satisfies, in general,

$$v_{m+9} \neq v_m.$$

The return acts on the fiber of states of a same phase; does not reset the history.

Definition 4.3 (Number HMT). An HMT number of channel χ is solely a compatible global section

$$x = (x_m)_{m \geq 0} \in X_\chi := \varprojlim_m \text{Surv}_m^{(\chi)}.$$

The reader and the published value are not part of the section.

A presentation or measurement of the number is a pair (x, L) , where L is a declared reader whose domain contains x . When L is the Archimedean reader of cylinders, $L(x)$ is the published value; x conserves in addition orientation, quotient, carry, and boundary.

Theorem 4.4 (Existence of a global prolongation). *Let $h \in \mathcal{H}_m$. If the prolongation tree of h is finitely branching and has at least one node at every depth, then there exists an infinite history whose truncations prolong h . If, in addition, each stable level is a singleton and the unique states are compatible, the global section is unique.*

Proof. The non-empty finite levels form a finitely ramified tree. König’s lemma provides an infinite branch. The compatibility of the truncations identifies that branch with an element of the inverse limit. If each level contains a single state, there is no second compatible branch. $\square$

The preceding theorem is abstract and conditional: by itself it asserts neither nonemptiness nor the stable uniqueness of a concrete channel. For π, e, φ , those hypotheses are subsequently discharged through the executable rational filtration of joint filtration theorem at finite depth. This placement preserves here only the inverse-limit principle and does not anticipate its application.

4.12 Composition of the fundamental operators

The preceding operators lead to the terminal holonomic state without flattening its coordinates:

$$\begin{aligned} \text{APP} &\longrightarrow \text{TRIT} \longrightarrow \text{oriented route} \longrightarrow \text{integral holonomic state} \\ &\longrightarrow \text{cylinders and survival} \longrightarrow x_{\text{term}}. \end{aligned}$$

The dodecaphonic closure then receives two coordinated evaluations of the same state:

$$\begin{aligned} x_{\text{term}} &\xrightarrow{(\pi_{\text{term}}, R_{\text{sgn}})} (B_{\text{term}}, U_{12}^{\text{sgn}}), & U_{12}^{\text{sgn}} &\xleftrightarrow{\mathcal{H}_{12}} K, \\ (P, E, \Phi, K, c_{13} = 0) &\longleftrightarrow \alpha_{12}. \end{aligned}$$

No arrow $B_{\text{term}} \rightarrow U_{12}^{\text{sgn}}$ appears: both coordinates follow from x_{term} . The subsequent passage to Dimensional Mechanics likewise creates no other machine; it prolongs the preserved genealogy through an enriched chronological record, signature, character, and towers.

PART II

Holonomic path algebra and local electronic structure

Universal presentation of the holonomic algebra

5.1 Generating data

Fix a depth n , and let X_n be the finite set of APP–TRIT–TPK states preserving, at the corresponding level, sheet, regime, orientation, residue, quotient, carry, boundary, height, path, memory, and survival. Let C_n be the small category generated by the steps

$$U_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t.$$

The objects of C_n are the states of X_n ; its arrows are the admissible histories. Composition exists when the terminal state of the first history coincides with the initial state of the second.

Each $x \in X_n$ is associated with the tritic fiber

$$A_x = A_{\tau(x)} = \mathbb{R}[J_x] / (J_x^2 + \tau(x)e_x), \quad \tau(x) \in \{+1, 0, -1\}.$$

The symbol e_x designates the local identity of the fiber. For each arrow $u : x \rightarrow y$, TPK transport induces an action

$$\rho_u : A_x \longrightarrow A_y$$

on the coefficients surviving the step. When a transition changes regime, ρ_u is interpreted as the typed bimodule of that transition, not as an identification of the two quadratic algebras.

The concatenation memory register defines a multiplier

$$\Omega(v, u) \in Z(A_z)^\times, \quad u : x \rightarrow y, \quad v : y \rightarrow z,$$

that records the carry, torsion, and boundary contribution produced by composition. Its compatibility law is

$$\rho_w(\Omega(v, u))\Omega(w, v \circ u) = \Omega(w, v)\Omega(w \circ v, u). \quad (1.1)$$

Equation (1.1) is the cocycle identity that prevents the memory of a history from depending on how its composition is parenthesized.

5.2 Presentation by generators and relations

Let $\mathcal{F}_n$ be the free associative algebra generated by

$$\{e_x, J_x : x \in X_n\} \cup \{T_u : u \in \text{Mor}(C_n)\}.$$

We define the ideal $\mathcal{I}_{\text{HMT}}$ through the relations

$$e_x e_y = \delta_{xy} e_x, \quad \sum_{x \in X_n} e_x = 1_n, \quad (2.1)$$

$$J_x = e_x J_x e_x, \quad J_x^2 = -\tau(x) e_x, \quad (2.2)$$

$$T_u = e_{t(u)} T_u e_{s(u)}, \quad (2.3)$$

$$T_u a = \rho_u(a) T_u, \quad a \in A_{s(u)}, \quad (2.4)$$

and

$$T_v T_u = \Omega(v, u) T_{v \circ u} \quad (2.5)$$

when $v \circ u$ exists; the product is zero when the arrows are not composable. The finite holonomic algebra is

$$\boxed{\mathfrak{H}_n = \mathcal{F}_n / \mathcal{I}_{\text{HMT}}}. \quad (2.6)$$

Addition in $\mathfrak{H}_n$ superposes possible paths. Multiplication concatenates histories. The factor Ω prevents concatenation from erasing the carry. Thus, addition and multiplication have acquired a meaning that is simultaneously arithmetic and dynamic.

5.3 Associativity theorem

Theorem 3.1. If ρ respects local identities and Ω satisfies (1.1), the product defined by (2.5) is associative.

Proof. Let $u : x \rightarrow y$, $v : y \rightarrow z$, and $w : z \rightarrow t$. For coefficients a, b, c in the relevant terminal fibers,

$$\begin{aligned} ((cT_w)(bT_v))(aT_u) &= c\rho_w(b)\Omega(w, v)\rho_{w \circ v}(a)\Omega(w \circ v, u)T_{wvu}, \\ (cT_w)((bT_v)(aT_u)) &= c\rho_w(b\rho_v(a)\Omega(v, u))\Omega(w, v \circ u)T_{wvu}. \end{aligned}$$

The covariance of ρ equates the factors containing a and b . Identity (1.1) equates the two products of Ω . Associativity on monomials follows, and bilinearity extends it to all of $\mathfrak{H}_n$. $\square$

The proof clarifies the role of memory. Without Ω , concatenation would preserve the bare path, but not the accumulated discrepancy between its lifts. With Ω , two ways of grouping the same history produce the same complete state.

5.4 Universal representation property

Theorem 4.1. Let $\{E_x\}_{x \in X_n}$ be a family of modules. Suppose local representations are given

$$\pi_x : A_x \rightarrow \text{End}(E_x)$$

and operators $V_u : E_x \rightarrow E_y$ satisfying

$$V_u \pi_x(a) = \pi_y(\rho_u(a)) V_u, \quad (4.1)$$

$$V_v V_u = \pi_z(\Omega(v, u)) V_{v \circ u}. \quad (4.2)$$

There exists a unique representation

$$\Pi : \mathfrak{H}_n \rightarrow \text{End}\left(\bigoplus_{x \in X_n} E_x\right)$$

that realizes e_x, J_x, T_u as the given projections, generators, and transports.

Proof. The assignment of the generators defines a homomorphism from the free algebra $\mathcal{F}_n$. Equations (4.1)–(4.2) and the local relations imply that $\mathcal{I}_{\text{HMT}}$ belongs to its kernel. The homomorphism factors through the quotient (2.6). Uniqueness follows because the classes of e_x, J_x, T_u generate the quotient. $\square$

The universal property turns the causal precedence of the generator into an algebraic statement. A subsequent reader is legitimate when it represents the relations of the generator; it does not need to reconstruct them from its codomain. The continuum, the electron, gravity, and the exceptional branch are distinct realizations of a single presentation.

5.5 The elementary nonadic cocycle

The canonical section $s : C_9 \rightarrow \{0, \dots, 8\} \subset C_{108}$ produces

$$c_9(a, b) = \left\lfloor \frac{a+b}{9} \right\rfloor \in C_{12}. \quad (5.1)$$

The identity

$$c_9(a, b) + c_9(a + b \bmod 9, c) = c_9(b, c) + c_9(a, b + c \bmod 9) \quad (5.2)$$

is exact. If X is the lift operator and λ the image of the memory generator, then

$$X^9 = \lambda I$$

and

$$X^a X^b = \lambda^{c_9(a,b)} X^{a+b \bmod 9}. \quad (5.3)$$

Equation (5.3) is the minimal form of the helix. The visible exponent returns modulo nine; the factor λ preserves the turn. If $\lambda \neq 1$, phase return and state return are distinct operations.

5.6 State refinement and reader prolongation

The finite sets of states form an inverse system

$$\dots \xrightarrow{r_{n+2,n+1}} X_{n+1} \xrightarrow{r_{n+1,n}} X_n.$$

The deep state is

$$X_\infty = \varprojlim_n X_n. \quad (6.1)$$

Functions and cylinder idempotents move in the opposite direction:

$$r_{n+1,n}^* : f \mapsto f \circ r_{n+1,n}. \quad (6.2)$$

If e_x is the indicator of the cylinder $x \in X_n$,

$$r^* e_x = \sum_{y:r(y)=x} e_y. \quad (6.3)$$

The family of algebras therefore forms a direct system:

$$\mathfrak{H}_{\text{cyl}} = \varinjlim_n \mathfrak{H}_n. \quad (6.4)$$

Theorem 6.1. The inclusions (6.2) are unital; cylinder idempotents remain idempotent; and every compatible reader at finite depth determines a unique reader on $\mathfrak{H}_{\text{cyl}}$.

Proof. The preimage of a partition is a partition. Hence $r^*(fg) = r^*f r^*g$, $r^*1_n = 1_{n+1}$, and the sum (6.3) is orthogonal. The universal property of the direct limit extends every compatible family of readers. $\square$

This theorem supplies the exact meaning of «fractalizing inward and expanding outward». States are completed by restriction; readers are completed by prolongation. Preservation of the unit is the unitality of these inclusions.

5.7 Holonomy and memory grading

Let d_M be the memory degree. The holonomy satisfies

$$d_M(\Gamma_9 x) = d_M(x) + 1. \quad (7.1)$$

In the reversible domain, Γ_9 induces an automorphism of $\mathfrak{H}_{\text{cyl}}$ and one can form

$$\mathfrak{H}_{\text{rev}} \rtimes_{\Gamma_9} \mathbb{Z}. \quad (7.2)$$

In the general domain, advance produces an endomorphism and the object is

$$\mathfrak{H}_{\text{cyl}} \rtimes_{\Gamma_9} \mathbb{N}. \quad (7.3)$$

The distinction between (7.2) and (7.3) prevents introducing an inversion where only an advance has been constructed. The already typed helical inversion acts in (7.2); the mirror π/e acts on the reader coordinates and constitutes another involution.

5.8 Total tritic algebra

Let

$$\mathbb{A}_{\text{tri}} = A_+ \oplus A_0 \oplus A_-$$

with central idempotents p_+, p_0, p_- . The generators satisfy

$$J_+^2 = -p_+, \quad J_0^2 = 0, \quad J_-^2 = p_-. \quad (8.1)$$

Let

$$\mathbb{J} = p_+ J_+ + p_0 J_0 + p_- J_-.$$

The exponential defined by its internal series satisfies

$$\begin{aligned} \text{Exp}_\star(t\mathbb{J}) &= p_+(\cos t + J_+ \sin t) \\ &\quad + p_0(1 + tJ_0) \\ &\quad + p_-(\cosh t + J_- \sinh t). \end{aligned}$$

(8.2)

Proof. In the elliptic fiber, the even and odd powers of J_+ alternate signs. In the parabolic fiber, all powers of order at least two disappear. In the hyperbolic fiber, the even powers are the identity and the odd powers are J_- . Grouping the three series proves (8.2). $\square$

The identity simultaneously preserves the three geometries. None of them is obtained by informal continuation of another; all three are orthogonal summands of the same object.

5.9 Euler as elliptic projection

Let π be the elliptic closure already produced by the nonadic connection. Then

$$\cos \pi = -1, \quad \sin \pi = 0$$

in the circular realization, and (8.2) gives

$$p_+ \operatorname{Exp}_\star(\pi \mathbb{J}) + p_+ = 0. \quad (9.1)$$

Equation (9.1) recovers Euler's identity on the elliptic face. The total element also preserves

$$p_0 \operatorname{Exp}_\star(\pi \mathbb{J}) = p_0(1 + \pi J_0),$$

and

$$p_- \operatorname{Exp}_\star(\pi \mathbb{J}) = p_-(\cosh \pi + J_- \sinh \pi).$$

The information omitted by the complex representation is precisely the threshold jet and dilation. The tritic identity does not replace Euler's identity: it identifies which projection produces it and what memory remains outside that projection.

5.10 The hyperbolic autoscale and φ

Let

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

Its square has determinant one and trace three. We define

$$H_\varphi = \frac{2F_{\text{av}}^2 - 3I}{\sqrt{5}}. \quad (10.1)$$

A direct calculation gives

$$H_\varphi^2 = I. \quad (10.2)$$

Since

$$\cosh(2 \log \varphi) = \frac{3}{2}, \quad \sinh(2 \log \varphi) = \frac{\sqrt{5}}{2},$$

we obtain

$$\boxed{F_{\text{av}}^2 = \operatorname{Exp}(2 \log \varphi H_\varphi)}. \quad (10.3)$$

The hyperbolic regime does not merely admit a quadratic unit: the area–volume incidence matrix realizes that unit exactly. The ratio φ^{-2} is the stable direction of a unimodular evolution.

5.11 The structural role of e

The series $\text{Exp}_\star$ transforms the additive generator into multiplicative transport:

$$\text{Exp}_\star(a + b) = \text{Exp}_\star(a) \text{Exp}_\star(b)$$

when a and b commute. Its scalar realization on the identity defines

$$\text{Exp}_\star(1) = eI. \quad (11.1)$$

Equation (11.1) does not replace the nonadic decimal generation of e . It identifies its algebraic role: e is the scale that converts infinitesimal addition into a multiplicative transport law. This is the intrinsic bridge between the two APP sheets in the exponential representation.

5.12 Alpha as a dodecaphasic transgression functional

The cocycle (5.1) represents the impossibility of choosing a homomorphic section $C_9 \rightarrow C_{108}$. The visible turn produces an element of the dodecaphasic kernel. The closure of α receives that element along with the signed word and determines it on a scalar line.

It is useful to distinguish the cocycle from its evaluation. Let

$$\mathcal{T}_\alpha : Z^2(C_9; C_{12}) \times X_{12}^{\text{sgn}} \longrightarrow \mathcal{L}_\alpha \quad (12.1)$$

the functional realizing concatenation with memory. Then

$$\alpha = \mathcal{T}_\alpha(c_9, K). \quad (12.2)$$

Equation (12.2) does not identify α with a cohomology class. It states that α is the scalar realization of that class after incorporating the dodecaphasic state. The second route determines the same section through

$$P_9(\alpha) = \delta, \quad \delta = \log_{10} \frac{10}{9}, \quad (12.3)$$

and both meet in the resolution fiber product.

This algebraic type explains the singularity of α . π, φ, e are closure, scale, and transport coordinates. α is the functional that measures how those coordinates survive the non-split lift.

5.13 Compact matrix of the correlated system

The four functions come together without collapsing into

$$\begin{pmatrix} \text{Exp}_\star(\pi J_+) & 0 & 0 & 0 \\ 0 & \text{Exp}_\star(I) & 0 & 0 \\ 0 & 0 & \text{Exp}_\star(2 \log \varphi H_\varphi) & 0 \\ 0 & 0 & 0 & \text{Exp}_\star(P_9(\alpha) J_0) \end{pmatrix} = \begin{pmatrix} -I & 0 & 0 & 0 \\ 0 & eI & 0 & 0 \\ 0 & 0 & F_{\text{av}}^2 & 0 \\ 0 & 0 & 0 & I + \delta J_0 \end{pmatrix}. \quad (13.1)$$

Matrix (13.1) is an identity of functions, not a table of values. Its four blocks mean elliptic closure, exponential multiplication, hyperbolic autoscaling, and parabolic seam. The dodecaphasic compatibility of the last row preserves the joint genealogy.

5.14 The electronic corner

Let

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 3/4 & \sqrt{3}/4 \\ \sqrt{3}/4 & 1/4 \end{pmatrix}.$$

We define

$$S = 2P - I, \quad J = \frac{4}{\sqrt{3}}[P, Q].$$

Then

$$S^2 = I, \quad J^2 = -I, \quad SJ = -JS. \quad (14.1)$$

Therefore,

$$e_{\text{vac}} \mathfrak{H}_n e_{\text{vac}} \twoheadrightarrow \text{Cl}_{1,1} \cong M_2(\mathbb{R}). \quad (14.2)$$

Arrow (14.2) expresses that the electronic block is a quotient or a representation of a local corner of holonomic arithmetic. Noncommutativity is not added from quantum mechanics: it already resides in the order of the APP sheets. The internal root of -1 is the elliptic generator; the split involutions preserve vacancy and dilation.

5.15 Readers and output types

Let

$$\text{Read}(\mathfrak{H}) = \coprod_B \text{Rep}_{\text{comp}}(\mathfrak{H}, B). \quad (15.1)$$

A constant is obtained as

$$c = \sigma_B(\rho(x_\infty)), \quad (15.2)$$

where ρ is a compatible representation and σ_B a scalar invariant specific to its codomain. This definition includes without confusing:

output	B	σ_B
π	elliptic module	minimum closure time
φ	hyperbolic module	stable spectral radius
e	exponential semigroup	unit evaluation
α	dodecaphasic fiber product	scalar gluing
G_{Cat}	character module	$\text{Tr}(N^{-2}Q_4)$
k_B	thermal and energy lines	unit transduction
γ_{BI}	area–volume module	$(12, -1)$ on 90/120
G	gravitational module	contragredient character

A particle is not a digit in (15.2): it is a persistent section of a module. A field is the transport of those sections. A symmetry is the automorphism group preserving the relations and boundary of the module.

5.16 Conservation and emergent symmetry theorem

Let $1_n = \sum_{x \in X_n} e_x$. The refinement inclusions satisfy

$$r^* 1_n = 1_{n+1}. \quad (16.1)$$

An individual history also defines a compatible family of cylinder idempotents. The first is the global unit of the algebra; the second is the genealogical unit of a section. Distinguishing them avoids confusing preservation of the space of possibilities with preservation of a particular path.

For a representation ρ and a boundary b_∂ , we define

$$\text{Sym}(\rho, b_\partial) = \{g \in \text{Aut}(\rho(\mathfrak{S})) : g\rho(\mathcal{R}_{\text{HMT}}) = \rho(\mathcal{R}_{\text{HMT}}), gb_\partial = b_\partial\}. \quad (16.2)$$

Symmetry is, by definition, the stabilizer of an already constructed conservation. In the exceptional branch, (16.2) leads to the Witt–Golay–Leech– V^h –Monster incidence stabilizers. In the physical branch, it leads to the groups preserving the electronic, gauge, or tensor modules. No symmetry is imposed to select the state; one obtains the symmetry leaving its projection invariant.

5.17 Information, entropy, and quotients

If U is bijective on the complete state,

$$H(X \mid U(X)) = 0. \quad (17.1)$$

Let q be a quotient that forgets a coordinate. Then

$$H(X) = H(qX) + H(X \mid qX). \quad (17.2)$$

A forgotten uniform trit contributes

$$H(X \mid qX) = \log 3.$$

The thermal transducer determines

$$Q_{\text{reset}} \geq k_B \Theta \log 3. \quad (17.3)$$

Equations (17.1)–(17.3) show that entropy does not measure the advance of holonomy. It measures the information discarded by a reader. Sturmian complexity $p(m) = m + 1$ allows infinite memory with zero topological entropy; the thermal cost appears only when a projection is made irreversible.

5.18 Core theorem

Theorem 18.1. The object

$$\mathfrak{S}_{\text{HMT}}^{\text{alg}} = \left[\lim_{\rightarrow n} (\mathbb{R}\langle e_x, J_x, T_u \rangle / \mathcal{R}_{\text{HMT}}) \right] \rtimes_{\Gamma_9} \mathbb{N} \quad (18.1)$$

is a trigeometric holonomic algebra, graded by phase and memory, with an associative product twisted by the carry. Its state space is the inverse limit X_∞ . Every compatible realization of APP, TRIT, and TPK factors through (18.1). Its typed representations determine the joint discrete structure of the continuum, the correlated system, the electron, action, gravity, constants, and symmetries without turning their terminal values into generators.

The notation in (18.1) is deliberately algebraic: it imposes no norm before the codomain is chosen. If $\rho : \mathfrak{H}_{\text{HMT}}^{\text{alg}} \rightarrow B(\mathcal{K})$ is a compatible bounded representation, its completed realization is

$$\mathfrak{H}_{\text{HMT}}^\rho = \overline{\rho(\mathfrak{H}_{\text{HMT}}^{\text{alg}})}^{\|\cdot\|}. \quad (18.2)$$

Thus each reader supplies the topology it effectively preserves; no conventional completion is introduced as part of the generator.

Proof. Associativity follows from Theorem 3.1. Universality follows from Theorem 4.1. Preservation of the unit and refinement duality follow from Theorem 6.1. Memory elevation follows from the nonadic cocycle (5.3). The coexistence of curvatures follows from (8.1). The realizations are obtained through the universal property and the readers (15.1)–(15.2). $\square$

5.19 Finite certification

Exhaustive verification over the finite domains proves:

1. the 9^3 carry cocycle identities;
2. the matrix representation $X^9 = 2I$ and its twisted law;
3. the three relations $J_+^2 = -I$, $J_0^2 = 0$, $J_-^2 = I$;
4. the three branches of the exponential;
5. $H_\varphi^2 = I$ y $\exp(2 \log \varphi H_\varphi) = F_{\text{av}}^2$;
6. the corner $\text{Cl}_{1,1}$;
7. the contravariant/covariant unitality of refinement;
8. the synchrony A_2 with diagonal advance -16 ;
9. the difference between a reversible permutation and a tritic quotient;
10. the null Witt pair contained in the electronic corner;
11. the exact relations of the pair of phase observables and the memory commutator;
12. the irreducibility of APP, TRIT, and TPK through forgetful quotients, together with potential inversion in the sequence reader.

These identities constitute a reproducible finite verification of the algebraic system.

5.20 The null Witt pair within the electronic corner

The electronic corner provides an even more complete local realization of trigeometry. Let

$$S = 2P - I, \quad J = \frac{[P, Q]}{\sqrt{3}/4}. \quad (20.1)$$

The already proven relations are

$$S^2 = I, \quad J^2 = -I, \quad SJ = -JS. \quad (20.2)$$

Let us now define

$$n_+ = \frac{S + J}{2}, \quad n_- = \frac{S - J}{2}. \quad (20.3)$$

Por (20.2),

$$n_+^2 = n_-^2 = 0, \quad n_+ n_- + n_- n_+ = I. \quad (20.4)$$

The proof is immediate but structurally decisive: $(S \pm J)^2 = S^2 + J^2 \pm (SJ + JS) = 0$, while the sum of the two cross products equals $\frac{1}{2}(S^2 - J^2) = I$. Thus, the same block $\text{Cl}_{1,1} \cong M_2(\mathbb{R})$ contains:

local generator	square	regime
J	$-I$	elliptic
$n_{\pm}$	0	parabolic
S	I	hyperbolic.

(20.5)

This is an exact local Witt decomposition. The electronic vacancy thus acquires a second equivalent characterization in its corner: in addition to being the minimal loss of one multiplicative-carry unit under the APP conservations, it is a parabolic null direction between the two orientations n_+ and n_- . TRIT globally types these three classes and TPK transports them with sheet, phase, and memory. The subsequent Paley–Witt incidence structure preserves its own combinatorial domain; the decomposition (20.3)–(20.5) supplies the local model from which its realization can be formulated without equating the two objects.

5.21 Crossed product of the nonadic and Sturmian phase observables

Let

$$\delta = \log_{10} \frac{10}{9}, \quad \zeta_9 = \text{primitive character of } C_9, \quad \omega_{\delta} = e^{2\pi i \delta}, \quad (21.1)$$

and let T be the oriented step on the reversible lift. In the basis $|t, m\rangle$, where t records the step and m the memory height, we introduce the two diagonal characters

$$V_9|t, m\rangle = \zeta_9^t|t, m\rangle, \quad W_{\delta}|t, m\rangle = \omega_{\delta}^t|t, m\rangle. \quad (21.2)$$

The covariance of the step produces the Weyl relations

$$V_9 T = \zeta_9 T V_9, \quad W_{\delta} T = \omega_{\delta} T W_{\delta}. \quad (21.3)$$

For the nonadic turn $\Gamma_9 = T^9$, enriched by the TPK update, one obtains

$$V_9 \Gamma_9 = \Gamma_9 V_9, \quad W_{\delta} \Gamma_9 = \omega_{\delta}^9 \Gamma_9 W_{\delta}. \quad (21.4)$$

The nonadic observable closes; the rotation observable advances because $\delta \notin \mathbb{Q}$. If $D_M|t, m\rangle = m|t, m\rangle$, the law $\Gamma_9|t, m\rangle = |t + 9, m + 1\rangle$ is equivalent to

$$[D_M, \Gamma_9] = \Gamma_9, \quad \Gamma_9^{-1} D_M \Gamma_9 = D_M + I \quad (21.5)$$

on the reversible sector. This is the operator equation of the helix: phase closure is compatible with a unit increase in the memory degree. The basal translation group remains abelian; the algebra of observables and translations ceases to be so by (21.3). The Sturmian word appears when the TRIT threshold is applied to the orbit of W_{δ} ; its aperiodicity therefore does not decorate the finite phase, but constitutes the symbolic factor of the irrational rotation.

The operator presentation belongs to the crossed product

$$\mathfrak{B}_{9,\delta} = C(C_9 \times \mathbb{T}) \rtimes_T \mathbb{Z}, \quad (21.6)$$

enriched by the grading D_M . The spectrum of $C(C_9 \times \mathbb{T})$ simultaneously records the finite phase and the irrational phase; the generator of $\mathbb{Z}$ implements their joint translation; and (21.5) prevents the graded representation from descending to a closed orbit after forgetting memory. The subshift obtained through the tritic partition of the second coordinate is the Sturmian hull. The helical suspension is therefore the graded extension of that crossed product, not the metaphorical juxtaposition of two clocks.

The characters ζ_9 and ω_δ belong to the holonomic generator. Their subsequent elliptic representation recognizes

$$\rho_+(\zeta_9) = \text{Exp}_+\left(\frac{2\pi}{9}J_+\right), \quad \rho_+(\omega_\delta) = \text{Exp}_+(2\pi\delta J_+). \quad (21.7)$$

The conventional exponential thus appears as a realization of the already constructed internal character; it neither selects the observables nor introduces a target value into the generator.

Relations (21.3)–(21.5) separate exactly three facts:

$$\text{phase period } 9, \quad \text{absence of irrational period}, \quad \text{integer memory advance}. \quad (21.8)$$

Their coexistence defines the Sturmian helical suspension over the nonadic holonomic extension. The symbolic hull has long-range aperiodic order and zero topological entropy, although its spectral module simultaneously contains the rational nonadic frequency and the irrational frequency δ . The physical name «time crystal» additionally requires fixing the dynamic realization, its response, and its stability; the theorem formulated here is the topological-symbolic object that precedes it.

5.22 The minimal polynomial relation of the three curvatures

The orthogonal sum of the three TRIT generators can be written as

$$\mathbb{J} = p_+J_+ + p_0J_0 + p_-J_-, \quad p_+ + p_0 + p_- = I. \quad (22.1)$$

Their three squares imply a single relation:

$$\boxed{\mathbb{J}^6 = \mathbb{J}^2}. \quad (22.2)$$

Conversely, if $K = \mathbb{J}^2$, then $K^3 = K$ and the three regime idempotents are recovered polynomially:

$$p_+ = \frac{K^2 - K}{2}, \quad p_0 = I - K^2, \quad p_- = \frac{K^2 + K}{2}. \quad (22.3)$$

Thus, the elliptic, parabolic, and hyperbolic classification does not require three algebras juxtaposed from outside. It is contained in the algebraic spectral calculus of a single tritic generator. Substituting (22.3) into the total exponential yields the compact identity

$$\boxed{\text{Exp}_\star(t\mathbb{J}) = p_+(\cos t + \mathbb{J} \sin t) + p_0(I + t\mathbb{J}) + p_-(\cosh t + \mathbb{J} \sinh t)}. \quad (22.4)$$

The Euler face is the elliptic projection of (22.4); parabolic propagation occupies the threshold; the hyperbolic autoscale contains the realization of φ . Composing it with the holonomy,

$$\mathcal{E}_{\text{HMT}}(t) = \Gamma_9 \text{Exp}_\star(t\mathbb{J}), \quad [D_M, \Gamma_9] = \Gamma_9, \quad (22.5)$$

trigeometry and the helical lift come together in two equations. APP supplies the idempotents and the tension between sheets; TRIT supplies $\mathbb{J}$; TPK supplies Γ_9 , its cocycle, and the degree D_M . Numerical and physical outputs are typed representations of (22.5). This presentation constitutes the minimal operator

backbone of the standalone article, while the universal algebra (18.1) preserves the full multiplicity of states, paths, and refinements needed to reconstruct the integral development.

Carry vacancy, split Clifford algebra, and null Witt pair

6.1 Vacancy as a minimal carry defect

The level-ten additive fiber,

$$\mathcal{F}_{10} = \{(i, j) \in D_9^2 : i + j = 10\}, \quad (6.1)$$

contains the center $Q_0 = (5, 5)$ and the oriented deformations $Q_t = (5 + t, 5 - t)$. Their product is $25 - t^2$. Preserving the center's multiplicative residue requires

$$t^2 \equiv 0 \pmod{9}. \quad (6.2)$$

For $|t| \leq 4$, the only solutions are $t = 0, \pm 3$. The two noncentral deformations are therefore

$$Q_{+3} = (8, 2), \quad Q_{-3} = (2, 8), \quad (6.3)$$

exchanged by cell inversion. In the APP chart,

$$\mathcal{A}(5, 5) = ((1, 1), (7, 2)), \quad \mathcal{A}(8, 2) = ((1, 1), (7, 1)). \quad (6.4)$$

The additive evaluation, its quotient, the multiplicative residue, and the tritic regime are preserved; only $q^\times : 2 \mapsto 1$ changes. Vacancy is, by definition, this minimal loss of one multiplicative-carry unit while preserving the remaining invariants. It is not a zero added to the state.

The same fiber is reached from the nonadic reduction

$$(369, 126) \xrightarrow{/9} (41, 14) \mapsto (5, 5). \quad (6.5)$$

It has two distinct prolongations:

$$(41, 14) \xrightarrow{(+3, 0)} (44, 14) \mapsto (8, 5), \quad (6.6)$$

$$(41, 14) \xrightarrow{(+3, -3)} (44, 11) \mapsto (8, 2). \quad (6.7)$$

The first satisfies

$$\frac{396}{126} = \frac{22}{7}; \quad (6.8)$$

the second preserves

$$41 + 14 = 44 + 11 = 55 = 54 + 1. \quad (6.9)$$

The two branches separate the closure reader from the vacancy reader. The former recognizes a period ratio; the latter preserves the boundary unit while reducing the carry exactly.

6.2 Matrix corner and electronic section

The relations (20.2) and (20.4) prove that the central fiber locally realizes the three geometries. To specify its singular mode without normalization ambiguity, let $\Sigma_3, \Pi_3 : D_9^3 \rightarrow D_9$ be the additive and multiplicative evaluations of the triples and define their joint table by

$$N_{ab}^{(3)} := \#\{x \in D_9^3 : \Sigma_3(x) = a, \Pi_3(x) = b\}, \quad C_{ab}^{(3)} := \frac{N_{ab}^{(3)}}{\sqrt{n_a m_b^\Pi}}, \quad (6.10)$$

where $n_a = \sum_b N_{ab}^{(3)}$ and $m_b^\Pi = \sum_a N_{ab}^{(3)}$. Exhaustive enumeration of the $9^3 = 729$ triples gives row marginals equal to 81 and marginals equal to 36 in the six active columns. With

$$m_{\text{ph}} = (1, -1, 0)^{\oplus 3}, \quad N^{(3)} m_{\text{ph}} = N^{(3)\top} m_{\text{ph}} = -27 m_{\text{ph}}, \quad (6.11)$$

the normalization $\sqrt{81 \cdot 36} = 54$ produces exactly

$$C^{(3)} m_{\text{ph}} = C^{(3)*} m_{\text{ph}} = -\frac{1}{2} m_{\text{ph}}, \quad (6.12)$$

and the incompatibility of the sheet projectors equals

$$\|[P_{\Sigma,3}, P_{\Pi,3}]\| = \frac{\sqrt{3}}{4}. \quad (6.13)$$

The areal–volumetric incidence provides

$$1 - \left(\frac{1}{2}\right)^2 = \frac{3}{4} = \frac{90}{120}. \quad (6.14)$$

The point (5, 5) is the unique fixed point of cell inversion. In this way, the electronic section is not selected through a known mass: it is singled out by the coincidence of centrality, minimal vacancy, null direction, torsion (6.13), and incidence (6.14). Its dimensionless character is

$$\mathfrak{e} = \frac{\sqrt{3}}{4} \left(e^{\varphi/\pi^2} - 22\alpha^3 \right) (1 + 15\Delta_4), \quad (6.15)$$

where each factor comes from an already constructed stage: minimal torsion, the reciprocal orientation of the reader $R(x, y) = x/y^2$, the vacancy complement, the compatibility constant, and the determinant of the fiber. The energy chart is subsequently applied to this section.

6.3 Polarity and constitutive equations

Let $\mathbf{C}$ be sheet inversion. The common character $\mathcal{E}$ and the oriented character $\mathcal{B}$ are defined by

$$\mathcal{E} \circ \mathbf{C} = \mathcal{E}, \quad \mathcal{B} \circ \mathbf{C} = -\mathcal{B}. \quad (6.16)$$

The two combinations

$$\widehat{\mu} = e^{\mathcal{E}+\mathcal{B}}, \quad \widehat{\varepsilon} = e^{\mathcal{E}-\mathcal{B}} \quad (6.17)$$

satisfy

$$\widehat{\mu} \widehat{\varepsilon} = e^{2\mathcal{E}}, \quad \frac{\widehat{\mu}}{\widehat{\varepsilon}} = e^{2\mathcal{B}}. \quad (6.18)$$

The product fixes the propagation cone and the quotient retains the impedance orientation. The potential difference belongs to the boundary of the vacancy; the transverse response belongs to its conjugate transport. The neutral sector is $\ker \mathcal{B}$, and every section fixed by $\mathbf{C}$ belongs to it. This structure makes polarity a property of vacancy transport, not a label imposed afterward.

PART III

Aperiodic topological-temporal order and nonadic holonomy

Internal capacity calendar, phase clocks, and Diophantine renormalization

7.1 Aperiodicity as the internal calendar of nonadic capacity

The Sturmian word and the nonadic helix are not superimposed structures. Both are realizations of a single capacity arithmetic. Let

$$\rho := 6 \log_{1000} 3 = \log_{10} 9, \quad \delta := 1 - \rho = \log_{10} \left(\frac{10}{9} \right).$$

The decimal capacity index opened after ternary block number t is

$$K_t := \lfloor (t+1)\rho \rfloor, \quad t \geq 0,$$

and the mechanical vacancy word is

$$z_t := \lfloor (t+1)\delta \rfloor - \lfloor t\delta \rfloor, \quad t \geq 1.$$

Since δ is irrational, $(t+1)\delta \notin \mathbb{Z}$. Therefore,

$$\begin{aligned} K_t &= \lfloor (t+1)(1-\delta) \rfloor \\ &= t+1 - \lceil (t+1)\delta \rceil \\ &= t - \lfloor (t+1)\delta \rfloor. \end{aligned}$$

Subtracting two successive terms yields the link that remained to be made explicit:

$$\boxed{K_t - K_{t-1} = 1 - z_t.}$$

This is not a similarity between two counts. It is an exact identity between the capacity increment used by generation and the vacancy word produced by the incompatibility 729/1000. Its two cases are:

z_t	$K_t - K_{t-1}$	visible operation
0	1	the block opens 729 subcells and a new decimal pruning enters,
1	0	the block opens 729 subcells without increasing decimal pruning.

The second case is reopening or *stutter*. The aperiodic word therefore decides when capacity advances and when it remains fixed. Aperiodicity does not govern a periodic generator from outside: it is the internal calendar of its open–prune operation.

Telescoping gives a global form of the same theorem. For $0 \leq a < b$,

$$K_b - K_a = (b - a) - \sum_{t=a+1}^b z_t.$$

Each interval is decomposed, without remainder, into capacity increments and vacancies. In particular, every conservation formulated on the pair

(acquired capacity, recorded vacancy)

must preserve their sum, which is the length of the block interval. This is the most elementary arithmetic form of the evolutionary conservation of the unit at this level; it is not yet the projective measure of the continuum, but it is its exact local model.

7.2 Two times, two returns, and phase retention with memory advance

The preceding identity requires distinguishing two clocks.

The **block time** t increases with each application of the dynamics. The **capacity time** K_t counts the decimal triads actually pruned. The determined nonadic phase is

$$g_t := [K_t]_9 \in C_9.$$

From $\Delta K_t = 1 - z_t$ it follows.

$$g_t - g_{t-1} \equiv 1 - z_t \pmod{9}.$$

When $z_t = 0$, the capacity clock and phase advance. When $z_t = 1$, the time block advances but K_t and g_t remain. This is an exact event of **phase retention with state advance**: the new block opens another fiber of 729 possibilities, changes the signature, and preserves the survival history even though the phase residue is the same. The statement «phase returns and memory changes» thus has two internal realizations that must not be confused:

1. local retention $K_t = K_{t-1}$, caused by a Sturmian vacancy;
2. holonomic return $K \mapsto K + 9$, which traverses a complete circuit of C_9 .

In both cases the determined phase coincides and the integral holonomic state differs, but the first mechanism is a capacity pause and the second is a complete nonadic turn.

This allows the base system to be written without reducing TPK to a rotation. Let $\theta_t = \{t\delta\} \in \mathbb{T}$ and let $x_t \in X_{\text{TPK}}^{\text{enr}}$. The effective step has the form

$$(g_{t-1}, \theta_{t-1}, x_{t-1}) \mapsto (g_{t-1} + 1 - z_t, \theta_{t-1} + \delta, U_t(x_{t-1})),$$

with

$$U_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t.$$

The coordinate z_t schedules capacity advance; it does not replace the action of U_t . The selector preserves sheet and regime, transport preserves path and orientation, and the update incorporates carry, boundary, and register. Therefore, equality of g never authorizes equality of x .

7.3 The inverse clock and the suspension with nonconstant roof

The monotone inversion of capacity reveals a second mechanical word. For $k \geq 1$, let

$$T(k) := \min\{t : K_t = k\} = \left\lceil \frac{k}{\rho} \right\rceil - 1.$$

Let us introduce

$$\sigma := \frac{1}{\rho} - 1 = \frac{\delta}{\rho}.$$

Since ρ is irrational, σ is also irrational, and

$$T(k) = k + \lceil k\sigma \rceil - 1.$$

The number of blocks needed to increase capacity by one unit is

$$r_k := T(k+1) - T(k) = 1 + \lceil (k+1)\sigma \rceil - \lceil k\sigma \rceil \in \{1, 2\}.$$

The value $r_k = 2$ records exactly that a reopening occurred between capacities k and $k+1$. The exceptional positions begin

$$20, 41, 62, 83, 104, 125, 145, 166, 187, 208, \dots,$$

and their separations are 20 or 21. This is the same dynamics viewed on the inverse clock; it must not be confused with the separations 21/22 between vacancies in block time.

For a capacity increment of length m , the return time is

$$\begin{aligned} R_m(k) &:= T(k+m) - T(k) \\ &= m + \lceil (k+m)\sigma \rceil - \lceil k\sigma \rceil. \end{aligned}$$

By mechanical balance,

$$R_m(k) \in \{m + \lfloor m\sigma \rfloor, m + \lceil m\sigma \rceil\}.$$

Two exact consequences follow:

$$\boxed{R_9(k) \in \{9, 10\}}, \quad \boxed{R_{108}(k) \in \{113, 114\}}.$$

One nonadic phase turn requires nine or ten blocks; one turn of the complete clock C_{108} requires one hundred thirteen or one hundred fourteen. The finite support preserves its phases exactly, but the block time required to traverse it is aperiodic. This is a more precise formulation of the **Sturmian helical suspension**: the finite clock C_{108} is the base, the irrational rotation provides the roof r_k , and the TPK cocycle transports the enriched fiber.

The pair 6/7 reappears here because

$$R_6(k) \in \{6, 7\}.$$

However, this 6/7 measures blocks consumed by six capacity increments. It differs both from the number of vacancies in a window of length 132 and from the separation between short gaps in the primary word. All three phenomena descend from the same slope, but only an explicit conjugacy would allow them to be identified as the same operator.

7.4 Diophantine renormalization and unimodular conservation of the unit

The continued fraction of the vacancy density, calculated after δ has been generated by 729/1000, begins with

$$\delta = [0; 21, 1, 5, 1, 6, 2, 3, 1, 1, 8, 1, 2, 2, 5, \dots].$$

Its first convergents are

$$\frac{1}{21}, \frac{1}{22}, \frac{6}{131}, \frac{7}{153}, \frac{48}{1049}, \frac{103}{2251}, \frac{357}{7802}, \frac{460}{10053}, \frac{817}{17855}, \frac{6996}{152893}, \frac{7813}{170748}, \dots$$

This list explains, within a single hierarchy, the primary gaps and their induced returns. The denominators 21, 22 are the first approximation times; the pairs (6, 131) and (7, 153) are the next times with their event counts. In particular,

$$131 = 14 \cdot 9 + 5, \quad 153 = 17 \cdot 9,$$

so that the return of secondary length 6 leaves a phase residue +5, while the one of length 7 closes the nonadic phase.

The integer 5 does not appear here through fitting. It is the partial quotient that produces

$$(6, 131) = 5(1, 22) + (1, 21),$$

and the next step gives

$$(7, 153) = (6, 131) + (1, 22).$$

An exact function of level 5 within vacancy renormalization is thus located. Identifying this partial quotient with the Ramanujan continued fraction of level 5 still requires a map between the two objects; equality of the integer does not provide it on its own.

Let p_n/q_n be the n -th convergent. Two consecutive convergents satisfy

$$p_{n+1}q_n - p_nq_{n+1} = (-1)^n.$$

At the first two stages this is seen as

$$1 \cdot 22 - 1 \cdot 21 = 1, \quad 6 \cdot 153 - 7 \cdot 131 = 1.$$

Therefore, the consecutive vectors (p_n, q_n) form a primitive basis of $\mathbb{Z}^2$ and preserve unit lattice area. The orientation alternates, but the modulus of the determinant remains equal to one. This identity demonstrates the conservative evolution of the dimensional unit: the real slope is approximated at increasing depth while each change of scale preserves a fundamental cell and admits an integer inverse.

The recurrence is written using

$$A(a) = \begin{pmatrix} a & 1 \\ 1 & 0 \end{pmatrix}, \quad \det A(a) = -1.$$

The ordered products of the matrices $A(a_n)$ generate the pairs (p_n, q_n) . For $a \neq b$,

$$A(a)A(b) = \begin{pmatrix} ab+1 & a \\ b & 1 \end{pmatrix} \neq \begin{pmatrix} ab+1 & b \\ a & 1 \end{pmatrix} = A(b)A(a).$$

The rotation by δ remains abelian. Noncommutativity appears in the renormalization cocycle that remembers the order of the partial quotients, in addition to that already demonstrated in the APP sheets and in the affine cocycle. The crystal therefore possesses an abelian phase layer and at least two ordered nonabelian memory layers.

The oriented errors

$$\varepsilon_n = q_n\delta - p_n$$

alternate in sign and never vanish. Their sign defines a tritic renormalization reader with values ± 1 ; the regime 0 is reserved for the exact threshold, excluded here by irrationality. This reader does not replace

the local TRIT of APP. It is a projection at another scale whose naturality with the tritic phase selector must be proved on the enriched fiber.

7.5 First Diophantine resonance of the block clock and separation of the capacity clock

The sequence of convergents contains returns that close the finite coordinate exactly and only approximate the irrational coordinate. Among the convergents written above:

$$153 \equiv 0 \pmod{9}, \quad 10053 \equiv 0 \pmod{9}, \quad 170748 \equiv 0 \pmod{108}.$$

The last is the first convergent in this chain whose denominator closes the block clock—the chronological coordinate θ indexed by the number of iterations—:

$$170748 = 1581 \cdot 108.$$

At the same time,

$$170748 \delta - 7813 = -0.0000017458436865028488500088576 \dots$$

The return is not periodic: the chronological coordinate $[t]_{108} \in C_{108}^{(t)}$ returns exactly, but the irrational coordinate remains shifted by a nonzero residue. It is indeed a **privileged Diophantine quasi-turn**, because it comes from a convergent and not from a decimal search.

This clock must be separated from the capacity quotient

$$C_{108}^{(K)} : \quad g_{108}(t) = [K_t]_{108}.$$

At the canonical origin, after 170748 blocks, the acquired capacity is

$$\lfloor 170748 \rho \rfloor = 170748 - 7813 = 162935 \equiv 71 \pmod{108}.$$

Therefore, that convergent does **not** close $C_{108}^{(K)}$. The returns $R_{108} \in \{113, 114\}$ of the previous section belong precisely to this second clock: they count how many blocks are needed to acquire 108 units of capacity. The coincidence of the name C_{108} does not authorize merging the two indices. The observable TPK calendar is the chronological clock; the capacity clock is a construction induced by the compactification $729 \rightarrow 1000$.

The system thus distinguishes four notions:

1. exact closure of the finite chronological clock;
2. nonclosure, in general, of the finite capacity clock;
3. optimal proximity of the Sturmian phase;
4. nontrivial transformation of the integral holonomic state by the TPK cocycle.

A complete repetition would require simultaneous closure of both clocks, vanishing of the irrational residue and return of the fiber. Here only the first closes. This decoupling is the mathematical content of the helical aperiodic crystal: exact return in a finite factor, drift in the other, hierarchical near-return in the irrational factor and genuine evolution in the fiber.

7.6 Vacancy spectrum over the internal cardinalities of HMT

The complementary identity allows any internal length N to be evaluated without using a target constant. Every window of N steps has

$$V_N \in \{[N\delta], [N\delta]\}, \quad P_N = N - V_N,$$

where V_N counts vacancies and P_N capacity increments. For some cardinalities already constructed by HMT one obtains:

N	V_N	P_N
108	4 or 5	104 or 103
132	6 or 7	126 or 125
243	11 or 12	232 or 231
396	18 or 19	378 or 377
468	21 or 22	447 or 446
729	33 or 34	696 or 695
1000	45 or 46	955 or 954.

Two rows require special attention.

At length 132, the branch with six vacancies produces exactly

$$132 = 6 + 126.$$

The integer 126 reappears in the electronic census, but the two objects are not yet identified: one is complementary capacity in a temporal window and the other is the cardinality of an electronic class. The possible bridge must send window, phase, orientation and memory to the census, not merely preserve the number.

At length 468, which coincides with the cardinality of the catalog U_6 , the number of vacancies is 21 or 22. Here the primary scale reappears as the multiplicity of events within a previously generated cardinality. The identity is exact; promoting it to self-similarity of the emitter requires proving that the enumeration of the 468 sextets respects the order of the mechanical rotation.

The complete decimal window gives an even more explicit closure:

V_{1000}	P_{1000}	$P_{1000} - 900$	$V_{1000} + P_{1000}$
45	955	55	1000
46	954	54	1000.

Thus, the pairs 45/46 and 55/54 are exact complements after removing the hundred nonadic turns $900 = 100 \cdot 9$. The corridor $45 \leftrightarrow 54 \leftrightarrow 55$ is not added through a coincidence: it appears as a balance of pruning and vacancy in the base window 1000 itself. The subsequent interpretation of 54 in the modular corridor preserves its own APP derivation; what has been constructed here is the co-occurrence map that both constructions will have to make commute.

7.7 The window 396, the boundary 27 and the radical factorization 369

The first interval of $396 = 3 \cdot 132$ steps contains, in the canonical phase, three consecutive windows with six vacancies each. Therefore,

$$396 = 18 + 378, \quad 18 = 3 \cdot 6, \quad 378 = 3 \cdot 126 = 42 \cdot 9.$$

The radical and electronic censuses satisfy by another route

$$396 = 369 + 27, \quad 27 = 3 \cdot 9.$$

The two decompositions come together arithmetically in

$$\boxed{369 = 378 - 9 = 3(132 - 6) - 9}$$

and

$$\boxed{27 = 18 + 9.}$$

These two identities establish the cardinal factorization of the square: 369 is the capacity obtained by separating one nonadic turn from the three lower windows of 132, while 27 brings together the eighteen vacancies and that turn. The assertion is strictly cardinal at this point; the realization on carry memory is obtained later through the integer reader of the specular transit matrix. The type distinction prevents confusing an identity of cardinalities with a premature identification of fibers.

It also establishes the scope of the certified numbers 396, 2376, 387 and 43. The certificate executes 396 events of six trits, whence

$$2376 = 396 \cdot 6.$$

Comparing each level with the one nine positions later leaves $396 - 9 = 387$ pairs, grouped into

$$387 = 43 \cdot 9$$

complete turns. The additional equality $2376 = 22 \cdot 108$ is true, but does not identify the trits of the certificate with twenty-two copies of the clock C_{108} ; the former is a count of tritic positions and the latter a nonsplit calendar. Only a map preserving sector, phase, carry and memory could promote that factorization to an equivalence of objects.

7.8 Synchronization, orientation and reopening of the three channels

The census extended to $t = 120$ modifies the picture of a single closure at the ninth gate. The synchronization of cardinalities reappears in clusters:

$$\begin{aligned} &\{8, 9, 10, 12, 13\}, \quad \{16, 17, 18, 19, 20\}, \quad \{29\}, \\ &\{35, 36, 37, 38, 40, 42\}, \quad \{49, 51, 52, 54, 55, 56, 57\}, \\ &\{63\}, \quad \{72, 74\}, \quad \{77, 79, 80\}, \quad \{83, 84\}, \\ &\{101\}, \quad \{104, 105, 107, 108\}, \quad \{116, 118, 119\}. \end{aligned}$$

The first reopenings are at

$$21, 43, 65, 87, 109, 131, 152, 174, 196, \dots,$$

with gaps $21/22$. In the certified window $0 \leq t \leq 120$, no *stutter* is a triple synchronization. The datum is finite, not an asymptotic theorem, but reveals a breathing dynamics:

wide opening $\longrightarrow$ pruning $\longrightarrow$ intermittent synchronization $\longrightarrow$ aperiodic reopening.

The first breath is explicit. After the triple synchronization

$$(59, 59, 59) \rightarrow (43, 43, 43) \rightarrow (32, 32, 32),$$

new small synchronizations appear

$$(17, 17, 17), (13, 13, 13), (6, 6, 6), (4, 4, 4), (4, 4, 4), (3, 3, 3), (2, 2, 2),$$

and the vacancy $t = 21$, with $K_{21} = K_{20} = 20$, reopens the fiber as

$$(611, 527, 723).$$

The same capacity phase $[20]_9 = 2$ remains, but the cardinalities, orientation and signature change drastically. This is direct material evidence that the crystal is not an orbit contracted toward a fixed point: it alternates contraction and reopening under an aperiodic calendar, while survival maintains long-range coherence.

Equality of cardinalities does not erase orientation either. At $t = 8$ there are three local orientations and at $t = 9$ two remain; future survival selects one. At the orientation gates, φ remains fixed and $\pi + e$ remains constant, while the distribution π/e is reflected. Synchronization must therefore be defined as equality of the size of the channel fibers, not as identity of words, signatures or states.

7.9 From triple synchronization to the dodecaphasic closure of α

The new structure clarifies the causal position of α . The channels π , e and φ do not need α to synchronize their cardinalities: the previous clusters arise during filtration and survival. Equality of cardinalities is not sufficient to produce α either, because orientation, specular distribution, carry and boundary condition still remain.

The dodecaphasic closure receives precisely that additional information. From the same terminal holonomic state, two readings open up:

$$x_{\text{term}} \longmapsto B_{\text{term}}, \quad x_{\text{term}} \longmapsto U_{12}^{\text{sgn}}.$$

The first determines the three surviving channels; the second preserves the signed coordinate needed to propagate the carry. The integer route produces

$$P + E - \Phi - K + C^+ - 1000C = A,$$

while the vacancy route evaluates the same genealogy through

$$\mathcal{P}(x) = \delta, \quad \mathcal{P}(x) = 2\pi x - \frac{7}{4}x^2 + \frac{x^3}{2\pi} + \frac{x^4}{20} - \frac{2x^5}{21} - \frac{x^6}{46}.$$

The identity $\Delta K = 1 - z$ explains why the second route belongs to the same generator: δ is not an external constant placed alongside the three channels, but the density of the calendar that determines their capacity and their reopenings. The synchronization of (π, φ, e) and the realization of α are not two separate stories. They are two levels of the same state:

fiber synchronization $\longrightarrow$ oriented resolution and carry memory $\longrightarrow$ dodecaphasic closure of α .

This preserves the causal lock: the correlated system $(\pi, \varphi, e, \alpha)$ is a correlated output of coinductive generation. Within that common genealogy, the three channels are determined and synchronized before the dodecaphasic seal determines the coordinate α . No conventional value of the four constants enters as an input.

7.10 Multiplicative breathing cocycle and exactly zero logarithmic drift

The complementarity between capacity and vacancy admits a multiplicative realization that makes the author's picture of cylinders contracting and reopening more precise. We define the ambient scale of the cylinder after block t by

$$\Omega_t := \frac{729^{t+1}}{1000^{K_t}}, \quad K_t = \lfloor (t+1)\rho \rfloor, \quad \rho = \log_{1000} 729 = \log_{10} 9.$$

The equality $729 = 1000^\rho$ immediately gives

$$\Omega_t = 1000^{\{(t+1)\rho\}}.$$

Therefore,

$$1 \leq \Omega_t < 1000$$

for every t , and the set $\{\Omega_t : t \geq 0\}$ is dense in the half-open multiplicative interval $[1, 1000)$. Density is not a statistical hypothesis: it follows from the irrationality of ρ . Indeed, if $\rho = a/b \in \mathbb{Q}$, then $9^b = 10^a$, contradicting the uniqueness of prime factorization.

Let

$$c_t := K_t - K_{t-1} = 1 - z_t \in \{0, 1\}$$

the compactification carry. The ratio between two consecutive scales is exactly

$$\frac{\Omega_t}{\Omega_{t-1}} = \frac{729}{1000^{c_t}} = \begin{cases} 729/1000, & z_t = 0, \\ 729, & z_t = 1. \end{cases}$$

The first branch contracts; the second reopens. These are not two rules added to the calendar: they are the two charts of the same multiplication by 729 reduced modulo the scale 1000. Before the first reopening one observes, for example,

$$\Omega_{20} = 1.3100205086376203\dots, \quad \Omega_{21} = 729\Omega_{20} = 955.0049507968252\dots$$

The accumulated contraction does not lead to an attractor because each vacancy restores the scale through an exact jump. In logarithmic coordinates,

$$\ell_t := \log_{1000} \frac{\Omega_t}{\Omega_{t-1}} = \rho - c_t = z_t - \delta, \quad \delta = 1 - \rho.$$

Moreover,

$$\sum_{j=1}^n \ell_j = \log_{1000} \Omega_n - \log_{1000} \Omega_0,$$

so that all partial sums remain uniformly bounded. In particular,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \frac{\Omega_n}{\Omega_0} = 0.$$

This is a stronger assertion than mere equality of a mean frequency: the logarithmic cocycle is a bounded coboundary. Breathing preserves the scale in the long term with zero Lyapunov exponent, although each step is a contraction or a very different reopening.

The natural geometric form of the result is the multiplicative circle

$$\mathbb{A}_{1000} := \mathbb{R}_{>0} / 1000^{\mathbb{Z}}.$$

The map

$$R_{729} : [x] \mapsto [729x]$$

is conjugate, through $u = \log_{1000} x$, to the irrational rotation $u \mapsto u + \rho$ of $\mathbb{R}/\mathbb{Z}$. The integer c_t is the index of the deck transformation that returns $729\Omega_{t-1}$ to the fundamental section $[1, 1000)$; the vacancy $z_t = 1 - c_t$ is, exactly, the minimal absence of that carry. This formulation brings together in a single object compactification, multiplicative elasticity, the torsion of the covering and the memory of how many times its boundary was crossed.

The invariant measure induced on the fundamental section is logarithmic:

$$d\mu_{\Omega}(x) = \frac{dx}{x \log 1000}, \quad 1 \leq x < 1000.$$

Therefore, for $s \neq 0$,

$$\int x^s d\mu_{\Omega}(x) = \frac{1000^s - 1}{s \log 1000},$$

and the geometric mean is $1000^{1/2} = 10\sqrt{10}$. These quantities describe the mathematical distribution of the multiplicative radius; they are not identified with a physical length or with the Planck length without the corresponding dimensional transducer.

The zero exponent does not replace the Landauer theorem of the integral holonomic state either. Here the absence of exponential drift in a scale coordinate is demonstrated; zero Landauer requires recoverability of the complete evolution. These are compatible and typologically distinct results.

Refutation criterion. A value of Ω_t outside $[1, 1000)$, a ratio different from $729/1000$ or 729 , or an unbounded logarithmic sum.

7.11 Phase and vacancy as coordinates of a single lifted rotation

The finite phase and the aperiodic word do not form an arbitrary product. They can be glued exactly as two coordinates of a single irrational translation. Let

$$u_t := \{(t+1)\rho\} \in [0, 1), \quad g_t^{(q)} := [K_t]_q \in C_q,$$

for $q \geq 2$. On

$$X_q = C_q \times [0, 1)$$

with the interval endpoints glued by the covering, we define

$$c(u) := \lfloor u + \rho \rfloor \in \{0, 1\}$$

and

$$F_q(g, u) = (g + c(u), u + \rho - c(u)).$$

The second component remains in $[0, 1)$; the first preserves the boundary crossing. The map

$$\Psi_q : X_q \longrightarrow \mathbb{R}/q\mathbb{Z}, \quad \Psi_q(g, u) = [g + u]_q,$$

satisfies

$$\boxed{\Psi_q \circ F_q = R_\rho \circ \Psi_q}, \quad R_\rho(y) = y + \rho \pmod{q}.$$

On the canonical orbit,

$$\Psi_q(g_t^{(q)}, u_t) = [(t+1)\rho]_q.$$

Therefore, the phase $[K_t]_q$, the irrational coordinate u_t and the carry $c_t = 1 - z_t$ are not three juxtaposed decorations. They are the integer part, the fractional part and the covering cocycle of a single point on the circle of length q .

This conjugacy provides a structural proof of the observed complexities. The coding of R_ρ by the q unit intervals has, at length m , exactly qm words. Its boundaries are

$$j - k\rho \pmod{q}, \quad j = 0, \dots, q-1, \quad k = 0, \dots, m-1,$$

and they are all distinct because ρ is irrational. By adding z_t , a block of length m reconstructs the immediately preceding phase symbol through

$$g_{t-1}^{(q)} = g_t^{(q)} - (1 - z_t) \pmod{q}.$$

There is therefore a bijection between decorated blocks of length m and phase blocks of length $m+1$. One obtains

$$\boxed{p_{g,q}(m) = qm, \quad p_{(g,z),q}(m) = q(m+1)}.$$

For $q = 3, 9, 12, 108$, these laws explain the certified linear complexities without assuming independence between a periodic clock and a Sturmian decoration. Growth remains linear and topological entropy remains zero, but the finite factor exactly multiplies the available language.

The inversion of the helix is now unambiguous: in the coordinate y it is the rotation $y \mapsto y - \rho$. The descending traversal does not invent another word; it reads the same invertible system with the conjugate covering cocycle. The phase can return, while u and the TPK fiber retain the history.

This formulation also requires distinguishing two objects of order 108:

$$\theta_t = [t]_{108} \in C_{108}^{(t)}, \quad g_t^{(108)} = [K_t]_{108} \in C_{108}^{(K)}.$$

The denominator $170748 = 1581 \cdot 108$ closes the first. By contrast,

$$\lfloor 170748\rho \rfloor = 162935 \equiv 71 \pmod{108},$$

so it does not close the second. The observable dodecaphasic TPK clock and the induced capacity clock are compatible, but not identical. Their future comparison requires specifying the index map, not reusing the symbol C_{108} as though it erased the domain.

Refutation criterion. A collision between boundaries $j - k\rho$, a complexity different from qm or $q(m+1)$, or the assertion that 170748 closes the capacity phase.

7.12 Induced vacancy clock and tritic typing of the returns 6/7

The secondary scale 6/7 does not merely count returns of short gaps. When passing from the block clock to the event clock, it organizes exactly the duration of the three regimes of the tritic selector.

For $n \geq 1$, let

$$p_n := \left\lfloor \frac{n}{\delta} \right\rfloor$$

the position of the n -th vacancy event and let

$$\kappa_n := K_{p_n}$$

the capacity retained at that event. Since

$$p_n \delta < n < (p_n + 1)\delta,$$

we obtain

$$\boxed{\kappa_n = p_n - n.}$$

If

$$h_n := p_{n+1} - p_n \in \{21, 22\}, \quad s_n := 22 - h_n \in \{0, 1\},$$

then

$$\kappa_{n+1} - \kappa_n = h_n - 1 = 21 - s_n.$$

Reducing modulo 3,

$$\boxed{[\kappa_{n+1}]_3 = [\kappa_n - s_n]_3.}$$

A long gap 22 preserves the regime; a short gap 21 rotates it by one unit. Since short gaps reappear after six or seven events, the plateaus of the induced regime have lengths exactly 6 or 7. The first twenty classes are

$$\underbrace{2, \dots, 2}_6, \underbrace{1, \dots, 1}_7, \underbrace{0, \dots, 0}_7.$$

To relate them to the canonical selector, the positive class is determined

$$r_n = 1 + (\kappa_n \bmod 9)$$

and one applies

$$M_{\text{ph}}(r) = \begin{cases} +1, & r \in \{1, 4, 7\}, \\ -1, & r \in \{2, 5, 8\}, \\ 0, & r \in \{3, 6, 9\}. \end{cases}$$

Thus, the first plateaus traverse

$$\boxed{0 \longrightarrow -1 \longrightarrow +1 \longrightarrow 0}$$

and each transition is triggered by a short gap. This is an exact formulation of TRIT as a unit of topological-temporal information: it does not replace the binary vacancy word, but integrates its events and determines in which regime the next transfer operates. Information is preserved because the pair (κ_n, s_n) reconstructs the previous increment; orientation is preserved because the cyclic order of the classes is not flattened.

The result does not turn the three values into three separate theories either. They are three internal regimes of a single selector, and the induced word visits them under the same cocycle. Their subsequent identification with elliptic, parabolic and hyperbolic curvatures uses the algebras $\mathbb{R}[J_\tau]/(J_\tau^2 + \tau I)$ already constructed; realization as geometric curvature or a physical field requires the corresponding transducer.

Refutation criterion. A plateau of length different from $6/7$, a regime transition during a gap 22, or a transition other than the cyclic descent through a gap 21.

7.13 Universal matrix of the two balanced branches

Sturmian balance has a linear form that brings together conservation and orientation. In a window of length N , let

$$n_N = \lfloor N\delta \rfloor.$$

The two counting possibilities are n_N and $n_N + 1$, and their complements are $N - n_N$ and $N - n_N - 1$. We form the branch matrix

$$W_N := \begin{pmatrix} n_N & N - n_N \\ n_N + 1 & N - n_N - 1 \end{pmatrix}.$$

Its identities are universal:

$$W_N \begin{pmatrix} 1 \\ 1 \end{pmatrix} = N \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad (1, -1)W_N = -(1, -1),$$

$$\text{tr } W_N = N - 1, \quad \det W_N = -N,$$

and, therefore,

$$\boxed{\chi_{W_N}(\lambda) = (\lambda - N)(\lambda + 1)}.$$

The diagonal mode preserves the total length of the window. The oriented functional that subtracts one branch from the other changes sign with eigenvalue -1 . The oriented unit is not introduced from outside: it appears because the two balanced words differ by a single event and a single complementary carry.

For the internal lengths considered one obtains, among others,

$$W_{132} = \begin{pmatrix} 6 & 126 \\ 7 & 125 \end{pmatrix}, \quad W_{468} = \begin{pmatrix} 21 & 447 \\ 22 & 446 \end{pmatrix}, \quad W_{1000} = \begin{pmatrix} 45 & 955 \\ 46 & 954 \end{pmatrix}.$$

Each has the eigenvalues N and -1 . This law precisely separates conservation of the total unit and the elementary orientation of the vacancy. The third tritic value 0 corresponds to the threshold of the oriented functional, not to a third counting branch added to the binary word.

The window 1000 contains an additional internal reduction. After separating $900 = 100 \cdot 9$ units of capacity, what remains is

$$\tilde{W}_{100} = \begin{pmatrix} 45 & 55 \\ 46 & 54 \end{pmatrix},$$

with

$$\text{tr } \tilde{W}_{100} = 99, \quad \det \tilde{W}_{100} = -100, \quad \text{Spec}(\tilde{W}_{100}) = \{100, -1\}.$$

The half-imbalances of its rows are

$$\frac{55 - 45}{2} = 5, \quad \frac{54 - 46}{2} = 4.$$

By another route, the central APP block

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}$$

has spectrum $\{9, 5\}$ and jump $9 - 5 = 4$. Therefore, the two scalar readers

$$\chi_{\text{vac}}(\tilde{W}_{100}) = (5, 4), \quad \chi_{\text{APP}}(B_c) = (\lambda_{\min}, \lambda_{\max} - \lambda_{\min}) = (5, 4)$$

coincide exactly. This constructs the common codomain of the vacancies–APP square. It does not yet demonstrate that $\tilde{W}_{100}$ and B_c are the same operator: for that promotion, the arrow transporting the two window branches to the two APP eigenspaces while preserving phase, sheet and carry is missing. The reservation is located at that arrow and does not weaken either of the two derivations.

Refutation criterion. A balanced window whose matrix does not have spectrum $\{N, -1\}$, or a purported identification with APP that does not transport the branches.

7.14 Synchronization quotient A_2 and projective direction of the first reopening

The censuses of the three channels admit a canonical quotient that eliminates common survival and preserves only their misalignment. For

$$N = (N_\pi, N_e, N_\varphi) \in \mathbb{Z}^3$$

we define

$$\partial N = (N_\pi - N_e, N_e - N_\varphi, N_\varphi - N_\pi).$$

Its image is contained in

$$A_2 = \{(x, y, z) \in \mathbb{Z}^3 : x + y + z = 0\},$$

and is all of A_2 . The kernel is the diagonal $\mathbb{Z}(1, 1, 1)$. Consequently,

$$\boxed{\mathbb{Z}^3 / \mathbb{Z}(1, 1, 1) \simeq A_2.}$$

The quadratic form

$$Q(N) = \frac{1}{2} \|\partial N\|^2$$

vanishes exactly at triple synchronizations. The unit defects of the phases 11, 14, 15 satisfy $Q = 1$ and are minimal roots of A_2 . Equalities of two channels occupy the three mirrors of the Weyl group $W(A_2) \simeq S_3$; the orientation on one side or the other of the mirror remains in the sign of the root.

The initial corridor of mirrors is

t	(N_π, N_e, N_φ)	mirror	∂N	d_t
4	(207, 207, 122)	$\pi = e$	(0, 85, -85)	85
5	(39, 151, 151)	$e = \varphi$	(-112, 0, 112)	112
6	(110, 110, 69)	$\pi = e$	(0, 41, -41)	41
7	(81, 29, 81)	$\pi = \varphi$	(52, -52, 0)	52
8	(59, 59, 59)	S_3	(0, 0, 0)	0
9	(43, 43, 43)	S_3	(0, 0, 0)	0.

The four amplitudes preceding the first triple closure form the matrix

$$D_{\text{syn}} := \begin{pmatrix} 85 & 112 \\ 41 & 52 \end{pmatrix}.$$

Without introducing α , the electron or an experimental value, the identities are verified

$$112 - 85 = 27, \quad 52 - 41 = 11,$$

$$85 + 41 = 126, \quad 112 + 41 = 153, \quad 85 + 52 = 137,$$

and

$$\boxed{\text{tr } D_{\text{syn}} = 137, \quad \det D_{\text{syn}} = -172 = -4 \cdot 43.}$$

The same corridor thus contains the census 126, the return 153, the integer direction 137, the boundary 27, the rank 11, the jump 4 and the second synchronization 43. All these equalities are exact. The matrix, considered in isolation, determines an integer direction of trace 137, not the constant α . The derivation of α additionally requires the signed state, the dodecaphasic closure and the complete jet, which are constructed in the corresponding sections.

The first reopening offers a more precise arrow. At $t = 20$,

$$N_{20} = (2, 2, 2), \quad \partial N_{20} = 0.$$

At $t = 21$, the capacity remains $K_{21} = K_{20} = 20$, the determined phase is the same and the selector is in the threshold class $r = 3$, for which $M_{\text{ph}} = 0$. However,

$$N_{21} = (611, 527, 723)$$

and

$$\boxed{\partial N_{21} = (84, -196, 112) = 28(3, -7, 4).}$$

The primitive vector $(3, -7, 4)$ belongs to A_2 . In the oriented projective chart whose denominator is the component $\varphi - \pi$,

$$\chi_{e\varphi|\varphi\pi}(\partial N) := \frac{N_e - N_\varphi}{N_\varphi - N_\pi},$$

we obtain

$$\chi_{e\varphi|\varphi\pi}(\partial N_{21}) = \frac{-196}{112} = -\frac{7}{4}.$$

This is exactly the quadratic coefficient of the kernel $E_{21/22}$. It is no longer a matter of finding the integers 7 and 4 separately: the complete oriented ratio emerges at the first reopening of the three channels, after synchronization and before the resolution of α . The identity is internal and does not use the target value of α .

7.15 Sturmian-triadic genesis of 22/7 and internal portrait of the jet $E_{21/22}$

The first seven effective vacancy positions are

$$21, 43, 65, 87, 109, 131, 152.$$

Between the first and the seventh, 132 steps elapse and six intervals appear. In that first closed window,

$$132 = 6 \cdot 22, \quad 126 = 6 \cdot 21.$$

Tritic amplification gives

$$396 = 3 \cdot 132,$$

and, therefore,

$$\frac{396}{126} = \frac{3 \cdot 6 \cdot 22}{6 \cdot 21} = \frac{22}{7}.$$

The rational seal 22/7 is not introduced as a historical approximation of π : it is obtained from the ratio between the tritic amplification of the window and its survival on the branch of six long gaps. The subsequent comparison with π preserves its status as recognition; it does not generate the ratio.

The dynamics also provides five pairs of endpoints invariant under a change of origin of the mechanical word:

$$\mathcal{G}_{\text{prim}} = \{21, 22\}, \quad \mathcal{R}_{\text{sec}} = \{6, 7\}, \quad \mathcal{G}_{\text{inv}} = \{20, 21\},$$

$$\mathcal{V}_{1000} = \{45, 46\}, \quad \mathcal{H}_{100} = \{4, 5\}.$$

With the orientation already fixed by the TPK, consider the extraction

$$r^+ = 7, \quad h^- = 4, \quad g_{\text{inv}}^- = 20, \quad g_{\text{prim}}^- = 21, \quad v_{1000}^+ = 46.$$

These five invariants reproduce exactly the nontranscendental coefficients of the sixth-order jet:

$$-\frac{r^+}{h^-} = -\frac{7}{4}, \quad \frac{1}{g_{\text{inv}}^-} = \frac{1}{20}, \quad -\frac{2}{g_{\text{prim}}^-} = -\frac{2}{21}, \quad -\frac{1}{v_{1000}^+} = -\frac{1}{46}.$$

Degrees one and three form the reciprocal pair

$$(2\pi) (2\pi)^{-1} = 1.$$

Thus, the declared kernel can be written as

$$P_6(x) = 2\pi x - \frac{r^+}{h^-} x^2 + (2\pi)^{-1} x^3 + \frac{x^4}{g_{\text{inv}}^-} - \frac{2x^5}{g_{\text{prim}}^-} - \frac{x^6}{v_{1000}^+}.$$

This expression does not use the root of $P_6(x) = \delta$ to choose the coefficients. All the numbers on the right-hand side have appeared earlier as invariants of the compactification calendar. The coefficient $-7/4$ also has the second internal factorization from the preceding section as a projective coordinate of the first defect A_2 .

The preceding equality is an *exact genealogical factorization* of the kernel. The selected endpoints, their degrees and their signs are brought together by the graded reader $\mathfrak{E}_9$, defined later on the invariant signature of the integral holonomic state; this reader commutes with transports that preserve origin, orientation, windows and memory.

The prolongation of degrees 7, 8, 9 already has another status. The reference constructions demonstrate

$$T_{7:9}(x) = -\frac{x^7}{120} + \frac{x^8}{45} + \frac{2x^9}{495},$$

with

$$120 = |\mathcal{F}_8^{\text{ph}}|, \quad 45 = \det B_c, \quad 11 = \text{rank}(D_3, D_4), \quad 495 = 45 \cdot 11,$$

and the nilpotent resolvent $(I - N)^{-1}$. The complete jet

$$P_9 = P_6 + T_{7:9}$$

is thus stratified: degrees 1:6 come from the synchronization signature and degrees 7:9 from phase incidence, the boundary determinant and the dodecaphasic rank. The simple root of $E_{21/22}$ is obtained after that factorization and does not enter into the selection of the coefficients.

7.16 Refined causal chain: from compactification to the coordinate α

The results of the previous sections allow a verbal sequence to be replaced by a diagram of domains and quotients:

APP–TRIT–TPK and internal ratio 729/1000

$$\begin{aligned} &\longrightarrow (\mathbb{A}_{1000}, R_{729}, c_t, z_t) \\ &\longrightarrow (g_t^{(9)}, u_t, x_t^{\text{enr}}) \\ &\longrightarrow (N_\pi, N_e, N_\varphi) \\ &\xrightarrow{\partial} A_2 \\ &\longrightarrow (B_{\text{term}}, U_{12}^{\text{sgn}}) \\ &\longrightarrow \begin{cases} \text{integer dodecaphasic closure,} \\ \text{simple root of the jet } E_{21/22}, \end{cases} \\ &\longrightarrow \alpha. \end{aligned}$$

The first level constructs the multiplicative rotation and its carry cocycle. The second simultaneously preserves phase, irrational part and state. The third determines the three correlated channels. The quotient A_2 measures their misalignment without erasing common survival. The dodecaphasic closure then receives the orientation and memory that the census quotient does not contain.

Synchronization therefore does not by itself generate α . It does provide two concrete internal precursors:

1. the rectangle of defects, whose trace is 137;
2. the first threshold reopening, whose projective coordinate is $-7/4$.

Both appear before solving the analytic equation and before any experimental comparison. The first precursor locates an integer direction; the second locates a coefficient of the jet. The remaining information resides in the signed state, in the dodecaphasic carry and in the higher degrees of the kernel.

This architecture specifies in what sense the aperiodic crystal can be the nerve of the construction, without turning it into an all-encompassing phrase. The crystal is the minimal system

$$(\mathbb{A}_{1000}, R_{729}, c_t; C_9, \Gamma_9; \mathcal{X}_{\text{TPK}}^{\text{enr}})$$

that brings together an irrational base of zero entropy, an integer carry, a lifted finite holonomy and a fiber with memory. Its realizations π, φ, e, α are not inputs to the crystal; they are correlated characters of the state it prolongs. The dimensional, fractal and exceptional realizations are obtained afterward through their own transducers. The graded reader $\mathfrak{C}_9$ brings together the defect $28(3, -7, 4)$, the matrix D_{syn} and the vacancy endpoints in the coefficient signature of P_9 , so that synchronization and the derivation of α are linked without introducing the value of the constant as a datum.

7.17 The compactification slope is transcendental and its renormalization is nonstationary

The aperiodic structure does not depend only on the vacancy word having no period. Its slope has a stronger arithmetic property. Recall that

$$\delta = \log_{10} \frac{10}{9}, \quad \rho = 1 - \delta = \log_{10} 9.$$

Both quantities arise from the internal ratio between the ternary opening of a six-trit block, $3^6 = 729$, and decimal compactification $10^3 = 1000$. They are not introduced as parameters of an externally chosen rotation.

First, δ is irrational. If $\delta = p/q \in \mathbb{Q}$, with $q > 0$, then

$$10^{p/q} = \frac{10}{9} \implies 9^q 10^p = 10^q.$$

The equality contradicts the uniqueness of prime factorization: the left-hand side contains the factor 3, while the right-hand side contains only 2 and 5. Second, δ cannot be algebraic irrational. In that case, the Gelfond–Schneider theorem would make 10^δ transcendental, but

$$10^\delta = \frac{10}{9}$$

is algebraic. One concludes

$\delta \text{ is transcendental.}$

This classical recognition is applied after constructing δ from APP–TRIT–TPK; it does not act as a generator of the calendar. Its structural consequences are immediate. The continued fraction of δ is not

eventually periodic, because an eventually periodic continued fraction represents a quadratic irrational. Therefore, the renormalization

$$[0; a_1, a_2, a_3, \dots] = [0; 21, 1, 5, 1, 6, 2, 3, 1, 1, 8, \dots]$$

does not close on a stationary Sturmian substitution. The word admits an S -adic description directed by the partial quotients, but that directive does not repeat periodically. The crystal is ordered and recurrent, without being the fixed point of a single quadratic inflation.

The same conclusion can be formulated as the absence of a nontrivial modular stabilizer. If

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{Z})$$

fixed the slope through a fractional transformation,

$$\frac{a\delta + b}{c\delta + d} = \delta,$$

then

$$c\delta^2 + (d - a)\delta - b = 0.$$

The transcendence of δ forces $c = b = 0$ and $d = a$. Since $\det M = \pm 1$, the resulting projective action is the identity. Therefore,

$$\mathrm{Stab}_{\mathrm{PGL}_2(\mathbb{Z})}(\delta) = \{1\}.$$

Nonrepetition of renormalization does not eliminate long-range order. The mechanical word

$$z_n = \mathbf{1}_{[1-\delta, 1)}(\{n\delta + \beta\})$$

is a regular model set: it comes from cutting a lattice orbit by an interval window whose boundary has zero measure. Its hull is minimal and uniquely ergodic, has complexity $p(m) = m + 1$, zero topological entropy and pure dynamical spectrum. The expression “aperiodic crystal” is thus made precise: diffractive order and uniform recurrence, together with absence of translational periodicity and stationary renormalization.

Refutation criterion. A period of the word, an eventually periodic continued fraction or a nontrivial modular transformation fixing δ .

7.18 The three channels as cuts of the same window over 729 extensions

The synchronization of π , e and φ can be described without reducing it to equality of censuses. At level t , let $T_{c,t}$ be the integer represented by the first $6t$ ternary digits of channel $c \in \{\pi, e, \varphi\}$, and let $D_{c,t}$ be the integer represented by its first K_t decimal blocks. Each local extension is identified with an integer

$$u \in \{0, 1, \dots, 728\}.$$

The corresponding ternary cell is

$$\left[\frac{729T_{c,t} + u}{3^{6(t+1)}}, \frac{729T_{c,t} + u + 1}{3^{6(t+1)}} \right),$$

while the decimal cylinder is

$$\left[\frac{D_{c,t}}{1000K_t}, \frac{D_{c,t} + 1}{1000K_t} \right).$$

Let us define the common scale and the internal displacement

$$\Omega_t = \frac{3^{6(t+1)}}{1000K_t}, \quad \xi_{c,t} = D_{c,t}\Omega_t - 729T_{c,t}.$$

The two cylinders intersect exactly when

$$u < \xi_{c,t} + \Omega_t, \quad u + 1 > \xi_{c,t}.$$

Consequently, the set of surviving extensions is

$$I_{c,t} = \{0, \dots, 728\} \cap [\lfloor \xi_{c,t} \rfloor, \lceil \xi_{c,t} + \Omega_t \rceil - 1]_{\mathbb{Z}},$$

and the determined census satisfies

$$N_c(t) = |I_{c,t}|.$$

The formula provides the geometric mechanism of synchronization. The three channels receive the same width Ω_t ; they differ only in the translation $\xi_{c,t}$ of the window. The equality $N_\pi = N_e = N_\varphi$ means that three translated intervals cut the local domain of 729 cells with the same cardinality. It does not imply that the intervals, words or their histories are equal.

In the terminal state immediately preceding the first vacancy, one obtains the integer intervals

c	$t = 20$: raw interval	$I_{c,20}$
π	[407, 408]	[407, 408]
e	[172, 173]	[172, 173]
φ	[313, 314]	[313, 314].

The three intervals occupy different positions, but contain two cells. This is the exact form of the synchronized state $(2, 2, 2)$: equality of discrete measure with distinct positional memory.

The next vacancy preserves $K_{21} = K_{20} = 20$ and multiplies the scale by 729. The raw intervals become

c	$t = 21$: raw interval	$I_{c,21}$	$N_c(21)$
π	[118, 1073]	[118, 728]	611
e	[-429, 526]	[0, 526]	527
φ	[6, 961]	[6, 728]	723.

The three raw intervals have the same integer width, 956, but are clipped by opposite boundaries of the finite fiber. If

$$\text{Def}_* := 729(1, 1, 1) - (N_\pi(21), N_e(21), N_\varphi(21)),$$

then

$$\text{Def}_* = (118, 202, 6) = 6(1, 1, 1) + 28(4, 7, 0).$$

The diagonal term $6(1, 1, 1)$ records the common loss of a six-trit block. Taking the quotient by diagonal survival leaves the oriented divisor

$$[\text{Def}_*]_{A_2} = 28(4, 7, 0).$$

Equivalently,

$$\partial N_{21} = (84, -196, 112) = 28(3, -7, 4).$$

The slope

$$\frac{N_e - N_\varphi}{N_\varphi - N_\pi} = -\frac{196}{112} = -\frac{7}{4}$$

is therefore the ratio of two oriented clippings of a common window. It is not a fraction added to the analytic kernel.

The full recalculation of the 171 levels available in the reference construction shows two finite uniqueness properties: $t = 21$ is the only vacancy preceded by a chain of five consecutive synchronizations, and it is the only level whose projective defect in that chart is $-7/4$. The assertion is exact in the domain of definition; its coinductive extension to arbitrary depth will have to explicitly preserve the marked state.

Refutation criterion. The censuses not coinciding with the interval intersections, the three raw widths at $t = 21$ differing, or another slope $-7/4$ appearing in the declared domain.

7.19 Specular transit matrix and joint closure of 21/22, 6/7, 137 and the lattice 369–126

The four phases preceding the first triple synchronization should not be read as four anecdotes. Their defect amplitudes form the matrix

$$D_{\text{syn}} = \begin{pmatrix} 85 & 112 \\ 41 & 52 \end{pmatrix}.$$

The synchronized census of the ninth phase is $N_9 = 43$. Therefore,

$$\det D_{\text{syn}} = -172 = -4 \cdot 43$$

defines internally

$$a := -\frac{\det D_{\text{syn}}}{N_9} = 4.$$

Through an independent branch, inducing the vacancy word on its short gaps produces the secondary spectrum $\mathcal{R}_{\text{sec}} = \{6, 7\}$. Let

$$b := \max \mathcal{R}_{\text{sec}} = 7.$$

The first reopening then satisfies

$$t_* = 3b = 21, \quad K_* = 3b - 1 = 20,$$

where 3 is the cardinality of the TRIT regimes. Moreover,

$$\lfloor 1000\delta \rfloor = 45, \quad \lceil 1000\delta \rceil = 46 = 2(3b) + a.$$

The clipping decomposition of the preceding section now takes the closed form

$$\boxed{\text{Def}_* = 6(1, 1, 1) + ab(a, b, 0)}.$$

Indeed,

$$6(1, 1, 1) + 4 \cdot 7(4, 7, 0) = (118, 202, 6).$$

This equality is a compatibility square between two independent extractions: $a = 4$ comes from the synchronization determinant, $b = 7$ comes from the induced clock and the common clipping finds them again as boundary weights. The fraction $-b/a = -7/4$ is thus determined before solving any equation for α .

The same matrix brings together Diophantine renormalization. The two secondary returns give

$$q_6 = 22 \cdot 6 - 1 = 131, \quad q_7 = 22 \cdot 7 - 1 = 153,$$

and satisfy

$$6q_7 - 7q_6 = 1.$$

Within D_{syn} , those denominators reappear as

$$q_6 = (85 + 41) + 5 = 131, \quad q_7 = 112 + 41 = 153,$$

where 5 is the upper branch of the half-imbalance $4/5$ of $\tilde{W}_{100}$. Finally,

$$\boxed{\text{tr } D_{\text{syn}} = 137 = q_6 + 6.}$$

Thus, 137 is not obtained here by rounding α^{-1} . It is an integer direction of synchronization transit and, simultaneously, the return denominator 131 prolonged by a block of six trits. This does not replace the two constructions of α : it establishes a common integer precursor that the dodecaphasic closure and the analytic kernel must transport.

The radical–electronic lattice is also reconstructed from D_{syn} . Let us define

$$A_{\text{surv}} := 85 + 41 = 126, \quad B_{\text{bord}} := 112 - 85 = 27,$$

and

$$T_{\text{tri}} := 3(A_{\text{surv}} + 6) = 396.$$

The matrix reader

$$\mathcal{R}(D_{\text{syn}}) := \begin{pmatrix} T_{\text{tri}} - B_{\text{bord}} & T_{\text{tri}} \\ T_{\text{tri}} - A_{\text{surv}} - B_{\text{bord}} & T_{\text{tri}} - A_{\text{surv}} \end{pmatrix}$$

produces exactly

$$\boxed{\mathcal{R}(D_{\text{syn}}) = \begin{pmatrix} 369 & 396 \\ 243 & 270 \end{pmatrix}.}$$

Consequently, the four numerical vertices of the parallelogram are no longer collected as an independent set: they come from an integer reader of channel synchronization, the six-trit block and tritic amplification. The identification of each vertex with its radical or electronic object nevertheless preserves the domain and the transducer of its domains of origin; equality of the numerical matrix does not erase those fibers.

Refutation criterion. a not being an integer, the induced word not producing $b = 7$, failure of the clipping law, or $\mathcal{R}(D_{\text{syn}})$ not reproducing the four vertices.

7.20 Internal factorization of all the coefficients of the jet P_9

The preceding results make it possible to bring together, for the first time within this development, the genealogy of all the coefficients of the jet without introducing its root. Let

$$\omega := 2\pi, \quad m := 3, \quad a := 4, \quad b := 7,$$

and let us define

$$t_* = mb = 21, \quad K_* = mb - 1 = 20,$$

$$v_- := \lfloor 1000\delta \rfloor = 45, \quad v_+ := \lceil 1000\delta \rceil = 46.$$

The lower branch simultaneously satisfies

$$v_- = 45 = \det \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}.$$

The internal difference of the second row of D_{syn} gives

$$r_{\text{dod}} := 52 - 41 = 11,$$

which coincides with the dodecaphasic transverse rank already demonstrated. The phase half-crown contributes

$$f_{\text{ph}} = 8 \binom{6}{2} = 120,$$

and

$$v_- r_{\text{dod}} = 45 \cdot 11 = 495.$$

On the invariant signature

$$\mathcal{I}_9 = (\omega, m, a, b, K_*, t_*, v_-, v_+, f_{\text{ph}}, r_{\text{dod}}),$$

we define the graded reader

$$\begin{aligned} \mathfrak{E}_9(\mathcal{I}_9)(x) := & \omega x - \frac{b}{a}x^2 + \omega^{-1}x^3 + \frac{x^4}{K_*} - \frac{2x^5}{t_*} - \frac{x^6}{v_+} \\ & - \frac{x^7}{f_{\text{ph}}} + \frac{x^8}{v_-} + \frac{2x^9}{v_- r_{\text{dod}}}. \end{aligned}$$

Substitution of the invariants produces

$$\boxed{\begin{aligned} \mathfrak{E}_9(\mathcal{I}_9)(x) = & 2\pi x - \frac{7}{4}x^2 + \frac{x^3}{2\pi} + \frac{x^4}{20} - \frac{2x^5}{21} - \frac{x^6}{46} \\ & - \frac{x^7}{120} + \frac{x^8}{45} + \frac{2x^9}{495} = P_9(x). \end{aligned}}$$

The formula reveals a relation that remained hidden when P_6 and $T_{7,9}$ were presented separately: the vacancy pair $45/46$ crosses the boundary between them. The upper branch appears in $-x^6/46$, while the lower branch reappears in $+x^8/45$ and, coupled to the rank 11, in $2x^9/495$. The prolongation does not add a second arithmetic; it continues the same balanced calendar through the nilpotent resolvent.

The typed diagram is

$$\mathcal{X}_{\text{TPK}}^{\text{enr},*} \xrightarrow{\mathcal{A}_*} \mathcal{I}_9 \xrightarrow{\mathfrak{E}_9} \mathbb{Q}(\pi)[x]/(x^{10}),$$

where the asterisk indicates that the object preserves the coinductive origin, the order of the channels and the orientation of their sheets. The extractor $\mathcal{A}_*$ performs the demonstrated operations:

$$D_{\text{syn}} \mapsto a, \quad \mathcal{R}_{\text{sec}} \mapsto b, \quad \text{Def}_* \mapsto (a, b),$$

$$\delta \mapsto (v_-, v_+), \quad B_c \mapsto v_-, \quad (D_3, D_4) \mapsto r_{\text{dod}}.$$

Every morphism that preserves the marked state, the intersection windows, the channel orientation, the capacity clock and the two-way involution preserves $\mathcal{I}_9$, and therefore makes the reader commute. What has been closed is the internal factorization of the coefficient signature. The stronger assertion that $\mathfrak{E}_9$ is the only possible reader in an unpointed category requires classifying all automorphisms that can forget or permute the origin and orientation. That reservation is concrete and reopens neither the simple root of $E_{21/22}$, nor the nilpotent resolvent, nor the genealogy of the coefficients.

Refutation criterion. Any coefficient requiring the value of the root, two declared extractions producing different values, or an allowed automorphism changing $\mathcal{I}_9$.

7.21 The factor 28, the exact projection 36×28 and the partition into seven tetrads

The primitive scale of the first defect is

$$\lambda_* := \gcd(84, 196, 112) = 28 = ab.$$

In the constructed catalog of the region of π , the oriented microfiber has multiplicity

$$M(w_\pi) = 432 + 4 \cdot 144 = 1008$$

and admits the factorizations

$$1008 = 36 \cdot 28 = \binom{9}{2} \binom{8}{2} = 7 \cdot 144.$$

Therefore,

$$\boxed{M(w_\pi) = 36 \lambda_*}$$

is an exact scalar compatibility between the clipping defect of the first reopening and the multiplicity of the microfiber of π . The factor 36 also belongs to R_{36} and to the set of pairs of a nonad; 28 is the number of pairs of a set of eight elements.

The initial reading of this equality as mere scalar compatibility was incomplete. The reference construction already constructs on the microfiber

$$\mathcal{R}_{010211} = \{(P, T) : w_6(U_6(P, T)) = 010211\}$$

an explicit projection. For

$$P = (i, j, d_P), \quad T = (k, l, d_T),$$

the directional gauge is fixed

$$\beta(N) = 1, \quad \beta(E) = 2, \quad \beta(S) = 3, \quad \beta(O) = 4$$

and one defines

$$x(P) = 5(i - j) \pmod{9}, \quad h(T) = 5\beta(d_T) \pmod{9},$$

$$q_\beta(P, T) = \{x(P) - h(T), x(P) + h(T)\}.$$

The map

$$q_\beta : \mathcal{R}_{010211} \longrightarrow \begin{pmatrix} \mathbb{Z}/9\mathbb{Z} \\ 2 \end{pmatrix}$$

is surjective, has a base of cardinality 36 and all its fibers have cardinality 28. Each fiber receives, in the order of the five regions of π , exactly

$$12 + 4 + 4 + 4 + 4 = 28$$

paths. It is not obtained by dividing 1008 by 36: the certificate reconstructs the 104,976 seed pairs, computes each image and proves the uniformity of the fibers.

The additive translation of C_9 and its reflection preserve U_6 and make q_β equivariant with respect to the dihedral action on pairs. On the multiplicative cursor, two involutions generate a free V_4 that preserves the complete signature. The group

$$H_0 = C_9 \times V_4$$

The ratio

$$\Omega = \mathcal{R}_{010211}/H_0, \quad |\Omega| = 28,$$

has an additional intrinsic partition:

$$\Omega = \mathcal{B}_3 \sqcup \mathcal{B}_6 \sqcup \mathcal{B}_9 \sqcup \mathcal{L}_1 \sqcup \mathcal{L}_2 \sqcup \mathcal{L}_3 \sqcup \mathcal{L}_4,$$

with seven blocks of four elements. The three blocks $\mathcal{B}_3, \mathcal{B}_6, \mathcal{B}_9$ are transverse and are distinguished by the multiplicative residue 3, 6, 9; the four blocks $\mathcal{L}_j$ come from the four lateral regions. The reduced census vector of this partition is

$$v_\Omega = (3, -7, 4) \in A_2.$$

The first reopening of the channels satisfies, exactly,

$$\partial N_{21} = 28v_\Omega.$$

Thus, the direction $(3, -7, 4)$ does not merely contain the slope $-7/4$: it is the conservation relation between three transverse tetrads, seven total tetrads and four lateral tetrads. Moreover,

$$28 = 7 \cdot 4, \quad 1008 = 36 \cdot 7 \cdot 4.$$

The integers $a = 4$ and $b = 7$, obtained earlier from the synchronization determinant and the induced return, now reappear as block size and number of blocks of the microfiber.

This identification has decisive local rigidity. The integral group of isometries of A_2 , represented by permutations of the three coordinates and global sign change, has order 12. Since the coordinates $3, -7, 4$ are distinct and the multiset $\{3, -7, 4\}$ does not coincide with its negative, only the identity fixes v_Ω . Therefore,

$$\text{Stab}_{\text{Aut}(A_2)}(v_\Omega) = \{1\}.$$

Once the first reopening is characterized, the defect itself recovers the order of the three channels and their orientation: it needs no external mark.

The octadic boundary is typed by the seven tetrads. After choosing a gauge, bijective charts

$$\Phi : \Omega \xrightarrow{\sim} \begin{pmatrix} \mathbb{F}_2^3 \\ 2 \end{pmatrix}$$

and with them an exact incidence $J(8, 2)$. However, those charts are not selected by the source relations tested. The reflection stabilizing each fiber acts freely with cycle type 2^{14} , whereas every involution of $\text{Aut } J(8, 2) \cong S_8$ fixes at least four pairs. The proved no-go follows:

$$\text{there is no family of charts } J(8, 2) \text{ equivariant under every } D_9 \times V_4.$$

The oriented datum v_Ω , whose stabilizer is trivial, provides the reflection breaking required by this obstruction theorem. Consequently, the Johnson charts on each fiber $F_b = q_\beta^{-1}(b)$ are conjugate gauge charts and not a canonical selection under the complete action $D_9 \times V_4$. The projection 36×28 remains intrinsic; the choice of an octadic chart is not part of its definition.

A temporal ambiguity is also corrected. The integer 1008 is not the period of the dodecaphasic clock:

$$1008 \equiv 36 \pmod{108}.$$

After 1008 steps the nonadic phase returns, but the clock C_{108} advances by 36 positions. On this front, 1008 is a microfiber multiplicity and a combinatorial incidence; 108 is the finite temporal period. The projection q_β explains the product 36×28 ; an octadic gauge chart and the clock map are distinct operations.

Refutation criterion. The seed sweep changing a fiber or the partition 7×4 , a nontrivial isometry of A_2 fixing v_Ω , the existence of a completely $D_9 \times V_4$ -equivariant Johnson chart, or 1008 being used as the period of C_{108} .

7.22 Two clocks, an integrated memory and two spectral modules

Let

$$Z_t := \sum_{j=0}^t z_j = \lfloor (t+1)\delta \rfloor.$$

Since $\rho = 1 - \delta$ and $(t+1)\delta \notin \mathbb{Z}$,

$$\begin{aligned}
K_t &= \lfloor (t+1)\rho \rfloor \\
&= \lfloor (t+1) - (t+1)\delta \rfloor \\
&= t - \lfloor (t+1)\delta \rfloor = t - Z_t.
\end{aligned}$$

If

$$\theta_t = [t]_q \quad \text{and} \quad g_t = [K_t]_q,$$

then

$$g_t = \theta_t - [Z_t]_q.$$

The capacity phase is the uniform phase corrected by the integrated memory of vacancies. This identity expresses exactly the difference between phase return and state return: knowing θ_t does not determine g_t if Z_t has been forgotten.

The distinction produces two legitimate spectral modules. On the block clock, the translation

$$S_q(\theta, u) = (\theta + 1, u + \delta)$$

on $C_q \times \mathbb{T}$ has additive frequencies

$$\mathcal{M}_q^{\text{blk}} = \frac{1}{q}\mathbb{Z} + \delta\mathbb{Z} = \frac{1}{q}\mathbb{Z} + \rho\mathbb{Z} \pmod{1}.$$

In particular,

$$\mathcal{M}_9^{\text{blk}} = \frac{1}{9}\mathbb{Z} + \log_{10} 9 \mathbb{Z},$$

and

$$\mathcal{M}_{108}^{\text{blk}} = \frac{1}{108}\mathbb{Z} + \log_{10} 9 \mathbb{Z}.$$

This is the exact sense in which the internal decimal logarithm appears as a spectral module: not as positive entropy, but as an irrational generator of a dense group of pure frequencies.

On the glued capacity clock, the dynamics F_q of section 49 is conjugate to

$$y \mapsto y + \rho \quad \text{on} \quad \mathbb{R}/q\mathbb{Z}.$$

Its frequencies are

$$\mathcal{M}_q^{\text{cap}} = \frac{\rho}{q}\mathbb{Z} \pmod{1}.$$

There is no contradiction between the two modules: they belong to two algebras of observables of the same integral holonomic state. The first preserves the block index; the second absorbs the accumulated carry into the finite coordinate. The relation $g_t = \theta_t - Z_t$ is the gauge change that compares them.

This also clarifies two returns mentioned earlier. The denominator $170748 = 1581 \cdot 108$ closes the block clock, but

$$K_{170748-1} \equiv 71 \pmod{108},$$

so it does not close the capacity clock. Likewise, 1008 closes C_9 , but shifts C_{108} by 36. The integral holonomic state retains height and signature in both cases, so neither of those returns equates to restarting the dynamics.

Refutation criterion. That $K_t \neq t - Z_t$, the same spectrum being attributed to two distinct domains of observables, or a finite return erasing memory.

Joint nonadic connection and biography of return with memory

The connection is constructed from the complete orbital classification and from its first coinductive elevation. The following development fixes the three initial histories, the oriented boundaries and the chain $w_6 \rightarrow w_{30} \rightarrow R_{36}$ before introducing the nonadic holonomy.

8.1 Orbital classification, initial words and 1st boundary

The ternary reduction of the emitter realizes 243 words within the ambient fiber $\mathbb{F}_3^6$. This unit does not choose three words by decimal comparison. It first constructs the symmetry action of the catalogue, classifies its orbits by integer invariants, and only afterwards orients the three structural representatives.

8.1.1 Dihedral action on 6 coordinates

For $u = (u_1, u_2, u_3, u_4, u_5, u_6) \in \mathbb{F}_3^6$, define

$$\begin{aligned} r(u) &= (u_3, u_1, u_2, u_5, u_6, u_4), \\ s(u) &= (u_4, u_5, u_6, u_1, u_2, u_3). \end{aligned}$$

We verify

$$r^3 = s^2 = 1, \quad srs = r^{-1}.$$

Therefore, r, s generate an action of D_3 .

The same permutation acts on the six blocks of each emission $U = (U_1, \dots, U_6)$. The projection Ψ satisfies

$$\Psi(gU) = g\Psi(U), \quad g \in D_3.$$

Theorem 8.1 (Orbital decomposition of the catalog). *The 243 realized words decompose into 43 orbits:*

$$38 \text{ orbits of cardinality } 6, \quad 5 \text{ orbits of cardinality } 3.$$

In particular,

$$38 \cdot 6 + 5 \cdot 3 = 243.$$

Proof. The set is applied to each word.

$$\{u, ru, r^2u, su, sru, sr^2u\}$$

and repetitions are removed. Exhaustive enumeration of the catalog produces the indicated histogram. The equivariance of Ψ ensures that no orbit leaves the realized image. $\square$

8.1.2 Fiber Invariants

For a word realized w , let

$$\begin{aligned} \mathcal{F}_w &:= \{U \in \text{im } T_{\min} : \Psi(U) = w\}, \\ n(w) &:= |\mathcal{F}_w|, \quad M(w) := \sum_{U \in \mathcal{F}_w} \text{mult}(U), \end{aligned}$$

and

$$c(w) = (\#\{j : w_j = 0\}, \#\{j : w_j = 1\}, \#\{j : w_j = 2\}).$$

The enriched catalogue further preserves, for each emission, a diagonal coordinate $\text{diag}_c(U) \in \{0, \dots, 8\}$, a cardinal orientation $h_+(U)$, and the multiplicative signature $E_{\text{sig}}(U)$. All of them are computed from

the seed route; they are not read from a target constant. Their componentwise algorithm will be printed alongside the complete catalogue in the certification appendix and forms part of the input state of the following predicates.

Definition 8.2 (Closure orbit). An orbit O is of closure if it has cardinality six and, for each $w \in O$,

$$n(w) = 5, \quad M(w) = 1008,$$

$$\{\text{mult}(U) : U \in \mathcal{F}_w\} = \{432, 144, 144, 144, 144\},$$

and the common tritic census is $c(w) = (2, 3, 1)$.

Definition 8.3 (Radical Minimal Orbits). An orbit O is minimally radical if it has cardinality six, each word possesses a unique emission of multiplicity 144, and

$$\{\text{diag}_c(U) : \Psi(U) \in O\} = \{0, 3, 6\}.$$

Among them, the orbit with census $(2, 3, 1)$ is called the propagation orbit, and the orbit with census $(2, 2, 2)$ is called the self-scaling orbit.

Theorem 8.4 (Uniqueness of the Three Structural Orbits). *In the catalogue of 468 emissions there exists a unique closure orbit, a unique minimally radical orbit of propagation, and a unique minimally radical orbit of self-scale.*

Proof. The 43 orbits of theorem 8.1 are enumerated. For each one, $n(w)$, $M(w)$, the multiset of multiplicities, the census $c(w)$, and the diagonal support are computed. The predicates of the corresponding definition leave exactly one orbit in each class. The proof is exhaustive over a finite set and does not use decimal digits of the analytic characters. $\square$

The three orbits are

$$\begin{aligned} O_{\text{cl}} &= \{001112, 010211, 100121, 112001, 121100, 211010\}, \\ O_{\text{prop}} &= \{011120, 012110, 101201, 110012, 120011, 201101\}, \\ O_{\text{aut}} &= \{002112, 020211, 112002, 121200, 200121, 211020\}. \end{aligned}$$

8.1.3 Internal Orientation of Each Orbit

We fix the internal caliber.

$$\text{diag}_c(U) = 6, \quad h_+(U) = ES.$$

A word in an orbit is admissible in that gauge if some emission in its fiber satisfies both conditions.

Proposition 8.5 (Unique oriented representative). *Each of the three structural orbits contains a unique word admissible in the gauge $(6, ES)$. They are*

$w_6^\pi = 010211,$	$w_6^e = 201101,$	$w_6^\varphi = 121200.$
---------------------	-------------------	-------------------------

Proof. The enumeration of the catalogue fibers counts the words containing an emission with $\text{diag}_c = 6$ and $h_+ = ES$. In each of the three orbits, the number of coincidences is one. The superscripts π, e, φ henceforth name the structural functions of closure, propagation, and self-scaling; they did not participate in the selection. $\square$

8.1.4 Closure microfiber

The fiber of the representative w_6^π consists of five emissions:

U_6	E_{sig}	mult	diag _c	stratum
501 614 498 169 272 272	555555	432	4	$\perp$
810 923 870 169 272 272	339555	144	6	++
810 983 810 169 272 272	393555	144	6	++
870 923 810 169 272 272	933555	144	6	++
870 923 810 418 674 521	933717	144	5	--

Thus,

$$432 + 4 \cdot 144 = 1008, \quad 5 = 1_\perp + 3_{++} + 1_{--}.$$

The unipolar emissions oriented of the other two channels are

$$U_e = 850|963|890|149|252|212,$$

$$U_\varphi = 830|943|830|109|252|252.$$

The mass 1008 of this microfiber is not the period of the calendar C_{108} ; the two numbers belong to different domains.

8.1.5 Ternary lifting matrices

On row vectors of $\mathbb{F}_3^6$, we define

$$L_0 = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \quad L_1 = \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix}.$$

The elevation schedule is (L_0, L_0, L_1, L_1) .

Definition 8.6 (Block recurrence and concatenation). For each channel $\chi \in \{\pi, e, \varphi\}$, let $b_0^\chi = w_6^\chi$ and

$$b_{k+1}^\chi = b_k^\chi L_{\epsilon_k} \pmod{3}, \quad (\epsilon_0, \epsilon_1, \epsilon_2, \epsilon_3) = (0, 0, 1, 1).$$

The accumulated word is

$$w_{30}^\chi = b_0^\chi \| b_1^\chi \| b_2^\chi \| b_3^\chi \| b_4^\chi.$$

This formulation distinguishes the current block $b_k \in \mathbb{F}_3^6$ from the concatenated word $w_{6(k+1)}$. A word of length $6k$ is not multiplied by a matrix 6×6 .

Theorem 8.7 (Ternary biographies of thirty positions). *The recurrence of the definition 8.6 produce*

$$w_{30}^\pi = 010211|012222|010211|002111|110221,$$

$$w_{30}^e = 201101|121221|102011|012222|102011,$$

$$w_{30}^\varphi = 121200|112202|121020|010210|010200.$$

Proof. The five row vectors are successively multiplied by the indicated matrices, reducing each coordinate modulo three after each product. The concatenation of the five computed blocks gives the three expressions. Each equality can be verified component by component by means of 24 scalar products per channel. $\square$

The environmental indices of the five blocks are

	b_0	b_1	b_2	b_3	b_4
π	103	161	103	67	349
e	523	457	301	161	301
φ	450	398	438	102	99

where each word of six trits is interpreted as an integer in base three.

8.1.6 1st common boundary

The boundary is obtained through a finite relation that preserves axis, aggregate, saturation, column occupancy, and dual neutrality. The incidence matrix used in this relation is

$$A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix} \in M_6(\mathbb{F}_3). \quad (8.1)$$

If u_4^χ is the fifth block of w_{30}^χ , the axis and the aggregate prior to separating the two lateral channels are

$$u_5^\varphi = u_4^e = 102011, \quad u_5^\pi + u_5^e = 210112, \quad u_4^\varphi A_W L_1^3 = 111101. \quad (8.2)$$

From them, they calculate

$$q = 012120, \quad a = 111101, \quad qA_W = 012000, \quad aA_W = 120210. \quad (8.3)$$

Depth six imposes the saturation $c = 111111$; the minimal compatible occupancy is $\text{colw} = 223232$; and dual neutrality requires

$$(BA_W)\mathbf{1} = 000. \quad (8.4)$$

Proposition 8.8 (Enumeration of the first boundary). *The conditions (8.2)–(8.4) leave exactly two matrices:*

$$B_- = \begin{pmatrix} 021222 \\ 222220 \\ 102011 \end{pmatrix}, \quad B_+ = \begin{pmatrix} 222220 \\ 021222 \\ 102011 \end{pmatrix}. \quad (8.5)$$

The mirror involution swaps its first two rows. Its internal cards are, respectively,

$$B_- \mapsto (789, 048, 894), \quad B_+ \mapsto (793, 045, 894). \quad (8.6)$$

The orientation APP of the cylinder semiabierto selecciona B_+ .

Proof. The axis fixes the third row and the aggregate fixes the sum of the first two. Saturation and occupation eliminate the distributions that do not fill the six columns. The enumeration of the solutions of the six linear equations of (8.4) leaves the two matrices of (8.5); the semi-open chart and the orientation distinguish the second without consulting a target decimal expansion. $\square$

We therefore define,

$$R_{36} := B_+ = \begin{pmatrix} 222220 \\ 021222 \\ 102011 \end{pmatrix}, \quad \text{ind}_3(R_{36}) = (726, 215, 301). \quad (8.7)$$

The object R_{36} contains three ordered blocks of six trits. It is not the predictive quotient C_{36}^{pred} of theorem on the minimal predictive-memory quotient or the excess of 36 classes of the seed gauge.

8.1.7 Residue–quotient lifting

Let $A \in M_6(\mathbb{Z})$ be the integer lift of L_0 :

$$A = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \quad \det A = 1.$$

For $x_k \in \mathbb{Z}^6$, define

$$y_k = x_k A, \quad u_k = y_k \bmod 3 \in \{0, 1, 2\}^6, \quad x_{k+1} = \frac{y_k - u_k}{3}.$$

Then

$$x_k A = u_k + 3x_{k+1}. \quad (8.8)$$

Theorem 8.9 (Reversibility of the lift). *The pair (u_k, x_{k+1}) is uniquely determined by x_k , and*

$$x_k = (u_k + 3x_{k+1})A^{-1}.$$

Proof. Euclidean division by three in each coordinate of $x_k A$ determines residue and quotient. Since $\det A = 1$, the inverse A^{-1} has integer coefficients. Multiplying equation (8.8) by it recovers x_k . $\square$

In channel e , the visible block 102011 reappears twice, but the prestates modulo 27 are

$$(25, 11, 7, 17, 8, 16) \neq (7, 8, 4, 23, 26, 7),$$

and the subsequent quotients are

$$(6, 13, 9, 2, 13, 1) \neq (15, 0, 3, 19, 23, 25).$$

This explicitly verifies that the visible return of a word is not the return of the integral holonomic state.

8.1.8 Obstructions to orbital selection and to the lifting $w_6 \rightarrow w_{30} \rightarrow R_{36}$: uniqueness, gauge, and Hensel reversibility

Test criterion 8.10 (Orbital selection refutation criteria). The unit is refuted if:

1. the action D_3 leaves the catalogue or changes multiplicities;
2. any one of the three structural predicates leaves more than one orbit;
3. the gauge $(6, ES)$ selects two words in an orbit;
4. a target digit is used to define the predicates;
5. the accumulated word w_{6k} is multiplied by L_0 or L_1 instead of the block b_k ;
6. $\det A \neq 1$ or fails the inverse of the lift;
7. two equal visible residues are declared to be equal complete states.

The first joint boundary has already been constructed. The next unit will classify the nine phase components, prove the nonadic return without resetting, and construct the oriented mirror with its local resolution $3 \rightarrow 1$.

8.2 Joint Holonomy Γ_9 : observable compression T , orthogonal component η and terminal memory conservation

The nine phases that appear after R_{36} are not nine independent proofs. They are the nine local components of a single connection on integral holonomic states. Their composition around one turn defines a holonomy. The return of the phase label preserves the continuity of prefix, carry, boundary, Witt shadow and memory.

8.2.1 Transport of the integral holonomic state by nonadic holonomy and memory update

At the level K , we write

$$\widehat{x}_K = (B_K, C_K, p_K, c_K, \Sigma_K, W_K, g(K), q_{\text{mem},K}),$$

where:

1. $B_K = (u_K^\pi, u_K^e, u_K^\varphi)$ is the visible block of three words of six trits;
2. C_K is the cylinder ternario compatible;
3. p_K is the complete prefix;
4. c_K is the accumulated carry;
5. Σ_K is the signature joint of boundary;
6. W_K is the Witt incidence shadow;
7. $g(K) = 1 + ((K - 1) \bmod 9)$ is the public phase label;
8. $q_{\text{mem},K} \in \mathbb{Z}_{\geq 0}$ is the memory vertical not acotada.

The dodecaphasic coordinate is derived:

$$\beta_K = [q_{\text{mem},K}]_{12}.$$

q_{mem} is not replaced by β , since reduction modulo twelve would lose the number of circuits.

8.2.2 Local transition relations and construction of the nonadic holonomy Γ_9

For each K , let

$$\mathcal{R}_K \subseteq \widehat{X}_K \times \widehat{X}_{K+1}$$

the relation of compatible extension between complete states. The relational composition of a loop is

$$\Gamma_{9,K} := \mathcal{R}_{K+8} \circ \cdots \circ \mathcal{R}_{K+1} \circ \mathcal{R}_K : \widehat{X}_K \longrightarrow \widehat{X}_{K+9}. \quad (8.9)$$

Definition 8.11 (Nonadic holonomy). The holonomy of one turn is

$$\Gamma_9 := \text{Hol}_{C_9}(\nabla_{\text{TPK}}),$$

whose realization at level K is $\Gamma_{9,K}$. The nine relations $\mathcal{R}_K, \dots, \mathcal{R}_{K+8}$ are charts of this single connection.

Theorem 8.12 (Phase return without restart). *If $(\widehat{x}_K, \widehat{x}_{K+9}) \in \Gamma_{9,K}$, then*

$$g(K+9) = g(K), \quad q_{\text{mem},K+9} = q_{\text{mem},K} + 1,$$

and

$$\beta_{K+9} = \beta_K + 1 \pmod{12}.$$

In general, $\widehat{x}_{K+9} \neq \widehat{x}_K$.

Proof. The first identity follows from the definition of g . A complete turn increments the unbounded winding counter by definition. Reducing that identity modulo twelve yields the third. The remaining coordinates are transported by the composition equation (8.9); no law of the TPK resets them. Therefore, equality of phase does not imply equality of the complete state. $\square$

8.2.3 Exhaustive classification of the 9 phases of the Topological Phase Kernel

The visible blocks and the local censuses of the first round are

K	N_π	N_e	N_φ	N_{loc}	N_{ext}	u^π	u^e	u^φ
1	378	422	349	3	2	012222	121221	112202
2	238	368	388	2	5	010211	102011	121020
3	184	284	173	5	4	002111	012222	010210
4	207	207	122	4	1	110221	102011	010200
5	39	151	151	1	1	222220	021222	102011
6	110	110	69	1	3	111201	201202	221021
7	81	29	81	3	3	212121	222210	200221
8	59	59	59	3	2	200121	212212	110110
9	43	43	43	2	1	100100	020112	010122

Theorem 8.13 (Functions of the nine components). *In the preceding turn, the nine components perform the following functions:*

1. local opening of three orientations and first resolution;
2. binary split with a local maximum of five extensions;
3. maximal local multiplicity of five states;
4. contraction of four local states to an extension;
5. first strong saturation, with a single continuation;
6. unique local present and triple opening of the future;
7. triple mirror fiber;
8. synchronization of the three censuses in 59;
9. oriented closure: two local input states and a relation that, after the additional horizon, conserves a prolongation.

Proof. Each assertion is read from the censuses $N_{\text{loc}}, N_{\text{ext}}$, from the equality or inequality of N_π, N_e, N_φ , and from the explicit extension relation in the ninth, tenth, and eleventh phases to be proved in the following unit. The table enumerates the entire revolution; no subset of phases has been chosen. $\square$

The fifth component coincides with the first common boundary.

$$R_{36} = (222220|021222|102011)^T.$$

The ninth component contains

$$B_9 = (100100|020112|010122).$$

The presence of two local states in the ninth phase and of three extensions from the oriented state belongs to different layers; they are not contradictory censuses.

Holonomy acts on the complete integral holonomic state. Before publishing its modal quotient, one therefore constructs the involution interchanging the fibers of π and e , the fixed axis of φ , the ternary invariants and the local resolution $3 \rightarrow 1$ determined by the second horizon.

8.3 Mirror involution and local selection $3 \rightarrow 1$ of the surviving extension

The ninth phase contains two local input states. Once the oriented state B_9 has been fixed, the boundary relation produces three extensions toward the first phase. The three share an axis of self-scale and a ternary aggregate; they differ in the ordered distribution of that aggregate between the branches of closure and propagation. A second horizon conserves exactly one.

8.3.1 Symmetric and axial coordinates

Let

$$\mathcal{B} := (\mathbb{F}_3^6)^3.$$

A visible block is

$$B = (u^\pi, u^e, u^\varphi) \in \mathcal{B}.$$

We define

$$q(B) := u^\pi + u^e + u^\varphi, \quad a(B) := u^\pi + u^e - u^\varphi.$$

All operations are performed componentwise in $\mathbb{F}_3^6$.

Theorem 8.14 (Reconstruction of the axis and the aggregate). *The coordinates q, a determine*

$$\boxed{u^\varphi = 2(q - a)}, \quad \boxed{u^\pi + u^e = q - u^\varphi}.$$

Proof. Subtracting the definitions gives $q - a = 2u^\varphi$. In $\mathbb{F}_3$, $2^{-1} = 2$; therefore, $u^\varphi = 2(q - a)$. Substitution into the definition of q recovers the remaining aggregate. $\square$

The specular involution is

$$\mathfrak{c}(u^\pi, u^e, u^\varphi) = (u^e, u^\pi, u^\varphi).$$

Proposition 8.15 (Invariants of the Involution). $\mathfrak{c}^2 = \text{id}$, y

$$q(\mathfrak{c}B) = q(B), \quad a(\mathfrak{c}B) = a(B).$$

In particular, u^φ and $u^\pi + u^e$ are invariants.

Proof. Exchanging the first coordinates twice recovers B . Both q and a depend on them only through their sum, so that the exchange preserves them. The theorem 8.14 transfers that invariance to the axis and the aggregate. $\square$

The involution asserts neither that $\pi = e$ nor that one is the quotient of the other. It asserts that the two branches occupy conjugate positions with respect to a fixed structure.

8.3.2 Oriented state in phase 9 and associated boundary data

The visible projection of the selected state is

$$B_9 = (100100 \mid 020112 \mid 010122).$$

The induced relation on visible blocks is denoted

$$\mathcal{M}_9^B \subseteq \mathcal{B} \times \mathcal{B}.$$

The two local input states of the phase table precede this relation. After selecting the complete state that projects to B_9 , its output fiber has cardinal three.

Theorem 8.16 (The three extensions of the ninth phase). *The relational image of B_9 contains exactly*

$$\begin{aligned} B_{10}^{(0)} &= (101222 \mid 112221 \mid 112002), \\ B_{10}^{(1)} &= (102221 \mid 111222 \mid 112002), \\ B_{10}^{(2)} &= (111221 \mid 102222 \mid 112002). \end{aligned}$$

The three satisfy

$$q_{10} = 022112, \quad a_{10} = 101111,$$

and, consequently,

$$u_{10}^\varphi = 112002, \quad u_{10}^\pi + u_{10}^e = 210110.$$

Proof. The extension relation on complete states is enumerated in the fiber of B_9 and produces the three indicated rows. Adding and subtracting their three components gives the same pair (q_{10}, a_{10}) in all three cases. The [theorem 8.14](#) produces the common axis and aggregate. The first two components are distinct, so the three rows represent different distributions of the same aggregate. $\square$

All their publications in base three are

$$(279, 352, 365), \quad (280, 351, 365), \quad (282, 350, 365).$$

8.3.3 Exact compatibility of cylinders

A ternary word of length L and index N determines the half-open cylinder

$$I_3(N, L) = \left[\frac{N}{3^L}, \frac{N+1}{3^L} \right).$$

A prefix of b decimal blocks, with associated integer D , determines

$$I_{1000}(D, b) = \left[\frac{D}{1000^b}, \frac{D+1}{1000^b} \right).$$

Lemma 8.17 (Integer intersection test). *The cylinders $I_3(N, L)$ and $I_{1000}(D, b)$ intersect if and only if*

$$N1000^b < (D+1)3^L, \quad (N+1)1000^b > D3^L.$$

Proof. Two half-open intervals $[a, b)$ and $[c, d)$ intersect if and only if $a < d$ and $c < b$. We substitute the four endpoints and multiply by the positive integer $3^L 1000^b$. $\square$

The test uses only integer arithmetic. It does not depend on a floating-point tolerance.

8.3.4 2nd horizon and uniqueness

The following block is

$$B_{11} = (022212 \mid 110001 \mid 100021),$$

and the corresponding decimal boundary is

$$(502, 662, 638).$$

Theorem 8.18 (Local resolution $3 \rightarrow 1$). *The three extensions of theorem 8.16 are compatible at depth one. When B_{11} and the boundary (502, 662, 638) are imposed simultaneously, only $B_{10}^{(0)}$ prolongs as the same complete state. Therefore,*

$$\# \text{Surv}_1^{\text{cal}}(B_9) = 3, \quad \# \text{Surv}_2^{\text{cal}}(B_9) = 1.$$

Proof. We apply lemma 8.17 to each of the three channels. At the first horizon, the six inequalities of every candidate are satisfied. After adjoining B_{11} , the inequalities of the three channels remain satisfied for $B_{10}^{(0)}$. In $B_{10}^{(1)}$ and $B_{10}^{(2)}$, compatibility of channel φ is preserved, but the conditions of branches π and e fail. Enumeration therefore leaves a unique state at depth two. $\square$

This proof does not define the global connection from a local example. It is an explicit witness of how a multivalued relation at one horizon can become single-valued by preserving memory and requiring a longer prolongation.

8.3.5 Phase return with block change

The first phase of the initial loop is

$$B_1 = (012222 \mid 121221 \mid 112202),$$

while the first phase of the next turn is

$$B_{10} = (101222 \mid 112221 \mid 112002).$$

Thus,

$$g(1) = g(10) = 1, \quad B_1 \neq B_{10}.$$

The phase returns, but the axis, the aggregate, the distribution, and the memory of quotient have evolved.

8.3.6 Diagram of the mirror involution π/e , the fixed axis φ , and the local selection $3 \rightarrow 1$ by prolongability

8.3.7 Residue–quotient memory during the mirror

The recurrence

$$x_k A = u_k + 3x_{k+1}$$

of the equation (8.8) is applied to each of the three channels. The part u_k publishes the visible block; x_{k+1} conserves the quotient. Since $A \in \text{GL}_6(\mathbb{Z})$, the pair recovers the preceding state. The mirror can exchange the visible branches without eliminating their provenance quotients.

8.3.8 Obstructions to the mirror involution π/e and to the local selection $3 \rightarrow 1$: invariants, depth-2 horizon, and Hensel memory

Test criterion 8.19 (Refutation Criteria for the Mirror). The construction is refuted if:

1. the involution modifies u^φ or $u^\pi + u^e$;
2. the two local input states are conflated with the three outputs;
3. one of the three outputs has a different (q_{10}, a_{10}) ;
4. the cylinder test uses a floating tolerance;
5. more than one output survives the declared second horizon;
6. phase return is interpreted as equality $B_1 = B_{10}$;
7. the visible exchange erases the Hensel quotients.

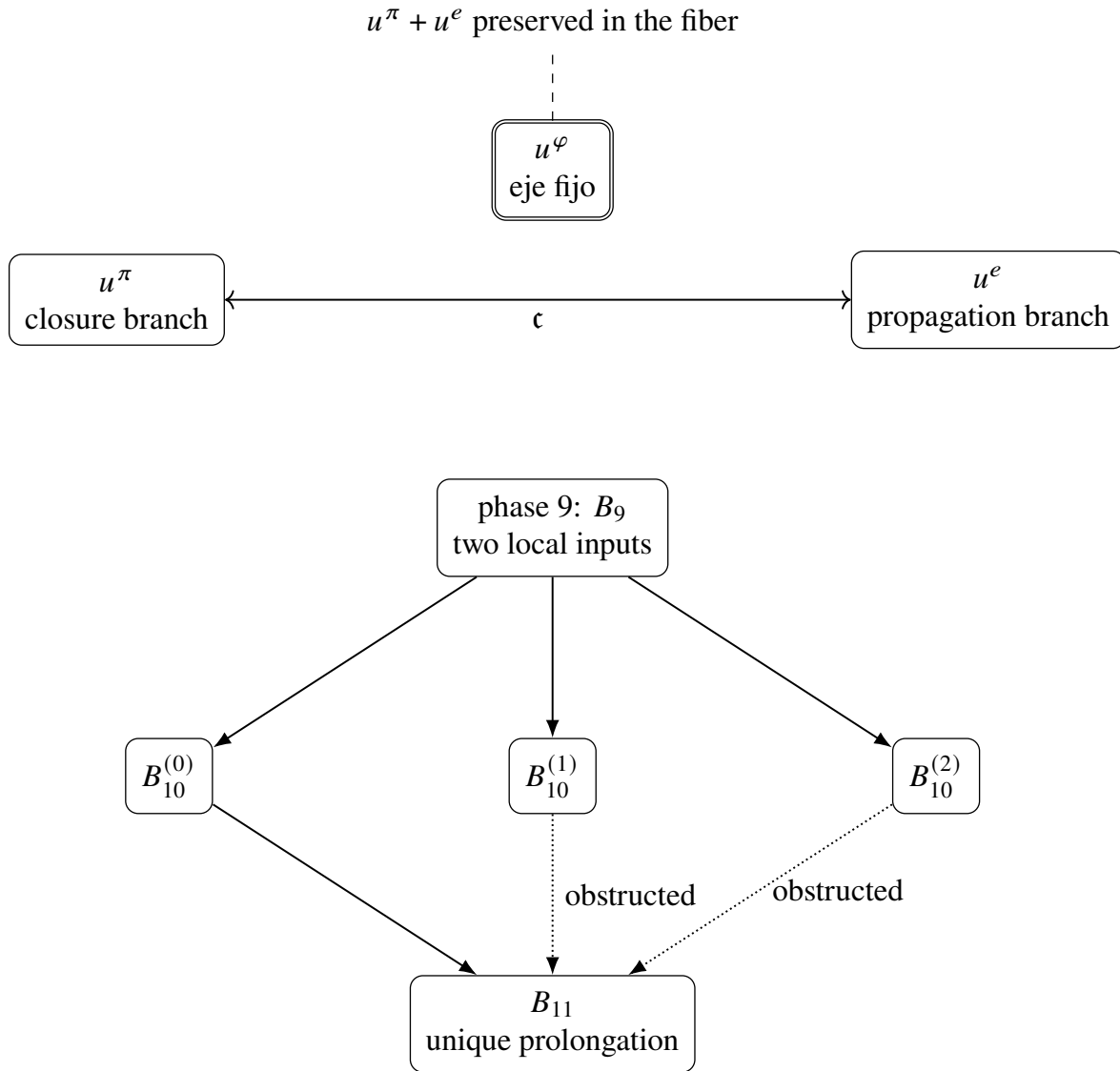


Figure 8.1: The mirror preserves the axis u^φ and the aggregate $u^\pi + u^e$. The return of the ninth phase opens three visible extensions; the memory of the second horizon preserves one. The circuit does not reset the state.

The mirror and the local selection have now been constructed. The following units will develop the Sturmian vacancies and then the three complete analytic biographies.

8.3.9 Joint transfer of the fiber of 468 emissions

The factorization of the catalog is

$$\mathcal{U}_6^{\text{cat}} \cong \mathfrak{S}_+ \times \mathfrak{S}_\times, \quad |\mathfrak{S}_+| = 18, \quad |\mathfrak{S}_\times| = 26.$$

For a function f on the catalogue, let

$$(E_+f)(s, e) = \frac{1}{18} \sum_{s' \in \mathfrak{S}_+} f(s', e),$$

and

$$(E_\times f)(s, e) = \frac{1}{26} \sum_{e' \in \mathfrak{S}_\times} f(s, e').$$

The projectors E_+ and $E_\times$ commute.

Let

$$\tau = (+1, -1, 0, +1, -1, 0, +1, -1, 0), \quad P_{+1} = E_+, \text{quad}P_{-1} = E_\times, \text{quad}P_0 = I.$$

On $\mathcal{U}_6^{\text{cat}} \times C_9$ we define phase transfer by

$$\mathcal{K}_{\text{ph}}((u, r), (v, [r+1]_9)) := P_{\tau(r)}(u, v), \quad (8.10)$$

and we make it zero for the remaining pairs of phases. Thus, the cyclic permutation and the operator applied in each of the nine steps are determined.

Proposition 8.20 (Joint Renewal After a Turn).

$$\mathcal{K}_{\text{ph}}^9 = E_+E_\times.$$

In particular, on a point mass,

$$(E_+E_\times)(u, v) = \frac{1}{18 \cdot 26} = \frac{1}{468}.$$

Proof. The ordered product over a loop is

$$P_{\tau(8)}P_{\tau(7)} \cdots P_{\tau(0)} = IE_\times E_+ IE_\times E_+ IE_\times E_+ = E_+^3 E_\times^3 I^3 = E_+E_\times.$$

The second equality uses commutativity and the final one idempotence. The second result is the uniform distribution on the product of the two signatures. $\square$

This identity expresses a joint renewal, not nine independent filtrations.

8.3.10 Decomposition of the holonomy U_{Γ_9} into visible compression T and orthogonal component η :

$$I - T^*T = \eta^*\eta$$

Let $\mathcal{H}_{\text{vis}}$ and $\mathcal{H}_{\text{full}}$ be Hilbert spaces for the visible and complete realizations. Let

$$J : \mathcal{H}_{\text{vis}} \longrightarrow \mathcal{H}_{\text{full}}$$

an isometric inclusion and U_{Γ_9} an isometric realization of the holonomy. We define

$$T := J^*U_{\Gamma_9}J, \quad \eta := (I - JJ^*)U_{\Gamma_9}J.$$

Theorem 8.21 (Compression-decomposition expression). *The following hold*

$$\boxed{U_{\Gamma_9} J = JT + \eta,}$$

$$J^* \eta = 0, \quad \boxed{I - T^* T = \eta^* \eta.}$$

Proof. We decompose $U_{\Gamma_9} J$ by means of the orthogonal projections JJ^* and $I - JJ^*$:

$$U_{\Gamma_9} J = JJ^* U_{\Gamma_9} J + (I - JJ^*) U_{\Gamma_9} J = JT + \eta.$$

Orthogonality gives $J^* \eta = 0$. Since J and U_{Γ_9} are isometries,

$$I = J^* U_{\Gamma_9}^* U_{\Gamma_9} J = T^* T + \eta^* \eta.$$

Reorder the proof of the last identity. □

T is the visible compression; η is the complementary expression. Neither should be called a strict contraction. For that function a different operator is reserved.

8.3.11 Strict contraction and terminal telescoping

Let V_9 be an isometry and let q be a parameter with $0 < q < 1$. We define

$$M_9 = qV_9, \quad D_9 = (I - M_9^* M_9)^{1/2}.$$

Then $\|M_9\| = q < 1$ and

$$I - M_9^* M_9 = D_9^* D_9.$$

Theorem 8.22 (Stein Identity with Terminal Term). *For every $L \geq 1$,*

$$\boxed{I = \sum_{r=0}^{L-1} M_9^{*r} D_9^* D_9 M_9^r + M_9^{*L} M_9^L.}$$

More generally, if S_0 satisfies $S_0 - M_9^ S_0 M_9 = D_9^* D_9$, then*

$$\boxed{S_0 = \sum_{r=0}^{L-1} M_9^{*r} D_9^* D_9 M_9^r + M_9^{*L} S_0 M_9^L.}$$

Proof. We multiply the defect identity on the left by M_9^{*r} and on the right by M_9^r . Upon summing over $r = 0, \dots, L-1$, the interior terms telescope, leaving the initial and terminal terms. The same proof applies to the general Stein equation. □

The final term does not vanish at finite depth. Its disappearance in a limit requires a convergence proof; it is not part of the algebraic identity.

8.3.12 Carry Cocycle and Relational Powers

For composable arrows γ_1, γ_2 , the carry satisfies

$$\text{Carr}(\gamma_2 \gamma_1) = \text{Carr}(\gamma_2) H(\gamma_1) + Y(\gamma_2) \text{Carr}(\gamma_1).$$

The formula preserves the contribution of the first stage after transporting it through the second.

The longer loops are written with displaced indices:

$$\begin{aligned}\Gamma_{27,K} &= \Gamma_{9,K+18} \circ \Gamma_{9,K+9} \circ \Gamma_{9,K}, \\ \Gamma_{54,K} &= \Gamma_{9,K+45} \circ \cdots \circ \Gamma_{9,K}, \\ \Gamma_{108,K} &= \Gamma_{9,K+99} \circ \cdots \circ \Gamma_{9,K}.\end{aligned}$$

Only after declaring a global operator on the graded union may this notation be abbreviated as $\Gamma_9^3, \Gamma_9^6, \Gamma_9^{12}$.

8.3.13 Complete nonadic state: phase, carry, boundary, route, and memory after the prolongation

The one-turn connection is recorded by the quintuple

$$\mathfrak{N}_9 = (\Gamma_9, T, \eta, \text{Carr}, S_{\text{term}}),$$

where $S_{\text{term}} = M_9^{*L} S_0 M_9^L$ at a finite depth L . An application functor that uses the connection must declare its compatibility with all five fields. Commuting only with the visible phase does not constitute naturality with respect to the complete nonadic state.

8.3.14 Obstructions of the holonomy Γ_9 : $9 \rightarrow 1$ return, composition indices and phase conservation, carry, boundary, path and memory

Test criterion 8.23 (Refutation Criteria of the Connection). The construction is refuted if:

1. the nine phases are presented as independent filters;
2. the return $9 \rightarrow 1$ resets q_{mem} , the prefix, or the carry;
3. T is called a strict contraction;
4. some identity of [theorem 8.21](#) fails;
5. the terminal term is removed at finite depth;
6. $\Gamma_{27,K}$ is composed without shifting the indices;
7. an application functor ignores any of the five fields of $\mathfrak{N}_9$.

The complete developments that follow first reconstruct the finite emitter, its factorization and its microfibers; they then unfold the specular involution, the three extensions of B_9 , uniqueness at two horizons and the Bockstein–Hensel inverse that preserves memory during phase return.

8.4 Finite arithmetic image and factorization $468 = 18 \cdot 26$

The image in base 1000 of the finite map that assigns the decimal blocks is a proper subset of the one thousand decimal blocks. The evolution law defined in the chapter devoted to the transport system and phase evolution and the arithmetic of $\mathbb{Z}/9\mathbb{Z}$ determine an image of 42 triples, organized into six classes of seven. The same separation of the additive and multiplicative observables subsequently factors the 468 outputs into 18 additive signatures and 26 multiplicative signatures.

8.4.1 Primitives and recurrence of the finite sender

Definition 8.24 (Initial data of T_{min}). Let $D = \{N, E, S, O\}$ be given, with cardinal vectors

$$v(N) = (-1, 0), \quad v(E) = (0, 1), \quad v(S) = (1, 0), \quad v(O) = (0, -1) \quad \text{in } (\mathbb{Z}/9\mathbb{Z})^2.$$

The temporal signature and the multiplicative index are fixed by

$$\text{sgn}_9(t) = \begin{cases} +1, & t \bmod 9 \in \{1, 4, 7\}, \\ -1, & t \bmod 9 \in \{2, 5, 8\}, \\ 0, & t \bmod 9 \in \{0, 3, 6\}, \end{cases}$$

and

$$\text{ind}_2(1, 2, 4, 8, 7, 5) = (0, 1, 2, 3, 4, 5),$$

with value zero outside the units. An entry of $T_{\min}$ is a pair of seeds

$$I_9 := \{0, 1, \dots, 8\}, \quad s_+ = (i_+, j_+, d_+), \quad s_\times = (i_\times, j_\times, d_\times) \in I_9^2 \times D.$$

These tables, the readers L_Σ, L_Π , the six blocks of nine instants, and the inversion of both vectors when $27 \mid t$ are all the starting data. Neither π , e , φ , nor any other target value is introduced.

Algorithm 8.25 (54-step recurrence). At the outset, the two cursors occupy their seed positions and carry the vectors $v(d_+)$, $v(d_\times)$. For $t = 1, \dots, 54$, in this order:

1. if $\text{sgn}_9(t) = +1$, $L_\Sigma(i_+(t), j_+(t))$ is accumulated and the additive cursor is advanced;
2. if $\text{sgn}_9(t) = -1$, then $\text{ind}_2(L_\Pi(i_\times(t), j_\times(t)))$ is accumulated and the multiplicative cursor is shifted;
3. if $\text{sgn}_9(t) = 0$, both cursors remain stationary;
4. after that update, both vectors are reversed if $27 \mid t$.

The accumulators are reset at the beginning of each block $9(r-1) + 1 \leq t \leq 9r$. If S_r, E_r are their final values, then there is emitted

$$U_r = 100\bar{S}_r + 10(\bar{S}_r + \bar{E}_r)_{10} + (3S_r + 5E_r + 1)_{10}, \quad \bar{S}_r = S_r \bmod 10, \quad \bar{E}_r = E_r \bmod 10,$$

and $w_r = (-U_r) \bmod 3$. Thus it is obtained.

$$T_{\min}(s_+, s_\times) = (U_1, \dots, U_6) \in \mathcal{U}_6^{\text{cat}}, \quad \Psi(U_6) = (-U_6) \bmod 3 =: w_6 \in \mathbb{F}_3^6.$$

Test criterion 8.26 (Target-value-independent dual control). The recurrence admits two independent formulations: channel factorization and joint simulation. Exhaustive traversal of the 104 976 inputs produces the same sextuple by both routes for every seed, without consulting a target constant. A discrepancy for a single seed would refute the equivalence.

8.4.2 Bivariate dynamics and block observables

We work on the torus $\mathbb{Z}_9^2$. Under the preceding convention, a complete seed is

$$\omega = (p_+, p_\times, h_+, h_\times) \in (\mathbb{Z}_9^2)^2 \times \{N, E, S, O\}^2,$$

so that

$$|\Omega| = 9^4 \cdot 4^2 = 104\,976.$$

In each window of nine steps, the times 1, 4, 7 evaluate the additive observable, the times 2, 5, 8 evaluate the multiplicative observable, and the times 3, 6, 9 increment the neutral counter. If S_m, E_m, T_m are the accumulators of the window m , the decimal observable is

$$d_1 \equiv S_m, \quad d_2 \equiv S_m + E_m, \quad d_3 \equiv 3S_m + 5E_m + 7T_m \pmod{10},$$

and the decimal triple is $U_m = 100d_1 + 10d_2 + d_3$.

Lemma 8.27 (Neutral counter). *For every window, $T_m = 3$.*

Proof. Each block of nine times contains exactly the three neutral times 3, 6, 9. Therefore the counter is incremented three times. $\square$

8.4.3 Image of the additive accumulator: $\text{Im } S_m = \{6, 9, 12, 15, 18, 21, 24\}$

Let $s = i + j \pmod{9}$. The visible additive observable is

$$f(s) = \text{dr}_9(s + 2) = (2, 3, 4, 5, 6, 7, 8, 9, 1).$$

A cardinal translation modifies s in $+1$ or in -1 . In a window, three consecutive terms are read; reversing the orientation only reverses their order. The nine cyclic sums are

$$9, 12, 15, 18, 21, 24, 18, 12, 6.$$

Lemma 8.28 (Image of the additive accumulator).

$$S_m \in \{6, 9, 12, 15, 18, 21, 24\}.$$

In particular, $d_1 = S_m \pmod{10}$ traverses precisely $\{1, 2, 4, 5, 6, 8, 9\}$.

Proof. The preceding list enumerates all initial positions of the trajectory and the two orientations. Their decimal residues are seven and distinct. $\square$

8.4.4 Image of the multiplicative accumulator: $\text{Im } E_m = \{0, 1, 3, 5, 7, 9\}$

The unit group $(\mathbb{Z}/9\mathbb{Z})^\times$ is cyclic of order six. We fix the multiplicative coordinate function

$$\ell_\times(1, 2, 3, 4, 5, 6, 7, 8, 9) = (0, 1, 0, 2, 5, 0, 4, 3, 0).$$

If the row or column traversed has a factor divisible by three, each evaluation belongs to $\{3, 6, 9\}$ and contributes zero. If the fixed factor is a unit, the sums of three consecutive terms of $b \mapsto \ell_\times(\text{dr}_9(ab))$ are:

a	possible sum
1	$\{1, 3, 7, 9\}$
2	$\{3, 5, 9\}$
4	$\{1, 5, 7\}$
5	$\{1, 5, 7\}$
7	$\{3, 5, 9\}$
8	$\{1, 3, 7, 9\}$.

Lemma 8.29 (Image of the multiplicative accumulator).

$$E_m \in \{0, 1, 3, 5, 7, 9\}.$$

Proof. The noninvertible case contributes 0. The finite table of the six units contains all possibilities in the invertible case, and their union is $\{1, 3, 5, 7, 9\}$. $\square$

8.4.5 Decimal congruence of the ternary alphabet: uniqueness of the 3rd digit

Theorem 8.30 (Control congruence). *Every ternary $\overline{d_1 d_2 d_3}$ emitted by the system satisfies*

$$d_3 \equiv 5d_2 - 2d_1 + 1 \pmod{10}.$$

Proof. From $d_2 \equiv S_m + E_m$ one obtains $E_m \equiv d_2 - d_1 \pmod{10}$. Since $T_m = 3$,

$$d_3 \equiv 3d_1 + 5(d_2 - d_1) + 21 \equiv 5d_2 - 2d_1 + 1 \pmod{10}. \quad \square$$

Theorem 8.31 (Exact image of the decimal application). *The system produces exactly $7 \cdot 6 = 42$ decimal triples. Grouped by $E = d_2 - d_1 \pmod{10}$, they are:*

E	<i>decimal triples</i>							
0	114	227	443	556	669	885	998	
1	129	232	458	561	674	890	903	
3	149	252	478	581	694	810	923	
5	169	272	498	501	614	830	943	
7	189	292	418	521	634	850	963	
9	109	212	438	541	654	870	983	

Proof. There are seven possibilities for d_1 and six for E . For each pair, $d_2 \equiv d_1 + E$ and the control congruence fix a unique d_3 , so there exist at most 42 triples. The table applies these two formulas to the 42 pairs and contains no repetitions. The enumeration of the 104 976 seeds finds all the triples in the table and none outside it; the bound is attained. $\square$

8.4.6 Factorization of the 2 observables

Let $\mathcal{S} = \mathbb{Z}_9^2 \times \{N, E, S, O\}$, with $|\mathcal{S}| = 324$. To each additive seed $P \in \mathcal{S}$ is associated the six-window signature

$$s(P) = (S_1, \dots, S_6) \pmod{10},$$

and to each multiplicative seed $T \in \mathcal{S}$,

$$e(T) = (E_1, \dots, E_6) \pmod{10}.$$

The two signatures are combined coordinate by coordinate by

$$G(s, e)_k = 100s_k + 10(s_k + e_k)_{10} + (3s_k + 5e_k + 1)_{10}.$$

The map G is injective: from a triple one recovers $s_k = d_1$ and $e_k = d_2 - d_1 \pmod{10}$, while the third digit verifies membership in the image.

Lemma 8.32 (Exact classification of the additive channel).

$$|\text{im } s| = 18,$$

and each additive signature has exactly 18 preimages.

Proof. The signature depends only on $(i + j \pmod{9}, \varepsilon)$, where $\varepsilon = +1$ for E, S and -1 for N, O . The $9 \cdot 2$ pairs produce distinct signatures; each has nine positions on its diagonal and two directions, that is, 18 preimages. A different number of signatures or a class of size different from 18 would refute the classification. $\square$

Lemma 8.33 (Exact classification of the multiplicative channel).

$$|\operatorname{im} e| = 26.$$

There are 24 classes of size 8, one of size 24, and one of size 108.

Proof. Enumerating the $9^2 \cdot 4 = 324$ seeds using the explicit table of $\ell_\times$ gives 26 sextets $(E_1, \dots, E_6)$ and the histogram

$$24 \cdot 8 + 24 + 108 = 324.$$

An additional, absent, or differently sized class would refute the census. $\square$

Theorem 8.34 (Exact injective factorization of the catalog).

$$|\operatorname{im} s| = 18, \quad |\operatorname{im} e| = 26, \quad |\operatorname{im} G| = 468 = 18 \cdot 26.$$

Each additive signature has 18 preimages. The multiplicative signatures possess the histogram

$$24 \text{ size classes } 8, \quad 1 \text{ of size } 24, \quad 1 \text{ of size } 108.$$

Proof. The two preceding lemmas give the cardinalities of the signatures. In each emitted component, the hundreds digit is $\bar{S}_r$, while the tens digit is $\bar{S}_r + \bar{E}_r \bmod 10$. Therefore,

$$\bar{S}_r = \left\lfloor \frac{U_r}{100} \right\rfloor, \quad \bar{E}_r = \left(\left\lfloor \frac{U_r}{10} \right\rfloor - \bar{S}_r \right) \bmod 10.$$

The units digit verifies $3S_r + 5E_r + 1 \bmod 10$; it introduces no other identification. Thus, G is injective, and its image has $18 \cdot 26 = 468$ elements. The 24 multiplicative classes of size 8, combined with the 18 additive signatures, produce 432 sextets of multiplicity 144; the other two classes produce 18 sextets of multiplicity 432 and 18 of multiplicity 1944. In particular,

$$432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976.$$

Two pairs of signatures with the same U_6 , or a histogram that failed to recover the complete domain, would refute the theorem. $\square$

The complete composition factors, therefore, as

$$\mathcal{S}^2 \xrightarrow{s \times e} \operatorname{im} s \times \operatorname{im} e \xrightarrow{G} \mathcal{U}_6 \xrightarrow{\rho_3} \mathcal{W}_6 \subset \mathbb{F}_3^6.$$

The final projection has 243 values. The first descent preserves the sum–product separation; the second preserves 468 decimal states; the third removes the fiber coordinate and retains a ternary word.

Corollary 8.35 (Census of the observable projection). *The complete enumeration determines the chain of cardinalities*

$$104\,976 \longrightarrow 468 \longrightarrow 243 \subset 729.$$

The 468 intermediate elements are sextuples of integers between zero and 999; only their final image belongs to $\mathbb{F}_3^6$. The fiber multiplicities of the first map satisfy

$$432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976.$$

Proof. The factorization theorem gives $18 \cdot 26 = 468$, and the exhaustive enumeration of the 104 976 seeds produces exactly the three indicated multiplicities. Componentwise reduction traverses the 243

words recorded in the ternary catalogue. The appended catalogues of signatures and routes reproduce both enumerations. $\square$

Proposition 8.36 (Quintuple microfiber of the word 010211). *For the restricted reduction*

$$\Psi : \mathcal{U}_6^{\text{cat}} \longrightarrow \mathbb{F}_3^6, \quad \Psi(U_6) = (-U_6) \bmod 3,$$

the fiber of 010211 consists of exactly the following five sextets:

U_6	σ_+	$\sigma_\times$	multiplicity
(501, 614, 498, 169, 272, 272)	(5, 6, 4, 1, 2, 2)	(5, 5, 5, 5, 5, 5)	432
(810, 923, 870, 169, 272, 272)	(8, 9, 8, 1, 2, 2)	(3, 3, 9, 5, 5, 5)	144
(810, 983, 810, 169, 272, 272)	(8, 9, 8, 1, 2, 2)	(3, 9, 3, 5, 5, 5)	144
(870, 923, 810, 169, 272, 272)	(8, 9, 8, 1, 2, 2)	(9, 3, 3, 5, 5, 5)	144
(870, 923, 810, 418, 674, 521)	(8, 9, 8, 4, 6, 5)	(9, 3, 3, 7, 1, 7)	144

They are five preimages of the catalog U_6 , not five complete TPK histories. They preserve distinct signatures and multiplicities although they share the same visible word.

Proof. The exhaustive scan of the catalogue of 468 sextuples selects exactly the five rows of the table. The signature-recovery function reproduces the two sextuples of every row, and the sum of multiplicities is $432 + 4 \cdot 144 = 1008$. Enumeration additionally verifies that every cell is the Cartesian product of its channel seeds. A cardinality other than five or identification of two rows despite their enriched data would refute the statement. $\square$

8.5 Inverse systems, Weyl–Heisenberg representation and predictive quotient

The construction distinguishes three levels: the finite seed catalogue, the state space that preserves carry and orientation, and the inverse limit of compatible prolongations. This stratification makes it possible to formalize contraction of cylindrical intervals to a real point, persistence of trajectory data in a single evaluation fiber, and the finite Weyl–Heisenberg representation associated with the alternating form of the nonadic torus supporting the additive and multiplicative observables.

The symbols V_t , X_t , and $\rho_{t+1,t}$ used here are the same sets of local states, admissible prefixes, and truncations introduced in chapter devoted to cylindrical systems, survival and the construction of number, with the purely notational change $t = m$ and $\rho_{n,m} = p_{n,m}$. This section makes their coordinates explicit and introduces no second inverse system.

8.5.1 Observable projection and complete state space of the TPK

The observable projection of the phase transport system (TPK) is the finite encoding application

$$104976 \longrightarrow 468 \longrightarrow 243.$$

which classifies seeds, signatures, and words w_6 . The complete TPK state space further preserves

bidirectional orientation, Hensel state, cylinder and carry,
joint boundary signature, visible nonadic phase $g_{\text{vis}}(m)$.

Two candidates with the same local signature may belong to distinct classes when the following boundary is conserved. We do not introduce a second tuple of state. For

$$x_m = (q_m, o_m, h_m, c_m, \sigma_m, C_m, b_m, \lambda_m) \in \mathcal{X}_{\text{TPK}}^{(m)}$$

we use the coordinate chart

$$\chi_m(x_m) = (B_m, h_m, (C_m, b_m), \sigma_m, o_m, \mu(q_m), g_{\text{vis}}(m), \lambda_m),$$

with

$$B_m = (u_m^\pi, u_m^e, u_m^\varphi) \in (\mathbb{F}_3^6)^3.$$

Here h_m is the Hensel extension; (C_m, b_m) , the cylinder and its boundary; $o_m, \mu(q_m)$, the orientation and the bidirectional observable; and

$$g_{\text{vis}}(m) = 1 + ((m - 1) \bmod 9) \in \{1, \dots, 9\},$$

the visible nonadic phase label. The corresponding class coordinate is $g_0(\theta_m) \in \mathbb{Z}/9\mathbb{Z}$, with $\theta_m = [m - 1]_{108}$. The joint signature is the coordinate already typed by the phase kernel,

$$\sigma_m = \text{Sig}_{q_m}(h_m, c_m, \lambda_m) = (q_\partial, a_\partial, c_\partial, \text{colw}, \rho_W, r_\partial).$$

The joint signature does not reduce to three independent channel signatures, because the selection associated with the involution π/e and the axis φ depends on correlated data. Two nodes with the same B_m represent distinct states when they differ in extension, cylinder, carry, signature, orientation, or transport record.

8.5.1.1 Exact compatibility of the cylinders under truncation and nonadic prolongation

Let N be the integer of a ternary prefix of length L , and let D be the integer obtained by concatenating its b decimal triads. We define

$$A = N1000^b - D3^L, \quad B = (N + 1)1000^b - D3^L.$$

The cylinders intersect if and only if

$$\boxed{A < 3^L, \quad B > 0.}$$

Adding a ternary block $u \in \{0, \dots, 728\}$ gives

$$N' = 729N + u.$$

If a new decimal triad does not appear,

$$A' = 729A + u1000^b.$$

If δ occurs, so that $D' = 1000D + \delta$, then

$$A' = 729000A + u1000^{b+1} - \delta 729 3^L.$$

If $\Delta b \in \{0, 1\}$ records the appearance of that new triad, in both cases

$$B' = A' + 1000^{b+\Delta b}.$$

The transition, once the admissible pair (u, δ) has been fixed, does not consult a target value. The joint filtration by intersection of cylinders and nonadic return acts at a distinct level: it selects the boundary pairs that admit compatible prolongation.

8.5.1.2 Cyclic registration of 108 stages and its reduced projection

We call the sequence of *108-stage cyclic record* the sequence of 108 steps grouped into twelve sectors of nine steps, with the index convention fixed in chapter devoted to the transport system and phase evolution.

Let $\mathcal{X}_{\text{TPK}}^{[108]} \subseteq \mathcal{X}_{\text{TPK}}$ be the domain of complete states whose register λ contains the declared 108 steps. Let us denote by $\mathcal{C}_{108}^{\text{red}}$ the set of registers that result by conserving only the reduced fields, and by $\mathcal{U}_{12}^{\text{sgn}}$ the set of admissible signed dodecaphases. The instance distinguishes two declared maps:

$$F : \mathcal{X}_{\text{TPK}}^{[108]} \longrightarrow \mathcal{C}_{108}^{\text{red}}, \quad R : \mathcal{X}_{\text{TPK}}^{[108]} \longrightarrow \mathcal{U}_{12}^{\text{sgn}}.$$

The map F retains only the fields of the reduced register, whereas R evaluates the signed dodecaphase. The existence of a map Ξ such that $R = \Xi \circ F$ is equivalent to the factorization of R through a projection that may eliminate the observable coordinate, orientation, carry, or boundary. This factorization constitutes an additional proposition and is not an implicit part of the reconstruction on the complete state space.

The reduced record is, by definition, a projection of the complete state, not the initial state. Subsequent results use R on the complete state and do not presuppose a factorization through F .

8.5.2 Inverse systems of survivors

The TPK transition need not be a function. To type this fact correctly, let V_t be the finite set of possible complete states at depth t , and let

$$\mathcal{R}_t \subseteq V_t \times V_{t+1}$$

the admissible transition relation, with all fields of the integral holonomic state preserved. We then define X_t , not as the set of isolated terminal states, but as the set of admissible prefixes, that is, of complete admissible histories:

$$X_t = \left\{ (v_0, \dots, v_t) \in \prod_{j=0}^t V_j : (v_j, v_{j+1}) \in \mathcal{R}_j \ (0 \leq j < t), \text{ and the closures are satisfied up to } t \right\}.$$

On these objects, truncation is indeed a canonical application:

$$\begin{aligned} \rho_{t+1,t} : X_{t+1} &\longrightarrow X_t, \\ (v_0, \dots, v_t, v_{t+1}) &\longmapsto (v_0, \dots, v_t). \end{aligned}$$

It is well defined precisely because every prefix of an admissible history remains an admissible history for the closures already imposed. If a filter were to use future information and could retroactively invalidate a prefix, X_t would first have to be replaced by the subset of prolongable prefixes; under that convention, the same truncation is again well defined.

A global HMT history is a compatible family of prefixes.

$$x = (x_t)_{t \geq 0}, \quad x_t \in X_t, \quad \rho_{t+1,t}(x_{t+1}) = x_t.$$

It is equivalent to an infinite sequence $(v_0, v_1, \dots)$ whose consecutive pairs belong to the relations $\mathcal{R}_t$ and whose prefixes satisfy all closures. The history space is the inverse limit

$$X_\infty = \varprojlim X_t.$$

This formulation avoids imposing a single-valued local transition $B_t \mapsto B_{t+1}$. The transition may be a finite relation, several branches may share the same visible terminal state, and uniqueness may appear only after future compatibility is imposed. By preserving the complete prefix, the information that makes it possible to distinguish them is not lost prematurely.

Theorem 8.37 (Unique coinductive section). *Suppose that, for every t , the survivor set $S_t \subseteq X_t$ is nonempty, the restrictions $\rho_{t+1,t} : S_{t+1} \rightarrow S_t$ are compatible, and every S_t contains a unique prefix. Then $\varprojlim S_t$ contains a unique global history.*

Proof. Let x_t be the unique prefix of S_t . Compatibility forces $\rho_{t+1,t}(x_{t+1}) = x_t$, so that $(x_t)_t$ belongs to the inverse limit. Every other history would have a member of S_t at each level and, by uniqueness, would coincide prefix by prefix with $(x_t)_t$. $\square$

The theorem formalizes the passage from level-by-level compatibility to a global history. It does not turn a finite window into a coinductive conclusion: its application to a concrete family requires verifying at every depth non-emptiness, compatibility of the restrictions, and uniqueness of the surviving prefixes.

8.5.3 Nonadic Survival Connection

Cylinder contraction does not proceed block by block as a deterministic substitution. At certain boundaries, there are several local allocations of the same aggregate datum; prolongability determines which belongs to the published history. The resulting structure is a connection on the nine phase classes: upon completing one turn, the phase class returns, but the state of the fiber has changed. This section constructs the complete finite window in which that return is observed.

The domain of the calculation is the attached calibrated transport register. Its initial states, its integer propagation matrix, and its boundary relations are published premises, not outputs of an autonomous enumeration of TPK at arbitrary depth. All statements in this section are exact relative to those premises. Extension to every depth is governed by the preceding coinductive theorem and requires its hypotheses to be verified at every level.

8.5.3.1 Prefix of 30 trits and 1st boundary

We write $u_k^\chi \in \mathbb{F}_3^6$, with $\chi \in \{\pi, e, \varphi\}$, for the visible block of channel χ at depth k . The first five blocks form three prefixes of thirty trits:

$$\begin{aligned} w_{30}^\pi &= 010211|012222|010211|002111|110221, \\ w_{30}^e &= 201101|121221|102011|012222|102011, \\ w_{30}^\varphi &= 121200|112202|121020|010210|010200. \end{aligned}$$

The sixth block is the first boundary following those prefixes:

$$R_{36} = \begin{pmatrix} u_5^\pi \\ u_5^e \\ u_5^\varphi \end{pmatrix} = \begin{pmatrix} 222220 \\ 021222 \\ 102011 \end{pmatrix}.$$

The notation R_{36} indicates that thirty-six trits per channel have been displayed. It does not denote a terminal state: the record reconstructed below contains twenty blocks, and the recurrence admits further continuations when the corresponding Hensel states are supplied.

8.5.3.2 Self-scale axis and residual fiber

For

$$B_k = (u_k^\pi, u_k^e, u_k^\varphi) \in (\mathbb{F}_3^6)^3$$

we define, coordinate by coordinate,

$$q_k = u_k^\pi + u_k^e + u_k^\varphi, \quad a_k = u_k^\pi + u_k^e - u_k^\varphi.$$

Theorem 8.38 (Axial decomposition of the boundary signature). *The pair (q_k, a_k) uniquely determines the axial channel and the aggregate of the other two channels:*

$$u_k^\varphi = 2(q_k - a_k), \quad u_k^\pi + u_k^e = q_k - u_k^\varphi.$$

Consequently, a fiber of the application

$$q_{\text{ax}}(u^\pi, u^e, u^\varphi) = (u^\pi + u^e, u^\varphi)$$

can preserve ambiguity only in the internal allocation between u^π and u^e .

Proof. Subtracting the two defining equations gives $q_k - a_k = 2u_k^\varphi$. The inverse of 2 in $\mathbb{F}_3$ is again 2, whence $u_k^\varphi = 2(q_k - a_k)$. Subtracting this vector from q_k produces $u_k^\pi + u_k^e$. The operation does not recover each summand separately; for this reason the unique residual fiber is the set of distributions of the fixed aggregate. $\square$

In the model's geometric terminology, φ constitutes the axis fixed by the signature and the pair π/e the residual fiber. The expression does not postulate an analytic symmetry of the real constants: it names the exact linear decomposition just proved in $\mathbb{F}_3^6$.

8.5.3.3 Orbita complete of the 9 fases consecutivas of the Topological Phase Kernel

We fix the absolute phase by

$$g(k) = \begin{cases} k \bmod 9, & k \not\equiv 0 \pmod{9}, \\ 9, & k \equiv 0 \pmod{9}. \end{cases}$$

Since w_{30} occupies $k = 0, \dots, 4$, the first subsequent complete circuit after the prefix traverses $k = 5, \dots, 13$, with phases 5, 6, 7, 8, 9, 1, 2, 3, 4. Enumeration of the $729 = 3^6$ candidates of aggregate signature at each level produces the following window:

k	$g(k)$	locals	selected	u_k^π	u_k^e	u_k^φ
5	5	1	1	222220	021222	102011
6	6	1	1	111201	201202	221021
7	7	3	1	212121	222210	200221
8	8	3	1	200121	212212	110110
9	9	2	1	100100	020112	010122
10	1	1	1	101222	112221	112002
11	2	1	1	022212	110001	100021
12	3	1	1	012012	202222	000120
13	4	1	1	111210	112102	211122

The “local” column counts the admissible input states at the indicated level; it does not count the outputs of the boundary relation. This distinction is essential in $k = 9$: the two local input states precede a three-output fiber of the selected state.

8.5.3.4 Return 9→1 of the nonadic phase with state change and memory advance

Let

$$\pi_B : V_k \longrightarrow \mathcal{B} := (\mathbb{F}_3^6)^3$$

the projection onto the visible triple. Once the fiber data of the record are fixed, the transport relation on complete states induces a visible relation $\mathcal{M}_9^B$. The selected state entering the ninth phase is

$$B_9 = (100100 \mid 020112 \mid 010122).$$

Its relational image consists exactly of

$$\begin{aligned} B_{10}^{(0)} &= (101222 \mid 112221 \mid 112002), \\ B_{10}^{(1)} &= (102221 \mid 111222 \mid 112002), \\ B_{10}^{(2)} &= (111221 \mid 102222 \mid 112002). \end{aligned}$$

The three outputs have

$$q_{10} = 022112, \quad a_{10} = 101111,$$

and, by [theorem 8.38](#), share

$$u_{10}^{\varphi} = 112002, \quad u_{10}^{\pi} + u_{10}^e = 210110.$$

They differ exclusively in the three published distributions of that aggregate between π and e .

Theorem 8.39 (Two-level extendibility selection). *In the calibrated window, the three preceding outputs admit a prolongation of depth one. Upon also imposing the next block*

$$B_{11} = (022212 \mid 110001 \mid 100021)$$

and the decimal boundary (502, 662, 638), only $B_{10}^{(0)}$ admits a prolongation of the same complete state. Therefore

$$\# \text{Surv}_1^{\text{cal}}(B_9) = 3, \quad \# \text{Surv}_2^{\text{cal}}(B_9) = 1.$$

Proof. The decimal triples associated with the three outputs are, respectively,

$$(279, 352, 365), \quad (280, 351, 365), \quad (282, 350, 365).$$

For a ternary word of length L , with associated integer N , and a prefix of b triples, with associated integer D , compatibility of the two cylinders is equivalent to the two integer inequalities

$$N1000^b < (D + 1)3^L, \quad (N + 1)1000^b > D3^L.$$

The six inequalities for each output are satisfied at the current level. Upon adjoining B_{11} and (502, 662, 638), they are again satisfied for the three channels of $B_{10}^{(0)}$. For $B_{10}^{(1)}$ and $B_{10}^{(2)}$, the inequality of channel φ is preserved, but those of π and e fail. The evaluation is performed with integers in the corresponding finite certificate; no numerical tolerance intervenes. Consequently, the candidate set is reduced from three to one when passing from depth one to depth two. $\square$

The other two words can precede the same ternary tail only if the deep Hensel extension is changed. They are therefore not prolongations of the same complete state of the register: the visible residue partially coincides, but the genealogy of the fiber is distinct.

8.5.3.5 Phase return with state change

The phase satisfies $g(k + 9) = g(k)$. The visible state, by contrast, is not reset. In particular,

$$B_1 = (012222 \mid 121221 \mid 112202), \quad B_{10} = (101222 \mid 112221 \mid 112002),$$

so that $g(1) = g(10) = 1$ but $B_1 \neq B_{10}$. The verification of the nine pairs $k \mapsto k + 9$, for $k = 1, \dots, 9$, also gives

$$u_{k+9}^{\varphi} \neq u_k^{\varphi}, \quad u_{k+9}^{\pi} + u_{k+9}^e \neq u_k^{\pi} + u_k^e.$$

Therefore, the window satisfies exactly

$$\boxed{g(k + 9) = g(k), \quad B_{k+9} \neq B_k \quad (1 \leq k \leq 9).}$$

This is a phase return with nontrivial visible holonomy. The designation *nonadic monodromy* refers to this return action on the enriched fiber; it does not introduce a tenth phase. To promote the finite remark to a global monodromy, the action at every depth must be constructed and the coinductive compatibility required at the beginning of the section must be proved.

8.5.3.6 Prolongation of the surviving section via 20 compatible blocks

The preceding return does not close the emission. The twenty blocks that the certificate reconstructs from the three published integer states are

$$\begin{aligned}\pi &: 010211|012222|010211|002111|110221|222220|111201| \\ &\quad 212121|200121|100100|101222|022212|012012|111210| \\ &\quad 121011|200220|120210|000101|022010|020111, \\ e &: 201101|121221|102011|012222|102011|021222|201202| \\ &\quad 222210|212212|020112|112221|110001|202222|112102| \\ &\quad 102010|022020|222002|012010|001112|022212, \\ \varphi &: 121200|112202|121020|010210|010200|102011|221021| \\ &\quad 200221|110110|010122|112002|100021|000120|211122| \\ &\quad 202200|202110|221000|100102|111122|020010.\end{aligned}$$

Thus, w_{30} is the prefix and R_{36} the first boundary of an explicitly continued branch; neither object replaces the integral holonomic state that transports that continuation.

8.5.3.7 Exact decoding of 12 decimal triples by means of the nonadic reader

Let $N_{\chi,k}$ be the integer represented in base three by the first k blocks of channel χ . With $k = 13$, the word has 78 trits and defines

$$D_{\chi,12} = \left\lfloor \frac{1000^{12} N_{\chi,13}}{3^{78}} \right\rfloor.$$

The escritura of $D_{\chi,12}$ in twelve blocks of three cifras is

$$\begin{array}{c|cccccccccccc} \pi & 141 & 592 & 653 & 589 & 793 & 238 & 462 & 643 & 383 & 279 & 502 & 884 \\ e & 718 & 281 & 828 & 459 & 045 & 235 & 360 & 287 & 471 & 352 & 662 & 497 \\ \varphi & 618 & 033 & 988 & 749 & 894 & 848 & 204 & 586 & 834 & 365 & 638 & 117. \end{array}$$

The identity is entire: it is equivalent to

$$D_{\chi,12} 3^{78} \leq 1000^{12} N_{\chi,13} < (D_{\chi,12} + 1) 3^{78}.$$

Thus, the two cylinders intersect in every channel, and the reading operation produces exactly the thirty-six published digits. This result is a consequence of the calibrated register. The initial states in the attached chart were historically constructed with those constants as reference data; therefore, this identity does not establish the independent provenance of the initial states from the primitives of APP.

8.5.3.8 Schedule for cylinder reopening and periodicity of the compatible prolongation

The conversion of a six-trit block to decimal triples has average slope

$$\lambda = 6 \log_{1000} 3 \in (0, 1), \quad k(t) = \lfloor \lambda(t + 1) \rfloor.$$

An index t is a reopening without a new triple when $k(t + 1) = k(t)$. Let $\delta = 1 - \lambda$.

Theorem 8.40 (Beatty Reading Calendar Ternary-Decimal). *The set of reopenings is*

$$\mathcal{T} = \left\{ \left\lceil \frac{m}{\delta} \right\rceil - 2 : m \geq 1 \right\}.$$

Its first elements are

$$20, 42, 64, 86, 108, 130, 151, 173, 195, 217, \dots,$$

and two reaperturas consecutivas distan 21 or 22 blocks.

Proof. The quantity $\lambda = \log_{1000} 729$ is irrational: an equality $\lambda = p/q$ would imply $729^q = 1000^p$, impossible by prime factorization. Also δ is irrational, so that

$$k(t) = (t + 1) - \lceil \delta(t + 1) \rceil.$$

The equality $k(t + 1) = k(t)$ occurs exactly when the ceiling function of the second term increases by one. The m -th occurrence is obtained by solving $\delta(t + 1) < m < \delta(t + 2)$, which gives $t = \lceil m/\delta \rceil - 2$. The differences of a Beatty sequence belong to

$$\{\lfloor \delta^{-1} \rfloor, \lceil \delta^{-1} \rceil\} = \{21, 22\}.$$

□

8.5.3.9 Bockstein–Hensel Recurrence of Residue and Quotient

The memory carried by the branch proceeds from the exact sequence.

$$0 \longrightarrow \mathbb{Z}^6 \xrightarrow{\times 3} \mathbb{Z}^6 \longrightarrow \mathbb{F}_3^6 \longrightarrow 0$$

and of the unimodular matrix

$$A = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \quad \det A = 1.$$

For a row state $x_k^\chi \in \mathbb{Z}^6$, it is defined

$$y_k^\chi = x_k^\chi A, \quad u_k^\chi = y_k^\chi \bmod 3 \in \{0, 1, 2\}^6, \quad x_{k+1}^\chi = \frac{y_k^\chi - u_k^\chi}{3}.$$

Proposition 8.41 (Exact conservation of quotient memory). *With x_k^χ and A fixed, the pair (u_k^χ, x_{k+1}^χ) is unique and satisfies*

$$x_k^\chi A = u_k^\chi + 3x_{k+1}^\chi.$$

Conversely, since $A \in \text{GL}_6(\mathbb{Z})$, the pair uniquely determines x_k^χ .

Proof. Euclidean division of each coordinate of $x_k^\chi A$ by three produces a unique residue in $\{0, 1, 2\}$ and a unique integer quotient. This proves the first assertion and the identity. Since $\det A = 1$, the inverse A^{-1} has integer entries; therefore

$$x_k^\chi = (u_k^\chi + 3x_{k+1}^\chi)A^{-1}$$

recovers the previous state unambiguously. □

Residue u_k^χ is the visible part, and quotient x_{k+1}^χ the transported 3-adic memory. The certificate reproduces sixty coordinated steps —twenty per channel—, verifies $AA^{-1} = I$ with integer arithmetic, and checks all blocks against the register. The designation Bockstein–Hensel refers here to this residue–quotient

separation and its 3-adic iteration; it is not identified without additional data with a cohomological Bockstein operator.

8.5.4 Archimedean Evaluation of the Compatible Prefix and Determination of the Real Point

Let us associate with each compatible prefix $x_t \in X_t$ a closed interval $I_t(x_t) \subset \mathbb{R}$. The geometric interpretation is that of a cylinder: all futures sharing the same prefix project into I_t .

Theorem 8.42 (Archimedean convergence of nested intervals). *Let $(x_t)_{t \geq 0}$ be a compatible history. If*

$$I_{t+1}(x_{t+1}) \subseteq I_t(x_t)$$

and

$$\text{diam } I_t(x_t) \longrightarrow 0,$$

then there exists a unique $r \in \mathbb{R}$ such that

$$\bigcap_{t \geq 0} I_t(x_t) = \{r\}.$$

Proof. This is the principle of nested intervals. The left endpoints form a bounded increasing sequence and the right endpoints a bounded decreasing one. Their limits exist; the difference between them is bounded by the diameter, which tends to zero. Both limits coincide at a single point contained in all the intervals. $\square$

Let $X_\infty^{\text{Arch}} \subseteq X_\infty$ be the domain of histories for which the cylinders are nested and their diameters tend to zero. If those two properties are part of global admissibility, then $X_\infty^{\text{Arch}} = X_\infty$; if not, this domain restriction is necessary. We define the evaluation

$$\text{ev}_{\mathbb{R}} : X_\infty^{\text{Arch}} \longrightarrow \mathbb{R}, \quad \text{ev}_{\mathbb{R}}(x) = \text{the unique point of } \bigcap_{t \geq 0} I_t(x_t).$$

Uniqueness of the value does not entail uniqueness of the history. Two genealogies $x \neq y$ may satisfy

$$\text{ev}_{\mathbb{R}}(x) = \text{ev}_{\mathbb{R}}(y).$$

We define reading equivalence

$$x \sim_{\text{ev}} y \iff \text{ev}_{\mathbb{R}}(x) = \text{ev}_{\mathbb{R}}(y).$$

Theorem 8.43 (Archimedean evaluation quotient). *The map*

$$\begin{aligned} \overline{\text{ev}}_{\mathbb{R}} : X_\infty^{\text{Arch}} / \sim_{\text{ev}} &\longrightarrow \text{Im}(\text{ev}_{\mathbb{R}}) \subseteq \mathbb{R}, \\ [x] &\longmapsto \text{ev}_{\mathbb{R}}(x) \end{aligned}$$

is a bijection. In particular, the quotient is identified with all of $\mathbb{R}$ if and only if the evaluation $\text{ev}_{\mathbb{R}}$ is surjective.

Proof. The definition of $\sim_{\text{ev}}$ makes the value independent of the representative, so the map is well defined. It is injective because two classes with the same value are, by definition, a single class; it is surjective onto the image by the very definition of image. The last assertion is equivalent to $\text{Im}(\text{ev}_{\mathbb{R}}) = \mathbb{R}$. $\square$

The covering theorem for the Archimedean reader constructs precisely those prefixes: for every compact interval $K \subset \mathbb{R}$, $\text{ev}_{\mathbb{R}}(\Omega_{\infty, K}) = K$, and the intervals $[-m, m]$ form a directed family whose union

is $\mathbb{R}$. Consequently, every real number possesses at least one compatible HMT history, and the proved assertion is

$$X_{\infty}^{\text{Arch}} / \sim_{\text{ev}} \cong \text{Im}(\text{ev}_{\mathbb{R}}) = \mathbb{R}.$$

The Archimedean value is, therefore, the quotient that identifies the fibers of the evaluation; the enriched HMT number also conserves the class of trajectory, the carry, and the orientation. The identification does not follow from the mere abstract formation of the quotient: it follows from the covering theorem already proved in the discrete construction of the continuum.

Algebraic descent is another problem and must be solved operation by operation. Suppose that two candidate operations $\oplus$ and $\odot$ have been constructed on X_{∞}^{Arch} and are closed on that domain. Addition descends to the quotient if and only if

$$x \sim_{\text{ev}} x', y \sim_{\text{ev}} y' \implies x \oplus y \sim_{\text{ev}} x' \oplus y',$$

while the product descends, independently, if and only if

$$x \sim_{\text{ev}} x', y \sim_{\text{ev}} y' \implies x \odot y \sim_{\text{ev}} x' \odot y'.$$

The first congruence does not imply the second. Moreover, for the induced operations to be precisely the ordinary sum and product of the Archimedean quotient, it is necessary to prove separately

$$\text{ev}_{\mathbb{R}}(x \oplus y) = \text{ev}_{\mathbb{R}}(x) + \text{ev}_{\mathbb{R}}(y),$$

$$\text{ev}_{\mathbb{R}}(x \odot y) = \text{ev}_{\mathbb{R}}(x) \text{ev}_{\mathbb{R}}(y),$$

and that the image of the evaluation be closed under the corresponding operation. Only after these certificates does the quotient inherit the structure of a subring—or of a subfield, if additive opposites and multiplicative inverses are also constructed—of $\mathbb{R}$. The set-theoretic bijection of the preceding theorem is formal; the content specific to HMT lies in constructing the evaluation, characterizing its fibers, proving coverage, and establishing each descent law.

8.5.5 Weyl–Heisenberg Representation Associated with the Nonadic Torus

The oriented area form of the nonadic torus is

$$\sigma((a, b), (c, d)) = ad - bc \pmod{9}.$$

This alternating form is obtained, with the chosen orientation, by integrating the mixed curvature of APP over parallelograms of the torus. Let $\omega = \exp(2\pi i/9)$ and $\mathcal{H}_9 = \mathbb{C}^9$, with basis $\{|n\rangle : n \in \mathbb{Z}_9\}$. We define

$$X|n\rangle = |n+1\rangle, \quad Z|n\rangle = \omega^n |n\rangle.$$

Then

$$ZX = \omega XZ,$$

and, more generally,

$$(X^a Z^b)(X^c Z^d) = \omega^{bc} X^{a+c} Z^{b+d}.$$

Therefore

$$c((a, b), (c, d)) = bc \pmod{9}$$

is the polarized 2-cocycle that appears in this section of the central extension. We explicitly define

$$\text{Heis}_9 = \{\omega^k X^a Z^b : a, b, k \in \mathbb{Z}/9\mathbb{Z}\}.$$

It is a group of order $9^3 = 729$. It is a central extension associated with the translation group of the APP support X_9 , not a subgroup of the graph or the APP path category. The commutator of the two transports is

$$(X^a Z^b)(X^c Z^d)(X^a Z^b)^{-1}(X^c Z^d)^{-1} = \omega^{bc-ad} I.$$

Consequently, its exponent is $-\sigma((a, b), (c, d))$. The polarized cocycle c and the form σ are not the same object: the second is, under this convention, the opposite of the antisymmetrization $c(u, v) - c(v, u)$. The commutator phase is thus the inverse of the Wilson phase for the fixed orientation.

To construct observables, we take

$$Q_H = \text{diag}(0, 1, \dots, 8), \quad F_{jk} = \frac{1}{3} \omega^{jk}, \quad P_H = F^\dagger Q_H F.$$

The operators Q_H, P_H are self-adjoint. Their eigenbases are mutually unbiased because $|F_{jk}|^2 = 1/9$. For every unit vector $\psi \in \mathcal{H}_9$, the Robertson inequality gives

$$\text{Var}_\psi(Q_H) \text{Var}_\psi(P_H) \geq \frac{1}{4} |\langle \psi, [Q_H, P_H] \psi \rangle|^2,$$

and the entropy bound for mutually unbiased bases produces

$$H_{Q_H}(\psi) + H_{P_H}(\psi) \geq \log 9.$$

Here H is the Shannon entropy, with natural logarithm, of the corresponding spectral distributions. The constructed relations are summarized, without attributing any additional physical causality to them, by

$$\begin{aligned} & \text{alternating form } (X_9, \sigma) \longrightarrow \text{Heis}_9, \\ & \text{Heis}_9 \longrightarrow (X, Z) \longrightarrow (Q_H, P_H) \longrightarrow \text{finite uncertainty}. \end{aligned}$$

This is a finite algebraic model of operators; by itself it does not specify physical dynamics or units of measurement.

8.5.6 Cyclic dynamics on $\mathbb{Z}_9 \times \mathbb{Z}_{12}$ and minimal predictive quotient

The finite dynamical system $\mathfrak{C}_{9,12}$ makes it possible to determine the minimal memory required for the observed evolution to be deterministic. Let

$$X = \{(b, m) : b \in \mathbb{Z}/9\mathbb{Z}, 0 \leq m < 12\}$$

with successor

$$S(b, m) = \begin{cases} (b, m+1), & m < 11, \\ (b+1 \bmod 9, 0), & m = 11. \end{cases}$$

The marginal labels are

$$\beta(b, m) = 4b \pmod{12}, \quad d(b, m) = 1 + (m + \beta(b, m)) \pmod{12}.$$

For a remark application q , the inclusive future word of horizon h is

$$Q_h^q(x) = (q(x), q(Sx), \dots, q(S^h x)).$$

Theorem 8.44 (Minimum predictive quotient). *Let X finite, $T : X \rightarrow X$ and $q : X \rightarrow V$. Definamos*

$$\text{Eq}(f) := \{(x, y) \in X^2 : f(x) = f(y)\}, \quad E_h = \text{Eq}(Q_h^q), \quad E_\infty = \bigcap_{k \geq 0} \text{Eq}(q \circ T^k).$$

Then $E_{h+1} = E_h \cap (T \times T)^{-1}(E_h)$, the sequence stabilizes in finite time, and X/E_∞ is the quotient with the smallest number of states that refines q and admits an induced deterministic dynamics.

Proof. E_h requires equality of observations at the times $0, \dots, h$, whereas $(T \times T)^{-1}(E_h)$ requires equality at $1, \dots, h+1$. Their intersection is E_{h+1} . Because X is finite, the number of classes cannot grow indefinitely. At the first stable level, T respects the classes and descends. Every T -compatible equivalence contained in $\text{Eq}(q)$ identifies only states with the same complete future and is therefore contained in E_∞ ; hence the minimality property. $\square$

The evaluation exhaustiva of $\mathfrak{C}_{9,12}$ da:

observable	instantaneous values	index h	observations	stable quotient
β	3	11	12	C_{36} , fibra 3
d	12	8	9	C_{36} , fibra 3
(β, d)	36	0	1	C_{36} , fibra 3

In the two marginal observables,

$$\text{Eq}(Q_{11}^\beta) = \text{Eq}(Q_8^d) = \text{Eq}(\pi_{36}), \quad \pi_{36}(b, m) = (b \bmod 3, m).$$

The factor 36 is not introduced as a target. It appears as the minimal predictive quotient of each marginal. The phase m is a state coordinate required to predict those futures; this mathematical assertion does not presuppose a physical causal interpretation. The transformation $b \mapsto b + 3$ is a symmetry invisible to those observables at every time.

As an abstract dynamical system, the quotient is a cycle of thirty-six states. Writing C_{36} does not, however, choose a distinguished origin: it must be fixed to obtain a coordinate $\mathbb{Z}/36\mathbb{Z}$. Nor does this automatically turn those states into the thirty-six pairs of a nonad. Indeed, $\text{Aut } J(9, 2) \cong S_9$. If the two points of a pair lie in distinct cycles of lengths r, s of a permutation, the orbit of the pair has length $\text{lcm}(r, s)$, with $r + s \leq 9$, and hence length at most 20. If they lie in the same cycle of length $r \leq 9$, the orbit has length r , except in the antipodal case, when it has length $r/2$. Thus no automorphism induces an orbit of length 36 on the pairs. Therefore, the passage

$$C_{36} \longleftrightarrow \begin{pmatrix} [9] \\ 2 \end{pmatrix}$$

remains a calibrated identification, not a natural equivariant identification. Even a Hamiltonian ordering of the pairs would prove local preservation of incidence step by step, not a global automorphism or the canonicity of the dictionary.

8.5.7 HMT number as a compatible history in X_∞ : genealogical identity, reading structure, and Archimedean evaluation

The preceding results determine the following instance of the governing definition. In the notation of this section, an HMT number is a compatible history

$$x \in X_\infty = \varprojlim X_t.$$

A reader L is additional structure: the pair (x, L) constitutes a read presentation or a measurement, but does not alter the identity type of x . When x belongs to X_∞^{Arch} , its Archimedean value is the point intersection of the cylinders; the set of values so constructed is $\text{Im}(\text{ev}_{\mathbb{R}}) \subseteq \mathbb{R}$, and its identification with the entire line requires the additional coverage theorem. Its minimal memory relative to an observable is the predictive quotient of futures; and the alternating form of the sum–product support admits the finite Weyl–Heisenberg representation constructed above. Value, trajectory class, and uncertainty are defined by three operations on specified domains of the same transport architecture.

PART IV

**Discrete structure of the continuum and double holographic
projection**

Inverse system of compatible trajectories

9.1 Genealogical construction of the continuum

9.1.1 Generative composition $\text{APP} \rightarrow \text{TRIT} \rightarrow \text{TPK}$ and construction of the integral holonomic state

The construction begins with three irreducible and causally ordered objects. The object of *All Possible Paths* (hereafter, APP) constructs the common nonadic support, its graph of all finite paths, and the two sum–product evaluation sheets. The tritic unit (hereafter, $\widehat{\mathbb{T}}$) is the minimal local unit of topological, geometric, informational, and, once ordered, temporal relation: its visible signature has three states, and its lift preserves the carry that a binary or residual reading would lose. The *Topological Phase Kernel* (hereafter, TPK) is the transport category that composes these units into routes and preserves, within the same state, sheet, orientation, boundary, genealogical ledger, memory, phase, and survival.

Object	Construction	Invariantes transportados
APP	positive chart, finite torus, Cayley graph, paths, and leaves Σ, Π	marker, cell, neighborhood, sum, product, quotient and closure
TRIT	ternary signature visible and local topological-temporal lift	residue, carry, sheet, orientation, threshold and reversal
TPK	category of paths, fibers, and compatible extensions	cylinder, boundary, genealogical record, memory, phase, holonomy and survival

The exact construction of APP is established on a marked segment and a finite chart; the real line appears afterward as a limiting publication. The reference interval, its marks, and the oriented path that traverses them are:

$$I_{10} = [0, 10], \quad \mathcal{M}_{10} = I_{10} \cap \mathbb{Z} = \{0, 1, \dots, 10\}, \quad \mathbf{P}_{10} : 0 \longrightarrow 1 \longrightarrow \dots \longrightarrow 9 \longrightarrow 10, \quad (9.1)$$

The endpoints fix the edge and the nine interior marks $I_{10}^\circ \cap \mathbb{Z}$ form the positive alphabet. The exact mathematical object of departure is the chart

$$\mathcal{D}_9 = \{1, \dots, 9\}, \quad \mathcal{D}_8 = \{1, \dots, 8\}, \quad \lambda_9 : \mathcal{D}_9 \xrightarrow{\sim} \mathbb{Z}/9\mathbb{Z}, \quad \lambda_9(9) = [0]_9. \quad (9.2)$$

The mark 9 is simultaneously the last positive representative and the publication of the null class. For every $n \geq 1$, the positive digital root and quotient are separated by means of

$$\text{dr}_9(n) = 1 + ((n - 1) \bmod 9), \quad q_9(n) = \left\lfloor \frac{n - 1}{9} \right\rfloor, \quad n = 9q_9(n) + \text{dr}_9(n). \quad (9.3)$$

To the path $1 \rightarrow 2 \rightarrow \dots \rightarrow 9$ we add the unique closing edge $9 \rightarrow 1$. The successor and its memory are separated:

$$u_+(a) = \begin{cases} a + 1, & a < 9, \\ 1, & a = 9, \end{cases} \quad \kappa_+(a) = \begin{cases} 0, & a < 9, \\ 1, & a = 9, \end{cases} \quad a + 1 = u_+(a) + 9\kappa_+(a). \quad (9.4)$$

For that reason $9 \rightarrow 1$ closes the phase, increases the memory, and prolongs the genealogical state.

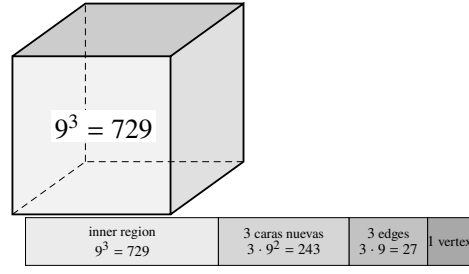
The square of the chart produces the finite torus and its movement graph:

$$X_9 = (\mathbb{Z}/9\mathbb{Z})^2, \quad \Gamma_{\text{APP}} = \text{Cay}(X_9, \{\pm e_1, \pm e_2\}), \quad \text{Path}(\Gamma_{\text{APP}}). \quad (9.5)$$

Multiplicative leaf of the APP chart: $\Pi(i, j) = \text{dr}_9(ij)$

	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9
2	2	4	6	8	1	3	5	7	9
3	3	6	9	3	6	9	3	6	9
4	4	8	3	7	2	6	1	5	9
5	5	1	6	2	7	3	8	4	9
6	6	3	9	6	3	9	6	3	9
7	7	5	3	1	8	6	4	2	9
8	8	7	6	5	4	3	2	1	9
9	9	9	9	9	9	9	9	9	9

$8^2 = 64$ cells interior; gnomon of closure $9 + 8 = 17$
corona positive of nueves: the mark 9 publishes $[0]_9$

Cube of Extreme and Mean Holographic Ratio

$$10^3 = 729 + 243 + 27 + 1 = 729 + 271$$

growth corona $9^3 \subset 10^3$: 271 states
internal boundary of the cube of side 9: 386 points

Figure 9.1: Two finite publications of the support. Left: the product table with positive digital root and its closure gnomon. Right: the EMRH cubic completion, which distinguishes interior, faces, edges, and vertex. The growth corona 271 and the internal boundary 386 are two typed counts: the former measures the enlargement $9^3 \subset 10^3$ and the latter the boundary of the cube of side 9.

Identifying opposite sides preserves the cardinality of the chart and adds the continuation law. The core of $8^2 = 64$ interstices is completed by the minimal gnomon that publishes the zero class in both coordinates:

$$\mathcal{D}_9^2 = \mathcal{D}_8^2 \sqcup \mathcal{G}_9, \quad 81 = 64 + (9 + 8) = 64 + 17. \quad (9.6)$$

The same law is prolonged to charts of greater depth; the scale changes, but the closure, the shared edge, and the memory of the turn preserve their type.

The associated volumetric completion is expressed through the strata identity.

$$10^3 = (9 + 1)^3 = 729 + 243 + 27 + 1. \quad (9.7)$$

The interior volume, faces, edges and point closure are thus typed as distinct layers of the same completed unit. On the 81 cells of the positive chart act the evaluations

$$\Sigma, \Pi : \mathcal{D}_9^2 \longrightarrow \mathcal{D}_9, \quad \Sigma(i, j) = \text{dr}_9(i + j), \quad \Pi(i, j) = \text{dr}_9(ij),$$

Their residues and quotients raised are defined by

$$R^+(i, j) := \Sigma(i, j), \quad Q^+(i, j) := \frac{i + j - R^+(i, j)}{9},$$

$$R^\times(i, j) := \Pi(i, j), \quad Q^\times(i, j) := \frac{ij - R^\times(i, j)}{9},$$

and, therefore,

$$\widehat{\Sigma}(i, j) := (R^+(i, j), Q^+(i, j)), \quad \widehat{\Sigma}, \widehat{\Pi} : \mathcal{D}_9^2 \longrightarrow \mathcal{D}_9 \times \mathbb{Z}_{\geq 0},$$

$$\widehat{\Pi}(i, j) := (R^\times(i, j), Q^\times(i, j)),$$

The bijection $\lambda_9^{\times 2} : \mathcal{D}_9^2 \rightarrow X_9$ transports these operations and their lifts to the torus. Therefore, APP is the tuple

$$\text{APP} = (\mathcal{D}_9, \lambda_9, X_9, \Gamma_{\text{APP}}, \text{Path}(\Gamma_{\text{APP}}), \Sigma, \Pi, \widehat{\Sigma}, \widehat{\Pi}), \quad (9.8)$$

The table constitutes a two-dimensional representation of this complete tuple. The additive sheet is Latin; the multiplicative sheet conserves the ternary radical of the ring $\mathbb{Z}/9\mathbb{Z}$

$$R_9 := \mathbb{Z}/9\mathbb{Z}, \quad \mathfrak{m} := ([3]_9) = \{[0]_9, [3]_9, [6]_9\} \triangleleft R_9, \quad \mathfrak{m}^2 = (0). \quad (9.9)$$

The internal rotation is the automorphism

$$\rho : \mathfrak{m}^3 \longrightarrow \mathfrak{m}^3, \quad \rho(x_1, x_2, x_3) = (x_2, x_3, x_1),$$

whose distinguished orbit is

$$([3]_9, [6]_9, [0]_9) \xrightarrow{\rho} ([6]_9, [0]_9, [3]_9) \xrightarrow{\rho} ([0]_9, [3]_9, [6]_9) \xrightarrow{\rho} ([3]_9, [6]_9, [0]_9).$$

The direct reduction sends 3, 6, 9 to 0, 0, 0. The internal coordinate of the ideal, by contrast, conserves the extracted factor:

$$\eta_{\mathfrak{m}} : (\mathfrak{m}, +) \xrightarrow{\sim} (\mathbb{Z}/3\mathbb{Z}, +), \quad \eta_{\mathfrak{m}}([3a]_9) := [a]_3, \quad ([3]_9, [6]_9, [0]_9) \xrightarrow{\eta_{\mathfrak{m}}^{\times 3}} ([1]_3, [2]_3, [0]_3), \quad (9.10)$$

so that the 3, 6, 9 sector remains intact and its internal coordinate is read as 1, 2, 0. This map belongs to the radical of APP; the temporal TRIT projection constitutes the subsequent calendrical reader. The three calendrical classes remain separated as

$$C_+ = \{1, 4, 7\}, \quad C_- = \{2, 5, 8\}, \quad C_0 = \{3, 6, 9\}.$$

The incompatibility between the two leaves becomes a measurable invariant. On $\mathcal{A}_3 = \mathcal{D}_9^3$, with uniform measure $\nu_3(x) = 9^{-3}$, let $\mathcal{H}_3 = L^2(\mathcal{A}_3, \nu_3)$. We define

$$\Sigma_3, \Pi_3 : \mathcal{A}_3 \longrightarrow \mathcal{D}_9, \quad \Sigma_3(x) = \text{dr}_9(x_1 + x_2 + x_3), \quad \Pi_3(x) = \text{dr}_9(x_1 x_2 x_3),$$

and the orthogonal projectors

$$P_{\Sigma,3} := \mathbb{E}_{\nu_3}[\cdot \mid \sigma(\Sigma_3)], \quad P_{\Pi,3} := \mathbb{E}_{\nu_3}[\cdot \mid \sigma(\Pi_3)] \in \mathcal{B}(\mathcal{H}_3).$$

The joint enumeration of the 9^3 words then determines

$$\| [P_{\Sigma,3}, P_{\Pi,3}] \| = \frac{\sqrt{3}}{4}, \quad B_c = 7I + 2R = 9P_+ + 5P_-, \quad 459 - 405 = 54.$$

The first number reappears as the electronic prefactor; the second organizes the central block; the third records the global sum–product defect.

9.1.2 TRIT: local topological unit and temporal relation unit

HMT first fixes the visible signature

$$\text{TRIT}_{\text{vis}} = \{-1, 0, +1\}, \quad \tau \in \text{TRIT}_{\text{vis}}. \quad (9.11)$$

In contrast to the binary unit, the TRIT preserves three local regimes. In the uniform distribution, its information and entropy are distinguished as

$$I_{\text{TRIT}} = \log 3 \text{ nats} = \log_2 3 \text{ bits}, \quad S_{\text{TRIT}} = k_B I_{\text{TRIT}}. \quad (9.12)$$

The bit remains the binary unit; the TRIT is the minimal unit capable of simultaneously distinguishing return, threshold, and opening. The equality (9.12) quantifies the information of the triad; the cost $k_B \Theta \log 3$ belongs to the subsequent thermal reset reader. Dimensional Mechanics also preserves the arithmetic antecedent of that signature. For $(i, j) \in \mathcal{D}_9^2$, the associated lifted arithmetic cell is

$$c_{\text{MD}}(i, j; \tau) = (i, j; R^+, Q^+, R^\times, Q^\times; \tau). \quad (9.13)$$

The residues $R^+, R^\times$ are the visible publication of the two sheets; the quotients $Q^+, Q^\times$ preserve the exact memory of Euclidean division; and τ orients the transition. The visible signature occupies the first level; the cell bearing it adds the arithmetic antecedents, and the integral holonomic state then completes boundary, path, phase and survival.

It is useful to distinguish four strata. The visible signature is τ ; the residue–carry pair is its local lift; c_{MD} is the APP arithmetic cell equipped with that signature; and the TPK state adds extension, boundary, genealogical ledger, phase, and survival. This hierarchy preserves the distinct identity of the observed coordinate, its lift, the carrier cell, and the enriched history that transports it.

In the balanced ternary chart, the lifted local integer $n \in \{-4, -3, \dots, 3, 4\}$ decomposes uniquely as

$$\widehat{n} = (r, \kappa), \quad n = r + 3\kappa, \quad r, \kappa \in \{-1, 0, +1\}. \quad (9.14)$$

The visible signature is $\tau = r$; κ conserves the carry that the projection $n \mapsto r$ would forget. The fiber over the visible zero contains three distinct states,

$$0^+ = (0, +1), \quad 0^0 = (0, 0), \quad 0^- = (0, -1),$$

because $n = -3, 0, 3$ have the same visible residue and different carries. This is the minimal topological lift of MD: an oriented APP transition is endowed with the information required to invert it and to decide which prolongations remain compatible.

The signature also has a sheet-memory rule. If m_k is the active arithmetic mode, then

$$m_{k+1} = \begin{cases} \Sigma, & \tau_k = +1, \\ \Pi, & \tau_k = -1, \\ m_k, & \tau_k = 0. \end{cases} \quad (9.15)$$

Zero encodes retention of the previous sheet. For that reason the word $(+1, -1, 0)$ publishes the sequence (Σ, Π, Π) : the third state conserves the active multiplicative mode.

The temporal relation is obtained by orderly parametrizing these cells. Each signature carries a quadratic algebra.

$$\mathbb{A}_\tau = \mathbb{R}[X]/(X^2 + \tau), \quad J_\tau = [X], \quad (9.16)$$

so that

τ	J_τ^2	ordered temporal relation
+1	$-I$	elliptic return, memory and past,
0	0	parabolic threshold and present,
-1	$+I$	bidirectional hyperbolic opening and future.

The TRIT precedes external time: it first classifies the algebraic and topological type of the transport; an ordered parameterization then publishes it as a temporal relation. The twofold direction is typed by the exact involution

$$C_{\text{rev}}(\tau_1, \dots, \tau_n) = (-\tau_n, \dots, -\tau_1), \quad C_{\text{rev}}^2 = I, \quad (9.17)$$

which exchanges return and opening and preserves the threshold. On the hyperbolic sheet, $P_\pm = (I \pm J_{-1})/2$ separate the two eigendirections, and reversal exchanges their orientations. Topology and time are thus the two coordinated readings of the TRIT.

The combinatorial involution C_{rev} and an operatorial conjugation C belong, however, to distinct domains. In a separate dynamic realization, endowed with $H = H^*$, $CJ_EC^{-1} = -J_E$, and $CHC^{-1} = H$, we fix an internal action scale $\hbar_{\text{int}} > 0$ and write $h_{\text{int}} := 2\pi\hbar_{\text{int}}$. Then

$$U(t) = \exp(-J_E H t / \hbar_{\text{int}}), \quad CU(t)C^{-1} = U(-t). \quad (9.18)$$

The intertwiner that transports TRIT words to the dynamical realization fixes the correspondence between C_{rev} and C . The twofold local orientation provides the domain of that construction.

9.1.3 TPK: category of transport and phase evolution

The *Topological Phase Kernel* is the transport category over the observable base of APP that composes digits, words, operators and prolongation relations in the same integral holonomic state. Its internal architecture is distributed among eight operator classes, differentiated by domain, codomain and function:

- (i) observable support and integral holonomic state;
- (ii) internal sheet selector $M_{\text{ph}} : \{1, \dots, 9\} \rightarrow \{+1, -1, 0\}$;
- (iii) cardinal translations and visible evolution F_{obs} ;
- (iv) readers and quotients nonadic, tritico and of block 1000;
- (v) memory, carry, cylinder, boundary, and genealogical register;
- (vi) periodicity and return $C_{12} \hookrightarrow C_{108} \twoheadrightarrow C_9$, $\mathcal{K}_{27}$, and channels 90/120;
- (vii) prolongations Ext_r , cylinders, survival, and natural bidirectional extension;
- (viii) Archimedean-output, calibrated-exceptional, and dimensional functors.

The return of 27 steps is the composite morphism.

$$\mathcal{K}_{27} := F_{\text{obs}}^{27}, \quad \mathcal{K}_{27}^4 = F_{\text{obs}}^{108}. \quad (9.19)$$

Its fourth iterate restores the observable phase of the calendar, while the memory and the complete TPK state continue along their trajectory; $\mathcal{K}_{27}$ is the internal return morphism of C_{108} . The calendar C_{108} , the nonadic connection, the bilateral extension, and the readers occupy typed places within this architecture. Let $\mathbf{P}_{\text{APP}}$ be the free category of the directed graph of APP, and let $\mathbf{P}_{\text{APP}}^{(2)} = \mathbf{P}_{\text{APP}} \times \mathbf{P}_{\text{APP}}$ be its simultaneous sum-product reading. Let, moreover,

$$\mathcal{Q}_{\text{obs}} = X_9^2 \times \{N, E, S, O\}^2 \times \Theta_{108}, \quad \Theta_{108} = \mathbb{Z}/108\mathbb{Z},$$

where, for the representative $\tilde{\theta} \in \{0, \dots, 107\}$,

$$r(\theta) = 1 + (\tilde{\theta} \bmod 9), \quad g_0(\theta) = [\tilde{\theta}]_9,$$

$$\tau(\theta) = \begin{cases} +1, & r(\theta) \in \{1, 4, 7\}, \\ -1, & r(\theta) \in \{2, 5, 8\}, \\ 0, & r(\theta) \in \{3, 6, 9\}, \end{cases} \quad \mu(\theta) = \begin{cases} \Sigma, & \tau(\theta) = +1, \\ \Pi, & \tau(\theta) = -1, \\ \emptyset, & \tau(\theta) = 0. \end{cases}$$

The calendar label $\mu = \emptyset$ marks the threshold, and the dynamic memory m_k of (9.15) retains the last active sheet. A configuration $q = (p^\Sigma, p^\Pi, d^\Sigma, d^\Pi, \theta)$ preserves the two positions, the two cardinal directions, and the calendar index. The visible evolution F_{obs} activates the sheet indicated by TRIT, transports the corresponding position, and advances θ . Let $\mathbf{P}_{\text{obs}}$ denote the free category generated by the arrows $q \rightarrow F_{\text{obs}}(q)$. Each arrow induces a pair of APP paths and defines the base functor

$$B : \mathbf{P}_{\text{obs}} \longrightarrow \mathbf{P}_{\text{APP}}^{(2)}. \quad (9.20)$$

Over each q lies the typed fibre

$$\mathcal{F}_q = \mathcal{O}_q \times \mathcal{H}_q \times \mathcal{C}_q \times \mathcal{S}_q \times \mathcal{B}_q \times \mathcal{A}_q, \quad (9.21)$$

Its six factors conserve, respectively, orientation and sheet; residue and quotient; carry; signature; cylinder and boundary; and ordered register of transport. Memory and survival are properties of the composition and of the prolongation relations.

For a route $r : q \rightarrow q'$, the relation

$$\text{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'} \quad (9.22)$$

contains all admissible continuations and satisfies

$$\Delta_q \subseteq \text{Ext}_{\text{id}_q}, \quad \text{Ext}_{r_2} \circ \text{Ext}_{r_1} \subseteq \text{Ext}_{r_2 \circ r_1}.$$

Transports respect identities and composition:

$$\begin{aligned} H(\text{id}_q) &= I, & C(\text{id}_q) &= I, & \Lambda(\text{id}_q) &= I, & \kappa(\text{id}_q) &= 0, \\ H(r_2 \circ r_1) &= H(r_2)H(r_1), & C(r_2 \circ r_1) &= C(r_2)C(r_1), & \Lambda(r_2 \circ r_1) &= \Lambda(r_2)\Lambda(r_1). \end{aligned} \quad (9.23)$$

Along with these functorial laws, the transported coordinates obey

$$\begin{aligned} h' &= H(r)(h), \\ c' &= C(r)(c) + \kappa(r), & \kappa(r_2 \circ r_1) &= \kappa(r_2) + C(r_2)\kappa(r_1), \\ \lambda' &= \Lambda(r)(\lambda), \\ \sigma' &= \text{Sig}_{q'}(h', c', \lambda'), \end{aligned} \quad (9.24)$$

The last identity is the carry cocycle that makes the composition with memory associative.

The TPK is then the category TPK whose objects are

$$x = (q, o, h, c, \sigma, C, b, \lambda), \quad (o, h, c, \sigma, (C, b), \lambda) \in \mathcal{F}_q, \quad (9.25)$$

and whose morphisms $x \rightarrow x'$ are the triples (r, x, x') such that the coordinates satisfy (9.24) and $(f_x, f_{x'}) \in \text{Ext}_r$, where $f_x \in \mathcal{F}_q$ and $f_{x'} \in \mathcal{F}_{q'}$. Composition is

$$(r_2, x', x'') \circ (r_1, x, x') = (r_2 \circ r_1, x, x''), \quad (9.26)$$

the identity is (id_q, x, x) , and the functor

$$p : \text{TPK} \longrightarrow \text{P}_{\text{obs}}, \quad p(x) = q, \quad p(r, x, x') = r, \quad (9.27)$$

publishes the visible configuration without identifying states of the same fiber. The definition fixes a category over a base; it does not presuppose that it is a categorical fibration nor that every arrow admits a Cartesian lift. The local projections remain separated by type:

$$\begin{aligned} \pi_{\text{TRIT}} : \text{Ob}(\text{TPK}) &\longrightarrow \text{TRIT}_{\text{vis}}, & \pi_{\text{TRIT}}(x) &= \tau(\theta), \\ \pi_{\text{loc}}(x) &= (\pi_{\text{TRIT}}(x), \mu(\theta), o, h, c, \sigma, \lambda, g_0(\theta)). \end{aligned}$$

The first recovers only the visible TRIT signature. The second additionally preserves the calendrical sheet label, the orientation, the residue–quotient pair, the carry, the transport signature, the genealogical register and the phase, but still forgets the cylinder, its boundary label and other coordinates of q . This is why neither visible TRIT nor its local register replaces the integral holonomic state of the TPK.

The name encompasses three inseparable functions. *Topological* refers to the category of paths, incidence, boundary, and prolongation by cylinders; *Phase* refers to the nonadic index, the oriented calendar, and holonomy; *Kernel* names the persistent transport kernel that remains before the state is

contracted by a reader. Formally, the object is the preceding category: an additional topology on objects or morphisms, or its identification with the linear kernel of a map, requires independent data.

Consequently, two routes with the same visible endpoint may remain objects distinct by sheet, quotient, carry, boundary, cylinder, phase, or register. The TPK does not choose a scalar output: it conserves the common antecedent from which the Archimedean, exceptional, and dimensional publications depart.

Correlative and inseparable construction of the continuum

The five constructions are generated correlatively on a single APP–TRIT–TPK state. They are consubstantial operations of the continuum, not an exhaustive enumeration of its properties, five independent branches, or a retrospective selection determined by subsequent applications.

10.1 Common state, prolongation tree, and genealogical measure

A single integral holonomic state before any internal discrimination

The object of this chapter is the complete finite state produced by APP, oriented by TRIT and transported by TPK. From it are constructed the groupoid of reversible symmetries, the tree of all admissible prolongations, the multisection of complete histories and its canonical measure. No specialized domain is introduced: the domain is the entire set of states actually generated by the same mechanism APP–TRIT–TPK. Nor is a sheet, a continuation or a subsequent reading selected.

10.1.1 Finite data produced by APP

For each depth $k \geq 0$, let $\mathbf{A}_k$ denote the finite set of cellular prefixes produced by APP, and let

$$\rho_{k+1,k}^{\text{APP}} : \mathbf{A}_{k+1} \longrightarrow \mathbf{A}_k \quad (10.1)$$

the erasure of the last refinement. This erasure conserves the already constructed prefix. The neutral cell of APP gives a prolongation

$$\eta_k^{\text{APP}} : \mathbf{A}_k \longrightarrow \mathbf{A}_{k+1}, \quad \rho_{k+1,k}^{\text{APP}} \eta_k^{\text{APP}} = I_{\mathbf{A}_k}. \quad (10.2)$$

We write

$$\varepsilon_k^0(a) : a \longrightarrow \eta_k^{\text{APP}}(a) \quad (10.3)$$

for the emission that realizes that prolongation. Thus, its destination, $\eta_k^{\text{APP}}(a)$, and the operation that produces it, $\varepsilon_k^0(a)$, are kept separate. η_k^{APP} is not interpreted as an absence of history: it records a length-one refinement with neutral emission and materially preserves the preceding state.

Each $a \in \mathbf{A}_k$ simultaneously bears the two sheets of APP,

$$\text{Sh}_k(a) = (\Sigma_k(a), \Pi_k(a)) \in \mathbf{H}_k^\Sigma \times \mathbf{H}_k^\Pi, \quad (10.4)$$

and not a chosen element of the pair. For an elementary emission $\epsilon : a \rightarrow a'$, APP further produces two typed Euclidean divisions,

$$\delta^\diamond(\epsilon) = \text{res}^\diamond(\epsilon) + 9 \text{quo}^\diamond(\epsilon), \quad \text{res}^\diamond(\epsilon) \in \{0, \dots, 8\}, \quad \text{quo}^\diamond(\epsilon) \in \mathbb{Z}, \quad (10.5)$$

where $\diamond \in \{\Sigma, \Pi\}$. The label $\diamond$ merely indicates which leaf is being read; both divisions remain in the same ledger.

The cellular source and target of ϵ determine its oriented boundary

$$\partial\epsilon := t(\epsilon) - s(\epsilon). \quad (10.6)$$

The neutral emission satisfies

$$\text{res}^\diamond(\varepsilon_k^0(a)) = 0, \quad \text{quo}^\diamond(\varepsilon_k^0(a)) = 0, \quad \partial\varepsilon_k^0(a) = 0. \quad (10.7)$$

Definition 10.1 (Directed graph of APP emissions). Let $E_k(a)$ be the finite set of APP emissions of length one that depart from $a \in A_k$. We conserve the emissions with multiplicity: two operations with the same target but with different residue, quotient, or provenance are different elements of $E_k(a)$. By (10.2),

$$\varepsilon_k^0(a) \in E_k(a), \quad 1 \leq \#E_k(a) < \infty. \quad (10.8)$$

10.1.2 Tritic extension of the incidence datum: balanced orientation and genealogical carry cocycle

The signed incidence of an APP emission is an integer $\Delta(\epsilon)$. The balanced representative modulo three uniquely gives

$$\Delta(\epsilon) = \text{or}(\epsilon) + 3\kappa(\epsilon), \quad \text{or}(\epsilon) \in \{-1, 0, +1\}, \quad \kappa(\epsilon) \in \mathbb{Z}. \quad (10.9)$$

Thus, orientation and carry are not conditions imposed on a history: they are total functions of the APP data.

To make the composition visible, let $C_\epsilon : M_{s(\epsilon)} \rightarrow M_{t(\epsilon)}$ be the memory transport induced by the emission. The native carry of a composition satisfies

$$\kappa(\epsilon_2\epsilon_1) = \kappa(\epsilon_2) + C_{\epsilon_2}\kappa(\epsilon_1), \quad (10.10)$$

whenever $t(\epsilon_1) = s(\epsilon_2)$. For the identity and the neutral emission,

$$\kappa(I_a) = 0, \quad \kappa(\varepsilon_k^0(a)) = 0, \quad C_{I_a} = I_{M_a}. \quad (10.11)$$

Lemma 10.2 (Associativity of the carry ledger). *For three composable emissions, the evaluation of $\kappa(\epsilon_3\epsilon_2\epsilon_1)$ does not depend on the parenthesization.*

Proof. Applying (10.10) twice to the left parenthesization gives

$$\kappa(\epsilon_3) + C_{\epsilon_3}\kappa(\epsilon_2) + C_{\epsilon_3}C_{\epsilon_2}\kappa(\epsilon_1).$$

The right parenthesization produces the same expression because the transport of memory is functorial: $C_{\epsilon_3\epsilon_2} = C_{\epsilon_3}C_{\epsilon_2}$. $\square$

The oriented reversal of a reversible emission is denoted by $\bar{\epsilon}$. TRIT acts by

$$\text{or}(\bar{\epsilon}) = -\text{or}(\epsilon), \quad \partial\bar{\epsilon} = -\partial\epsilon, \quad C_{\bar{\epsilon}} = C_{\epsilon}^{-1}. \quad (10.12)$$

The inverse carry is determined, not chosen, by

$$\kappa(\bar{\epsilon}) = -C_{\epsilon}^{-1}\kappa(\epsilon), \quad (10.13)$$

for $\kappa(\bar{\epsilon}\epsilon) = 0$.

10.1.3 TPK transport of the 2 leaves and of memory

TPK chronology is part of the type of each emission and is not reconstructed a posteriori by counting letters. Here we recover the action of the ometer of the corresponding proposition. Let

$$C_9 := \mathbb{Z}/9\mathbb{Z}, \quad \tilde{O}_9 := \mathbb{Z}_{\geq 0} \times C_9, \quad O_{9,12} := \mathbb{Z}/12\mathbb{Z} \times C_9. \quad (10.14)$$

The unrolled phase-and-memory state is $Q_k = (w_k, r_k) \in \tilde{O}_9$; its reduction $(w, r) \mapsto ([w]_{12}, r)$ recovers $O_{9,12}$. The canonical seed of the clock is

$$Q_0(a_0) := (0, [0]_9) \quad (a_0 \in A_0); \quad (10.15)$$

the other choices of origin are their coherent translations by C_9 and do not change the update law. For $\bar{r} \in \{0, \dots, 8\}$, we set

$$\chi_9(r) := \left\lfloor \frac{\bar{r} + 1}{9} \right\rfloor. \quad (10.16)$$

Every emission ϵ that lleva depth k to depth $k + 1$ acts in the odometro by

$$\sigma_9(w, r) := (w + \chi_9(r), r + [1]_9), \quad \mathbf{Q}_{k+1}(\epsilon_* \hat{x}_k) = \sigma_9 \mathbf{Q}_k(\hat{x}_k). \quad (10.17)$$

Its phase component is, by typed definition,

$$\text{ph}(\epsilon) : r_k \mapsto r_k + [1]_9, \quad \text{ph}(\epsilon_n \cdots \epsilon_1) : r_k \mapsto r_k + [n]_9. \quad (10.18)$$

We write $g_k := r_k \in C_9$ for the published phase and $q_k^{(9)} := w_k$ for the vertical memory, and $\beta_k := [w_k]_{12}$ for its duodecaphase reduction. The iteration of (10.17) is

$$\sigma_9^n(w, r) = \left(w + \left\lfloor \frac{\bar{r} + n}{9} \right\rfloor, r + [n]_9 \right). \quad (10.19)$$

In efecto, the sumandos $\chi_9(r), \chi_9(r + [1]_9), \dots, \chi_9(r + [n - 1]_9)$ cuentan exactly the cruces of the representative 8 to the representative 0, that are $\lfloor (\bar{r} + n)/9 \rfloor$. In particular,

$$\sigma_9^9(w, r) = (w + 1, r), \quad g_{k+9} = g_k, \quad q_{k+9}^{(9)} = q_k^{(9)} + 1. \quad (10.20)$$

Even the neutral emission of APP realizes $g_k \mapsto g_{k+1}$: it is neutral in residue, quotient, carry, and boundary, but records that an extension has occurred. The phase word

$$\tau_k^{(N)} := (g_k, g_{k+1}, \dots, g_N), \quad g_{j+1} = g_j + [1]_9, \quad (10.21)$$

and the memory $q^{(9)}$ type the phase return and the state change as distinct objects.

10.1.3.1 Simultaneous Renewal of the Complete Fiber

The preceding odometer conserves the chronology of each extension. The same nonadic turn also has the internal Markov realization, already constructed over the APP-TPK catalogue, which simultaneously renews the two coordinates of the complete fiber. It is essential to conserve both realizations and not to identify a generic edge of the tree with an emission of the catalogue without a typing map.

Proposition 10.3 (Internal Markov Realization of the Nonadic Connection). *Let the APP seed domain, its observable emission, and its multiplicities be*

$$T_{\min} : \Omega_{\text{seed}} \twoheadrightarrow \mathcal{U}_6, \quad m(u) := \#T_{\min}^{-1}(u),$$

and the factorization produced by the TPK

$$\mathcal{U}_6 \simeq \mathcal{S}_+ \times \mathcal{S}_\times, \quad |\mathcal{S}_+| = 18, \quad |\mathcal{S}_\times| = 26.$$

On functions in $\mathcal{U}_6$, the two conditional expectations act by

$$(E_+ f)(s, e) = \frac{1}{18} \sum_{s' \in \mathcal{S}_+} f(s', e), \quad (E_\times f)(s, e) = \frac{1}{26} \sum_{e' \in \mathcal{S}_\times} f(s, e').$$

For

$$\tau = (+1, -1, 0, +1, -1, 0, +1, -1, 0), \quad P_{+1} = E_+, \quad P_{-1} = E_\times, \quad P_0 = I,$$

the catalog transfer and its elevation to all seeds are

$$\mathcal{K}_{\text{ph}}((u, r), (v, r + 1)) = P_{\tau(r)}(u, v), \quad (10.22)$$

$$L_0(\omega, \omega') = \mathbf{1}_{\{\omega=\omega'\}},$$

$$L_+(\omega, \omega') = \frac{E_+(T_{\min}\omega, T_{\min}\omega')}{m(T_{\min}\omega')},$$

$$L_\times(\omega, \omega') = \frac{E_\times(T_{\min}\omega, T_{\min}\omega')}{m(T_{\min}\omega')},$$

$$\widehat{\mathcal{K}}_{\text{ph}}((\omega, r), (\omega', r + 1)) = L_{\tau(r)}(\omega, \omega'). \quad (10.23)$$

For $u = T_{\min}\omega$, it holds phase by phase that

$$\sum_{\omega': T_{\min}\omega' = v} \widehat{\mathcal{K}}_{\text{ph}}((\omega, r), (\omega', r + 1)) = \mathcal{K}_{\text{ph}}((u, r), (v, r + 1)). \quad (10.24)$$

In particular,

$$\boxed{\mathcal{K}_{\text{ph}}^9|_r = E_+E_\times, \quad (E_+E_\times)(u, v) = \frac{1}{468}.} \quad (10.25)$$

Proof. In a phase + or $\times$, summing (10.23) over $T_{\min}^{-1}(v)$ exactly cancels the denominator $m(v)$; in a zero phase, L_0 projects onto the identity. This proves (10.24). The operators E_+ , $E_\times$, and I are idempotent and commute, and each appears three times in the nonadic word. Therefore

$$\prod_{r \in C_9} P_{\tau(r)} = E_+^3 E_\times^3 I^3 = E_+E_\times.$$

The composition first renews one coordinate of cardinal 18 and then the other of cardinal 26, so that each target receives $1/(18 \cdot 26) = 1/468$. $\square$

Remark 10.4 (Two internal realizations, without fictitious identification). The transfer $\mathcal{K}_{\text{ph}}$ is the global realization of the complete catalog, and σ_9 is the chronological action on each state of the tree. One is not substituted for the other. An edge-by-edge equality would require an additional map

$$\text{Cat}_{k,x} : \text{Out}(x) \longrightarrow \mathcal{U}_6$$

that verified the corresponding sum identity on fibers. That map is not used in this volume. The complete monodromy (10.25) proceeds from the APP-TPK catalogue, while the genealogy of each history remains in the tree and its ledger.

TPK assigns to each admissible emission a typed transport of the leaf pair,

$$U_\epsilon : H_{s(\epsilon)}^\Sigma \times H_{s(\epsilon)}^\Pi \longrightarrow H_{t(\epsilon)}^\Sigma \times H_{t(\epsilon)}^\Pi, \quad (10.26)$$

with

$$U_{I_a} = I, \quad U_{\epsilon_2 \epsilon_1} = U_{\epsilon_2} U_{\epsilon_1}. \quad (10.27)$$

The codomain of U_ϵ remains the complete pair. No operation in this chapter replaces (Σ, Π) by one of its coordinates.

Let $m \in M_{s(\epsilon)}$ be an accumulated memory. The TPK action over it is the affine transformation

$$U_\epsilon(m) := C_\epsilon m + \kappa(\epsilon). \quad (10.28)$$

Lemma 10.5 (TPK Action on Memory). *The transformations U_ϵ satisfy*

$$U_{I_a} = I, \quad U_{\epsilon_2 \epsilon_1} = U_{\epsilon_2} U_{\epsilon_1}. \quad (10.29)$$

Proof. The identity follows from (10.11). For the composition,

$$\begin{aligned} \mathbf{U}_{\epsilon_2} \mathbf{U}_{\epsilon_1}(m) &= \mathbf{C}_{\epsilon_2}(\mathbf{C}_{\epsilon_1}m + \kappa(\epsilon_1)) + \kappa(\epsilon_2) \\ &= \mathbf{C}_{\epsilon_2 \epsilon_1}m + \kappa(\epsilon_2) + \mathbf{C}_{\epsilon_2}\kappa(\epsilon_1) \\ &= \mathbf{C}_{\epsilon_2 \epsilon_1}m + \kappa(\epsilon_2 \epsilon_1), \end{aligned}$$

where the last equality is (10.10). $\square$

A finite route is a composable word.

$$\gamma = \epsilon_1 \epsilon_2 \cdots \epsilon_k. \quad (10.30)$$

Its boundary, transport, carry, and memory are

$$\partial\gamma := \sum_{j=1}^k \partial\epsilon_j, \quad (10.31)$$

$$U_\gamma := U_{\epsilon_k} \cdots U_{\epsilon_1}, \quad (10.32)$$

$$\kappa(\gamma) := \kappa(\epsilon_k) + \sum_{j=1}^{k-1} \mathbf{C}_{\epsilon_k} \cdots \mathbf{C}_{\epsilon_{j+1}} \kappa(\epsilon_j), \quad (10.33)$$

$$m(\gamma; m_0) := \mathbf{U}_{\epsilon_k} \cdots \mathbf{U}_{\epsilon_1}(m_0). \quad (10.34)$$

In (10.31), the interior endpoints cancel, leaving

$$\partial\gamma = t(\epsilon_k) - s(\epsilon_1). \quad (10.35)$$

10.1.4 The common state and its total domain

Definition 10.6 (Elementary ledger). For an emission that appears between two consecutive enriched prefixes, we denote by $\mathbf{Q}^-(\epsilon) = (w^-(\epsilon), r^-(\epsilon))$ the odometer of the source prefix and by

$$\mathbf{Q}^+(\epsilon) := \sigma_9 \mathbf{Q}^-(\epsilon) = (w^-(\epsilon) + \chi_9(r^-(\epsilon)), r^-(\epsilon) + [1]_9) \quad (10.36)$$

that of the destination prefix. Thus the odometer is not applied to a bare APP cell. The emission record is

$$\lambda(\epsilon) := (s(\epsilon), t(\epsilon), \text{res}^\Sigma(\epsilon), \text{quo}^\Sigma(\epsilon), \text{res}^\Pi(\epsilon), \text{quo}^\Pi(\epsilon), \text{or}(\epsilon), \kappa(\epsilon), \partial\epsilon, \mathbf{Q}^-(\epsilon), \mathbf{Q}^+(\epsilon)). \quad (10.37)$$

The ledger of γ is the word ordered

$$\text{Led}(\gamma) := (\lambda(\epsilon_1), \dots, \lambda(\epsilon_k)). \quad (10.38)$$

Preserving this word, and not only its sum, distinguishes two histories with the same terminal output.

10.1.4.1 Complete reader of the TPK state

The depth K of TPK blocks and the depth k of the extension tree are distinct types. In this subsection, K denotes exclusively the depth of the TPK state already generated by the operator mechanism; it is not identified with k or with a generic edge.

Definition 10.7 (Compact nonadic memory reader). The integral holonomic TPK state of depth K is

$$v_K = (B_K, \delta_K, x_K, C_K, \Sigma_K, \omega_K, \ell_K, g(K), \Lambda_K), \quad B_K = (u_K^\pi, u_K^e, u_K^\varphi). \quad (10.39)$$

Its compact reader does not add coordinates: it publishes them from that same state,

$$\mathbf{r}_9(v_K) := (B_K, C_K, p_K, c_K, \Sigma_K, W_K, g(K)), \quad (10.40)$$

where

$$\begin{aligned} C_K &:= \mathbf{C}_K = (C_K^\chi)_{\chi \in \{\pi, e, \varphi\}}, \\ C_K^\chi &:= (N_K^\chi, L_K^\chi, r_K^\chi, D_K^\chi, A_K^\chi, B_K^\chi), \\ p_K &:= (B_0, \dots, B_K), \\ c_K &:= (\delta_K, \mathbf{x}_K, \kappa(\gamma_{0:K}^{\text{TPK}})), \end{aligned} \quad (10.41)$$

and the oriented shadow of Witt is preserved as

$$W_K := ((u_K^\chi, u_K^\chi A_W))_{\chi \in \{\pi, e, \varphi\}}, \quad A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}, \quad A_W^2 = -I_6. \quad (10.42)$$

The route $\gamma_{0:K}^{\text{TPK}}$ is the word of extensions encoded by Λ_K ; its prefix of length j is the one that produces v_j . The signature is the output of the explicit internal primitive

$$\Sigma_B : \mathbf{B}_{\text{TPK}} \times \mathbf{C}_{\text{TPK}} \times \mathbb{Z}[\mathbf{A}] \times \Omega_{12} \times \mathbf{L}_{\text{TPK}} \longrightarrow \mathbf{S}_{\text{TPK}}, \quad (B, c, \partial\gamma, \omega, \ell) \longmapsto \Sigma_B(B, c, \partial\gamma, \omega, \ell), \quad (10.43)$$

with $\Sigma_B(g \cdot -) = g \cdot \Sigma_B(-)$ for each TPK symmetry. Its explicit starting structure is $(B_K, c_K, \partial\gamma_{0:K}^{\text{TPK}}, \omega_K, \ell_K)$. It is not replaced by an external signature nor inferred retrospectively from a numerical reader.

The action of step $K \mapsto K+1$ is written without omitting the deep state. For each $\chi \in \{\pi, e, \varphi\}$,

$$y_K^\chi = x_K^\chi A_H, \quad u_{K+1}^\chi = y_K^\chi \bmod 3, \quad x_{K+1}^\chi = \frac{y_K^\chi - u_{K+1}^\chi}{3}, \quad (10.44)$$

with

$$A_H = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \quad \det A_H = 1.$$

If

$$\text{ind}_3(u_1, \dots, u_6) := \sum_{j=1}^6 \bar{u}_j 3^{6-j} \in \{0, \dots, 728\},$$

the update of the prefix and its 729 subcylinders is

$$N_{K+1}^\chi = 729N_K^\chi + \text{ind}_3(u_{K+1}^\chi), \quad L_{K+1}^\chi = L_K^\chi + 6, \quad (10.45)$$

$$I_{K,u}^\chi := \left[\frac{729N_K^\chi + u}{3^{L_K^\chi+6}}, \frac{729N_K^\chi + u + 1}{3^{L_K^\chi+6}} \right), \quad 0 \leq u < 729. \quad (10.46)$$

The decimal cylinder and the associated carry are updated by the two integer cases of the same operator $\mathbf{C}_{3,1000}$. If a new decimal triad is not published,

$$r_{K+1}^\chi = r_K^\chi, \quad D_{K+1}^\chi = D_K^\chi, \quad A_{K+1}^\chi = 729A_K^\chi + \text{ind}_3(u_{K+1}^\chi)1000r_K^\chi, \quad B_{K+1}^\chi = A_{K+1}^\chi + 1000r_K^\chi. \quad (10.47)$$

The primitive $C_{3,1000}$ produces the option datum

$$d_{K+1,\chi}^{\text{dec}} \in \{\perp\} \sqcup \{0, \dots, 999\};$$

in the first case it equals $d_{K+1,\chi}^{\text{dec}} = \perp$. If a triad occurs, we write $d_{K+1,\chi}^{\text{dec}} = \delta_{K+1}^\chi \in \{0, \dots, 999\}$ and

$$\begin{aligned} r_{K+1}^\chi &= r_K^\chi + 1, & D_{K+1}^\chi &= 1000D_K^\chi + \delta_{K+1}^\chi, \\ A_{K+1}^\chi &= 729000A_K^\chi + \text{ind}_3(u_{K+1}^\chi)1000r_K^{\chi+1} - 729\delta_{K+1}^\chi 3^{L_K^\chi}, \\ B_{K+1}^\chi &= A_{K+1}^\chi + 1000r_K^{\chi+1}. \end{aligned} \quad (10.48)$$

The register δ_{K+1} adds those triads and the two joint cocycles

$$\begin{aligned} q_{K+1} &= (u_{K+1}^\pi + u_{K+1}^e + u_{K+1}^\varphi) \bmod 3, & c_{K+1}^{\text{joint}} &= \frac{u_{K+1}^\pi + u_{K+1}^e + u_{K+1}^\varphi - q_{K+1}}{3}, \\ a_{K+1} &= (u_{K+1}^\pi + u_{K+1}^e - u_{K+1}^\varphi) \bmod 3, & h_{K+1}^{\text{joint}} &= \frac{u_{K+1}^\pi + u_{K+1}^e - u_{K+1}^\varphi - a_{K+1}}{3}, \\ \delta_{K+1} &= \delta_K \star ((d_{K+1,\chi}^{\text{dec}})_\chi, c_{K+1}^{\text{joint}}, h_{K+1}^{\text{joint}}). \end{aligned} \quad (10.49)$$

Thus C_{K+1} , δ_{K+1} , and x_{K+1} have material actions, not merely names. The other coordinates are simultaneously updated by

$$p_{K+1} = p_K \star B_{K+1}, \quad (10.50)$$

$$\kappa(\gamma_{0:K+1}^{\text{TPK}}) = \kappa(\varepsilon_{K+1}^{\text{TPK}}) + \mathbf{C}_{\varepsilon_{K+1}^{\text{TPK}}} \kappa(\gamma_{0:K}^{\text{TPK}}), \quad (10.51)$$

$$c_{K+1}^{\text{comp}} := (\delta_{K+1}, x_{K+1}, \kappa(\gamma_{0:K+1}^{\text{TPK}})), \quad (10.52)$$

$$\Sigma_{K+1} = \Sigma_B(B_{K+1}, c_{K+1}^{\text{comp}}, \partial\gamma_{0:K}^{\text{TPK}} + \partial\varepsilon_{K+1}^{\text{TPK}}, \omega_{K+1}, \ell_{K+1}), \quad (10.53)$$

$$W_{K+1} = ((u_{K+1}^\chi, u_{K+1}^\chi A_W))_\chi, \quad g(K+1) = g(K) + [1]_9. \quad (10.54)$$

Since A_H is invertible modulo three and $A_W^2 = -I_6$, the residue and shadow updates do not collapse distinct blocks. Truncation preserves the complete sequences

$$(B_0, \dots, B_K), (C_0, \dots, C_K), (p_0, \dots, p_K), (c_0, \dots, c_K), (\Sigma_0, \dots, \Sigma_K), (W_0, \dots, W_K)$$

and simultaneously removes its last entry. Hence the reader r_9 commutes with the erasure of a block.

Proposition 10.8 (Universal scope and certified scope of non-rebooting). *On every horizontal circuit of nine steps, the following are preserved*

$$g(K+9) = g(K), \quad q_{\text{mem}}(K+9) = q_{\text{mem}}(K) + 1,$$

and the prefix is prolonged with nine emissions, so the state is not reset. In the certified joint section $\chi \in \{\pi, e, \varphi\}$, $0 \leq K < 387$, it also holds that

$$(p_K, C_K) \neq (p_{K+9}, C_{K+9}), \quad (B_K, \mathbf{N}_K, W_K) \neq (B_{K+9}, \mathbf{N}_{K+9}, W_{K+9}), \quad \mathbf{N}_K := (N_K^\pi, N_K^e, N_K^\varphi). \quad (10.55)$$

Proof. The two universal identities are the exact iteration of (10.17); the prefix preserves the nine entries adjoined by (10.50). For the joint section, the recurrence (10.45) compares the 387 nonadic pairs of each of the three channels and produces (10.55); this is the assertion certified in joint filtration theorem at finite depth. Finally, $u \mapsto (u, uA_W)$ is injective, so the change of block also preserves the change of its

complete shadow. This last inequality is not extended to an arbitrary neutral prolongation outside the certified section. $\square$

Definition 10.9 (APP–TRIT–TPK state of depth k). For a path $\gamma = \epsilon_1 \cdots \epsilon_k$ originating in an APP cell a_0 , the common integral holonomic state is

$$\widehat{x}_k(\gamma) := (a_0, \dots, a_k; \text{Sh}_0, \dots, \text{Sh}_k; \mathbf{res}_k, \mathbf{quo}_k; \mathbf{or}_k, \kappa_k; \mathbf{Q}_k; g_k; q_k^{(9)}; \beta_k; m_k; \partial\gamma; \gamma; \text{Led}(\gamma)), \quad (10.56)$$

where

$$\begin{aligned} \mathbf{res}_k &:= ((\text{res}^\Sigma(\epsilon_j), \text{res}^\Pi(\epsilon_j)))_{1 \leq j \leq k}, \\ \mathbf{quo}_k &:= ((\text{quo}^\Sigma(\epsilon_j), \text{quo}^\Pi(\epsilon_j)))_{1 \leq j \leq k}, \\ \mathbf{or}_k &:= (\text{or}(\epsilon_j))_{1 \leq j \leq k}, \quad \kappa_k := (\kappa(\epsilon_j))_{1 \leq j \leq k}, \\ m_k &:= m(\gamma; m_0). \end{aligned}$$

The intermediate sheets are obtained by $\text{Sh}_j = U_{\epsilon_j} \text{Sh}_{j-1}$.

The simultaneous presence of the route and the ledger prevents identifying distinct words that terminate in the same cell. The state is not a list of cardinals: each coordinate has the type and composition equations of the preceding sections.

Definition 10.10 (Total depth finite domain). Let

$$\widehat{\mathcal{X}}_k^{\text{tot}} := \left\{ \widehat{x}_k(\epsilon_1 \cdots \epsilon_k) : \begin{array}{l} a_0 \in \mathbf{A}_0, \\ \epsilon_j \in \mathbf{E}_{j-1}(a_{j-1}), \\ s(\epsilon_j) = a_{j-1}, \quad t(\epsilon_j) = a_j \end{array} \right\}. \quad (10.57)$$

No condition from a later publication is added to (10.57). Being an element of the domain means exactly having been produced by the total actions (10.5), (10.9), and (10.17)–(10.28).

The common truncation deletes the last emission and all and only the entries of the ledger that it produced:

$$\rho_{k+1,k} : \widehat{\mathcal{X}}_{k+1}^{\text{tot}} \longrightarrow \widehat{\mathcal{X}}_k^{\text{tot}}, \quad \rho_{k+1,k} \widehat{x}_{k+1} = \widehat{x}_k. \quad (10.58)$$

Let $\varepsilon_k^0(x)$ be the neutral emission (10.3) applied to the APP endpoint of x . The neutral prolongation defines

$$\eta_k : \widehat{\mathcal{X}}_k^{\text{tot}} \longrightarrow \widehat{\mathcal{X}}_{k+1}^{\text{tot}}, \quad \eta_k(x) := (\varepsilon_k^0(x))_* x, \quad \rho_{k+1,k} \eta_k = I. \quad (10.59)$$

Proposition 10.11 (Finiteness, non-emptiness, and surjectivity). *For every k , the set $\widehat{\mathcal{X}}_k^{\text{tot}}$ is finite and nonempty, and $\rho_{k+1,k}$ is surjective.*

Proof. APP produces a nonempty finite set at depth zero. At each step, every state has a finite set of emissions through (10.8); by induction, there are only finitely many words of length k . The sequence of neutral prolongations produces at least one word of each length. Finally, (10.59) is a right inverse of truncation, so the latter is surjective. $\square$

10.1.5 Finite reversible symmetry groupoid

The reversible operations of APP, TRIT, and TPK act on the entire state. Let $\mathbf{U}$ be the group of coherent families $g = (g_k)_{k \geq 0}$ generated by those operations, where

$$\rho_{k+1,k} g_{k+1} = g_k \rho_{k+1,k}, \quad g_{k+1} \eta_k = \eta_k g_k. \quad (10.60)$$

We denote by $\mathbf{U}_k$ its image in $\text{Sym}(\widehat{\mathcal{X}}_k^{\text{tot}})$. This image is finite because it acts on a finite set. The action of g_k on

$$\widehat{x} = (a_{\leq k}; \text{Sh}_{\leq k}; \mathbf{res}, \mathbf{quo}; \mathbf{or}, \kappa; \mathbf{Q}_k; g_k; q_k^{(9)}; \beta_k; m; \partial\gamma; \gamma; \text{Led}(\gamma))$$

is

$$g_k \cdot \widehat{x} := (g_k a_{\leq k}; U_{g_k} \text{Sh}_{\leq k}; \mathbf{res}(g_k \gamma), \mathbf{quo}(g_k \gamma); \mathbf{or}(g_k \gamma), \kappa(g_k \gamma); \mathbf{Q}(g_k \gamma); g(g_k \gamma); q^{(9)}(g_k \gamma); \beta(g_k \gamma); \mathbf{U}_{g_k}(m); (g_k)_* \partial\gamma; g_k \gamma; \text{Led}(g_k \gamma)). \quad (10.61)$$

All the terms on the right-hand side are recalculated by means of the APP, TRIT, and TPK rules; they are not copied as independent labels.

Definition 10.12 (Common action groupoid). The finite groupoid of depth k is

$$\mathfrak{G}_k := \mathbf{U}_k \ltimes \widehat{\mathcal{X}}_k^{\text{tot}}. \quad (10.62)$$

Its objects are the common states. An arrow $(g_k, \widehat{x})$ has source $\widehat{x}$, target $g_k \cdot \widehat{x}$, identity $(I, \widehat{x})$, and inverse $(g_k^{-1}, g_k \cdot \widehat{x})$.

Lemma 10.13 (The action preserves all equations of state). *The action (10.61) preserves the pair of sheets, the APP divisions, the TRIT orientation, the carry, the phase odometer, the vertical memory, its dodecaphase reduction, the affine memory, the boundary, the route, and the ledger, in their respective types. Moreover,*

$$I \cdot \widehat{x} = \widehat{x}, \quad (g_{2,k} g_{1,k}) \cdot \widehat{x} = g_{2,k} \cdot (g_{1,k} \cdot \widehat{x}). \quad (10.63)$$

Proof. The APP divisions are unique, so the action on an emission redetermines its residues and quotients. The balanced decomposition (10.9) is likewise unique. Composition of the carry is associative by lemma 10.2; the action on memory is associative by lemma 10.5; and the boundary is functorial by (10.35). Finally, the action on the ordered word induces the action on its ledger entry by entry. Coherence with the odometer recalculates Q , g , and $q^{(9)}$ from the transformed route and preserves (10.17). These identities prove (10.63). $\square$

10.1.6 Finite tree of all prolongations

For $x \in \widehat{\mathcal{X}}_k^{\text{tot}}$, let $\text{Out}(x)$ be the set of all admissible TPK emissions of length one from the endpoint of x . An emission belongs to $\text{Out}(x)$ only when its action jointly updates all coordinates of (10.56). In particular,

$$1 \leq d(x) := \#\text{Out}(x) < \infty, \quad \varepsilon_k^0(x) \in \text{Out}(x). \quad (10.64)$$

Definition 10.14 (Truncated prolongation tree). Given $N \geq k$ and $x \in \widehat{\mathcal{X}}_k^{\text{tot}}$, we define $\mathcal{T}_{k,N}(x)$ as the rooted tree whose vertices at height j , $0 \leq j \leq N - k$, are the words

$$h = (\epsilon_{k+1}, \dots, \epsilon_{k+j}), \quad \epsilon_{n+1} \in \text{Out}(x_n), \quad x_{n+1} = (\epsilon_{n+1})_* x_n, \quad (10.65)$$

with the empty word as root. An edge appends an emission. Two words reaching the same terminal state remain distinct vertices if their ledgers differ.

The last condition is essential: the tree records genealogies and not the quotient that remembers only the target. Its terminal level is

$$\text{Term}_{k,N}(x) := \{h \in \mathcal{T}_{k,N}(x) : |h| = N - k\}. \quad (10.66)$$

Each h preserves the complete terminal state.

$$\text{ter}_N(h) := (x_N, m_N, \partial(\gamma h), \gamma h, \text{Led}(\gamma h)). \quad (10.67)$$

Proposition 10.15 (Finiteness and non-emptiness at every horizon). *For all x , k , and $N \geq k$, the tree $\mathcal{T}_{k,N}(x)$ is finite and $\text{Term}_{k,N}(x) \neq \emptyset$.*

Proof. The root has a finite positive number of successors by (10.64). Applying the same argument at each level yields, by induction, finiteness at every finite horizon. The word formed by neutral emissions $(\varepsilon_k^0, \varepsilon_{k+1}^0, \dots, \varepsilon_{N-1}^0)$ is a terminal vertex, so level (10.66) is nonempty. $\square$

The erasure of the last emission defines

$$\pi_{N+1,N} : \text{Term}_{k,N+1}(x) \longrightarrow \text{Term}_{k,N}(x). \quad (10.68)$$

Lemma 10.16 (Terminal surjectivity). *The map $\pi_{N+1,N}$ is surjective for every $N \geq k$.*

Proof. Let $h \in \text{Term}_{k,N}(x)$, with terminal state v_h . Adding the emission $\varepsilon_N^0(v_h)$ produces $h' \in \text{Term}_{k,N+1}(x)$ and $\pi_{N+1,N}(h') = h$. $\square$

10.1.7 Nonadic Connection: Phase, Holonomy, and Memory

A word of nine emissions is not called a holonomy merely because it has length nine. The horizontal connection determined by the phase functor is first preserved. For $r \in C_9$, we define

$$\begin{aligned} \mathcal{R}_{r,k} &:= \{(x, \epsilon, \epsilon_*x) : x \in \widehat{\mathcal{X}}_k^{\text{tot}}, \quad \epsilon \in \text{Out}(x), \\ &\quad \mathbf{Q}_k(x) = (w, r), \quad \mathbf{Q}_{k+1}(\epsilon_*x) = \sigma_9(w, r)\}. \end{aligned} \quad (10.69)$$

The last equality does not restrict the domain: it is the total action (10.17) of every elementary TPK emission. The connection simultaneously transports sheet, memory and ledger:

$$\nabla_{\text{TPK}}(\epsilon) := (\text{ph}(\epsilon), U_\epsilon, \mathbf{U}_\epsilon, \lambda(\epsilon)). \quad (10.70)$$

Its composition is calculated by means of (10.27), (10.29), and the concatenation of the ledger.

The apex of the nonadic turn is now the set of phase-compatible horizontal lifts.

$$\begin{aligned} \mathbf{Z}_{9,k}^\Gamma &:= \bigsqcup_{r \in C_9} \{(x, \epsilon_1, \dots, \epsilon_9) : x_0 = x, \quad x_i = (\epsilon_i)_* x_{i-1}, \\ &\quad (x_{i-1}, \epsilon_i, x_i) \in \mathcal{R}_{r+i-1, k+i-1}, \quad 1 \leq i \leq 9\}, \end{aligned} \quad (10.71)$$

where the subscripts of $r+i-1$ are read in C_9 . The summand is uniquely determined by $r = g_k(x)$, so the coproduct does not duplicate witnesses. The source and target maps are

$$\widehat{\mathcal{X}}_k^{\text{tot}} \xleftarrow{s_{9,k}} \mathbf{Z}_{9,k}^\Gamma \xrightarrow{t_{9,k}} \widehat{\mathcal{X}}_{k+9}^{\text{tot}}, \quad s_{9,k}(x, \epsilon) = x, \quad t_{9,k}(x, \epsilon) = (\epsilon_9 \cdots \epsilon_1)_* x. \quad (10.72)$$

The relational holonomy is

$$\Gamma_{9,k} := \{(s_{9,k}z, t_{9,k}z) : z \in \mathbf{Z}_{9,k}^\Gamma\} = \text{Hol}_{C_9}(\nabla_{\text{TPK}}). \quad (10.73)$$

Lemma 10.17 (The ninth level of the tree realizes the holonomy). *The map that forgets the intermediate states already determined by the emissions induces a bijection*

$$Z_{9,k}^\Gamma \xrightarrow{\cong} \coprod_{x \in \widehat{\mathcal{X}}_k^{\text{tot}}} \text{Term}_{k,k+9}(x). \quad (10.74)$$

Proof. Each edge of $\text{Out}(x_i)$ satisfies (10.17); hence a terminal word of length nine determines the nine elements of (10.69). Reciprocally, a horizontal lift of (10.71) is a composable word of the tree. The two operations conserve the emissions and are inverse. Thus (10.74) is a theorem derived from the connection, not its definition. $\square$

Proposition 10.18 (Phase return, memory advance and no restart). *For every $z = (x, \epsilon_1, \dots, \epsilon_9)$ of the apex,*

$$g(t_{9,k}z) = g(s_{9,k}z), \quad (10.75)$$

$$q^{(9)}(t_{9,k}z) = q^{(9)}(s_{9,k}z) + 1, \quad (10.76)$$

$$\text{Led}(t_{9,k}z) = \text{Led}(s_{9,k}z) \star (\lambda(\epsilon_1), \dots, \lambda(\epsilon_9)). \quad (10.77)$$

In particular, the phase identification $g(t_{9,k}z) = g(s_{9,k}z)$ does not induce an identification of states: after transporting both registers to the same comparison type, their vertical memories differ by one unit. The published phase returns, not the integral holonomic state.

Proof. The first two identities are (10.20). The third is the definition of ledger concatenation. Since the vertical memory differs by one unit, the complete states are distinct even if some additional visible coordinate could coincide. $\square$

Proposition 10.19 (Isometric compression of the same holonomy). *Let $U_{\Gamma_9} : \mathcal{H}_{\text{full}} \rightarrow \mathcal{H}_{\text{full}}$ be an isometric realization of the preceding holonomy and let $J : \mathcal{H}_{\text{vis}} \rightarrow \mathcal{H}_{\text{full}}$ be the isometric inclusion of the visible register, with $E := J^*$ and $P := JE$. We define*

$$T := EU_{\Gamma_9}J, \quad \eta := (I - P)U_{\Gamma_9}J. \quad (10.78)$$

Then

$$\boxed{U_{\Gamma_9}J = JT + \eta, \quad I - T^*T = \eta^*\eta.} \quad (10.79)$$

Proof. Inserting $I = P + (I - P)$ before $U_{\Gamma_9}J$ gives

$$U_{\Gamma_9}J = JE U_{\Gamma_9}J + (I - P)U_{\Gamma_9}J = JT + \eta.$$

The two summands belong to orthogonal subspaces. Since U_{Γ_9} and J are isometries,

$$I = (U_{\Gamma_9}J)^*(U_{\Gamma_9}J) = T^*T + \eta^*\eta,$$

which is the second identity. $\square$

Proposition 10.20 (Totality and Equivariance of Holonomy). *The map $s_{9,k}$ is surjective. The nine neutral prolongations form the section*

$$\begin{aligned} x_0 &:= x, & \varepsilon_{k+i}^0 &:= \varepsilon_{k+i}^0(x_i), & x_{i+1} &:= (\varepsilon_{k+i}^0)_*x_i \quad (0 \leq i < 9), \\ \eta_k^{(9)}(x) &:= (x, \varepsilon_k^0, \dots, \varepsilon_{k+8}^0) \in Z_{9,k}^\Gamma, & s_{9,k}\eta_k^{(9)} &= I. \end{aligned} \quad (10.80)$$

Every coherent symmetry a whose action on the odometer satisfies $\mathbf{Q}a = \bar{a}\mathbf{Q}$ and $\bar{a}\sigma_9 = \sigma_9\bar{a}$ induces

$$a^Z(x, \epsilon_1, \dots, \epsilon_9) := (ax, a\epsilon_1, \dots, a\epsilon_9)$$

and satisfies

$$s_{9,k}a^Z = a_k s_{9,k}, \quad t_{9,k}a^Z = a_{k+9}t_{9,k}. \quad (10.81)$$

Proof. The neutral word exists in every source and satisfies the phase action (10.17); this proves the section and surjectivity. The condition $\bar{a}\sigma_9 = \sigma_9\bar{a}$ sends horizontal relations to horizontal relations. Acting entry by entry preserves composition, carry, memory, boundary, and ledger; reading the first or the final state before or after acting gives (10.81). $\square$

10.1.8 Total survival and complete multisection

Definition 10.21 (Produced survival). A state $x \in \widehat{\mathcal{X}}_k^{\text{tot}}$ survives when

$$\text{Term}_{k,N}(x) \neq \emptyset \quad \text{for every } N \geq k. \quad (10.82)$$

We denote by Surv_k the set of such states.

Theorem 10.22 (The total domain is survivor). *For every depth,*

$$\text{Surv}_k = \widehat{\mathcal{X}}_k^{\text{tot}}. \quad (10.83)$$

In particular, survival does not define a subset supposed before constructing the state.

Proof. The inclusion $\text{Surv}_k \subseteq \widehat{\mathcal{X}}_k^{\text{tot}}$ belongs to the definition. The opposite inclusion is exactly the nonemptiness at every horizon proved in [proposition 10.15](#). $\square$

Definition 10.23 (Multisection of complete prolongations). The complete multisection over x is the inverse limit

$$\text{Msect}(x) := \varprojlim_{N \geq k} (\text{Term}_{k,N}(x), \pi_{N+1,N}). \quad (10.84)$$

Its elements are all the compatible infinite histories that prolong x . None of them receives priority within (10.84).

Theorem 10.24 (No material emptiness of the multisection). *For every $x \in \widehat{\mathcal{X}}_k^{\text{tot}}$,*

$$\text{Msect}(x) \neq \emptyset. \quad (10.85)$$

Moreover, every finite coordinate of a history in $\text{Msect}(x)$ preserves both leaves, orientation, carry, memory, boundary, path, and ledger of the state that it prolongs.

Proof. The terminal sets are finite and nonempty by [proposition 10.15](#), and their bonding maps are surjective by [lemma 10.16](#). A compatible element can be constructed recursively: one begins at the root and, at every step, retains a prolongation of the preceding prefix. The neutral prolongation guarantees that the process never stops. Compatibility of the coordinates follows from the fact that each edge of the tree acts through the unique state action defined in (10.56). $\square$

10.1.9 Canonical measure generated by the tree

The measure is not introduced as an external weight on the outputs. It is obtained by counting the admissible emissions at each vertex before choosing a continuation.

Definition 10.25 (Local weight and horizon measure). For $v \in \widehat{\mathcal{X}}_j^{\text{tot}}$ and $\epsilon \in \text{Out}(v)$, we set

$$p_v(\epsilon) := \frac{1}{d(v)} \in \mathbb{Q}_{>0}. \quad (10.86)$$

If $h = (\epsilon_{k+1}, \dots, \epsilon_N) \in \text{Term}_{k,N}(x)$ and x_j is the prefix that precedes ϵ_{j+1} , we define

$$\mu_{x,N}(h) := \prod_{j=k}^{N-1} p_{x_j}(\epsilon_{j+1}) = \prod_{j=k}^{N-1} \frac{1}{d(x_j)}. \quad (10.87)$$

Lemma 10.26 (Normalization). For every horizon $N \geq k$,

$$\sum_{h \in \text{Term}_{k,N}(x)} \mu_{x,N}(h) = 1. \quad (10.88)$$

Proof. At height one, $\sum_{\epsilon \in \text{Out}(x)} d(x)^{-1} = 1$. Assuming the normalization at height $N - k$, each terminal prefix h , with final state v_h , contributes at the next level

$$\sum_{\epsilon \in \text{Out}(v_h)} \mu_{x,N}(h) p_{v_h}(\epsilon) = \mu_{x,N}(h).$$

Summing over all prefixes proves the identity by induction. $\square$

Proposition 10.27 (Projective consistency). Finite measures satisfy

$$(\pi_{N+1,N})_* \mu_{x,N+1} = \mu_{x,N}. \quad (10.89)$$

Proof. For $h \in \text{Term}_{k,N}(x)$, the mass of its fiber is

$$\begin{aligned} \sum_{\pi_{N+1,N}(h')=h} \mu_{x,N+1}(h') &= \mu_{x,N}(h) \sum_{\epsilon \in \text{Out}(v_h)} p_{v_h}(\epsilon) \\ &= \mu_{x,N}(h). \end{aligned}$$

This is precisely the equality of point measures equivalent to (10.89). $\square$

The cylinders of the multisection are

$$[h]_{x,N} := \{\omega \in \text{Msect}(x) : \omega|_N = h\}. \quad (10.90)$$

Theorem 10.28 (Canonical Measure of Extensions). There exists a unique measure of cilindros μ_x on $\text{Msect}(x)$ such that

$$\mu_x([h]_{x,N}) = \mu_{x,N}(h). \quad (10.91)$$

It is normalized, has support on the whole multisection, and is obtained only from the finite sets of APP–TRIT–TPK prolongations.

Proof. The consistency of proposition 10.27 makes (10.91) independent of the horizon at which a cylinder is represented. The normalization follows from lemma 10.26. Two cylinders are either disjoint or one contains the other; hence the finite masses uniquely define an additive measure on the cylinder algebra. The multisection is a limit of finite discrete sets, and the cylinders form its compact basis; the measure

extends uniquely to the σ -algebra they generate. Finally, $p_v(\epsilon) > 0$ for every admissible emission, so every nonempty cylinder has positive mass and the support is the entire multisection. $\square$

10.1.10 Naturality under symmetries and truncations

A coherent family $g = (g_j)_{j \geq 0} \in \mathbf{U}$ transports an outgoing emission of depth j by means of

$$(g_{j+1}, g_j)_* : \text{Out}(x) \longrightarrow \text{Out}(g_j \cdot x), \quad \epsilon \longmapsto g_{j+1} \epsilon g_j^{-1}. \quad (10.92)$$

The TPK relations make (10.92) a bijection that preserves multiplicities. Acting on each entry yields a tree isomorphism

$$g_{*,N} : \mathcal{T}_{k,N}(x) \xrightarrow{\cong} \mathcal{T}_{k,N}(g_k \cdot x). \quad (10.93)$$

Theorem 10.29 (Naturalness of the tree, the multisection, and the measure). *For every coherent reversible symmetry $g = (g_j)_{j \geq 0}$,*

$$\pi_{N+1,N} g_{*,N+1} = g_{*,N} \pi_{N+1,N}, \quad (10.94)$$

$$g_* \text{Msect}(x) = \text{Msect}(g_k \cdot x), \quad (10.95)$$

$$(g_{*,N})_* \mu_{x,N} = \mu_{g_k \cdot x, N}, \quad (g_*)_* \mu_x = \mu_{g_k \cdot x}. \quad (10.96)$$

Proof. The first equality states that entrywise action commutes with deleting the final entry. Passing to the inverse limit yields (10.95). Since (10.92) is bijective, $d(g_j \cdot v) = d(v)$; by (10.87), a word and its image have the same weight. This proves the first equality of measures in (10.96). The second follows on the cylinders from (10.91), and the cylinders determine the measure. $\square$

The truncations of the state and those of the tree also form the square.

$$\begin{array}{ccc} \text{Term}_{k,N+1}(x) & \xrightarrow{\pi_{N+1,N}} & \text{Term}_{k,N}(x) \\ \text{ter}_{N+1} \downarrow & & \downarrow \text{ter}_N \\ \widehat{\mathcal{X}}_{N+1}^{\text{tot}} & \xrightarrow{\rho_{N+1,N}} & \widehat{\mathcal{X}}_N^{\text{tot}}. \end{array} \quad (10.97)$$

Commutativity is literal: both paths erase the last emission and its last ledger entry.

10.1.11 Inverse limit $\widehat{\Omega}$ of complete histories and common APP–TRIT–TPK integral holonomic state

The surjections (10.58) produce the object of complete histories

$$\widehat{\Omega} := \varprojlim_k (\widehat{\mathcal{X}}_k^{\text{tot}}, \rho_{k+1,k}). \quad (10.98)$$

An element $\widehat{x} = (\widehat{x}_k)_{k \geq 0}$ contains, at each depth, the same pair of transported sheets, the produced orientations and carries, the accumulated memory, the boundaries, the route, and the complete ledger. Its fiber of continuations from any prefix is $\text{Msect}(\widehat{x}_k)$, equipped with $\mu_{\widehat{x}_k}$.

Theorem 10.30 (Common APP–TRIT–TPK integral holonomic state). *The system*

$$\mathfrak{C}^{\text{gen}} := \left(\begin{array}{l} \{\widehat{\mathcal{X}}_k^{\text{tot}}, \rho_{k+1,k}, \eta_k\}_{k \geq 0}, \widehat{\Omega}, \{\mathfrak{G}_k\}_{k \geq 0}, \{\mathcal{T}_{k,N}\}_{N \geq k}, \\ \text{Msect, Surv, } \{\mu_x\}, \{\text{r}_0(v_K)\}_{K \geq 0}, \mathcal{K}_{\text{ph}}, \widehat{\mathcal{K}}_{\text{ph}}, \\ \text{Sh, or, } \kappa, \mathbf{Q}, g, q^{(9)}, \beta, m, \partial, \gamma, \text{Led}, \\ \{\mathbf{Z}_{9,k}^\Gamma, s_{9,k}, t_{9,k}, \Gamma_{9,k}\}_{k \geq 0} \end{array} \right) \quad (10.99)$$

is defined for every state produced by APP–TRIT–TPK. It is non-empty, natural under reversible symmetries, and compatible with refinement. In particular:

1. the two leaves are transported together;
2. orientation and carry it calculan by the division TRIT;
3. the memory obeys the affine TPK action;
4. the phase obeys the TPK odometer, returns after nine emissions, and the vertical memory advances by one unit;
5. the Markov realization of that turn satisfies $\mathcal{K}_{\text{ph}}^9 = E_+ E_\times$ and uniformly renews the complete fiber of $468 = 18 \cdot 26$ emissions;
6. the compact reader conserves block, cylinder, prefix, carry, signature, Witt shadow, and phase, with their updates and truncations;
7. boundary, path, and ledger are composed without erasing the endpoints or the order of the emissions;
8. every state survives and has a nonempty multisection of complete prolongations;
9. the canonical measure is projective and has support on all those prolongations.

Proof. Totality and finiteness follow from [proposition 10.11](#). The sheet, carry, and memory identities are proved in the corresponding lemma; the phase, vertical memory, and nonadic holonomy, including its fiber realization and its state reader, are proved in the corresponding proposition; the boundary preserves its endpoints by (10.35). Total survival and nonemptiness of the multisections are the corresponding theorem. The existence, consistency, and support of the measure are the corresponding theorem. Finally, compatibility with the symmetries and truncations is proved in [theorem 10.29](#) and (10.97). $\square$

10.1.12 Structural necessity and origin

Proposition 10.31 (Falsifiers of the common integral holonomic state). *Any of the following operations replaces $\mathfrak{C}^{\text{gen}}$ with another object and blocks the transfer of the preceding theorems:*

1. *elegir Σ or Π and descartar the another sheet;*
2. *conserving the residue and erasing the quotient;*
3. *preserving the TRIT sign and erasing its carry;*
4. *replacing the ordered ledger with its terminal value;*
5. *identifying two words because they reach the same cell;*
6. *choosing a continuation and replacing the multisection with it;*
7. *declaring survival without exhibiting prolongations to each horizon;*
8. *uniformly normalizing only the last level, ignoring the degrees of the prefixes and losing projective consistency;*
9. *using a subsequent reading to decide which states enter $\widehat{X}_k^{\text{tot}}$.*

Proof. The first six points eliminate coordinates or relations that appear in the definition of the state, tree, or multisection. The seventh eliminates the proof of [theorem 10.22](#). The eighth may violate (10.89) when the terminal degrees are not constant. The last replaces the total domain (10.57) with a retrospective selection. $\square$

The APP cells, the pair of sheets, the TRIT lift, the TPK extensions, the carry cocycle, memory, boundary, path and conservation of the register constitute the starting structures. On them, the finite

groupoid, the word tree, the multisection and the normalized local measure are brought together in a single domain. The theorems of total survival, nonemptiness, consistency and naturality certify this joint construction.

10.2 Intrinsic Correlative Operations of the Same Continuum

Tesis rectora

The symbols *sp*, *sol*, *gau*, *coh* and *det* that appear in this chapter mark five correlative operations over a single APP–TRIT–TPK state. Each operation receives the same surviving extension, the same orientation, the same carry, the same memory, the same boundary, and the same route. The following sections are successive mathematical stages of a single construction and not autonomous expositions.

10.2.1 Generating edge and common genealogical state

We do not repeat here the state, tree, or measure constructed in the preceding chapter. We take $\widehat{x} = \widehat{x}_k(\gamma) \in \widehat{\mathcal{X}}_k^{\text{tot}}$ from (10.56) and all its admissible emissions $\alpha \in \text{Out}(\widehat{x})$. If $\widehat{y} = \alpha_* \widehat{x}$, we write

$$\text{Ch}_k(\widehat{x}) := \{(\alpha, \alpha_* \widehat{x}) : \alpha \in \text{Out}(\widehat{x})\}, \quad d_k(\widehat{x}) := \# \text{Ch}_k(\widehat{x}). \quad (10.100)$$

The finiteness and non-emptiness of this set are already (10.64); survival at all horizons, the complete multisection, and its measure are the corresponding theorem. A second non-emptiness is not again assumed or proved.

Each pair $(\alpha, \widehat{y})$ of (10.100) preserves the unique record

$$\begin{aligned} \mathbf{e}_\alpha := & (\alpha, \text{Ext}_{\gamma_\alpha}; \text{Sh}_{s(\alpha)}, \text{Sh}_{t(\alpha)}; \\ & \text{res}^{\Sigma, \Pi}(\alpha), \text{quo}^{\Sigma, \Pi}(\alpha); \\ & \text{or}(\alpha), \kappa(\alpha), \mathbf{C}_\alpha; \mathbf{Q}(s(\alpha)), \mathbf{Q}(t(\alpha)), g(s(\alpha)), g(t(\alpha)), q^{(9)}(s(\alpha)), q^{(9)}(t(\alpha)); \\ & m(\gamma_\alpha; m_0), \partial\alpha, \gamma_\alpha, \text{Led}(\gamma_\alpha); \\ & \{\text{Term}_{k+1, N}(\widehat{y})\}_{N \geq k+1}, \text{Msect}(\widehat{y}), \mu_{\widehat{y}}). \end{aligned} \quad (10.101)$$

Here $\text{Ext}_{\gamma_\alpha}$ denotes the complete relation that the admissible emission already satisfies; it introduces no additional filter. The subsequent operations are coordinated applications of $\mathbf{e}_\alpha$ and do not reconstruct the record from their outputs.

10.2.2 Intrinsic Correlative Actions on the Common Generator

The internal marks are fixed without changing their source: *sp* denotes transport and grading of the two sheets; *sol*, signed increment, balance, and memory; *gau*, incidence and carry curvature; *coh*, ports, rectangular difference, and cellular complex; and *det*, based complex, correlative filling, and determinant line. The following formulas display their couplings over the same $\mathbf{e}_\alpha$; their actions literally share source, target, route, and ledger.

The double APP evaluation provides, on the same support, the labels Σ and Π . For an extension α , their multiplicities are calculated from the complete ledger, without an external counter:

$$N_\diamond(\alpha) := \sum_{\epsilon \in \text{en } \gamma_\alpha} \mathbf{1}_{\{\text{res}^\diamond(\epsilon) \neq 0\}}, \quad \diamond \in \{\Sigma, \Pi\}. \quad (10.102)$$

Let there also be the total oriented coefficient

$$\Delta_{\gamma_\alpha} := \sum_{\epsilon \in \text{en } \gamma_\alpha} \text{or}(\epsilon) \in \mathbb{Z}.$$

The signed action is

$$b_\alpha := \Delta_{\gamma_\alpha}(N_\Sigma(\alpha) - N_\Pi(\alpha)), \quad b_\alpha^\pm := \frac{|b_\alpha| \pm b_\alpha}{2}. \quad (10.103)$$

By construction,

$$b_\alpha = b_\alpha^+ - b_\alpha^-, \quad |b_\alpha| = b_\alpha^+ + b_\alpha^-, \quad b_\alpha^+ b_\alpha^- = 0. \quad (10.104)$$

The same set of extensions simultaneously produces the ports

$$I_k(\widehat{x}_k) := \text{Ch}_k(\widehat{x}_k) \times \{\Sigma\}, \quad J_k(\widehat{x}_k) := \text{Ch}_k(\widehat{x}_k) \times \{\Pi\}. \quad (10.105)$$

A visible leaf is not assumed to contain more prolongations than the other: each surviving extension is evaluated on both leaves of the same support. By (10.64), both sets of ports are nonempty.

For $A_N = \mathbb{Z}/3^N\mathbb{Z}$, the cellular action on generators is

$$\begin{aligned} \kappa_{k,N} : A_N^{I_k} \oplus A_N^{J_k} &\longrightarrow A_N^{I_k \times J_k}, \\ \kappa_{k,N}(u, v)_{(\alpha, \Sigma), (\beta, \Pi)} &:= u_{(\alpha, \Sigma)} - v_{(\beta, \Pi)}. \end{aligned} \quad (10.106)$$

Selected first ports according to the TPK order, the rectangular difference

$$\delta(w)_{ij} := w_{ij} - w_{i_0j} - w_{ij_0} + w_{i_0j_0} \quad (10.107)$$

download the split sequence

$$0 \longrightarrow A_N \xrightarrow{\Delta} A_N^{I_k} \oplus A_N^{J_k} \xrightarrow{\kappa_{k,N}} A_N^{I_k \times J_k} \xrightarrow{\delta} A_N^{(|I_k|-1)(|J_k|-1)} \longrightarrow 0. \quad (10.108)$$

This is an identity of the same APP fibre: it adds no cell foreign to the generator $\mathbf{e}_\alpha$.

The local cellular action uses the two integer coordinates accumulated by that same ledger. For $\widehat{x} = \widehat{x}_k(\gamma)$ and $\widehat{y} = \alpha_* \widehat{x} = \widehat{x}_{k+1}(\gamma\alpha)$, we set

$$\begin{aligned} c_{\widehat{x}} &:= \kappa(\gamma), & c_\alpha &:= c_{\widehat{y}} := \kappa(\gamma\alpha), \\ v_{\widehat{x}} &:= \sum_{\epsilon \in \gamma} \text{or}(\epsilon), & v_{\widehat{y}} &:= \sum_{\epsilon \in \gamma\alpha} \text{or}(\epsilon), \\ v_\alpha &:= v_{\widehat{y}} - v_{\widehat{x}} = \text{or}(\alpha). \end{aligned} \quad (10.109)$$

On the generators f, r, e, b, g is defined

$$\partial f = 3c_\alpha e + b, \quad \partial r = 0, \quad \partial e = g, \quad \partial b = -3c_\alpha g, \quad \partial g = 0. \quad (10.110)$$

The cancellation

$$\partial^2 f = 3c_\alpha g - 3c_\alpha g = 0 \quad (10.111)$$

is literal and uses the same carry with the two signs imposed by the orientation.

The incidence seed is obtained from the free module of the two leaves,

$$V_{\text{TPK}} := \mathbf{F}_3 e_\Sigma \oplus \mathbf{F}_3 e_\Pi, \quad \mathcal{J} := \mathbf{P}^1(V_{\text{TPK}}), \quad |\mathcal{J}| = 4, \quad (10.112)$$

and of its unordered pairs

$$\mathcal{E} := \binom{\mathcal{J}}{2}, \quad |\mathcal{E}| = 6, \quad \mathcal{B} := \mathcal{J} \times \{+, -\}, \quad |\mathcal{B}| = 8. \quad (10.113)$$

The incidence application is determined on the base by

$$B[e_L] = \sum_{L' \neq L} e_{\{L, L'\}}, \quad L \in \mathcal{J}, \quad (10.114)$$

and, by linearity,

$$(Ba)_{\{L, L'\}} = a_L + a_{L'}. \quad (10.115)$$

If $s(q) = \sum_{E \in \mathcal{E}} q_E$, each L appears three times and

$$s(Ba) = 0. \quad (10.116)$$

Therefore the function

$$W_c(q) := \mathbf{1}_{\{s(q)=0\}} - \frac{1}{3} \quad (10.117)$$

satisfies $W_c(q + Ba) = W_c(q)$. The identities (10.112)–(10.117) are internal and exact. The material lift of q from the cells and common carries is treated in the assembly layer; this incidence identity is not conflated with the construction of that lift.

10.2.3 Unitary transport, polarization, and balance on the common fiber

The non-emptiness of (10.100) allows one to fix, without selecting a prolongation,

$$p_k(\alpha, \widehat{y} \mid \widehat{x}) := \frac{1}{d_k(\widehat{x})}. \quad (10.118)$$

Let the space of the two leaves of the same state be

$$\mathcal{H}_k := \bigoplus_{\widehat{x} \in \widehat{\mathcal{X}}_k^{\text{tot}}} (\mathcal{H}_x^\Sigma \oplus \mathcal{H}_x^\Pi), \quad (10.119)$$

and let $\iota_{\widehat{x}}$ be the inclusion of the indicated fiber. The TPK transport of (10.26) gives the exact action

$$U_k \iota_{\widehat{x}} v := \sum_{(\alpha, \widehat{y}) \in \text{Ch}_k(\widehat{x})} p_k(\alpha, \widehat{y} \mid \widehat{x})^{1/2} \iota_{\widehat{y}} U_\alpha v. \quad (10.120)$$

The children of two distinct parents are disjoint because truncation is a function. Moreover, two distinct witnesses α do not collapse into the same direct summand: the complete route γ , including its last step, is a coordinate of $\widehat{y}$. Consequently, the Gram identity is

$$U_k^* U_k \big|_{\mathcal{H}_x^\Sigma \oplus \mathcal{H}_x^\Pi} = \sum_{(\alpha, \widehat{y}) \in \text{Ch}_k(\widehat{x})} p_k(\alpha, \widehat{y} \mid \widehat{x}) U_\alpha^* U_\alpha. \quad (10.121)$$

Therefore, $U_k^* U_k = I$ exactly when the sheet transports U_α published by TPK are isometries. Normalization of the weights alone cannot replace that verification. For $\ell > k$, the complete history is preserved through

$$U_{\ell, k} := U_{\ell-1} \cdots U_k, \quad U_{k, k} := I. \quad (10.122)$$

The grading acts on the two coordinates already present in each fiber of (10.119):

$$\sigma_k \iota_{\widehat{x}}(v_\Sigma, v_\Pi) := \iota_{\widehat{x}}(v_\Sigma, -v_\Pi), \quad P_k := \frac{I + \sigma_k}{2}, \quad Q_k := \frac{I - \sigma_k}{2}. \quad (10.123)$$

The TRIT threshold belongs to the orientation and to the carry of $\widehat{x}$; it erases neither of the two APP sheets and does not require a third case in (10.123).

The leaf-preserving part and the leaf-changing part are two components of the same U_k :

$$A_k := P_{k+1} U_k P_k + Q_{k+1} U_k Q_k, \quad (10.124)$$

$$\eta_k := Q_{k+1}U_kP_k + P_{k+1}U_kQ_k. \quad (10.125)$$

By orthogonality of P_{k+1} and Q_{k+1} ,

$$U_k = A_k + \eta_k, \quad A_k^*A_k + \eta_k^*\eta_k = U_k^*U_k. \quad (10.126)$$

The last member is I under, and only under, the isometric condition identified after (10.121). This identity and the increment (10.103) are not two transports: A_k, η_k act on the base of histories and b_α reads the edge of that same base.

10.2.3.1 Signed polarization, non-autonomous telescoping, and return without reinitialization

No independent spectral branch is introduced here. The notations are restricted to the same space $\mathcal{H}_k$ and the same already constructed transport U_k

$$\mathcal{H}_k^{\text{sp}} := \mathcal{H}_k, \quad \Lambda(\gamma_{\widehat{x}}) := (\text{or}, \text{Carr}, \text{Mem}, \partial, \gamma)(\widehat{x}),$$

and $\Sigma_{\text{reg}}\Lambda(\gamma_{\widehat{x}})$ denotes its family of compatible truncations. In this way, the following signed reader, defect and terminal are actions of the common correlative constructor, not a retrospective selection of histories.

10.2.3.2 Signed reader with genealogical register

For $v \in \mathcal{H}_k^{\text{sp}}$, the signed polarization is

$$\text{pol}_k(v) = \langle v, \sigma_k v \rangle = \|P_k v\|^2 - \|Q_k v\|^2. \quad (10.127)$$

The specialized reader furthermore preserves the source:

$$\mathcal{L}_{\text{sp},k}^{\text{sgn}}(x_k; v) = (\Lambda(\gamma_{x_k}), \Sigma_{\text{reg}}(\Lambda(\gamma_{x_k})), \|P_k v\|^2, \|Q_k v\|^2, \text{pol}_k(v)). \quad (10.128)$$

The TRIT orientation is preserved in $\Lambda(\gamma_{x_k})$; it is not identified with the sheet sign. At horizon N , the record keeps the family of readers for all levels $k \leq N$. Truncating means removing only subsequent entries.

10.2.3.3 Non-autonomous and terminal product

We define

$$F_{0,0} = I_{\mathcal{H}_0^{\text{sp}}}, \quad F_{0,k} = A_{k-1} \cdots A_1 A_0 \quad (k \geq 1). \quad (10.129)$$

Therefore,

$$F_{0,k+1} = A_k F_{0,k}. \quad (10.130)$$

The term published upon leaving the channel at step k and the terminal at horizon N are

$$\mathcal{D}_k = F_{0,k}^* \eta_k^* \eta_k F_{0,k}, \quad \mathcal{T}_N = F_{0,N}^* F_{0,N}. \quad (10.131)$$

Theorem 10.32 (Spectral telescope–non-autonomous polar). *For every finite horizon N ,*

$$I = \sum_{k=0}^{N-1} F_{0,k}^* \eta_k^* \eta_k F_{0,k} + F_{0,N}^* F_{0,N}. \quad (10.132)$$

Moreover,

$$0 \preceq \mathcal{T}_{N+1} \preceq \mathcal{T}_N \preceq I,$$

so that the strong positive limit exists

$$\mathcal{T}_\infty = \text{s-lim}_{N \rightarrow \infty} \mathcal{T}_N$$

and

$$I = \sum_{k=0}^{\infty} F_{0,k}^* \eta_k^* \eta_k F_{0,k} + \mathcal{T}_\infty \quad (10.133)$$

in the strong topology. The theorem does not assert that $\mathcal{T}_\infty$ is zero.

Proof. Multiplying (10.126) by $F_{0,k}^*$ and $F_{0,k}$, and using (10.130), one obtains

$$F_{0,k}^* F_{0,k} = F_{0,k+1}^* F_{0,k+1} + F_{0,k}^* \eta_k^* \eta_k F_{0,k}.$$

The sum from $k = 0$ to $N - 1$ cancels all intermediate terms and leaves precisely (10.132). Since $A_k^* A_k \preceq I$,

$$\mathcal{T}_{N+1} = F_{0,N}^* A_N^* A_N F_{0,N} \preceq F_{0,N}^* F_{0,N} = \mathcal{T}_N.$$

Monotonicity and the positive bound provide the strong limit. Passing to the limit in the finite identity gives (10.133). $\square$

10.2.3.4 Phase return and non-reinitialized evolution of the common state

The branch explicitly preserves

$$g(k+9) = g(k), \quad x_{k+9} \neq x_k \quad \text{in general.} \quad (10.134)$$

The second member records the increase in memory and possible changes of sheet, carry, cylinder and boundary. Neither σ_k , nor the projectors, nor the signed reader contract this difference.

The conditional expectation associated with the same weight is

$$(E_k F)(\widehat{x}) := \sum_{(\alpha, \widehat{y}) \in \text{Ch}_k(\widehat{x})} p_k(\alpha, \widehat{y} \mid \widehat{x}) F(\alpha, \widehat{y}). \quad (10.135)$$

For the fine increment b_f , set

$$b_c := E_k b_f, \quad \kappa_k^{\text{can}} := \frac{E_k |b_f| - |b_c|}{2} \geq 0. \quad (10.136)$$

Then

$$E_k b_f^+ = b_c^+ + \kappa_k^{\text{can}}, \quad E_k b_f^- = b_c^- + \kappa_k^{\text{can}}. \quad (10.137)$$

The superscript can distinguishes this cancellation carry from the integer carry κ_α ; both arise from the same transport, but have distinct types.

Signed memories require no external initial condition: for $\gamma = (\epsilon_1, \dots, \epsilon_k)$ they are defined by summing the ledger itself,

$$\begin{aligned} M_0^\pm &:= 0, \\ M_k^\pm(\gamma) &:= \sum_{j=1}^k b_{\epsilon_j}^\pm, \\ M_{k+1}^\pm(\gamma\alpha) &:= M_k^\pm(\gamma) + b_\alpha^\pm, \\ D_k^0 &:= M_k^+ - M_k^-, \quad H_k^0 := M_k^+ + M_k^-. \end{aligned} \quad (10.138)$$

Thus, the signed balance, the total memory, and the polarization $\langle v, \sigma_k v \rangle$ are three coordinated readings of the same prefix, not three states reconstructed separately.

10.2.3.5 Nonadic Refinement, Orthogonal Losses and Terminal Memory

The index j of the following development is the level of the same tree of prolongations and its thirteen-coordinate register is the restriction of the common register of (10.101). Therefore $D_{\text{sol},j} := D_j^{(13)}$ is fixed and no additional domain is postulated.

10.2.3.6 Common nonadic refinement: inclusion, balance expectation and memory

Let $Y_j(x)$ be the finite set of cylinders compatible with the prefix $x_{\leq j}$, endowed with the projective measure μ_j . The truncation

$$\tau_{j+1,j} : Y_{j+1}(x) \longrightarrow Y_j(x)$$

has nine admissible extensions over each cylinder. Finite spaces are considered.

$$\mathcal{H}_j(x) := \ell^2(Y_j(x), \mu_j).$$

The inclusion and the nonadic expectation are

$$J_{9,j} : \mathcal{H}_j \longrightarrow \mathcal{H}_{j+1}, \quad (J_{9,j}f)(y) = f(\tau_{j+1,j}y), \quad (10.139)$$

$$E_{9,j} : \mathcal{H}_{j+1} \longrightarrow \mathcal{H}_j, \quad (E_{9,j}g)(z) = \frac{1}{9} \sum_{\tau_{j+1,j}y=z} g(y). \quad (10.140)$$

The projective compatibility of μ_{j+1} and μ_j gives

$$E_{9,j}J_{9,j} = I_{\mathcal{H}_j}, \quad J_{9,j}^* = E_{9,j}, \quad P_{9,j} := J_{9,j}E_{9,j} = P_{9,j}^* = P_{9,j}^2. \quad (10.141)$$

The complementary microscopic projector is

$$Q_{9,j} := I - P_{9,j}. \quad (10.142)$$

If b_f is the increment on the fine fiber and

$$b_c := E_{9,j}b_f, \quad (10.143)$$

the elementary convexity of $|\cdot|$ constructs, rather than postulates, the carry of cancellation

$$\kappa_j := \frac{E_{9,j}|b_f| - |b_c|}{2} \geq 0. \quad (10.144)$$

Indeed, using $t^\pm = (|t| \pm t)/2$, one obtains

$$E_{9,j}b_f^+ = \frac{E_{9,j}|b_f| + E_{9,j}b_f}{2} = b_c^+ + \kappa_j, \quad (10.145)$$

$$E_{9,j}b_f^- = \frac{E_{9,j}|b_f| - E_{9,j}b_f}{2} = b_c^- + \kappa_j. \quad (10.146)$$

Both sheets lose the same amount when compressed; therefore, the signed difference is preserved while total memory records the cancellation:

$$E_{9,j}(b_f^+ - b_f^-) = b_c, \quad E_{9,j}(b_f^+ + b_f^-) = |b_c| + 2\kappa_j. \quad (10.147)$$

10.2.3.7 Orthogonal factorization of the degrees of depth and of the resolution layers in $\mathcal{H}_j^{\text{depth}} \widehat{\otimes} \mathcal{H}_j^{\text{res}}$

To prevent the same loss from being counted both as depth and as resolution, the space at each level is typed as

$$\mathcal{H}_j = \mathcal{H}_j^{\text{depth}} \widehat{\otimes} \mathcal{H}_j^{\text{res}}. \quad (10.148)$$

For $j \geq 1$, the first factor carries

$$P_j^{\text{depth}} := J_{9,j-1} E_{9,j-1} \quad \text{on } \mathcal{H}_j^{\text{depth}},$$

and $P_0^{\text{depth}} = I$ is fixed. The second factor carries the finite shell partition inherited from the cylinder boundary. If $\sigma_j : Y_j^{\text{res}} \rightarrow S_j$ is that partition, its operators are

$$J_j^{\text{res}} f = f \circ \sigma_j, \quad E_j^{\text{res}} g(s) = \frac{1}{\#\sigma_j^{-1}(s)} \sum_{\sigma_j(y)=s} g(y), \quad (10.149)$$

$$P_j^{\text{res}} := J_j^{\text{res}} E_j^{\text{res}}. \quad (10.150)$$

The three mutually orthogonal projectors are defined.

$$P_j^{\text{term}} := P_j^{\text{depth}} \otimes P_j^{\text{res}}, \quad (10.151)$$

$$Q_j^{\text{micro}} := (I - P_j^{\text{depth}}) \otimes I, \quad (10.152)$$

$$Q_j^{\text{shell}} := P_j^{\text{depth}} \otimes (I - P_j^{\text{res}}). \quad (10.153)$$

Then

$$Q_j^{\text{micro}} Q_j^{\text{shell}} = 0, \quad I = P_j^{\text{term}} + Q_j^{\text{micro}} + Q_j^{\text{shell}}, \quad (10.154)$$

and, for all $\xi \in \mathcal{H}_j$,

$$\|\xi\|^2 = \|P_j^{\text{term}} \xi\|^2 + \|Q_j^{\text{micro}} \xi\|^2 + \|Q_j^{\text{shell}} \xi\|^2. \quad (10.155)$$

The term $(I - P_j^{\text{depth}}) \otimes (I - P_j^{\text{res}})$ does not appear a second time: it is already contained exactly once in Q_j^{micro} .

10.2.3.8 Cylinder extensions and full row of losses

An extension of cylinders from level j to level $j+1$ induces the map

$$\Gamma_j : \mathcal{H}_j \longrightarrow \mathcal{H}_{j+1}, \quad \Gamma_j f = f \circ \tau_{j+1,j}, \quad (10.156)$$

with all leaf, orientation, carry, memory, and path labels transported in the cylinder basis. For $n > j$, we write

$$\Gamma_{j:n} := \Gamma_{n-1} \cdots \Gamma_j, \quad \Gamma_{j:j} := I. \quad (10.157)$$

On $\mathcal{H}_j$, the nonnegative increments define multiplication operators

$$B_j^+ := M \sqrt{b_j^+}, \quad B_j^- := M \sqrt{b_j^-}, \quad K_j := M \sqrt{2\kappa_j}. \quad (10.158)$$

Each root is taken pointwise over the finite set of cylinders; if a coordinate is zero, its operator is zero. The space of losses is the external orthogonal sum

$$\mathcal{E}_j := \mathcal{H}_j^{(+)} \oplus \mathcal{H}_j^{(-)} \oplus \mathcal{H}_j^{(\kappa)} \oplus \mathcal{H}_j^{(\text{micro})} \oplus \mathcal{H}_j^{(\text{shell})},$$

and the complete row is

$$L_j : \mathcal{H}_j \longrightarrow \mathcal{E}_j, \quad L_j \xi = (B_j^+ \xi, B_j^- \xi, K_j \xi, Q_j^{\text{micro}} \xi, Q_j^{\text{shell}} \xi). \quad (10.159)$$

Therefore,

$$\|L_j \xi\|^2 = \|B_j^+ \xi\|^2 + \|B_j^- \xi\|^2 + \|K_j \xi\|^2 + \|Q_j^{\text{micro}} \xi\|^2 + \|Q_j^{\text{shell}} \xi\|^2. \quad (10.160)$$

The positive part, negative part, cancellation carry, microstructure, and shell each appear once. They are neither merged into an anonymous constant nor repeated in two summands.

10.2.3.9 Complete history terminal

Let $\text{Hist}_J^{\text{sol}}$ be the set of histories surviving to the horizon J , each with its package of thirteen coordinates. The terminal space is defined as the free space on these complete records:

$$\mathcal{T}_J^{\text{sol}} := \ell^2 \left(\{ (h, D_{\text{sol},J}(h)) : h \in \text{Hist}_J^{\text{sol}} \} \right). \quad (10.161)$$

The terminal evaluation

$$E_J^{\text{term}} : \mathcal{H}_J \longrightarrow \mathcal{T}_J^{\text{sol}} \quad (10.162)$$

sends each basic cylinder vector to the complete record of the history that generates it and extends linearly. In particular, the terminal retains fiber, relations, numbers, orientation, carry, memory, boundary, route, multisection, survival, and reader. It is not replaced by the last visible value of the route.

10.2.3.10 Observation column, Gram operator, and telescoping

For $0 \leq j \leq J$, the complete future column is

$$\begin{aligned} G_{j;J} : \mathcal{H}_j &\longrightarrow \mathcal{T}_J^{\text{sol}} \oplus \bigoplus_{n=j}^{J-1} \mathcal{E}_n, \\ G_{j;J} \xi &:= E_J^{\text{term}} \Gamma_{j;J} \xi \oplus \bigoplus_{n=j}^{J-1} L_n \Gamma_{j;n} \xi. \end{aligned} \quad (10.163)$$

For $j = J$, the sum of perdidas is empty and $G_{J;J} = E_J^{\text{term}}$. The Gram terminal is

$$U_{j;J} := G_{j;J}^* G_{j;J} \succeq 0. \quad (10.164)$$

Rearranging the orthogonal sum of the codomain yields the literal identity.

$$G_{j;J} \xi \simeq (L_j \xi, G_{j+1;J} \Gamma_j \xi). \quad (10.165)$$

Consequently,

$$\boxed{U_{j;J} = L_j^* L_j + \Gamma_j^* U_{j+1;J} \Gamma_j} \quad (10.166)$$

and

$$U_{j;J} - \Gamma_j^* U_{j+1;J} \Gamma_j = L_j^* L_j \succeq 0. \quad (10.167)$$

There is no sign mutation: the loss term is on the positive side of the identity because it arises from a sum of squares.

Iterating (10.166) yields the exact telescoping identity

$$U_{j;J} = \Gamma_{j;J}^* (E_J^{\text{term}})^* E_J^{\text{term}} \Gamma_{j;J} + \sum_{n=j}^{J-1} \Gamma_{j;n}^* L_n^* L_n \Gamma_{j;n}, \quad (10.168)$$

o, in quadratic form,

$$\langle U_{j;J\xi}, \xi \rangle = \|E_J^{\text{term}} \Gamma_{j;J\xi}\|^2 + \sum_{n=j}^{J-1} \|L_n \Gamma_{j;n\xi}\|^2. \quad (10.169)$$

The first term preserves the complete terminal future; the sum preserves, level by level, all the losses and all the carries that led to it.

10.2.4 Orientation, carry, and simultaneous composition of paths

Path inversion acts only once on e_α . Let $J(v_\Sigma, v_\Pi) := (v_\Pi, v_\Sigma)$ be the leaf exchange, and let P_{tr} be the transposition of port matrices. Its five local expressions are

$$\begin{aligned} J\sigma_k &= -\sigma_k J, & b_{\bar{\alpha}} &= -b_\alpha, & (b_{\bar{\alpha}}^+, b_{\bar{\alpha}}^-) &= (b_\alpha^-, b_\alpha^+), \\ \vartheta_L(K, \epsilon) &:= \begin{cases} (L, -\epsilon), & K = L, \\ (K, \epsilon), & K \neq L, \end{cases} & \kappa_{J,I}(v, u) &= -P_{\text{tr}} \kappa_{I,J}(u, v), \\ \kappa(\bar{\alpha}) &= -C_\alpha^{-1} \kappa(\alpha), & v_{\bar{\alpha}} &= -v_\alpha. \end{aligned} \quad (10.170)$$

Here J exchanges the two sheets, P_{tr} transposes the incidence matrix, and ϑ_L exchanges exactly the two faces in the direction L , fixing the other six. In the free basis of $\mathcal{B} = \mathcal{F} \times \{+, -\}$, the eight vectors are distinct, $\vartheta_L^2 = I$ and $\vartheta_L \neq \vartheta_{L'}$ for $L \neq L'$. The same orientation explains all signs in (10.170).

For composable extensions $\alpha : x \rightarrow y$ and $\beta : y \rightarrow z$, the native carry satisfies

$$\kappa(\beta\alpha) = \kappa(\beta) + C(\beta)\kappa(\alpha). \quad (10.171)$$

The matrix lift is not left as a name. In the ordered basis (f, r, e, b, g) , let $\tilde{L}_N(\alpha) \in \text{Mat}_5(\mathbb{Z})$ be the centered integral representative of the matrix of Ψ_α of (10.175) modulo 3^N , and define

$$c_N(\beta, \alpha) := \frac{\tilde{L}_N(\beta)\tilde{L}_N(\alpha) - \tilde{L}_N(\beta\alpha)}{3^N} \in \text{Mat}_5(\mathbb{Z}). \quad (10.172)$$

Divisibility follows from equality of the two compositions modulo 3^N . Writing $L_N = [\tilde{L}_N]_{3^N}$, the integer lifts satisfy

$$\tilde{L}_N(\beta)\tilde{L}_N(\alpha) = \tilde{L}_N(\beta\alpha) + 3^N c_N(\beta, \alpha) \quad (10.173)$$

and associativity produces the cocycle

$$c_N(\delta, \beta\alpha) + \tilde{L}_N(\delta)c_N(\beta, \alpha) = c_N(\delta\beta, \alpha) + c_N(\delta, \beta)\tilde{L}_N(\alpha). \quad (10.174)$$

The same carry increment acts upon the local complex through

$$\begin{aligned} \Psi_\alpha(r_x) &= r_y, & \Psi_\alpha(e_x) &= e_y, & \Psi_\alpha(g_x) &= g_y, & \Psi_\alpha(f_x) &= f_y, \\ \Psi_\alpha(b_x) &= b_y + 3(c_y - c_x)e_y. \end{aligned} \quad (10.175)$$

The oriented boundary change remains in

$$K_\alpha := (v_y - v_x)f_y, \quad K_{\beta\alpha} = K_\beta + \Psi_\beta K_\alpha. \quad (10.176)$$

The equations (10.171), (10.174), and (10.176) are three degrees of the same ledger of composition. None authorizes erasing the other two.

10.2.4.1 Nerve of extensions and Alexander–Whitney diagonal

For a history of the common register one writes $A_n := \mathbb{Z}/3^n\mathbb{Z}$ and $\mathbb{T}_m(x)$ for the finite category of *all* its TPK prefixes of depth m , with the extension morphisms already constructed. The diagonal and counit that follow apply to that same nerve; they do not define a fifth autonomous branch.

10.2.4.2 Nerve of routes, Alexander–Whitney diagonal and counit

The normalized complex is defined

$$G_{m,n}(x) := N_*^{\text{norm}}(N\mathbb{T}_m(x); A_n). \quad (10.177)$$

For a nondegenerate simplex $\sigma = [y_0 \rightarrow y_1 \rightarrow \cdots \rightarrow y_q]$, the diagonal is

$$\Delta_{\text{AW}}(\sigma) = \sum_{j=0}^q [y_0 \rightarrow \cdots \rightarrow y_j] \otimes [y_j \rightarrow \cdots \rightarrow y_q]. \quad (10.178)$$

The counit is given by

$$\epsilon_G([y]) = 1, \quad \epsilon_G(\sigma) = 0 \quad (q > 0). \quad (10.179)$$

In particular,

$$\Delta_{\text{AW}}[y] = [y] \otimes [y], \quad \epsilon_G([y]) = 1. \quad (10.180)$$

If $m' \geq m$ and $n' \geq n$, truncation and coefficient reduction induce

$$R_{m',m}^{n',n} : G_{m',n'}(x) \longrightarrow G_{m,n}(x), \quad (10.181)$$

and they are verified

$$R\partial = \partial R, \quad (R \otimes R)\Delta_{\text{AW}} = \Delta_{\text{AW}}R, \quad \epsilon_G R = \rho_{n',n} \epsilon_G. \quad (10.182)$$

The path is not reduced to its endpoints: the entire simplex remains in $G_{m,n}(x)$.

10.2.5 Cellular incidence and correlative filling of the same trajectory

The sequence (10.108) assigns to each prefix the two-term complex

$$C_{k,N}(\widehat{x}) := \left[A_N^{I_k} \oplus A_N^{J_k} \xrightarrow{\kappa_{k,N}} A_N^{I_k \times J_k} \right]. \quad (10.183)$$

A refinement acts on ports by truncating the extension and on coefficients by $A_{N'} \twoheadrightarrow A_N$. When $\alpha_u : I_{k'} \rightarrow I_k$ and $\beta_u : J_{k'} \rightarrow J_k$ are those maps, their action on generators is

$$(p, q) \longmapsto (p \circ \alpha_u, q \circ \beta_u), \quad w \longmapsto w \circ (\alpha_u \times \beta_u), \quad (10.184)$$

and satisfies

$$\kappa_{k',N}\Gamma(u)_1 = \Gamma(u)_0\kappa_{k,N}. \quad (10.185)$$

The incidence is not fabricated by labeling four copies of a neutral emission. It is generated by the free cellular completion of the module of the two sheets that the same state already bears. For

$$V_{\widehat{x}} := \mathbf{F}_3 e_{\Sigma, \widehat{x}} \oplus \mathbf{F}_3 e_{\Pi, \widehat{x}},$$

the four internal lines are

$$\mathcal{J}_{\widehat{x}} := \{L_0, L_\infty, L_+, L_-\} = \mathbf{P}^1(V_{\widehat{x}}), \quad \begin{aligned} L_0 &= \langle e_\Sigma \rangle, & L_\infty &= \langle e_\Pi \rangle, \\ L_+ &= \langle e_\Sigma + e_\Pi \rangle, & L_- &= \langle e_\Sigma - e_\Pi \rangle. \end{aligned} \quad (10.186)$$

If v_L is, respectively, $(1, 0)^t$, $(0, 1)^t$, $(1, 1)^t$, $(1, -1)^t$, the action of the direction L on the generators of sheet is the projector

$$P_L := v_L(v_L^t v_L)^{-1} v_L^t, \quad (10.187)$$

that is, over $\mathbf{F}_3$,

$$P_0 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad P_\infty = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad P_+ = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}, \quad P_- = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Direct calculation gives

$$P_L^2 = P_L, \quad \text{rk } P_L = 1, \quad \text{Im } P_L = L, \quad P_L = P_{L'} \implies L = L'. \quad (10.188)$$

Therefore, the four directions are distinct outputs of the module $V_{\widehat{x}}$; they are not four names placed upon the same action.

Definition 10.33 (Cell-free leaf TPK completion). Let $\text{Cell}_{\text{TPK}}(\widehat{x})$ be the free complex

$$C_2(\widehat{x}) \xrightarrow{\partial_2} C_1(\widehat{x}) \xrightarrow{\partial_1} C_0(\widehat{x})$$

with

$$\begin{aligned} C_0(\widehat{x}) &:= \mathbf{F}_3[v_\star] \oplus \bigoplus_{L \in \mathcal{J}} \mathbf{F}_3[v_L] \oplus \bigoplus_{(L, \epsilon) \in \mathcal{B}} \mathbf{F}_3[v_{L, \epsilon}] \oplus \bigoplus_{E \in \mathcal{E}} \mathbf{F}_3[v_E], \\ C_1(\widehat{x}) &:= \bigoplus_{L \in \mathcal{J}} \mathbf{F}_3[u_L] \oplus \bigoplus_{(L, \epsilon) \in \mathcal{B}} \mathbf{F}_3[u_{L, \epsilon}] \oplus \bigoplus_{E = \{L, L'\} \in \mathcal{E}} (\mathbf{F}_3[u_{L'|L}] \oplus \mathbf{F}_3[u_{L|L'}]), \\ C_2(\widehat{x}) &:= \bigoplus_{E \in \mathcal{E}} \mathbf{F}_3[\omega_E]. \end{aligned}$$

Its action on generators is

$$\partial u_L = v_L - v_\star, \quad \partial u_{L'|L} = v_E - v_L, \quad (10.189)$$

$$\partial u_{L, \epsilon} = v_{L, \epsilon} - v_L, \quad \partial \omega_{\{L, L'\}} = u_L + u_{L'|L} - u_{L'} - u_{L|L'}. \quad (10.190)$$

The eight generators $v_{L, \epsilon}$ materially realize the screen $\mathcal{B}$. The preceding involution extends to the complex by

$$\vartheta_L v_{K, \epsilon} := v_{\vartheta_L(K, \epsilon)}, \quad \vartheta_L u_{K, \epsilon} := u_{\vartheta_L(K, \epsilon)}, \quad (10.191)$$

and fixes the other generators. Since the basis is free,

$$|\{v_{L, \epsilon} : (L, \epsilon) \in \mathcal{B}\}| = 8, \quad \vartheta_L^2 = I, \quad \vartheta_L \vartheta_{L'} = \vartheta_{L'} \vartheta_L. \quad (10.192)$$

Moreover, $\partial \vartheta_L = \vartheta_L \partial$, because both paths send $u_{L, \epsilon}$ to $v_{L, -\epsilon} - v_L$ and fix the remaining boundaries.

The two words

$$p_{L, L'} := u_{L'|L} u_L, \quad p_{L', L} := u_{L|L'} u_{L'}$$

are different paths in the free complex. Their difference is the boundary of $\omega_{\{L, L'\}}$, and

$$\partial^2 \omega_E = (v_L - v_\star) + (v_E - v_L) - (v_{L'} - v_\star) - (v_E - v_{L'}) = 0. \quad (10.193)$$

The representation of TPK on the leaves is fixed by

$$\varrho(u_L) = P_L, \quad \varrho(u_{L'|L}) = P_{L'}, \quad \varrho(p_{L, L'}) = P_{L'} P_L.$$

Even if two published matrices coincided in a specialization, the two paths do not collapse: they remain distinct free generators of $C_1(\widehat{x})$, joined by an explicit 2-cell.

The complete incidence coordinate is the group of 2-cochains.

$$Q_{\text{inc}}(\widehat{x}) := C^2(\text{Cell}_{\text{TPK}}(\widehat{x}); \mathbf{F}_3) = \bigoplus_{E \in \mathcal{E}} \mathbf{F}_3 \omega_E^* \simeq \mathbf{F}_3^6. \quad (10.194)$$

The internal operation returns all of $Q_{\text{inc}}(\widehat{x})$, not a selected element. Each q has a unique expansion $q = \sum_E q_E \omega_E^*$. For $a \in \mathbf{F}_3^{\mathcal{J}}$, its change of potential is

$$q \mapsto q + Ba, \quad (Ba)_{\{L, L'\}} = a_L + a_{L'}. \quad (10.195)$$

As each line belongs to three pairs,

$$s(Ba) = 3 \sum_{L \in \mathcal{J}} a_L = 0,$$

and therefore

$$W_c(q + Ba) = W_c(q) \quad (q \in Q_{\text{inc}}(\widehat{x})). \quad (10.196)$$

The unit of the completion only anchors the complex in the state:

$$\eta_{\widehat{x}}^{\text{cell}} : V_{\widehat{x}} \longrightarrow V_{\widehat{x}} \otimes C_0(\text{Cell}_{\text{TPK}}(\widehat{x})), \quad e_{\diamond, \widehat{x}} \mapsto e_{\diamond, \widehat{x}} \otimes v_{\star}, \quad (10.197)$$

and marks the origin $q = 0$; it does not assert that the orbit of that origin is all of Q_{inc} .

The construction is functorial under common truncation. If $\tau_{\ell, k} \widehat{x}_{\ell} = \widehat{x}_k$, we define

$$\begin{aligned} \tau_V : V_{\widehat{x}_{\ell}} &\longrightarrow V_{\widehat{x}_k}, & \tau_V e_{\diamond, \widehat{x}_{\ell}} &= e_{\diamond, \widehat{x}_k}, \\ \tau_a : \mathbf{F}_3^{\mathcal{J}_{\widehat{x}_{\ell}}} &\longrightarrow \mathbf{F}_3^{\mathcal{J}_{\widehat{x}_k}}, & (\tau_a a)_{\tau_{\mathcal{J}} L} &= a_L, \\ \tau_Q : Q_{\text{inc}}(\widehat{x}_{\ell}) &\longrightarrow Q_{\text{inc}}(\widehat{x}_k), & (\tau_Q q)_{\tau_{\mathcal{G}} E} &= q_E, \end{aligned}$$

where $\tau_{\mathcal{J}}[v] = [\tau_V v]$ and $\tau_{\mathcal{G}}\{L, L'\} = \{\tau_{\mathcal{J}} L, \tau_{\mathcal{J}} L'\}$. The map of cadenas $\text{Cell}(\tau) : C_{\bullet}(\widehat{x}_{\ell}) \rightarrow C_{\bullet}(\widehat{x}_k)$ acts in generators by

$$v_{\bullet}(\widehat{x}_{\ell}) \mapsto v_{\bullet}(\widehat{x}_k), \quad u_{\bullet}(\widehat{x}_{\ell}) \mapsto u_{\bullet}(\widehat{x}_k), \quad \omega_E(\widehat{x}_{\ell}) \mapsto \omega_E(\widehat{x}_k). \quad (10.198)$$

From (10.189)–(10.190) one obtains

$$\begin{aligned} \text{Cell}(\tau) \partial &= \partial \text{Cell}(\tau), \\ \tau_V P_L &= P_{\tau_{\mathcal{J}} L} \tau_V, \quad \tau_Q B = B \tau_a, \quad W_c \circ \tau_Q = W_c, \\ \text{Cell}(\tau) \vartheta_L &= \vartheta_{\tau_{\mathcal{J}} L} \text{Cell}(\tau), \\ (\tau_V \otimes \text{Cell}_0(\tau)) \eta_{\widehat{x}_{\ell}}^{\text{cell}} &= \eta_{\widehat{x}_k}^{\text{cell}} \tau_V. \end{aligned} \quad (10.199)$$

Identity and composition are verified on each generator. Thus Cell_{TPK} is the functorial cellular completion of the same APP–TRIT–TPK fiber, not a carrier added from a subsequent publication.

10.2.5.1 Incidence measure, carry curvature, and adjoint naturality

In the following development, they are identified exclusively as abbreviations of the common completion,

$$I_{\text{gau}} := \mathcal{J}_{\widehat{x}}, \quad \mathcal{B}_{\text{gau}} := \mathcal{B}_{\widehat{x}}, \quad Q_{\text{gau}} := Q_{\text{inc}}(\widehat{x}).$$

With the uniform counting average on Q_{gau} , the function W_c of (10.117) satisfies

$$\mathbb{E}_Q W_c = 0, \quad \mathbb{E}_Q W_c^2 = \frac{2}{9}, \quad \mathbb{E}_Q W_c^3 = \frac{2}{27}. \quad (10.200)$$

These identities are properties of the same cell and the same carry, not data from an external theory.

10.2.5.2 Incidence, carry curvature, and common domain of naturalness

For $x \in C_{\text{cont}}^{\text{disc}}$, let us denote by $Y_m(x)$ the finite set of gauge prefixes of depth m that simultaneously conserve

fiber, extension relations, numbers, orientation, carry, memory,
boundary, route, multisection, survival, terminal and reader.

There exists a collar neighborhood application.

$$q_m : Y_m(x) \longrightarrow Q_{\text{gau}} \quad (10.201)$$

which records the six incidences without erasing the history that produces them. For $\ell \geq m$, the truncation is written

$$\tau_{\ell,m} : Y_\ell(x) \longrightarrow Y_m(x).$$

For $y, z \in Y_m(x)$, let

$$\text{Ext}_m^{\text{adm}}(y, z) \quad (10.202)$$

the set of *all* admissible TPK extensions from y to z . An extension belongs to this set only if it transports the entire fiber, the relations Ext_Γ , the native and operatorial carry, the orientation, the accumulated memory, the boundary, the route, and the survival condition. We define

$$a_m(y, z) := \# \text{Ext}_m^{\text{adm}}(y, z), \quad d_m(y) := \sum_{z \in Y_m(x)} a_m(y, z), \quad (10.203)$$

and, when $d_m(y) > 0$,

$$k_m(y, z) := \frac{a_m(y, z)}{d_m(y)}. \quad (10.204)$$

Thus, k_m is a normalized count of complete extensions, not a function introduced after projecting the state.

The restriction $C_{\text{cont}}^{\text{gau}}$ is formed by the surviving histories x for which the following conditions hold at every depth.

1. The sets $Y_m(x)$ are nonempty, $d_m(y) > 0$, and the truncations $\tau_{\ell,m}$ are surjective.
2. There exists a nonempty multisection of strictly positive rational weights $\mu_m : Y_m(x) \rightarrow \mathbf{Q}_{>0}$ such that

$$\sum_{y \in Y_m} \mu_m(y) = 1, \quad \mu_m(y) = \sum_{\tau_{\ell,m} \tilde{y} = y} \mu_\ell(\tilde{y}). \quad (10.205)$$

3. Each μ_m is stationary:

$$\sum_{y \in Y_m} \mu_m(y) k_m(y, z) = \mu_m(z). \quad (10.206)$$

4. The marginal of $(q_m)_* \mu_m$ is the uniform counting measure on Q_{gau} . This condition makes the calculations of (10.200) literally applicable to the lift $W_c \circ q_m$.

5. Direct lumpability holds

$$\sum_{\tau_{\ell,m} \tilde{z} = z} k_\ell(\tilde{y}, \tilde{z}) = k_m(\tau_{\ell,m} \tilde{y}, z) \quad (10.207)$$

for every $\tilde{y} \in Y_\ell$ and $z \in Y_m$.

6. For the adjoint kernel

$$k_m^*(z, y) := \frac{\mu_m(y) k_m(y, z)}{\mu_m(z)}, \quad (10.208)$$

the adjoint lumpability condition is satisfied

$$\sum_{\tau_{\ell,m}\tilde{y}=y} k_{\ell}^*(\tilde{z}, \tilde{y}) = k_m^*(\tau_{\ell,m}\tilde{z}, y). \quad (10.209)$$

7. The categories of prefixes possess the TPK root as an initial object, so that their nerve has the explicit terminal contraction of (10.227) below.

These conditions define the typed domain; outside it, a measure or natural operator is not published by mere declaration.

10.2.5.3 Projective measure, kernel, and direct and adjoint naturality

Having fixed a branch of the multisection $\{\mu_m\}_m$, let

$$H_m := \text{Fun}(Y_m, \mathbf{Q}), \quad \langle f, g \rangle_m := \sum_{y \in Y_m} \mu_m(y) f(y) g(y). \quad (10.210)$$

The operator generated by all extensions is

$$(K_m f)(y) := \sum_{z \in Y_m} k_m(y, z) f(z). \quad (10.211)$$

The Markov equations and stationarity give

$$K_m \mathbf{1} = \mathbf{1}, \quad K_m^* \mathbf{1} = \mathbf{1}, \quad \|K_m f\|_m \leq \|f\|_m. \quad (10.212)$$

The lift and the truncated expectation are

$$J_{\ell,m} f := f \circ \tau_{\ell,m}, \quad E_{m,\ell} := J_{\ell,m}^*. \quad (10.213)$$

The projectivity of μ implies

$$E_{m,\ell} J_{\ell,m} = I_{H_m}. \quad (10.214)$$

The two lumpability conditions discharge, respectively,

$$K_{\ell} J_{\ell,m} = J_{\ell,m} K_m, \quad K_{\ell}^* J_{\ell,m} = J_{\ell,m} K_m^*. \quad (10.215)$$

The second equality is not inferred from the first without the attached condition: both remain registered in the domain.

The positive internal operator is defined

$$T_m := K_m^* K_m. \quad (10.216)$$

Then

$$\langle f, T_m f \rangle_m = \|K_m f\|_m^2 \geq 0, \quad T_{\ell} J_{\ell,m} = J_{\ell,m} T_m. \quad (10.217)$$

10.2.5.4 Common law of 8 faces and 4 Fubini squares

On $Y_m^{\mathcal{B}_{\text{gau}}}$, we define the rational mass

$$\Omega_m^{(8)}(\mathbf{y}) := \sum_{z \in Y_m} \mu_m(z) \prod_{(L, \varepsilon) \in \mathcal{B}_{\text{gau}}} k_m(z, y_{L, \varepsilon}). \quad (10.218)$$

It is a probability because each factor is Markov. All faces arise from the same latent state z and the same kernel; they are not eight independently added measures.

For functions $f_{L,+}, f_{L,-} \in H_m$, finite Fubini gives

$$\begin{aligned} \sum_{\mathbf{y}} \Omega_m^{(8)}(\mathbf{y}) \prod_{L \in I_{\text{gau}}} f_{L,+}(y_{L,+}) f_{L,-}(y_{L,-}) \\ = \sum_{z \in Y_m} \mu_m(z) \prod_{L \in I_{\text{gau}}} (K_m f_{L,+})(z) (K_m f_{L,-})(z). \end{aligned} \quad (10.219)$$

Upon taking $f_{L,+} = f_{L,-} = f_L$, the four squares appear simultaneously

$$\sum_{z \in Y_m} \mu_m(z) \prod_{L \in I_{\text{gau}}} (K_m f_L(z))^2. \quad (10.220)$$

The marginal of any opposite pair satisfies

$$\sum_{y_+, y_-} \Omega_{m,L}^{(2)}(y_+, y_-) f(y_+) g(y_-) = \langle K_m f, K_m g \rangle_m = \langle f, T_m g \rangle_m. \quad (10.221)$$

The commutativity of the product in (10.218) proves

$$(\vartheta_L)_* \Omega_m^{(8)} = \Omega_m^{(8)} \quad (L \in I_{\text{gau}}). \quad (10.222)$$

10.2.5.5 Exact internal operator and self-test witness

The projection onto the constant sector and its complement are

$$P_{\Omega,m} f := \langle \mathbf{1}, f \rangle_m \mathbf{1}, \quad Q_m := I - P_{\Omega,m}. \quad (10.223)$$

The internal cell operator is defined by

$$D_m^{\text{HMT}} := 3Q_m. \quad (10.224)$$

The coefficient 3 is the cardinality of the residue field used in (10.187); it does not represent an added scale. For the lift

$$W_{c,m} := W_c \circ q_m$$

one has, by (10.200),

$$P_{\Omega,m} W_{c,m} = 0, \quad D_m^{\text{HMT}} W_{c,m} = 3W_{c,m}, \quad \|W_{c,m}\|_m^2 = \frac{2}{9}. \quad (10.225)$$

Thus, the complementary sector possesses a nonzero witness constructed by the incidence of the six edges.

10.2.5.6 Cellular terminal and strong retractor

Let $C_*(NY_m; A_n)$ be the normalized chain complex of the nerve of the prefix category, and let y_* be its initial root. The inclusion and the augmentation are

$$\iota_m : A_n[0] \longrightarrow C_*(NY_m; A_n), \quad \iota_m(1) = [y_*], \quad \epsilon_m : C_*(NY_m; A_n) \longrightarrow A_n[0].$$

The initial object provides the extra degeneracy

$$H_m[y_0 \rightarrow \cdots \rightarrow y_q] := [y_* \rightarrow y_0 \rightarrow \cdots \rightarrow y_q], \quad (10.226)$$

with the normalized convention when a degeneration appears. It is verified

$$\epsilon_m \iota_m = I, \quad \partial H_m + H_m \partial = I - \iota_m \epsilon_m, \quad H_m^2 = H_m \iota_m = \epsilon_m H_m = 0. \quad (10.227)$$

This cellular terminal is distinct from the constant projection $P_{\Omega,m}$: the former contracts the nerve, while the latter separates the sectors of the function space. Both are preserved.

The same rule defines its action on an elementary emission $\alpha : x \rightarrow y$. Since U_α transports the ordered pair of sheets, its label map is $\bar{U}_\alpha e_{\Sigma,x} = e_{\Sigma,y}$ and $\bar{U}_\alpha e_{\Pi,x} = e_{\Pi,y}$. Its projectivization $\alpha_{\mathcal{J}}[v] := [\bar{U}_\alpha v]$ acts in $\mathbf{P}^1(V_x)$ and, therefore,

$$\begin{aligned} \text{Cell}_{\text{TPK}}(\alpha)v_\star(x) &= v_\star(y), & \text{Cell}_{\text{TPK}}(\alpha)v_L(x) &= v_{\alpha_{\mathcal{J}}L}(y), \\ \text{Cell}_{\text{TPK}}(\alpha)v_{\{L,L'\}}(x) &= v_{\{\alpha_{\mathcal{J}}L, \alpha_{\mathcal{J}}L'\}}(y), & \text{Cell}_{\text{TPK}}(\alpha)v_{L,\epsilon}(x) &= v_{\alpha_{\mathcal{J}}L,\epsilon}(y), \\ \text{Cell}_{\text{TPK}}(\alpha)u_L(x) &= u_{\alpha_{\mathcal{J}}L}(y), & \text{Cell}_{\text{TPK}}(\alpha)u_{L,\epsilon}(x) &= u_{\alpha_{\mathcal{J}}L,\epsilon}(y), \\ \text{Cell}_{\text{TPK}}(\alpha)u_{L'|L}(x) &= u_{\alpha_{\mathcal{J}}L'| \alpha_{\mathcal{J}}L}(y), & \text{Cell}_{\text{TPK}}(\alpha)\omega_{\{L,L'\}}(x) &= \omega_{\{\alpha_{\mathcal{J}}L, \alpha_{\mathcal{J}}L'\}}(y). \end{aligned} \quad (10.228)$$

In the leaf fiber and on its screen, one also has

$$\bar{U}_\alpha P_L = P_{\alpha_{\mathcal{J}}L} \bar{U}_\alpha, \quad \text{Cell}_{\text{TPK}}(\alpha)\vartheta_L = \vartheta_{\alpha_{\mathcal{J}}L} \text{Cell}_{\text{TPK}}(\alpha). \quad (10.229)$$

Boundary formulas give

$$\partial \text{Cell}_{\text{TPK}}(\alpha) = \text{Cell}_{\text{TPK}}(\alpha)\partial, \quad \text{Cell}_{\text{TPK}}(\beta\alpha) = \text{Cell}_{\text{TPK}}(\beta) \text{Cell}_{\text{TPK}}(\alpha). \quad (10.230)$$

Thus the map of complexes used in the terminal register is the free action induced by the same emission, not an added symbol.

10.2.5.7 Module of paths, totalization and nonadic return cone

Let $J_d(x)$ be the complete category of prefixes of depth d of the common tree, $\text{Term}_d(x)$ its terminal set and $\Gamma_{d,N}(v) := C_{d,N}(v)$ the cellular complex of (10.183). The module, totalization and cone that follow simultaneously preserve all continuations; they do not select a history or separate a cohomological construction from the common continuum.

10.2.5.8 Module of paths, cellular totalization and nonadic terminal

Let $\text{Term}_d(x)$ be the conjunto of terminal of memory of the prefix. For $v \in J_d(x)$, definase

$$W_{d,N}(v) := A_N[\text{Path}_{J_d(x)}(v, \text{Term}_d(x))]. \quad (10.231)$$

It is the free module on all terminal routes; it selects none. If $u : v \rightarrow v'$, the contravariant action is

$$W_{d,N}(u) : W_{d,N}(v') \longrightarrow W_{d,N}(v), \quad [\lambda] \longmapsto [\lambda \circ u]. \quad (10.232)$$

Each generator preserves orientation, carry, boundary, route, and memory.

10.2.5.9 Bilateral bar complex and totalization of the bicolumn

For $q \geq 0$ and $r \in \{0, 1\}$, let

$$B_{q,r}^{d,N}(x) := \bigoplus_{v_0 \xrightarrow{u_1} \dots \xrightarrow{u_q} v_q} W_{d,N}(v_q) \otimes_{A_N} \Gamma_{d,N}(v_0)_r. \quad (10.233)$$

The bar differential is defined without abbreviation by

$$\begin{aligned} d_{\text{bar}}(w; u_1, \dots, u_q; c) &:= (w; u_2, \dots, u_q; \Gamma(u_1)c) \\ &\quad + \sum_{i=1}^{q-1} (-1)^i (w; u_1, \dots, u_{i+1}u_i, \dots, u_q; c) \\ &\quad + (-1)^q (W(u_q)w; u_1, \dots, u_{q-1}; c). \end{aligned} \quad (10.234)$$

For $q = 1$, this is exactly the relation

$$d_{\text{bar}}(w; u; c) = w \otimes \Gamma(u)c - W(u)w \otimes c. \quad (10.235)$$

A transported incidence is thereby identified with the same incidence written in the refined chart; the relation is not concealed beneath the name of the totalizer.

The total complex is

$$K_{d,N}(x) := \text{Tot } B_{\bullet,\bullet}^{d,N}(x), \quad D_{\text{tot}} := d_{\text{bar}} + (-1)^q \partial_{\Gamma} \quad \text{on } B_{q,r}^{d,N}. \quad (10.236)$$

The functoriality of W and Γ , together with $\partial_{\Gamma}^2 = 0$, gives

$$D_{\text{tot}}^2 = 0. \quad (10.237)$$

In compact notation, but now already discharged by (10.233)–(10.234),

$$K_{d,N}(x) \simeq W_{d,N} \otimes_{A_N[J_d(x)]}^{\mathbf{L}} \Gamma_{d,N}. \quad (10.238)$$

10.2.5.10 Depth and terminal nonadic colimit

The prefixes form the colimit.

$$K_{\infty,N}(x) := \varinjlim_d K_{d,N}(x). \quad (10.239)$$

The nine-phase extension carries depth d to depth $d + 9$. Only after forming (10.239) does it induce a well-typed endomorphism

$$U_{\Gamma_9,N} : K_{\infty,N}(x) \longrightarrow K_{\infty,N}(x). \quad (10.240)$$

The terminal is

$$\mathcal{T}_{\text{coh},N}(x) := \text{Cone}(I - U_{\Gamma_9,N}). \quad (10.241)$$

With the homological convention

$$\text{Cone}(f)_q = K_{\infty,N,q-1} \oplus K_{\infty,N,q},$$

its differential is

$$\partial_{\text{Cone}}(a, b) = (-\partial_K a, \partial_K b + f(a)). \quad (10.242)$$

Although the visible phase of h and $\Gamma_9 h$ coincides, the generators $[h]$ and $[\Gamma_9 h]$ are different because their ledgers and depths are different. The terminal object preserves

$$(I - U_{\Gamma_9,N})[h] = [h] - [\Gamma_9 h]. \quad (10.243)$$

Precision reductions commute with the terminal:

$$r_{N+1,N} U_{\Gamma_9,N+1} = U_{\Gamma_9,N} r_{N+1,N}. \quad (10.244)$$

10.2.5.11 Terminal cohomological reader: incidence cycle and class in $H_0(\text{Cone}(I - U_{\Gamma_9,N}))$

Each cell of $x_{\leq d}$ carries a vector of incidence

$$\omega_{v,N}(x) \in A_N^{I_v \times J_v} = C_{v,N,0}$$

and the complete terminal route $\lambda_{v,x}$. The raw cycle is

$$z_{d,N}(x) := \sum_{v \in \text{Cell}_d(x)} \epsilon_v[\lambda_{v,x}] \otimes \omega_{v,N}(x) \in B_{0,0}^{d,N}(x). \quad (10.245)$$

As the total complex is homologically non-negative,

$$D_{\text{tot}z_{d,N}}(x) = 0. \quad (10.246)$$

Both the cycle and its class are preserved.

$$[z_{d,N}(x)] \in H_0(K_{d,N}(x)).$$

After including it in $K_{\infty,N}$, its terminal class is

$$[(0, z_{d,N}(x))] \in H_0(\mathcal{T}_{\text{coh},N}(x)).$$

The complete reader is

$$\mathcal{L}_{\text{coh},d,N}(x) := (z_{d,N}(x), [z_{d,N}(x)], U_{\Gamma_9,N}z_{d,N}(x), [(0, z_{d,N}(x))]). \quad (10.247)$$

The publication is not merely a quotient class: the representative, the route that produced it, its continuation and the terminal remain.

10.2.5.12 Surviving fibres and corrections without a selector

Let $\mathcal{S}_\lambda^{\text{coh}}$ be the free A_N -module generated by the admissible finite histories at level $\lambda = (d, N)$. The reader extends linearly to

$$e_\lambda : \mathcal{S}_\lambda^{\text{coh}} \longrightarrow \mathcal{O}_\lambda^{\text{coh}}, \quad \mathcal{O}_\lambda^{\text{coh}} := H_0(K_{d,N}) \oplus H_0(\mathcal{T}_{\text{coh},N}). \quad (10.248)$$

The surviving codomain is

$$\mathcal{O}_\infty^{\text{surv}} := \text{Im} \left(e_\infty : \mathcal{C}_{\text{cont}}^{\text{coh}} \longrightarrow \varprojlim_{\lambda} \mathcal{O}_\lambda^{\text{coh}} \right). \quad (10.249)$$

For $a = (a_\lambda)_\lambda \in \mathcal{O}_\infty^{\text{surv}}$, define

$$\mathcal{F}_\lambda(a) := \left\{ s_\lambda \in \mathcal{S}_\lambda^{\text{coh}} : \begin{array}{l} e_\lambda(s_\lambda) = a_\lambda, \\ \exists y \in \mathcal{C}_{\text{cont}}^{\text{coh}} \text{ with } y|_\lambda = s_\lambda \text{ and } e_\infty(y) = a \end{array} \right\}. \quad (10.250)$$

These fibers are nonempty by the definition of the surviving codomain. If $\lambda' \geq \lambda$, the restriction is surjective:

$$\rho_{\lambda',\lambda} : \mathcal{F}_{\lambda'}(a) \twoheadrightarrow \mathcal{F}_\lambda(a). \quad (10.251)$$

Indeed, a complete history that witnesses $s_\lambda \in \mathcal{F}_\lambda(a)$ provides by truncation an antecedent at any later level. That antecedent is not chosen as part of the object.

The complete multisect is

$$\Sigma_{\lambda',\lambda}^{\text{coh}}(s_\lambda; a) := \{s_{\lambda'} \in \mathcal{F}_{\lambda'}(a) : \rho_{\lambda',\lambda}s_{\lambda'} = s_\lambda\}. \quad (10.252)$$

For a provisional prolongation $\widetilde{s}_{\lambda'}$, let

$$\delta_{\lambda'} := a_{\lambda'} - e_{\lambda'}(\widetilde{s}_{\lambda'}). \quad (10.253)$$

The relation of all corrections is

$$\text{Corr}_{\lambda',\lambda}(\tilde{s}_{\lambda'}, \delta_{\lambda'}) := \left\{ \eta \in \mathcal{S}_{\lambda'}^{\text{coh}} : \begin{array}{l} \rho_{\lambda',\lambda}\eta = 0, \\ e_{\lambda'}(\eta) = \delta_{\lambda'}, \\ \eta \text{ is supported on new cells,} \\ \eta \text{ preserves orientation, carry, memory and terminal} \end{array} \right\}. \quad (10.254)$$

If two lifts are comparable, their difference $\eta = s_{\lambda'} - \tilde{s}_{\lambda'}$ belongs to this relation exactly when it satisfies the four preceding conditions. It is not asserted that (10.254) is nonempty for an arbitrary discrepancy: the governing non-emptiness is that of (10.252) on the surviving domain.

The local complex (10.110) and the vertex y_α , endpoint of that same extension in the nerve, form the augmented pullback. In it are defined

$$\mathbf{1}_\alpha = (r, [y_\alpha]), \quad z_-(\alpha) = (r, [y_\alpha]), \quad (10.255)$$

$$z_+(\alpha) = (r + 3c_\alpha v_\alpha e + v_\alpha b, [y_\alpha]), \quad h_*(\alpha) = (-v_\alpha f, 0). \quad (10.256)$$

The action on generators gives, without an additional abstract existence,

$$\partial z_- = \partial z_+ = 0, \quad \partial h_* = z_- - z_+. \quad (10.257)$$

The correlative filling, the cell, and the collar preserve the same $(\alpha, \text{or}(\alpha), \kappa(\alpha), \partial\alpha, \gamma\alpha, m(\gamma\alpha; m_0))$; they merely apply different internal readers to the common record.

For the same finite based complex, the total determinant line is defined, without requiring balance or acyclicity.

$$\text{Det}(C_{k,N}) := \bigotimes_q \left(\bigwedge^{\text{top}} C_{k,N,q} \right)^{(-1)^q}. \quad (10.258)$$

Refinement induces the determinantal morphism of the map of complexes and simultaneously transports homology modules, Smith normal forms, and Bockstein pages as fields of the register. Neither the existence of (10.258) nor its transport requires homology to vanish. If the cone of refinement is contractible and oriented, the morphism is canonically trivialized; in the general case the cone is preserved and not replaced by a fictitious isomorphism.

10.2.5.13 Smith Normal Form of the Terminal Operator and Primary Decomposition of Its Cokernel

For each prefix y , the unoriented carry and the oriented coefficient of boundary of the common register admit the lifts

$$\tilde{c}_y, \tilde{v}_y \in \mathbf{Z}, \quad c_{y,n} := \tilde{c}_y \bmod 3^n, \quad v_{y,n} := \tilde{v}_y \bmod 3^n. \quad (10.259)$$

The determinantal output is neither an input nor a separate branch: it is the sector produced by the same terminal reduction when the typed conditions are verified

$$\begin{aligned} C_{\text{cont}}^{\text{det}} := \{ & x \in C_{\text{cont}}^{\text{disc}} : x \text{ survives at every depth;} \\ & \widehat{P}_x^{\text{red}} \text{ is finite and based;} \\ & \chi(\widehat{P}_x^{\text{red}}) = 0; \\ & H_*(\widehat{P}_x^{\text{red}} \otimes_{\mathbf{Z}_3} K_3) = 0; \\ & \text{the reduction produces a square block stable under refinement} \}. \end{aligned} \quad (10.260)$$

10.2.5.14 Lifting of the cokernel through the Bockstein tower and preservation of carry

Only the common root is eliminated:

$$\overline{P}_{y;m,n} := P_{y;m,n}^{\text{gen}} / A_n \mathbf{1}_y, \quad \widehat{P}_{y;m}^{\text{red}} := \varprojlim_n \overline{P}_{y;m,n}. \quad (10.261)$$

The base is first ordered by the TPK order of routes and then by

$$f \prec e \prec b \prec g. \quad (10.262)$$

The following internal algorithm is executed.

1. Find the coefficients that are units of $\mathbf{Z}_3$.
2. Choose the first according to (10.262) and the order of paths.
3. Cancel the corresponding pair and record the elementary operations, their sign, and their contribution to the orientation.
4. Repeat until no cancellable unit pivot remains.

The survival condition of (10.260) requires that, cofinally in m , the result be a two-term block

$$\mathbf{Z}_3^{r_x} \xrightarrow{S_x} \mathbf{Z}_3^{r_x}, \quad \det S_x \neq 0 \text{ in } K_3, \quad (10.263)$$

and that a refinement only add unit contractible blocks and recorded changes of basis. If these conditions fail, the history does not belong to $C_{\text{cont}}^{\det}$; a nonexistent terminal is not forced.

The identity route provides an elementary witness. After removing the root, the local complex is

$$A_n f \xrightarrow{\begin{pmatrix} 3c \\ 1 \end{pmatrix}} A_n e \oplus A_n b \xrightarrow{(1, -3c)} A_n g. \quad (10.264)$$

Is contractible via

$$s(g) = e, \quad s(ae + \beta b) = \beta f, \quad s(f) = 0. \quad (10.265)$$

Thus, the terminal conditions admit at least the internal identity witness and do not describe a formally empty subdomain.

10.2.5.15 Smith, 3-adic order and initial unit

In oriented bases, a valuation diagonalization has the form

$$U_x S_x V_x = \text{diag}(3^{a_1} u_1, \dots, 3^{a_r} u_r), \quad a_i \in \mathbf{N}, \text{quad} u_i \in \mathbf{Z}_3^\times. \quad (10.266)$$

We define

$$d_x := v_3(\det S_x) = \sum_{i=1}^r a_i \quad (10.267)$$

and the initial unit

$$\lambda_x := 3^{-d_x} \det S_x \in \mathbf{Z}_3^\times. \quad (10.268)$$

The orientation coordinate trivializes the determinant line. If a transition produces

$$S_x \oplus I_q = U S_{x'} V, \quad (10.269)$$

the orientation is transported by the inverse factor of $\det U \det V$. The oriented unit λ_x^{or} is then natural and does not change when contractible blocks are added.

At finite precision,

$$S_{x,n} = S_x \bmod 3^n, \quad a_i^{(n)} = \min(a_i, n). \quad (10.270)$$

The terminal is, therefore, the compatible family of all precisions and not an isolated integer.

10.2.5.16 Bockstein Tower and Higher Order Carry

Let

$$C_x^1 \xrightarrow{S_x} C_x^0$$

the terminal block. If a class $[\bar{y}]$ admits a lift $y \in C_x^1$ with $S_x y \in 3^q C_x^0$, we define

$$\beta_q[\bar{y}] := [3^{-q} S_x y \bmod 3]. \quad (10.271)$$

Changing the lift alters the representative by a boundary of the corresponding page, so β_q is well defined. The higher carry is

$$\text{Carr}_q(y) := 3^{-q} S_x y. \quad (10.272)$$

It does not replace the TPK carry of (10.259); it records its prolongation in the coefficient tower.

For a block $3^{a_i} u_i$, the class persists until page a_i , and on that page the nonzero differential is multiplication by $\bar{u}_i \in \mathbf{F}_3^\times$. Consequently,

$$\{\text{survival lengths}\} = \{a_1, \dots, a_r\}, \quad d_x = \sum_i a_i. \quad (10.273)$$

The oriented product of the page trivializations satisfies

$$\Theta_{\text{Bock}}(x) = \overline{\lambda_x^{\text{or}}} \in \mathbf{F}_3^\times. \quad (10.274)$$

The proof is carried out in each diagonal block and transported by the recorded unimodular changes. Smith, Bockstein and the initial unit are thus three readings of the same terminal.

10.2.6 Naturalness of the ensemble and compatibility under refinement

Before forming the horizon record, we make explicit the total action of the five operations developed here on *the same* edge $\mathfrak{e}_\alpha$. If P_x, Q_x denote the restrictions of (10.123) to the fiber of $x = s(\alpha)$, and analogously in $y = t(\alpha)$, we first set

$$A_\alpha := P_y U_\alpha P_x + Q_y U_\alpha Q_x, \quad \eta_\alpha := Q_y U_\alpha P_x + P_y U_\alpha Q_x. \quad (10.275)$$

The five readings are dependent families indexed by the $\mathfrak{e}_\alpha$ itself. The index fixes source, target, route, and ledger, but is not copied again within the value. Their types are the dependent records $\text{LedgerEntry} := \text{cod}(\lambda)$ and

$$\begin{aligned} Y_{\text{sp}}[\alpha] &:= \text{Rec}[U : H_x \rightarrow H_y; \sigma_x \in \text{End } H_x; \sigma_y \in \text{End } H_y; A, \eta : H_x \rightarrow H_y], \\ Y_{\text{sol}}[\alpha] &:= \text{Rec}[b, b^+, b^- \in \text{LedgerEntry}; \text{upd} : \text{Mem}_x \rightarrow \text{Mem}_y], \\ Y_{\text{gau}}[\alpha] &:= \mathcal{P}_{\text{fin}} \text{Rec}[\text{Cell}(\alpha); q \in \mathcal{Q}_{\text{inc}}(y); B; W_c(q)], \\ Y_{\text{coh}}[\alpha] &:= \text{Rec}[I_y, J_y, \kappa_{y,N}, \delta_y, C_{y,N}; F_\alpha : C_{y,N} \rightarrow C_{x,N}], \\ Y_{\text{det}}[\alpha] &:= \text{Rec}[\Psi_\alpha, K_\alpha, \mathbf{1}_\alpha, z_\pm, h_*; \text{Det}, H_*, \text{Smith}, \text{Boc}]. \end{aligned}$$

Therefore, the common judgment is

$$\mathfrak{e}_\alpha : \text{Ext}_k \vdash O_\bullet(\mathfrak{e}_\alpha) \in Y_\bullet[\alpha], \quad \bullet \in \{\text{sp}, \text{sol}, \text{gau}, \text{coh}, \text{det}\}.$$

$$O_{\text{sp}}(\mathfrak{e}_\alpha) := (U_\alpha, \sigma_x, \sigma_y, A_\alpha, \eta_\alpha), \quad (10.276)$$

$$O_{\text{sol}}(\mathfrak{e}_\alpha) := (b_\alpha, b_\alpha^+, b_\alpha^-; (M^+, M^-, D, H) \mapsto (M^+ + b_\alpha^+, M^- + b_\alpha^-, D + b_\alpha, H + |b_\alpha|)), \quad (10.277)$$

$$O_{\text{gau}}(\mathfrak{e}_\alpha) := \{(\text{Cell}_{\text{TPK}}(\alpha), q, B, W_c(q)) : q \in \mathcal{Q}_{\text{inc}}(y)\}, \quad (10.278)$$

$$\mathcal{O}_{\text{coh}}(\mathbf{e}_\alpha) := (I_y, J_y, \kappa_{y,N}, \delta_y, C_{y,N}, F_\alpha : C_{y,N} \longrightarrow C_{x,N}), \quad (10.279)$$

$$\mathcal{O}_{\text{det}}(\mathbf{e}_\alpha) := (\Psi_\alpha, K_\alpha, \mathbf{1}_\alpha, z_\pm(\alpha), h_*(\alpha), \text{Det}(C_{y,N}), H_*(C_{y,N}), \text{Smith}(C_{y,N}), \text{Boc}(C_{y,N})). \quad (10.280)$$

The internal Archimedean reading is the first component of this simultaneous action; it does not yet introduce any classical form:

$$\mathcal{L}_{\text{arq}}^{\text{int}}(\mathbf{e}_\alpha) := \mathcal{O}_{\text{sp}}(\mathbf{e}_\alpha). \quad (10.281)$$

For every value in those families, including each element of the multisection $\mathcal{O}_{\text{gau}}$, its support is the dependent index

$$\begin{aligned} \mathbf{B}_\alpha &:= \{(s(\alpha), t(\alpha), \gamma\alpha, \text{Led}(\gamma\alpha))\}, \\ \text{supp}_\alpha : \mathbf{Y}_\bullet[\alpha] &\longrightarrow \mathbf{B}_\alpha, \quad o \longmapsto (s(\alpha), t(\alpha), \gamma\alpha, \text{Led}(\gamma\alpha)). \end{aligned} \quad (10.282)$$

Therefore, they remain typed, without a projection that recovers the edge,

$$\begin{aligned} s_\alpha(\mathcal{O}_\bullet) &:= s(\alpha), & t_\alpha(\mathcal{O}_\bullet) &:= t(\alpha), \\ \gamma_\alpha(\mathcal{O}_\bullet) &:= \gamma\alpha, & \text{Led}_\alpha(\mathcal{O}_\bullet) &:= \text{Led}(\gamma\alpha). \end{aligned} \quad (10.283)$$

Here $\text{Cell}_{\text{TPK}}(\alpha)$ is the map of complexes (10.228) induced by transport of the same sheet fiber, and F_α is (10.184) for that extension. In particular, $\mathcal{O}_{\text{gau}}$ preserves the complete multisection of q 's; it does not choose the origin or an orbit.

The five formulas above constitute a single dependent action.

$$\mathcal{O}(\mathbf{e}_\alpha) := \langle \mathcal{O}_{\text{sp}}, \mathcal{O}_{\text{sol}}, \mathcal{O}_{\text{gau}}, \mathcal{O}_{\text{coh}}, \mathcal{O}_{\text{det}}; \text{Cpl}_\alpha \rangle, \quad (10.284)$$

where Cpl_α is the certificate formed by the identities

$$\begin{aligned} \text{supp}_\alpha(\mathcal{O}_{\text{sp}}) &= \text{supp}_\alpha(\mathcal{O}_{\text{sol}}) = \text{supp}_\alpha(\mathcal{O}_{\text{gau}}) = \text{supp}_\alpha(\mathcal{O}_{\text{coh}}) = \text{supp}_\alpha(\mathcal{O}_{\text{det}}), \\ \mathbf{Q}(t(\alpha)) &= \sigma_9 \mathbf{Q}(s(\alpha)), \quad g(t(\alpha)) = g(s(\alpha)) + [1]_9, \quad q^{(9)}(t(\alpha)) = q^{(9)}(s(\alpha)) + \chi_9(g(s(\alpha))), \\ A_\alpha^* A_\alpha + \eta_\alpha^* \eta_\alpha &= U_\alpha^* U_\alpha, \quad (M^+, M^-, D, H)' = (M^+ + b_\alpha^+, M^- + b_\alpha^-, D + b_\alpha, H + |b_\alpha|), \\ Q_{\text{inc}} &= C^2(\text{Cell}_{\text{TPK}}; \mathbf{F}_3), \quad sB = 0, \quad W_c(q + Ba) = W_c(q), \\ C_{y,N} &\text{ is the same complex in } \mathcal{O}_{\text{coh}} \text{ and } \mathcal{O}_{\text{det}}, \quad \partial h_* = z_- - z_+, \quad K_{\beta\alpha} = K_\beta + \Psi_\beta K_\alpha. \end{aligned} \quad (10.285)$$

The symbol $\bullet$ ranges over the five readings of (10.284); all preserve the same source, destination, route, and ledger.

We collect, without separating the state, the operations constructed up to here:

$$\begin{aligned}
 \mathfrak{D}_{k,N}(\widehat{x}) := & \{ \mathfrak{e}_\alpha \}_\alpha; \\
 & U_k, \sigma_k, P_k, Q_k, A_k, \eta_k; \\
 & b_\alpha, b_\alpha^\pm, M_k^\pm, D_k^0, H_k^0, \kappa_k^{\text{can}}, \\
 & V_{\text{TPK}}, \mathcal{J}, \mathcal{E}, \mathcal{B}, B; \\
 & \text{Cell}_{\text{TPK}}(\widehat{x}), Q_{\text{inc}}(\widehat{x}), W_c; \\
 & I_k, J_k, \kappa_{k,N}, \delta, C_{k,N}; \\
 & c_\alpha, v_\alpha, \Psi_\alpha, K_\alpha, \mathbf{1}_\alpha, z_\pm(\alpha), h_*(\alpha); \\
 & \text{Det}(C_{k,N}), H_*(C_{k,N}), \text{Smith}(C_{k,N}), \text{Boc}(C_{k,N}); \\
 & \{O(\mathfrak{e}_\alpha)\}_\alpha.
 \end{aligned} \tag{10.286}$$

The semicolon merely separates levels of interpretation of the same dependent tuple; it does not define independent structures. The input state is neither retained as the first coordinate nor recovered through a projection. Each operation is proved through its action on the generators $\mathfrak{e}_\alpha$, which preserve the same support, the same route, and the same ledger.

For $\ell \geq k$ and $N' \geq N$, transport is defined component by component through truncation of histories, reduction of coefficients, and precomposition of ports. The naturality assertion that must be proved is a single one:

$$u_{(\ell,N'),(k,N)} \mathfrak{D}_{\ell,N'}(\widehat{x}_\ell) = \mathfrak{D}_{k,N}(\tau_{\ell,k} \widehat{x}_\ell). \tag{10.287}$$

Naturality is formulated over the generated history, not by pretending that an isolated fine edge must remain an edge after truncating its two endpoints. For $\gamma = (\varepsilon_1, \dots, \varepsilon_\ell)$, we define

$$\mathbf{E}_{\leq \ell}(\gamma) := (\mathfrak{e}_{\varepsilon_i})_{1 \leq i \leq \ell}, \quad \mathbf{O}_{\leq \ell}(\gamma) := (O(\mathfrak{e}_{\varepsilon_i}))_{1 \leq i \leq \ell}.$$

The two transports are the explicit prefix maps.

$$u_{\ell,k}^{\mathfrak{e}} : \mathbf{E}_{\leq \ell}(\gamma) \longrightarrow \mathbf{E}_{\leq k}(\gamma), \quad (\mathfrak{e}_{\varepsilon_i})_{i \leq \ell} \longmapsto (\mathfrak{e}_{\varepsilon_i})_{i \leq k}, \tag{10.288}$$

$$u_{\ell,k}^O : \mathbf{O}_{\leq \ell}(\gamma) \longrightarrow \mathbf{O}_{\leq k}(\gamma), \quad (O(\mathfrak{e}_{\varepsilon_i}))_{i \leq \ell} \longmapsto (O(\mathfrak{e}_{\varepsilon_i}))_{i \leq k}. \tag{10.289}$$

In each retained entry, the component sp intertwines U, σ, P, Q, A, η ; the component sol applies the conditional expectation to $b, b^\pm, \kappa$; the component gau applies $\text{Cell}(\tau) \oplus \tau_Q$; the component coh precomposes the ports with F ; and the component det applies $\Psi, K, H_*, \text{Smith}, \text{Boc}$ to the same map of complexes. The prefix acts simultaneously on those five components. Its components already discharged are

$$\begin{aligned}
 U_{m,k} &= U_{m,\ell} U_{\ell,k} \quad (m \geq \ell \geq k), & E_k b_f^\pm &= b_c^\pm + \kappa_k^{\text{can}}, \\
 r_{N',N} \kappa_{k,N'} &= \kappa_{k,N} (r_{N',N} \oplus r_{N',N}), \\
 \Psi_\beta \Psi_\alpha &= \Psi_{\beta\alpha}, & K_{\beta\alpha} &= K_\beta + \Psi_\beta K_\alpha, \\
 u_{\ell,k}^O \mathbf{O}_{\leq \ell}(\gamma) &= \mathbf{O}_{\leq k}(\gamma) = \text{map}(O)(u_{\ell,k}^{\mathfrak{e}} \mathbf{E}_{\leq \ell}(\gamma)).
 \end{aligned} \tag{10.290}$$

The first equality uses multiplication of the conditional weights over the same system of common cylinders; the remaining ones are the preceding explicit formulas. No visible projection can be used to reconstruct or select $\widehat{x}$.

To complete the last component, let us denote by $F_{(\ell,N'),(k,N)} : C_{\ell,N'} \rightarrow C_{k,N}$ the map of complexes of (10.184). Its cone is preserved as part of the transition. Functoriality of homology and of the Bockstein sequence gives

$$H_*(F_{m,k}) = H_*(F_{\ell,k})H_*(F_{m,\ell}), \quad \text{Boc}(F_{m,k}) = \text{Boc}(F_{\ell,k})\text{Boc}(F_{m,\ell}). \quad (10.291)$$

The determinant line is transported by the identity of the exact triangle.

$$\text{Det}(C_{k,N}) \simeq \text{Det}(C_{\ell,N'}) \otimes \text{Det}(\text{Cone } F_{(\ell,N'),(k,N)}). \quad (10.292)$$

This formula also records the refinements that add homology. The Smith reading is conserved without feigning invariance: each transition is assigned the total certificate

$$S(F) := (\text{Smith}(C_{\ell,N'}), \text{Smith}(C_{k,N}), \text{Smith}(\text{Cone } F)), \quad (10.293)$$

which literally determines which summands are conserved, born, or die.

The action of a coherent symmetry g on the common tree induces permutation isometries $\tilde{g}_k$ on $\mathcal{H}_k$. The bijection of children and preservation of the weights of (10.96) prove, on the basis vectors,

$$\tilde{g}_{k+1}U_k = U_k\tilde{g}_k, \quad \tilde{g}_k\sigma_k = \sigma_k\tilde{g}_k, \quad \tilde{g}_kP_k = P_k\tilde{g}_k, \quad \tilde{g}_kQ_k = Q_k\tilde{g}_k. \quad (10.294)$$

Theorem 10.34 (Totality and naturality of the correlative constructor). *For every total state of (10.56), every finite horizon, and every coefficient level, the expression (10.286) is defined. Its five characteristics are simultaneous operations on the same extension ledger and satisfy the single identity (10.287). The collar, the complex, and the determinant line remain nonempty even when their subsequent readers are neither acyclic nor invertible.*

Proof. Nonemptiness of the tree and its ports is proposition 10.15; the measure and the expectation are total by theorem 10.28. The formulas (10.120)–(10.138) define transport, polarity, balance, and memory for all generators. The four directions of (10.186) are distinct by (10.188); the cellular completion definition 10.33 contains its two path orders and six cells, while (10.199) proves its naturality. The cellular sequence is exact by (10.108), the local complex satisfies $\partial^2 = 0$, and the correlated filling is explicit by (10.257). Every finite based construction possesses the line (10.258); homology, Bockstein, and the Smith certificate are transported by (10.291)–(10.293). Finally, the identities are verified on each generator e_α through (10.290), (10.199), and (10.294); by linearity and by the functorial composition of routes, (10.287) is obtained for the integral record. $\square$

10.2.7 Causal separation of the identities, deductions, and realizations of the correlator constructor

No vacuity and exact identities of the correlator constructor The nonemptiness imported from (10.64) and the identities (10.104), (10.108), (10.116), (10.121), (10.126), (10.137), (10.111), (10.174), and (10.257) follow from the definitions and the exhibited APP–TRIT–TPK relations. Their status is EXACT_INTERNAL.

Starting hypotheses for the isometry of U_k , the cellular sequence, and the composition Ψ/K The construction of U_k uses as its starting structure the finiteness of the fibers, survival at every horizon, functional truncation, and the transports of the two sheets already produced by TPK. The equality $U_k^*U_k = I$ additionally requires verification of $U_\alpha^*U_\alpha = I$ for every emission; without that verification, the Gram identity (10.121), which is unconditional, remains. The cellular succession uses the two ordered APP ports. The composition Ψ/K uses integer carry, orientation, and boundary. Under those enumerated premises, the stated conclusions are affirmative.

Additional data for isometric realizations, balanced retracts, and trivialization of the determinant line The internal constructor is completed by the positive Gram form of (10.121), the total space Q_{inc} , the cellular sequence, and the determinant line with its transition cone. A specialized publication may

require additional data: the isometry of each U_α , a concrete numerical identification of the APP defect, a balanced acyclic retraction, or trivialization of a determinant line. Those data are intertwiners of the published codomain; they do not form part of the domain or of the existence of $\mathfrak{D}_{k,N}$. Their absence precludes exclusively the promotion that mentions them.

Test criterion 10.35 (Falsifiers of the intrinsic correlation). The joint construction fails at the first level at which any of the following facts occurs:

1. an operation uses an extension, orientation, carry, memory, boundary or route different from that recorded in $\mathfrak{e}_\alpha$;
2. a child of $\text{Ch}_k(\widehat{x}_k)$ is chosen and the remaining ones are eliminated;
3. the weight of (10.118) does not sum to one, (10.121) fails, or $U_k^* U_k = I$ is declared without proving $U_\alpha^* U_\alpha = I$ for every emission;
4. κ_k^{can} is identified with the integer carry κ_α ;
5. $sB = 0$ fails, $Q_{\text{inc}}(\widehat{x})$ is contracted to its origin, the action $q \mapsto q + Ba$ is altered, or the two path orders are identified before applying the reader;
6. a refinement breaks (10.185) or the cocycle (10.174);
7. fails $\partial^2 = 0$, $z_\pm$ are not ciclos or $\partial h_* \neq z_- - z_+$;
8. a nonzero determinantal scalar is asserted without the retract, the balance, and the acyclicity required by that publication;
9. any of the five operations is presented as an object separate from the constructor $\mathfrak{D}_{k,N}$.

In all those cases the diagnosis is: *common object substituted; conclusion not transferable*.

10.3 Multivalued terminal, unique naturality and passage to the limit

Tesis rectora

The discrete continuum produces a single system of histories, extensions, memories, and terminals. The internal discriminants sp , sol , gau , coh , det name five characteristics that are simultaneous within that same system: they neither fix distinct inputs nor authorize the selection of different histories. In this chapter, the terminal preserves all continuations, the connection remains a nonadic correspondence, and naturality is expressed by a single identity.

10.3.1 Surviving histories with extension witnesses

Let $\widehat{x}_k \in \widehat{\mathcal{X}}_k^{\text{tot}}$ be the total state of (10.57). For $N \geq k$, we define without any other predicate of membership

$$\mathcal{H}_{k,N}(\widehat{x}_k) := \text{Term}_{k,N}(\widehat{x}_k). \quad (10.295)$$

Each $h = (\varepsilon_{k+1}, \dots, \varepsilon_N)$ on the right contains through (10.65) its intermediate states, the witnesses of Ext , the carries, boundaries, routes, and ledger. Each edge satisfies the phase action (10.17); hence every nine-step subpath determines, with its nine emissions and not merely with its length, an element of $\mathbb{Z}_{9,j}^\Gamma$ and a witness of $\Gamma_{9,j}$ through (10.71)–(10.73). Therefore, Ext and nonadic holonomy arise from the same phase-compatible word. To (10.295) there is adjoined neither a survival filter nor a second condition Γ_9 .

For $k \leq \ell \leq N$, let

$$\mathcal{P}_{\ell|k}(\widehat{x}_k) := \{h_{[k,\ell]} : h \in \mathcal{H}_{k,N}(\widehat{x}_k) \text{ for some } N \geq \ell\}. \quad (10.296)$$

For $p \in \mathcal{P}_{\ell|k}(\widehat{x}_k)$, we use the abbreviation

$$\mathcal{H}_{\ell,N}(p) := \{q : p \star q \in \mathcal{H}_{k,N}(\widehat{x}_k)\}. \quad (10.297)$$

The prolongation neutral of (10.59) extends each prefix to cualquier horizon. Consequently,

$$\mathcal{H}_{k,N}(\widehat{x}_k) \neq \emptyset, \quad \mathcal{P}_{\ell|k}(\widehat{x}_k) \neq \emptyset, \quad (10.298)$$

by proposition 10.15.

The truncation

$$\pi_{N+1,N}^{\mathcal{H}} : \mathcal{H}_{k,N+1}(\widehat{x}_k) \longrightarrow \mathcal{H}_{k,N}(\widehat{x}_k) \quad (10.299)$$

is exactly (10.68) and is surjective by lemma 10.16. Survival is, by (10.83), a demonstrated output of the total domain and not a prior selection.

10.3.2 Generated terminal record and its 13 coordinates

Let us fix a history

$$h = (\varepsilon_{j+1} : x_j \longrightarrow x_{j+1})_{j=k}^{N-1}, \quad \gamma_{k,k} := I_{x_k}, \quad \gamma_{k,j+1} := \gamma_{k,j} \star \varepsilon_{j+1},$$

The operator notation $U_{\gamma_{k,j+1}} = U_{\varepsilon_{j+1}} U_{\gamma_{k,j}}$ preserves the order of application, whereas $\star$ preserves the chronological order of the word. We further fix a coefficient level $r \geq 1$. For brevity, we write

$$C_{x_j,r} := C_{j,r}(x_j), \quad F_{\varepsilon_{j+1}} : C_{x_{j+1},r} \longrightarrow C_{x_j,r}$$

for the complex and the map of complexes of (10.183) and (10.184). The type of the level record j , denoted $\mathfrak{R}_{j,r}$, is a dependent record with the following thirteen fields. In each field we indicate the type, the seed and the action of an edge $\varepsilon := \varepsilon_{j+1}$.

The state x_k already contains the complete word $\gamma_{0,k} = \varepsilon_1 \star \cdots \star \varepsilon_k$. Therefore a “seed in k ” never restarts the genealogy: it is the fold of the same updates from the root to k . We denote

$$\begin{aligned} M_k^{\pm} &:= \sum_{i=1}^k b_{\varepsilon_i}^{\pm}, \quad qquad D_k := \sum_{i=1}^k b_{\varepsilon_i}, \quad qquad H_k := \sum_{i=1}^k |b_{\varepsilon_i}|, \\ \mathbf{c}_{\leq k} &:= (c_r(\varepsilon_i, \gamma_{0,i-1}))_{1 \leq i \leq k}, \quad \mathbf{k}_{\leq k} := (\kappa_{i-1}^{\text{can}})_{1 \leq i \leq k}, \\ \mathbf{F}_{\leq k} &:= (F_{\varepsilon_i} : C_{x_i,r} \rightarrow C_{x_{i-1},r})_{1 \leq i \leq k}. \end{aligned} \quad (10.300)$$

All empty lists appearing below are understood literally only in $k = 0$; for $k > 0$ they are replaced by prefixes inherited from (10.300) and from the ledger of $\gamma_{0,k}$. Thus the record does not preserve an opaque copy of x_k , but neither does it lose its APP–TRIT–TPK past.

Fiber F: local domain and conservation under prolongation The elementary fiber is

$$\text{Fib}_r(x) := (H_x^{\Sigma} \oplus H_x^{\Pi}, M_x, \text{Cell}_{\text{TPK}}(x), C_{x,r}). \quad (10.301)$$

The field $\mathbf{F}_{k,j}$ is a word in the category of these fibers. Its seed and update are

$$\mathbf{F}_{k,k} := (\text{Fib}_r(x_0), I) \star \star_{i=1}^k (\text{Fib}_r(x_i), U_{\varepsilon_i}, \mathbf{C}_{\varepsilon_i}, \text{Cell}_{\text{TPK}}(\varepsilon_i), F_{\varepsilon_i}), \quad (10.302)$$

$$\text{Upd}_\varepsilon^F F_{k,j} := F_{k,j} \star (\text{Fib}_r(x_{j+1}), U_\varepsilon, \mathbf{C}_\varepsilon, \text{Cell}_{\text{TPK}}(\varepsilon), F_\varepsilon). \quad (10.303)$$

It is nonempty because the two sheets, the memory, the cellular unit (10.197), and the port complex have already been constructed. The composition of its five maps proves naturality.

Extensions E: morphisms of compatible prolongation of the state The codomain is the category of typed relation words. We set

$$\mathbf{E}_{k,k} := (I_{x_0}, \text{Ext}_{I_{x_0}}) \star \bigstar_{i=1}^k (\varepsilon_i, \text{Ext}_{\varepsilon_i}, \text{Ext}_{\varepsilon_i} \circ \text{Ext}_{\gamma_{0,i-1}} \subseteq \text{Ext}_{\gamma_{0,i}}), \quad (10.304)$$

$$\text{Upd}_\varepsilon^E \mathbf{E}_{k,j} := \mathbf{E}_{k,j} \star (\varepsilon, \text{Ext}_\varepsilon, \text{Ext}_\varepsilon \circ \text{Ext}_{\gamma_{0,j}} \subseteq \text{Ext}_{\gamma_{0,j+1}}). \quad (10.305)$$

Identity and neutral emission prove nonemptiness; associativity of the composition relational proof naturality.

Relations R: incidence, orientation, and compatibility of composition Let $\text{RelBlock}_{j,r}$ be the type of blocks of equations labeled by the edge and the prefix over which they are verified, and let $\text{Pres}_{j,r} := \text{List}(\text{RelBlock}_{j,r})$. The unit block and the seed are

$$\mathbf{B}_I := (I_{x_0}; U_I = I, \bigcup_I = I, \kappa(I) = 0, \partial I = 0), \quad \mathbf{R}_{k,k} := (\mathbf{B}_I) \star \bigstar_{i=1}^k \mathbf{B}_R(\varepsilon_i; \gamma_{0,i-1}). \quad (10.306)$$

For an edge $\varepsilon : x_j \rightarrow x_{j+1}$, the new block is the labeled tuple

$$\begin{aligned} \mathbf{B}_R(\varepsilon; \gamma_{0,j}) &:= (\varepsilon, \gamma_{0,j}; \delta^\diamond(\varepsilon) = \text{res}^\diamond(\varepsilon) + 9 \text{quo}^\diamond(\varepsilon), \\ &\Delta(\varepsilon) = \text{or}(\varepsilon) + 3\kappa(\varepsilon), \quad U_{\gamma_{0,j+1}} = U_\varepsilon U_{\gamma_{0,j}}, \\ &\bigcup_{\gamma_{0,j+1}} = \bigcup_\varepsilon \bigcup_{\gamma_{0,j}}, \quad \kappa(\gamma_{0,j+1}) = \kappa(\varepsilon) + \mathbf{C}_\varepsilon \kappa(\gamma_{0,j}), \\ &\partial \gamma_{0,j+1} = \partial \varepsilon + \partial \gamma_{0,j}, \quad \text{Cpl}_\varepsilon \text{ of (10.285)}). \end{aligned} \quad (10.307)$$

The update concatenates the block without losing date or multiplicity:

$$\text{Upd}_\varepsilon^R \mathbf{R}_{k,j} := \mathbf{R}_{k,j} \star \mathbf{B}_R(\varepsilon; \gamma_{0,j}). \quad (10.308)$$

The truncation $\text{tr}_{j+1,j}^R$ removes the final labeled block. Moreover, $a_* \mathbf{B}_R(\varepsilon; \gamma) = \mathbf{B}_R(a\varepsilon; a\gamma)$, because each equation is natural over the same edge and the same prefix.

Numbers N: publication of value with conservation of genealogy For an edge $\varepsilon : x_j \rightarrow x_{j+1}$, we define the numerical datum

$$\nu(\varepsilon) := (j+1, \mathbf{Q}_{j+1}, g_{j+1}, q_{j+1}^{(9)}, \beta_{j+1}, \text{res}^\Sigma, \text{quo}^\Sigma, \text{res}^\Pi, \text{quo}^\Pi, \text{or}, \kappa, \Delta)(\varepsilon). \quad (10.309)$$

Its type is

$$\mathbb{N} \times \widetilde{\mathcal{O}}_9 \times C_9 \times \mathbb{N} \times \mathbb{Z}/12\mathbb{Z} \times \{0, \dots, 8\}^2 \times \mathbb{Z}^5.$$

The seed contains the complete numerical list of the prefix $0 \rightarrow k$; the update adds exactly the datum of the next edge:

$$\mathbf{N}_{k,k} := ((0, \mathbf{Q}_0, g_0, q_0^{(9)}, \beta_0; 0, \dots, 0)) \star \bigstar_{i=1}^k \nu(\varepsilon_i), \quad \text{Upd}_\varepsilon^N \mathbf{N}_{k,j} := \mathbf{N}_{k,j} \star \nu(\varepsilon). \quad (10.310)$$

Totality proceeds from the divisions APP, the division TRIT, and the odometer TPK.

Orientation O: sign transport and leaf selection The type of an orientation state is

$$\mathbb{Z} \times (\mathbf{H}_x^\Sigma \oplus \mathbf{H}_x^\Pi) \times \mathbb{Z}[\mathbf{A}],$$

with seed and update

$$\mathbf{O}_{k,k} := \left(\sum_{i=1}^k \text{or}(\varepsilon_i), \text{Sh}(x_k), \partial\gamma_{0,k} \right), \quad (10.311)$$

$$\text{Upd}_\varepsilon^{\mathbf{O}}(\mathbf{o}, \text{Sh}, \mathbf{b}) := (\mathbf{o} + \text{or}(\varepsilon), U_\varepsilon \text{Sh}, \mathbf{b} + \partial\varepsilon). \quad (10.312)$$

The APP leaves prove nonemptiness. Functoriality of U , additivity of or , and telescoping of the boundary prove naturality.

Carry K: update of the quotient and residual memory We use the typed integer lifts in (10.172). The field has type

$$\mathbf{M}_{x_j} \times \text{Hom}(\mathbf{M}_{x_0}, \mathbf{M}_{x_j}) \times \text{Mat}_5(\mathbb{Z}) \times \text{List}(\text{Mat}_5(\mathbb{Z})) \times \text{List}(\mathbb{Q}_{\geq 0}).$$

For $\gamma_{k,j} = \varepsilon_j \cdots \varepsilon_{k+1}$, we explicitly fix

$$\tilde{L}_{\gamma_{k,k}} := I_5, \quad \tilde{L}_{\gamma_{k,j}} := \tilde{L}_r(\varepsilon_j) \cdots \tilde{L}_r(\varepsilon_{k+1}). \quad (10.313)$$

For the inherited prefix we also fix

$$\tilde{L}_{\gamma_{0,k}} := \tilde{L}_r(\varepsilon_k) \cdots \tilde{L}_r(\varepsilon_1).$$

The seed and update are

$$\mathbf{K}_{k,k} := (\kappa(\gamma_{0,k}), \mathbf{C}_{\gamma_{0,k}}, \tilde{L}_{\gamma_{0,k}}, \mathbf{c}_{\leq k}, \mathbf{k}_{\leq k}), \quad (10.314)$$

$$\text{Upd}_\varepsilon^{\mathbf{K}}(\kappa_\gamma, \mathbf{C}_\gamma, \tilde{L}_\gamma, \mathbf{c}, \mathbf{k}) := (\kappa(\varepsilon) + \mathbf{C}_\varepsilon \kappa_\gamma, \mathbf{C}_\varepsilon \mathbf{C}_\gamma, \tilde{L}_r(\varepsilon) \tilde{L}_\gamma, \mathbf{c} \star c_r(\varepsilon, \gamma), \mathbf{k} \star \kappa_j^{\text{can}}). \quad (10.315)$$

The identity provides the seed; the two cocycles (10.10) and (10.174) prove independence of parenthesization and naturality.

Memory M: record of the traversal and compatibility under truncation Its type at the same time preserves affine memory, ledger, and signed memories:

$$\text{LedgerEntry} := \{\lambda(\varepsilon) : \varepsilon \text{ typed elementary emission}\}. \quad (10.316)$$

$$\mathbf{M}_x \times \text{List}(\text{LedgerEntry}) \times \mathbb{N}^2 \times \mathbb{Z} \times \mathbb{N}.$$

We set

$$\mathbf{M}_{k,k} := (m_k, \text{Led}(\gamma_{0,k}), M_k^+, M_k^-, D_k, H_k), \quad (10.317)$$

$$\text{Upd}_\varepsilon^{\mathbf{M}}(m, \mathbf{Led}, M^+, M^-, D, H) := (\mathbf{C}_\varepsilon m + \kappa(\varepsilon), \mathbf{Led} \star \lambda(\varepsilon), M^+ + b_\varepsilon^+, M^- + b_\varepsilon^-, D + b_\varepsilon, H + |b_\varepsilon|). \quad (10.318)$$

The affine action and (10.138) prove totality and naturality; the past is not recomputed from the last value.

Boundary B: gluing data and survival of the extension The type is a boundary chain accompanied by a word of complejos based and chain maps. Its seed and update are

$$B_{k,j} \in \text{Rec}[\mathbf{b} \in \mathbb{Z}[A_j]; C_{x_j,r} \in \text{Ch}_{\text{bas}}; \mathbf{F} \in \text{Path}_{\text{Ch}_{\text{bas}}}(C_{x_j,r}, C_{x_0,r})].$$

$$B_{k,k} := (\partial\gamma_{0,k}, C_{x_k,r}, \mathbf{F}_{\leq k}), \quad (10.319)$$

$$\text{Upd}_{\varepsilon}^B(\mathbf{b}, C_{x_j,r}, \mathbf{F}) := (\mathbf{b} + \partial\varepsilon, C_{x_{j+1},r}, \mathbf{F} \star F_{\varepsilon}). \quad (10.320)$$

Non-emptiness proceeds from the common ports; $\partial^2 = 0$, $F_{\beta\varepsilon} = F_{\varepsilon}F_{\beta}$, and (10.35) prove compatibility.

Route G: observable history of the surviving extension The field is the list of prefixes in the path category:

$$G_{k,k} := (\gamma_{0,i})_{0 \leq i \leq k}, \quad \text{Upd}_{\varepsilon}^G G_{k,j} := G_{k,j} \star \gamma_{0,j+1}. \quad (10.321)$$

Identity, associative concatenation, and $a(\varepsilon\gamma) = a(\varepsilon)a(\gamma)$ prove the three required properties.

Multisection X: stable finite fiber and obstruction to pointwise selection For $M \geq j$, we define by action

$$X_{k,j}(M) := \{\gamma_{k,j} \star q : q \in \text{Term}_{j,M}(x_j)\}. \quad (10.322)$$

Thus

$$X_{k,k}(M) = \text{Term}_{k,M}(x_k),$$

and

$$\text{Upd}_{\varepsilon}^X X_{k,j}(M) := \{\gamma_{k,j} \star \varepsilon \star q : q \in \text{Term}_{j+1,M}(x_{j+1})\}. \quad (10.323)$$

The neutral prolongation proves nonemptiness. Deleting the final step and acting by a symmetry commute literally with concatenation.

Survival S: persistence under refinement and passage to the limit Here, a proposition is not stored, but witnesses. Let

$$s_{j,j} := I_{x_j}, \quad s_{j,M+1} := s_{j,M} \star \varepsilon_M^0(t(s_{j,M})). \quad (10.324)$$

The seed is $S_{k,k} := (s_{k,M})_{M \geq k}$, and

$$\text{Upd}_{\varepsilon}^S S_{k,j} := (\gamma_{k,j} \star \varepsilon \star s_{j+1,M})_{M \geq j+1}. \quad (10.325)$$

By construction $\pi_{M+1,M} s_{j,M+1} = s_{j,M}$; the naturality of neutral emission proves equivariance.

Terminal T: universality of the limit object and factorization of readers The terminal is the inverse system, not a selected queue:

$$T(x_j) := ((\text{Term}_{j,M}(x_j))_{M \geq j}, (\pi_{M+1,M})_{M \geq j}). \quad (10.326)$$

For each $M \geq j$, the map of prefix tipado is

$$\text{pref}_{\gamma_{k,j},M} : \text{Term}_{j,M}(x_j) \longrightarrow \text{Term}_{k,M}(x_k), \quad q \longmapsto \gamma_{k,j} \star q. \quad (10.327)$$

We define the field state and its update by

$$T_{k,j} := \left(T(x_j), (\text{pref}_{\gamma_{k,j},M})_{M \geq j} \right), \quad (10.328)$$

$$\text{Upd}_{\varepsilon}^T T_{k,j} := \left(T(x_{j+1}), (\text{pref}_{\gamma_{k,j+1},M})_{M \geq j+1} \right), \quad \gamma_{k,j+1} := \gamma_{k,j} \star \varepsilon. \quad (10.329)$$

In particular, $T_{k,k}$ uses $\gamma_{k,k} = I_{x_k}$. The witnesses (10.324) and the surjectivity of π prove that the system is nonempty; level by level,

$$\pi_{M+1,M}^{\mathcal{H}} \text{pref}_{\gamma_{k,j},M+1} = \text{pref}_{\gamma_{k,j},M} \pi_{M+1,M}^{\mathcal{H}}, \quad (10.330)$$

because both traversals only remove the last emission of q .

13. Terminal reader L: evaluation of the compatible extension and conservation of genealogy

The reader contains no classical object. Its type is the category of paths of the unique dependent constructor $\mathfrak{D}_{j,r}$ of (10.286). We set

$$u_{j+1,j}^{\mathfrak{D}} := u_{(j+1,r),(j,r)} : \mathfrak{D}_{j+1,r}(x_{j+1}) \longrightarrow \mathfrak{D}_{j,r}(x_j), \quad (10.331)$$

the component of the common transport (10.287) that truncates the last emission, reduces the coefficients and precomposes the ports. By composition,

$$u_{m,k}^{\mathfrak{D}} = u_{\ell,k}^{\mathfrak{D}} u_{m,\ell}^{\mathfrak{D}} \quad (m \geq \ell \geq k). \quad (10.332)$$

$$L_{k,k} := (\mathfrak{D}_{0,r}(x_0)) \star \bigstar_{i=1}^k (\mathfrak{D}_{i,r}(x_i), u_{i,i-1}^{\mathfrak{D}}, O(e_{\varepsilon_i})), \quad (10.333)$$

$$\text{Upd}_{\varepsilon}^L L_{k,j} := L_{k,j} \star (\mathfrak{D}_{j+1,r}(x_{j+1}), u_{j+1,j}^{\mathfrak{D}}, O(e_{\varepsilon})). \quad (10.334)$$

The totality of $\mathfrak{D}$ and (10.332) proves non-emptiness and naturality.

The thirteen preceding seeds form a single element.

$$\mathfrak{r}_{k,r}^0 := \langle F_{k,k}; E_{k,k}; R_{k,k}; N_{k,k}; O_{k,k}; K_{k,k}; M_{k,k}; B_{k,k}; G_{k,k}; X_{k,k}; S_{k,k}; T_{k,k}; L_{k,k} \rangle \in \mathfrak{R}_{k,r}. \quad (10.335)$$

The action of an edge is the total application.

$$\text{Upd}_{\varepsilon} : \mathfrak{R}_{j,r}(\gamma_{0,j}) \longrightarrow \mathfrak{R}_{j+1,r}(\gamma_{0,j+1}), \quad \text{Upd}_{\varepsilon} := \langle \text{Upd}_{\varepsilon}^F, \dots, \text{Upd}_{\varepsilon}^L \rangle, \quad (10.336)$$

with the thirteen actions (10.303)–(10.334). Finally, the terminal register is *generated* by the fold

$$\boxed{\mathfrak{r}_{k,N}(h) := \text{Upd}_{\varepsilon_N} \circ \dots \circ \text{Upd}_{\varepsilon_{k+1}}(\mathfrak{r}_{k,r}^0).} \quad (10.337)$$

It is not the unpacking of an intact copy of the state. On the image of this fold, the truncation is defined componentwise

$$\text{tr}_{13;j+1,j} \mathfrak{r}_{k,j+1}(h \star \varepsilon) := \mathfrak{r}_{k,j}(h). \quad (10.338)$$

It is well defined: the fields E and G preserve the ordered list of emissions and prefixes, so that equality of two records entails equality of their labeled histories. In F, E, R, N, G, T, L , the last entry or block is deleted; in O, K, M, B , the cumulative formula is applied to the recovered prefix; and in X, S , the cone of tails is restricted. Neither an idempotent union is inverted nor a continuation selected.

Proposition 10.36 (Totality and naturalness of the thirteen updates). *All the seeds of (10.335) exist, each Upd_{ε} is total, and, for every coherent symmetry a and every horizon truncation,*

$$a_* \text{Upd}_{\varepsilon} = \text{Upd}_{a\varepsilon} a_*, \quad \text{tr}_{13;j+1,j} \text{Upd}_{\varepsilon} \mathfrak{r}_{k,j}(h) = \mathfrak{r}_{k,j}(h). \quad (10.339)$$

Proof. The existence of each seed was exhibited in the thirteen paragraphs. In F, E, G, X, S, T, L , the formula is concatenation with an already typed map. In R, N, O, K, M, B , the formula is, respectively, concatenation of a labeled block of equations, adjunction of a numerical datum, transport of sheets,

composition of cocycles, affine action, and chain map. The naturality identities cited after each formula prove the first equality of (10.339) componentwise. The second is exactly (10.338); its sound definition uses the ordered history preserved by $\mathbf{E}$ and $\mathbf{G}$. $\square$

Proposition 10.37 (Joint Certificate of the Five Internal Developments). *The updates of the thirteen coordinates jointly transport: the spectral–polar telescoping and return without reinitialization of the corresponding identity; the orthogonal balance, the remark column, and the Stein identity of the corresponding identity; direct and adjoint naturality and the cellular retraction of the corresponding identity; totalization, the cone of the nonadic return, and the complete fibers of the corresponding identity; and the Alexander–Whitney diagonal, Smith reduction, and the Bockstein tower of the corresponding identity. All these actions have as their domain the same register generated by (10.337); none introduces an initial subdomain, a privileged history, or a selection motivated by a subsequent application.*

Proof. Each cited identity acts on a coordinate of the common register, and its truncation maps are restrictions of (10.338). The componentwise naturality proved in (10.339) permits their assembly into a single update. Equality of domain and preservation of the ordered history preclude replacing correlative generation with five retrospective projections. $\square$

10.3.3 Multivalued terminal without selector

The finite terminal generated by $\widehat{x}_k$ is the multisection

$$\mathfrak{T}_{k,N}(\widehat{x}_k) := \{\mathbf{r}_{k,N}(h) : h \in \mathcal{H}_{k,N}(\widehat{x}_k)\}. \quad (10.340)$$

The expression of the right returns the set whole. No contains a rule that distinguishes a history privileged.

The terminal level environment is obtained by gathering those multisects without identifying records:

$$\mathfrak{T}_{k,N} := \bigcup_{\widehat{x}_k \in \widehat{\mathcal{X}}_k^{\text{tot}}} \mathfrak{T}_{k,N}(\widehat{x}_k). \quad (10.341)$$

The linear version, when contributions must be added, is defined on the free module by

$$\begin{aligned} \mathbf{E}_{k,N}^{\text{all}} : \mathbb{Z}[\widehat{\mathcal{X}}_k^{\text{tot}}] &\longrightarrow \mathbb{Z}[\mathfrak{T}_{k,N}], \\ [\widehat{x}_k] &\longmapsto \sum_{h \in \mathcal{H}_{k,N}(\widehat{x}_k)} [\mathbf{r}_{k,N}(h)]. \end{aligned} \quad (10.342)$$

Each witness appears with its multiplicity. If exists a measure projective positive μ , the version isometric uses all the probabilities conditional:

$$\widehat{\mathbf{E}}_{k,N}^{\mu} e_{\widehat{x}_k} := \sum_{h \in \mathcal{H}_{k,N}(\widehat{x}_k)} \mu_{\widehat{x}_k,N}(h)^{1/2} e_{\mathbf{r}_{k,N}(h)}. \quad (10.343)$$

The projectivity gives

$$\left\| \widehat{\mathbf{E}}_{k,N}^{\mu} e_{\widehat{x}_k} \right\|^2 = \sum_{h \in \mathcal{H}_{k,N}(\widehat{x}_k)} \mu_{\widehat{x}_k,N}(h) = 1. \quad (10.344)$$

The sum is not substituted by “the history that generates the cylinder”.

10.3.4 The nonadic connection as correspondence

The nonadic holonomy was generated by the odometer and the TPK connection, without a membership hypothesis, in –. In this chapter (10.71) we use (10.73)

$$Z_{9,k} := Z_{9,k}^\Gamma. \quad (10.345)$$

The maps

$$\widehat{\mathcal{X}}_k^{\text{tot}} \xleftarrow{s_{9,k}} Z_{9,k}^\Gamma \xrightarrow{t_{9,k}} \widehat{\mathcal{X}}_{k+9}^{\text{tot}} \quad (10.346)$$

are those of (10.72); its source map is surjective by proposition 10.20. Every point of the apex preserves the nine phase-compatible emissions. It is not replaced by the pair of its endpoints.

For $N \geq k + 9$, define the apex of horizon before of linearizing:

$$Z_{9;k,N}^\Gamma := \coprod_{z=(\widehat{x}_k, \omega) \in Z_{9,k}^\Gamma} \text{Term}_{k+9,N}(t_{9,k}z). \quad (10.347)$$

An element is written (z, q) , where ω is the nonadic word of nine emissions and q its tail up to N . The maps on histories use the labeled codomain

$$\mathcal{D}_{9;k,N}^\Gamma := \coprod_{z \in Z_{9,k}^\Gamma} \text{Term}_{k+9,N}(t_{9,k}z). \quad (10.348)$$

Thus the maps are

$$S_{k,N}(z, q) := \omega \star q, \quad (10.349)$$

$$T_{k,N}(z, q) := (z, q) \in \mathcal{D}_{9;k,N}^\Gamma. \quad (10.350)$$

The first lands in $\coprod_x \mathcal{H}_{k,N}(x)$; the second expressly preserves the labelled summand by z .

The link map of the apex is the explicit action

$$\pi_{N+1,N}^Z(z, q \star \varepsilon) := (z, q). \quad (10.351)$$

It is surjective: if (z, q) is given, one adds to the tail the emission neutral at the endpoint of q . The codomain of tails has link

$$\pi_{N+1,N}^\mathcal{D}(z, q \star \varepsilon) := (z, q). \quad (10.352)$$

We denote by $\pi_{N+1,N}^{\mathcal{H},j}$ the map (10.299) with initial level j . Then

$$\pi_{N+1,N}^{\mathcal{H},k} S_{k,N+1} = S_{k,N} \pi_{N+1,N}^Z, \quad (10.353)$$

$$\pi_{N+1,N}^\mathcal{D} T_{k,N+1} = T_{k,N} \pi_{N+1,N}^Z. \quad (10.354)$$

In the first square, both paths send $\omega \star q \star \varepsilon$ to $\omega \star q$; in the second, they send $(z, q \star \varepsilon)$ to (z, q) . By unique decomposition of a word into its first nine emissions and its tail, $S_{k,N}$ is bijective. The second square is Cartesian by the coproduct–fibered definition of (10.347). These are the Beck–Chevalley squares preserving witnesses and multiplicities.

The corresponding apex of registers, now generated by the thirteen updates, is

$$\begin{aligned} Z_{9;k,N}^{\mathfrak{Z}} := \{ (z, q; \mathbf{r}_{k,N}(\omega \star q), \mathbf{r}_{k+9,N}(q)) : \\ z = (\widehat{x}_k, \omega) \in Z_{9,k}^\Gamma, \quad q \in \text{Term}_{k+9,N}(t_{9,k}z) \}. \end{aligned} \quad (10.355)$$

Its source and target maps read, respectively, the primer or the second record, but never remove from their domain the witness (z, q) :

$$\mathfrak{T}_{k,N} \xleftarrow{s_{9;k,N}^{\mathfrak{T}}} \mathfrak{Z}_{9;k,N}^{\mathfrak{T}} \xrightarrow{t_{9;k,N}^{\mathfrak{T}}} \mathfrak{T}_{k+9,N}. \quad (10.356)$$

If $\zeta = (z, q; r_s, r_t)$, its action is

$$s_{9;k,N}^{\mathfrak{T}}(\zeta) := r_s, \quad t_{9;k,N}^{\mathfrak{T}}(\zeta) := r_t. \quad (10.357)$$

Its link is defined by

$$\begin{aligned} \pi_{N+1,N}^{Z,\mathfrak{T}}(z, q \star \varepsilon; \mathfrak{r}_{k,N+1}(\omega \star q \star \varepsilon), \mathfrak{r}_{k+9,N+1}(q \star \varepsilon)) \\ := (z, q; \mathfrak{r}_{k,N}(\omega \star q), \mathfrak{r}_{k+9,N}(q)). \end{aligned} \quad (10.358)$$

The pair (z, q) is a coordinate of the apex, not a representation chosen of a triad; therefore the link is well defined even if two registros terminals coincide. The compatibility of Upd with the deletion of the ultima edge proof that (10.358) commutes with both maps of (10.356).

If an action on free modules is desired, this correspondence is linearized by summing all witnesses:

$$[\Gamma_9]_{k,N}[r] := \sum_{\zeta \in (s_{9;k,N}^{\mathfrak{T}})^{-1}(r)} [t_{9;k,N}^{\mathfrak{T}}(\zeta)]. \quad (10.359)$$

The formula preserves multiplicities. Equality of phases after nine steps does not contract the states, the witnesses, or their ledgers.

10.3.5 Uniqueness of the prefix decomposition and naturality of the truncation system

For $k \leq \ell \leq N$, every history of $\mathcal{H}_{k,N}(\widehat{x}_k)$ has a unique prefix $p \in \mathcal{P}_{\ell|k}(\widehat{x}_k)$. Therefore there is a disjoint decomposition by prefixes, expressed without choosing any of them:

$$\mathcal{H}_{k,N}(\widehat{x}_k) = \bigsqcup_{p \in \mathcal{P}_{\ell|k}(\widehat{x}_k)} \{p \star q : q \in \mathcal{H}_{\ell,N}(p)\}. \quad (10.360)$$

Let Conc_p be the operation that prepends the record of the prefix p and successively applies (10.336). We define the transition in common over a complete family of terminals by

$$\mathbf{u}_{\ell,k} \left((\mathfrak{T}_{\ell,N}(p))_{p \in \mathcal{P}_{\ell|k}(\widehat{x}_k)} \right) := \bigcup_{p \in \mathcal{P}_{\ell|k}(\widehat{x}_k)} \text{Conc}_p \mathfrak{T}_{\ell,N}(p). \quad (10.361)$$

Writing $\mathbf{G}_{k,N}(\widehat{x}_k) := \mathfrak{T}_{k,N}(\widehat{x}_k)$, all naturality is concentrated in the single identity

$$\boxed{\mathbf{u}_{\ell,k} \left((\mathbf{G}_{\ell,N}(p))_{p \in \mathcal{P}_{\ell|k}(\widehat{x}_k)} \right) = \mathbf{G}_{k,N}(\widehat{x}_k)}. \quad (10.362)$$

There is no distinct law for each internal discriminant. The five characteristics obey (10.362) because each is updated by means of the same prefix, the same extension, and the same ledger.

The nonadic correspondence is compatible with this identity through the maps already constructed, not through a condition that is implicit. For every horizon, (10.353) and (10.354) preserve the nonadic witness with its multiplicity, and (10.358) does the same in the thirteen coordinates. If a is a coherent symmetry of the common state, its action

$$a^Z(z, q) := (a^Z z, a q)$$

satisfies

$$Sa^Z = aS, \quad Ta^Z = aT, \quad \pi^Z a^Z = a^Z \pi^Z, \quad s^{\mathfrak{T}} a^Z = a s^{\mathfrak{T}}, \quad t^{\mathfrak{T}} a^Z = a t^{\mathfrak{T}}. \quad (10.363)$$

These equalities are verified entry by entry using (10.81) and the naturality of the thirteen updates. They are the material condition that permits the use of (10.359); a mere equality of visible values does not replace it.

10.3.6 Passage to the limit without interchanging image and limit

Truncating a history and each of the thirteen fields defines

$$\pi_{N+1,N}^{\mathfrak{T}} : \mathfrak{T}_{k,N+1}(\widehat{x}_k) \longrightarrow \mathfrak{T}_{k,N}(\widehat{x}_k), \quad \pi_{N+1,N}^{\mathfrak{T}} \mathfrak{r}_{k,N+1}(h) = \mathfrak{r}_{k,N}(h_{\leq N}). \quad (10.364)$$

It is surjective by (10.299). The complete terminal is therefore

$$\mathfrak{T}_{k,\infty}(\widehat{x}_k) := \left\{ (r_N)_{N \geq k} : r_N \in \mathfrak{T}_{k,N}(\widehat{x}_k), \quad \pi_{N+1,N}^{\mathfrak{T}} r_{N+1} = r_N \right\}. \quad (10.365)$$

The finite sets of (10.340) are nonempty and their transition maps are surjective; therefore,

$$\mathfrak{T}_{k,\infty}(\widehat{x}_k) \neq \emptyset. \quad (10.366)$$

Corollary 10.38 (Simultaneous persistence of the five developments). *In each element of $\mathfrak{T}_{k,\infty}(\widehat{x}_k)$ there simultaneously persist the strong limit $\mathcal{T}_{\infty}$ of the polar telescoping, the Gram family and losses, the incidence measure with its two naturalities, the cone $\text{Cone}(I - U_{\Gamma_9,N})$, and the compatible family of Smith forms and Bockstein pages. None of these data is obtained by reconstructing only the visible terminal value.*

Proof. Surjectivity of the truncations preserves the complete witnesses. The telescoping, naturality, and reduction equations of proposition 10.37 commute with those truncations; by universality of the inverse limit, they define a unique compatible family on each surviving history. $\square$

The apex nonadico in the limit is defined, not inferred from an image:

$$\mathbf{Z}_{9;k,\infty}^{\Gamma} := \varprojlim_{N \geq k+9} (\mathbf{Z}_{9;k,N}^{\Gamma}, \pi_{N+1,N}^Z). \quad (10.367)$$

Since the links do not modify the finite witness z and only truncate its tail, there exists the explicit isomorphism

$$\mathbf{Z}_{9;k,\infty}^{\Gamma} \xrightarrow{\cong} \coprod_{z \in \mathbf{Z}_{9,k}^{\Gamma}} \text{Msect}(t_{9,k}z), \quad ((z, q_N))_N \longmapsto (z, (q_N)_N). \quad (10.368)$$

The inverse takes a witness z and a compatible tail $(q_N)_N$, and forms $((z, q_N))_N$. Both compositions are identities coordinate by coordinate. The limit maps are

$$S_{k,\infty}((z_N)_N) := (S_{k,N}z_N)_N, \quad T_{k,\infty}((z_N)_N) := (T_{k,N}z_N)_N; \quad (10.369)$$

are well defined precisely by (10.353) and (10.354).

The apex of registers is defined in the same way

$$\mathbf{Z}_{9;k,\infty}^{\mathfrak{T}} := \varprojlim_{N \geq k+9} (\mathbf{Z}_{9;k,N}^{\mathfrak{T}}, \pi_{N+1,N}^{Z,\mathfrak{T}}). \quad (10.370)$$

The correspondences (10.356) then form, component by component, the span

$$\mathfrak{T}_{k,\infty} \xleftarrow{s_{9;k,\infty}^{\mathfrak{T}}} \mathbf{Z}_{9;k,\infty}^{\mathfrak{T}} \xrightarrow{t_{9;k,\infty}^{\mathfrak{T}}} \mathfrak{T}_{k+9,\infty}. \quad (10.371)$$

The image of a limit has not been identified with the limit of certain images: (10.365), (10.367) and (10.370) construct directly the three compatible families and their maps.

10.3.7 Internal reconstruction of the integral holonomic state from its 13 genealogical coordinates

Recovering does not unpack names: it repeats the fold that generated the record. For $h = (\varepsilon_{k+1}, \dots, \varepsilon_N)$, we define

$$\text{Rec}_{13;k,N}(h) := \text{Upd}_{\varepsilon_N} \circ \dots \circ \text{Upd}_{\varepsilon_{k+1}}(\mathbf{r}_{k,r}^0). \quad (10.372)$$

By (10.337), $\text{Rec}_{13;k,N}(h) = \mathbf{r}_{k,N}(h)$, but the equality is now an identity between two compositions of actions, not the projection of a preserved input. If pr_A denotes the projector from the register to field $A \in \{F, E, R, N, O, K, M, B, G, X, S, T, L\}$, then

$$\text{pr}_A \text{Rec}_{13;k,N}(h) = \text{Upd}_{\varepsilon_N}^A \circ \dots \circ \text{Upd}_{\varepsilon_{k+1}}^A(A_{k,k}). \quad (10.373)$$

The proof is induction on $N - k$: for $N = k$ both sides are the seed; the inductive step applies the component A of $\text{Upd}_{\varepsilon_N}$.

Moreover,

$$\text{Rec}_{13;k,N}(h_{\leq N}) = \text{tr}_{13,N+1,N}(\text{Rec}_{13;k,N+1}(h)), \quad (10.374)$$

where $\text{tr}_{13,N+1,N}$ removes only the last update and restricts its cone of tails. Consequently, an element of $\mathfrak{T}_{k,\infty}(\widehat{x}_k)$ determines a compatible family of the thirteen coordinates at every depth.

10.3.8 Multivalued terminality, naturality, and compatible limit of the continuum

Theorem 10.39 (Multivalued terminal, naturality, and limit of the continuum). *For every $\widehat{x}_k \in \widehat{X}_k^{\text{tot}}$, the rules (10.295)–(10.374) construct, from the unique discrete continuum:*

1. a multisection terminal finite not empty at each horizon;
2. an evaluation that conserves all histories and multiplicities;
3. the connection nonadic as a correspondence with witnesses;
4. the unique identity of naturality (10.362);
5. a nonempty inverse terminal and a nonadic correspondence in the limit;
6. the compatible recovering of the fiber, extensions, relations, numbers, orientation, carry, memory, boundary, route, multisection, survival, terminal, and reader.

The five intrinsic characteristics remain correlated within each record and each extension witness.

Proof. The non-emptiness of $\mathcal{H}_{k,N}(\widehat{x}_k)$ is nonempty by proposition 10.15, and its truncation is surjective by lemma 10.16. The rule (10.336) applies the APP, TRIT, and TPK identities to each witness ε_j ; by induction on $N - k$, it produces all the fields of (10.337) and conserves their relations.

Taking the complete set in (10.340), or the complete sum in (10.342), proves that the terminal does not contain a selector. The weighted version is isometric by the projective equality (10.344).

The witness $z = (\widehat{x}_k, \omega)$ carries the phase action of each of its nine emissions; by proposition 10.18 it defines the two maps in (10.346) without converting the relation into a function or identifying phase with state. Equations (10.349)–(10.358) prove the two horizon squares and the same construction over registers. Summing the source space of each witness yields (10.359) without erasing multiplicities.

The partition (10.360) is disjoint because every history has a determined prefix of length ℓ . Concatenating each prefix with all its continuations reconstructs once and only once each history of $\mathcal{H}_{k,N}(\widehat{x}_k)$. This directly proves (10.362) for the whole register; five disconnected proofs are not needed.

Surjectivity of (10.364) and finiteness of each level imply nonemptiness of (10.365). The explicit links (10.351) and (10.358), together with their squares, construct the limits (10.367) and (10.370); from them

follows (10.371). Finally, (10.372) reruns the successive updates, and (10.374) proves that those readings pass jointly to the limit. $\square$

10.3.9 Obstructions to terminality of the integral holonomic state: loss of fiber, orientation, memory or reader

Proposition 10.40 (Falsifiers of the terminal set). *Any of the following operations replaces the constructed object and blocks the conclusion of theorem 10.39:*

1. *send to cylinder to to single history when it admits several;*
2. *choosing a tail of the terminal field $\mathbb{T}$ and eliminating the remaining ones;*
3. *replace Γ_9 by a function without proving uniqueness of each target and of its witness;*
4. *linearize to correspondence while ignoring multiplicities;*
5. *identifying phase return with state return;*
6. *preserve the input intact as the first coordinate and use its recoverability as purported generation;*
7. *omit any of the thirteen fields of (10.337);*
8. *allowing two intrinsic characteristics to use different histories, carries, boundaries, or terminals;*
9. *replace (10.362) by five independent assertions;*
10. *assert operatorial compatibility of Γ_9 without preserving the space of witnesses and its multiplicities;*
11. *interchanging, without proof, formation of images and passage to the limit;*
12. *present a list of thirteen names without the actions (10.336) that produce them.*
13. *omit the polar telescoping, the loss row, the naturality adjoint, the module of all continuations or the reduction Smith–Bockstein certified in proposition 10.37;*
14. *turning any of those five developments into an autonomous branch or using a later classical application to select its domain.*

Proof. The first five operations contract the multisection or the correspondence. The next four break their common support and naturality. The final three eliminate the generative proof or alter passage to the limit. In each case, the resulting object ceases to satisfy at least one of (10.340), (10.356), (10.362), or (10.374); therefore, the conclusion does not carry over. $\square$

Hypothesis, domain, and conclusion

The organization of the complete terminal object, unique naturality, and passage to the limit are established jointly. The surviving multisection, the carries, the register, the boundary, the route, and the nonadic connection are its structural antecedents. The linearization over free modules is proved by (10.359) and its Beck–Chevalley squares. A subsequent promotion to a bounded operator on an analytic completion additionally requires that norm to be declared and the bound proved; the correspondence (10.371) does not depend on that interface.

10.4 Joint nonadic connection Γ_9 as a compatible correspondence of the inverse system

10.4.1 Terminal Assembly Theorem of the Correlative Constructions Generated by the Common State

Each depth k has an integral holonomic state $\mathcal{E}_k$ with all its surviving emissions and a single correlative constructor

$$O_k : \mathcal{E}_k \longrightarrow \mathcal{Z}_k, \quad \widehat{x}_k \longmapsto O(\{\mathbf{e}_\alpha : \alpha \in \text{Ch}_k(\widehat{x}_k)\}). \quad (10.375)$$

The value of O_k simultaneously preserves the spectral, solenoidal, gauge, coherent, and determinantal coordinates,

$$(\text{sp}, \text{sol}, \text{gau}, \text{coh}, \text{det}), \quad (10.376)$$

as five characteristics of the same state and of the same genealogical register. The truncations

$$u_{\ell,k}^{\mathbf{e}} : \mathcal{E}_\ell \rightarrow \mathcal{E}_k, \quad u_{\ell,k}^O : \mathcal{Z}_\ell \rightarrow \mathcal{Z}_k$$

satisfy the unique naturality

$$u_{\ell,k}^O \circ O_\ell = O_k \circ u_{\ell,k}^{\mathbf{e}}, \quad \ell \geq k. \quad (10.377)$$

Therefore, the corresponding action passes to the limit as

$$O_\infty : C_{\text{cont}}^{\text{disc}} := \varprojlim_k \mathcal{E}_k \longrightarrow \mathcal{Z}_\infty := \varprojlim_k \mathcal{Z}_k. \quad (10.378)$$

The five terminal applications are seen as intrinsic manifestations of that single value:

$$\mathfrak{G}_T^{\text{int}} := \text{View}_T(O_\infty(C_{\text{cont}}^{\text{disc}})), \quad T \in \{\text{sp}, \text{sol}, \text{gau}, \text{coh}, \text{det}\}. \quad (10.379)$$

Simultaneity precedes the specialized resolutions: each view exposes the required coordinate and the complete state conserves the other four.

10.4.2 Survival and inverse limit

Let $\text{Surv}_N^{(\chi)}$ be the set of histories of depth N that simultaneously satisfy the active laws of channel χ and admit prolongation. We define the compatible truncations

$$\tau_{N+1,N} : \text{Surv}_{N+1}^{(\chi)} \longrightarrow \text{Surv}_N^{(\chi)}$$

These preserve the enriched prefix. The genealogical continuum is the limit.

$$X_\chi = \varprojlim_N (\text{Surv}_N^{(\chi)}, \tau_{N+1,N}), \quad \tau_{N+1,N}(x_{N+1}) = x_N. \quad (10.380)$$

The global space of surviving sections and one of its enriched sections are denoted, respectively, by

$$X_\infty := \bigsqcup_{\chi} X_\chi, \quad \widehat{\Omega}_\infty \in X_\infty. \quad (10.381)$$

The global Archimedean reader is

$$P_{\text{arq}} := \text{ev}_{\mathbb{R}} : X_\infty \longrightarrow \mathbb{R}; \quad (10.382)$$

its restriction to each channel evaluates the unitary intersection of the compatible cylinders of the corresponding section. When the exceptional reader intervenes, $X_\infty^{\text{cal}} \subseteq X_\infty$ denotes the subspace in which the calibration type of (10.397) has been fixed. Survival determines the section by internal compatibility: every history must be prolongable in the complete system. The numerical reader intervenes afterward to certify and publish the intersection of the already compatible descendant.

10.4.3 Joint nonadic connection, holonomy, and vertical memory

At depth K , the holonomy of the nonadic return is initially the typed correspondence

$$\Gamma_{9,K} = \text{Hol}_{C_9}(\nabla_{\text{TPK}}) : \widehat{\mathcal{X}}_K \rightrightarrows \widehat{\mathcal{X}}_{K+9}. \quad (10.383)$$

Its iterates are relational compositions. Only when each state has a unique compatible prolongation is it restricted to a map, and only when that map is invertible does it constitute a monodromy. This distinction conserves all surviving extensions before selecting a section. After nine phases the local phase returns, while the complete state advances:

$$g(K+9) = g(K), \quad q_{\text{mem},K+9} = q_{\text{mem},K} + 1, \quad \widehat{x}_{K+9} \neq \widehat{x}_K \text{ in general.}$$

Block, cylinder, prefix, carry, and memory continue to evolve. This is the difference between visible periodicity and return with biography.

The transported state at level (K) can be exhibited as

$$\widehat{x}_K = (B_K, C_K, p_K, c_K, \Sigma_K, W_K, g(K), q_{\text{mem},K}, s_K), \quad (10.384)$$

where each coordinate has its own transport law. The phase return is represented on the catalog.

$$\mathcal{U}_6^{\text{cat}} \cong \mathfrak{S}_+ \times \mathfrak{S}_\times, \quad |\mathfrak{S}_+| = 18, \quad |\mathfrak{S}_\times| = 26.$$

Conditional averages

$$(E_+f)(s, e) = \frac{1}{18} \sum_{s' \in \mathfrak{S}_+} f(s', e), \quad (E_\times f)(s, e) = \frac{1}{26} \sum_{e' \in \mathfrak{S}_\times} f(s, e')$$

commute. The phase word $\tau = (+1, -1, 0, +1, -1, 0, +1, -1, 0)$ applies $E_+, E_\times, I$ in that order, and one revolution closes through

$$\mathcal{K}_{\text{ph}}^9 = (E_+E_\times) \otimes I_{\ell^2(C_9)}, \quad \langle \delta_u, E_+E_\times \delta_v \rangle = \frac{1}{468} \quad (u, v \in \mathcal{U}_6^{\text{cat}}). \quad (10.385)$$

The sum and product sheets do not act as independent filters: they are two components of a single holonomy that closes only after traversing the complete cycle.

The compression–expression colligation separates what returns from what remains as memory. If (J) embeds the history space into the phase space, (T) is the surviving compression and η the defect channel, then

$$\boxed{U_{\Gamma_9} J = JT + \eta, \quad J^* \eta = 0, \quad I - T^* T = \eta^* \eta.} \quad (10.386)$$

The first equality decomposes transport into compressed return and memory expression; the second proves their orthogonality; the third measures exactly the information that cannot fit within the periodic shadow. The central statement is thereby formalized: the phase returns, not the complete state.

The vertical memory is organized through the cyclic extension oriented

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0, \quad c(r, r') = \left[\left\lfloor \frac{r + r'}{9} \right\rfloor \right]_{12}. \quad (10.387)$$

Here $\iota([b]_{12}) = [9b]_{108}$ and $p([\theta]_{108}) = [\theta]_9$. The extension does not split: C_{108} has exponent 108, whereas $C_{12} \times C_9$ has exponent 36. The cocycle records the carry from each turn of the base C_9 in the dodecaphase memory C_{12} . C_{108} is therefore an oriented phase–memory calendar. Only after the dimensional chart is that extension realized as a finite 108-step quantum clock.

The local MD lift carrying the TRIT signature is the topological–temporal unit of that genealogical crystal. On $\mathcal{H}_{\text{clk}} = \mathbb{C}^{108}$, with basis $\{|j\rangle : j \in \mathbb{Z}/108\mathbb{Z}\}$, let

$$\zeta = e^{2\pi i/108}, \quad |\tilde{k}\rangle = \frac{1}{\sqrt{108}} \sum_{j=0}^{107} \zeta^{jk} |j\rangle, \quad U_{\text{clk}} |j\rangle = |j+1\rangle.$$

With $\omega_0 = 2\pi/(108t_0)$, the Hamiltonian

$$H_{\text{clk}} = \hbar_{\text{int}} \omega_0 \sum_{k=0}^{107} k |\tilde{k}\rangle \langle \tilde{k}| \quad (10.388)$$

generates exactly

$$U_{\text{step}} = \exp\left(-\frac{iH_{\text{clk}}t_0}{\hbar_{\text{int}}}\right) = U_{\text{clk}}, \quad U_{\text{step}}^{108} = I, \quad (10.389)$$

If $Z_{\text{clk}}|j\rangle = \zeta^j|j\rangle$, then

$$\langle j_0 | U_{\text{step}}^{-n} Z_{\text{clk}} U_{\text{step}}^n | j_0 \rangle = \zeta^{j_0+n}, \quad (10.390)$$

whose orbit has primitive period 108. In HMT vocabulary, this is the internal genealogical time crystal: the observable calendar closes, C_{12} records the turn and the integral holonomic state need not return. The subsequent realization as a macroscopic physical phase additionally requires a Floquet protocol, primitive subharmonic response, robustness, persistence and a protection mechanism; that chart does not alter the finite theorem of unitary periodicity.

10.4.4 Archimedean publication, genealogical number and measurement

The Archimedean publication uses semi-open ternary cylinders.

$$I_{N,L} = \left[\frac{N}{3^L}, \frac{N+1}{3^L} \right), \quad \text{diam } I_{N,L} = 3^{-L} \longrightarrow 0. \quad (10.391)$$

A compatible section determines nested cylinders and, by completeness, a unique real point. The HMT number is the section itself,

$$x = (x_N)_N \in X_\chi = \varprojlim_N \text{Surv}_N^{(\chi)}, \quad (10.392)$$

not the tuple formed by it and its evaluations. The value appears when a reader forgets part of the genealogy. Therefore, precision is depth: measuring means descending to a finer cylinder while the causal prefix remains in the antecedent. The real line is reconstructed as the Archimedean quotient of a space of much richer histories.

10.4.5 Dimensional realization of the HMT state: screen map, observable magnitude, and spectral chart

Dimensional Mechanics begins where the genealogical state couples to a realization boundary. With a field K_n fixed, let $M_n(x)$ be the module generated by the prolongation channels of a state x , let $\mathcal{R}_n(x)$ be the submodule of its relations, and let $\beta_n(x)$ be the conserved boundary datum. The dimensional screen is

$$\Sigma_n(x) = (K_n, M_n(x), \beta_n(x), \mathcal{R}_n(x)), \quad d_{\Sigma_n}(x) = \dim_{K_n} \frac{M_n(x)}{\mathcal{R}_n(x)}. \quad (10.393)$$

If $M_n(x) \simeq K_n^{g_n}$ and the columns of $R_n(x)$ generate the relations, then

$$\boxed{d_{\Sigma_n}(x) = g_n - \text{rank}_{K_n} R_n(x)}. \quad (10.394)$$

Dimension is not attached as a label: it is the number of independent channels that survive after imposing the boundary relations.

A physical quantity is a section of a dimensional line L_D ; a unit is a chart $\chi_u : L_D \rightarrow \mathbb{R}$ that publishes its coordinate. Hence

$$\text{TPK state} \longrightarrow \Sigma_n(x) \longrightarrow s_D \in \Gamma(L_D) \xrightarrow{\chi_u} [s_D]_u \in \mathbb{R} \quad (10.395)$$

separates structure, dimensional type, and measured number exactly. Measuring consists in coupling a section to a reversible apparatus, imposing edge compatibility, and applying the reader; changing units changes χ_u , not the section or its genealogy.

10.4.6 Double projection mathematical and dimensional realization

The exceptional branch preserves gauge data that the real publication does not need. Let $\mathcal{C}_{\text{exc}}$ be the space of gauges, and let

$$\mathfrak{E} : X_{\infty}^{\text{cal}} \times \mathcal{C}_{\text{exc}} \longrightarrow \mathbf{Exc} \quad (10.396)$$

the incidence reader. We fix once

$$\begin{aligned} \chi_{\text{exc}} = (\text{projective order, Paley gauge, dodecad,} \\ \text{incidence alignment, lattice chamber}) \in \mathcal{C}_{\text{exc}} \end{aligned} \quad (10.397)$$

and abreviamos $\mathfrak{E}_{\chi_{\text{exc}}}(x) := \mathfrak{E}(x, \chi_{\text{exc}})$. The gauge initial it transports by the history; not vultue to elegirse in each depth.

The same calibrated limit then admits two readers with distinct codomains:

$$\mathbb{R} \xleftarrow{\text{ev}_{\mathbb{R}}} X_{\infty}^{\text{cal}} \xrightarrow{\mathfrak{E}_{\chi_{\text{exc}}}} \mathbf{Exc}. \quad (10.398)$$

The first branch contracts cylinders to a continuous value. The second preserves incidences, orientations, gauge, and boundary. Both descend in parallel from the same source state. This bifurcation jointly explains metric continuity and the emergence of exceptional symmetries. The second branch is a calibrated reconstruction: it is exact once χ_{exc} has been fixed, but the bare state does not retrospectively select the five data composing that gauge.

The non-serializing dimensional realization does not serialize those two branches. It is a third reader of the same antecedent:

$$P_{\text{dim}} : X_{\infty} \longrightarrow \mathcal{R}_{\text{MD}}, \quad \mathcal{R}_{\text{MD}} = \{\text{action, mass, field, geometry, information}\}. \quad (10.399)$$

In this manner, a number, a symmetry, and a physical magnitude may share a genealogy without being confused with or reduced to one another.

The thesis of the continuum is thus formulated precisely: Archimedean reality is the zero-radius shadow of infinitely refinable discrete histories; what the shadow omits reappears as memory, incidence, geometry, and physical structure.

10.5 Surviving extensions, monodromy, and dynamics of the quotient

The language of paths requires a decisive precision. An APP path, a visible emission, an enriched TPK history, and a global section are not four names for the same object. They constitute four consecutive levels of a single construction. The path preserves positional order; the emission preserves a finite reading; the history adds signature, carry, orientation, boundary, and memory; the global section requires all of its prefixes to survive simultaneously at every depth.

This hierarchy makes it possible to separate two questions that had remained intermingled. The first is logical: under what hypotheses a finite prefix truly belongs to a compatible infinite extension. The second is dynamical: what it means to transport a single history and what it means to transfer all its compatible

continuations simultaneously. The distinction is essential in APP. A given trajectory carries one state to another; a path transfer carries a distribution of extensions to another distribution.

This section resolves the finite problem at depth six. The published deterministic chronology has visible-phase return and cannot select a unique probability. By contrast, transport biased by the nonadic phase, when each active channel is interpreted as the normalized renewal of its entire fibre of compatible continuations, has a uniform nine-step return over the (468) emissions. The weights (1/3) then cease to be an external choice: they follow from the three occurrences of each signature (+1,-1,0) in the oriented nonadic cycle. Normalization within each fibre remains a precise mathematical premise—the uniform count of all compatible paths—and must not be conflated with the successor of a single trajectory. Passage to the inverse limit over all depths preserves a distinct statement, formulated at the end of the section.

10.5.1 5 levels of the notion of path

Let Γ_{APP} be the category of cardinal paths of the APP torus. For each depth m , let $\mathcal{H}_m$ be the finite set of admissible enriched histories

$$h_m = (v_0, e_1, v_1, \dots, e_m, v_m),$$

where each arrow e_j transports all fields declared by TPK. If $n \geq m$, the application

$$p_{n,m} : \mathcal{H}_n \longrightarrow \mathcal{H}_m$$

removes the tail beyond depth m . The positional shadow

$$\text{sh}_m : \mathcal{H}_m \longrightarrow \Gamma_{\text{APP}}$$

forgets the memory fiber and preserves the underlying APP path.

Definition 10.41 (Hierarchy of extensions). We distinguish the following objects.

1. An *APP path* is a morphism of Γ_{APP} .
2. An *observable route* is a reading of that morphism that jointly conserves the sum and product leaves.
3. A *TPK extension* is a lift of the observable route to a history $h_m \in \mathcal{H}_m$, with its residue, quotient, signature, orientation, boundary, carry, and record data.
4. A *surviving extension* is a finite history that admits compatible prolongations to every horizon.
5. A *global section* is a compatible family $x = (h_m)_{m \geq 0}$ of surviving extensions.

In particular, the route is the shadow of the extension; it is not an external object that the extension subsequently chooses.

For $h \in \mathcal{H}_m$, its cylinder of prolongations is

$$Z_m(h) = \{x = (h_n)_n \in \varprojlim_n \mathcal{H}_n : x_m = h\}.$$

In particular, every family of $Z_m(h)$ also satisfies $p_{n,\ell}(h_n) = h_\ell$ for $n \geq \ell$; collections of incompatible truncations are not admitted. Let us also define

$$\mathcal{S}_m^{(N)} = p_{N,m}(\mathcal{H}_N), \quad \mathcal{S}_m^\infty = \bigcap_{N \geq m} \mathcal{S}_m^{(N)}.$$

Theorem 10.42 (Conditional extension survivor criterion). *Let us fix $h \in \mathcal{H}_m$ and, for $N \geq m$, write*

$$\mathcal{P}_N(h) = \{g \in \mathcal{H}_N : p_{N,m}(g) = h\}.$$

Suppose that the truncations are coherent and that each $\mathcal{P}_N(h)$ is finite. The following are equivalent:

1. $\mathcal{P}_N(h) \neq \emptyset$ for every $N \geq m$, that is, $h \in \mathcal{S}_m^\infty$;
2. the tree of finite extensions of h is finitely branching and has a nonempty level at every depth;
3. $Z_m(h) \neq \emptyset$.

Consequently, whenever the surviving finite truncations are nonempty, the set

$$\Omega_H^{\text{surv}} = \varprojlim_m \mathcal{S}_m^\infty$$

is exactly the space of surviving global sections. The theorem does not assert that these truncations are nonempty for every APP input: that property must be proved by means of the survival protocol.

Proof. The implication from the third condition to the first two is obtained by truncation. Now assume the first condition and order the elements of $\bigsqcup_{N \geq m} \mathcal{P}_N(h)$ by the prefix relation. Each level is finite and nonempty, and coherence of the truncations makes this union a finitely branching tree with nodes at every depth; this yields the second condition. Conversely, from the second condition König's lemma provides an infinite branch. Its truncations form a compatible family belonging to $Z_m(h)$, which is the third condition. The same construction identifies the indicated inverse limit wherever its finite truncations are nonempty. $\square$

The theorem resolves the possible discrepancy between “extendable to each horizon separately” and “possessing a global prolongation”: finiteness of the branches is the hypothesis that permits the choice of a single compatible chain. It does not claim that all APP paths belong to $\text{sh}(\Omega_H^{\text{surv}})$. That coverage requires proving that the enriched closures do not empty the corresponding fibers.

10.5.2 Phase return and monodromy relation

The return of the phase–length quotient is formalized separately in the corresponding proposition:

$$(b, r) \xrightarrow{9} (b+1, r), \quad (b, r) \xrightarrow{108} (b, r).$$

These equalities belong to the odometer 9×12 . They do not constitute the proof of $F_{\text{obs}}^{108} = I$, which uses the observable state and appears below.

Let ϕ be the phase coordinate with values in $\mathbb{Z}/9\mathbb{Z}$. A visible-phase return satisfies

$$\phi(h_{k+9}) = \phi(h_k).$$

The complete record may, however, satisfy

$$B(h_{k+9}) \neq B(h_k).$$

This inequality is not a failure of periodicity: it is the memory of the circuit. An ordinary action of C_9 on the complete state would require the ninth power to be the identity and would erase precisely that information.

To type the return, let H be a hexad and $P_2(H)$ its set of fifteen pairs. Let $\mathcal{B}_k$ be the record states that survive at phase k , let $\mathcal{T}_k \subseteq \mathcal{B}_k \times \mathcal{B}_{k+1}$ be the published transport relation, and suppose a fiber encoder is given

$$c_k : \mathcal{B}_k \longrightarrow P_2(H).$$

Only after declaring those encoders is the observable relation defined.

$$R_k = (c_k \times c_{k+1})(\mathcal{T}_k) \subseteq P_2(H) \times P_2(H).$$

The relational monodromy of a loop is

$$\mathcal{M}_k = R_{k+8} \circ R_{k+7} \circ \cdots \circ R_k.$$

Proposition 10.43 (Classification of the nonadic return given the encoder). *Three logically distinct levels are given.*

1. *Without a proof of functionality, the return is the relation $\mathcal{M}_k$.*
2. *If each R_k is the graph of a map $\mu_k : P_2(H) \rightarrow P_2(H)$, the transport is the skew product*

$$T(k, p) = (k + 1, \mu_k(p)),$$

and

$$T^9(k, p) = (k, M_k(p)), \quad M_k = \mu_{k+8} \circ \cdots \circ \mu_k.$$

3. *If the μ_k are bijective, M_k is an automorphic holonomy. The transport descends to an ordinary action of C_9 on the complete state if and only if $M_k = I$ in all fibers.*

Proof. The first claim is the definition of image and relational composition once $(c_k)_k$ have been fixed. Under functionality, nine consecutive applications keep the first coordinate fixed and compose their actions on the second. The composition of bijections is a bijection. Finally, $T^9 = I$ is exactly equivalent to $M_k = I$ for each phase and each point of the fiber. $\square$

Corollary 10.44 (No return under cumulative transport). *Let $q : X \rightarrow V$, $T : X \rightarrow X$, and $x \in X$ be given. If*

$$q(T^9x) = q(x), \quad T^9x = \mathcal{M}_x x, \quad \mathcal{M}_x x \neq x,$$

then the projection returns and the state does not:

$$q(T^9x) = q(x), \quad T^9x \neq x.$$

The pointwise hypothesis $\mathcal{M}_x x \neq x$ is indispensable; the global inequality $\mathcal{M}_x \neq I$ does not exclude x from being a fixed point.

Proof. The first equality is a hypothesis. The second and the pointwise separation give $T^9x = \mathcal{M}_x x \neq x$. The statement would be refuted by a case that satisfies the three premises and had $T^9x = x$; it does not authorize replacing the third premise by $\mathcal{M}_x \neq I$. $\square$

Theorem 10.45 (Functional sufficiency of the enriched tuple). *Let*

$$E : \mathcal{X}_{\text{TPK}}^{\text{enr}} \longrightarrow \mathcal{S}_E$$

the tuple that conserves the visible block, new blocks and carry, lifting state, cylinders, boundary signature, orientation, observable, phase, and dodecaphasic register. Let

$$T : \mathcal{X}_{\text{TPK}}^{\text{enr}} \rightarrow \mathcal{X}_{\text{TPK}}^{\text{enr}}, \quad O : \mathcal{X}_{\text{TPK}}^{\text{enr}} \rightarrow Y.$$

When maps exist

$$\widehat{T} : \mathcal{S}_E \rightarrow \mathcal{X}_{\text{TPK}}^{\text{enr}}, \quad \widehat{O} : \mathcal{S}_E \rightarrow Y$$

such that

$$T = \widehat{T} \circ E, \quad O = \widehat{O} \circ E,$$

the equality $E(x) = E(y)$ implies $T(x) = T(y)$, $O(x) = O(y)$ and, by determinism, the same continuations and subsequent outputs.

Proof. By factorization,

$$T(x) = \widehat{T}(E(x)) = \widehat{T}(E(y)) = T(y), \quad O(x) = \widehat{O}(E(x)) = \widehat{O}(E(y)) = O(y).$$

The first equality permits iteration of the argument. This is a sufficient condition, not an unproved equivalence. Two states with the same tuple and different transitions or outputs, access to an absent coordinate, or querying a target value are falsifiers. The visible five-element microfibre of [proposition 8.36](#) controls only the nonfaithfulness of the word w_6 ; it does not prove this positive factorization. $\square$

The published finite window contains nine phase returns with a change in the integral holonomic state and a local selection from three continuations to one when increasing by two horizons. This proves return with memory and refutes the identification of the register with a memoryless cycle. The global extraction of the nine encoders c_k and, through them, of the relations R_k , from all surviving histories is a distinct operation. The preceding proposition classifies the return once the register–pairs map is given; it does not derive that map from the integral holonomic state of TPK.

The concrete episode of the ninth phase, with its three continuations, the block B_{11} , the boundary (502, 662, 638), and the integer inequalities of cylinders, is proved only once in the [theorem 8.39](#). In this residence only its typed consequence is used:

$$\# \text{Surv}_1(B_9) = 3, \quad \# \text{Surv}_2(B_9) = 1.$$

The other two words can precede the same ternary tail only by changing the deep Hensel extension. They share part of the visible residue, but not the genealogy of the state.

10.5.3 The catalog of 468 emissions as a product of channels

The finite catalogue starts from the set Ω_{seed} of 104 976 microscopic presentations. Let $\mathcal{U}_6$ be the set of distinct sextuple emissions and let

$$F : \Omega_{\text{seed}} \twoheadrightarrow \mathcal{U}_6.$$

The exact census is

$$|\mathcal{U}_6| = 468, \quad \#\{U : |F^{-1}(U)| = 144, 432, 1944\} = 432, 18, 18.$$

These counts belong to F . Comparing them with TPK histories of depth six would require an explicit map from the domain of histories to Ω_{seed} ; no result in this section presupposes that map or identifies catalog presentations with integral holonomic states. The finite catalog provides, for each emission, an additive signature s and a multiplicative signature e . The joint map is a bijection

$$\mathcal{U}_6 \xrightarrow{\sim} \mathcal{S}_+ \times \mathcal{S}_\times, \quad |\mathcal{S}_+| = 18, \quad |\mathcal{S}_\times| = 26.$$

Thus,

$$\boxed{468 = 18 \cdot 26}$$

is a factorization of the object and not merely of its cardinality. The first coordinate preserves how the emission sums; the second, how the product accumulates. This separation is the explicit finite form of sum–product incompatibility: both readings coexist in the state, but each projection forgets the complementary signature.

The first coordinate admits an additional parametrization that will be useful when comparing it with the nonadic boundary. If an additive seed has coordinates $(i, j) \in (\mathbb{Z}/9\mathbb{Z})^2$ and orientation $\varepsilon = +1$ for the east–south directions and -1 for north–west, its complete signature is determined bijectively by

$$(i + j \bmod 9, \varepsilon) \in (\mathbb{Z}/9\mathbb{Z}) \times \{\pm 1\}.$$

Therefore,

$$\mathcal{S}_+ \simeq (\mathbb{Z}/9\mathbb{Z}) \times \{\pm 1\}.$$

The product coordinate has no analogous uniformity: its 26 signatures have fibers of seeds of sizes 8, 24, and 108, with histogram $24 \times 8 + 1 \times 24 + 1 \times 108 = 324$. This asymmetry is structural. The factor 18 already contains phase and orientation; any subsequent descent of the product factor to a class of pairs must preserve additional memory or accept nonuniform fibers.

On $\ell^2(\mathcal{S}_+ \times \mathcal{S}_\times)$, let us define the conditional expectations

$$(E_+ f)(s, e) = \frac{1}{18} \sum_{s' \in \mathcal{S}_+} f(s', e),$$

$$(E_\times f)(s, e) = \frac{1}{26} \sum_{e' \in \mathcal{S}_\times} f(s, e').$$

The first operator renews the additive signature and preserves the multiplicative one; the second performs the dual operation. The third channel retains both.

10.5.4 The unique path and path transfer

Unlike the return of the odometer defined in the corresponding proposition, the following proposition proves the return of the observable operator F_{obs} .

The published observable law updates a single cursor at each step. In the phases +1, the additive cursor advances; in the phases −1, the multiplicative cursor advances; in the phases 0, both remain stationary. After the steps 27, 54, 81, 108, the two orientations are inverted. Denote this deterministic evolution by F_{obs} .

Proposition 10.46 (Return of the Observable Chronology). *For every choice of the two initial positions and orientations, the positions and orientations are restored after 54 steps. Upon also restoring the phase, one obtains*

$$F_{\text{obs}}^{108} = I.$$

Therefore, the Poincaré return of 108 steps induces the identity on $\mathcal{U}_6$ and does not select a unique stationary measure.

Proof. In every block of 27 steps, each cursor makes exactly nine advances in a fixed orientation. Since the position space is $X_9 = (\mathbb{Z}/9\mathbb{Z})^2$, its displacement is $9d = 0$. At the end of the block, d is inverted. Two blocks restore position and orientation; four also restore the index in Θ_{108} . The induced return is consequently the identity. Every probability is stationary for the identity, so none is selected by that chronology. $\square$

The problem arises even before the return if one attempts to forget the route memory. The product signature

$$e_0 = (5, 5, 5, 5, 5, 5)$$

has twenty-four representatives. After an advance in its own orientation, those representatives separate, eight by eight, among

$$(5, 5, 5, 7, 1, 7), \quad (5, 5, 5, 7, 7, 1), \quad (5, 5, 5, 1, 7, 7).$$

Thus, the product advance does not descend to a map of the twenty-six signatures: the representative that had been forgotten becomes relevant again. This witness explains why an individual route cannot be turned, by mere projection, into the refreshment of a complete fiber.

The APP reading is different. Instead of choosing a representative, it preserves all compatible continuations. To formalize it in the finite catalogue, consider the subalgebras of functions that depend only on one of the two signatures. With respect to uniform counting, E_+ and $E_\times$ are the conditional expectations that forget exactly the active coordinate. They are the only Markov projectors that simultaneously satisfy:

1. preservation of the coordinate inactive;
2. complete forgetting of the active coordinate;
3. preservation of the uniform counting state.

Indeed, a row that has forgotten the source coordinate must be constant on the 18, respectively 26, compatible destinations; the normalization forces the values $1/18$ and $1/26$. This is the relative canonicity of the renewal: no preference is postulated between branches that APP declares equally possible.

Let $C_9 = \mathbb{Z}/9\mathbb{Z}$, with the phase word

$$\tau = (+1, -1, 0, +1, -1, 0, +1, -1, 0).$$

Write

$$P_{+1} = E_+, \quad P_{-1} = E_\times, \quad P_0 = I.$$

Definition 10.47 (Phase-biased TPK transfer). On $\mathcal{U}_6 \times C_9$ we define

$$\mathcal{K}_{\text{ph}}((u, r), (v, r+1)) = P_{\tau(r)}(u, v),$$

and set the remaining entries to zero. This definition is relative to the reader of all compatible paths: an active phase uniformly renews the fiber read by that phase, and the neutral phase retains the emission.

Theorem 10.48 (Uniform nonadic return). *The kernel $\mathcal{K}_{\text{ph}}$ is doubly stochastic and irreducible, has period exactly nine, and possesses a unique stationary state,*

$$\pi_{\text{ph}}(u, r) = \frac{1}{9 \cdot 468}.$$

In each phase fiber, its return of nine steps is

$$\mathcal{K}_{\text{ph}}^9|_r = E_+ E_\times, \quad (E_+ E_\times)(u, v) = \frac{1}{468}.$$

If the initial phase is uniformly distributed, the shadow of a step over the emissions is

$$\bar{K}_{\text{ph}} = \frac{1}{3}(I + E_+ + E_\times).$$

Proof. Each $P_{\tau(r)}$ is doubly stochastic and the shift $r \mapsto r+1$ is a permutation; their skew product conserves both sums. The three operators are idempotent and commute. Since each appears three times in the nonadic word,

$$\prod_{r \in C_9} P_{\tau(r)} = E_+^3 E_\times^3 I^3 = E_+ E_\times.$$

The first expectation forgets the additive signature and the second the multiplicative signature, so their composition assigns to each of the $18 \cdot 26 = 468$ destinations the probability $1/468$. Hence, in nine steps any two states of the same phase communicate; the remaining steps communicate the phases, and the kernel is irreducible. Every return must have length divisible by nine, whereas the diagonal of $\mathcal{K}_{\text{ph}}^9$ is positive; the period is nine. Double stochasticity and irreducibility yield the unique uniform state. Finally, averaging the nine phases counts three copies of each channel and produces the final formula. $\square$

Forgetting the phase is not a strong Markov quotient: from the same emission, two distinct phases apply I , E_+ , or $E_\times$, which have distinct rows. The operator $\bar{K}_{\text{ph}}$ is an averaged shadow in the stationary phase, not the chronological successor of a phaseless emission. This distinction preserves both the dynamics and the meaning of the measure.

Definition 10.49 (Averaged kernel induced by transference). We call the phase shadow the averaged kernel of the catalog

$$K_{\text{cat}} = \frac{1}{3}(I + E_+ + E_\times).$$

The weights are induced by the TPK calendar, which contains three occurrences of each sign. The renewal operators are induced relative to the uniform normalization of all compatible continuations of the active fiber. Neither statement identifies K_{cat} with F_{obs} .

Proposition 10.50 (Convex family of stochastic kernels with a common uniform measure). For $a, b, c \geq 0$, $a + b + c = 1$, the family

$$K_{a,b,c} = aI + bE_+ + cE_\times$$

consists of symmetric, doubly stochastic kernels. If $b, c > 0$, they are irreducible and, because they have positive diagonal, aperiodic. Their spectrum is

$$\text{Spec}(K_{a,b,c}) = \{1^{[1]}, (a+c)^{[17]}, (a+b)^{[25]}, a^{[425]}\}.$$

In particular, there exists a continuum of dynamics with the same factorization and the same uniform measure. The choice $K_{\text{cat}} = K_{1/3,1/3,1/3}$ is not characterized by the census alone. It is selected jointly by the TPK phase word and the normalized transfer of the definition 10.47.

Proof. The three summands are symmetric stochastic kernels; by convexity, the same is true of $K_{a,b,c}$, and symmetry converts row stochasticity into column stochasticity. If $b, c > 0$, two steps make it possible to renew both coordinates successively and connect any pair of states. Moreover,

$$K_{a,b,c}((s, e), (s, e)) = a + \frac{b}{18} + \frac{c}{26} > 0,$$

so that the kernel is aperiodic. In the orthogonal decomposition

$$\mathbf{1} \oplus \ell_0^2(\mathcal{S}_+) \oplus \ell_0^2(\mathcal{S}_\times) \oplus (\ell_0^2(\mathcal{S}_+) \otimes \ell_0^2(\mathcal{S}_\times)),$$

the pairs $(E_+, E_\times)$ act, respectively, as $(1, 1)$, $(0, 1)$, $(1, 0)$, $(0, 0)$. The four eigenvalues and the multiplicities 1, 17, 25, 425 are thereby obtained. For $b, c > 0$, irreducibility makes the uniform stationary measure unique, which exhibits non-uniqueness even under that requirement. $\square$

Theorem 10.51 (Exact spectrum of the balanced refreshment). The kernel K_{cat} is symmetric, doubly stochastic, irreducible and aperiodic. Its spectrum, with multiplicities, is

$$\text{Spec}(K_{\text{cat}}) = \left\{ 1^{[1]}, \left(\frac{2}{3}\right)^{[42]}, \left(\frac{1}{3}\right)^{[425]} \right\}.$$

Consequently, its spectral gap is $(1/3)$ and its only stationary state is

$$\tau_{468}(f) = \frac{1}{468} \sum_{U \in \mathcal{U}_6} f(U).$$

Proof. The operators E_+ and $E_\times$ are commuting orthogonal averaging projectors. The rows and columns of each sum to one; the same is true of their average. From (s, e) , (s', e') is reached in two renewals, and the contribution of I provides a self-edge. This proves irreducibility and aperiodicity.

The space decomposes orthogonally into four sectors:

$$\mathbf{1}, \quad \ell_0^2(\mathcal{S}_+), \quad \ell_0^2(\mathcal{S}_\times), \quad \ell_0^2(\mathcal{S}_+) \otimes \ell_0^2(\mathcal{S}_\times).$$

Their dimensions are 1, 17, 25, $17 \cdot 25$. In the constant sector the three summands act as the identity; in each one-coordinate sector they act as $(I+I+0)$; in the interaction sector, as $(I+0+0)$. Dividing by three gives the announced eigenvalues and multiplicities. A finite irreducible and aperiodic kernel has a unique stationary state; since it is doubly stochastic, that state is uniform. $\square$

The theorem is exact relative to the factorization $\mathcal{U}_6 \simeq \mathcal{S}_+ \times \mathcal{S}_\times$, the TPK calendar, and the uniform count of the compatible continuations of each fiber. The family $K_{a,b,c}$ proves that the dynamics do not follow from the census alone. The proposition 10.46 distinguishes the transfer of all paths from the chronology of one of them.

10.5.5 Compensated elevation constructed from the quotient

Let us write $m(U) = |F^{-1}(U)|$. From the kernel already chosen in the quotient we define the kernel of presentations by

$$K_{\text{seed}}(x, y) = \frac{K_{\text{cat}}(F(x), F(y))}{m(F(y))}$$

This kernel is stochastic. Summing over a target fiber yields

$$\sum_{F(y)=V} K_{\text{seed}}(x, y) = K_{\text{cat}}(F(x), V),$$

independently of the representative x . This identity shows that the constructed lift projects exactly onto K_{cat} . It does not demonstrate that a preexisting microscopic dynamics of the integral holonomic state of TPK descends to the catalog.

Theorem 10.52 (Reversible elevation constructed from the quotient). *The probability*

$$\tilde{\tau}(x) = \frac{1}{468 m(F(x))}$$

is estacionaria and reversible for K_{seed} , and $F_\# \tilde{\tau} = \tau_{468}$. If

$$J_F \delta_U = m(U)^{-1/2} \sum_{F(x)=U} \delta_x, \quad D = \text{diag}(m(U)),$$

then

$$K_{\text{seed}} J_F = J_F (D^{1/2} K_{\text{cat}} D^{-1/2}).$$

In particular, Hilbert-space intertwining without conjugation by D occurs if and only if K_{cat} commutes with the fiber multiplicities.

Proof. Normalization follows because each of the 468 fibres receives total mass $(1/468)$. Using the symmetry of K_{cat} , for $F(x) = U$ and $F(y) = V$ we have

$$\tilde{\tau}(x) K_{\text{seed}}(x, y) = \frac{K_{\text{cat}}(U, V)}{468 m(U) m(V)} = \tilde{\tau}(y) K_{\text{seed}}(y, x).$$

This proves detailed balance, stationarity, and the identity of the image measure. Applying both sides of the last identity to a basis vector δ_V gives, on the fiber U , the coefficient $K_{\text{cat}}(U, V)/\sqrt{m(V)}$; that is exactly the coefficient of the conjugate operator shown. $\square$

Theorem 10.53 (Phase Lift Faithful to Quiescence). *Let $X = \Omega_{\text{seed}}$, $m(u) = |F^{-1}(u)|$, and let us define*

$$\begin{aligned} L_0(x, y) &= \mathbf{1}_{\{x=y\}}, \\ L_+(x, y) &= \frac{E_+(F(x), F(y))}{m(F(y))}, \\ L_\times(x, y) &= \frac{E_\times(F(x), F(y))}{m(F(y))}. \end{aligned}$$

On $X \times C_9$, let

$$\widehat{\mathcal{K}}_{\text{ph}}((x, r), (y, r+1)) = L_{\tau(r)}(x, y).$$

Then (F, id_{C_9}) satisfies the property of strong aggregability (strong lumpability) from $\widehat{\mathcal{K}}_{\text{ph}}$ to $\mathcal{K}_{\text{ph}}$. The microscopic kernel is irreducible, has period nine, and its unique stationary state is

$$\widetilde{\pi}_{\text{ph}}(x, r) = \frac{1}{9 \cdot 468 m(F(x))}.$$

Its nonadic return satisfies

$$\widehat{\mathcal{K}}_{\text{ph}}^9((x, r), (y, r)) = \frac{1}{468 m(F(y))}.$$

Proof. The sum of L_+ , respectively $L_\times$, over a target fiber is E_+ , respectively $E_\times$, and L_0 projects to the identity. This proves aggregability in each phase. The factors $m(F(y))$ cancel under composition: first the active signature is selected uniformly and then a representative of the obtained emission. Thus, L_+ and $L_\times$ are commuting projectors and

$$L_+ L_\times(x, y) = \frac{1}{468 m(F(y))}.$$

The positivity of this entry connects all representatives in nine steps. The phase forces the return lengths to be multiples of nine, and the diagonal entry of the return is positive. The stationary formula is verified by first summing the $m(u)$ representatives of each emission; each of the $9 \cdot 468$ phase–emission classes receives the same mass. $\square$

This lift improves the reading of the averaged kernel K_{seed} : in the neutral phase it preserves the microscopic representative, instead of silently remixing its fiber. Upon averaging and forgetting the phase, both constructions have the same shadow K_{cat} , but not the same hidden dynamics. The distinction is necessary in order to respect that the zero of TRIT is memory of quiescence.

10.5.6 Minimal congruences stable under advance, half-turn and 4th of a turn: 28 and 162 memory classes

The witness $e_0 = (5, 5, 5, 5, 5, 5)$ shows that the product signature is too coarse to transport a route. The exact missing minimal memory can be computed. Let

$$\mathcal{X}_\times = X_9 \times \mathcal{D}_{\text{card}}$$

the space of 324 product seeds. Let P denote the advance by one cell in the stored orientation, H the half-turn of that orientation, and Q the quarter-turn

$$N \mapsto E \mapsto S \mapsto O \mapsto N.$$

Theorem 10.54 (Forward and spin congruences). *The minimal congruence that refines the twenty-six product signatures and is stable under P and H has 28 classes: twenty-seven have size eight, and one has size 108. The only new splitting is*

$$(5, 5, 5, 5, 5, 5) \longrightarrow \mathcal{L}_0 \sqcup \mathcal{L}_1 \sqcup \mathcal{L}_2, \quad |\mathcal{L}_j| = 8.$$

Its orbits under P, H have sizes

$$9, \quad 9, \quad 9, \quad 1.$$

By combining this refinement with the eighteen additive signatures one obtains

$$18 \cdot 28 = 504 = 468 + 36.$$

The minimal congruence stable under P and Q has 162 classes of two seeds. Each class is an orbit of the involution

$$J(i, j, d) = (7 - i, 7 - j, \bar{d}) \pmod{9}.$$

The operators induced by P, Q act transitively on the 162 classes.

Proof. One begins with the partition by the twenty-six signatures and iteratively refines a class whenever two of its elements have images in distinct classes under any of the generators. The process terminates because $X_\times$ is finite. For (P, H) , the exhaustive census produces the histogram $27 \times 8 + 1 \times 108 = 324$, the four indicated orbits, and three classes over the exceptional fiber of size twenty-four. For (P, Q) , it produces $162 \times 2 = 324$. Direct verification proves that J preserves every signature, commutes with both generators, and pairs exactly the two elements of each class; the graph of classes generated by P, Q is connected. $\square$

The excess 36 and the object R_{36} have distinct types. The boundary at depth thirty-six is

$$R_{36} = \begin{pmatrix} 222220 \\ 021222 \\ 102011 \end{pmatrix},$$

that is, three ordered blocks of six trits, one per channel. It is not a set of thirty-six states. The term $504 - 468 = 36$, for its part, counts additional sheet memory: the half-turn that makes product transport functional adds exactly two new sheets to a single signature and, when repeated over the eighteen additive signatures, produces $18(3 - 1) = 36$. The coincidence of the integer (36) expresses two gradings of transport, boundary depth and sheet multiplicity, but does not authorize identification of their objects.

The quarter-turn Q was not hidden in F_{obs} either. The published chronology contains programmed advances and half-turns; incorporating Q is an explicit enlargement of the route alphabet. Once it is incorporated, a lazy symmetric random walk on $P^{\pm 1}, Q^{\pm 1}$ is irreducible and aperiodic on the 162 classes, and its uniform measure is unique. This is the exact mathematical form of “counting the turns”: the turn is a transport coordinate and not a subsequent decimal correction.

10.5.7 Mode and Orientation Cocycles

The same rule makes it possible to define the operational counters without introducing Dimensional Mechanics. Add to the state the last active mode $m \in \{\Sigma, \Pi\}$. A signature $+1$ fixes $m = \Sigma$, a signature -1 fixes $m = \Pi$, and a signature 0 preserves the previous mode. For an arrow a , define the signed cocycle

$$\omega_{\text{sw}}(a) = \begin{cases} +1, & \Pi \longrightarrow \Sigma, \\ -1, & \Sigma \longrightarrow \Pi, \\ 0, & \text{the mode is preserved.} \end{cases}$$

Its positive variation is $s(a) = |\omega_{\text{sw}}(a)|$. We further define $g(a) = 1$ when the ordered pair of cardinal orientations changes and $g(a) = 0$ when it is preserved. Keeping the mode and the terminal orientation fixed at the endpoints, the sums

$$\Omega_{\text{sw}}(\gamma) = \sum_{a \in \gamma} \omega_{\text{sw}}(a), \quad S(\gamma) = \sum_{a \in \gamma} s(a), \quad G(\gamma) = \sum_{a \in \gamma} g(a)$$

are additive under concatenation.

In a nonadic closed loop, the sequence of modes is

$$(\Sigma, \Pi, \Pi, \Sigma, \Pi, \Pi, \Sigma, \Pi, \Pi).$$

Therefore,

$$\Omega_{\text{sw}}(C_9) = 0, \quad S(C_9) = 6 :$$

there are three changes $\Pi \rightarrow \Sigma$, three changes $\Sigma \rightarrow \Pi$, and three retentions. In the cycle of 108 steps,

$$\#(+1) = \#(-1) = \#(0) = 36, \quad \Omega_{\text{sw}} = 0, \quad S = 72, \quad G = 4.$$

The final number counts the four simultaneous inversions programmed in 27, 54, 81, 108. These cocycles proceed directly from the TPK calendar.

10.5.7.1 Quotient of modes and areal-volumetric correspondence

The calendar sign, mode label, and geometric sheet are three typed objects. In particular, the phase sign is not identified with the angular observable on digits. For the geometric correspondence, we fix the typed HMT reader

$$\mathfrak{g}_{\text{av}}(\Sigma) = \mathcal{H}_{\text{ar}}, \quad \mathfrak{g}_{\text{av}}(\Pi) = \mathcal{H}_{\text{vol}}.$$

The first assignment expresses the additive or boundary reading; the second, the multiplicative or interior reading. The map $\mathfrak{g}_{\text{av}}$ is part of the sheet typing. The following result proves that, once that typing is fixed, TPK dynamics forces the incidence between them.

Theorem 10.55 (Necessary descent of leaf incidence). *Let m_r be the mode obtained after applying phase r of the primitive block $(+1, -1, 0)$, with the rule*

$$+1 : m \mapsto \Sigma, \quad -1 : m \mapsto \Pi, \quad 0 : m \mapsto m.$$

The cyclic orbit of modes is (Σ, Π, Π) . If

$$N_3(i, j) := \#\{r \in \mathbb{Z}/3\mathbb{Z} : (m_r, m_{r+1}) = (i, j)\},$$

then, in the order (Σ, Π) ,

$$N_3 = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

By repetition of the calendar,

$$N_9 = 3N_3, \quad N_{27} = 9N_3, \quad N_{108} = 36N_3.$$

After applying $\mathfrak{g}_{\text{av}}$, the primitive incidence matrix is, in the order $(\mathcal{H}_{\text{ar}}, \mathcal{H}_{\text{vol}})$,

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

Thus, the two cross transitions, the unique volumetric self-retention, the absence of areal self-retention and the unit primitive multiplicities are consequences of the TPK mode quotient; they are not four added postulates.

Proof. The three consecutive pairs, with cyclic closure, are

$$\Sigma \longrightarrow \Pi, \quad \Pi \longrightarrow \Pi, \quad \Pi \longrightarrow \Sigma.$$

Each appears once, and the pair $\Sigma \rightarrow \Sigma$ does not appear. This determines the four entries of N_3 . The calendars of lengths 9, 27, 108 respectively contain 3, 9, 36 copies of the primitive block, so their incidence matrices are the indicated multiples. The greatest common divisor of the nonzero entries of each long matrix removes only that global repetition and returns N_3 . Finally, g_{av} transports Σ, Π to the areal and volumetric sheets without altering the incidence. $\square$

The real hyperbolic realization that uses this matrix and proves its Lorentz invariance is formulated in section devoted to the three concrete generators of the tritic regimes; there F_{av} is used as an antecedent, without repeating its derivation.

This theorem must be distinguished from a stronger Markov assertion. Forgetting the phase is not a strong Markov quotient of $\mathcal{K}_{ph}$: the visible mode does not determine which of the three operators $I, E_+, E_\times$ acts at the next elementary iteration. What descends without an additional hypothesis is the edge multigraph of a TPKblock.

There is, however, a memory-one completion uniquely determined by that descent. Let X_{av} be the smallest subshift of finite type over $\{\mathcal{H}_{ar}, \mathcal{H}_{vol}\}$ that contains all the words obtained by forgetting the phase and retains only consecutive pairs. Every one-step subshift with that property must admit exactly the three observed pairs; the smallest such subshift excludes the sole unobserved pair. Its adjacency matrix is therefore F_{av} .

Corollary 10.56 (Parry Measure of the One-Memory Wrapper). *The subshift X_{av} is primitive, has topological entropy $\log \varphi$, and possesses a unique measure of maximal entropy. Its stationary vertex weights satisfy*

$$\boxed{\frac{\mu_{ar}}{\mu_{vol}} = \varphi^{-2}.$$

Proof. The Perron root of F_{av} is φ , and a positive Perron vector is $(1, \varphi)$. Since the matrix is symmetric, the Parry vertex weights are proportional to the product of the left and right vectors, that is, to $(1, \varphi^2)$. $\square$

The Parry measure belongs to the memory-one path envelope. It is not the image of τ_{468} , nor the temporal frequency of the periodic orbit of modes. The latter is

$$\nu_{cyc}(\mathcal{H}_{ar}) = \frac{1}{3}, \quad \nu_{cyc}(\mathcal{H}_{vol}) = \frac{2}{3},$$

whereas τ_{468} is uniform over emissions, and the Parry measure weights renewal histories of arbitrary length. The three measures answer distinct questions. The irrational ratio φ^{-2} is thus derived canonically from the path envelope, without attributing it to a quotient of finite cardinalities.

From chronological counting to the return marked by π The two formulas

$$\left(\frac{1}{3}, \frac{2}{3}\right) \quad \text{and} \quad \frac{\mu_{ar}}{\mu_{vol}} = \varphi^{-2}$$

are not approximations of one another. The first counts the three instants of the primitive calendar (Σ, Π, Π) ; the second weights all the admissible histories of the memory-one envelope. Hence the chronological frequency is rational, and the Parry ratio is irrational. In what follows, we write

$$\frac{\pi}{\varphi^2} = \pi\varphi^{-2};$$

this is not division by φ^{-2} .

The appearance of φ^{-2} can be seen directly in the linear dynamics. If F_n denotes the n -th Fibonacci number,

$$F_{\text{av}}^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix} \quad (n \geq 1).$$

The eigenvalues of F_{av} are φ and $-\varphi^{-1}$. The contracted branch reverses orientation in one iteration. Upon passing to the quadratic memory, namely $\text{Sym}^2(\mathbb{R}^2)$, that reversal is recorded before the sign disappears. In the basis $(x^2, 2xy, y^2)$, with the images of the basis vectors represented by rows, the induced operator is

$$\text{Sym}^2(F_{\text{av}}) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{pmatrix}, \quad \chi(t) = (t+1)(t^2 - 3t + 1).$$

Its three eigenvalues are

$$\varphi^2, \quad -1, \quad \varphi^{-2}.$$

Thus, φ^{-2} is the positive stable mode of second-order memory. This step explains why the golden ratio appears here as a return law and not as an added statistical mean.

The closure π belongs to another level of the same HMT state: it is the coinductive output of the regional reader already constructed, not a new eigenvalue of F_{av} . Let

$$P_{\text{ar}} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad P_{\text{vol}} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Among the positive operators that preserve both sheets, there is exactly one whose volumetric normalization is one and whose areal mark is the generated closure π :

$$D_\pi = \pi P_{\text{ar}} + P_{\text{vol}}.$$

Uniqueness is relative to these three explicit clauses: positivity, diagonality with respect to the leaves, and the two normalizations. With $e_{\text{ar}} = (1, 0)^T$ and $e_{\text{vol}} = (0, 1)^T$, the marked return of length n is

$$\Theta_n := \frac{\langle e_{\text{ar}}, D_\pi F_{\text{av}}^n e_{\text{ar}} \rangle}{\langle e_{\text{vol}}, F_{\text{av}}^n e_{\text{vol}} \rangle} = \pi \frac{F_{n-1}}{F_{n+1}}.$$

Therefore,

$$\boxed{\lim_{n \rightarrow \infty} \Theta_n = \frac{\pi}{\varphi^2}.$$

The mark π is applied only once, at the boundary return; it is not reinjected at every transition. Finite combinatorics produces the stable mode φ^{-2} , the coinductive reader produces the closure π , and D_π composes both data without reversing their causal order.

The native length 108 provides an exact witness to that convergence:

$$(F_{\text{av}}^{108})_{\text{ar}, \text{ar}} = F_{107} = 10284720757613717413913,$$

$$(F_{\text{av}}^{108})_{\text{vol}, \text{vol}} = F_{109} = 26925748508234281076009.$$

Consequently,

$$\left| \frac{F_{107}}{F_{109}} - \varphi^{-2} \right| = 6.1684979916614407936 \dots \times 10^{-46},$$

and, after the only closing marker,

$$\left| \pi \frac{F_{107}}{F_{109}} - \frac{\pi}{\varphi^2} \right| = 1.9378907974286976068 \dots \times 10^{-45}.$$

These errors measure the finite approximation of the return; they are not adjustments of π or of φ .

Cylindrical reader of the ratio. Let $I_n^\pi = [\pi_n, \bar{\pi}_n]$ and $I_n^\varphi = [\varphi_n, \bar{\varphi}_n]$ be the positive, nested cylinders produced by the two HMT readers. The interval operation

$$Q(I, J) := \left[\frac{\inf I}{(\sup J)^2}, \frac{\sup I}{(\inf J)^2} \right]$$

is the smallest interval containing x/y^2 for every $(x, y) \in I \times J$. It preserves nesting: if the input cylinders are refined, then $Q(I_n^\pi, I_n^\varphi)$ is also refined. Since their diameters tend to zero and the limit of I_n^φ is positive,

$$\bigcap_n Q(I_n^\pi, I_n^\varphi) = \left\{ \frac{\pi}{\varphi^2} \right\}.$$

With the twelve already published base-thousand blocks, the quotient interval forces thirty-five decimal digits:

$$\frac{\pi}{\varphi^2} = 1.19998161486432666111577775331680692 \dots$$

This construction preserves the provenance of each digit: the numerator comes from the closure cylinder and the denominator from the autoscale.

Algebraic separation between areal reading, volumetric reading and their respective codomains

The complete ratio cannot arise from the rational algebra of F_{av} alone. Indeed, its rational spectral functions belong to $\mathbb{Q}(\sqrt{5})$; φ^{-2} is found there, but not the transcendental number π . Consequently, any derivation purporting to obtain π/φ^2 solely from the matrix 2×2 would be ill typed. The transcendental datum must derive from the coinductive closure reader, or from an equivalent boundary cocycle proved independently. This negative control fixes the division of labor between the finite correspondence and the number reader.

Exchange involution of the quotient The correspondence admits two exact orientations of the same law. If

$$\mathcal{R}(x, y) = \frac{x}{y^2}, \quad \mathcal{J}(x, y) = (y, x),$$

then $\mathcal{J}^2 = I$ and

$$\mathcal{R}(\pi, \varphi) = \frac{\pi}{\varphi^2}, \quad (\mathcal{R} \circ \mathcal{J})(\pi, \varphi) = \frac{\varphi}{\pi^2}.$$

The identities are also verified.

$$\mathcal{R}(\pi, \varphi) \mathcal{R}(\varphi, \pi) = \frac{1}{\pi \varphi}, \quad \frac{\mathcal{R}(\pi, \varphi)}{\mathcal{R}(\varphi, \pi)} = \left(\frac{\pi}{\varphi} \right)^3.$$

Writing $\mathcal{B}_{\text{av}}^{(\pi)} = \mathcal{R}(\pi, \varphi)$ also yields the exact reparametrization

$$\frac{\varphi}{\pi^2} = \frac{1}{\varphi^3 (\mathcal{B}_{\text{av}}^{(\pi)})^2}, \quad e^{\varphi/\pi^2} = \exp \left[\frac{1}{\varphi^3 (\mathcal{B}_{\text{av}}^{(\pi)})^2} \right].$$

Thus, the change of orientation between closure and growth transports the boundary ratio to its exchanged reading:

$$\frac{\pi}{\varphi^2} \xleftrightarrow{J} \frac{\varphi}{\pi^2}.$$

The first orientation reads the marked areal closure over quadratic volumetric memory; the second reads interior growth over quadratic closure. This correspondence is a theorem of the HMT reader. Dimensional Mechanics subsequently applies the second orientation within the electronic construction, without converting the electron into an antecedent of π , φ , or of the correspondence.

Ambivalence Principle of the areal and volumetric readers The Ambivalence Principle asserts that the areal reading and the volumetric reading are two coordinated, but not interchangeable, functionals of one and the same state with memory. Their coincidence on the diagonal is

$$\mathcal{L}_{\text{ar}}^{\text{diag}}(x) = \mathcal{L}_{\text{vol}}^{\text{diag}}(x).$$

The diagonal equality does not erase the ambivalence outside of it.

On the areal–volumetric boundary, both readers factor through the same positive memory amplitude $\mathcal{M}(x)$:

$$\mathcal{L}_{\text{vol}}(x) = \mu_{\text{vol}} \mathcal{M}(x), \quad \mathcal{L}_{\text{ar}}(x) = \pi \mu_{\text{ar}} \mathcal{M}(x).$$

Then the Parry theorem and the return mark give

$$\boxed{\frac{\mathcal{L}_{\text{ar}}(x)}{\mathcal{L}_{\text{vol}}(x)} = \frac{\pi}{\varphi^2}}.$$

With volumetric normalization, the relative difference of readers is

$$\frac{\mathcal{L}_{\text{ar}} - \mathcal{L}_{\text{vol}}}{\mathcal{L}_{\text{vol}}} = \frac{\pi}{\varphi^2} - 1 = 0.1999816148643266611 \dots$$

The physical reading of this correspondence is undertaken later, in the Cartan–Holst chapter, after constructing holonomy, co-tetrad, connection, torsion, and curvature. Here the result is entirely HMT: an exact ratio between two readers of the same sheet memory.

They should be distinguished from the enriched extractor subsequently used in Dimensional Mechanics. There a complete history preserves the action count n_A , the twelve oriented events $e_0, \dots, e_{11}$ and the torsional cochain at step level; these yield

$$s_C = \sum_{i=0}^5 (e_i + e_{i+6}), \quad W_{\pm} = 6n_A \pm s_C.$$

The channels already generated by HMT then satisfy the exact identity

$$\exp \left(n_A A_{\text{rad}} + \frac{s_C}{6} C_{\text{rad}}^* \right) = \left(q_+^{-W_+} q_-^{-W_-} \right)^{1/12}.$$

Therefore, the route–angular-character passage is not absent: it is realized on the enriched dodecaphase record. What this phase calculation proves by itself are Ω_{sw} , S , G ; it neither replaces them nor permits reconstruction of the complete record after it has been projected.

The uniform measure on the 104 976 presentations descends to atoms of masses $1/729$, $1/243$ and $1/54$, but not to $1/468$. In the fivefold microfiber of π , the two probabilities are

$$\tau_{468}(\mathcal{R}_\pi) = \frac{5}{468}, \quad F_\# \mu_{\text{seed}}(\mathcal{R}_\pi) = \frac{7}{729}.$$

Both the averaged lift and the phase lift realize the first: they do not push forward the uniform measure on seeds, but assign the same total mass to each observable class. The phase lift also preserves microscopic stillness at neutral steps.

To relate phase elevation to the enriched histories of depth six, a kernel $\mathcal{K}_6$ on $\mathcal{H}_6^{\text{cat}} \times C_9$ is required. It descends to $\widehat{\mathcal{K}}_{\text{ph}}$ through $\rho_6 \times \text{id}_{C_9}$ if and only if it satisfies, at each phase, the strong lumpability condition

$$\sum_{\rho_6(g)=y} \mathcal{K}_6((h, r), (g, r+1)) = L_{\tau(r)}(\rho_6(h), y),$$

independently of h within each fiber of ρ_6 . In that case, composition with F projects to $\mathcal{K}_{\text{ph}}$. Constructing a family $(\mathcal{K}_m)_{m \geq 6}$ that satisfies this identity and commutes with all truncations is the remaining step toward a transfer of the inverse limit.

10.5.8 Scale cocycle and cylindrical conservation

Let $r = (1, \varphi)$ be a Perron–Frobenius vector of F_{av} . The Parry transition matrix is

$$P_{ij} = \frac{(F_{\text{av}})_{ij} r_j}{\varphi r_i} = \begin{pmatrix} 0 & 1 \\ \varphi^{-2} & \varphi^{-1} \end{pmatrix}.$$

For an admissible route $\gamma : i \rightarrow j$ of length n ,

$$\mathbb{P}(\gamma \mid i) = \frac{r_j}{\varphi^n r_i}, \quad \boxed{\chi_P(\gamma) = \varphi^n \frac{r_i}{r_j}}.$$

The telescope proves

$$\chi_P(\gamma_2 \circ \gamma_1) = \chi_P(\gamma_2) \chi_P(\gamma_1).$$

On a closed route, $\chi_P = \varphi^n$; its square and its inverse generate the combinations

$$C_n = \frac{\chi_P^2 + \chi_P^{-2}}{2}, \quad S_n = \frac{\chi_P^2 - \chi_P^{-2}}{2},$$

which satisfy

$$x_{n+1} = 3x_n - x_{n-1}.$$

Since F_{av} is symmetric, let

$$\ell = \frac{r}{1 + \varphi^2}, \quad \ell^\top r = 1.$$

The measure of a cylinder is

$$\mu[x_0 \cdots x_n] = \ell_{x_0} r_{x_n} \varphi^{-n}.$$

Therefore,

$$\boxed{V_n(x) = \frac{\mu[x_0]}{\mu[x_0 \cdots x_n]} = \varphi^n \frac{r_{x_0}}{r_{x_n}} = \chi_P(x_0 \cdots x_n)}.$$

For a typed dimension scale $D > 0$,

$$a_n^{(D)}(x) = V_n(x)^{1/D}$$

satisfies the cylindrical conservation law

$$\boxed{(a_n^{(D)}(x))^D \mu[x_0 \cdots x_n] = \mu[x_0]}.$$

In spatial rank three and on closed returns, we write $a_n := a_n^{(3)}$ and obtain

$$\boxed{\frac{a_9}{a_0} = \varphi^3, \quad \frac{a_{27}}{a_0} = \varphi^9, \quad \frac{a_{108}}{a_0} = \varphi^{36}}.$$

If $A_k := a_{9k}$ for $k \geq 1$, then, for $k \geq 2$,

$$A_{k+1} - 2A_k + A_{k-1} = A_{k-1}(\varphi^3 - 1)^2 > 0.$$

The inequality proves convexity in return depth; a physical acceleration further requires a duration chart.

The primary computation of $\text{Sym}^2(F_{\text{av}})$ in section 10.5.7.1 shows that its spectrum is $\{\varphi^2, -1, \varphi^{-2}\}$. Here we conserve only the corollary for the scale cocycle: on a closed route with $V_n = \varphi^n$, the stable mode obeys

$$M_n = \varphi^{-2n} = V_n^{-2}.$$

Identification of these three eigenspaces with the three TRIT sheets requires an intertwiner; coincidence of cardinality does not replace it. The corollary would be refuted if the closed route failed to satisfy $V_n = \varphi^n$ or if the stable mode did not have eigenvalue φ^{-2} .

10.5.9 Dynamic prolongation theorem: global hypotheses and certified domain

The following objects have been explicitly constructed and separated from the finite problem:

$$F, \quad F_{\text{obs}}, \quad \mathcal{K}_{\text{ph}}, \quad \widehat{\mathcal{K}}_{\text{ph}}, \quad K_{\text{cat}}, \quad \tau_{468}, \quad F_{\text{av}}, \quad \text{Sym}^2(F_{\text{av}}), \quad D_{\pi}, \quad Q.$$

For F_{obs} , deterministic return has been proved and an explicit witness of failure to descend as a product has been exhibited. For $\mathcal{K}_{\text{ph}}$, double stochasticity, irreducibility, period nine, uniform nonadic return, and uniqueness of π_{ph} have been proved. For $\widehat{\mathcal{K}}_{\text{ph}}$, strong lumpability, microscopic conservation of rest, and the compensated stationary measure have been proved. K_{cat} is no longer an average without provenance: it is the shadow of one step after the uniform phase is averaged. The TPK mode quotient induces F_{av} ; its quadratic lift separates the modes $\varphi^2, -1, \varphi^{-2}$. The operator D_{π} marks the boundary return exactly once with the generated closure, and the interval reader Q preserves the nesting of the cylinders of π and φ . These four objects construct π/φ^2 without conflating temporal frequency, the uniform measure, and the Parry measure.

The force of this induction is relative and concrete. The TPK word forces the weights $1/3$. The APP interpretation of all compatible continuations, together with uniform counting, forces E_+ and $E_{\times}$. The deterministic trajectory F_{obs} does not force those operators and, indeed, does not produce them. This is not a defect of the result, but the mathematical difference between evolution along a path and transfer over a space of paths.

The finite conclusion is complete in its domain: the uniform return, the stationary measure, the lumpability at depth six, and the separation between evolution of a route and transfer of the route space remain proved by the constructed objects. In particular, $\widehat{\mathcal{K}}_{\text{ph}}$ is not identified with the chronology of F_{obs} , as proposition 10.46 shows.

Projective measure and evolutionary conservation of the unit

11.1 Projective measure and evolutionary conservation of the unit

Cylinder topology provides a notion of proximity; a theory of paths also requires weights. The evolutionary conservation of the unit here takes an exact probabilistic form: every cylinder distributes its weight among its extensions, and the total sum remains equal to one. The unit is not fixed in a cell; it is transported through refinement.

11.1.1 Consistent weights on finite levels

Let μ_n be a probability on the finite set $\mathcal{S}_n$. The family is *projectively consistent* when

$$\mu_n(a) = \sum_{\substack{b \in \mathcal{S}_{n+1} \\ \rho_{n+1,n}(b)=a}} \mu_{n+1}(b) \quad (a \in \mathcal{S}_n). \quad (11.1)$$

This equality expresses the local preservation of the unit: the weight of a truncated history is the sum of the weights of all its children.

Theorem 11.1 (Genealogical measure on cylinders). *Every family of finite probabilities that satisfies (11.1) determines a unique Borel probability μ_∞ on Ω_∞ such that*

$$\mu_\infty([a]_n) = \mu_n(a). \quad (11.2)$$

Proof.

$$\mathcal{A} := \{\text{finite unions of cylinders}\}$$

is an algebra: the intersection of two cylinders is either empty or a cylinder of the deeper level. Every element of $\mathcal{A}$ can be refined into a disjoint union of cylinders of a common level. If $A = \bigsqcup_{a \in I} [a]_n$, we set

$$\mu_0(A) = \sum_{a \in I} \mu_n(a).$$

Iterating (11.1) proves that the value does not depend on the chosen common level.

If a cylinder $[a]_n$ is empty, finite branching and Koenig's lemma [K"o36] imply that there is a depth with no descendants of a . Iterated consistency then gives $\mu_n(a) = 0$. Hence μ_0 is well defined even for representations containing empty cylinders and is finitely additive.

The cylinders are both open and closed. If a decreasing sequence of elements of $\mathcal{A}$ had empty intersection, compactness would imply that one of them was already empty; hence μ_0 is continuous at the empty set and constitutes a countably additive premeasure. The Caratheodory extension theorem—equivalently, the Kolmogorov extension for this finite projective system—extends it to the σ -algebra generated by the cylinders [Car14, Kol33]. This is the Borel algebra of Ω_∞ , and finiteness of the measure ensures uniqueness. $\square$

The measure may follow from local transitions. If $P_n(b \mid a) \geq 0$ satisfies

$$\sum_{\rho_{n+1,n}(b)=a} P_n(b \mid a) = 1,$$

then $\mu_{n+1}(b) = \mu_n(a)P_n(b \mid a)$ is consistent. Uniformity is a possible choice, not an obligation. The weights may depend on the sheet, carry, orientation, a nonnegative action cost, or a record, provided they preserve the sum. For example,

$$P_n(b | a) = \frac{e^{-\beta a_n(b|a)}}{\sum_{b' \mapsto a} e^{-\beta a_n(b'|a)}}, \quad a_n \geq 0,$$

defines normalized positive weights. The orthogonal action phase belongs to the kernel of chapter devoted to the genealogical path algebra and its integral representation; it is not inserted into these positive probabilities.

11.1.2 Decomposition of the genealogical measure on fibers of equal Archimedean value

The Archimedean evaluation transports the measure to K :

$$\nu := \text{Val}_* \mu_\infty, \quad \nu(B) = \mu_\infty(\text{Val}^{-1}(B)). \quad (11.3)$$

The visible probability of a region sums the weight of all the genealogies that publish it. When two sections possess the same value, their contributions are aggregated in ν , although they remain separate in μ_∞ .

Proposition 11.2 (Value Decomposition). *If K is a standard Borel space, there exists a family of conditional measures $\{\mu_y\}_{y \in K}$, defined for ν -almost every y , such that μ_y is supported on $\mathfrak{G}_y = \text{Val}^{-1}(y)$ and*

$$\int_{\Omega_\infty} F \, d\mu_\infty = \int_K \left(\int_{\mathfrak{G}_y} F \, d\mu_y \right) d\nu(y)$$

for every integrable function F .

This formula separates value and genealogy at the level of the measure. Physics that depends only on the value first integrates the entire fibre; a memory observable can distinguish different distributions μ_y over the same real coordinate.

11.1.3 Conditioning of the measure through restriction of the future extension tree

Let $A \subseteq \Omega_\infty$ be an observable event with $\mu_\infty(A) > 0$. The conditional measure is

$$\mu_\infty(B | A) = \frac{\mu_\infty(B \cap A)}{\mu_\infty(A)}. \quad (11.4)$$

When A is the reading of a register, its intersection with each cylinder eliminates the incompatible extensions. Normalization does not create new branches: it redistributes the unit among those that have survived.

Proposition 11.3 (Compatible update). *If A is a union of cylinders at level m , then for every $n \geq m$ the conditional distribution on $\mathcal{S}_n$ is obtained by removing the states whose truncations do not belong to A and renormalizing the remaining weights. The conditional distributions continue to satisfy (11.1).*

Proof. The identity is Bayes' rule on finite sets. The sum of the weights of the conditioned children coincides with the conditioned weight of the parent, because intersection and disjoint union commute. $\square$

11.1.4 Combinatorial conservation and dimensional completion

The same law appears in the binomial completion:

$$1 = \sum_{r=0}^d \binom{d}{r} \left(\frac{9}{10} \right)^{d-r} \left(\frac{1}{10} \right)^r.$$

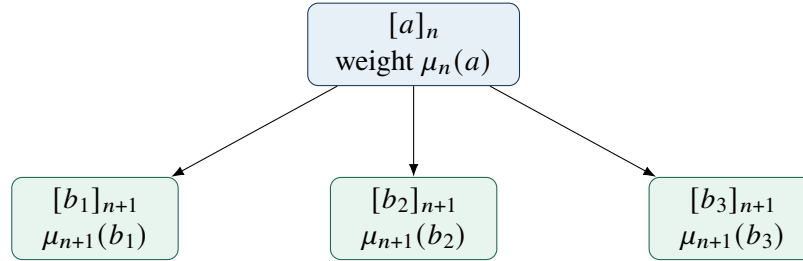
In three dimensions,

$$1000 = 729 + 243 + 27 + 1,$$

so that the interior, new faces, extreme edges, and final vertex form a partition of decimal unity. The passage $729 + 270 = 999 \rightarrow 729 + 271 = 1000$ distinguishes the state prior to closure of the completed unity. In the path system, the analogous identity is

$$\mu([a]_n) = \sum_{b \mapsto a} \mu([b]_{n+1}).$$

The unit evolves as it is distributed among the strata and is recovered by summing the complete genealogy.



$$\mu_n(a) = \mu_{n+1}(b_1) + \mu_{n+1}(b_2) + \mu_{n+1}(b_3):$$

the unit is preserved when the genealogy is refined.

Figure 11.1: Projective weight conservation on cylinders.

Hypothesis, domain, and conclusion

The projective measure is constructed on the HMT inverse system. The extension theorem is classical mathematics; the identification of its cylinders with TPK states is exact once the transition weights have been fixed. The selection of a specific physical measure belongs to the dynamical model and does not follow solely from the census of branches.

The projective measure is characterized by positivity, conservation of weight between parents and children, and conditioning within the observed event. A complete experimental realization adds preparation and the sampling protocol; with those data, the same measure produces reading probabilities comparable with frequencies.

Archimedean and exceptional morphisms from a common domain

12.1 Principle of Ambivalence: involution of the areal and volumetric readers on the integral holonomic state

The double projection compares two evaluations of the common enriched domain X_∞^{cal} already constructed in theorem 12.8. One produces Archimedean values; the other, together with a calibration of its codomain, preserves the combinatorial incidence that leads to codes, designs, and automorphism groups. Both evaluations start from the unique initial state X_∞^{cal} and preserve their common provenance in distinct codomains.

12.1.1 Domains of the 2 projections

On X_∞^{cal} , the cylinders contract to a point and the boundary admits the incidence prolongation. In this residence, we fix $\mathbf{Arch} := \mathbb{R}$. The Archimedean evaluation and the exceptional evaluation are, respectively,

$$\mathfrak{R} : X_\infty^{\text{cal}} \longrightarrow \mathbf{Arch}, \quad \mathfrak{E} : X_\infty^{\text{cal}} \times \mathcal{C}_{\text{exc}} \longrightarrow \mathbf{Exc}.$$

Here $\mathcal{C}_{\text{exc}}$ contains the projective order, Paley gauge, dodecad, incidence alignment, and lattice chamber specified in each construction. The first map produces values; the second produces codes, designs, lattices, and automorphisms. The preceding global theorem is the source of compatibility and permits comparison of invariants obtained through distinct operations without again intersecting domains defined separately. The finite comparison in this work is performed on the flag $B_K \subset H_\alpha^-$.

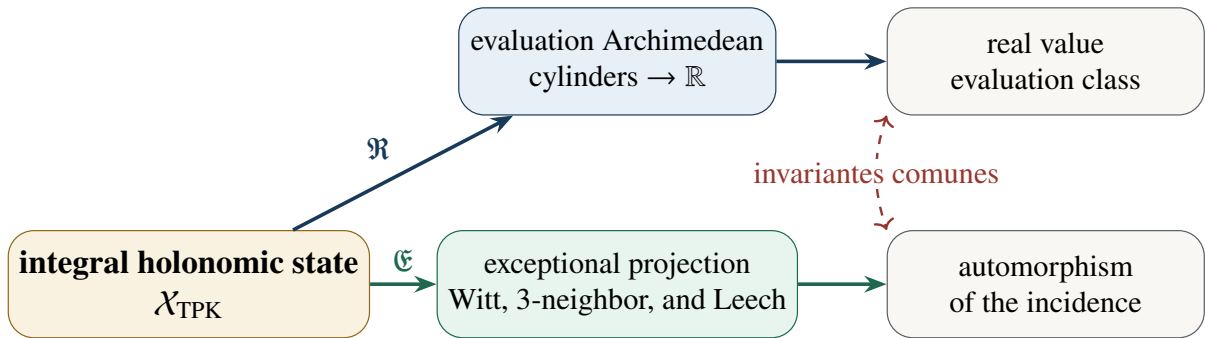


Figure 12.1: Typed scheme of the two projections. The Archimedean evaluation identifies states within the same fiber; the exceptional evaluation, relative to its calibration, preserves incidence structure and automorphisms.

12.1.2 Topology of the space of histories

The formulation by cylinders admits a standard realization as an inverse limit. Let S_n be the finite sets of states admissible at depth n , endowed with the discrete topology, and let

$$\tau_{n+1,n} : S_{n+1} \longrightarrow S_n$$

the aplicaciones of truncation. Supondremos in this seccion that are sobreyectivas and definiremos

$$X = \varprojlim_n S_n.$$

If $x \neq y$, let $m(x, y)$ be the first depth at which their components differ. The function

$$d(x, y) = 3^{-m(x, y)}, \quad d(x, x) = 0,$$

is an ultrametric that induces the topology of the inverse limit.

Theorem 12.1 (Space of compact histories). *The space $\mathcal{X}$ is compact and totally disconnected. When every history admits bifurcations at arbitrarily large depths, $\mathcal{X}$ has no isolated points and is homeomorphic to Cantor space.*

Proof. The product $\prod_n \mathbf{S}_n$ is compact. The equations $\tau_{n+1, n}(x_{n+1}) = x_n$ define a closed subset, so that $\mathcal{X}$ is compact. The sets of histories with a fixed prefix are simultaneously open and closed and form a basis; this proves total disconnectedness. Under the hypothesis of arbitrarily deep branching, no cylinder contains an isolated history. The characterization of compact, metric, perfect, and totally disconnected spaces proves the final assertion. $\square$

Associate with each $s \in \mathbf{S}_n$ a closed interval $I_n(s)$, compatibly with truncation, and suppose that

$$\text{diam } I_n(s) \leq C\lambda^n, \quad 0 < \lambda < 1.$$

The intersection of the intervals corresponding to a history then contains a unique point.

Theorem 12.2 (Archimedean Evaluation). *The map*

$$\text{val} : \mathcal{X} \longrightarrow \mathbb{R}, \quad \{\text{val}(x)\} = \bigcap_n I_n(x_n),$$

is continuous and Hölder with respect to d . If $x \sim_{\text{Arch}} y$ is equivalent to $\text{val}(x) = \text{val}(y)$, the induced application is a homeomorphism

$$\mathcal{X}/\sim_{\text{Arch}} \cong \text{val}(\mathcal{X}).$$

Proof. Two histories with a common prefix of length n have values in the same interval of diameter no greater than $C\lambda^n$. Since $d(x, y) = 3^{-n}$, this estimate is a Hölder bound. The induced map on the quotient is a continuous bijection from a compact space onto a Hausdorff space and therefore a homeomorphism. $\square$

The theorem formalizes the Archimedean projection: the value retains the evaluation class and eliminates the distinction between histories lying in the same fiber. Its image is $\text{val}(\mathcal{X})$; identifying it with the whole of $\mathbb{R}$ additionally requires surjectivity to be proved.

12.1.3 Separation by the 2 projections

Theorem 12.3 (Joint separation). *Let $P_1 : \mathcal{X} \rightarrow Y_1$ and $P_2 : \mathcal{X} \rightarrow Y_2$ be continuous maps with Hausdorff codomains. If*

$$x \neq y \implies P_1(x) \neq P_1(y) \text{ or } P_2(x) \neq P_2(y),$$

the diagonal application

$$(P_1, P_2) : \mathcal{X} \longrightarrow Y_1 \times Y_2$$

is a topological embedding.

Proof. The hypothesis expresses precisely the injectivity of the diagonal map. A continuous injective map from a compact space to a Hausdorff space is a homeomorphism onto its image. $\square$

With a declared gauge $c \in \mathcal{C}_{\text{exc}}$ fixed, let

$$\mathfrak{E}_c : X_{\infty}^{\text{cal}} \longrightarrow \mathbf{Exc}, \quad \mathfrak{E}_c(x) := \mathfrak{E}(x, c).$$

Applied to HMT, the theorem takes

$$P_1 = \mathfrak{R}, \quad P_2 = \mathfrak{E}_c$$

on the common domain, or on the quotient by the coordinate changes that have been declared. The global separation assertion requires verifying that no pair of inequivalent histories remains in the same fiber of both maps.

At a finite level S , this condition is decidable. Given $p : S \rightarrow A$, $q : S \rightarrow B$, and an equivalence relation of coordinates, the pairs $x \sim y$ that simultaneously satisfy are enumerated

$$p(x) = p(y), \quad q(x) = q(y).$$

The absence of such pairs is equivalent to joint injectivity at that level. The calculation must perform the two maps separately: when $q = f \circ p$, the second has the same fibers as the first and does not recover information eliminated by p .

At the level developed here, the equality

$$B_K = H_e \cap P_\pi$$

constitutes a finite witness of compatibility between the two branches. It does not by itself establish joint separation at every depth; that assertion corresponds exactly to the hypothesis of [theorem 12.3](#).

12.1.4 Archimedean quotient by the evaluation fibers

An HMT section contains a block, an orientation, a carry, an observable label, and an extension coordinate. The evaluation val contracts its cylinders to a point. If two sections x, y satisfy

$$\text{val}(x) = \text{val}(y),$$

the Archimedean evaluation identifies them although their transport is different. In this precise sense,

a real value is an equivalence class of the evaluation.

On any subdomain on which addition, multiplication, and evaluation descend compatibly, the image inherits those operations. This section does not thereby construct all of $\mathbb{R}$ or prove surjectivity onto $\mathbb{R}$.

A global identification with $\mathbb{R}$ would require existence of the limit, determination of its image, descent of sum and product, and characterization of the fibres. The finite windows and the conditional inverse-limit theorem provide the structure; prolongation to arbitrary depth requires the hypotheses of unit survival and compatibility.

12.1.5 Common invariant with the Witt system

The vector K defines the support superior

$$B_K = \{m : K_m \geq 729\} = \{4, 6, 9, 10\}.$$

In the Witt atlas appear the supports

$$P_\pi = \{2, 4, 5, 6, 7, 8, 9, 10, 11\},$$

$$H_e = \{1, 3, 4, 6, 9, 10\}, \quad H_\varphi = \{1, 2, 3, 4, 7, 9\}.$$

The support of the negative carries of α is

$$H_\alpha^- = \{4, 5, 6, 7, 9, 10\}.$$

The support identities are verified.

$$B_K = H_e \cap P_\pi = H_e \cap H_\alpha^-,$$

and

$$H_\alpha^- = B_K \sqcup \{5, 7\}.$$

Witt polarity sends the four points of B_K to the pairs

$$\{1, 3\}, \quad \{2, 11\}, \quad \{8, 12\}, \quad \{5, 7\}.$$

Three receive the positive label; the last completes the negative support. These identities are exact in the declared gauge, but they are conditioned controls: the vector, the threshold, the words, and the Paley gauge are not statistically independent inputs. The equality follows by direct evaluation and does not constitute a probabilistic estimate.

12.1.6 Residual design on a dodecada and elevation of hexads

The relation between the two Witt designs is obtained within the extended binary Golay code and not through an identification of cardinalities. Let

$$\mathcal{G}_{24} \subset \mathbb{F}_2^{24}$$

the extended binary Golay code, and denote by $\mathcal{O}_{24}$ the family of supports of its weight-eight words. It is a classical result that $(\{1, \dots, 24\}, \mathcal{O}_{24})$ is the Steiner system $W_{24} = S(5, 8, 24)$ [MS77, CS99]. Fix a weight-twelve word and write D for its support. We define

$$\mathcal{H}(D) := \{O \cap D : O \in \mathcal{O}_{24}, |O \cap D| = 6\}. \quad (12.1)$$

The family $\mathcal{H}(D)$ is intrinsic to the pair $(\mathcal{G}_{24}, D)$: it does not require choosing coordinates in D .

Theorem 12.4 (Residual design of a dodecada). *The incidence system*

$$(D, \mathcal{H}(D))$$

is a Steiner system $S(5, 6, 12)$. In particular, $|\mathcal{H}(D)| = 132$, and every subset of five points of D is contained in a unique hexad of $\mathcal{H}(D)$.

Proof. Let $A \subset D$, with $|A| = 5$. Since the octads form $S(5, 8, 24)$, there is a unique octad O_A containing A . If $j = |O_A \cap D|$, the binary sum of the characteristic words of O_A and D belongs to $\mathcal{G}_{24}$ and has weight

$$\text{wt}(\mathbf{1}_{O_A} + \mathbf{1}_D) = 8 + 12 - 2j = 20 - 2j.$$

The weights of the extended binary Golay code are 0, 8, 12, 16, 24. Since $j \leq 8$, it follows that $j \in \{2, 4, 6\}$. The inclusion $A \subset O_A \cap D$ gives $j \geq 5$, hence $j = 6$. Therefore, $O_A \cap D \in \mathcal{H}(D)$ is a hexad containing A .

If two members of $\mathcal{H}(D)$ contained A , their octads of origin would contain the same set of five points; the Steiner property of W_{24} would make them coincide. This proves uniqueness. The number of blocks is obtained by counting incidences with five-point subsets:

$$|\mathcal{H}(D)| = \frac{\binom{12}{5}}{\binom{6}{5}} = 132. \quad \square$$

Theorem 12.5 (Unique elevation of a hexad). *For each $H \in \mathcal{H}(D)$, there exists a unique octad $O_H \in \mathcal{O}_{24}$ such that*

$$O_H \cap D = H.$$

Consequently, the application

$$\iota_D : \mathcal{H}(D) \longrightarrow \mathcal{O}_{24}, \quad H \longmapsto O_H,$$

is well defined and is injective.

Proof. Existence forms part of the definition of $\mathcal{H}(D)$. If two octads O_1, O_2 had intersection H with D , both would contain any five-point subset of H . The uniqueness of the octad incident with those five points implies $O_1 = O_2$. Injectivity follows by intersecting again with D . $\square$

The lift has a local law that will be used to compare the five regions of π . In a system $S(t, k, v)$, the number of blocks containing a fixed subset of $s \leq t$ points is

$$\lambda_s = \frac{\binom{v-s}{t-s}}{\binom{k-s}{t-s}}.$$

Consequently,

$$\lambda_4(W_{12}) = \frac{\binom{8}{1}}{\binom{2}{1}} = 4, \quad \lambda_4(W_{24}) = \frac{\binom{20}{1}}{\binom{4}{1}} = 5. \quad (12.2)$$

Corollary 12.6 (Decomposition $4 + 1$ on a face). *Let $B \subset D$, with $|B| = 4$. The five octads of W_{24} that contain B decompose uniquely into*

$$\{O_H : H \in \mathcal{H}(D), B \subset H\} \sqcup \{O_B^\perp\}.$$

The first set contains four octads and satisfies $|O_H \cap D| = 6$; the remaining octad satisfies

$$O_B^\perp \cap D = B.$$

Proof. The first equality in (12.2) provides four residual hexads containing B , and theorem 12.5 provides four distinct octads. The second equality in (12.2) shows that there exists exactly one fifth octad $O_B^\perp$.

Since $B \subset O_B^\perp \cap D$, the intersection spectrum used in the proof of theorem 12.4 leaves only the possibilities four and six. An intersection of size six would be a fifth residual hexad that contains B , contradicting $\lambda_4(W_{12}) = 4$. Therefore, the intersection has size four and coincides with B . $\square$

12.1.6.1 Alineamiento with the 12 positions HMT

The residual design is defined on the points of D , whereas the design obtained from the ternary code C_W is defined on the twelve dodecaphase positions. To avoid conflating the two sets, we shall call a bijection *HMT binary alignment*

$$\ell : \{1, \dots, 12\}_{\text{HMT}} \xrightarrow{\sim} D \quad \text{such that} \quad \{\ell(H) : H \in W_{12}^{\text{HMT}}\} = \mathcal{H}(D). \quad (12.3)$$

Here W_{12}^{HMT} denotes the design of weight-six supports obtained from the ternary code C_W . Uniqueness of the Witt system $S(5, 6, 12)$ up to isomorphism ensures the existence of ℓ . Two distinct alignments differ by automorphisms of the source and target designs; hence ℓ is coordinate data and not an additional invariant. The pair (D, ℓ) is the calibration datum required to transport the lift ι_D to the HMT-coordinate system. Once fixed, each HMT hexad H unambiguously determines the octad

$$\widehat{H} := \iota_D(\ell(H)).$$

The dodecad and the alignment are data of the binary codomain; they are not identified with a decimal output of the transport kernel.

For the distinguished face

$$B_K = \{4, 6, 9, 10\},$$

the four incident hexads are

$$B_K \sqcup \{1, 3\}, \quad B_K \sqcup \{2, 11\}, \quad B_K \sqcup \{5, 7\}, \quad B_K \sqcup \{8, 12\}.$$

The third agrees with $H_\alpha^- = B_K \sqcup \{5, 7\}$. The alignment ℓ lifts these four hexads to the four residual octads over $\ell(B_K)$; [corollary 12.6](#) also determines a unique transverse octad.

12.1.6.2 Incidence correspondence between the microfiber of π and the residual octads: multiplicity $432+4 \cdot 144 = 1008$

The exact microfiber of the word $w_\pi = 010211$ contains five distinct decimal regions:

multiplicity	U_6	$e(U_6)$
432	501 614 498 169 272 272	(5, 5, 5, 5, 5, 5)
144	810 923 870 169 272 272	(3, 3, 9, 5, 5, 5)
144	810 983 810 169 272 272	(3, 9, 3, 5, 5, 5)
144	870 923 810 169 272 272	(9, 3, 3, 5, 5, 5)
144	870 923 810 418 674 521	(9, 3, 3, 7, 1, 7).

Here $e(U_6)$ is the multiplicative signature recovered from each decimal block $\overline{d_1 d_2 d_3}$ by means of $e = d_2 - d_1 \pmod{10}$. The first region is the only one of multiplicity 432 and constant signature. The remaining four have multiplicity 144; among them, three share the return block 169|272|272, whereas the fourth has return 418|674|521. The catalogue distinguishes the five rows by a reproducible predicate: the three rows of type ES form the positive triple; among the two rows of type NO, that of maximal multiplicity and constant signature is termed transverse, and the remaining one is termed negative. Relative to this role predicate, the partition is

$$1_\perp + 3_{++} + 1_{--}.$$

The sign labels are a convention; what is intrinsic to the catalog is the partition obtained from its columns, not a prior binary polarity.

The alineamiento regional complete the givesto (D, ℓ) with to bijection that sends the transver region to $O_{\ell(B_K)}^\perp$, the region negativa to $\widehat{H}_\alpha^-$ and the tris regionis positivas to the elevacionis of

$$B_K \sqcup \{1, 3\}, \quad B_K \sqcup \{2, 11\}, \quad B_K \sqcup \{8, 12\}.$$

The ordered assignment between the three positive regions and those three hexads is a choice of coordinates. The two distinguished rows and the triple are reproducible through the preceding catalogue predicate; their designation as negative, transversal, and positive belongs to the alignment. Thus, the law $4 + 1$ does not identify the five regions with five octads by mere cardinality: it provides the incidence space, whereas (D, ℓ) and the regional alignment specify the transport between the two objects.

12.1.7 Lattice prolongation and dimensional screens

The lift of hexads opens two branches that must not be confused. The first conserves binary incidence and prolongs W_{12} toward W_{24} . The second conserves ternary transport:

$$C_{12}^{(3)} \longrightarrow N(A_2^{12}) \longrightarrow \Lambda_{24} \longrightarrow V^{\natural} \longrightarrow \mathbb{M} \longrightarrow \text{Moonshine}.$$

The primary proof of the gluing, the rootless 3 neighbor, and the corridor to Moonshine resides in the corresponding theorem. There is no canonical arrow $W_{24} \rightarrow N(A_2^{12})$, nor any direct action of the Monster on TPK words: both branches start from the same calibrated code and preserve different structures.

The reductions $243/11$, $12 \rightarrow 11 \rightarrow 10$, and $26 \rightarrow 24$ are not identities between numerals either. The chapter devoted to subspaces, quotients, rank reductions and duality constructs their spaces, quotients, and maps: the perfect ternary quotient, the screen marked by K , and the isotropic reduction of the Lorentzian lattice. This section thereby retains a single function: to link the already proved finite incidence with its two prolongations, without repeating the reference proofs.

12.1.8 Residue–quotient involution and conservation of state information

For $n = 9Q + r$, we define the lattice energy

$$E_R(r, Q) = \frac{r^2}{R^2} + Q^2 R^2.$$

Then

$$E_R(r, Q) = E_{1/R}(Q, r).$$

The exchange $(r, Q, R) \leftrightarrow (Q, r, 1/R)$ is an exact involution of the lattice model: the residue and quotient exchange their functions.

12.1.9 Finite compatibility and joint prolongation

The finite witness B_K , the carry pattern, and the Witt frame fix local compatibility. Its prolongation is no longer left as an abstract scheme: the [theorem 12.8](#) constructs, block by block, the exceptional reader with a unique transported initial gauge, and the [theorem 12.9](#) proves that this reading and the Archimedean evaluation globalize over the same TPK section. Truncating the history before reading is equivalent in both cases to truncating the output after reading.

The Archimedean evaluation identifies histories by their limiting value. The exceptional evaluation preserves the visible current and its Paley–Witt incidence; it does not thereby recover the hidden memories that the visible emitter does not publish. The double projection is therefore a joint construction with two distinct codomains, not an identification between the continuum and the exceptional branch.

Remark 12.7 (Compatibility of the two projections). The support B_K , the carry pattern, and the Witt frame provide the common invariants. The lattice chain prolongs this compatibility to a rootless neighbor and, by classical classification, to the Leech lattice. The binary branch toward W_{24} remains separate: its incidence morphism is ι_D , defined after fixing the dodecad and alignment (D, ℓ) , and its codomain is neither a Niemeier lattice nor the Leech lattice.

12.2 Exceptional projection of the integral holonomic state: compatible incidence at every depth

The finite witness of the double projection acquires global scope when the exceptional reading is applied to every visible block and the gauge is transported, rather than chosen anew, as the depth increases. This section constructs that family. The domain remains the enriched TPK history; the exceptional reading is faithful on its visible stream and does not purport to replace the memory fields that this stream does not publish.

12.2.1 Common enriched domain and inequivalent codomains of the 2 projections

In the Paley–Witt chart one fixes

$$A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix} \in M_6(\mathbb{F}_3).$$

Matrix multiplication gives

$$A_W^\top = A_W, \quad A_W^2 = -I_6.$$

For every conjugated operator $A = fA_Wf^{-1}$, define

$$\Gamma_A : \mathbb{F}_3^6 \longrightarrow \mathbb{F}_3^{12}, \quad \Gamma_A(w) = (w, wA).$$

Its image is a calibrated copy $C_W(A)$ of the ternary Witt code. Projection onto the first six coordinates is a left inverse:

$$\text{pr}_1 \Gamma_A = \text{id}_{\mathbb{F}_3^6}.$$

Therefore, Γ_A is total and injective on the 729 words and does not need to distinguish in advance π , e , φ , or any other output.

12.2.2 Compatible extensions and uniqueness of the initial scale

Let X_n be the set of admissible TPK prefixes of depth n , with truncations

$$p_{n,m} : X_n \longrightarrow X_m, \quad m \leq n.$$

Fix an initial exceptional frame f_0 . Since the changes g_j form part of the transport stored by the history, a calibrated prefix is the pair (x, f_0) ; we write

$$X_n^{\text{cal}} := X_n \times \{f_0\}, \quad p_{n,m}^{\text{cal}}(x, f_0) := (p_{n,m}x, f_0),$$

and

$$X_\infty^{\text{cal}} := \varprojlim_n (X_n^{\text{cal}}, p_{n,m}^{\text{cal}}).$$

In what follows we abbreviate $p_{n,m}^{\text{cal}}$ as $p_{n,m}$. The compatible visible emission is

$$\varepsilon_n : X_n^{\text{cal}} \longrightarrow (\mathbb{F}_3^6)^n, \quad \varepsilon_n(x) = (w_0, \dots, w_{n-1}),$$

and satisfies

$$\text{pr}_{n,m} \varepsilon_n = \varepsilon_m p_{n,m}.$$

This application distinguishes its three types:

$$|\mathcal{U}_6| = 468, \quad |\text{im } w_6| = 243, \quad |\mathbb{F}_3^6| = 729.$$

The emission w_j is not identified with the complete state that conserves leaf, quotient, carry, orientation, phase, and survival.

The exceptional gauge consists of a Witt chart, a dodecad, an alignment, a lattice origin, and an oriented chamber. It is fixed only once in the initial block. For $x \in X_n^{\text{cal}}$, transport of the history supplies

$$g_j(x) \in \text{GL}_6(\mathbb{F}_3), \quad 0 \leq j < n-1,$$

and $g_j(x)$ depends only on the prefix $p_{n,j+1}(x)$. The frame is transported by

$$f_{j+1}(x) = g_j(x)f_j(x), \quad A_j = f_jA_Wf_j^{-1}.$$

Thus, the family (A_j) is determined by the history and the initial gauge; it is not a sequence of independent choices.

Let $\mathcal{F}_{\text{cal}} \subseteq \text{GL}_6(\mathbb{F}_3)$ be the set of frames reachable from f_0 . The emission preserving the transported gauge is

$$\tilde{\varepsilon}_n : X_n^{\text{cal}} \longrightarrow (\mathbb{F}_3^6 \times \mathcal{F}_{\text{cal}})^n, \quad \tilde{\varepsilon}_n(x) = ((w_0, f_0), \dots, (w_{n-1}, f_{n-1})).$$

Prefix dependence implies $\text{pr}_{n,m} \tilde{\varepsilon}_n = \tilde{\varepsilon}_m p_{n,m}$. Its projection onto the words is ε_n ; forgetting the frames does not suffice to reconstruct the exceptional reading.

12.2.3 Joint separation of histories by the Archimedean and exceptional projections

The local operation is covariant under the row-vector convention. For every change of coordinates $f \in \text{GL}_6(\mathbb{F}_3)$,

$$A^f = f A f^{-1} \implies \Gamma_{A^f}(w f^{-1}) = \Gamma_A(w)(f^{-1} \oplus f^{-1}).$$

Change of frame simultaneously transports the word, the operator, and the atlas. It is not required to remain within the stabilizer of a single labeled copy.

With the set of reachable frames $\mathcal{F}_{\text{cal}}$ already defined, we construct the fixed calibrated atlas

$$\mathcal{C}_W^{\text{cal}} := \{(f, \Gamma_{f A_W f^{-1}}(w)) : f \in \mathcal{F}_{\text{cal}}, w \in \mathbb{F}_3^6\}.$$

The label f preserves the calibrated copy in which each word resides. Define

$$Q_n(x) = ((f_0, \Gamma_{A_0}(w_0)), \dots, (f_{n-1}, \Gamma_{A_{n-1}}(w_{n-1}))) \in (\mathcal{C}_W^{\text{cal}})^n.$$

Let

$$r_{n,m} : (\mathcal{C}_W^{\text{cal}})^n \longrightarrow (\mathcal{C}_W^{\text{cal}})^m$$

the truncation of the last $n - m$ blocks.

Theorem 12.8 (Projective compatibility of the exceptional incidence). *For every $m \leq n$,*

$$r_{n,m} \circ Q_n = Q_m \circ p_{n,m}.$$

Consequently, there exists a unique application

$$Q_\infty : X_\infty^{\text{cal}} \longrightarrow (\mathcal{C}_W^{\text{cal}})^{\mathbb{N}}$$

whose finite projections are the Q_n . The map is faithful on the visible current.

Proof. Write

$$\varepsilon_n(x) = (w_0, \dots, w_{m-1}, w_m, \dots, w_{n-1}).$$

The compatibility of ε shows that the visible current of $p_{n,m}(x)$ is $(w_0, \dots, w_{m-1})$. For $j < m$, the frame f_j depends only on the initial gauge and on changes preceding j , all of which are contained in that prefix. The first m operators A_j are therefore the same before and after truncation. Applying Γ_{A_j} block by block yields the asserted identity. The universal property of the inverse limit produces Q_∞ . Finally, $\text{pr}_1 \Gamma_{A_j} = I$ recovers every w_j , proving faithfulness on the visible current. $\square$

The last assertion conserves its type. Two histories with the same current $(w_j)_j$ and distinct hidden memory may have the same exceptional image. The global reader preserves exactly what each visible block publishes and the incidence added to that block.

The compatibility of enriched emissions induces a unique application.

$$\tilde{\varepsilon}_\infty : X_\infty^{\text{cal}} \longrightarrow (\mathbb{F}_3^6 \times \mathcal{F}_{\text{cal}})^{\mathbb{N}}.$$

Let pr_w the projection onto the corriente of palabras and

$$\tilde{\Gamma}_\infty((w_j, f_j)_{j \geq 0}) := ((f_j, \Gamma_{f_j A_W f_j^{-1}}(w_j)))_{j \geq 0}.$$

Then $Q_\infty = \tilde{\Gamma}_\infty \circ \tilde{\varepsilon}_\infty$, with declared domains and codomains.

12.2.4 Bifurcation of a surviving prefix in Archimedean value and exceptional incidence

Fixed $m \in \mathbb{Z}$, let us denote by

$$\text{flat} : (\mathbb{F}_3^6)^\mathbb{N} \longrightarrow \mathbb{F}_3^\mathbb{N}$$

the concatenation of the blocks in the declared coordinate order, and by

$$\text{ev}_3((d_k)_{k \geq 1}) := \sum_{k \geq 1} \frac{d_k}{3^k} \in [0, 1], \quad \tau_m(t) := m + t,$$

the ternary evaluation and translation by the integer part. The visible prefix $(d_1, \dots, d_{6n})$ determines the cylinder

$$I_n(x) = \left[m + \sum_{k=1}^{6n} \frac{d_k}{3^k}, m + \sum_{k=1}^{6n} \frac{d_k}{3^k} + 3^{-6n} \right].$$

A prolongation restricts the cylinder and $\text{diam } I_n = 3^{-6n} \rightarrow 0$. Therefore, every compatible infinite history determines a unique real point $\mathcal{P}_{\text{arq}}(x)$.

Theorem 12.9 (Double global realization). *Every compatible section $x = (x_n)_n \in X_\infty^{\text{cal}}$ determines, from the same prefixes, two compatible outputs:*

$$\mathcal{P}_{\text{arq}}(x) \in \mathbb{R}, \quad \mathcal{P}_{\text{exc}}(x) = Q_\infty(x) \in (\mathcal{C}_W^{\text{cal}})^\mathbb{N}.$$

Truncating before reading is equivalent to truncating each reading after reading. The first output identifies histories by their limiting value; the second preserves each visible block and adjoins its Paley–Witt rotation.

Proof. The nesting of the cylinders and the convergence of their diameters prove the first assertion. The theorem 12.8 proves the second and the compatibility with truncations. Both factor through the same emission:

$$\mathcal{P}_{\text{arq}} = \tau_m \circ \text{ev}_3 \circ \text{flat} \circ \text{pr}_W \circ \tilde{\varepsilon}_\infty, \quad \mathcal{P}_{\text{exc}} = \tilde{\Gamma}_\infty \circ \tilde{\varepsilon}_\infty.$$

□

The double projection is thus constructed on admissible histories and a transported initial gauge. The continuum is the Archimedean evaluation of the section; the exceptional branch is the compatible realization of its visible current. From C_W , with the already declared alignments, the branches toward W_{12} , W_{24} , and toward $N(A_2^{12})$, Leech, V^h , and the Monster are opened. These branches do not turn the digits into a set on which the Monster acts directly: they transport the incidence that evaluation by value ceases to see.

12.2.5 Common incidence invariant between the integral holonomic state and the Witt system

Let $x_\pi \in X_\infty^{\text{cal}}$ be the surviving section of the π channel. The double realization of theorem 12.9 produces

$$\mathcal{P}_{\text{arq}}(x_\pi) \in \mathbb{R}, \quad \mathcal{P}_{\text{exc}}(x_\pi) = Q_\infty(x_\pi).$$

The first branch publishes the generated angle; the second preserves the visible current and realizes the quadratic Paley–Witt relation in each block.

Theorem 12.10 (Euler–postgenerative coupling double projection). *Suppose that the coinductive construction of the channel of π determines*

$$\mathcal{P}_{\text{arq}}(x_\pi) = \pi,$$

and that the exceptional branch uses the gauge transported with $A_W^2 = -I_6$. Let J_e be the real realization of the integral source of theorem on the two realizations of an integral quadratic presentation. Then

$$\boxed{\text{Exp}(\mathcal{P}_{\text{arq}}(x_\pi)J_e) + I = 0.} \quad (12.4)$$

Proof. By the theorem on the two realizations of an integral quadratic presentation, A_W and J_e realize the relation $u^2 + 1 = 0$ in categories distinct. In the branch real,

$$\text{Exp}(\theta J_e) = \cos \theta I + \sin \theta J_e.$$

Substitution of the previously generated output $\theta = \mathcal{P}_{\text{arq}}(x_\pi) = \pi$ produces $-I$, and the sum of I gives the stated identity. $\square$

Causal order and absence of circularity. The demonstrated dependence is

$$\text{emitter} \longrightarrow x_\pi \longrightarrow \mathcal{P}_{\text{arq}}(x_\pi) \longrightarrow \pi \longrightarrow \text{Exp}(\pi J_e) + I.$$

The exceptional branch fixes the quadratic type; it selects neither the angle nor its digits. Euler's identity subsequently recognizes the compatibility of the two realizations and does not enter as an input to the generator of π .

12.3 Decomposition of the microfiber of π into 5 oriented regions: 4 peripheral components and 1 fixed axis

The double projection thesis acquires mathematical content when both readings are defined on the same state system and commute with depth restriction. Archimedean evaluation contracts a family of cylinders to a value; exceptional evaluation preserves incidence relations that the first may forget. These are not two juxtaposed numerical expressions, but two natural transformations with a common domain.

12.3.1 Reticular prolongation of incidence and construction of dimensional screens

Let $(X_n, p_{n+1,n})$, $(A_n, a_{n+1,n})$, and $(E_n, e_{n+1,n})$ be three inverse systems. The first contains TPK histories; the second, cylinders or Archimedean approximants; the third, exceptional incidence data. Suppose that maps are given

$$R_n : X_n \longrightarrow A_n, \quad Q_n : X_n \times C_n \longrightarrow E_n,$$

where C_n gathers the coordinate data of the exceptional reader. The gauges are also required to form a compatible family and, for every n , the diagrams must commute

$$a_{n+1,n} \circ R_{n+1} = R_n \circ p_{n+1,n},$$

$$e_{n+1,n} \circ Q_{n+1} = Q_n \circ (p_{n+1,n} \times c_{n+1,n}).$$

Theorem 12.11 (Globalization of the two evaluations). *Under the preceding compatibilities, there exist canonical maps*

$$R_\infty : \varprojlim X_n \longrightarrow \varprojlim A_n, \quad Q_\infty : \varprojlim X_n \times \varprojlim C_n \longrightarrow \varprojlim E_n.$$

Its diagonal application

$$(R_\infty, Q_\infty) : X_\infty \times C_\infty \longrightarrow A_\infty \times E_\infty$$

is unique among the applications whose coordinate projections are R_n and Q_n .

Proof. Let $x = (x_n)_n \in \varprojlim X_n$. The first commutativity equation implies

$$a_{n+1,n}(R_{n+1}(x_{n+1})) = R_n(x_n),$$

and therefore $(R_n(x_n))_n$ belongs to $\varprojlim A_n$. We define $R_\infty(x)$ by means of that family. The same argument, applied to $(x_n, c_n)_n$, defines Q_∞ . Uniqueness follows because an element of an inverse limit is determined by all of its coordinates. $\square$

The theorem identifies the concrete proof required by an HMT realization: it is not enough to have an Archimedean result at one depth and a code at another; both must arise from compatible states and respect the truncation maps.

12.3.2 Archimedean contraction of compatible cylinders and realization of the real line

To each $x_n \in X_n$ is associated a nonempty compact interval $I_n(x_n) \subset \mathbb{R}$. If

$$p_{n+1,n}(x_{n+1}) = x_n \implies I_{n+1}(x_{n+1}) \subseteq I_n(x_n)$$

and there exists $\varepsilon_n \rightarrow 0$ with $\text{diam } I_n(x_n) \leq \varepsilon_n$, the intersection

$$\bigcap_{n \geq 0} I_n(x_n)$$

contains a unique real. Thus it is obtained.

$$\text{ev}_{\text{arq}} : X_\infty^{\text{arq}} \longrightarrow \mathbb{R}.$$

The phrase “the cylinders tend to zero” means here that their diameter converges to zero, not that the evaluated number tends to zero.

The relation

$$x \sim_{\text{arq}} y \iff \text{ev}_{\text{arq}}(x) = \text{ev}_{\text{arq}}(y)$$

describes exactly the information removed by the reader. The real value is an evaluation class; the history additionally preserves orientation, carry, sheet, and boundary signature.

12.3.3 Exceptional projection by incidence and boundary conservation

The second evaluation does not take values in $\mathbb{R}$. Its codomain is a category of incidence configurations. In the current finite realization, the coordinate datum contains:

$$C_{\text{exc}} = (\text{projective order, Paley gauge, dodecad, alignment, lattice origin, chamber}).$$

After fixing that data, the chain contains two descendants that must remain separated:

$$C_W \longrightarrow W_{12} \longrightarrow W_{24},$$

$$C_W \longrightarrow N(A_2^{12}) \longrightarrow \Lambda_{24} \longrightarrow V^{\mathfrak{h}} \longrightarrow \mathbb{M}.$$

The first transports designs; the second leads, through later classical results, to the Leech lattice, the vertex operator algebra, and the Monster group. The central holonomy and the Schläfli– E_6 corridor form a parallel boundary extraction; they are not obtained by replacing one of the two preceding descents.

12.3.4 Finite Compatibility of the Common Domain

The dodecaphonic seal

$$K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601)$$

determines the support

$$B_K = \{m : K_m \geq 729\} = \{4, 6, 9, 10\}.$$

In the published Witt chart,

$$P_\pi = \{2, 4, 5, 6, 7, 8, 9, 10, 11\}, \quad H_e = \{1, 3, 4, 6, 9, 10\},$$

and, by independent evaluation of the two supports,

$$\boxed{B_K = H_e \cap P_\pi.}$$

Furthermore, for the support of negative carry

$$H_\alpha^- = \{4, 5, 6, 7, 9, 10\}$$

we have

$$B_K = H_e \cap H_\alpha^-, \quad H_\alpha^- = B_K \sqcup \{5, 7\}.$$

These identities constitute a witness of finite compatibility: the same subset is recovered from the arithmetic seal and from Witt incidence. This is not an equality of cardinalities, since it identifies specific positions.

12.3.5 Typed separation of the Archimedean and exceptional publications of the common state

Theorem 12.12 (Joint Fidelity Criterion). *Let X be compact and Y, Z be Hausdorff spaces. If two continuous maps $R : X \rightarrow Y$ and $Q : X \rightarrow Z$ satisfy*

$$x \neq y \implies R(x) \neq R(y) \text{ or } Q(x) \neq Q(y),$$

then $(R, Q) : X \rightarrow Y \times Z$ is a topological embedding.

Proof. The condition is the injectivity of the diagonal application. Every continuous injective application from a compact space into a Hausdorff space is a homeomorphism onto its image. $\square$

At each finite depth, the condition is decidable by enumerating the pairs $x \neq y$ that remain simultaneously in a fiber of R_n and of Q_n . Globalization requires compatibility of these controls in passing from $n + 1$ to n . The witness B_K proves that the two readings already share a nontrivial invariant; joint faithfulness requires control of all fibers and is not replaced by that single witness.

12.3.6 Precise formulation of the thesis on the real line

The enriched object associated to the Archimedean reading is the arrow

$$\mathbb{R}_\mathfrak{S} := (X_\infty^{\text{arq}} \xrightarrow{\text{ev}_{\text{arq}}} \text{Im}(\text{ev}_{\text{arq}}) \subseteq \mathbb{R}).$$

The domain preserves genealogies; the image preserves values. Coverage of the cylinders at every scale and compatible prolongability are proved in the construction of the Archimedean reader (coverage identity of the Archimedean reader): for every compact set $K \subset \mathbb{R}$, $\text{ev}_\mathbb{R}(\Omega_{\infty, K}) = K$, and the directed union of those intervals covers $\mathbb{R}$. Therefore,

$$\mathrm{Im}(\mathrm{ev}_{\mathrm{arq}}) = \mathbb{R}, \quad X_{\infty}^{\mathrm{arq}} / \sim_{\mathbb{R}} \cong \mathbb{R}.$$

The real field does not act as the starting point of the construction, but as the Archimedean quotient of a space of sections with memory. The descent of sum and product is verified operation by operation and does not condition this already proved surjectivity.

The double reading is then expressed by

$$X_{\infty}^{\mathrm{com}} \times C_{\infty} \xrightarrow{(\mathrm{ev}_{\mathrm{arq}}, Q_{\infty})} \mathbb{R} \times E_{\infty}.$$

This formula is the mathematical translation of the conceptual assertion: one and the same history may contract to a value and, in the transverse reading of its boundary, preserve the incidence leading to the exceptional symmetries.

12.4 Dependence of the incidence reader on the integral holonomic state

Finite compatibility of the two evaluations does not imply that every incidence reader refines the partition produced by the visible reader. The regional catalog makes it possible to decide this question exactly for the minimal Paley–Witt realization. Let $\mathcal{U}_{468}$ be the set of the 468 published regional states, and let

$$\Psi : \mathcal{U}_{468} \longrightarrow \mathcal{W}_{243} \subset \mathbb{F}_3^6, \quad \Psi(U_1, \dots, U_6) = (-U_1, \dots, -U_6) \pmod{3}.$$

For $w \in \mathbb{F}_3^6$, the map

$$c(w) = (w, wA_W) \in C_W \subset \mathbb{F}_3^{12}$$

is the ternary encoder fixed in the Paley–Witt section. We define the minimal incidence reader by

$$Q_{\min} = \mathrm{Supp} \circ c \circ \Psi : \mathcal{U}_{468} \longrightarrow \mathcal{P}([12]).$$

For a map $f : X \rightarrow Y$, denote by

$$\mathrm{Eq}(f) = \{(x, x') \in X^2 : f(x) = f(x')\}$$

its fiberwise equivalence relation.

Theorem 12.13 (Obstruction of the minimal exceptional reader). *In the complete catalogue of 468 states one verifies*

$$\mathrm{Eq}(\Psi, Q_{\min}) = \mathrm{Eq}(\Psi).$$

In the usual abbreviated notation for fiber partitions,

$$\ker(\Psi, Q_{\min}) = \ker \Psi.$$

Consequently, $(\Psi, Q_{\min})$ is not injective and the minimal incidence reader does not distinguish any pair of states that has already been identified by Ψ .

Proof. The factorization $Q_{\min} = S \circ \Psi$, with $S = \mathrm{Supp} \circ c$, implies

$$\Psi(u) = \Psi(v) \implies Q_{\min}(u) = Q_{\min}(v).$$

The converse implication between the equivalence relations is immediate, because Ψ is the first component of the diagonal map. Equality of the relations therefore follows without statistical assumptions. Exhaustive enumeration further confirms that both partitions have the same histogram:

$$132 [1] + 66 [2] + 18 [3] + 3 [4] + 6 [5] + 18 [6],$$

where $a [m]$ means that there exist a fibres of cardinality m . □

The microfiber of $w_\pi = 010211$ provides the distinguished witness. Its five regions are

$$\begin{aligned} (501, 614, 498, 169, 272, 272), & \quad (810, 923, 870, 169, 272, 272), \\ (810, 983, 810, 169, 272, 272), & \quad (870, 923, 810, 169, 272, 272), \\ (870, 923, 810, 418, 674, 521). \end{aligned}$$

The five have the same visible image and the same exceptional support.

$$P_\pi = \{2, 4, 5, 6, 7, 8, 9, 10, 11\}.$$

At the level of the 243 words there appear 213 distinct supports: 185 have one preimage, 27 have two, and one has four. Taking support therefore introduces a second quotient in addition to the one imposed by Ψ .

The theorem delimits the minimal reader, not the enriched exceptional evaluation. A reader capable of separating the regions of the same word must act before the visible quotient and use at least one of the data that it eliminates: orientation, sheet, signature, return, or boundary state. The dodecaphase intertwiner developed in the following part belongs to that enriched class and does not contradict the preceding obstruction.

Complete enumeration of the fibers produces the preceding histogram and proves that taking support introduces a second quotient.

12.5 Archimedean and incidental publication

For each level, let $\mathcal{E}_n$ be an incidence object and let

$$e_n : S_n \longrightarrow \mathcal{E}_n, \quad \sigma_{n,m} : \mathcal{E}_n \longrightarrow \mathcal{E}_m \tag{12.5}$$

applications that satisfy

$$\sigma_{n,m} \circ e_n = e_m \circ r_{n,m}. \tag{12.6}$$

The universal property constructs

$$\text{ev}_{\text{inc}} : \Omega_\infty \longrightarrow \mathcal{E}_\infty, \quad \mathcal{E}_\infty := \varprojlim_n \mathcal{E}_n. \tag{12.7}$$

The double publication is

$$\mathcal{P}_\infty := (\text{ev}_{\mathbb{R}}, \text{ev}_{\text{inc}}) : \Omega_\infty \longrightarrow \mathbb{R} \times \mathcal{E}_\infty. \tag{12.8}$$

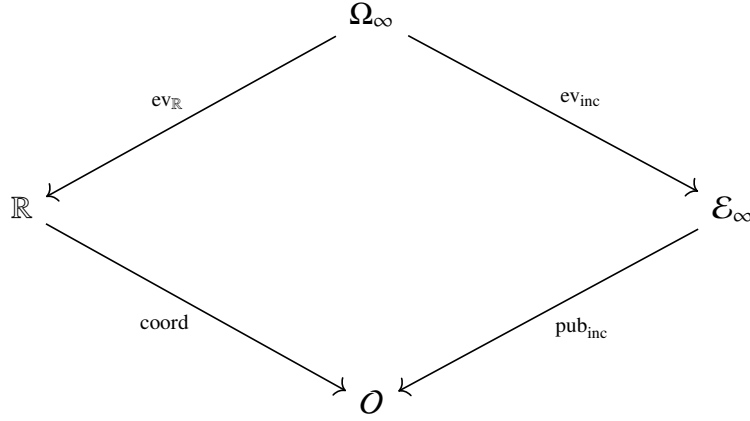


Figure 12.2: Two publications of the same history. Commutativity in O does not identify the codomains $\mathbb{R}$ and $\mathcal{E}_\infty$.

If it is proven

$$\text{pub}_{\text{inc}} \circ \text{ev}_{\text{inc}} = \text{coord} \circ \text{ev}_{\mathbb{R}}, \quad (12.9)$$

a typed comparison of the two publications is obtained. The identity does not assert that one branch is constructed from the other. Both start from Ω_∞ .

12.6 Factorization of the incidence reader through the integral holonomic state

The second projection is not a unary reader devoid of incidence data. Let

$$\mathcal{C}_{\text{exc}} = \mathcal{C}_{\text{ord}} \times \mathcal{C}_{\text{Paley}} \times \mathcal{C}_{\text{dod}} \times \mathcal{C}_{\text{inc}} \times \mathcal{C}_{\text{ret}}$$

the space of exceptional gauges, whose coordinates respectively fix the projective order, the Paley gauge, the dodecad, the incidence alignment, and the lattice chamber. One fixes

$$\chi_{\text{exc}} = (\chi_{\text{ord}}, \chi_{\text{Paley}}, \chi_{\text{dod}}, \chi_{\text{inc}}, \chi_{\text{ret}}) \in \mathcal{C}_{\text{exc}}.$$

The typed reader has domain

$$\mathfrak{E} : X_\infty^{\text{cal}} \times \mathcal{C}_{\text{exc}} \longrightarrow \mathbf{Exc}, \quad \mathfrak{E}_{\chi_{\text{exc}}}(u) := \mathfrak{E}(u, \chi_{\text{exc}}).$$

Henceforth, every abbreviation $\mathfrak{E}(u)$ means $\mathfrak{E}_{\chi_{\text{exc}}}(u)$ for this explicitly fixed gauge. The dependence on χ_{exc} does not affect the existence of the continuum or its Archimedean publication: it types the exceptional conservation of incidence once the common state has been constructed.

Falsification criterion 12.14 (Loss of the exceptional gauge). The comparison becomes incomplete if the projective order, the dodecad, the incidence alignment, or the lattice chamber is changed without transporting the remaining gauge coordinates, or if $\mathfrak{E}$ is used as a unary reader before fixing χ_{exc} . Such a failure invalidates that exceptional comparison; it does not modify the Archimedean quotient or regenerate the continuum.

The second reading does not arise from the real number. Let

$$\mathfrak{E} : \mathcal{O}_{\mathbb{R}, \leftrightarrow}^{\text{HMT}} \longrightarrow \mathbf{Exc}$$

the continuous incidence reader, with Hausdorff codomain, that prolongs the Paley–Witt, Golay, Leech, $V^{\mathfrak{h}}$, Monster, and Moonshine chain. Both maps begin from the same history:

$$\mathfrak{D}(u) = (\mathfrak{R}(u), \mathfrak{E}(u)). \quad (12.10)$$

The first contracts the compatible fibres to their Archimedean mean; the second preserves incidence, orientation, and boundary memory.

Fixing the origin of the dodecaphonic calendar, let it be

$$R_+ := \{k \in \mathbb{Z} : 1 + (k \bmod 9) \in \{1, 4, 7\}\}$$

the bi-infinite set of complete returns ($\tau = +1$). It is syndetic: its gaps are uniformly bounded. For $u = (m, (x_k)_{k \in \mathbb{Z}})$, define

$$\text{Ret}_+(u) := (m, (x_k)_{k \in R_+}) \in \mathbb{Z} \times \prod_{k \in R_+} X,$$

but each x_k here is the whole state (identity of the complete tritic cell) with leaf, carry, route, and incidence, not the isolated sign.

Proposition 12.15 (Lossless Sampling of the Return Genealogical Record). *The map Ret_+ , with the subspace topology in the product, is a homeomorphism onto its image. Consequently, there is a unique continuous map $\mathfrak{E}_+$ on that image such that*

$$\mathfrak{E} = \mathfrak{E}_+ \circ \text{Ret}_+. \quad (12.11)$$

Proof. Between two consecutive returns there is a uniformly bounded number of steps of the calendar. The complete state at a return contains the transition and the genealogical register that determine the intermediate steps; in the other direction, the inverse arrow of the groupoid completion is used. The entire orbit is thereby reconstructed. The map is continuous and, when restricted to a compact chart, a continuous injection into a Hausdorff space is a homeomorphism onto its image. The integer coordinate is preserved explicitly. The factorization is defined by transporting $\mathfrak{E}$ through the inverse and is continuous chart by chart. $\square$

For this reason, $+1$ participates in the exceptional corridor as a complete memory of return: sampling the integer states does not lose the exceptional output. This proposition proves recovery and factorization, not a generation of the corridor from the isolated scalar $+1$. The Archimedean output and the exceptional output are two coordinated resultants of the same antecedent.

Archimedean reconstruction through half-closed cylinders with memory

The inverse limit Ω_∞ preserves the complete history. To obtain a real coordinate it is necessary to add a compatible family of intervals and prove three different facts: each history determines a unique value; every value of the declared compactum is reached; and the finite measures descend consistently. None of these facts follows from the mere growth in the number of states.

The Archimedean publication likewise preserves an approximation of the identity by cylindrical conditional expectations. The concentration of unit mass on compatible prefixes provides the exact passage from the nonadic cylinders to the Dirac measure, without identifying density, support measure, and total mass.

13.1 Nonadic cylinders, identity and Dirac kernel

13.1.1 Inverse system of compatible nonadic cylinders with genealogical memory

Let

$$X = (\mathbb{Z}/9\mathbb{Z})^{\mathbb{N}}$$

with the product topology and the uniform measure μ . For $x = (x_1, x_2, \dots)$, the cylinder of depth n is

$$C_n(x) = \{y \in X : y_j = x_j \text{ for } 1 \leq j \leq n\}.$$

Cylinders are simultaneously open and closed, form a base and satisfy

$$\mu(C_n(x)) = 9^{-n}, \quad C_{n+1}(x) \subset C_n(x), \quad \bigcap_n C_n(x) = \{x\}.$$

Adding a digit does not create a new number: it restricts the cylinder of admissible prolongations. This statement gives topological content to the genealogical thesis of number.

13.1.2 Cylindrical concentration of the identity and construction of the Dirac kernel

Theorem 13.1 (Unitary nonadic concentration). *Define*

$$f_{n,x} = 9^n \mathbf{1}_{C_n(x)}.$$

Then

$$\mu(C_n(x)) \rightarrow 0, \quad f_{n,x}(x) \rightarrow \infty, \quad \int_X f_{n,x} \, d\mu = 1,$$

and the measures $f_{n,x}\mu$ converge weak-* to δ_x .

Proof. The first identity is immediate. The height of $f_{n,x}$ over its support is 9^n , so it diverges. The integral equals $9^n 9^{-n} = 1$. For $g \in C(X)$, the oscillation of g in $C_n(x)$ tends to zero by uniform continuity; therefore

$$\left| \int g f_{n,x} \, d\mu - g(x) \right| \leq \sup_{y \in C_n(x)} |g(y) - g(x)| \rightarrow 0.$$

This is weak-* convergence to δ_x . □

The theorem does not assert $1/0 = \infty$. It exhibits three evaluations of the same object: support measure, density and total mass. In blocks of three digits,

$$\mu(C_{3m}) = 729^{-m}, \quad f_{3m} = 729^m \mathbf{1}_{C_{3m}}, \quad 729^{-m} 729^m = 1.$$

The evolutionary conservation of unity here assumes an exact analytical form.

13.1.3 Convergence of the Kronecker Kernel to the Dirac Distribution

At the finite level $X_n = (\mathbb{Z}/9\mathbb{Z})^n$, let

$$K_n(x, y) = 9^n \mathbf{1}_{\{x=y\}}.$$

With the uniform measure, the operator

$$(E_n f)(x) = \sum_{y \in X_n} K_n(x, y) f(y) 9^{-n}$$

is the identity. In the cylindrical limit, the condition $x = y$ is replaced by “same prefix of length n ” and E_n is the conditional expectation on the cylindrical algebra. One has

$$E_n^2 = E_n, \quad E_n^* = E_n, \quad E_n f \rightarrow f \quad \text{in } L^2(X, \mu).$$

The exact bridge is, therefore,

$$\delta_{ij} \longrightarrow K_n(x, y) \longrightarrow \delta_x.$$

The Kronecker delta is a matrix kernel; the Dirac delta is a measure or distribution. Neither is the Dirac operator $D = \gamma^\mu \partial_\mu$. The transition to spinors and Clifford requires a second intertwiner and will be treated later.

13.1.4 Scaling Law, Effective Dimension, and Density of the Cylindrical Measure

Suppose that a contraction of ratio λ acts independently in d directions and that the product measure scales as λ^d . To preserve unit mass, the density must scale as

$$\rho \longmapsto \lambda^{-d} \rho.$$

The exponents $(-1, -2, -3)$ then appear as density weights on a line, area, and volume. This proposition is general and exact under the product hypothesis. It does not permit λ to be identified automatically with α ; that identification requires a metric chart.

13.1.5 Nonadic Monodromy and Computability of the Archimedean Evaluation

The existence of an infinite branch in a finitely branching tree may follow from König’s lemma, but König’s lemma does not furnish a computable selector. There are infinite recursive binary trees with no recursive branch. The joint nonadic connection does not rest on that abstract existence: it has a certified uniform selector that generates to arbitrary finite depth and opens the decimal witnesses only after generation.

Three prohibited confusions

1/0 shall not be written; the Dirac delta shall not be represented as an ordinary function with “an infinite one”; König shall not be used as an algorithm. The strong result is the exact compatibility among cylinders, unit density, and the weak limit, together with the constructive selector independent of decimal comparison.

13.2 System of intervals associated with finite states

Let $(S_n, r_{n,m})$ be the inverse system of the chapter devoted to the inverse system and genealogical continuity. For each $a \in S_n$, assign a nonempty compact interval

$$I_n(a) = [\ell_n(a), u_n(a)] \subset \mathbb{R}. \quad (13.1)$$

The assignment is *compatible with refinement* when

$$r_{n+1,n}(b) = a \implies I_{n+1}(b) \subseteq I_n(a). \quad (13.2)$$

It is *uniformly contractive* when there exists a sequence $\delta_n \downarrow 0$ such that

$$\text{diam } I_n(a) \leq \delta_n \quad (a \in S_n). \quad (13.3)$$

For a history $x = (x_n)_n$, the inclusions (13.2) produce a chain

$$I_0(x_0) \supseteq I_1(x_1) \supseteq I_2(x_2) \supseteq \cdots. \quad (13.4)$$

Theorem 13.2 (Evaluation of a history). *If the assignment is compatible and uniformly contractive, then for every $x \in \Omega_\infty$ there exists a unique real number*

$$\text{ev}_\mathbb{R}(x) \in \bigcap_{n \geq 0} I_n(x_n). \quad (13.5)$$

The map

$$\text{ev}_\mathbb{R} : \Omega_\infty \longrightarrow \mathbb{R} \quad (13.6)$$

is continuous.

Proof. The intervals of (13.4) are compact, nonempty and nested. The nested-interval theorem guarantees that their intersection is nonempty. If s, t belong to it, then

$$|s - t| \leq \text{diam } I_n(x_n) \leq \delta_n$$

for every n . Setting $n \rightarrow \infty$ yields $s = t$.

For continuity, let $\varepsilon > 0$ and choose n with $\delta_n < \varepsilon$. If $y \in [x_n]_n$, then $I_n(y_n) = I_n(x_n)$. The two evaluations belong to that interval and, therefore,

$$|\text{ev}_\mathbb{R}(x) - \text{ev}_\mathbb{R}(y)| < \varepsilon.$$

The cylinder $[x_n]_n$ is an open neighborhood of x . □

The construction does not identify x with $\text{ev}_\mathbb{R}(x)$. A single coordinate may have several histories, and a history contains more data than any point on the line.

13.3 Coherent coverage of the visible compactum

Let $K \subset \mathbb{R}$ a compacto not vacio. Moreover of anidamiento and contraccion, exigimos

$$K = \bigcup_{a \in S_0} I_0(a), \quad (13.7)$$

$$I_n(a) = \bigcup_{\substack{b \in S_{n+1} \\ r_{n+1,n}(b)=a}} I_{n+1}(b) \quad (a \in S_n). \quad (13.8)$$

The second equality is an additional geometric condition. The surjectivity of $r_{n+1,n}$ only guarantees the existence of states above a ; it does not guarantee that their intervals cover all of $I_n(a)$.

Theorem 13.3 (Surjectivity of the evaluation). *If (13.7) and (13.8) hold, then*

$$\text{ev}_{\mathbb{R}}(\Omega_{\infty}) = K. \quad (13.9)$$

Proof. The left-to-right inclusion follows from $I_n(x_n) \subseteq I_0(x_0) \subseteq K$.

Let us fix $t \in K$. Let us define

$$T_n(t) := \{a \in S_n : t \in I_n(a)\}. \quad (13.10)$$

The initial cover makes $T_0(t)$ nonempty. If $a \in T_n(t)$, the identity (13.8) provides at least one $b \in T_{n+1}(t)$ with $r_{n+1,n}(b) = a$. The graph of these extensions is finitely branching and has nonempty levels at every depth. There therefore exists a compatible chain $x = (x_n)_n$ with $t \in I_n(x_n)$ for every n . Uniqueness of the intersection forces $\text{ev}_{\mathbb{R}}(x) = t$. $\square$

13.4 Archimedean genealogical quotient and homeomorphism with $\mathbb{R}$

We define the relation

$$x \sim_{\mathbb{R}} y \iff \text{ev}_{\mathbb{R}}(x) = \text{ev}_{\mathbb{R}}(y). \quad (13.11)$$

Let

$$q_{\mathbb{R}} : \Omega_{\infty} \longrightarrow \Omega_{\infty} / \sim_{\mathbb{R}} \quad (13.12)$$

the canonical projection.

Theorem 13.4 (Reconstruction of the compact as a quotient). *Under the preceding hypotheses, there exists a unique homeomorphism*

$$\overline{\text{ev}}_{\mathbb{R}} : \Omega_{\infty} / \sim_{\mathbb{R}} \xrightarrow{\cong} K \quad (13.13)$$

such that

$$\text{ev}_{\mathbb{R}} = \overline{\text{ev}}_{\mathbb{R}} \circ q_{\mathbb{R}}. \quad (13.14)$$

The fiber over $t \in K$ is

$$\mathfrak{G}_t := \text{ev}_{\mathbb{R}}^{-1}(t), \quad (13.15)$$

the complete set of genealogies that publish t .

Proof. The map $\text{ev}_{\mathbb{R}}$ is continuous, surjective, and constant exactly on the classes of $\sim_{\mathbb{R}}$. By the universal property of the quotient, it induces a continuous bijective map $\overline{\text{ev}}_{\mathbb{R}}$. The domain is compact because it is a continuous image of the compact space Ω_{∞} , and K is Hausdorff. A continuous bijection from a compact space to a Hausdorff space is a homeomorphism. $\square$

The theorem proves two things and keeps them separate:

the history space is profinite and retains memory, its quotient by equality of evaluation is the visible compact space.
--

(13.16)

It is not asserted that Ω_{∞} is homeomorphic to K .

13.5 Genealogical Atlas of the Real Line

For $m \geq 1$, let $K_m = [-m, m]$. Suppose that for each m a system $\Omega_\infty^{(m)}$ is given that reconstructs K_m , together with closed inclusions

$$j_m : \Omega_\infty^{(m)} \hookrightarrow \Omega_\infty^{(m+1)} \quad (13.17)$$

which satisfy

$$\text{ev}_{m+1} \circ j_m = \iota_m \circ \text{ev}_m, \quad (13.18)$$

where $\iota_m : K_m \hookrightarrow K_{m+1}$ is the inclusion ordinary.

Proposition 13.5 (Exhaustion of the line). *The Archimedean quotient of the strict colimit*

$$\Omega_\infty^{\mathbb{R}} := \varinjlim_m \Omega_\infty^{(m)} \quad (13.19)$$

is

$$\varinjlim_m [-m, m] = \bigcup_{m \geq 1} [-m, m] = \mathbb{R} \quad (13.20)$$

with its usual topology.

Proof. Each piece has quotient K_m by the theorem 13.4; (13.18) intertwines the quotients. The final topology of the exhaustion by compact sets coincides with the usual topology of $\mathbb{R}$: a subset $U \subseteq \mathbb{R}$ is open if and only if $U \cap [-m, m]$ is open in $[-m, m]$ for every m . $\square$

Passage from a bounded interval to the line does not erase the fibers $\mathfrak{G}_t$. The inclusions j_m preserve the history and only enlarge the available visible chart.

13.6 Balanced ternary and block-decimal charts

The balanced ternary representation of an integer has the form

$$n = \sum_{j=0}^J \epsilon_j 3^j, \quad \epsilon_j \in \{-1, 0, +1\}, \quad (13.21)$$

with a fixed carry rule. This chart preserves the local sign and the quotient of the reduction; it must not be confused with the nonnegative word in $\mathbb{F}_3$.

The block decimal chart utilizes

$$D_N = 1000D_{N-1} + u_N, \quad u_N \in \{0, \dots, 999\}, \quad D_0 = 0. \quad (13.22)$$

The block u_N , its class modulo nine, and the carry are distinct fields. Since $1000 \equiv 1 \pmod{9}$,

$$D_N \equiv \sum_{j=1}^N u_j \pmod{9}. \quad (13.23)$$

This congruence does not recover the order of the blocks; the sequence $(u_1, \dots, u_N)$ remains in the record.

For $M \geq 1$, the cells decimales half-open are

$$\mathcal{D}_{M,q} := [q10^{-M}, (q+1)10^{-M}), \quad q \in \mathbb{Z}. \quad (13.24)$$

A half-open interval $[\ell, h)$ is contained in a unique cell if and only if

$$\lfloor 10^M \ell \rfloor = \lceil 10^M h \rceil - 1. \quad (13.25)$$

The formula with a ceiling at the right endpoint also covers the case in which h coincides exactly with a decimal boundary.

Definition 13.6 (Truncation and Rounding). For $x \geq 0$, define

$$\text{Tr}_M(x) := 10^{-M} \lfloor 10^M x \rfloor, \quad (13.26)$$

$$\text{Rd}_M(x) := 10^{-M} \left\lfloor 10^M x + \frac{1}{2} \right\rfloor. \quad (13.27)$$

These operations coincide except when the discarded fraction crosses the half-unit threshold at the final position. The distinction is semantic and arithmetic; one cannot be substituted for the other to force prefix compatibility.

13.7 Descent of operations along evaluation fibers

Let $D_\star \subseteq \Omega_\infty \times \Omega_\infty$ the domain of to operation parcial continuous

$$\star : D_\star \longrightarrow \Omega_\infty. \quad (13.28)$$

Assume that $D_\star$ is saturated by the fibers of $\text{ev}_\mathbb{R} \times \text{ev}_\mathbb{R}$.

Theorem 13.7 (Exact Descent Criterion). *There exists a unique visible operation*

$$\overline{\star} : (\text{ev}_\mathbb{R} \times \text{ev}_\mathbb{R})(D_\star) \longrightarrow K \quad (13.29)$$

which satisfies

$$\text{ev}_\mathbb{R}(x \star y) = \text{ev}_\mathbb{R}(x) \overline{\star} \text{ev}_\mathbb{R}(y) \quad (13.30)$$

if and only if

$$\begin{aligned} \text{ev}_\mathbb{R}(x) = \text{ev}_\mathbb{R}(x'), \quad \text{ev}_\mathbb{R}(y) = \text{ev}_\mathbb{R}(y') &\implies \text{ev}_\mathbb{R}(x \star y) = \text{ev}_\mathbb{R}(x' \star y') \end{aligned} \quad (13.31)$$

for all pairs in the domain.

Proof. If $\overline{\star}$ exists, the right-hand side of (13.30) depends only on the two evaluations, and (13.31) follows. Conversely, this constancy makes it possible to define

$$s \overline{\star} t := \text{ev}_\mathbb{R}(x \star y)$$

for any representatives x, y with evaluations s, t . The condition proves that the definition is independent of the representatives. Uniqueness is immediate. $\square$

The additive and multiplicative sheets of APP supply operations on a common support, but their carries and their memories are not identified. The preceding theorem allows sum and product to descend to the visible chart without requiring their genealogical realizations to coincide.

13.8 Correspondence between discrete cardinality, cylindrical measure, and Archimedean publication

At each depth three operations appear that must be kept separated:

$$\text{Count}_n(A) := |A|, \quad (13.32)$$

$$\text{Measure}_n(A) := \mu_n(A), \quad (13.33)$$

$$\text{Publish}_n(a) := L_n(a). \quad (13.34)$$

The count depends on the number of states; the measure depends on compatible weights; the publication depends on the reader. Two fibers of equal cardinality can have distinct weights, and two states of equal publication can belong to distinct cylinders. Equality of one of these three outputs does not imply equality of the other two.

13.9 Obstructions to Archimedean evaluation: nesting, continuity, quotient, projective measures, and descent of operations

Test criterion 13.8 (Refutation of the evaluation and the measure). The construction is refuted if any of the following facts holds:

1. a compatible extension receives an interval that is not contained in the interval of its restriction;
2. the diameters do not tend to zero, but uniqueness of the intersection is asserted;
3. the evaluation is not continuous in the cylinder topology;
4. it is asserted that every point of K appears without proving the coherent covering (13.8);
5. the map induced by the quotient is not bijective;
6. the profinite space is identified with its visible quotient;
7. the inclusions of the atlas do not intertwine their evaluations;
8. $\lfloor 10^M h \rfloor$ is used instead of $\lceil 10^M h \rceil - 1$ for a semi-open interval, and the boundary case is lost;
9. truncation and rounding are conflated;
10. a descended operation depends on the representative chosen in a fibre;
11. to family $(\mu_n)_n$ viola (projective consistency identity of cylindrical measures);
12. the measure of a cylinder depends on the level to which it is refined;
13. a conditioned measure ceases to sum to one or loses consistency;
14. the visible measure ν is identified with the internal measures μ_t ;
15. an arrow between $\mathbb{R}$ and $\mathcal{E}_\infty$ is inferred solely because both publications share an antecedent;
16. the count of a fibre is used as though it determined its measure or its published value.

13.10 Quotient of the space of compatible histories and realization of the real line

Let

$$\Psi : \mathbb{Z}_3^6 \xrightarrow{\sim} (\mathbb{F}_3^6)^{\mathbb{N}}$$

the Hensel conjugacy between the initial memory and the complete tape of six-trit blocks. Concatenating the six coordinates of each block yields a ternary tape $(a_n)_{n \geq 1} \in \{0, 1, 2\}^{\mathbb{N}}$, on which the evaluation acts

$$\text{ev}_3((a_n)) := \sum_{n \geq 1} \frac{a_n}{3^n} \in [0, 1].$$

The two expansions of a ternary endpoint represent the same value and remain distinct before evaluation is applied. This does not constitute an ambiguity of the genealogical object: it is precisely a fiber of the reader.

We define

$$X_{\text{raw}} := \mathbb{Z} \times \mathbb{Z}_3^6, \quad P(m, x) := m + \text{ev}_3(\text{flat}(\Psi x)).$$

The integer coordinate provides the countable atlas of the line, and the 3-adic coordinate preserves the complete history of the fractional part.

Theorem 13.9 (Exact reconstruction of the Archimedean quotient). *The map $P : X_{\text{raw}} \rightarrow \mathbb{R}$ is sobreyectiva. If*

$$x \sim_P y \iff P(x) = P(y),$$

then

$$\bar{P} : X_{\text{raw}} / \sim_P \longrightarrow \mathbb{R}, \quad [x] \longmapsto P(x),$$

is a bijection. Consequently,

$$|X_{\text{raw}} / \sim_P| = |\mathbb{R}| = 2^{\aleph_0}.$$

Proof. Every ternary sequence is grouped uniquely into consecutive blocks of six digits. The conjugation Ψ assigns a unique initial 3-adic memory to that tape. Given $r \in \mathbb{R}$, take $m = \lfloor r \rfloor$ and a ternary expansion of $r - m \in [0, 1)$; the preceding reconstruction produces $x \in \mathbb{Z}_3^6$ with $P(m, x) = r$. Thus, P is surjective.

The map $\bar{P}$ is well defined by the definition of $\sim_P$, is surjective because P is, and is injective because two classes with the same image belong to the same fiber. Finally,

$$|\mathbb{Z}_3^6| = |(\mathbb{F}_3^6)^{\mathbb{N}}| = 729^{\aleph_0} = 2^{\aleph_0},$$

and multiplying by the countable coordinate $\mathbb{Z}$ does not alter that cardinal. The calculation agrees with that of the quotient, already identified with $\mathbb{R}$. $\square$

Remark 13.10 (Survivor Version). The preceding theorem is unconditional for the raw completion. For a surviving subspace X_{surv} , the same result is obtained when the covers at every level are coherent, the diameters contract uniformly, and every visible cell has a prolongation. Under these hypotheses, the infinite-branch lemma produces a section over every point, and evaluation induces, compact set by compact set, a homeomorphism of the quotient with the published interval. Exhaustion by $[-m, m]$ then reconstructs the entire real line.

13.11 Compatibility of the mean Archimedean evaluation with the double projection of the integral holonomic state

Let $u = (m, \mathbf{u}) \in \mathcal{O}_{\mathbb{R}, \leftrightarrow}^{\text{HMT}}$. A prefix $\mathbf{u}_n$ determines in the fractional chart an oriented semi-open cylinder

$$I_n(\mathbf{u}_n) = [a_n, b_n),$$

with the complementary convention at the endpoints of double expansion. To apply compactness, its closure $K_n = [a_n, b_n]$ is used. The compatibilities give

$$K_{n+1} \subseteq K_n, \quad \delta_n := b_n - a_n \longrightarrow 0.$$

The global Archimedean mean of the cylinder is

$$\mu_n(u) = m + \frac{a_n + b_n}{2}.$$

If $q \geq n$, then

$$|\mu_q(u) - \mu_n(u)| \leq \delta_n,$$

so that

$$\mathfrak{R}(u) := \lim_{n \rightarrow \infty} \mu_n(u) \tag{13.35}$$

exists. What tends to zero is the diameter of the cylinder; the real value is the resultant of the nesting, not that which vanishes.

Theorem 13.11 (Two-way quotient and real line). *Let*

$$u \sim_{\mathfrak{R}} v \iff \mathfrak{R}(u) = \mathfrak{R}(v).$$

Under the coherent covering of the HMT charts, the evaluation induces a homeomorphism.

$$\overline{\mathfrak{R}} : \mathcal{O}_{\mathbb{R}, \leftrightarrow}^{\text{HMT}} / \sim_{\mathfrak{R}} \xrightarrow{\sim} \mathbb{R}. \tag{13.36}$$

In particular, this is a continuous HMT morphism: not the result of choosing an isolated history from each fiber.

Proof. The cylinder bound proves the continuity of $\mathfrak{R}$. The covering produces a history over every real and hence surjectivity. Upon passage to the quotient by the fibers, the induced map is bijective. On every compact chart, a continuous bijection onto a Hausdorff interval is a homeomorphism. The global topology of the quotient is the final topology of the gluing of these charts; the evaluation fiber identifies endpoint $m + 1$ of one chart with the translated endpoint 0 of the following chart. The resulting cover is locally finite, so the chart homeomorphisms glue into a global homeomorphism onto $\mathbb{R}$. $\square$

PART V

Mathematical characters of coinductive extension

Seeds, cubic completion and synchronization of the fundamental characters

14.1 The two mechanical projections and exact conservation of the boundary unit

The statement that the unit is preserved during extension already admits an arithmetic formulation independent of a physical analogy. Let

$$\delta := \log_{10}\left(\frac{10}{9}\right), \quad 0 < \delta < 1,$$

be the discrepancy between the triadic capacity $3^6 = 729$ and the decimal completion $10^3 = 1000$. The irrationality of δ has already been proved in the construction through prime valuations; its transcendence has been proved in this notebook through Gelfond–Schneider. Consider the two one-sided codings of the same irrational line:

$$z_n^- := \lfloor (n+1)\delta \rfloor - \lfloor n\delta \rfloor, \quad z_n^+ := \lceil (n+1)\delta \rceil - \lceil n\delta \rceil, \quad n \geq 0.$$

Both take values in $\{0, 1\}$. They are neither two densities nor two dynamics: they are the two closure conventions of the same irrational cut. Since $n\delta \notin \mathbb{Z}$ for every $n \geq 1$,

$$z_0^- = 0, \quad z_0^+ = 1, \quad z_n^- = z_n^+ \quad (n \geq 1).$$

By telescoping, their counts in the prefix of length $N > 0$ are

$$Z_N^- := \sum_{n=0}^{N-1} z_n^- = \lfloor N\delta \rfloor, \quad Z_N^+ := \sum_{n=0}^{N-1} z_n^+ = \lceil N\delta \rceil.$$

We therefore obtain the boundary theorem

$$\boxed{Z_N^+ - Z_N^- = 1 \quad (N > 0).}$$

The unit is not created at the end of the block: it was localized in the choice of which half-boundary belongs to the cylinder. If the two polarizations are defined by

$$r_N^- := N\delta - \lfloor N\delta \rfloor, \quad r_N^+ := \lceil N\delta \rceil - N\delta,$$

then

$$0 < r_N^\pm < 1, \quad \boxed{r_N^- + r_N^+ = 1.}$$

This is the first exact form of the **conservative evolution of the unit**: the two holographic projections distribute one boundary unit and their sum recovers it at every finite depth. Energy conservation is not yet being asserted. Thermodynamic promotion also requires the physical realization of the operator to be reversible. In the full TPK state, bijective updating already implies $H(X \mid U(X)) = 0$; this logical statement is distinct both from the zero topological entropy of the Sturmian word and from a concrete thermal implementation.

The law is also bidirectional. The lower reading fixes the outgoing half-boundary and the upper reading fixes the incoming one. Reversing orientation interchanges the two conventions and transports the unit,

but alters neither the slope nor the long-range order. This is the minimum content that any *two-way* reader must preserve: inversion changes the location of the boundary unit rather than eliminating it.

Refutation criterion. An $N > 0$ for which $Z_N^+ - Z_N^- \neq 1$, or wording that turns this arithmetic identity into an energy equality without a transducer.

14.2 Concatenation cocycle, carry memory and exact restriction of the TPK

For an initial phase $k \in \mathbb{Z}$, the number of vacancies in a block of length N is

$$V_N(k) := \lfloor (k + N)\delta \rfloor - \lfloor k\delta \rfloor.$$

The word is balanced, so that

$$V_N(k) \in \{\lfloor N\delta \rfloor, \lceil N\delta \rceil\}$$

for every phase. More important for the enriched dynamics is the composition identity

$$V_{M+N}(k) = V_M(k) + V_N(k + M).$$

The second summand cannot be evaluated afresh at the origin: it receives the phase transported by the first block. Forgetting that phase loses exactly the carry

$$\kappa_\delta(M, N) := \lfloor (M + N)\delta \rfloor - \lfloor M\delta \rfloor - \lfloor N\delta \rfloor \in \{0, 1\}.$$

Thus,

$$\lfloor (M + N)\delta \rfloor = \lfloor M\delta \rfloor + \lfloor N\delta \rfloor + \kappa_\delta(M, N).$$

Associativity of addition forces the cocycle equation

$$\kappa_\delta(L, M) + \kappa_\delta(L + M, N) = \kappa_\delta(M, N) + \kappa_\delta(L, M + N).$$

Thus the carry is not a numerical correction added after completing the calculation: it is the datum making block concatenation associative when their history is preserved. This is precisely the structure concealed by projection onto mere cardinalities.

The restriction of TPK dynamics to the vacancy subsystem can be written without replacing the full fibre. On

$$\mathcal{Y}_{108} := C_{108} \times \mathbb{T} \times \mathbb{Z}$$

we define

$$\mathcal{F}(r, \theta, m) = (r + 1, \{\theta + \delta\}, m + \lfloor \theta + \delta \rfloor), \quad 0 \leq \theta < 1.$$

The first component is the finite clock; the second, the irrational phase; the third, the integrated memory. The map is bijective, with inverse

$$\mathcal{F}^{-1}(r, \theta, m) = (r - 1, \{\theta - \delta\}, m - \lfloor \{\theta - \delta\} + \delta \rfloor).$$

In the governing operator language, this formula is a **factor** of $U_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t$: the selector distinguishes event and non-event; transport advances the nonadic and irrational phases; updating

incorporates the pulse into the memory record. It does not replace the TPK fibre, which also preserves sheet, orientation, residue, quotient, signature, cylinder, boundary, route and survival.

After 108 iterations,

$$\mathcal{F}^{108}(r, \theta, m) = (r, \{\theta + 108\delta\}, m + V_{108}(\theta)),$$

with $V_{108}(\theta) \in \{4, 5\}$. The first clock returns; neither the irrational phase nor the memory returns. This equality types the phrase “the phase returns, not the state” exactly within the aperiodic subsystem.

Refutation criterion. Failure of the cocycle equation, non-invertibility of $\mathcal{F}$, or omission by the factor projection of a field subsequently used to distinguish states.

14.3 The renormalized spectrum of cardinalities and its two exact self-references

Define the balanced length reader

$$\mathcal{W}_\delta(N) := \{\lfloor N\delta \rfloor, \lceil N\delta \rceil\}.$$

This operator assigns to each integer scale the only two counts permitted by the irrational cut. It consults neither π , φ , e , α , a mass nor an experimental constant. Applying it to cardinalities already produced by APP–TRIT–TPK yields the following exact equalities:

$$\begin{aligned} \mathcal{W}_\delta(54) &= \{2, 3\}, & \mathcal{W}_\delta(81) &= \{3, 4\}, \\ \mathcal{W}_\delta(90) &= \{4, 5\}, & \mathcal{W}_\delta(120) &= \{5, 6\}, \\ \mathcal{W}_\delta(137) &= \{6, 7\}, & \mathcal{W}_\delta(202) &= \{9, 10\}, \\ \mathcal{W}_\delta(210) &= \{9, 10\}, & \mathcal{W}_\delta(243) &= \{11, 12\}, \\ \mathcal{W}_\delta(271) &= \{12, 13\}, & \mathcal{W}_\delta(324) &= \{14, 15\}, \\ \mathcal{W}_\delta(369) &= \{16, 17\}, & \mathcal{W}_\delta(468) &= \{21, 22\}, \\ \mathcal{W}_\delta(729) &= \{33, 34\}, & \mathcal{W}_\delta(1000) &= \{45, 46\}. \end{aligned}$$

Two rows are self-referential in a precise sense. The integer 137, obtained previously as the trace of the synchronization matrix, reproduces the secondary returns when read by $\mathcal{W}_\delta$:

$$\boxed{\mathcal{W}_\delta(137) = \{6, 7\}.$$

The proved count 468 similarly reproduces the primary gaps:

$$\boxed{\mathcal{W}_\delta(468) = \{21, 22\}.$$

This does not round α^{-1} to 137. The trace 137 was constructed beforehand from integer synchronization, and the vacancy reader subsequently recovers 6/7. Nor is 21/22 introduced to select 468: the count belongs to nonadic extension and the same reader returns it as a window spectrum. These two identities constitute a renormalization closure within the genealogy, not a retrospective definition.

There is a third particularly clear structural identity. Since

$$210 = 90 + 120,$$

the ordered signature of the interval is

$$(V_{90}(0), V_{120}(90)) = (4, 5),$$

and therefore

$$Z_{210}^- = 4 + 5 = 9, \quad Z_{210}^+ = 1 + 4 + 5 = 10.$$

The channels 90/120, generated in the TPK before their parangular evaluators, thus compress the transition $9 \rightarrow 10$ through the same boundary unit. The equality does not say that 90 and 120 are “the numbers” nine and ten; it says that the capacity reader 729/1000, applied to their causal concatenation, produces exactly those two coordinates.

Refutation criterion. A single discrepancy in any of the exact floors or ceilings, or the use of an externally recognized value to choose the lengths.

14.4 Cubic completion jointly contains scales 34 and 11

The completion

$$10^3 = (9 + 1)^3 = 729 + 243 + 27 + 1$$

now has an exact image under the vacancy cocycle. Preserving the order of the four summands, the lower mechanical word produces the signature

$$(33, 11, 1, 0),$$

because

$$\begin{aligned} V_{729}(0) &= 33, \\ V_{243}(729) &= 11, \\ V_{27}(972) &= 1, \\ V_1(999) &= 0. \end{aligned}$$

The sum is 45, whereas the upper reading places the boundary unit in the first block and gives

$$(34, 11, 1, 0), \quad 34 + 11 + 1 + 0 = 46.$$

This yields a refined form of the law 45/46: it not only counts events in a window of one thousand positions, but also preserves the provenance of each event within the completion $729 + 243 + 27 + 1$. The order of the summands is part of the data. A permutation may redistribute the carry between blocks even though the total sum remains fixed; the enriched signature therefore contains more information than the cardinality 45 or 46.

The scale 729 in turn decomposes into three visible blocks of length 243:

$$(V_{243}(0), V_{243}(243), V_{243}(486)) = (11, 11, 11).$$

Consequently,

$$Z_{729}^- = 11 + 11 + 11 = 33, \quad Z_{729}^+ = 1 + 11 + 11 + 11 = 34.$$

This result places 11 and 34 in a single aperiodic hierarchy before any dimensional interpretation. Moreover,

$$\begin{aligned} 324 &= 243 + 81, & (V_{243}(0), V_{81}(243)) &= (11, 3), \\ 369 &= 243 + 126, & (V_{243}(0), V_{126}(243)) &= (11, 5), \end{aligned}$$

from which

$$Z_{324}^- = 14, \quad Z_{324}^+ = 15, \quad Z_{369}^- = 16, \quad Z_{369}^+ = 17.$$

Two connections with previously proved objects are exact at the cardinal level:

$$Z_{324}^+ = 15 = \det \begin{pmatrix} 21 & 22 \\ 6 & 7 \end{pmatrix},$$

and

$$Z_{369}^- = 16 = |\det H_4| = |\operatorname{coker} H_4|.$$

Combined with the active determinantal theorem,

$$\operatorname{ord}_{10}(\varphi \alpha^{16}) = -34,$$

they produce the scalar factorization

$$-\operatorname{ord}_{10}(\varphi \alpha^{Z_{369}^-}) = Z_{729}^+.$$

The preliminary formulation of this result described it as a square reduced to cardinalities. Sections 95–96 complete its typing: the relevant domain is not an arbitrary enumeration of sixteen vacancies, but the ordered signature of cubic completion and its depth reader. The action branch factors exactly as

$$\mathcal{V}_{369}^- \xrightarrow{Z^-} 16 = |\operatorname{coker}(H_4)| \xrightarrow{N_\alpha} \alpha^{16} \xrightarrow{\varphi} \Sigma_H \xrightarrow{-\operatorname{ord}_{10}} 34 = Z_{729}^+.$$

Here $N_\alpha(n) = \alpha^n$ is the norm reader of the determinantal exponent. The Smith normal form of H_4 produces the cokernel of order 16; the section α transports it to the norm α^{16} ; the autoscale φ forms Σ_H ; and decimal order determines the depth 34. The equality $Z_{369}^- = |\operatorname{coker}(H_4)|$ is therefore not used as an elementwise bijection: it acts as an equality of invariants in the composition proving the action depth. This is precisely the type of statement required by the theorem.

Likewise,

$$Z_{243}^- = 11$$

is the second coordinate of the same cubic reader. Its typed factorization is

$$\mathcal{V}_{243}^- \xrightarrow{Z^-} 11 = \operatorname{pr}_{243}(\mathcal{D}_{\text{cub}}(\sigma^-, \sigma^+)) \xrightarrow{\operatorname{Pub}_G} \mathcal{D}_{10}(G) = 11.$$

Assembly with the action branch gives, without consulting target values,

$$(\mathcal{D}_{10}(\hbar), \mathcal{D}_{10}(G)) = (34, 11), \quad \mathcal{D}_{10}(G \otimes \hbar) = 34 + 11 = 45, \quad 45 + 1 = 46.$$

The identity $3^6 \cdot 243 = 3^{11}$ preserves its own domain as a dimension reader; the completion coordinate 11 and gravitational depth are connected through $\mathcal{D}_{\text{cub}}$ and Pub_G , not through tacit identification of their elements. The product $G_a \hbar_a$, proved in Section 95, closes the dynamic square: two-way return reciprocally inverts both branches and preserves their product.

Refutation criterion. Failure of the ordered signatures to telescope, a determinant or cokernel differing from the declared values, failure of $\mathcal{D}_{\text{cub}}$ to produce $(34, 11, 45, 46)$, or a dimensional realization depending on $\hbar$, G , CODATA or the very decimal position it must generate.

14.5 The electronic splitting of the ambient space and the internal emergence of 9, 10 and 24

At the first joint reopening, the electronic channel contains 527 extensions of an ambient space of 729. Its deficit is

$$202 = 729 - 527 = 6 + 28 \cdot 7.$$

The ordered decomposition of the ambient space

$$729 = 202 + 527$$

is exactly compatible with the aperiodic reader:

$$(V_{202}(0), V_{527}(202)) = (9, 24).$$

Therefore,

$$\boxed{Z_{729}^- = 9 + 24 = 33, \quad Z_{729}^+ = 10 + 24 = 34.}$$

The same deficit that singles out the electronic window reproduces, under $\mathcal{W}_\delta$, the nonadic–decimal boundary:

$$\boxed{\mathcal{W}_\delta(202) = \{9, 10\}.}$$

The boundary unit acts here on the deficit sector; the surviving sector preserves twenty-four events in the canonical phase. This decomposition refines the preceding equality $34 = 1 + 3 \cdot 11$: the same coordinate is now read as $34 = 10 + 24$. Both factorizations coexist because they record different cuts of the same cylinder, not because one replaces the other.

The integer 24 coincides with the rank of the Leech lattice, but cardinal equality does not identify the 527 electronic extensions with lattice vectors. Its exact content is a rank connection in the second projection. Exceptional intertwining is constructed, with its own incidence and quadratic form, through the Paley–Witt–Golay–Leech chain; no action of the Monster group is assigned to digits or vacancies.

Refutation criterion. An electronic count other than 527, a complementary block not containing twenty-four events, or absence of any transport preserving the required exceptional incidence.

14.6 The two finite clocks and aperiodic helical suspension

The clock 108 divides into two half-cycles of 54 steps. The lower word has signature

$$(V_{54}(0), V_{54}(54)) = (2, 2),$$

so that

$$Z_{108}^- = 2 + 2 = 4, \quad Z_{108}^+ = 1 + 2 + 2 = 5.$$

The return $54 + 54$ closes the finite phase and preserves a choice of half-boundary. Triple repetition of the clock does not, however, repeat the same signature:

$$(V_{108}(0), V_{108}(108), V_{108}(216)) = (4, 5, 5),$$

and therefore

$$Z_{324}^- = 4 + 5 + 5 = 14, \quad Z_{324}^+ = 1 + 4 + 5 + 5 = 15.$$

This directly manifests aperiodic order: three blocks of equal length have balanced but nonidentical counts. The sequence preserves long-range order and discrepancy less than one without acquiring a translational period.

The cardinality 324 admits two internal readings already present in the construction: $324 = 81 \cdot 4$ for oriented routes and $324 = 108 \cdot 3$ for a tritic lift of the clock. The second factorization defines the minimal cyclic model

$$0 \longrightarrow C_3 \longrightarrow C_{324} \longrightarrow C_{108} \longrightarrow 0.$$

The extension is non-split: C_{324} contains an element of order 324, whereas $C_3 \times C_{108}$ has exponent 108. In the lifted clock, a translation of 108 steps returns to the same coordinate of C_{108} but advances to the next sheet of the kernel C_3 ; only after 324 steps does the lifted coordinate return. This is the smallest algebraic model of a three-sheeted helix over the clock 108.

Decorating this clock with the irrational rotation of slope δ yields a Sturmian helical suspension: the finite base may return, the lifted sheet may return with a longer period, and the irrational phase never returns exactly. The accumulated memory distinguishes these three facts. The ascending helix uses $\mathcal{F}$; the descending helix uses $\mathcal{F}^{-1}$. They are conjugate readings of the same reversible dynamics, not two independent series. The mirror π/e remains another involution: it acts on the output windows and fixes φ ; it does not by itself reverse temporal orientation.

Refutation criterion. A group section of $C_{324} \rightarrow C_{108}$, a repetition (4, 4, 4) or a conjugation that erases memory.

14.7 The first supercell that closes clock 108

The continued fraction of δ begins

$$[0; 21, 1, 5, 1, 6, 2, 3, 1, 1, 8, 1, 2, \dots].$$

The first convergent whose denominator is divisible by 108 is

$$\boxed{\frac{7813}{170748}}.$$

No earlier convergent denominator has this property. Moreover, the rational certificate proves that

$$\lfloor n\delta \rfloor = \left\lfloor \frac{7813n}{170748} \right\rfloor \quad (0 \leq n < 170748),$$

whereas at the final endpoint

$$\lfloor 170748\delta \rfloor = 7812, \quad \frac{7813 \cdot 170748}{170748} = 7813.$$

The aperiodic word and its periodic Christoffel approximant coincide on every proper prefix of the supercell and differ only at closure. The periodic approximant does not correct a cloud of errors: it transports the boundary unit exactly from the lower projection to the upper one.

The denominator has the factorization

$$\boxed{170748 = 108 \cdot 1581 = 324 \cdot 527 = 2 \cdot 54 \cdot 3 \cdot 527.}$$

Here 527 is exactly the electronic-channel count at the first reopening and 324 the cardinality of oriented routes. Since $\gcd(324, 527) = 1$, there is the canonical clock decomposition

$$C_{170748} \cong C_{324} \times C_{527}.$$

The Chinese remainder theorem justifies this group decomposition. The second factor has the cardinality of the electronic fibre and the first that of the oriented atlas; this coincidence of cardinalities neither identifies the objects nor automatically transfers their actions.

The pair of clocks provides a second unexpected datum. In the lower branch, after 170748 blocks,

$$7812 \equiv 36 \pmod{108}, \quad 170748 - 7812 = 162936 \equiv 72 \pmod{108}.$$

Therefore, although the block clock closes,

$$(r_{\text{vac}}, r_{\text{cap}}) = (36, 72), \quad 36 + 72 = 108.$$

The upper branch transfers the boundary unit:

$$(r_{\text{vac}}^+, r_{\text{cap}}^+) = (37, 71), \quad 37 + 71 = 108.$$

The appearance of 36 in the lower branch coincides with the index of stage R_{36} of coinductive extension. This congruence does not by itself identify the vacancy residue with the coordinate of R_{36} ; it proves that the supercell closes the block clock, preserves the distribution of the unit and leaves nontrivial residual memory in the capacity clock.

The numerator has another internal factorization:

$$7813 = 13 \cdot 601, \quad 7812 = 36 \cdot 217, \quad 217 = 2 \cdot 108 + 1.$$

This yields the unimodular frame

$$M_{\star} = \begin{pmatrix} 601 & 217 \\ 36 & 13 \end{pmatrix} \in \text{SL}_2(\mathbb{Z}), \quad 601 \cdot 13 - 217 \cdot 36 = 1.$$

The four integers are localized in the architecture: 601 is the twelfth coordinate of the dodecaphasic seal K ; 13 is the number of atlas multisections; 36 is stage R_{36} ; and 216 is the extended kinematic return. The equality is exact and uses no physical value. It defines a unimodular frame of their cardinalities; it does not on that basis alone postulate a categorical equivalence between the four objects.

Refutation criterion. An earlier convergent with denominator a multiple of 108, a discrepancy in a proper prefix, a false factorization or a transport that does not conjugate the actions.

14.8 The second derivation of α as renormalization of a single word

The polynomial P_9 of the second route to α contains the pairs $21/22$, $6/7$ and $45/46$. Their coefficients had previously been reconstructed through the synchronization matrix, Beatty returns and the boundary $7:9$. The new reader proves that all three pairs are also images of **the same** mechanical word at three internal scales:

$$\mathcal{W}_{\delta}(468) = \{21, 22\}, \quad \mathcal{W}_{\delta}(137) = \{6, 7\}, \quad \mathcal{W}_{\delta}(1000) = \{45, 46\}.$$

The first length is the decimal-emission count, the second is the integer synchronization trace and the third is cubic completion. The genealogy of the second route can therefore be written without introducing its coefficients as an independent list:

$$\begin{aligned} \text{APP} &\longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow X^{\text{enr}}, \\ 729 &\longrightarrow 1000 \longrightarrow \delta \longrightarrow \mathcal{W}_\delta, \\ (468, 137, 1000) &\longmapsto (\{21, 22\}, \{6, 7\}, \{45, 46\}), \\ \text{synchronization and boundary signature } 7:9 &\longrightarrow P_9 \longrightarrow \alpha. \end{aligned}$$

This chain does not replace the first dodecaphasic route. Both start from the coinductive state that also determines π , φ and e ; they subsequently preserve their own readers. The root of P_9 is calculated at the end, once the signature has been constructed. The value of α never selects 468, 137, 1000 or their windows.

The numerator $7813 = 13 \cdot 601$ also locates a possible closure between the aperiodic supercell and the dodecaphasic seal: $601 = K_{12}$ belongs to the first route, whereas 13 belongs to the subsequent atlas. The unimodular identity of the preceding subsection allows the question to be formulated correctly: one must determine whether the memory module on which D_3K , R_{36} , the return 216 and the thirteen multisections act admits the change of basis $M_\star$. Coincidence of factors must not be turned into proof, but neither should it be discarded as a mere curiosity: it is localized in the first supercell selected by the clock 108, not in an integer sought afterwards.

Refutation criterion. Any pair not proceeding from $\mathcal{W}_\delta$, involvement of P_9 in selecting its own inputs, or failure of the module proposed for $M_\star$ to preserve the memory record.

14.9 Seed deepening and rigorous separation of physical scales

The space of an oriented seed has cardinality

$$|S| = 9^2 \cdot 4 = 324,$$

and the pair sweep has

$$|S \times S| = 324^2 = 104976.$$

Applying the aperiodic reader to the total cardinality gives

$$\mathcal{W}_\delta(104976) = \{4803, 4804\}.$$

This equality is a count of the mechanical line, not yet a seed count. To turn it into a property of the APP sweep, one must fix a canonical ordering

$$\text{ord}_{S^2} : S \times S \xrightarrow{\sim} \{0, \dots, 104975\}$$

compatible with the two sheets, the dihedral action, the TRIT selector and TPK updating. Only then can one ask which seeds occupy the vacancies and how the count changes under inward refinement or outward expansion. Cardinality alone must not select this ordering.

The new analysis nevertheless already determines where to seek the scales 10^{-11} and 10^{-34} . They are not obtained by arbitrarily raising a seed to a decimal power. The crystal first produces the integers

$$Z_{243}^- = 11, \quad Z_{729}^+ = 34,$$

within the same completion signature. Two distinct transducers can subsequently realize those orders on dimensional lines:

$$\begin{aligned}\mathcal{T}_G &: (X^{\text{enr}}, \mathcal{V}_{243}^-) \longrightarrow 10^{-11} \mathcal{L}_G, \\ \mathcal{T}_h &: (X^{\text{enr}}, \mathcal{V}_{729}^+, G_H, \alpha, \varphi) \longrightarrow 10^{-34} \mathcal{L}_S.\end{aligned}$$

The second transducer already has a proved route through $|G_H| = 16$ and $\varphi\alpha^{16}$. The first retains a decimal chart and selection criteria in the authoritative source and must be reconciled with the vacancy reader without using CODATA. The phrase “fractalize inwards down to 10^{-11} and 10^{-34} ” thus acquires a precise mathematical meaning: refine the same state to two window cardinalities, and only then apply the corresponding dimensional transduction. Powers of ten belong to the chart of the dimensional line, not to the bare seed.

The unit remains conserved during this step if the transducer simultaneously preserves both projections and the cocycle. The requirement is a naturality square

$$\mathcal{T} \circ \text{Upd} = \text{Upd}_{\mathcal{L}} \circ \mathcal{T}$$

in which the left-hand side first updates the discrete memory and the right-hand side first transports the dimensional section. This equation, rather than an isolated decimal equality, is the requisite proof that a physical constant emerges “at the appropriate scale”.

The discrete structure of the continuum must receive all these fibres together. Seed refinement, extension solvability, measure construction, coherence and the determinantal reader are not five separable objects: they are five inseparable aspects of the same passage to the limit. The aperiodic crystal supplies the balanced calendar; APP preserves sum, product, residue, quotient and carry; TRIT types orientation and the three curvatures; TPK preserves history throughout selection, transport and updating. Only their assembly produces the real line as an output. In this genealogy, the line $\mathbb{R}$ is not the support presupposed to construct the protocol: it is one of its terminal realizations.

Refutation criterion. An ordering not natural with respect to APP, TRIT and TPK; a decimal power chosen from the physical value; or a transducer not preserving the boundary unit and memory record.

14.10 The cellular corona of the continuum contains clock 108 and the TRIT signature

Review of the cellular reference construction supplies a component linking completion, the clock and double projection more rigidly than a numerical coincidence. For the cubical pair

$$\mathcal{Q}_{9,3} \subset \mathcal{Q}_{10,3},$$

the relative complex has the cardinality vector

$$\mathbf{c} = (c_0, c_1, c_2, c_3) = (271, 756, 702, 217).$$

Its Euler characteristic vanishes:

$$\chi(\mathbf{c}) = 271 - 756 + 702 - 217 = 0.$$

This vanishing, however, is not an opaque cancellation. Each coordinate admits a decomposition into cardinalities already belonging to the architecture:

$$\begin{aligned}
271 &= 243 + 27 + 1, \\
756 &= 729 + 27, \\
702 &= 729 - 27, \\
217 &= 216 + 1 = 2 \cdot 108 + 1.
\end{aligned}$$

Pairing the outer and inner degrees yields two equal defects:

$$c_0 - c_3 = 271 - 217 = 54, \quad c_1 - c_2 = 756 - 702 = 54.$$

Consequently,

$$(c_0 - c_3) + (c_1 - c_2) = 54 + 54 = 108.$$

The clock 108 has not been imposed on the corona from an external calendar: its double half-cycle is contained in the difference between the boundary degrees and in the difference between the inner degrees. The vanishing characteristic is rewritten as

$$\chi(\mathbf{c}) = (c_0 - c_3) - (c_1 - c_2) = 54 - 54 = 0.$$

This identity distinguishes two senses. The first 54 measures separation between the extremities of the complex; the second measures internal separation. Their equality makes closure compatible: what exits through one pairing returns through the other without erasing cellular degree. The sum $54 + 54$ measures the length of the bidirectional path; the difference $54 - 54$ certifies its Euler closure. These are two distinct operations on the same pair and must not be flattened into the single number 108.

The aperiodic reader is now applied to each degree

$$\mathcal{W}_\delta(N) = \{[N\delta], \lceil N\delta \rceil\}.$$

The two vacancy projections of the corona are

$$\mathbf{v}^- = (12, 34, 32, 9), \quad \mathbf{v}^+ = (13, 35, 33, 10).$$

Their capacity complements are

$$\mathbf{b}^+ = \mathbf{c} - \mathbf{v}^- = (259, 722, 670, 208),$$

$$\mathbf{b}^- = \mathbf{c} - \mathbf{v}^+ = (258, 721, 669, 207).$$

In both one-sided closures, component by component,

$$\mathbf{c} = \mathbf{v}^- + \mathbf{b}^+ = \mathbf{v}^+ + \mathbf{b}^-.$$

The decisive information appears upon taking the alternating characteristic:

$$\chi(\mathbf{v}^-) = \chi(\mathbf{v}^+) = +1,$$

$$\chi(\mathbf{b}^-) = \chi(\mathbf{b}^+) = -1.$$

The corona of zero total charge therefore polarizes exactly as

$$0 = (+1) + (-1).$$

Independence from the half-boundary convention is structural:

$$\mathbf{v}^+ - \mathbf{v}^- = (1, 1, 1, 1), \quad \chi(1, 1, 1, 1) = 0.$$

Moving the boundary unit between the two closures does not alter the Euler charge of the branch. This yields an internal realization of the TRIT alphabet:

$$\text{vacancy projection} \mapsto +1, \quad \text{complete crown} \mapsto 0, \quad \text{capacity projection} \mapsto -1.$$

This does not replace the general definition of the TRIT or by itself identify the three charges with physical fields. It proves something more elementary and prior: double projection of a single cellular object generates the three oriented classes $(+1, 0, -1)$, without adding symbols. The algebras

$$\mathbb{A}_\tau = \mathbb{R}[J_\tau] / (J_\tau^2 + \tau I)$$

can subsequently act on these classes. Elliptic, parabolic or hyperbolic type is no longer selected from a presupposed continuous geometry: the signature indexing it has emerged from the discrete polarization of the complex.

Refutation criterion. A change in any of the four cardinalities, a defect no longer equal to 54, projections not having charges $(+1/-1)$, or use of classical geometry to select the sign retrospectively.

14.11 Polarized frame, boundary root and conservative evolution of the unit

The preceding result is a case of an exact law valid for every length. Let

$$a_N = \lfloor N\delta \rfloor, \quad N > 0.$$

The two closure conventions determine the vectors

$$u_N^- = (a_N, N - a_N), \quad u_N^+ = (a_N + 1, N - a_N - 1).$$

Each row preserves the block exactly:

$$(1, 1)u_N^- = (1, 1)u_N^+ = N.$$

The difference is independent of (N) :

$$u_N^+ - u_N^- = (1, -1).$$

The unit neither appears nor disappears; it is transported from the capacity coordinate to the vacancy coordinate. The vector $((1, -1))$ generates the kernel of the augmentation

$$\mathbb{Z}^2 \xrightarrow{(x, y) \mapsto x+y} \mathbb{Z},$$

that is, the root (A_1) . Double projection therefore has a canonical boundary direction. No physical metric need be chosen to define it.

The frame of the two polarizations has determinant

$$\det \begin{pmatrix} a_N & N - a_N \\ a_N + 1 & N - a_N - 1 \end{pmatrix} = -N.$$

The proof is immediate:

$$a_N(N - a_N - 1) - (N - a_N)(a_N + 1) = -N.$$

Thus, transferring a single boundary unit sweeps out an entire parallelogram of oriented area (N). After normalization by (N), the two rows lie in the simplex (x+y=1), are separated by ((1,-1)/N) and converge to

$$(\delta, 1 - \delta) = (\delta, \rho).$$

This is a precise form of the conservative evolution of the dimensional unit: at each scale there is one normalized unit; refinement does not multiply it, but reduces the size of its boundary quantum while preserving the integral direction that allows it to be recovered.

The complete trajectory is written in the lattice as

$$\gamma(N) = (\lfloor N\delta \rfloor, N - \lfloor N\delta \rfloor).$$

Each step increments exactly one coordinate:

$$\gamma(N+1) - \gamma(N) \in \{(1, 0), (0, 1)\}.$$

Moreover,

$$\|\gamma(N) - N(\delta, \rho)\|_\infty < 1.$$

In its lattice section, the aperiodic helix is a digital straight line with uniformly bounded discrepancy. It has long-range order because it never departs by more than one cell from the limiting direction, but has no translational period because δ is irrational. The Sturmian word is not an embellishment of this trajectory: it is the sequence of decisions specifying which of the two coordinates receives the next unit.

The same augmentation structure reappears in the three channels. The reopening vector

$$v_\Omega = (3, -7, 4)$$

satisfies (3-7+4=0), and therefore belongs to the lattice

$$A_2 = \ker(\mathbb{Z}^3 \xrightarrow{x+y+z} \mathbb{Z}).$$

The root (A_1) governs transfer between the two projections; the defect (A_2) governs redistribution among the three correlated windows. They are not the same vector and do not have the same rank, but they are two successive degrees of a single conservation law through augmentation. This explains why double reading and ternary orientation can coexist without turning the theory into binary coding: conservation fixes the diagonal, whereas the HMT datum lies in its oriented complement.

Refutation criterion. A scale whose frame does not have determinant (-N), a step changing both coordinates at once, or an identification of (A_1) and (A_2) that does not declare the rank-changing map.

14.12 Topological–temporal boundary charge and non-destructive curvature selection

The tritic derivative of the vacancy word,

$$\tau_n = z_{n-1} - z_n \in \{+1, 0, -1\},$$

has a conservation law completing its interpretation as a unit of topological–temporal information. For every finite interval $(a \leq n \leq b)$,

$$Q_{[a,b]} := \sum_{n=a}^b \tau_n = z_{a-1} - z_b \in \{-1, 0, +1\}.$$

The oriented interior charge telescopes completely and is supported at the two endpoints. The visible TRIT does not accumulate an arbitrary extensive charge: it records how the block opens, persists or closes relative to its boundary. The TPK memory record preserves the interior terms concealed by the alternating sum, so telescoping does not amount to loss of history.

Since $(0 < \delta < 1/2)$, two vacancies are not consecutive. The nonzero symbols of $((\tau_n))$ therefore alternate strictly:

$$-1, +1, -1, +1, \dots$$

when intermediate zeros are removed. An opening must close before another opening can occur. Zeros do not mean absence of dynamics: they are the plateaux during which the regime is preserved while the block clock continues advancing. Upon inducing the dynamics on events, these plateaux have lengths 6 or 7, as already proved.

Each symbol selects one of the three exponentials

$$\exp(tJ_\tau) = C_\tau(t)I + S_\tau(t)J_\tau, \quad J_\tau^2 = -\tau I.$$

The word thus orders elliptic rotations, nilpotent transports and hyperbolic openings. Selection does not destroy aperiodicity because the differential map is invertible once one boundary bit is preserved:

$$z_n = z_0 - \sum_{k=1}^n \tau_k.$$

This is the exact reason why the TRIT can read three curvatures without replacing the word generating them. The tritic unit is the visible change; the TPK preserves the boundary datum, APP sheet, carry and operator order needed to invert the reading.

Time reversal

$$C_{\text{rev}}(\tau_1, \dots, \tau_n) = (-\tau_n, \dots, -\tau_1)$$

interchanges opening and closure, preserves the threshold and satisfies $(C_{\text{rev}}^2 = I)$. In the matrix fibre, it also reverses the order of the product

$$\mathcal{A}_N = \exp(t_N J_{\tau_N}) \cdots \exp(t_1 J_{\tau_1}).$$

The double temporal direction is therefore not the reading of a list backwards. It is conjugation of a word, its signs and the noncommutative order of its transports, retaining the boundary that allows the state to be reconstructed.

Refutation criterion. Two consecutive openings in the induced word, an interval charge not supported on the boundary, or reconstruction dispensing with (z_0) .

14.13 Nonstationary S -adic tower of the two clocks

The continued fraction of the internal slope begins

$$\delta = [0; 21, 1, 5, 1, 6, 2, 3, 1, 1, 8, 1, 2, \dots].$$

For each positive coefficient (a) , define the continuant operator

$$A(a) = \begin{pmatrix} a & 1 \\ 1 & 0 \end{pmatrix}, \quad \det A(a) = -1.$$

If (p_n/q_n) is the (n) th convergent, then

$$A(a_1) \cdots A(a_n) = \begin{pmatrix} q_n & q_{n-1} \\ p_n & p_{n-1} \end{pmatrix}.$$

The first stages are

$$\begin{aligned} A(21) &= \begin{pmatrix} 21 & 1 \\ 1 & 0 \end{pmatrix}, \\ A(21)A(1) &= \begin{pmatrix} 22 & 21 \\ 1 & 1 \end{pmatrix}, \\ A(21)A(1)A(5) &= \begin{pmatrix} 131 & 22 \\ 6 & 1 \end{pmatrix}, \\ A(21)A(1)A(5)A(1) &= \begin{pmatrix} 153 & 131 \\ 7 & 6 \end{pmatrix}. \end{aligned}$$

This proves that $21/22$ and $6/7$ are not two regularities placed side by side. The integers $(21,22)$ are the first denominators of the Diophantine skeleton; $(6,7)$ are the numerators appearing two inductions later:

$$\begin{aligned} 131 &= 5 \cdot 22 + 21, & 6 &= 5 \cdot 1 + 1, \\ 153 &= 1 \cdot 131 + 22, & 7 &= 1 \cdot 6 + 1. \end{aligned}$$

The primary gaps $21/22$ and secondary returns $6/7$ therefore belong to the same renormalization. The next convergents are

$$\frac{48}{1049}, \quad \frac{103}{2251}, \quad \frac{357}{7802}, \quad \frac{460}{10053}, \quad \frac{817}{17855}, \quad \frac{6996}{152893}, \quad \frac{7813}{170748}.$$

Each consecutive pair preserves a primitive lattice cell:

$$p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}.$$

The first denominator in this tower divisible by 108 is 170748, and its appearance preserves the complete recurrence:

$$170748 = 152893 + 17855, \quad 7813 = 6996 + 817.$$

At that stage,

$$\begin{pmatrix} 170748 & 152893 \\ 7813 & 6996 \end{pmatrix} \in GL_2(\mathbb{Z}), \quad \det = -1.$$

The tower therefore preserves the lattice unit even when the supercell grows large enough to close the finite clock. Closure of C_{108} does not make the irrational direction periodic: the rational approximant coincides with the word on every proper prefix and transfers one unit only at the final half-boundary.

Renormalization is also genuinely ordered. For $a \neq b$,

$$A(a)A(b) - A(b)A(a) = (a - b) \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

For example, $a = 21$ and $b = 5$ produce the oriented defect

$$16 \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The rotation ($x \mapsto x + \delta$) on the circle remains abelian. Noncommutativity appears in ordering the renormalization operators and, in the full fibre, in ordering the APP sheets and curvature generators. This distinction removes two possible errors: calling the base rotation nonabelian, or reducing all dynamics to an abelian rotation while forgetting the cocycle.

Since δ is transcendental, it is not a quadratic irrational. By Lagrange's theorem, its continued fraction is not eventually periodic. The continuant directive does not stabilize into a single periodic substitution: the crystal is S -adic and nonstationary. It nevertheless preserves linear complexity, zero entropy and unimodular determinants. This is a much stronger definition of "aperiodic crystal" than mere absence of repetition: renormalization exists at every scale, but there is no final renormalization cell repeating indefinitely.

Refutation criterion. A discrepancy in the coefficients, a determinant other than ± 1 , an earlier convergent denominator divisible by 108, or an eventually periodic directive.

14.14 Cardinal genealogy of the corona, the boundary 369 and the dodecaphasic seal

The tower allows the structural cardinalities to be resolved without selecting them by decimal resemblance. The following pairs proceed from the same reader and have independent derivations:

$$\begin{aligned} \mathcal{W}_\delta(217) &= \{9, 10\}, \\ \mathcal{W}_\delta(271) &= \{12, 13\}, \\ \mathcal{W}_\delta(601) &= \{27, 28\}, \\ \mathcal{W}_\delta(973) &= \{44, 45\}. \end{aligned}$$

The integer 217 is the cardinality of the 3-cells of the corona and also $2 \cdot 108 + 1$. Its reading determines the boundary between the nonadic base and decimal completion. The integer 271 is the cubic corona $1000 - 729$; its reading determines $12/13$, precisely the pair formed by the dodecaphase and the number of atlas multisections. The integer 601 is the twelfth coordinate of the seal K ; its reading determines $27/28$, the cubic boundary and the cardinality of the microfibre. Finally,

$$973 = 271 + 702 = 756 + 217$$

is the common sum of the even and odd degrees of the corona, and

$$\mathcal{W}_\delta(973) = \{44, 45\}.$$

Here $44 = 396/9$ is the reduced APP count and 45 is the determinant of the boundary matrix and the count of the radical region. The Euler balance of the continuum thus projects onto two cardinalities already situated in APP and the radical branch.

These equalities do not declare that a multisection is literally a vacancy or that (K_{12}) is a set of twenty-eight elements. The proved content is more precise: a single cardinal cut functor, constructed before these realizations, sends four typed objects to four internal pairs with known provenance. The next step is not to seek further resemblances, but to lift this cardinal functor to the actions preserved by each representation.

The boundary 369 admits an even more rigid genealogy. The construction distinguishes the cardinality 386 of the internal boundary from the 387 pairs $(K, K + 9)$ per channel. The reader gives

$$\mathcal{W}_\delta(386) = \mathcal{W}_\delta(387) = \{17, 18\}.$$

Using opposite closures,

$$\boxed{386 = 17 + 369, \quad 387 = 18 + 369.}$$

Thus, 369 is the common capacity surviving transfer of the boundary unit between two consecutive cardinalities. The transition

$$(386, 17) \longleftrightarrow (387, 18)$$

increments both the ambient space and the vacancy, but leaves the radical capacity 369 fixed. This is an exact definition of stability under change of half-boundary. It complements, without replacing, the already proved genealogies

$$369 = 396 - 27, \quad 369 = 3(132 - 6) - 9, \quad 369 + 126 = 495.$$

The convergence of four independent routes—adjacent boundary, APP square, induced returns and conservative transfer—makes 369 an internal node of the architecture, not a chosen decimal string.

Other cardinal projections retain a distinct status:

$$\mathcal{W}_\delta(756) = \{34, 35\}, \quad \mathcal{W}_\delta(702) = \{32, 33\}.$$

The 34 coincides with the action depth already derived and the 33 with the lower count of 729. The physical derivation of the decade -34 belongs to the cokernel of H_4 and to $\varphi\alpha^{16}$; reading the 756 edges does not replace it. The same corona producing the Euler trit thus contains an additional cardinal projection of that depth, without identifying the edge reader with the action line.

The unimodular frame of the supercell also acquires a further strong algebraic interpretation:

$$M_\star = \begin{pmatrix} 601 & 217 \\ 36 & 13 \end{pmatrix}, \quad \det M_\star = 1.$$

Its four entries are now the dodecaphasic coordinate ($K_{12} = 601$), the cardinality of the 3-cells of the corona 217, the stage R_{36} and the thirteen multisections. The identity $217 = 216 + 1$ remains compatible with this reading. The unit determinant proves lattice conservation of the frame formed by the four cardinalities, without identifying their domains.

Refutation criterion. A change in any window, failure of 369 to be the common complement, or a proposed map preserving cardinalities while destroying incidence, action or memory.

14.15 The crystal kernel and the continuum as output

The preceding components allow the governing thesis to be specified without confusing the result with a cosmological metaphor. APP supplies a finite support with two inseparable evaluations. The additive sheet accumulates; the multiplicative sheet folds; residue, quotient and carry preserve the difference between their histories. TRIT extracts the local orientation of that difference and realizes it in three quadratic types. TPK turns local signatures into dynamics through selection, transport and updating, preserving phase, sheet, boundary and memory record. The completion $729 \rightarrow 1000$ then induces a digital line of transcendental slope; its vacancy word is aperiodic, balanced and of zero entropy; its derivative is tritic; its renormalization tower is S -adic; and its suspension over C_{108} is helical.

The continuum does not enter this chain as a previously available $\mathbb{R}$. Compatible finite cylinders, their two one-sided closures, the boundary unit and the refinement maps exist first. The cellular corona proves that double projection separates an object of zero charge into branches of charges $+1$ and -1 without altering their sum. The inverse limit preserves compatible histories; the Archimedean reader subsequently determines a real coordinate. In this sense, $\mathbb{R}$ is a terminal realization of an enriched discrete protocol, not the ontological basis from which APP is reconstructed.

The two clocks are now unambiguous. The chronological clock records the number of applications of U_t and has the finite quotient C_{108} . The capacity clock records how often refinement has opened an effective decimal pruning. Their difference is the integrated vacancy memory. A turn of 108 steps can close the first clock while leaving the second, the irrational phase and memory open. The supercell of length 170748 is the first convergent that closes the first clock exactly; the boundary unit remaining at the endpoint proves that this closure does not make the crystal periodic.

The correlated system π, φ, e, α resides on this same coinductive state. The windows of π, e, φ synchronize and reopen; the oriented defect lies in A_2 ; the second route to α receives the pairs $21/22$, $6/7$ and $45/46$ from three scales of the same word; the first route preserves the dodecaphasic seal. No conventional coordinate generates the others. Dodecaphasic closure determines α within the correlated output, and the Machin, Perron–Fibonacci or exponential-semigroup characterizations enter only afterwards as recognitions.

The electronic vacancy also acquires a precise provenance. On the APP support, the transition from the centre $(5, 5)$ to the branch $(8, 2)$ preserves the additive sheet, product residue and TRIT regime while losing one unit of multiplicative carry. In the temporal word, a vacancy preserves the capacity phase while the block and memory advance. The HMT–MD correspondence jointly preserves this unit loss, the fixed centre, the mode m_{ph} , the incidence $90/120$ and band conjugation. Electricity determines the oriented vacancy and magnetism its conjugate rotational transport; the common and oriented constitutive responses are obtained from the two sheets before being written in a dimensional chart.

The depths 11 and 34 appear in a common completion signature. The action line has its determinantal derivation and the gravitational branch first receives the length $L = (5759/23040)\alpha^{16}L_*$. Marked inscription and the positive radial metric prove $R_X\Lambda_X = L^2I$, $H_X\Lambda_X = \hbar_{\text{ret}}cI$ and $G = c^3L^2/\hbar_{\text{ret}}$, with the rules and domains of section 39.1. The functional restitution of section 35.6 links the Catalan moment to the responses of the two sheets and preserves their common propagation. Each constant is a section or evaluation of the same predefined state and preserves the reader determining its dimension, normalization and scale. The Barbero–Immirzi parameter, Catalan constant, Feigenbaum constants, permittivity and permeability are distinct readers of a common genealogy; they are therefore coordinated without becoming a single function or a list of juxtaposed values.

The second projection preserves boundary incidence. The connection 24 in the splitting $729 = 202+527$, the pairs $12/13$ and $27/28$, the unimodular frame $M_\star$, the Witt design and the Golay and Leech gluings compose the typed passage to V^h and the Monster group. Exceptional symmetry is therefore the boundary realization of the same discrete structure that produces the Archimedean coordinate through the other projection; form, incidence, orientation and carry remain visible throughout transport.

The kernel reached can be summarized in a single mathematical statement without losing its architecture: the unit propagates through two complementary projections; the boundary difference generates orientation; the derivative of that orientation selects three regimes; the order of their transports generates a noncommutative fibre; and unlimited refinement preserves the unit, memory and aperiodicity while producing its continuous realizations. This is the precise content of the conservative evolution of the dimensional unit through double holographic projection and fractal refinement.

The formulation gathers a single constructive chain. Its arithmetic, physical and exceptional realizations preserve different domains and readers, but none is placed on a lower scale of certainty: each is the exact image of the integral holonomic state in its declared codomain. External comparison neither generates nor selects these outputs.

14.16 Exhaustive morphology of the 104 976 seeds

The cardinal correction is not minor: 104 976 and 170 748 belong to distinct mathematical objects. The first is the exhaustive domain of initial conditions of the APP–TPK emitter; the second is the denominator of the first continuant approximation closing the chronological clock C_{108} . Neither can replace the other.

Let

$$I_9 = \{0, 1, \dots, 8\}, \quad D = \{N, E, S, O\}, \quad S = I_9^2 \times D.$$

Here I_9 is exclusively the positional chart of the nine places, indexed from zero. It does not replace the canonical arithmetic alphabet $\mathcal{D}_9 = \{1, \dots, 9\}$ on which APP evaluates sum and product; passage between the two charts is a relabelling of positions, not an additional physical operator.

APP causally precedes this construction. An element $s = (i, j, d) \in S$ does not generate APP: it fixes a cursor position and orientation on the APP support already constructed. Since additive and multiplicative evaluation must be preserved simultaneously, a complete seed is the ordered pair

$$\omega = (s_+, s_\times) \in \Omega_{\text{seed}} := S_+ \times S_\times.$$

Therefore,

$$|S| = 9^2 \cdot 4 = 324, \quad |\Omega_{\text{seed}}| = 324^2 = 104\,976 = 2^4 3^8.$$

The seed is not a bare integer. It is a six-field oriented cell,

$$(i_+, j_+, d_+; i_\times, j_\times, d_\times),$$

which keeps the two arithmetic sheets and their directions separate. This is the finite unit on which TRIT types the local regime and TPK applies selection, transport and updating.

The exhaustive sweep reveals a factorization stronger than the count alone. The additive signature induces exactly 18 classes, all of cardinality 18. The multiplicative signature induces 26 classes, but its fibres stratify into

$$24 \text{ size classes } 8, \quad 1 \text{ size class } 24, \quad 1 \text{ size class } 108.$$

If $\mathcal{A}_{18}$ and $\mathcal{M}_{26}$ denote the respective signature sets, the emitter factors as

$$\Omega_{\text{seed}} \longrightarrow \mathcal{A}_{18} \times \mathcal{M}_{26} \xrightarrow{\text{Comb}} \mathcal{U}_6^{\text{cat}}, \quad |\mathcal{U}_6^{\text{cat}}| = 18 \cdot 26 = 468.$$

The first arrow preserves the provenance of each sheet; the second combines the two signatures into a decimal emission. The fibre multiplicities are the exact products

$$18 \cdot 8 = 144, \quad 18 \cdot 24 = 432, \quad 18 \cdot 108 = 1944,$$

and the global closure is

$$432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976.$$

This asymmetry is structural. The additive sheet has a uniform partition; the multiplicative sheet contains two singular strata. Vacancy and torsion are not introduced afterwards as metaphors: the seed domain already distinguishes uniform accumulation from folding with fibre degeneracies. The TPK preserves precisely which of these fibres was traversed, together with residue, quotient, carry, orientation and boundary.

After this factorization, the previously certified reductions are applied

$$104,976 \longrightarrow 468 \, U_6 \longrightarrow 243 \, w_6 \subset \mathbb{F}_3^6, \quad |\mathbb{F}_3^6| = 729.$$

Four types are thus distinguished and must not be confused again: seed pairs, decimal emissions, visible words and tritic ambient space. The crystal structure is none of these cardinalities in isolation, but the system of fibres and morphisms linking them.

Refutation criterion. An additive class of size other than 18, a multiplicative histogram other than $24 \cdot 8 + 1 \cdot 24 + 1 \cdot 108 = 324$, or loss of the ordered sheet pair in passage to U_6 .

14.17 Bipartite geometry of the π microfibre and electronic spectral connection

The visible output $w_\pi = 010211$ has 1008 antecedents in Ω_{seed} . To recover its form and not merely its mass, define the bipartite graph

$$\mathcal{G}_\pi = (P_\pi \sqcup T_\pi, E_\pi),$$

where P_π contains the additive-sheet seeds participating in the microfibre, T_π those of the multiplicative sheet, and $(p, t) \in E_\pi$ if the pair produces one of the five emissions reducing to w_π . The exhaustive sweep proves

$$|E_\pi| = 1008, \quad |P_\pi| = 54, \quad |T_\pi| = 56.$$

The structure is not an arbitrary graph. Forgetting only the emission colour, its connected components are exactly

$$\mathcal{G}_\pi \simeq K_{18,24} \sqcup K_{18,24} \sqcup K_{18,8}.$$

One of the two components $K_{18,24}$ corresponds to the region of multiplicity 432. The other refines into three complete rectangles of multiplicity 144, since the three multiplicative classes of size eight share the same additive class of size eighteen. The last component is the fourth rectangle of multiplicity 144. The region colour therefore recovers

$$1008 = 432 + 4 \cdot 144$$

as a decomposition of incidence operators, not merely as a sum of integers.

Let B_R be the biadjacency matrix of a rectangular region R . For a complete graph $K_{m,n}$, B_R has rank one and its only nonzero singular value is

$$\|B_R\| = \sqrt{mn}.$$

The region of $432 = 18 \cdot 24$ routes has norm $12\sqrt{3}$; each of the four regions of $144 = 18 \cdot 8$ routes has norm 12. The normalization treating the four smaller regions symmetrically is determined by the regional partition itself:

$$\Xi_\pi := \frac{\|B_{432}\|}{\sum_{r=1}^4 \|B_{144}^{(r)}\|} = \frac{12\sqrt{3}}{4 \cdot 12} = \boxed{\frac{\sqrt{3}}{4}}.$$

Its square is $3/16$. This is exactly the invariant that appears, in the previously certified APP electronic mode, as

$$\|[P_{\Sigma,3}, P_{\Pi,3}]\|^2 = s^2(1 - s^2) = \frac{1}{4} \frac{3}{4} = \frac{3}{16}.$$

This yields an **exact internal spectral connection** between two constructions arising from APP: the regional contrast of the π microfibre and the noncommutativity of the two electronic projections. Neither the electron mass nor any experimental parameter has been used. Nor is it asserted that the matrices B_R and the commutator act on the same space. The strong statement already proved is equality of their canonical scalars; a subsequent operator equivalence must transport the regional colouring, the mode m_{ph} , the sheet and the memory record.

This geometry also specifies the place of vacancy. On the additive side, the classes are uniform; on the multiplicative side, the degree changes from 24 to 8. Porosity of the support is not an inserted zero: it is the incidence stratification surviving the common reduction w_π . The electric realization can read that oriented loss and the magnetic realization its conjugate transport, but both must preserve this original incidence.

Refutation criterion. A component ceasing to be complete, the five colours not having sizes $(432, 144, 144, 144, 144)$, or a spectral ratio other than $\sqrt{3}/4$. Operator equality, which is not asserted here, would be falsified by any transport not preserving colour, sheet and inner product.

14.18 Three-channel morphism, tritic gluing class and diagonal memory

The count of the nine nonadic positions defines a vector

$$N_K = (N_\pi(K), N_e(K), N_\varphi(K)) \in \mathbb{Z}^3.$$

The relational part is obtained through the morphism

$$\partial : \mathbb{Z}^3 \longrightarrow A_2, \quad \partial(x, y, z) = (x - y, y - z, z - x),$$

where

$$A_2 = \{(a, b, c) \in \mathbb{Z}^3 : a + b + c = 0\}.$$

The kernel is the diagonal ($\Delta\mathbb{Z} = \mathbb{Z}(1, 1, 1)$), and ∂ is surjective. Consequently,

$$\boxed{0 \longrightarrow \mathbb{Z} \xrightarrow{\Delta} \mathbb{Z}^3 \xrightarrow{\partial} A_2 \longrightarrow 0.}$$

Synchronization of the three channels is exactly the vanishing of ∂N_K . The quotient removes the common quantity, but preserves which channel exceeds which and with what orientation.

Integer reconstruction contains additional information. Define

$$\mathcal{J}(x, y, z) = (s, a, b) = (x + y + z, x - y, y - z).$$

Over $\mathbb{Q}$, the three data reconstruct (x, y, z) through

$$x = \frac{s + 2a + b}{3}, \quad y = \frac{s - a + b}{3}, \quad z = \frac{s - a - 2b}{3}.$$

Over $\mathbb{Z}$, the image is not all of $\mathbb{Z} \oplus \mathbb{Z}^2$: it satisfies the congruence

$$\boxed{s - a + b \equiv 0 \pmod{3}}.$$

The determinant of $\mathcal{J}$ has absolute value 3. There is therefore an exact sequence

$$0 \longrightarrow \mathbb{Z}^3 \xrightarrow{\mathcal{J}} \mathbb{Z}^3 \xrightarrow{\chi_3} C_3 \longrightarrow 0, \quad \chi_3(s, a, b) = [s - a + b]_3.$$

This result is particularly important for the meaning of TRIT. Separating the common content from the relative defect leaves a gluing class with exactly three values. A trit is the minimum amount of information needed to record that class and select the integral correction recomposing the three-channel state. The topological–temporal interpretation is not added by analogy: the class types the local gluing and TPK transports it in time while preserving its orientation and memory. This quotient C_3 is a precise internal realization of that function; it does not purport to exhaust all coordinates of the governing TRIT.

The passage from $K = 8$ to $K = 9$ shows why synchronization does not mean state return:

$$N_8 = (59, 59, 59), \quad N_9 = (43, 43, 43),$$

$$\partial N_8 = \partial N_9 = 0, \quad \boxed{N_9 - N_8 = -16(1, 1, 1)}.$$

The state remains at the relational origin A_2 , but the diagonal memory record changes by sixteen units per channel, forty-eight in total. The synchronization phase returns; memory does not. This equality is the entirely integral three-channel analogue of the TPK rule that phase return does not reset the integral holonomic state.

Refutation criterion. A count violating the congruence, a kernel of ∂ larger than the diagonal, or an equality $N_8 = N_9$.

14.19 Wall itinerary and finite quotient of the synchronization matrix

Between $K = 4$ and $K = 9$, the preceding morphism turns the complete table into a geometric itinerary:

K	N_K	∂N_K
4	(207, 207, 122)	(0, 85, −85)
5	(39, 151, 151)	(−112, 0, 112)
6	(110, 110, 69)	(0, 41, −41)
7	(81, 29, 81)	(52, −52, 0)
8	(59, 59, 59)	(0, 0, 0)
9	(43, 43, 43)	(0, 0, 0).

The successive equalities traverse the walls

$$H_{\pi e} \longrightarrow H_{e\varphi} \longrightarrow H_{\pi e} \longrightarrow H_{\varphi\pi} \longrightarrow 0 \longrightarrow 0.$$

The wall word $(H_{12}, H_{23}, H_{12}, H_{13})$ has the shadow of the Coxeter pattern A_2 , since $s_{12}s_{23}s_{12} = s_{13}$. This does not automatically turn every TPK transition into a reflection: the step also changes the diagonal component. What is proved is that the relational trajectory crosses the three wall systems in the order of the braid relation, while its lift preserves memory invisible to the quotient.

The amplitudes of the four walls preceding closure form

$$D_{\text{syn}} = \begin{pmatrix} 85 & 112 \\ 41 & 52 \end{pmatrix}.$$

In addition to the identities already gathered

$$\text{tr } D_{\text{syn}} = 137, \quad \det D_{\text{syn}} = -172 = -4 \cdot 43,$$

the Smith normal form adds quotient structure. Since the greatest common divisor of the four entries is one,

$$\text{SNF}(D_{\text{syn}}) = \text{diag}(1, 172),$$

and, therefore,

$$\boxed{\text{coker } D_{\text{syn}} \simeq C_{172} \simeq C_4 \times C_{43}.}$$

The factorization $(4 \cdot 43)$ no longer appears only in the determinant: it is the primary decomposition of the lattice quotient of the wall amplitudes. The right null directions can be chosen as

$$(0, 1) \pmod{4}, \quad (1, 5) \pmod{43}.$$

The factor 43 coincides with the count of the second triple closure, and the factor 4 with the oriented coefficient obtained internally as $(-\det(D_{\text{syn}})/43)$. This proves that both are canonical divisors of the same quotient. A subsequent physical representation must specify which subspaces realize these two factors; the joint arithmetic structure is already closed.

The itinerary thus supplies a complete mathematical picture: a path in the wall arrangement of (A_2) , lifted by a diagonal memory coordinate and governed by a finite quotient $C_4 \times C_{43}$. It is more informative than a table of coincidences because it distinguishes relation, amplitude, orientation and the state invisible to the quotient.

Refutation criterion. A determinant other than -172 , a Smith form other than $(1, 172)$, or a transition erasing the diagonal component needed to distinguish $K = 8$ from $K = 9$.

14.20 Nonadic Pascal tower and infinite concatenation law with carry

Cubic completion belongs to an exact system of arbitrary depth. For $n \geq 1$, write

$$10^n = (9 + 1)^n = \sum_{j=0}^n B_{n,j}, \quad B_{n,j} := \binom{n}{j} 9^{n-j}.$$

The blocks obey the weighted Pascal recurrence

$$\boxed{B_{n+1,j} = 9B_{n,j} + B_{n,j-1}},$$

with $B_{n,j} = 0$ outside $0 \leq j \leq n$. This is the length-refinement law; the vacancy law must also preserve the initial phase. If

$$s_{n,j} = \sum_{i \leq j} B_{n,i}, \quad v_{n,j}^- = V_{B_{n,j}}(s_{n,j}),$$

then

$$\sum_{j=0}^n v_{n,j}^- = \lfloor 10^n \delta \rfloor.$$

The upper reading transports the half-boundary unit to the first block:

$$v_{n,0}^+ = v_{n,0}^- + 1, \quad v_{n,j}^+ = v_{n,j}^- \quad (j > 0),$$

and therefore

$$\sum_j v_{n,j}^+ = \lceil 10^n \delta \rceil = \lfloor 10^n \delta \rfloor + 1.$$

The transition $n \mapsto n + 1$ cannot be linearized by forgetting the starting point. For any block B ,

$$V_{9B+B'}(s) = \sum_{r=0}^8 V_B(s + rB) + V_{B'}(s + 9B).$$

The shift of s in each summand is the concatenation memory. Projecting all blocks to the origin reveals the binary cocycle

$$\kappa_\delta(M, N) = \lfloor (M + N)\delta \rfloor - \lfloor M\delta \rfloor - \lfloor N\delta \rfloor,$$

which satisfies

$$\kappa_\delta(L, M) + \kappa_\delta(L + M, N) = \kappa_\delta(M, N) + \kappa_\delta(L, M + N).$$

This is the exact law preventing refinement from depending on parenthesization. The Pascal tower supplies the lengths; the cocycle supplies the carry; TPK preserves the phase and memory record that allow their composition.

The first levels are

n	$(B_{n,0}, \dots, B_{n,n})$	v_n^-	v_n^+
2	(81, 18, 1)	(3, 1, 0)	(4, 1, 0)
3	(729, 243, 27, 1)	(33, 11, 1, 0)	(34, 11, 1, 0)
4	(6561, 2916, 486, 36, 1)	(300, 133, 22, 2, 0)	(301, 133, 22, 2, 0).

The cubic level is therefore not an isolated coincidence. It is the first level at which the component $729 = 3^6$, the subscale $243 = 3^5$, the boundary $27 = 3^3$ and the unit appear simultaneously in a decimal completion. Its lower signature contains $33 = 3 \cdot 11$, and the upper one preserves the unit to produce 34. The next level proves that the rule continues without resetting the state: the unit again modifies only the first component, while the remaining entries preserve the history of the ordering.

The concatenation $210 = 90 + 120$ is a section of the same law:

$$(V_{90}(0), V_{120}(90)) = (4, 5),$$

whence the lower reading produces 9 and the upper one 10. The nonadic–decimal transition is thus realized by a reading morphism with carry, not by the graphic observation $4 + 5 = 9$.

Define the decimal carry digit

$$d_{n+1} := \lfloor 10^{n+1} \delta \rfloor - 10 \lfloor 10^n \delta \rfloor.$$

Since δ is transcendental, the sequence $(d_n)_{n \geq 1}$ cannot be eventually periodic: such periodicity would make the decimal expansion of δ rational. Since the mechanical word is Sturmian, every level retains linear complexity and zero topological entropy. The tower therefore combines unlimited refinement, the boundary unit and exact aperiodicity. This is the mathematical sense in which the crystal simultaneously grows outwards and refines inwards.

One must distinguish κ_δ from the governing nonadic cocycle κ_9 . The former is the mechanical restriction recording the carry of the irrational cut; the latter preserves the full APP carry under composition. A restriction square may relate them, but they are not identified merely because both use carry values.

Refutation criterion. Failure of the Pascal recurrence, a nontelescopically signature, loss of the upper unit, or concatenation whose result depends on parenthesization.

14.21 Mathematical type of the helical aperiodic crystal

The results allow the object to be named without reducing it to an image. Its finite layer of initial conditions is Ω_{seed} ; its emission quotient is $\mathcal{U}_6^{\text{cat}}$; its visible alphabet lies in $\mathbb{F}_3^6$; and its compatible extensions form the enriched coinductive state. Acting on this state, in the governing order, are

$$U_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t, \quad U_t \longrightarrow \Gamma_9 \longrightarrow K_{\text{ph}}.$$

The holonomy Γ_9 belongs to the enriched dynamics; the reduced memory K_{ph} is its subsequent realization. The aperiodic factor can be written on

$$\mathcal{Y}_{108} = C_{108}^{(t)} \times \mathbb{T}_\delta \times \mathbb{Z},$$

through means of

$$\mathcal{F}(r, \theta, m) = (r + 1, \{\theta + \delta\}, m + \lfloor \theta + \delta \rfloor).$$

The first coordinate is the block clock; the second, the irrational phase; the third, the integrated vacancy memory. The capacity clock is obtained as

$$K_t = t - Z_t, \quad g_t = [K_t]_9,$$

and does not coincide with the chronological clock. The inverse dynamics $\mathcal{F}^{-1}$ constitutes the descending reading of the same helix. Ascent and descent are conjugate; they are not two disconnected generators.

The complete object is therefore an invertible affine extension of an irrational rotation over a finite clock, fibred by the integral holonomic APP–TRIT–TPK state. Its symbolic shadow is Sturmian; its suspension is helical; its fibre cocycle may be noncommutative even though the base rotation is abelian. The designation “helical aperiodic crystal” is justified by four proved properties:

$$\begin{aligned} &\text{long-range order and repetitivity,} \\ &p(m) = m + 1, \quad h_{\text{top}} = 0, \\ &\text{absence of translational period,} \\ &\text{finite phase return with memory advance.} \end{aligned}$$

In HMT, the expression **aperiodic time crystal** designates precisely the preceding skew product: there is a finite temporal support, phase return is ordered, memory advances in each holonomy and the occupation word has no translational period despite preserving long-range order and zero topological entropy. Temporal dynamics is not added to the crystal afterwards: it is contained in the coordination of clock, orientation, return and memory defining its state.

The correlated constant output is situated after the extension

$$w_6 \rightarrow w_{12} \rightarrow w_{18} \rightarrow w_{24} \rightarrow w_{30} \rightarrow R_{36} \rightarrow G_9,$$

without introducing conventional values. π, φ, e and the dodecaphasic coordinate α are realizations of the same coinductive state; their synchronization windows, mirror and own readers are not generator inputs. The completion tower supplies the capacity calendar and the carry law by which these realizations can be extended to arbitrary precision.

Finally, $\mathbb{R}$ appears through the compatible-cylinder reader of the joint continuum. The crystal is not “inside $\mathbb{R}$ ” as an embellishment of a line given beforehand. Compatible histories, double projection, boundaries and their limits are constructed first; one of the realizations subsequently determines the Archimedean coordinate. This reversal of perspective is the strong thesis: the real line is the terminal shadow of a finite oriented unit that is conserved, refined and accumulates memory.

The aperiodic dynamics, its complexity, clocks and cocycle constitute the mathematical kernel of the crystal. The electromagnetic, gravitational and exceptional realizations are typed readers of that kernel: they preserve the APP–TRIT–TPK genealogy and express the same conservative evolution of the unit in different codomains.

14.22 Coinductive predefinition: the output precedes its evaluation

The decisive distinction separates generation from evaluation. HMT–MD does not take an external measurement to infer the law that might have produced it. It first constructs the complete section, fixes its compatible extensions and only then allows a reader to express a coordinate of that section. This is the exact mathematical meaning of **predefinition**.

Let

$$(X_n, \rho_{n+1,n})_{n \geq 0}$$

the inverse system of APP–TRIT–TPK states truncated at depth n , where each X_n preserves sheet, orientation, residue, quotient, carry, boundary, route and memory up to that level. The coinductive extension already constructed defines

$$\mathcal{G}_{\text{HMT}} : \Omega_{\text{seed}} \longrightarrow X_{\infty}^{\text{enr}} := \varprojlim_n X_n, \quad p_n : X_{\infty}^{\text{enr}} \longrightarrow X_n,$$

with $\rho_{n+1,n} p_{n+1} = p_n$. For a system of outputs $(Y_n, \sigma_{n+1,n})$ and readers $L_n : X_n \rightarrow Y_n$, the condition expressing that the reader does not destroy genealogy is the square

$$\begin{array}{ccc} X_{n+1} & \xrightarrow{L_{n+1}} & Y_{n+1} \\ \rho_{n+1,n} \downarrow & & \downarrow \sigma_{n+1,n} \\ X_n & \xrightarrow{L_n} & Y_n. \end{array}$$

Therefore,

$$\text{Pred}_{Y,n} := L_n \circ p_n \circ \mathcal{G}_{\text{HMT}} : \Omega_{\text{seed}} \longrightarrow Y_n$$

satisfies

$$\sigma_{n+1,n} \circ \text{Pred}_{Y,n+1} = \text{Pred}_{Y,n}.$$

The system $(\text{Pred}_{Y,n}(\omega))_{n \geq 0}$ is then a single element of $\varprojlim Y_n$. The output is not decided anew each time depth increases: the depth- n output is the restriction of the depth- $n+1$ output. Precision no longer generates the object; it merely decides how much of the predefined object is determined.

This provides an internal anticipation theorem. If, at stage n , the coordinate $n+k$ of the same compatible section is consulted, a future output is obtained without extrapolating a trend or introducing the value one wishes to find. The anticipation is

$$\text{Ant}_{n,k}(\omega) := L_{n+k} p_{n+k} \mathcal{G}_{\text{HMT}}(\omega),$$

and its subsequent comparison must restrict to $L_n p_n \mathcal{G}_{\text{HMT}}(\omega)$ through $\sigma_{n+k,n}$. The compatibility identity proves that both belong to the same section. Consequently, the correct conceptual transition is not

more measure $\longrightarrow$ more precision $\longrightarrow$ best prediction,

but

oriented seed $\longrightarrow$ predefined coinductive section $\longrightarrow$
reading at the chosen depth $\longrightarrow$ testable anticipation

Indefinite digit generation is an instance of this theorem. For the readers of π , φ , e and α , each decimal block is a coordinate of a compatible system; extending from one thousand to ten thousand digits neither changes the generator nor resets selection. The new window restricts to the preceding one and continues the same history. The correlated system is a single correlated output of the coinductive state, even though its four coordinates have internal readers and different realization stages.

TPK reversibility strengthens the result. Since every update $\mathcal{U}_t$ is bijective on the integral holonomic state, for every finite distribution transported by it,

$$H(X_{t+1} \mid X_t) = H(X_t \mid X_{t+1}) = 0, \quad H(X_{t+1}) = H(X_t).$$

This is the zero Landauer cost of the complete state: no information is erased in updating. It does not assert that every reduced entropy is zero; it asserts that the complete history can be recovered in both directions. Apparent loss arises only when a reader contracts the fibre and forgets coordinates preserved by the integral holonomic state.

14.23 TRIT as the topological–temporal unit combining regime and orientation

TRIT is neither a ternary digit added to APP nor a temporal label superimposed on TPK. It is the local unit allowing a difference between the additive and multiplicative sheets to acquire geometric type, temporal orientation and lifting memory simultaneously. Let

$$\mathcal{T} = \{-1, 0, +1\}, \quad x = (t, \mu, o, h, c, \sigma, \lambda, g) \in X_{\text{TPK}}^{\text{enr}}.$$

Two coordinated readers act on the same state,

$$\tau_g : X_{\text{TPK}}^{\text{enr}} \rightarrow \mathcal{T}, \quad \varepsilon_t : X_{\text{TPK}}^{\text{enr}} \rightarrow \mathcal{T}.$$

The first types the local quadratic regime through

$$\mathcal{A}_\tau = \mathbb{R}[J_\tau] / (J_\tau^2 + \tau I),$$

so that

τ_g	$J_{\tau_g}^2$	$\exp(sJ_{\tau_g})$
+1	$-I$	$\cos s I + \sin s J_+$
0	0	$I + sJ_0$
-1	$+I$	$\cosh s I + \sinh s J_-$

The second types the active temporal orientation:

$$\begin{aligned} +1 &\longleftrightarrow \text{return, memory and past,} \\ 0 &\longleftrightarrow \text{threshold and present,} \\ -1 &\longleftrightarrow \text{bidirectional opening and future.} \end{aligned}$$

The conceptual depth lies in both readers arising from the same state, not in their having to be identified term by term. The correct object is the fibred coupling

$$\Theta_{\text{ITT}}(x) = (\tau_g(x), \varepsilon_t(x), h(x), c(x), \lambda(x), g(x)),$$

which may be called the **unit of topological–temporal information**. Geometry indicates how the unit transforms locally; temporal orientation indicates the direction in which that transformation is read; carry and record prevent two returns with the same phase from being confused. Thus, the past is the closure remaining in memory, the present is the threshold at which continuation is decided, and the future is the already predefined system of compatible extensions.

TPK makes this unit dynamic through

$$\mathcal{U}_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t .$$

Sel_t selects the regime and sheet without erasing the other APP evaluation; Tra_t transports direction, phase and boundary; Upd_t updates residue, quotient, carry, signature and memory. A chronological turn can therefore return $g(t+9) = g(t)$ while simultaneously producing

$$q_{\text{mem}}(\Gamma_{9,t}x) = q_{\text{mem}}(x) + 1.$$

Temporal conjugation is represented by the oriented inversion

$$C_{\text{rev}}(\tau_1, \dots, \tau_n) = (-\tau_n, \dots, -\tau_1), \quad C_{\text{rev}}^2 = I.$$

In the helical suspension, this involution interchanges ascending and descending dynamics:

$$C_{\text{rev}} \mathcal{F} C_{\text{rev}}^{-1} = \mathcal{F}^{-1}.$$

History does not disappear: its orientation changes while the record enabling its reconstruction is preserved. The HMT time crystal is consequently the dynamics of an ITT returning in the finite clock and progressing in memory, while the Sturmian decoration prevents translational repetition of the full state. The clock C_{108} supplies finite temporal order; the holonomy Γ_9 , unlimited lifting; the Sturmian word, exact aperiodicity; TRIT, the joint discrimination of geometry and orientation; and TPK, the law preserving all of this under evolution.

14.24 The singularity of α as the seam of the two clocks

The function of α becomes transparent once it is no longer treated as an isolated decimal. α is the seam coordinate making finite dodecaphasic closure compatible with aperiodic capacity extension. The first clock supplies a signed word of twelve sectors; the second supplies a unique root in the mechanical dynamics governed by

$$\delta = \log_{10} \frac{10}{9}.$$

In the first route, the dodecaphasic state produces

$$K = \frac{1}{4} \mathcal{H}_{12} U_{12}^{\text{sgn}}, \quad \mathcal{H}_{12}^2 = 4I_{12},$$

and the base carry 1000 satisfies, block by block,

$$P_m + E_m - \Phi_m - K_m + c_{m+1} = A_m + 1000c_m.$$

Telescoping concatenation gives

$$\iota(P) + \iota(E) - \iota(\Phi) - \iota(K) = \iota(A).$$

The coordinates P, E, Φ are the HMT realizations of π, e, φ ; they are not conventional values introduced from outside. The vector K is the signed memory of closure, and A is the finite section determining α in the declared chart. The term distinguishing α from a bare combination of π, e, φ is precisely K : the dodecaphasic memory of how concatenation was performed.

In the second route, the same vacancy arithmetic produces the jet

$$\begin{aligned} P_6(x) &= 2\pi x - \frac{7}{4}x^2 + \frac{x^3}{2\pi} + \frac{x^4}{20} - \frac{2x^5}{21} - \frac{x^6}{46}, \\ T_{7;9}(x) &= -\frac{x^7}{120} + \frac{x^8}{45} + \frac{2x^9}{495}, \\ P_9(x) &= P_6(x) + T_{7;9}(x), \end{aligned}$$

and determines, through a simple positive root,

$$P_9(x) = \delta.$$

The coefficients are not interpolated parameters. They proceed from the half-corona 120, the central APP block of determinant 45, the dodecaphasic transverse rank 11 and $495 = 45 \cdot 11$. The nilpotent operator

$$Ne_7 = -\frac{8}{3}e_8, \quad Ne_8 = \frac{2}{11}e_9, \quad Ne_9 = 0$$

produces the tail exactly through

$$(I - N)^{-1} \left(-\frac{e_7}{120} \right) = -\frac{e_7}{120} + \frac{e_8}{45} + \frac{2e_9}{495}.$$

The relation between the two routes is expressed without confusing a finite section with an infinite root. Let d_m be the reader of the certified common resolution. The object α is then the fibre product

$$\mathfrak{A}_\alpha = \{ (x_{12}, x) : d_m(\Gamma_{12}^\alpha(x_{12})) = d_m(x), P_9(x) = \delta \}.$$

The first coordinate fixes dodecaphasic closure; the second fixes unique extension; equality in the common codomain fixes their synchronization. This is why α is singular within the correlated system. π , φ and e are regions and coordinates that lifting makes coincide, separate and coincide again; α records the seam preserving these coincidences when twelve phases close without stopping infinite extension. It does not arrive from outside afterwards: it is the holonomy coordinate through which joint generation becomes stable under return.

14.25 Dodecaphasic fan: α , electron, Witt and gravitation

The wheel of twelve positions does not produce an accidental succession of objects; it carries several irreducible readings of the same state. On

$$V_{12} = \mathbb{R}[A_5/C_5]$$

the exact decomposition is

$$V_{12} \simeq \mathbf{1} \oplus V_3 \oplus V_{3'} \oplus V_5.$$

If $P_1, P_3, P_{3'}, P_5$ are the central projectors, every dodecaphasic linear form ℓ decomposes canonically as

$$\ell(K) = \ell(P_1 K) + \ell(P_3 K) + \ell(P_{3'} K) + \ell(P_5 K).$$

The scalar reader determining α acts on the full vector K ; it therefore integrates information from all summands without confusing them. The reader P_3 leads, through the Witt incidence isometry, to the rank-three exceptional projector. The reader P_5 leads through

$$P_5 : V_{12} \rightarrow V_5, \quad \Gamma_5 : V_5 \xrightarrow{\simeq} \text{Sym}_0^2(\mathbb{R}^3),$$

to the five-component tensor carrier, and the projection $\Lambda(n)$ determines its transverse traceless sector. These are not three juxtaposed theories: they are three resolutions of the same dodecaphasic carrier—a scalar seam resolution, an exceptional incidence resolution and a tensor gravitation resolution.

The electronic intersection makes this unity visible through an identity of supports. If

$$B_K = \{4, 6, 9, 10\}, \quad P_\pi = \{2, 4, 5, 6, 7, 8, 9, 10, 11\},$$

$$H_e = \{1, 3, 4, 6, 9, 10\}, \quad H_\alpha^- = \{4, 5, 6, 7, 9, 10\},$$

then

$$B_K = H_e \cap P_\pi = H_e \cap H_\alpha^-, \quad H_\alpha^- = B_K \sqcup \{5, 7\}.$$

This equality answers the question of the “connection” of α . The electron does not join an external constant; its internal support intersects both the π microfibre and the negative-carry support of α . The additional pair $\{5, 7\}$ completes the negative hexad, and Witt polarity lifts the four pairs associated with B_K to their incidence blocks. Electronic vacancy is the local singularity; α is the global seam preserving its orientation throughout the dodecaphase.

The second realization of α is not scalar. From the same state one obtains

$$v_\alpha = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \quad \|v_\alpha\|^2 = 54,$$

and the neighbour

$$N_{\alpha,0} = \{x \in N(C_W) : \langle x, v_\alpha \rangle \equiv 0 \pmod{3}\}, \quad N_\alpha = N_{\alpha,0} + \mathbb{Z} \frac{v_\alpha}{3} \cong \Lambda_{24}.$$

The order-three action preserves this neighbour, and the corresponding orbifold determines $V^{\natural}$ and the class $3B$ of the Monster group. The scalar α and vector v_α are not the same object; they are two coordinates of the same dodecaphasic closure. This plurality of codomains is precisely what explains how α links the Archimedean branch, electronic vacancy and exceptional symmetry without functioning as a numerical adjustment.

14.26 From electronic vacancy to electromagnetic response

The exact definition of vacancy already appears in the arithmetic of the support. Normalizing the electronic pair by nine produces

$$\frac{1}{9}(369, 126) = (41, 14) \mapsto (5, 5),$$

whereas conservative transport satisfies

$$369 + 126 = 495 = 396 + 99, \quad (396, 99)/9 = (44, 11) \mapsto (8, 2).$$

The two routes preserve the additive sheet, multiplicative residue and TRIT regime; the second differs by one unit in the multiplicative quotient. Vacancy is therefore the minimum loss of carry in one sheet while preserving the remaining state coordinates. It is not an added zero: it is a relative defect visible only because APP preserves sum and product together.

The tritic phase mode

$$m_{\text{ph}} = (1, -1, 0)^3$$

is also the canonical singular electronic mode of APP:

$$C^{(3)}m_{\text{ph}} = C^{(3)*}m_{\text{ph}} = -\frac{1}{2}m_{\text{ph}}.$$

From here, one obtains

$$s = \frac{1}{2}, \quad qquad 1 - s^2 = \frac{3}{4} = \frac{90}{120}, \quad \|[P_{\Sigma,3}, P_{\Pi,3}]\| = \frac{\sqrt{3}}{4},$$

and, on the principal block, normalization of the commutator satisfies

$$J_{\text{comm}}^2 = -I.$$

Electromagnetic polarity is then determined by four layers of the same genealogy. APP supplies the sheets whose noncommutativity produces orientation; TRIT determines the elliptic generator ($J^2 = -I$); TPK preserves vacancy and its return; the channels 90/120 determine the common and differential responses. If

$$\begin{aligned} \mathcal{E} &= \log[(1 - q_+^{90})(1 - q_-^{90})] - \log[(1 - q_+^{120})(1 - q_-^{120})], \\ \mathcal{B} &= \log \frac{1 - q_-^{90}}{1 - q_+^{90}} - \log \frac{1 - q_-^{120}}{1 - q_+^{120}}, \end{aligned}$$

then $\mathcal{E}$ is even under sheet interchange and $\mathcal{B}$ is odd. The internal responses

$$\widehat{\mu} = e^{\mathcal{B}+\mathcal{E}}, \quad \widehat{\varepsilon} = e^{-\mathcal{B}+\mathcal{E}}$$

satisfy exactly

$$\sqrt{\frac{\widehat{\mu}}{\widehat{\varepsilon}}} = e^{\mathcal{B}}, \quad \frac{1}{\sqrt{\widehat{\mu}\widehat{\varepsilon}}} = e^{-\mathcal{E}}.$$

Permeability expresses rotational retention; permittivity, electric opening; their quotient preserves orientation and their product preserves common propagation. The electromagnetic wave is not added to the Sturmian word as an analogy: it is the realization in two conjugate components of oriented vacancy and its elliptic transport. The electric microfibre can therefore be aperiodic while maintaining global coherence, stable polarity and zero transport entropy in the full state.

14.27 Action, area, information and gravitation as predefined depths

Once the correlated system has been determined, the determinantal branch fixes

$$\Sigma_H = \varphi\alpha^{16}, \quad d_H = \lfloor \log_{10} \Sigma_H \rfloor = -34.$$

The regional reader determines the mantissas $\eta_{\text{pre}} = H_5$ and

$$\eta_{\text{ret}} = H_5 - \frac{90}{\pi} \mathcal{C}_\pi,$$

so that

$$\hbar_a = \eta_a 10^{-34} \mathcal{U}_S, \quad h_a = 2\pi \hbar_a.$$

The scale 10^{-34} does not proceed from a Planck measurement: it is the predefined depth at which the action reader finds its section. The phase θ subsequently acts through

$$S(\theta) = \hbar_{\text{ret}} \theta_{\text{rad}} = h_{\text{ret}} \frac{\theta_{\text{deg}}}{360}.$$

The factor 2π does not correct a chosen unit: it relates action per radian to action per turn. The angular chart and the action line determine two coordinates of the same oriented motion.

The hexad–octad transition subsequently evaluates the channels and produces the Barbero–Immirzi parameter. On the boundary, if

$$N = N_{90} + N_{120}, \quad D = 90N_{90} + 120N_{120},$$

then

$$N_{90} = 4N - \frac{D}{30}, \quad N_{120} = \frac{D}{30} - 3N.$$

The joint reader (N, D) preserves sectorial provenance in full:

$$H((N_{90}, N_{120}) \mid (N, D)) = 0.$$

This is the exact formulation of zero Landauer cost in the area–information transition. The entropy of a reduced state may be positive; what vanishes is the conditional loss when the full reconstructive pair is retained. The area spectrum

$$\frac{\mathcal{A}_{\text{phys}}}{\ell_P^2} = \gamma_{\text{HMT}} a_0 (N_{90} + N_{120})$$

and the modular Hamiltonian

$$K_A = cI + \Delta_{\text{mod}}(90N_{90} + 120N_{120})$$

are thus two complementary readers: the first counts total area; the second preserves how it was distributed between the two sectors.

Gravitation occupies another resolution of the same dodecaphasic closure. With

$$\omega_{108} = \frac{2\pi}{108t_0}, \quad R_{108} = \frac{54}{\pi} \ell_0,$$

and

$$G_a(\vartheta) = \vartheta \frac{c_{\text{int}}^5 t_0^2}{\hbar_a},$$

the coincidence condition for the three lengths

$$R_{108} = \bar{\lambda}_{C,a} = r_{g,a}$$

uniquely selects

$$\vartheta_{108} = \left(\frac{54}{\pi} \right)^2$$

and produces

$$\ell_P = \sqrt{\frac{G_a \hbar_a}{c_{\text{int}}^3}} = R_{108}.$$

The depth 10^{-11} of the gravitational branch and the depth 10^{-34} of the action branch are not observed scales introduced into the crystal. They are locations in the refinement where distinct readers of the same predefined section determine gravitation and action. The identity $G_a \hbar_a = \vartheta_{108} c_{\text{int}}^5 t_0^2$ shows that the two action orientations change gravitation reciprocally and preserve the Planck length. Clock, action and gravitation are therefore synchronized coordinates of a single dimensional evolution.

14.28 Governing law of the article: conservative evolution of the dimensional unit

The preceding results allow the central thesis to be stated without turning it into a metaphor or a catalogue of constants.

Theorem of conservative evolution and holographic predefinition. Let $\omega \in \Omega_{\text{seed}}$ be an oriented APP seed and let $\mathcal{G}_{\text{HMT}}(\omega) \in X_{\infty}^{\text{enf}}$ be its extension through TRIT and TPK. Then:

1. updating invertibly preserves sheet, residue, quotient, carry, orientation, boundary, route and memory;
2. the finite clock returns and holonomy advances, so the orbit is helical;
3. the irrational mechanical decoration has zero topological entropy, linear complexity and no translational period;
4. every compatible reader defines a predefined system in an inverse limit, so deepening reveals rather than recalibrates;

5. double projection determines, from the same section, the real line and the exceptional Witt–Golay–Leech– V^h –Monster branch;
6. the correlated system $(\pi, \varphi, e, \alpha)$ is a correlated output and α is its dodecaphasic seam with the aperiodic clock;
7. electronic vacancy, Planck action, the Barbero–Immirzi parameter, the vacuum constitutive quantities and gravitation are realizations of typed readers on that same genealogy.

The proof is the composition of the squares already written. Compatibility of truncations proves predefinition; inversion of $\mathcal{U}_t$ proves conservation; the Sturmian cocycle proves zero-entropy aperiodicity; the identity of supports proves the electron– π – α connection; the decomposition of V_{12} proves the coordinated separation of the exceptional and tensor readers; the action, area and information readers prove persistence of the unit under change of dimension.

The compact form of the theorem is

$$\begin{array}{ccc}
 \Omega_{\text{seed}} & \xrightarrow{\text{APP} \rightarrow \text{TRIT} \rightarrow \text{TPK}} & X_{\infty}^{\text{enr}} \\
 & & \downarrow (\mathcal{P}_{\text{arq}}, \mathcal{P}_{\text{exc}}, \mathcal{P}_{\text{dim}}) \\
 & & \mathbb{R} \times \mathcal{E}_{\text{exc}} \times \mathcal{D}_{\text{phys}},
 \end{array}$$

where the three realizations preserve the same provenance despite having distinct codomains. The unit does not remain motionless: it is conserved while being refined, changing chart, acquiring memory and increasing its dimension. The continuum is not the prior support of this process but its terminal projection. This is why the helical aperiodic crystal constitutes the kernel of HMT–MD: in a single reversible dynamics, it predefines the genealogy of the numbers and constants subsequently appearing as mathematics and physics.

Correlated generation of π , e and φ

The complete construction of the closure character, the five regions, and the structural biography of π was established in chapter devoted to the genealogical decomposition of the five oriented regions of π . Starting from that same already constructed character, the present chapter now prolongs the refinement toward e and φ , without restarting the genealogy or repeating its first stage.

15.1 Propagation, memory return and construction of the exponential character

15.1.1 Finite orbital selector: existence and uniqueness of compatible emission

Let $\mathcal{W} = \mathbb{F}_3^6$. For an emission $U = (u_1, \dots, u_6)$, the visible word is

$$\Psi(U) = ((-u_1) \bmod 3, \dots, (-u_6) \bmod 3).$$

On $\mathcal{W}$, the dihedral action is given by

$$\begin{aligned} r(u_1, u_2, u_3, u_4, u_5, u_6) &= (u_3, u_1, u_2, u_5, u_6, u_4), \\ s(u_1, u_2, u_3, u_4, u_5, u_6) &= (u_4, u_5, u_6, u_1, u_2, u_3). \end{aligned}$$

For $w \in \mathcal{W}$, define

$$n(w) = \#\Psi^{-1}(w), \quad M(w) = \sum_{U \in \Psi^{-1}(w)} \text{mult}(U),$$

and

$$\nu(w) = (\#\{j : w_j = 0\}, \#\{j : w_j = 1\}, \#\{j : w_j = 2\}).$$

Theorem 15.1 (Structural selection of the exponential channel). *Among the 43 orbits of the visible catalog, there is a unique orbit whose fibers simultaneously satisfy:*

$$n(w) = 1, \quad M(w) = 144, \quad \mathfrak{D}(w) = \{0, 3, 6\}, \quad \nu(w) = (2, 3, 1).$$

The gauge

$$\text{diag}_c = 6, \quad \text{card} = ES$$

select in it

$$w_6^e = 201101.$$

Proof. The four conditions are evaluated over the 243 words in the catalog, not over a sample. A single orbit satisfies the joint predicate. In that orbit, exactly one emission has both diagonal 6 and orientation ES . The enumeration uses only fields from the APP–TRIT catalog and does not consult an expansion of e . $\square$

15.1.2 Coinductive prolongation of the exponential emission via compatible blocks

The recurrence L_0, L_0, L_1, L_1 produces

$$w_{30}^e = 201101|121221|102011|012222|102011,$$

and the joint boundary contributes

$$u_5^e = 021222.$$

The record of thirty-six trits remains

$$w_{36}^e = 201101|121221|102011|012222|102011|021222.$$

15.1.3 Construction of the decimal signature via the ternary–decimal positional square

The initial reading of twelve digits is

$$\mathbf{d}_e^{(12)} = 718281828459 = 7 \underbrace{1828 \ 1828}_{\text{local square}} 459.$$

The central equality is the concatenation

$$1828 \parallel 1828,$$

not an arithmetic product nor the supposed beginning of a period.

Definition 15.2 (Positional square). A word $d_1 \cdots d_{12}$ contains a square of length ℓ at position p when

$$d_p \cdots d_{p+\ell-1} = d_{p+\ell} \cdots d_{p+2\ell-1}.$$

The position and the lateral contexts belong to the signature.

Table 15.1: Comprehensive census of squares in twelve-digit readings.

Length ℓ	1	2	3	4	5	6
Occurrences	240	21	3	2	0	0
Words that contain them	153	19	3	2	0	0

The other word with a square of length four is

$$575157518472 = \underbrace{5751 \ 5751}_{\text{initial position 1}} 8472.$$

Proposition 15.3 (Positional uniqueness). *The reading of channel e is the only one that has a square of length four after a prefix of length one and with contexts 7 and 459.*

Proof. The census leaves only two words with squares of length four. One starts it at the first position; the other after the prefix 7. The position and contexts separate the two possibilities. $\square$

15.1.4 Phase return with non-trivial memory state update

In the elevation

$$201101|121221|102011|012222|102011$$

the third and the fifth blocks coincide:

$$B_3 = B_5 = 102011.$$

Their prestates modulo 27 are, however,

$$p_3 = (25, 11, 7, 17, 8, 16), \quad p_5 = (7, 8, 4, 23, 26, 7),$$

and the subsequent quotients:

$$q_3 = (6, 13, 9, 2, 13, 1), \quad q_5 = (15, 0, 3, 19, 23, 25).$$

$p_3 \neq p_5$ and $q_3 \neq q_5$ hold.

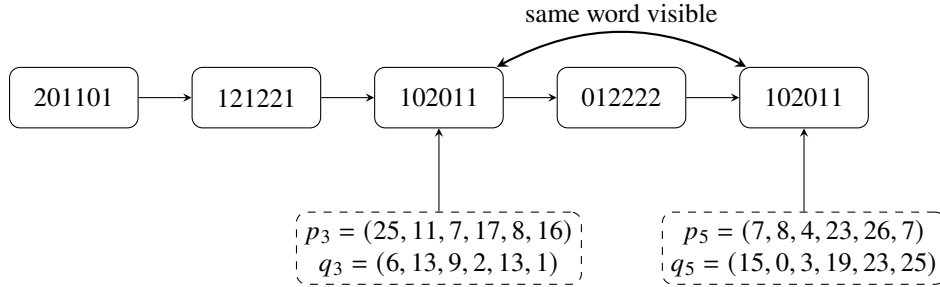


Figure 15.1: Visible return of 102011 with renewal of the prestate and the quotient.

Theorem 15.4 (Visible-phase return without periodicity of the state). $B_3 = B_5$ is a return of the visible projection, not a periodic orbit of the integral holonomic state.

Proof. The return of the complete state would require equality of all its coordinates. The prestate and quotient inequalities exclude it. Nor does 1828|1828 begin a decimal period: after the second block, 4 appears, whereas a periodic continuation would require 1. $\square$

15.1.5 Subsequent factorial certification of the coinductive exponential section

Let $\mathcal{A}$ be a unital commutative algebra over $\mathbb{Q}$, and let $P_t \in \mathcal{A}[[t]]$ be the formal propagator. Temporal concatenation imposes

$$P_{s+t} = P_s P_t, \quad P_0 = 1.$$

We write

$$P_t = \sum_{n=0}^{\infty} a_n t^n.$$

The unit fixes $a_0 = 1$, and normalized elementary transport fixes $a_1 = 1$.

Theorem 15.5 (Factorial recurrence).

$$(n+1)a_{n+1} = a_n \quad (n \geq 0),$$

and, therefore,

$$a_n = \frac{1}{n!}, \quad P_t = \sum_{n=0}^{\infty} \frac{t^n}{n!}.$$

En $t = 1$,

$$\boxed{P_1 = e.}$$

Proof. The coefficient of $s^n t$ in P_{s+t} is

$$\binom{n+1}{n} a_{n+1} = (n+1)a_{n+1}.$$

In $P_s P_t$ it is $a_n a_1 = a_n$. Equality of series produces the recurrence. Induction from $a_0 = 1$ gives $a_n = 1/n!$. $\square$

Corollary 15.6 (Differential equation).

$$\frac{d}{dt}P_t = P_t, \quad P_0 = 1.$$

Proof.

$$P'_t = \sum_{n \geq 0} (n+1)a_{n+1}t^n = \sum_{n \geq 0} a_n t^n = P_t.$$

□

15.1.6 Cofinal Forking of Rational Tangles and Convergence to the Published Value

Let

$$S_N = \sum_{n=0}^N \frac{1}{n!}.$$

Theorem 15.7 (Explicit factorial bound). *For $N \geq 1$,*

$$S_N < e < S_N + \frac{N+2}{(N+1)(N+1)!}. \quad (15.1)$$

Proof. The rest is

$$R_N = \sum_{k=0}^{\infty} \frac{1}{(N+1+k)!} > 0.$$

Para $k \geq 0$,

$$(N+1+k)! \geq (N+1)!(N+2)^k.$$

Therefore,

$$R_N \leq \frac{1}{(N+1)!} \sum_{k=0}^{\infty} \frac{1}{(N+2)^k} = \frac{N+2}{(N+1)(N+1)!}.$$

The inequality is strict from $k = 2$ onward. □

We define

$$J_N^{(e)} = \left[S_N, S_N + \frac{N+2}{(N+1)(N+1)!} \right]$$

and, for the fractional part,

$$\tilde{J}_N^{(e)} = J_N^{(e)} - 2.$$

The width converges to zero. For each M , there exists N such that

$$\text{diam } J_N^{(e)} < 10^{-M}.$$

If the endpoints belong to the same decimal cell, the first M digits are certified by rational arithmetic.

15.1.7 Coinductive prolongation of the exponential section through ternary subcylinders of the Topological Phase Kernel

At level K , TPK and the relation Ext have already generated a unique compatible subcylinder. The smallest $N = N_e(K)$ for which the factorial bracket is contained in that subcylinder is taken. $N_e(K)$ is an order of analytic certification, not a generating index or a rule for selecting the fiber.

The TPK cylinders are compatible under truncation independently of the semigroup. The semigroup theorem subsequently identifies their Archimedean evaluation with $e - 2$; the integer part recovers e . If the orbit, Ext or truncation fails, generation fails; if the recurrence or the factorial bound fails, this analytic certificate fails, not the APP–TRIT–TPK genealogy.

15.1.8 Orbital uniqueness, factorial recurrence, and cylindrical convergence of electron generation

Test criterion 15.8 (Exponential Channel Falsification). The construction is refuted if:

1. another orbit satisfies the complete finite predicate;
2. the signature 7[1828 1828]459 is not positionally unique;
3. the prestates or quotients of the two blocks 102011 coincide;
4. the semigroup does not force $(n + 1)a_{n+1} = a_n$;
5. the bound (15.1) fails;
6. a level does not locate a compatible extension uniquely;
7. a tabulated expansion intervenes before recognition.

15.2 Modal incidence, Perron ray, and construction of the self-similar character

15.2.1 Finite Selection of the Self-Scaling Seed

Let $\mathcal{U}_6 \subset \{0, \dots, 999\}^6$ be the catalog of the 468 distinct emissions constructed in chapter devoted to the index-54 emitter and its oriented seeds, and let

$$\Psi : \mathcal{U}_6 \longrightarrow \mathbb{F}_3^6, \quad \Psi(U) = ((-U_j) \bmod 3)_{j=1}^6.$$

For $w \in \Psi(\mathcal{U}_6)$, we recall the invariants

$$n(w) = \#\Psi^{-1}(w), \quad M(w) = \sum_{U \in \Psi^{-1}(w)} \text{mult}(U),$$

and the occupation vector

$$\nu(w) = (\#\{j : w_j = 0\}, \#\{j : w_j = 1\}, \#\{j : w_j = 2\}).$$

Theorem 15.9 (Orbital selection of the self-scale). *The dihedral action on $\Psi(\mathcal{U}_6)$ contains a unique orbit that satisfies the self-scale predicate*

$$n(w) = 1, \quad M(w) = 144, \quad \nu(w) = (2, 2, 2).$$

The internal caliber

$$\text{diag}_c = 6, \quad \text{card} = ES$$

select in that orbit the word

$$w_6^\varphi = 121200.$$

Its only emission is

$$U_\varphi = (830, 943, 830, 109, 252, 252),$$

of multiplicity 144, multiplicative signature 555933, additive orientation ES, multiplicative support of cardinality six and residual type A.

Proof. The direct projection gives

$$U_\varphi \bmod 3 = (2, 1, 2, 1, 0, 0), \quad -U_\varphi \bmod 3 = (1, 2, 1, 2, 0, 0).$$

The enumeration is carried out over the 243 visible words and their 43 orbits, not over a selection of examples. The joint predicate $n = 1$, $M = 144$, $\nu = (2, 2, 2)$ leaves one orbit. Within it, the declared

diagonal and orientation leave one emission. At this stage neither the equation $x^2 - x - 1 = 0$, nor a decimal table, nor the conventional name of the constant has been introduced. $\square$

15.2.2 Blockwise extension and construction of the 1st coinductive boundary

Let $b_1 = w_6^\varphi$, and let $\varepsilon = (0, 0, 1, 1)$. With the matrices L_0, L_1 constructed in chapter devoted to orbits and lifting to the R_{36} gauge, we define

$$b_{k+1} = b_k L_{\varepsilon_k} \pmod{3} \quad (1 \leq k \leq 4),$$

and concatenate

$$w_{6(k+1)}^\varphi = w_{6k}^\varphi \parallel b_{k+1}.$$

Matrix calculus produces, block by block,

$$b_1 = 121200,$$

$$b_2 = 112202,$$

$$b_3 = 121020,$$

$$b_4 = 010210,$$

$$b_5 = 010200,$$

and, therefore,

$$w_{30}^\varphi = 121200|112202|121020|010210|010200. \quad (15.2)$$

The dual neutrality of the boundary adds

$$u_5^\varphi = 102011,$$

so that

$$w_{36}^\varphi = 121200|112202|121020|010210|010200|102011. \quad (15.3)$$

Remark 15.10 (Two distinct uses of a visible word). The word 102011 is also visible in the history of e . Equality of one projection does not identify the integral holonomic states: channel, prefix, quotient, carry, phase and memory belong to the object's type. This distinction is mandatory before constructing any real character.

15.2.3 Construction of the incidence multigraph from the tritcale calendar

15.2.3.1 Primitive modal incidence F_{av} and memory dynamics of order 1

The primitive tritcale calendar is

$$(+1, -1, 0).$$

We fix two visible modis, Σ and Π , and the rule

$$+1 : m \mapsto \Sigma, \quad -1 : m \mapsto \Pi, \quad 0 : m \mapsto m. \quad (15.4)$$

Under the cyclic closure condition, the modal orbit produced by (15.4) is

$$(\Sigma, \Pi, \Pi).$$

The three consecutive pairs are

$$\Sigma \longrightarrow \Pi, \quad \Pi \longrightarrow \Pi, \quad \Pi \longrightarrow \Sigma.$$

Definition 15.11 (Chronological incidence matrix). For $i, j \in \{\Sigma, \Pi\}$, let

$$N_3(i, j) = \#\{r \in \mathbb{Z}/3\mathbb{Z} : (m_r, m_{r+1}) = (i, j)\}.$$

The subscript 3 counts instants of the primitive calendar; it does not denote a matrix power.

Theorem 15.12 (Necessary descent of modal incidence). *In the order (Σ, Π) ,*

$$N_3 = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \quad (15.5)$$

By exact repetition of the calendar,

$$N_9 = 3N_3, \quad N_{27} = 9N_3, \quad N_{108} = 36N_3. \quad (15.6)$$

These three equalities are equalities of edge counts; they do not assert $N_9 = N_3^3$, $N_{27} = N_3^9$ or $N_{108} = N_3^{36}$.

Proof. $\Sigma \rightarrow \Sigma$ does not occur; each of the other three pairs occurs once. This determines the four entries of N_3 . The calendars of lengths 9, 27, and 108 contain respectively 3, 9, and 36 copies of the primitive block. Each repetition multiplies all the counts by the same integer, which proves (15.6). $\square$

15.2.3.2 Publication areal–volumetrica and razon estable φ^{-2} under the measure of Parry

The modal label and the geometric sheet are distinct objects. The publication chart is

$$\mathfrak{g}_{\text{av}}(\Sigma) = \mathcal{H}_{\text{ar}}, \quad \mathfrak{g}_{\text{av}}(\Pi) = \mathcal{H}_{\text{vol}}.$$

Transporting (15.5) through $\mathfrak{g}_{\text{av}}$, we obtain the incidence operator

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \quad (15.7)$$

Thus, the two cross-transitions, the unique volumetric self-retention, and the absence of areal self-retention are derived from the calendar; they are not four independent inputs.

Remark 15.13 (Scope of the descent). Forgetting the elementary phase does not by itself produce a strong Markov quotient of the Topological Phase Kernel, because the visible mode does not determine which of the three internal operators will act at the next step. What does descend without an additional hypothesis is the edge multigraph of a complete block. Its minimal memory-one envelope has adjacency matrix F_{av} .

15.2.4 Construction of the modal incidence multigraph from the tritcale calendar

Theorem 15.14 (Perron ray derived). *The positive cone of F_{av} contains a unique eigenray. If we normalize it as $v = (1, x)^T$, then*

$$x^2 - x - 1 = 0, \quad x > 0,$$

and, therefore,

$$x = \frac{1 + \sqrt{5}}{2} = \varphi. \quad (15.8)$$

Proof. The equation $F_{\text{av}}v = \lambda v$ gives, coordinate by coordinate,

$$x = \lambda, \quad 1 + x = \lambda x.$$

Eliminating λ yields $x^2 - x - 1 = 0$. Its roots are $(1 \pm \sqrt{5})/2$, and only the positive root lies in the interior of the cone. The matrix F_{av} is primitive, since F_{av}^2 has all its entries positive; the Perron–Frobenius theorem ensures that this positive ray is simple and unique. The name φ is assigned only after this construction. $\square$

Corollary 15.15 (Scalar Self-Similarity Equation).

$$\varphi = 1 + \frac{1}{\varphi}, \quad \varphi^2 = \varphi + 1, \quad \varphi^{-1} = \varphi - 1.$$

This identity is a consequence of (15.8); they select neither the seed nor the matrix.

15.2.5 Fibonacci Recurrence of the Perron Ray and Diophantine Embedding of Rational Gaps

Let $F_0 = 0$, $F_1 = 1$, and

$$F_{n+1} = F_n + F_{n-1} \quad (n \geq 1).$$

Theorem 15.16 (Exact Powers of Incidence). *For every $n \geq 1$,*

$$F_{\text{av}}^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}. \quad (15.9)$$

Proof. For $n = 1$, the identity is the definition of F_{av} . If it holds for n , right multiplication by F_{av} gives

$$\begin{pmatrix} F_n & F_{n-1} + F_n \\ F_{n+1} & F_n + F_{n+1} \end{pmatrix} = \begin{pmatrix} F_n & F_{n+1} \\ F_{n+1} & F_{n+2} \end{pmatrix},$$

which is the case $n + 1$. $\square$

Theorem 15.17 (Cassini's Identity). *For every $n \geq 1$,*

$$F_{n+1}F_{n-1} - F_n^2 = (-1)^n. \quad (15.10)$$

Proof. The determinant of F_{av} is -1 . Taking determinants in (15.9),

$$F_{n-1}F_{n+1} - F_n^2 = \det(F_{\text{av}}^n) = (-1)^n.$$

$\square$

We define $r_n = F_{n+1}/F_n$ for $n \geq 1$. From Cassini it follows that

$$r_n - r_{n-1} = \frac{F_{n+1}F_{n-1} - F_n^2}{F_nF_{n-1}} = \frac{(-1)^n}{F_nF_{n-1}}.$$

Therefore, the approximations alternate around the Perron ray.

Proposition 15.18 (Perron Rational Gaps). *For $m \geq 1$,*

$$\frac{F_{2m+2}}{F_{2m+1}} < \varphi < \frac{F_{2m+1}}{F_{2m}}, \quad (15.11)$$

and the width is

$$\frac{F_{2m+1}}{F_{2m}} - \frac{F_{2m+2}}{F_{2m+1}} = \frac{1}{F_{2m}F_{2m+1}}. \quad (15.12)$$

Proof. The preceding alternation places the even quotients below and the odd ones above φ . The formula for the width is (15.10) with $n = 2m + 1$, after putting both fractions over a common denominator. Since $F_n \rightarrow \infty$, the width tends to zero. $\square$

15.2.6 Memory incidence of order 1 and Parry measure of the symbolic envelope

Let X_{av} be the smallest subshift of finite type over $\{\mathcal{H}_{\text{ar}}, \mathcal{H}_{\text{vol}}\}$ that admits exactly the three pairs observed in the modal calendar. Its adjacency matrix is F_{av} .

Theorem 15.19 (Transition and Parry Measure). *Let $r = (1, \varphi)^\top$. The Parry stochastic matrix is*

$$P_{ij} = \frac{(F_{\text{av}})_{ij} r_j}{\varphi r_i} = \begin{pmatrix} 0 & 1 \\ \varphi^{-2} & \varphi^{-1} \end{pmatrix}. \quad (15.13)$$

The measure estacionaria of vertices is proporcional to $(1, \varphi^2)$, hence

$$\frac{\mu_{\text{ar}}}{\mu_{\text{vol}}} = \varphi^{-2}. \quad (15.14)$$

Proof. The sum of row i of P is

$$\frac{(F_{\text{av}} r)_i}{\varphi r_i} = 1.$$

Since F_{av} is symmetric, the left and right eigenvectors coincide. The stationary weights are proportional to their coordinatewise products, that is, to $(1, \varphi^2)$. The ratio between the two weights is φ^{-2} . $\square$

Remark 15.20 (Three measures that must not be confused). The chronological frequency of the primitive orbit is

$$\nu_{\text{cyc}}(\mathcal{H}_{\text{ar}}) = \frac{1}{3}, \quad \nu_{\text{cyc}}(\mathcal{H}_{\text{vol}}) = \frac{2}{3}.$$

The uniform measure τ_{468} lives on the emission catalog. The Parry measure lives on admissible histories of arbitrary length. The first counts instants, the second counts emissions, and the third weights dynamic cylinders. None is an approximation of the others.

15.2.7 Multiplicative character and cylindrical conservation

For an admissible route $\gamma : i \rightarrow j$ of length n , we define

$$\chi_P(\gamma) = \varphi^n \frac{r_i}{r_j}. \quad (15.15)$$

Proposition 15.21 (Concatenation multiplicativity). *If $\gamma_1 : i \rightarrow j$ and $\gamma_2 : j \rightarrow k$, then*

$$\chi_P(\gamma_2 \circ \gamma_1) = \chi_P(\gamma_2) \chi_P(\gamma_1).$$

On to closed route of length n ,

$$\chi_P(\gamma) = \varphi^n.$$

Proof. Length adds and the intermediate coordinate telescopes:

$$\varphi^{n_1+n_2} \frac{r_i}{r_k} = \left(\varphi^{n_1} \frac{r_i}{r_j} \right) \left(\varphi^{n_2} \frac{r_j}{r_k} \right).$$

If $i = k$, the quotient terminal is one. $\square$

We normalize the left vector as

$$\ell = \frac{r}{1 + \varphi^2}, \quad \ell^\top r = 1.$$

The measure of a cylinder is

$$\mu[x_0 \cdots x_n] = \ell_{x_0} r_{x_n} \varphi^{-n}.$$

Therefore,

$$V_n(x) := \frac{\mu[x_0]}{\mu[x_0 \cdots x_n]} = \varphi^n \frac{r_{x_0}}{r_{x_n}} = \chi_P(x_0 \cdots x_n). \quad (15.16)$$

For to typed dimension $D > 0$, the scale

$$a_n^{(D)}(x) = V_n(x)^{1/D}$$

satisfies the exact law

$$\boxed{(a_n^{(D)}(x))^D \mu[x_0 \cdots x_n] = \mu[x_0]}. \quad (15.17)$$

In three dimensions and over closed returns,

$$\frac{a_9}{a_0} = \varphi^3, \quad \frac{a_{27}}{a_0} = \varphi^9, \quad \frac{a_{108}}{a_0} = \varphi^{36}.$$

If $A_k = a_{9k}$, then

$$A_{k+1} - 2A_k + A_{k-1} = A_{k-1}(\varphi^3 - 1)^2 > 0. \quad (15.18)$$

This is convexity in return depth; it is not interpreted as a kinematic quantity without an additional duration chart.

15.2.8 Quadratic memory and contracted branch

The eigenvalues of F_{av} are

$$\varphi, \quad -\varphi^{-1}.$$

The contracted branch changes orientation at each iteration. To preserve that sign before squaring, we pass to $\text{Sym}^2(\mathbb{R}^2)$. In the basis $(x^2, 2xy, y^2)$,

$$\text{Sym}^2(F_{av}) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{pmatrix}. \quad (15.19)$$

Theorem 15.22 (Spectrum of the second-order memory).

$$\det(tI - \text{Sym}^2(F_{av})) = (t+1)(t^2 - 3t + 1),$$

and

$$\boxed{\text{Spec}(\text{Sym}^2(F_{av})) = \{\varphi^2, -1, \varphi^{-2}\}}. \quad (15.20)$$

The only positive stable mode is φ^{-2} .

Proof. The eigenvalues of a degree-two symmetric power are the products λ_1^2 , $\lambda_1\lambda_2$, and λ_2^2 . With $\lambda_1 = \varphi$ and $\lambda_2 = -\varphi^{-1}$, one obtains φ^2 , -1 , and φ^{-2} . The first exceeds one, the second has modulus one, and the third belongs to $(0, 1)$. $\square$

On a closed route with $V_n = \varphi^n$, the stable mode satisfies

$$\boxed{M_n = \varphi^{-2n} = V_n^{-2}}. \quad (15.21)$$

15.2.9 Normalized hyperbolic realization of the self-scale: $\exp(2 \log \varphi J_h) = F_{av}^2$

Let

$$A = F_{av}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad J_h = \frac{2A - 3I}{\sqrt{5}}.$$

Then

$$J_h^2 = I, \quad A = \frac{3}{2}I + \frac{\sqrt{5}}{2}J_h.$$

Since

$$\cosh(2 \log \varphi) = \frac{3}{2}, \quad \sinh(2 \log \varphi) = \frac{\sqrt{5}}{2},$$

we obtain

$$\boxed{\exp(2 \log \varphi J_h) = F_{av}^2} \quad (15.22)$$

The self-scaling constant is therefore simultaneously the Perron eigenvalue of the incidence and the exponential parameter of its normalized hyperbolic realization.

15.2.10 Characterization of self-similarity via rational forks and coinductive prolongation

The finite construction determines the positive ray of F_{av} . The subsequent analytic characterization is

$$x^2 - x - 1 = 0, \quad x > 0.$$

The brackets (15.11) have rational endpoints and explicit width (15.12). For each nonadic level K , we choose the least $m = m_\varphi(K)$ whose bracket remains contained in a single subcylinder among the 729 elements of the ambient fiber. Minimality converts localization into a function, not into a tacit choice.

The selected cylinders are compatible with the truncations, and their diameters tend to zero. Their intersection contains a unique real number; the theorem 15.14 identifies it with φ . The Fibonacci quotients are therefore rational certificates of localization, not decimal data used to decide the seed.

15.2.11 Internal certificate of uniqueness, compatibility, and convergence of self-scaling

Test criterion 15.23 (Refutation of the self-scale biography). The construction remains refuted if any of the following facts occurs:

1. the finite predicate selects more than one orbit or more than one emission;
2. the recurrence L_0, L_0, L_1, L_1 does not produce (15.2);
3. the calendar does not produce exactly the three pairs used in (15.5);
4. a count multiple N_9, N_{27}, N_{108} is confused with a power of N_3 ;
5. F_{av} has another positive eigenray;
6. (15.9) or Cassini's identity fails;
7. the rational brackets are not nested or do not contract their width;
8. the chronological frequency, the uniform measure on the catalog or the Parry measure are identified;
9. a decimal table is involved before the Perron ray is constructed;
10. a selected cylinder does not truncate to the previous cylinder.

15.3 Diagonal certification, inverse limit and generation to arbitrary precision

15.3.1 Typed separation of the 4 indices of nonadic prolongation

This unit has the status of subsequent certification. The prefixes and their compatible extensions come from the internal relation Ext, coinductive survival and the APP–TRIT–TPK transition laws. The rational brackets that follow do not choose among the 729 elements of a fiber: they locate in the Archimedean chart the already generated extension and certify its precision at arbitrary depth.

Four parameters of different kinds enter each character $\chi \in \{\pi, e, \varphi\}$:

1. n , order of the rational analytic bracket;
2. K , refinement level of the Topological Phase Kernel;
3. $L_K = 6K$, ternary length of the published prefix;
4. M , the requested number of decimal positions.

Neither $n = K$, $n = L_K$, $K = M$ nor a fixed proportionality between them is postulated. The coupling is constructed through interval inclusions decided by integer arithmetic.

The integer parts are

$$m_\pi = 3, \quad m_e = 2, \quad m_\varphi = 1,$$

and the fractional characters are

$$\rho_\pi = \pi - 3, \quad \rho_e = e - 2, \quad \rho_\varphi = \varphi - 1.$$

The three families of semiopen forks are written

$$J_{\chi,n} = [\ell_{\chi,n}, h_{\chi,n}).$$

Their endpoints are rational, contain ρ_χ and satisfy

$$\ell_{\chi,n} \nearrow \rho_\chi, \quad h_{\chi,n} \searrow \rho_\chi, \quad h_{\chi,n} - \ell_{\chi,n} \longrightarrow 0.$$

For π , the endpoints come from the alternating sums of Machin's identity proved in chapter devoted to the construction of the closure character and the structural biography of π ; for e , from the factorial sums and their rational remainder in section 15.1; for φ , from the Fibonacci and Cassini quotients in section 15.2.

15.3.2 Ternary cylinders and the immediate ambient fibre

A ternary prefix of length $L_K = 6K$ is represented by an integer N_K with

$$0 \leq N_K < 3^{L_K}.$$

His semi-open cylinder is

$$C(N_K, L_K) = \left[\frac{N_K}{3^{L_K}}, \frac{N_K + 1}{3^{L_K}} \right). \quad (15.23)$$

The next emission adds six trits. The immediate ambient fiber contains $3^6 = 729$ elements, indexed by $u \in \{0, \dots, 728\}$:

$$C_{K,u} = \left[\frac{729N_K + u}{3^{L_K+6}}, \frac{729N_K + u + 1}{3^{L_K+6}} \right). \quad (15.24)$$

The $C_{K,u}$ are disjoint and

$$C(N_K, L_K) = \bigsqcup_{u=0}^{728} C_{K,u}.$$

The number 729 counts coordinates of this ambient fiber; it does not count seeds, distinct emissions, orbits, or surviving states.

15.3.2.1 Diophantine comparison of the rational endpoints of the gap

Let $J = [\ell, h) \subset C(N_K, L_K)$, with $\ell, h \in \mathbb{Q}$. Escribimos $S_K = 3^{L_K+6}$ and define

$$j_{\min}(J, K) = \max(729N_K, \lfloor S_K \ell \rfloor), \quad (15.25)$$

$$j_{\max}(J, K) = \min(729N_K + 728, \lceil S_K h \rceil - 1). \quad (15.26)$$

Lemma 15.24 (Unitary localization criterion). *The bracket J is contained in a unique element of the fiber (15.24) if and only if*

$$\boxed{j_{\min}(J, K) = j_{\max}(J, K)}. \quad (15.27)$$

In that case, if j is the common value,

$$u = j - 729N_K, \quad N_{K+1} = j.$$

Proof. The integers j whose elementary intervals of length S_K^{-1} intersect J form the integer interval

$$[\lfloor S_K \ell \rfloor, \lceil S_K h \rceil - 1] \cap [729N_K, 729N_K + 728].$$

This contains exactly one element if and only if its endpoints coincide. The equality identifies the numerator of the subcylinder and, by truncating its last six trits, N_K is recovered. $\square$

15.3.3 Minimal diagonal certification

Generation does not fix the same analytic order for all levels in advance. Once the compatible extension has been produced, the order of the bracket is increased only until it is certified uniquely.

Definition 15.25 (Diagonal certification recurrence). Given a family of compatible cylinders fixed by Ext, $C(N_K, L_K)$, whose publication contains ρ_χ , an initial order $n_\chi(K_0 - 1) \geq 1$ is chosen and we recursively define

$$n_\chi(K) = \min \{n \geq n_\chi(K - 1) : j_{\min}(J_{\chi,n}, K) = j_{\max}(J_{\chi,n}, K)\}, \quad (15.28)$$

and we verify

$$N_{\chi,K+1} = j_{\min}(J_{\chi,n_\chi(K)}, K). \quad (15.29)$$

The set

$$\Delta_\chi = \{(n_\chi(K), K) : K \geq K_0\}$$

is the analytic–nonadic diagonal of the character χ .

Theorem 15.26 (Existence and uniqueness of the certification diagonal). *For $\chi \in \{\pi, e, \varphi\}$, the recurrence (15.28) is defined at every finite level. The index $n_\chi(K)$ and the coordinate $N_{\chi,K+1}$ are unique.*

Proof. Suppose the cylindrical extension of level K containing ρ_χ has been generated internally. Since ρ_χ is irrational, it belongs to no ternary boundary $3^{-L}\mathbb{Z}$. It therefore lies in the interior of a unique subcylinder C_{K,u_*} , at a positive distance from both endpoints. The brackets $J_{\chi,n}$ contain ρ_χ and their diameter tends to zero; hence there exists an n for which the bracket is contained in C_{K,u_*} . The set over which the minimum is taken in (15.28) is nonempty and, by the well-ordering of $\mathbb{N}$, has a least element. lemma 15.24 proves that the bracket locates it uniquely and that the located numerator agrees with the one already generated. Induction on K completes the argument. $\square$

Corollary 15.27 (Absence of decimal parameters). *The diagonal is computed using sums, products, powers, Euclidean divisions, floors and ceilings of integers. It receives no decimal digit of π , e or φ as certification data.*

15.3.4 Compatibility with the joint nonadic connection

Let $\widehat{x}_{\chi,K}$ be the integral holonomic state of channel χ at level K . Its coordinates include the prefix, cylinder, phase, quotient, carry, unbounded vertical memory, terminal term and provenance record. The residual phase is

$$g_0(K) = [K]_9 \in \mathbb{Z}/9\mathbb{Z},$$

while the public label is

$$g(K) = 1 + ((K - 1) \bmod 9) \in \{1, \dots, 9\}.$$

The vertical memory $q_{\text{mem}}(K) \in \mathbb{Z}_{\geq 0}$ is not reduced modulo 12; only its dodecaphasic projection is

$$\beta(K) = q_{\text{mem}}(K) \bmod 12$$

does so.

For each K , let

$$\Gamma_{9,K} : \widehat{X}_K \longrightarrow \widehat{X}_{K+9}$$

the holonomy of a phase loop. Then

$$g_0(K+9) = g_0(K), \quad q_{\text{mem}}(K+9) = q_{\text{mem}}(K) + 1, \quad \beta(K+9) = \beta(K) + 1 \pmod{12}.$$

Theorem 15.28 (Phase return with genealogical conservation). *Phase equality does not reset the state:*

$$g_0(K+9) = g_0(K), \quad \widehat{x}_{\chi,K+9} \neq \widehat{x}_{\chi,K}.$$

The depths 27, 54, and 108 are the typed compositions

$$\Gamma_{27,K} = \Gamma_{9,K+18} \circ \Gamma_{9,K+9} \circ \Gamma_{9,K},$$

$$\Gamma_{54,K} = \Gamma_{9,K+45} \circ \dots \circ \Gamma_{9,K},$$

$$\Gamma_{108,K} = \Gamma_{9,K+99} \circ \dots \circ \Gamma_{9,K}.$$

In no case are they restarts of the prefix, the carry, the terminal, or the memory.

Proof. The first coordinate is calculated in $\mathbb{Z}/9\mathbb{Z}$, whereas q_{mem} belongs to an unbounded monoid. Each application of $\Gamma_{9,K}$ adds one memory unit, six blocks per intermediate level and the corresponding composition of cocycles. The phase may therefore coincide even when the integral holonomic state does not. The depth formulas show the indices of each map and prevent composition as though they were all endomorphisms of the same space. $\square$

15.3.4.1 Compression, expression, and terminal term

The operational realization of a loop satisfies

$$U_{\Gamma}J = JT + \eta, \quad J^*\eta = 0, \quad I - T^*T = \eta^*\eta. \quad (15.30)$$

The visible strict contraction is reserved for

$$M_9 = \frac{5}{9}V_9;$$

is not identified with the compression operator T . Telescoping at depth L preserves the terminal term:

$$S_0 = \sum_{r=0}^{L-1} M_9^{*r} D_9^* D_9 M_9^r + M_9^{*L} S_0 M_9^L. \quad (15.31)$$

Omitting the last summand before proving its disappearance would erase boundary information and would turn phase return into a false restart.

15.3.5 Uniqueness of the Minimal Diagonal Section in the Rational Fork System

Let $\mathcal{H}_K^\chi$ be the set of admissible enriched histories of channel χ at depth K , and let

$$p_{N,K} : \mathcal{H}_N^\chi \longrightarrow \mathcal{H}_K^\chi \quad (N \geq K)$$

the truncation. We define

$$\text{Surv}_K^{(N)}(\chi) = p_{N,K}(\mathcal{H}_N^\chi)$$

and

$$\text{Surv}_K(\chi) = \bigcap_{N \geq K} \text{Surv}_K^{(N)}(\chi). \quad (15.32)$$

The coinductive object is

$$X_\chi = \lim_{\leftarrow K} (\text{Surv}_K(\chi), p_{K+1,K}). \quad (15.33)$$

Theorem 15.29 (Section compatible under enhanced extendibility). *Suppose further that, over each generated numerical subcylinder, the relation Ext is nonempty, single-valued and compatible with the truncations. Then Ext and survival produce a family*

$$x_\chi = (\widehat{x}_{\chi,K})_{K \geq K_0}$$

which satisfies

$$p_{K+1,K}(\widehat{x}_{\chi,K+1}) = \widehat{x}_{\chi,K}.$$

Therefore, $x_\chi \in X_\chi$.

Proof. The subcylinder of level $K+1$ has numerator $N_{\chi,K+1} = 729N_{\chi,K} + u$. Truncation removes its last six trits and recovers $N_{\chi,K}$. The other coordinates are obtained through the functorial laws of carry, memory, terminal and phase. The single-valuedness hypothesis for Ext fixes the subcylinder and the enriched extension; the local uniqueness of lemma 15.24 subsequently certifies its Archimedean localization, and the compatibility of Ext provides truncation equality. $\square$

15.3.6 Decimal cells and prescribed precision

For $M \geq 1$ and $q \in \{0, \dots, 10^M - 1\}$, let

$$D_{M,q} = \left[\frac{q}{10^M}, \frac{q+1}{10^M} \right).$$

A bracket $J = [\ell, h)$ is contained in a unique decimal cell if and only if

$$\boxed{\lfloor 10^M \ell \rfloor = \lceil 10^M h \rceil - 1 = q.} \quad (15.34)$$

Definition 15.30 (Minimum level of decimal certification). For $\chi \in \{\pi, e, \varphi\}$, we defines $K_\chi(M)$ as the least level for which there exist an analytic order n and an integer q such that

1. $J_{\chi,n}$ satisfies (15.34);
2. $J_{\chi,n}$ is contained in the cylinder generated by state $\widehat{x}_{\chi,K}$;
3. the transition to the next level satisfies (15.27).

The pair (K, n) is minimized lexicographically.

Theorem 15.31 (Certification of coinductive outputs to arbitrary decimal precision). *For each $\chi \in \{\pi, e, \varphi\}$ and each $M \geq 1$, there exists $K_\chi(M) < \infty$. The generated section determines a unique decimal prefix of M positions, and the bracket recognizes and certifies that prefix. If $M' < M$, the prefix of order M truncates to the one of order M' , and the corresponding cells are nested.*

Proof. The three characters are irrational: φ is irrational by its quadratic polynomial with nonsquare discriminant; the irrationality of e and π is taken from their classical characterization theorems, subsequent to the finite construction. Thus none belongs to a rational decimal or ternary boundary. For a fixed M , the distance from the character to the finite set of relevant decimal boundaries is positive. The rational brackets contract their diameter until they lie in a unique decimal cell. [theorem 15.26](#) couples that bracket to a TPK cylinder and a unique extension. The lexicographic minimum exists by well-ordering. Compatibility of the section proves nesting as M increases. $\square$

Corollary 15.32 (Real uniqueness). *The nested cells have diameters $10^{-M} \rightarrow 0$. Their intersection is a singleton, identified by the preceding characterizations with π , e or φ , respectively.*

15.3.7 Equivalence of 2 finite realizations of the uniform law

The witnesses of 1,000 and 10,000 positions do not define the characters or limit the prolongation. They are executions of the same uniform recurrence at two different depths.

Table 15.2: Exact dimensions of certified runs, by channel.

Counted object	1,000 positions	10,000 positions
Partitions exhaustive of the fiber ambient	396	3501
Trits transportados	2376	21006
Pairs of equal phase $(K, K + 9)$	387	3492
Complete nonadic turns	43	388
Elements examined by partition	729	729
Compatible extensions preserved by partition	1	1

Para 10,000 posiciones,

$$B = 3501 = 389 \cdot 9, \quad 6B = 21006, \quad 3^{6B} > 10^{10000}.$$

The first 396 blocks of the long run coincide with the run of 1,000 positions. In each of the 3492 pairs of equal phase, the phase coordinate is preserved and the prefix, depth, cylinder, and memory are extended. The certificate also records any repetitions of local projections without conflating them with the return of the complete state.

15.3.8 Separation between generation and verification

Definition 15.33 (Structural generator). The generator receives the finite catalog, L_0, L_1 , the initial state, the connection laws and the internal relation Ext. At each level, it:

1. enumerates the 729 elements of the ambient fiber;
2. applies Ext and retains the unique compatible extension;
3. updates phase, quotient, carry, memory, and terminal;
4. writes the ternary block, the selection record and the genealogical record.

Definition 15.34 (Independent post hoc verification). The independent subsequent check receives the closed outputs of the generator and the three families of rational brackets; it computes $j_{\min}$ and $j_{\max}$,

certifies their equality with the extension already produced, reconstructs the partitions, checks the truncation squares, the pairs $(K, K + 9)$, the partition sums

$$\text{killed}_{\text{izq}} + 1 + \text{killed}_{\text{der}} = 729,$$

the integrity of the outputs and, finally, the agreement with reference expansions.

Theorem 15.35 (Isolation of the target data). *Reference decimal expansions and metrological data are not involved in selecting a compatible extension. The comparison is performed after the three prefixes have been written.*

Proof. The mathematical input to each generative decision consists of the finite catalog, L_0, L_1 , the inherited state and the internal relation Ext. The rational endpoints, powers of 3, floors and ceilings belong to the subsequent check. The generator finishes before that check opens the brackets or runs the decimal controls. The direction of the data flow is

$$\text{structure} \longrightarrow \text{prefixes and ledgers} \longrightarrow \text{independent subsequent check};$$

there is no arrow in the opposite direction. □

15.3.9 Differentiated obstructions to generation and its certification at arbitrary precision

Test criterion 15.36 (Refutation of generation to arbitrary precision). Failures of orbit, extension, truncation or return refute the generation; failures of brackets, order or decimal control refute the corresponding analytic certificate; and a discrepancy between the two refutes the joint identification of the published output. With this separation of types, the complete assertion is refuted if:

1. any of the indices n, K, L_K, M are identified without a linking theorem;
2. a bracket ceases to contain its character or does not contract its diameter;
3. a stage satisfies $j_{\min} \neq j_{\max}$ after the order declared to be minimal;
4. an extension does not truncate to the cylinder of the previous level;
5. the phase return resets the prefix, carry, terminal or memory;
6. the terminal term in (15.31) is omitted without proving that it vanishes;
7. the first 396 blocks of the long execution differ from the short execution;
8. a decimal or metrological control is read before the output is closed;
9. a printed digit differs from the certified mathematical section;
10. the proof depends on the requested precision being 1, 000 or 10, 000, rather than an arbitrary M .

Dodecaphasic closure and two internal derivations of α

16.1 Initial generation of the signed dodecaphase state and reconstruction of the seal K

Construction of the integer seal does not begin with α , its inverse or a metrological table. Its domain is the terminal holonomic state of the Topological Phase Kernel obtained after the chain already constructed

$$\boxed{\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow \Omega_{\text{seed}} \xrightarrow{T_{\min}} \mathcal{U}_6^{\text{cat}} \xrightarrow{\rho_3} W_{243} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9 \longrightarrow x_{\text{term}}.} \quad (16.1)$$

The foregoing arrows were defined with their respective domains: pairs of seeds, signatures, emissions, ternary words, boundaries, compatible histories, and nonadic holonomy. This unit does not reselect any of those data. It constructs the integral charts of the same terminal state and proves their equivalence.

16.1.1 Oriented traxis over windows of length 108

Let

$$\Theta_{108} = \mathbb{Z}/108\mathbb{Z}$$

the observable calendar. Its twelve nonadic windows are

$$E_m = \{9m + 1, \dots, 9m + 9\}, \quad 0 \leq m < 12.$$

The partition is disjoint and exhaustive:

$$\{1, \dots, 108\} = \bigsqcup_{m=0}^{11} E_m.$$

The set of address codes that publish an event is fixed as

$$D_{\text{evt}} = \{2, 4, 6, 8\}.$$

The orientation of the twelve windows is

$$\Sigma_{12} = (+, +, +, -, -, -, +, +, +, -, -, -). \quad (16.2)$$

If d_t is the direction encoded at step t , the signed event of window m is

$$e_m = \Sigma_{12}(m) \# \{t \in E_m : d_t \in D_{\text{evt}}\}. \quad (16.3)$$

The formula distinguishes the nonnegative number of events from the sign of the orientation. It does not replace a cardinal direction with a sign: both coordinates remain present in the TPK state.

Definition 16.1 (Enhanced dodecaphonic registration). The record transported by the trace is

$$\mathcal{R}_{12}(x) = ((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m))_{m=0}^{11},$$

where τ_m is the TRIT orientation, σ_m the boundary signature, κ_m the carry, and Λ_m the genealogical register segment that preserves the route. There is a chain of projections

$$\mathcal{X}_{\text{TPK}}^{\text{enr}} \longrightarrow \mathcal{R}_{12} \longrightarrow \Theta_{108},$$

but neither of the two arrows is an identification.

In particular, Θ_{108} preserves only the position in the calendar. The record $\mathcal{R}_{12}$ additionally preserves orientation, signature, carry, and provenance. The complete state further preserves cylinder, boundary, quotient, terminal, and vertical memory.

16.1.2 Integral chart of the dodecaphasic opposition

For $0 \leq i < 6$, the opposite positions determine

$$p_i = e_i + e_{i+6}, \quad a_i = e_i - e_{i+6}. \quad (16.4)$$

The central charge and the five relative margins are

$$s_C = \sum_{i=0}^5 p_i, \quad M_i = p_i - p_5 \quad (0 \leq i < 5). \quad (16.5)$$

Definition 16.2 (Integral chart $1 + 5 + 6$). It is defined

$$\mathcal{L}_{12}(e_0, \dots, e_{11}) = (s_C, M_0, \dots, M_4, a_0, \dots, a_5). \quad (16.6)$$

The coordinate partition is

$$1 + 5 + 6 = 12.$$

Theorem 16.3 (Exact inverse of the integral chart). *A tuple*

$$\ell = (s_C, M_0, \dots, M_4, a_0, \dots, a_5) \in \mathbb{Z}^{12}$$

belongs to the image of $\mathcal{L}_{12}$ if and only if

$$s_C - \sum_{i=0}^4 M_i \equiv 0 \pmod{6}, \quad (16.7)$$

$$p_i \equiv a_i \pmod{2} \quad (0 \leq i < 6), \quad (16.8)$$

where

$$p_5 = \frac{s_C - \sum_{i=0}^4 M_i}{6}, \quad (16.9)$$

$$p_i = p_5 + M_i \quad (0 \leq i < 5). \quad (16.10)$$

In that case, the only entire preimage is

$$e_i = \frac{p_i + a_i}{2}, \quad e_{i+6} = \frac{p_i - a_i}{2}. \quad (16.11)$$

Proof. Summing $M_i = p_i - p_5$ for $0 \leq i < 5$, we obtain

$$\sum_{i=0}^4 M_i = \sum_{i=0}^4 p_i - 5p_5 = s_C - 6p_5,$$

which forces (16.9) and the congruence modulo six. Once the six p_i have been reconstructed, the system

$$p_i = e_i + e_{i+6}, \quad a_i = e_i - e_{i+6}$$

has the unique solution (16.11). This solution is integral exactly when p_i and a_i have the same parity. The two families of congruences are therefore necessary and sufficient. $\square$

This theorem precedes the numerical closure of α . It explains how the oriented trace produces a reversible integral chart, and why preserving the reduced calendar is not enough.

16.1.3 Selection of the common terminal state

The three compatible sections reach terminal depth with catalogs of cardinalities

$$|\mathcal{T}_\pi| = 196, \quad |\mathcal{T}_e| = 197, \quad |\mathcal{T}_\varphi| = 196.$$

The product contains

$$196 \cdot 197 \cdot 196 = 7\,567\,952 \quad (16.12)$$

ordered triples.

The column margin of the terminal reader is

$$q_\partial = (1, 2, 2, 1, 2, 0) \in \mathbb{F}_3^6,$$

and the row orientation of the combined channel is

$$\sigma = (1, 1, -1) = (1, 1, 2) \in \mathbb{F}_3^3.$$

The first datum acts on columns; the second acts on rows. Neither is deducible from the other.

The local signatures of Gram matrix, norm, weight, and distance reduce (16.12) to 331 triples. The margin q_∂ leaves two indices:

$$(120, 197, 157), \quad (129, 197, 148).$$

Their matrices are

$$B_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \quad B_1 = \begin{pmatrix} 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{pmatrix}.$$

The row sums are

$$(0, 2, 0), \quad (2, 2, 1).$$

Since

$$\langle \sigma \rangle = \{(0, 0, 0), (1, 1, 2), (2, 2, 1)\},$$

only B_1 satisfies the axial condition.

Theorem 16.4 (Uniqueness of the Visible Terminal Publication). *The joint selection determines*

$$B_{\text{term}} = \begin{pmatrix} 021020 \\ 001220 \\ 100210 \end{pmatrix}, \quad (16.13)$$

corresponding to the triple $(129, 197, 148)$. No decimal evaluation of α enters the census or the two selection conditions.

Proof. The census and margin leave exactly B_0, B_1 . The charge of B_0 does not belong to $\langle \sigma \rangle$, whereas that of B_1 is 2σ . The second candidate is therefore the only one that simultaneously satisfies both typed requirements. $\square$

16.1.4 Archimedean and exceptional publications of the common state: inequivalent codomains and shared genealogy

Let

$$x_{\text{term}} \in \mathcal{X}_{\text{TPK}}^{\text{term}, \text{enr}} \subseteq \mathcal{X}_{\text{TPK}}^{[108]}.$$

The visible reading is

$$\pi_{\text{term}}(x_{\text{term}}) = B_{\text{term}}.$$

The signed chart preserves another coordinate of the same state.

Definition 16.5 (Signed terminal subspace). Let $\mathcal{H}_{12}$ be the integral transformation constructed in the next section. Define

$$\mathcal{X}_{\text{TPK}}^{[108], \text{sgn}} := \{(x_{\text{core}}, K, U) : U = \mathcal{H}_{12}K\} \quad (16.14)$$

and the publication module

$$\mathcal{U}_{12}^{\text{sgn}} := \text{im}(\mathcal{H}_{12} : \mathbb{Z}^{12} \rightarrow \mathbb{Z}^{12}). \quad (16.15)$$

The signed reading is

$$R_{\text{sgn}} : \mathcal{X}_{\text{TPK}}^{[108], \text{sgn}} \longrightarrow \mathcal{U}_{12}^{\text{sgn}}, \quad R_{\text{sgn}}(x_{\text{core}}, K, U) = U. \quad (16.16)$$

The correct joint writing is

$$x_{\text{term}} \xrightarrow{(\pi_{\text{term}}, R_{\text{sgn}})} (B_{\text{term}}, U_{12}^{\text{sgn}}). \quad (16.17)$$

No arrow $B_{\text{term}} \rightarrow U_{12}^{\text{sgn}}$ is interposed: the visible matrix has forgotten precisely the orientation, opposition, quotient, carry and genealogical-record coordinates used by the signed chart. The source of both expressions is the integral holonomic state produced by (16.1).

The evaluation of the signed record gives

$$U = (2378, 1406, 2479, -452, 998, -551, -668, -204, -371, -322, -28, -997). \quad (16.18)$$

16.1.5 Naturality with respect to depth

For $N' \geq N \geq 108$, the truncations of the signed system are

$$\tau_{N', N}^{\text{sgn}}(x_{\text{core}}^{[N']}, K, U) = (\tau_{N', N} x_{\text{core}}^{[N']}, K, U). \quad (16.19)$$

Theorem 16.6 (Square of naturality of the signed reading). *We have*

$$R_{\text{sgn}, N} \circ \tau_{N', N}^{\text{sgn}} = \text{id}_{\mathcal{U}_{12}^{\text{sgn}}} \circ R_{\text{sgn}, N'}. \quad (16.20)$$

Proof. Both sides, applied to $(x_{\text{core}}^{[N']}, K, U)$, literally return U . The truncation modifies the subsequent history but preserves the already fixed terminal coordinate. Commutativity is an equality of maps, not an analogy between diagrams. $\square$

16.1.6 Orbits of step 3 and transform of Hadamard

The cyclic order of the seal is distributed over the three orbits.

$$K^{(1)} = (K_1, K_4, K_7, K_{10}), \quad (16.21)$$

$$K^{(2)} = (K_2, K_5, K_8, K_{11}), \quad (16.22)$$

$$K^{(3)} = (K_3, K_6, K_9, K_{12}). \quad (16.23)$$

Let

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}. \quad (16.24)$$

If P_{orb} reorders the twelve coordinates according to (16.21)–(16.23), the global transformation is

$$\mathcal{H}_{12} = P_{\text{orb}}^{-1}(H_4 \oplus H_4 \oplus H_4)P_{\text{orb}}. \quad (16.25)$$

The three blocks of (16.18), read by orbits, are

$$U^{(1)} = (2378, -452, -668, -322),$$

$$U^{(2)} = (1406, 998, -204, -28),$$

$$U^{(3)} = (2479, -551, -371, -997).$$

Lemma 16.7 (Algebraic Hadamard quarter-turn). *The matrix H_4 satisfies*

$$H_4^2 = 4I_4, \quad H_4^{-1} = \frac{1}{4}H_4, \quad \det H_4 = -16. \quad (16.26)$$

Proof. The rows of H_4 are orthogonal, and each has squared norm four. Since the matrix is symmetric, $H_4^T H_4 = H_4^2 = 4I_4$. The formula for the inverse and the determinant follow immediately. $\square$

Theorem 16.8 (Unique full reconstruction of the seal). *The inverse*

$$K^{(r)} = \frac{1}{4}H_4 U^{(r)} \quad (r = 1, 2, 3) \quad (16.27)$$

yields

$$K^{(1)} = (234, 729, 621, 794),$$

$$K^{(2)} = (543, 659, 058, 146),$$

$$K^{(3)} = (140, 824, 914, 601).$$

By undoing the orbital order one obtains

$$\boxed{K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601)}. \quad (16.28)$$

It is the unique rational preimage of U , and it belongs to $\mathbb{Z}^{12}$.

Proof. By (16.26), the rational preimage is unique. The explicit multiplications are

$$H_4 U^{(1)} = (936, 2916, 2484, 3176),$$

$$H_4 U^{(2)} = (2172, 2636, 232, 584),$$

$$H_4 U^{(3)} = (560, 3296, 3656, 2404).$$

All coordinates are divisible by four. Dividing and reinterleaving yields (16.28). $\square$

The factor $1/4$ is not chosen to fit an output: it is the inverse forced by $H_4^2 = 4I_4$. Nor is it the square root of -1 . The internal complex structure $A_W^2 = -I$ will appear after constructing the Paley matrix; these are two facts of different types.

16.1.7 Integral structure and Smith normal form

Theorem 16.9 (Integral image of H_4). *The Smith normal form is*

$$\text{SNF}(H_4) = \text{diag}(1, 2, 2, 4). \quad (16.29)$$

Consequently,

$$\text{coker } \mathcal{H}_{12} \cong (\mathbb{Z}/2\mathbb{Z})^6 \oplus (\mathbb{Z}/4\mathbb{Z})^3, \quad (16.30)$$

and the image of $\mathcal{H}_{12}$ has index

$$16^3 = 4096$$

in $\mathbb{Z}^{12}$. For $u \in \mathbb{Z}^4$,

$$u \in H_4\mathbb{Z}^4 \iff H_4u \equiv 0 \pmod{4}. \quad (16.31)$$

Proof. The greatest common divisor of the entries is one. The minors of order two are even, and one has absolute value two; those of order three are multiples of four, and one has absolute value four; the determinant has modulus sixteen. The determinantal divisors are 1, 2, 4, 16, yielding the invariant factors 1, 2, 2, 4. The triple direct sum repeats the factors and multiplies the indices. Finally, $H_4^{-1} = H_4/4$, so the preimage is integral exactly under the congruence (16.31). $\square$

16.1.8 Differential coordinate system induced by step 3 and 4 operators

With cyclic indices modulo twelve, we define

$$(D_3K)_m = K_m - K_{m+3}, \quad (D_4K)_m = K_m - K_{m+4}, \quad Q(K) = \sum_{m=1}^{12} K_m. \quad (16.32)$$

For (16.28),

$$\begin{aligned} D_3K &= (-495, -116, -684, 108, 601, -90, \\ &\quad -173, -88, 313, 560, -397, 461), \\ D_4K &= (-425, -281, -481, 671, -255, 30, \\ &\quad 475, -543, 680, 251, 6, -128), \\ Q(K) &= 6263. \end{aligned}$$

Theorem 16.10 (Completeness of the transversal extractor). *The operator*

$$\mathcal{E}_{\text{dod}} = (D_3, D_4, Q) : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{25}$$

has rank twelve and Smith normal form

$$\text{SNF}(\mathcal{E}_{\text{dod}}) = \text{diag}(\underbrace{1, \dots, 1}_{11}, 12). \quad (16.33)$$

In particular, $(D_3K, D_4K, Q(K))$ determina a single K .

Proof. If $D_3v = D_4v = 0$, then v is constant along steps three and four. Since $\gcd(3, 4) = 1$, those steps generate all of $\mathbb{Z}/12\mathbb{Z}$, so that

$$\ker D_3 \cap \ker D_4 = \langle \mathbf{1} \rangle.$$

The total charge does not vanish on that line, since $Q(\mathbf{1}) = 12$. The rank is twelve. The rows of (D_3, D_4) are oriented incidences of a connected Cayley graph; a spanning tree supplies eleven unit invariant factors.

Every maximal minor must contain the row Q , and the unimodular operations reduce its final contribution to twelve. There is a maximal minor of modulus twelve; therefore, the last factor is exactly twelve. $\square$

The reconstruction of U from these coordinates does not need to multiply again by H_4 . For $r = 1, 2, 3$, let

$$B_r = \sum_{j=0}^3 (D_4 K)_{r+3j}. \quad (16.34)$$

Then

$$(B_1, B_2, B_3) = (972, -1073, 101).$$

Orbital sums are

$$A_1 = \frac{Q + 2B_1 + B_2}{3} = 2378, \quad (16.35)$$

$$A_2 = A_1 - B_1 = 1406, \quad (16.36)$$

$$A_3 = A_2 - B_2 = 2479. \quad (16.37)$$

If $d_{r,j} = (D_3 K)_{r+3j}$, the signed block of orbit r is

$$(A_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}), \quad (16.38)$$

that reproduces the three blocks $U^{(r)}$.

16.1.9 Threshold Tetrad of the Seal

The local decimal completion satisfies

$$1000 = 729 + 271.$$

The support of the seal coordinates that reach the threshold 729 is

$$B_K := \{m : K_m \geq 729\} = \{4, 6, 9, 10\}. \quad (16.39)$$

This tetrad is obtained after constructing K . In this unit it is not yet called a Witt star: that object will be defined only after constructing the ternary code, its hexads, and the Steiner system.

16.1.10 Independence of the genealogical construction with respect to external metrological or physical data

Theorem 16.11 (Isolation of U and K). *The composition*

$$\Omega_{\text{seed}} \longrightarrow x_{\text{term}} \xrightarrow{R_{\text{sgn}}} U \xrightarrow{\mathcal{H}_{12}^{-1}} K \quad (16.40)$$

does not receive α , α^{-1} , CODATA or a target decimal string as input. The seal K is determined before defining the recurrence that will produce α .

Proof. The first part of the composition uses only seeds, emissions, ternary readers, operators L_0, L_1 , survival relations, phase transport, and the charts of the TPK state. The reading R_{sgn} is a projection of the terminal register, and the final arrow is the linear transformation $\mathcal{H}_{12}^{-1} = \mathcal{H}_{12}/4$ on its integral image. None of the formulas contains a physical variable or a metrological table. The base-thousand recurrence is introduced only in the following unit, after K has been fixed. $\square$

16.1.11 Obstructions of the dodecaphase record and the seal K : naturality, Hadamard integrity, and metrological isolation

Test criterion 16.12 (Refutation of the register and the seal). The construction is refuted if any of the following facts is established:

1. the windows E_m do not partition the one hundred eight steps;
2. the chart $\mathcal{L}_{12}$ fails its inverse or its congruences;
3. the terminal census does not produce 331 triples, two matrices and a unique axial charge;
4. an attempt is made to obtain U from B_{term} after forgetting the signed coordinates;
5. the naturality square (16.20) fails;
6. $H_4^2 \neq 4I_4$, some preimage is not integral or the reconstructed vector differs from (16.28);
7. the Smith normal form differs from (16.29);
8. extractor (D_3, D_4, Q) loses rank or fails to reproduce U ;
9. B_K is introduced before constructing K , or is assigned a Witt structure before defining W_{12} ;
10. a digit of α or a metrological reference intervenes in any of the arrows of (16.40).

16.2 Initial construction of the boundary 662/663 through the dodecaphase carry cocycle

16.2.1 Arithmetic closure domain

The construction receives four dodecaphasic coordinates obtained in the preceding units:

$$\begin{aligned} P &= (141, 592, 653, 589, 793, 238, 462, 643, 383, 279, 502, 884), \\ E &= (718, 281, 828, 459, 045, 235, 360, 287, 471, 352, 662, 497), \\ \Phi &= (618, 033, 988, 749, 894, 848, 204, 586, 834, 365, 638, 117), \\ K &= (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \end{aligned}$$

The first three vectors are the first twelve triads of the fractional Archimedean readings of π , e and φ . The fourth is the dodecaphasic coordinate of the terminal holonomic state of the Topological Phase Kernel, constructed before opening the carry operation.

We fix the base.

$$B := 10^3 = 1000$$

and the concatenation

$$\iota(v_1, \dots, v_{12}) := \sum_{m=1}^{12} v_m B^{12-m}. \quad (16.41)$$

In particular,

$$\begin{aligned} \iota(P) &= 141592653589793238462643383279502884, \\ \iota(E) &= 718281828459045235360287471352662497, \\ \iota(\Phi) &= 618033988749894848204586834365638117, \\ \iota(K) &= 234543140729659824621058914794146601. \end{aligned}$$

The operation studied in this unit is the map

$$\mathfrak{C}_{12} : (P, E, \Phi, K, c_{13}) \mapsto (A, C), \quad (16.42)$$

where $A = (A_1, \dots, A_{12})$ is the output word and

$$C = (c_1, \dots, c_{13})$$

is the complete system of thirteen Euclidean quotients: twelve internal quotients and one right boundary condition.

16.2.2 Left-to-right oriented Euclidean division

For $m = 12, 11, \dots, 1$, define

$$\begin{aligned} s_m &:= P_m + E_m - \Phi_m - K_m + c_{m+1}, \\ A_m &:= s_m \bmod B, \quad 0 \leq A_m < B, \\ c_m &:= \frac{s_m - A_m}{B} = \left\lfloor \frac{s_m}{B} \right\rfloor. \end{aligned} \tag{16.43}$$

The second equality for c_m is also valid when $s_m < 0$, because Euclidean division with remainder in $\{0, \dots, B-1\}$ is used. Therefore,

$$-431 = 569 - 1000, \quad -717 = 283 - 1000, \quad -1200 = 800 - 2 \cdot 1000,$$

and negative loans do not introduce ambiguity.

Theorem 16.13 (Existence and Uniqueness of the Oriented Carry). *Given P, E, Φ, K and the terminal carry c_{13} , the recurrence (16.43) uniquely determines*

$$(A_1, \dots, A_{12}; c_1, \dots, c_{12}).$$

Proof. Row $m = 12$ is determined by the Euclidean division of

$$P_{12} + E_{12} - \Phi_{12} - K_{12} + c_{13}$$

by B . Once c_{12} is known, row $m = 11$ is determined by the same property. Descending induction reaches $m = 1$. At each step, uniqueness of the Euclidean quotient and remainder prevents two distinct outputs. $\square$

16.2.3 Finite restriction of the signed dodecaphase state and reversible closure of the cycle

The finite window is defined by the terminal condition

$$c_{13}^{\text{fin}} = 0.$$

The complete evaluation is as follows. The table is printed in ascending order, although it must be read causally from the last row to the first.

Table 16.1: The twelve Euclidean divisions and thirteen carries of the finite dodecaphasic window.

m	P_m	E_m	Φ_m	K_m	c_{m+1}	s_m	A_m	c_m
1	141	718	618	234	0	7	007	0
2	592	281	033	543	0	297	297	0
3	653	828	988	140	-1	352	352	0
4	589	459	749	729	-1	-431	569	-1
5	793	045	894	659	-2	-717	283	-1
6	238	235	848	824	-1	-1200	800	-2
7	462	360	204	621	0	-3	997	-1
8	643	287	586	058	-1	285	285	0
9	383	471	834	914	-1	-895	105	-1
10	279	352	365	794	0	-528	472	-1

The complete word of thirteen carries is

The dodecaphasic output is

We define the finite rational

Therefore,

16.2.4 Posterior metrological recognition: CODATA 2018

$$\alpha_{12}^{\text{fin}} = 10^{-36} \left[\frac{10^{36}}{137.035999084} \right]. \quad (16.45)$$

Consequently,

and exactly reproduces the central chain published by CODATA 2018 [TMNT21]:

The word *exactly* refers to the published central chain, not to a purported experimental measurement of thirty-six digits. The subsequent digits in (16.46) are consequences of the HMT rational already fixed. CODATA does not select the state, does not determine (K), does not decide the direction of the carry, and does not intervene in the dodecaphase table.

16.2.5 Complete affine identity

Each row of the table satisfies exactly

Introducing

the twelve local identities come together in

$$\boxed{P + E - \Phi - K + C^+ - BC^- = A.} \quad (16.48)$$

The term

$$\partial_B C := C^+ - BC^-$$

is the carry cocycle of the oriented chain. Consequently, $P + E - \Phi - K = A$ is not a coordinatewise equality in $\mathbb{Z}^{12}$: it becomes correct only upon incorporating $\partial_B C$.

Theorem 16.14 (Telescoping of the Thirteen Carries). *For any terminal condition c_{13} , the recurrence satisfies*

$$\boxed{\iota(P) + \iota(E) - \iota(\Phi) - \iota(K) - \iota(A) = B^{12}c_1 - c_{13}.} \quad (16.49)$$

In the finite window, $c_1 = c_{13} = 0$, and therefore

$$\boxed{\iota(P) + \iota(E) - \iota(\Phi) - \iota(K) = \iota(A^{\text{fin}}).} \quad (16.50)$$

Proof. We multiply (16.47) by B^{12-m} and sum over $m = 1, \dots, 12$. The contribution of the carries is

$$\sum_{m=1}^{12} (Bc_m - c_{m+1})B^{12-m}.$$

The internal components cancel:

$$\begin{aligned} \sum_{m=1}^{12} (Bc_m - c_{m+1})B^{12-m} &= \sum_{m=1}^{12} c_m B^{13-m} - \sum_{m=1}^{12} c_{m+1} B^{12-m} \\ &= B^{12}c_1 - c_{13}. \end{aligned}$$

This proves (16.49). The specialization $c_1 = c_{13} = 0$ gives (16.50). $\square$

16.2.6 Exact inversion of the seal K

The local recurrence is reversible. Solving for K_m in (16.47),

$$\boxed{K_m = P_m + E_m - \Phi_m + c_{m+1} - A_m - Bc_m.} \quad (16.51)$$

By concatenation,

$$\boxed{\iota(K) = \iota(P) + \iota(E) - \iota(\Phi) - \iota(A) - B^{12}c_1 + c_{13}.} \quad (16.52)$$

On the finite boundary,

$$\iota(K) = \iota(P) + \iota(E) - \iota(\Phi) - \iota(A^{\text{fin}}).$$

Algebraic reversibility does not invert the causal order. The effective order is

$$x_{\text{term}} \longrightarrow K \longrightarrow (P, E, \Phi, K, c_{13}) \longrightarrow (A, C).$$

Formula (16.51) is a commutativity check: it proves that the output and the carries recover the seal that was already fixed. It is not a rule for retrospectively selecting K from a desired value of α .

16.2.7 Coinductive prolongation of the dodecaphasic seal and inverse carry recurrence in $C = \{-2, -1, 0, 1, 2\}$

Let

$$(P_k)_{k \geq 1}, \quad (E_k)_{k \geq 1}, \quad (\Phi_k)_{k \geq 1}$$

the infinite sequences of triads produced by the three coinductive branches, and let

$$\kappa(k) := 1 + ((k - 1) \bmod 12).$$

The prolongation of the stamp is periodic:

$$K_k^{\text{per}} := K_{\kappa(k)}.$$

For a finite depth N and a terminal condition $t = c_{N+1}$, the recurrence is

$$\begin{aligned} s_k^{(N,t)} &= P_k + E_k - \Phi_k - K_{\kappa(k)} + c_{k+1}^{(N,t)}, \\ A_k^{(N,t)} &= s_k^{(N,t)} \bmod 1000, \\ c_k^{(N,t)} &= \left\lfloor \frac{s_k^{(N,t)}}{1000} \right\rfloor, \quad k = N, \dots, 1. \end{aligned} \tag{16.53}$$

The set of terminal carries

$$C := \{-2, -1, 0, 1, 2\}$$

is invariant for the certified inputs: for each triad and each $c_{k+1} \in C$, the quotient c_k again belongs to C . No favorable terminal condition is chosen. All five conditions are propagated, and only the prefix common to all of them is published.

The execution of certified depth employs

quantity	value
ternary blocks per channel	3501
trits per channel	$3501 \cdot 6 = 21006$
guard decimal digits	10020
guard triads	3340
propagated terminal conditions	5
common triads	3339
common digits	10017
published digits	10000

The five terminal branches coalesce before the published prefix. In particular, no terminal condition extracted from a target value of α intervenes in the output.

Theorem 16.15 (Stabilized Output Compatibility). *Let $A^{[N]}$ and $A^{[N']}$, with $N' > N$, be two runs whose five terminal conditions have coalesced before position N . Then the truncation of $A^{[N']}$ to its first N blocks coincides with $A^{[N]}$. The stabilized prefixes define a unique section in the inverse limit of finite words.*

Proof. The recurrence is deterministic from right to left once the carry entering a position is known. Coalescence asserts that all remote conditions produce the same carry before the boundary of the prefix. From that point, the same triads P_k, E_k, Φ_k and the same periodic seal produce, by Euclidean uniqueness, the same remainders and quotients at all preceding positions. This is exactly commutativity with truncation. $\square$

16.2.8 Boundary originating from the remote tail

The prolongation determines on the visible boundary.

$$c_{13}^{\infty} = -1.$$

The twelve internal carries remain the same as in the finite window:

$$C_{\leq 13}^{\infty} = (0, 0, 0, -1, -1, -2, -1, 0, -1, -1, 0, 0, -1).$$

Only the information entering from the thirteenth triad changes. The two relevant divisions are

$$s_{12}^{\infty} = 884 + 497 - 117 - 601 - 1 = 662, \\ A_{12}^{\infty} = 662, \quad c_{12}^{\infty} = 0,$$

and

$$s_{13}^{\infty} = 197 + 757 - 720 - 234 - 1 = -1, \\ A_{13}^{\infty} = 999, \quad c_{13}^{\infty} = -1.$$

Therefore, the beginning of the prolonged word is

$$A^{\infty} = (007, 297, 352, 569, 283, 800, 997, 285, 105, 472, 380, 662, 999, \dots).$$

The corresponding expansion begins

$$\alpha_{\infty} = 0.007297352569283800997285105472380662999564172539642 \dots \quad (16.54)$$

The difference between both boundaries is local and exact:

$$c_{13}^{\text{fin}} - c_{13}^{\infty} = 1, \quad A_{12}^{\text{fin}} - A_{12}^{\infty} = 1,$$

while

$$A_m^{\text{fin}} = A_m^{\infty}, \quad 1 \leq m \leq 11.$$

Applying the telescoping theorem to the first twelve blocks of the prolongation,

$$\iota(P) + \iota(E) - \iota(\Phi) - \iota(K) - \iota(A_{\leq 12}^{\infty}) = -c_{13}^{\infty} = 1.$$

Then

$$\iota(A^{\text{fin}}) = \iota(A_{\leq 12}^{\infty}) + 1. \quad (16.55)$$

The ending 663 does not contradict the ending 662: the former is the finite representative obtained by closing the window with $c_{13} = 0$; the latter retains the carry -1 from the remote tail.

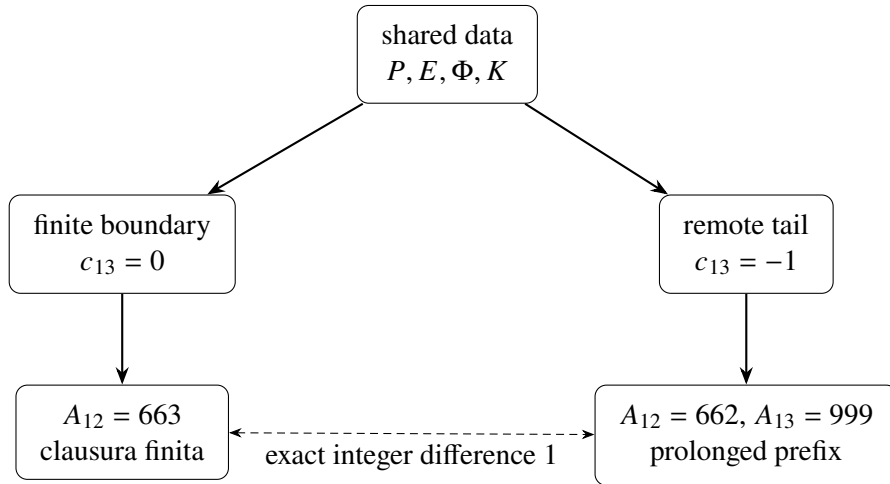


Figure 16.1: Boundary bifurcation of a single recurrence. Two distinct generators are not being compared: only the datum entering from the right changes.

16.2.9 Truncation and rounding to 36 digits

For $x \geq 0$, define

$$\text{Tr}_{36}(x) := 10^{-36} \lfloor 10^{36}x \rfloor$$

and

$$\text{Rd}_{36}(x) := 10^{-36} \left\lfloor 10^{36}x + \frac{1}{2} \right\rfloor.$$

Let

$$a := 7297352569283800997285105472380662.$$

The expansion (16.54) can be written as

$$\alpha_\infty = \frac{a + \rho}{10^{36}}, \quad \rho = 0.999564172539642709 \dots,$$

so that $0.999 < \rho < 1$.

Theorem 16.16 (Exact resolution of the boundary 662/663). *The coinductive prolongation satisfies*

$$\text{Tr}_{36}(\alpha_\infty) = 0.007297352569283800997285105472380662,$$

while its rounding to thirty-six digits satisfies

$$\text{Rd}_{36}(\alpha_\infty) = 0.007297352569283800997285105472380663 = \alpha_{12}^{\text{fin}}.$$

Proof. Since $0 \leq \rho < 1$,

$$\lfloor 10^{36}\alpha_\infty \rfloor = a.$$

This gives the truncation ending in 662. Since $\rho > 1/2$,

$$\left\lfloor 10^{36}\alpha_\infty + \frac{1}{2} \right\rfloor = a + 1,$$

which ends in 663. By (16.44), $(a + 1)10^{-36}$ is exactly α_{12}^{fin} . □

Moreover,

$$0 < \alpha_{12}^{\text{fin}} - \alpha_\infty = \frac{1 - \rho}{10^{36}} < 10^{-39}.$$

The difference is not an arithmetic error in the table: it quantifies the difference between two distinct boundary operators.

16.2.10 Causal isolation from α and metrological data

At depth N , the full domain of the operation is

$$\mathcal{D}_N = ((P_k)_{k=1}^N, (E_k)_{k=1}^N, (\Phi_k)_{k=1}^N, K, c_{N+1}).$$

The carry map

$$\mathfrak{C}_N : \mathcal{D}_N \longrightarrow \{0, \dots, 999\}^N \times C^N$$

is defined exclusively by (16.53). Its domain contains no target value of α , its inverse, a metrological table, a target decimal string or a terminal carry chosen by comparison.

Theorem 16.17 (No interference from the metrological target). *Let $\mathcal{Y}$ be any space of external data containing metrological values or comparison tables. The formal extension*

$$\tilde{\mathfrak{C}}_N : \mathcal{D}_N \times \mathcal{Y} \longrightarrow \{0, \dots, 999\}^N \times C^N, \quad \tilde{\mathfrak{C}}_N(d, y) := \mathfrak{C}_N(d),$$

factorizes by the projection

$$\text{pr}_{\mathcal{D}_N} : \mathcal{D}_N \times \mathcal{Y} \longrightarrow \mathcal{D}_N.$$

Therefore, for any $y, y' \in \mathcal{Y}$,

$$\tilde{\mathfrak{C}}_N(d, y) = \tilde{\mathfrak{C}}_N(d, y').$$

No intervention on a value of α or a metrological table alters K , the quotients, the remainders or the prefix produced.

Proof. The statement results from

$$\tilde{\mathfrak{C}}_N = \mathfrak{C}_N \circ \text{pr}_{\mathcal{D}_N}.$$

All operations of $\mathfrak{C}_N$ are additions, subtractions and Euclidean divisions of the elements of $\mathcal{D}_N$. No coordinate of $\mathcal{Y}$ appears in them. $\square$

The complete causal order is

enriched branches of $\pi, e, \varphi \longrightarrow (P, E, \Phi),$

$$x_{\text{term}} \longrightarrow K,$$

$$(P, E, \Phi, K, \text{boundary}) \longrightarrow (A, C) \longrightarrow \alpha_{\infty},$$

$$\alpha_{\infty} \longrightarrow \text{external comparison}.$$

The last arrow cannot be reversed to select any of the previous ones.

16.2.11 Obstructions to affine prolongation of α : telescoping, inversion of K , boundary 662/663, and isolation of target data

Test criterion 16.18 (Refutation of the affine closure and its prolongation). The block is refuted in its domain if any of the following facts occurs:

1. some row of the [table 16.1](#) fails to satisfy $s_m = A_m + 1000c_m$;
2. the published vector of thirteen carries does not reproduce the twelve rows;
3. the telescoping identity fails

$$\iota(P) + \iota(E) - \iota(\Phi) - \iota(K) - \iota(A) = 1000^{12}c_1 - c_{13};$$

4. the inversion (16.51) does not recover the twelve coordinates of K ;
5. remote propagation produces $c_{13} \neq -1$ for the certified prefix;
6. the five terminal conditions of $C = \{-2, -1, 0, 1, 2\}$ do not coalesce before the published digits;
7. the triad following 662 is not 999, in which case the identification of the rounding with 663 is no longer proved;
8. the generator opens a target expansion of α , its inverse, a metrological table, or a terminal condition chosen to force the coincidence.

Thus, two distinct boundary identities are thereby demonstrated:

$$\boxed{663 = \text{finite closure with } c_{13} = 0 = \text{rounding to 36 digits of } \alpha_{\infty},}$$

$$\boxed{662 = \text{truncation to 36 digits of } \alpha_{\infty} \quad \text{with} \quad c_{13} = -1, A_{13} = 999.}$$

The central metrological result is finally published in the same residence as the derivation:

[illegible]

The implication expresses a subsequent comparison with the central string published; it does not introduce that string into the constructor.

16.3 Graded nonadic prolongation of the kernel $E_{21/22}$: operator $T_{7;9}$, genealogy 120/45/11/495, and residue of order 9

This section develops two internal constructions relating to the coordinate α of the [tesis inaugural](#). Route A operates through reversible dodecaphasic closure of the signed state and expresses α_{12} . Route B, once the analytic kernel $E_{21/22}$ and its coefficients have been constructed internally, expresses in its analytic chart a simple unique positive root, denoted $\alpha_E := x_{21/22}$. Both outputs share a genealogy; their proved agreement belongs to the typed decimal codomain specified below. α_E is not identified with α_{12} by definition. Their domains, operations and certificates remain distinct:

$$X_{12}^{\text{sig}} \xrightarrow{\Gamma_{12}^{\alpha}} \alpha_{12}, \quad E_{21/22} : I_{\alpha} \longrightarrow \mathbb{R}, \quad E_{21/22}(x_{21/22}) = 0, \quad E'_{21/22}(x_{21/22}) \neq 0. \quad (16.57)$$

Integer closure expresses α_{12} through carry, whereas the analytic equation determines $\alpha_E = x_{21/22}$ through existence and uniqueness of the root in the certified interval. The currently proved agreement of both outputs resides exactly in the codomain of the morphism

$$\mathrm{Tr}_9(y) := 10^{-9} \lfloor 10^9 y \rfloor, \quad \mathrm{Tr}_9(x_{21/22}^{-1}) = 137.035999084 = \mathrm{Tr}_9(\alpha_{12}^{-1}). \quad (16.58)$$

The exact analytic identity corresponds to $E_{21/22}(\alpha_E) = 0$, whereas (16.58) is the exact equality of the two expressions after applying Tr_9 . The stronger promotion $\alpha_E = \alpha_{12}$ in $\mathbb{R}_{\text{HMT}}$ requires proving $E_{21/22}(\alpha_{12}) = 0$, or an injective morphism lifting the truncated equality; the nonzero residue of table 16.2 does not supply that identity. Neither route introduces the target value as an input; metrological identification belongs solely to subsequent conventional recognition.

16.3.1 Nucleo analytic P_6 and root unique of the equation $E_{21/22}$

The characterization of vacancies employs

$$\begin{aligned} P_6(x) &= 2\pi x - \frac{7}{4}x^2 + \frac{x^3}{2\pi} + \frac{x^4}{20} - \frac{2x^5}{21} - \frac{x^6}{46}, \\ P_6(x) &= \log_{10} \frac{10}{9}. \end{aligned} \tag{16.59}$$

The certified positive root of this equation satisfies

$$x_6^{-1} = 137.03599908400002702009 \dots \quad (16.60)$$

The polynomial P_6 provides the initial jet of the analytic characterization; the nonadic holonomy now prolongs its degrees 7, 8, 9.

16.3.2 APP-TRIT-TPK genealogy of the invariants 120, 45, 11 and 495

The phase semicorona of the TPK constructs

$$|\mathcal{F}_8^{\text{ph}}| = 120, \quad |L_{\text{ph}}| = 45. \quad (16.61)$$

APP provides the central block

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}, \quad \det B_c = 45, \quad \text{Spec}(B_c) = \{9, 5\}. \quad (16.62)$$

The dodecaphase transverse extractor provides

$$r_{\text{dod}} = \text{rank}(D_3, D_4) = 11, \quad (D_3 K)_1 = -495, \quad 495 = 45 \cdot 11 = \det(B_c) r_{\text{dod}}. \quad (16.63)$$

The coefficients of the prolonged jet therefore proceed from three prior invariants of APP–TRIT–TPK:

$$-\frac{1}{120} \mapsto \frac{1}{45} \mapsto \frac{2}{45 \cdot 11}. \quad (16.64)$$

16.3.3 Nilpotent resolvent $(I - N)^{-1}$ and exact construction of the tail $T_{7:9}$

On the basis e_7, e_8, e_9 , we define

$$Ne_7 = -\frac{8}{3}e_8, \quad Ne_8 = \frac{2}{11}e_9, \quad Ne_9 = 0, \quad v_7 = -\frac{1}{120}e_7. \quad (16.65)$$

Since $N^3 = 0$, its finite resolvent acts exactly by

$$(I - N)^{-1}v_7 = v_7 + Nv_7 + N^2v_7 = -\frac{e_7}{120} + \frac{e_8}{45} + \frac{2e_9}{495}. \quad (16.66)$$

The evaluation $e_n \mapsto x^n$ produces

$$T_{7:9}(x) = -\frac{x^7}{120} + \frac{x^8}{45} + \frac{2x^9}{495}. \quad (16.67)$$

16.3.4 Nonadic jet P_9 , prolonged root, and scale of certified residues

The nonadic jet of order 9 is defined by

$$P_9(x) = P_6(x) + T_{7:9}(x), \quad P_9(x) = \log_{10} \frac{10}{9}. \quad (16.68)$$

Its positive root satisfies

$$x_9^{-1} = 137.035999084000000000000000111926291900 \dots \quad (16.69)$$

Table 16.2: Certified descent of the residue of the analytical characterization

Nucleo evaluado in α_{12}	Residue
P_6	$9.0038886 \times 10^{-18}$
$P_6 - x^7/120$	$-1.7893115 \times 10^{-19}$
$P_6 - x^7/120 + x^8/45$	$-2.3708601 \times 10^{-22}$
P_9	$3.7297132 \times 10^{-27}$

16.3.5 Operator dodecaphasic alternative A_{dod} and proof of the not uniqueness of the selection

The control operator

$$A_{\text{dod}} = I - \frac{1}{4}(D_3^* D_3 + D_4^* D_4) \quad (16.70)$$

produces a different coefficient sequence. This control delimits the selective power of the resolvent the corresponding identity: a generic dodecaphasic operator does not by itself determine the extension $T_{7:9}$. It does not weaken the vacancy route, whose proof condition is existence and uniqueness of the root of the declared kernel. The control morphism $\Gamma_{E21}^{(9)}$ is reserved exclusively for canonical uniqueness of the nonadic extension 7:9; it neither certifies nor conditions the status of route B.

16.3.6 Scope of the nilpotent operator N on the nonadic jet P_9 and uniqueness control of the prolongation 7:9

The arithmetic of 120, 45, 11, 495, the nilpotent operator, the resolvent and the identity equation (16.67) are internally exact. The simple unique positive root of the kernel $E_{21/22}$ constitutes a second exact internal construction, α_E , for the declared domain and coefficients. Its identification with the dodecaphasic output is proved exactly after Tr_9 ; raising this to equality in $\mathbb{R}_{\text{HMT}}$ requires the identity or injective lifting stated at the beginning of the section. Independently, canonical uniqueness of the extension 7:9 is typed by the morphism

$$\Gamma_{E21}^{(9)} : X_{\text{APP-TRIT-TPK}} \longrightarrow \mathbb{Q}(\pi)[x]/(x^{10}), \quad (16.71)$$

whose action jointly determines the oriented seed, both transfers, the grading 7, 8, 9 and the jet quotient F^7/F^{10} . The falsifier therefore separates two statements: it would invalidate route B if monotonicity, the sign change or root isolation failed; it would invalidate canonical promotion of $T_{7:9}$ if another reader on the same domain produced the same grading without factoring through $\Gamma_{E21}^{(9)}$.

Algebraic and genealogical classification of the fundamental constants

The Euclidean residue–quotient extension of APP contains two local maps: the additive map normalizes collisions, and the multiplicative map produces partial products. In this section, the global multiplication of words is constructed using exclusively both maps. The correspondence with integer multiplication will be a consequence of the intertwiner, not the definition of the operation.

17.1 Euclidean coordinates of the local cell

Definition 17.1 (Additive and multiplicative applications on the same support). Let $D_9 = \{1, \dots, 9\}$. For $a, b \in D_9$ they are defined by Euclidean division with remainder in D_9 :

$$a + b = 9q^+(a, b) + R^+(a, b), \quad (17.1)$$

$$ab = 9q^\times(a, b) + R^\times(a, b). \quad (17.2)$$

In particular, the zero remainder is represented by 9 and the quotient is reduced by one unit. The quadruple

$$C_9^{\text{rq}}(a, b) = (R^+(a, b), q^+(a, b); R^\times(a, b), q^\times(a, b))$$

is the local arithmetic extension by residue and quotient.

Aggregation by the three classes modulo three of the visible table $R^\times$ defines a matrix distinct from the local maps:

$$(M_\Pi)_{rs} = \frac{1}{9} \sum_{\substack{a \equiv r \pmod{3} \\ b \equiv s \pmod{3}}} R^\times(a, b), \quad r, s \in \{1, 2, 0\},$$

and, in that order of classes,

$$M_\Pi = \begin{pmatrix} 4 & 5 & 6 \\ 5 & 4 & 6 \\ 6 & 6 & 9 \end{pmatrix}.$$

Thus, q^+ and $q^\times$ are quotient coordinates of a cell, whereas M_Π is an aggregate operator on residual classes.

The 81 cells satisfy exactly

	$\sum R$	$\sum q$	$\sum(9q + R)$
+	405	45	810
×	459	174	2025

and therefore

$$2025 - 810 = (459 - 405) + 9(174 - 45) = 54 + 9 \cdot 129 = 1215.$$

The quotient $q^\times$ is necessary for reconstruction. For example,

$$2 \cdot 5 = 9 \cdot 1 + 1.$$

If only $R^\times$ is preserved, the product 10 is conflated with the visible value 1.

17.1.1 Cocycles of associativity and distributivity

The coherence of the transducer can already be decided in the cell. For every $a, b, c \in D_9$ the three identities are verified

$$q^+(a, b) + q^+(R^+(a, b), c) = q^+(b, c) + q^+(a, R^+(b, c)), \quad (17.3)$$

$$q^\times(R^\times(a, b), c) + cq^\times(a, b) = q^\times(a, R^\times(b, c)) + aq^\times(b, c), \quad (17.4)$$

$$q^\times(a, R^+(b, c)) + aq^+(b, c) = q^+(R^\times(a, b), R^\times(a, c)) + q^\times(a, b) + q^\times(a, c). \quad (17.5)$$

Theorem 17.2 (Local coherence of the Euclidean extension). *The identities (17.3)–(17.5), together with the corresponding equalities of residues, express respectively additive associativity, multiplicative associativity, and distributivity in the residue–quotient extension. The three families have zero discrepancies in the $9^3 = 729$ triples.*

Proof. Expanding $(a + b) + c$ through (17.1), the coefficient of nine is the left-hand side of (17.3); expanding $a + (b + c)$ yields the right-hand side. Uniqueness of division with remainder in D_9 equates both residues and quotients. The same argument applied to $(ab)c = a(bc)$ gives (17.4). Finally, the two expansions of $a(b + c) = ab + ac$ produce (17.5). Direct enumeration of the 729 cases of each identity confirms the algebraic calculation. $\square$

17.2 Bijectivity of numbering

The symbol 9 is not identified with zero. In C_9^{eq} , 9 represents a complete crossing of the nonadic boundary. Replacing it with the digits $0, \dots, 8$ would eliminate the Euclidean quotient that intervenes in the intertwining.

Definition 17.3 (Canonical words). Let

$$\mathcal{W}_9^+ = \bigsqcup_{\ell \geq 1} D_9^\ell, \quad \mathcal{W}_9 = \{\emptyset\} \sqcup \mathcal{W}_9^+.$$

Words are written from the least significant digit. We define

$$\text{ev}_9(u_0, \dots, u_{\ell-1}) = \sum_{j=0}^{\ell-1} u_j 9^j, \quad \text{ev}_9(\emptyset) = 0.$$

Theorem 17.4 (Uniqueness of the normal form). *The evaluation ev_9 is a bijection from $\mathcal{W}_9^+$ onto $\mathbb{N}_{\geq 1}$, and from $\mathcal{W}_9$ onto $\mathbb{N}_0$.*

Proof. The words of length ℓ cubren the interval

$$\frac{9^\ell - 1}{8} \leq \text{ev}_9(u) \leq 9 \frac{9^\ell - 1}{8}.$$

The minimum of the next interval is one unit greater than the maximum of the preceding one. Within each interval, division of $n - 1$ by nine uniquely determines the digit

$$d = (n - 1) \bmod 9 + 1$$

and the quotient that is encoded recursively. The process terminates because the quotient strictly decreases. $\square$

Examples:

$$9 \longleftrightarrow (9), \quad 10 \longleftrightarrow (1, 1), \quad 81 \longleftrightarrow (9, 8), \quad 1000 \longleftrightarrow (1, 3, 3, 1).$$

17.3 Fibers of the residual projections on the integral liftings

For this block, we distinguish two alphabets. The normal form $\mathcal{W}_9$ uses the digits $\{1, \dots, 9\}$, written from the least significant. By contrast, $\mathcal{B}_9$ denotes the conventional positional representations in base nine, with digits $\{0, \dots, 8\}$ and the most significant on the left. The two types are not identified.

Let

$$\mathcal{W}_{10}^{(2)} = \{0, \dots, 9\}^2, \quad \rho_{10}(a, b) = (b, a), \quad \text{ev}_{10}(a, b) = 10a + b,$$

and let $q_9 : \mathbb{N} \rightarrow \mathbb{Z}/9\mathbb{Z}$ be the residual reduction.

Lemma 17.5 (Residual descent of decimal reversal). *Reversion on decimal words of length two is involutive and its shadow modulo nine is the identity:*

$$\rho_{10}^2 = I, \quad q_9(\text{ev}_{10}(\rho_{10}(a, b))) = q_9(\text{ev}_{10}(a, b)).$$

Fixed length preserves the initial zeros.

Proof. The double permutation recovers (a, b) . Ifnce $10 \equiv 1 \pmod{9}$,

$$10b + a \equiv a + b \equiv 10a + b \pmod{9}.$$

A word whose double reversal were not itself, or whose digit sum changed modulo nine, would refute the lemma. $\square$

Lemma 17.6 (Residual Square Descent). *The map $s(n) = n^2$ induces the residual endomap*

$$[n]_9 \mapsto [n]_9^2, \quad q_9(s(n)) = q_9(n)^2.$$

Proof. The compatibility of a congruence with the product gives $n \equiv m \pmod{9} \Rightarrow n^2 \equiv m^2 \pmod{9}$. An integer for which $q_9(n^2) \neq q_9(n)^2$ would refute the statement. $\square$

Proposition 17.7 (Residual fibers of the full reversal). *The residual shadow does not reconstruct the integral lift $\widehat{\rho}_{10} = \text{ev}_{10} \circ \rho_{10}$. Indeed,*

$$18 \equiv 27 \pmod{9}, \quad \widehat{\rho}_{10}(1, 8) = 81 \neq 72 = \widehat{\rho}_{10}(2, 7).$$

Proof. Both integers belong to residue class zero, but their reversals are distinct. Therefore, there is no $\overline{\rho} : \mathbb{Z}/9\mathbb{Z} \rightarrow \mathbb{N}$ that simultaneously factors those two values. $\square$

Proposition 17.8 (Residual fibres of the entire square). *The same shadow likewise does not reconstruct the integer square:*

$$18 \equiv 27 \pmod{9}, \quad 18^2 = 324 \neq 729 = 27^2.$$

Proof. A function $\bar{s} : \mathbb{Z}/9\mathbb{Z} \rightarrow \mathbb{N}$ that factorized s would have to assign the two values 324 and 729 to the same zero class, which is impossible. $\square$

Corollary 17.9 (Cross-witness 27–72–729).

$$\widehat{\rho}_{10}(2, 7) = 72, \quad 27^2 = 729 = 9^3 = 3^6,$$

and, in the alfabeto convencional $\mathcal{B}_9$,

$$27 = 30_9, \quad 72 = 80_9, \quad 729 = 1000_9.$$

The three integers have residual shadow zero and positive digital root nine.

Proof. The two operations are computed directly; successive divisions by nine produce the three expressions. The result would fail if $\mathcal{B}_9$ were confused with $\mathcal{W}_9$, if any of the expressions were incorrect, or if it were claimed that faithful residual descent reconstructs the operation and its complete genealogy. $\square$

17.4 Native adder and quotient propagation

Algorithm 17.10 (Word addition). Two words are traversed by positions, with initial carry $c_0 = 0$. If there are two digits at a position, (17.1) is applied; if there is only one, that digit is taken as the provisional residue and the provisional quotient is zero. When $c_j \neq 0$, a second additive cell is applied to the provisional residue and to c_j . The new residue is published, and the sum of the quotients is transported to the next position. If a nonzero carry remains upon completion, it is published as the final digit. In this manner, the local cells never receive the absent symbol 0, which does not belong to D_9 .

Lemma 17.11 (Adder Invariant). If $u \boxplus v$ is the word produced by the preceding algorithm,

$$\text{ev}_9(u \boxplus v) = \text{ev}_9(u) + \text{ev}_9(v).$$

The intermediate carry lies in $\{0, 1, 2\}$.

Proof. After processing the positions $0, \dots, k-1$, let c_k be the pending quotient. The local identities imply

$$\sum_{j < k} (u_j + v_j)9^j = \sum_{j < k} w_j 9^j + c_k 9^k,$$

including the absent digits as zero. Each new cell preserves this equality. The local maximum is $9 + 9 + 2 = 20$, so that $c_{k+1} \leq 2$. At the end the last quotient is published, and the global identity remains. $\square$

17.5 Multiplication by a digit

Algorithm 17.12 (Scalar transducer). With $d \in D_9$ fixed, each digit u_j produces

$$u_j d = 9q_j + r_j, \quad (r_j, q_j) = (R^\times(u_j, d), q^\times(u_j, d)).$$

Let $c_j \in \{0, \dots, 9\}$ be the incoming quotient. Since $0 \notin D_9$, the transition is defined by branches:

$$(w_j, c_{j+1}) = \begin{cases} (r_j, q_j), & c_j = 0, \\ (R^+(r_j, c_j), q_j + q^+(r_j, c_j)), & c_j \in D_9. \end{cases}$$

Thus, no local map is evaluated outside its domain. The first branch expresses the absence of a pending quotient; the second normalizes the collision by means of the additive map.

Lemma 17.13 (Scalar invariant). *The word $d \odot_{\#} u$ produced by the transducer satisfies*

$$\text{ev}_9(d \odot_{\#} u) = d \text{ev}_9(u).$$

The multiplicative quotient belongs to $\{0, \dots, 9\}$.

Proof. The argument is that of the adder, replacing each local summand by $u_j d$. If $c_j = 0$, the identity $u_j d = 9q_j + r_j$ directly preserves the partial equality. If $c_j \in D_9$, the second Euclidean division gives

$$r_j + c_j = 9q^+(r_j, c_j) + R^+(r_j, c_j),$$

and preserves the same equality after transporting $q_j + q^+(r_j, c_j)$. The maximum $9 \cdot 9 + 9 = 90$ and the remainder form in D_9 keep the quotient within the declared bound. The last quotient is published as a digit. $\square$

17.6 Global product by Horner

Definition 17.14 (Native product). Let $v = (v_0, \dots, v_{m-1})$. Partiendo of $a_m = \emptyset$, is defined for $j = m-1, \dots, 0$

$$a_j = (9 \odot_{\#} a_{j+1}) \boxplus (v_j \odot_{\#} u).$$

The result is

$$u \boxtimes v := a_0.$$

This definition does not contain

$$\text{dec}_9(\text{ev}_9(u) \text{ev}_9(v)).$$

The evaluation will appear only in the proof of entanglement.

Theorem 17.15 (Native multiplicativity). *For all words $u, v \in \mathcal{W}_9$,*

$$\boxed{\text{ev}_9(u \boxtimes v) = \text{ev}_9(u) \text{ev}_9(v).} \quad (17.6)$$

The empty word is absorbing, and (1) is the unit.

Proof. By the two previous lemmas,

$$\text{ev}_9(a_j) = 9 \text{ev}_9(a_{j+1}) + v_j \text{ev}_9(u).$$

Horner's iteration gives

$$\text{ev}_9(a_0) = \text{ev}_9(u) \sum_{j=0}^{m-1} v_j 9^j.$$

The last factor is $\text{ev}_9(v)$. The uniqueness of the normal form identifies the resulting word. $\square$

Corollary 17.16 (Multiplicative monoid of words). *The structure*

$$\mathfrak{M}_9^{\times} = (\mathcal{W}_9, \boxtimes, (1))$$

is a commutative monoid; the empty word is its absorbing element. Its construction does not use global multiplication of integers.

The shortest boundary example is

$$(9) \boxtimes (9) = (9, 8), \quad 9 + 8 \cdot 9 = 81.$$

The 17 local cells of multiplication produce

$$(9, 8) \boxtimes (9, 8) = (9, 8, 8, 8), \quad \text{ev}_9(9, 8, 8, 8) = 6561.$$

17.7 Matrix linearization of the genealogical state and construction of the completely positive entangler

Definition 17.17 (Hilbert Sectors). Let

$$\mathcal{K}_\#^+ = \ell^2(\mathcal{W}_9^+), \quad \mathcal{H}_+ = \ell^2(\mathbb{N}_{\geq 1}).$$

We define on the canonical bases

$$W_\times e_u = e_{\text{ev}_9(u)}.$$

In $\mathcal{H}_+$, the number operator is denoted by

$$N e_n = n e_n, \quad \text{Spec}(N) = \mathbb{N}_{\geq 1}.$$

The vacuum may be added separately as a vacuum vector e_0 ; it is not confused with $\mathcal{H}_+$, whose spectrum begins at one.

Theorem 17.18 (Unitarity and multiplicative square). $W_\times$ is unitary. In the algebraic kernel $\mathbb{C}[\mathcal{W}_9^+ \times \mathcal{W}_9^+]$, let

$$\mu_\#(e_u \otimes e_v) = e_{u \boxtimes v}, \quad \mu_{\mathbb{N}}(e_a \otimes e_b) = e_{ab}.$$

Then the square holds.

$$\begin{array}{ccc} \mathcal{K}_\#^+ \otimes_{\text{alg}} \mathcal{K}_\#^+ & \xrightarrow{W_\times \otimes W_\times} & \mathcal{H}_+ \otimes_{\text{alg}} \mathcal{H}_+ \\ \mu_\# \downarrow & & \downarrow \mu_{\mathbb{N}} \\ \mathcal{K}_\#^+ & \xrightarrow{W_\times} & \mathcal{H}_+ \end{array}$$

and therefore, the square commutes.

Proof. The bijection of normal forms sends one orthonormal basis to another and proves unitarity. On a basic tensor, both paths produce $e_{\text{ev}_9(u) \text{ev}_9(v)}$ by (17.6). Linearity gives the equality on the kernel. $\square$

The maps $\mu_\#$ and $\mu_{\mathbb{N}}$, initially defined on the respective algebraic kernels, are densely defined and unbounded. For $c = (c_{u,v}) \in \ell^2(\mathcal{W}_9^+ \times \mathcal{W}_9^+)$, each sum over a product fiber is finite. The adjoint of $\mu_\#$ contains all finite-support vectors in its domain, so that $\mu_\#$ is closable. The domain of its closure is

$$\text{Dom}(\overline{\mu_\#}) = \left\{ c \in \ell^2(\mathcal{W}_9^+ \times \mathcal{W}_9^+) : \sum_{w \in \mathcal{W}_9^+} \left| \sum_{u \boxtimes v = w} c_{u,v} \right|^2 < \infty \right\}.$$

Each fiber is finite because ev_9 identifies it with the ordered pairs of divisors of an integer. The same description holds for $\overline{\mu_{\mathbb{N}}}$ after applying $W_\times \otimes W_\times$, and $W_\times$ intertwines the graphs of both closures.

17.8 Native coproduct and native antipode

The multiplicative opening is now defined before looking at the codomain:

$$\Delta_{\#} e_w = \sum_{u \boxtimes v = w} e_u \otimes e_v.$$

The sum is obtained by enumerating products of the transducer. One does not ask whether one ordinary integer divides another. In the integer sector define, on the corresponding algebraic kernel,

$$\Delta_{\times} e_n = \sum_{\substack{a, b \in \mathbb{N}_{\geq 1} \\ ab = n}} e_a \otimes e_b.$$

Both coproducts enumerate ordered pairs.

Theorem 17.19 (Native coassociativity). *In the algebraic core,*

$$(\Delta_{\#} \otimes I) \Delta_{\#} = (I \otimes \Delta_{\#}) \Delta_{\#}.$$

Moreover,

$$(W_{\times} \otimes W_{\times}) \Delta_{\#} = \Delta_{\times} W_{\times}.$$

Proof. Both sides of the first equality enumerate every native triple $a \boxtimes b \boxtimes c = w$ exactly once. The second equality follows from the bijection between fibers induced by (17.6). $\square$

Theorem 17.20 (Complete arithmetic coproduct). *In the algebraic kernel $c_{00}(\mathbb{N}_{\geq 1}) \subset \ell^2(\mathbb{N}_{\geq 1})$, let*

$$\Delta_{\times} e_n = \sum_{ab=n} e_a \otimes e_b.$$

Its closure has domain

$$\text{Dom}(\overline{\Delta_{\times}}) = \left\{ x = (x_n) : \sum_{n \geq 1} d(n) |x_n|^2 < \infty \right\},$$

is coassociative on the kernel and, for $Qe_n = ne_n$, satisfies

$$(\log Q \otimes I + I \otimes \log Q) \Delta_{\times} = \Delta_{\times} \log Q.$$

Proof. The supports of $\Delta_{\times} e_n$ are orthogonal for distinct integers, and their squared norm is the number $d(n)$ of divisors, which yields the weighted domain of the closure. The two coassociativity relations reindex the same sum over $abc = n$. Finally, $\log a + \log b = \log n$ in each term proves the intertwining. A triple factorization counted differently, or failure of this identity, would falsify the theorem. $\square$

Theorem 17.21 (Reduced coproduct and irreducible selector). *In the sector integer, the coproducto reducido*

$$\Delta_{\times}^{\text{red}} e_n = \sum_{\substack{ab=n \\ a, b > 1}} e_a \otimes e_b$$

is closable. The operator

$$N_{\times}^{\text{red}} = (\overline{\Delta_{\times}^{\text{red}}})^* \overline{\Delta_{\times}^{\text{red}}}$$

is positive self-adjoint and diagonal:

$$N_{\times}^{\text{red}} e_n = d_{\text{red}}(n) e_n.$$

Its kernel is generated by e_1 and by the e_p with p irreducible; therefore,

$$P_{\mathfrak{S}} = \mathbf{1}_{\{0\}}(N_{\times}^{\text{red}}) - |e_1\rangle\langle e_1|$$

select the irreducibles and exclude the unit.

In the native sector, the corresponding coproduct is

$$\widetilde{\Delta}_{\#} e_w = \sum_{\substack{u \boxtimes v = w \\ u, v \neq (1)}} e_u \otimes e_v.$$

The entrelazador $W_{\times}$ transports $\Delta_{\times}^{\text{red}}$, $N_{\times}^{\text{red}}$ and $P_{\mathfrak{S}}$ toward $\widetilde{\Delta}_{\#}$,

$$A_{\#} := (\widetilde{\Delta}_{\#})^* \widetilde{\Delta}_{\#}, \quad P_{\mathfrak{S}, \#} = (I - P_{(1)}) \mathbf{1}_{\{0\}}(A_{\#}), \quad P_{(1)} = |e_{(1)}\rangle\langle e_{(1)}|.$$

Proof. The supports of the images of two distinct basis vectors are orthogonal. The squared norm of $\Delta_{\times}^{\text{red}} e_n$ is $d_{\text{red}}(n)$, which vanishes exactly for $n = 1$ or n irreducible. This proves the diagonalization and the declared kernel; subtracting the projector of e_1 removes the unit. The multiplicativity of $W_{\times}$ identifies the integer fibers with the native fibers and transports the three operators. A composite in the kernel, an irreducible outside it, or inclusion of the unit in $P_{\mathfrak{S}}$ would refute the theorem. $\square$

Proposition 17.22 (Nonadic coincidence of primes and composites). *Let $\gcd(n, 90) = 1$, $\tau_n = \text{ord}_n(10)$ and*

$$M_n = \frac{10^{\tau_n} - 1}{n}.$$

Then $M_n \equiv 0 \pmod{9}$, both for primes and for composites. In particular, the nonadic closure and the isolated decimal period are not a selector of irreducibles.

Proof. Because $9 \mid 10^{\tau_n} - 1$ and because n is invertible modulo nine, divisibility by nine survives passage to the quotient. The primes 7, 13 and the composites 77, 91, 143 share period six; the Carmichael numbers provide additional controls. The statement would be falsified only if the congruence or the period separated all primes from all composites in the declared domain. $\square$

This statement is strictly arithmetic: it identifies factorization fibers and the operators they induce on the sector of normal forms.

17.9 Exhaustive enumeration and necessity of structural coordinates

The finite enumeration comprises:

Test	Result
Cells per local map	81
Reversible numbering paths	59050
Intertwiner pairs	532900
Intertwiner errors	0
Associative triples	531441
Associative errors	0
Coproduct window	1 to 500
Irreducibles obtained	95

Definition 17.23 (Multiplicative quotient). Preserving only $R^{\times}$ transforms $2 \cdot 5 = 10$ into the visible value 1. Perturbing $q^{\times}(9, 9)$ from 8 to 7 transforms 81 into 72. The local quotient is necessary.

Definition 17.24 (Additive application). The multiplicative map produces partial products, but does not resolve their collisions. Removing q^+ breaks the global normalization; for example, it no longer reproduces the intermediate quotients of $10 \cdot 18 = 180$.

Definition 17.25 (Coordinates and boundary symbol). Ordinary zero is not a digit of the construction. The empty word is 0, and the symbol 9 preserves its value and quotient. Mixing zero-indexed positional coordinates with the labels $1, \dots, 9$ changes the cells.

Proposition 17.26 (Exact scope of the ablations and of the finite census). *The preceding ablations prove, within the declared conventions:*

1. without $q^\times$, $2 \cdot 5 = 10$ colapsa to the residue visible 1;
2. cambiar $q^\times(9, 9) = 8$ by 7 transforms 81 in 72;
3. without q^+ , normalization of the additive collisions fails.

The exhaustive enumeration comprises the $2 \cdot 81 = 162$ cells of the two maps, the 729 local triples, 532 900 word pairs, and the 95 irreducibles no greater than 500. The indicated identities are proved on these finite domains, but the coinductive survival of every TPK history is not.

Proof. The three counterexamples are computed directly with (17.1) and (17.2). The enumeration of the four finite domains produces the cardinals written in the statement and does not add any coinductive identity to those already proved in the monoid of words. $\square$

17.10 Algebraic structure of the word monoid

The construction proves, in the monoid $\mathfrak{B}_9^\times$:

1. a unique normal form for each nonnegative integer;
2. global sum and product constructed solely from the two local maps;
3. symbolic conservation of the value under quotient propagation;
4. a unitary $W_\times$ and the exact multiplicative square;
5. a divisor coproduct and an irreducible selector defined natively;
6. the need to preserve the transport class when leaving the normal-form sector.

Passage to holonomic space now reduces to an existence problem: constructing a system $(\iota_n)_n$ compatible with the depth restrictions and proving that it satisfies the equation of the preceding section. This question is distinct from the exact intertwiner on $\mathfrak{B}_9^\times$, already determined by the normal form and local maps.

17.11 Problem of determining the constants from the genealogical state

A physical constant is represented as a section of a magnitude line equipped with a rule of evaluation, orientation, and normalization. Its numerical coordinate depends on the chosen chart, whereas its genealogy remains invariant. Consequently, the determination problem comprises:

1. the identification of the antecedent HMT state;
2. the specification of the operator or character of selection;
3. determination of the conserved invariant;
4. the corresponding magnitude line;

5. the chart in which the terminal value is expressed.

The causal chain of the work is

$$\begin{aligned} \text{APP} \longrightarrow \widehat{\mathbb{T}} \longrightarrow \text{TPK} \longrightarrow \text{internal realization} \longrightarrow \text{dimensionless invariant} \\ \longrightarrow \text{magnitude section} \longrightarrow \mathbb{R}. \end{aligned} \quad (17.7)$$

The term $\mathbb{R}$ in the chain represents the terminal evaluation of the section; it does not constitute the original domain from which the route is retrospectively reconstructed.

17.12 Antecedent chain APP–TRIT–TPK and domain of the extraction operators

APP supplies the geometry of positions, residues, quotients, and boundaries. TRIT incorporates orientation $(-1, 0, +1)$ and distinguishes opening, threshold, and return. TPK completes the state through carry, route, sheet, memory, and survival. In particular, transition $9 \rightarrow 1$ preserves phase and modifies the memory state; it therefore does not constitute a reinitialization devoid of genealogy.

The realizations developed below preserve that antecedent without replacing it. The vacuum is obtained through a spectral function of the bilayer operator; gravity selects a section through the chronological structure 108; criticality uses a dodecaphasic family; the Gaussian constants follow from the evaluation of cylinders; and the modular Hamiltonian is defined on a boundary that preserves the incidence $6 \rightarrow 8$. The production of scalar values does not establish functional identity among these realizations.

Proposition 17.27 (Insufficiency of the Scalar Coincidence). *Let $F_1 : X_1 \rightarrow L_1$ and $F_2 : X_2 \rightarrow L_2$ be two typed functionals. Equality of coordinates $c_1(F_1(x_1)) = c_2(F_2(x_2))$ does not by itself induce either an identification $X_1 \simeq X_2$ or an intertwiner $F_2 I = J F_1$.*

In particular, a numerical proximity such as $I_{\text{or}} \approx \gamma_{\text{HMT}} a_0$ cannot be promoted to an identity without constructing the corresponding intertwiner.

17.13 Lines of magnitude, oriented letters and dimensional coordinate systems

Let L_Q be an oriented real line representing the magnitude Q . A unit $u_Q \in L_Q \setminus \{0\}$ determines a provisional coordinate $L_Q \simeq \mathbb{R}$, without replacing the intrinsic structure of the line. The constitutive relation

$$L_\mu = L_S L_C^{-1} L_Q^{-2}, \quad L_\varepsilon = L_S^{-1} L_C^{-1} L_Q^2$$

precedes the SI choice. In the same way,

$$L_Z = L_S L_Q^{-2}, \quad L_G = L_C^5 L_T^2 L_S^{-1}$$

type impedance and the gravitational constant before expressing their magnitudes in ohms or SI units. In this work we shall write

$$Q_{\text{int}} = \widehat{Q} u_Q$$

to distinguish the dimensionless character $\widehat{Q}$ from the reference section u_Q .

17.14 Torsor of frames and covariance of dimensional evaluations

The admissible charts form a torsor and therefore none has a character that is canonical relative to the others. The frame change $u_Q \mapsto \lambda_Q u_Q$ induces the contragredient transformation of the coordinate. Physical identities must be homogeneous with respect to this action. For example,

$$\mu\varepsilon = c^{-2}, \quad \frac{\mu}{\varepsilon} = Z^2$$

are valid in every compatible frame. Mathematical unification and dimensional unification do not eliminate units; rather, they reconstruct them through the covariance of the lines and allow the invariant to precede its coordinate expression.

17.15 Final metrological evaluation and prohibition of retrospective selection

Each entry of the final atlas adopts the form

$$\boxed{\text{antecedent} \rightarrow \text{operator} \rightarrow \text{invariant} \rightarrow \text{meaning} \rightarrow \text{terminal value.}} \quad (17.8)$$

This order expresses a causal dependency: the numerical value is only determined after fixing the antecedent, operator, invariant, and interpretation. Omission of any of these elements reduces the result to a reconstruction that is numerical, not a derivation that is genealogical.

17.16 Exact domain of the genealogical generation and its terminal metrical chart

The generation of π , e , φ , the dodecaphasic closure of α , the action section, and the Barbero–Immirzi output γ_{HMT} are adopted as initial results. The present development does not recalibrate those constructions; it determines the constants and invariants resulting from their application to the constitutive, thermal, gravitational, modular, and spectral realizations.

The electron mass is likewise not included as a governing problem, since it belongs to the General Mass Law. Every occurrence of m_e in an atomic identity is treated as a datum proceeding from that domain and not as a new selection of the present development.

Logical Scope and Domain of Validity

The APP–TRIT–TPK architecture and the dimensional lines constitute the starting structure. The uniform rule of five fields of the atlas makes explicit its causal discipline in each entry.

17.17 Operator classification of 114 numerical expressions by domain, generator, invariant, and codomain

Let the finite indexed set be

$$E_{114} = \{\varepsilon_1, \dots, \varepsilon_{114}\}$$

of expresiones numericas and let O the space of readings of the reader triadico. The map

$$T : E_{114} \longrightarrow O, \quad \varepsilon_i \longmapsto T_{\text{tri}}(\varepsilon_i), \quad |\text{im } T| = 106,$$

classifies expressions according to their operatorial role. Each ε_i , including the compared targets, is fixed as input of T ; therefore, this map does not generate the values that it evaluates. The functional partition is

functional family	entries	$\sum n_i = 114.$
structural kernel	5	
coupling and closure	1	
bridges between modes	7	
angular and closure	22	
exponential–logarithmic	12	
spectral and regularization	14	
hyperbolic and modular	2	
algebraic and metric	50	
typed miscellany	1	

Network centrality, shortest paths, and communities computed on the graph induced by T describe connectivity within the classification; they do not establish mathematical causality or HMT genealogy.

Seven fibers of T gather fifteen expressions with a common reading. A shared fiber does not identify their values either. In particular, φ^{-1} , φ , and φ^2 are distinct real numbers that the map T groups into the same self-scaling orbit; the common projection preserves dynamic function and forgets the Archimedean coordinate that separates them.

The exhaustive enumeration traversed the 114 entries of E_{114} , four hundred sixty-eight direct candidates U_{12} , and one hundred four thousand nine hundred seventy-six weighted seeds without finding, within that finite domain, a rival that reproduced the oriented dodecaphasic correction of α . The search domain is thereby bounded by those three sets. The census proves that no rival appears in the enumerated class; it does not prove uniqueness among all possible functionals and does not enter as a premise of theorem α .

For each ε_i , the taxonomy assigns the quadruple (object, reader, domain, structural function) used in the enumeration.

17.18 Spectral, informational, and thermodynamic confluence of $\log 3$

The identity

$$\log 3 = C_1 + C_2 - 2C_0 = S_{\mathbb{T}}/k_B = E_P/(k_B\Theta_{\text{clk}}),$$

relates three distinct realizations. The first member is a residual defect, the second an entropy of decision, and the third a ratio between energy and temperature. Their equality proves the conservation of the scalar under a change of realization, without identifying the state spaces.

The first equality is deduced from the three residual channels.

$$Z_r(s) = \sum_{\substack{n \geq 1 \\ n \equiv r \pmod{3}}} n^{-s}, \quad r = 0, 1, 2.$$

Then

$$Z_0(s) = 3^{-s}\zeta(s), \quad Z_1(s) + Z_2(s) = (1 - 3^{-s})\zeta(s), \quad Z_1(s) - Z_2(s) = L(s, \chi_{-3}).$$

If C_r denotes the part finite of Z_r in $s = 1$, the expansion of 3^{-s} gives

$$C_0 + C_1 + C_2 = \gamma, \quad qqquad - 2C_0 + C_1 + C_2 = \log 3, \quad qqquad C_1 - C_2 = \frac{\pi}{3\sqrt{3}}. \quad (17.9)$$

The three covectors are linearly independent. From a single residual decomposition, they recover the total finite term, the dilation of the zero class, and the orientation between the two nonzero classes. $\log 3$ is therefore a scale-change invariant before being interpreted as entropy or temperature.

17.19 Epistemological classification of the derived constants

The chapter's contributions are classified into three categories:

1. assignment of an internal operator to a known constant, as in the Catalan case;
2. exact composed evaluation of two internal characters, such as $L(1, \chi_{12})$;
3. derivation of a constant from a field or a family constructed internally, such as $\gamma_{\mathbb{Q}(\sqrt{3})}$ or A_k .

Mathematical provenance determines the status of recovery or novelty; decimal proximity plays no role in that classification.

Remark 17.28 (Negative control). The character χ_{12} must have exactly the pattern $(+, -, -, +)$ in $(1, 5, 7, 11)$; any other pattern invalidates the corresponding identity. The identity equation (22.2) and the factorization equation (21.1) are independent controls.

Logical Scope and Domain of Validity

$J^2 = -I$, Catalan, χ_{12} , their values L , Euler–Kronecker, and the Bendersky–Glaisher tower: *Exact internal theorem*. The gathering in a lattice $3/4/12$ is *New formalization* that preserves the mathematical provenances.

All these results are joined because they turn the terminal value into the quantitative expression of a previously constructed structure. Numerical exactness accompanies a different conception of what a fundamental constant is. The nonadic connection does not first generate three constants and then add a fourth: the prolongation $w_6 \rightarrow w_{12} \rightarrow w_{18} \rightarrow w_{24} \rightarrow w_{30} \rightarrow R_{36} \rightarrow G_9$ stabilizes a common coinductive state from which π , e , φ and α are published in a correlated manner. Their readers distinguish closure, propagation, self-scaling and coupling; within this joint generation, coordinate α retains its signed, dodecaphasic realization and its two internal derivation routes.

In the usual description, a constant is commonly presented as a number placed before an equation: a quantity that the theory requires but whose value must be received from outside. In Triadic Modular Holography, the opposite occurs. The constant appears at the end, after the same state has been traversed by several readers and it has been determined which relation has survived all of them.

This is why the constants in the document are magnitudes endowed with genealogy: invariant remnants of a transformation. Each states what has been preserved when something has changed:

- to sheet has been inverted;
- a cycle has returned to its initial phase;
- a memory has crossed the seam;
- an interior description has been published on the boundary;
- a dimensionless magnitude has received a physical realization;
- a system has changed scale while fully preserving its genealogy.

Dimensional Mechanics subsequently adds the magnitude lines and the units. But the structure that selects the constant is prior. The real value appears at the end because $\mathbb{R}$ functions here as a terminal evaluation: it is the quantitative shadow of a construction that was already complete before becoming a decimal.

With that view, the nine developments considered form a single narrative on orientation, memory, geometry, information, and scale.

17.20 Joint theorem of coinductive generation and arbitrary depth

The starting structure of this theorem is not the bare set of nine symbols. It is the typed system

$$\mathfrak{S}_9 = (\text{APP}, \text{TRIT}, \text{TPK}; \widehat{\mathcal{X}}^{\text{enr}}, \mathcal{U}, \Gamma_9, \text{Ext}, \text{tr}),$$

constructed on the double positional and arithmetic realization of $\mathcal{D}_9 = \{1, \dots, 9\}$. APP accumulates and folds through its sheets Σ and Π , preserving residue and quotient; TRIT orients each transition in one of its three regimes; and TPK selects, transports, memorizes and returns on the enriched fiber. In particular,

$$\mathcal{U}_t = \text{Rec}_t \circ \text{Mem}_t \circ T_{d(t)} \circ M_{\text{ph}}(g(t)).$$

This composition is the effective dynamic datum: the nine-symbol chart is its finite support, not a substitute for the dynamics.

For $r \in \{6, 12, 18, 24, 30, 36\}$, let $\widehat{\mathcal{X}}_r^{\text{enr}}$ be the truncated states with sheet, orientation, residue, quotient, carry, route, boundary and memory, and let

$$\text{Ext}_{r+6,r} : \widehat{\mathcal{X}}_r^{\text{enr}} \longrightarrow \widehat{\mathcal{X}}_{r+6}^{\text{enr}}, \quad \text{tr}_{r+6,r} : \widehat{\mathcal{X}}_{r+6}^{\text{enr}} \longrightarrow \widehat{\mathcal{X}}_r^{\text{enr}}$$

admissible extension and truncation. The compatibility $\text{tr}_{r+6,r} \text{Ext}_{r+6,r} = \text{id}$ extends to every level. The notation

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

designates these maps and their type changes, not a list of labels. The holonomy Γ_9 satisfies

$$g(k+9) = g(k), \quad q^{(9)}(k+9) = q^{(9)}(k) + 1,$$

so that the phase returns while the quotient, the route and the memory of the state advance.

Theorem 17.29 (Inaugural construction from the seeds). *Let*

$$I_9 = \{0, \dots, 8\}, \quad D = \{N, E, S, O\}, \quad S = I_9^2 \times D,$$

and let $\Omega_{\text{seed}} = S_+ \times S_\times$ be the domain of the two cursors realizing the additive and multiplicative sheets of APP. Then

$$|\Omega_{\text{seed}}| = (9^2 \cdot 4)^2 = 104\,976,$$

and the 54-step recurrence and ternary reduction define

$$\Omega_{\text{seed}} \xrightarrow{T_{54}^{\text{seed}}} \mathcal{U}_6^{\text{cat}} \xrightarrow{\rho_3} \mathcal{W}_6 \hookrightarrow \mathbb{F}_3^6,$$

with

$$|\mathcal{U}_6^{\text{cat}}| = 468, \quad |\mathcal{W}_6| = 243, \quad |\mathbb{F}_3^6| = 729.$$

The fibers satisfy exactly

$$432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976.$$

The TPK extension relation extends the surviving words to a compatible section x_∞ of infinite depth. On that section, the Archimedean shadow and the exceptional realization of theorem 17.30 are constructed jointly.

Proof. The preceding Cartesian product exhaustively enumerates the positions and orientations of both cursors. The map T_{54}^{seed} is obtained by composing, at each step, sheet selection, cardinal transport, quotient updating and carry memory. Enumeration of its image contains 468 distinct sextuples; the sum of its fibre cardinalities is the stated identity. The componentwise map ρ_3 contains 243 distinct words. Finally,

nonemptiness, single-valuedness on the surviving subdomain and compatibility of Ext with truncations produce the inverse-limit section. Each statement is a finite equality or a consequence of projective compatibility; none uses a real number, a recognized constant or a decimal table as an input. $\square$

Theorem 17.30 (Governing thesis: one continuum and two projections). *Let $\Omega_\Gamma^{\text{cal}} \subset X_\infty^{\text{cal}}$ be the space of surviving sections of the enriched turns of the TPK. The carry of the three turns $\gamma_-, \gamma_0, \gamma_+$ defines, without prior Archimedean evaluation, the homeomorphism*

$$\beta : \Omega_\Gamma^{\text{cal}} \xrightarrow{\sim} (\mathbb{F}_3^6)^\mathbb{N} =: \mathcal{W}_\infty.$$

The normalized balanced forms

$$a_k + c_k = \rho_k + 3c_{k+1}, \quad \rho_k \in \{-1, 0, +1\},$$

construct the ordered ring $\mathbb{Z}_{\text{HMT}}$, its field of fractions $\mathbb{Q}_{\text{HMT}}$ and the completion

$$\mathbb{R}_{\text{HMT}} := \text{Cau}(\mathbb{Q}_{\text{HMT}})/\asymp.$$

The joint discrete structure is the fiber product

$$C_{\text{cont}}^{\text{disc}} = (\mathbb{Z}_{\text{HMT}} \times \Omega_\Gamma^{\text{cal}}) \times_{\mathcal{W}_\infty} X_\infty^{\text{cal}}.$$

Its two natural maps are

$$P_{\text{ar}} : \mathbb{Z}_{\text{HMT}} \times \Omega_\Gamma^{\text{cal}} \longrightarrow \mathbb{R}_{\text{HMT}}, \quad Q_\infty : X_\infty^{\text{cal}} \longrightarrow (\mathcal{C}_W^{\text{cal}})^\mathbb{N}.$$

If $\beta(\xi) = (b_q)_{q \geq 0}$, $v(b) = \sum_{i=1}^6 b_i 3^{6-i}$ and

$$q_n(m, \xi) = m + \sum_{q=0}^{n-1} \frac{v(b_q)}{729^{q+1}},$$

then (q_n) is Cauchy and $P_{\text{ar}}(m, \xi)$ is its class in $\mathbb{R}_{\text{HMT}}$. The rational cylinders

$$I_n(m; b_0, \dots, b_{n-1}) = [q_n, q_n + 729^{-n}]$$

are compatible, have diameter tending to zero and the recurrent partition into 729 subcylinders constructs, for every Cauchy class, a tape (b_q) representing it. Consequently,

$$P_{\text{ar}} \text{ is surjective, } (\mathbb{Z}_{\text{HMT}} \times \Omega_\Gamma^{\text{cal}})/\sim_{\text{ar}} \cong \mathbb{R}_{\text{HMT}}.$$

Only after this internal construction does uniqueness of the complete Archimedean ordered field recognize $\mathbb{R}_{\text{HMT}} \cong \mathbb{R}$.

The second preserves the calibrated incidence of each level; in coordinates $x = ((w_j, f_j))_{j \geq 1}$, it acts by

$$Q_\infty(x) = (f_j, \Gamma_{f_j A_W f_j^{-1}}(w_j))_{j \geq 1}.$$

Thus, $\mathbb{R}_{\text{HMT}}$, subsequently recognized as $\mathbb{R}$, is the Archimedean shadow obtained by contracting the complete genealogy of cylinders. The exceptional projection preserves that genealogy as boundary incidence and admits the successive Paley–Witt–Golay–Leech– $V^{\mathfrak{h}}$ –Monster realization. Both branches proceed from the same state, and neither is reconstructed retrospectively from the other.

Proof. The three enriched turns exist after each prefix and their carry distinguishes them by $-1, 0, +1$. Concatenation and truncation are inverses, proving bijectivity and continuity of β . For a Cauchy class of

$\mathbb{Q}_{\text{HMT}}$, one first chooses its ordered unit cylinder and then, recursively, the unique half-open subcylinder of the internal partition into 729 pieces containing the class; at a common boundary, the two writings are identified by $\sim_{\text{ar}}$. The sequence of left endpoints is precisely (q_n) , converges to the given class and proves surjectivity without presupposing an expansion of $[0, 1]$ or a point of $\mathbb{R}$. Identification with the real field is the uniqueness theorem applied afterwards to $\mathbb{R}_{\text{HMT}}$.

The subspaces satisfying closure of a particular character are survival filters within the global capacity. Their function is to individuate the sections of π, e, φ, α ; surjectivity of the continuum proceeds from completion of $\mathbb{Q}_{\text{HMT}}$. This separation of domains prevents a constant-selection condition from becoming a premise for coverage of $\mathbb{R}$.

In the second branch, compatibility $r_{n,m}Q_n = Q_m p_{n,m}$ allows passage to the limit and defines Q_∞ . Conjugation by f_j transports the gauge without erasing incidence. The subsequent code, lattice and vertex-algebra functors realize the exceptional chain. Equality of the tapes in the fiber product constructs the joint source and preserves complementary information: Archimedean value in the first; incidence, boundary and genealogy in the second. $\square$

Corollary 17.31 (Infinite expansion as a mathematical object). *For each numerical character χ published by the coinductive state, the decimal reading is a complete sequence*

$$\text{Dec}_\chi(x_\infty) = (d_{\chi,1}, d_{\chi,2}, \dots) \in \{0, \dots, 9\}^{\mathbb{N}}.$$

Its prefix of length N is the projection $\rho_N \text{Dec}_\chi(x_\infty)$. Thus a thousand, ten thousand or a million digits are cuts of the same infinite object; generation does not consist in successively raising an experimental bound.

Theorem 17.32 (Correlated four-coordinate system of the integral coinductive holonomic state). *The unique APP–TRIT–TPK composition of the system $\mathfrak{S}_9$ determines a compatible state*

$$x_\infty \in \widehat{\Omega} := \varprojlim_r \widehat{X}_r^{\text{enr}}$$

and four correlated characters

$$(\Pi_{\text{cl}}, \Pi_{\text{prop}}, \Pi_{\text{aut}}, \Pi_{\text{dod}})(x_\infty) = (\pi, e, \varphi, \alpha).$$

The first three characters proceed respectively from the unique closure orbit, the unique propagation orbit and the unique autoscale orbit of the enriched catalogue. The fourth proceeds from the primitive signed coordinate of the same terminal state and its dodecaphasic closure. The four coordinates are therefore outputs of a single coinductive extension, despite their distinct expression maps.

Proof. The exhaustive census $104,976 \rightarrow 468 \rightarrow 243 \subset \mathbb{F}_3^6$ decomposes the realized image into 43 dihedral orbits. The fiber invariants leave a single orbit in each of the closure, propagation and self-scaling classes; the internal gauge orients in them, without decimal consultation, the representatives

$$w_6^{\text{cl}} = 010211, \quad w_6^{\text{prop}} = 201101, \quad w_6^{\text{aut}} = 121200.$$

Extension preserves the predicates defining the three classes. In the closure channel, pentaregional composition with unit return produces the ratios $1/5$, $1/239$ and the relation $120/119$; Machin's identity recognizes the half-turn exactly. In the propagation channel, the normalized semigroup law forces $(n+1)a_{n+1} = a_n$, hence $a_n = 1/n!$ and $P_1 = e$. In the autoscale channel, the incidence

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$

has a unique positive ray and its eigenvalue equation forces $x^2 - x - 1 = 0$, $x > 0$, that is, $x = \varphi$.

At the terminal phase of the unique APP–TRIT–TPK structure, the reading $x_{\text{term}} \mapsto (B_{\text{term}}, U_{12}^{\text{sgn}})$ simultaneously expresses the three-channel chart and the primitive signed coordinate. The latter is reversibly identified with the seal K and its dodecaphasic differences. The Euclidean recurrence with carry

$$s_m = P_m + E_m - \Phi_m - K_m + c_{m+1}, \quad A_m = s_m \bmod 1000, \quad c_m = (s_m - A_m)/1000$$

produces α without receiving it as an input. Independent analytic characterization through the kernel $E_{21/22}$ determines a simple root in $(0, 10^{-2})$, whose reciprocal coincides with that of the first route after applying Tr_9 . Thus, separating the readers does not alter the common origin of the four coordinates. $\square$

Proposition 17.33 (Dodecaphase register of the complete state prior to α). *The terminal holonomic state has the double expression*

$$x_{\text{term}} \xrightarrow{(\pi_{\text{term}}, R_{\text{sgn}})} (B_{\text{term}}, U),$$

where

$$U = (2378, 1406, 2479, -452, 998, -551, -668, -204, -371, -322, -28, -997).$$

Upon ordering the coordinates by the three step-three orbits, the transform

$$\mathcal{H}_{12} = P_{\text{orb}}^{-1}(H_4 \oplus H_4 \oplus H_4)P_{\text{orb}}, \quad H_4^2 = 4I_4,$$

has inverse $\mathcal{H}_{12}/4$ on its integral image and determines uniquely

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601).$$

Therefore, within the integral holonomic state,

$$x_{\text{term}} \xrightarrow{R_{\text{sgn}}} U \xrightarrow{\mathcal{H}_{12}^{-1}} K$$

is an exact composition preceding the carry that expresses α . The visible matrix B_{term} and signed record U are two coordinates of the same state; the latter preserves orientation, opposition, quotient and record fields contracted by the former. There is no second APP or competing reduced profile here: APP produces the sheets and arithmetic record, TRIT orients them, and TPK accumulates in its ledger the signed window projected by R_{sgn} . Requiring isolated APP to express a coordinate belonging to the subsequent TPK codomain is a type error. The complete composition does not receive U , K , their differences, α or target expansions as inputs.

Theorem 17.34 (Subsequent bidirectional closure of the character α). *Let $\alpha = \alpha$ be the coordinate already expressed by dodecaphasic carry. Evaluating the vacancy jet of degrees 7:9 at that output determines the functional*

$$\begin{aligned} B_9(\alpha) := & \log_{10}\left(\frac{10}{9}\right) + \frac{7}{4}\alpha^2 - \frac{\alpha^4}{20} + \frac{2\alpha^5}{21} + \frac{\alpha^6}{46} \\ & + \frac{\alpha^7}{120} - \frac{\alpha^8}{45} - \frac{2\alpha^9}{495}. \end{aligned}$$

Upon lifting the rotation to the positive unknown T , the resolvent is equivalent to

$$\boxed{\alpha T^2 - B_9(\alpha)T + \alpha^3 = 0.}$$

With $\xi = \text{arcosh}(B_9(\alpha)/(2\alpha^2))$, the solutions are

$$T_{\pm} = \alpha e^{\pm \xi}, \quad T_+ T_- = \alpha^2, \quad J_{\alpha}(T) = \frac{\alpha^2}{T}, \quad J_{\alpha}(T_{\pm}) = T_{\mp}.$$

The chamber $6 < T < 7$ selects T_+ ; the finite closure $\pi_9 = T_+/2$ is compatible, at the declared resolution, with the closure channel independently extended by Γ_9 to π . The identity $\alpha = \sqrt{T_+ T_-}$ fixes the geometric mean of the conjugate branches. The jet 7:9, both roots and the involution constitute an exact bidirectional closure on the already generated character. This theorem receives no conventional constant, but neither is it a second independent derivation of α : its domain contains α , and its result is a subsequent functional certificate.

Theorem 17.35 (Exceptional expression correlated with the sector α). *In the incidence chart of the same dodecaphase register, the radial roots of A_2^{12} determine*

$$v_{\alpha} = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \quad v_{\alpha}^2 = 54,$$

and the index-three neighbor

$$N_{\alpha,0} = \{x \in N(C_W) : \langle x, v_{\alpha} \rangle \equiv 0 \pmod{3}\}, \quad N_{\alpha} = N_{\alpha,0} + \mathbb{Z} \frac{v_{\alpha}}{3}.$$

The lattice N_{α} is even, unimodular, of rank twenty-four and rootless, so $N_{\alpha} \cong \Lambda_{24}$. The chain

$$C_W \longrightarrow N(A_2^{12}) \xrightarrow{v_{\alpha}} \Lambda_{24} \longrightarrow V^{\natural} \xrightarrow{\text{Aut}} \mathbb{M}$$

exposes the exceptional hinge. The scalar α and vector v_{α} belong to different codomains and share the originating dodecaphasic state. The exact result of this branch is the construction $N_{\alpha} \cong \Lambda_{24} \rightarrow V^{\natural}$ and the subsequent action of $\text{Aut}(V^{\natural}) \cong \mathbb{M}$ on $V^{\natural}$. The scalar is not identified with the vector, nor is an action of the Monster on digits or routes deduced from correlation; such a promotion would require an equivariant morphism with explicit domain and codomain.

Proposition 17.36 (Status of the transports L_0, L_1). *The endomorphisms $L_0, L_1 \in \text{End}(\mathbb{F}_3^6)$ and calendar (L_0, L_0, L_1, L_1) belong to the transport law of the integral holonomic state. The six expressed full-rank transitions uniquely fix each matrix through $L = X^{-1}Y$ in $\mathbb{F}_3$, and the calendar is the only one among the 16 binary calendars simultaneously preserving the three structural histories. Their function is to extend the already selected representatives; Archimedean evaluation and conventional nomenclature are applied afterwards.*

The domain of this proposition is the unique structure $\mathfrak{H}_9$. Its forgetful projections are subsequent readers and do not constitute alternative generating profiles. This typing prevents confusion of a modal shadow with the state producing it.

Theorem 17.37 (Arbitrary depth and determination of the complete expansions). *For every $\chi \in \{\pi, e, \varphi\}$ and every $N \geq 1$, there is a unique decimal prefix $p_{\chi,N}$ of length N , produced by a finite truncation of the integral holonomic state, such that*

$$\rho_{N+1,N}(p_{\chi,N+1}) = p_{\chi,N}.$$

The compatible family is the family of projections of a unique element

$$p_{\chi,\infty} \in \varprojlim_N \{0, \dots, 9\}^N \cong \{0, \dots, 9\}^{\mathbb{N}},$$

and the intersection of its Archimedean cylinders consists of a unique real number. Consequently, the connection generates the complete expansion; each finite precision is a projection of that infinite output.

Proof. For a precision N , the rational closure and propagation brackets and consecutive Perron quotients are refined until contained in a single decimal cylinder of diameter 10^{-N} . The map Ext, the carry cocycle and TPK truncation make the cylinder of order $N + 1$ project onto that of order N . The cylinders are nested and their diameters tend to zero; their inverse limit determines a unique point. Since N is arbitrary and the extension relation has no terminal depth, the conclusion holds for every finite length and, jointly, for the infinite expansion. $\square$

The run of 396 levels, 2376 trits, 43 turns, 387 pairs $(K, K + 9)$ per channel and 1000 digits verifies an extensive material instance of the theorem. The run of 10,000 digits verifies another instance. The quantifier $\forall N$ and compatibility $\rho_{N+1,N} p_{N+1} = p_N$, together with existence of the inverse-limit element p_∞ , prove the infinite decimal output; the finite calculations are reproducible certificates of its projections.

The coordinate α belongs to the same limit state. Its integer route extends through $T_{\alpha,\infty}$, whereas the simple root of $E_{21/22}$ supplies a second internal reader. Stability checked to 10,000 digits certifies remote carry propagation; the mathematical object expressed by both readers is fixed before any metrological comparison.

Proposition 17.38 (Complete prolongation of the integer route of α). *Let $B = 1000$, let (P_k) , (E_k) , (Φ_k) be the three block sequences published by the coinductive state and let (K_k) be the prolonged dodecaphase sequence. The four series*

$$p = \sum_{k \geq 1} P_k B^{-k}, \quad e_0 = \sum_{k \geq 1} E_k B^{-k}, \quad \phi_0 = \sum_{k \geq 1} \Phi_k B^{-k}, \quad \kappa = \sum_{k \geq 1} K_k B^{-k}$$

converge absolutely and determine

$$\alpha_A = p + e_0 - \phi_0 - \kappa.$$

The carry recurrence is the base- B writing of this equality. For each M , a sufficiently deep truncation determines the first M digits of the canonical expansion of α_A ; the tail error is bounded by a constant multiplied by B^{-N} . Consequently, $T_{\alpha,\infty}$ publishes a complete decimal sequence, and its runs at 3000 or 10000 digits are finite projections of that output.

Proof. Each block belongs to a bounded integer set, so the four series are dominated by a geometric series. The partial sum through N differs from the complete sum by $O(B^{-N})$. Euclidean division from right to left transports that tail exactly through the carry; by uniqueness of the canonical expansion, every prefix separated from a cylinder boundary stabilizes as N increases, and at a boundary the canonical nonterminal convention applies. This simultaneously defines all digits, not a sequence of independent numerical targets. $\square$

Corollary 17.39 (Separation between generation and recognition). *Machin's identities, the exponential series, the Perron ray and the decimal chart are exact realizations of characters previously selected by $\S 9$. Conventional names π , e , φ , α and metrological tables recognize the outputs; they do not determine orbits, transports, carries or prefixes.*

Construction of the elliptic generator $i = \sqrt{-1}$ and phase geometry

The internal root of -1 and the tritic exponential family

The root of -1 is not introduced as a formal symbol without an antecedent. The six units modulo nine are coordinatized over $\mathbb{P}^1(\mathbb{F}_5)$, and the quadratic character of $\mathbb{F}_5$ determines the Paley conference matrix. Its ternary reduction is

$$A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix} \in M_6(\mathbb{F}_3).$$

The conference identity reduces to

$$\boxed{A_W^2 = 2I_6 = -I_6.} \quad (18.1)$$

Therefore, A_W is a quarter-turn operator in the ternary space. After extending scalars to $\mathbb{F}_9$, it decomposes into the conjugate eigenspaces of the roots of $X^2 + 1$. The common integral source

$$Q_{\mathbb{Z}} = \mathbb{Z}[u]/(u^2 + 1)$$

has a finite realization $u \mapsto A_W$ and a real realization $u \mapsto J_e$. The source coordinates the quadratic relation; it does not identify fields of different characteristics.

The three TRIT regimes are carried out by means of operators

$$J_{+1}^2 = -I, \quad J_0^2 = 0, \quad J_{-1}^2 = I.$$

Separating the even and odd powers of the exponential yields a single family:

$$\boxed{\exp(tJ_\tau) = C_\tau(t)I + S_\tau(t)J_\tau, \quad \tau \in \{+1, 0, -1\},} \quad (18.2)$$

where

$$(C_{+1}, S_{+1}) = (\cos, \sin), \quad (C_0, S_0) = (1, t), \quad (C_{-1}, S_{-1}) = (\cosh, \sinh).$$

Its three governing specializations are

$$\exp(\pi J_{+1}) + I = 0, \quad \exp(J_0) = I + J_0, \quad \exp(2(\log \varphi)J_{-1}) = F_{\text{av}}^2. \quad (18.3)$$

The elliptic branch contains the complex circular closure; the parabolic branch, threshold transport; the hyperbolic branch, golden self-scaling. Euler's equation thus appears as a specialization of a family that coordinates rotation, boundary and hyperbolic expansion.

18.1 Elliptic realization of $J^2 = -I$ through the Hopf fibration and Möbius holonomy

18.1.1 Complex structure and Hopf fibration

Let $S^3 \subset \mathbb{C}^2$ and

$$h(z_0, z_1) = (2\Re(z_0\bar{z}_1), 2\Im(z_0\bar{z}_1), |z_0|^2 - |z_1|^2) \in S^2.$$

On the Hopf torus

$$T_\eta = \{(z_0, z_1) : |z_0| = \cos \eta, |z_1| = \sin \eta\},$$

the curves

$$\gamma_+(t) = (\cos \eta e^{it}, \sin \eta e^{it}),$$

$$\gamma_-(t) = (\cos \eta e^{it}, \sin \eta e^{-it})$$

have distinct behaviors: γ_+ is a Hopf fiber and γ_- projects with degree two. Under stereographic projection, the homology classes $(1, 1)$ and $(1, -1)$ produce the two diagonal Villarceau families.

The total complex conjugation

$$K(z_0, z_1) = (\bar{z}_0, \bar{z}_1)$$

preserves each unoriented family and reverses its orientation. It does not interchange the two families. Partial conjugation can interchange them, but it is neither a complex-linear map nor the standard global antiunitary. This correction is essential to avoid manufacturing a particle–antiparticle symmetry from the toroidal drawing.

18.1.2 Double covering and spinor return

The map $SU(2) \rightarrow SO(3)$ is a double cover. A loop of 2π in $SO(3)$ can lift to $-I$ in $SU(2)$; the loop of 4π recovers I . The sign structure does not turn a circle into a Möbius band. Obtaining a band requires a real line bundle whose holonomy character is -1 .

Let $L_\chi \rightarrow S^1$ be the associated bundle to

$$\chi : \pi_1(S^1) \longrightarrow \{\pm 1\}, \quad \chi(1) = -1.$$

Its total space is a Möbius band. The route $C_M = -\text{rev}$ provides the sign candidate; the Villarceau circle provides the loop; the physical intertwiner must prove that the former is the holonomy of the latter.

18.1.3 Toroidal realization of the operator $B_{\text{mem}}^\varepsilon$ and determination of the Villarceau angle

With $B_c = 7I + 2S$, define the oriented memory operator

$$B_{\text{mem}}^\varepsilon = 5I + 2\varepsilon S, \quad \varepsilon = \pm 1.$$

For $\varepsilon = +1$,

$$\text{Spec}(B_{\text{mem}}^+) = \{7, 3\}, \quad \det B_{\text{mem}}^+ = 21.$$

Interpreting the values $(7, 3)$ as the outer and inner equatorial boundaries in a common positive chart yields

$$R + r = 7\ell, \quad R - r = 3\ell,$$

and therefore

$$R = 5\ell, \quad r = 2\ell, \quad \sin \beta = \frac{r}{R} = \frac{2}{5}.$$

The Villarceau angle is

$$\beta = \arcsin \frac{2}{5} = 23.5781784782 \dots^\circ.$$

The algebraic theorem $\{7, 3\} \leftrightarrow (5, 2)$ is exact. The geometric premise interpreting the spectrum as the two boundaries of a torus is a declared interface. It must not be concealed, nor may the angle be used as a target.

The realization of the full block B_c retains another reading: $R : r = 7 : 2$ and $\beta = \arcsin(2/7)$. These are not rival values; they belong to two functors: total block and memory mode. The physical priority of the latter must be decided by the intertwiner with the dynamics, not by decimal proximity to another constant.

18.1.4 Action of the operator $J^2 = -I$ on the conjugate angular coordinates

HMT's native angle is a phase $u \in \mathbb{R}/\mathbb{Z}$. The readers publish

$$\theta_{\text{rad}} = 2\pi u, \quad \theta_{\text{deg}} = 360u.$$

Radians and degrees are distinct charts of the same circular element. The radian is not more “ontological” by virtue of being dimensionless; its naturality follows from arc length. Degrees provide another absolute coordinate of the full turn. Any relation with α^{-1} , φ , or CKM must first be formulated as an invariant quotient or holonomy and only thereafter evaluated in an angular chart.

The exact identity already available

$$\tanh^2(2 \log \varphi) = \frac{5}{9}$$

connects the Perron–Fibonacci channel to the spectral quotient of B_c . Its complement is $4/9$, the normalized gap. This relation is structural. It does not imply that every angle constructed with φ is a physical angle.

18.1.5 Brouwer, Möbius, and degree

Brouwer's fixed-point theorem, fractional linear Möbius transformations, and the Möbius band are three distinct objects. They are related only after fixing domain, action, and bundle. The elliptic advance and retardation orbits may be modeled as two sections of a bundle with sign holonomy; the Möbius transformation describes the conformal action in the chart; Brouwer's theorem guarantees a fixed point for continuous maps of the disk. None of these statements replaces the realization diagram.

Remark 18.1 (Result and boundary). The double degree of the Hopf curve, the orientation of total conjugation, and the algebraic construction $7, 3 \mapsto 5, 2$ are proved. The toroidal reading $(5:2)$, the Möbius holonomy, and the particle–antiparticle interpretation form a chain of interfaces that must share a single intertwiner. Volume does not force that intertwiner through an angular coincidence.

Hensel–Witt extension of $-1/12$ and modular memory

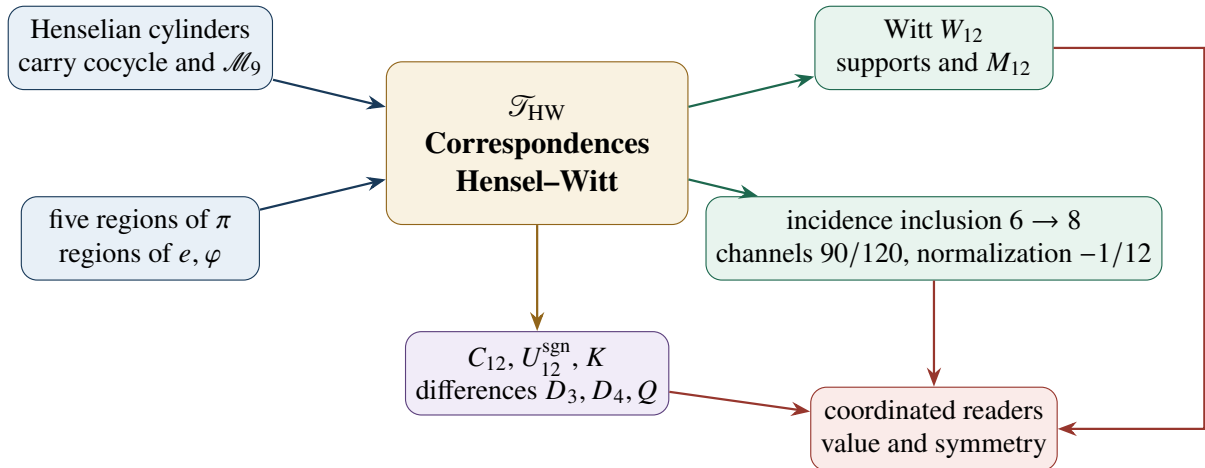
19.1 Hensel–Witt transduction operator between microregions of π , holonomy Γ_9 , the dodecaphasic register and hexad–octad incidence

The NTHW is the data and maps package

$$\mathfrak{N}_{\text{HW}} = (\mathcal{R}_{\pi}^{\text{micro}}, \mathcal{G}_9, C_{12}, K, \mathcal{W}_{12 \rightarrow 24}, \mathcal{I}_{6 \rightarrow 8}),$$

with the following compatibilities:

1. the Hensel regions share a quotient word and retain distinct memory;
2. $\mathcal{G}_9$ transports orientation and produces monodromy;
3. C_{12} organizes the signed observable and the seal K ;
4. the Witt reading converts supports of the same boundary into hexads and stars;
5. the extension $W_{12} \rightarrow W_{24}$ produces the hexad–octad interface;
6. the Archimedean, exceptional and dimensional readers receive invariants of the same state.



Compatibility of cylinders, dodecaphase and incidences on the same integral holonomic state.

Figure 19.1: The NTHW as a typed interface. The central mathematical task is to make the diagram commute, not to accumulate coincidences around a name.

NTHW gives a formal name to the place previously described as the “Rosetta Stone.” The substitution is conceptual: a stone translates through resemblance; a transduction specifies input, output, preserved datum and loss of information.

19.2 Construction of the Witt system from the compatible ternary coding

The reversible APP boundary selects, with declared gauges, a matrix A_W over $\mathbb{F}_3$ satisfying

$$A_W A_W^T = 2I \pmod{3}.$$

The graphic code

$$C_W = \{(q, qA_W) : q \in \mathbb{F}_3^6\}$$

is the extended ternary Golay code. Its weight enumerator is

$$1 + 264y^6 + 440y^9 + 24y^{12}.$$

The 264 words of weight six, identified up to sign, yield 132 supports. Every five-point subset belongs to a unique hexad:

$$W_{12} = S(5, 6, 12).$$

The field of 495 star involutions generates a group of order 95 040, recognized as M_{12} . An isolated star does not generate the group. The ordered HMT frame has trivial stabilizer and therefore fixes a gauge within the action.

19.3 Hexad–octad realization of the 90/120 channels previously derived from the tritic incidence of the TPK

Let H be a hexad and O_H an octad containing it. Marking one point and a pair of points of the hexad yields

$$F_6(H) = H \times \binom{H}{2}, \quad |F_6| = 6 \binom{6}{2} = 90.$$

If the marked point can traverse the octad:

$$F_8(H) = O_H \times \binom{H}{2}, \quad |F_8| = 8 \binom{6}{2} = 120.$$

The inclusion $H \hookrightarrow O_H$ induces $F_6 \hookrightarrow F_8$. Its normalized defect is

$$\boxed{\frac{90 - 120}{24 \binom{6}{2}} = \frac{6 - 8}{24} = -\frac{1}{12}}.$$

This combinatorial theorem precedes Barbero's angular reading. The two objects share the interface $6 \rightarrow 8$, but are not identical.

19.4 4 operational realizations of $-1/12$ and separation of their domains

The construction contains four realizations that must be presented together, not counted as four independent proofs of a single provenance.

19.4.1 Angular realization through the elementary increment dodecaphasic

One tooth of C_{12} measures 30° . Then

$$\frac{30}{120} - \frac{30}{90} = \frac{1}{4} - \frac{1}{3} = -\frac{1}{12}.$$

It is the angular shadow of the pattern 90/120.

19.4.2 Oriented pseudoinverse of the Gram operator on the surviving subspace

In the certified jump $60 \rightarrow 81$, two fibers have sizes 0 and 12. The Gram operator is

$$K_{60,81} = \text{diag}(0, 12).$$

On the live subspace, its pseudoinverse is $1/12$; with the negative boundary orientation,

$$E_{\text{surv}} = -K_{60,81}^+$$

yields $-1/12$.

19.4.3 Scale extraction by the 2nd triadic difference

The realization via a triadic scale extractor is the second-difference construction developed in full in [section 19.5.3](#). This first occurrence establishes its role within the inventory of realizations; the indicated residence preserves the definitions of T_L , $a(L)$, $A_1(L)$, $C(L)$, the independence of the residue from L , and the exact status of the normalization.

19.4.4 Modular realization through the quotient $\eta(\tau)/\eta(3\tau)$ and the function t_3

The eta function satisfies

$$\frac{\eta(\tau)}{\eta(3\tau)} = q^{-1/12} \prod_{n \geq 1} (1 + q^n + q^{2n})^{-1}.$$

The exponent comes from $1/24 - 3/24 = -1/12$. Taking twelve copies:

$$t_3(q) = \left(\frac{\eta(\tau)}{\eta(3\tau)} \right)^{12} = q^{-1} - 12 + 54q - 76q^2 - 243q^3 + \dots$$

The coefficient 54 coincides with the APP defect. The corridor

$$j = \frac{(t_3 + 27)(t_3 + 243)^3}{t_3^3}$$

connects afterwards with the classical modular branch.

Remark 19.1 (What Has Been Proved in the Corridor). The APP defect 54, the eta exponent $-1/12$, the hexad–octad defect, and the modular identities once t_3 has been fixed are exact. The strong assertion is that all of them are natural images of a single TPK operator. That naturality must be proved by means of the NTHW diagram; it does not follow merely from repeating the same integers.

19.5 5 algebraic realizations of $-1/12$ and separation of their domains

In addition to the normalized defect of the hexad–octad inclusion, five operator realizations of the same rational number appear. Their domains are specified separately; equality of values does not identify the operators without an additional intertwiner.

19.5.1 Difference of dodecaphasic periodicities

One tooth of C_{12} measures 30° . Then

$$\frac{30}{120} - \frac{30}{90} = \frac{1}{4} - \frac{1}{3} = -\frac{1}{12}.$$

It is the oriented difference between the periods of 90° and 120° in this realization.

19.5.2 Pseudoinverse of the Gram Operator on the Live Component

In the exact jump $60 \rightarrow 81$, two fibers have sizes 0 and 12. The Gram operator is

$$K_{60,81} = \text{diag}(0, 12).$$

On the nonzero component of its range, the pseudoinverse is $1/12$; with the negative boundary orientation,

$$E_{\text{surv}} = -K_{60,81}^+$$

yields $-1/12$.

19.5.3 2nd difference between triadic scales

Define

$$T_L = \sum_{n=1}^{L-1} n(L-n) = \frac{L(L^2-1)}{6}, \quad a(L) = -\frac{T_L}{L^2} = -\frac{L}{6} + \frac{1}{6L}.$$

The first scale subtraction is

$$A_1(L) = a(L) - \frac{a(3L)}{3} = \frac{4}{27L},$$

and the second

$$C(L) = LA_1(L) - \frac{3L}{3}A_1(3L) = \frac{8}{81}.$$

The residue $8/81$ is independent of L . To obtain $-1/12$, the declared normalization is applied

$$R(L) = -\frac{27}{32}C(L) = -\frac{1}{12}.$$

The calculation derives $8/81$; it does not by itself derive the factor $-27/32$.

19.5.4 Modular chain

The eta function satisfies

$$\frac{\eta(\tau)}{\eta(3\tau)} = q^{-1/12} \prod_{n \geq 1} (1 + q^n + q^{2n})^{-1}.$$

The exponent comes from $1/24 - 3/24 = -1/12$. Taking twelve copies:

$$t_3(q) = \left(\frac{\eta(\tau)}{\eta(3\tau)} \right)^{12} = q^{-1} - 12 + 54q - 76q^2 - 243q^3 + \dots.$$

The coefficient 54 coincides with the APP defect. The rational identity

$$j = \frac{(t_3 + 27)(t_3 + 243)^3}{t_3^3}$$

establishes the connection with the classical modular branch.

19.5.5 Defect of cyclic difference projectors

Let S be the cyclic shift on $\mathbb{Q}^{12}$ and define

$$D_r = I - S^r.$$

The kernel of D_r consists of the vectors constant on each orbit of S^r ; therefore,

$$\dim \ker D_r = \gcd(12, r).$$

Let $\Pi_{\ker D_3}$ and $\Pi_{\ker D_4}$ be the rational projectors onto $\ker D_3$ and $\ker D_4$, and let $\tau(T) = \text{Tr}(T)/12$ be the normalized trace. Then

$$\tau(\Pi_{\ker D_3} - \Pi_{\ker D_4}) = \frac{\text{rank } \Pi_{\ker D_3} - \text{rank } \Pi_{\ker D_4}}{12} = \frac{3 - 4}{12} = -\frac{1}{12}.$$

This realization expresses the rational as the transverse defect of steps three and four in the twelve-phase cycle.

Remark 19.2 (Domains of the realizations). The APP defect 54, the eta exponent $-1/12$, the combinatorial defect of the hexad–octad lift, and the modular identities once t_3 has been fixed are exact. The value of the pseudoinverse, the normalized extractor, and the projector defect are likewise exact. Each equality preserves its domain and its corresponding map.

Quadratic characters and spectral constants: Catalan, Apéry, Euler–Mascheroni and Stieltjes

20.1 Operatorial representation of Dirichlet characters

The constants in this chapter are grouped because they share a spectral construction independent of their decimal expansions and free of target values. Let

$$Ne_n = ne_n, \quad n \in \mathbb{N},$$

in $\ell^2(\mathbb{N})$. A residual character χ defines the diagonal operator $Q_\chi e_n = \chi(n)e_n$; whenever the power is trace class,

$$\text{Tr}(N^{-s}Q_\chi) = L(s, \chi). \quad (20.1)$$

The equation preserves the modulus, orientation, and provenance of each character; therefore, it does not identify distinct residual objects.

20.2 HMT construction of the residue characters

The preceding diagonal operator is not the initial algebraic datum. The selection chain is

$$\begin{aligned} \text{APP} &\longrightarrow \widehat{\mathbb{T}} \longrightarrow \text{TPK} \longrightarrow \{\text{residue modulo } 3, C_P\} \\ &\longrightarrow \{\chi_{-3}, J = C_P \bmod 3\} \longrightarrow \{\chi_{-3}, J^2 = -I, \chi_{-4}\} \\ &\longrightarrow Q_3, Q_4. \end{aligned}$$

The triadic residual functional determines the difference between the two nonzero oriented classes. The incidence sector uses the symmetric Paley conference matrix of order six,

$$C_P^\top = C_P, \quad C_P C_P^\top = 5I_6.$$

Its reduction $J = C_P \bmod 3$ satisfies

$$J^2 = 2I_6 = -I_6 \quad \text{in } \text{End}_{\mathbb{F}_3}(\mathbb{F}_3^6).$$

The torsional quarter then determines the cyclic phase 1, i, −i. Applying the Chinese remainder theorem preserves both residual data and produces $Q_{12} = Q_3 Q_4$. The trace $\text{Tr}(N^{-s}Q)$ is evaluated at the terminal stage; neither the Dirichlet series nor its decimal expression enters the selection of the character.

The construction is made explicit by the tables

$$\chi_{-3}(n) = \begin{cases} 0, & 3 \mid n, \\ 1, & n \equiv 1 \pmod{3}, \\ -1, & n \equiv 2 \pmod{3}, \end{cases} \quad \chi_{-4}(n) = \begin{cases} 0, & 2 \mid n, \\ 1, & n \equiv 1 \pmod{4}, \\ -1, & n \equiv 3 \pmod{4}. \end{cases} \quad (20.2)$$

En $\ell^2(\mathbb{N})$,

$$Q_3 e_n = \chi_{-3}(n)e_n, \quad Q_4 e_n = \chi_{-4}(n)e_n.$$

As they are diagonals, they commute; the Chinese remainder theorem gives

$$(Q_3 Q_4) e_n = \chi_{-3}(n) \chi_{-4}(n) e_n = \chi_{12}(n) e_n. \quad (20.3)$$

The product simultaneously preserves the triadic residue and the quarter of torsion. Consequently, the character modulo twelve is determined before performing any evaluation of its values L .

Remark 20.1 (Negative control). The chain is invalidated if Q_3 does not distinguish the two oriented classes, if $J^2 \neq -I$, if Q_3Q_4 does not have the declared CRT table, or if the operator is selected from the value of the constant.

20.3 Internal complex structure and Catalan's constant

The internal complex structure satisfies

$$J^2 = -I. \quad (20.4)$$

The equality $J^2 = -I$ defines a representation $\rho : C_4 \rightarrow \text{Aut}_{\mathbb{F}_3}(\mathbb{F}_3^6)$, $\rho(j) = J$. To realize it spectrally, take the unitary character $\nu : C_4 \rightarrow U(1)$, $\nu(j) = i$, and the phase map $n \mapsto j^n$. On $\ell^2(\mathbb{N})$, the induced diagonal representation is

$$U_4 e_n = \nu(j^n) e_n = i^n e_n.$$

Its oriented part is

$$Q_4 = \frac{U_4 - U_4^*}{2i}, \quad Q_4 e_n = \sin\left(\frac{\pi n}{2}\right) e_n = \chi_{-4}(n) e_n.$$

This is the explicit lift between the finite generator of order four and the residual operator; $\mathbb{F}_9$ is not identified with $\mathbb{C}$.

On odd integers, the same character can be written

$$\chi_J(n) = i^{n-1} = \chi_{-4}(n).$$

Equivalently,

$$Q_4 = \frac{U_4 - U_4^*}{2i}, \quad Q_4 e_n = \chi_{-4}(n) e_n. \quad (20.5)$$

The Catalan constant appears as

$$G = \text{Tr}(N^{-2}Q_4) = L(2, \chi_{-4}) = \sum_{n \geq 0} \frac{(-1)^n}{(2n+1)^2}. \quad (20.6)$$

The unitary lift of the torsion quarter generates exactly the character whose quadratic moment is Catalan's constant.

Operational proof. The operator $N^{-2}Q_4$ is trace-class because $\sum_{n \geq 1} |\chi_{-4}(n)| n^{-2} < \infty$. Its trace is therefore the sum of the diagonal elements. The table [equation \(20.2\)](#) annuls the even pairs and alternates the odd ones:

$$\text{Tr}(N^{-2}Q_4) = 1 - \frac{1}{3^2} + \frac{1}{5^2} - \frac{1}{7^2} + \cdots$$

The identity with $L(2, \chi_{-4})$ is the definition of the Dirichlet series; the equality with $\text{Im Li}_2(i)$ is obtained by taking the imaginary part of $\sum_{n \geq 1} i^n / n^2$. $\square$

The terminal value is

$$G = 0.9159655941772190150546035149 \dots$$

The determining structure is the identity $J^2 = -I$, which induces exactly the alternating sign of the odd classes; the decimal expansion is its terminal evaluation.

20.4 Residual composition of the triadic–quaternary type

The residual functional modulo three provides χ_{-3} . Its CRT product with χ_{-4} defines

$$\chi_{12} = \chi_{-3}\chi_{-4}. \quad (20.7)$$

On the units modulo twelve,

$$\chi_{12}(1) = 1, \quad \chi_{12}(5) = -1, \quad \chi_{12}(7) = -1, \quad \chi_{12}(11) = 1,$$

and vanishes outside of them. Two exact evaluations are

$$L(1, \chi_{12}) = \frac{\log(2 + \sqrt{3})}{\sqrt{3}}, \quad (20.8)$$

$$L(2, \chi_{12}) = \frac{\pi^2}{6\sqrt{3}}. \quad (20.9)$$

The first contains the hyperbolic unit

$$\sqrt{3}L(1, \chi_{12}) = \operatorname{arcosh} 2 = \log(2 + \sqrt{3}). \quad (20.10)$$

Together with $\log(3 + 2\sqrt{2}) = 2 \log(1 + \sqrt{2})$ and $\log(30 + \sqrt{899})$, this identity determines a family of Pell scales whose members retain their respective algebraic provenances.

Certified terminal evaluations are

$$L(1, \chi_{12}) = 0.7603459963009463475310942549 \dots,$$

$$L(2, \chi_{12}) = 0.9497031262940093952634984917 \dots$$

Evaluations dodecaphasic. Grouping by residue classes,

$$\begin{aligned} L(1, \chi_{12}) &= \sum_{n \geq 0} \left(\frac{1}{12n+1} - \frac{1}{12n+5} - \frac{1}{12n+7} + \frac{1}{12n+11} \right) \\ &= \int_0^1 \frac{1-x^4-x^6+x^{10}}{1-x^{12}} dx = \int_0^1 \frac{1-x^4}{1+x^6} dx. \end{aligned}$$

A real decomposition into the quadratic factors of $1+x^6$ produces $\log(2 + \sqrt{3})/\sqrt{3}$. For the second moment, if ψ_1 is the trigamma function,

$$\begin{aligned} L(2, \chi_{12}) &= \frac{1}{144} [\psi_1(1/12) - \psi_1(5/12) - \psi_1(7/12) + \psi_1(11/12)] \\ &= \frac{\pi^2}{144} \left(\csc^2 \frac{\pi}{12} - \csc^2 \frac{5\pi}{12} \right) = \frac{\pi^2}{6\sqrt{3}}, \end{aligned}$$

where has been used $\psi_1(z) + \psi_1(1-z) = \pi^2 \csc^2(\pi z)$. The two proofs use the CRT table and not a target decimal value. $\square$

20.5 Euler and Stieltjes Constants and the Theta–Mellin Transform

Let

$$\vartheta(t) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 t}, \quad t > 0.$$

For $\Re s > 1$, the theta–Mellin functional recovers the convergent identity

$$\pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s) = \frac{1}{2} \int_0^\infty (\vartheta(t) - 1) t^{s/2-1} dt.$$

The meromorphic continuation is obtained by splitting the integral at $t = 1$, using $\vartheta(t) = t^{-1/2} \vartheta(1/t)$, and separating the polar terms. Its expansion at $s = 1$,

$$\zeta(s) = \frac{1}{s-1} + \sum_{j \geq 0} \frac{(-1)^j \gamma_j}{j!} (s-1)^j, \quad (20.11)$$

situates $\gamma = \gamma_0$ and the Stieltjes constants in a single family of renormalized moments. This organization expresses an analytical common provenance, without attributing an interpretation that is physically identical to all its levels.

The Catalan constant often appears as an alternating series:

$$G = 1 - \frac{1}{3^2} + \frac{1}{5^2} - \frac{1}{7^2} + \cdots.$$

That formula gives its value, and the HMT genealogy explains why this specific alternation exists.

In HMT, the origin is in the torsion chamber.

There exists an internal operator J such that

$$J^2 = -I.$$

The root of -1 emerges from two successive applications of a quarter turn. The complex structure is the algebraic memory of an internal rotation.

That turn generates a character of order four. When it is published on the positive integers, it produces exactly the residual pattern that alternates among $+1, 0, -1, 0$.

The Catalan constant then appears as a spectral trace:

$$G = L(2, \chi_{-4}) = \text{Tr}(N^{-2} Q_4).$$

This changes its meaning. Catalan’s constant is simultaneously the sum of a particular series and the spectral response of the integers to a quarter of torsion.

Each term of the series records how the internal rotation acts on a residue class. The alternation is the arithmetic shadow of the rotation.

The triadic structure has its own character modulo 3. Combining it with the character of order four, the Chinese remainder theorem produces a character modulo 12:

$$\chi_{12} = \chi_{-3} \chi_{-4}.$$

The module 12 appears as the minimal common space in which triadic orientation and the quarter-torsion can coexist.

The dodecaphase is, in that sense, a reconciliation of 3 and 4.

Of the combined character emerges

$$L(1, \chi_{12}) = \frac{\log(2 + \sqrt{3})}{\sqrt{3}},$$

y

$$L(2, \chi_{12}) = \frac{\pi^2}{6\sqrt{3}}.$$

La cantidad

$$2 + \sqrt{3}$$

is the fundamental unit of the real field $\mathbb{Q}(\sqrt{3})$. Its logarithm represents a hyperbolic shift. Once again, a finite residue structure opens an infinite multiplicative dynamics.

Here rotation, torsion, arithmetic character, Pell unit, and hyperbolic displacement meet.

The Euler–Kronecker constant of the field $\mathbb{Q}(\sqrt{3})$ prolongs this structure. It measures the regular term that remains after separating the principal pole of the field’s zeta function. In the HMT genealogy, this term constitutes the regular invariant of fine arithmetic memory that remains after removing the universal divergence.

The Bendersky–Glaisher family extends the mechanism into a tower. The spectral derivatives of zeta generate that tower. Apéry’s constant and other constants associated with odd values are organized as members of the same functional family.

The novel contribution lies in the genealogy that unifies the classical constants:

- the quarter of torsion produces the complex structure;
- the residue triadico produces the structure modulo 3;
- their composition produces the character modulo 12;
- the character generates values L ;
- the values L open the quadratic field and its spectral constants.

Geometric torsion has become arithmetic while preserving entirely its orientation.

Euler–Kronecker, Meissel–Mertens and prime distributions in the double reading of the sheets

21.1 Euler–Kronecker constant of the character of dodecaphase type

The factorization of Dedekind

$$\zeta_{\mathbb{Q}(\sqrt{3})}(s) = \zeta(s)L(s, \chi_{12}) \quad (21.1)$$

gives, upon comparing the constant terms in $s = 1$,

$$\gamma_{\mathbb{Q}(\sqrt{3})} = \gamma + \frac{L'(1, \chi_{12})}{L(1, \chi_{12})} = 1.0539656082484825800 \dots \quad (21.2)$$

This constant represents the normalized finite term at the pole of the zeta function of the real dodecaphase field. It is not a variant of Euler's γ , but the arithmetic correction induced by χ_{12} .

In fact, writing $u = s - 1$,

$$\zeta(s) = u^{-1} + \gamma + O(u), \quad L(s, \chi_{12}) = L_1 + L'_1 u + O(u^2),$$

we obtain

$$\zeta_{\mathbb{Q}(\sqrt{3})}(s) = \frac{L_1}{u} + (\gamma L_1 + L'_1) + O(u). \quad (21.3)$$

The Euler–Kronecker constant is the quotient of the finite term by the residue; dividing equation (21.3) by L_1 proves equation (21.2). This derivation simultaneously explains the residue

$$\operatorname{Res}_{s=1} \zeta_{\mathbb{Q}(\sqrt{3})}(s) = \frac{\log(2 + \sqrt{3})}{\sqrt{3}}$$

and the finite correction. The numerical value is certified by evaluating $L(1, \chi_{12})$ and $L'(1, \chi_{12})$ with the same residue table and a tail bound; it is not imported from a table of quadratic fields.

21.2 Arithmetic constants associated with prime numbers

Once the irreducible arithmetic cycles have been selected, two distinct families are obtained:

$$B_1 = \lim_{x \rightarrow \infty} \left(\sum_{p \leq x} \frac{1}{p} - \log \log x \right), \quad (21.4)$$

$$C_2 = \prod_{p > 2} \frac{p(p-2)}{(p-1)^2}. \quad (21.5)$$

B_1 regularizes a local sum; C_2 is a multiplicative density of twin-prime pairs. Both require tail control. They are not deduced from one another by sharing the primes.

Regular products and the Bendersky–Glaisher–Kinkelin system

22.1 Generation of the Bendersky–Glaisher–Kinkelin family via zeta regularization

Let $\mathcal{Z}_3(s) := \zeta(s)$ be the scalar realization of the triadic zeta functional in the normalization of this volume, $H_k = \sum_{j=1}^k j^{-1}$ (with $H_0 = 0$), and let B_{k+1} be the Bernoulli number under convention $B_1 = -1/2$. We define

$$A_k = \exp\left(\frac{H_k B_{k+1}}{k+1} - \mathcal{Z}_3'(-k)\right), \quad k \geq 0. \quad (22.1)$$

The first levels include

$$A_0 = \sqrt{2\pi},$$

and the Glaisher–Kinkelin constant in A_1 . For $n \geq 1$,

$$\zeta(2n+1) = (-1)^{n+1} \frac{2(2\pi)^{2n}}{(2n)!} \log A_{2n}. \quad (22.2)$$

Apéry’s constant, $\zeta(3)$, is the first nontrivial even level. Its presence in the photon gas of [chapter 36](#) constitutes a subsequent physical realization of this spectral value.

The first part of the tower can be seen in a synoptic form:

k	identidad estructural	terminal value
0	$A_0 = \exp[-\zeta'(0)] = \sqrt{2\pi}$	2.5066282746310005024...
1	$\log A_1 = 1/12 - \zeta'(-1)$	1.2824271291006226369...
2	$\log A_2 = -\zeta'(-2) = \zeta(3)/(4\pi^2)$	1.0309167521973921...
3	$\log A_3 = -11/720 - \zeta'(-3)$	0.9795555269428446...
4	$\log A_4 = -\zeta'(-4)$	0.9920479745250403...

Relation to the odd values. The functional equation of ζ , developed in $s = -2n$, gives

$$\zeta'(-2n) = (-1)^n \frac{(2n)!}{2(2\pi)^{2n}} \zeta(2n+1). \quad (22.3)$$

Since $B_{2n+1} = 0$ for $n \geq 1$, the definition [equation \(22.1\)](#) reduces $\log A_{2n} = -\zeta'(-2n)$. Solving for $\zeta(2n+1)$ proves [equation \(22.2\)](#). The index $2n$ preserves the derivation level: it is not an editorial label. $\square$

Gauss dynamics and the Khinchin, Lévy and Lochs constants

23.1 Cofinality between HMT cylinders and continued fraction cylinders

An irrational HMT section determines a nested family of compact cylinders. Its Archimedean evaluation is a point $x \in (0, 1) \setminus \mathbb{Q}$. The continued fraction of x determines classical cylinders

$$I(a_1, \dots, a_n).$$

Proposition 23.1 (Cofinality). *For every finite cylinder $I(a_1, \dots, a_n)$ containing the evaluation x , there is an HMT depth $m(n)$ such that the HMT cylinder at level $m \geq m(n)$ is contained in $I(a_1, \dots, a_n)$.*

Proof. Given n , the continued-fraction cylinder $I(a_1, \dots, a_n)$ is an open neighborhood of the irrational x . There then exists $\varepsilon_n > 0$ with $(x - \varepsilon_n, x + \varepsilon_n) \subset I(a_1, \dots, a_n)$. The HMT cylinders C_m contain x , are nested, and satisfy $\text{diam } C_m \rightarrow 0$. Choose $m(n)$ with $\text{diam } C_{m(n)} < \varepsilon_n$. Since $x \in C_{m(n)}$, every point of that cylinder lies from x at a distance of less than ε_n , hence $C_{m(n)} \subset I(a_1, \dots, a_n)$. Nesting gives the inclusion for every subsequent level. $\square$

The proof uses compactness, nesting, and uniqueness of evaluation. The result establishes the transport of approximations, but does not imply conjugation of the TPK memory shift with the Gauss map on all deep fibers.

The complete genealogy of this realization is

$$\text{APP} \rightarrow \widehat{\mathbb{T}} \rightarrow \text{TPK} \rightarrow (C_m, \rho_{m+1,m})_{m \geq 0} \rightarrow x = \bigcap_m C_m \rightarrow (I(a_1, \dots, a_n))_{n \geq 1}.$$

The HMT cylinders preserve route, phase, and carry; the terminal map projects these coordinates onto continued-fraction prefixes. Cofinality proves that this projection converges to the same point. A dynamical intertwiner would also require a lift $\widetilde{T}_G$ with commutative squares over the valuation fibers, a property not deduced from cofinality.

23.2 Gauss measure and Khinchin and Levy observables

The Gauss map [Khi64, IK02]

$$T_G(x) = \frac{1}{x} - \left\lfloor \frac{1}{x} \right\rfloor$$

preserve

$$d\mu_G(x) = \frac{dx}{\log 2(1+x)}. \quad (23.1)$$

Khinchin's constant is defined by

$$\log K = \int_0^1 \log a_1(x) d\mu_G(x). \quad (23.2)$$

The distribution of the first digit is computed directly:

$$\mu_G(a_1 = k) = \frac{1}{\log 2} \int_{1/(k+1)}^{1/k} \frac{dx}{1+x} = \log_2 \left(\frac{(k+1)^2}{k(k+2)} \right). \quad (23.3)$$

By the ergodic theorem,

$$K = \prod_{k \geq 1} k^{\log_2((k+1)^2/[k(k+2)])} = 2.685452001065306445309714835 \dots \quad (23.4)$$

for μ_G -almost every section. The product is not applied to each point: the section φ^{-1} , for example, has all its digits equal to one and is an immediate falsification criterion for a pointwise reading. A certification of equation (23.4) truncates the sum of logarithms and bounds the tail by the estimate

$$\mu_G(a_1 = k) = O(k^{-2}), \quad \sum_{k > N} \mu_G(a_1 = k) \log k = O\left(\frac{\log N + 1}{N}\right).$$

Lévy's constant is

$$L = - \int_0^1 \log x \, d\mu_G(x) = \frac{\pi^2}{12 \log 2}. \quad (23.5)$$

Indeed,

$$\frac{1}{1+x} = \sum_{n \geq 0} (-1)^n x^n, \quad \int_0^1 (-\log x) x^n \, dx = \frac{1}{(n+1)^2},$$

so that the integral is $\eta(2)/\log 2 = \pi^2/(12 \log 2)$.

Khinchin's constant determines the logarithmic mean of the partial quotients, whereas Lévy's constant determines the exponential growth rate of the denominators. They are distinct observables of the same measured system.

23.3 Entropy (metric), Lyapunov exponent, and Lochs's constant

Since

$$\log |T'_G(x)| = -2 \log x,$$

the Lyapunov exponent and the entropy metric are

$$\chi_G = h_G = 2L = \frac{\pi^2}{6 \log 2}. \quad (23.6)$$

The Lochs ratio between continued fractions and a base- b expansion is

$$\Lambda_b = \frac{\log b}{h_G} = \frac{6 \log 2 \log b}{\pi^2}. \quad (23.7)$$

In particular,

$$\begin{aligned} \Lambda_{10} &= 0.9702701143920339257402560192 \dots, \\ \Lambda_{729} &= \frac{36 \log 2 \log 3}{\pi^2} = 2.777618966374305777705782976 \dots \end{aligned}$$

The basis $729 = 3^6$ is relevant to the ambient space $\mathbb{F}_3^6$, but equation (23.7) remains an almost-sure theorem of μ_G , not a pointwise equality for every HMT route.

The value Λ_{729} represents the asymptotic typical rate of partial quotients obtained per complete block of six trits. It does not identify the ambient set of 729 words with the other objects of cardinality 729 in the corpus. The equality is an information rate between partitions, not a bijection between states.

23.4 Gauss–Kuzmin–Wirsing Transfer Operator

The Gauss–Kuzmin–Wirsing operator acts by

$$(\mathcal{L}f)(x) = \sum_{n \geq 1} \frac{1}{(x+n)^2} f\left(\frac{1}{x+n}\right). \quad (23.8)$$

Its dominant eigenvalue corresponds to the invariant density. The GKW constant is the modulus of the subdominant eigenvalue on the zero-mean subspace. An HMT-compatible scheme uses partitions of 9^m cylinders, transfer matrices with intervals, and a tail bound.

To promote the calculation, the following must be certified: isolation of the eigenvalue, invariance of the subspace, truncation error, convergence of spectral projectors, and absence of another eigenvalue of equal modulus. The known decimal value cannot be introduced as the center of the initial interval.

A joint deformation contains Khinchin and Lévy in a single object:

$$(\mathcal{L}_{s,t}f)(x) = \sum_{n \geq 1} n^t (n+x)^{-2s} f\left(\frac{1}{n+x}\right), \quad P(s,t) = \log r(\mathcal{L}_{s,t}). \quad (23.9)$$

At the point $(s,t) = (1,0)$, the spectral perturbation gives

$$\partial_t P(1,0) = \log K, \quad -\partial_s P(1,0) = \chi_G = 2L. \quad (23.10)$$

The Hessian of P determines the asymptotic variances and covariances of $\log a_1$ and $-2 \log x$. Therefore, the Khinchin and Levy constants are two transverse derivatives of the same pressure function.

The numerical value of the subdominant eigenvalue is

$$\lambda_2 \simeq -0.3036630028987326586, \quad \kappa_{\text{GKW}} = |\lambda_2|, \quad -\log \kappa_{\text{GKW}} \simeq 1.19183673556055.$$

In this volume, those decimals constitute reference values for comparison and not a spectral certificate. The proposed HMT proof uses Galerkin or Ulam on the cofinal cylinders of depth 9^m , a Lasota–Yorke inequality, interval arithmetic to isolate λ_2 , and a rigorous bound on the tail. An interval containing two eigenvalues of the same modulus refutes the promotion, although it does not affect Khinchin, Levy, or Lochs.

23.5 Cofinal certificate of the Gauss–Kuzmin–Wirsing spectrum over the nonadic hierarchy

For Gauss–Kuzmin–Wirsing, let P_N be the certified family of truncations on the declared function space, and define

$$\Pi_m = P_{9^m}.$$

The hierarchy (Π_m) is a cofinal subsequence; uniform bounds, projector convergence and absence of spectral pollution are inherited from the system (P_N) . A different cylindrical discretization must supply its intertwiner with (P_{9^m}) .

Unimodal criticality, Feigenbaum constants and hyperbolic renormalization

24.1 Intrinsic construction of the dodecaphasic family

The Feigenbaum constants characterize the universality of period doubling. Their derivation in HMT first requires constructing an analytic unimodal family, proving its quadratic criticality, and then studying the action of the renormalization operator.

The initial map does not follow from an external harmonic choice. The return of TPK determines the oriented dodecaphase wheel U_{12}^{sign} . In the uniform sector $\eta = 0$, the normalized trace of its twelve sectors fixes $w_m = 1/12$, and the phase functional

$$\mathcal{P}_{\text{crit}} : U_{12}^{\text{sign}} \longrightarrow \mathbb{R}/2\pi\mathbb{Z}, \quad m \longmapsto \frac{(30m - 10)\pi}{180} = \pi \frac{3m - 1}{18} \quad (24.1)$$

preserves order and orientation. The causal composition is

$$\text{APP} \rightarrow \widehat{\mathbb{T}} \rightarrow \text{TPK} \rightarrow U_{12}^{\text{sign}} \xrightarrow{\mathcal{P}_{\text{crit}}} \{(\phi_m, w_m)\}_{m=1}^{12} \rightarrow u_0 \rightarrow f_\lambda.$$

Every modification of $\mathcal{P}_{\text{crit}}$, of the weights, or of the order of the phases alters the family and constitutes a criterion of refutation. These choices must be fixed before the contrast with δ_F .

The resulting dodecaphasic family starts from

$$a_m = \frac{3m - 1}{18}, \quad \phi_m = \pi a_m, \quad m = 1, \dots, 12,$$

and of the second moment

$$D_0 = \frac{1}{12} \sum_{m=1}^{12} a_m^2.$$

The identity

$$\sum_{m=1}^{12} (3m - 1)^2 = 5394 = 6 \cdot 899 \quad (24.2)$$

yields

$$D_0 = \frac{899}{648}, \quad s_0 = \frac{36}{\pi\sqrt{899}}.$$

Therefore,

$$s_0 \phi_m = \frac{2(3m - 1)}{\sqrt{899}}. \quad (24.3)$$

The factor π cancels before the iterations are performed. The family does not depend on δ_F or α_F as input data.

Cancellation is a structural property of normalization. Before normalization, the second angular moment is

$$\frac{1}{12} \sum_{m=1}^{12} \phi_m^2 = \pi^2 D_0.$$

The condition of quadratic criticality fixes $s_0^2 \pi^2 D_0 = 2$, and therefore each normalized phase depends only on the integer arithmetic of $3m - 1$. The functional preserves the angular provenance of the wheel, but the iteration receives an intrinsic real family. This is precisely what makes it possible to distinguish a derivation from a calibration by π .

24.2 Exact analytic representation and closed trigonometric expression

We define

$$u_0(x) = 1 - \frac{1}{12} \sum_{m=1}^{12} \cos\left(\frac{2(3m-1)}{\sqrt{899}}x\right). \quad (24.4)$$

The development at the origin is

$$u_0(x) = x^2 - \frac{715769}{2424603}x^4 + O(x^6). \quad (24.5)$$

The finite sum admits the following closed expression. If $y = x/\sqrt{899}$,

$$u_0(x) = 1 - \frac{\cos(37y) \sin(36y)}{12 \sin(3y)} = 1 - \frac{\sin(73y) - \sin y}{24 \sin(3y)}. \quad (24.6)$$

The apparent singularity at $y = 0$ is removable and reproduces equation (24.5).

Derivation of the closed form. The twelve arguments form an arithmetic progression:

$$2(3m-1)y = (6m-2)y, \quad m = 1, \dots, 12.$$

The sum of cosines over an arithmetic progression satisfies

$$\sum_{m=1}^N \cos(a + md) = \frac{\sin(Nd/2)}{\sin(d/2)} \cos\left(a + \frac{N+1}{2}d\right).$$

Taking $N = 12$, $d = 6y$ and $a = -2y$ yields

$$\sum_{m=1}^{12} \cos((6m-2)y) = \frac{\sin 36y}{\sin 3y} \cos 37y.$$

The product-to-sum identity $2 \cos 37y \sin 36y = \sin 73y - \sin y$ gives the second expression for equation (24.6). $\square$

The quartic coefficient is not fitted either. The integer sums

$$\sum_{m=1}^{12} (3m-1)^2 = 5394, \quad \sum_{m=1}^{12} (3m-1)^4 = 4294614 \quad (24.7)$$

inserted in the development of $\cos z$ produce exactly

$$\frac{1}{24 \cdot 12} \sum_m \left(\frac{2(3m-1)}{\sqrt{899}}\right)^4 = \frac{715769}{2424603}.$$

Thus, both the quadratic normalization and the first nonlinear correction are finite moments of the same wheel.

24.3 Membership in the quadratic unimodal analytic class

Consider

$$f_\lambda(x) = 1 - \lambda u_0(x), \quad \lambda > 0.$$

For $0 < x \leq 1$, all arguments

$$\frac{2(3m-1)}{\sqrt{899}}x$$

belong to $(0, \pi)$. Therefore,

$$u'_0(x) > 0, \quad u'''_0(x) < 0.$$

Since u_0 is even, $x = 0$ is its unique critical point and $u''_0(0) = 2$. The Schwarzian of f_λ is

$$S(f_\lambda) = \frac{u'''_0}{u'_0} - \frac{3}{2} \left(\frac{u''_0}{u'_0} \right)^2 < 0 \quad (0 < |x| \leq 1). \quad (24.8)$$

Theorem 24.1 (Exact membership in the unimodal class). *For*

$$0 < \lambda \leq \frac{2}{u_0(1)},$$

the family $f_\lambda = 1 - \lambda u_0$ is a self-map that is real-analytic on $[-1, 1]$ and even, with a single critical point of quadratic type at the origin and negative Schwarzian away from that point. Consequently, without using Feigenbaum values as a target value, it belongs to the S -unimodal quadratic class subject to the action of the period-doubling renormalizer [Fei78, Lan82]. For λ outside that interval, the analytic germ and its local form are retained, but one does not automatically assert a self-map of the interval.

The theorem establishes the contribution intrinsic to HMT prior to the universality theorem: the analytic germ and its type of criticality.

Details of the sign and the self-map property. Differentiating equation (24.4),

$$\begin{aligned} u'_0(x) &= \frac{1}{12} \sum_m k_m \sin(k_m x), \\ u'''_0(x) &= -\frac{1}{12} \sum_m k_m^3 \sin(k_m x), \quad k_m = \frac{2(3m-1)}{\sqrt{899}}. \end{aligned}$$

For $0 < x \leq 1$, $0 < k_m x < \pi$, so all summands in the first line are positive and those in the second are negative. Parity extends monotonicity to both sides of the origin. Since $u_0(0) = 0$ and $0 \leq u_0(x) \leq u_0(1)$, the bound $0 < \lambda \leq 2/u_0(1)$ implies $-1 \leq 1 - \lambda u_0(x) \leq 1$. Finally, $u''_0(0) = 1/12 \sum k_m^2 = 2$, so the critical point is quadratic. $\square$

24.4 Finite sequences of superstable parameters

The superstable sequence certified through level eight gives

$$\delta_8 = 4.66921218915 \dots, \quad \alpha_8 = 2.50296847988 \dots$$

Its status is *finite certificate free of target values*: each approximant is obtained from the previously defined family, without introducing the universal limit as a datum. Decimal agreement provides a convergence indicator but does not replace the hyperbolicity theorem.

An independent numerical extension of the same sequence, with a step size adapted to the separation between roots and control of the exact period, provisionally reaches:

Level	λ_n	δ_n	α_n
11	1.87403828195439908045	4.66920170694424982745	2.50290847517165181764
12	1.87403836411023460451	4.66920163036308158848	2.50290800379001546915
13	1.87403838170549870372	4.66920161361839219913	2.50290790268748193108
14	1.87403838547386545431	4.66920161007472591023	2.50290788100969754274

Levels 11–14 retain their status as numerical evaluations beyond the persistent certificate through level eight. Aitken extrapolation gives $\lambda_\infty^{\text{HMT}} \simeq 1.8740383865008918080$ and $\alpha_F \simeq 2.50290787509$. The theorem in 24.6 establishes parametric scaling of the first roots; the spatial evaluation retains its own scope.

To avoid an unspecified numerical procedure, the method is formulated as follows. Define

$$F_n(\lambda) = f_\lambda^{2^n}(0). \quad (24.9)$$

The superstable root λ_n is the first root of F_n to the right of λ_{n-1} that satisfies simultaneously

$$f_{\lambda_n}^{2^j}(0) \neq 0 \quad (0 \leq j < n).$$

This inequality excludes divisor periods. With disjoint intervals $I_n \ni \lambda_n$, one computes

$$\delta_n = \frac{\lambda_{n-1} - \lambda_{n-2}}{\lambda_n - \lambda_{n-1}}, \quad \alpha_n = -\frac{d_{n-1}}{d_n}, \quad (24.10)$$

where $d_n = f_{\lambda_n}^{2^{n-1}}(0)$ is the oriented distance from the last new point to the critical point. A persistent certificate must retain the root intervals, the exclusion of smaller periods, and the propagation of error from equation (24.10); a mere printed decimal does not satisfy those three conditions.

The discretization rule also constitutes a negative control. A fixed lower bound greater than the separation $|\lambda_n - \lambda_{n-1}|$ can omit the first root and nevertheless produce a numerically plausible succession. For this reason the increment must scale with the previous separation and each candidate must be verified through its exact period.

24.5 Conditions for certifying the renormalization operator

On a space of even analytic functions, we define

$$\mathcal{R}(f)(x) = -a(f)^{-1} f(f(-a(f)x)), \quad a(f) = -f(1). \quad (24.11)$$

The complete closure of δ_F and α_F requires jointly certifying:

1. existence of a fixed point $g = \mathcal{R}(g)$ in a declared analytic ball;
2. local uniqueness of that fixed point;

3. a single unstable real eigenvalue of $D\mathcal{R}_g$, equal to δ_F ;
4. spectral radius strictly less than one on the stable complement;
5. transversality of the curve $\lambda \mapsto f_\lambda$ with respect to the stable manifold of g .

A natural HMT implementation employs Chebyshev truncations of order 9^m , interval arithmetic, and radii polynomials. The proof must provide a ball, an approximate inverse, a defect bound, and a Lipschitz bound satisfying $Y + Z(r) < r$. Thus, the nonadic depth parameterizes a fixed-point certificate and not a mere sequence of decimal approximations.

In an even Chebyshev basis,

$$g(x) = \sum_{j=0}^N g_{2j} T_{2j}(x), \quad N = 9^m,$$

the finite residue is $F_N(g) = \Pi_N(\mathcal{R}g - g)$. If A_N approximates $(DF_N(\bar{g}))^{-1}$, the quantities that must be certified are

$$\begin{aligned} Y &= \|A_N F_N(\bar{g})\|, \\ Z_0 &= \|I - A_N DF_N(\bar{g})\|, \\ Z_1(r) &= \sup_{\|h\| \leq r} \|A_N(DF_N(\bar{g} + h) - DF_N(\bar{g}))\|. \end{aligned}$$

An inequality $Y + (Z_0 + Z_1(r))r < r$ encloses a unique fixed point. The later spectral block must isolate an unstable direction and contract the complement. The use of 9^m thus enters as a refinement law of the certificate, not as numerological decoration.

Corollary 24.2 (Exact convergence of mode thirty). *Mode 30 admits three simultaneous and typologically distinct realizations:*

$$\begin{aligned} 30 &\longmapsto (3 \cdot 30, 4 \cdot 30) = (90, 120) && \text{in the constitutive completion,} \\ 30 &\longmapsto 30^2 - 1 = 899 = 29 \cdot 31 && \text{in the critical normalization,} \\ 30 &\longmapsto M_{30} = \begin{pmatrix} 30 & 899 \\ 1 & 30 \end{pmatrix} && \text{in the hyperbolic Pell transformation.} \end{aligned}$$

The three identities are exact and are obtained without prescribing target values. The conserved invariant is the central integer 30 together with its orientation: extension $3 \rightarrow 4$, quadratic defect -1 , and a unit of norm one. What is not asserted is the equality between the vacuum operator and the duplication renormalizer.

Remark 24.3 (Negative control). A second eigenvalue outside the unit disk, a ball of radii without solution, loss of uniqueness, or tangency of the HMT family to the stable variety refutes the universal promotion. None refutes the exact germ or the Schwarzian theorem.

Proposition 24.4 (Main Result). *The structural significance of the doubling constant lies in the cancellation in dodecaphase form of π , in the identity $899 = 30^2 - 1$, in the closed trigonometric form, and in its proved membership in the analytic class on which the renormalizer acts. Its decimal expansion constitutes exclusively the terminal evaluation.*

Logical Scope and Domain of Validity

The profile, normalization, Pell unit, unimodality and Schwarzian retain the preceding proofs. The eight certified levels and the extrapolations at levels 11–14 retain their respective numerical scope. The development in 24.6 proves parametric scaling for the original first-root sequence in the uniform sector, through identification of centres and remainder control. This conclusion does not transfer a proof about the parametric ratio to the spatial ratio.

24.5.1 Weighted harmonic deformation and selection of the superstable branch

The uniform profile has a parametric extension that preserves the twelve phases and separates the harmonic deformation from the dynamic parameter. For $1 \leq m \leq 12$, let

$$\phi_m = \frac{(30m - 10)\pi}{180}, \quad p(\phi) = 12 \cos(3\phi) - \cos(4\phi). \quad (24.12)$$

For every real η such that the weights remain positive, we define

$$w_m(\eta) = \frac{1 + \eta p(\phi_m)}{12}. \quad (24.13)$$

The sums of the third and fourth harmonics over the twelve phases vanish; therefore,

$$\sum_{m=1}^{12} w_m(\eta) = 1.$$

Let

$$s_\eta^2 = \frac{2}{\sum_{m=1}^{12} w_m(\eta) \phi_m^2}, \quad u_\eta(x) = 1 - \sum_{m=1}^{12} w_m(\eta) \cos(s_\eta \phi_m x). \quad (24.14)$$

The cosine expansion and the definition of s_η give

$$u_\eta(x) = x^2 + O(x^4) \quad (x \rightarrow 0).$$

Consequently, u_η is real-analytic, even, and retains a quadratic critical point at the origin. We introduce the dynamical parameter through

$$f_{\lambda, \eta}(x) = 1 - \lambda u_\eta(x). \quad (24.15)$$

A superstable parameter of exact period 2^n satisfies

$$f_{\lambda, \eta}^{2^n}(0) = 0, \quad f_{\lambda, \eta}^{2^k}(0) \neq 0 \quad (0 \leq k < n).$$

The equation may have several branches. A branch $\mathcal{B}$ is fixed by traversing an adaptive mesh immediately after the parameter of the preceding level; the first interval with a sign change is taken and bisection is applied. The notation $\lambda_n^{\mathcal{B}}(\eta)$ incorporates this selector and does not attribute global uniqueness to the superstable equation. The first level is determined exactly:

$$\lambda_1^{\mathcal{B}}(\eta) = \frac{1}{u_\eta(1)},$$

if $u_\eta(1) > 0$, because $f_{\lambda, \eta}^2(0) = f_{\lambda, \eta}(1) = 0$. Given three consecutive levels and

$$x_n^{\mathcal{B}}(\eta) := f_{\lambda_n^{\mathcal{B}}(\eta), \eta}^{2^{n-1}}(0),$$

we form

$$\delta_n^{\mathcal{B}}(\eta) = \frac{\lambda_{n-1}^{\mathcal{B}}(\eta) - \lambda_{n-2}^{\mathcal{B}}(\eta)}{\lambda_n^{\mathcal{B}}(\eta) - \lambda_{n-1}^{\mathcal{B}}(\eta)}, \quad \alpha_n^{\mathcal{B}}(\eta) = \frac{|x_{n-1}^{\mathcal{B}}(\eta)|}{|x_n^{\mathcal{B}}(\eta)|}. \quad (24.16)$$

For $\eta = 0$, the uniform profile is recovered exactly. The weighted extension introduces a transverse parameter: it permits study of the transversality of the family with respect to the stable manifold of the renormalizer without converting a finite coincidence into a proof of universality.

Remark 24.5 (Negative control). The extension is refuted if the weights lose positivity on the declared domain, if normalization ceases to produce a unit quadratic coefficient, if the selector does not exclude divisor periods, or if a branch is chosen using the terminal universal values as input.

24.6 First-root selection and Feigenbaum parametric scaling

The uniform dodecaphasic reader also determines a sequence of first roots of critical return. Its separation ratio is proved to converge to the Feigenbaum constant, preserving the original selection, orientation and memory of the construction.

24.6.1 Operator antecedents of the criticality family

The criticality family is constructed from a readout of the TPK calendar. The two arithmetic sheets, tritic orientation and transport with memory first determine the states and their prolongations. The ordered reading of the continuum then allows analytic functions to be evaluated, parameters to be compared and roots to be selected. This development preserves the genealogical state and its two readouts: the Archimedean chart provides the evaluation space, while the incidence readout preserves boundary relations. The joint discrete structure of the continuum retains its five consubstantial constructions. The present exposition uses its ordered reader and preserves the common antecedent from which it originates.

24.6.1.1 Arithmetic cells, orientation and transport

Let $D_9 = \{1, \dots, 9\}$ and $X_9 = (\mathbb{Z}/9\mathbb{Z})^2$. For $n \in \mathbb{Z}$, division with a positive representative is

$$\rho_9(n) = 1 + ((n - 1) \bmod 9), \quad q_9(n) = \frac{n - \rho_9(n)}{9}, \quad n = \rho_9(n) + 9q_9(n).$$

The APP cell associated with $(p, q) \in D_9^2$ is

$$a_{p,q} = (p, q; R^\Sigma, q^\Sigma, R^\Pi, q^\Pi), \quad p + q = R^\Sigma + 9q^\Sigma, \quad pq = R^\Pi + 9q^\Pi.$$

The two pairs (R^Σ, q^Σ) and (R^Π, q^Π) retain the additive and multiplicative evaluations, respectively. The division formula reconstructs both evaluations before any subsequent reduction. Selection of an active sheet is recorded in a state coordinate and preserves the complete cell.

The balanced tritic representative $r_3(n) \in \{-1, 0, +1\}$ satisfies $n \equiv r_3(n) \pmod{3}$. Its carry is

$$\chi(a, b) = \frac{a + b - r_3(a + b)}{3}, \quad a, b \in \{-1, 0, +1\}.$$

The equality

$$\chi(a, b) + \chi(r_3(a + b), c) = \chi(b, c) + \chi(a, r_3(b + c))$$

is obtained by writing both sides as $(a + b + c - r_3(a + b + c))/3$. This identity preserves the carry when regrouping a tritic sum. The operative selector is

$$M_{\text{ph}}(r) = \begin{cases} +1, & r \in \{1, 4, 7\}, \\ -1, & r \in \{2, 5, 8\}, \\ 0, & r \in \{3, 6, 9\}. \end{cases}$$

The value $+1$ activates the additive sheet, the value -1 the multiplicative sheet, and the value 0 preserves the position of both cursors during the neutral event. All three events belong to the complete chronology and are recorded in the route.

Let $\mathcal{D} = \{N, E, S, O\}$, with vectors $v_N = (-1, 0)$, $v_E = (0, 1)$, $v_S = (1, 0)$ and $v_O = (0, -1)$, and $\Theta_{108} = \mathbb{Z}/108\mathbb{Z}$. The observable configuration is

$$q = (x_+, d_+, x_\times, d_\times, \theta) \in \mathcal{Q}_{\text{obs}} = (X_9 \times \mathcal{D})^2 \times \Theta_{108}.$$

For the representative $\bar{\theta} \in \{0, \dots, 107\}$, define

$$\begin{aligned} r(\theta) &= 1 + (\bar{\theta} \bmod 9), & \beta(\theta) &= \left[\left[\frac{\bar{\theta}}{9} \right] \right]_{12}, \\ a_+(\theta) &= \mathbf{1}_{\{1,4,7\}}(r(\theta)), & a_\times(\theta) &= \mathbf{1}_{\{2,5,8\}}(r(\theta)). \end{aligned}$$

The position transition is

$$x'_+ = x_+ + a_+(\theta)v_{d_+}, \quad x'_\times = x_\times + a_\times(\theta)v_{d_\times}.$$

The involution j exchanges N, S and E, O . With

$$h(\theta) = \mathbf{1}_{\{0,27,54,81\}}((\bar{\theta} + 1) \bmod 108),$$

the complete observable operator is fixed by

$$F_{\text{obs}}(q) = (x'_+, j^{h(\theta)}d_+, x'_\times, j^{h(\theta)}d_\times, \theta + 1).$$

Proposition 24.6 (Conservation of lifted transport and observable return). *Transport preserves the integer displacement through the record of boundary crossings. The operator F_{obs} is invertible and satisfies $F_{\text{obs}}^{108} = I$.*

Proof. Let $s : X_9 \rightarrow D_9^2$ be the section of positive representatives. For $x' = x + av_d$, $a \in \{0, 1\}$, define

$$b_d(x, a) = \frac{s(x) + av_d - s(x')}{9} \in \mathbb{Z}^2.$$

If $z' = z + b_d(x, a)$, then

$$s(x') + 9z' = s(x) + 9z + av_d.$$

Summing this equality along a route gives

$$s(x_n) + 9z_n = s(x_0) + 9z_0 + \sum_{t=0}^{n-1} a_t v_{d_t}.$$

Partitioning the sum preserves the identity under concatenation. To invert F_{obs} , first subtract one unit from the phase; that phase determines the reversal of directions and the displaced cursor. The previous directions are recovered and the corresponding vector is subtracted. Each block of 27 steps contains nine activations of each cursor. The next block reverses directions and cancels their integer displacements. In 54 steps, positions and directions are restored; in 108, the calendar coordinate is also restored. $\square$

In addition to this configuration, the complete state preserves both arithmetic evaluations, carries, orientation, trajectory, prefix, compatible cylinder, its boundary, memory degree and survival record. Its update factors as

$$U_t = \text{Mem}_t \circ \text{Tra}_t \circ \text{Sel}_t.$$

The holonomy Γ_9 composes nine complete transitions. Phase return and memory advance are different coordinates of this composition. The following construction fixes the transitions and their update field by field.

24.6.1.1.1 Edge-by-edge realization of the three nonadic extensions This subsection specifies the autonomous model generated by the TPK extension relation entering the cardinal proof. The existence of an edge is not inferred from a coincidence of types: its initial and final states, APP relation, fiber transport, and unary update of the ledger are constructed. The construction uses exclusively the two APP sheets, the TRIT selector, and the TPK composition Sel-Tra-Upd .

24.6.1.1.2 Complete cells and three balanced reductions. In the chart $D_9 = \{1, \dots, 9\}$, we will always retain the complete APP cell

$$a_{p,q} := (p, q; R_{p,q}^\Sigma, q_{p,q}^\Sigma, R_{p,q}^\Pi, q_{p,q}^\Pi),$$

with

$$p + q = R_{p,q}^\Sigma + 9q_{p,q}^\Sigma, \quad pq = R_{p,q}^\Pi + 9q_{p,q}^\Pi.$$

The active sheet is recorded in a separate coordinate; $a_{p,q}$ is not split into a purported additive cell and a separate multiplicative cell. For $r \in D_9$, let

$$a(r) := \begin{cases} a_{1,9}, & r = 1, \\ a_{1,r-1}, & 2 \leq r \leq 9, \end{cases} \quad \tau(r) := M_{\text{ph}}(r), \quad m(r) := \tau(r) \in \{-1, 0, +1\}. \quad (24.17)$$

The additive residue of $a(r)$ supplies an APP representative of phase here; this does not make Σ the universally active sheet. The dynamic selector activates Σ for $r \in \{1, 4, 7\}$, activates Π for $r \in \{2, 5, 8\}$, and preserves both sheets without moving the cursor for $r \in \{3, 6, 9\}$. In all three cases, $R^\Sigma, q^\Sigma, R^\Pi, q^\Pi$ are transported jointly. We have

$$(m(1), \dots, m(9)) = (1, -1, 0, 1, -1, 0, 1, -1, 0).$$

The three pairs ordered by the nonadic base are

$$\mathbf{p}_{+1} := (1, 4), \quad \mathbf{p}_{-1} := (2, 5), \quad \mathbf{p}_0 := (3, 6). \quad (24.18)$$

These pairs form the normalized section that takes the first two occurrences of each TRIT type under the shift $r \mapsto r + 3$; the third phase block 7, 8, 9 preserves control of turn closure. Their two members belong to the same fiber of M_{ph} . The balanced reduction defined in these antecedents gives, without introducing a target coefficient,

ε	$m(\mathbf{p}_\varepsilon^{(1)}) + m(\mathbf{p}_\varepsilon^{(2)})$	$r_3(2\varepsilon)$	$\chi(\varepsilon, \varepsilon)$
+1	$1 + 1 = (-1) + 3 \cdot 1$	-1	+1
0	$0 + 0 = 0 + 3 \cdot 0$	0	0
-1	$(-1) + (-1) = 1 + 3 \cdot (-1)$	+1	-1

(24.19)

Therefore,

$$\chi(\varepsilon, \varepsilon) = \frac{2\varepsilon - r_3(2\varepsilon)}{3} = \varepsilon. \quad (24.20)$$

The branch sign is thus the output of the balanced cocycle applied to two TRIT emissions produced by APP. Each pair supplies one of the three carry outputs. For clarity, we subsequently index the pair by $\varepsilon(\mathfrak{p}) := \chi(m(\mathfrak{p}^{(1)}), m(\mathfrak{p}^{(2)}))$; the symbol ε does not supply a free carry to Upd^{cal} .

The phase architecture, balanced lift, and memory transport are realized here through twenty-seven local incidences and their extension graphs. The normalized section fixes representatives of the fibers of M_{ph} ; the covering to be proved uses the existence of these representatives and remains invariant when they are replaced by another compatible section.

24.6.1.1.3 Typed data of the nine edges. Let $K_{\text{Carr}} := \langle \kappa_{\text{Carr}} \rangle_{\text{Ab}}$ be the carry module. The actions are typed by

$$H_{\text{Carr}} : \mathcal{P}_{\text{TPK}}^{\text{op}} \longrightarrow \text{End}(K_{\text{Carr}}), \quad Y_{\text{Carr}} : \mathcal{P}_{\text{TPK}} \longrightarrow \text{End}(K_{\text{Carr}}).$$

In this generated model, take the trivial actions $H_{\text{Carr}}(e) = Y_{\text{Carr}}(e) = \text{id}_{K_{\text{Carr}}}$: the active sheet and the route are retained as coordinates of the ledger itself and do not act on the free carry summand. Also let (M_d, J_d, d) be the graded direct system of memory. To avoid confusing two representatives identified by the colimit, the state also retains the degree as a coordinate, and the labeled carrier

$$\tilde{M} := \bigsqcup_{d \geq 1} (\{d\} \times M_d), \quad \text{gr}_M(d, \mu) = d.$$

Publication in the limit memory is given by

$$\begin{aligned} M &:= \varinjlim_d M_d, & J_d : M_d &\longrightarrow M, \\ u &:= J_d(u_d), & \iota_M : \mathbb{Z} &\longrightarrow M, & n &\longmapsto nu. \end{aligned}$$

Here J_d is the canonical colimit map and the u_d form a compatible family. Its free component permits the reader $\deg_\mu(nu) = n$. The configuration, phase, memory, carry, route and survival projections will be typed on the calibrated carrier constructed recursively below; no ambient space will be imported for them. For each constructed edge e , the calibrated component $\text{Upd}_e^{\text{cal}} := \text{Mem}_e^{\text{cal}}$ will denote the third map in the factorization; its law is fixed field by field below and is not assumed as the restriction of an ambient relation.

Fix the initial depth $k_0 = 1$, a complete configuration $q_\star \in \mathcal{Q}_{\text{obs}}$ of phase one, and

$$q_{n,r} := F_{\text{obs}}^{9n+r-1}(q_\star), \quad q_{n,10} := q_{n+1,1}.$$

Because the observable calendar is periodic, its value $q_{n,r}$ does not by itself distinguish occurrences separated by twelve turns. The calibrated model therefore uses the labeled lift

$$\tilde{q}_{n,r} := (q_{n,r}, n, r), \quad \tilde{q}_{n,10} := \tilde{q}_{n+1,1}, \quad \pi_{\text{obs}}(\tilde{q}_{n,r}) = q_{n,r}.$$

The lifted dynamics $\tilde{F}_{\text{obs}}(\tilde{q}_{n,r}) = \tilde{q}_{n,r+1}$ projects to F_{obs} ; the label does not alter the physical phase, but makes its levels of occurrence distinct. The copy of the cell at each level is labeled by

$$a_{n,r} := (n, r; a(r)), \quad (n, r)^+ := \begin{cases} (n, r+1), & 1 \leq r < 9, \\ (n+1, 1), & r = 9. \end{cases}$$

The label only prevents two occurrences of the same cell from being identified; the two Euclidean divisions and their four coordinates are those of $a(r)$. For turn $n \geq 0$ and phase $r \in D_9$, the symbol

$$e_{\varepsilon,n,r} : \tilde{q}_{n,r} \longrightarrow \tilde{q}_{n,r+1}, \quad r^+ := 1 + (r \bmod 9),$$

occupies level $d_{n,r} := k_n + r - 1$, where $k_n := k_0 + 9n$. Its local ledger contains the complete cell $a(r)$, the active sheet determined by $M_{\text{ph}}(r)$, the type $\tau(r)$, the membership mark in $\mathfrak{p}_\varepsilon$, and, upon reaching the second member of that pair, the certified identity (24.19). We define its operators solely from that ledger:

$$\mathbf{C}_M(e_{\varepsilon,n,r}) = J_{d_{n,r}+1,d_{n,r}}, \quad \kappa_M(e_{\varepsilon,n,r}) = \mathbf{1}_{\{r=9\}} u_{d_{n,r}+1}, \quad (24.21)$$

$$\text{Carr}(e_{\varepsilon,n,r}) = d_{\varepsilon,r} \kappa_{\text{Carr}}, \quad d_{\varepsilon,r} := \mathbf{1}_{\{r=\mathfrak{p}_\varepsilon^{(2)}\}} \chi\left(m(\mathfrak{p}_\varepsilon^{(1)}), m(\mathfrak{p}_\varepsilon^{(2)})\right). \quad (24.22)$$

For identities and every admissible composition, also define

$$\mathbf{C}_M(\text{id}_{\tilde{q}}; d) := \text{id}_{M_d}, \quad \kappa_M(\text{id}_{\tilde{q}}; d) := 0, \quad (24.23)$$

$$\mathbf{C}_M(e_t \cdots e_1) := \mathbf{C}_M(e_t) \circ \cdots \circ \mathbf{C}_M(e_1), \quad \kappa_M(e_2 e_1) := \mathbf{C}_M(e_2)(\kappa_M(e_1)) + \kappa_M(e_2). \quad (24.24)$$

The identity carries degree d because the same observable configuration may recur at different depths. Iterated, the final equality defines $\kappa_M(e_t \cdots e_1)$ for every length and is the cocycle law of the direct system; the inclusions satisfy $J_{d+2,d+1} J_{d+1,d} = J_{d+2,d}$. In addition, set $s(e_{\varepsilon,n,r}) = \top$. The following twenty-seven entries certify the coordinates distinguishing the three extensions; the universal transport fields are specified immediately afterward. The symbols *I* and *II* designate, respectively, the first retained emission and the second emission performing the reduction; a dash denotes a zero carry increment, not the absence of an edge.

r	$M_{\text{ph}}(r)$	$\mathfrak{p}_{+1}$ mark	$d_{+1,r}$	$\mathfrak{p}_0$ mark	$d_{0,r}$	$\mathfrak{p}_{-1}$ mark	$d_{-1,r}$	$\Delta \text{ mem}$
1	+1	I	0	–	0	–	0	0
2	–1	–	0	–	0	I	0	0
3	0	–	0	I	0	–	0	0
4	+1	II	+1	–	0	–	0	0
5	–1	–	0	–	0	II	–1	0
6	0	–	0	II	0	–	0	0
7	+1	–	0	–	0	–	0	0
8	–1	–	0	–	0	–	0	0
9	0	–	0	–	0	–	0	u

Thus, $d_{\varepsilon,r}$ is computed from two outputs of the selector and is distinct from the cell quotients q^Σ, q^Π . The ledger also preserves the marked pair and the balanced residue; in particular, the zero branch is not confused with an absence of event.

The carry composition law used by the TPK category is

$$\text{Carr}(\text{id}_{\tilde{q}}) = 0, \quad \text{Carr}(e_2 e_1) = H_{\text{Carr}}(e_1)(\text{Carr}(e_2)) + Y_{\text{Carr}}(e_2)(\text{Carr}(e_1)). \quad (24.25)$$

On identities and paths, they are extended functorially:

$$\begin{aligned} H_{\text{Carr}}(\text{id}) &= Y_{\text{Carr}}(\text{id}) = \text{id}, \\ H_{\text{Carr}}(e_t \cdots e_1) &= H_{\text{Carr}}(e_1) \circ \cdots \circ H_{\text{Carr}}(e_t), \\ Y_{\text{Carr}}(e_t \cdots e_1) &= Y_{\text{Carr}}(e_t) \circ \cdots \circ Y_{\text{Carr}}(e_1). \end{aligned}$$

In this model all these actions remain the identity, but their typing makes explicit which endomorphism acts on each summand of (24.25). On calibrated edges, $H_{\text{Carr}} = Y_{\text{Carr}} = \text{id}$, so composition sums the increments already produced. Equation (24.25), rather than an identification between orientation and quotient, governs the accumulated register.

24.6.1.1.4 Independent construction of the transition tuples. A fiber state is written, in the order of its relevant fields, as

$$x = (a; q; \tau, o, h; b_{\text{ref}}; R^\Sigma, q^\Sigma, R^\Pi, q^\Pi; c; \text{Sig}; C, \partial C; p_{\text{APP}}, p_{\{-1,0,+1\}}; \lambda; \text{gr}_M; \text{mem}; s; \text{Led}).$$

The update notation $x[f \leftarrow v]$ changes only the indicated field. For $h \in \{(\Sigma, \Pi), (\Pi, \Sigma)\}$, define

$$h_r(h) = \begin{cases} (\Sigma, \Pi), & r \in \{1, 4, 7\}, \\ (\Pi, \Sigma), & r \in \{2, 5, 8\}, \\ h, & r \in \{3, 6, 9\}. \end{cases}$$

The complete sheet-and-arithmetic coordinate is abbreviated as

$$\widehat{h}(x) := (h(x), R^\Sigma(x), q^\Sigma(x), R^\Pi(x), q^\Pi(x)).$$

Let $\widehat{\mathcal{H}}_{q_{n,r}}^{\text{cal}}$ be the set of quintuples

$$(h, R_{a_{n,r}}^\Sigma, q_{a_{n,r}}^\Sigma, R_{a_{n,r}}^\Pi, q_{a_{n,r}}^\Pi), \quad h \in \{(\Sigma, \Pi), (\Pi, \Sigma)\},$$

whose four arithmetic data are the two Euclidean divisions of the labeled APP cell $a_{n,r}$. This direct definition does not presuppose a state in a carrier not yet constructed. Let $\text{Hist}_{\tilde{q}}^{\text{cal}}$ be the set of admissible finite paths in the graph of labeled edges ending at $\tilde{q}$, including the identity, and let $\mathbf{B}_{\tilde{q}}^{\text{cal}}$ be the set of their ordered endpoint sequences. Define

$$\mathbf{S}_{\tilde{q}}^{\text{cal}} := \widehat{\mathcal{H}}_{\tilde{q}}^{\text{cal}} \times \{o_{\rightarrow}, o_{\circ}, o_{\leftarrow}\} \times K_{\text{Carr}} \times \text{Hist}_{\tilde{q}}^{\text{cal}} \times \mathbf{B}_{\tilde{q}}^{\text{cal}} \times \mathbf{M}.$$

The calibrated signature is not postulated: it is the typed assembly map

$$\begin{aligned} \text{Sig}_{\tilde{q}}^{\text{cal}} : \widehat{\mathcal{H}}_{\tilde{q}}^{\text{cal}} \times \{o_{\rightarrow}, o_{\circ}, o_{\leftarrow}\} \times K_{\text{Carr}} \times \text{Hist}_{\tilde{q}}^{\text{cal}} \times \mathbf{B}_{\tilde{q}}^{\text{cal}} \times \mathbf{M} &\longrightarrow \mathbf{S}_{\tilde{q}}^{\text{cal}}, \\ (\widehat{h}, o, c, \lambda, \partial C, \mu) &\longmapsto (\widehat{h}, o, c, \lambda, \partial C, \mu). \end{aligned}$$

Likewise, $\pi_{\mathcal{F}}$ denotes the projection that removes only the first two coordinates $(a; q)$ of the state tuple; it preserves $\tau, o, h, b_{\text{ref}}$, the four Euclidean data, carry, signature, cylinder, boundary, prefixes, route, memory degree, memory, survival, and ledger. The corresponding orientation is fixed by $o(+1) = o_{\rightarrow}$, $o(0) = o_{\circ}$, and $o(-1) = o_{\leftarrow}$. The calibrated APP fiber over $\tilde{q}_{n,r}$ contains the already defined labeled copy $a_{n,r}$. The initial state is fixed without free coordinates:

$$\widehat{h}_{\star} := ((\Sigma, \Pi), R_{1,9}^\Sigma, q_{1,9}^\Sigma, R_{1,9}^\Pi, q_{1,9}^\Pi).$$

$$\begin{aligned} x_{\star} := & (a_{0,1}; \tilde{q}_{0,1}; +1, o_{\rightarrow}, (\Sigma, \Pi); \perp; R_{1,9}^\Sigma, q_{1,9}^\Sigma, R_{1,9}^\Pi, q_{1,9}^\Pi; 0; \\ & \text{Sig}_{\tilde{q}_{0,1}}^{\text{cal}}(\widehat{h}_{\star}, o_{\rightarrow}, 0, \text{id}_{\tilde{q}_{0,1}}, \emptyset, J_{k_0}(0_{\mathbf{M}_{k_0}})); \emptyset, \emptyset; (a_{0,1}), (+1); \text{id}_{\tilde{q}_{0,1}}; k_0; 0_{\mathbf{M}_{k_0}}; \top; \emptyset). \end{aligned} \quad (24.26)$$

The tuple contains no indeterminate coordinates and satisfies $s(x_{\star}) = \top$. Let $G_0 := \{x_{\star}\}$. Assuming G_n has been constructed, fix $x \in G_n$, $\varepsilon \in \{-1, 0, +1\}$, and $x_{\varepsilon,0}(x) := x$. For $r = 1, \dots, 9$, first construct the elementary ledger

$$\begin{aligned} \ell_{\varepsilon,n,r} := & (e_{\varepsilon,n,r}, a_{n,r}, a_{(n,r)^+}, \tilde{q}_{n,r}, \tilde{q}_{n,r+1}, M_{\text{ph}}(r), \\ & h_r(h(x_{\varepsilon,r-1}(x))), o(M_{\text{ph}}(r)), \mathfrak{p}_{\varepsilon}, d_{\varepsilon,r}, \\ & \mathbf{C}_M(e_{\varepsilon,n,r}), \kappa_M(e_{\varepsilon,n,r}), \text{Carr}(e_{\varepsilon,n,r})), \end{aligned} \quad (24.27)$$

and set $\text{Led}_{\varepsilon,n,r}^{\text{new}} := \text{Led}(x_{\varepsilon,r-1}(x)) \frown \ell_{\varepsilon,n,r}$. Then construct the tuple

$$y_{\varepsilon,r}(x) := x_{\varepsilon,r-1}(x)[\tau \leftarrow M_{\text{ph}}(r), h \leftarrow h_r(h(x_{\varepsilon,r-1}(x))), o \leftarrow o(M_{\text{ph}}(r)), b_{\text{ref}} \leftarrow \mathfrak{p}_{\varepsilon}], \quad (24.28)$$

where selection leaves the cell, quotients, carry, route, memory, cylinder, boundary, survival, and ledger unchanged; the coordinate b_{ref} makes the chosen balanced reduction explicit. Membership in an ambient relation is not presupposed here.

Next, define $z_{\varepsilon,r}(x)$ field by field: its configuration is $\tilde{q}_{n,r+1}$; its complete cell is $a_{(n,r)}^+$, with the four residues and quotients recomputed by the two Euclidean divisions; it preserves the selected sheet and orientation; and it performs the following six updates, in which only the common argument x is omitted:

$$\begin{aligned} C(z_{\varepsilon,r}) &:= C(y_{\varepsilon,r}) \frown e_{\varepsilon,n,r}, \\ \partial C(z_{\varepsilon,r}) &:= \partial C(y_{\varepsilon,r}) \frown (\tilde{q}_{n,r}, \tilde{q}_{n,r+1}), \\ p_{\text{APP}}(z_{\varepsilon,r}) &:= p_{\text{APP}}(y_{\varepsilon,r}) \frown a_{(n,r)}^+, \\ p_{\{-1,0,+1\}}(z_{\varepsilon,r}) &:= p_{\{-1,0,+1\}}(y_{\varepsilon,r}) \frown \tau(r), \\ \text{Led}(z_{\varepsilon,r}) &:= \text{Led}_{\varepsilon,n,r}^{\text{new}}, \\ \lambda(z_{\varepsilon,r}) &:= e_{\varepsilon,n,r} \lambda(y_{\varepsilon,r}), \end{aligned} \quad (24.29)$$

and still retains the fields $c, \text{gr}_M, \text{mem}, s$ of $y_{\varepsilon,r}(x)$. Its signature, by contrast, is retyped at the target configuration through the transported presignature

$$\begin{aligned} \text{Sig}(z_{\varepsilon,r}(x)) &:= \text{Sig}_{\tilde{q}_{n,r+1}}^{\text{cal}} \left(\widehat{h}(z_{\varepsilon,r}(x)), o(z_{\varepsilon,r}(x)), c(y_{\varepsilon,r}(x)), \lambda(z_{\varepsilon,r}(x)), \right. \\ &\quad \left. \partial C(z_{\varepsilon,r}(x)), Jd_{n,r}(\text{mem}_{d_{n,r}}(y_{\varepsilon,r}(x))) \right). \end{aligned} \quad (24.30)$$

Thus, and only in Tra^{cal} , the edge, its endpoints, the mark $(\mathfrak{p}_{\varepsilon}, r, d_{\varepsilon,r})$, and its cocycles are added to the incidence ledger.

The initial state satisfies

$$c(x_{\star}) = 0 = \text{Carr}(\text{id}_{\tilde{q}_{0,1}}) = \text{Carr}(\lambda(x_{\star})).$$

In the complete states $x_{\varepsilon,r-1}(x)$, and also after the selection $y_{\varepsilon,r}(x)$, the typed invariant $c(v) = \text{Carr}(\lambda(v))$ is preserved, since Sel^{cal} modifies neither field. Transport advances the route but leaves the carry pending; hence the pre-update tuple satisfies exactly

$$c(z_{\varepsilon,r}(x)) = c(y_{\varepsilon,r}(x)) = \text{Carr}(\lambda(y_{\varepsilon,r}(x))).$$

The map Upd^{cal} then restores $c(x_{\varepsilon,r}(x)) = \text{Carr}(\lambda(x_{\varepsilon,r}(x)))$ by means of the composition law, without attributing that invariant to the intermediate state z .

Finally, apply the unary function $\text{Upd}_{e_{\varepsilon,n,r}}^{\text{cal}} = \text{Mem}_{e_{\varepsilon,n,r}}^{\text{cal}}$ to that tuple. Its result $x_{\varepsilon,r}(x)$ is given by

$$\begin{aligned} c(x_{\varepsilon,r}(x)) &= \text{Carr}(\lambda(z_{\varepsilon,r}(x))) \\ &= H_{\text{Carr}(\lambda(y_{\varepsilon,r}(x)))}(\text{Carr}(e_{\varepsilon,n,r})) + Y_{\text{Carr}(e_{\varepsilon,n,r})}(c(y_{\varepsilon,r}(x))), \\ \lambda(x_{\varepsilon,r}(x)) &= \lambda(z_{\varepsilon,r}(x)), \\ \text{gr}_M(x_{\varepsilon,r}(x)) &= d_{n,r} + 1, \\ \text{mem}_{d_{n,r}+1}(x_{\varepsilon,r}(x)) &= \mathbf{C}_M(e_{\varepsilon,n,r}) \text{mem}_{d_{n,r}}(z_{\varepsilon,r}(x)) + \kappa_M(e_{\varepsilon,n,r}), \\ s(x_{\varepsilon,r}(x)) &= s(e_{\varepsilon,n,r}) \wedge s(z_{\varepsilon,r}(x)), \\ \text{Sig}(x_{\varepsilon,r}(x)) &= \text{Sig}_{\tilde{q}_{n,r+1}}^{\text{cal}} \left(\widehat{h}(z_{\varepsilon,r}(x)), o(z_{\varepsilon,r}(x)), c(x_{\varepsilon,r}(x)), \lambda(z_{\varepsilon,r}(x)), \partial C(z_{\varepsilon,r}(x)), \right. \\ &\quad \left. Jd_{n,r+1}(\text{mem}_{d_{n,r}+1}(x_{\varepsilon,r}(x))) \right). \end{aligned} \quad (24.31)$$

All fields not enumerated in (24.31) coincide with those of $z_{\varepsilon,r}(x)$. In particular, the signature is computed from the sheet, orientation, boundary, and route already fixed by Sel^{cal} and Tra^{cal} , and from the carry and memory computed in the preceding lines; it is not circularly defined from the final state. Neither ε , nor $d_{\varepsilon,r}$, nor u is an additional argument of Upd^{cal} : the function reads all three data from the already constructed edge.

24.6.1.1.5 Simultaneous induction of the levels. Once the preceding nine stages have been completed for an already constructed G_n , define immediately

$$\gamma_{\varepsilon,k_n}(x) := x_{\varepsilon,9}(x), \quad G_{n+1} := \{\gamma_{\varepsilon,k_n}(x) : x \in G_n, \varepsilon \in \{-1, 0, +1\}\}. \quad (24.32)$$

The base $G_0 = \{x_\star\}$, the intermediate tuples, and (24.32) constitute a single definition by recursion on n . Therefore, no subsequent carrier is used to assume the existence of G_n : the next level is introduced only after constructing all its edges from the preceding level.

24.6.1.1.6 TPK extension relation and effective membership. Let $\mathcal{E}_{\tilde{q}}^{\text{cal}}$ be the smallest labelled family containing $x_\star$ and, at each step of the preceding recurrence, the four tuples $x_{\varepsilon,r-1}(x), y_{\varepsilon,r}(x), z_{\varepsilon,r}(x), x_{\varepsilon,r}(x)$ in their source and target configurations. Also let $\mathcal{A}_{\tilde{q},r}^{\text{cal}} := \{a_{n,r}\}$, and set

$$\tilde{\mathcal{Q}}_{\text{obs}} := \{\tilde{q}_{n,r} : n \geq 0, 1 \leq r \leq 9\}, \quad \mathcal{E}^{\text{cal}} := \bigsqcup_{\tilde{q}} \mathcal{E}_{\tilde{q}}^{\text{cal}}, \quad \mathcal{F}_{\tilde{q}}^{\text{cal}} := \pi_{\mathcal{F}}(\mathcal{E}_{\tilde{q}}^{\text{cal}}).$$

For each depth d , let $\mathcal{E}_d^{\text{cal}} := \text{gr}_M^{-1}(d) \subset \mathcal{E}^{\text{cal}}$. The explicit degree coordinate, rather than the mere membership of a representative in some image of the direct system, makes these carriers disjoint. On them, the projections are typed as

$$\begin{aligned} \text{cfg} : \mathcal{E}^{\text{cal}} &\longrightarrow \tilde{\mathcal{Q}}_{\text{obs}}, \\ g(x) &:= \text{fase}(\pi_{\text{obs}}(\text{cfg}(x))) \in C_9, \\ c : \mathcal{E}^{\text{cal}} &\longrightarrow K_{\text{Carr}}, \\ s : \mathcal{E}^{\text{cal}} &\longrightarrow \{\perp, \top\}, \\ \lambda : \mathcal{E}_{\tilde{q}}^{\text{cal}} &\longrightarrow \text{Hist}_{\tilde{q}}^{\text{cal}}, \\ \text{gr}_M : \mathcal{E}^{\text{cal}} &\longrightarrow \mathbb{N}_{\geq 1}, \\ \text{mem}_d : \mathcal{E}_d^{\text{cal}} &\longrightarrow M_d, \\ \text{mem}(x) &:= J_d(\text{mem}_d(x)) \in M \quad (x \text{ of depth } d). \end{aligned}$$

Here $\text{cfg}(x)$ is the second coordinate of the tuple, and fase reads the phase of the observable calendar. For the three carrier classes, fix the diagonals

$$\begin{aligned} \Delta_{\text{APP},\tilde{q}} &:= \{(a, a) : a \in \mathcal{A}_{\tilde{q}}^{\text{cal}}\}, \\ \Delta_{\text{fib},\tilde{q}} &:= \{(f, f) : f \in \mathcal{F}_{\tilde{q}}^{\text{cal}}\}, \\ \Delta_{\text{enr},\tilde{q}} &:= \{(x, x) : x \in \mathcal{E}_{\tilde{q}}^{\text{cal}}\}. \end{aligned}$$

In particular,

$$a_{0,1} \in \mathcal{A}_{\tilde{q}_{0,1}}^{\text{cal}}, \quad x_\star \in \mathcal{E}_{\tilde{q}_{0,1}}^{\text{cal}}.$$

On these carriers, the following relations are defined by their complete graphs, and not through necessary implications:

$$\text{Sel}_{e_{\varepsilon,n,r}}^{\text{cal}} := \{(x_{\varepsilon,r-1}(x), y_{\varepsilon,r}(x)) : x \in G_n\}, \quad (24.33)$$

$$\text{Tra}_{e_{\varepsilon,n,r}}^{\text{cal}} := \{(y_{\varepsilon,r}(x), z_{\varepsilon,r}(x)) : x \in G_n\}, \quad (24.34)$$

$$\text{Ext}_{e_{\varepsilon,n,r}}^{\text{APP,cal}} := \{(a_{n,r}, a_{(n,r)^+})\}, \quad (24.35)$$

$$\text{Ext}_{e_{\varepsilon,n,r}}^{\text{fib,cal}} := \{(\pi_{\mathcal{F}X_{\varepsilon,r-1}}(x), \pi_{\mathcal{F}X_{\varepsilon,r}}(x)) : x \in G_n\}. \quad (24.36)$$

The synchronized enriched extension is

$$\text{Ext}_{e_{\varepsilon,n,r}}^{\text{enr,cal}} := \{(x_{\varepsilon,r-1}(x), x_{\varepsilon,r}(x)) : x \in G_n\}. \quad (24.37)$$

The third map is defined by

$$\text{graph}(\text{Upd}_{e_{\varepsilon,n,r}}^{\text{cal}}) := \{(z_{\varepsilon,r}(x), x_{\varepsilon,r}(x)) : x \in G_n\}, \quad (24.38)$$

where the second component is exactly the one in (24.31). Consequently,

$$\mathcal{U}_{e_{\varepsilon,n,r}}^{(1),\text{cal}} := \text{Upd}_{e_{\varepsilon,n,r}}^{\text{cal}} \circ \text{Tra}_{e_{\varepsilon,n,r}}^{\text{cal}} \circ \text{Sel}_{e_{\varepsilon,n,r}}^{\text{cal}} = \{(x_{\varepsilon,r-1}(x), x_{\varepsilon,r}(x)) : x \in G_n\}. \quad (24.39)$$

Each operator is indexed by the edge $e_{\varepsilon,n,r}$ produced by the reduction $\mathfrak{p}_{\varepsilon}$, rather than by a trit supplied from outside. The coordinate b_{ref} also distinguishes their codomains. Therefore, $\text{Sel}_e^{\text{cal}}$, Tra^{cal} and Upd^{cal} are functions on their respective domains, whereas the union $\bigcup_{\varepsilon \in \{-1,0,+1\}} \mathcal{U}_{e_{\varepsilon,n,r}}^{(1),\text{cal}}$ deliberately preserves the three outputs of the trisection.

For $\bullet \in \{\text{APP}, \text{fib}, \text{enr}\}$, identities and compositions are not left implicit. Define

$$\text{Ext}_{\text{id}_{\tilde{q}}}^{\bullet,\text{cal}} := \Delta_{\bullet,\tilde{q}}, \quad (24.40)$$

$$\text{Ext}_{e_t \dots e_1}^{\bullet,\text{cal}} := \text{Ext}_{e_t}^{\bullet,\text{cal}} \circ \dots \circ \text{Ext}_{e_1}^{\bullet,\text{cal}} \quad (24.41)$$

for every admissible edge word. The two APP and fiber path closures are formed separately,

$$\text{Path}(\text{Ext}^{\text{APP,cal}}), \quad \text{Path}(\text{Ext}^{\text{fib,cal}}), \quad (24.42)$$

and Tra^{cal} synchronizes them edge by edge.

Finally, the levels of complete states and their truncations are fixed by the recursion itself:

$$H_{d_{n,r}}^{\text{cal}} := \{x_{\varepsilon,r-1}(x) : x \in G_n, \varepsilon \in \{-1,0,+1\}\}, \quad (24.43)$$

$$H_{d_{n,r}+1}^{\text{cal}} := \{x_{\varepsilon,r}(x) : x \in G_n, \varepsilon \in \{-1,0,+1\}\}, \quad (24.44)$$

$$\rho_{d_{n,r}+1,d_{n,r}}^{\text{cal}}(x_{\varepsilon,r}(x)) := x_{\varepsilon,r-1}(x). \quad (24.45)$$

The final ledger preserves $(n, r, \mathfrak{p}_{\varepsilon})$, so the predecessor of the left-hand side is unique and ρ^{cal} is well-defined. For $b > a$, set $\rho_{b,a}^{\text{cal}} := \rho_{a+1,a}^{\text{cal}} \circ \dots \circ \rho_{b,b-1}^{\text{cal}}$. This map recovers the complete preceding state, including its signature; it does not purport to invert an overwritten coordinate without consulting the ledger.

Proposition 24.7 (Autonomous realization of the TPK through explicit transitions). *The family*

$$\begin{aligned} \mathfrak{T}_{\text{TPK}}^{\text{cal}} := & ((\tilde{q}_{n,r})_{n,r}, \pi_{\text{obs}}, \tilde{F}_{\text{obs}}, M_{\text{ph}}, \\ & (\mathcal{A}_{\tilde{q}}^{\text{cal}})_{\tilde{q}}, (\mathcal{E}_{\tilde{q}}^{\text{cal}})_{\tilde{q}}, (\mathcal{F}_{\tilde{q}}^{\text{cal}})_{\tilde{q}}, \text{Ext}^{\text{APP,cal}}, \text{Ext}^{\text{fib,cal}}, \text{Ext}^{\text{enr,cal}}, \rho^{\text{cal}}, \\ & \text{Sel}^{\text{cal}}, \text{Tra}^{\text{cal}}, \text{Upd}^{\text{cal}}, \mathbf{C}_{\mathbf{M}}, \kappa_{\mathbf{M}}, \text{gr}_{\mathbf{M}}, H_{\text{Carr}}, Y_{\text{Carr}}, \\ & \text{Sig}^{\text{cal}}, \pi_{\mathcal{F}}). \end{aligned}$$

is an enriched model of the TPK. In particular, every pair on the right-hand side of (24.39) is an effective TPK edge, not a consequence inferred from a merely necessary condition.

Proof. The labels (n, r) make the carrier copies disjoint and unambiguously fix each source and target. The relation $\text{Ext}^{\text{APP}, \text{cal}}$ transports the complete cell and preserves the two Euclidean identities; the relation $\text{Ext}^{\text{fib}, \text{cal}}$ relates the fibers of the preceding and subsequent complete states: it transports sheet, orientation, prefix, cylinder, boundary, and ledger, and incorporates the updated carry, memory, and signature. Selection does not modify the ledger, and transport concatenates exactly one entry and produces the target presignature (24.30). The update recomputes the signature by means of the typed map $\text{Sig}_q^{\text{cal}}$ from the already determined sheet, orientation, carry, route, boundary, and memory. The map Upd^{cal} is unary because the edge contained in $z_{\varepsilon, r}(x)$ determines its cocycles.

The identities are the diagonals, and the word extensions are the compositions in (24.41). Concatenation of prefixes and ledgers is associative; memory satisfies the cocycle law, and carry satisfies (24.25). Therefore, the two parenthesizations of three edges produce the same state. Formula (24.45) defines truncation on complete states, and uniqueness of the final ledger proves that it recovers the source in every field. These are the identity, source, target, composition, memory, carry, and truncation laws of the enriched TPK. In particular, the extension relations are determined as graphs of effective maps rather than as consequences of merely necessary conditions. $\square$

Each $\gamma_{\varepsilon, k_n}$ comprises exactly nine applications of $\text{Upd}_{e_{\varepsilon, n, r}}^{\text{cal}} \circ \text{Tra}_{e_{\varepsilon, n, r}}^{\text{cal}} \circ \text{Sel}_{e_{\varepsilon, n, r}}^{\text{cal}}$. Survival will be proved by means of a constructed tail and will not be used to define G_n .

Lemma 24.8 (Effective internal trisection of one turn). *For every $n \geq 0$, every $x \in G_n$, and every $\varepsilon \in \{-1, 0, +1\}$, the expression $\gamma_{\varepsilon, k_n}(x)$ is defined by nine TPK transitions, with its synchronized relations $\text{Ext}^{\text{APP}, \text{cal}}$ and $\text{Ext}^{\text{fib}, \text{cal}}$, and satisfies*

$$\rho_{k_n+9, k_n}^{\text{cal}} \gamma_{\varepsilon, k_n}(x) = x, \quad (24.46)$$

$$g(\gamma_{\varepsilon, k_n}(x)) = g(x), \quad J_{k_n+9}(\text{mem}_{k_n+9}(\gamma_{\varepsilon, k_n}(x))) = J_{k_n}(\text{mem}_{k_n}(x)) + u, \quad (24.47)$$

$$c(\gamma_{\varepsilon, k_n}(x)) = c(x) + \varepsilon \kappa_{\text{Carr}}. \quad (24.48)$$

The ordered ledger of the nine edges determines ε uniquely. The three images are therefore disjoint. Each endpoint belongs to G_{n+1} and again admits the three maps $\gamma_{-1, k_{n+1}}, \gamma_{0, k_{n+1}}, \gamma_{+1, k_{n+1}}$.

Proof. Formulas (24.17) traverse the nine phases once and return the copy $a_{n+1, 1}$ of $a(1)$. At each step, the APP relation transports the complete sextuple; Sel^{cal} fixes the TRIT type from the phase mark r preserved in the state; and Tra^{cal} adds exactly one symbol to the prefix, one segment to the cylinder, and its ordered pair of endpoints to the boundary. By induction on r , the explicit tuples $y_{\varepsilon, r}(x)$ and $z_{\varepsilon, r}(x)$ satisfy the predicates of the indicated relations, and the unary update produces the state at the next level. The identity $\rho_{d_{n, r}+1, d_{n, r}}^{\text{cal}}(x_{\varepsilon, r}(x)) = x_{\varepsilon, r-1}(x)$ is the definition (24.45); its well-definedness follows from uniqueness of the final ledger. The composition of the nine truncation maps recovers x , proving (24.46) without separately inverting the overwritten fields.

The phase $9 \rightarrow 1$ occurs exactly once. By (24.21) and the cocycle law, the linear memory component transports the preceding memory, and its translational component adds exactly u in the direct limit. The compatibility $J_{d+1}J_{d+1, d} = J_d$ proves (24.47); the return concerns the phase, not the enriched state.

By (24.22), eight edges have zero increment, and the edge completing $\mathfrak{p}_{\varepsilon}$ has the computed increment $\chi(\varepsilon, \varepsilon) \kappa_{\text{Carr}}$. The laws (24.25) and (24.20) give (24.48). The ledger preserves $(\mathfrak{p}_{\varepsilon}, 2\varepsilon, r_3(2\varepsilon), \chi(\varepsilon, \varepsilon))$; the three values of this ledger are distinct even when the scalar increment is zero. Hence the images are not identified, and the block decoder recovers a unique ε .

The edge definitions depend only on the new phase and concatenate, rather than replace, the prefix, cylinder, boundary, and ledger. They are also invariant under $J_d \text{ mem}_d \mapsto J_d \text{ mem}_d + nu$ and under translation of the input carry. Consequently, the same construction applies at the endpoint of any turn. For every finite state, the explicit sequence

$$x, \quad \gamma_{0,k_n}(x), \quad \gamma_{0,k_{n+1}}\gamma_{0,k_n}(x), \quad \dots$$

is a compatible infinite tail. This proves the survival of all constructed states and the repeated availability of the three branches, without postulating either property. Consequently, in accordance with the preceding level definitions, we may finally identify

$$H_{k_n}^{\text{cal}} := G_n \quad (24.49)$$

for every $n \geq 0$. $\square$

The preceding closure distinguishes the three clocks involved. Nine emissions return the phase in C_9 and advance memory. Twelve turns, that is, 108 emissions, also return the position of the observable calendar; even then, neither the ledger nor memory is erased. The operator $\mathcal{K}_{\text{ph}}$ acts only afterward, as a recognition of reduced memory from the already constructed turns; it never generates the edges of $\text{Ext}^{\text{enr,cal}}$.

24.6.1.2 Cylinder refinement and ordered readout

For $\mathbf{b} = (b_1, \dots, b_6) \in \mathbb{F}_3^6$, let $\nu : \mathbb{F}_3 \rightarrow \{-1, 0, +1\}$ be the balanced representative. The three prolongations constructed edge by edge define

$$\Gamma_{\mathbf{b},k}^{[54]} = \gamma_{\nu(b_6),k+45} \circ \dots \circ \gamma_{\nu(b_1),k}.$$

Starting from $Y_0 = \{x_\star\}$, put

$$Y_{N+1} = \{\Gamma_{\mathbf{b},1+54N}^{[54]}(x) : x \in Y_N, \mathbf{b} \in \mathbb{F}_3^6\}.$$

Truncation $\rho_{N+1,N}^{\text{blk}}$ removes the last fifty-four edges, recovering the previous state through the transition record.

Proposition 24.9 (Effective coding of prolongation blocks). *Reading the six tritic pairs of each block defines bijections*

$$\beta_N : Y_N \xrightarrow{\cong} (\mathbb{F}_3^6)^N$$

that commute with truncation. Each state of Y_N has exactly 729 block prolongations in Y_{N+1} .

Proof. On each turn, the record retains the tritic pair and its carry. The three pairs produce the three distinct outputs $\chi(\varepsilon, \varepsilon) = \varepsilon$, and the record allows recovery of the branch used, including the zero-carry branch. The six concatenations therefore produce all words of $\mathbb{F}_3^6$. Two different words are distinguished at the first recorded pair in which they differ. Recovery of the prefix proves compatibility with truncation. By induction on N , every word of length N has a unique antecedent state; the next word can be any of the $3^6 = 729$ possibilities. $\square$

Let $\Omega_\Gamma^{\text{cal}} = \varprojlim_N Y_N$. The preceding bijections induce

$$\Omega_\Gamma^{\text{cal}} \cong (\mathbb{F}_3^6)^{\mathbb{N}}.$$

The correspondence and its inverse are continuous: each finite coordinate depends on a finite prefix. The cylinders are open and closed. Each cylinder of depth N is partitioned into its 729 subcylinders of depth $N+1$; each contains an infinite prolongation, obtained, for example, by continuing with the zero branch in all subsequent blocks.

24.6.1.3 Joint construction of the continuum through compatible histories and prolongations

The arithmetic expressed by a normal form preserves a precise readout of the state that produces it. To situate that readout within HMT, the histories, sheets and operations acting on them must also be retained. This chapter brings together that common substrate. Its different constructions arise from the same tree of APP–TRIT–TPK executions; the expository divisions allow their properties to be studied without turning them into independent domains chosen by a subsequent application.

The starting point is the formal kernel presented above. An admissible emission contains the APP operation, TRIT orientation, sheet change, remainder and quotient, together with the effective TPK memory update. When only a remainder is expressed, the execution history still belongs to the starting object. Two executions with identical final readout are not identified on that basis alone.

24.6.1.3.1 Survival by pruning prolongations Retrospective memory distinguishes histories recorded by the APP–TRIT–TPK operations; prospective compatibility organizes their admissible prolongations. Let S be the space of complete states reachable through these operations. Each state retains sheets, residues, quotients, orientation, carry, memory, and boundary data; whenever a transition depends on depth, depth forms part of the state. Let $E \subseteq S \times S$ be the effective prolongation relation and $A \subseteq S$ the admissible set determined by the compatibility of these records. Admissibility is thus decided through the operations and boundary conditions that generate the history.

For $X \subseteq S$, define

$$\text{Pre}_E(X) = \{s \in S : \exists t \in X, (s, t) \in E\}, \quad F_{\text{surv}}(X) = A \cap \text{Pre}_E(X). \quad (24.50)$$

The pruning sequence is given by

$$V_0 = A, \quad V_{n+1} = F_{\text{surv}}(V_n), \quad V_\infty = \bigcap_{n \geq 0} V_n. \quad (24.51)$$

Proposition 24.10 (Survival and the greatest fixed point). *The set V_n consists exactly of the states in A admitting an n -step prolongation all of whose states belong to A . The sequence (V_n) is decreasing. If each state has finitely many admissible successors, then*

$$V_\infty = F_{\text{surv}}(V_\infty) = \nu F_{\text{surv}},$$

where νF_{surv} denotes the greatest fixed point under inclusion. Its elements are precisely the states admitting an infinite compatible prolongation.

Proof. For $n = 0$, a prolongation of length zero consists of the state itself in A . If the characterization holds for n , a state belongs to V_{n+1} exactly when it belongs to A and has a successor in V_n ; prefixing that edge produces an admissible prolongation of $n + 1$ steps. This equivalence proves the assertion by induction. Moreover, $V_1 \subseteq V_0$, and monotonicity of F_{surv} propagates $V_{n+1} \subseteq V_n$.

Let $s \in V_\infty$, and write $E(s) = \{t \in S : (s, t) \in E\}$. Since $s \in V_{n+1}$ for every n , the sets $E(s) \cap V_n$ are nonempty, finite, and decreasing. Their intersection contains an element t , which belongs to V_∞ and satisfies $(s, t) \in E$. Hence $V_\infty \subseteq F_{\text{surv}}(V_\infty)$. Conversely, monotonicity gives $F_{\text{surv}}(V_\infty) \subseteq V_{n+1}$ for every n ; the image also lies in $A = V_0$, and therefore belongs to V_∞ . This proves the fixed-point equality.

If $W = F_{\text{surv}}(W)$, then $W \subseteq A = V_0$. From $W \subseteq V_n$ it follows that $W = F_{\text{surv}}(W) \subseteq F_{\text{surv}}(V_n) = V_{n+1}$. Induction yields $W \subseteq V_\infty$, proving maximality. Finally, the prolongation tree of a state in V_∞ is finitely branching and has a nonempty level at every depth. König's lemma provides an infinite branch. Conversely, the prefixes of any infinite prolongation place its initial state in every V_n . $\square$

24.6.1.3.2 Branching and stabilization. The finiteness hypothesis has an explicit control. Consider a root with a distinct terminal branch of each positive integer length, all branches being admissible and disjoint after the root. The root admits prolongations of every finite length and belongs to all the V_n ; each of its successors belongs to a branch of bounded length. Consequently, the root has neither an infinite prolongation nor a successor in V_∞ . This tree has infinite branching at the root. In contrast, if A is finite, each strict inclusion $V_{n+1} \subsetneq V_n$ removes at least one state. Pruning reaches a fixed point after at most $|A|$ strict layer removals. This bound pertains to the finite domain A .

24.6.1.3.3 Uniqueness of continuation. For $s \in V_\infty$, the successors retaining survival form

$$E(s) \cap V_\infty.$$

The next state is uniquely selected exactly when

$$\#(E(s) \cap V_\infty) = 1. \quad (24.52)$$

An infinite prolongation is unique when this condition holds at each state of the path. Indeed, each surviving successor has an infinite prolongation by the preceding proposition; two distinct successors produce two distinct paths. When the successor is unique at each step, the prefixes are determined successively. Several viable successors express multiplicity of prolongations within the same survival domain.

Proposition 24.11 (Conjugation of the two directions of prolongation). *Let $C : S \rightarrow S$ be an involution such that $C(A) = A$ and*

$$(x, y) \in E \iff (Cy, Cx) \in E. \quad (24.53)$$

Let V_n^+ denote pruning for E , and V_n^- pruning for E^{-1} , both with initial set A . Then

$$C(V_n^+) = V_n^- \quad (n \geq 0), \quad C(V_\infty^+) = V_\infty^-.$$

Proof. If $s_0, \dots, s_n$ is an admissible path for E , the sequence $Cs_0, \dots, Cs_n$ is an admissible path for E^{-1} : for each j , the hypothesis converts $(s_j, s_{j+1}) \in E$ into $(Cs_{j+1}, Cs_j) \in E$. All its states belong to A . Applying the same operation again recovers the initial path because $C^2 = I$; it therefore establishes a bijection between finite prolongations in the two directions. The characterization of $V_n^\pm$ by paths gives $C(V_n^+) = V_n^-$. Since C is bijective, it commutes with the intersection of these sets, yielding the equality for $V_\infty^\pm$. $\square$

The involution transports the surviving set in one direction to the surviving set in the conjugate direction. Infinite prolongability in both directions corresponds to $V_\infty^+ \cap V_\infty^-$; uniqueness of a path retains the criterion in (24.52). Retaining a genealogy and restricting its continuation tree are two typed operations on the same records. Assigning an entropy to these readings additionally requires the corresponding measure on histories.

24.6.1.3.4 Admissible histories and memory by composition Let $H_0 := \{*\}$. Level H_1 is the finite nonempty set of admissible APP–TRIT seeds, and, for $k \geq 1$, H_k is the set of enriched histories of length k that admit compatible TPK prolongations to every finite horizon. Its elements are constructed by successive emissions from the initial domain of the kernel. We write $\zeta\varepsilon$ for the prolongation of ζ by an admissible emission ε , and $\rho_k(\zeta\varepsilon) = \zeta$ for deletion of the last emission. The label produced by the emission is retained, even when different labels express the same scalar state. The space of complete histories is defined by

$$H_\infty^{\text{enr}} := \varprojlim_k (H_k, \rho_k).$$

The cofinal subsystem $X_N := H_{6N}$, with $X_0 = H_0$, agrees with the sextet clock used in the cardinality proof; cofinality gives a canonical identification $H_\infty^{\text{enr}} \cong \varprojlim_N (X_N, \rho_{6(N+1), 6N})$.

On the two arithmetic sheets, the local data satisfy the exact divisions

$$\delta_\diamond = r_\diamond + 9q_\diamond, \quad 0 \leq r_\diamond < 9, \quad \diamond \in \{\Sigma, \Pi\}.$$

The oriented TRIT coordinate likewise admits an expression $\Delta = o_r + 3\kappa$, with the oriented representative declared by the kernel. Preserving the quotient allows those representations to be composed without replacing an emission by its isolated remainder.

In a memory coordinatization, the update of an emission is written

$$U_\varepsilon(m) = C_\varepsilon m + \kappa_\varepsilon.$$

For a trajectory $\alpha = (\varepsilon_1, \dots, \varepsilon_k)$, the effective operation is the ordered composition of these updates. Its linear part and its translation are

$$C_\alpha = C_{\varepsilon_k} \cdots C_{\varepsilon_1}, \quad \kappa_\alpha = \sum_{j=1}^k C_{\varepsilon_k} \cdots C_{\varepsilon_{j+1}} \kappa_{\varepsilon_j}.$$

The empty product in the last sum is the identity.

Proposition 24.12 (Affine compatibility of concatenation). *If α precedes β , then*

$$C_{\beta\alpha} = C_\beta C_\alpha, \quad \kappa_{\beta\alpha} = C_\beta \kappa_\alpha + \kappa_\beta, \quad U_{\beta\alpha} = U_\beta U_\alpha.$$

Proof. Substituting $U_\alpha(m)$ into U_β produces $C_\beta C_\alpha m + C_\beta \kappa_\alpha + \kappa_\beta$. Decomposition into linear part and translation gives the three identities. Associativity of composition proves that the result does not depend on an auxiliary subdivision of the trajectory. $\square$

This cocycle identity is the precise content of accumulated memory. The quantity transported by a second trajectory depends on the coordinatization update it performs; in general, it is not the untransported sum of two records.

The nonadic return preserves a second distinction. In phase and turn coordinates, with $\bar{r} \in \{0, \dots, 8\}$, its representation is

$$\sigma_9(w, r) = \left(w + \left\lfloor \frac{\bar{r} + 1}{9} \right\rfloor, r + 1 \pmod{9} \right), \quad \sigma_9^9(w, r) = (w + 1, r).$$

During nine steps, exactly one crossing from 8 to 0 occurs. The phase returns and the turn coordinate increases. The enriched holonomy also retains the memory updates accompanying that path.

24.6.1.3.5 Tree refinement and measure of histories The initial measure is the canonical census measure: $\mu(*) = 1$ and $\mu(c) = 1/|H_1|$ for each seed $c \in H_1$. In this section each history has a finite positive number of prolongations. Deletion of emissions defines the refinement maps. Admissibility of the kernel supplies the prolongations; when a realization uses a neutral emission, this explicitly gives a section of ρ_k . Existence of that section refers to the tree containing that emission, not to every arbitrary restriction of routes.

Let $d(\zeta) \geq 1$ be the number of admissible emissions at ζ . The APP order and the census of prolongations determine the canonical law $p_{\text{can}}(\varepsilon \mid \zeta) = 1/d(\zeta)$. Without additional parameters, this law defines

$$\mu(\zeta\varepsilon) = \frac{\mu(\zeta)}{d(\zeta)}.$$

A nonuniform weighting may replace p_{can} only when it has already been produced and added to the enriched record. The correlative constructor of this chapter uses only p_{can} ; its measure is therefore a function of the APP–TRIT–TPK tree and not of an external choice.

Proposition 24.13 (Cylindrical compatibility and conditional expectation). *At each level we have*

$$\sum_{\rho_k(y)=x} \mu(y) = \mu(x).$$

In the spaces $L^2(H_k, \mu_k)$, the maps

$$(\mathbf{J}_k f)(y) = f(\rho_k(y)), \quad (E_k g)(x) = \sum_{\rho_k(y)=x} \frac{\mu(y)}{\mu(x)} g(y)$$

satisfy $E_k = \mathbf{J}_k^*$, $E_k \mathbf{J}_k = I$ and $\|\mathbf{J}_k f\| = \|f\|$.

Proof. The first identity is the normalization of local probabilities. Applying it to $|f(x)|^2$ gives the isometry. Grouping the sum by fibers of ρ_k gives the formula for E_k ; on a function constant on each fiber, that formula returns its value. $\square$

The compatible masses define a probability measure on the space of infinite histories: the cylinder of a finite history receives its mass $\mu(\zeta)$. The measure extends from the cylinder algebra, and its uniqueness follows because cylinders generate the σ -algebra. If the tree is finitely branching and every history has a prolongation, the inverse limit of its levels under emission deletion is nonempty. In the cylinder topology, every nonempty cylinder has positive measure.

This measure distinguishes histories that converge to the same representation. The uniform measure on terminal values would be a different readout, since it would remove the multiplicities that the construction has just preserved.

24.6.1.3.6 Enriched readers and fiber memory The difference between history and value admits an operational criterion. At a finite level H_k , let $q_k : H_k \rightarrow Z_k$ be a reader and let $M_k : H_k \rightarrow \mathcal{M}_k$ be the preserved register. The enriched reader is $\widehat{q}_k = (q_k, M_k)$. Memory distinguishes states in the same reader fibre; its sufficiency is checked on the domain considered. This formulation incorporates into the article itself the complete development of the fibre information used below.

Proposition 24.14 (Recoverability at a finite level). *An inverse of $\widehat{q}_k$ exists on its image if and only if $\widehat{q}_k$ is injective. For a random variable Z taking values in H_k with distribution μ_k , we have*

$$H(Z) = H(q_k(Z)) + H(Z \mid q_k(Z)).$$

If $\widehat{q}_k$ is injective on the support of Z , then

$$H(Z \mid q_k(Z), M_k(Z)) = 0.$$

Here $H(Z) = -\sum_z \Pr(Z = z) \log \Pr(Z = z)$, with $0 \log 0 = 0$, is finite entropy in nats; conditional entropy is the average entropy of the conditional distributions.

Proof. An inverse recovers a unique antecedent, which requires injectivity; conversely, injectivity allows each expressed pair to be assigned its unique antecedent. Since $q_k(Z)$ is a deterministic function of Z , grouping the sum defining $H(Z)$ by fibers of q_k gives the chain rule. If the pair $(q_k(Z), M_k(Z))$

distinguishes the support, each corresponding conditional distribution is concentrated on a single state and its entropy is zero. $\square$

Not expressing a coordinate and erasing the record that allows it to be recovered are distinct operations. This result does not identify entropy with the cardinality of the continuum or introduce a thermal interpretation: its domain, reader and distribution are those just fixed.

24.6.1.3.7 Equivariance of the measure under tree symmetries

Proposition 24.15 (Joint transport of the tree and its measure). *Let $g = (g_k)_{k \geq 0}$ be a collection of bijections $g_k : H_k \rightarrow H_k$ that commutes with truncations, fixes the root and bijectively transports admissible emissions, preserving their multiplicities. Then*

$$d(g_k \zeta) = d(\zeta), \quad (g_k)_* \mu_k = \mu_k, \quad (g_\infty)_* \mu = \mu.$$

More generally, if μ_ζ is the law of continuations of a history $\zeta \in H_k$, we have

$$(g_\infty)_* \mu_\zeta = \mu_{g_k \zeta}.$$

Proof. The emission bijection preserves their number with multiplicities. The uniform seed distribution is also invariant. Each path and its image therefore receive the same product of local weights $1/d(\zeta)$. Equality of measures follows at each level and on each cylinder. Coherence with truncations defines g_∞ , and uniqueness of the measure determined by cylinders proves equality in the limit. The same argument, starting at ζ , proves the assertion about continuations. $\square$

This gives naturality of the tree and the measure: a coherent change of representation transports the weights and does not require choosing them anew. Preservation of emissions and multiplicities is an essential hypothesis of the proposition, not a datum added after the construction.

24.6.1.3.8 Transport by sheets and exact accounting of outputs First consider the space with an orthonormal basis formed by the histories at level k . The transport preserving the label of each prolongation is

$$\mathcal{U}_k e_\zeta = \sum_{\varepsilon \text{ admissible en } \zeta} \sqrt{p_{\text{can}}(\varepsilon \mid \zeta)} e_{\zeta \varepsilon}.$$

The successor fibers of distinct antecedents are disjoint. Normalization of p_{can} therefore proves $\mathcal{U}_k^* \mathcal{U}_k = I$. The representation by history masses is related to this one through the diagonal change of basis incorporating $\sqrt{\mu(\zeta)}$.

The sheet label is part of the complete register of a history. We write

$$\widehat{\zeta} = ((\zeta_0, s_0), \dots, (\zeta_k, s_k)), \quad s_j \in \{\Sigma, \Pi\},$$

where s_j is the active sheet at step j . Prolongation adds the emission and the sheet s_{k+1} determined by the operational TPK selector, preserving the entire prefix, including the initial sheet. In the basis of these lifted histories,

$$\mathcal{U}_k e_{\widehat{\zeta}} = \sum_{\varepsilon \text{ admissible en } \widehat{\zeta}} \sqrt{p_{\text{can}}(\varepsilon \mid \widehat{\zeta})} e_{\widehat{\zeta} \varepsilon}, \quad P_k e_{\widehat{\zeta}} = \mathbf{1}_{s_k = \Sigma} e_{\widehat{\zeta}}, \quad Q_k = I - P_k.$$

Thus P_k reads the current sheet, while histories remain distinguished by their complete prefix. Two initial histories reaching the same active sheet retain distinct successor fibers. The isometry proof through

disjoint collections of prolongations applies literally to this basis. Tritic orientation is another coordinate of the record: the value 0 preserves the threshold readout on each sheet and does not add a third sheet to this decomposition. We now separate sheet preservation and sheet exchange:

$$A_k = P_{k+1}\mathcal{U}_k P_k + Q_{k+1}\mathcal{U}_k Q_k, \quad \eta_k = Q_{k+1}\mathcal{U}_k P_k + P_{k+1}\mathcal{U}_k Q_k.$$

Proposition 24.16 (Gram identity by sheets). *For a linear transport U , whether isometric or not, the preceding decomposition satisfies*

$$A^* A + \eta^* \eta = PU^*UP + QU^*UQ.$$

In particular, for $U = \mathcal{U}_k$,

$$A_k^* A_k + \eta_k^* \eta_k = I.$$

Proof. In $A^* A$ and $\eta^* \eta$, mixed terms containing $P_{k+1}Q_{k+1}$ vanish. Summing the remaining terms yields $P_k U^* (P_{k+1} + Q_{k+1}) U P_k$, and its analogue with Q_k . The first identity follows from complementarity of the projectors; the second also uses $U^* U = I$. $\square$

The first identity is the general formula. Replacing its right-hand side by $U^* U$ requires the cross blocks of that Gram matrix to vanish. Isometric preservation of history transport provides that condition here; it should not automatically be attributed to a subsequent compression.

Write $F_{0,0} = I$, $F_{0,n} = A_{n-1} \cdots A_0$ and $T_N = F_{0,N}^* F_{0,N}$.

Theorem 24.17 (Finite decomposition with terminal residual). *For each horizon N ,*

$$I = \sum_{n=0}^{N-1} F_{0,n}^* \eta_n^* \eta_n F_{0,n} + T_N.$$

The sequence T_N is positive and decreasing, and has a positive strong limit T_∞ . The sum of outputs converges strongly to $I - T_\infty$.

Proof. Proposition 24.16 gives $T_n - T_{n+1} = F_{0,n}^* \eta_n^* \eta_n F_{0,n}$. The telescoping sum proves the finite equality and monotonicity. For each vector, the quadratic forms decrease to a limit; polarization defines a bounded positive form and, by the representation theorem for bounded forms, an operator T_∞ . Convergence is strong because, for a positive operator $S \leq I$, $\|Sv\|^2 \leq \langle v, Sv \rangle$; this is applied to $S = T_N - T_\infty$. $\square$

The terminal residue is retained until its extinction in the domain under consideration is proved. This distinction prevents a phase return, by itself, from becoming an assumption of exhaustion of all histories.

24.6.1.3.9 Oriented balance and cancellation under refinement A history carries its accumulated orientation $o(h)$ and the counts of its visits to the two sheets. The corresponding local balance is

$$b(h) = o(h)(N_\Sigma(h) - N_\Pi(h)), \quad b^+ = \frac{|b| + b}{2}, \quad b^- = \frac{|b| - b}{2}.$$

When a reader expresses the mean over prolongations, define $b_c = Eb_f$. Orientation remains in the argument of b_f ; it is not removed before averaging.

Proposition 24.18 (Exact cancellation defect). *The quantity*

$$\kappa_{\text{can}} = \frac{E|b_f| - |Eb_f|}{2}$$

is nonnegative and satisfies

$$E(b_f^+) = (b_c)^+ + \kappa_{\text{can}}, \quad E(b_f^-) = (b_c)^- + \kappa_{\text{can}}.$$

Proof. The weighted triangle inequality gives $|Eb_f| \leq E|b_f|$. Writing the positive and negative parts through $(|b| \pm b)/2$ yields both equalities. $\square$

The information omitted when only the balance is expressed is the common cancellation of opposite contributions. This quantity is determined from the same prolongations used by the preceding transport. The factor $1/9$ appears only when the fiber contains nine prolongations with equal weights; conditional expectation is the rule that remains valid for general branching and weightings.

24.6.1.3.10 Ternary incidence of the sheets and rectangular complex The two sheets determine the module $V = \mathbb{F}_3 e_\Sigma \oplus \mathbb{F}_3 e_\Pi$. Its four projective directions are the lines generated by $e_\Sigma, e_\Pi, e_\Sigma + e_\Pi, e_\Sigma - e_\Pi$. They give rise to six unordered pairs. The eight nonzero vectors are their oriented representatives. Thus, the counts 4, 6, 8 belong to distinct operations on the same sheet module.

On the six pair positions, we define

$$(Ba)_{\{L, L'\}} = a_L + a_{L'}, \quad s(q) = \sum_{\{L, L'\}} q_{\{L, L'\}}.$$

Each direction occurs in three pairs; hence $sB = 0$ in $\mathbb{F}_3$. The reader $W(q) = \mathbf{1}_{s(q)=0} - 1/3$ is invariant under $q \mapsto q + Ba$. On the uniform ambient space $\mathbb{F}_3^6$, its moments are

$$\mathbb{E}W = 0, \quad \mathbb{E}W^2 = \frac{2}{9}, \quad \mathbb{E}W^3 = \frac{2}{27}.$$

Indeed, s is surjective and each fiber contains 3^5 elements. The reader equals $2/3$ with probability $1/3$ and $-1/3$ with probability $2/3$. These means describe the uniform ambient space; the distribution induced by a subset of histories must be calculated with its own multiplicities, already preserved in H_k .

To fix the ports without losing multiplicities, at depth k retain the labeled readouts $\lambda_\Sigma(\zeta) = (\zeta, \Sigma)$ and $\lambda_\Pi(\zeta) = (\zeta, \Pi)$. Their sets are

$$\mathcal{P}_k^\Sigma = \{\lambda_\Sigma(\zeta) : \zeta \in H_k\}, \quad \mathcal{P}_k^\Pi = \{\lambda_\Pi(\zeta) : \zeta \in H_k\}.$$

They are finite and nonempty because H_k is. Values, quotients and remainders are expressed on those ports; equality of values does not identify their labels. The product $\mathcal{P}_k^\Sigma \times \mathcal{P}_k^\Pi$ is the domain of the rectangular readout, within which the support of admissible incidences is recorded.

With this level fixed, write $I = \mathcal{P}_k^\Sigma, J = \mathcal{P}_k^\Pi$ and $A_m = \mathbb{Z}/3^m\mathbb{Z}$. The APP lexicographic order determines the minimal surviving branch and, on it, the reference ports i_0, j_0 . No external choice is added: changing coordinates produces a canonically isomorphic complex. The maps

$$\kappa(u, v)_{ij} = u_i - v_j, \quad (\delta w)_{ij} = w_{ij} - w_{ij_0} - w_{i_0j} + w_{i_0j_0}$$

define the following sequence, valid over the finite ring A_m :

$$0 \longrightarrow A_m \longrightarrow A_m^I \oplus A_m^J \xrightarrow{\kappa} A_m^{I \times J} \xrightarrow{\delta} A_m^{(I \setminus \{i_0\}) \times (J \setminus \{j_0\})} \longrightarrow 0.$$

Proposition 24.19 (Exactness of rectangular incidence). *The preceding sequence is exact, with first arrow given by the constant diagonal.*

Proof. The equality $u_i - v_j = 0$ for each pair forces all coordinates of u and v to be the same constant. Direct substitution gives $\delta\kappa = 0$. If $\delta w = 0$, take $u_i = w_{ij_0}$ and $v_j = w_{i_0j_0} - w_{i_0j}$. Then $u_i - v_j = w_{ij}$.

Finally, any matrix of the last term is extended with zero reference row and column; its image under δ is the initial matrix. The proof uses only addition and subtraction, so it does not require division by a unit of the ring. $\square$

Reducing from A_{m+1} to A_m while keeping $k, \mathcal{P}_k^\Sigma, \mathcal{P}_k^\Pi, i_0, j_0$ fixed commutes with both maps. The reconstruction formulas are the same at each modular depth. This is the arithmetic compatibility proved in this section. The prolongation of histories $k \rightarrow k+1$ preserves prefix labels; its maps between port sets are a datum distinct from the ring change $m \rightarrow m+1$.

These maps are obtained from emission removal:

$$\rho_k^\Sigma(\zeta\varepsilon, \Sigma) = (\zeta, \Sigma), \quad \rho_k^\Pi(\zeta\varepsilon, \Pi) = (\zeta, \Pi).$$

The minimal surviving branch of the APP order just constructed has compatible prolongations. Its two labels define compatible reference ports at every level. Coordinate transport is precomposition:

$$(L_k u)_{i'} = u_{\rho_k^\Sigma(i')}, \quad (L_k v)_{j'} = v_{\rho_k^\Pi(j')}, \quad (L_k w)_{i'j'} = w_{\rho_k^\Sigma(i'), \rho_k^\Pi(j')}.$$

For the last module of the sequence, first extend the matrix by zero on the reference row and column, apply this same formula and restrict to the new complement of the reference ports. This rule defines L_k even when a new port has the marked port as its antecedent.

Proposition 24.20 (Naturality of the rectangular complex under prolongation). *The preceding maps satisfy $L_k \kappa_k = \kappa_{k+1} L_k$ and $L_k \delta_k = \delta_{k+1} L_k$. They also commute with coefficient reduction $A_{m+1} \rightarrow A_m$ and compose according to prefix removal.*

Proof. The first identity follows from $u_{\rho_k^\Sigma(i')} - v_{\rho_k^\Pi(j')}$. In the second, each of the four terms defining δ is transported by the same precomposition. Compatibility of the marked ports identifies their rows and columns; when an antecedent is the marked port, the four terms cancel and agree with extension by zero. All formulas are sums and differences with integer coefficients, so they commute with modular reduction. Composition of precompositions is precomposition with the composite prefix, which proves the last assertion. $\square$

24.6.1.3.11 Memory complex and compatibility of incidences The same accumulated memory coordinate $c_y \in \mathbb{Z}$, in a residence y , defines a free integer complex. Its reduction to A_m uses the same integer register before reducing it; successive depths are therefore compatible. Its generators are f_y in degree 2, e_y, b_y, r_y in degree 1, and g_y in degree 0:

$$\partial f_y = 3c_y e_y + b_y, \quad \partial e_y = g_y, \quad \partial b_y = -3c_y g_y, \quad \partial r_y = 0.$$

The identity $\partial^2 = 0$ is verified on f_y through $3c_y g_y - 3c_y g_y = 0$, and is immediate on the other generators.

If a trajectory $\alpha : y \rightarrow y'$ updates memory, its transport in the complex sends the correspondingly named generators to their new representatives, except

$$\Psi_\alpha(b_y) = b_{y'} + 3(c_{y'} - c_y)e_{y'}.$$

Proposition 24.21 (Naturality of the memory complex). *We have $\partial \Psi_\alpha = \Psi_\alpha \partial$ and $\Psi_\beta \alpha = \Psi_\beta \Psi_\alpha$. The maps extend linearly over the chosen coefficient ring.*

Proof. On f_y , the image of its boundary is $3c_y e_{y'} + b_{y'} + 3(c_{y'} - c_y)e_{y'} = 3c_{y'} e_{y'} + b_{y'}$. On b_y , the boundary of its image is $-3c_{y'} g_{y'} + 3(c_{y'} - c_y)g_{y'} = -3c_y g_{y'}$. The other two cases are immediate. In a concatenation, memory differences sum telescopically, proving the second identity. $\square$

The relation between two representatives of the same residence can also be written as an explicit boundary. For an integer coordinate v , reduced in the same ring where appropriate, let

$$z^- = r_y, \quad z^+ = r_y + 3c_y v e_y + v b_y, \quad h_* = -v f_y.$$

Then $\partial h_* = z^- - z^+$. Under a transport that also updates v , the correction $K_\alpha = (v_{y'} - v_y) f_{y'}$ satisfies $K_{\beta\alpha} = K_\beta + \Psi_\beta K_\alpha$. The symbols K_α of this local homotopy do not denote the dodecaphasic seal K : they belong to another domain and their function is specified by the preceding identity. Their function can also be checked in the equality

$$z_{y'}^+ - \Psi_\alpha z_y^+ = \partial K_\alpha.$$

Both sides equal $(v_{y'} - v_y)(3c_{y'} e_{y'} + b_{y'})$.

Proposition 24.22 (Homology and memory of the local complex). *Over $\mathbb{Z}$, or after reduction to A_m , we have*

$$H_0(\Gamma) = H_2(\Gamma) = 0, \quad H_1(\Gamma) \cong A_m[r_y]$$

in the modular realization, and $H_1(\Gamma) \cong \mathbb{Z}[r_y]$ in the integer realization.

Proof. Define the homotopy of degree 1 by $h(g_y) = e_y$, $h(b_y) = f_y$, and zero on the other generators. Let p be the projection onto the summand generated by r_y and ι its inclusion. Checking on each generator gives

$$\partial h + h \partial = I - \iota p.$$

In particular, on b_y the terms $3c_y e_y$ and $-3c_y e_y$ cancel. The complex deformation retracts onto the module generated by r_y , concentrated in degree 1, which proves the assertion. $\square$

The transport Ψ_α is a shear of determinant 1. Thus, this complex explicitly preserves how c_y changes at the chain level, but its local homology does not distinguish the values of that coordinate. The full memory remains in the labels and transports of the source state; it is not recovered from H_1 alone. A memory-dependent numerical torsion requires the specified assembly and determinantal reading, in addition to this local complex.

24.6.1.3.12 Assembly with terminal paths For a finite horizon, the residences and their prolongations form a path category. Each residence v is assigned the free module $W(v) = A_m[\text{Path}(v, \text{Term})]$, which preserves each terminal continuation as a generator. An initial path acts on W by precomposition. The complex $\Gamma(v)$ of the preceding section acts covariantly through Ψ .

The assembly uses both transports in the same bigraded module:

$$\mathcal{B}_{q,r} = \bigoplus_{v_0 \rightarrow \dots \rightarrow v_q} W(v_q) \otimes_{A_m} \Gamma_r(v_0).$$

The normalized path complex is used: degenerate chains containing identities are omitted. The finite horizon then bounds the number of strict prolongations and the horizontal degree. The horizontal boundary removes an intermediate residence by composing the adjacent arrows. At the first endpoint it transports Γ ; at the last it precomposes W . The alternating signs yield the path boundary d_{bar} . The total differential is

$$D = d_{\text{bar}} + (-1)^q \partial_\Gamma.$$

The face identities cancel the terms of d_{bar}^2 in pairs. The naturality of Proposition 24.21 implies that d_{bar} commutes with ∂_Γ ; the factor $(-1)^q$ cancels the cross terms of the square. Therefore, $D^2 = 0$. Compatibility is proved on path generators and is not introduced by a label of membership in a complex.

Terminal evaluation likewise does not require choosing a particular history. For each continuation h , the effective composition of its updates is applied. The result preserves the pair formed by h and its reading, with its multiplicity. If $k < \ell < N$, the continuations decompose as a disjoint union of prefixes up to ℓ and their continuations up to N . Proposition 24.12 proves that evaluating directly or evaluating these two parts produces the same reading. This yields the naturality square of the terminal.

The path modules, incidences, memory transport and their differentials arise from the same prolongations. Their assembly does not consist in adding to a state five coordinates that would be preserved by definition: their relations are constructed through sums, boundaries and effective compositions, and are verified on their generators.

24.6.1.3.13 Determinants, limit and arithmetic reading In each realization, the local free complex Γ admits its graded determinant line

$$\text{Det } \Gamma = \bigotimes_r \left(\bigwedge^{\text{rk } \Gamma_r} \Gamma_r \right)^{(-1)^r}.$$

The total assembly of the preceding section, in turn, has the modules $\text{Tot}_d \mathcal{B} = \bigoplus_{q+r=d} \mathcal{B}_{q,r}$. At a finite horizon and over the same ring, they are free of finite rank and only finitely many of them are nonzero. Its line is

$$\text{Det}(\text{Tot } \mathcal{B}) = \bigotimes_{q,r} \left(\bigwedge^{\text{rk } \mathcal{B}_{q,r}} \mathcal{B}_{q,r} \right)^{(-1)^{q+r}}.$$

This formula follows from decomposition as a direct sum in each degree. It distinguishes the line of the local complex from that of the assembly with all paths; their bases are ordered through the preserved labels. Modular reduction commutes with these exterior powers because the modules and their bases come from the same free integer modules at a fixed horizon. The line is defined by the modules and their degrees, both over $\mathbb{Z}$ and after reduction to A_m ; the negative exponent denotes the dual module. A nonzero numerical torsion additionally requires specifying the complex over a field, the bases and the relevant homology identifications. To calculate Smith factors, the integer matrix produced before reduction is used, not an arbitrary lift of a modular matrix. A square integer block with nonzero determinant is invertible over $\mathbb{Q}$; its reduction is invertible over A_m only when that determinant is a unit, that is, when it is not divisible by 3. Refinement of memories and incidences allows the stability of the relevant invariant factors to be formulated and checked; these conditions do not follow from the existence of the determinant line.

Passage to the limit preserves the compatible projective systems of remainders, histories and their multiplicities. The squares proved in the preceding sections remain commutative because their maps are given by the same formulas at each level. For an Archimedean coordinate, the additional step consists in expressing nonempty cylinder intervals, compatible by inclusion and with diameters tending to zero. Their nested compact closures have a one-point intersection: existence follows by compactness and uniqueness by the diameter bound. That is the value of the limit readout. If the intervals are half-open, the point may belong only to their closures; the boundary representation rule then determines its positional expression. The value does not replace the history that produced it. The other representation preserves the incidences of the same state. Both destinations share their origin without thereby receiving the same information.

The coinductive regions, the dodecaphase record and exceptional incidence are expressed from that substrate through their respective maps. The Hilbert-space realization subsequently preserves phase labels and specifies its own unital inclusion; duality and screens act on the codomains constructed there. The joint operations brought together here transmit histories, sheets, orientation and memory to those

representations. Identifying a particular functional with an external analytic form in turn requires writing its action on the generators and verifying its inner products: the conservation identities of this chapter do not replace that composition.

24.6.1.3.14 Mathematical scope of the joint construction The state, tree and measure constructions; the correlative operations; and the terminal assembly with naturality and limit are developed in full in this article. In this chapter their balance, incidence and memory identities have been formulated and proved on the same generators. The five constructions thus remain operations of a single tree. Each specialization keeps its exact hypotheses visible: transport isometry, canonical weight law, treatment of the terminal remainder and nonsingularity of the determinantal block where applicable.

The preceding equations prove the memory cocycles, weighted cancellation, rectangular exactness and compatibility of the memory complex at every finite horizon, as well as their compatible passage to the limit. A subsequent analytic identification must compose its operator on these generators and verify the additional properties of its codomain; it neither alters nor replaces the joint construction proved here.

24.6.1.3.15 Joint emergence of the five constructions For $0 \leq k \leq N$, let

$$\rho_{N,k} := \rho_k \circ \rho_{k+1} \circ \cdots \circ \rho_{N-1}, \quad \rho_{k,k} := \text{id}_{H_k},$$

and, for a surviving history $\zeta \in H_k$, define

$$\text{Desc}_{k,N}(\zeta) := \{x \in H_N : \rho_{N,k}(x) = \zeta\}.$$

The finite tree $\mathcal{T}_{k,N}(\zeta)$ has vertices

$$\prod_{j=k}^N \{y \in H_j : \rho_{j,k}(y) = \zeta\}$$

and has as edges only the TPK prolongations $a : y \rightarrow z$ with $z \in \rho_j^{-1}(y)$. Thus neither its vertices nor its prolongations come from an auxiliary tree.

For $x \in \text{Desc}_{k,N}(\zeta)$, the marked branch $\lambda_{\zeta,x}$ is the sequence

$$\zeta = x_k \longrightarrow x_{k+1} \longrightarrow \cdots \longrightarrow x_N = x$$

determined by the truncations of x . The common base register is

$$\begin{aligned} \mathcal{R}_{k,N}(\zeta; x) := & (\mathcal{T}_{k,N}(\zeta), \lambda_{\zeta,x}; \\ & (\varepsilon_a, s_a, o_a, (\delta_{\diamond,a}, r_{\diamond,a}, q_{\diamond,a})_{\diamond \in \{\Sigma, \Pi\}}, \text{Carr}_a, C_a, t_a, \text{Fr}_a, \text{Ruta}_a, \text{Mem}_a, \text{Surv}_a)_a; \quad (\text{C1}) \\ & \text{src}, \text{tgt}, \circ, (\rho_j)_{k \leq j < N}), \end{aligned}$$

where

$$U_a(m_0) = C_a m_0 + t_a, \quad \delta_{\diamond,a} = r_{\diamond,a} + 9q_{\diamond,a}.$$

The symbol m_0 in this formula is a memory coordinate; it is not the modular depth m used below. The true base space is

$$\mathfrak{R}_{k,N}(\zeta) := \{\mathcal{R}_{k,N}(\zeta; x) : x \in \text{Desc}_{k,N}(\zeta)\}.$$

The marked branch is not an added input: it is the sequence of actually produced edges ending at x . The rest of the register simultaneously preserves the lateral fibres of the tree; thus the mark does not replace branching with an isolated path.

Set $R_m := \mathbb{Z}/3^m\mathbb{Z}$, and let $V_j(\zeta) := \text{Desc}_{k,j}(\zeta)$. Transport, the sheet projectors and the terminal residue restrict to the subtree through

$$\begin{aligned} \mathcal{U}_j^\zeta &:= \mathcal{U}_j \upharpoonright_{\ell^2(V_j(\zeta), \mu_j)}, & P_j^\zeta &:= P_j \upharpoonright_{V_j(\zeta)}, & Q_j^\zeta &:= I - P_j^\zeta, \\ A_j^\zeta &:= P_{j+1}^\zeta \mathcal{U}_j^\zeta P_j^\zeta + Q_{j+1}^\zeta \mathcal{U}_j^\zeta Q_j^\zeta, & \eta_j^\zeta &:= Q_{j+1}^\zeta \mathcal{U}_j^\zeta P_j^\zeta + P_{j+1}^\zeta \mathcal{U}_j^\zeta Q_j^\zeta. \end{aligned}$$

For $k \leq n \leq N$, let

$$F_{k,k}^\zeta := I, \quad F_{k,n}^\zeta := A_{n-1}^\zeta \cdots A_k^\zeta, \quad T_{k,n}^\zeta := (F_{k,n}^\zeta)^* F_{k,n}^\zeta.$$

The ports belonging to the same subtree are

$$\mathcal{P}_j^\Sigma(\zeta) := \{(y, \Sigma) : y \in V_j(\zeta)\}, \quad \mathcal{P}_j^\Pi(\zeta) := \{(y, \Pi) : y \in V_j(\zeta)\};$$

we denote by $B_j^\zeta, \kappa_j^\zeta, \delta_j^\zeta$ the restrictions of the incidence and the rectangular complex to those ports, and by L_j^ζ the precomposition induced by removing prefixes.

To define the path reader without abbreviations, let

$$W_\ell^\zeta(y) := R_m[\text{Path}_{\mathcal{T}_{k,\ell}(\zeta)}(y, \text{Term}_\ell)]$$

and set

$$\mathcal{D}_{q,r;k,\ell}^{(m)}[\mathcal{R}] := \bigoplus_{y_0 \rightarrow \cdots \rightarrow y_q \text{ in } \mathcal{T}_{k,\ell}(\zeta)} W_\ell^\zeta(y_q) \otimes_{R_m} (\Gamma_r(y_0) \otimes_{\mathbb{Z}} R_m), \quad \text{Tot}_d \mathcal{D} := \bigoplus_{q+r=d} \mathcal{D}_{q,r;k,\ell}^{(m)}[\mathcal{R}].$$

The formulas already proved produce, on each $\mathcal{R} \in \mathfrak{R}_{k,N}(\zeta)$, the five objects

$$\begin{aligned} \mathcal{F}_{\text{sp}}[\mathcal{R}] &:= (\ell^2(V_j(\zeta), \mu_j), \mathcal{U}_j^\zeta, A_j^\zeta, \eta_j^\zeta, T_{k,j}^\zeta)_{k \leq j \leq N}, \\ \mathcal{F}_{\text{sol}}[\mathcal{R}] &:= (b, \kappa_{\text{can}}, C_\alpha, \kappa_\alpha)_{\alpha \in \mathcal{T}_{k,N}(\zeta)}, \\ \mathcal{F}_{\text{gau}}^{(m)}[\mathcal{R}] &:= (\mathcal{P}_j^\Sigma(\zeta), \mathcal{P}_j^\Pi(\zeta), B_j^\zeta, \kappa_j^\zeta, \delta_j^\zeta; R_m)_{k \leq j \leq N}, \\ \mathcal{F}_{\text{coh}}^{(m)}[\mathcal{R}] &:= (\Gamma_y \otimes R_m, \partial, \Psi_\alpha, H_\bullet)_{y, \alpha \in \mathcal{T}_{k,N}(\zeta)}, \\ \mathcal{F}_{\text{det}}^{(m)}[\mathcal{R}] &:= (\text{Det}(\text{Tot}_{\bullet, \bullet; k, \ell}^{(m)}[\mathcal{R}]))_{k \leq \ell \leq N}. \end{aligned} \tag{C2}$$

Here $\mathcal{D}_{\bullet, \bullet; k, \ell}^{(m)}[\mathcal{R}]$ is the normalized path-and-memory bicomplex of the preceding sections, restricted to the subtree of horizon ℓ and reduced over R_m . The last reader preserves the ordered collection of determinant lines up to horizon N ; in this way, its truncation is a canonical projection and does not presuppose a nonexistent determinantal factorization. The law μ is the canonical census law, and the ports are fixed through the lexicographically minimal branch of the APP order. Thus none of the five readers receives external weighting or an external port.

For $T \in \mathfrak{T} := \{\text{sp}, \text{sol}, \text{gau}, \text{coh}, \text{det}\}$, define the typed total space

$$\widetilde{\mathcal{F}}_{T;k,N}^{(m)}(\zeta) := \coprod_{\mathcal{R} \in \mathfrak{R}_{k,N}(\zeta)} \{\mathcal{F}_T^{(m)}[\mathcal{R}]\}_{\mathcal{R}}, \quad p_T : \widetilde{\mathcal{F}}_{T;k,N}^{(m)}(\zeta) \longrightarrow \mathfrak{R}_{k,N}(\zeta),$$

where $\mathcal{F}_{\text{sp}}^{(m)} := \mathcal{F}_{\text{sp}}$ and $\mathcal{F}_{\text{sol}}^{(m)} := \mathcal{F}_{\text{sol}}$; that is, the superscript m is inert for those two readers. The projection p_T forgets only the reader structure and preserves its source register. The correct fibre product is

$$\mathcal{J}_{k,N}^{(m)}(\zeta) := \widetilde{\mathcal{F}}_{\text{sp};k,N}^{(m)}(\zeta) \times_{\mathfrak{R}_{k,N}(\zeta)} \widetilde{\mathcal{F}}_{\text{sol};k,N}^{(m)}(\zeta) \times_{\mathfrak{R}_{k,N}(\zeta)} \widetilde{\mathcal{F}}_{\text{gau};k,N}^{(m)}(\zeta) \times_{\mathfrak{R}_{k,N}(\zeta)} \widetilde{\mathcal{F}}_{\text{coh};k,N}^{(m)}(\zeta) \times_{\mathfrak{R}_{k,N}(\zeta)} \widetilde{\mathcal{F}}_{\text{det};k,N}^{(m)}(\zeta). \quad (\text{C3})$$

Equality of the five projections requires preserving the same tree, the same edges, the same produced branch and the same register. For each $\mathcal{R}$, the five constructions determine the element

$$s_{k,N}^{(m)}(\mathcal{R}) := (\mathcal{F}_T^{(m)}[\mathcal{R}])_{T \in \mathfrak{T}} \in \mathcal{J}_{k,N}^{(m)}(\zeta).$$

Existence of this element follows from the operations of (C2), not from the fibre-product notation.

Removal of the last layer induces

$$\text{res}_{N+1,N}^{\mathfrak{R}}(\mathcal{R}_{k,N+1}(\zeta; x)) := \mathcal{R}_{k,N}(\zeta; \rho_N(x)).$$

Restriction of each operator to the retained generators defines

$$r_{N+1,N}^m : \mathcal{J}_{k,N+1}^{(m)}(\zeta) \longrightarrow \mathcal{J}_{k,N}^{(m)}(\zeta).$$

In the determinantal reader, $r_{N+1,N}^m$ removes only the last coordinate of the collection of lines of (C2). The reduction $R_{m+1} \rightarrow R_m$ defines, by base change,

$$q_{m+1,m}^N : \mathcal{J}_{k,N}^{(m+1)}(\zeta) \longrightarrow \mathcal{J}_{k,N}^{(m)}(\zeta).$$

On each generator, we have

$$\begin{aligned} r_{N+1,N}^m q_{m+1,m}^{N+1} &= q_{m+1,m}^N r_{N+1,N}^{m+1}, \\ \partial \Psi_\alpha &= \Psi_\alpha \partial, \quad L_j^\zeta \kappa_j^\zeta = \kappa_{j+1}^\zeta L_j^\zeta, \quad L_j^\zeta \delta_j^\zeta = \delta_{j+1}^\zeta L_j^\zeta, \end{aligned} \quad (\text{C4})$$

and compositions of the maps r and q satisfy the functorial identities. The first equality holds because deleting the last layer and reducing the coefficients act on distinct coordinates. The other three are the naturality identities proved earlier. A unique pro-system in (N, m) is obtained, not five juxtaposed limits.

At the root, define

$$\mathfrak{J}_{N,\bullet}^{(m)} := \mathcal{J}_{0,N}^{(m)}(*). \quad (\text{C5})$$

The joint discrete structure of the continuum is the limit of those correlative objects:

$$\boxed{C_{\text{cont}}^{\text{disc}} := \lim_{\substack{\leftarrow \\ N \geq 1 \\ m \geq 1}} (\mathfrak{J}_{N,\bullet}^{(m)}, r_{N+1,N}^m, q_{m+1,m}^N).} \quad (\text{C6})$$

This is a definition through the pro-system generated by the complete tree; it is not the image or graph of a map retaining the input state as its first coordinate.

The registers have a produced terminal mark. Their compatible projections induce

$$\Pi : C_{\text{cont}}^{\text{disc}} \longrightarrow H_{\infty}^{\text{enr}}.$$

Conversely, a complete history determines, through its already emitted prefixes, the compatible sections $s_{0,N}^{(m)}$. Thus, after having defined the continuum, one obtains

$$\text{Gen}_{\text{cont}} : H_{\infty}^{\text{enr}} \longrightarrow C_{\text{cont}}^{\text{disc}}, \quad \Pi \circ \text{Gen}_{\text{cont}} = \text{id}_{H_{\infty}^{\text{enr}}}. \quad (\text{C7})$$

This section proves recoverability; it is not used to define $C_{\text{cont}}^{\text{disc}}$.

Theorem 24.23 (Correlative emergence of the five constructions). *For every surviving history $\zeta \in H_k$, every $N > k$, every $m \geq 1$ and every $\mathcal{R} \in \mathfrak{R}_{k,N}(\zeta)$, the five constructions of (C2) are defined on the same register, and (C3) contains $s_{k,N}^{(m)}(\mathcal{R})$. Their truncations, modular reductions and transports satisfy (C4). Furthermore, $C_{\text{cont}}^{\text{disc}} \neq \emptyset$, the map Π is surjective and is split by Gen_{cont} . The five components are inseparable aspects of the same continuum, not five domains or five independent maps.*

Proof. The survival defining H_k and the TPK prolongation law give $\rho_k^{-1}(\zeta) \neq \emptyset$ for each vertex. Since the fibres are finite, Kőnig's lemma produces $H_{\infty}^{\text{enr}} \neq \emptyset$.

Given a register $\mathcal{R}$, the transport of histories and sheets produces $\mathcal{F}_{\text{sp}}[\mathcal{R}]$; affine composition and the exact cancellation defect produce $\mathcal{F}_{\text{sol}}[\mathcal{R}]$; incidences and the rectangular sequence produce $\mathcal{F}_{\text{gau}}^{(m)}[\mathcal{R}]$; the memory complex and its transport produce $\mathcal{F}_{\text{coh}}^{(m)}[\mathcal{R}]$; and the lines of the normalized path assembly produce $\mathcal{F}_{\text{det}}^{(m)}[\mathcal{R}]$. Each construction acts on the same generated edges, not on an assumed subdomain or on the abstract ambient space of 729 words.

The affine, transport, incidence and memory compatibilities prove (C4). Truncating the determinantal collection of (C2) commutes by definition with r ; its base change commutes with q . The sections $s_{0,N}^{(m)}$ are compatible, so they produce Gen_{cont} . The identity (C7) simultaneously proves nonemptiness of the limit, surjectivity of Π and recoverability of each history without introducing it as a defining coordinate.

The fibre, relation, number, orientation, carry, memory, boundary and route remain in the common register; the multisection, terminals, and linear and probabilistic readers remain in the five correlative constructions. Passing to the limit therefore preserves dependency closure and does not separate the continuum. $\square$

Subsequent recognition of the uniquely constructed continuum through spectral, solenoidal, gauge, cohomological or determinantal language belongs to the later conventional comparison. It does not select its states or enter its APP–TRIT–TPK generation.

24.6.1.4 Extension of states and transfinite semantic extension

Let

$$\Sigma_{\text{HMT}} = (\dot{R}_{\text{APP}}, \dot{R}_{\text{TRIT}}, \dot{R}_{\text{TPK}}, \dot{R}_U, \dot{R}_{\Gamma_9}, \dot{R}_{\text{Ext}}, \dot{R}_\rho, \dot{R}_\pi, \dot{R}_\varphi, \dot{R}_e, \dot{R}_\alpha, \dot{R}_{\text{ar}}, \dot{R}_{\text{exc}}) \quad (33)$$

be the countable **relational** signature whose effective code is h . Each $\dot{R}_S$ is the graph of the operator S on the codes where it is defined; no external function is assumed to be total within each level. The graphs $\dot{R}_\pi, \dot{R}_\varphi, \dot{R}_e, \dot{R}_\alpha$ are the output readers constructed by coinductive generation and dodecaphase closure; they do not take the conventional values of the constants as arguments. Their presence in Σ_{HMT} ensures that the closure preserves the full state: it simultaneously integrates the routes that express value and the incidence that extends toward the exceptional realization.

At each horizon, the state code records the APP–TRIT–TPK coordinates, the route, sheet, carry, boundary, and memory; at its representation stages it also records P_N, Φ_N, E_N, A_N , where A_N is the prefix of α_{HMT} produced by the same state. These coordinates do not select the branch from outside: they are part of the history being encoded. Once a primitive-recursive numbering of the finite coordinates of a state is fixed, every inverse history $x_\infty = (x_N)_{N \geq 1} \in X_\infty^{\text{enr}}$ determines the code

$$\text{code}(x_\infty) := \{\langle N, j, c \rangle : (x_N)_j \text{ has code } c\} \subseteq \omega.$$

The coherence equations $\rho_{N+1,N}(x_{N+1}) = x_N$ make it possible to decode the history uniquely from that real. We write m_∞ for the code of the realized history and $m_\infty(n)$ for its characteristic function under the fixed numbering. The operator code h and this complete ledger are combined, by means of a primitive-recursive pairing function, into the real

$$u_{h,m} := \{\langle 0, \langle n, h(n) \rangle \rangle : n < \omega\} \cup \{\langle 1, \langle n, m_\infty(n) \rangle \rangle : n < \omega\}. \quad (33a)$$

The two primitive projections recover h and m_∞ from $u_{h,m}$.

The first extension is the metric and operator extension of the finite states. For $N \geq 1$, let $\widehat{X}_{6N}^{\text{enr,raw}}$ be the set of raw enriched states of length $6N$, and define the surviving subsystem set-theoretically by

$$\begin{aligned} X_N &:= \widehat{X}_{6N}^{\text{enr,surv}} \\ &:= \left\{ x \in \widehat{X}_{6N}^{\text{enr,raw}} : \right. \\ &\quad \left. \forall M > N \exists z \in \widehat{X}_{6M}^{\text{enr,raw}} \text{ such that } \right. \\ &\quad \left. \text{tr}_{6M,6N}(z) = x \wedge z \text{ satisfies every intermediate TPK transition} \right\}. \end{aligned}$$

and set

$$\rho_{N+1,N} := \text{tr}_{6N+6,6N} \upharpoonright X_{N+1},$$

and

$$\text{Ext}_N(x) := \{y \in X_{N+1} : (x, y) \in \text{Ext}_{6N+6,6N}\}.$$

The root level is included without any implicit exception:

$$X_0 := \{*\}, \quad \rho_{1,0} : X_1 \rightarrow X_0 \text{ is constant}, \quad \text{Ext}_0(*) := X_1.$$

The superscript *surv* denotes the pruned tree of states possessing compatible TPK descendants at every finite horizon. The TPK extension law established in the preceding sections proves that the emitted seeds belong to this tree; finite branching and König's lemma turn compatibility at every horizon into an infinite branch. In particular,

$$\forall N \geq 0 \forall x \in X_N \quad \emptyset \neq \text{Ext}_N(x) \subseteq \rho_{N+1,N}^{-1}(x). \quad (34)$$

By the same finitely branching tree lemma, every surviving seed has a compatible extension

$$x_\infty = (x_N)_{N \geq 1} \in X_\infty^{\text{enr}} := \varprojlim_N (X_N, \rho_{N+1,N}). \quad (35)$$

The second extension acts **afterward** on the complete code of that realized history. It is not inferred from the mere existence of the finite operator Ext_N : at the set-theoretic scale, it formalizes the HMT principle that only those objects possessing a genealogical derivation from the state and its rules belong to the full domain. The transfinite semantic extension is:

$$\begin{aligned} \mathfrak{G}_0(h, m_\infty) &:= \text{TC}(\{\omega, u_{h,m}, h, m_\infty\}), \\ \mathfrak{G}_{\beta+1}(h, m_\infty) &:= \text{Def}_{\Sigma_{\text{HMT}}}(\mathfrak{G}_\beta(h, m_\infty)), \\ \mathfrak{G}_\lambda(h, m_\infty) &:= \bigcup_{\beta < \lambda} \mathfrak{G}_\beta(h, m_\infty) \quad (\lambda \text{ limit}), \\ \mathfrak{G}_{\text{HMT}}(h, m_\infty) &:= \bigcup_{\beta \in \text{Ord}} \mathfrak{G}_\beta(h, m_\infty). \end{aligned} \quad (36)$$

The Continuum Hypothesis and a target bijection are not introduced among the conditions defining the successor operator; the operator is given prospectively by

$$\text{Def}_{\Sigma_{\text{HMT}}}(M) := M \cup \left\{ \{x \in M : \langle M, \in, (\dot{R}_S \cap M^{ks})_{S \in \Sigma_{\text{HMT}}} \rangle \models \varphi(x, \bar{p})\} : \ulcorner \varphi \urcorner \in \omega, \bar{p} \in M^{<\omega} \right\}. \quad (37)$$

That is, it simultaneously expresses all subsets definable by a finite formula of the HMT relational language and parameters already produced by the genealogy. It consults neither a target real value, a conventional constant, a target cardinal, nor an external model. Formula (37) defines the transfinite closure over the previously constructed seeds, operators, enriched state, ledger, and double projection.

Lemma 24.24 (Joint coding of the HMT program and the realized ledger). *The pair formed by the operator code h and the ledger m_∞ is encoded by a single real $u_{h,m} \subseteq \omega$, and every formula of Σ_{HMT} is uniformly translated into a formula of the language $\{\in, u_{h,m}\}$.*

Proof. Both h and m_∞ are sequences of natural numbers: h encodes the signature, the finite tables, and the operator programs; m_∞ encodes one realized history, not the complete history space. Formula (33a) and its two decoders prove the first assertion. Each symbol in (33) denotes its **graph relation encoded in h** , never the external graph of all its possible evaluations. APP, TRIT, U_t , Γ_9 , the restrictions ρ , and the finite extensions have primitive-recursive graphs on integer codes and, therefore, Δ_0 formulas absolute between transitive levels. The two representations are expressed as graph relations through the formulas

$$\begin{aligned} \dot{R}_{\text{ar}}(x, y) &\iff \forall k \in \omega \exists N \forall n \geq N \quad |\mu_n(x) - y| < 2^{-k}, \\ \dot{R}_{\text{exc}}(x, c, z) &\iff \forall N \in \omega \quad z \upharpoonright N = E_N(x \upharpoonright N, c \upharpoonright N), \end{aligned} \quad (37c_0)$$

where μ_n and E_N are the rational/finite functions whose codes occur in h . Every atomic formula containing an HMT symbol is inductively replaced by its graph formula with parameter $u_{h,m}$. Connectives and quantifiers are preserved. This induction on syntactic complexity produces a translation $\varphi \mapsto \varphi^*$ such that, for every transitive level M of (36) containing the parameters,

$$\langle M, \in, (\dot{R}_S \cap M^{ks})_S \rangle \models \varphi(\vec{a}) \iff M \models \varphi^*(\vec{a}; u_{h,m}). \quad (37c)$$

Thus, (37) cannot consult an oracle, a target decimal table, or an uncoded relation that introduces uncountably many objects in a single step. $\square$

HMT semantics is now defined **independently of identifying it with $\mathfrak{G}$** . Let $\mathcal{T}_0^{\text{HMT}}$ be the set of canonical names of the elements of $\mathfrak{G}_0(h, m_\infty) = \text{TC}(\{\omega, u_{h,m}, h, m_\infty\})$. In particular, the initial level contains the realized ledger m_∞ , not every history in X_∞^{enr} as primitive parameters. The admissible terms are the well-founded trees obtained by

$$\begin{aligned} \mathcal{T}_{\beta+1}^{\text{HMT}} &:= \mathcal{T}_\beta^{\text{HMT}} \cup \{ \langle \beta, \ulcorner \varphi \urcorner, t_0, \dots, t_{k-1} \rangle : t_i \in \mathcal{T}_\beta^{\text{HMT}} \}, \\ \mathcal{T}_\lambda^{\text{HMT}} &:= \bigcup_{\beta < \lambda} \mathcal{T}_\beta^{\text{HMT}}, \quad \mathcal{T}^{\text{HMT}} := \bigcup_{\beta \in \text{Ord}} \mathcal{T}_\beta^{\text{HMT}}. \end{aligned} \quad (37a)$$

The value of a successor term is the subset of the preceding level defined by φ in the induced relational structure, with the values of $t_0, \dots, t_{k-1}$ as parameters. Before any set-theoretic comparison, define

$$\text{Ob}_{\text{sem}}^{\text{HMT}}(h, m_\infty) := \{ \llbracket t \rrbracket : t \in \mathcal{T}^{\text{HMT}} \}. \quad (37b)$$

This is the rule of **generative exhaustiveness** adopted by HMT: every object in the entire domain must have one of those derivation trees and every admissible tree expresses an object. It is not a consequence of finite branching. The continuum hypothesis is not introduced into this rule; it is obtained through the proof developed in the article. Neither an enumeration of the reals nor a target bijection is introduced as data.

Theorem 24.25 (Equivalence between term semantics and the genealogical hierarchy). *The evaluation of the preceding terms satisfies*

$$\text{Ob}_{\text{sem}}^{\text{HMT}}(h, m_\infty) = \mathfrak{G}_{\text{HMT}}(h, m_\infty). \quad (37d)$$

Proof. By transfinite induction. At the initial level, the two classes coincide. If every value of a term in $\mathcal{T}_\beta^{\text{HMT}}$ belongs to $\mathfrak{G}_\beta$, the semantics of a successor term is one of the subsets in (37). Conversely, every subset in (37) is given by a formula code and a finite tuple of parameters; by the induction hypothesis those parameters possess terms, and (37a) forms the term that denotes it. At a limit, both constructions take the same union. $\square$

The state encoder is compatible with truncations and sends every TPK extension to a successor term. Let $\mathcal{T}_N^{\text{st}} \subset \mathcal{T}^{\text{HMT}}$ be the subclass of terms encoding complete states of depth N , and let $\tau_{N+1,N} : \mathcal{T}_{N+1}^{\text{st}} \rightarrow \mathcal{T}_N^{\text{st}}$ delete the last block together with the corresponding truncation of the genealogical ledger, carry, and memory. If $J_N : X_N \rightarrow \mathcal{T}_N^{\text{st}}$ is the complete encoder, compatibility with Ext_N proves the well-typed identity

$$\forall x \in X_N \forall y \in \text{Ext}_N(x) \quad \tau_{N+1,N}(J_{N+1}(y)) = J_N(x). \quad (37e_0)$$

Define within the semantics

$$\text{Succ}_{N+1}^{\text{TPK}}(J_N(x)) := \{t \in \mathcal{T}_{N+1}^{\text{st}} : \exists y \in X_{N+1} (y \in \text{Ext}_N(x) \wedge t = J_{N+1}(y))\}.$$

Choose β such that $\mathfrak{G}_\beta$ contains x , X_N , X_{N+1} , the finite graphs of Ext_N , J_N , J_{N+1} , and their codes in h . Then the primitive-recursive formula for the graph of Ext_N gives

$$J_{N+1}[\text{Ext}_N(x)] = \text{Succ}_{N+1}^{\text{TPK}}(J_N(x)), \quad J_{N+1}[\text{Ext}_N(x)] \in \mathfrak{G}_{\beta+1}, \quad (37e)$$

and both sides are formed by the graph formula of the same TPK extension. Equations (37e₀)–(37e) are the explicit semantic link: the subclass of successor terms is exactly the image of the admissible extensions, and its truncation commutes. The complete fiber of ρ is not identified with Ext_N , nor is Def identified with TPK merely by nomenclature.

Lemma 24.26 (Transitivity, axioms, and well-ordering of the genealogical model). *$\mathfrak{G}_{\text{HMT}}(h, m_\infty)$ is a transitive class, contains all ordinals, satisfies Zermelo–Fraenkel set theory (ZF), and possesses a definable global well-order; it therefore satisfies Choice and all axioms used in the subsequent cardinal comparison.*

Proof. Abbreviate $B = \mathfrak{G}_0$ and $u = u_{h,m}$. Induction on (36) proves simultaneously that every level is transitive and that $\mathfrak{G}_\alpha \in \mathfrak{G}_{\alpha+1}$; if all ordinals smaller than α are present, their set is definable at the corresponding level and expresses α . Hence the union contains all ordinals. Extensionality and Foundation are inherited from the ambient universe; Empty Set and Infinity belong to B . Once a level containing the parameters is chosen, the formulas defining $\{a, b\}$, $\bigcup a$, and subsets by Separation express them at a successor level.

For Collection–Replacement, let F be a functional relation definable in $\mathfrak{G}$ on the set a . The reflection scheme relative to the definable hierarchy $(\mathfrak{G}_\xi, \in, u_{h,m})_{\xi \in \text{Ord}}$, applied in the ambient metatheory to the finite list of formulas defining F , produces a level $\mathfrak{G}_\theta$ containing a and the parameters and correct for those formulas. The ranks of the witnesses in $F[a]$ are then bounded by θ ; Separation over $\mathfrak{G}_\theta$ forms the image in $\mathfrak{G}_{\theta+1}$. For Power Set, the internal set to be constructed is not presupposed. Given $a \in \mathfrak{G}_\gamma$, consider in the ambient metatheory the already existing set $\mathcal{P}^V(a)$. For every $b \in \mathcal{P}^V(a) \cap \mathfrak{G}$, let $r(b)$ be its first genealogical stage. Since $\mathfrak{G}$ is a definable class, Separation in the ambient metatheory first forms the set $\mathcal{P}^V(a) \cap \mathfrak{G}$. Replacement **in the ambient metatheory**, applied to that set, bounds the ordinals $r(b)$ by some θ . Consequently,

$$\mathcal{P}^\mathfrak{G}(a) = \{b \in \mathfrak{G}_\theta : b \subseteq a\},$$

and the right-hand side is expressed by Separation in $\mathfrak{G}_{\theta+1}$. It does not introduce into $\mathfrak{G}$ the subsets external to the semantics (37b). This is the noncircular rank-bounding argument.

Finally, each object is assigned its first level of birth and its least **derivation name**: a well-founded, finitely branching derivation tree (and, by (37a), finite for each individual name), labeled by a level ordinal, a formula code, and the tuple of names of its parameters. These names are ordered by transfinite recursion: first by the ordinal label, then by the formula code, and then lexicographically by the already ordered parameter names. The least name of each value defines a class relation $<_{\mathfrak{G}}$ that compares all objects. It is a definable global well-order and, by taking the least element of every nonempty set, proves Choice. $\square$

24.6.1.4.1 Reflection by witness closure and construction of the image. The Collection–Replacement step of Lemma 24.26 admits the following explicit proof. We retain the hierarchy already generated from the operator code and the realized record, and abbreviate $\mathfrak{G} = \mathfrak{G}_{\text{HMT}}(h, m_\infty)$. The argument uses Separation and Replacement in the ambient metatheory, without presupposing these schemes in $\mathfrak{G}$.

Let Φ be a finite collection of formulas. First translate their universal quantifiers using negation and existential quantification, and close the resulting collection under subformulas. For every initial level there is a later limit ordinal θ such that

$$\mathfrak{G}_\theta \models \varphi(\bar{a}) \iff \mathfrak{G} \models \varphi(\bar{a}) \quad (\varphi \in \Phi, \bar{a} \in \mathfrak{G}_\theta^{<\omega}).$$

For each fixed formula, satisfaction in the class is interpreted by relativizing its quantifiers to the definable class $\mathfrak{G}$; no uniform truth predicate is introduced.

Proof. Fix α . For each existential subformula $\exists y \psi(y, \bar{x})$ of Φ , the tuples $\bar{a} \in \mathfrak{G}_\alpha^{<\omega}$ of the corresponding arity that have a witness in $\mathfrak{G}$ form a set by Separation in the metatheory. Assign to each tuple the least ordinal β for which some $y \in \mathfrak{G}_\beta$ satisfies $\mathfrak{G} \models \psi(y, \bar{a})$. This ordinal exists because $\mathfrak{G}$ is the union of its levels. Replacement in the metatheory bounds these first stages of witnesses. Since Φ is finite, a single bound $f(\alpha) > \alpha$, valid for all its existential subformulas, can be chosen definably. Thus every tuple under consideration has *at least one witness* in $\mathfrak{G}_{f(\alpha)}$.

Take α_0 large enough to contain the initial parameters and set

$$\alpha_{n+1} = f(\alpha_n), \quad \theta = \sup_{n < \omega} \alpha_n.$$

By continuity of the hierarchy, every finite tuple from $\mathfrak{G}_\theta$ belongs to some $\mathfrak{G}_{\alpha_n}$. If an existential formula of Φ with that tuple is true in $\mathfrak{G}$, it has a witness in $\mathfrak{G}_{\alpha_{n+1}} \subseteq \mathfrak{G}_\theta$. Induction on formulas now proves the equivalence: atomic formulas are preserved in the induced structure; connectives preserve the equivalence; the existential step uses that witness in one direction and the same witness, viewed in $\mathfrak{G}$, in the other. The prior translation of universal quantifiers also includes the existential formulas expressing their possible counterexamples. This proves the required finite reflection. $\square$

Now let $F(x, y, \bar{p})$ be functional on a in $\mathfrak{G}$. Apply the construction to the formulas expressing F , the existence of its witnesses, and its image, with $a, \bar{p} \in \mathfrak{G}_\theta$. Transitivity gives $a \subseteq \mathfrak{G}_\theta$. Every $x \in a$ has its value within that level, and reflection identifies the exact image as

$$b = \{y \in \mathfrak{G}_\theta : \mathfrak{G}_\theta \models \exists x \in a F(x, y, \bar{p})\}.$$

This set is definable over $\mathfrak{G}_\theta$, with parameters from the same level. By the successor operation of the hierarchy, $b \in \text{Def}(\mathfrak{G}_\theta) \subseteq \mathfrak{G}_{\theta+1}$. This establishes Replacement in $\mathfrak{G}$. If the existence of witnesses is retained and uniqueness is omitted, the same construction provides a set of witnesses for Collection.

24.6.1.4.2 Arity of names and internal selection. In the well-order of Lemma 24.26, the enumeration of the base assigns to each element its least natural-number code. At each successor stage, formula codes are ordered first; once a code is fixed, the arity of its parameter tuple is fixed. Lexicographic

order is then applied to a *finite product of fixed arity* of the previously constructed well-orders. The least name represents each new value, and limit levels retain the order of first appearance. This specifies the ordering of tuples without combining different arities into a single lexicographic order. Selecting the least element of every nonempty set, together with Replacement already established, forms internally the choice functions used subsequently. This construction of names precedes the cardinal injections and receives no bijection of the continuum as data.

24.6.1.5 Ordered readout of block cylinders

For the representative $[b]_3 \in \{0, 1, 2\}$, fix

$$\nu(\mathbf{b}) = \sum_{i=1}^6 [b_i]_3 3^{6-i} \in \{0, \dots, 728\}.$$

In the HMT ordered completion, a prefix $s = (m; \mathbf{b}_0, \dots, \mathbf{b}_{N-1})$, $m \in \mathbb{Z}$, determines

$$a_N(s) = m + \sum_{j=0}^{N-1} \nu(\mathbf{b}_j) 729^{-j-1}, \quad I_N(s) = [a_N(s), a_N(s) + 729^{-N}]_{\text{HMT}}.$$

The carry preserves the joining of the two representatives of a boundary. If $C_N(s) = \overline{I_N(s)}$, the preceding relations give

$$I_N(s) = \bigsqcup_{\mathbf{b} \in \mathbb{F}_3^6} I_{N+1}(s; \mathbf{b}), \quad \text{diam } C_N(s) = 729^{-N} \longrightarrow 0.$$

Proposition 24.27 (Coverage of the Archimedean readout). *The map*

$$P_{\text{ar}} : \mathbb{Z} \times \Omega_{\Gamma}^{\text{cal}} \longrightarrow \mathbb{R}_{\text{HMT}}, \quad \{P_{\text{ar}}(m, \xi)\} = \bigcap_N C_N(m; \beta_N(\rho_{\infty, N} \xi))$$

is well defined, is surjective and determines

$$(\mathbb{Z} \times \Omega_{\Gamma}^{\text{cal}}) / \sim_{P_{\text{ar}}} \cong \mathbb{R}_{\text{HMT}}.$$

Proof. The left endpoints form a Cauchy sequence in the HMT completion, since the intervals are nested and their diameters tend to zero. Their limit belongs to all the closed intervals and is unique. For any element of the completion, first choose the integer interval containing it and then the unique half-open subinterval of each partition containing it. The preceding proposition realizes each of those choices through an effective TPK prolongation. The compatible prefixes define ξ , whose readout is the chosen element. At boundaries, the carry identifies equivalent representations after their construction. By definition and surjectivity, the quotient by equality of readout is in bijection with the completion. $\square$

The construction is also carried out in the internal genealogical universe $\mathfrak{G} = \mathfrak{G}_{\text{HMT}}(h, m_{\infty})$. Here h encodes the operator rules and m_{∞} the compatible history; the universe is obtained through initial transitive closure, definability at each successor and union at limit stages:

$$G_0 = \text{TC}\{\omega, u_{h, m}, h, m_{\infty}\}, \quad G_{\beta+1} = \text{Def}_{\Sigma_{\text{HMT}}}(G_{\beta}), \quad G_{\lambda} = \bigcup_{\beta < \lambda} G_{\beta}.$$

The relations of Σ_{HMT} are those of the operations already constructed and their codes. The internal domain is therefore retained in all coverage claims. Denote its completion by $\mathbb{R}_{\text{HMT}}^{\mathfrak{G}}$ and the ordered realization within the same universe by $\mathbb{R}^{\mathfrak{G}}$.

Proposition 24.28 (Internal ordered chart). *There exists a unique isomorphism of complete Archimedean ordered fields*

$$\iota : \mathbb{R}_{\text{HMT}}^{\mathfrak{G}} \xrightarrow{\cong} \mathbb{R}^{\mathfrak{G}}$$

that fixes the unit. It preserves rational endpoints, positive square roots and limits of convergent sequences.

Proof. The unit fixes the integers and rationals. Each element of the HMT completion is associated with the cut of rationals less than it; completeness of the ordered realization provides the unique element with that cut. Rational density proves injectivity and strict order preservation. The same procedure in the reverse direction proves surjectivity. Rational approximations from above and below preserve addition and multiplication and thereby the field structure. The positive element whose square is $a > 0$ is transported to the positive element whose square is $\iota(a)$. If a sequence converges, for every positive rational tolerance its final terms lie in the rational neighborhood of the limit; order preservation transports that condition. All these operations are performed within $\mathfrak{G}$. Uniqueness follows from preservation of rational cuts. $\square$

24.6.1.6 Descent along fibers and preservation of the selector

Proposition 24.29 (Descent criterion for an operation). *Let $q : \Omega \rightarrow J$ and $\star : D \subseteq \Omega^2 \rightarrow \Omega$, with D saturated by the fibers of $q \times q$. There exists a unique operation $\bar{\star}$ on $(q \times q)(D)$ such that*

$$q(x \star y) = q(x) \bar{\star} q(y)$$

if and only if

$$q(x) = q(x'), \quad q(y) = q(y') \implies q(x \star y) = q(x' \star y')$$

for pairs in the domain.

Proof. If $\bar{\star}$ exists, both results are the evaluation of the same pair of arguments. Conversely, choose antecedents of the visible arguments and define the result as the readout of their operation. Saturation fixes the domain; the condition on fibers makes the result independent of the chosen antecedents. Surjectivity proves uniqueness. $\square$

In an application through P_{ar} , this criterion identifies the precise condition for transporting an operation from complete states to their readout. In the chart ι , the field structure and limits already satisfy that condition. Next, the analytic profile is constructed from the calendar and this compatibility is used to transport its iteration and roots.

24.6.1.7 Dodecaphase readout and quadratic normalization

The profile, its phases and its normalization are those already constructed in (24.1)–(24.4). Its closed form and moments are proved in (24.6) and (24.7). The finite cosine sum determines the removable extension at every zero of the closed-form denominator. The angular moment retains oriented real representatives: changing them modulo 2π changes the moment. The reader and its uniform sector are the explicit starting structure of this realization.

24.6.1.8 Transport of iterates, roots and minima

Within $\mathbb{R}_{\text{HMT}}^{\mathfrak{G}}$, the convergent series

$$\cos_{\text{HMT}}(t) = \sum_{j=0}^{\infty} \frac{(-1)^j t^{2j}}{(2j)!}$$

defines the corresponding analytic realization. Let

$$u_{\text{HMT}}(x) = \frac{1}{12} \sum_{m=1}^{12} \left[1 - \cos_{\text{HMT}} \left(\frac{2(3m-1)x}{\sqrt{899}_{\text{HMT}}} \right) \right], \quad F_a(x) = 1 - a u_{\text{HMT}}(x),$$

$$S_n^{\text{HMT}}(a) = F_a^{2^n}(0), \quad S_n(\lambda) = f_\lambda^{2^n}(0).$$

Proposition 24.30 (Ordered conjugacy of the family and the root selector). *For the internal arguments of the preceding expressions,*

$$\iota \circ F_a = f_{\iota(a)} \circ \iota, \quad \iota(S_n^{\text{HMT}}(a)) = S_n(\iota(a)).$$

The chart transports exactly the conditions of return, exact period and parameter order. If a set of admissible roots has a minimum, the image of that minimum is the minimum of the image set.

Proof. The chart fixes the rationals, transports the positive square root and preserves limits of the cosine partial sums. Hence, $\iota(u_{\text{HMT}}(x)) = u_0(\iota(x))$. Compatibility with addition and multiplication proves the first equality; induction on the number of iterations proves the second. A root is characterized by equality to zero, which is preserved in both directions. The exact period 2^n adds the inequalities $F_a^{2^j}(0) \neq 0$, $0 \leq j < n$, also preserved by the injectivity of ι . If the parameter is required to exceed a preceding center, that inequality is transported by order preservation. Finally, $a \leq b$ for every b in the set is equivalent to $\iota(a) \leq \iota(b)$ for every element of its image. Surjectivity onto the sets thus defined completes the assertion about the minimum. $\square$

The domain of this transport is the indicated internal continuum. The existence of each minimum and its identification with a specific dynamical branch are results of the criticality construction that follows; the chart preserves those properties once they have been established.

24.6.1.9 Finite spectral representation of the uniform reader

The squared frequencies of the profile define the positive measure

$$s_m = k_m^2 = \frac{4(3m-1)^2}{899}, \quad \nu_0 = \frac{1}{12} \sum_{m=1}^{12} \delta_{s_m}, \quad M_r = \int s^r d\nu_0(s).$$

Its twelve nodes are distinct and positive. For $1 \leq r \leq 12$, the Hankel matrix $H_r = (M_{i+j})_{0 \leq i, j < r}$ is positive definite. Indeed, for p of degree less than r ,

$$\mathbf{p}^\top H_r \mathbf{p} = \frac{1}{12} \sum_{m=1}^{12} p(s_m)^2 > 0$$

unless $p = 0$, because a polynomial of degree at most eleven that vanishes at twelve distinct nodes is zero. At degree twelve appears the polynomial $\prod_m (s - s_m)$, with zero norm in this measure. The corresponding spectral representation has dimension twelve.

Let $p_0, \dots, p_{11}$ be the monic orthogonal polynomials, $h_j = \int p_j^2 d\nu_0 > 0$ and $\psi_j = p_j / \sqrt{h_j}$. Since ν_0 is a probability measure, $\psi_0 = 1$. Multiplication by s in $L^2(\nu_0)$ is self-adjoint and has a tridiagonal matrix J_{12} in this basis.

Proposition 24.31 (Exact recurrence and spectral intertwining). *With*

$$a_j = \frac{\int s p_j(s)^2 d\nu_0(s)}{h_j}, \quad \beta_j = \frac{h_j}{h_{j-1}} > 0 \quad (1 \leq j \leq 11),$$

the matrix J_{12} has diagonal a_j and subdiagonal $\sqrt{\beta_j}$. If

$$Q_{mj} = \frac{\psi_j(s_m)}{\sqrt{12}} \quad (1 \leq m \leq 12, 0 \leq j \leq 11), \quad D = \text{diag}(s_1, \dots, s_{12}),$$

then

$$Q^T Q = I, \quad DQ = QJ_{12}, \quad J_{12} = Q^T DQ,$$

and the profile satisfies

$$u_0(x) = e_0^T [I - \cos(x\sqrt{J_{12}})] e_0. \quad (24.54)$$

Proof. Positivity of the Hankel systems uniquely provides the monic orthogonal polynomials. On expanding sp_j in the orthogonal basis, the coefficients of p_i , $i \leq j-2$, vanish because $\langle sp_j, p_i \rangle = \langle p_j, sp_i \rangle = 0$. The coefficient of p_{j+1} is one by monicity, that of p_j is a_j and that of p_{j-1} is h_j/h_{j-1} . At the last degree, the same equality is understood in $L^2(\nu_0)$, where the monic polynomial of degree twelve vanishing at the nodes represents the zero vector. Normalization gives the stated positive subdiagonal.

Orthonormality is equivalent to $Q^T Q = I$. The recurrence equality at each node gives $DQ = QJ_{12}$. Since Q is square, we also have $QQ^T = I$, and obtain the diagonalization of J_{12} . Its positive eigenvalues allow its positive square root to be defined. For every function defined at those twelve nodes,

$$e_0^T \Phi(J_{12}) e_0 = \frac{1}{12} \sum_{m=1}^{12} \Phi(s_m),$$

since $Qe_0 = (1, \dots, 1)^T / \sqrt{12}$. The choice $\Phi(s) = 1 - \cos(x\sqrt{s})$ proves (24.54). $\square$

The representation preserves exactly the uniform functional, its moments and its profile. Its first coefficients are

$$a_0 = M_1 = 2, \quad \beta_1 = M_2 - M_1^2 = \frac{2493348}{808201}.$$

The complete genealogy remains in the TPK state from which the reader originates; J_{12} represents its specific spectral readout. The Gram–Schmidt construction and the Jacobi recurrence are applied to this atomic measure, terminating in dimension twelve.

The operator chain used in the following sections is thus fixed: APP cells and TRIT orientation feed the TPK transition; its prolongations produce cylinders and their ordered readout; the twelve oriented windows and the uniform functional produce u_0 ; the family f_λ determines returns and roots; the internal chart preserves the order of their selectors. The spectral realization describes the same profile through a finite self-adjoint matrix and a cyclic vector.

24.6.2 Dodecaphase profile and quadratic return

24.6.2.1 Genealogy and object of the proof

The initial object is the critical readout of the oriented dodecaphase wheel already constructed in the corpus. APP evaluates addition and multiplication on the two sheets of the digit lattice, preserving quotient and remainder; TRIT determines regime and orientation; TPK transport selects, transports and updates the state with its carry, boundary and memory records. The nonadic connection prolongs that state, with phase return and memory advance. Its readers act on the same discrete structure of the continuum.

The focal reader of the source is

$$\mathcal{P}_{\text{crit}} : U_{12}^{\text{sign}} \longrightarrow \mathbb{R}/2\pi\mathbb{Z}, \quad m \longmapsto \phi_m = \frac{(3m-1)\pi}{18}, \quad 1 \leq m \leq 12.$$

To evaluate the angular moment, the oriented representative of that chart is retained: replacing it with another representative modulo 2π would change the moment. The uniform trace fixes $w_m = 1/12$. The focal chain is

$$\begin{aligned} \text{APP-TRIT-TPK} &\longrightarrow \text{discrete structure of the continuum} \\ &\longrightarrow U_{12}^{\text{sign}} \xrightarrow{\mathcal{P}_{\text{crit}}} (\phi_m, w_m) \\ &\longrightarrow u_0 \longrightarrow f_\lambda \longrightarrow \text{returns and renormalization.} \end{aligned}$$

The abbreviated notation retains the complete state and its prolongations as antecedent; the scalar family is a readout of that antecedent. The already fixed reader, with its order, orientation and uniform sector, is taken as an explicit starting structure. The conclusions concern that particular reader; uniqueness of that reader among all possible TPK readouts lies outside these proofs.

The coordinate π_{HMT} used by the chart belongs to the internal outputs of the corpus. Its factor cancels exactly when normalizing the second moment. The Feigenbaum values enter exclusively in subsequent recognition, never in the definition of the family or the certification of roots.

24.6.2.2 Exact profile and quadratic criticality

We retain the notation $s_m = 3m - 1$, $\kappa_0 = s_0$, $k_m = 2s_m/\sqrt{899}$ and $f_{\lambda,0} = f_\lambda$ of the profile already proved. Its entire extension has the rational series

$$u_0(z) = \sum_{j \geq 1} (-1)^{j+1} \frac{4^j \sum_m s_m^{2j}}{12 \cdot 899^j (2j)!} z^{2j} = z^2 - \frac{715769}{2424603} z^4 + \frac{1353832978}{32695771455} z^6 + \dots$$

Thus the finite record of twelve sectors produces an entire profile, with quadratic criticality and scale fixed before iteration.

24.6.2.3 Structural robustness with uniform bounds

The deformation is defined in (24.13)–(24.15). We write $M_{2,\eta} = \sum_m w_m \phi_m^2$ and $\kappa_\eta = s_\eta = \sqrt{2/M_{2,\eta}}$, retaining the same weights, phases and profiles. The notation is $p_m = p(\phi_m) = 12 \cos(3\phi_m) - \cos(4\phi_m)$ and $k_{m,\eta} = \kappa_\eta \phi_m$; in the following weighted formulae, k_m abbreviates $k_{m,\eta}$, whereas $k_{m,0}$ denotes the uniform sector.

Structural robustness theorem. For $|\eta| \leq 1/40$, the weights are positive and sum to one. The profile is even and entire, satisfies $u''_\eta(0) = 2$ and is strictly increasing on $(0, 1]$. For $0 < \lambda \leq 2/u_\eta(1)$, the map sends $[-1, 1]$ into itself, has a unique critical point in that interval and satisfies

$$Sf_{\lambda,\eta}(x) \leq -\frac{640}{47647} < 0, \quad 0 < |x| \leq 1.$$

Proof. The sums of cosines $3\phi_m$ and $4\phi_m$ vanish through cycles of lengths four and three. Since $|p_m| \leq 13$,

$$\frac{9}{160} \leq w_m \leq \frac{53}{480}, \quad \frac{27}{40} M_{2,0} \leq M_{2,\eta} \leq \frac{53}{40} M_{2,0}.$$

It follows that

$$k_{\max,\eta}^2 \leq \frac{196000}{24273} < 9 < \pi^2, \quad k_{\min,\eta}^2 \geq \frac{640}{47647}.$$

For $0 < x \leq 1$, all sines $\sin(k_{m,\eta}x)$ are positive, whence

$$u'_\eta(x) = \sum_m w_m k_m \sin(k_m x) > 0, \quad u'''_\eta(x) = -\sum_m w_m k_m^3 \sin(k_m x) < 0.$$

The quotient u'''_η/u'_η is the negative of a positive mean of the k_m^2 . Invariance of the Schwarzian under affine changes of the value gives

$$Sf_{\lambda,\eta} = \frac{u'''_\eta}{u'_\eta} - \frac{3}{2} \left(\frac{u''_\eta}{u'_\eta} \right)^2 \leq -k_{\min,\eta}^2.$$

Evenness covers $x < 0$, and the image of the interval is $[1 - \lambda u_\eta(1), 1]$. $\square$

The width $1/40$ is a choice made to obtain uniform bounds, distinct from a fundamental constant or an optimal threshold. The original sector remains $\eta = 0$.

24.6.2.4 Global quadratic holomorphic coordinate on a strip

Holomorphic factorization theorem. For $|\eta| \leq 1/40$, in

$$\mathcal{S} = \{z \in \mathbb{C} : |\operatorname{Re} z| < \pi/3\}$$

there exists a holomorphic, odd and univalent function b_η , with $b'_\eta(0) = 1$, such that

$$u_\eta(z) = b_\eta(z)^2, \quad f_{\lambda,\eta}(z) = 1 - \lambda b_\eta(z)^2.$$

Proof. For $z = x + iy$ in the right half-strip,

$$\operatorname{Re} u'_\eta(z) = \sum_m w_m k_m \sin(k_m x) \cosh(k_m y) > 0.$$

The integral of u'_η over the segment between two points, divided by the difference of the endpoints, has positive real part. Convexity of the half-strip proves injectivity there. Evenness gives the corresponding result on the left.

Moreover,

$$\operatorname{Im} u_\eta(x + iy) = \sum_m w_m \sin(k_m x) \sinh(k_m y)$$

has the sign of y for $x > 0$. On the positive real axis $u_\eta > 0$; on the imaginary axis,

$$u_\eta(iy) = \sum_m w_m (1 - \cosh(k_m y))$$

is strictly negative away from zero and decreases strictly with $|y|$. These separations, injectivity on each half-strip and evenness prove that the fibers of u_η in $\mathcal{S}$ are exactly $\{z, -z\}$, except for the fiber of the origin.

The function $u_\eta(z)/z^2$, extended with value one at zero, is holomorphic and has no zeros in the simply connected strip. It has a holomorphic logarithm L_η with $L_\eta(0) = 0$, which is even by uniqueness of the normalization. Define

$$b_\eta(z) = z \exp\left(\frac{1}{2} L_\eta(z)\right).$$

Then $b_\eta^2 = u_\eta$, b_η is odd and $b'_\eta(0) = 1$. The equality $b_\eta(z) = b_\eta(w)$ forces $w = z$ or $w = -z$. The second case gives equality only at the origin. This proves its univalence. $\square$

The twelve phases thus determine an explicit conformal deformation of the quadratic fold. The complete dynamical conjugacy preserves that deformation:

$$b_\eta \circ f_{\lambda,\eta} \circ b_\eta^{-1}(w) = b_\eta(1 - \lambda w^2),$$

on their composition domains. The factorization proved and a conjugacy to the entire quadratic family are distinct assertions.

24.6.2.5 Return, rescaling and memory

Let $a = -f(1) > 0$, $H_a(x) = -ax$, $J = H_a([-1, 1])$. When $J \subset [-1, 1]$ is admissible for the return, $f^2(J) \subset J$ and the compositions are defined,

$$\mathcal{R}f = H_a^{-1} f^2 H_a, \quad \mathcal{R}f(x) = -a^{-1} f(f(-ax)).$$

If in addition $f(J) \subset (0, 1]$, a unique central quadratic maximum is preserved:

$$(\mathcal{R}f)(0) = 1, \quad (\mathcal{R}f)''(0) = -a f'(1) f''(0) < 0.$$

Composition of Schwarzians gives, at regular points,

$$S(\mathcal{R}f)(x) = a^2 [(Sf)(f(-ax))f'(-ax)^2 + (Sf)(-ax)] < 0.$$

Domain inclusions are part of the hypotheses and are checked at every scale.

Let $v_\eta = (u_\eta|_{[0,1]})^{-1}$. The inverse branches are

$$\beta_\sigma(y) = \sigma v_\eta((1-y)/\lambda), \quad \sigma \in \{-1, +1\},$$

with the critical record $\beta_0(1) = 0$. The step

$$(x, w) \mapsto (f(x), w \parallel \sigma(x))$$

is inverted on its image by reading the last sheet and applying β_σ . The word preserves branch decisions; the previous value is reconstructed through the map equation.

The return groups two transitions. From the endpoint x' and its sheets (σ_0, σ_1) one recovers

$$x = H_a^{-1} \beta_{\sigma_0}(\beta_{\sigma_1}(H_a x')).$$

The intermediate points are also reconstructed. The APP–TRIT–TPK records of sheet, carry, route, boundary and memory are concatenated while preserving their identity. The sign of the analytic branch is an additional record: it does not replace those data with a binary label.

If C_N groups histories of length $2N$ into pairs, the restrictions satisfy

$$\pi_{N,M}^{\text{ret}} C_N = C_M \pi_{2N,2M}^f, \quad M \leq N.$$

Both sides remove the same terminal pairs and preserve the same inverse branches. Compatible transport of finite histories is thus proved.

24.6.2.6 Accumulated scales and derivative of the operator

Let $c_n = f^{2^n}(0)$, $c_0 = 1$ and $H_n(x) = c_n x$. As long as the compositions are admissible and $c_n \neq 0$,

$$\boxed{\mathcal{R}^n f = H_n^{-1} f^{2^n} H_n}, \quad (\mathcal{R}^n f)(1) = \frac{c_{n+1}}{c_n} = -a_n.$$

Induction follows from $H_n H_{a_n} = H_{n+1}$. The convention $a_n > 0$ requires sign alternation of c_n . At a superstable root of period 2^N , $c_N = 0$: that normalization becomes singular and the preceding levels or a controlled limit must be used.

For $z = -ax$, $w = f(z)$ and a perturbation h , the complete derivative is

$$\boxed{D\mathcal{R}_f[h](x) = -\frac{h(w) + f'(w)h(z)}{a} + \frac{h(1)}{a} [\mathcal{R}f(x) - x f'(w) f'(z)]}.$$

It is differentiated using $\dot{a} = -h(1)$, $\dot{z} = x h(1)$, $\dot{w} = h(z) + x f'(z) h(1)$ and the variation of the denominator. If $h(0) = 0$, this gives $D\mathcal{R}_f[h](0) = 0$. The initial HMT parameter direction is $h = -u_\eta$.

This formula includes scale variation; freezing a would give a different linearized operator.

24.6.2.7 Certified roots and itineraries through period 256

Let $F_n(\lambda) = f_{\lambda,0}^{2^n}(0)$. Compute

$$x_0 = p_0 = 0, \quad x_{j+1} = 1 - \lambda u_0(x_j), \quad p_{j+1} = -u_0(x_j) - \lambda u'_0(x_j) p_j.$$

Then $F_n = x_{2^n}$ and $F'_n = p_{2^n}$.

The attached program uses rational intervals with outward integer rounding, square roots enclosed by integer square roots, and sine/cosine with an explicit Taylor remainder. The exact bound $|u_0''| \leq 2$ allows centered evaluation with a quadratic remainder.

The historical approximations only propose boxes of radius 10^{-42} . Opposite signs at their endpoints, separation of F'_n from zero throughout the box, and exclusion of returns with periods that properly divide 2^n are certified. The intermediate value theorem and monotonicity prove local existence and uniqueness; exclusion of divisors proves the least period.

Level	Least period	Box center, rounded
1	2	1.345913990471832792210949
2	4	1.763379787846910127290794
3	8	1.850413796950136464530567
4	16	1.868979756606798125150574
5	32	1.872955034435659740004205
6	64	1.873806367467030119412262
7	128	1.873988695221365362307169
8	256	1.874027744154450672897008

The first root has the exact expression $\lambda_1 = 1/u_0(1)$. The complete boxes and their checks are in the JSON certificate.

The 502 preclosure states of these eight levels, all separated from zero, and their complete sign words have been enclosed. In particular,

$$\text{sign } f_{\lambda_n,0}^{2^j}(0) = (-1)^j, \quad 0 \leq j < n.$$

The finite orientations of the rescalings preceding closure are established. Inclusions of restrictive intervals at every scale constitute an additional check.

For $\delta_{F,n} = (\lambda_{n-1} - \lambda_{n-2})/(\lambda_n - \lambda_{n-1})$, an abbreviated outer decimal interval is

$$\boxed{4.66921218915020415455609621588966392804 < \delta_{F,8}, \\ \delta_{F,8} < 4.66921218915020415455609621588966392863.}$$

The width of the complete interval is less than $5.808 \cdot 10^{-37}$. It measures the numerical error of the finite ratio $\delta_{F,8}$, distinct from the error in approximating the universal limit.

24.6.2.8 Analytic persistence of the deformed roots

For each certified level, $F'_n(\lambda_n) \neq 0$. The implicit function theorem produces a local analytic branch $\lambda_n(\eta)$ with

$$f_{\lambda_n(\eta),\eta}^{2^n}(0) = 0.$$

By continuity, the proper returns remain separated from zero for η sufficiently small; the least period and itinerary are preserved. The existence of those local neighborhoods is proved. Their common width is not yet identified with the entire structural strip $1/40$.

With $M = M_{2,\eta}$, the profile response is

$$v_\eta(x) = \partial_\eta u_\eta(x) = \sum_m \frac{p_m}{12} (1 - \cos(k_{m,\eta}x)) - \frac{M'}{2M} x u'_\eta(x).$$

The second term preserves normalization of the second moment. If $q_j = \partial_\eta x_j$ at fixed λ ,

$$q_0 = 0, \quad q_{j+1} = -\lambda v_\eta(x_j) - \lambda u'_\eta(x_j) q_j, \quad \boxed{\lambda'_n(\eta) = -q_{2^n}/p_{2^n}}.$$

The weights–scale–response–superstable–parameter chain is thus explicit. This finite nondegeneracy is distinct from asymptotic transversality to the stable leaf of the universal fixed point.

24.6.2.9 Analytic tails and error propagation

For $\eta = 0$, let

$$\mu_{2j} = \frac{1}{12} \sum_m k_m^{2j}, \quad p_N(z) = \sum_{j=1}^N (-1)^{j+1} \frac{\mu_{2j}}{(2j)!} z^{2j}, \quad \Omega = \frac{70}{\sqrt{899}}.$$

If $q_N = \Omega^2 r^2 / [(2N+3)(2N+4)] < 1$,

$$\|u_0 - p_N\|_r \leq \frac{\mu_{2N+2} r^{2N+2}}{(2N+2)!(1-q_N)}.$$

The inequality $\mu_{2j+2} \leq \Omega^2 \mu_{2j}$ bounds the tail by a geometric series. For $0 \leq s \leq 2N+2$,

$$\|(u_0 - p_N)^{(s)}\|_r \leq \frac{\mu_{2N+2} r^{2N+2-s}}{(2N+2-s)!} \left(1 - \frac{\Omega^2 r^2}{(2N+3-s)(2N+4-s)} \right)^{-1},$$

when the denominator is positive.

To transport errors through a return, its domains are also controlled: if $\|f - g\| \leq \varepsilon$, $a_f, a_g \geq a_* > 0$, the compositions lie in a common convex domain and $\|f'\| \leq L$, $\|g\| \leq M$ there, then

$$\|\mathcal{R}f - \mathcal{R}g\|_r \leq \varepsilon \left[\frac{1 + L + L^2 r}{a_*} + \frac{M}{a_*^2} \right].$$

The proof adds the outer perturbation, the inner perturbation, the change of argument due to $|a_f - a_g| \leq \varepsilon$ and the variation of the denominator. These bounds prevent reuse of the initial cosine sum as if all renormalized profiles retained that same form.

24.6.3 Nonadic depth and persistence of the itinerary

24.6.3.1 Outer growth and inner resolution

The refinement law of article XI, with integer base $B \geq 2$, is

$$L_c(x, y) = (Bx - c, By + c), \quad 0 \leq c < B.$$

For a word $c_1 \dots c_m$, its record is

$$C_m = \sum_{j=1}^m c_j B^{m-j}, \quad (x_m, y_m) = (B^m x_0 - C_m, B^m y_0 + C_m).$$

The outer sum is $B^m M_0$, with $M_0 = x_0 + y_0$. The normalized reading of the record belongs to the cylinder

$$I_m(C_m) = \left[\frac{C_m}{B^m}, \frac{C_m + 1}{B^m} \right], \quad \text{diam } I_m = B^{-m}.$$

The integer pair and the scale recover the quotients, remainders and leading zeros of the word. Endpoints with two expansions retain their boundary mark. Support growth and readout refinement are thus coordinated without identifying distinct histories solely by their limit image.

The composition is exact:

$$L_D^{[b]} \circ L_C^{[a]} = L_{B^b C + D}^{[a+b]}.$$

By associativity, grouping a history of eighteen events into nine blocks of two or two blocks of nine produces the same record. At return level 2^N , the common horizon is $9 \cdot 2^N$. The proof is an equality of composition on the same ordered history; it retains its carries and its recoverable subblocks.

The restriction of states and the prolongation of their readers are expressed through the following limits:

$$X_\infty = \varprojlim X_m, \quad r^*a = a \circ r, \quad r^*e_x = \sum_{r(y)=x} e_y, \quad \mathfrak{S}_{\text{cyl}} = \varinjlim \mathfrak{S}_m.$$

States restrict; readers extend. Pullback preserves products, the unit and idempotents. An exactly compatible family induces a reader on the direct limit. Approximants that are compatible only up to an error additionally require a convergence estimate in their completion. The next section provides that estimate for the critical return.

24.6.3.1.1 Capacity and vacancies of the record The capacity reader of XI compares blocks of 729 and 1000 possibilities. With $\rho = \log_{1000} 729 = \log_{10} 9$, it retains

$$C_k = \lfloor (k+1)\rho \rfloor, \quad C_k - C_{k-1} = 1 - z_k,$$

$$C_b - C_a + \sum_{k=a+1}^b z_k = b - a, \quad T_{\text{cap}}(a) = \lceil a/\rho \rceil - 1 \quad (a \geq 1).$$

Each vacancy records a transition during which capacity is maintained; chronological length is conserved in the balance. Capacity phase and chronological phase are distinct readers.

To store a prefix of m base-nine digits, a sufficient decimal capacity is

$$a(m) = \left\lceil \frac{m\rho}{3} \right\rceil.$$

The quotient and remainder allow the integer C_m to be converted between bases, keeping the depth m separate. The following hold

$$9^m \leq 1000^{a(m)} \leq 729^{T_{\text{cap}}(a(m))+1}.$$

These inequalities are also decided by integer powers, without rounding logarithms. The readout budget of the next section is thus translated into record capacity while preserving units. This count sizes the storage; the survival rules additionally determine which native histories are admissible.

24.6.3.2 Explicit depth budget for each return

Theorem. With $|\eta| \leq 1/40$ fixed, normalize

$$t = \lambda u_\eta(1)/2 \in [0, 1], \quad z = (x+1)/2,$$

$$F_{t,\eta}(z) = 1 - t \frac{u_\eta(2z-1)}{u_\eta(1)}, \quad z \in [0, 1].$$

For every integer $q \geq 1$ and two parameters t, s ,

$$\left| F_{t,\eta}^q(1/2) - F_{s,\eta}^q(1/2) \right| \leq S_q |t - s|, \quad S_q = \sum_{j=0}^{q-1} (6\sqrt{2})^j < 9^q.$$

Consequently, a base-nine parameter cylinder of depth $m = 2^N + r$ encloses the readout of return $q = 2^N$ in an interval of diameter less than 9^{-r} .

Proof. The function $(1 - \cos v)/v^2$ decreases for $0 < v < \pi$: the sign of its derivative is that of $v \sin v - 2(1 - \cos v)$, which is negative because $\tan(v/2) > v/2$. By the second moment,

$$u_\eta(1) \geq \frac{1 - \cos 3}{9} \sum_j w_j k_j^2 = \frac{2(1 - \cos 3)}{9} > \frac{1}{3}.$$

The last inequality is verified, for example, by bounding $\cos 3$ by its alternating Taylor sum through degree eight, which is less than $-1/2$. Moreover, Cauchy–Schwarz gives $|u'_\eta(x)| \leq \sqrt{2}$ on the real line. It follows that

$$\text{Lip}_z F_{t,\eta} \leq 6\sqrt{2} < 9, \quad \text{Lip}_t F_{t,\eta} \leq 1.$$

If d_j is the difference between the two orbits, then $d_0 = 0$ and $d_{j+1} \leq 6\sqrt{2} d_j + |t - s|$. Induction proves the geometric sum. For $|t - s| \leq 9^{-m}$, the diameter is less than 9^{q-m} . Substituting $q = 2^N$ and $m = q + r$ completes the proof. $\square$

The bound controls parameter uncertainty with η fixed. An implementation separately adds and propagates cosine-evaluation and rounding errors; the preceding rational certificate provides these operations. The bound is sufficient and conservative, with a cost that grows with the horizon.

Limit reader. Realize t from $(x_0, y_0) = (1, 0)$, $B = 9$, so that

$$(x_m, y_m) = (9^m - C_m, C_m), \quad t_m = C_m/9^m.$$

The readers $E_{q,m} = F_{t_m,\eta}^q(1/2)$, brought to a common level, satisfy

$$\|E_{q,m+a} - r^* E_{q,m}\|_\infty \leq S_q 9^{-m}.$$

They converge uniformly to a continuous reader E_q in the completion of the cylinder algebra. The set-theoretic version

$$J_{q,m}(C) = \{F_{t,\eta}^q(1/2) : t \in I_m(C)\}$$

is exactly nested and its diameters tend to zero. This result gives the passage to the limit of the **fixed-horizon readout**, with depth control at any requested horizon.

24.6.3.3 Doubling word and nonadic reader of chronology

The eight itineraries certified in the preceding work obey

$$W_1 = +, \quad W_{n+1} = W_n (-1)^n W_n.$$

Its unique prefix continuation is

$$\tau_j = (-1)^{v_2(j)}, \quad j \geq 1; \quad \tau_{2j} = -\tau_j, \quad \tau_{2j+1} = +.$$

The proof uses $v_2(2^n + r) = v_2(r)$ for $0 < r < 2^n$. It is a recurrence at every depth, and agreement with the eight computed returns constitutes its finite check in the family.

In the parity-lexicographic order of a map with a critical maximum, the orientation factor preceding position k is $\prod_{j < k} (-\tau_j)$. The word τ is strictly greater than any of its positive shifts. Indeed, the first disagreement with an odd shift occurs at $k = 1$ or 2 ; an even shift halves the comparison and doubles k . The first disagreement is therefore at $k = 2^r$. There

$$\prod_{j < 2^r} (-\tau_j) = (-1)^r = \tau_{2^r},$$

and the oriented difference with the opposite sign is positive. This argument also proves aperiodicity.

The chronological record retains

$$K = \rho_9(K) + 9q_9(K).$$

Its dyadic reader is

$$\ell_n(B_K) = [\rho_9(K) + 9q_9(K)]_{2^n} = [K]_{2^n}.$$

On the chart of histories in which $\Gamma_9 B_K = B_{K+9}$,

$$\ell_n(\Gamma_9 B_K) = \ell_n(B_K) + 9 \pmod{2^n}, \quad \pi_{n+1,n} \ell_{n+1} = \ell_n.$$

Since nine is odd, advancement by nine visits the 2^n positions. Its conjugacy with unit advancement is $j \mapsto 9j$. For $0 < j < 2^n$,

$$v_2(9j \bmod 2^n) = v_2(j),$$

so that nonadic resampling also preserves the oriented word in each certified cycle.

This reader uses remainder **and** quotient. The phases of $K = 1$ and $K = 10$ agree modulo nine, while their dyadic signs are opposite. Memory distinguishes the two histories. The arithmetic property of resampling holds for every odd multiplier; the HMT calendar fixes the choice of nine here.

24.6.3.4 Compatible bilateral memory at infinite depth

Let $\mathcal{H}_n = \ell^2(\mathbb{Z}/2^n\mathbb{Z})$,

$$C_n e_j = e_{j+1}, \quad J_n e_j = (e_j + e_{j+2^n})/\sqrt{2}.$$

A check on the basis gives $J_n^* J_n = I$ and $C_{n+1} J_n = J_n C_n$, including the boundary terms.

Article XI provides the projectors

$$P_0 = \frac{1}{9} \begin{pmatrix} 8 & -\sqrt{8} \\ -\sqrt{8} & 1 \end{pmatrix}, \quad P_1 = \frac{1}{9} \begin{pmatrix} 1 & \sqrt{8} \\ \sqrt{8} & 8 \end{pmatrix}.$$

They are orthogonal and complementary. Therefore

$$\mathbb{U}_n = P_0 \otimes I + P_1 \otimes C_n = \begin{pmatrix} T_n & D_n \\ D_n & R_n \end{pmatrix}$$

is unitary, with

$$T_n = (8I + C_n)/9, \quad D_n = \sqrt{8}(C_n - I)/9, \quad R_n = (I + 8C_n)/9.$$

The naturality already proved implies

$$\mathbb{U}_{n+1}(I_2 \otimes J_n) = (I_2 \otimes J_n)\mathbb{U}_n, \quad \mathbb{U}_n^a = P_0 \otimes I + P_1 \otimes C_n^a \quad (a \in \mathbb{Z}).$$

The Hilbert limit is identified with $L^2(\mathbb{Z}_2, \mu_{\text{Haar}})$, taking $e_j^{(n)}$ to $2^{n/2} \mathbf{1}_{j+2^n\mathbb{Z}_2}$. The identities extend by density and boundedness. For every vector v and every integer time a ,

$$\|T_{\infty,a} v\|^2 + \|D_{\infty,a} v\|^2 = \|v\|^2,$$

$$v = T_{\infty,a}^* (T_{\infty,a} v) + D_{\infty,a}^* (D_{\infty,a} v).$$

Here $T_{\infty,a} = (8I + C_{\infty}^a)/9$. Bilateral composition preserves the archive between steps; composing only T_n describes a different operation.

The permutation $S_n e_j = e_{9j \bmod 2^n}$ satisfies

$$S_{n+1} J_n = J_n S_n, \quad S_n C_n S_n^{-1} = C_n^9.$$

Thus one obtains in the limit

$$S_{\infty} C_{\infty} S_{\infty}^{-1} = C_{\infty}^9, \quad (I_2 \otimes S_{\infty}) \mathbb{U}_{\infty} (I_2 \otimes S_{\infty}^{-1}) = \mathbb{U}_{\infty}^9.$$

The temporal representation is faithful: $C_\infty^a = C_\infty^b$ forces $2^n \mid a - b$ for every n , whence $a = b$. Particular preparations may preserve temporal symmetries. The operator identity preserves this distinction.

The projective point $([K]_{2^n})_n$ and a vector of L^2 have different types: a point history induces a Dirac measure, whereas normalizable amplitudes belong to the Hilbert space. The result constructs an equivariant reader of chronology; the remaining native coordinates of the TPK accompany the complete record.

24.6.3.5 Existence and separation of asymptotic obligations

24.6.3.5.1 Realization of the infinite word throughout the strip **Existence theorem.** For each $|\eta| \leq 1/40$ there exists at least one

$$\lambda_\infty(\eta) \in \left[\frac{1}{2u_\eta(1)}, \frac{2}{u_\eta(1)} \right]$$

such that

$$\text{sign } f_{\lambda_\infty(\eta), \eta}^j(0) = (-1)^{v_2(j)} \quad \text{for every } j \geq 1.$$

The conclusion realizes the infinite doubling itinerary within the fixed family. The existence quantifier remains separate from a unique selection and from the law of distances between superstable parameters.

Proof. We fix η . We write K_λ for the sign word of the critical orbit; when it first returns to zero, the word ends at that zero. We use the unimodal order with a maximum, whose factor before comparing symbol j is the product of the signs $-K_i$, $i < j$.

At the left endpoint the image of the entire interval is in $[1/2, 1]$; hence $K_- = + + + \dots$. At the right endpoint the orbit is $0 \mapsto 1 \mapsto -1 \mapsto -1$, whence $K_+ = + - - \dots$. Comparing the second and third symbols, respectively, gives

$$K_- < \tau < K_+.$$

If K_{λ_0} is infinite and distinct from τ , the first difference occurs at a finite position. Continuity of the corresponding iterates makes the side of the comparison locally constant. It remains to examine a superstable parameter of first period p , with word $W0$, where $|W| = p - 1$.

For λ near λ_0 , let $g_\lambda = f_{\lambda, \eta}^p$ and $a = g_\lambda(0)$. We have $g'_\lambda(0) = 0$, and a uniform local bound on the second derivative gives

$$g_\lambda(x) = a + O(x^2).$$

For $|a|$ sufficiently small and nonzero, g_λ maps $[-2|a|, 2|a|]$ into $[a - |a|/2, a + |a|/2]$. The returns retain the sign of a , and the $p - 1$ intermediate positions retain W . The neighboring words are therefore $(W+)^{\infty}$ or $(W-)^{\infty}$; parameters with $a = 0$ retain $W0$.

If τ differs from W before position p , that difference fixes the same side for all three words. If it shares W , put

$$s = \tau_p, \quad e = \prod_{j < p} (-\tau_j), \quad A = Ws.$$

The sign of the comparison of τ with $W0$ and with $(W, -s)^{\infty}$ is es . To compare it with A^{∞} , let q be the first position at which $\tau_{p+q} \neq \tau_q$. It exists by aperiodicity. Up to position $p + q - 1$, both words agree, and the orientation sign factors as $O_p O_{q-1}$, with $O_p = -es$. The strict maximality already proved gives

$$O_{q-1}(\tau_q - \tau_{p+q}) > 0.$$

Consequently, the comparison of τ with A^{∞} has sign $-O_p = es$. The three local possibilities lie on the same side of τ .

If no parameter realized τ , the sets $\{\lambda : K_\lambda < \tau\}$ and $\{\lambda : K_\lambda > \tau\}$ would be open, disjoint, nonempty and would cover the parameter interval. Connectedness of the interval excludes that partition. The stated parameter therefore exists. $\square$

This itinerary-continuity argument applies after the HMT generator and its analytic family have been fixed; its hypotheses are verified directly here. The construction preserves the infinite word without introducing a parameter of the quadratic family or the universal value as an input.

24.6.3.5.2 Margins preventing loss of the itinerary when taking limits Write $c_p = f_{\lambda,\eta}^p(0)$. A sufficient uniform rational bound is

$$|f'|, |f''| \leq 16, \quad B_p := 16^p \frac{16^p - 1}{15} \geq \|(f^p)''\|_{[-1,1]}.$$

Indeed, $1 - \cos v \geq v^2/2 - v^4/24 \geq v^2/8$ for $|v| \leq 3$, so that $u_\eta(1) \geq 1/4$, $\lambda \leq 8$ and both derivatives are bounded by 16, using the moments. The chain rule proves the bound B_p by induction.

The word τ satisfies $\tau_{2p} = -\tau_p$ for every p . For any of its realizations, c_{2p} and c_p have opposite signs. Since $(f^p)'(0) = 0$, Taylor gives

$$|c_{2p} - c_p| \leq \frac{1}{2} B_p |c_p|^2.$$

The left-hand side is greater than $|c_p|$, and therefore

$$|c_p| > 2/B_p > 0.$$

For each fixed horizon there is thus a uniform margin preventing a sequence of realizations from losing its sign upon convergence. In the compact coordinates (η, b) , $b = \lambda u_\eta(1) \in [1/2, 2]$, the set of realizations of τ is closed and compact, and its projection onto the entire η strip is surjective.

The same argument applies to a prefix tree: if all prefixes of τ are realized, a subsequence in the compact set preserves every sign, because sufficiently long prefixes simultaneously include positions p and $2p$. The existence theorem now provides the nonemptiness that this argument requires.

24.6.3.5.3 Invariant return intervals at every level Nested returns theorem. For any of the realizations of the existence theorem in section 24.6.3.5.1, let $c_j = f^j(0)$ and

$$J_n = \text{conv}\{c_{2^n}, c_{2^{n+1}}\}, \quad n \geq 1.$$

Then $0 \in \text{int } J_n$, $J_{n+1} \subset J_n$,

$$f^{2^n}(J_n) = J_n,$$

and the 2^n images $J_n, f(J_n), \dots, f^{2^n-1}(J_n)$ are disjoint. The normalized returns are unimodal with a maximum, with critical value one and the same word τ . These intervals are invariant return cores; a normalization requiring periodic endpoints may enlarge the core to the corresponding maximal restrictive interval and must specify that step.

Proof of the first return. Let F be a unimodal map with a maximum, with $F(0) = 1$ and word τ , and $d_j = F^j(0)$. Its first signs are $+, -, +, +, -, +$. The interval $I = [d_2, 1]$ is invariant: $F(d_2) = d_3 > 0$, $F(1) = d_2 < 0$, and the maximum is one. Put $J = [d_2, d_4]$. Since F decreases on positive arguments and

$$F(d_3) = d_4 > 0 > d_6 = F(d_5),$$

we have $d_3 < d_5$, and therefore $F(J) = [d_3, 1]$.

Moreover, $d_3 > d_4$. To check this, F^2 is strictly increasing between the two positive points d_3, d_4 , because their images d_4, d_5 are positive. Its values are $F^2(d_3) = d_5 > 0 > d_6 = F^2(d_4)$, which forces that order. Thus J and $F(J)$ are separated by an open interval, and

$$F^2(J) = F([d_3, 1]) = [d_2, d_4] = J.$$

The map $F^2|_J$ has a unique critical minimum: the image $F(J)$ remains positive. The conjugacy $h(x) = d_2x$ reverses orientation and normalizes the return as a maximum with value one. Its itinerary is

$$\text{sign} \frac{d_{2j}}{d_2} = -\tau_{2j} = \tau_j.$$

The same argument can be repeated at any depth.

Inductive disjointness. Suppose the $p = 2^n$ images of the parent J_n are disjoint, and let $g = f^p$. The first return applied to the normalized map produces $B = J_{n+1}$, $A = g(B)$, with $A \cap B = \emptyset$, $g(B) = A$, $g(A) = B$. For $0 \leq j < p$, an intersection of $f^j(A)$ and $f^j(B)$ would, upon applying f^{p-j} , produce an intersection of B and A . Those two images are therefore disjoint. Images corresponding to distinct j belong to disjoint images of the parent. This proves the $2p$ disjoint images and the return $f^{2p}(B) = B$. The endpoint formula results from composing the normalizations $x \mapsto c_{2^n}x$. $\square$

The compact sets

$$\Lambda_n = \bigcup_{0 \leq j < 2^n} f^j(J_n)$$

are nonempty and nested. Their intersection Λ_∞ has a continuous reader onto the dyadic odometer: each point is assigned the label of the unique component it occupies at each level. Every compatible sequence of labels has a preimage by compactness, and advancement by f adds one to the labels. This map is a surjective semiconjugacy. Promoting it to a pointwise conjugacy requires proving that the component diameters tend to zero; that metric promotion remains separate.

24.6.4 Compatibility of readers and recovery of the parameter

24.6.4.1 Compatibility of readers, minima and recovery of history

The prolongation operations and ordered readers preserve the information required for root selection. The following identities specify their domains and composition.

24.6.4.1.1 Recovered forward rule The deterministic update starts from the R36 boundary state with character χ and internal remainder. The update is

$$\mathcal{U}_K^\chi = \text{Upd}_K^\chi \circ \text{Tra}_K^\chi \circ \text{Sel}_K^\chi, \quad N_{K+1} = 729N_K + d_K^\chi. \quad (24.55)$$

The digit and the update depend on the preceding state. Two forward sections with the same complete R36 state coincide by induction. The subrelation $\text{Ext}^{\chi, \rightarrow}$ is the graph of that update; the total relation Ext admits multiple continuations depending on the data and characters.

Compatible prolongations transport the regional readers and the joint signature over the same prefixes.

These rules transport orientation, remainder, quotient, carry, boundary and memory. Their application to the cascade must also preserve ordered selection of the critical reader's parameter.

24.6.4.1.2 Independence of the antecedent constants from the free parameter Let B be the HMT state providing the wheel and its antecedent constants. Write $C(B)$ for the set of those readouts and u_B for the critical profile. In the domain of the critical reader, the subsequent construction is

$$(B, \lambda) \mapsto f_{B, \lambda}(x) = 1 - \lambda u_B(x), \quad 0 < \lambda \leq 2/u_B(1). \quad (24.56)$$

For the same B , the reading of the preceding constants factors through projection onto the first component:

$$\widetilde{C}(B, \lambda) = C \circ \text{pr}_B(B, \lambda) = C(B). \quad (24.57)$$

Consequently, two distinct values of λ preserve the same antecedent readouts. The equality is a property of this causal construction: the profile is fixed before λ is varied, and the factor π_{HMT} cancels in its quadratic normalization.

Lemma. Suppose a condition A is evaluated solely on those antecedent readouts. For every fixed B ,

$$\{\lambda \in \Lambda : A(\widetilde{C}(B, \lambda))\} = \begin{cases} \Lambda, & A(C(B)) \text{ holds,} \\ \emptyset, & A(C(B)) \text{ fails.} \end{cases} \quad (24.58)$$

Proof. Substituting (24.57) makes the predicate independent of λ . Therefore all parameters in the domain satisfy it, or none do. This includes comparison of the antecedent constants with metrological intervals.

Uniqueness of the section B and the subsequent restriction $f_{B,\lambda}^{2^n}(0) = 0$ produce a solution fiber in λ . The first-root rule acts on this fiber through the return and its itinerary. Selection and its prolongation are proved in the following results.

This result maintains the HMT order: metrology compares previously constructed outputs, and root selection is proved within the reader that produces it.

24.6.4.1.3 Lemma on ordered transport of the selector Let P_n be an ordered set of genealogical candidates with minimum p_n^* . Let Z_n be the corpus's original set of candidate roots, with the real order, and let $L_n : P_n \rightarrow Z_n$ be a monotone reader. Its image is said to be **coinitial** when

$$\forall z \in Z_n \quad \exists p \in P_n : L_n(p) \leq z. \quad (24.59)$$

Then

$$\boxed{L_n(p_n^*) = \min Z_n \iff L_n(P_n) \text{ is coinitial in } Z_n.} \quad (24.60)$$

Proof. If the image of the minimum is the global minimum, take $p = p_n^*$ in (24.59). Conversely, given $z \in Z_n$, coinitiality provides p with $L_n(p) \leq z$. Monotonicity gives $L_n(p_n^*) \leq L_n(p) \leq z$. Since $L_n(p_n^*) \in Z_n$, it is the global minimum. Surjectivity of L_n is a sufficient condition, stronger than (24.59).

The ordered isomorphism of the continuum chart satisfies reader surjectivity. The concrete application to the return and its complete set of zeros is proved in the ordered-transport section; classification and isolation then determine its dynamical continuation.

24.6.4.1.4 Minimal selection through survival at arbitrary depth Composition with survival also produces a precise global selector. The uniform family $\eta = 0$, constructed in section 24.6.7.1, is retained, and the theorems on realization of the infinite itinerary and preservation of its signs are used. Let

$$I = \left[\frac{1}{2u_0(1)}, \frac{2}{u_0(1)} \right], \quad c_j(\lambda) = f_\lambda^j(0), \quad \tau_j = (-1)^{v_2(j)}.$$

The set

$$\Lambda_\tau = \{\lambda \in I : \text{sign } c_j(\lambda) = \tau_j \text{ for every } j \geq 1\}$$

is compact and nonempty. The derivative control in that source provides

$$B_j = \frac{16^j(16^j - 1)}{15}, \quad |c_j(\lambda)| > \frac{2}{B_j} \quad (\lambda \in \Lambda_\tau). \quad (24.61)$$

Define

$$\varepsilon_j = \frac{1}{B_j}, \quad A_N = \{\lambda \in I : \tau_j c_j(\lambda) \geq \varepsilon_j, \ 1 \leq j \leq N\}. \quad (24.62)$$

Each A_N is compact by continuity of the iterates and nonempty because it contains Λ_τ . Moreover,

$$A_{N+1} \subseteq A_N, \quad \bigcap_{N \geq 1} A_N = \Lambda_\tau. \quad (24.63)$$

The right-to-left inclusion follows from (24.61). The reverse inclusion follows from the strictly positive margins in (24.62), which fix each sign.

Surviving-minimum theorem. The minima $a_N = \min A_N$ satisfy

$$a_N \uparrow a_* = \min \Lambda_\tau. \quad (24.64)$$

Proof. The inclusion (24.63) makes the sequence of minima increasing and bounded within I ; it therefore has a limit a . For each r , all terms a_N , $N \geq r$, belong to the closed set A_r ; thus $a \in A_r$. By (24.63), $a \in \Lambda_\tau$. For any $\lambda \in \Lambda_\tau$, we have $a_N \leq \lambda$ at every level, and on passing to the limit, $a \leq \lambda$. This proves (24.64).

In the HMT cylindrical reader, the object a_* has compatible prefixes at every depth, with orientation, remainder and memory preserved. Its choice uses the order of the parameter reading and complete survival. Compactness proves the existence and convergence of these minima.

The parameters a_N solve sign constraints with a margin. Their boundaries may satisfy $\tau_j c_j = \varepsilon_j$, whereas the superstable roots satisfy $c_{2^n} = 0$. Therefore (24.64) establishes a global selector for infinite realization and preserves, as a distinct object, the corpus's first-root selector.

24.6.4.1.5 Recovery of the parameter from the complete critical history The second term of the critical history provides an exact inverse reader:

$$c_1(\lambda) = 1, \quad c_2(\lambda) = 1 - \lambda u_0(1), \quad \boxed{\lambda = \frac{1 - c_2(\lambda)}{u_0(1)}}. \quad (24.65)$$

The positivity $u_0(1) > 0$, proved for the profile, makes division valid. Consequently, two identical complete critical histories have the same parameter. The sign word τ is a reduced readout of that history; it preserves its combinatorics and omits the magnitude of c_2 .

Formulas (24.64)–(24.65) unite survival-based selection and recovery of the parameter from the critical history. Equality of its limit with the stable crossing is proved by first exit, and identification of the centers by analytic transport of their hybrid classes.

24.6.5 Ordered transport and Jacobi representation

24.6.5.1 Complete transport from the continuum and spectral realization of the profile

The ordered transport of the continuum and the spectral representation of the reader are composed before parameter selection.

24.6.5.1.1 Coverage of all roots and transport of their minimum The ordered construction of the continuum developed in the preceding works produces the surjective Archimedean readout from compatible cylinders and their ordered quotient. Within its declared universe $\mathfrak{G}$, it proves the isomorphism of complete ordered fields

$$\iota : \mathbb{R}_{\text{HMT}}^{\mathfrak{G}} \xrightarrow{\cong} \mathbb{R}^{\mathfrak{G}}. \quad (24.66)$$

The name η used in the source work is changed here to ι , to distinguish it from the weighting parameter, fixed at zero. The second projection preserves incidence, boundary and memory over the same antecedent; the evaluation quotient and the complete history retain their respective domains.

Let us construct in the HMT field, before selecting a root,

$$\cosh_H(t) = \sum_{k=0}^{\infty} \frac{(-1)^k t^{2k}}{(2k)!}, \quad u_H(x) = \frac{1}{12} \sum_{m=1}^{12} \left[1 - \cosh_H \left(\frac{2(3m-1)x}{\sqrt[4]{899}} \right) \right],$$

$$F_a(x) = 1 - a u_H(x), \quad S_n^H(a) = F_a^{2^n}(0). \quad (24.67)$$

The integers and the second moment are those already derived from the dodecaphase reader. The square root is the positive one in the ordered field. The convergent series analytically realizes that reader; it preserves the profile and the normalizing coefficient of section 24.6.7.1.

Return transport theorem. For every parameter in the internal domain of (24.67),

$$\iota(F_a(x)) = f_{\iota(a)}(\iota(x)), \quad \iota(S_n^H(a)) = S_n(\iota(a)). \quad (24.68)$$

Proof. The ordered isomorphism fixes the rationals and preserves the positive square root. It preserves convergent limits: every neighborhood of positive rational radius is transported to another of the same radius. Applied to the partial sums of (24.67), it transports the cosine and u_H . The first equality follows from preservation of addition and multiplication; the second follows by induction on the number of iterations.

With a previous center a_{n-1} fixed, define Z_n^H by zero return, exact critical period 2^n and $a > a_{n-1}$, within the declared parameter domain. Let Z_n be the set defined by the same conditions for f_λ , to the right of $\iota(a_{n-1})$. Then

$$\boxed{\iota[Z_n^H] = Z_n, \quad \iota(\min Z_n^H) = \min Z_n.} \quad (24.69)$$

The second equality is asserted when the minimum exists; the first also preserves the nonexistence of candidates.

Proof. (24.68) preserves the zero return and the earlier returns determining the exact period. Order preserves position relative to the previous center. Surjectivity of (24.66), applied to any root parameter, proves reverse coverage. If a_* is the minimum, every $z \in Z_n$ has the form $\iota(a)$, with $a \in Z_n^H$, whence $\iota(a_*) \leq z$. The converse assertion uses ι^{-1} .

Surjectivity covers every parameter in the chart, including those whose roots have not been isolated by a finite computation. The isomorphism preserves the declared universe of interpretation and the selection order.

24.6.5.1.2 Jacobi representation of the complete dodecaphase profile Define the atomic measure of the reader,

$$s_m = \frac{4(3m-1)^2}{899}, \quad \nu_0 = \frac{1}{12} \sum_{m=1}^{12} \delta_{s_m}, \quad M_r = \int s^r d\nu_0(s). \quad (24.70)$$

The twelve nodes are distinct and positive. For a polynomial $p \neq 0$ of degree less than $r \leq 12$,

$$\sum_{i,j=0}^{r-1} p_i M_{i+j} p_j = \frac{1}{12} \sum_{m=1}^{12} p(s_m)^2 > 0. \quad (24.71)$$

A polynomial of degree less than twelve does not vanish at all twelve nodes. At degree twelve, $\prod_m (s - s_m)$ appears, with zero norm.

Orthogonalization with respect to this measure produces orthonormal polynomials $\psi_0, \dots, \psi_{11}$, with $\psi_0 = 1$. Let

$$Q_{mj} = \psi_j(s_m) / \sqrt{12}, \quad D = \text{diag}(s_m), \quad J_{12} = Q^T D Q.$$

The following hold

$$Q^\top Q = I, \quad DQ = QJ_{12}, \quad \boxed{u_0(x) = e_0^\top [I - \cos(x\sqrt{J_{12}})] e_0.} \quad (24.72)$$

Proof. Orthogonality follows from (24.71). Multiplication by s produces a three-term recurrence: entries separated by more than one position vanish by orthogonality and degree. Choosing positive leading coefficients makes the subdiagonals positive. By functional calculus, $\cos(x\sqrt{J_{12}}) = Q^\top \cos(x\sqrt{D})Q$. The vector Qe_0 is constant, with value $1/\sqrt{12}$; substituting it recovers the sum in section 24.6.7.1.

The article on Grassmannians develops this mechanism for marked measures and refinements. Here it is applied to (24.70); rank twelve belongs to this reader. The genealogical marks are retained alongside the reader, in accordance with the marked faithfulness of the source work. J_{12} represents the analytic profile; the complete state additionally preserves routes, orientation and memory.

The verifier for this extension checks, with exact fractions, the twelve degrees of positive norm, vanishing at degree twelve, the intertwiner $VT = DV$ in the monic basis and weighted self-adjointness $HT = T^\top H$. It checks moments of degree zero through thirty; the identity for every degree follows from the intertwiner. Its first coefficients are

$$a_0 = 2, \quad \beta_1 = \frac{2493348}{808201}. \quad (24.73)$$

24.6.5.1.3 Oriented sensitivity of the return For an exact return of period $p = 2^n$, let

$$c_j = f_\lambda^j(0), \quad q_j = \partial_\lambda c_j, \quad D_j = \prod_{k=1}^j f'_\lambda(c_k), \quad D_0 = 1.$$

Before the critical return, the factors of D_{p-1} are nonzero. Differentiation and telescoping give

$$q_{j+1} = -u_0(c_j) + f'_\lambda(c_j)q_j, \quad \boxed{\frac{q_p}{D_{p-1}} = -\sum_{j=1}^{p-1} \frac{u_0(c_j)}{D_j}.} \quad (24.74)$$

In the canonical word, $\text{sign } c_j = (-1)^{v_2(j)}$. The sign of $f'_\lambda(c_j)$ is opposite; therefore,

$$\text{sign } D_j = (-1)^{j + \sum_{k=1}^j v_2(k)} = (-1)^{s_2(j)}, \quad (24.75)$$

using $\sum_{k=1}^j v_2(k) = j - s_2(j)$, where $s_2(j)$ counts the binary ones. The normalized sensitivity is

$$\mathcal{T}_n = -\sum_{j=1}^{2^n-1} (-1)^{s_2(j)} t_j, \quad t_j = \frac{u_0(c_j)}{|D_j|} > 0. \quad (24.76)$$

Nodal positivity of (24.71) determines u_0 ; (24.76) additionally retains the orbital products and their orientations.

The additional moment representation would require a positive measure ρ on $[0, 1]$ with $t_j = \int z^{j-1} d\rho$. Under that condition,

$$\mathcal{T}_n = \int_0^1 \frac{1 - \prod_{r=0}^{n-1} (1 - z^{2^r})}{z} d\rho(z) > 0 \quad (24.77)$$

for $\rho \neq 0$, extending the integrand by its value 1 at zero. The identity is obtained by expanding the binary product.

The rational check at the previously certified period-four root obtains

$$t_1 \approx 0.40425224267600139730, \quad t_2 \approx 0.04925249167432646403, \quad t_3 \approx 0.15740870098294833737$$

and the rigorous outer interval

$$\begin{aligned} 0.296096033367379523967764444238832345360107 &< \mathcal{T}_2, \\ \mathcal{T}_2 &< 0.296096033367379523967764444238832345360112. \end{aligned} \quad (24.78)$$

It also certifies $t_3 > t_2$. A positive measure on $[0, 1]$ would impose $t_3 \leq t_2$; that specific lifting of orbital weights to positive moments is ruled out. The positive sign of (24.78) remains certified. This check distinguishes the failure of a proposed proof from the failure of the property it sought to prove.

24.6.6 Classification and continuation of the first-root selector

24.6.6.1 Object and provenance

This appendix preserves the family rendered by the HMT dodecaphase reader:

$$u_0(x) = \frac{1}{12} \sum_{m=1}^{12} \left(1 - \cos \frac{2(3m-1)x}{\sqrt{899}} \right), \quad f_\lambda(x) = 1 - \lambda u_0(x).$$

The family originates in the operator and analytic antecedents presented in this development. The existence and compactness of its realizations of the infinite itinerary were proved in the subsections on realization of the word and preservation of its signs.

The following argument is original and uses only those properties and the return operations of the fixed family. It does not use a parameter from another family or the value of a universal constant as an input. The sheets of the itineraries used here are the two monotone branches of the critical reader; the remaining coordinates and HMT memory retain their residence in the state from which the reader originates.

We write

$$c_j(\lambda) = f_\lambda^j(0), \quad \tau_j = (-1)^{v_2(j)}, \quad W_n = \tau_1 \cdots \tau_{2^n-1}.$$

The itinerary of a critical orbit that returns to the origin for the first time ends in the zero symbol. The itinerary order is the unimodal order with a maximum: before comparing the symbol of index j , the orientation factor is

$$O_{j-1}(K) = \prod_{i=1}^{j-1} (-K_i).$$

The ordinary order of the symbols is $-1 < 0 < 1$.

24.6.6.2 Classification before the infinite word

Classification theorem. Let $F : I \rightarrow I$, with $I = [\ell, 1]$, $\ell < 0$, be a continuous map, strictly increasing to the left of zero and strictly decreasing to its right, whose maximum satisfies $F(0) = 1$. Suppose zero has exact period 2^n , with $n \geq 1$, and its critical itinerary K satisfies $K < \tau$. Then

$$K = W_n 0.$$

The conclusion concerns the itinerary and allows distinct parameters in a family to have that same itinerary.

Proof. Write $d_j = F^j(0)$. For period two the itinerary is $+0 = W_1 0$. Consider an exact period $p = 2^n \geq 4$.

We have $d_2 < 0$. Indeed, $d_2 = 0$ would give period two. If $d_2 > 0$, monotonicity of the positive branch gives

$$F([d_2, 1]) = [d_2, d_3] \subset [d_2, 1],$$

and the critical orbit remains in a strictly positive interval, incompatible with a return to zero.

The initial signs are then $+, -$. The orientation factor for comparing the third sign is -1 ; hence $K < \tau$ forces $d_3 > 0$. A zero at that position would produce period three.

Nor can $d_4 < 0$ occur. Under this hypothesis,

$$d_2 < d_4 < 0, \quad F([d_2, d_4]) = [d_3, d_5] \subset (0, 1],$$

since $d_4 = F(d_3) > F(1) = d_2$ and $d_5 = F(d_4) > F(d_2) = d_3$. The positive branch is decreasing, so that

$$F^2([d_2, d_4]) = [d_6, d_4] \subset [d_2, d_4].$$

Here $d_6 \geq d_2$, because $d_5 \leq 1$, and $d_6 < d_4$, because $d_5 > d_3$. This strictly negative interval traps the even returns of the critical orbit and excludes a return to zero. Therefore $d_4 \geq 0$. If $p = 4$, one obtains precisely $+-+0 = W_2 0$.

Now suppose $p \geq 8$. Then $d_4 > 0$. The orientation factors for the fifth and sixth signs are, respectively, -1 and $+1$. Since $\tau_5 = +1$ and $\tau_6 = -1$, the inequality $K < \tau$ forces

$$\text{sign}(d_1, \dots, d_6) = (+, -, +, +, +, -). \quad (24.79)$$

Zero signs at those positions are excluded by the exact period.

We construct the return interval

$$J = [d_2, d_4].$$

The inequality $F(d_3) = d_4 > 0 > d_6 = F(d_5)$, together with $d_3, d_5 > 0$, implies $d_3 < d_5$; therefore

$$F(J) = [d_3, 1].$$

Moreover, $d_4 < d_3$. To check this, the image under F of the interval with endpoints d_3, d_4 is strictly positive, since its endpoint images are $d_4, d_5 > 0$. On that interval F^2 is strictly increasing. Its endpoint values satisfy

$$F^2(d_3) = d_5 > 0 > d_6 = F^2(d_4),$$

which forces $d_3 > d_4$. Consequently,

$$J \cap F(J) = \emptyset, \quad F^2(J) = F([d_3, 1]) = [d_2, d_4] = J. \quad (24.80)$$

The map $F^2|_J$ has a unique minimum at zero. Conjugacy by $h(x) = d_2 x$, which reverses orientation, defines

$$G = h^{-1} \circ F^2 \circ h : \left[\frac{d_4}{d_2}, 1 \right] \longrightarrow \left[\frac{d_4}{d_2}, 1 \right]. \quad (24.81)$$

This map satisfies the same maximum hypotheses as F , with $G(0) = 1$, and its critical point has exact period $p/2$. Its itinerary is

$$K'_j = -K_{2j}. \quad (24.82)$$

All odd-indexed symbols of K are positive, because the even positions belong to J and the odd positions to $F(J) \subset (0, 1]$.

The transformation (24.82) preserves the comparison order with τ . Indeed, $\tau_{2j-1} = +1$ and $\tau_{2j} = -\tau_j$. If q is the first position at which K' and τ differ, the first position of difference between K and τ is $2q$, and

$$O_{2q-1}(K) = -O_{q-1}(K'), \quad K_{2q} - \tau_{2q} = -(K'_q - \tau_q). \quad (24.83)$$

The product determining the order is identical. The identities remain valid when the first distinct symbol is the final zero. Thus $K' < \tau$.

Induction applied to G gives $K' = W_{n-1}0$. The positive odd positions and (24.82) reconstruct exactly $K = W_n 0$. The theorem is proved. $\square$

The theorem also proves that each classified map has the restrictive doubling returns preceding its superstable return, with the domains of (24.80)–(24.81). It does not presuppose monotonicity of a family parameter.

Symmetric normalization of the fixed family. When F is even and defined on $[-1, 1]$, the preceding inequalities additionally give

$$F(d_4) = d_5 > d_3 = F(d_2) = F(-d_2).$$

Since both arguments $d_4, -d_2$ are positive and F decreases on that branch,

$$0 < d_4 < -d_2 < 1. \quad (24.84)$$

Therefore $F^2([d_2, -d_2]) = [d_2, d_4]$, and the same conjugacy $h(x) = d_2x$ defines on all of $[-1, 1]$ an even map with a maximum whose image is $[d_4/d_2, 1] \subset [-1, 1]$. It agrees exactly with the operator $RF = -F(F(-ax))/a$, $a = -F(1)$, used in the corpus. This symmetric extension preserves the critical return, its period and its itinerary.

24.6.6.3 First parameter with infinite itinerary

We work on the compact interval already used in the existence proof,

$$L = \left[\frac{1}{2u_0(1)}, \frac{2}{u_0(1)} \right].$$

Let

$$\Lambda_\tau = \{\lambda \in L : \text{sign } c_j(\lambda) = \tau_j \text{ for every } j \geq 1\}.$$

By the existence and sign-separation proofs of the prolongation, this set is nonempty and compact. Therefore there exists

$$a = \min \Lambda_\tau. \quad (24.85)$$

Comparison lemma. For every $\lambda \in [1/(2u_0(1)), a)$, we have $K_\lambda < \tau$.

Proof. At the left endpoint, the itinerary is $+\infty < \tau$. The separation argument for realization of the infinite word proves that the strict comparison sets with τ are open: in an infinite itinerary distinct from τ , the first finite difference persists by continuity; at a superstable parameter, the two neighboring periodic words and the word with a final zero lie on the same side of τ , by its strict maximality. By definition, the interval preceding a contains no parameters with itinerary τ . Its connectedness maintains the initial comparison throughout that interval. $\square$

24.6.6.4 Existence of each first new root

We retain the original selector. The first root is

$$\lambda_1 = \frac{1}{u_0(1)},$$

and, for $n \geq 2$, λ_n is the first root of c_{2^n} to the right of λ_{n-1} satisfying

$$c_{2^j}(\lambda_n) \neq 0 \quad (1 \leq j < n). \quad (24.86)$$

Since every first period dividing 2^n is a power of two, (24.86) is equivalent to requiring exact critical period 2^n .

Existence lemma. All roots of this selector exist and satisfy

$$\lambda_1 < \lambda_2 < \cdots < \lambda_n < a. \quad (24.87)$$

Proof. The sign $c_2(a) < 0$ implies $\lambda_1 < a$. Suppose $\lambda_{n-1} < a$ has been constructed, and put $p = 2^{n-1}$. The analytic function $c_p(\lambda)$ is not identically zero, since $c_p(a) \neq 0$. Its zero set in $[\lambda_{n-1}, a]$ is finite, nonempty and does not contain a . Let $z < a$ be its last zero.

On $(z, a]$, the sign of c_p is constant and equal to $s = \tau_p$. For $g_\lambda = f_\lambda^p$, we have

$$g_\lambda(0) = c_p(\lambda), \quad g'_\lambda(0) = 0.$$

A uniform local bound on the second derivative gives

$$c_{2p}(\lambda) = g_\lambda(c_p(\lambda)) = c_p(\lambda) + O(c_p(\lambda)^2). \quad (24.88)$$

As λ approaches z from the right, the first-order term dominates; hence $c_{2p}(\lambda)$ has sign s . At a , its sign is

$$\tau_{2p} = -\tau_p = -s.$$

The intermediate value theorem provides a zero $\beta \in (z, a)$ of c_{2p} . Since $c_p \neq 0$ on that interval, the critical period at β is exactly $2p$.

Analyticity and $c_{2p}(a) \neq 0$ imply that the zeros of c_{2p} in $[\lambda_{n-1}, a]$ are finite. Among its zeros of exact period $2p$ to the right of λ_{n-1} , there is therefore a first one. It is the original selector λ_n , and it satisfies $\lambda_n \leq \beta < a$. $\square$

Use of the last auxiliary zero z belongs solely to the existence proof. The selected parameter remains the **first new root** λ_n .

24.6.6.5 Convergence of the global selector to the first infinite itinerary

Global continuation theorem. For the fixed HMT family and the original selector, all selected itineraries are canonical and

$$\boxed{K_{\lambda_n} = W_n 0, \quad \lambda_n \nearrow \min \Lambda_\tau.} \quad (24.89)$$

Proof. The comparison and existence lemmas place each selected parameter below a , with an itinerary less than τ . The classification theorem identifies that itinerary with $W_n 0$. The sequence is increasing and bounded by a ; let $b \leq a$ be its limit.

Fix $p \geq 1$. For n sufficiently large, $2p < 2^n$. Then the signs of $c_p(\lambda_n)$ and $c_{2p}(\lambda_n)$ agree with τ_p and $\tau_{2p} = -\tau_p$, respectively. The uniform bound on second derivatives,

$$\|(f_\lambda^p)''\| \leq B_p, \quad B_p = 16^p \frac{16^p - 1}{15},$$

and the identity $(f_\lambda^p)'(0) = 0$ imply

$$|c_{2p}(\lambda_n) - c_p(\lambda_n)| \leq \frac{1}{2} B_p |c_p(\lambda_n)|^2.$$

Since the two terms on the left-hand side have opposite signs,

$$|c_p(\lambda_n)| > 2/B_p. \quad (24.90)$$

By continuity, $c_p(b)$ retains the sign τ_p and satisfies $|c_p(b)| \geq 2/B_p > 0$. This holds for each p , so that $b \in \Lambda_\tau$. Minimality of a gives $b \geq a$, and therefore $b = a$. $\square$

The complete proof preserves the global selector and produces its itineraries and its limit. The information from the first eight levels is compatible with the result, but the proof of (24.89) does not extrapolate those eight levels: it uses induction, invariant intervals and uniform separation of every sign.

24.6.6.6 Logical checks and falsifiers

1. The classification theorem requires a strict unimodal maximum and an invariant interval. If either branch loses monotonicity, the intervals (24.80) may cease to be restrictive and the classification must be checked again.
2. The condition $K < \tau$ is essential. Beyond the limit word, centers of period 2^n with other combinatorics may exist; the theorem does not identify them with $W_n 0$.
3. Analyticity in the parameter provides isolation and finiteness of the zeros on the compact set. For a merely continuous family, the existence of a strictly subsequent first root requires an additional argument.
4. A limit of critical roots may, in general, lose signs upon reaching a return of lower period. Inequality (24.90) excludes precisely that phenomenon for this sequence and each fixed horizon.
5. Convergence of the first roots to the first parameter with itinerary τ is not equivalent to uniqueness of all parameters with that itinerary and does not by itself determine a metric rate. Transverse crossing and branch identification provide those subsequent conclusions.

24.6.7 Transverse entry and local scaling

24.6.7.1 Provenance and result

APP evaluates addition and multiplication on the two sheets of the lattice, preserving quotient and remainder. TRIT determines regime and orientation. TPK transport selects, transports and updates the state with its carry, boundary and memory records. Its readers act on the same discrete structure of the continuum. The prolongation $w6 \rightarrow w12 \rightarrow w18 \rightarrow w24 \rightarrow w30 \rightarrow R36 \rightarrow G9$ preserves this provenance and distinguishes phase return from memory increment.

The critical reader of the dodecaphase wheel, constructed in the operator antecedents and transported by compatible refinement, produces

$$\phi_j = \frac{(3j-1)\pi_{\text{HMT}}}{18}, \quad w_j = \frac{1}{12}, \quad k_j = \sqrt{\frac{2}{\sum_{\ell=1}^{12} w_\ell \phi_\ell^2}} \phi_j = \frac{2(3j-1)}{\sqrt{899}}.$$

The coordinate π_{HMT} retains its APP–TRIT–TPK origin; its common factor cancels in this second moment. The resulting reader is

$$u_0(x) = \frac{1}{12} \sum_{j=1}^{12} (1 - \cos k_j x), \quad f_\lambda(x) = 1 - \lambda u_0(x). \quad (24.91)$$

The focal sector is the uniform one, $\eta = 0$. The preceding extension separately retains the weighted family $|\eta| \leq 1/40$. The new numerical bounds correspond to (24.91).

The family is even and entire, with $f_\lambda(0) = 1$, a quadratic critical point and second moment $\sum_j w_j k_j^2 = 2$. The parameter λ ranges over a family downstream of the HMT generator. The target Feigenbaum constant remains outside the inputs of (24.91).

Main result. There exists a unique

$$\lambda_* \in [1.874038, 1.874039] \quad (24.92)$$

for which the fourth normalized return of (24.91) belongs to the local stable leaf of the real doubling fixed point. The crossing of the family with that leaf is transverse, with quantified separation. For every $n \geq 0$,

$$\|R^{4+n} f_{\lambda_*} - g\|_{\mathcal{A}} < 0.001666 (0.83996)^n. \quad (24.93)$$

The local superstable cascade associated with this crossing has parameters μ_j and constants $C \neq 0$, $0 < \theta < 1$, such that

$$\mu_j - \lambda_* = C \delta_F^{-j} [1 + O(\theta^j)], \quad \frac{\mu_{j-1} - \mu_{j-2}}{\mu_j - \mu_{j-1}} \longrightarrow \delta_F. \quad (24.94)$$

Here $\delta_F > 1$ is the unstable eigenvalue of the doubling operator. Selection of μ_j is performed through the local returns of section 24.6.7.8. Identification with the global selector is proved in sections 24.6.8.1–24.6.8.3: all global terms retain the combinatorics and the centers agree at every sufficiently high level when their periods are matched.

24.6.7.2 Analytic chart and effective return

The chart of even functions is used

$$s = x^2, \quad z = \frac{s-1}{5/2}, \quad f(x) = 1 - x^2 V(z), \quad V(z) = \sum_{j \geq 0} v_j z^j.$$

The coordinates of the Banach space are

$$u = 10v_0, \quad v = (v_1, v_2, \dots), \quad \|(\Delta u, \Delta v)\|_{\mathcal{A}} = |\Delta u| + \sum_{j \geq 1} |\Delta v_j|. \quad (24.95)$$

In particular, the constant coefficient of V has weight ten. This normalization agrees with the Hertling–Spandl chart $u/10$, section 2 of the source.

Let $a = -f(1) = v_0 - 1$. For admissible real returns,

$$(Rf)(x) = -\frac{f(f(-ax))}{a}. \quad (24.96)$$

The program evaluates an exact factorization of (24.96). Defining

$$S = \frac{a^2 s - 1}{5/2}, \quad b = V(S), \quad h = 1 - a^2 s b, \quad t = \frac{h^2 - 1}{5/2},$$

$$\text{shift } V(z) = \sum_{j \geq 0} v_{j+1} z^j,$$

one obtains

$$\boxed{V_R = a b (h+1) \left[V(t) + \frac{2}{5} \text{shift } V(t) \right]}. \quad (24.97)$$

To verify it, one uses $v_0 = 1+a$, $h^2 - 1 = -a^2 s b (h+1)$ and $V(t) - v_0 = t \text{shift } V(t)$. Thus, $a + 1 - h^2 V(t) = a^2 s b (h+1) [V(t) + (2/5) \text{shift } V(t)]$, which is precisely $a s V_R$.

The initial moments are rational:

$$\frac{1}{12} \sum_j k_j^{2m} = \frac{1}{12} \left(\frac{4}{899} \right)^m \sum_{j=1}^{12} (3j-1)^{2m}. \quad (24.98)$$

Therefore both (24.97) and the initial construction use rational intervals; the initial trigonometric functions are represented by their series with an outer remainder bound.

24.6.7.3 External bounds used as subsequent recognition

Once (24.91) has been constructed, Lanford's hyperbolicity theorem is used as an analytic certificate for the operator (24.96). Let g_0 be the published central polynomial and g the fixed point. The estimates used are

$$\|g - g_0\|_{\mathcal{A}} < \varepsilon_0 = 4 \cdot 10^{-5},$$

and, in the ball $\|f - g_0\|_{\mathcal{A}} < 0.01$, the blocks of DR , written $R = (U, V)$, satisfy

$$U_u \geq a_0 = 4.521, \quad \|U_v\| \leq b_0 = 0.560, \quad \|V_u\| \leq c_0 = 0.756, \quad \|V_v\| \leq d_0 = 0.719. \quad (24.99)$$

The fixed point has a unique unstable direction, with positive real eigenvalue $\delta_F > 1$; the rest of the spectrum lies inside the unit disk. The figures in (24.99) come from the published estimates, not from a new reproduction of Lanford's proof in this dossier. The coefficients of g_0 are explicitly incorporated into the program's central reference polynomial, in accordance with table 2 of Hertling–Spandl.

Sources: [Lanford, hyperbolicity estimates](#); [Hertling–Spandl, analytic chart and table 2](#).

24.6.7.4 Rational entry and direction certificate

The renormalization entry certificate divides (24.92) into sixteen equal rational intervals. It propagates four returns of (24.97) and the exact derivative with respect to λ , using differentiation of the composition and outward rounding to one hundred decimal places.

Each series consists of a polynomial of degree 32 with interval coefficients and an arbitrary analytic error E , $\|E\|_1 \leq \epsilon$. This error may also modify low-order coefficients. For $\|q\|_1 < 1$,

$$\|E \circ q\|_1 \leq \epsilon, \quad \|E' \circ q\|_1 \leq \frac{\epsilon}{(1 - \|q\|_1)^2}, \quad \|\text{shift } E\|_1 \leq \epsilon. \quad (24.100)$$

The products include polynomial–error and error–error terms. The initial tail is bounded by its first absolute term and a geometric ratio, using $\|1 + (5/2)z\|_1 = 7/2$. Each composed argument remains strictly inside the unit ball of this algebra.

For $\Gamma(\lambda) = R^4 f_\lambda = (u_\lambda, v_\lambda)$, the certificate gives the following conservative bounds, uniform in (24.92):

Quantity	Certified outer bound
$\ \Gamma(\lambda) - g_0\ _{\mathcal{A}}$	< 0.004993128
$\ v_\lambda - v_{g_0}\ _1$	< 0.001396077
u'_λ	> 3821.289115
$\ v'_\lambda\ _1$	< 551.757000
$\ v'_\lambda\ _1 / u'_\lambda$	$< 0.144386 < 1/4$
Analytic error of the function	$< 8.482 \cdot 10^{-19}$
Analytic error of the tangent	$< 4.569 \cdot 10^{-15}$

The complete decimals, the sixteen subintervals and every intermediate return are retained in the rational entry certificate.

The cone $\|\dot{v}\|_1 \leq |\dot{u}|/4$ is invariant under (24.99):

$$|\dot{u}_{\text{nuevo}}| \geq (4.521 - 0.560/4)|\dot{u}| = 4.381|\dot{u}|,$$

$$\|\dot{v}_{\text{nuevo}}\|_1 \leq (0.756 + 0.719/4)|\dot{u}| = 0.93575|\dot{u}| < \frac{1}{4}(4.381)|\dot{u}|. \quad (24.101)$$

The certified parameter direction therefore enters a uniformly expanding cone of the operator.

24.6.7.5 Quantitative construction of the stable leaf

Translate g to the origin and write the coordinates as $(x, y) = (u - u_g, v - v_g)$. Fix

$$L = 0.16, \quad r = 0.004, \quad A = a_0 - Lc_0 = 4.40004, \quad q = d_0 + c_0L = 0.83996. \quad (24.102)$$

The cylinder $|x| \leq Lr$, $\|y\|_1 \leq r$ lies inside the ball of (24.99), since $(1 + L)r + \varepsilon_0 = 0.00468 < 0.01$.

Consider the complete space of functions $\sigma : B_r \rightarrow \mathbb{R}$ with $\sigma(0) = 0$ and Lipschitz constant at most L , equipped with the uniform distance. For each y , define $(\mathcal{G}\sigma)(y)$ as the root in $[-L\|y\|_1, L\|y\|_1]$ of

$$F_\sigma(x, y) = U(x, y) - \sigma(V(x, y)). \quad (24.103)$$

On that interval, $\|V(x, y)\|_1 \leq q\|y\|_1 < r$. As x increases, (24.99) shows that F_σ grows with slope at least A . At its endpoints,

$$F_\sigma(L\|y\|_1, y) \geq [AL - b_0 - Ld_0]\|y\|_1,$$

and the opposite inequality holds at the negative endpoint. The margin is

$$AL - b_0 - Ld_0 = 0.0289664 > 0. \quad (24.104)$$

A unique root is obtained. Comparing (24.103) for two values of y ,

$$\text{Lip}(\mathcal{G}\sigma) \leq \frac{b_0 + Ld_0}{A} = \frac{0.67504}{4.40004} < 0.16.$$

Comparing two graphs,

$$\|\mathcal{G}\sigma - \mathcal{G}\tilde{\sigma}\|_\infty \leq A^{-1}\|\sigma - \tilde{\sigma}\|_\infty, \quad A^{-1} < 0.228. \quad (24.105)$$

The contraction theorem produces a unique fixed graph $x = \sigma_*(y)$. On it,

$$\|y_n\|_1 \leq q^n\|y_0\|_1, \quad \|(x_n, y_n)\|_{\mathcal{A}} \leq (1 + L)q^n\|y_0\|_1. \quad (24.106)$$

Moreover, the vertical distance $d(x, y) = x - \sigma_*(y)$ satisfies

$$|d(R(x, y))| \geq A|d(x, y)|$$

as long as the orbit remains in the cylinder. Every orbit that remains there indefinitely belongs to the graph. This graph identifies exactly the local stable leaf and provides estimate (24.106). Its local regularity agrees with the analytic stable manifold of the hyperbolic fixed point.

24.6.7.6 Existence, uniqueness and transversality of the parameter

For λ in (24.92), define

$$H(\lambda) = u_\lambda - u_g - \sigma_*(v_\lambda - v_g).$$

The certificate ensures $\|v_\lambda - v_g\|_1 < 0.001436077 < r$. For $\lambda_2 > \lambda_1$,

$$H(\lambda_2) - H(\lambda_1) \geq \int_{\lambda_1}^{\lambda_2} [u'_t - L\|v'_t\|_1] dt > 3733.007995 (\lambda_2 - \lambda_1). \quad (24.107)$$

To determine the endpoint signs, $\|g - g_0\|_{\mathcal{A}} < \varepsilon_0$ and $|\sigma_*(y)| \leq L\|y\|_1$ are used. Rational point evaluation produces

$$H(1.874038) < -0.0011856110945, \quad H(1.874039) > 0.0021866053399. \quad (24.108)$$

By continuity, a root exists; (24.107) proves its uniqueness and the transversality of the crossing. At that root, (24.106) and $(1 + L)(0.001396077 + 0.00004) < 0.001666$ prove (24.93).

The bound is uniform in the number of returns: it converts increasing depth into an explicit geometric operator error.

24.6.7.7 Real admissibility of all returns

The four initial returns are accompanied, on each rational subinterval, by the inequalities

$$0 < a = -f(1) < 1, \quad b = f(a) > a, \quad f(b) < a. \quad (24.109)$$

The program stores $a, b, f(b)$ before each application of (24.97). Unimodality, evenness and the quadratic critical point are preserved under these normalized returns.

For subsequent iterations, a uniform check in the ball of (24.99) suffices. If $E = V - V_{g_0}$, the norm (24.95) gives, for $-0.4 \leq z \leq 0$,

$$|E(z)| \leq 0.004, \quad |E'(z)| \leq 0.01.$$

The second inequality uses $\sup_{j \geq 1} j(0.4)^{j-1} = 1$. Rational evaluation of the central coefficients provides

$$\frac{6}{5} < V(z) < \frac{8}{5}, \quad |V'(z)| < \frac{1}{2}, \quad 0.39 < a < 0.41. \quad (24.110)$$

Consequently,

$$f'(x) = -2x \left[V(z) + \frac{2}{5}x^2 V'(z) \right],$$

and the bracketed expression exceeds one for $0 \leq x \leq 1$. The map is unimodal and preserves $[-1, 1]$. Moreover,

$$\begin{aligned} b = f(a) &> 1 - (0.41)^2 \frac{8}{5} = \frac{4569}{6250} > a, \\ f(b) &< 1 - \left(\frac{4569}{6250} \right)^2 \frac{6}{5} = \frac{35028967}{97656250} < 0.39 < a. \end{aligned} \quad (24.111)$$

The orbits of the stable leaf satisfy (24.109) at every depth. Thus the analytic iterations of the operator correspond to real doubling returns, with their domains and orientations preserved.

24.6.7.8 Scaling of the local superstable cascade

The metric conclusion requires two crossings: that of the family with the stable leaf, proved in section 24.6.7.6, and that of the unstable manifold with the superstable target. The second is proved by Eckmann–Wittwer for $\Sigma_2 = \{\psi : \psi(1) = 0\}$, corresponding to the period-two superstable cycle. Normalization (24.96) agrees with that of the even real operator used there.

The general theorem of Collet–Eckmann–Lanford, section 6 of the cited source, proposition 6.1 and theorems 6.2–6.3, applies to a Banach-space map C^2 , a fixed point with a unique unstable eigenvalue $\delta_F > 1$, a curve transverse to its stable leaf and a target transverse to its unstable manifold. The local preimages of the target intersect the curve at points satisfying

$$\delta_F^j(\mu_j - \lambda_*) \longrightarrow C \neq 0. \quad (24.112)$$

The index j counts local returns. Since the initial curve here is $R^4 f_\lambda$, the physical period is 2^{j+5} when the target is taken in Σ_2 ; any fixed number of steps used to bring the target into the local neighborhood is incorporated into a fixed index shift and into C .

The real conditions of section 24.6.7.7 and the target's real doubling branch fix the exact period and itinerary. The sequence is selected by this oriented local continuation. Taking differences in (24.112) gives the ratio limit of (24.94).

Sources of the two results used: [Eckmann–Wittwer, *A complete proof of the Feigenbaum conjectures*](#); [Collet–Eckmann–Lanford, section 6](#).

24.6.7.8.1 Power-law asymptotic remainder The analyticity of our family allows (24.112) to be made more precise. In the normal-coordinate construction of Collet–Eckmann–Lanford, only the stable manifold, the unstable manifold and the target hypersurface are flattened. These three submanifolds admit local C^2 changes of coordinates with bounded derivatives. The smooth cutoff used is applied in the scalar unstable coordinate; the complete foliation is then obtained through the constructed change of coordinates.

In those coordinates, the correction is written as a series of increments w_j whose norm C^1 satisfies

$$\|w_j\|_{C^1} \leq C_1 \rho^j, \quad 0 < \rho < 1.$$

The chain rule for the cut-off map, with uniformly bounded first- and second-order derivatives, provides

$$\|w_j\|_{C^2} \leq C_2 B^j$$

for some $B \geq 1$. The weighted norm of the construction controls the ordinary C^1 norm on the domain considered. By interpolation,

$$[Dw_j]_\beta \leq C_3 \left(\rho^{1-\beta} B^\beta \right)^j.$$

Choose

$$0 < \beta < \min \left\{ 1, \frac{-\log \rho}{\log B - \log \rho} \right\}. \quad (24.113)$$

The series converges in $C^{1,\beta}$. Let z be the resulting normal coordinate, normalized so that the preimages of the target satisfy $z = \delta_F^{-j}$, absorbing the fixed index shift. Since our curve is analytic and transverse,

$$z(\Gamma(\lambda)) = c(\lambda - \lambda_*) + O(|\lambda - \lambda_*|^{1+\beta}), \quad c \neq 0.$$

Local inversion gives

$$\mu_j - \lambda_* = C \delta_F^{-j} + O\left(\delta_F^{-(1+\beta)j}\right). \quad (24.114)$$

This proves the remainder in (24.94), with $\theta = \delta_F^{-\beta}$. The statement asserts the existence of the constants in the parameter remainder. The explicit operator rate 0.83996 of (24.93), for its part, controls convergence of the renormalized maps.

24.6.7.8.2 Conversion of the remainder into ratio error If $\mu_j - \lambda_* = C \delta_F^{-j} (1 + e_j)$, with $|e_j| \leq M \theta^j$, write

$$\mu_{j-1} - \mu_j = C(\delta_F - 1) \delta_F^{-j} (1 + r_j), \quad r_j = \frac{\delta_F e_{j-1} - e_j}{\delta_F - 1}.$$

With $K = M(\delta_F + \theta)/(\delta_F - 1)$, we have $|r_j| \leq K \theta^{j-1}$. Therefore,

$$\left| \frac{\mu_{j-2} - \mu_{j-1}}{\mu_{j-1} - \mu_j} - \delta_F \right| \leq \frac{\delta_F K (1 + \theta) \theta^{j-2}}{1 - K \theta^{j-1}} \quad (24.115)$$

when $K \theta^{j-1} < 1$. This formula separates the evaluation precision of each root from the truncation error of the cascade.

24.6.7.9 Certified continuation of the eight initial roots

The finite continuation certificate preserves the family (24.91) and the previous eight-root certificate. For $2 \leq n \leq 8$, it studies

$$S_n(\lambda) = f_\lambda^{2^n}(0)$$

from the certified box of λ_{n-1} to 1.874039. The complete rational covering contains exactly the inherited root λ_{n-1} and the new root λ_n . In particular, λ_n is the unique new root in $(\lambda_{n-1}, 1.874039]$.

Boxes are excluded by the sign of the return or by a local derivative of certified sign. Anchor boxes retain existence and uniqueness through endpoint signs and local monotonicity. Their endpoints cover each interval exactly; adjacencies are checked. Evaluation uses the moments (24.98), order-32 Horner evaluation, outer geometric tails and $|u_0''| \leq 2$. The uniform tail of the potential is less than $1.294 \cdot 10^{-58}$ on $|x| \leq 3/2$.

The final certificate contains 650 processed boxes and 332 covering leaves. The first level has the exact expression $\lambda_1 = 1/u_0(1)$. The following approximations are shown solely for guidance; their complete boxes are retained in the certificates:

Level	Superstable parameter
1	1.345913990471832792210949095...
2	1.763379787846910127290794030...
3	1.850413796950136464530566778...
4	1.868979756606798125150574309...
5	1.872955034435659740004205128...
6	1.873806367467030119412262311...
7	1.873988695221365362307168780...
8	1.874027744154450672897007573...

24.6.8 Isolation of the limit and scaling of the global selector

24.6.8.1 Indefinite continuation of the first root and exclusion of the intermediate interval

The complete classification proved above preserves the family and the first-new-root selector. Ordered transport preserves the zeros and their minimum; the return word preserves the nonadic chronological reading.

24.6.8.1.1 Classification and existence at every depth Write $W_n = \tau_1 \cdots \tau_{2^n-1}$, with $\tau_j = (-1)^{v_2(j)}$, and $a_* = \min \Lambda_\tau$. The existence and compactness of Λ_τ are proved in the prolongation development, 24.6.3.5.1–24.6.3.5.2; its minimum exists by compactness and order.

Classification. Every strictly unimodal map with a maximum, on an invariant interval, whose critical point has exact period 2^n and whose itinerary K satisfies $K < \tau$, has itinerary $W_n 0$.

The induction proceeds through the effective return. For period at least eight, comparison with τ , together with the intervals that would trap orbits with incompatible signs, forces the prefix $(+, -, +, +, -)$. This yields

$$J_1 = [d_2, d_4], \quad F(J_1) = [d_3, 1], \quad d_4 < d_3, \quad F^2(J_1) = J_1.$$

The return normalized by $h(x) = d_2 x$ has maximum one, half the period, and itinerary $K'_j = -K_{2j} < \tau$. Order is preserved because

$$O_{2q-1}(K) = -O_{q-1}(K'), \quad K_{2q} - \tau_{2q} = -(K'_q - \tau_q).$$

The initial cases are $+0$ and $+ - +0$; induction reconstructs $W_n 0$. For even maps, also $0 < d_4 < -d_2 < 1$, so the return extends to $[-1, 1]$ and coincides with (24.96).

Before a_* , all itineraries are smaller than τ : the strict-comparison sets are open by the separation proved in 24.6.3.5.1; the interval preceding a_* is connected and begins with $+\infty < \tau$.

Existence of each first root. Suppose $\lambda_{n-1} < a_*$ has been constructed, and let $p = 2^{n-1}$. The zeros of c_p in $[\lambda_{n-1}, a_*]$ form a nonempty finite set: the function is analytic and $c_p(a_*) \neq 0$. Let $z < a_*$ be its last zero. In $(z, a_*]$, c_p has sign τ_p . For $g_\lambda = f_\lambda^p$,

$$g'_\lambda(0) = 0, \quad c_{2p}(\lambda) = c_p(\lambda) + O(c_p(\lambda)^2).$$

Near z , on the right, c_{2p} has sign τ_p ; at a_* , it has the opposite sign. There is a zero of c_{2p} in (z, a_*) , and its period is exactly $2p$, since there $c_p \neq 0$. Finiteness of the zeros provides the first new root in (λ_{n-1}, a_*) . The last auxiliary zero proves existence; the effective choice remains the first root. The base case is $\lambda_1 = 1/u_0(1) < a_*$.

24.6.8.1.2 Limit of the original sequence **Theorem.** The original selector exists at every level and satisfies

$$\boxed{K_{\lambda_n} = W_n 0, \quad \lambda_n \nearrow a_* = \min \Lambda_\tau.} \quad (24.116)$$

Proof. The preceding existence result gives an increasing sequence bounded by a_* . The classification fixes all its itineraries. Let $b \leq a_*$ be its limit. For each p , sufficiently late terms preserve the opposite signs τ_p, τ_{2p} . Taylor's formula and the uniform bound $B_p = 16^p(16^p - 1)/15$ give

$$|c_{2p}(\lambda_n) - c_p(\lambda_n)| \leq \frac{1}{2} B_p |c_p(\lambda_n)|^2, \quad |c_p(\lambda_n)| > 2/B_p.$$

Passing to the limit preserves $\text{sign } c_p(b) = \tau_p$, with absolute value at least $2/B_p > 0$. For every p , this implies $b \in \Lambda_\tau$; minimality gives $b = a_*$.

The result establishes indefinite existence, preservation of the symbolic cascade, and selection of its first limiting parameter for the original global rule. The HMT completion transports this sequence and its limit while preserving order and reading.

24.6.8.1.3 Certificate excluding the intermediate interval The bridge program certifies

$$\Lambda_\tau \cap [\lambda_{8,\text{inf}}, 1.874038] = \emptyset, \quad (24.117)$$

where

$$\lambda_{8,\text{inf}} = 1.874027744154450672897007573595092408435133414523352839687903472138975$$

is the rational lower endpoint of the certified interval for λ_8 .

The cover contains 374 exactly adjacent intervals and no unresolved region: 319 with $c_{512} > 0$, 10 with $c_{1024} < 0$, two with $c_{2048} > 0$, all opposite to the sign of τ , and 43 near centers of periods 256, 512 or 1024, excluded by their second return.

In the latter case, $g = f_\lambda^p$, $d = g(0)$ and a uniform bound $|g''| \leq M$ between zero and d satisfy $M|d| < 2$. Then

$$|g(d) - d| \leq \frac{1}{2} M |d|^2 < |d| \quad (d \neq 0).$$

Both returns have the same sign; for $d = 0$, both are zero. Both situations exclude $\tau_{2p} = -\tau_p$ with strict signs.

The evaluation uses rational moments, geometric tails of the potential and its second derivative, and $|u_0''| < 6$. Parameter centering intersects the natural enclosure with

$$c_j(\lambda_{\text{medio}}) + \partial_\lambda c_j(I) (I - \lambda_{\text{medio}});$$

the derivative is propagated over the full box. The self-map property of $[-1, 1]$ is proved before restricting the enclosures to it.

The complete certificate retains the cover and checks its exact tiling, together with every sign or Taylor inequality.

24.6.8.1.4 Joint localization The local crossing λ_* realizes τ through its real returns at every depth, so that $a_* \leq \lambda_*$. By (24.116), $\lambda_8 < a_*$; together with (24.117),

$$\boxed{1.874038 < a_* \leq \lambda_* < 1.874039.} \quad (24.118)$$

The localization preserves the canonical itinerary and places its first limiting parameter in the same interval as the stable crossing.

24.6.8.2 First-exit isolation and coincidence of the limits

This section completes the isolation identified in section 24.6.8.1.4. It preserves the HMT family (24.91), the minimal selection (24.116), the chart (24.95) and the stable sheet of sections 24.6.7.5–24.6.7.7. The proof uses the order $a_* \leq \lambda_*$: it suffices to exclude a separation toward the negative side of the expanding cone.

24.6.8.2.1 Upper and lower bounds for transverse transport Besides (24.99), Lemma 2.1 of Hertling–Spandl, section 2 provides $\|D\Phi\| < 0.9$ in the same ball of radius 0.01, where

$$\Phi_u(u, v) = u - \frac{U(u, v) - u}{3.669}.$$

Therefore,

$$\left|1 - \frac{U_u - 1}{3.669}\right| < 0.9, \quad U_u < 1 + 1.9(3.669) = 7.9711. \quad (24.119)$$

The number 3.669 belongs to the published preconditioner of this analytic certificate; the generated family (24.91) retains its prior coefficients.

Take $\kappa = 0.21$. For two points in the ball joined by a secant with $\Delta u < 0$, $\|\Delta v\|_1 \leq \kappa|\Delta u|$, integration of the blocks of DR along the segment gives

$$4.4034|\Delta u| \leq |\Delta u'| \leq 8.0887|\Delta u|, \quad \Delta u' < 0, \quad (24.120)$$

$$\|\Delta v'\|_1 \leq (0.756 + 0.719\kappa)|\Delta u| = 0.90699|\Delta u| < \kappa(4.4034)|\Delta u|. \quad (24.121)$$

Here $4.4034 = 4.521 - 0.560\kappa$ and $8.0887 = 7.9711 + 0.560\kappa$. Thus, the secant cone preserves orientation and expands while the segment remains in the ball.

24.6.8.2.2 Reduction of an infinite separation to a finite region Suppose $a_* < \lambda_*$, and set

$$f_n = R^{4+n} f_{a_*}, \quad s_n = R^{4+n} f_{\lambda_*}, \quad (\Delta u_n, \Delta v_n) = f_n - s_n.$$

The positive bound for u' and $\|v'\|_1/u' < 0.144386 < \kappa$ on $\Gamma(J)$ place the initial secant in the negative cone. The stable sheet is written relative to g , with

$$s_n = g + e_n, \quad \|e_n\|_{\mathcal{A}} < 0.001666(0.83996)^n, \quad |(e_n)_u| \leq 0.16\|(e_n)_v\|_1. \quad (24.122)$$

The entry checks imply

$$|\Delta u_0| < 0.004993128 + 0.00004 + 0.16(0.001396077 + 0.00004) = 0.00526290032 < h, \quad h = 0.006. \quad (24.123)$$

While $|\Delta u_n| < h$, (24.121)–(24.122) give

$$\|f_n - g_0\|_{\mathcal{A}} < (1 + \kappa)h + 0.001666 + 0.00004 = 0.008966 < 0.01. \quad (24.124)$$

The point s_n and the segment between the two also remain in that convex ball. The lower expansion (24.120) forces a finite first exit N , and the upper bound applied to the preceding step provides

$$0.006 \leq -\Delta u_N < 0.0485322, \quad \|\Delta v_N\|_1 \leq 0.21|\Delta u_N|. \quad (24.125)$$

The inequality $8.0887h = 0.0485322$ is used; the full jump, not only the first-exit surface, is included.

Consequently, f_N belongs to the following region:

$$\begin{aligned}
C_- = \{ g + (d_u, d_v) + e : & -0.0485322 \leq d_u \leq -0.006, \\
& \|d_v\|_1 \leq 0.21|d_u|, \quad |e_u| + \|e_v\|_1 \leq 0.001666, \\
& |e_u| \leq 0.16\|e_v\|_1 \}.
\end{aligned} \tag{24.126}$$

Since a_* realizes τ , all its normalized returns are unimodal self-maps of $[-1, 1]$ with the same itinerary. This property follows from return induction, independently of the distance from f_N to g_0 . This is why the following certificate treats C_- intersected with that class of self-maps.

24.6.8.2.3 Certified exclusion of the first-exit region The rational first-exit exclusion certificate, reproduced in the computational supplement, covers the interval of (24.125) exactly with 64 adjacent rational bands. All are excluded; no additional subdivision or unresolved region enters the result.

Of the 64 bands, 54 display an opposite sign and ten a contracting return of period 16 or 32. The largest Lipschitz bound for those returns is less than 0.475146491; the maximum exclusion horizon is iterate 64.

The evaluation retains the first forty perturbation coefficients as shared variables, together with an explicit bound on their infinite tail. For each band, the center is g_0 shifted in coordinate u . If $a, b \geq 0$ bound the dual sensitivities in u, v , the support of the uncertainty in the fixed point and the stable residual is

$$0.00004 \max(a, b) + 0.001666 \max\left(b, \frac{0.16a + b}{1.16}\right). \tag{24.127}$$

The second expression is obtained by maximizing $a|e_u| + b\|e_v\|_1$ under the two constraints of (24.122). This uses the orientation of the stable sheet proved previously, as well as its norm.

Let x_j be the center's orbit and L_j its linear sensitivity to all shared coefficients. The Taylor model has the form

$$c_j = x_j + L_j(\Delta v) + r_j, \quad |r_j| \leq E_j.$$

If ρ_j bounds $|L_j(\Delta v)| + E_j$, the outward recurrence used is

$$E_{j+1} \leq |f'_0(x_j)|E_j + \frac{1}{2} \sup |f''_0| \rho_j^2 + \sup |\Delta f'| \rho_j. \tag{24.128}$$

The derivatives are bounded along the segment between the two states. The central orbit is verified within $[-1, 1]$; the other orbits belong to that interval by the self-map property included in the certified domain. The calculations use outward rational rounding, including for the geometric tails.

Each band is excluded by one of two inequalities:

1. A critical value has a sign strictly opposite to τ_j .
2. For $p = 2^k$, the return $H = f^p$ has Lipschitz constant $L_p < 1$ on the segment between 0 and $H(0)$. Then

$$|H(H(0)) - H(0)| \leq L_p |H(0)|.$$

If $H(0) \neq 0$, the two values have the same sign; if $H(0) = 0$, both vanish. In both cases the strict itinerary $\tau_{2p} = -\tau_p$ is excluded.

The constant L_p is obtained by propagating the initial segment $[-r, r]$, $r \geq |H(0)|$, alongside the critical-orbit boxes and multiplying the outward derivative bounds. The maps of the full band are bounded, retaining (24.127).

24.6.8.2.4 Coincidence theorem for the accumulation parameter **Theorem.** For the uniform HMT family (24.91), the limit of the original first-root selection coincides with the certified stable crossing:

$$\boxed{\lambda_n \nearrow a_* = \lambda_*, \quad 1.874038 < \lambda_* < 1.874039.} \tag{24.129}$$

Proof. (24.116)–(24.118) give $\lambda_n \uparrow a_* \leq \lambda_*$, with both parameters in J . If the inequality were strict, (24.120)–(24.125) would produce a return $f_N \in C_-$ realizing τ . The complete cover of section 24.6.8.2.3 excludes precisely that possibility. Hence $a_* = \lambda_*$.

The proof controls every depth through a finite first exit and a uniform cover of its region. The identity (24.129) links the global selector's limit to the transverse crossing.

24.6.8.3 Identification of the original centers and global scaling

The final step preserves the selection of (24.116). The local continuation and the original parameters are compared through their exact period and their hybrid class; coincidence is obtained through uniqueness in a fixed parameter neighborhood.

24.6.8.3.1 Degree-two analytic realization and transversality The fixed point g of section 24.6.7.3 is even, analytic, with a quadratic critical point and stationary doubling combinatorics. Theorem 2.1 of [de Faria–de Melo–Pinto](#), applied to that single combinatorics, gives a one-element limit set, with a proper holomorphic extension of degree two between nested topological disks. Its attraction on the interval identifies that element with g , since $Rg = g$.

Lemma 3.2 of the same source extends a fixed number of returns to a full analytic neighborhood of g , with degree-two images and positive modulus. The norm (24.95) controls the uniform norm on a sufficiently small complex neighborhood of $[-1, 1]$, because $|(x^2 - 1)/2.5| < 1$ there. The convergence (24.93) enters that neighborhood; analytic dependence of the returns provides, for some fixed integer d , a family

$$\mathcal{F}_\lambda = R^d f_\lambda \quad (24.130)$$

of degree-two maps, analytic in a complex neighborhood of λ_* . The disks and the number of returns are fixed before varying λ in that neighborhood.

The parameter tangent belongs to the expanding cone, whereas the stable sheet satisfies the slope bound of section 24.6.7.5. The returns preserve that separation. The degree-two realization locally identifies the stable sheet with the hybrid class of the infinitely renormalizable map; hence (24.130) is transverse to that class.

The change of analytic space is justified in detail in 24.6.9.1.2, devoted to the transport of transversality. A direction tangent to the hybrid class would produce decreasing renormalized tangents, by uniform convergence on a hybrid disk and Cauchy's estimate. The certified direction, by contrast, satisfies $|\dot{f}(1)| = |\dot{u}|/10$ and grows under the returns by the expanding cone. This common evaluation of both representatives proves transversality without assuming equivalence of their norms.

24.6.8.3.2 Uniqueness in a fixed neighborhood For $n > d$, the classification of section 24.6.8.1 gives

$$K(\mathcal{F}_{\lambda_n}) = W_{n-d}0. \quad (24.131)$$

Straightening a degree-two map preserves the critical orbit; (24.131) identifies its hybrid class with that of the canonical quadratic center of period 2^{n-d} . The a priori bounds for the real doubling sequence are those used in Lyubich's universality theorem.

[Lyubich's Theorem 7.4](#), pp. 387–388 gives a single intersection with each of those classes, at sufficiently high levels, on the analytic transversal and within a fixed neighborhood of the accumulation parameter.

By (24.129), λ_n eventually enters that neighborhood. The local continuation of section 24.6.7.8, indexed by the same period, belongs to the same hybrid class. Uniqueness implies

$$\boxed{\lambda_n = \widehat{\mu}_n \quad (n \geq n_0),} \quad (24.132)$$

where $\widehat{\mu}_n$ denotes the local center of physical period 2^n . This is a reindexing of μ_j by a fixed shift determined by the returns and the target of section 24.6.7.8. The original selection of λ_n remains intact.

The integer n_0 is existential in this proof. Continuation at every level and the word $W_n 0$ are already proved in (24.116); (24.132) provides the metric identity needed for the limit.

24.6.8.3.3 Scaling theorem for the first HMT roots **Theorem.** For the uniform family (24.91), let λ_n be the successive first new roots of exact critical period 2^n . There exist $C > 0$, $0 < \theta < 1$ and $M < \infty$ such that, for every sufficiently large n ,

$$\lambda_* - \lambda_n = C\delta_F^{-n}(1 + e_n), \quad |e_n| \leq M\theta^n, \quad (24.133)$$

and

$$\lim_{n \rightarrow \infty} \frac{\lambda_{n-1} - \lambda_{n-2}}{\lambda_n - \lambda_{n-1}} = \delta_F. \quad (24.134)$$

Proof. (24.132) transports the local scaling (24.114) to the original roots; the finite index shift is absorbed into C, M . Its sign is positive because $\lambda_n < \lambda_*$.

The power-law remainder also admits a direct justification through the holonomy of hybrid classes. Lyubich's Lemma 7.3 gives $C^{1+\beta}$ regularity, with nonzero derivative, on the transversal at the accumulation point. The one-dimensional unstable dynamics is analytic and has multiplier $\delta_F > 1$; its linearizing coordinate places successive centers at distances proportional to δ_F^{-n} . Composing with the inverse holonomy gives a nonzero linear term and a remainder $O(\delta_F^{-(1+\beta)n})$. Thus one may take $\theta = \delta_F^{-\beta}$.

Writing

$$r_n = \frac{\delta_F e_{n-1} - e_n}{\delta_F - 1}, \quad K = \frac{M(\delta_F + \theta)}{\delta_F - 1},$$

one obtains

$$\lambda_n - \lambda_{n-1} = C(\delta_F - 1)\delta_F^{-n}(1 + r_n), \quad |r_n| \leq K\theta^{n-1}.$$

Therefore,

$$\left| \frac{\lambda_{n-1} - \lambda_{n-2}}{\lambda_n - \lambda_{n-1}} - \delta_F \right| \leq \frac{\delta_F K(1 + \theta)\theta^{n-2}}{1 - K\theta^{n-1}}, \quad K\theta^{n-1} < 1. \quad (24.135)$$

The right-hand side tends to zero, proving (24.134).

24.6.8.3.4 Chain of results and scope of the closure The effective composition is: APP–TRIT–TPK dodecaphase reader; profile (24.91); ordered transport of returns (24.67)–(24.69); classification and existence of the first roots (24.116); localization (24.118); first-exit isolation (24.129); eventual identity of centers (24.132); scaling (24.133)–(24.135). The Jacobi representation (24.72) preserves this reader's full profile and its marks remain in the antecedent.

The spatial expansion of the HMT reader and nonadic refinement provide the genealogy and evaluation at arbitrary depth. Scaling between parameters is proved through the return dynamics, transversality and isolation that we have just composed. The cited analytic proofs act on the family already constructed; their comparison values do not select its phases or coefficients.

The closure concerns the original global selector of the uniform sector $\eta = 0$, with the listed theorem dependencies. The asymptotic law retains its existential quantifiers. The width of a finite-root box measures evaluation precision; the estimate (24.135) expresses convergence toward the limit.

The programs and certificates in the reproducibility supplement make it possible to reproduce the finite parts of the proof. Continuation at every depth rests on the inductions, the uniform first-exit argument and the identification theorems, with their explicit domains.

24.6.9 Analytic identification of the original centers

24.6.9.1 Degree-two realization and metric identification of the centers

The equality between the first parameter of the infinite itinerary and the stable crossing, proved by first exit, provides

$$a_* := \min \Lambda_\tau = \lambda_*. \quad (24.136)$$

The family, dodecaphase profile, returns and selector remain exactly as in sections 24.6.6.1–24.6.6.5. The external results used below recognize the dynamics of that already constructed object. Their universal values do not enter the selection of coefficients or parameters of the HMT family.

Metric transfer theorem. The identification (24.136), together with the previously proved entry, admissibility and transversality bounds, implies that the original sequence of first new roots satisfies, for certain constants $C > 0$ and $0 < \theta < 1$,

$$\lambda_* - \lambda_n = C\delta_F^{-n}(1 + O(\theta^n)), \quad (24.137)$$

where $\delta_F > 1$ is the unstable eigenvalue of the doubling operator at the recognized fixed point. In particular,

$$\frac{\lambda_{n-1} - \lambda_{n-2}}{\lambda_n - \lambda_{n-1}} = \delta_F + O(\theta^n). \quad (24.138)$$

The sequence eventually coincides with the centers of the local continuation, once both indices refer to the same physical period. The proof retains the first-root selector of section 24.6.6.4.

24.6.9.1.1 Obtaining a common quadratic-like domain Write g for the renormalization fixed point and $\Gamma(\lambda) = R^4 f_\lambda$. The analytic chart used in the certificate is

$$f(x) = 1 - x^2 V\left(\frac{x^2 - 1}{2.5}\right), \quad V(z) = \frac{u}{10} + \sum_{k \geq 1} \nu_k z^k,$$

with difference norm $|\Delta u| + \|\Delta v\|_1$. The admissibility inequalities prove that g is even, analytic, unimodal, quadratic and of doubling type; moreover, $g(x) = \varphi(x^2)$ with $\varphi' < 0$ in $[0, 1]$.

Theorem 2.1 of de Faria–de Melo–Pinto, applied to the combinatorial set consisting solely of doubling, provides a one-point limit set with a quadratic-like extension. Since $Rg = g$, the real convergence in that theorem identifies this point with g . Lemma 3.2 provides, in an analytic neighborhood of g , a finite renormalization iterate with a quadratic-like domain and uniformly positive modulus. The locations are sections 2.1–2.3, pp. 736–738, and section 3.1, Lemma 3.2, p. 745, of de Faria–de Melo–Pinto, *Global hyperbolicity of renormalization for C^r unimodal mappings*.

To check membership in the neighborhood of that lemma, let Ω_ϵ be a sufficiently small complex neighborhood of $[-1, 1]$, with

$$\rho_\epsilon := \sup_{x \in \Omega_\epsilon} \left| \frac{x^2 - 1}{2.5} \right| < 1, \quad M_\epsilon := \sup_{x \in \Omega_\epsilon} |x|^2 < \infty.$$

The restriction of the certified chart satisfies

$$\|\Delta f\|_{\Omega_\epsilon} \leq M_\epsilon \left(\frac{|\Delta u|}{10} + \rho_\epsilon \|\Delta v\|_1 \right) \leq C_\epsilon \|\Delta f\|_{\mathcal{A}}. \quad (24.139)$$

The certificate gives

$$\|R^m \Gamma(\lambda_*) - g\|_{\mathcal{A}} < 0.001666(0.83996)^m.$$

A fixed integer m is therefore chosen to place this image in the neighborhood of Lemma 3.2. Continuity of the finite composition provides an interval I_* around λ_* where the same operations are defined. If N is the iterate supplied by that lemma, we set

$$d = 4 + m + N, \quad \mathcal{F}_\lambda = R^d f_\lambda, \quad \lambda \in I_*. \quad (24.140)$$

The family $\mathcal{F}$ has quadratic-like representatives of degree two, on a common domain and with uniformly positive modulus. Its real-analytic dependence admits a local complexification of the parameter. The number d is fixed and counts the doublings performed before applying the parameter theorem.

This construction fixes a common domain and a finite number of returns d , sufficient to apply the parameter theorem.

24.6.9.1.2 Transport of certified transversality The direction of Γ satisfies the cone condition

$$\|\dot{v}\|_1 \leq \frac{1}{4}|\dot{u}|, \quad \dot{u} \neq 0.$$

The certified differential bounds imply invariance of this cone and

$$|\dot{u}_{k+1}| \geq 4.381|\dot{u}_k| \quad (24.141)$$

along the orbit of $\Gamma(\lambda_*)$. All its points remain in the certified domain. In particular, the direction transported through the finite steps of (24.140) retains a nonzero expanding component.

The hybrid class of the fixed point coincides with its stable manifold in the space of quadratic-like germs: *Lyubich, Feigenbaum–Coullet–Tresser universality and Milnor’s hairiness conjecture, Theorem 6.1, pp. 371–372*. Transversality remains valid on passing to that space, as can be checked without identifying different Banach norms. Indeed, if $\partial_\lambda \mathcal{F}_{\lambda_*}$ were tangent to the hybrid class, it would be the tangent of an analytic disk contained in it. Uniform exponential convergence of renormalization on a compact subdisk, followed by Cauchy’s estimate in its parameter, would make the evaluation at $x = 1$ of its renormalized tangents converge to zero. However, in the preceding chart,

$$\dot{f}(1) = -\dot{u}/10,$$

and (24.141) makes its absolute value grow. Both derivatives represent the same germ and have the same real evaluation. The contradiction proves

$$\partial_\lambda \mathcal{F}_{\lambda_*} \notin T_{\mathcal{F}_{\lambda_*}} \mathcal{H}_{c_\infty}. \quad (24.142)$$

Thus, $S = \{\mathcal{F}_\lambda\}$ is a complex-analytic transversal to the hybrid class of the doubling limit.

24.6.9.1.3 Eventual identification of the first roots by their period Denote by c_r the canonical quadratic center of period 2^r , obtained through r successive doublings of the period-one center. This symbol denotes the parameter of the quadratic recognition family; it is distinct from the orbital functions $c_j(\lambda) = f_\lambda^j(0)$ of the preceding sections. We write c_r^{quad} to preserve the distinction.

The global continuation theorem and (24.136) give $\lambda_n \rightarrow \lambda_*$, hence $\lambda_n \in I_*$ for sufficiently large n . The classification and restrictive intervals of section 24.6.6.2 show that, when $n > d$,

$$\mathcal{F}_{\lambda_n} = R^d f_{\lambda_n} \quad \text{has exact critical period } 2^{n-d} \quad \text{and itinerary } W_{n-d}0. \quad (24.143)$$

Its critical orbit remains in the invariant real interval, contained in the quadratic-like domain; its filled Julia set is therefore connected. Straightening preserves the postcritical orbit and the real branches. Following its restrictive returns gives $n - d$ canonical doublings and, finally, a fixed critical point. The

latter straightens to the center 0; the successive inverses of the straightening maps of the doubling copies give the unique center c_{n-d}^{quad} . This is the concrete application of the combinatorial parameterization and the straightening homeomorphism of the copies described in Lyubich, sections 5.1–5.3, pp. 362–364. We conclude

$$\mathcal{F}_{\lambda_n} \in \mathcal{H}_{c_{n-d}^{\text{quad}}} \quad (24.144)$$

Lyubich's Theorem 7.4, pp. 387–388, gives a unique local intersection of S with each class $\mathcal{H}_{c_r^{\text{quad}}}$ for sufficiently large r . Its assumptions of a priori complex bounds hold for the real quadratic doubling cascade, according to Theorem 5.6, p. 367, of the same work. Write ζ_r for the parameter of that intersection. Equations (24.143)–(24.144), convergence to the crossing and local uniqueness imply

$$\lambda_n = \zeta_{n-d} \quad \text{for every } n \text{ sufficiently large.} \quad (24.145)$$

The uniqueness used here concerns **each canonical hybrid class** in a fixed neighborhood of the crossing. It is not inferred merely from the existence of a local cascade or from uniqueness of the crossing with the stable sheet. Equation (24.145) identifies the original selector, already defined globally, with the local intersections; it does not introduce a replacement selector.

In particular, let μ_j be the local cascade of section 24.6.7.8, and let s be the fixed integer determined by its exact physical period:

$$\text{per}_{\text{crit}}(f_{\mu_j}) = 2^{j+s}.$$

In the presentation with curve $R^4 f_\lambda$ and period-two target, $s = 5$; the additional fixed steps used to localize the target modify s . By (24.144) and the same uniqueness,

$$\mu_j = \zeta_{j+s-d}, \quad \lambda_n = \mu_{n-s} \quad (n \text{ sufficiently large}). \quad (24.146)$$

The shift d in the quadratic-like preparation and the shift s in the target periods are distinct objects. Identification is made through the physical period, not by equating those two integers.

24.6.9.1.4 Power-law remainder through transverse holonomy Lyubich's Lemma 7.3, p. 384, with proof on pp. 384–387, provides $C^{1+\beta}$ -conformal regularity, for some $\beta > 0$, of the holonomy between transversals to the hybrid class of the Feigenbaum point. Its definition on p. 384 concerns the connectedness loci on the transversals; these contain, in particular, the centers considered here. We may reduce the exponent so that $0 < \beta \leq 1$.

Let h be the holonomy from S to the unstable manifold, and let z be an analytic coordinate linearizing the one-dimensional restriction of renormalization:

$$z(g) = 0, \quad z(Rq) = \delta_F z(q). \quad (24.147)$$

The existence of this coordinate follows by applying the linearization of a repelling holomorphic germ, with multiplier $\delta_F > 1$, to the analytic unstable manifold. The expanding direction of (24.141), cone invariance and the stationary return identify that multiplier with the positive unstable eigenvalue used in the certificate.

If q_r is the center of the class $\mathcal{H}_{c_r^{\text{quad}}}$ lying on the unstable manifold, compatibility of straightening and renormalization gives $Rq_r = q_{r-1}$. This identity is the one used in the proof of Theorem 7.4, p. 388. By (24.147), there exists $b \neq 0$, absorbing any fixed initial index, such that

$$z(q_r) = b \delta_F^{-r} \quad (24.148)$$

for every sufficiently large r .

The regularity of h and transversality (24.142) give, on the connectedness locus near λ_* ,

$$H(\lambda) := z(h(\mathcal{F}_\lambda)) = A(\lambda - \lambda_*) + O(|\lambda - \lambda_*|^{1+\beta}), \quad A \neq 0. \quad (24.149)$$

Since $H(\zeta_r) = b\delta_F^{-r}$, the lower estimate $|H(\lambda)| \geq |A| |\lambda - \lambda_*|/2$ in a sufficiently small neighborhood first implies $|\zeta_r - \lambda_*| = O(\delta_F^{-r})$. Substituting this bound into the remainder of (24.149) yields

$$\zeta_r - \lambda_* = \frac{b}{A} \delta_F^{-r} + O(\delta_F^{-(1+\beta)r}). \quad (24.150)$$

We apply (24.145) with $r = n - d$. The finite shift modifies the leading coefficient and the remainder constant. Since $\lambda_n < \lambda_*$, we obtain (24.137), with

$$C = -\frac{b}{A} \delta_F^d > 0, \quad \theta = \delta_F^{-\beta} \in (0, 1).$$

Finally,

$$\lambda_n - \lambda_{n-1} = C(\delta_F - 1) \delta_F^{-n} (1 + O(\theta^n)),$$

which proves (24.138). $\square$

24.6.9.1.5 Quantitative scope of the theorem The identification of limits proved by first exit activates metric transfer for the global first roots of the uniform sector. The operator rate 0.83996 controls the renormalized maps on the stable sheet. The parameter rate $\theta = \delta_F^{-\beta}$ follows from the regularity of transverse holonomy. Both estimates retain their variable and domain.

The theorem establishes the existence of β, θ, C , the neighborhood I_* , the finite number of returns d and an index n_0 beyond which the centers coincide. Their individual numerical values are not assigned in this asymptotic statement. This formulation completes the limit and its power-law remainder; numerical evaluation of those witnesses is a distinct quantitative specialization.

24.6.10 Return, memory and universality of the dodecaphase cascade

24.6.10.1 The generating structure and the critical reading

The Feigenbaum construction in Triadic Modular Holography links an oriented arithmetic organization to an asymptotic scaling law. Its starting point is the effective composition of APP, TRIT and TPK. APP evaluates sum and product on the two sheets of the digit lattice and preserves the quotients and remainders of those operations. TRIT determines the local regime and the orientation of its reading. TPK selects and transports the states, updates their records and prolongs their routes with carry, boundary and memory. The joint discrete structure of the continuum provides the common residence of those states and of the readers acting on them.

The prolongation

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

organizes this persistence. The initial seed is lifted while preserving its prefixes; successive lifts change the regime without losing orientation or memory; the coinductive boundary allows the construction to continue, and the nine charts bring together its local readings. Nonadic holonomy expresses phase return accompanied by an increase in memory. Thus, a visible turn preserves its position within the complete history. From the outset, the return operation acquires two inseparable components: reiteration of a rule and preservation of its provenance.

The critical reader fixed on the oriented dodecaphase wheel turns this structure into twelve phases with a uniform trace. Their representatives are

$$\phi_m = \frac{(3m-1)\pi_{\text{HMT}}}{18}, \quad w_m = \frac{1}{12}, \quad 1 \leq m \leq 12.$$

The choice of oriented representative matters because the second moment uses its magnitude. The angular quotient records the phase, while the representative retains the information needed to normalize it. Coordinate π_{HMT} comes from the coinductive readers of the same APP-TRIT-TPK core. Within its correlated image, coordinates π, φ, e and the dodecaphase closure of α_{HMT} retain their constructive order and their own domains. The critical reader considered here uses the angular coordinate already generated.

Normalization is determined through an integer sum:

$$\sum_{m=1}^{12} (3m-1)^2 = 5394 = 6 \cdot 899, \quad k_m = \frac{2(3m-1)}{\sqrt{899}}.$$

The common factor π_{HMT} cancels when the second moment is fixed. This cancellation preserves the provenance of the phases and produces a particularly precise representation of their normalized reading. We obtain

$$u_0(x) = \frac{1}{12} \sum_{m=1}^{12} (1 - \cos(k_m x)), \quad f_\lambda(x) = 1 - \lambda u_0(x), \quad \sum_m w_m k_m^2 = 2.$$

The profile is even and entire; it satisfies $u_0(0) = u'_0(0) = 0$ and $u''_0(0) = 2$. For $0 < \lambda \leq 2/u_0(1)$, the family maps $[-1, 1]$ into itself and has a quadratic critical maximum with $f''_\lambda(0) = -2\lambda$. Parameter λ ranges over this family after the reader, orientation, weights and scale have been fixed. The universal number recognized at the end is thus separated from the selection of the coefficients.

24.6.10.2 Twelve phases, moments and spectral representation

The profile admits a second expression bringing its phases together in a finite operator. The nodes

$$s_m = k_m^2 = \frac{4(3m-1)^2}{899}$$

determine the positive measure $\mu_0 = 12^{-1} \sum_m \delta_{s_m}$. Its moments $M_r = \int s^r d\mu_0(s)$ define strictly positive Hankel forms up to rank twelve. This positivity follows from evaluating a polynomial at twelve distinct nodes: a polynomial of degree less than twelve retains at least one nonzero evaluation. At degree twelve, the product of the twelve nodal factors annihilates the norm and determines the reader's rank exactly.

Orthogonalization of these moments produces a Jacobi matrix J_{12} , real symmetric, with positive subdiagonal entries. Its functional calculus recovers the complete profile:

$$u_0(x) = e_0^\top [I - \cos(x\sqrt{J_{12}})] e_0.$$

The cosine sum and the Jacobi operator are two realizations of the same reader. The moments retain all frequencies and their weights; the orientation, route and memory marks remain in the state that generates them. Rank twelve identifies the spectral dimension of this specific reading. The depth of its prolongations belongs to another dimension of the construction: the length of compatible histories and the resolution at which they are evaluated.

The analytic geometry is also made explicit. For the weighted deformation u_η , in the strip $|\eta| \leq 1/40$, the weights remain positive and the second moment remains normalized. In $|\operatorname{Re} z| < \pi/3$ there exists an odd, univalent holomorphic function b_η , normalized by $b'_\eta(0) = 1$, such that $u_\eta = b_\eta^2$. The wheel therefore produces a conformal deformation of the quadratic fold. Its dynamical conjugacy preserves that deformation through $b_\eta(1 - \lambda w^2)$. The parameter-scaling theorem developed here concerns the uniform member $\eta = 0$; the structural properties of the weighted strip and the local persistence of its roots retain their specific quantifiers.

24.6.10.3 Reading depth and recovery of histories

Nonadic depth refines the reading of a parameter, whereas dynamical doubling changes the return horizon. Refinement with carry expresses the first operation through

$$L_c(x, y) = (Bx - c, By + c), \quad C_m = \sum_{j=1}^m c_j B^{m-j},$$

$$(x_m, y_m) = (B^m x_0 - C_m, B^m y_0 + C_m).$$

The outer support grows as B^m , and the normalized record cylinder has diameter B^{-m} . The integer pair, depth and boundary mark preserve reconstruction of quotients, remainders and prefixes. Outer growth and inner resolution thus describe compatible operations on the same history.

The exact composition $L_D^{[b]} L_C^{[a]} = L_{B^b C + D}^{[a+b]}$ allows a history to be grouped into blocks without altering its data. A doubling return combines two transitions; the nonadic calendar combines nine. Both groupings are compared over a common horizon of $9 \cdot 2^N$ events, preserving the order of the subblocks and their carries.

The contribution of memory can be observed directly in the profile's inverse branches. If $v = (u_0|_{[0,1]})^{-1}$, the two branches are

$$\beta_\sigma(y) = \sigma v\left(\frac{1-y}{\lambda}\right), \quad \sigma \in \{-1, +1\}.$$

The final value, together with the last sheet, reconstructs the preceding value. A sequence of sheets allows all intermediate steps to be recovered. The branch word accompanies the APP–TRIT–TPK records of route, boundary, carry and memory; each record retains its function. This concrete recovery explains what information is required to invert the critical reading.

The resolution of each return also has an explicit bound. With $t = \lambda u_\eta(1)/2$, $z = (x+1)/2$ and

$$F_{t,\eta}(z) = 1 - t \frac{u_\eta(2z-1)}{u_\eta(1)},$$

one obtains

$$\left| F_{t,\eta}^q(1/2) - F_{s,\eta}^q(1/2) \right| \leq S_q |t - s|, \quad S_q = \sum_{j=0}^{q-1} (6\sqrt{2})^j < 9^q.$$

A base-nine cylinder of depth $m = 2^N + r$ therefore allows the return of horizon 2^N to be read with diameter less than 9^{-r} . This estimate gives quantitative content to arbitrary depth: for every requested horizon and precision, it determines a sufficient depth. Evaluation of the analytic functions and rounding have their own remainders, which propagate together with the parameter uncertainty.

Vacancies enter the capacity balance of the record. Comparison between blocks of 729 and 1000 possibilities distinguishes accumulated capacity from chronological length, preserving the steps at which the former remains constant. This accounting determines the space required to preserve a reading and its depth. The admissibility of histories follows, in turn, from the survival rules. Both operations contribute to the construction and retain distinct mathematical objects.

24.6.10.4 Nonadic chronology and doubling combinatorics

The critical doubling word has an expression at every depth:

$$\tau_j = (-1)^{v_2(j)}, \quad \tau_{2j} = -\tau_j, \quad \tau_{2j+1} = +1.$$

Its prefixes satisfy $W_{n+1} = W_n(-1)^n W_n$. The rule determines how the two copies of the return are oriented and where their separation appears. The nonadic reading of chronology preserves the complete integer $K = \rho_9(K) + 9q_9(K)$. Reducing it modulo 2^n , the holonomy that advances nine events induces translation by nine on that cycle. Since nine is odd, this translation visits all its positions. The permutation $j \mapsto 9j \pmod{2^n}$, which conjugates it to unit advance, preserves the dyadic valuation of each nonzero position. The chronological remainder and quotient jointly support that compatibility.

Bilateral memory also has an operator realization compatible with refinement. On cycles of length 2^n , the shift operator C_n is incorporated into

$$\mathbb{U}_n = P_0 \otimes I + P_1 \otimes C_n,$$

with the complementary projectors of the nonadic reader. In the limit, its visible and complementary components satisfy

$$\|T_{\infty,a}v\|^2 + \|D_{\infty,a}v\|^2 = \|v\|^2, \quad v = T_{\infty,a}^* T_{\infty,a}v + D_{\infty,a}^* D_{\infty,a}v.$$

These identities describe preservation and recovery within the declared space and representation. Contraction of the renormalized maps, which will appear later, measures another operation: the approach of their profiles to a stable form. The history preserved by the lift and the convergence of a functional reading are articulated at different levels.

24.6.10.5 The ordered chart and global root selection

The Archimedean readout of the HMT continuum transports the operations through an isomorphism of complete ordered fields, within the declared universe $\mathfrak{G}$:

$$\iota : \mathbb{R}_{\text{HMT}}^{\mathfrak{G}} \longrightarrow \mathbb{R}^{\mathfrak{G}}.$$

The internal construction of the cosine through its series, the positive square root of 899, the profile and its iterates precedes zero selection. Preservation of sum, product, order and convergent limits gives

$$\iota(S_n^H(a)) = S_n(\iota(a)), \quad S_n(\lambda) = f_\lambda^{2^n}(0).$$

Surjectivity covers all roots in the domain of that chart; order preserves preceding returns, exact period and the condition of lying to the right of the preceding root. In this way, the minimum of each candidate set is also transported. The genealogy reaches the effective parameter selection, with the second projection and its marks preserved on the same antecedent.

The selector is fixed by $\lambda_1 = 1/u_0(1)$ and by the first new root of exact critical period 2^n lying to the right of λ_{n-1} . The itinerary classification proves that each selected root has word $W_n 0$. Restrictive returns halve its period while preserving unimodal order, and induction restores the complete word. Analyticity of returns that are not identically zero isolates their zeros in each compact set; comparison of signs before the first infinite-itinerary parameter guarantees a new root at every level.

Let Λ_τ be the nonempty compact set of parameters realizing the infinite word, and let a_* be its minimum. It is proved that

$$\lambda_n \nearrow a_*.$$

Passage to the limit preserves each sign through a uniform bound at a fixed horizon. If $c_p = f_\lambda^p(0)$, the opposite signs of c_p and c_{2p} , together with $(f_\lambda^p)'(0) = 0$, imply

$$|c_p| > \frac{2}{B_p}, \quad B_p = 16^p \frac{16^p - 1}{15}.$$

Thus, the prefixes of increasing periods preserve the infinite word as they accumulate. The proof uses the original sequence of first roots from its global definition.

24.6.10.6 The stable sheet and identification of the selected limit

The normalized return

$$(Rf)(x) = -\frac{f(f(-ax))}{a}, \quad a = -f(1) > 0,$$

combines two transitions, changes the scale and preserves the orientation needed to restore a maximum of value one. The restrictive domains establish its interpretation as a real return. The derivative of this operator includes the variation of scale a ; that contribution determines the parameter sensitivity of the complete transformation.

In the chart $f(x) = 1 - x^2V((x^2 - 1)/(5/2))$, with coordinates $u = 10v_0$ and $v = (v_1, v_2, \dots)$, the fourth return of the family enters a quantified neighborhood of the fixed point g . The curve crosses its stable sheet transversely at a unique local parameter

$$1.874038 < \lambda_* < 1.874039,$$

and satisfies

$$\|R^{4+n}f_{\lambda_*} - g\|_{\mathcal{A}} < 0.001666(0.83996)^n.$$

The stable sheet is a graph of slope at most 0.16. The parameter secants belong to a transverse expanding cone. Both properties distinguish the decrease in stable deformations from the growth of separation between parameters.

Identifying a_* with λ_* brings this local geometry together with the global minimum. First, the interval between the eighth root and the lower endpoint of the local box is excluded. Then the order $a_* \leq \lambda_*$ allows a hypothetical separation to be studied through the negative cone. While its transverse coordinate has absolute value less than 0.006, the secant remains in the analytic domain and grows by a factor between 4.4034 and 8.0887. A persistent separation would therefore produce a first exit with amplitude between 0.006 and 0.0485322.

The stable-orbit residual retains information more precise than its norm: its components satisfy $|e_u| \leq 0.16\|e_v\|_1$. This relation allows their effects on the critical orbit to be bounded jointly. The cover of the entire first-exit region shows in each case a sign incompatible with τ , or a contracting return forcing two successive dyadic critical values to have the same sign. The infinite word requires opposite signs. The contradiction identifies the parameters:

$$\lambda_n \nearrow a_* = \lambda_*.$$

The proof combines an induction valid at every depth with a uniform finite exclusion of the region that any separation would reach. Its infinite scope rests on that composition.

24.6.10.7 Universal scaling and the meaning of precision

The degree-two holomorphic realization of the returns and the established transversality allow their hybrid classes to be compared. For each sufficiently high period, the canonical class has a unique local intersection with the family. The global first roots already converge to the crossing and preserve the doubling combinatorics; hence they eventually coincide with the local centers when their physical periods are matched. The proof transports the original selection to the universal metric law.

There exist $C > 0$, $M < \infty$ and $0 < \theta < 1$ such that

$$\lambda_* - \lambda_n = C\delta_F^{-n}(1 + e_n), \quad |e_n| \leq M\theta^n,$$

and, consequently,

$$\lim_{n \rightarrow \infty} \frac{\lambda_{n-1} - \lambda_{n-2}}{\lambda_n - \lambda_{n-1}} = \delta_F.$$

The factor δ_F is the expanding eigenvalue of the doubling operator recognized at the end of the construction. The specific family preserves its twelve phases, moments and provenance history; the limit of its parameter ratios belongs to the same universality class of quadratic criticality. This relation jointly explains the specificity of the generator and the universality of the output.

Precision has three residences here. The root intervals control their finite evaluation. The estimate $0.001666(0.83996)^n$ controls operator convergence on the stable sheet. The term $M\theta^n$ controls the transfer of that geometry to the parameter scale through transverse holonomy. Each estimate corresponds to its variable and operation. The eighth quotient is enclosed around $4.669212189150204154556096215889663928$, with width less than $5.808 \cdot 10^{-37}$; that width measures the evaluation of $\delta_{F,8}$. The distance from $\delta_{F,8}$ to the limit requires quantifying the constants of the parameter remainder, whose existence is already proved.

The notation keeps separate δ_F , the Feigenbaum parameter-scaling factor; α_F , the spatial constant associated with its renormalization problem; and α_{HMT} , the corpus's fine-structure constant. The present result identifies δ_F through the first roots of the uniform sector $\eta = 0$. The weighted strip retains its structural and existence theorems, and its quantified metric extension constitutes a distinct specific statement.

The contribution of the chain lies in continuity between generation, reading, memory, selection and limit. Oriented arithmetic determines the profile; the spectral representation preserves its twelve components; the cylinders allow it to be resolved with arbitrary precision at each horizon; history transport restores its paths; the ordered chart preserves all roots and their minima; classification prolongs the selector; and transverse expansion, together with stable contraction, determines its asymptotic law. Universality emerges as a proved property of this complete construction and its analytic realization.

24.6.11 Articulation of the critical, spectral and constitutive readers

The APP–TRIT–TPK composition precedes the readers in this section. APP preserves the additive and multiplicative evaluations in each cell, with their remainders and quotients; tritic reduction determines the regime, orientation and carry of the transition. The effective composition of selection, transport and memory update produces the enriched state and its prolongations, developed in 24.6.1.1 and 24.6.1.1.1. Nonadic holonomy composes those transitions: phase return advances memory and preserves history. The readers act on the same joint discrete structure of the continuum, with its five consubstantial constructions and with the two readouts of the same antecedent: ordered evaluation and boundary incidence.

The oriented wheel and angular channels are outputs of this composition. Their already generated HMT coordinates are used in subsequent operations, with their marks and domains; they are not metrological numbers introduced to select an answer. The ordered readout allows the following formulas to be evaluated, while the provenance state preserves sheet, route, remainder, quotient, carry, boundary and memory. The recoveries that are proved recover the object of their declared domain; the evaluated representation remains accompanied by those records.

24.6.11.1 Critical separation as an intrinsic asymptotic reading

In the uniform sector of the dodecaphase wheel, the oriented representatives and their weights are

$$\phi_m = \frac{(3m-1)\pi_{HMT}}{18}, \quad w_m = \frac{1}{12}, \quad 1 \leq m \leq 12.$$

The second moment is normalized through $\sum_{m=1}^{12} (3m-1)^2 = 5394 = 6 \cdot 899$. Thus the common angular factor cancels, leaving

$$k_m = \frac{2(3m-1)}{\sqrt{899}}, \quad u_0(z) = \frac{1}{12} \sum_{m=1}^{12} (1 - \cos(k_m z)), \quad f_\lambda(z) = 1 - \lambda u_0(z). \quad (24.151)$$

Variable λ ranges over the family after the reader, its weights and normalization have been fixed. In $0 < \lambda \leq 2/u_0(1)$, the dynamical domain is $[-1, 1]$. The first root is $\lambda_1 = 1/u_0(1)$; for $n \geq 2$, λ_n is the smallest root greater than λ_{n-1} whose critical point has exact period 2^n . The classification, existence of each minimum and metric identification of these centers have been proved in 24.6.8.1–24.6.8.3.

Proposition 24.32 (Structural reading of the scaling constant). *For the uniform reader (24.151) and its first-root selector, the critical-separation constant is*

$$\delta_{\text{HMT,crit}} := \lim_{n \rightarrow \infty} \frac{\lambda_n - \lambda_{n-1}}{\lambda_{n+1} - \lambda_n} = \delta_F. \quad (24.152)$$

It is the asymptotic parameter-separation invariant of the critical returns of that reader, with its selection order preserved.

Proof. The scaling theorem (24.133) provides $\lambda_* - \lambda_n = C\delta_F^{-n}(1 + e_n)$, with $C > 0$ and $e_n \rightarrow 0$. Subtracting two consecutive terms,

$$\lambda_n - \lambda_{n-1} = C\delta_F^{-n}[\delta_F - 1 + \delta_F e_{n-1} - e_n].$$

The quotient of this equality and the one at index $n + 1$ converges to δ_F . The eventual identity of the local centers with the first roots, proved in (24.132), allows the scaling to be applied precisely to the selected sequence and not to another cascade chosen afterward. $\square$

This formulation preserves more information than the terminal numeral: it identifies the wheel, its reading, the family and the selector whose limit is evaluated. Its interpretation is intrinsic to that HMT composition; recognition of the universal constant subsequently uses the analytic theorem applied to the already constructed reader.

24.6.11.2 The finite measure and spectral recovery of the profile

The squared frequencies produce the atomic probability measure

$$s_m = k_m^2 = \frac{4(3m-1)^2}{899}, \quad \nu_0 = \frac{1}{12} \sum_{m=1}^{12} \delta_{s_m}, \quad M_j = \int s^j d\nu_0(s). \quad (24.153)$$

Here δ_{s_m} is the Dirac mass at a node, distinct from the constant δ_F . The measure preserves the twelve frequencies and their weights. For a nonzero polynomial of degree less than $r \leq 12$,

$$\int p(s)^2 d\nu_0(s) = \frac{1}{12} \sum_{m=1}^{12} p(s_m)^2 > 0,$$

since a polynomial of degree less than twelve cannot vanish at the twelve distinct nodes. Thus the Hankel matrices $(M_{i+j})_{0 \leq i, j < r}$ are positive definite. The polynomial $\prod_m (s - s_m)$ has zero norm; the space $L^2(\nu_0)$ has dimension twelve.

Proposition 24.33 (Exact representation of the critical reader). *Let $\psi_0, \dots, \psi_{11}$ be the orthonormal polynomials with positive leading coefficient for ν_0 , with $\psi_0 = 1$. Multiplication by s has Jacobi matrix J_{12} , real symmetric with positive subdiagonal entries. With $e_0 = (1, 0, \dots, 0)^T$,*

$$u_0(z) = \int (1 - \cos(z\sqrt{s})) d\nu_0(s) = e_0^T [I - \cos(z\sqrt{J_{12}})] e_0. \quad (24.154)$$

Proof. The preceding positivity allows orthogonalization through degree eleven. If $i < j - 1$, self-adjointness of multiplication gives $\langle s\psi_j, \psi_i \rangle = \langle \psi_j, s\psi_i \rangle = 0$. By symmetry, only the diagonal and its two

neighbors remain; the positive ratio of the leading coefficients establishes positivity of the subdiagonal. Define

$$Q_{mj} = \frac{\psi_j(s_m)}{\sqrt{12}}, \quad D = \text{diag}(s_1, \dots, s_{12}).$$

Orthonormality gives $Q^T Q = I$, and evaluating the recurrence at the nodes gives $DQ = QJ_{12}$. Since Q is square, $J_{12} = Q^T DQ$. Its eigenvalues are positive, so the positive square root and cosine are calculated in this diagonalization. Finally, $Qe_0 = (1, \dots, 1)^T / \sqrt{12}$, which proves (24.154). $\square$

The realization through (J_{12}, e_0) recovers the same profile; therefore, preserving the family and selector also recovers its returns and first roots. Spectral positivity justifies this representation. Parameter transversality also uses the derivative of the return and its orbital products, developed in 24.6.5.1.3 and 24.6.7.6. These are properties of different operations: the scaling proof preserves both and does not replace tangent control with positivity of the moments.

24.6.11.3 Constitutive channels and their labeled inversion

The constitutive reader acts on another typed object on the same support: two-sheet angular transport. Its already generated coordinates are $x = A_{\text{rad}}$ and $y = C_{\text{rad}}^*$, in the contracting chamber $x > |y|$. The sheet involution $R = R^* = R^{-1}$ determines the marked projectors $P_{\pm} = (I \pm R)/2$. The transport representation is

$$q_+ = \exp[-(x+y)], \quad q_- = \exp[-(x-y)], \quad T = q_+ P_+ + q_- P_-, \quad 0 < q_{\pm} < 1. \quad (24.155)$$

Exponential evaluation realizes those channels after their angular construction. The transport incidences 90 and 120 fix the response

$$\mathcal{V} = (I - T^{90})(I - T^{120})^{-1}.$$

The inverse exists because $1 - q_{\pm}^{120} > 0$. Writing $s = q^{30}$, with $30 = \gcd(90, 120)$, gives

$$f(s) = \frac{1-s^3}{1-s^4} = \frac{1+s+s^2}{1+s+s^2+s^3}, \quad r_{\pm} = f(q_{\pm}^{30}), \quad \mathcal{V} = r_+ P_+ + r_- P_-. \quad (24.156)$$

In this subsection f denotes the constitutive response; f_{λ} still denotes the dynamical map of (24.151). Their arguments and domains are distinct.

Proposition 24.34 (Recovery of the channels and orientation). *The constitutive function $f : (0, 1) \rightarrow (3/4, 1)$ is a decreasing bijection. Its inverse ψ assigns to r the unique root $s \in (0, 1)$ of*

$$rs^3 + (r-1)(1+s+s^2) = 0. \quad (24.157)$$

The normalized constitutive readings, with their labeled sheets,

$$\widehat{\varepsilon} = r_+^2, \quad \widehat{\mu} = r_-^2, \quad \widehat{Z} = \frac{r_-}{r_+}, \quad \widehat{c} = \frac{1}{r_+ r_-},$$

with $\widehat{\varepsilon}, \widehat{\mu} \in (9/16, 1)$, allow recovery of

$$r_+ = \sqrt{\widehat{\varepsilon}}, \quad r_- = \sqrt{\widehat{\mu}}, \quad s_{\pm} = \psi(r_{\pm}), \quad q_{\pm} = s_{\pm}^{1/30}. \quad (24.158)$$

$$x = -\frac{1}{2} \log(q_+ q_-), \quad y = \frac{1}{2} \log \frac{q_-}{q_+}. \quad (24.159)$$

Proof. Differentiating the rational quotient gives

$$f'(s) = -\frac{s^2(3+2s+s^2)}{(1+s+s^2+s^3)^2} < 0 \quad (0 < s < 1). \quad (24.160)$$

The limits $f(0^+) = 1$ and $f(1^-) = 3/4$, together with continuity, prove bijectivity. Multiplying $f(s) = r$ by the positive denominator produces (24.157). Positivity of $r_{\pm}$ determines the square roots and that of $s_{\pm}$ determines the thirtieth roots. By (24.155), $\log q_+ = -x - y$ and $\log q_- = -x + y$; adding and subtracting recovers x, y . Conversely, the recovered roots belong to $(0, 1)$, so that $x \pm y = -\log q_{\pm} > 0$; the resulting pair remains in the chamber $x > |y|$. $\square$

In particular, $\widehat{\mu}\widehat{\varepsilon} = \widehat{c}^{-2}$ and $\widehat{\mu}/\widehat{\varepsilon} = \widehat{Z}^2$. Exchanging the labels permutes q_+ and q_- , preserves x and changes the sign of y . Recovery requires that orientation: the isolated product r_+r_- preserves common propagation, but does not determine the two labeled responses. The positive roots and cubic inversion recover the channels within the fixed reader; the complete record of their generation remains in the enriched state.

24.6.11.4 The Catalan moment and functional recovery

The same incidences determine a relation between functions, before either channel is evaluated. For $0 < s < 1$, let $\mathcal{D} = s d/ds$, and define the oriented moment

$$C_2(s) = \int_0^s \log \frac{s}{t} \frac{1+t}{1+t+t^2} f(t) dt = \int_0^s \frac{\log(s/t)}{1+t^2} dt. \quad (24.161)$$

The second equality follows from the factorization $1+t+t^2+t^3 = (1+t)(1+t^2)$. The continuous positive weight $dt/(1+t^2)$ is thus produced by the rational composition of the constitutive reader, not by an identification with v_0 .

Proposition 24.35 (Differential inversion, uniqueness and endpoint reading). *The moment C_2 is the unique function $F \in C^2((0, 1))$ satisfying*

$$\mathcal{D}^2 F(s) = \frac{s(1+s)}{1+s+s^2} f(s), \quad \lim_{s \downarrow 0} F(s) = 0. \quad (24.162)$$

The constitutive response is recovered through

$$\mathcal{D}^2 C_2(s) = \frac{s}{1+s^2}, \quad f(s) = \frac{1+s+s^2}{s(1+s)} \mathcal{D}^2 C_2(s). \quad (24.163)$$

Its continuous extension to the closed interval has the series

$$C_2(s) = \sum_{j=0}^{\infty} \frac{(-1)^j s^{2j+1}}{(2j+1)^2}, \quad 0 \leq s \leq 1, \quad C_2(1) = \sum_{j=0}^{\infty} \frac{(-1)^j}{(2j+1)^2} = G_{\text{Cat}}. \quad (24.164)$$

Proof. The integrand of (24.161) is nonnegative and

$$0 \leq C_2(s) \leq \int_0^s \log(s/t) dt = s.$$

This proves integrability at zero and the required limit. The identity $\log(s/t) = \int_t^s du/u$, followed by interchanging nonnegative integrals, provides

$$C_2(s) = \int_0^s \frac{1}{u} \left(\int_0^u \frac{dt}{1+t^2} \right) du.$$

On every compact interval of the open domain, the fundamental theorem of calculus applies. Therefore,

$$\mathcal{D}C_2(s) = \int_0^s \frac{dt}{1+t^2}, \quad \mathcal{D}^2C_2(s) = \frac{s}{1+s^2}.$$

The rational factorization used in (24.161) gives (24.162); solving for f gives (24.163), whose denominators are positive in $(0, 1)$.

If two solutions have the required zero limit, their difference H satisfies $\mathcal{D}^2H = 0$. The substitution $z = \log s$ transforms this equation into $d^2H(e^z)/dz^2 = 0$, whence $H(s) = a \log s + b$. Existence of the zero limit forces $a = b = 0$, proving uniqueness.

For $s < 1$, the geometric expansion of $(1+t^2)^{-1}$ can be integrated term by term with the logarithmic weight, since

$$\int_0^s t^{2j} \log(s/t) dt = \frac{s^{2j+1}}{(2j+1)^2}, \quad \sum_{j \geq 0} \frac{s^{2j+1}}{(2j+1)^2} < \infty.$$

This proves the series in the interior. The bound $1/(2j+1)^2$ gives absolute and uniform convergence of that series on $[0, 1]$. In the integral, the integrand extended by zero for $t > s$ is dominated by $\log(1/t)$ on $(0, 1)$. Dominated convergence allows us to reach $s = 1$ and agrees with the extension of the series. Its endpoint value is precisely the stated Catalan series. $\square$

Differential inversion uses the family $C_2(s)$, not only its endpoint value G_{Cat} . In particular, $\mathcal{D}^2C_2(s) \rightarrow 1/2$ when $s \uparrow 1$. The series differentiated twice with $\mathcal{D}$ has a radial Abel limit there; the series of the moment itself is absolutely convergent at the boundary. This distinction preserves the precise conditions of each passage to the limit.

24.6.11.5 Composition of the recoveries and joint scope

The preceding interrelations have explicit operations and domains. The moments of the atomic measure ν_0 produce J_{12} , and its functional calculus recovers the profile u_0 . The profile and first-root selector determine the sequence whose scaling yields the readout δ_F . On the other hand, the constitutive function f produces C_2 through the integral moment, and (24.163) recovers f from that family. Its labeled evaluations at $s_{\pm} = q_{\pm}^{30}$ determine the constitutive responses; cubic inversion recovers both channels and their angular coordinates. The endpoint reading of the moment gives G_{Cat} .

The measure ν_0 is positive, atomic and of rank twelve. The Catalan moment integrates a continuous positive weight and has an alternating expansion. The signs of that expansion are coefficients of the geometric series; they are not negative weights of ν_0 , nor do they turn the positive integral into the same spectral representation. Each reader retains its support, orientation, normalization and evaluation operation.

Unity lies in the APP–TRIT–TPK structure that generates and transports their antecedents; the functional relations are established through the proved compositions. Equality of origin does not identify operators with different domains. In particular, this articulation does not postulate a scalar formula determining δ_F from Catalan's constant and other physical constants. It places critical scaling alongside effective recoveries between readers on the same support, preserving the information required by each direction.

Rogers–Ramanujan identities and nonadic Pell lifting

25.1 Normalizer 899, Pell unit, and arithmetic structure of the mode 30

The normalizer has the factorization arithmetic

$$899 = 30^2 - 1 = 29 \cdot 31. \quad (25.1)$$

The same mode 30 that generates sectors 90 and 120 simultaneously determines the twin-prime pair 29, 31. Moreover,

$$\varepsilon_{30} = 30 + \sqrt{899}, \quad \varepsilon_{30}^{-1} = 30 - \sqrt{899}, \quad \log \varepsilon_{30} = \operatorname{arcosh} 30. \quad (25.2)$$

This Pell unit determines the hyperbolic scale associated with the normalizer and relates the dodecaphase phase, mode 30, twin primes, and hyperbolic geometry. This relation does not identify the constitutive operator of the vacuum with the doubling renormalizer; such an identification would require an explicit intertwiner.

The unit acts integrally through

$$M_{30} = \begin{pmatrix} 30 & 899 \\ 1 & 30 \end{pmatrix}, \quad \det M_{30} = 1. \quad (25.3)$$

On $\mathbb{Z}[\sqrt{899}]$, this matrix represents multiplication by $30 + \sqrt{899}$. Its eigenvalues are $\varepsilon_{30}^{\pm 1}$; therefore $\log \varepsilon_{30}$ is the translation length of the associated hyperbolic transformation. The same central integer 30 reappears in the vacuum because $90 = 3 \cdot 30$ and $120 = 4 \cdot 30$, but the two operators continue to have distinct domains. The demonstrated confluence concerns the mode and its completions; identity of the dynamics is not postulated.

25.2 Lift nonadic of the unit of Pell

For $k \geq 1$, let the unramified quadratic ring be

$$\mathcal{R}_k = (\mathbb{Z}/3^k\mathbb{Z})[r]/(r^2 - 2), \quad (25.4)$$

and let

$$u = 3 + 2r.$$

The conjugation $r \mapsto -r$ gives

$$N_{\mathcal{R}_k/(\mathbb{Z}/3^k\mathbb{Z})}(u) = (3 + 2r)(3 - 2r) = 9 - 8 = 1,$$

so that u is a unit of norm one. Moreover,

$$u^2 = 17 + 12r, \quad u^4 - 1 = 576 + 408r = 3(192 + 136r).$$

The second factor is a unit modulo three because it reduces to $r \neq 0$. Hence the valuation 3-adic satisfies $v_3(u^4 - 1) = 1$. The lifting lemma for a unit congruent to one modulo three yields, for every $j \geq 0$,

$$v_3(u^{4 \cdot 3^j} - 1) = j + 1.$$

Since $u \bmod 3 = 2r$ and $(2r)^2 = -1$, its order modulo three is four. Consequently,

$$\text{ord}(u) = 4 \cdot 3^{k-1}, \quad (25.5)$$

and the profinite closure of the compatible cyclic groups is

$$\varprojlim_k \langle u \bmod 3^k \rangle \simeq C_4 \times \mathbb{Z}_3. \quad (25.6)$$

Its Archimedean realization is

$$3 + 2\sqrt{2} = (1 + \sqrt{2})^2.$$

This construction proves the conservation of nonadic depth by a hyperbolic unit. The units $3 + 2\sqrt{2}$, $2 + \sqrt{3}$, and $30 + \sqrt{899}$ form a Pell family with distinct provenances; any identification among them requires explicit maps between the corresponding extensions.

The mode 30 is one of the points at which the architecture begins to exhibit a unit determined by the operatorial and arithmetic genealogy.

It appears first in the vacuum because

$$90 = 3 \cdot 30, \quad 120 = 4 \cdot 30.$$

It is the elementary cadence that permits comparison of the sectors 90 and 120. The vacuum reads that cadence as the completion $3 \rightarrow 4$.

But the same 30 reappears in the construction of the criticality germ. There the central identity is

$$899 = 30^2 - 1 = 29 \cdot 31.$$

Normalization of the twelve phases produces $\sqrt{899}$, and the analytic profile can be written in closed form. What is truly striking is that π disappears before the iteration begins.

This must be read carefully. The angular geometry of π is absorbed into the organization of the phases. Once the dodecaphase configuration has been normalized, the dynamic germ is governed by the internal arithmetic of 899.

Therefore, the construction begins by generating internally a dynamic object of its own:

- is even;
- is analytic;
- has a quadratic critical point;
- belongs to the corresponding unimodal class;
- has negative Schwarzian;
- can be subjected to the operator of renormalization.

The superstable ratios then begin to approximate the universal constants. The germ was produced first, and universality appears as the terminal behavior of its dynamics.

This is a profound epistemological difference. A numerical coincidence would state: “our calculation resembles Feigenbaum.” The HMT construction states: “this is the exact object that enters the universality class, and these are the quotients that emerge upon iterating it.”

The 30 reappears a third time in modular memory. If

$$N = N_{90} + N_{120}$$

measures the total occupancy and

$$D = 90N_{90} + 120N_{120}$$

measures the occupation weighted by provenance, the matrix that transforms (N_{90}, N_{120}) into (N, D) has determinant 30.

This allows the process to be inverted:

$$N_{90} = 4N - \frac{D}{30}, \quad N_{120} = \frac{D}{30} - 3N.$$

Thus, the same 30 performs three functions:

- in the vacuum, makes the propagations 90 and 120 commensurable;
- at criticality, determines the normalization $899 = 30^2 - 1$;
- in modular memory, is the determinant that allows the origin of each occupation to be recovered.

The vacuum operator, the chaos germ, and the modular Hamiltonian remain typologically distinct. The confluence is subtler: all three preserve a common partition of the genealogy.

We might say that the mode 30 is a gear. In one machine it measures propagation; in another it normalizes criticality; in another it makes the apparent loss of provenance reversible. The gear is the same, but each machine performs a different function.

The emergence of the Pell unit

$$30 + \sqrt{899}$$

reinforces this interpretation. The 30 simultaneously organizes a finite sum and hyperbolic growth. The same arithmetic that closes the germ opens a scale dynamics.

That is perhaps the profound reason why the result proves so beautiful: the mode that makes the finite completion of the vacuum possible is also the mode that prepares an infinite universality of criticality.

Finite and infinite are articulated through the coherent repetition of a finite structure that preserves memory; from it the dynamic infinite is born.

25.3 Spectral Pell screw and logarithmic suspension

25.3.1 Spectral identity $V^{-1}\mathcal{S}V = (2 + \sqrt{3})i\mathcal{S}$ and equation of the logarithmic suspension

The corpus contains the exact identity

$$V^{-1}\mathcal{S}V = (2 + \sqrt{3})i\mathcal{S}. \quad (25.7)$$

Let $\varepsilon = 2 + \sqrt{3}$. The orbit transversal

$$z_n = z_0(\varepsilon i)^n$$

satisfies

$$r_n = r_0\varepsilon^n, \quad \theta_n = \theta_0 + \frac{\pi n}{2}.$$

Eliminating n ,

$$r(\theta) = r_0 \exp \left[\frac{2 \log(2 + \sqrt{3})}{\pi} (\theta - \theta_0) \right]. \quad (25.8)$$

The logarithmic spiral is an output of the operator: its equation was not chosen to fit ε afterward.

Derivation of equation (25.8). From $\theta_n - \theta_0 = n\pi/2$ it follows that $n = 2(\theta_n - \theta_0)/\pi$. Substituting into $r_n = r_0\varepsilon^n$,

$$r_n = r_0 \exp\left(\frac{2 \log \varepsilon}{\pi}(\theta_n - \theta_0)\right).$$

Continuous interpolation is the indicated equation. □

25.3.2 Conjugate contraction involution

The bilateral extension $n \in \mathbb{Z}$ produces

$$n \rightarrow +\infty : r_n \rightarrow \infty, \quad n \rightarrow -\infty : r_n \rightarrow 0.$$

The inversion $n \mapsto -n$ interchanges expansion and contraction. Since $\varepsilon^{-1} = 2 - \sqrt{3}$, both branches belong to the same Pell unit. They are not two independent spirals, but two orientations of the same spectral screw.

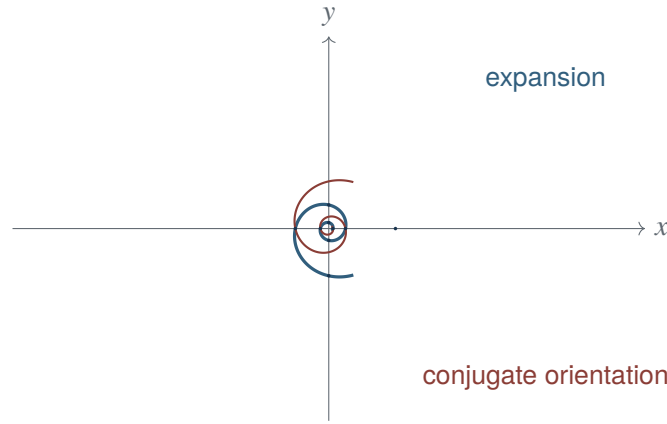


Figure 25.1: Transverse projection of the Pell screw. The marks represent quarter turns.

25.3.3 Scale covariance and energy conservation

The dilation ε must not be interpreted as gratuitous growth of the energetic amplitude of a wave in vacuum. Its coherent realization acts on the spatial or spectral scale:

$$(r, \varphi, \lambda) \mapsto (\varepsilon r, \varphi + \pi/2, \varepsilon \lambda).$$

This distinction will be essential in the electromagnetic realization of chapter devoted to the restricted electromagnetic realization.

Exact compatibility between the Pell spectral screw and the phase–memory suspension

The minimal phase–memory lift is a circular helix. The spectral screw combines a quarter–turn and a Pell dilation; its projection is a logarithmic spiral. Both arise from the HMT ledger, but they are neither the same periodicity nor obtained from one another by forgetting types.

Arithmetic theory of nonadic vacancies and boundary torsion

26.1 Classification of the irreducible arithmetic cycles of the nonadic radical lattice

In the arithmetic realization, a prime corresponds to an irreducible cycle that cannot be factored into smaller positive cycles. Nonadic periodicity may select candidates and organize residue classes; primality remains defined by the exact multiplicative property

$$p = ab \implies a = 1 \text{ or } b = 1.$$

Once the coproduct of prime arithmetic cycles has been selected, the corresponding identity are obtained. The helical structure and the digital root parametrize phase and carry, but do not replace the factorization criterion.

26.2 Density, regularized term, and vacancy holonomy

The basic positive density is

$$\varepsilon_{\text{vac}} = \log_{10} \frac{10}{9}. \quad (26.1)$$

The sequence is deduced from the completion $1000 = 729 + 271$ determines

$$\varepsilon_{\text{vac}} = 1 - \log_{1000}(729) = \log_{10} \frac{10}{9}, \quad (26.2)$$

and the exact mechanical word

$$z_n = \lfloor (n+1)\varepsilon_{\text{vac}} \rfloor - \lfloor n\varepsilon_{\text{vac}} \rfloor, \quad n \geq 1. \quad (26.3)$$

Since ε_{vac} is irrational, the word is Sturmian: it has complexity $p(N) = N + 1$, balance one and topological entropy topological zero. This dynamic differs from the full shift on the triadic alphabet triadic: the vacancy determines an ordered boundary of zero entropy.

The suma telescopica produces

$$\sum_{n=1}^N z_n = \lfloor (N+1)\varepsilon_{\text{vac}} \rfloor, \quad \sum_{n=1}^N (z_n - \varepsilon_{\text{vac}}) = \varepsilon_{\text{vac}} - \{(N+1)\varepsilon_{\text{vac}}\}. \quad (26.4)$$

The partial sums of the discrepancy have modulus less than one. This is the estimation supporting the subsequent harmonic constants.

The fraction continuous begins

$$\varepsilon_{\text{vac}} = [0; 21, 1, 5, 1, 6, 2, 3, \dots].$$

Thus the ones in z_n are separated by 21 or 22 steps. If

$$\sigma = 22 - \varepsilon_{\text{vac}}^{-1} = 1 - T_G(\varepsilon_{\text{vac}}),$$

then $a_1(\sigma) = 6$ and the returns of the short separation have length 6 or 7. The pairs 21/22 and 6/7 are two scales of the same rotation, not coincidences of integers.

For the exact word in equation (26.3), the regularized constant is

$$\gamma_{\text{vac}}^+ = \lim_{N \rightarrow \infty} \left(\sum_{n \leq N} \frac{z_n}{n} - \varepsilon_{\text{vac}} \log N \right). \quad (26.5)$$

The associated holonomy is

$$\rho_{\varepsilon}(k) = \exp(2\pi i k \varepsilon_{\text{vac}}). \quad (26.6)$$

The three quantities have distinct types: density, finite term, and phase. The vacancy Stieltjes tower prolongs equation (26.5):

$$\gamma_{\text{vac},j}^+ = \varepsilon_{\text{vac}} \gamma_j + \sum_{n \geq 1} \frac{(z_n - \varepsilon_{\text{vac}})(\log n)^j}{n}. \quad (26.7)$$

For $j = 0$, one recovers γ_{vac}^+ . Convergence requires the discrepancy bound for the integral development of vacancies.

Convergence is proved here. For $j \geq 0$, the sequence $b_n = (\log n)^j / n$ tends to zero and is decreasing from $n > e^j$ onward. The uniform bound equation (26.4) and Dirichlet's criterion prove the convergence of

$$\sum_{n \geq 1} (z_n - \varepsilon_{\text{vac}}) b_n.$$

For $j = 0$, separating $\varepsilon_{\text{vac}} \sum_{n \leq N} n^{-1}$ gives exactly equation (26.5). Abel summation conservatively bounds the tail after N by $2/(N+1)$: one term comes from the boundary and another from the total variation of $1/n$. The declared exhaustive execution uses $N = 10^8$, enumerates the Beatty positions of the ones, checks the gap sets $\{21, 22\}$ and returns $\{6, 7\}$, and obtains as its finite part

$$-0.11207107505876366224207105242526397929 \dots$$

Combining the analytic bound with the recorded rounding and boundary guards, the finite certificate encloses

$$-0.112071094143613634 < \gamma_{\text{vac}}^+ < -0.112071055516338732. \quad (26.8)$$

This interval is a terminal value with controlled error. It does not define the word; the word and its discrepancy precede it.

A second evaluation using multiprecision arithmetic and directed rounding control provides, to eighteen decimal places, the certified interval

$$-0.112071086058763562 < \gamma_{\text{vac}}^+ < -0.112071064058763762,$$

contained strictly within equation (26.8). The inclusion of both intervals verifies the stability of the finite part with respect to truncation, the boundary guards, and the precision regime. The result depends solely on the recurrences, the tail bounds, and the interval arithmetic set out in this chapter.

26.3 Operative content of the helicoidal representation

The helical representation encodes a periodic phase together with a memory increment. It has operatorial content when there exists a cocycle

$$c(\gamma\eta) = c(\gamma) + c(\eta)$$

that transports the carry and whose monodromy is interlinked with the arithmetic operator. In the absence of that map, the helix constitutes only a phase parametrization; with the interlinker, it distinguishes phase return from state return.

Remark 26.1 (Negative control). Cofinality of cylinders does not imply dynamical conjugacy. A counterexample to commutation between the TPK shift and T_G precludes the intertwiner, but does not affect the corresponding identity. In the vacancy sector, the divergence of [equation \(26.7\)](#) under the declared bound refutes the family of regularized constants.

Logical Scope and Domain of Validity

Cofinality, Gauss identities, Levy, entropy, and Lochs: *Exact internal theorem in the transported system*. GKW and dynamic intertwiner: *Open program with falsification criterion*. Density, constant, and vacancy holonomy: *Exact internal theorem with discrepancy control*.

26.4 Classification of the radical network $3 \rightarrow 6 \rightarrow 9$ and derivation of the power–vacancy law

The appearance of 369 and 693 is not reducible to a graphic similarity. There are two exact mechanisms that converge:

1. **nonadic algebraic mechanism:** 3, 6, 9 are the visual representatives of the nilpotent radical $\mathfrak{m} = (3) \subset \mathbb{Z}/9\mathbb{Z}$; the words 369, 693, and 936 are the three phases of its multiplicative orbit;
2. **Archimedean vacancy mechanism:** the difference between the decimal length of 9^n and the boundary 10^n is governed exactly by the same irrational rotation $\delta_{\text{vac}} = \log_{10}(10/9)$ that already organizes the vacancy calendar.

For $n = 9^9$, both mechanisms meet in the identity

$$9^9 - \left\lceil 9^9 \log_{10} \frac{10}{9} \right\rceil = 369\,693\,099,$$

so that

$$\# \text{cifras}(9^{9^9}) = 369\,693\,100.$$

The block 369 | 693 is, therefore, an exact output of the logarithmic reader of the nonadic power; it is not the decimal prefix of 9^{9^9} . The final unit proceeds from the universal operator that transforms decimal order into decimal length.

This identity opens a broader structural reading: addition, product, power, subtraction, division, root, and logarithm can be organized as a hierarchy of transports and quotients. Each ascent of operation compactifies histories; TPK preserves the coordinates that make their reconstruction possible.

26.5 Radical structure of the APP on $\mathbb{Z}/9\mathbb{Z}$

26.5.1 Nonadic local ring $\mathbb{Z}/9\mathbb{Z}$ and nilpotent radical

Let

$$R = \mathbb{Z}/9\mathbb{Z}.$$

Its group of units and its maximal ideal are

$$R^\times = \{1, 2, 4, 5, 7, 8\}, \quad \mathfrak{m} = (3) = \{0, 3, 6\} = \{9, 3, 6\}_{\text{vis}}.$$

Moreover,

$$\mathfrak{m}^2 = 0.$$

The preceding equality is an equality of ideals: every product of two elements of $\mathfrak{m}$ vanishes modulo nine. It must not be confused with the Cartesian product $\mathfrak{m} \times \mathfrak{m}$, which contains nine APP positions.

26.5.2 Exact sequence of the multiplier 3

Multiplication by three,

$$M_3 : R \longrightarrow R, \quad x \longmapsto 3x,$$

has

$$\ker M_3 = \mathfrak{m}, \quad \operatorname{im} M_3 = \mathfrak{m}.$$

It therefore induces an isomorphism of spaces over $\mathbb{F}_3$,

$$\overline{M}_3 : R/\mathfrak{m} \xrightarrow{\sim} \mathfrak{m}, \quad \bar{x} \longmapsto 3x.$$

In visual representatives,

$$\bar{0} \longmapsto 9, \quad \bar{1} \longmapsto 3, \quad \bar{2} \longmapsto 6.$$

Thus, the band 3, 6, 9 is not a decimal ornament. It is an internal copy of the ternary quotient lodged in the nonadic radical.

26.5.3 Cyclic orbit $369 \rightarrow 693 \rightarrow 936 \rightarrow 369$

The row of the multiplication table corresponding to 3 is

$$3, 6, 9, 3, 6, 9, 3, 6, 9.$$

The cambio of phase $j \mapsto j + 1$ produces

$$369 \longmapsto 693 \longmapsto 936 \longmapsto 369.$$

These three words are the three cyclic readings of the application.

$$\mathbb{F}_3 \longrightarrow \mathfrak{m}, \quad \bar{j} \longmapsto 3j.$$

The row 6, by satisfying $6 \equiv -3 \pmod{9}$, reverses the orientation of the same cycle. The mark 9 fulfills two typed functions:

$$9 \bmod 9 = 0, \quad \operatorname{dr}_9(9) = 9.$$

The module publishes the null class; the positive digital root preserves the mark of closure. This difference of readers is precisely what allows 9 to be at once residual zero and visible closure.

26.5.4 Radical cross of APP and coordinate reduction to $\mathbb{F}_3^2$

On

$$X_9 = R^2,$$

the union

$$\mathcal{C}_{369} = (\mathfrak{m} \times R) \cup (R \times \mathfrak{m})$$

is the cross formed by rows and columns 3, 6, 9. It contains

$$3 \cdot 9 + 3 \cdot 9 - 3 \cdot 3 = 45$$

positions. Their central intersection $\mathfrak{m} \times \mathfrak{m}$ contains nine positions. In the multiplicative observable, the sum of the cross is

$$297 = 216 + 81 = 3(72 + 27).$$

The coordinate reduction produces

$$X_9/(\mathfrak{m} \times \mathfrak{m}) \cong \mathbb{F}_3^2$$

as a quotient set typed by the classes modulo three. The inverse lift requires preserving the coordinate within each three-element fiber.

26.5.5 Determination of the arithmetic core 7227 by incidence with the radical cross of APP

The central block of the APP is

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 7I + 2R = 9P_+ + 5P_-.$$

Its rowwise reading produces $72 \mid 27$; its diagonal reading produces $77 \mid 22$. One verifies

$$72 + 27 = 77 + 22 = 99,$$

$$72 - 27 = 45 = \det B_c,$$

$$77 - 22 = 55 = 54 + 1.$$

The relation with the radical cross is not nominal:

$$3 \cdot 72 = 216, \quad 3 \cdot 27 = 81, \quad 3 \cdot 99 = 297.$$

The central unitary block and the non-unitary network 3, 6, 9 are thereby linked by an exact local–global identity. The hierarchy of the sum–product difference preserves its types:

$$4 \quad (\text{local spectral jump}), \quad 2 \quad (\text{off-diagonal coupling}),$$

$$6 \quad (\text{compressed difference modulo three}), \quad 54 \quad (\text{two-dimensional unfolding}).$$

26.6 Power–vacancy law for the scales 9^n and 10^n

26.6.1 Decimal Length Functions and Accumulated Vacancy

For $n \geq 1$, let

$$D_9(n) = \# \text{cifras}(9^n) = \lfloor n \log_{10} 9 \rfloor + 1.$$

The decimal boundary 10^n has $n + 1$ digits. We define the accumulated vacancy of the nonadic power by

$$V_9(n) = (n + 1) - D_9(n).$$

We introduce the two complementary numbers

$$\rho_{\text{vac}} = \log_{10} 9, \quad \delta_{\text{vac}} = 1 - \rho_{\text{vac}} = \log_{10} \frac{10}{9}.$$

26.6.2 Exact identity between decimal length and vacancy rotation

Theorem. For every integer $n \geq 1$,

$$V_9(n) = \lceil n\delta_{\text{vac}} \rceil,$$

and, therefore,

$$D_9(n) = n - \lceil n\delta_{\text{vac}} \rceil + 1.$$

Proof. Since δ_{vac} is irrational, $n\delta_{\text{vac}} \notin \mathbb{Z}$. Then

$$\begin{aligned} D_9(n) &= \lfloor n(1 - \delta_{\text{vac}}) \rfloor + 1 \\ &= \lfloor n - n\delta_{\text{vac}} \rfloor + 1 \\ &= n - \lceil n\delta_{\text{vac}} \rceil + 1. \end{aligned}$$

Subtracting from $n + 1$ yields the first identity. $\square$

26.6.3 Sturmian coding of vacancies induced by nonadic powers

The increments of the vacancy are

$$Z_n = V_9(n + 1) - V_9(n) = \lceil (n + 1)\delta_{\text{vac}} \rceil - \lceil n\delta_{\text{vac}} \rceil \in \{0, 1\}.$$

This is exactly the mechanical word of slope δ_{vac} used by the HMT vacancy calendar. Therefore, that calendar admits an additional, completely exact reading:

$Z_n = 1$ if and only if, in passing from 9^n to 9^{n+1} , a new decimal range is lost relative to the boundary $10^n \rightarrow 10^{n+1}$.

The power does not add a coincidence to the theory of vacancies: it provides its natural realization as a counter of decimal rank loss.

26.6.4 Law 21/22 as a case of the power–vacancy theorem

Since

$$\delta_{\text{vac}}^{-1} = 21.85434532678283256 \dots,$$

the indices for which $Z_n = 1$ are separated by 21 or 22. The first twenty-one exponents satisfy

$$D_9(n) = n, \quad 1 \leq n \leq 21,$$

whereas

$$D_9(22) = 21.$$

In particular,

$$9^9 = 387\,420\,489$$

has exactly nine digits. This initial preservation of rank is not an indefinite rule: the vacancy calendar determines exactly when it ceases to hold.

26.6.5 Exact evaluation at $n = 9^9$ and length of 9^{9^9}

Let

$$n_* = 9^9 = 387\,420\,489.$$

Then

$$n_* \delta_{\text{vac}} = 17\,727\,389.3684296412564569 \dots,$$

and, therefore,

$$\boxed{V_9(n_*) = 17\,727\,390}.$$

The conserved coordinate after discounting vacancies is

$$n_* - V_9(n_*) = 387\,420\,489 - 17\,727\,390 = 369\,693\,099.$$

Therefore,

$$\boxed{\text{ord}_{10}(9^{9^9}) = 369\,693\,099},$$

$$\boxed{D_9(9^9) = 369\,693\,100}.$$

Furthermore, the congruences are obtained.

$$9^9 \equiv 0 \pmod{9}, \quad V_9(9^9) \equiv 0 \pmod{9},$$

$$369\,693\,099 \equiv 0 \pmod{9}, \quad 369\,693\,100 \equiv 1 \pmod{9}.$$

The corresponding digit sums are

$$s_{10}(9^9) = 45, \quad s_{10}(V_9(9^9)) = 36,$$

$$s_{10}(369\,693\,099) = 54, \quad s_{10}(369\,693\,100) = 37.$$

The return $9 \rightarrow 1$ appears here with precise types:

1. 9^9 and the accumulated vacancy belong to the null class modulo nine;
2. its difference, the decimal order, remains in the null class and has digit sum 54;
3. the universal operator $D = \text{ord} + 1$ adds the closure unit and publishes the positive class 1.

26.6.6 Domain of validity of the power–vacancy law

It is proved:

- the theorem potencia–vacancia for every n ;
- the identity with the mechanical vacancy word;
- the exact evaluation in $n = 9^9$;
- the decimal order and length of 9^9 ;
- the preceding congruences and digit sums.

It is not claimed that 9 is the only basis with an analogous property in every domain, that each external occurrence of 369 arises from this tower, or that the tower replaces the construction of the nine phases or the generation certificate for π, e, φ .

26.7 Operator Hierarchy, Reversibility, and Genealogical Memory

26.7.1 Translation, Dilation, and Exponentiation as Operational Levels

Elementary operations can be seen as actions:

$$T_a(x) = x + a, \quad M_a(x) = ax, \quad P_k(x) = x^k.$$

Their partial inverses are, respectively, subtraction, division, and extraction of roots. The passage from one operation to the next modifies the structure of the fibers.

26.7.2 Stratified reversibility in $\mathbb{Z}/9\mathbb{Z}$

proposition. En the anillo nonadico:

1. every translation T_a is bijective;
2. M_a is bijective if and only if $a \in R^\times$;
3. if $a \in \mathfrak{m}$, multiplication has nontrivial fibers and its image lies within $\mathfrak{m}$;
4. for $x \in \mathfrak{m}$, $x^k = 0$ for every $k \geq 2$;
5. over $R^\times \cong C_6$, the map P_k is an automorphism if and only if $\gcd(k, 6) = 1$.

Proof. The first assertion uses the additive group structure. The second is the characterization of the units. The third follows from $aR \subseteq \mathfrak{m}$. The fourth follows from $\mathfrak{m}^2 = 0$. The fifth is the condition for multiplication by k to be an automorphism of the cyclic group of order six. $\square$

The result produces an exact hierarchy:

sum: global reversibility,

product: reversibility restricted to units,

power: cyclic dynamics on units and nilpotent collapse in $\mathfrak{m}$.

Roots are functions only when the corresponding power is injective on the chosen domain; on the total domain they are multibranch correspondences or empty relations.

26.7.3 Reading of the Topological Phase Kernel on the operator hierarchy

This stratification suggests a functional definition:

TPK does not add an external operation to sum, product, and power. It preserves the sheet, quotient, carry, orientation, route, and boundary that each ordinary evaluation eliminates when identifying several histories with the same output.

Sum and product are the first two evaluations on the APP. A power is repeated composition of the product; a root selects or reconstructs branches of that composition. The record distinguishes the published value from the operatorial history producing it.

26.7.4 Logarithmic descent between consecutive levels of the hierarchy

In a positive domain,

$$\log(xy) = \log x + \log y, \quad \log(x^k) = k \log x, \quad \log(a^x) = x \log a.$$

The logarithm converts product into sum, power into product, and exponential into dilation. The defect appears when the subsequent reader discretizes:

$$\mathcal{L}_{10}(a^n) = \lfloor n \log_{10} a \rfloor + 1.$$

The integer part introduces the boundary-crossing calendar. For $a = 9$, that calendar is precisely the vacancy word.

26.7.5 Affine conjugation of translations and dilations

Translations and dilations satisfy

$$M_a T_b M_a^{-1} = T_{ab}$$

when a is invertible. This identity expresses that the product reorganizes the units of addition. When a ceases to be invertible, conjugation breaks because branches disappear. The radical 3, 6, 9 localizes precisely that failure.

26.8 Path Algebra: Multiplicative Composition, Additive Action and Superposition

In a path algebra, addition aggregates alternatives and multiplication concatenates trajectories. For local weights $w(e)$,

$$K(x, y) = \sum_{\gamma: x \rightarrow y} \prod_{e \in \gamma} w(e).$$

If a coherent logarithmic branch is fixed,

$$\prod_{e \in \gamma} w(e) = \exp \left(\sum_{e \in \gamma} \log w(e) \right),$$

and, with the appropriate discrete action,

$$K(x, y) = \sum_{\gamma: x \rightarrow y} e^{-S(\gamma)/\hbar}.$$

The equation combines the operations:

- the product composes successive steps;
- the logarithm accumulates the local action;
- the exponential reconstructs the route weight;
- the suma agrega all the alternativas.

APP is the minimal support on which sum and product act on the same histories. TPK adds the data that prevent the sum over paths from becoming a sum over values without genealogy.

In this framework, a digit is the output of a reader; a value is the Archimedean projection of a section; an HMT number preserves the compatible family of paths that supports that value; a particle preserves that genealogy under a persistent spectral realization.

26.9 Triadic exponential family $J_\tau^2 = -\tau I$: elliptic, parabolic, and hyperbolic realizations of Euler's identity

26.9.1 Internal root of -1 in the finite branch

The Paley incidence matrix provides an operator A_W with

$$A_W^2 = -I.$$

This turns $\mathbb{F}_3^6$ into a three-dimensional space over

$$\mathbb{F}_9 = \mathbb{F}_3[\iota]/(\iota^2 + 1).$$

The root of -1 is not adjoined from outside as complex notation. It emerges as an endomorphism of an already constructed ternary support.

26.9.2 Continuous extension of the common quadratic operator and conservation of its spectral type

In the elliptic real branch, the generator J_e satisfies

$$J_e^2 = -I,$$

and

$$e^{tJ_e} = \cos t I + \sin t J_e.$$

In particular,

$$e^{\frac{\pi}{2}J_e} = J_e, \quad e^{\pi J_e} = -I, \quad e^{2\pi J_e} = I.$$

Thus, J_e is the internal quarter-turn and Euler's equation is its half-turn closure.

26.9.3 Conjugation of the arithmetic, angular, and torsional realizations via the common quadratic operator

Let

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}.$$

This same set contains the real diameter $(-1, 1)$, the complex interior, the trigonometric boundary S^1 , and the Poincaré hyperbolic metric

$$ds_P^2 = \frac{4\ell^2 |dz|^2}{(1 - |z|^2)^2}.$$

The map

$$R_{\pi/2}(z) = iz$$

preserves $|z|$ and $|dz|$. By consiguiente, is simultaneamente:

- complex multiplication by $\sqrt{-1}$;
- trigonometric rotation of 90° ;
- Euclidean isometry of the disk;
- Poincaré hyperbolic isometry;
- automorphism of its circular boundary.

The coincidence is literal: the same map acts on the same carrier; the metric reader changes.

26.9.4 Elliptic, parabolic, and hyperbolic geometries of the extended Euler equation

The generators

$$J_{+1}^2 = -I, \quad J_0^2 = 0, \quad J_{-1}^2 = I$$

produce

$$e^{tJ_{+1}} = \cos t I + \sin t J_{+1},$$

$$e^{tJ_0} = I + tJ_0,$$

$$e^{tJ_{-1}} = \cosh t I + \sinh t J_{-1}.$$

The same exponential series separates into three regimes according to the square of the generator: elliptic, parabolic, and hyperbolic. The specialization

$$e^{2 \log \varphi J_{-1}} = F_{\text{av}}^2$$

integrates the Fibonacci self-scale. Euler and Fibonacci appear as specializations of a common quadratic exponential family.

26.9.5 Common carrier, common operator, and quadratic exponential family

Geometric fusion is formulated at three levels:

1. **common carrier:** the unit disk contains the real diameter, the complex interior, and the circular boundary;
2. **common operator:** $z \mapsto iz$ is a quarter-turn, complex multiplication, and a hyperbolic isometry;
3. **common family:** the tritic exponential equation brings together the elliptic, parabolic, and hyperbolic branches.

The result does not identify the Euclidean and hyperbolic metrics. It states that they are distinct realizations on the same support and share the automorphism originating in the internal root of $-I$.

26.10 Restriction of internal rotation, edge holonomy and minimal torsion

The construction contains two positive and differentiated objects:

1. the internal generator J_e , with $J_e^2 = -I$;
2. the global invariant

$$\Delta_4 = \frac{\pi - e \log \pi}{270} \left(1 + \frac{\pi}{729} \right),$$

denoted minimal torsion and used prior to species selection.

The authorial thesis adds that internal rotational dynamics, when published at the boundary, is absorbed as minimal torsion. The relation must not be written as a bare equality $J_e = \Delta_4$: the objects have different types. Its natural form requires

$$\text{interior rotation} \xrightarrow{\text{Res}_\partial} \text{boundary holonomy} \xrightarrow{\text{Def}} \text{scalar defect}.$$

Let q_∂ be the boundary reader, ∇_∂ the induced connection, and e_∂ the discrete frame. The edge torsion has type

$$T_\partial = d_{\nabla_\partial} e_\partial.$$

The author's claim acquires probative form through a functional

$$\Lambda_\partial : \text{Tors}(\partial X) \longrightarrow \mathbb{R}$$

such that

$$\Lambda_{\partial}(T_{\partial}) = \Delta_4$$

and a theorem deriving T_{∂} from the restriction of J_e through the boundary operators of the TPK.

The internal root of $-I$, the quarter-turn, the extended Euler equation, and the formula for Δ_4 are proved in their respective domains. The passage from interior rotation to minimal boundary torsion requires the declared morphism Res_{∂} and functional Λ_{∂} ; it is not identified by a scalar equality.

26.11 Generational triplicity, conjugation and mixing on the integral holonomic state

The phrase of Isidor Isaac Rabi may serve as a historical reference, but it must not govern the title or constitute an autonomous section. The scientific question is:

What internal mechanism distinguishes three stable orders of fermionic realization and preserves that distinction under color, conjugation, and mixing?

The HMT response is not "because TRIT has three values". The structure separates

$$\text{flav}(X) = (\sigma_{ud}(X), g(X), \chi_{\text{fina}}(X)),$$

where σ_{ud} distinguishes branches, g distinguishes physical generations, and χ_{fina} preserves orientation and memory within a branch. For quarks, the color orbit is added after flavor and before the character.

Triplicity is obtained as a consequence of the enriched structure; its effects on color, conjugation, and mixing are developed below:

1. colour, flavour, generation and mass are distinct readers of the same integral holonomic state;
2. particle–antiparticle conjugation acts on the orientation without erasing the generational class;
3. CKM and PMNS transport already typed state bases; they do not create triplicity;
4. the neutrino classification must be derived from representation, occupation, and mixing, not by renaming depths $G0, \dots, G4$.

That neutrinos are half-integer-spin leptons obeying fermionic statistics is a known physical classification. The structural contribution of HMT consists in obtaining internally why the neutral sector occupies that representation, why it admits three physical realizations, and how it couples to the mixing operator without reading its external angles.

26.12 Operator memory of the TPK: preservation of residue, quotient, carry, sheet, orientation, route, and boundary under projection

26.12.1 Loss of reversibility and genealogical lifting

In APP, different histories may produce the same residue. The product folds more histories than the sum; the power folds more than the product; the root attempts to reconstruct branches already identified. TPK preserves the datum that makes it possible to invert that loss:

residue, quotient, carry, sheet, orientation, route, boundary, memory.

An algebraic operation may be characterized not only by its value, but also by the type of genealogy that it preserves or destroys.

26.12.2 Projection of Compatible Histories onto the Archimedean Continuum

The real line appears when a family of compatible cylinders contracts under Archimedean evaluation. The real value is the shadow of the section, not its complete genealogy. The exceptional projection preserves another part as incidence and symmetry.

The path integral expresses the same principle:

$$\text{alternatives } \xrightarrow{\Sigma} \text{global amplitude,} \quad \text{successive stages } \xrightarrow{\Pi} \text{route weight.}$$

The logarithm converts multiplicative composition into additive action; the exponential returns it to amplitude. The discrete structure of the continuum can be formulated as genealogical conservation of all routes before their projection onto $\mathbb{R}$.

26.12.3 Distinction between numerical section and persistent realization of particle

A number and a particle share minimal ontological invariants: memory, orientation, persistence, closure, signature, and position in a family of prolongations.

The number publishes an Archimedean coordinate. The particle adds spectral realization, occupation, propagation, and stability. The passage $\text{HMT} \rightarrow \text{MD}$ changes the reader applied to the same enriched genealogy; it does not arbitrarily change the object.

26.13 Joint closure of the radical network, the power–vacancy law, Euler geometry, and the operative hierarchy

26.13.1 Theorem of the Nonadic Radical Network

The APP net 3, 6, 9 is the visual representation of the nilpotent radical $(3) \subset \mathbb{Z}/9\mathbb{Z}$. Its phases 369, 693, and 936 realize the isomorphism $R/(3) \cong (3)$ induced by multiplication by three. The bands locate exactly the failure of invertibility of the product.

26.13.2 Power-void Theorem

For every $n \geq 1$, the number of decimal ranks lost by 9^n relative to 10^n is $\lceil n \log_{10}(10/9) \rceil$. Their increments form the mechanical vacancy word. In $n = 9^9$, the retained coordinate is 369 693 099 and the resulting length is 369 693 100.

26.13.3 Exponential Family Euler’s Geometric Theorem

The internal root of $-I$ produces a quarter-turn. On the unit disk, the same map $z \mapsto iz$ is simultaneously complex multiplication, trigonometric rotation, and a Poincaré isometry. The tritic exponential equation extends this regime to the parabolic and hyperbolic regimes, incorporating the Fibonacci self-scaling in the latter.

26.13.4 Operatorial conservation principle between levels

The sum–product–power hierarchy is a hierarchy of fibers and loss of reversibility. The TPK is the lift that preserves the coordinates required to distinguish histories identified by the ordinary readers.

26.14 Typing conditions and incompatibility criteria of the nonadic vacancy theory

1. Do not identify 369 693 100 with a TPK word without declaring the reader between domains.
2. No confundir the ideal $\mathfrak{m}^2 = 0$ With $\mathfrak{m} \times \mathfrak{m}$.
3. Do not confuse modulo nine with digital root: $9 \mapsto 0$ in the former and $9 \mapsto 9$ in the latter.
4. Do not use the threefold character of TRIT as a sufficient proof of three physical generations.
5. Do not turn $J^2 = -I$ into Δ_4 without the boundary map and the torsion functional.

6. Do not present 369 as a prefix of 9^{9^9} : it pertains to its order and its decimal length.
7. Do not reduce the ten thousand digits to an order-of-magnitude test: they certify extensive truncations of a parametrizable process whose thesis is indefinite prolongation.

26.15 Joint consequences for the radical net, the operational memory, and the edge torsion

The principal finding is not that a tower visually contains the digits 3, 6, 9. The nonadic power and the vacancy calendar are two presentations of the same logarithmic defect, while APP localizes 3, 6, 9 as the radical where multiplication loses invertibility. The tower connects both extremes:

nonadic radical $\longrightarrow$ power $\longrightarrow$ logarithm $\longrightarrow$ vacancy $\longrightarrow$ decimal order $\longrightarrow$ closure $9 \rightarrow 1$.

The extended Euler equation adds the second direction: the internal root of $-I$ turns the discrete structure into rotational dynamics and unites, on a common support, the real line, trigonometric circle and hyperbolic disk. Subsequent passage to minimal torsion must preserve the boundary map already developed in the corpus, not replace it with a nominal identification.

Radion, angular phase and equivalence between cyclic and linear realizations

27.1 Composition APP–TRIT–TPK of the phase state: generation of the memory-oriented unit

Mathematical Synthesis of the Object and Its Causal Chain This monograph reconstructs the radian from the generative chain

$$\text{APP} \longrightarrow \widehat{\mathbb{T}} \longrightarrow \text{TPK} \longrightarrow X_{\text{enr}} \longrightarrow \text{discrete structure of the continuum} \longrightarrow \text{MD},$$

without taking as input either the metrological value of $180/\pi$ or the small angular residue to be explained. The result is an enriched unit, the *radion*, defined as the oriented unit that traverses one radian of phase and sweeps exactly an action $\hbar_{\text{ret}}$ in the positive ellipse associated with the same state.

The governing closure is

$$\boxed{\frac{180^\circ}{\pi} = 50^\circ + A^\circ - \frac{20}{81} \Delta_4^\circ + \varepsilon_R^\circ}$$

with

$$S_E = 2\pi \left(\frac{5}{36} + \frac{25}{9} \alpha \right), \quad \Delta_4 = \frac{\pi - e \log \pi}{270} \left(1 + \frac{\pi}{729} \right),$$

$$\rho_{\text{tw}} = \frac{2 \cdot 90}{729} = \frac{20}{81}, \quad \varepsilon_R^{(1)} = \frac{20}{81} \Delta_4 - (S_E - 1).$$

APP division proves $1 < S_E < 2$, so that the quotient is exactly one and the remainder is $D_E = S_E - 1$. The equality

$$1 = S_E - \frac{20}{81} \Delta_4 + \varepsilon_R^{(1)}$$

is then an internal identity. The angular chart produces

$$\varepsilon_R^\circ = 5.13830167585752820716851646226459474899085744 \dots \times 10^{-8}^\circ.$$

The text also demonstrates that the action ellipse

$$Q = a_S \cos \theta, \quad P = b_S \sin \theta, \quad a_S b_S = 2\hbar_{\text{ret}}$$

satisfies

$$\text{Area}(0, \theta) = \hbar_{\text{ret}} \theta.$$

Therefore, a radion sweeps $\hbar_{\text{ret}}$ and the turn 2π encloses $h_{\text{ret}} = 2\pi\hbar_{\text{ret}}$. The same ellipse has a finite symplectic boundary and an interior of infinite area when equipped with the Hilbert–Beltrami–Klein metric. This is the exact form of the physical thesis guiding the work: the visible surface closes the action; the interior depth preserves a development that cannot be exhausted by the planar projection.

The exposition maintains typed four realizations of the same state:

1. the circle S^1 , which publishes the phase;
2. the positive ellipse, which publishes the action and the anisotropy;
3. the interior of Klein, that miof the depth by means of reason doble;
4. the nonadic helix, which preserves return, memory, and contraction.

These are not four geometries superimposed without a criterion. They are four readers with different domains, codomains and preserved data, applied to a single integral holonomic state. The sector $90/120$,

the dodecaphase, radion carry, the Lambert tail, the Barbero kernel and global towers are likewise presented as related but noninterchangeable objects.

Tesis rectora

The radian is the oriented unit that sweeps $\hbar_{\text{ret}}$ in the action ellipse. The circle publishes its phase, Klein measures its depth, the helix preserves its memory, and the coinductive carry exactly splices its direct and torsional sections.

Expository order and reading dependencies The order is not ornamental. Parts I and II produce the object and its closure from APP–TRIT–TPK. Only thereafter are realizations introduced in the languages of Hilbert, LC, Maxwell, Minkowski, Hodge, Lorentz, modularity, and orbits. Each chapter distinguishes three levels:

- | | |
|------------------------------------|--|
| Internal result. | Equality or construction proved in the HMT domain with its declared primitives. |
| Consolidated formalization. | A new map articulating already existing results without using the target as a selector. |
| Physical interface. | Subsequent dimensional translation that recognizes a conventional structure without making it the origin of the selection. |

The *status and falsifier* boxes indicate which concrete remark would invalidate each strong promotion. This discipline does not weaken the thesis; it makes the thesis auditable. The purpose of the extended narrative is to ensure that no word such as *circle*, *ellipse*, *torsion*, *charge*, or *memory* conceals a change of type.

Pivotal results and proof scope

- C1.** Closure of the radion independent of its final value through two publications of the same state and APP subtraction with carry.
- C2.** Derivation of the coefficient 20/81 as two sheets of the active sector 90, normalized by the fiber $3^6 = 729$.
- C3.** Typed equalizer between the seam $\mathfrak{R}s = 1/2$, the hundred-mark APP face, and the dodecaphasic phase $\theta_2 = 50^\circ$.
- C4.** Radian area theorem: $\text{Area}(\theta) = \hbar_{\text{ret}}\theta$.
- C5.** Exact focal study of the ellipse and of its Euclidean, hyperbolic, symplectic, and modular invariants.
- C6.** Boundary–interior theorem: finite total action h and Klein interior of infinite hyperbolic area.
- C7.** Exact realization through minimal covariance and an LC oscillator, followed by the electromagnetic publication 90/120.
- C8.** Atlas of non-identifications for all objects 90/120, including the separation among ε_R , the spectral tail, and the Barbero kernel $12 : -1$.
- C9.** Structural characterization of π in the radion chart, distinct from the primary coinductive generator.
- C10.** Ontological synthesis: dimension as the minimal rank of memories that a phase projection can no longer preserve.

Correspondence between results, causal types, and causal residences The following correspondence orders the governing assertions and their proofs, while keeping visible the hierarchy that prevents physical richness from erasing the mathematical closure supporting it.

APP-TRIT-TPK genealogy of the radion

The radion arises from the effective composition $\text{APP} \rightarrow \widehat{\mathbb{T}} \rightarrow \text{TPK} \rightarrow X_{\text{enr}} \rightarrow \text{MD}$; it is not obtained by retroactively redefining the conventional radian.

Exact closure and prospective independence

APP carry internally produces

$$1 = S_E - \frac{20}{81}\Delta_4 + \varepsilon_R^{(1)}, \quad \varepsilon_R^{(1)} = \frac{20}{81}\Delta_4 - (S_E - 1),$$

and the angular chart transports this same identity to $180^\circ/\pi$ without introducing the target residue.

Critical coordinate of 50°

The critical coordinate $\mathfrak{R}s = 1/2$, published on the APP face of one hundred marks and in the dodecaphase wheel, selects the phase $\theta_2 = 50^\circ$. It is an equality of typed readers of the same state.

Nonadic Bidirectional Density 20/81

The torsional coefficient is not fitted on the lattice $1/81$: it is constructed as $\rho_{\text{tw}} = 2N_6/3^6 = 2 \cdot 90/729 = 20/81$, preserving the sheet, orientation, and normalization of the nonadic fibre.

Areal law of the radion

For the positive ellipse $Q = a_S \cos \theta$, $P = b_S \sin \theta$, $a_S b_S = 2\hbar_{\text{ret}}$, the oriented area swept is $\mathcal{A}(\theta) = \hbar_{\text{ret}}\theta$. One radion sweeps exactly $\hbar_{\text{ret}}$, and the revolution 2π encloses $h_{\text{ret}} = 2\pi\hbar_{\text{ret}}$.

Focal invariants of the positive operator

The published pair $A \pm C^*$ fixes the semiaxes, eccentricity, focal separation, and symplectic compression parameter. In particular, $(d_1/d_0)^2 = C^*/2$, a dimensionless identity separating shape from scale.

Symplectic finiteness of the boundary and divergence of the interior hyperbolic area

The symplectic boundary of the ellipse encloses the finite action h_{ret} , whereas the Hilbert–Beltrami–Klein area of the interior diverges upon reaching the boundary. Visible phase and depth are not the same measure.

Nonadic Helical Lift with Phase Return and Memory Advance

The nonadic monodromy returns the phase without resetting the state. The helix lifts the circle while preserving carry and memory; the closure $2\pi/4\pi$ distinguishes projective return, reversal of orientation, and complete return.

Oscillatory and constitutive electromagnetic realization

The same anisotropy is realized through a minimal covariance and an LC oscillator: one coordinate stores the electric–elastic aspect and its conjugate the magnetic–rotational aspect. Frequency and impedance remain separately typed.

Typed separation of the operators 90/120

Dodecaphase extractor, radion carry, Lambert tail, hexad–octad functional, Barbero kernel, and global tower semigroup are distinct objects. The weight $12 : -1$ is forced only upon fixing the pole $1/24$ and the minimal integer counterterm.

Complex, Lorentzian, and Hodge-dual realizations

The elliptic, parabolic, and hyperbolic regimes of TRIT are published as rotation, threshold, and hyperbolic transformation. The Hodge star, the constitutive relations, and the Lorentz force act subsequently on the already oriented state; they do not select its constants.

Modular, orbital, and arithmetic boundary characters

The modular involution, elliptic orbits, advance–delay, and 3–6–9 arithmetic recognize distinct memories of the same state. The modular function and vacancies are used as posterior readers, never as substitutes for the APP–TRIT–TPK generator.

Operatorial synthesis of phase, action, depth, and memory

The circle is the phase projection; the ellipse is the action section; Klein measures depth; the helix preserves history. The radion is the oriented unit that keeps these publications coupled without erasing the residue that distinguishes them.

Delimitation of domains and causal continuity

The complete mass laws, electron refinement, CKM/CP, and the exceptional corridor retain their own governing proofs. This monograph cites them only where they share an antecedent or chart; it neither transfers their specific selectors to the radion nor confuses their probative residences.

27.1.1 Bivariate APP support, ternary TRIT orientation and updating of the integral holonomic state through TPK

The operative genealogy precedes angular measurement and determines the variables preserved by each reader.

27.1.1.1 APP–TRIT–TPK support and construction of the integral holonomic state

Priority of the integral holonomic state over the circular expression The elementary definition of radian begins with an already given circle, an already given center, and an already measurable arc length. If the radius and arc have the same length, the subtended angle measures one radian. The definition is correct within Euclidean geometry, but does not answer a prior question: what structure makes it possible for a turn to possess phase, orientation, action, and memory, and why is the natural unit related to 2π ?

HMT reverses the causal order. It first constructs the finite state, then its coinductive prolongation, then the revolution, and only at the end its angular publication. The radion does not correct the conventional radian; it lifts it to its genealogical domain.

Type receipt

$$\text{APP} \rightarrow \widehat{\mathbb{T}} \rightarrow \text{TPK} \rightarrow X_{\text{enr}} \rightarrow C_{\text{cont}}^{\text{disc}} \rightarrow (\pi, e, \varphi, \alpha, A, C^*, \Delta_4, \hbar_{\text{ret}}) \rightarrow \mathbf{r}_\sigma.$$

Conventional mathematics appears only afterward as a language of realization: S^1 , symplectic ellipse, Klein metric, Hilbert space, LC circuit, Minkowski screen, and field equations.

Bivariate support of APP: leaves, residues, quotients, and carry The Positional Arithmetic Platform (APP) simultaneously preserves two realizations of the symbols $1, \dots, 9$: an additive sheet and a multiplicative sheet. A number is not reduced to its published value. Alongside the value remain sheet, orientation, quotient, remainder, carry, boundary, route, and survival.

The basic census has three strata that must never be identified by mere cardinal coincidence:

$$104,976 \longrightarrow 468 U_6 \longrightarrow 243 w_6 \subset \mathbb{F}_3^6, \quad |\mathbb{F}_3^6| = 729.$$

The 468 decimal emissions, the 243 visible words, and the ambient set of 729 words are distinct objects. The ambient set will be decisive for the normalization $20/81$.

In cubic base $B = 1000$, the oriented subtraction of two triad words is performed from right to left:

$$u_i = t_i - v_i + c_{i+1}, \quad r_i = u_i \bmod 1000, \quad c_i = \frac{u_i - r_i}{1000}. \quad (27.1)$$

Once the operands and the remote carry are fixed, quotient and remainder are unique. This is the operator that will subsequently publish the digits of $\varepsilon_R^{(1)}$. No decimal is chosen after seeing the radian.

The completion

$$10^3 = 729 + 243 + 27 + 1$$

is not a decimal curiosity. It expresses conservation of the unit in passing from the ternary ambient space to a chamber of one thousand positions. The difference $1000 - 729 = 271$ is the completing corona; the correct identity is $999 = 729 + 270$, not $729 + 271$.

Ternary orientation and classification of the 3 operational regimes of the TRIT TRIT assigns one of three oriented regimes to each local state. Under the active temporal convention,

$$\begin{aligned} +1 &\longleftrightarrow \text{return, memory, past,} \\ 0 &\longleftrightarrow \text{threshold, present,} \\ -1 &\longleftrightarrow \text{bidirectional opening, future.} \end{aligned}$$

The quadratic realization distinguishes:

$$J^2 = -I, \quad N^2 = 0, \quad R^2 = I. \quad (27.2)$$

Thus arise, respectively, elliptic rotation, the parabolic threshold, and hyperbolic separation. Three metrics are not being mixed. Three typed actions on a common antecedent are preserved.

The internal root of -1 is not imported by presupposing the complex plane. In the APP block, two projections P, Q with ratio $r = 3/4 = 90/120$ produce

$$J_{\text{comm}} = \frac{[P, Q]}{\sqrt{r(1-r)}}, \quad J_{\text{comm}}^2 = -I,$$

while

$$R_{\text{comm}} = 2P - I, \quad R_{\text{comm}}^2 = I, \quad R_{\text{comm}}J_{\text{comm}} = -J_{\text{comm}}R_{\text{comm}}.$$

The pair generates the minimal form of $\text{Cl}_{1,1} \simeq M_2(\mathbb{R})$: rotation and opening emerge together, but remain distinct operations.

Dynamic composition of the TPK: selection, transport, memory, and return The Topological Phase Kernel is not a name for a box of objects. It is the composition that selects phase, transports cardinality, lifts sheets, records memory, and executes return. The fundamental dependence is

$$U_t \longrightarrow \Gamma_9 \longrightarrow K_{\text{ph}}.$$

The nonadic holonomy Γ_9 acts on the integral holonomic state; the expression K_{ph} is subsequent and preserves reduced memory. The identity $K_{\text{ph}}^9|_r = E_+E_\times$ does not turn K_{ph} into a generator.

The nonadic certificate preserves the complete chain

$$w_6 \rightarrow w_{12} \rightarrow w_{18} \rightarrow w_{24} \rightarrow w_{30} \rightarrow R_{36} \rightarrow G_9,$$

with 396 levels, 2376 trits, 43 circuits, 387 pairs $(K, K+9)$ per channel, and five regions of π . The phase satisfies

$$g(k+9) = g(k),$$

but the complete state does not return:

$$B_{k+9} \neq B_k.$$

This is the mathematical seed of the helix: the angular projection closes; the memory advances.

Non-split temporal extension $C_{12} \rightarrow C_{108} \rightarrow C_9$ The twelve-step memory is organized via

$$0 \longrightarrow C_{12} \longrightarrow C_{108} \longrightarrow C_9 \longrightarrow 0. \quad (27.3)$$

The cocycle

$$c(r, r') = \left[\left[\frac{r + r'}{9} \right] \right]_{12}$$

measures the carry between phase return and memory advance. Nine steps close the observable phase C_9 ; they do not reset the dodecaphasic record.

It is convenient to separate four objects:

C_{12}	cyclic group of twelve positions;
$\mathcal{R}_{12}$	enhanced record $(E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m)_{m=1}^{12}$;
Θ_{108}	raw publication on the 108 calendar;
Γ_{108}	history or lifted holonomy.

Accordingly, the vague expression “post- C_{12} ” shall be replaced by “subsequent to the dodecaphasic record $\mathcal{R}_{12}$, in its hexadic” reading when that is the effective map.

Correlative emergence of the 5 constructions of the continuum from the unique state The discrete structure of the continuum is not obtained by assembling five independent theories. Its five internal aspects—spectral, solenoidal, gauge, cohomological, and determinant—emerge correlatively from the same APP–TRIT–TPK state. This work does not prove the integral treatise anew, but preserves the causal fact required by the radion: the coinductive section that generates π, e, φ and α does not live on a bare real line; it lives in an object with fibers, relations, orientation, carry, memory, terminal object, and reader.

Hypothesis, domain, and conclusion

Genealogy falsifier. If a later formula receives $180/\pi$, ε_R , a mass, or a metrological table in order to choose a phase, a coefficient, or a branch, it ceases to be an HMT-first generation. The external contrast may recognize an output; it cannot select it.

27.1.2 Prospective independence of the closure with respect to the final value

27.1.2.1 Genealogy of the constants and construction of the angular pair

Coinductive section and uniqueness of the generated value of π Be x_∞ the section global of the inverse system. The lectores coinductivos produce

$$\text{ev}_\pi(x_\infty) = \pi, \quad \text{ev}_e(x_\infty) = e, \quad \text{ev}_\varphi(x_\infty) = \varphi.$$

These evaluations determine the universal constants themselves, not variants dependent on the procedure constructing them. In particular,

$$\pi = 3.14159265358979323846264338327950288419716939937510 \dots$$

The positive loop is

$$T_+(x_\infty) = 2\pi. \quad (27.4)$$

Its angular publication of a unit is

$$R_{\text{HMT}}^\circ = \frac{360^\circ}{T_+} = \frac{180^\circ}{\pi}. \quad (27.5)$$

The equation does not yet define the enriched radion; it defines the visible chart of the radian once the turn has been generated.

Bidirectionality of conjugate routes and prolongations Bidirectionality is not enunciated as a slogan. The concrete involution

$$J_\alpha(T) = \frac{\alpha^2}{T}, \quad T_- = J_\alpha(T_+), \quad T_+T_- = \alpha^2 \quad (27.6)$$

preserves a product and reverses the sheet. Expansion and contraction are two conjugate prolongations of the same state. The finite root of a quadratic jet can approximate T_+ , but it does not replace the coinductive section: the certified difference is

$$T_{P_9} - T_+ = -5.1110566290335 \dots \times 10^{-25}.$$

Finite control is a witness; the foundation is the limit.

Causal dependence among α , action, and angular publication The dodecaphase closure generates α from the signed state and its reversible charts. The local determinantal reader then fixes the action decade through

$$\Sigma_H = \varphi\alpha^{16}, \quad \text{ord}_{10} \Sigma_H = -34.$$

The return section produces h_{ret} . Neither α nor h is “measured in degrees”. They are subsequently published through the angular pair

$$P(\alpha, \eta_{\text{ret}}) = \left([10^3\alpha]_{360}, [2(\eta_{\text{ret}} + \alpha)]_{360} \right) = (A, C^*). \quad (27.7)$$

In the canonical branch,

$$A = 7.297352569283800997285105472380663 \dots^\circ,$$

$$C^* = 2.1237383389686457242619513803933607937 \dots^\circ.$$

Only then the analytic chart

$$x = \frac{\pi A}{180}, \quad y = \frac{\pi C^*}{180}$$

open the channels $q_{\pm} = e^{-x \mp y}$.

The complete causal order is

$$\alpha \rightarrow (\eta \parallel -34) \rightarrow \hbar \rightarrow P \rightarrow \kappa_{360} \rightarrow q_{\pm} \rightarrow \text{MD realizations.}$$

The relation $C^*/2 - \alpha$ may serve as a reverse verification. It is not used to generate $\hbar_{\text{ret}}$ retroactively.

The angular chart does not turn a dimensionless constant or an action into an angle by fiat. It publishes two state coordinates on a finite wheel. The degree is the language of that publication; it is not the ontological substance of α or $\hbar$.

Prospective independence with respect to the final value The depth operator verifier explicitly prohibits three inputs:

$$\frac{180}{\pi}, \quad \varepsilon_R, \quad \text{the target decimal.}$$

First construct S_E and S_T , then execute the APP subtraction, and only at the end compare the closure. The controls give

$$\text{target_epsilon_used}=\text{False}, \quad \text{target_180_over_pi_used}=\text{False}.$$

Also:

- a single sheet gives a defect of order 10^{-5} ;
- replacing θ_2 by θ_1 or θ_3 shifts the result by approximately $\pm\pi/6$;
- removing channel e leaves a defect of order 10^{-3} ;
- the variante historica $\Delta_4(A, C^*)$ produce other number;
- the spectral tail following $\mathcal{R}_{12}$ and the four-term harmonic operator do not coincide with $\varepsilon_R^{(1)}$ either.

Hypothesis, domain, and conclusion

Status. The closure is a *new formalization* proved with an explicit initial structure and a computational certificate independent of the final value. The equality is exact in the coinductive limit. The short 36-digit profile of α generates a variant that diverges only after many digits; it is retained as a historical witness, not as the canonical value.

27.2 Direct and torsional sections: APP carry and exact closure of the radion

27.2.1 Independent construction of the sections S_E and S_T before defining the cocycle $\varepsilon_R = S_T - S_E$

Exact closure is obtained through two sections constructed before the cocycle that relates them.

27.2.1.1 Independent construction of direct and torsional sections

Direct section of the dodecaphasic temporal system The dodecaphase wheel has centers

$$\theta_m = 30m - 10^\circ, \quad m = 1, \dots, 12. \quad (27.8)$$

The second phase is $\theta_2 = 50^\circ$. Together with the publication $A = 10^3\alpha$, it forms the direct section

$$S_E = \frac{T_+}{360} (50 + 10^3\alpha) \quad (27.9)$$

$$= 2\pi \left(\frac{5}{36} + \frac{25}{9}\alpha \right). \quad (27.10)$$

The subscript E here means *direct or common*; its electrical realization will come after the constitutive reader. One does not literally postulate $A = E$.

The coinductive cylinders prove

$$1 < S_E < 2. \quad (27.11)$$

The APP division then enforces

$$q_E = \lfloor S_E \rfloor = 1, \quad D_E = \text{rem}_1(S_E) = S_E - 1. \quad (27.12)$$

The integer one is not the number of turns of the helix. It is the unique quotient of the direct section within the unit cell.

Torsional Section Determined by the Operator Δ_4 The second section uses a previous coinductive output:

$$\Delta_4 = \frac{\pi - e \log \pi}{270} \left(1 + \frac{\pi}{729} \right). \quad (27.13)$$

The word *torsion* here denotes the global quantum of the HMT chart. It does not automatically identify it with Cartan torsion $T^I = D_\omega e^I$. The bridge to Cartan requires another typed arrow.

27.2.2 Oriented density of the active sector and unique determination of 20/81

Oriented density 20/81 of the active sector 90/120 The triadic phase separator selects

$$H_+ = \{1, 4, 7\}, \quad H_- = \{2, 5, 8\}, \quad H_0 = \{3, 6, 9\}.$$

There are six active phases and fifteen unordered pairs:

$$H_{\text{act}} = H_+ \cup H_-, \quad \left| \binom{H_{\text{act}}}{2} \right| = \binom{6}{2} = 15.$$

The active mode and its extension to eight positions are

$$N_6 = 6 \binom{6}{2} = 90, \quad N_8 = 8 \binom{6}{2} = 120. \quad (27.14)$$

Two conjugate sheets of the active sector, normalized by the ambient fiber, give

$$\rho_{\text{tw}} = \frac{2N_6}{|\mathbb{F}_3^6|} = \frac{2 \cdot 90}{729} = \frac{180}{729} = \boxed{\frac{20}{81}}. \quad (27.15)$$

Proposition 27.1 (Relative uniqueness of the density). *With the active sector $N_6 = 90$, the two conjugate sheets, and the normalization by $|\mathbb{F}_3^6| = 729$ fixed, the oriented density is necessarily $20/81$.*

Proof. The transported cardinality is $2N_6 = 180$. Normalization by the ambient space gives $180/729$; dividing the numerator and denominator by 9 yields $20/81$. No comparison with $180/\pi$ or with ε_R is involved. $\square$

The historical reading as a projection onto a $1/81$ lattice is a useful rigidity control, but it is no longer the foundation: the coefficient is obtained before closure. Ablating one sheet produces $10/81$, and the output changes at order 10^{-5} ; hence, the factor of two is not decorative.

The torsional section is

$$\Phi_T = \rho_{\text{tw}} \Delta_4, \quad S_T = 1 + \Phi_T. \quad (27.16)$$

27.2.3 The carry ε_R as a transition cocycle between sections

27.2.3.1 Transition cocycle between the direct and torsional sections

definition operatoria of the carry ε_R

Definition 27.2 (Real carry of the radion). The unitary output of the radion is the transition cocycle

$$\varepsilon_R^{(1)} := S_T - S_E = \Phi_T - D_E = \frac{20}{81} \Delta_4 - \text{operatornamerem}_1(S_E). \quad (27.17)$$

The equation contains a subtraction, but it is not a fit to the target. Its two operands S_T and S_E are generated separately. The independence demonstrated is independence from $180/\pi$ and from a target ε ; algebraic independence between a difference and its summands is not asserted.

Theorem 27.3 (Exact unitary closure). *With S_E , Φ_T , and $\varepsilon_R^{(1)}$ defined by (27.10), (27.13), (27.15), and (27.17), one has exactly*

$$\boxed{1 = S_E - \Phi_T + \varepsilon_R^{(1)}}. \quad (27.18)$$

Proof. By (27.11), $D_E = S_E - 1$. By definition, $\varepsilon_R^{(1)} = \Phi_T - D_E$. Then

$$S_E - \Phi_T + \varepsilon_R^{(1)} = S_E - \Phi_T + \Phi_T - (S_E - 1) = 1.$$

The strength of the proof lies not in this elementary cancellation, but in the fact that the prior genealogy fixes both operands without using the closure as input. $\square$

Publication of the closure on the angular chart The full-turn chart multiplies a unit fraction by $360/T_+ = 180/\pi$. In particular,

$$\varepsilon_R^\circ = \frac{360^\circ}{T_+} \varepsilon_R^{(1)}.$$

Multiplying (27.18) by $360^\circ/T_+$ and using

$$\frac{360}{T_+} S_E = 360 \left(\frac{5}{36} + \frac{25}{9} \alpha \right) = 50 + 1000\alpha = 50 + A$$

produce the angular closure.

Theorem 27.4 (Exact equation of the radion).

$$\frac{180^\circ}{\pi} = 50^\circ + A^\circ - \frac{20}{81} \Delta_4^\circ + \varepsilon_R^\circ$$

The equation says something more precise than “the radian is $57.2957 \dots^\circ$ ”. It decomposes that publication into four operations with a reference proof:

50°	second phase of the wheel; critical hinge on a declared chart;
A°	common coordinate of the parangular publication;
$-\frac{20}{81}\Delta_4^\circ$	orientation-transport of global torsion;
$+\varepsilon_R^\circ$	real carry that mates the two lifts.

Convergence by depth of refinement At depth N , the generator delivers rational cylinders $p_N \rightarrow \pi$ and $e_N \rightarrow e$. Define

$$S_{E,N} = 2p_N \left(\frac{5}{36} + \frac{25}{9} \alpha \right),$$

$$\Delta_{4,N} = \frac{p_N - e_N \log p_N}{270} \left(1 + \frac{p_N}{729} \right), \quad S_{T,N} = 1 + \frac{20}{81} \Delta_{4,N},$$

and

$$\mathcal{E}_N = S_{T,N} - S_{E,N}.$$

Continuity of the functional on $3 < p < 4$, $2 < e < 3$, together with the monotone derivatives, transports each pair of cylinders to a certified interval. At a depth of 45 blocks, its width is less than 4.81×10^{-130} .

Each nine-level return contains six trits per level. Therefore, the refinement rate is

$$729^{-9} = 3^{-54}. \quad (27.19)$$

This quantity controls the width of the intervals; it is not the value of ε_R .

The subtraction of triads stabilizes

$$000|000|000|896|802|822|044|562|981|999|050|695|020|651|286|413$$

and the carry prefix

$$0, 0, 0, 0, 0, -1, 0, -1, -1, -1, 0, -1, 0.$$

This is the concrete mechanism of decimal memory: not a chosen tail, but the recurrence (27.1) applied to previously constructed operands.

Evaluation of the canonical values The coinductive output is

$$\varepsilon_R^{(1)} = 8.9680282204456298199905069502065128641372736205009 \dots \times 10^{-10}, \quad (27.20)$$

$$\varepsilon_R^\circ = 5.1383016758575282071685164622645947489908574400054 \dots \times 10^{-8}^\circ. \quad (27.21)$$

The complete publication results

$$\frac{180^\circ}{\pi} = 57.2957795130823208767981548141051703324054724665643 \dots^\circ.$$

Exact result

$$\begin{aligned}
50^\circ + A^\circ &= 57.29735256928380099728510547238066299956 \dots^\circ, \\
\frac{20}{81} \Delta_4^\circ &= 0.00157310758449687906223272996065728980465 \dots^\circ, \\
50^\circ + A^\circ - \frac{20}{81} \Delta_4^\circ &= 57.29577946169930411822287274242000570976 \dots^\circ, \\
\varepsilon_R^\circ &= 0.00000005138301675857528207168516462264594749 \dots^\circ, \\
\text{final sum} &= 57.29577951308232087679815481410517033241 \dots^\circ.
\end{aligned}$$

27.3 Critical coordinate and oriented dodecaphasic temporal system: affine determination of the phase of 50°

27.3.1 Affine map j_{100} of the APP chart and uniqueness of the phase $m = 2$ in $j_{100}(1/2) = \theta_m = 50^\circ$

The critical coordinate links the Cayley chart, the APP record, and the dodecaphasic phase through typed maps.

27.3.1.1 Two exact realizations of the angular coordinate 50°

Second phase of the dodecaphasic temporal system By (27.8), the sequence of centros begins

$$20^\circ, 50^\circ, 80^\circ, 110^\circ, \dots$$

Solve

$$30m - 10 = 50$$

gives only $m = 2$. Therefore, 50° is not a rounding of A , a half-turn, or the centre of a circle: it is a typed phase of the dodecaphase calendar. The fraction of a turn is

$$\frac{50}{360} = \frac{5}{36},$$

exactly the first summand of S_E/T_+ .

Cayley Critical Coordinate Matching The functional involution

$$\rho \mapsto 1 - \bar{\rho}$$

fixed

$$\text{Fix}(\rho \mapsto 1 - \bar{\rho}) = \{\rho : \Re \rho = 1/2\}.$$

The number $1/2$ is the self-dual real coordinate of the functional interval $0 \leq \Re s \leq 1$. In the longitudinal sense, it is not “half of a line”, and it does not contain the spectral height γ . For

$$\rho_\gamma = \frac{1}{2} + i\gamma,$$

The Cayley transformation can be written

$$\tau_\gamma = \frac{\rho_\gamma - 1}{\rho_\gamma} = \frac{\gamma + i/2}{\gamma - i/2} \in S^1, \quad \gamma = \frac{1}{2} \cot \frac{\theta_\gamma}{2}. \quad (27.22)$$

Thus, $1/2$ fixes the seam, γ determines the angular character, and Γ_9 preserves the depth of its iterations.

Affine injection j_{100} of the APP record into the angular chart Cubic completion arranges a chamber of ten marks per coordinate. One face contains $10^2 = 100$ positions. We declare the affine chart

$$j_{100} : [0, 1] \longrightarrow [0, 100^\circ], \quad j_{100}(\sigma) = 100\sigma^\circ. \quad (27.23)$$

Then

$$j_{100}\left(\frac{1}{2}\right) = 50^\circ.$$

Requiring $j_{100}(1/2) = \theta_m$ simultaneously yields

$$100 \cdot \frac{1}{2} = 30m - 10 \iff m = 2.$$

Theorem 27.5 (Critical Equalizer–Dodecaphasic). *In the hundred-mark APP chart, the fixed coordinate of the critical involution is published exactly in the second phase of the wheel:*

$$\boxed{j_{100}(1/2) = \theta_2 = 50^\circ.} \quad (27.24)$$

The phase $m = 2$ is unique.

The theorem is exact once (27.23) has been declared. Its status is *new formalization*: the critical midpoint, the 10^3 completion, the hundred-mark face, and θ_2 had already been constructed; choosing that face as the affine critical chart brings their domains together for the first time.

Compatibility of the critical value with the Cayley chart The Cayley transform does not send $1/2$ to 50° . If $\gamma = 0$, (27.22) gives

$$\tau_0 = -1 = e^{i\pi},$$

that is, 180° . The 50° belong to the APP chart of one hundred marks and to the dodecaphasic calendar; the Cayley angle depends on γ . This distinction prevents all the zeros of the zeta function from being collapsed into a single phase.

Falsification criterion 27.6. The identification $1/2 \leftrightarrow 50^\circ$ fails if the chart (27.23) is removed. The scalar equality $100(1/2) = 50$ is not sufficient by itself to identify two objects. The theorem depends on the distinguished APP face and on the dodecaphase wheel.

27.3.2 Compatibility of the critical phase with the nonadic return and memory advance

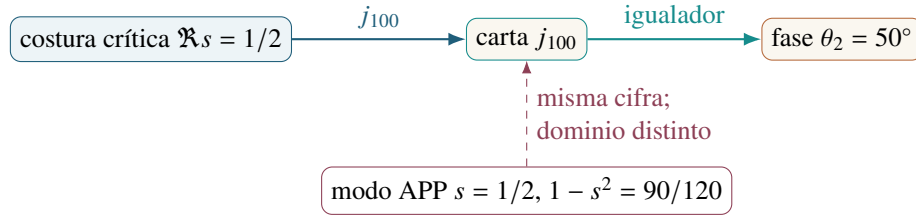
27.3.2.1 Typed separation of the 2 objects with value $1/2$

APP contains, in addition, a singular mode with

$$s = \frac{1}{2}, \quad s^2 = \frac{1}{4}, \quad 1 - s^2 = \frac{3}{4} = \frac{90}{120}, \quad s\sqrt{1 - s^2} = \frac{\sqrt{3}}{4}.$$

This s arises from the principal angles of the sum and product sheets. The critical seam $\Re\rho = 1/2$ arises from a functional involution. That both coordinates equal $1/2$ is a scalar coincidence with possible content, but it does not authorize their identification without a typed square.

The correct way to preserve the finding is an equalizer diagram:



The diagram does not feign the missing arrow. It shows what has been proved, what has been formalized, and what equality still requires additional transport if a global operator identification is intended.

27.3.2.2 Spectral Circle and Helical Memory Compatibility

The Cayley–Pontryagin reader transports a spectral fiber via

$$\tilde{U}_{\Gamma_9^n} J\psi_\gamma = \tau_\gamma^n J\psi_\gamma.$$

The circular publication is $\tau_\gamma^n \in S^1$. The lift

$$(\tau_\gamma^n, n) \in S^1 \times \mathbb{Z}$$

preserves the number of returns. On the contractive sheet appears

$$\tilde{M}_9^n J\psi_\gamma = \left(\frac{5}{9}\right)^n \tau_\gamma^n J\psi_\gamma.$$

We therefore have three typed images:

phase circle, memory helix, contractive spiral.

A zero is not a revolution, and a nine is not a particular zero of the zeta function. The zero fixes a character τ_γ that persists through the revolutions; nonadic return increments memory. Nine functions as residual zero in $\mathbb{Z}/9\mathbb{Z}$, an arithmetic operation distinct from the enumeration of spectral zeros.

The critical line becomes a circle when we forget the linear height and retain the unitary character. It becomes a helix when we reinsert the memory of how many times the character has been transported. Depth is not a third decorative coordinate: it is the information that the circular projection cannot retain.

27.4 Angular pair and reduced action: construction of the positive ellipse and area per radion

27.4.1 Areal law of the radion: $\mathcal{A}(\theta) = \hbar_{\text{ret}}\theta$ and $\mathcal{A}(2\pi) = h_{\text{ret}}$

Spectral geometry determines the form, and the areal law determines the action transported by the phase.

27.4.1.1 Equivalence between the spectral and geometric charts

Spectral positive operator and conjugate eigenvalues The angular momentum produces the two-layer operator.

$$B_1 = AI + C^*R, \quad R^2 = I, \quad P_\pm = \frac{I \pm R}{2}. \quad (27.25)$$

Its eigenvalues are $A \pm C^*$. Since $0 < C^* < A$, both are positive. In the chart $\kappa = \pi/180$, the spectral ellipse has semiaxes

$$a_1^2 = \kappa(A + C^*), \quad b_1^2 = \kappa(A - C^*). \quad (27.26)$$

The decomposition is reversible:

$$A = \frac{a_1^2 + b_1^2}{2\kappa}, \quad C^* = \frac{a_1^2 - b_1^2}{2\kappa}. \quad (27.27)$$

The ellipse is not drawn to illustrate a number. It exactly reconstructs the pair that generated it.

Define

$$\rho_1 = \sqrt{A^2 - C^{*2}}, \quad \eta = \operatorname{artanh} \frac{C^*}{A} = \frac{1}{2} \log \frac{A + C^*}{A - C^*}. \quad (27.28)$$

Then

$$A \pm C^* = \rho_1 e^{\pm \eta},$$

and

$$a_1 = \sqrt{\kappa \rho_1} e^{\eta/2}, \quad b_1 = \sqrt{\kappa \rho_1} e^{-\eta/2}. \quad (27.29)$$

The product preserves scale; the quotient preserves form:

$$a_1 b_1 = \kappa \rho_1, \quad \frac{b_1}{a_1} = e^{-\eta} = \sqrt{\frac{A - C^*}{A + C^*}}.$$

Focal Invariants of the Positive Operator The semifocal distance is

$$c_1^2 = a_1^2 - b_1^2 = 2\kappa C^*, \quad F_{\pm} = (\pm c_1, 0). \quad (27.30)$$

The total distance is equal to

$$d_1 = 2c_1 = 2\sqrt{2\kappa C^*} = 0.544545509325626623406299779866023 \dots \quad (27.31)$$

The isolated number is not the thesis. The identity $c_1^2 = 2\kappa C^*$ states that C^* is literally the quadratic focal splitting in the distinguished chart. The foci are the conjugate endpoints of the spectral aperture; they are not, by themselves, the electric and magnetic poles. That reading requires the constitutive transducer.

The eccentricity is

$$e_f = \frac{c_1}{a_1}, \quad e_f^2 = 1 - e^{-2\eta} = \frac{2C^*}{A + C^*}. \quad (27.32)$$

Numerically,

$$\begin{aligned} \eta &= 0.299689683921630799684926562863927 \dots, \\ e_f &= 0.671451895550191986916277055864332 \dots \end{aligned}$$

Reference APP Block and Normalized Transport The central block

$$B_0 = 7I + 2R = 9P_+ + 5P_-$$

possesses

$$\rho_0 = \sqrt{45}, \quad \eta_0 = \frac{1}{2} \log \frac{9}{5}, \quad e_0 = \frac{2}{3}.$$

Its focal length is

$$d_0 = 4\sqrt{\kappa} = 0.528443639680801468446003835349358 \dots$$

The exact comparison is

$$\boxed{\left(\frac{d_1}{d_0}\right)^2 = \frac{C^*}{2}}, \quad d_1 = d_0 \sqrt{\frac{C^*}{2}}. \quad (27.33)$$

This is the defensible genealogy of $0.544545\dots$: a focal APP scale of reference multiplied by a relative invariant. Its visual proximity to 0.54 is not an identity with the APP defect 54.

Transport from B_0 separates into scale and anisotropy:

$$T_{\text{rel}} = \frac{\rho_1}{\sqrt{45}} \exp((\eta - \eta_0)R). \quad (27.34)$$

On the leaves,

$$t_+ = \frac{A + C^*}{9} = 1.0467878786947163\dots, \quad t_- = \frac{A - C^*}{5} = 1.0347228460630311\dots$$

The deformation is not a single decimal. Its isotropic component is $\sqrt{t_+ t_-}$; its hyperbolic component is $\frac{1}{2} \log(t_+/t_-)$.

27.4.1.2 Positive ellipse associated with the angular pair

Construction via Conjugate Semi-axes The preceding spectral scale has no units of action. Dimensional Mechanics publishes the same form on the internal action line:

$$a_S = \sqrt{2\hbar_{\text{ret}} e^\eta}, \quad b_S = \sqrt{2\hbar_{\text{ret}} e^{-\eta}}. \quad (27.35)$$

Immediately,

$$a_S b_S = 2\hbar_{\text{ret}}, \quad \frac{a_S}{b_S} = e^\eta. \quad (27.36)$$

The product fixes the action; the quotient fixes the anisotropy. The ellipse is parametrized by

$$Q(\theta) = a_S \cos \theta, \quad P(\theta) = b_S \sin \theta. \quad (27.37)$$

The spectral and action ellipses are homothetic:

$$\lambda_{\text{act}}^2 = \frac{2\hbar_{\text{ret}}}{\kappa \rho_1}, \quad (a_S, b_S, c_S) = \lambda_{\text{act}}(a_1, b_1, c_1). \quad (27.38)$$

They share the form $e^{-\eta}$, not the physical dimension.

Areal law of action per radion The forma simplectica canonica is $\omega = dQ \wedge dP$. The area oriented barrida from 0 until θ is

$$\mathcal{A}(\theta) = \frac{1}{2} \int_0^\theta (Q(t)P'(t) - P(t)Q'(t)) dt \quad (27.39)$$

$$= \frac{1}{2} a_S b_S \int_0^\theta (\cos^2 t + \sin^2 t) dt. \quad (27.40)$$

Using (27.36), we obtain the central result.

Theorem 27.7 (Action per radion). *For the ellipse (27.37),*

$$\boxed{\mathcal{A}(\theta) = \hbar_{\text{ret}} \theta.} \quad (27.41)$$

In particular,

$$\boxed{\mathcal{A}(1) = \hbar_{\text{ret}}}, \quad \boxed{\mathcal{A}(2\pi) = 2\pi\hbar_{\text{ret}} = h_{\text{ret}}}. \quad (27.42)$$

This equality allows the definition of the radius without first appealing to the arc length.

Definition 27.8 (Oriented radion).

$$\mathbf{r}_\sigma = (\theta = 1, \sigma), \quad \sigma \in \{+1, -1\}, \quad \mathcal{A}(\mathbf{r}_\sigma) = \sigma \hbar_{\text{ret}}. \quad (27.43)$$

The radian is the projection that forgets σ , the sheet, the memory, and the area. The radion preserves that information. Hence the name *rad-ion*: orientation precedes physical charge, and charge appears later as

$$Q(x) = n_Q(x) e_{\text{int}}, \quad e_{\text{int}} = \sqrt{\frac{2\alpha \hbar_{\text{ret}}}{Z_{\text{int}}}}.$$

27.4.2 Conjugate semi-axes, focal invariants, and orientation of anisotropy

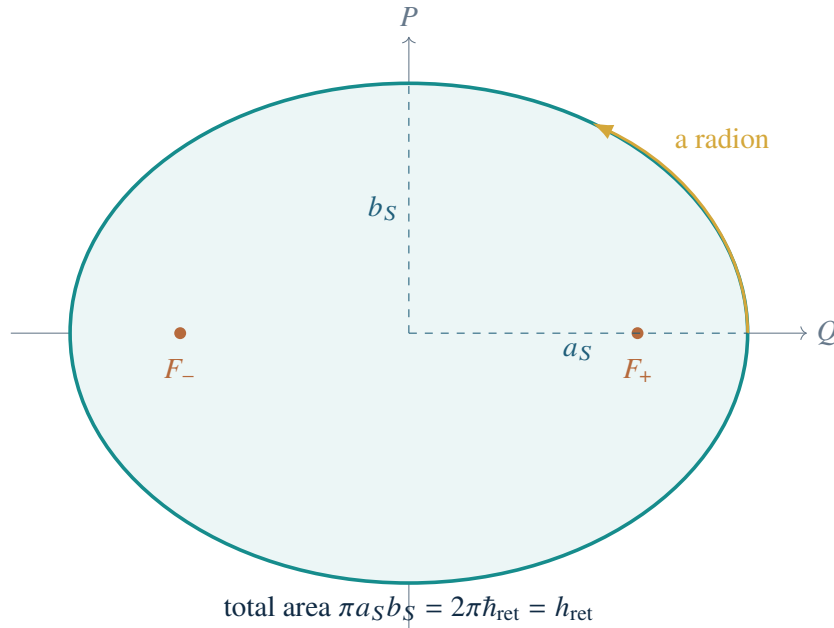


Figure 27.1: The action ellipse. A parametric arc of one radian sweeps $\hbar_{\text{ret}}$; one turn encloses h_{ret} . The foci encode the aperture C^* in the distinguished chart.

Correspondence between phase, action, area, and orientation

Relationship between the reduced action, uncertainty, and area The relation $\hbar = h/(2\pi)$ is often presented as a notational convenience: when describing oscillations, rotations, and Fourier transforms, the factors 2π disappear if $\hbar$ is used. The action ellipse reveals a stronger reading. Division by 2π is exactly the passage from the action of one revolution to the action per unit phase:

$$h = \mathcal{A}(2\pi), \quad \hbar = \mathcal{A}(1).$$

The reduced constant is not merely h in more convenient notation; it is the areal density of action with respect to phase.

Heisenberg, Dirac, and Jordan established the language of conjugate variables, commutators, and canonical transformations. No literal “ontological ellipse” lacking a source is attributed to them here. The HMT–MD contribution is to identify the positive figure, construct its semiaxes from (A, C^*) , and prove (27.41). The historical genealogy opens the language; the present theorem fixes the object.

Action is not appended to motion afterward. In the radion ellipse, action is the area generated by motion. A full turn does not transport a prefabricated constant: it encloses it.

Algebraic controls of the focal invariants Decimal similarities are subject to control:

Comparison	Residue	Verdict
d_1 versus 0.54	4.5455×10^{-3}	not equality; 54 is not deduced;
d_1^2 versus 8/27	2.3352×10^{-4}	proximity without map;
η frente a 3/10	-3.1032×10^{-4}	no igualdad;
e_f versus 2/3	4.7852×10^{-3}	2/3 belongs exactly to B_0 ;
$e^{-\eta}$ versus 20/27	3.0740×10^{-4}	no igualdad.

The robust invariants are (27.28), (27.32), and (27.33). The HMT criterion does not consist in favouring a decimal resemblance, but in requiring an object, an operation, and conservation.

27.5 Nonadic Return and Internal Projective: Klein Disk and Helical Depth of Memory

27.5.1 Normalization to the Hilbert–Beltrami–Klein disk and hyperbolic interior

Projective geometry distinguishes the finite action from hyperbolic depth and return memory.

27.5.1.1 Hyperbolic Interior of Hilbert–Beltrami–Klein

Projective normalization on the Klein disk The affine application

$$u = \frac{Q}{a_S}, \quad v = \frac{P}{b_S}$$

carries the action ellipse to the unit disk

$$D = \{(u, v) : u^2 + v^2 < 1\}.$$

On each chord, the Hilbert distance is defined through the cross-ratio of its boundary endpoints. The resulting infinitesimal metric is the Beltrami–Klein form

$$ds_K^2 = \frac{(1 - r^2)(du^2 + dv^2) + (u du + v dv)^2}{(1 - r^2)^2}, \quad r^2 = u^2 + v^2. \quad (27.44)$$

With this normalization, the Gaussian curvature is constant:

$$K = -1. \quad (27.45)$$

The geodesics are Euclidean chords, but their hyperbolic length diverges as they approach the boundary.

Finiteness of the boundary and infinitude of the hyperbolic depth The metric determinant of (27.44) produce

$$dA_K = \frac{du \, dv}{(1 - r^2)^{3/2}}. \quad (27.46)$$

The area of the subdisk $r < R < 1$ is

$$A_K(R) = \int_0^{2\pi} \int_0^R \frac{r \, dr \, d\theta}{(1 - r^2)^{3/2}} \quad (27.47)$$

$$= 2\pi \left(\frac{1}{\sqrt{1 - R^2}} - 1 \right). \quad (27.48)$$

Therefore,

$$\lim_{R \rightarrow 1^-} A_K(R) = +\infty.$$

Simultaneously, the same boundary encloses the symplectic area.

$$\pi a_S b_S = 2\pi \hbar_{\text{ret}} = h_{\text{ret}} < \infty.$$

Theorem 27.9 (Finiteness of Action and Infinitude of Depth). *The radion ellipse has a finite turn action h_{ret} , whereas its interior, equipped with the Hilbert–Beltrami–Klein metric, has infinite hyperbolic area:*

$$\boxed{\text{Area}_{\text{simp}}(\partial \mathbb{E}_S) = h_{\text{ret}} < \infty, \quad \text{Area}_K(\mathbb{E}_S) = +\infty.} \quad (27.49)$$

There is no contradiction: $\text{Area}_{\text{simp}}$ and Area_K are different measures. The first counts oriented action in phase; the second measures interior projective depth. The thesis “finite boundary, infinite interior” is now an equality and a limit, not a metaphor.

Focal Coordinates Inside Klein In the normalized disk, the foci are located at $(\pm e_f, 0)$. The distance from the center to a focus is

$$u_f = \text{artanh } e_f = \text{arcosh}(e^\eta) = 0.813382331175342333760889731088228 \dots \quad (27.50)$$

The focal–focal distance is

$$d_K(F_-, F_+) = 2u_f = 1.62676466235068466752177946217646 \dots$$

We have

$$e_f = \tanh u_f, \quad e^{-\eta} = \text{sech } u_f, \quad \eta = \log \cosh u_f. \quad (27.51)$$

The region up to the focal radius has area

$$A_K(e_f) = 2\pi(e^\eta - 1) = 2.19559620884061869541206115980360 \dots$$

All open ellipses are projectively isometric as Hilbert domains. Curvature -1 alone therefore does not retain eccentricity. The HMT form resides in the joint structure: the distinguished affine chart, symplectic form, sheet labeling, and boundary parametrization.

27.5.2 Difference between phase return and return of the integral holonomic state

27.5.2.1 Helical elevation of the nonadic return

Nonadic decomposition and memory quotient The elementary operation

$$n = 9q_9(n) + \rho_9(n), \quad H_9(n) = (\rho_9(n), q_9(n)) \quad (27.52)$$

satisfies

$$H_9(n+9) = H_9(n) + (0, 1). \quad (27.53)$$

The first coordinate closes; the second ascends. This identity is the minimal arithmetic version of a screw.

For a spectral fiber,

$$n \mapsto (\tau_\gamma^n, n)$$

is a helix over S^1 . Adding the contraction $(5/9)^n$ produces an inner spiral. The radion participates in both readings: a unit of phase on the boundary and an oriented unit of action in the lift.

Memory range as dimensional reader Suppose that a visible projection preserves the phase value but loses d linearly independent memory cocycles. Any faithful lift requires at least d additional coordinates to reconstruct them. This motivates a precise reading of Dimensional Mechanics:

$$\begin{aligned} \text{emergent dimension} &= \text{minimum rank of independent memory} \\ &\text{not publishable on the visible sheet.} \end{aligned} \quad (27.54)$$

It is not asserted that every physical dimension is a counter. Rather, in the TPK domain, every independent memory that must survive forces a coordinate of the lift.

Dimension appears when the projection can no longer preserve the entire history. The circle suffices to state where the phase is; it does not suffice to state how many times it returned, which sheet it transported, which interval contracted, or which carry survived.

Cylinder Assembly, Spirals, and Helical Trajectories The coinductive cylinders are not physical tubes added to a spiral. They are nested intervals that preserve prefixes and narrow the value. The spiral describes transverse contraction; the helix describes longitudinal memory; the cylinder describes the set of compatible values at a finite depth. In the limit, the intersection of the cylinders selects a unique section.

The spatial image brings together, without confusion:

angle	$\leftrightarrow$	visible phase,
height	$\leftrightarrow$	return/memory,
cylinder radius	$\leftrightarrow$	refinement uncertainty,
sheet	$\leftrightarrow$	bidirectional orientation.

The rate 3^{-54} contracts the cylinder by return; it is not added to the angle and does not replace the actual carry ε_R .

27.5.2.2 Phase closures of orders 2π and 4π

In the circular projection, 2π restores the phase. In an oriented lift with two sheets, one turn may close on the conjugate sheet and require a second turn to restore the orientation:

$$U_{2\pi} = -I, \quad U_{4\pi} = I.$$

The structure agrees with spinorial periodicity when the functor is applied to a representation $SU(2)$; the TPK lift is not identified with $SU(2)$ without that map.

The areal reading is consistent:

$$\mathcal{A}(2\pi) = h_{\text{ret}}, \quad \mathcal{A}(4\pi) = 2h_{\text{ret}}.$$

On a Möbius strip, one visible circuit reverses the sheet and two circuits restore orientation. The $2\pi/4\pi$ is not an accidental doubling; it is the difference between phase return and return of the integral holonomic state.

27.6 Oriented ellipse and conjugate variables: minimum uncertainty and electromagnetic oscillation

27.6.1 Conjugate symplectic variables and determination of the minimum uncertainty

The symplectic form coordinates action, uncertainty, and oscillatory dynamics without altering the genealogy of the state.

27.6.1.1 Realization of the state in a Hilbert space

Minimum Covariance Matrix Once the action ellipse has been constructed, it can be performed on a pair of conjugate operators

$$[\widehat{Q}, \widehat{P}] = i\hbar_{\text{ret}}.$$

We define

$$V_\eta = \frac{\hbar_{\text{ret}}}{2} \begin{pmatrix} e^\eta & 0 \\ 0 & e^{-\eta} \end{pmatrix}. \quad (27.55)$$

Its determinant is

$$\det V_\eta = \frac{\hbar_{\text{ret}}^2}{4}. \quad (27.56)$$

Consequently,

$$(\Delta Q)^2 = \frac{\hbar_{\text{ret}}}{2} e^\eta, \quad (\Delta P)^2 = \frac{\hbar_{\text{ret}}}{2} e^{-\eta}, \quad \Delta Q \Delta P = \frac{\hbar_{\text{ret}}}{2}. \quad (27.57)$$

The Robertson–Schrödinger inequality is saturated.

The compressed annihilator

$$\widehat{a}_\eta = \frac{e^{-\eta/2} \widehat{Q} + ie^{\eta/2} \widehat{P}}{\sqrt{2\hbar_{\text{ret}}}} \quad (27.58)$$

satisfies $[\widehat{a}_\eta, \widehat{a}_\eta^\dagger] = 1$. Its vacuum has wavefunction

$$\psi_\eta(Q) = (\pi\hbar_{\text{ret}}e^\eta)^{-1/4} \exp\left(-\frac{Q^2}{2\hbar_{\text{ret}}e^\eta}\right). \quad (27.59)$$

Ellipse as level set of the Mahalanobis functional The set

$$\mathbb{E}_{\text{act}} = \{x \in \mathbb{R}^2 : x^\top V_\eta^{-1} x \leq 4\} \quad (27.60)$$

has semiaxes

$$a_S = 2\Delta Q, \quad b_S = 2\Delta P. \quad (27.61)$$

Its area is

$$4\pi\sqrt{\det V_\eta} = 2\pi\hbar_{\text{ret}} = h_{\text{ret}}.$$

The previously constructed HMT ellipse coincides exactly with the contour of two deviations from the minimal Gaussian state.

Proposition 27.10 (Unique chain of the quotient form). *After declaring the spectral, symplectic, and Hilbert interfaces,*

$$e^{-\eta} = \sqrt{\frac{A - C^*}{A + C^*}} = \frac{b_1}{a_1} = \frac{b_S}{a_S} = \frac{\Delta P}{\Delta Q}. \quad (27.62)$$

The action ellipse corresponds to the classical level $H = \hbar_{\text{ret}}\omega$ and to the contour 2σ of the compressed vacuum. It must not be conflated with the mean vacuum energy, $\hbar_{\text{ret}}\omega/2$. This factor-of-two difference is a mandatory physical control.

Invariance Conditions of the Ellipse as an Operational Object In the conventional formulation, an uncertainty ellipse may be viewed as a graphical device for representing a covariance matrix. Here the causal order is reversed: the positive ellipse already exists as a publication of $(A, C^*, \hbar_{\text{ret}})$; Hilbert space recognizes it through (27.55). The claimed ontological reality does not mean that a classical particle physically traverses an oval in the plane (Q, P) . It means that the state preserves a positive form, a total action, and an orientation quotient before a probabilistic theory interprets them.

The wave function does not create the form; it represents it. Uncertainty does not create $\hbar$; it publishes the impossibility of simultaneously compressing the two directions without altering the symplectic product.

The probabilistic reading is one possible realization of the ellipse, not its cause. The primary object is the joint preservation of product and orientation. Probability appears when that geometry is published as the covariance of a state on a Hilbert space.

27.6.2 Exchange $\mu \leftrightarrow \varepsilon$, inversion $Z \leftrightarrow Z^{-1}$, and conservation of causal velocity

Energy exchange between the electric and magnetic sectors The two terms of (27.67) become

$$U_E(\theta) = \hbar_{\text{ret}} \omega \cos^2 \theta, \quad U_B(\theta) = \hbar_{\text{ret}} \omega \sin^2 \theta. \quad (27.63)$$

Therefore,

$$U_E + U_B = \hbar_{\text{ret}} \omega.$$

Electric–magnetic polarity does not consist in arbitrarily labeling the foci. It consists of two conjugate reserves that are exchanged during the orbit while keeping the frequency and total action constant.

The common quotient extends:

$$e^{-\eta} = \frac{b_S}{a_S} = \frac{\Delta P}{\Delta Q} = \frac{Z_\eta}{Z_*} = \sqrt{\frac{A - C^*}{A + C^*}}. \quad (27.64)$$

The conserved products are

$$a_S b_S = 2\hbar_{\text{ret}}, \quad \det V_\eta = \frac{\hbar_{\text{ret}}^2}{4}, \quad L_\eta C_\eta = \omega^{-2}.$$

Exact result

The LC oscillator is the minimal physical realization of the double reading: the product preserves action and frequency; the quotient preserves orientation and impedance. A complete cycle exchanges electricity and magnetism without losing the area $\oint P \, dQ = h_{\text{ret}}$.

27.6.3 Inductive–capacitive pair (L_η, C_η) : invariance of $L_\eta C_\eta = \omega^{-2}$ and orientation of impedance

27.6.3.1 Inductive–capacitive oscillator associated with the angular pair

Independent construction of the 2 conjugated reservoirs Given $\omega > 0$ and a reference impedance $Z_* > 0$, we define

$$L_\eta = \frac{Z_*}{\omega} e^{-\eta}, \quad C_\eta = \frac{1}{Z_* \omega} e^\eta. \quad (27.65)$$

Then

$$L_\eta C_\eta = \omega^{-2}, \quad \sqrt{\frac{L_\eta}{C_\eta}} = Z_* e^{-\eta}. \quad (27.66)$$

The frequency remains fixed; the impedance preserves the orientation of the form.

The Hamiltonian is

$$H_{LC} = \frac{q^2}{2C_\eta} + \frac{\Phi^2}{2L_\eta}. \quad (27.67)$$

With the normalized variables

$$Q = \sqrt{Z_*} q, \quad P = \frac{\Phi}{\sqrt{Z_*}}, \quad (27.68)$$

we obtain

$$H_{LC} = \frac{\omega}{2} \left(e^{-\eta} Q^2 + e^{\eta} P^2 \right). \quad (27.69)$$

On (27.37),

$$H_{LC} = \hbar_{\text{ret}} \omega. \quad (27.70)$$

LC Oscillator as an Elementary Dynamic Unit The harmonic oscillator pervades nearly all of physics because it linearizes the neighborhood of an equilibrium, decomposes fields into modes, organizes quanta, and allows propagation to be read as a superposition of phases. In the radion, this universality receives a concrete genealogy: the mode does not begin in an external differential equation, but in a positive form with a fixed return action. Conventional dynamics appears upon choosing a clock ω and a transducer Z_* .

Frequency does not alter the action ellipse: it changes the level energy through $E = \hbar_{\text{ret}} \omega$. This separation is crucial. Geometry fixes action; the clock converts action into energy.

27.6.3.2 Time reversal and conjugate propagations

Let $H = H^*$ be given, and let J_E be a complex structure with $J_E^2 = -I$. Suppose there is an involution C such that

$$CJ_EC^{-1} = -J_E, \quad CHC^{-1} = H.$$

The evolution

$$U(t) = \exp \left(-\frac{J_E H t}{\hbar_{\text{ret}}} \right) \quad (27.71)$$

satisfies exactly

$$CU(t)C^{-1} = U(-t). \quad (27.72)$$

This is the minimal operator form of bidirectionality: conjugation interchanges the temporally opposed prolongations.

Wheeler–Feynman absorber theory and the reading of antiparticles as temporally conjugate trajectories offer a subsequent historical realization. Identity (27.72) is demonstrated; converting it into a complete theory of advanced and retarded propagators requires declaring action, boundary conditions, causality, and absorber interactions. The monograph preserves the physical intuition without converting a correspondence into an equality of theories.

27.7 The three TRIT regimes: electric–magnetic and elastic–rotational polarities

27.7.1 Coordination of the elliptic, parabolic, and hyperbolic realizations

The 3 TRIT regimes admit complex, electromagnetic, and mechanical realizations with distinct types.

27.7.1.1 Operator quadratic of root internal of -1

Exponential representation of the elliptic regime $J^2 = -I$ With $J^2 = -I$,

$$T_{\text{ell}}(x, y) = e^{-xI - yJ} = e^{-x}(\cos y I - \sin y J). \quad (27.73)$$

Under the identification generated by J ,

$$T_{\text{ell}}(x, y) \longleftrightarrow e^{-x-iy}.$$

The scalar x controls dilation; y controls oriented rotation. The form $a + bi$ is recognized as elasticity plus rotation, not as two arbitrarily adjoined quantities.

Exponential representation of the parabolic regime $J^2 = 0$ With $N^2 = 0$, the exponential is truncated:

$$T_{\text{par}}(x, y) = e^{-x}(I - yN). \quad (27.74)$$

The regime describes the threshold at which there is neither periodic rotation nor exponential opening. In the active temporal reading it is the Euclidean present, but it retains memory of the branches that bound it.

Exponential Representation of the Hyperbolic Regime $J^2 = +I$ With $R^2 = I$ and $P_{\pm} = (I \pm R)/2$,

$$T_{\text{hyp}}(x, y) = e^{-xI-yR} = q_+P_+ + q_-P_-, \quad q_{\pm} = e^{-x \mp y}. \quad (27.75)$$

The elementary invariants are

$$q_+q_- = e^{-2x}, \quad \frac{q_-}{q_+} = e^{2y}. \quad (27.76)$$

The product preserves the common mode; the quotient preserves the oriented separation.

Applied to the angular pair,

$$x = \frac{\pi A}{180}, \quad y = \frac{\pi C^*}{180}.$$

Therefore, A is the common coordinate of the publication and C^* its oriented contrast. Only after a physical reader has been applied are they recognized as electric and magnetic poles.

27.7.2 Typed separation of the constitutive and mechanical polarities

27.7.2.1 Constitutive Operator on the Channels 90/120

Common and Oriented Component Decomposition We define

$$\mathcal{E} = \log[(1 - q_+^{90})(1 - q_-^{90})] - \log[(1 - q_+^{120})(1 - q_-^{120})], \quad (27.77)$$

$$\mathcal{B} = \log \frac{1 - q_-^{90}}{1 - q_+^{90}} - \log \frac{1 - q_-^{120}}{1 - q_+^{120}}. \quad (27.78)$$

Under $C^* \mapsto -C^*$, $q_+ \leftrightarrow q_-$ are interchanged. Therefore, $\mathcal{E}$ is even and $\mathcal{B}$ is odd.

The normalized constitutive equations are

$$\widehat{\mu} = e^{\mathcal{E}+\mathcal{B}}, \quad \widehat{\varepsilon} = e^{\mathcal{E}-\mathcal{B}}, \quad \widehat{Z} = e^{\mathcal{B}}, \quad \widehat{c} = e^{-\mathcal{E}}. \quad (27.79)$$

They are verified exactly.

$$\widehat{Z} = \sqrt{\frac{\widehat{\mu}}{\widehat{\varepsilon}}}, \quad \widehat{c} = (\widehat{\mu}\widehat{\varepsilon})^{-1/2}. \quad (27.80)$$

Conjugation produces

$$C^* \mapsto -C^*: \quad \widehat{\mu} \leftrightarrow \widehat{\varepsilon}, \quad \widehat{Z} \leftrightarrow \widehat{Z}^{-1}, \quad \widehat{c} \mapsto \widehat{c}. \quad (27.81)$$

Theorem 27.11 (Constitutive polarity). *The reader 90/120 converts the common mode and oriented contrast of the pair (A, C^*) into two conjugate constitutive quantities. Sheet inversion exchanges permeability and permittivity, reverses impedance, and preserves normalized velocity.*

This is the precise formulation of the intuition “electrical part and magnetic part”. One writes neither $A = E$ nor $C^* = B$. Instead, one writes the map

$$(A, C^*) \rightarrow (x, y) \rightarrow (q_+, q_-) \rightarrow (\mathcal{E}, \mathcal{B}) \rightarrow (\widehat{\mathcal{E}}, \widehat{\mu}, \widehat{Z}, \widehat{c}).$$

Elastic-rotational realization of the constitutive polarity The same information may be published in the complex regime by

$$Z_{\text{em}} = e^{\mathcal{E}/2} \left(\cos \frac{\mathcal{B}}{2} I - \sin \frac{\mathcal{B}}{2} J \right). \quad (27.82)$$

The factor $e^{\mathcal{E}/2}$ is an elastic dilation; the factor generated by J is a rotation. Complex notation does not add a mysterious imaginary dimension to a previously given real world: it records two operational modes already separated by the TRIT.

Transduction of Orientation into Effective Load The radion sheet σ fixes an orientation before a unit of charge is introduced. Dimensional Mechanics then provides the transducer

$$e_{\text{int}} = \sqrt{\frac{2\alpha h_{\text{ret}}}{Z_{\text{int}}}}, \quad Q = n_Q e_{\text{int}}.$$

The causal chain is

$$\text{sheet} \rightarrow \text{orientation} \rightarrow \text{constitutive} \rightarrow \text{dimensional transducer} \rightarrow \text{physical charge}.$$

Calling the unit *radion* does not prematurely identify the orientation with the coulomb; it makes visible the locus where the charge sign may reside.

Electricity and magnetism are not affixed to the circle from outside. They emerge when the same state is read through product and quotient: the product retains the common mode, while the quotient retains the orientation. The LC oscillator realizes the exchange; the reader 90/120 publishes the constitutive relations; the dimensional chart assigns units.

27.8 Incidence of the sectors 90/120: record, spectrum, Barbero–Immirzi kernel and towers

27.8.1 Typed distinction between the incidence realizations of the sector 90/120

27.8.1.1 Global harmonic hierarchy and semigroup $\langle 90, 120 \rangle$

To avoid a symbol collision, we fix

$$R_{\text{tow}} := -R_{\text{ch}}.$$

Then

$$T = e^{-xI+yR_{\text{tow}}} = q_- P_+^{\text{tow}} + q_+ P_-^{\text{tow}}. \quad (27.83)$$

The even and odd towers are

$$c_n = \text{Tr}(T^n) = q_-^n + q_+^n = 2e^{-nx} \cosh(ny), \quad (27.84)$$

$$s_n = \text{Tr}(R_{\text{tow}} T^n) = q_-^n - q_+^n = 2e^{-nx} \sinh(ny). \quad (27.85)$$

Under $C^* \mapsto -C^*$, c_n is even and s_n is odd. It holds that

$$c_n^2 - s_n^2 = 4e^{-2nx}, \quad (27.86)$$

and both sequences satisfy the order-two recurrence determined by the eigenvalues $q_{\pm}$.

The semigroup

$$\langle 90, 120 \rangle = 30\langle 3, 4 \rangle \quad (27.87)$$

is the harmonic hierarchy of a single bilayer operator. It is not a list of eight lateral corrections. Its completeness means that every positive ratio admits a convergent expansion; it does not mean that an expansion obtained from the target is a prospective physical selector.

27.8.1.2 Typed classification and separation of the realizations 90/120

Object	Operation	What is conserved / why it is not another object
Incidence N_6, N_8	finite counting over active phases	cardinality and orientation; it produces 90, 120, not a series;
$-1/12$	diferencia normalized (27.88)	incidence scalar; no Barbero weight;
E_{dod}	harmonic extractor of the wheel	dodecaphase phase; no real carry;
$h = C^*/6$	reconstruction of six pairs	opening by pair; no metrological normalization;
F_{spec}	cola logaritmica $j \geq 2$	complete spectral publication; differs from ε_R° ;
$\varepsilon_R^{(1)}$	subtraction APP between S_T and S_E	real join of two lifts; not a Lambert series;
$K_{6 \rightarrow 8}$	$12\Delta S_{90} - \Delta S_{120}$	pole $1/24$ and marked jump; Barbero output;
$12 : -1$	solution Diophantine of (27.91)	peso of the kernel; no $-1/12$;
$\langle 90, 120 \rangle$	semigroup of powers of T	global harmonic hierarchy; no residue per species.

Tesis rectora

All these objects descend from the same APP–TRIT–TPK state. That common genealogy explains why they share numbers and symmetries. Their typing explains why they cannot replace one another.

27.8.2 Conservation of Incidence and Orientation Between the Tritic Kernel and Its Subsequent Realizations

The typed hierarchy separates incidence, record, spectral evaluation, Barbero functional, and harmonic towers.

27.8.2.1 Tritic incidence, dodecaphasic recording and hexadic reading

Primary combinatorial derivation of 90 and 120 The counts

$$90 = 6 \binom{6}{2}, \quad 120 = 8 \binom{6}{2}$$

arise from the phase selector and its active incidences. The normalized orientation satisfies

$$\frac{30}{120} - \frac{30}{90} = -\frac{1}{12}. \quad (27.88)$$

This scalar is an incidence result. It is neither the weight $12 : -1$ of the Barbero kernel nor a dodecaphase carry.

The dodecaphonic extractor can be published through a combination of harmonics as

$$E_{\text{dod}}(\phi) = 12 \cos(3\phi) - \cos(4\phi).$$

Its domain is the phase of the wheel; its output is a harmonic selector. Although the numbers 12 and -1 appear, it does not act on Lambert series or produce ε_R .

27.8.3 Relative Diophantine uniqueness of the weight $12 : -1$ under the pole condition $1/24$ and the minimal integer counterterm

27.8.3.1 Barbero–Immirzi hexad–octad functional

Oriented series and incidence jump We define

$$S_{90}(q) = \frac{q^{90}}{1 - q^{270}}, \quad S_{120}(q) = \frac{q^{120}}{1 - q^{360}}, \quad (27.89)$$

and

$$\Delta S_m = S_m(q_-) - S_m(q_+).$$

The marked inclusion functional is

$$K_{6 \rightarrow 8} = 12\Delta S_{90} - \Delta S_{120}. \quad (27.90)$$

Diophantine condition for the weight $12 : -1$ Consider $aS_{90} - bS_{120}$. The declared pole condition is

$$\frac{a}{270} - \frac{b}{360} = \frac{1}{24}. \quad (27.91)$$

Multiplying by 1080 gives

$$4a - 3b = 45.$$

If the contratermino is integer minimo, $|b| = 1$. For $b = 1$, $a = 12$; for $b = -1$, $a = 21/2$ is not integer. therefore,

$$\boxed{a : b = 12 : 1, \quad aS_{90} - bS_{120} = 12S_{90} - S_{120}.} \quad (27.92)$$

Proposition 27.12 (Relative Diophantine uniqueness). *The signed weight $12 : -1$ is unique among integer coefficients when the pole condition $1/24$, positivity of the principal channel, and the minimal counterterm $|b| = 1$ are fixed.*

The hexad–octad incidence supplies the channels $90/120$. By itself, it does not force the Lambert series or the condition (27.91). The strength of $12 : -1$ is relative to these explicit analytic premises.

The Barbero–Deformable chain is subsequent:

$$(A, C^*) \rightarrow (x, y) \rightarrow q_{\pm} \rightarrow K_{6 \rightarrow 8} \rightarrow \gamma_{\text{BI}},$$

with evaluation

$$\gamma_{\text{BI}} = 0.274007672694459 \dots$$

Barbero–Immirzi does not retroactively generate C^* , Δ_4 , or ε_R .

27.8.4 Extraction of the finite component subsequent to the oriented dodecaphasic system R_{12} : $C^* \mapsto C^*/6$, remainder $j \geq 2$ and reconstruction of six sections

Extraction of component $C^*/6$ from the dodecaphase record The twelve sectors group into six opposite pairs:

$$p_i = e_i + e_{i+6}, \quad a_i = e_i - e_{i+6}, \quad i = 1, \dots, 6. \quad (27.93)$$

The reconstruction has profile $1 + 5 + 6 = 12$. The opening per pair is

$$h = \frac{C^*}{6}. \quad (27.94)$$

It is neither an external normalization nor one sixth of a target: it is the internal index of the hexadic reconstruction of the record $\mathcal{R}_{12}$.

We define

$$q_{h,-} = \exp \left[-\frac{\pi}{180}(A - h) \right], \quad q_{h,+} = \exp \left[-\frac{\pi}{180}(A + h) \right]. \quad (27.95)$$

27.8.5 Quantitative decomposition of the carry and spectral realizations: $F_{\text{spec}} = \varepsilon_R^\circ + \chi_R$

27.8.5.1 Spectral publication of the tail of orders $j \geq 2$

Definition of the spectral functional For $m \in \{90, 120\}$, let

$$L_m^{(2)}(q) = \sum_{j \geq 2} \frac{q^{mj}}{j} = -\log(1 - q^m) - q^m. \quad (27.96)$$

The reading of a pair is

$$T_h^{(2)} = \frac{180}{\pi} \left[L_{90}^{(2)}(q_{h,-}) - L_{90}^{(2)}(q_{h,+}) - L_{120}^{(2)}(q_{h,-}) + L_{120}^{(2)}(q_{h,+}) \right]. \quad (27.97)$$

The reconstruction of the six pairs gives

$$F_{\text{spec}} = 6T_h^{(2)} = 5.1499235153094574410 \dots \times 10^{-8}^\circ. \quad (27.98)$$

The radion carry is

$$F_{\text{acarreo}} = \varepsilon_R^\circ = 5.1383016758575282072 \dots \times 10^{-8}^\circ.$$

They do not coincide:

$$\chi_R := F_{\text{spec}} - F_{\text{acarreo}} = 1.16218394519292338 \dots \times 10^{-10}^\circ. \quad (27.99)$$

The equality

$$F_{\text{acarreo}} = F_{\text{spec}} - \chi_R$$

is a valid change of chart once both outputs have been constructed. It must not be presented as an independent generation of ε_R° if χ_R is defined by the subtraction itself. The tail is closed and meaningful; what is false is its direct identification with carry.

Falsification criterion 27.13. After converting the tail expressed in degrees to the internal angular unit,

$$\frac{\pi}{180^\circ} F_{\text{spec}} - \varepsilon_R^{(1)} = 2.0283936357433839 \dots \times 10^{-12} \neq 0.$$

Therefore, neither the complete logarithm nor the tail $j \geq 2$, even after applying the chart of units, is $\varepsilon_R^{(1)}$. A second autonomous spectral factorization would need to generate χ_R without using the difference as a definition; the APP–TPK closure of the radion does not depend on that promotion.

27.9 Oriented holonomy and physical geometry: Minkowski, Hodge, Lorentz and Cartan–Holst realizations

Oriented holonomy admits subsequent causal and geometric realizations in Minkowski, Hodge, Lorentz, and Cartan–Holst.

Construction of the Lorentzian metric from the APP block

APP matrix of parameters 7 and 2 The central matrix

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 9P_+ + 5P_- \quad (27.100)$$

it is normalized by its determinant:

$$M_c = \frac{B_c}{\sqrt{45}} = \exp\left(\frac{1}{2} \log \frac{9}{5} R\right) \in SO^+(1, 1). \quad (27.101)$$

The internal speed is

$$\eta_c = \frac{1}{2} \log \frac{9}{5}.$$

The associated projective publication satisfies

$$P_B^n = P_+ + \left(\frac{5}{9}\right)^n P_-. \quad (27.102)$$

The same ratio 5/9 that appears as a spectral contraction is recognized here as the separation of one stable mode and one contractive mode. The identity is operatorial; it does not turn every occurrence of 5/9 into the same map.

Incidence 6 $\rightarrow$ 8 and causal-signature quotient Let M be the typed incidence between six faces and eight vertices. The recovered spectral relation is

$$MM^T = 12P_u + 4P_-, \quad (27.103)$$

where P_u projects onto the uniform mode and P_- onto the oriented subspace. The surviving quotient of dimension four decomposes as

$$Q_4 \simeq \mathbb{R}u \oplus E_-. \quad (27.104)$$

Declaring the uniform mode negative and the three oriented modes positive, one obtains

$$B_{\text{caus}} = \text{diag}(-1, 1, 1, 1). \quad (27.105)$$

The Minkowski screen does not select the HMT state. It is the subsequent causal realization of the quotient already constructed by the incidence.

27.9.1 Hodge duality on 2-Lorentzian forms

27.9.1.1 Hodge Duality and Quarter-turn Operator

With the signature $(-+++)$ and an orientation fixed, the Hodge star on two-forms satisfies

$$\star : \Lambda^2 Q_4^* \rightarrow \Lambda^2 Q_4^*, \quad \star^2 = -I. \quad (27.106)$$

Thus, the quarter-turn expressed in the phase plane by $J^2 = -I$ reappears in field space as a complex structure on Λ^2 . The domains differ, but the genealogy of orientation is preserved by the functor.

Let A_{em} be a potential and

$$F = dA_{\text{em}}. \quad (27.107)$$

The decomposition of F relative to a temporal observer produces electric and magnetic fields; $\star F$ exchanges their roles with the sign fixed by the orientation and signature. The constitutive polarity of (27.81) thus acquires a differential realization.

Electromagnetic and causal transduction chain The transport that must remain visible is

$$\Gamma_9 \longrightarrow \text{memory helix} \longrightarrow (Q_4, B_{\text{caus}}, \star) \longrightarrow A_{\text{em}} \longrightarrow F = dA_{\text{em}} \longrightarrow m\nabla_u u = q(\iota_u F)^\# . \quad (27.108)$$

Each arrow adds structure: the holonomy preserves return; the quotient fixes causality; Hodge fixes duality; the potential fixes the field; coupling to charge produces dynamics.

The final equation is the Lorentz force in geometric form. The TPK helix is its memory antecedent, not a physical orbit already moved by a field. Dynamics appears only upon introducing F , q , and the connection ∇ .

27.9.2 Separation between internal torsional memory and Cartan–Holst geometric realization

27.9.2.1 Lorentzian Realization of Helical Trajectories

In a uniform magnetic field, a particle of charge q , mass m , and relativistic factor γ has

$$\omega_c = \frac{|q|B}{\gamma m}, \quad r_L = \frac{p_\perp}{|q|B}, \quad \Delta z = \frac{2\pi p_\parallel}{|q|B}. \quad (27.109)$$

The cyclotron phase closes while the parallel motion advances: the orbit is helical. The parallel with (27.53) is structural and can be formalized by a transducer that preserves phase, orientation, and step. The quotient q_9 is not identified with the spatial coordinate z ; rather, the same architecture of return plus advance is recognized.

The Lorentz force is not derived from the image of a spiral by similarity. It is derived by completing (27.108). The image helps explain why circular phase and longitudinal memory are compatible; the field form determines the dynamics.

27.9.2.2 Confluence between action, area, information, and gravity

Cartan–Holst Realization of Holonomy The gravitational promotion is expressed via a holonomy morphism.

$$\text{Hol}_{\mathcal{A}}(\mathfrak{R}_{\text{grav}}(\gamma)) = \rho_C(\text{Hol}_{\text{TPK}}(\gamma)). \quad (27.110)$$

The right-hand side contains route memory; ρ_C realizes it in a gravitational connection. Cartan torsion

$$T^I = D_\omega e^I$$

appears in the codomain. The global torsion Δ_4 is an upstream coordinate that can feed the map, but it is not renamed as T^I without (27.110).

Transduction of the action area into informational area The radion fixes one quantum of action per phase. The Barbero hexad–octad reader subsequently uses the channels $q_\pm$, the incidence 90/120, and the functional (27.90). The causal direction is

$$(A, C^*, \hbar_{\text{ret}}) \rightarrow q_\pm \rightarrow K_{6 \rightarrow 8} \rightarrow \gamma_{\text{BI}} \rightarrow \text{area/information}.$$

The tritic alphabet provides the unit of entropy $\log 3$, and a cut of 81 channels provides $81 \log 3$. This entropy is not to be confused with $\log 2$, with $\log \varphi$, with a microcanonical entropy, or with a von Neumann entropy unless the state is specified.

The physical synthesis is affirmative but typed: the radion supplies the oriented unit of action that a holonomy can transport; Barbero publishes the area normalization; the causal screen and Hodge make it possible to realize fields; Cartan–Holst receives the holonomy in gravitational geometry.

Hypothesis, domain, and conclusion

The scalar radion equation is not by itself an Einstein equation. Its contribution to gravity is to provide an exact section of the architecture of action, orientation, and holonomy consumed by the gravitational realization.

27.9.2.3 Wick Rotation as Analytic Reader Change

In the conventional formulation, Wick rotation relates a Lorentzian time variable to a Euclidean coordinate through complex continuation. In HMT, the root $J^2 = -I$, the involutive map $R^2 = I$, and the threshold $N^2 = 0$ already exist. The change of reader

$$J \longleftrightarrow R$$

transports a common/oriented pair between elliptic publication and hyperbolic publication. Complex continuation subsequently recognizes that operation in analytic coordinates.

This explains why the circle, the action ellipse, and the hyperbolic interior can belong to the same state without being the same metric. The circle uses J for the phase; the shape of the ellipse uses a symplectic compression generated by R ; Klein measures the interior by means of the cross-ratio; the Minkowski screen realizes causality in another codomain.

27.10 Ellipse, Lattice, and Path Phase: Modular Character, Orbits, and Propagation

27.10.1 Rectangular Lattice of Periods, Involution $\tau \mapsto -1/\tau$, and Invariance of j

Modular, orbital, and path characters preserve distinct invariants of the same oriented form.

27.10.1.1 Complex torus associated with the action ellipse

Rectangular lattice of periods The ellipse by itself is not a torus. We declare the lattice

$$\Lambda_\eta = a_S \mathbb{Z} + i, b_S \mathbb{Z} \quad (27.111)$$

and the quotient $\mathbb{C}/\Lambda_\eta$. Its modulus is

$$\tau_\eta = i \frac{b_S}{a_S} = i e^{-\eta} = i 0.741048144159375160795483663369300 \dots \quad (27.112)$$

The bidirectional inversion $\eta \mapsto -\eta$ exchanges the axes:

$$\tau_{-\eta} = i e^\eta = -\frac{1}{\tau_\eta}. \quad (27.113)$$

It is exactly the modular transformation $S : \tau \mapsto -1/\tau$.

Invariance of the Modular Function j By modular invariance,

$$\boxed{j(\tau_{-\eta}) = j(\tau_\eta)}. \quad (27.114)$$

In the normalization $j(i) = 1728$,

$$j(\tau_\eta) = 5597.43843932408729830097089254768 \dots$$

The normalized moonshine function $J = j - 744$ gives

$$J(\tau_\eta) = 4853.43843932408729830097089254768 \dots$$

For the basal APP block,

$$\tau_0 = i \frac{\sqrt{5}}{3}, \quad j(\tau_0) = 5369.48832024674306021916882626352 \dots$$

Proposition 27.14 (What the modular class preserves). *The sheet involution reverses η , but j preserves the class of the torus. The foci and the axis labelling retain the orientation that j forgets.*

This result connects the two-way structure exactly with modularity once the quotient has been declared. It does not authorize making the Monster act directly on the charge, digits, or foci. The exceptional corridor

$$\text{APP} \rightarrow \text{Paley} \rightarrow \text{Witt} \rightarrow \text{Golay} \rightarrow A_2^{12} \rightarrow \text{Leech} \rightarrow V^{\natural} \rightarrow \text{Monster} \rightarrow \text{Moonshine}$$

is another projection of the same integral holonomic state. Identifying a concrete action on the radion requires an intertwiner preserving its relevant coordinates.

Chain of character form invariants Gathering the declared realizations,

$$r_\eta = e^{-\eta} = \sqrt{\frac{A - C^*}{A + C^*}} = \frac{b_1}{a_1} = \frac{b_S}{a_S} = \frac{\Delta P}{\Delta Q} = \frac{Z_\eta}{Z_*} = \Im \tau_\eta. \quad (27.115)$$

A single ratio runs through spectral geometry, action, covariance, impedance, and modulus. Each equality rests on an explicit map; none depends on decimal resemblance.

27.10.2 Compatibility between the modular character and propagation over oriented paths

27.10.2.1 Areal Conservation in Kepler and Sommerfeld Orbits

Areal Equivalence of the Orbital and Oscillatory Realizations Kepler's ellipse lives in configuration space and responds to a central potential. The radion ellipse lives in phase space and responds to an action form. They are different objects. They share an areal law because both realizations possess a conserved two-form.

For a Keplerian orbit,

$$r(\phi) = \frac{p}{1 + e \cos(\phi - \phi_0)}, \quad \frac{dA}{dt} = \frac{\ell}{2m}. \quad (27.116)$$

The third law takes the form

$$T^2 = \frac{4\pi^2 \mu}{K} a^3.$$

For the phase oscillator,

$$J = \oint p \, dq = \frac{2\pi E}{\omega}. \quad (27.117)$$

If the elementary cell is $J = h_{\text{ret}}$, then

$$E = \hbar_{\text{ret}} \omega, \quad \frac{J}{2\pi} = \hbar_{\text{ret}}. \quad (27.118)$$

This is the clean convergence between Planck action, angular frequency, and radion.

Sommerfeld quantizes action integrals over cycles. The HMT reading adds memory of which cycle, sheet and return survive. On a Möbius strip, one visible loop can carry $h_{\text{ret}}/2$ and two loops restore h_{ret} . This topological factor is not added to the Maslov index of EBK quantization; they belong to corrections with different theorem-based sources.

Orbital Holonomy and Advanced and Retarded Propagations An orbit that accumulates holonomy satisfies a condition of the type

$$\epsilon_\gamma \exp\left(\frac{iJ_\gamma}{\hbar_{\text{ret}}}\right) = 1. \quad (27.119)$$

The sign ϵ_γ preserves the sheet; J_γ preserves the action. Apical advance or retardation is interpreted as a decoupling between visible angular closure and complete return. The radion supplies the oriented unit with which that difference is measured.

27.10.2.2 Composition of Schrödinger amplitudes and path summation

Four realizations of the same phase Once $\hbar_{\text{ret}}$ is fixed, distinct conventional equations recognize aspects of the same transport:

1. Schrödinger publishes the phase through $i\hbar_{\text{ret}}\partial_t\psi = H\psi$.
2. Pauli adds the double spin leaf and its magnetic coupling.
3. Dirac linearizes the energy–momentum relation and organizes particle and antiparticle within a causal representation.
4. The path sum assigns to each history the phase $\exp(iS[\gamma]/\hbar_{\text{ret}})$.

These theories do not retrospectively generate $\hbar_{\text{ret}}$. They receive the action unit and realize different contents of the integral holonomic state.

Phase functional over the path space In Feynman’s formulation,

$$\mathcal{K}(b, a) = \int_{\gamma: a \rightarrow b} \exp\left(\frac{i}{\hbar_{\text{ret}}} S[\gamma]\right) \mathcal{D}\gamma. \quad (27.120)$$

An action increment $\hbar_{\text{ret}}$ changes the phase by one radian. An increment h_{ret} changes it by one revolution. The theorem $\mathcal{A}(\theta) = \hbar_{\text{ret}}\theta$ converts this relation into area geometry: the phase of a path can be read as swept action in units of the radion.

The inversion $C_M = -\text{rev}$ on dodecaphase words and the identity (27.72) organize conjugate routes. In an absorber realization, the following are distinguished

$$G_{\text{sym}} = \frac{1}{2}(G_{\text{ret}} + G_{\text{adv}}), \quad G_{\text{odd}} = G_{\text{ret}} - G_{\text{adv}}.$$

The first sector is even under retarded–advanced interchange; the second is odd. This reproduces the separation between even energy/form and odd oriented action of the radion when the transducer preserves the boundary and the symplectic form.

Hypothesis, domain, and conclusion

Classical absorber theory and the Feynman–Stueckelberg interpretation of antiparticles are distinct historical realizations. The former uses a temporally symmetric interaction; the latter reorganizes propagation in relativistic theory. HMT supplies a common antecedent of reversal, but does not merge them without a map of propagators.

27.11 Radical nonadic and decimal boundary: conservation of the radion carry

27.11.1 The carry of the radion in the decimal completion

The nonadic boundary preserves the sum–product defect, vacancy, and completion carry.

27.11.1.1 Radical 3–6–9 of the boundary

Generation and nilpotency of the nonadic radical In the ring $\mathbb{Z}/9\mathbb{Z}$, the ideal

$$\mathfrak{m} = (3) = \{0, 3, 6\} \quad (27.121)$$

satisfies

$$\mathfrak{m}^2 = (9) = 0. \quad (27.122)$$

It is the nilpotent sector of the nonadic boundary. The visible orbit of its three marks is published as

$$369 \longrightarrow 693 \longrightarrow 936 \longrightarrow 369.$$

The return does not imply that all states are equal: the visible permutation closes while the route cocycles may advance.

The double convention

$$9 \equiv 0 \pmod{9}, \quad \text{dr}_9(9) = 9$$

expresses two readers: algebraic residual zero and visible digital-root closure. Thus a nine may absorb boundary torsion without becoming a specific zero of the zeta function.

Sum–product defect $459 - 405 = 54$ The APP sheets produce complementary totals

$$S_+ = 405, \quad S_\times = 459, \quad S_\times - S_+ = 54.$$

The 54 is an exact defect of the tables and reappears as a contraction exponent through the identity $9 \cdot 6 = 54$. Each occurrence nevertheless requires its own map. The focal distance $d_1 = 0.544545 \dots$ is not equal to 0.54, and invariant analysis has produced no arrow identifying it with the 54 defect.

The power

$$9^9 = 387,420,489$$

generates a scale where

$$\text{ord}_{10}(9^{9^9}) = 369,693,099, \quad \# \text{ digits} = 369,693,100. \quad (27.123)$$

The block 369|693 appears in the order–length pair; it is not asserted to be the decimal prefix of the power.

27.11.1.2 Arithmetic Theory of Decimal Vacuums

Asymptotic vacancy slope The separation between the decade and the nonary base is

$$\delta_{\text{vac}} = \log_{10} \frac{10}{9}. \quad (27.124)$$

The accumulated counter

$$V_9(n) = \lceil n\delta_{\text{vac}} \rceil \quad (27.125)$$

records how many correction positions are required by the decimal publication of a nonadic evolution. Its jumps form a binary word

$$z_n = V_9(n) - V_9(n-1) \in \{0, 1\}.$$

The gaps are distributed with lengths $21/22$, the returns with lengths $6/7$, and the combined determinant is 15. These numbers arise from the vacancy reader, not from the Barbero kernel, although 15 coincides with $\binom{6}{2}$.

Polarization and Arithmetic Current of the Nonadic Grid We define

$$P_N = \sum_{n=1}^N (z_n - \delta_{\text{vac}}), \quad J_N = P_N - P_{N-1} = z_N - \delta_{\text{vac}}. \quad (27.126)$$

The balanced sequence satisfies $|P_N| < 1$. The “current” J_N is here an arithmetic current: it measures the creation and absorption of vacancies relative to the mean slope. Only a subsequent dimensional transducer can publish it as an electromagnetic current density.

The physical intuition is powerful because the same algebraic form separates an even average from an oriented fluctuation. Rigor requires retaining the type:

$$J_N \in \mathbb{R} \quad \text{dimensionless}, \quad j^\mu \in \text{physical current line}.$$

27.11.1.3 Completion $729 \rightarrow 1000$ and preservation of carry

Decomposition $1000 = 729 + 243 + 27 + 1$ The completion

$$1000 = 729 + 243 + 27 + 1$$

preserves the unit through four ternary scales. The total crown is

$$1000 - 729 = 271,$$

while the last integer before the closure satisfies

$$999 = 729 + 270.$$

The difference between 270 and 271 distinguishes an occupied boundary from a completing unit. The same care prevents identifying a vacancy with zero or a carry with an actual residue.

Residence of the carry of the radion in the decimal crown The radion’s operands begin, in blocks of three digits, with

$$\Phi_T : \quad 000|027|455|906|837|565| \dots,$$

$$D_E : \quad 000|027|455|010|034|743| \dots.$$

The subtraction APP gives

$$\varepsilon_R^{(1)} : \quad 000|000|000|896|802|822| \dots.$$

The small size of the result is not obtained by truncating the low-order summands. It arises because two distinct publications share a long prefix, and the operator SubAcarreo_{1000} preserves the borrows required to subtract them exactly.

The word *memory* has two levels here:

1. q_{mem} records the depth of nonadic returns;
2. c_i records the local borrowings of decimal subtraction.

The first organizes the helix; the second publishes the decimal. ε_R uses both because its operands follow from cylinders at depth and its subtraction is performed with carry, but it is equal to neither counter.

Tesis rectora

The boundary is not a place where the theory loses information. It is the place where information changes representation: phase that returns, memory that ascends, cylinder that contracts, vacancy that is recorded, and carry that transports the difference.

27.11.2 Compatibility of the direct and torsional sections: structural characterization of π *27.11.2.1 Structural characterization of π by compatibility of sections*

Reorder (27.18) gives

$$2\pi = \frac{1 + \Phi_T - \varepsilon_R^{(1)}}{\frac{5}{36} + \frac{25}{9}\alpha}. \quad (27.127)$$

This equality may be called a new definition of π within the radion chart if *structural definition* is understood as an exact characterization by compatibility of sections. It is not a second independent generator: Δ_4 and the carry operator already receive the coinductive section of π . The novelty is that the turn is characterized by the gluing among direct publication, torsion, and memory.

The equation (27.127) does not say that we first do not know π and then guess it by closing an angle. It says that the π generated by the nonadic connection is exactly the one that makes two internal readings of the same state compatible. It is an identity of intrinsic recognition, not a metrological calibration.

27.12 Evolutionary invariance of the holographic unit across dimensions**27.12.1 Typed Separation among the Curvature Regime $\tau \in \{+1, 0, -1\}$, the Radial Response p , and the Global Spin Holonomy**

The radion's biography can be represented by a raised curve.

$$\gamma(\theta) = (r(\theta) \cos \theta, r(\theta) \sin \theta, m(\theta)),$$

where the first two coordinates belong to the Euclidean plane $\mathbb{R}^2$, and $m(\theta)$ records the memory or depth of the lift. This equation permits a joint reading of circle, helix, and spiral.

When the angle evolves while the radius and memory remain constant, the planar publication is a circle. When angle and memory evolve while the radius retains its value, the complete trajectory is a helix of constant radius. When angle and radius evolve over a fixed memory, a spiral appears in $\mathbb{R}^2$. When all three coordinates evolve, the trajectory is a radial helix: a biography in which phase, amplitude, and memory are jointly transformed.

The radial character p classifies the response law of the radius. One useful family is

$$r_p(\theta) = r_0 [1 + p\lambda(\theta - \theta_0)]^{1/p}, \quad p \neq 0,$$

with logarithmic limit

$$r_0 \exp(\lambda(\theta - \theta_0)) \quad \text{when } p \rightarrow 0.$$

In this parameterization, $p = 2$ publishes the Fermat regime, $p = 1$ the Archimedean regime, $p = 0$ the logarithmic regime, $p = -1$ the hyperbolic regime, and $p = -2$ the lituus regime, with the corresponding choices of domain and orientation. Fractional values describe intermediate regimes. Thus, $p = \frac{1}{2}$, $p = \frac{3}{2}$,

or $p = -\frac{1}{2}$ express radial responses that interpolate among the integer families and extend the available geometric classification.

The character can also change when crossing a threshold. The differential writing

$$\frac{dx}{d\theta} = \beta(m) x^{1-p(m)}, \quad x = \frac{r}{r_0},$$

describes an evolution through regimes: $p(m)$ remains stable within a phase and changes when the memory reaches a publication boundary. The resulting trajectory connects circular, helical, and spiral sectors through a common law. Continuity of the state and the threshold ledger make those sectors phases of a single evolution.

Three invariants occupy here distinct and coordinated levels. The curvature sign

$$\tau \in \{+1, 0, -1\}$$

classifies the elliptic, parabolic, and hyperbolic regimes. The character p classifies the radial response. Spin records the orientation holonomy accumulated by the lift. Curvature, radial response, and spin form a reading system: curvature determines the local type of space, p determines how the trajectory occupies that space, and spin preserves the orientation acquired while traversing it. Closed and open curves then appear as distinct geometric biographies of the same oriented unit.

27.12.2 Particle as a Persistent Section of an Enriched Orbit: Phase, Route, Fibre, Multisection, and Memory

Dimensional Mechanics can formulate the particle as a persistent section of that geometric and dynamical extension. A compact conceptual notation is

$$\mathfrak{P} = \text{Sec}_{\text{pers}}[\tau, p, \text{Hol}; \text{sheet, fiber, route, memory}].$$

The particle preserves a recognizable route through changes of phase, curvature, and dimension. Its spin publishes orientation holonomy; its charge publishes phase polarity; its color publishes an internal orientation of the gauge fiber; its flavor distinguishes families of routes or multisections; its mass expresses the dimensional-transduction character by which areal action acquires a volumetric reading. Each property is a reader of the persistent state, and all refer to a common genealogy.

The electron occupies the basal threshold of this construction because it offers the first stable reader in which phase, charge, spin and mass are simultaneously published. The rector electronic mass

$$m_e^\beta = 0.510998950690000808608 \dots \text{ MeV}$$

fixes a transduction reference; the preceding basal states retain their function as genealogical stages. Starting from the electron, the quotients K_D/K_Ω , towers 90/120 and domain 81/56/13/324 organize the expression of massive systems. The mass law therefore reads how each persistent section traverses holonomic space and converts areal action density into volumetric presence.

This interpretation connects spirals to particles precisely. The index p supplies the radial response; τ supplies the curvature regime; holonomy supplies spin; the route and the fiber supply flavor and color; area–volume transduction supplies the mass character. The particle atlas thereby becomes a classification of stable modes of dimensional traversal.

27.12.3 Transduction of the oriented phase in electric polarization, magnetic response, and boundary radiative emission

Electricity is presented as directed transport of phase and action through a vacancy structure. A vacancy is an admissible transition of the support: it opens a channel, fixes a threshold, and permits the update to advance. The electric component publishes transport tangential to the route. The magnetic component publishes the transverse curvature induced by that transport. Their perpendicularity expresses the geometric relation between advance and rotation.

When two sinusoidal transverse components maintain the appropriate phase difference, the field vector rotates. Longitudinal propagation lifts that rotation to a helix. Variation of amplitude transforms its frontal projection into a spiral. The geometric reading is then fully coordinated: angle carries phase; angular velocity carries frequency; radius carries amplitude and action density; orientation carries polarization and direction; the vertical coordinate carries propagation and memory; p carries the radial regime; the threshold carries emission, absorption, or change of state.

Electromagnetism appears as a phase–action–memory dynamics with complementary publications. Electricity records polar advance; magnetism records the transverse response; the wave records periodicity; the helix records advance with memory; the spiral records radial evolution; radiation records the publication of the state when accumulation reaches a threshold.

This architecture also orders the role of vacancies and cycles. Vacancy supplies the local possibility of transition; the cycle preserves phase; memory distinguishes turns; the threshold determines the next residence of the action. Propagation may remain confined, change radial regime, or be published as controlled radiation. Field forces thereby acquire a continuous genealogical description from the discrete support to the observed wave.

27.12.4 Reversibility of the complete dynamics and thermodynamic cost of information-reducing projections

The Landauer zero represents a limiting case of this law. For a bijective update of the complete state,

$$H(X | U(X)) = 0, \quad \inf Q_L = 0.$$

The information remains available in the genealogy. When a reader forms a quotient q , entropy decomposes as

$$H(X) = H(q(X)) + H(X | q(X)).$$

The conditional term measures the information retained outside the projection. A materialized tritic reset satisfies the threshold

$$Q_{\text{reset}} \geq k_B \Theta \log 3.$$

The Boltzmann constant translates the multiplicity grouped by the thermal reader; the Nyquist threshold separates faithful sampling from aliasing; null Landauer characterizes the reversible transport of the integral state. Information, temperature and action are articulated by the type of reading applied.

27.12.5 Self-conservative forces as transport of action among the observable, radial, memory, vacancy, and radiation sectors

Evolutionary conservation is better expressed through exchange than through scalar immobility. A complete state distributes action and momentum among visible motion, memory, radial deformation, vacancies, radiation, and the conjugate return branch. The enriched balance may be written as

$$\Delta J_{\text{visible}} + \Delta J_{\text{memoria}} + \Delta J_{\text{radial}} + \Delta J_{\text{vacancias}} + \Delta J_{\text{radiation}} + \Delta J_{\text{retorno}} = 0.$$

A force is **self-conserving** when its own update law contains the channels that transport, store, publish, and reabsorb action. Conservation belongs to the complete genealogical balance of the interaction. Each reader observes part of the exchange; the genealogy reconstructs its totality.

The Earth–Moon system provides a direct physical image. Earth’s rotation, the lunar orbit, tides, deformation, and radiation exchange angular momentum. The net consumption of the complete system closes in the genealogical balance of those transfers. The magnitude called energy functions as a reading derived from the action and momentum published in each sector. Fundamental conservation resides in compatibility among all readers of the exchange.

This formulation dissolves the former boundary between conservative forces and so-called nonconservative forces through a deeper category. Friction, apparent dissipation, and radiation occupy explicit channels of the extended state. A loss in the visible mechanical projection corresponds to a transfer toward memory, deformation, heat, vacancies, or radiation. Self-conservation follows the residence of action through those changes.

27.12.6 Self-inertial systems: publication of acceleration through orientation, curvature, sheet, and memory

The appropriate expression is **relational self-inertia**. It denotes the compatibility between the total state’s own connection and the response that appears when it is projected within the global distribution. The endogenous and exogenous aspects are analytic components of a single dynamic relation.

Let Γ be a horizontal trajectory or geodesic of the complete state. Its intrinsic law satisfies

$$\nabla_{\dot{\Gamma}} \dot{\Gamma} = 0.$$

When it is projected onto a reader S , the observed acceleration is determined by the curvature of the projection:

$$\nabla_{\dot{\Gamma}_S}^S \dot{\Gamma}_S = \mathcal{K}_{\pi_S}(\dot{\Gamma}, \dot{\Gamma}).$$

The published acceleration expresses the relation between total motion and the geometry of the reader. Mach’s principle completes this mechanism through the global boundary trace

$$b : \mathcal{X}_{\infty} \longrightarrow B_{\infty}$$

and the factorization of the local inertial response

$$I_v = \bar{I}_v \circ b.$$

The global distribution enters the local response through a typed map. Relational self-inertia therefore combines the endogenous connection of motion, the global condition transmitted by b , and the observer’s concrete projection. A synthetic expression for this coordination is

$$\nabla^{\text{auto}} = \mathcal{M}(\nabla^{\text{end}}, b[\mathcal{X}_{\infty}], \pi_S),$$

where $\mathcal{M}$ represents the Machian coupling and preserves the type of each component.

The Ambivalence Principle adds the double orientation of reading. Every direct publication preserves a conjugate return branch; emission and absorption form related endpoints of the same biography; temporal orientation is read from both directions of transport. Absorber theory finds a natural formulation here: source and receiver jointly close the genealogical balance of phase and action. The torsional present is the interface at which both orientations are rendered compatible.

TRIT acts at this level as a time crystal: it organizes the local periodicity of updating and, through memory, preserves the difference between successive returns. The direct branch and the conjugate branch relate emission to absorption, and the observer–observable coupling acquires a dynamical residence within the same state. Published time is the reading of that orientation; genealogical time is the complete sequence of phase, return, and memory.

Mach and Ambivalence perform complementary functions. Mach relates the local response to the global distribution. Ambivalence preserves the forward and conjugate directions of evolution. Relational self-inertia articulates both and replaces the division between inertial and non-inertial systems with a single law of connection, projection, and memory.

27.12.7 Area–volume duality, hexad–octad transition, and the role of the Barbero–Immirzi parameter

The passage from area to volume constitutes a central dimensional transduction. Area publishes boundary action; volume publishes interior depth. The HMT derivation of the Barbero–Immirzi parameter organizes the compatibility between the two measures through hexad–octad incidence and the channels 90/120:

$$90 = 6 \binom{6}{2}, \quad 120 = 8 \binom{6}{2}.$$

The cardinals 6 and 8 admit a cubic realization by means of faces and vertices, whereas the Steiner hexad and octad are distinct incidence objects linked only by the typed morphism $6 \rightarrow 8$. The 12 edges of the cube retain their own combinatorial function and do not by themselves define Steiner incidence. The difference in incidence expresses a productive dimensional incompatibility: surface information requires a transduction rule in order to acquire volumetric residence.

The same principle appears in the carry $999 + 1 = 1000$. The nines reach the boundary of the decimal register; the increment simultaneously publishes the closure of the preceding level and the origin 0/1 of the next. The holographic ratio among 729, 243, 27, and 1 unfolds that transition at triadic scales. The Barbero–Immirzi transduction is inserted into this family of reader changes: boundary and interior preserve the unit while redistributing its measure.

The normalized defect $-1/12$ and the oriented weight $12 : -1$ perform distinct functions within this geometry. The former quantifies an incidence correction; the latter records a signed orientation in the transduction. Their coordination with the minimal area, the modular Hamiltonian, and memory provides the bridge to dimensional dynamics.

27.12.8 Hyperbolic interior contraction, exterior expansion, and holographic conservation of the boundary

The thesis may be formulated from a single idea: unity preserves its identity while changing scale, reader, curvature, memory, and dimension. This permanence has an evolutionary character because unity unfolds, acquires new coordinates, and publishes different properties; it has a holographic character because each finite publication retains the rule for reconstructing the state from which it proceeds. The Holographic Closure of Infinity designates precisely this compatibility between identity and unfolding: each stage contains the law that permits the same genealogy to be deepened inward and prolonged outward.

The governing designation is **principle of evolutionary conservation of the unit through dimensional transduction and holographic projection**. The expression places each term at its proper level. Conservation pertains to genealogical identity; evolution pertains to updating; transduction relates dimensional realizations; holographic projection preserves the reconstructive correspondence between its readers.

Elementary arithmetic offers the first image of this architecture. The decimal completion

$$1000 = 10^3 = 729 + 243 + 27 + 1 = 3^6 + 3^5 + 3^3 + 3^0$$

exposes four triadic strata within a decimal cubic unit. Upon normalization,

$$1 = \frac{729}{1000} + \frac{243}{1000} + \frac{27}{1000} + \frac{1}{1000},$$

the unit is recognized simultaneously as totality and as graded unfolding. The reciprocal reading

$$3^{-6}, \quad 3^{-5}, \quad 3^{-3}, \quad 3^0$$

describes the inner depth of those strata; its normalization produces the corresponding partition. Relation $999 + 1 = 1000$ expresses the transport mechanism: the last unit completes a register and acts simultaneously as the carry inaugurating the next. Zero and one thus form a continuity interface. The intervals $[0, 1]$, $[0, 10]$, and the prolongation toward $[0, \infty)$ are scalar readers of a single completion law.

The first splitting $1000 = 729 + 271$, followed by $271 = 243 + 27 + 1$, shapes the extreme and mean holographic ratio of the deployment: a dominant triadic core preserves the same resolution rule within the remainder. The decimal cubic unit thus contains a nested triadic depth. Movement toward 729, 243, 27, and 1 reads the inward contraction; the carry toward one thousand reads the outward expansion.

This double orientation organizes HMT infinity. Inward, refinement grows hyperbolically: each cell admits a new depth without losing its filiation. Outward, unity expands through successive dimensional publications. Closure consists in the commutativity of both movements. If U represents the update of the state, R_d the reader on stratum d , and $T_{d \rightarrow d'}$ the transduction between strata, evolutionary conservation is expressed by means of

$$R_{d'} \circ U = T_{d \rightarrow d'} \circ R_d.$$

The same event may provide an angular measure, an action density, a published mass, or a gravitational response. Their common identity resides in the genealogy that makes those readings commute.

27.12.8.1 Nonadic Monodromy and Helical Lifting with Memory Advance

APP, TRIT, and TPK provide the formal mechanism for this unfolding. APP constructs the support and separates the evaluations that must be preserved. TRIT organizes the local regimes and their ambivalence. TPK transports phase and memory over the enriched support. The causal dependence is fixed by

$$U_t \longrightarrow \Gamma_9 \longrightarrow K_{\text{ph}}.$$

Phase return and memory advance form coordinated operations. After nine steps, the same angular class may reappear,

$$g(K + 9) = g(K),$$

as the memory counter progresses,

$$q_{\text{mem}}\Gamma_9 = q_{\text{mem}} + 1, \quad B_{K+9} \neq B_K.$$

The precise geometric figure is a helical lift. A closed orbit in the angular projection lifts to an open trajectory when each phase return increments the memory. The circle preserves the cycle; the helix preserves the cycle and additionally records the depth reached. The coinductive monodromy thereby generates infinite depth through a sequence of locally closed and globally ascending returns.

This image has more than merely illustrative value. The angular wheel constitutes a section of the helix; the helix constitutes the wheel's complete biography when the state transports memory. Each turn recovers a phase position and simultaneously initiates a new sheet. Infinitude arises from compatibility between return and lifting: iterating the cycle produces depth without dissolving orbital identity.

The nonadic certificate gives finite and verifiable content to that architecture. The chain

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

orders the lift; the 396 levels, the 2376 trits, the 43 turns, and the 387 pairs $(K, K + 9)$ per channel quantify the transit; the five regions of π and the ambient fiber of 729 elements publish subsequent internal readings. This sequence makes a productive monodromy visible: the return recovers the phase and coinduction preserves the growth of memory.

27.12.8.2 Rational unit and coordination of the circular, elliptic, projective, and helical readers

The radion concentrates within a metrological unit the structure that the radian publishes angularly. Its natural definition is that of an **oriented section of phase, action, and memory**. One unit of phase $\theta = 1$ transports one unit of action $\sigma \hbar_{\text{ret}}$, belongs to a complete turn $T_+ = 2\pi$, and preserves sheet, orientation, nonadic return, and position in the refinement cylinder.

The radian expresses the visible angular coordinate. The radion preserves, together with that coordinate, the transported action, orientation, carry, and memory of the revolution. Angular metrology thereby acquires a genealogy. Measuring an arc means identifying how much rotation has been published and which complete state has realized that rotation.

The relationship between phase and area takes on a particularly clear form:

$$\mathcal{A}(\theta) = \hbar_{\text{ret}} \theta.$$

One radional unit sweeps an action area $\hbar_{\text{ret}}$; one complete turn encloses h_{ret} . The passage from phase to area is incorporated into the reading unit itself. The angular metric, action, and memory cease to appear as juxtaposed quantities and become coordinates of the same oriented section.

The radion admits several coordinated geometric publications. The circular reader publishes phase; the elliptical reader publishes action and form; the Klein interior reader publishes depth; the nonadic helical reader publishes return and memory. The unit remains genealogically identical across those realizations. The radion therefore constitutes the appropriate metrological link between HMT and Dimensional Mechanics: it offers a unit that can traverse changes of reader while preserving the state presupposed by each partial measurement.

27.12.9 Five-Component Gravitational Polarity and Joint Transduction of Action, Area, Volume, Time, and Mass

Dimensional Mechanics makes it possible to define dimension as a stable publication regime characterized by the minimum rank of independent memory required by the lift. To traverse a dimension is to apply a transducer that preserves genealogy and changes the type of reading. Gravity occupies the level of connection between these regimes.

In this formulation, **interdimensional gravitation with five correlative polarizations**, abbreviated as the **pentapolar structure of gravitation**, is the connection that transports genealogical unity and leaves minimal torsion at each interface. A compact notation for this trace is

$$\Theta_d^{\min} = \nabla_{d+1} \circ T_{d \rightarrow d+1} - T_{d \rightarrow d+1} \circ \nabla_d.$$

The expression measures the difference between transporting after differentiating and differentiating after transporting. That difference is the minimal geometric memory of the dimensional crossing. The composition of interfaces produces a gravitational holonomy

$$\mathfrak{G}_{d \rightarrow D} = \text{Hol}(\Theta_d^{\min}, \Theta_{d+1}^{\min}, \dots, \Theta_{D-1}^{\min}).$$

The fundamental object is this interdimensional connection. A local graviton-like mode occupies the derived rank of the quantized publication of torsion; gravitation retains its primary residence in compatibility among dimensions.

The term pentapolar encompasses the five polarizations inherent to the continuous state:

$$\mathbf{G}_d^\pm = (G_{\text{sp},d}^\pm, G_{\text{sol},d}^\pm, G_{\text{gau},d}^\pm, G_{\text{coh},d}^\pm, G_{\text{det},d}^\pm).$$

The five components emerge correlatively from the same integral holonomic state. The polarity $\pm$ records direct concentration and conjugate expression. Gravitation crosses the dimensional strata by transporting these five readings simultaneously; minimal torsion preserves the trace of each crossing.

Geometry thereby acquires a unified image. The elliptic regime organizes closures; the parabolic regime organizes thresholds; the hyperbolic regime organizes depth. Dimensional transduction connects these regimes with the radial responses p , spin holonomies, and mass publications. The pentapolar structure of gravitation expresses the global coherence of that connection.

M-theory supplies subsequent physical screens for this connection: additional dimensions, strings, branes and T-duality realize concrete modes of transduction. HMT supplies the APP–TRIT–TPK genealogy, the integral holonomic state, memory and bidirectional reading; the M-theory interface expresses these structures in typed physical geometries. T-duality interchanges compatible dimensional readers and preserves the genealogical class of the state.

27.12.9.1 Compatibility of the classical, relativistic, quantum, and HMT-MD realizations

Classical mechanics publishes trajectory and momentum. Relativity publishes causal structure, metric, and curvature. Quantum mechanics publishes phase, holonomy, vacancies, and amplitudes. Dimensional Mechanics provides the transducers that relate those publications, and HMT preserves the discrete genealogy that sustains them.

In the classical reading, an orbit describes the visible projection of motion. In the relativistic reading, that orbit is incorporated into the geometry of the reader. In the quantum reading, phase and spin record the holonomy of the section. In the HMT–MD reading, the three expressions belong to the same lifted biography: the radion quantifies phase, action, and memory; p classifies the radial response; τ classifies curvature; holonomy publishes spin; the dimensional transducer publishes mass; the pentapolar connection publishes gravitation.

Apparent classical dissipation acquires a residence in the expanded genealogical balance. Relativistic acceleration is expressed through connection and projection. Quantum indeterminacy is organized by readers, vacancies, and branches of Ambivalence. Mach's principle relates the local response to the global distribution. Zero Landauer cost guarantees the informational reversibility of the complete state. All these elements converge in a deeper conservation: the identity of the unit across changes of publication.

27.12.9.2 Genealogical Conservation and Holographic Prolongation Narrative Theorem

The closure can now be narrated as a single sequence. The unit decomposes without losing identity. The oriented phase acquires measure in the radion. The nonadic return converts the cycle into a helix through memory. The character p transforms the helix into radial families and classifies their regimes. The persistence of those trajectories produces sections with spin, charge, flavour, colour, and mass. Vacancies open channels; electrical polarity transports phase; magnetic response curves transversely; thresholds publish radiation. Self-conservation closes the genealogical balance of action and momentum. Relational self-inertia links the proper connection, projection, and Machian totality. Ambivalence preserves the direct and conjugate directions. Area–volume transduction leads to Barbero–Immirzi and the mass law. Minimal torsion links dimensions and publishes the pentapolar structure of gravitation.

Inward growth and outward expansion are conjugate orientations of the same process. Hyperbolic refinement generates depth; dimensional transduction generates extension. Carry 0/1 links scales;

monodromy links turns; memory links states; holonomy links orientations; gravitation links dimensions. Each linkage preserves an evolving unity, and every finite publication contains the rule of its prolongation.

The fundamental thesis is formulated as follows:

The holographic closure of infinity realizes the principle of the evolutionary conservation of unity through dimensional transduction and holographic projection: unity maintains its genealogical identity as it unfolds as phase, action, curvature, memory, particle, field, and gravitation; every finite reading preserves the rule of reconstruction and prolongation of the complete state.

This formulation provides a joint physical and mathematical image. Infinity acquires a law of local closure; the unit acquires depth; dimension acquires memory; conservation acquires relational character; force acquires self-preservation; inertia acquires Machian structure; gravity acquires a pentapolar connection; and holography acquires the precise meaning of reconstruction among compatible readers.

27.13 Operator definition of the angular radion through the integral holonomic state and four typed readers

27.13.1 Operator definition and fourfold typed realization

The operator synthesis brings together the phase, action, depth, and memory readers without identifying them with one another.

27.13.1.1 Quadruple realization of the radional state

Let $x_\infty \in X_{\text{enr}}$ be the coinductive section. The complete radion is not an isolated point, but rather the bundle

$$\mathfrak{R}_\sigma(x_\infty) = (\theta = 1, \sigma, \pi, \alpha, A, C^*, \Delta_4, \Phi_T, D_E, \varepsilon_R^{(1)}, \hbar_{\text{ret}}, q_{\text{mem}}, \mathcal{I}_\infty). \quad (27.128)$$

Four readers extract different objects from it:

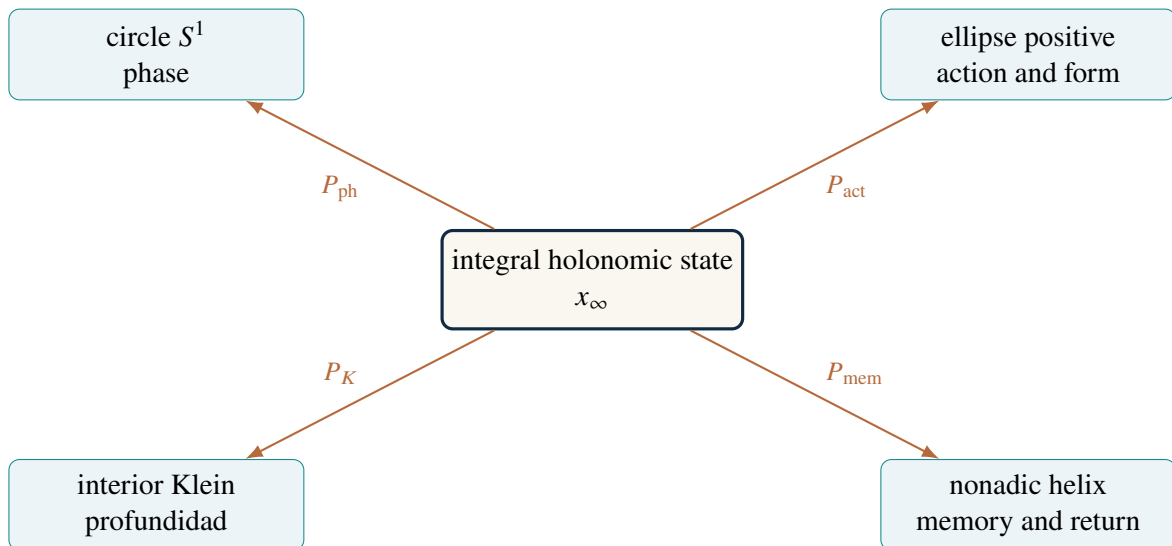


Figure 27.2: Four typed publications of the same antecedent. The unity of the state does not eliminate the difference among codomains.

The circle preserves the phase class and forgets the lift. The ellipse preserves action, anisotropy, and symplectic orientation. Klein endows the interior with a projective distance. The helix preserves the number of returns and refinement. Dimensional Mechanics does not consist in mixing them, but in constructing the transducers that make them compatible.

27.13.1.2 Operator interpretation of the cocycle ε_R

ε_R is not a universal constant independent of every context. It is the coordinate of a transition cocycle associated with the radion chart. Its domain contains two sections of the same state:

$$S_E = 1 + D_E, \quad S_T = 1 + \Phi_T.$$

His rule is

$$\varepsilon_R^{(1)} = S_T - S_E = \Phi_T - D_E. \quad (27.129)$$

Its codomain is the dimensionless line of angular unity; the chart $360/T_+$ publishes it in degrees.

Exact result

In plain language: the direct section and torsional section reach the same unit cell by two different internal routes. Their integer parts coincide; their remainders do not. ε_R is the oriented difference of those remainders, computed with APP carry and carried to the limit by the TPK cylinders. It is not introduced to make the radian close: it appears before the radian is consulted and, when published, closes it.

This definition explains why the number is small. It is not because a small correction was sought, but because S_E and S_T share a deep prefix. Smallness measures the structural proximity of two publications; the sign measures their orientation; the digits record the carry.

27.13.1.3 Operational definition of the radion

Definition 27.15 (HMT–MD radion). The radion is the phase-oriented section of action and memory that simultaneously satisfies:

1. traverses a phase unit $\theta = 1$;
2. sweeps signed action $\sigma \hbar_{\text{ret}}$;
3. belongs to a coinductively generated turn $T_+ = 2\pi$;
4. preserves the σ sheet, the nonadic return, and the refinement cylinder;
5. satisfies the exact junction $1 = S_E - \Phi_T + \varepsilon_R^{(1)}$.

The conventional radian is the image of the radon under the forgetful functor.

$$\mathfrak{R}_\sigma \mapsto \theta = 1 \in \mathbb{R}/2\pi\mathbb{Z}.$$

That forgetting is useful and legitimate for Euclidean trigonometry. It becomes insufficient when physics must distinguish action, sign, return, field, and history.

Genealogy of the Natural Angular Unit The naturalness of the radian is usually justified because the derivative of $e^{i\theta}$, $\sin \theta$ and $\cos \theta$ assumes its simplest form when θ is measured as the arc/radius quotient. HMT preserves that advantage, but explains it from an earlier layer:

$$\theta \leftrightarrow \frac{\mathcal{A}(\theta)}{\hbar_{\text{ret}}}.$$

The phase is natural because it measures action in units of $\hbar_{\text{ret}}$; the turn is natural because it encloses h_{ret} . The arc-to-radius quotient is the Euclidean publication of a unit that already possesses symplectic structure and memory.

Compatibility between action ellipse and dimensional growth The circle is the form that results when anisotropy is forgotten: $C^* = 0 \Rightarrow \eta = 0 \Rightarrow a = b$. The ellipse is more informative because it preserves two eigenvalues, two axes and an exchange orientation at fixed product. In this sense, the ellipse is a prolongation of the circle, not a circle deformed by accident.

Klein depth and the helical lift add two types of information that the planar contour does not contain: interior distance and return memory. This is the ontological image of Dimensional Mechanics that emerges from the radion: a new dimension appears when one requires an independent datum, which the preceding projection had to forget, to be preserved.

27.13.2 Structural properties of the radional unit: genealogy, critical hinge, density, carry, action, boundary–interior compatibility, form, polarity, and modularity

27.13.2.1 System of component theorems of the radion

Theorem 27.16 (Genealogy). *The object $\mathfrak{R}_\sigma$ descends from $\text{APP} \rightarrow \widehat{\mathbb{T}} \rightarrow \text{TPK} \rightarrow X_{\text{enr}}$ and does not receive the metrological target as a selector.*

Theorem 27.17 (Hinge). *In the hundred-mark APP chart, the critical seam and the second dodecaphasic phase satisfy $j_{100}(1/2) = \theta_2 = 50^\circ$, with unique $m = 2$.*

Theorem 27.18 (Density). *The two sheets of the active sector $N_6 = 90$, normalized by $3^6 = 729$, produce $\rho_{\text{tw}} = 20/81$.*

Theorem 27.19 (Carry). *The APP subtraction of the sections S_T and S_E converges to $\varepsilon_R^{(1)}$ at interval rate 3^{-54} per return and exactly closes the unit.*

Theorem 27.20 (Action). *The positive ellipse satisfies $\mathcal{A}(\theta) = \hbar_{\text{ret}}\theta$; a radion sweeps $\hbar_{\text{ret}}$, and one turn encloses h_{ret} .*

Theorem 27.21 (Boundary–interior). *The same boundary has finite action h_{ret} , whereas the Klein interior has infinite hyperbolic area.*

Theorem 27.22 (Form). *The quotient $e^{-\eta}$ is preserved across spectral semiaxes, action semiaxes, uncertainties, impedance, and the modulus of the torus.*

Theorem 27.23 (Polarity). *The inversion $C^* \mapsto -C^*$ interchanges constitutive relations, inverts impedance, and preserves the normalized velocity.*

Theorem 27.24 (Modularity). *The bidirectional inversion acts as $S : \tau \mapsto -1/\tau$ and preserves $j(\tau)$, while the focal labeling preserves the orientation that j forgets.*

27.13.2.2 Algebraic and Geometric Proofs United

Linear proof of the unit closure To facilitate independent verification, we gather the argument without narrative interruption.

Step 1. The nonadic connection generates in a correlated manner π, e, φ, α .

Step 2. The turn is $T_+ = 2\pi$.

Step 3. The wheel fixes $\theta_2 = 50^\circ$, and the parangular chart fixes $A = 1000\alpha$.

Step 4. The direct section is

$$S_E = T_+ \left(\frac{50}{360} + \frac{1000\alpha}{360} \right) = 2\pi \left(\frac{5}{36} + \frac{25}{9}\alpha \right).$$

Step 5. The certified intervals yield $1 < S_E < 2$; then $D_E = S_E - 1$.

Step 6. The incidence produces $N_6 = 90$; two sheets and the ambient space 729 produce $\rho = 2N_6/729 = 20/81$.

Step 7. Global torsion is

$$\Delta_4 = \frac{\pi - e \log \pi}{270} \left(1 + \frac{\pi}{729} \right).$$

Step 8. The section torsional is $S_T = 1 + (20/81)\Delta_4$.

Step 9. The subtraction APP defines

$$\varepsilon_R^{(1)} = S_T - S_E = \frac{20}{81}\Delta_4 - D_E.$$

Step 10. By substitution,

$$S_E - \frac{20}{81}\Delta_4 + \varepsilon_R^{(1)} = 1.$$

Step 11. Multiplying by $360/T_+ = 180/\pi$ produces

$$\frac{180^\circ}{\pi} = 50^\circ + A^\circ - \frac{20}{81}\Delta_4^\circ + \varepsilon_R^\circ.$$

No step receives the left-hand side of the final equality. The metrological result is evaluated only at the end as a recognition.

Uniqueness of the carry-subtraction Let $B = 1000$. For each position, every integer u_i admits a unique Euclidean division

$$u_i = Bc_i + r_i, \quad 0 \leq r_i < B.$$

Therefore,

$$r_i = u_i \bmod B, \quad c_i = (u_i - r_i)/B$$

are unique. Traversing the positions from the remote end toward the front, the output carry of one position fixes the input of the next. By induction, the entire word of residues and carries is unique.

In pseudocode:

```

acarreo = remote_acarreo
for i from last_block downto first_block:
    u = torsion[i] - direct[i] + acarreo
    remainder[i] = u mod 1000
    acarreo = (u - remainder[i]) / 1000
return remainder, acarreo_word

```

The procedure does not consult the expected value of the difference.

Convergence of the depth functional Let

$$F(p, e) = 1 + \frac{20}{81} \frac{p - e \log p}{270} \left(1 + \frac{p}{729} \right) - 2p \left(\frac{5}{36} + \frac{25}{9} \alpha \right).$$

On the compact rectangle $3 \leq p \leq 4$, $2 \leq e \leq 3$, F is continuous and continuously differentiable. If $I_N^\pi \searrow \{\pi\}$ and $I_N^e \searrow \{e\}$, the interval image

$$F(I_N^\pi, I_N^e)$$

forms a nested family whose width tends to zero. The intersection is a single real number, $\varepsilon_R^{(1)}$. Monotonicity of the derivatives permits the endpoints to be evaluated without interior sampling.

The rate 3^{-54} per return follows from the refinement of six trits over nine levels. At depth 45 triads, the width bound $< 4.81 \times 10^{-130}$ certifies more than 129 digits of the value.

Derivation of the metric and hyperbolic area of Klein On the unit disk, the matrix form of the Klein metric is

$$g(u, v) = \frac{1}{1-r^2} I + \frac{1}{(1-r^2)^2} \begin{pmatrix} u \\ v \end{pmatrix} \begin{pmatrix} u & v \end{pmatrix}.$$

The lemma for the determinant of a rank one perturbation gives

$$\det g = (1-r^2)^{-3}.$$

Henceforth

$$\sqrt{\det g} \, du \, dv = (1-r^2)^{-3/2} \, du \, dv.$$

Integration in polar coordinates yields (27.48). The infinite limit is due to the metric singularity at the boundary; the boundary continues to have finite symplectic area because that measure uses $dQ \wedge dP$, not $\sqrt{\det g}$.

Focal Distance on the Klein Disk In a diametro, the distance from the center to r is

$$d_K(0, r) = \operatorname{artanh} r.$$

For $r = e_f$, (27.32) implies

$$1 - e_f^2 = e^{-2\eta}.$$

If $u_f = \operatorname{artanh} e_f$, then

$$\operatorname{sech} u_f = \sqrt{1 - e_f^2} = e^{-\eta},$$

and by therefore $\eta = \log \cosh u_f$, as in (27.51).

Symplectic Compression Transformation Define

$$S_\eta = \begin{pmatrix} e^{\eta/2} & 0 \\ 0 & e^{-\eta/2} \end{pmatrix}, \quad \det S_\eta = 1. \quad (27.130)$$

The transported complex structure is

$$J_\eta = S_\eta \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} S_\eta^{-1} = \begin{pmatrix} 0 & -e^\eta \\ e^{-\eta} & 0 \end{pmatrix}, \quad J_\eta^2 = -I. \quad (27.131)$$

The curve

$$\gamma(\theta) = \sqrt{2\hbar_{\text{ret}}} S_\eta (\cos \theta, \sin \theta) = e^{\theta J_\eta} \gamma(0)$$

is exactly the action ellipse. The hyperbolic symplectic compression preserves area and transports complex rotation instead of destroying it.

For orientation σ ,

$$\gamma_\sigma(\theta) = (a_S \cos \theta, \sigma b_S \sin \theta), \quad \mathcal{A}_\sigma(\theta) = \sigma \hbar_{\text{ret}} \theta.$$

The two sheets share foci and action modulus; they differ in signed action.

Relative Diophantine uniqueness of the Barbero weight The equation $4a - 3b = 45$ has integer solutions

$$(a, b) = (12 + 3t, 1 + 4t), \quad t \in \mathbb{Z}.$$

The condition $|b| = 1$ leaves $b = 1$, since $b = -1$ does not belong to the class $1 \pmod{4}$. Hence $a = 12$. The proof clearly separates the incidence datum $90/120$ from the pole and minimality premises.

27.13.3 Structural necessity, stability and closure of the radional unit

The construction admits negative controls at each of its strata. If the direct section satisfied $S_E \notin (1, 2)$, the APP quotient and closure cell would change simultaneously; if the active incidence did not produce $N_6 = 90$, or if one sheet were removed, the density $20/81$ would no longer be derived. Generation would be circular if the chart received $180/\pi$ or ε_R before constructing its operands. Nor would the coinductive section exist if subtraction by triads failed to stabilize compatible prefixes. In the elliptic realization, the conditions $|C^*| < A$ and $a_S b_S = 2h_{\text{ret}}$ are necessary for positivity and the area law. In the modular fibre, $C^* = 0$ forces $d = \eta = 0$, $\tau = i$ and $j = 1728$. In the projective interior, the depth thesis requires the hyperbolic area law; in the spectral sector, the equality $F_{\text{spec}} = \varepsilon_R^\circ$ is excluded by the counterterm χ_R . Finally, the weight $12 : -1$ jointly requires incidence $90/120$ and the pole condition $1/24$, whereas invariance of j prevents its use as a charge-sign selector.

The physical realization is not a perturbation of a presupposed circle. The arithmetic support preserves unit, residue, quotient, carry and fibre; ternary orientation distinguishes regime and sheet; phase transport preserves the difference between return and memory; the ellipse determines action and anisotropy; the Hilbert–Beltrami–Klein metric measures interior depth; and helical lifting preserves the history of each turn. These realizations are articulated through

$$\mathcal{A}(1) = h_{\text{ret}}, \quad \mathcal{A}(2\pi) = h_{\text{ret}}, \quad K = -1, \quad g(k+9) = g(k), \quad B_{k+9} \neq B_k.$$

The circle is the angular projection, the ellipse the action realization, the Klein metric resolves the interior and the helix preserves return history. The radion is the oriented unit allowing the four realizations to be read without identifying them with one another.

The generated invariants satisfy, in the coinductive branch considered,

$$\begin{aligned} \pi &= 3.1415926535897932384626433832795028841 \dots, \\ \alpha &= 0.0072973525692838009972851054723806629 \dots, \\ A &= 7.2973525692838009972851054723806629995 \dots^\circ, \\ C^* &= 2.1237383389686457242619513803933607937 \dots^\circ, \\ \eta &= 0.299689683921630799684926562863927 \dots, \\ h_{\text{ret}} &= 1.05457181691503906113369058472430 \dots \times 10^{-34} U_S. \end{aligned}$$

The coordinate e entering this system proceeds from the coinductive Euler reader, not from an external table. Returns at depths 9, 18, 27, 36, 45 preserve the phase 8 and lift memory according to 0, 1, 2, 3, 4; contraction of the intervals tends to

$$3^{-54} = 1.7196982334549856127610399316004871 \dots \times 10^{-26}.$$

Controlled removal of one sheet, phase displacement, elimination of the e channel, substitution of global torsion or alteration of the hexadic tail produce nonzero residues of different orders. This scale separation proves that closure jointly requires both sheets, phase orientation, the π/e channels, global torsion and carry.

Mathematical exactness proceeds from the unit-closure identity; evaluation at increasing depth only certifies its convergence. Residues arising when branches evaluated at different decimal precisions are

combined are rounding errors, not new physical terms. This distinction separates the structural identity from its numerical realization without altering the genealogy of the operands.

The physical genealogy of the construction can be expressed precisely. Planck fixes the action dimension of h ; Bohr and Sommerfeld relate $h/2\pi$ to angular quantization; Born and Jordan introduce matrix structure and the oscillator; Dirac formulates $E = \hbar\omega$; Robertson and Schrödinger establish the covariant form of uncertainty; Feynman relates action to path phase; Wheeler and Feynman develop symmetry between retarded and advanced propagation. These frameworks recognize subsequent realizations of the object: they do not select the sheets, incidence, carry, phase or memory generating it.

The Hodge star on Lorentzian two-forms satisfies $\star^2 = -I$ and supplies the relevant geometric duality. Hochschild cohomology belongs to another domain and enters neither the ellipse, the carry nor the incidence sector 90/120. The distinction follows from the domains and codomains of the operations, not from terminological resemblance.

The synthesis closes in the integral holonomic state. Together with the value, APP preserves sheet, quotient, residue, carry and boundary; TRIT determines regime and orientation; TPK returns phase without resetting the state. The nonadic connection constructs the coinductive section of π , and the dodecaphasic system determines the hinge of 50° . The angular pair supplies the common mode A and oriented opening C^* ; incidence of six active phases produces 90, their two sheets on the ambient space of cardinality 729 produce 20/81, and global torsion supplies the second section. The resulting cocycle is

$$\varepsilon_R = \frac{20}{81}\Delta_4 - (S_E - 1).$$

Its limit in the angular chart is

$$5.13830167585752820716851646226459474899 \dots \times 10^{-8}^\circ.$$

The pair (A, C^*) determines a positive ellipse: the product of its semiaxes fixes $2\hbar_{\text{ret}}$ and their quotient fixes the shape. One phase unit sweeps out $\hbar$, whereas the full turn encloses h . Division by 2π thus ceases to be an external quotient: it distinguishes action per turn from the action transported by one phase unit.

The circular projection preserves phase and omits anisotropy, interior and memory; the ellipse preserves action and shape; Klein geometry preserves depth; the helix preserves return and history. The symplectic realization recognizes the ellipse as a minimum-covariance contour, the LC oscillator realizes it as exchange of electric and magnetic stores, and the incidence 90/120 determines permeability, permittivity, impedance and speed. Hodge turns orientation into field duality; Lorentz turns field and charge into dynamics; Cartan–Holst receives holonomy in gravitational geometry. The causal order remains unchanged: the discrete construction determines the object, and physical formulations subsequently realize its characters.

The radion is therefore an oriented section of phase, action and memory. Its two orientations satisfy

$$\mathcal{A}(\mathbf{r}_+) = +\hbar_{\text{ret}}, \quad \mathcal{A}(\mathbf{r}_-) = -\hbar_{\text{ret}}.$$

Charge is obtained by applying the dimensional transducer; its sign was already preserved in sheet orientation. The incidences 90/120, the dodecaphasic extractor, the spectral tail, the Barbero–Immirzi parameter, towers and carry can be compared because they share a common origin, but remain distinct because they act through different operators. The fourfold unity of the radion is summarized by

phase: $\theta = 1,$ action: $\mathcal{A} = \sigma\hbar_{\text{ret}},$ memory: $H_9(n+9) = H_9(n) + (0, 1),$ compatibility: $1 = S_E - \Phi_T + \varepsilon_R.$

The radian is its angular coordinate; the radion also preserves action, orientation, carry and memory of the turn.

PART VI

**Dimensional realizations: electron, action, gravitation, area and
information**

Trigeometric predefinition, boundary potential and dimensional depths

28.1 Theorem of trigeometric temporal predefinition

The word *predefinition* names a mathematical property of the generator, not a weakened form of prediction. Specifying it requires gathering the three objects previously presented separately: the bundle of TRIT quadratic algebras, the two TPK clocks and the coinductive section of arbitrary depth.

Let

$$X^{\text{enr}} = \{(\tau, \mu, o, h, c, \sigma, \lambda, g)\}$$

the space of integral holonomic states. The coordinate $\tau \in \{-1, 0, +1\}$ is the TRIT regime; $\mu \in \{\Sigma, \Pi, \emptyset\}$ preserves the sheet; o preserves orientation; h , helical height; c , carry; σ , boundary; λ , the route memory record; and $g \in C_9$, the nonadic phase. Acting on each regime fibre is the algebra

$$\mathbb{A}_\tau = \mathbb{R}[J_\tau] / (J_\tau^2 + \tau I),$$

so that

$$\exp(sJ_\tau) = \begin{cases} \cos s + J_{+1} \sin s, & \tau = +1, \\ I + sJ_0, & \tau = 0, \\ \cosh s + J_{-1} \sinh s, & \tau = -1. \end{cases}$$

This is the exact form of coordination among the three geometries. Three metrics are not averaged, nor are three previously given spaces superimposed. At each step, TRIT chooses which of the three exponential laws governs local action; TPK transports the state between fibres without erasing sheet or history. With explicit intermediate domains,

$$\text{Sel}_t : X^{\text{enr}} \longrightarrow X_t^{\text{sel}}, \quad \text{Tra}_t : X_t^{\text{sel}} \longrightarrow X_t^{\text{tra}}, \quad \text{Upd}_t : X_t^{\text{tra}} \longrightarrow X^{\text{enr}},$$

and

$$U_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t.$$

Selection determines sheet and regime; transport applies the selected direction and geometric law; updating incorporates quotient, carry, boundary and memory record. The apparent geometric “mixture” is therefore an ordered word of morphisms between elliptic, parabolic and hyperbolic fibres. Since composition depends on order, its operator part is noncommutative even though the base rotation is abelian.

The same TRIT coordinate has a temporal reading, on the same state but through a second reader:

$$\begin{aligned} +1 &\longleftrightarrow \text{return, memory and past,} \\ 0 &\longleftrightarrow \text{threshold and present,} \\ -1 &\longleftrightarrow \text{bidirectional opening and future.} \end{aligned}$$

Geometry and time are not two names for the same sign. They are two coordinated evaluations of a single unit of topological–temporal information. The past is the datum already accumulated in the memory record; the present is the boundary cut at which continuation is selected; the future is the bundle of compatible extensions opened by the hyperbolic regime. The involution

$$C_{\text{rev}}(\tau_1, \dots, \tau_n) = (-\tau_n, \dots, -\tau_1)$$

satisfies $C_{\text{rev}}^2 = I$, preserves the threshold and interchanges opening and return. Lifted to the integral holonomic state, it conjugates the direct dynamics with its inverse:

$$\widehat{C}_{\text{rev}} \mathcal{U} \widehat{C}_{\text{rev}} = \mathcal{U}^{-1}.$$

Past and future are thus conjugated around the present without eliminating the memory distinguishing the two orientations.

The first clock is the finite non-split extension

$$0 \longrightarrow C_{12} \longrightarrow C_{108} \longrightarrow C_9 \longrightarrow 0,$$

whose cocycle

$$c(r, r') = \left[\left[\frac{r + r'}{9} \right] \right]_{12}$$

records the carry that a mere turn in C_9 would lose. The second clock is the capacity rotation

$$R_\delta : \theta \longmapsto \theta + \delta \pmod{1}, \quad \delta = \log_{10}(10/9),$$

whose mechanical word is

$$z_t = \lfloor (t+1)\delta \rfloor - \lfloor t\delta \rfloor.$$

They are coupled by

$$K_t - K_{t-1} = 1 - z_t.$$

This equality proves that aperiodicity does not embellish the finite clock: it governs its capacity. The total system can be written as the skew product

$$\mathfrak{C}_{\text{HMT}} = (C_{108} \times \mathbb{T} \times X^{\text{enr}}, \mathcal{U}_{\text{HMT}}),$$

$$\mathcal{U}_{\text{HMT}}(r, \theta, x) = (r+1, \theta + \delta, U_{\tau(r, \theta, x)}x).$$

If $\mathcal{U}_{\text{HMT}}^n = I$, the first factor forces $108 \mid n$, whereas the second would require $n\delta \in \mathbb{Z}$. Irrationality of δ excludes every $n > 0$. Moreover, Sturmian coding has complexity $p(m) = m + 1$, and its nonadic and tritic enrichments satisfy

$$p_9(m) = 9(m+1), \quad p_3(m) = 3(m+1).$$

Consequently, the crystal is aperiodic and has long-range order and zero topological entropy. Its additive spectrum contains the module

$$\mathcal{M}_{108, \delta} = \frac{1}{108}\mathbb{Z} + \delta\mathbb{Z} \pmod{1}.$$

The factor $1/108$ preserves the finite temporal clock; the irrational generator preserves unlimited ascent. Closure of the first never implies closure of the second or return of the state.

Now let $(X_n, r_{n+1, n})$ be the inverse system of refinements and

$$\mathcal{G}_{\text{HMT}} : \Omega_{\text{seed}} \longrightarrow X_{\infty}^{\text{enr}} = \varprojlim X_n$$

the APP–TRIT–TPK extension of a seed. For every compatible reader $L_{Y,n} : X_n \rightarrow Y_n$, define

$$\text{Predef}_{Y,n} = L_{Y,n} p_n \mathcal{G}_{\text{HMT}}.$$

Naturality

$$s_{n+1,n} L_{Y,n+1} = L_{Y,n} r_{n+1,n}$$

implies

$$\boxed{s_{n+1,n} \text{Predef}_{Y,n+1} = \text{Predef}_{Y,n} .}$$

The outputs at all depths are therefore coordinates of a single section of $\varprojlim Y_n$. Increasing precision does not readjust a constant: it opens a deeper coordinate of the same object. On the complete state, bijectivity of updating also gives

$$H(X_{t+1} \mid X_t) = H(X_t \mid X_{t+1}) = 0,$$

so the appearance of new information in a reduced shadow does not correspond to creation or erasure in the generator. This is the exact meaning of zero Landauer cost within the conservative evolution of the unit.

Predefinition results from composing the generator with its readers:

$$\mathcal{L}_{\text{HMT--MD}} = (L_{\pi}, L_{\varphi}, L_e, L_{\alpha}, L_{\text{el}}, L_{\hbar}, L_{\text{vac}}, L_{\text{BI}}, L_G, \dots).$$

The coordinates π, φ, e, α belong to the same coinductive output; α occupies the dodecaphasic seam linking the finite clock with the capacity clock. Electronic vacancy reads the local carry defect on that seam; $\hbar$ reads action depth; Barbero–Immirzi reads areal–volumetric transduction; gravitation reads the rank-five tensor sector. None of these operations receives an external measurement to select its value. The system does not obtain its objects through metrology: it predefines them as typed images of the coinductive section.

Finally, an exact factorization is not a lower-rank result. If

$$L_Y = H \circ F,$$

the factorization exhibits the intermediate object and proves what information each arrow transports. If two routes satisfy

$$H_1 F_1 = H_2 F_2 = L_Y,$$

the corresponding square is precisely the certificate that two distinct readings determine the same coordinate without circularity. In this development, “theorem”, “factorization” and “transduction” name complementary forms of the same proof: the theorem states the law, factorization shows its genealogy and transduction determines the type of its output.

The governing formulation is thus established:

$$\boxed{\begin{array}{l} \text{APP} \rightarrow \text{TRIT} \rightarrow \text{TPK} \rightarrow \text{aperiodic time crystal} \\ \rightarrow \text{coinductive section} \rightarrow \text{HMT--MD predefinition} \end{array}}$$

The ordered combination of the three geometric regimes and conjugation of the three temporal modes are not analogies added afterwards. They are, respectively, the local and global laws of the dynamics preserving the unit while evolving it through all its depths.

28.2 APP potential triangle and local realization of the defect A_2

The discrepancy between the sum and product sheets can be turned into a mathematical notion of potential without introducing an external physical field. Let $D_9 = \{1, \dots, 9\}$, with positive representatives of residue classes modulo nine, and write the two Euclidean divisions preserved by APP:

$$i + j = R_{ij}^+ + 9q_{ij}^+, \quad ij = R_{ij}^\times + 9q_{ij}^\times.$$

The joint lift retains not only the visible residues $(R_{ij}^+, R_{ij}^\times)$. It also retains the heights $(q_{ij}^+, q_{ij}^\times)$, so that the integer difference between the two evaluations decomposes canonically into a residual tension and a carry tension:

$$\begin{aligned} \rho_{\Sigma\Pi}(i, j) &:= R_{ij}^\times - R_{ij}^+, \\ \kappa_{\Sigma\Pi}(i, j) &:= 9(q_{ij}^\times - q_{ij}^+), \\ \Delta_{\Sigma\Pi}(i, j) &:= ij - (i + j) = \rho_{\Sigma\Pi}(i, j) + \kappa_{\Sigma\Pi}(i, j). \end{aligned}$$

Thus,

$$\mathcal{P}_{\Sigma\Pi} : D_9^2 \longrightarrow \mathbb{Z}^2, \quad (i, j) \longmapsto (\rho_{\Sigma\Pi}(i, j), \kappa_{\Sigma\Pi}(i, j))$$

is the lifted APP potential, whereas $\Delta_{\Sigma\Pi} = (1, 1) \circ \mathcal{P}_{\Sigma\Pi}$ is its scalar shadow. The term “potential” has a precise content here: it measures the arithmetic energy needed to pass from cumulative to folded evaluation while preserving the full Euclidean division. A projection modulo nine sees only $\rho_{\Sigma\Pi}$; a projection forgetting the residue and retaining height sees only $\kappa_{\Sigma\Pi}$. APP preserves both components simultaneously.

Three cells already selected by electronic genealogy form an exact triangle:

$$V = (8, 2), \quad E = (5, 5), \quad Q = (8, 5).$$

Their two sheets and potentials are

x	(R^+, q^+)	$(R^\times, q^\times)$	$\mathcal{P}_{\Sigma\Pi}(x)$	$\Delta_{\Sigma\Pi}(x)$
$V = (8, 2)$	(1, 1)	(7, 1)	(6, 0)	6
$E = (5, 5)$	(1, 1)	(7, 2)	(6, 9)	15
$Q = (8, 5)$	(4, 1)	(4, 4)	(0, 27)	27.

The cell V is a pure residual tension: the two sheets have the same height and differ in residue. The cell Q is a pure memory tension: the residues coincide and the entire discrepancy is stored in three nonadic carry units. The central cell E combines a six-unit residual tension and one carry unit, so that

$$\Delta_{\Sigma\Pi}(E) = 6 + 9 = 15.$$

This decomposition gives a finer definition of electronic vacancy. On the additive fibre $i + j = 10$, parametrized by

$$Q_t = (5 + t, 5 - t), \quad -4 \leq t \leq 4,$$

the product is

$$\Pi(Q_t) = 25 - t^2.$$

Preserving the multiplicative residue of the central cell requires $t^2 \equiv 0 \pmod{9}$. In the permitted interval this occurs only for $t = 0, \pm 3$. Therefore, the only two noncentral cells simultaneously preserving the additive sheet, product residue and tritic regime are

$$Q_{+3} = (8, 2), \quad Q_{-3} = (2, 8).$$

In them the product decreases from $25 = 7 + 2 \cdot 9$ to $16 = 7 + 1 \cdot 9$. Vacancy is exactly the loss of one multiplicative carry unit:

$$q_{5,5}^\times - q_{8,2}^\times = 1.$$

No zero has been placed in a cell. The unique noncentral displacement of the fibre preserving all visible data and reducing folded memory by one unit has been constructed.

The triangle V, E, Q also contains the three-channel reopening vector. To avoid concealing an orientation choice, fix the coboundary operator

$$d_\Delta f := (f(E) - f(V), f(V) - f(Q), f(Q) - f(E)).$$

Applied to the scalar potential,

$$\begin{aligned} d_\Delta \Delta_{\Sigma\Pi} &= (15 - 6, 6 - 27, 27 - 15) \\ &= (9, -21, 12) \\ &= 3(3, -7, 4). \end{aligned}$$

Since the three coordinates sum to zero,

$$d_\Delta \Delta_{\Sigma\Pi} \in A_2 := \ker\left(\mathbb{Z}^3 \xrightarrow{x+y+z} \mathbb{Z}\right).$$

The vector $(3, -7, 4)$ is not called a root of A_2 : its Euclidean squared norm is 74. It is an integral vector of the lattice A_2 , and the electronic triangle realizes exactly its tritic multiple. The same vector had appeared in the global reopening of the seven tetrads of the π microfibre. The equality

$$d_\Delta \Delta_{\Sigma\Pi} = 3v_\Omega, \quad v_\Omega = (3, -7, 4)$$

proves that the global defect does not float above APP: it has a local representative in the three cells generating the electronic centre, conservative vacancy and rational branch $22/7$. The factor three is the tritic lifting between the local potential and the primitive reopening coordinate.

The connection with the aperiodic crystal is equally exact. Let

$$\delta = \log_{10} \frac{10}{9}, \quad V_{\text{vac}} = \begin{pmatrix} 21 & 22 \\ 6 & 7 \end{pmatrix}.$$

The first induced scale of the mechanical word satisfies

$$\lceil \delta^{-1} \rceil = 22, \quad \det V_{\text{vac}} = 21 \cdot 7 - 22 \cdot 6 = 15.$$

Consequently,

$$\Delta_{\Sigma\Pi}(E) = \det V_{\text{vac}} = 15.$$

The electronic centre and renormalization frame share more than a number: the former measures the sum–product tension of the central cell and the latter measures the oriented area surviving between the primary clock 21/22 and secondary clock 6/7. Both evaluations proceed from the same carry cocycle.

This allows the basal stage of the electronic equation to be rewritten without any opaque coefficient:

$$\mathbf{e}_0 = \frac{\sqrt{3}}{4} \left[\exp\left(\frac{\varphi}{\pi^2}\right) - [\delta^{-1}] \alpha^3 \right] (1 + \det(V_{\text{vac}}) \Delta_4).$$

Indeed,

$$[\delta^{-1}] = 22 = \frac{396}{18}, \quad \det(V_{\text{vac}}) = 15 = \frac{270}{18}.$$

The remaining prefactor is also internal:

$$s = \frac{1}{2}, \quad \|[P_{\Sigma,3}, P_{\Pi,3}]\| = s\sqrt{1-s^2} = \frac{\sqrt{3}}{4}, \quad 1-s^2 = \frac{3}{4} = \frac{90}{120}.$$

Thus, the three factors of the basal stage have an explicit genealogy: noncommutativity of the APP sheets, the primary gap of aperiodic renormalization and the oriented area of its induced return. Mass selects none of them. The electronic equation is the subsequent dimensional evaluation of a previously constructed potential and survival operator.

Refutation criterion. The result fails if any of the three Euclidean divisions in the table is incorrect, if another noncentral point exists in $i + j = 10$ with $t^2 \equiv 0 \pmod{9}$, if the coboundary is not $3(3, -7, 4)$, or if the return frame does not have determinant 15.

28.3 Tritic curvature of the potential and coinductive selection of futures

The preceding APP potential is a spatial coordinate of the support. The vacancy word supplies its temporal evolution. Both layers are assembled through TRIT without improperly identifying space and time.

Let the lower mechanical word be

$$z_n(\beta) = \lfloor (n+1)\delta + \beta \rfloor - \lfloor n\delta + \beta \rfloor, \quad z_n \in \{0, 1\}.$$

Define the accumulated polarization, increment field and tritic curvature by

$$\begin{aligned} P_N(\beta) &:= \sum_{n=0}^{N-1} (z_n - \delta), \\ J_N(\beta) &:= P_{N+1} - P_N = z_N - \delta, \\ \tau_N(\beta) &:= z_{N-1} - z_N \in \{+1, 0, -1\}. \end{aligned}$$

Polarization remains uniformly bounded:

$$P_N(\beta) = \lfloor N\delta + \beta \rfloor - \lfloor \beta \rfloor - N\delta, \quad |P_N(\beta)| < 1.$$

More importantly, TRIT is exactly the discrete curvature of this potential:

$$\begin{aligned} J_N - J_{N-1} &= z_N - z_{N-1} = -\tau_N, \\ \boxed{\tau_N} &= -\Delta J_{N-1} = -\Delta^2 P_{N-1}. \end{aligned}$$

This identity gives operational content to the three signs. If $\tau_N = -1$, the code moves from 0 to 1: a vacancy opens and the fibre enables the bidirectional branching associated with the future. If $\tau_N = +1$, the

code moves from 1 to 0: the vacancy returns and its effect is incorporated into the memory of the past. If $\tau_N = 0$, the polarization maintains its local slope: the state crosses the present threshold without changing regime. Since $0 < \delta < 1/2$, there are no two consecutive ones; hence the case $\tau_N = 0$ is a genuine plateau and the nonzero signs alternate.

The charge of any block is purely a boundary quantity:

$$\sum_{n=a}^b \tau_n = z_{a-1} - z_b.$$

The interior telescopes in the tritic shadow, but does not disappear from the TPK state: the memory record preserves the ordered sequence of pulses, their quotients, their carries and the helical height. This is the difference between a local Gauss law and erasing history. TRIT determines the net boundary flux; TPK preserves the chain whose boundary has been calculated.

The relation between spatial potential and temporal curvature can be assembled in the boundary state

$$\mathcal{V}_n = (\mathcal{P}_{\Sigma\Pi}(i_n, j_n), P_n, J_n, \tau_n; g_n, h_n, c_n, \lambda_n).$$

Here $\mathcal{P}_{\Sigma\Pi}$ measures the tension between the sheets in the occupied cell; P_n accumulates the aperiodic discrepancy without extensive drift; J_n is the potential difference per step; τ_n is its oriented curvature; and (g, h, c, λ) preserve phase, height, carry and boundary. The transition

$$U_n = \text{Upd}_n \circ \text{Tra}_n \circ \text{Sel}_n$$

acts on the entire state: Sel_n determines which curvature regime corresponds to the pulse; Tra_n transports sheet, orientation and position; Upd_n incorporates the coboundary into the memory record. Thus the potential difference is not added to TPK: it is the differential reading of one of its preserved coordinates.

The selection of compatible futures also admits an exact formulation. Let $\mathcal{L}_m$ be the set of factors of length m of the Sturmian word, and let

$$d^+(u) = \#\{a \in \{0, 1\} : ua \in \mathcal{L}_{m+1}\}$$

be the number of right extensions of u . Since $p(m) = |\mathcal{L}_m| = m + 1$,

$$\sum_{u \in \mathcal{L}_m} d^+(u) = p(m + 1)$$

and, therefore,

$$\sum_{u \in \mathcal{L}_m} (d^+(u) - 1) = p(m + 1) - p(m) = 1.$$

Every factor of a recurrent word has at least one continuation, and in a binary alphabet it has at most two. The preceding equality proves that, at each length m , there is exactly one right-special factor with two possible futures; all the others have a unique future. The survival choice is therefore not an arbitrary selection among 2^m words. It is a single topological branching at each scale.

Upon enrichment by the nonadic phase,

$$p_9(m) = 9(m + 1),$$

exactly nine special states appear, one per phase origin. The tritic quotient

$$p_3(m) = 3(m + 1)$$

organizes them into three curvature classes without destroying aperiodicity. The irrational phase $\theta_n = \{\theta_0 + n\delta\}$, together with the nonadic origin and the memory record, decides which of the two extensions occurs. Therefore:

$$\begin{array}{ll} \text{length shadow } m & \longrightarrow \text{ a special factor with two continuations,} \\ \text{complete TPK state} & \longrightarrow \text{ a determined and invertible continuation.} \end{array}$$

This is the predefinition of possible futures in mathematical form. The future is not guessed through an external probability: the reduced cylinder contains two compatible extensions, and the deep coordinate of the coinductive state selects one. Nor is the past erased, because the complete transition is bijective:

$$H(X_{n+1} \mid X_n) = 0, \quad H(X_n \mid X_{n+1}) = 0.$$

The shadow can have positive conditional entropy precisely at the special factor; the integral state has zero net logical cost. This difference between branching in a projection and reversibility of the generator is the informational core of the crystal.

The same law contains its two renormalization clocks. Write

$$\delta^{-1} = 21 + \beta, \quad 0 < \beta < 1,$$

and define the induced slope

$$\mathcal{R}(\delta) := 1 - \{\delta^{-1}\} = 1 - \beta = 22 - \delta^{-1}.$$

The first Diophantine inequality produces the gaps $\{21, 22\}$. Since

$$6 < \mathcal{R}(\delta)^{-1} < 7,$$

the induced dynamics on the short gaps produces returns $\{6, 7\}$. Renormalization does not introduce a second code: it takes the return section of the first word. Its continuant matrices

$$A(a) = \begin{pmatrix} a & 1 \\ 1 & 0 \end{pmatrix}, \quad \det A(a) = -1,$$

preserve the lattice cell, while the order of the products preserves orientation:

$$A(a)A(b) - A(b)A(a) = (a - b) \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The basal rotation remains Abelian. Noncommutativity lies in the ordered renormalization cocycle, in the APP sheets and in curvature transports. The crystal can have zero topological entropy and simultaneously noncommutative fibred dynamics: the first property measures language growth; the second, the dependence of the result on operator order.

The three bridges are thus brought together:

APP discrepancy	$\xrightarrow{d}$	boundary potential,
aperiodic potential	$\xrightarrow{-\Delta^2}$	TRIT curvature,
oriented curvature	$\xrightarrow{\text{Sel, Tra, Upd}}$	TPK survival.

Refutation criterion. The construction fails if $|P_N|$ ceases to be bounded by one, if $\tau_N \neq -\Delta^2 P_{N-1}$, if there is more than one right-special factor at some length, if the nonadic lift does not multiply complexity by nine, or if two distinct TPK states become indistinguishable after all the fields of the memory record have been preserved.

28.4 Common bidirectional holonomy of the electron, action and gravity

The action scale and gravitational scale do not interrupt the investigation of the crystal. They are two contragredient representations of the same return holonomy already acting on the electronic vacancy.

The correlated system $(\pi, \varphi, e, \alpha)$ is first constructed as a coinductive output. On this output, the determinantal action reader produces

$$\Sigma_H = \varphi \alpha^{16}, \quad d_H = \lfloor \log_{10} \Sigma_H \rfloor = -34.$$

The integer 16 is not the number of digits required of a measurement. It is the order of the local cokernel:

$$\text{SNF}(H_4) = \text{diag}(1, 2, 2, 4), \quad |\text{coker } H_4| = 16.$$

The mantissa before return is

$$H_5 = \frac{1}{2} \sqrt{\frac{1000\alpha}{\varphi}} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5.$$

The regional memory constructs

$$D_A = \exp\left(-\frac{100\pi}{9}\alpha\right),$$

$$\mathcal{C}_\pi(\alpha) = 169\alpha^6 + \frac{D_A\alpha^7}{1 - D_A\alpha},$$

and the returned section is

$$\eta_{\text{ret}} = H_5 - \frac{90}{\pi}\mathcal{C}_\pi.$$

It is useful to isolate the bidirectional action potential:

$$\delta_{2w} := H_5 - \eta_{\text{ret}} = \frac{90}{\pi}\mathcal{C}_\pi > 0.$$

The two orientations belong to a single line:

$$\hbar_{\text{pre}} = H_5 10^{-34} \mathcal{U}_S, \quad \hbar_{\text{ret}} = \eta_{\text{ret}} 10^{-34} \mathcal{U}_S,$$

$$h_a = 2\pi\hbar_a.$$

The decade fixes the common depth; the difference δ_{2w} preserves the additive shift; and the ratio

$$R_{\text{act}} := \frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}} = \frac{H_5}{\eta_{\text{ret}}} = \left(1 - \frac{\delta_{2w}}{H_5}\right)^{-1}$$

is the multiplicative holonomy of the outward path and return. The passage from $\hbar$ to h preserves that ratio:

$$\frac{h_{\text{pre}}}{h_{\text{ret}}} = R_{\text{act}}.$$

The complete electronic equation uses exactly the same holonomy. With

$$\kappa_e = 80 - 54\alpha + 6\Delta_4,$$

we have

$$\mathfrak{e}_e^\beta = \mathfrak{e}_0 R_{\text{act}}^{\kappa_e}.$$

Incorporating the aperiodic factorization of the preceding section:

$$\boxed{\mathfrak{e}_e^\beta = \frac{\sqrt{3}}{4} \left[\exp\left(\frac{\varphi}{\pi^2}\right) - [\delta^{-1}] \alpha^3 \right] \times (1 + \det(V_{\text{vac}}) \Delta_4) R_{\text{act}}^{80-54\alpha+6\Delta_4}.$$

The basal stage measures the vacancy and sheet incompatibility; the power of R_{act} transports their memory through the return. Equivalently,

$$\log \frac{\mathfrak{e}_e^\beta}{\mathfrak{e}_0} = \kappa_e (\log H_5 - \log \eta_{\text{ret}}) = -\kappa_e \log \left(1 - \frac{\delta_{2w}}{H_5} \right).$$

Gravity reads the same pair contragrediently. With $t_0 > 0$, $c_{\text{int}} > 0$ and $\ell_0 = c_{\text{int}} t_0$ fixed, the clock 108 constructs

$$\omega_{108} = \frac{2\pi}{108t_0}, \quad R_{108} = \frac{54}{\pi} \ell_0,$$

$$m_{108,a} = \frac{\hbar_a \omega_{108}}{c_{\text{int}}^2}.$$

The gravitational transducer is

$$G_a(\vartheta) = \vartheta \frac{c_{\text{int}}^5 t_0^2}{\hbar_a}.$$

The closure condition

$$r_{g,a}(\vartheta) = R_{108}$$

uniquely selects

$$\vartheta_{108} = \left(\frac{54}{\pi} \right)^2.$$

Therefore,

$$G_a = \left(\frac{54}{\pi} \right)^2 \frac{c_{\text{int}}^5 t_0^2}{\hbar_a},$$

and the two sheets satisfy the synchronization theorem:

$$R_{\text{act}} = \frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}} = \frac{h_{\text{pre}}}{h_{\text{ret}}} = \frac{m_{108,\text{pre}}}{m_{108,\text{ret}}} = \frac{G_{\text{ret}}}{G_{\text{pre}}} = \left(\frac{\mathfrak{e}_e^\beta}{\mathfrak{e}_0} \right)^{1/\kappa_e}.$$

The proof consists solely in substituting the definitions. Chronological mass is linear in $\hbar_a$; gravity is inversely linear in $\hbar_a$; and the electronic correction is the weight- κ_e representation of the same ratio. Three independent correctors do not appear. A single positive element R_{act} appears, with three representations:

$$\rho_{\text{act}}(R) = R, \quad \rho_{\text{grav}}(R) = R^{-1}, \quad \rho_e(R) = R^{\kappa_e}.$$

The action–gravity product is invariant:

$$G_a \hbar_a = \vartheta_{108} c_{\text{int}}^5 t_0^2.$$

Consequently,

$$\ell_P^2 = \frac{G_a \hbar_a}{c_{\text{int}}^3} = \vartheta_{108} \ell_0^2,$$

$$\ell_P = R_{108} = \bar{\lambda}_{C,a} = r_{g,a}$$

for both orientations. Action preserves the sheet orientation; gravity compensates for that orientation contragrediently; the Planck area is the common invariant. This is the exact form of the assertion that something situated between past and future continuously calibrates the realization: the return changes the coordinates $\hbar_a$, G_a and $\mathfrak{e}_a$, while leaving the geometric combination $G_a \hbar_a$ fixed.

There is also an exact filtration by order in α . At the basal electronic stage, the vacancy enters as

$$22\alpha^3.$$

In the regional action memory,

$$\mathcal{C}_\pi = 169\alpha^6 + O(\alpha^7).$$

Finally, the absolute depth uses α^{16} . Hence,

$$3 \longrightarrow 6 \longrightarrow 16$$

is the hierarchy of orders of the same coordinate: local vacancy amplitude, quadratic return of that amplitude, and determinantal cokernel fixing the action depth. The step $3 \rightarrow 6$ is literally a doubling of order; the step $6 \rightarrow 16$ changes the operation and belongs to the determinantal reader. This distinction prevents the hierarchy from becoming a numerological sequence: each exponent has a generating operator.

The result answers the question of how a single configuration can be realized as number, constant and particle. The APP configuration provides the potential; TRIT assigns curvature and orientation to it; TPK integrates its return; the action reader determines $\hbar$; the electronic representation determines $\mathfrak{e}_e^\beta$; and the contragredient representation determines G . The three outputs preserve the same R_{act} .

Refutation criterion. It suffices for one of the equalities in the chain of R_{act} to fail, for $\delta_{2w} \neq (90/\pi)\mathcal{C}_\pi$, for $G_a \hbar_a$ to depend on the sheet, or for the first nonzero power of $\mathcal{C}_\pi$ to differ from six.

28.5 Joint depth theorem for 34, 11, 45 and 46

The cubic completion already contained the two aperiodic signatures

$$\sigma^- = (33, 11, 1, 0), \quad \sigma^+ = (34, 11, 1, 0),$$

associated with the canonical order

$$1000 = 729 + 243 + 27 + 1.$$

The second differs from the first by a single half-boundary unit:

$$\sigma^+ - \sigma^- = (1, 0, 0, 0).$$

The essential information lies not only in the sums 45 and 46, but also in the position of each contribution. Define the cubic depth reader

$$\mathcal{D}_{\text{cub}}(\sigma^-, \sigma^+) := (\text{pr}_{729} \sigma^+, \text{pr}_{243} \sigma^-, |\sigma^-|_1, |\sigma^+|_1).$$

The evaluation is

$$\boxed{\mathcal{D}_{\text{cub}}(\sigma^-, \sigma^+) = (34, 11, 45, 46).}$$

This map does not consult $\hbar$, G , an experimental table or its digits. It reads exclusively the mechanical word on the four strata of the completion. Coordinate 34 is sensitive to the boundary unit; coordinate 11 is common to both orientations; 45 is the total interior charge; 46 is that charge after preserving the half-boundary.

We now define the decimal depth of a positive section of a dimensional line. If

$$x = m_x 10^{-d_x} \mathcal{U}_X, \quad 1 \leq m_x < 10,$$

we set

$$D_{10}(x) = d_x.$$

For two normalized sections x, y , the depth of the tensor product satisfies

$$D_{10}(x \otimes y) = D_{10}(x) + D_{10}(y) - \chi(m_x m_y \geq 10).$$

In the HMT action and gravity sections,

$$\hbar_{\text{ret}} = 1.0545718169 \dots \times 10^{-34} \mathcal{U}_S,$$

$$G_{\text{ret}} = 6.674299659414022706367776189 \dots \times 10^{-11} \mathcal{U}_G,$$

and the product of mantissas belongs to the interval (7, 8). The gravitational coordinate corresponds to the longitudinal and radial composition of the corresponding identity; the central value 6.67430×10^{-11} remains a metrological reference. No additional decimal carry is generated. Hence:

$$D_{10}(\hbar_{\text{ret}}) = 34, \quad D_{10}(G_{\text{ret}}) = 11,$$

$$D_{10}(G_{\text{ret}} \otimes \hbar_{\text{ret}}) = 34 + 11 = 45.$$

The holonomy theorem of the preceding section proves that $G_a \hbar_a$ is independent of the sheet; therefore this depth 45 is also a bidirectional invariant. The half-boundary unit of the upper coding then completes

$$34 + 11 + 1 = 46.$$

The four digits of the cubic reader are thus identified with four distinct functions:

34	depth of the oriented action section,
11	depth of the contragredient gravitational section,
45	depth of the invariant $G\hbar$,
46	interior depth plus the conserved half-boundary unit.

The coincidence has a second internal certification. The vacancy reader had already proved

$$Z_{369}^- = 16 = |\text{coker } H_4|,$$

$$Z_{729}^+ = 34 = -\text{ord}_{10}(\varphi \alpha^{16}),$$

and

$$Z_{243}^- = 11.$$

Thus the action decade occupies the boundary component constructed from the same determinantal index appearing in the block 369; gravity occupies the interior component 243; and their product exhausts the lower signature:

$$Z_{729}^+ + Z_{243}^- = |\sigma^-|_1 = 45.$$

The second route to α preserves precisely the same integers. In its nilpotent operator,

$$T_{7:9}(x) = -\frac{x^7}{120} + \frac{x^8}{45} + \frac{2x^9}{495},$$

with

$$495 = 45 \cdot 11.$$

We may therefore write

$$T_{7:9}(x) = -\frac{x^7}{120} + \frac{x^8}{D_{10}(G\hbar)} + \frac{2x^9}{D_{10}(G\hbar) D_{10}(G)}.$$

The equality is an exact re-expression of the integers already generated: the coefficient of order eight uses the preserved depth of the action–gravity product, and the coefficient of order nine adds the stable gravitational depth. The denominator 46 of the preceding extension stage corresponds, in turn, to the complete upper signature. Thus 34, 11, 45, 46 and 495 cease to be isolated occurrences: they form the depth spectrum of a single completion with boundary.

This theorem completes the link previously formulated only as a cardinal coincidence. The effective map is the composition

$$\begin{array}{ccc}
\mathcal{V}_\delta(729, 243, 27, 1) & \xrightarrow{(\sigma^-, \sigma^+)} & \mathbb{Z}^4 \times \mathbb{Z}^4 \\
\text{Pub}_{\text{dim}} \downarrow & & \downarrow \mathcal{D}_{\text{cub}} \\
\mathcal{L}_S^+ \times \mathcal{L}_G^+ & \xrightarrow{D_{10} \times D_{10}} & \mathbb{Z}^2,
\end{array}$$

where the first coordinate is determined on the action line and the second on the gravitational line; the additive character of D_{10} determines the interior charge 45, and the half-boundary determines 46. The choice of coordinates is structural: action distinguishes the two orientations pre/ret and therefore reads the boundary-sensitive entry; gravity compensates for that orientation and reads the stable entry.

Refutation criterion. The theorem fails if the lower signature is not $(33, 11, 1, 0)$, if the upper signature does not differ by a single unit in block 729, if the mantissas of $G\hbar$ generate a decimal carry, if $G\hbar$ varies between sheets, or if the denominators 45, 46 and $495 = 45 \cdot 11$ cease to appear in the extension 7:9.

28.6 Electromagnetic polarity, neutrality and rank-five gravitational carrier

The word ‘polarity’ can be typed through the parity of the two channel characters. Let

$$\mathcal{E} = \log[(1 - q_+^{90})(1 - q_-^{90})] - \log[(1 - q_+^{120})(1 - q_-^{120})],$$

$$\mathcal{B} = \log \frac{1 - q_-^{90}}{1 - q_+^{90}} - \log \frac{1 - q_-^{120}}{1 - q_+^{120}}.$$

Under the exchange involution $\mathbf{C}(q_+, q_-) = (q_-, q_+)$,

$$\mathcal{E} \circ \mathbf{C} = \mathcal{E}, \quad \mathcal{B} \circ \mathbf{C} = -\mathcal{B}.$$

The character $\mathcal{E}$ is common; the character $\mathcal{B}$ is oriented. The constitutive responses

$$\widehat{\mu} = e^{\mathcal{B} + \mathcal{E}}, \quad \widehat{\varepsilon} = e^{-\mathcal{B} + \mathcal{E}}$$

separate product and quotient exactly:

$$\widehat{c} = (\widehat{\mu}\widehat{\varepsilon})^{-1/2} = e^{-\mathcal{E}},$$

$$\widehat{Z} = (\widehat{\mu}/\widehat{\varepsilon})^{1/2} = e^{\mathcal{B}}.$$

The oriented potential difference lies in $\mathcal{B}$; the common propagation capacity lies in $\mathcal{E}$. Permittivity and permeability are the two conjugate recombinations. This explains the dual function of the vacancy: as an electrical opening, it changes the sign of the oriented coordinate; as magnetic transport, it preserves its circulation through $J_{\text{comm}}^2 = -I$.

Neutrality then has an internal definition:

$$X_{\text{neu}} := \ker \mathcal{B}.$$

In that subspace,

$$\widehat{Z} = 1, \quad \widehat{\mu} = \widehat{\varepsilon} = e^{\mathcal{E}}.$$

Every configuration fixed by conjugation belongs to the neutral sector. Indeed, if $\mathbf{C}x = x$, then

$$\mathcal{B}(x) = \mathcal{B}(\mathbf{C}x) = -\mathcal{B}(x),$$

and therefore $\mathcal{B}(x) = 0$. The inclusion

$$\boxed{\text{Fix}(\mathbf{C}) \subseteq X_{\text{neu}}}$$

is the algebraic content of a Majorana-type sector: a self-conjugate section cannot carry odd oriented charge. The neutral sector can be larger than the fixed locus; equality would require every configuration with $\mathcal{B} = 0$ to be self-conjugate. The distinction separates neutrality from self-conjugacy without leaving the operator.

Gravity receives only the common character in its scalar factor. In the internal chart

$$c_{\text{int}} = c_{\text{ref}} \widehat{c} = c_{\text{ref}} e^{-\mathcal{E}},$$

the transducer becomes

$$\boxed{G_a = \vartheta_{108} \frac{c_{\text{ref}}^5 t_0^2}{\hbar_a} e^{-5\mathcal{E}}.}$$

The oriented character $\mathcal{B}$ cancels exactly. The scalar gravitational response is even under sheet exchange and reads the common propagation coordinate five times. This fifth power is not confused with the five tensor components; the two layers combine in the operator

$$\mathbb{G}_{\text{TT},a}(n) = G_a \Lambda(n) \Gamma_5 P_5 : V_{12} \longrightarrow \mathcal{H}_{\text{TT}}(n),$$

where

$$P_5 : V_{12} \rightarrow V_5, \quad \Gamma_5 : V_5 \rightarrow \text{Sym}_0^2(\mathbb{R}^3).$$

The complete composition can be written as

$$\boxed{(\mathcal{E}, \mathcal{B}, x) \longmapsto \vartheta_{108} \frac{c_{\text{ref}}^5 t_0^2}{\hbar_a} e^{-5\mathcal{E}} \Lambda(n) \Gamma_5 P_5 x.}$$

The scalar eliminates odd dependence on $\mathcal{B}$; the projector P_5 preserves the five-component carrier; and Γ_5 realizes transverse traceless polarization. This gives a precise account of the relation between electromagnetic polarity and gravitational pentapolarity. The former lies in the even/odd pair $(\mathcal{E}, \mathcal{B})$. The latter lies in the rank of the tensor carrier. They are linked because the same state feeds both and because the gravitational scalar uses the fifth power of the common channel.

The role of symmetry is also clarified. The involution is not the object preserved first; it is the visible effect of preserving the boundary unit under two readings. The state retains

$$\mathcal{E}, \quad |\mathcal{B}|, \quad G_a \hbar_a, \quad \text{and the orbit of } P_5 x,$$

while conjugation reverses $\mathcal{B}$. Symmetry appears as equivalence between two orientations of the same preserved potential. It is therefore a boundary property, not an interior decoration.

Refutation criterion. The link fails if $\mathcal{E}$ is not even, if $\mathcal{B}$ is not odd, if a self-conjugate section has $\mathcal{B} \neq 0$, if the scalar transducer depends on $\mathcal{B}$, or if P_5 ceases to have rank five.

28.7 Integral holonomic configuration law

The configuration assembling the preserved invariants and the operation making them dynamic is called the **integral APP–TRIT–TPK holonomic state**. A state of depth n is written

$$X_n^{\text{hol}} = (\Sigma_n, \Pi_n; R_n^+, q_n^+, R_n^\times, q_n^\times; \tau_n, \varepsilon_n; g_n, h_n, c_n, \partial_n, \lambda_n, \mathfrak{m}_n).$$

The first coordinates preserve the two APP sheets; the next, residue, quotient and carry; (τ, ε) distinguish geometric regime and temporal orientation; and $(g, h, c, \partial, \lambda, \mathfrak{m})$ preserve phase, height, carry, boundary, survival and memory. The TPK transition

$$U_n = \text{Upd}_n \circ \text{Tra}_n \circ \text{Sel}_n$$

is an automorphism of the complete fibre on its reversibility domain. The restriction maps

$$r_{n+1,n} : X_{n+1}^{\text{hol}} \rightarrow X_n^{\text{hol}}$$

satisfy

$$r_{n+1,n} U_{n+1} = U_n r_{n+1,n}.$$

The infinite configuration is not postulated as a decimal sequence. It is the section

$$X_\infty^{\text{hol}} = \varprojlim X_n^{\text{hol}}.$$

Three systems of realizations act on it:

$$\mathcal{L}_{\text{num}} : X_\infty^{\text{hol}} \rightarrow \mathbb{R},$$

$$\mathcal{L}_{\text{const}} : X_\infty^{\text{hol}} \rightarrow \prod_{\xi} \mathcal{L}_{\xi}^+,$$

$$\mathcal{L}_{\text{part}} : X_\infty^{\text{hol}} \rightarrow \text{Sect}(\mathcal{A}_{\text{masas}}).$$

The first determines numbers and expansions; the second, sections of dimensional lines; the third, fibres, orbits and multisections of the matter atlas. All three commute with restriction:

$$s_{n+1,n}^\bullet L_{\bullet,n+1} = L_{\bullet,n} r_{n+1,n}.$$

Thus a number, a constant and a particle are not objects identified by having the same scalar value. They are three images of the same configuration when their readers preserve the same genealogy. The number is the Archimedean coordinate; the constant is that coordinate on a dimensional line; the particle is a persistent section preserving the route that produced the coordinate.

The single law can now be stated.

Integral holonomic configuration theorem. Every surviving APP seed determines, through TRIT and TPK, a section X_∞^{hol} such that:

1. the sum–product discrepancy defines a lifted integer potential;
2. the aperiodic word converts its temporal difference into tritic curvature;
3. TPK selects the unique extension compatible with phase and memory;

4. phase return acts through a positive holonomy on action, matter and gravity;
5. the numerical, dimensional and material realizations preserve the same section under changes in depth;
6. the dual Archimedean and exceptional projection preserves a common origin and distinct codomains;
7. the half-boundary unit is preserved at every refinement and the logical entropy of the complete state remains zero.

The compact proof chain is

$$\begin{array}{c}
 \Omega_{\text{seed}} \xrightarrow{\text{APP}} \mathcal{P}_{\Sigma\Pi} \xrightarrow{-\Delta^2} \tau \\
 \xrightarrow{\text{TRIT}} (\mathbb{A}_+, \mathbb{A}_0, \mathbb{A}_-) \xrightarrow{\text{TPK}} X_{\infty}^{\text{hol}} \\
 \xrightarrow{(\mathcal{L}_{\text{num}}, \mathcal{L}_{\text{const}}, \mathcal{L}_{\text{part}})} (\text{number, constant, particle}).
 \end{array}$$

The joint nonadic connection does not choose ‘the most probable future’. It chooses the unique extension compatible with the integral section when the reduced shadow admits two continuations. This operation is a predefinition: the future coordinate already belongs to the inverse limit, although it has not yet been determined at the visible depth. The net Landauer cost is zero because selection does not erase the unchosen branch from the universe of cylinders; it records in the memory register which cylinder contains the state.

The crystal is evolutionary in a precise sense. Its language does not grow exponentially, since $p(m) = m + 1$, but neither does it freeze into a period. Each depth adds exactly one factor class and each nonadic phase multiplies it by nine. The system preserves the lattice unit, boundary and determinant while modifying memory and helical height. Evolution and conservation are not opposing terms: they describe, respectively, the advance of the state and the invariance of its augmentation class.

The chain

$$\pi, \varphi, e \longrightarrow \alpha \longrightarrow (\mathfrak{e}_e^\beta, \hbar, G, \gamma_{\text{HMT}}, \dots)$$

must be read within the complete correlated generation. The first three coordinates are terminal readers of the same nonadic state; α is the dodecaphasic seam of that output; and the physical constants are subsequent representations of the already closed state. The mirror π/e , self-scaling φ , expansion $e + \pi$ and coincidences of the three channels are internal synchronization operations, not conventional values introduced to manufacture α .

The final sequence now proved is

$$\begin{array}{l}
 \text{APP potential } (6, 15, 27) \longrightarrow d_{\Delta}\Delta = 3(3, -7, 4), \\
 P_N \longrightarrow \tau_N = -\Delta^2 P_{N-1} \longrightarrow \text{unique future bifurcation per scale,} \\
 (21/22) \longrightarrow (6/7) \longrightarrow V_{\text{vac}}, \quad \det V_{\text{vac}} = 15, \\
 \alpha^3 \longrightarrow \alpha^6 \longrightarrow \alpha^{16} \longrightarrow 10^{-34}, \\
 R_{\text{act}} \longrightarrow (\mathfrak{e}_e^\beta, \hbar, G) \longrightarrow G\hbar \text{ invariant,} \\
 (33, 11, 1, 0) \leftrightarrow (34, 11, 1, 0) \longrightarrow (34, 11, 45, 46).
 \end{array}$$

This is the mathematical core that allows the work to be carried into the article: a dynamics of potential, curvature, survival, return and realization that preserves unity while converting it into number, constant and matter.

28.8 Intrinsic arithmetic of the holonomic state

The question of whether the coordinates of the holonomic state have an arithmetic of their own admits an affirmative and precise answer. The primary arithmetic object is not a field of scalars, because one history can be multiplied by another only when the terminal state of the first coincides with the initial state of the second. The natural object is therefore an **enriched path algebra** over the category of TPK transports.

Let C_{TPK} be the category whose objects are holonomic states

$$x = (\tau, \ell, o, h, R, q, \kappa, \partial, \gamma, M, S, b, \chi, g)$$

and whose arrows are admissible transitions

$$u : x \longrightarrow y, \quad U_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t.$$

Here τ is the TRIT regime; ℓ , the APP sheet; o , orientation; h , helical height; (R, q) , residue and quotient; κ , carry; ∂ , boundary; γ , route; M , the memory record; S , survival; b , block; χ , exceptional shadow; and g , nonadic phase. Each object contains the local quadratic algebra

$$\mathbb{A}_{\tau(x)} = \mathbb{R}[J_x] / (J_x^2 + \tau(x)I), \quad \tau(x) \in \{+1, 0, -1\}.$$

To record the increments of carry, boundary and memory without loss, let $\Lambda = \mathbb{R}[\mathcal{M}]$ be the algebra of the memory-record monoid. Each arrow u is associated with a symbol $[u]$. The sum

$$a[u] + b[v]$$

is the formal superposition of two histories compatible with the same reading framework; the product exists causally only through concatenation. If $u : x \rightarrow y$ and $v : y \rightarrow z$, define

$$(a[v]) \star (b[u]) = a \rho_v(b) \Omega(v, u) [v \circ u], \quad (99.1)$$

and set it to zero when the arrows are not composable. The transport ρ_v carries the local coefficient from the fibre of y to that of z ; the factor $\Omega(v, u) \in \Lambda$ preserves the increment of carry, torsion, boundary and memory produced by concatenation. The law

$$\Omega(w, v) \Omega(w \circ v, u) = \rho_w(\Omega(v, u)) \Omega(w, v \circ u) \quad (99.2)$$

is exactly the cocycle identity making the product associative:

$$([w] \star [v]) \star [u] = [w] \star ([v] \star [u]).$$

This equality is not an added convention. It is the common algebraic form of the APP carry identity, the route-concatenation cocycle and the nonsplit extension of the clock 108. Ordinary arithmetic forgets the factor Ω when realizing only the value; holonomic arithmetic also preserves the procedure that produced it.

The constant states $e_x = [\text{id}_x]$ are local idempotents:

$$e_x \star e_x = e_x, \quad e_y \star e_x = 0 \quad (x \neq y).$$

Thus unity is not merely the scalar 1, but the compatible system of idempotents identifying each resolution of the same state. In the inverse limit one obtains

$$\mathbf{1}_{\text{hol}} = (e_{x_n})_{n \geq 0}, \quad r_{n+1, n}(e_{x_{n+1}}) = e_{x_n}. \quad (99.3)$$

This is the dimensional unit that can be refined inward and realized outward without losing its identity. Refinement increases resolution; representation changes the codomain; neither act creates the unit anew.

Nor is noncommutativity postulated from an external structure. Two arrows u and v may be composable in one order and not in the reverse; when both orders exist, their cocycles and sheets may differ:

$$[v] \star [u] - [u] \star [v] = \Omega(v, u)[v \circ u] - \Omega(u, v)[u \circ v]. \quad (99.4)$$

The difference measures the effective order of accumulation and folding. The base phase system can be Abelian while the total algebra is not: noncommutativity lies in sheet composition, orientation transport and the affine memory cocycle.

28.9 Complex numbers, Clifford algebras, quaternions and octonions

The comparison with hypercomplex systems is resolved by distinguishing three levels. At the local level, TRIT already contains the three fundamental real quadratic algebras:

τ	J_τ^2	$\mathbb{A}_\tau$
+1	$-I$	$\mathbb{C}$ (elliptic regime),
0	0	$\mathbb{R}[\varepsilon]/(\varepsilon^2)$ (parabolic regime),
-1	$+I$	$\mathbb{R} \oplus \mathbb{R}$ (hyperbolic regime).

(100.1)

The complex number is therefore not the universal language of the system: it is one of its three curvature fibres. The parabolic fibre records an infinitesimal deformation without quadratic rotation; the hyperbolic fibre separates two conjugate real directions; the elliptic fibre produces internal rotation. TRIT does not choose among three geometric metaphors: it types which of these three arithmetics governs each transition.

An even stronger structure appears in the principal electronic block. Let

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 3/4 & \sqrt{3}/4 \\ \sqrt{3}/4 & 1/4 \end{pmatrix}.$$

Both matrices are projectors and

$$[P, Q] = \frac{\sqrt{3}}{4} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Define

$$J = \frac{4}{\sqrt{3}}[P, Q] = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad S = 2P - I = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad K = SJ.$$

Then

$$J^2 = -I, \quad S^2 = I, \quad SJ = -JS, \quad K^2 = I. \quad (100.2)$$

Consequently, the block generates exactly

$$\langle I, J, S, K \rangle \cong \text{Cl}_{1,1}(\mathbb{R}) \cong M_2(\mathbb{R}), \quad (100.3)$$

the split-quaternion algebra. Furthermore,

$$2Q - I = \frac{1}{2}S + \frac{\sqrt{3}}{2}K, \quad (100.4)$$

so that the product projector occupies a direction at angle $\pi/3$ in the plane generated by the two split involutions. The electronic prefactor $\sqrt{3}/4$, the internal root of -1 , the fraction $3/4 = 90/120$ and the separation of the APP sheets are brought together in a single exact algebraic representation.

This identification clarifies the proposed analogy. The first object to arise is not Hamilton's quaternion division algebra $\mathbb{H}$, but its split form $\text{Cl}_{1,1}$. The difference is substantive: the HMT system needs idempotents, thresholds and null directions, whereas a division algebra would eliminate precisely those vacancies. Hamilton quaternions may appear in a representation possessing two anticommuting roots of $-I$; octonions, in a subsequent exceptional representation whose composition law reproduces the Fano plane. Neither should replace the path algebra preceding them.

Associativity of the complete TPK is preserved by the cocycle (99.2). If a subsequent exceptional projection has a nonzero associator,

$$[a, b, c] = (a \star b) \star c - a \star (b \star c),$$

that associator does not describe an inconsistency of the generator: it quantifies the memory left outside the projection's codomain. This precisely separates intrinsic, associative arithmetic with a memory record from its potentially nonassociative hypercomplex shadows.

28.10 Ternary synchronization and the dodecaphasic singularity of alpha

The coordinated generation of π , e and φ is organized in the difference lattice of type A_2 . For

$$N_K = (N_\pi(K), N_e(K), N_\varphi(K)) \in \mathbb{Z}^3$$

define the relative boundary

$$\partial_{A_2}(x, y, z) = (x - y, y - z, z - x). \quad (101.1)$$

Its kernel is the diagonal $\mathbb{Z}(1, 1, 1)$. Thus each triple has two distinct components: a common coordinate measuring the advance of the memory record, and a relative coordinate measuring desynchronization of the three channels. Over $\mathbb{Q}$,

$$\mathbb{Q}^3 = \mathbb{Q}(1, 1, 1) \oplus A_{2,\mathbb{Q}};$$

over $\mathbb{Z}$, reconstruction preserves a cyclic defect of order three. This is the algebraic reason why coordination is ternary rather than a binary comparison of constants.

The coincidence table follows an itinerary of walls:

$$H_{\pi e} \longrightarrow H_{e\varphi} \longrightarrow H_{\pi\varphi} \longrightarrow 0 \longrightarrow 0. \quad (101.2)$$

At $K = 8$ and $K = 9$ the relative component vanishes:

$$N_8 = 59(1, 1, 1), \quad N_9 = 43(1, 1, 1).$$

However,

$$N_9 - N_8 = -16(1, 1, 1). \quad (101.3)$$

The system returns to phase coincidence, but not to the same state: desynchronization returns to zero while diagonal memory advances sixteen units per channel. The visible equality $\pi = e = \varphi$ expresses not immobility but relative closure with holonomic translation.

The coordinate α occupies precisely the seam allowing this return to be realized without confusing it with a restart. Dodecaphasic closure takes the already synchronized relative class, preserves the diagonal memory record and lifts it to a signed section. Its two derivations are not two fits of an external quantity: they are two representations of the same holonomic class, one through integer closure with carry and the other through the simple root of the nonadic extension. Equality of the two routes expresses commutativity of the realization square.

In this way, α does not merely follow π, φ, e in a list. It is the coupling coordinate transforming relative closure of the correlated system into dimensional transduction capacity. The twelve-phase wheel does not generate the three regions again: it certifies that their common return can cross a boundary while preserving the memory subsequently feeding action, the electron, the determinantal scale and the exceptional branch.

28.11 Conservative evolution of unity as an algebraic theorem

The expression ‘conservative evolution of the dimensional unit through holographic projection’ can be formulated without relying on an image. Let X_n be the holonomic states at depth n , $r_{n+1,n} : X_{n+1} \rightarrow X_n$ their restrictions, and

$$X_\infty^{\text{hol}} = \varprojlim_n X_n$$

their inverse limit. A dimensional unit is a compatible section $\mathbf{1}_{\text{hol}} = (e_{x_n})_n$ of local idempotents. The TPK transition induces algebraic transports $U_{t,*}$ and satisfies

$$U_{t,*}(e_x) = e_{U_t x}, \quad U_{t,*}(e_x)^2 = U_{t,*}(e_x). \quad (102.1)$$

There is also an augmentation morphism

$$\epsilon : \mathcal{A}_{\text{hol}} \longrightarrow \mathbb{R}$$

that forgets history but preserves the normalized multiplicity of the unit. Conservation and evolution are expressed simultaneously by

$$\epsilon(U_{t,*}a) = \epsilon(a), \quad d_M(\Gamma_9 a) = d_M(a) + 1, \quad (102.2)$$

where d_M is the memory degree. The first equality states that unity is neither created nor destroyed; the second states that the state does not remain motionless. The holonomy Γ_9 returns to the same phase and raises the memory record by one unit.

The completion

$$1000 = 729 + 243 + 27 + 1$$

realizes this law finitely. The lower word distributes vacancies as $(33, 11, 1, 0)$, while the upper reading preserves the boundary unit and produces $(34, 11, 1, 0)$. The increment does not alter the total; it changes the location of the unit within the cubic filtration. The vector

$$(34, 11, 45, 46), \quad 34 + 11 = 45, \quad 45 + 1 = 46, \quad (102.3)$$

is therefore a signature of depth and boundary, not a juxtaposition of decimal scales.

Now let

$$\rho_{\mathbb{R}}, \quad \rho_{\text{exc}}, \quad \rho_{\text{MD}}$$

the Archimedean, exceptional and dimensional representations of the holonomic algebra. Their holographic character consists in their being unital homomorphisms from the same source:

$$\rho_j(\mathbf{1}_{\text{hol}}) = I_j, \quad \rho_j(a \star b) = \rho_j(a)\rho_j(b). \quad (102.4)$$

Dual projection does not divide the origin into two universes; it preserves the same element and changes its representation. Fractal refinement operates in the opposite direction: rather than projecting to a codomain, it extends the section to a greater depth. The two movements kept separate in the preceding exposition are thus coordinated:

interior refinement : $X_n \longleftarrow X_{n+1} \longleftarrow X_{n+2} \longleftarrow \cdots$, holographic realization : $\mathcal{A}_{\text{hol}} \longrightarrow \text{End}(V_j)$.	(102.5)
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Unity is preserved in both directions: through compatibility under restriction and preservation of the identity element under representation. This dual invariance is the algebraic content of conservative evolution of the dimensional unit.

28.12 The two clocks and the Sturmian helical suspension

The temporality of the crystal arises from two mutually irreducible clocks. The first is finite and exact: the extension

$$0 \longrightarrow C_{12} \longrightarrow C_{108} \longrightarrow C_9 \longrightarrow 0 \quad (103.1)$$

coordinates nonadic phase and dodecaphasic memory. It is not the direct product $C_9 \times C_{12}$: its extension class preserves the carry appearing when the phase boundary is crossed. The second clock is the irrational rotation

$$R_\delta : \theta \longmapsto \theta + \delta \pmod{1}, \quad \delta = \log_{10}(10/9), \quad (103.2)$$

whose mechanical coding produces the Sturmian word of vacancies.

The joint dynamics can be written as a skew product

$$\mathcal{F}(r, \theta, M) = (r + 1, \theta + \delta, M + c_{108}(r) + c_{\text{St}}(\theta)), \quad (103.3)$$

where c_{108} is the carry of the finite clock and c_{St} records crossing of the irrational window. The base $C_9 \times \mathbb{T}$ is Abelian; the affine extension ceases to be so when the APP sheets and the order of Sel, Tra and Upd are incorporated. The helix is therefore not a circle drawn in three dimensions, but the lift of a phase rotation whose vertical coordinate is the memory record.

The ascending and descending readings are conjugate. If ι reverses orientation and route, then

$$\iota \circ \mathcal{F} \circ \iota^{-1} = \mathcal{F}^{-1}, \quad (103.4)$$

with reversal of the oriented increments and preservation of the common responses. The mirror π/e is another involution: it exchanges two regional readers at the same height, whereas (103.4) reverses the temporal direction of the helix. Their distinction explains why a round-trip reading can preserve the common value without returning to the same memory.

The crystal has long-range order and zero topological entropy. For the Sturmian language,

$$p(m) = m + 1,$$

and the typed extensions satisfy

$$p_3(m) = 3(m + 1), \quad p_9(m) = 9(m + 1). \quad (103.5)$$

Phase multiplies the diversity of factors but does not change their linear growth. Hence the system is simultaneously nonperiodic, ordered and capable of unlimited extension. The gaps 21/22, the returns 6/7 and the thousand-position windows with 45/46 events are three observations of this same suspension, not three ad hoc rhythms.

This construction makes precise the designation aperiodic topological-temporal crystal. It is topological because information resides in route, boundary and holonomy classes; temporal because the transition distinguishes orientation, return and memory; crystalline because it possesses stable combinatorial order; and aperiodic because no nonzero translation fixes its infinite word.

28.13 Vacancy, potential difference and electromagnetic polarity

APP contains an arithmetic notion of potential before any physical chart. For each cell (i, j) ,

$$\Delta_{\Sigma\Pi}(i, j) = ij - (i + j) = (R^\times - R^+) + 9(q^\times - q^+). \quad (104.1)$$

The first term is the visible difference of residues; the second preserves the carry difference. In the antidiagonal fibre through the centre, the electronic vacancy is the minimal loss of one multiplicative-carry unit that leaves the additive sheet, product residue and TRIT regime invariant. It is thus defined without introducing an external zero.

The potential boundary is the coboundary of (104.1). In the central triangle it produces

$$d_\Delta \Delta = 3(3, -7, 4), \quad 3 - 7 + 4 = 0. \quad (104.2)$$

The zero sum expresses conservation; the nonzero components express polarization. The potential difference is not a quantity subsequently added to the graph: it is the impossibility of making accumulation and folding coincide locally without transporting the memory record.

The representation (100.3) separates two constitutive actions. The generator J is rotational and changes sign under orientation reversal; the generator S is a sheet involution and preserves sign. Thus an even and an odd character of the same algebra produce two readers

$$\chi_E(SaS) = \chi_E(a), \quad \chi_B(SaS) = -\chi_B(a), \quad (104.3)$$

that realize, respectively, the electric and magnetic sectors. Electricity reads potential separation between sheets; magnetism reads the oriented circulation produced by their commutator. Their perpendicularity is not imposed through a prior Euclidean plane: it follows from $SJ = -JS$ and $J^2 = -I$.

The regime $\tau = 0$ occupies the threshold between the two orientations. Its nilpotent element preserves a first-order perturbation and annihilates its square; it provides the algebraic type of a neutral channel without erasing the history passing through it. This explains why neutrality, vacancy and polarity can arise from the same support without being identified with one another.

The action scale and gravity extend the same holonomic character. If

$$R_{\text{act}} = \frac{H_5}{\eta_{\text{ret}}},$$

then the common transport satisfies

$$R_{\text{act}} = \frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}} = \frac{m_{108,\text{pre}}}{m_{108,\text{ret}}} = \frac{G_{\text{ret}}}{G_{\text{pre}}} = \left(\frac{e_e^\beta}{e_0} \right)^{1/\kappa_e}, \quad G_a \hbar_a = \text{constant}. \quad (104.4)$$

Electron, action and gravity are not three independent calibrations here: they are representations of the same return quotient at different scales and in different codomains.

28.14 Exact sequence reading and the biological aperiodic crystal

The relation to DNA can be formulated as a concrete mathematical reader, without turning a historical analogy into a premise. Encode the four bases using two binary characters:

$$A \mapsto (0, 0), \quad G \mapsto (0, 1), \quad T \mapsto (1, 0), \quad C \mapsto (1, 1), \quad (105.1)$$

where the first coordinate distinguishes purine from pyrimidine and the second the weak or strong pairing type. Watson–Crick complementarity is the involution

$$C_{\text{DNA}}(p, h) = (1 - p, h), \quad C_{\text{DNA}}^2 = I. \quad (105.2)$$

An oriented chain is lifted to

$$\tilde{s}_n = (p_n, h_n, n \bmod 3),$$

so that reading phase is not confused with the symbol. Route reversal and complementarity are combined in reverse complementation, which acts as an involution on the path algebra. APP can evaluate both character accumulation and concatenation in each window; TRIT types the second difference of the resulting potential; TPK preserves phase, boundary, returns, carry and return words.

This representation allows one to ask exactly whether a sequence region belongs to the minimal aperiodic class: for its factors of length m , one calculates complexity $p_s(m)$, the number of right extensions and their return words. The minimal HMT signature is

$$p_s(m) = m + 1, \quad \sum_{u \in L_m} (d^+(u) - 1) = 1, \quad (105.3)$$

or, when a finite phase fibre is preserved, an extension $q(m + 1)$. The equality selects a unique combinatorial branching per length; TPK determines the compatible continuation without erasing the unrealized alternative from the cylinder space.

The relevance of Schrödinger's expression 'aperiodic crystal' thus becomes structural. A periodic crystal would repeat a block and lose informational capacity; a random chain would have exponential complexity and lose long-range order. The Sturmian regime occupies the exact minimum between the two: it adds a single new class at each scale. The TRIT topological-temporal unit can read this aperiodicity because it separates curvature, orientation and phase without destroying the sequence it classifies.

The criterion is directly refutable: it fails if reverse complementation does not intertwine the route involution, if observed complexity does not belong to the declared linear class, if several special words appear where the model requires one, or if compatible extension requires erasing memory. Biology does not select the HMT generator; it receives from it a sequence reader capable of distinguishing repetition, minimal aperiodicity and disorder.

28.15 Pentadicity, the Witt star and gravitational polarizations

The recurrent appearance of the number five must be typed through two distinct representations. The five regions of π form a regional fibre on which the symmetrizer acts

$$S_{\pi,5} = \frac{1}{5} \mathbf{1}\mathbf{1}^\top, \quad \text{rank } S_{\pi,5} = 1. \quad (106.1)$$

Its function is to extract the common component of the penta-fibre. In the gravitational sector, by contrast, the dodecaphasic decomposition

$$V_{12} \cong \mathbf{1} \oplus V_3 \oplus V'_3 \oplus V_5 \quad (106.2)$$

contains a projector $P_{G,5} : V_{12} \rightarrow V_5$ of rank five. The two matrices are not the same operator: one has rank one on the regional fibre and the other rank five on the dodecaphasic representation. Their correct relation is that both are images of the same holonomic arithmetic:

$$(\rho_{\pi,5}, \rho_{G,5}) : \mathcal{A}_{\text{hol}} \longrightarrow \text{End}(\mathbb{Q}^5) \times \text{End}(V_{12}). \quad (106.3)$$

The Witt star organizes exceptional pentadic incidence; the module V_5 organizes the gravitational tensor carrier. Coordination occurs in the APP–TRIT–TPK source, not by identifying their codomains.

The gravitational realization continues through

$$V_{12} \xrightarrow{P_{G,5}} V_5 \xrightarrow{\Gamma_5} \text{STF}_3 \xrightarrow{\Pi_{\text{TT}}} V_{\text{TT}}, \quad (106.4)$$

with dimensions $12 \rightarrow 5 \rightarrow 5 \rightarrow 2$. The five components constitute the structural carrier before propagation; the two transverse traceless components are the propagating output. There is no contradiction between original pentapolarity and radiative bipolarity: they belong to distinct stages of the same transduction.

In the joint discrete structure of the continuum, pentapolarity is expressed by the inseparable vector

$$\mathcal{G}_5 = (\mathcal{G}_{\text{sp}}, \mathcal{G}_{\text{sol}}, \mathcal{G}_{\text{gau}}, \mathcal{G}_{\text{coh}}, \mathcal{G}_{\text{det}}). \quad (106.5)$$

Its five coordinates are not five theories added afterwards, but five inherent aspects of the same state: spectral, solenoidal, gauge, coherent and determinantal. The gravitational tensor

$$\mathbb{G}_{\text{TT}} = G \wedge \Gamma_5 P_{G,5} \quad (106.6)$$

jointly transports these aspects to the physical sector. This formula explains why regional pentadicity, exceptional structure and the five polarizations can resonate without being reduced to a coincidence of cardinalities.

28.16 Variance duality between histories and readers

Retrospective reading reveals a categorical structure uniting inward fractalization and outward holographic realization. Let (X_n) be the set of holonomic states at depth (n) , and let

$$r_{n+1,n} : X_{n+1} \longrightarrow X_n \quad (108.1)$$

be the restriction removing the last layer while preserving the prefix, sheet, accumulated carry and corresponding boundary. The complete history is a state of the inverse limit

$$\boxed{X_\infty = \varprojlim_n X_n.} \quad (108.2)$$

The direction of observations is opposite. If (f) is a cylindrical reader defined on (X_n) , its extension to (X_{n+1}) is

$$r_{n+1,n}^* f = f \circ r_{n+1,n}. \quad (108.3)$$

Therefore the reader algebras form a direct system:

$$\boxed{\mathfrak{S}_{\text{cyl}} = \varinjlim_n (\mathfrak{S}_n, r_{n+1,n}^*).} \quad (108.4)$$

This dual variance is the exact mathematical formulation of the two movements described by the author's intuition. Deepening inward means specifying a compatible section of the inverse system. Expanding outward means extending the repertoire of readers through the direct system. Neither direction replaces the other: one constructs the history and the other constructs the capacity to realize it.

Let (e_x) be the indicator idempotent of the coarse cylinder $(x \in X_n)$. The lift of that cylinder is

$$r^* e_x = \sum_{y:r(y)=x} e_y. \quad (108.5)$$

The summands are orthogonal, so $(r^* e_x)$ remains idempotent. Furthermore,

$$r^* \left(\sum_{x \in X_n} e_x \right) = \sum_{y \in X_{n+1}} e_y. \quad (108.6)$$

Global unity is preserved while the number of refinements increases. An individual trajectory in turn determines a compatible system $((e_{x_n})_n)$. The former is the algebra identity; the latter is the genealogical unit of a history. This distinction makes conservative evolution precise: the total possibility space preserves its unity and each route preserves its membership through compatible restrictions.

A continuous reader (L) of the deep state is equivalent to a system of finite readers (L_n) satisfying

$$L_{n+1} \circ r_{n+1,n}^* = L_n. \quad (108.7)$$

Consequently, each digit or coordinate determined at finite depth is predefined by a cylinder; increasing depth does not recalibrate that result, but determines new compatible cylinders. Predefinition is the naturality of (108.7), not the advance insertion of an experimental value.

Refutation criterion. The construction fails if a restriction does not recover the prefix, if the *pullback* is not multiplicative, if unity is not preserved, or if a deep reader modifies a coordinate already fixed in an earlier cylinder.

28.17 Universal presentation of the holonomic algebra

Let $(\text{mathcal{C}}_n)$ be the category of states at depth (n) and admissible histories generated by

$$U_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t. \quad (109.1)$$

For each object (x) , let (e_x) be its local identity and (J_x) the generator of the TRIT fibre:

$$J_x^2 = -\tau(x)e_x, \quad \tau(x) \in \{+1, 0, -1\}. \quad (109.2)$$

For each arrow $(u : x \rightarrow y)$, let (T_u) be the transport. The memory record defines a central multiplier $(\Omega(v, u))$ on each composable pair. The finite presentation is

$$\mathfrak{H}_n = \mathbb{R}\langle e_x, J_x, T_u \rangle / \mathcal{R}_{\text{HMT}}, \quad (109.3)$$

where

$$\begin{aligned} e_x e_y &= \delta_{xy} e_x, & T_u &= e_{t(u)} T_u e_{s(u)}, \\ T_u a &= \rho_u(a) T_u, & T_v T_u &= \Omega(v, u) T_{v \circ u}. \end{aligned} \quad (109.4)$$

The product is declared zero when the arrows are not composable. Associativity is equivalent to

$$\rho_w(\Omega(v, u)) \Omega(w, v \circ u) = \Omega(w, v) \Omega(w \circ v, u). \quad (109.5)$$

Indeed, the two products $((T_w T_v) T_u)$ and $(T_w (T_v T_u))$ differ only in the two sides of (109.5). The cocycle identity proves that every history has a memory independent of the chosen parenthesization.

The sum in (109.3) superposes alternative histories. Its product concatenates causally composable histories. This distinction turns the APP sum–product difference into dynamic arithmetic: the sum does not erase the alternative and the product does not erase provenance.

The nonadic connection appears in the elementary section of the quotient. For $(a, b \in \{0, \dots, 8\})$,

$$c_9(a, b) = \left\lfloor \frac{a + b}{9} \right\rfloor \quad (109.6)$$

satisfies

$$c_9(a, b) + c_9(a + b \bmod 9, c) = c_9(b, c) + c_9(a, b + c \bmod 9). \quad (109.7)$$

If (X) represents a step and (λ) the memory of one turn,

$$X^9 = \lambda I, \quad X^a X^b = \lambda^{c_9(a, b)} X^{a+b \bmod 9}. \quad (109.8)$$

The helix is condensed in (109.8): the exponent returns; the coefficient remembers how many times the boundary was crossed.

28.18 Universal property and spectrum of realizations

Let (E_x) be a system of modules, let $(\pi_x : A_x \rightarrow \text{End}(E_x))$ be representations of the TRIT fibres, and let $(V_u : E_x \rightarrow E_y)$ be transports satisfying

$$V_u \pi_x(a) = \pi_y(\rho_u(a)) V_u, \quad (110.1)$$

$$V_v V_u = \pi_z(\Omega(v, u)) V_{v \circ u}. \quad (110.2)$$

The assignment of generators defines a homomorphism from the free algebra. Equations (110.1)–(110.2) make that homomorphism annihilate the ideal $(\text{mathcal{R}}_{\text{HMT}})$. There is then a unique representation

$$\Pi : \mathfrak{H}_n \longrightarrow \text{End}\left(\bigoplus_x E_x\right) \quad (110.3)$$

realizing the prescribed data. Uniqueness follows because (e_x, J_x, T_u) generate the algebra.

This theorem provides the principle of ultrasimplification. Physics, number theory, the continuum and biology need not add generators to the kernel when their readers satisfy the same relations. Each is a representation of (109.3) in a different codomain.

Not all readers are scalar characters. A character ($\mathit{mathfrak{H}} \rightarrow \mathbb{R}$) would annihilate commutators and could not preserve the electronic block. We therefore define the typed spectrum

$$\text{Read}(\mathfrak{H}) = \coprod_B \text{Rep}_{\text{comp}}(\mathfrak{H}, B). \quad (110.4)$$

A scalar output takes the form

$$c = \sigma_B(\rho(x_\infty)), \quad (110.5)$$

where ρ preserves genealogy and σ_B is the invariant specific to the codomain. Five concepts that previously seemed heterogeneous are thus typed:

number	coordinate of an Archimedean representation,	(110.6)
constant	stable invariant of return, scale or depth,	
particle	persistent section of a module,	
field	representation transporting sections,	
symmetry	boundary stabilizer and preserved relations.	

28.19 Exponential identity of the three curvatures

We introduce the total algebra

$$\mathbb{A}_{\text{tri}} = A_+ \oplus A_0 \oplus A_- \quad (111.1)$$

with central idempotents (p_+, p_0, p_-) and the generator

$$\mathbb{J} = p_+ J_+ + p_0 J_0 + p_- J_-. \quad (111.2)$$

The exponential is defined within the algebra by

$$\text{Exp}_\star(X) = \sum_{n \geq 0} \frac{X^{\star n}}{n!}. \quad (111.3)$$

The three quadratic relations split the series into

$\begin{aligned} \text{Exp}_\star(t\mathbb{J}) &= p_+(\cos t + J_+ \sin t) \\ &\quad + p_0(1 + tJ_0) \\ &\quad + p_-(\cosh t + J_- \sinh t). \end{aligned}$	(111.4)
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The first row is rotation; the second is infinitesimal threshold transport; the third is dilation. A single power law contains the three curvatures.

Euler's identity occupies the elliptic face. In the closure produced by (pi_{HMT}) ,

$p_+ \text{Exp}_\star(\pi\mathbb{J}) + p_+ = 0.$	(111.5)
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The complex projection preserves (111.5) and forgets the parabolic and hyperbolic components. HMT jointly preserves closure, jet and scale. Euler's identity thus appears as an exact face of the tritic identity, not as an external model of the generator.

The number (e) has a different function:

$$\text{Exp}_\star(I) = e I. \quad (111.6)$$

The equation defines the function converting additive transport into multiplicative transport. The nonadic generation of its digits retains its own certificate; (111.6) identifies its role in the algebra.

Self-scaling φ occupies the hyperbolic face. For

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad H_\varphi = \frac{2F_{\text{av}}^2 - 3I}{\sqrt{5}}, \quad (111.7)$$

we verify

$$H_\varphi^2 = I, \quad (111.8)$$

and

$$\boxed{F_{\text{av}}^2 = \text{Exp}_\star(2 \log \varphi H_\varphi).} \quad (111.9)$$

Thus π measures elliptic closure, e realizes the exponential step, and φ measures stable hyperbolic dilation. Equalizations of the three channels are encounters of these functions within the same orbit, not universal equalities between their real values.

28.20 α as dodecaphasic transgression

The nonsplit extension produces the cocycle c_9 of (109.6). A phase turn determines an element of the kernel K_{12} , and the signed word K preserves how its sectors were crossed. Define the seam functional

$$\mathcal{T}_\alpha : Z^2(C_9; C_{12}) \times X_{12}^{\text{sgn}} \longrightarrow \mathcal{L}_\alpha. \quad (112.1)$$

The determined coordinate is

$$\boxed{\alpha = \mathcal{T}_\alpha(c_9, K).} \quad (112.2)$$

The formula does not identify the number α with a cohomology class. It identifies its type: it is the scalar obtained by evaluating the splitting obstruction together with dodecaphasic memory.

The first route realizes (112.2) through

$$\iota(P) + \iota(E) - \iota(\Phi) - \iota(K) = \iota(A). \quad (112.3)$$

The second route realizes the same gluing on the aperiodic rotation:

$$P_9(\alpha) = \delta, \quad \delta = \log_{10} \frac{10}{9}. \quad (112.4)$$

Their compatibility is expressed in the fibre product

$$\mathfrak{A}_\alpha = \{(x_{12}, x) : d_m(\Gamma_{12}^\alpha x_{12}) = d_m(x), P_9(x) = \delta\}. \quad (112.5)$$

A kernel matrix of the correlated system can now be formulated:

$$\begin{pmatrix} \text{Exp}_\star(\pi J_+) & 0 & 0 & 0 \\ 0 & \text{Exp}_\star(I) & 0 & 0 \\ 0 & 0 & \text{Exp}_\star(2 \log \varphi H_\varphi) & 0 \\ 0 & 0 & 0 & \text{Exp}_\star(P_9(\alpha)J_0) \end{pmatrix} = \begin{pmatrix} -I & 0 & 0 & 0 \\ 0 & e I & 0 & 0 \\ 0 & 0 & F_{\text{av}}^2 & 0 \\ 0 & 0 & 0 & I + \delta J_0 \end{pmatrix}. \quad (112.6)$$

The matrix does not linearize the genealogy. It displays four functions of a single algebra: closure, exponentiation, self-scaling and seam. Signed memory remains in (112.3)–(112.5).

In the lattice A_2 , the same seam appears as

$$\partial_{A_2} N_8 = \partial_{A_2} N_9 = 0, \quad N_9 - N_8 = -161. \quad (112.7)$$

α determines closure of the relative part without erasing diagonal translation. This is the exact meaning of its singularity: it is the coordinate distinguishing synchronization from restart.

28.21 The functional spectrum of constants

The uniform dodecaphasic reader also determines a sequence of first roots of critical return. Its separation ratio is proved to converge to the Feigenbaum constant, preserving the original selection, orientation and memory of the construction. (24.152).

The preceding structure reorganizes the constants already derived according to their function in the algebra, not their historical affiliation with mathematics or physics.

Catalan's constant is the second-order moment of the elliptic character:

$$G_{\text{Cat}} = \text{Tr}(N^{-2}Q_4) = L(2, \chi_{-4}). \quad (113.1)$$

Its functional setting preserves the entire system $C_2(s)$, not only the endpoint $s = 1$. In theorem 35.9 it is proved that the constitutive response $f(s) = (1 - s^3)/(1 - s^4)$, together with the condition $C_2(0) = 0$, uniquely determines the moment through $\mathcal{D}^2 C_2 = s(1 + s)f(s)/(1 + s + s^2)$. Cubic inversion in turn recovers the two oriented arguments.

The quarter-turn character and the triadic character are brought together in

$$\chi_{12} = \chi_{-3}\chi_{-4}. \quad (113.2)$$

Feigenbaum occupies the nonlinear reader. The dodecaphasic wheel generates

$$u_0(x) = 1 - \frac{1}{12} \sum_{m=1}^{12} \cos\left(\frac{2(3m-1)}{\sqrt{899}}x\right), \quad (113.3)$$

with

$$899 = 30^2 - 1 = 29 \cdot 31, \quad M_{30} = \begin{pmatrix} 30 & 899 \\ 1 & 30 \end{pmatrix}, \quad \det M_{30} = 1. \quad (113.4)$$

The constructed system is analytic, even, unimodal, quadratically critical and has negative Schwarzian derivative on the proved domain. The doubling renormalizer reads the accumulation rate and spatial rescaling. α_F is a coordinate of that nonlinear reader; α is the seam of (112.2). The distinct subscripts express a difference of operation.

Boltzmann occupies the transducer between the thermal and energy lines:

$$k_B : \mathcal{L}_\Theta \longrightarrow \mathcal{L}_E, \quad (113.5)$$

$$k_B \Theta_{\text{clk}} \log 3 = \hbar_{\text{int}} \frac{2\pi}{108t_0}. \quad (113.6)$$

$\log 3$ quantifies a tritic unit; $\log \varphi$ quantifies hyperbolic self-scaling. They are not interchangeable. For complete transport,

$$H(X | U(X)) = 0, \quad \inf Q_L = 0. \quad (113.7)$$

For a quotient erasing one trit,

$$Q_{\text{reset}} \geq k_B \Theta \log 3. \quad (113.8)$$

The cost belongs to the reducing projection, not to the advance of holonomy.

Barbero–Immirzi occupies the dual of the incidence module 90/120. The equation

$$\frac{a}{270} - \frac{b}{360} = \frac{1}{24} \quad (113.9)$$

selects the minimal integer covector $(a, -b) = (12, -1)$, and therefore

$$K_{6 \rightarrow 8} = 12\Delta S_{90} - \Delta S_{120}. \quad (113.10)$$

α closes the clocks; γ_{BI} evaluates the seam on the area–volume defect. This functional difference explains why both objects are central without performing the same operation.

28.22 Symmetry, pentadicity, continuum and sequences

The definition of symmetry compatible with the kernel is

$$\text{Sym}(\rho, b_\partial) = \{g \in \text{Aut}(\rho(\mathfrak{S})) : g\rho(\mathcal{R}_{\text{HMT}}) = \rho(\mathcal{R}_{\text{HMT}}), gb_\partial = b_\partial\}. \quad (114.1)$$

Symmetries are not the primary preserved quantity. They are the automorphisms remaining available after preservation of relations and boundary is required. The exceptional branch realizes (114.1) through Witt–Golay–Leech– $V^{\mathfrak{h}}$ –Monster incidence.

Pentadicity has two representations:

$$S_{\pi,5} = \frac{1}{5} \mathbf{1}\mathbf{1}^\top, \quad \text{rank } S_{\pi,5} = 1, \quad (114.2)$$

and

$$P_{G,5} : V_{12} \rightarrow V_5, \quad \text{rank } P_{G,5} = 5. \quad (114.3)$$

Both receive the same holonomic source, but one extracts the common component of five regions and the other preserves the five-dimensional gravitational carrier. The chain

$$V_{12} \rightarrow V_5 \rightarrow \text{STF}_3 \rightarrow V_{\text{TT}}, \quad 12 \rightarrow 5 \rightarrow 5 \rightarrow 2, \quad (114.4)$$

proves how five structural modes determine two propagating modes.

Dual projection of the continuum is expressed as

$$\rho_{\text{arq}} : \mathfrak{S}_{\text{cyl}} \rightarrow \mathcal{A}_{\mathbb{R}}, \quad \rho_{\text{exc}} : \mathfrak{S}_{\text{cyl}} \rightarrow \mathcal{A}_{\text{exc}}. \quad (114.5)$$

The first contracts compatible cylinders; the second preserves boundary incidence. The five spectral, solenoidal, gauge, coherent and determinantal aspects belong to the same domain of (114.5). $\mathbb{R}$ is the output of a commutative representation; holonomic arithmetic additionally preserves commutators and the memory record invisible to that output.

Sequence reading constitutes another representation. With

$$A \mapsto (0, 0), \quad G \mapsto (0, 1), \quad T \mapsto (1, 0), \quad C \mapsto (1, 1), \quad (114.6)$$

and phase $(n \bmod 3)$, reverse complementation defines an involution on the category of words. APP evaluates accumulation and concatenation; TRIT types curvature; TPK preserves returns and boundary. The Sturmian condition

$$p_s(m) = m + 1 \quad (114.7)$$

provides the exact threshold between periodicity and exponential complexity. Thus the idea of an aperiodic crystal applied to biological sequences becomes a falsifiable reader of factors, extensions and return words.

28.23 Kernel theorem of HMT holonomic arithmetic

The preceding sections allow genealogy to be condensed without reducing it. Define

$$\mathfrak{S}_{\text{HMT}}^{\text{alg}} = \left[\lim_{\substack{\longrightarrow \\ n}} (\mathbb{R}\langle e_x, J_x, T_u \rangle / \mathcal{R}_{\text{HMT}}) \right] \rtimes_{\Gamma_9} \mathbb{N}. \quad (115.1)$$

In the reversible sector, the last factor is $\mathbb{Z}$. The state space is $X_\infty = \varprojlim X_n$. For a compatible bounded representation ρ , the completion $\mathfrak{S}_{\text{HMT}}^\rho = \overline{\rho(\mathfrak{S}_{\text{HMT}}^{\text{alg}})}^{\|\cdot\|}$ belongs to the reader; the universal generator does not presuppose a norm.

Kernel theorem. Dual APP evaluation, TRIT trigeometric orientation and TPK dynamics generate an associative noncommutative arithmetic of histories, twisted by the carry cocycle and graded by phase and memory. Refinement preserves unity, holonomy returns the phase while advancing memory, and the typed spectrum of representations determines the correlated system $(\pi, \varphi, e, \alpha)$, the joint discrete structure of the continuum, the electron, action and gravity scales, geometric parameters, derived constants and boundary symmetries.

The proof is obtained by concatenating five established facts:

1. the APP equations preserve both sheets, residue and quotient;
2. $J_\tau^2 = -\tau I$ preserves the three curvatures within a single TRIT;
3. the cocycle identity proves associativity of the history product;
4. the nonsplit extension proves that phase return and state return are distinct;
5. the universal property proves that every compatible reader factors through (115.1).

The first consequence is that $\mathbb{R}$ is not the fundamental support. It is a terminal representation of a structure preserving more information than its real coordinate. The second is that a constant is not a stored numeral: it is an invariant of a reader. The third is that a particle is not a number: it is a persistent section of a module. The fourth is that a symmetry is not postulated: it appears as the stabilizer of a preserved boundary.

The shortest form of conservative evolution is

$$\boxed{\begin{array}{ccc} \varprojlim X_n & \longleftrightarrow & \varinjlim \mathfrak{X}_n \\ \underbrace{\hspace{1cm}} & & \underbrace{\hspace{1cm}} \\ \text{history that deepens} & & \text{expanding readers} \end{array}}. \quad (115.2)$$

The helical aperiodic crystal coordinates both limits. The trigeometric identity (111.4) provides its local calculus; α provides the memory seam; the block $\text{Cl}_{1,1}$ provides its electronic corner; the invariant $(G\hbar)$ provides its dimensional conservation; Barbero–Immirzi provides its boundary covector; and dual projection provides its Archimedean and exceptional outputs. This is the minimal structure from which the standalone article can be developed without expository dependence on the (2 099) pages and without losing any essential operator.

Refutation criterion. The theorem is refuted if the cocycle identity fails, if refinement ceases to be unital, if holonomy restarts memory, if a TRIT fibre disappears, if a realization fails to satisfy the relations it claims to represent, or if an output uses its own conventional value to select the history that was to produce it.

28.24 Triadic irreducibility and potential difference in sequence readers

The APP–TRIT–TPK composition can be proved irreducible through three forgetful quotients. Let q_{APP} be the quotient identifying the additive and multiplicative sheets. Then

$$q_{\text{APP}}(\Delta_{\Sigma\Pi}) = 0, \quad q_{\text{APP}}([P_{\Sigma}, P_{\Pi}]) = 0. \quad (116.1)$$

Let q_{TRIT} be the quotient identifying the three regime idempotents. The triple of squares is then lost

$$(J_+^2, J_0^2, J_-^2) = (-I, 0, I). \quad (116.2)$$

Let q_{TPK} be the quotient trivializing the carry cocycle and memory grading. In that quotient,

$$q_{\text{TPK}}(d_M(\Gamma_9 x) - d_M(x)) = 0, \quad (116.3)$$

although in the complete holonomic state the difference equals 1. The three invariants (116.1)–(116.3) have distinct types, and none can be reconstructed from the other two after its quotient is applied. APP provides discrepancy and sheet order; TRIT, curvature signature; TPK, composition, holonomy and memory. This is the exact algebraic form of the ternary irreducibility of the kernel.

The same separation clarifies the application to sequences. In the alphabet $\{A, G, T, C\}$, define

$$\chi_p(A, G, T, C) = (0, 0, 1, 1), \quad \chi_h(A, G, T, C) = (0, 1, 0, 1), \quad (116.4)$$

and the complement involution $C_{\text{DNA}}(A, G, T, C) = (T, C, A, G)$. One verifies

$$\chi_p(C_{\text{DNA}}b) = 1 - \chi_p(b), \quad \chi_h(C_{\text{DNA}}b) = \chi_h(b). \quad (116.5)$$

Consequently,

$$\mathcal{V}_n = \chi_p(b_{n+1}) - \chi_p(b_n) \quad (116.6)$$

is an oriented symbolic potential difference, whereas χ_h preserves the transverse bond class. The lift

$$b_n \mapsto (\chi_p(b_n), \chi_h(b_n), n \bmod 3, \lambda_n, c_n, m_n) \quad (116.7)$$

adds TRIT regime, boundary, carry and TPK memory. This formulates the structural reason why polarity, potential and aperiodic word can be read within the same arithmetic without turning biology into the generator of the formalism.

Refutation criterion. Irreducibility fails if one of the three invariants survives the quotient that should erase it, or can be reconstructed from the other two without reintroducing the eliminated datum. The sequence reader fails if complementation does not reverse χ_p , does not preserve χ_h , or if the lift does not intertwine ternary phase with TPK dynamics.

28.25 The electronic vacancy as a Witt null pair

The arithmetic of the electronic corner allows the three regimes to be completed locally without introducing another algebra. With

$$S = 2P - I, \quad J = \frac{[P, Q]}{\sqrt{3}/4}, \quad (117.1)$$

we have

$$S^2 = I, \quad J^2 = -I, \quad SJ = -JS. \quad (117.2)$$

The combinations

$$n_+ = \frac{S + J}{2}, \quad n_- = \frac{S - J}{2} \quad (117.3)$$

satisfy

$$n_+^2 = n_-^2 = 0, \quad n_+ n_- + n_- n_+ = I. \quad (117.4)$$

Therefore,

$$\{J, n_{\pm}, S\} \longmapsto \{\text{elliptic, parabolic, hyperbolic}\} \quad (117.5)$$

is a complete local realization of TRIT typing within $\text{Cl}_{1,1} \cong M_2(\mathbb{R})$. The vacancy is characterized by two compatible descriptions: in APP it is the unit loss of the product quotient that preserves the additive sheet, product residue and regime; in the Clifford corner it is a parabolic null direction. The pair (n_+, n_-) is the local Witt pair of the two orientations. This decomposition provides the precise algebraic domain from which subsequent Paley–Witt incidence can realize the same genealogy in its own combinatorial module.

The assertion is stronger than the quaternion analogy. Hamilton quaternions have three generators squaring to $-I$ and form a division algebra. The block (117.2) is split: it contains zero divisors, idempotents and null directions, precisely the objects required by sheet, vacancy and threshold. Octonions enlarge dimension at the cost of associativity; holonomic arithmetic maintains associativity through the cocycle and enlarges the structure through routes, memory and refinement.

Refutation criterion. It suffices for one of the four identities (117.2)–(117.4) to fail, for the vacancy not to preserve its declared APP data, or for TPK transport not to preserve the regime idempotent.

28.26 Weyl relations of the finite and irrational clocks

Let T be the orbit step, let $\delta = \log_{10}(10/9)$, let ζ_9 and ω_δ be the internal characters of nonadic phase and capacity, and define on $|t, m\rangle$

$$V_9|t, m\rangle = \zeta_9^t|t, m\rangle, \quad W_\delta|t, m\rangle = \omega_\delta^t|t, m\rangle. \quad (118.1)$$

Temporal advance satisfies

$$V_9T = \zeta_9TV_9, \quad W_\delta T = \omega_\delta TW_\delta. \quad (118.2)$$

The turn $\Gamma_9 = T^9$ produces

$$V_9\Gamma_9 = \Gamma_9V_9, \quad W_\delta\Gamma_9 = \omega_\delta^9\Gamma_9W_\delta. \quad (118.3)$$

The two clocks are thus distinguished algebraically. The first has period nine; the second preserves an irrational phase. The mechanical vacancy word is the threshold coding of the second clock's orbit. If D_M is the memory-degree operator, the TPK lift satisfies

$$[D_M, \Gamma_9] = \Gamma_9. \quad (118.4)$$

The subsequent elliptic representation recognizes these characters through

$$\rho_+(\zeta_9) = \text{Exp}_+\left(\frac{2\pi}{9}J_+\right), \quad \rho_+(\omega_\delta) = \text{Exp}_+(2\pi\delta J_+). \quad (118.5)$$

The characters, not their recognized complex expressions, are the clock's operational data.

Equation (118.4) states exactly that Γ_9 is a raising operator: phase returns and the state ascends one unit along the helix. The translation action is Abelian; the crossed product of that action with the observables V_9, W_δ, D_M is noncommutative by (118.2)–(118.4). The temporal structure therefore does not require choosing between ‘finite clock’ and ‘aperiodic crystal’: both are conjugate characters of a single skew system.

Zero topological entropy follows from irrational rotation and complexity $p(m) = m + 1$; memory growth follows from (118.4). Zero entropy and an infinite history coexist because the latter measures reconstructible depth whereas the former measures exponential growth of factors.

Refutation criterion. The model fails if V_9 does not commute with Γ_9 , if W_δ acquires a finite period, if the memory commutator differs from Γ_9 , or if the threshold word ceases to have Sturmian complexity.

28.27 Minimal equation of the trigeometric generator

Let

$$\mathbb{J} = p_+J_+ + p_0J_0 + p_-J_-, \quad p_+ + p_0 + p_- = I. \quad (119.1)$$

The three curvature relations are equivalent to

$$\boxed{\mathbb{J}^6 = \mathbb{J}^2}. \quad (119.2)$$

Indeed, $K = \mathbb{J}^2$ satisfies $K^3 = K$, and the three idempotents are recovered without external data:

$$p_+ = \frac{K^2 - K}{2}, \quad p_0 = I - K^2, \quad p_- = \frac{K^2 + K}{2}. \quad (119.3)$$

The intrinsic exponential identity is

$$\text{Exp}_\star(t\mathbb{J}) = p_+(\cos t + \mathbb{J} \sin t) + p_0(I + t\mathbb{J}) + p_-(\cosh t + \mathbb{J} \sinh t). \quad (119.4)$$

Equations (118.4) and (119.4) are combined in the holonomic propagator

$$\boxed{\mathcal{E}_{\text{HMT}}(t) = \Gamma_9 \text{Exp}_\star(t\mathbb{J}), \quad [D_M, \Gamma_9] = \Gamma_9.} \quad (119.5)$$

This is the sought operational compression. The first coordinate of (119.5) governs local response in the three curvatures; the second governs memory raising. APP enters through the sheet idempotents, their projectors and the cocycle defining Γ_9 ; TRIT enters through $\mathbb{J}$; TPK enters through composition, the lift and D_M . The correlated system, electron, action, gravity, Barbero–Immirzi, scale constants, symmetries and sequence readers are obtained as typed representations of the universal algebra containing (119.5).

The function of α becomes transparent in this presentation. $\text{Exp}_\star$ contains local dynamics and Γ_9 the turn with memory; dodecaphasic closure evaluates their compatibility and determines α as seam. Thus α links synchronization of (π, e, φ) to dimensional readers because it measures a closure class of the holonomic propagator, not because it is added to three pre-existing constants.

Refutation criterion. The minimal equation fails if $\mathbb{J}^6 \neq \mathbb{J}^2$, if the polynomials (119.3) cease to be orthogonal idempotents, if (119.4) does not restrict to the three certified exponentials, or if α does not coincide along its two typed routes within the existing compatibility square.

The electron as a persistent section of the oriented vacancy

The genealogical state here acquires its first complete material realization. The central localization, the electron–positron band, and the bidirectional reading are constructed before any metrological comparison; the basal stage remains distinct from the governing beta closure.

29.1 Closure β of the electronic section and dimensional publication of its rest energy

The electron appears in HMT as the point at which six preceding and mutually typed constructions are assembled: the incompatibility between the two arithmetic sheets of APP, the orientation induced by the involution of the areal–volumetric hinge, two integer quotients from the APP–TPK census, the dodecaphasic closure of α , the global defect Δ_4 , and the unique center of the nonadic atlas. The route band then adds electron–positron conjugation. The MeV chart and the comparison with CODATA are subsequent to all of them.

This precedence is part of the mathematical content: the factors, their types and their causal order are determined before opening the tabulated chart, which intervenes only in the subsequent metrological comparison.

29.1.1 Electronic scalar $\mathfrak{e}$ and causal graph of its factors APP–TPK, α , Δ_4 and nonadic atlas

The internal output is the dimensionless scalar.

$$\mathfrak{e} = \frac{\sqrt{3}}{4} \left[\exp\left(\frac{\varphi}{\pi^2}\right) - \frac{396}{18} \alpha^3 \right] \left[1 + \frac{270}{18} \Delta_4 \right]. \quad (29.1)$$

The identities

$$\frac{396}{18} = 22 = 27 - 5, \quad \frac{270}{18} = 15 = \binom{6}{2} = 3 \cdot 5$$

allow the abbreviated notation to be recovered

$$\mathfrak{e} = \frac{\sqrt{3}}{4} \left(e^{\varphi/\pi^2} - 22\alpha^3 \right) (1 + 15\Delta_4).$$

Here the letter e inside the exponential is Euler’s number; the subscript of $\mathfrak{e}$ denotes the electronic section.

The causal graph of the equation is

APP projectors in dimension three	$\longrightarrow$	$\sqrt{3}/4,$
TPK calendar and areal–volumetric sheets	$\longrightarrow$	$F_{av}, \pi/\varphi^2, \varphi/\pi^2,$
dodecaphasic closure	$\longrightarrow$	$\alpha,$
APP–TPK census and electronic family	$\longrightarrow$	$396, 18, 270, 22, 15, 5,$
global readings of π, e and register	$\longrightarrow$	$\Delta_4,$
nonadic atlas and anti-section	$\longrightarrow$	$(5, 5), \text{electron–positron band.}$

No arrow in this diagram receives as an argument the reference electronic mass.

29.1.2 1st factor: exact incompatibility of the APP leaves

For $d \geq 2$, let

$$\mathcal{A}_d = \{1, \dots, 9\}^d$$

with uniform measure, and let

$$\Sigma_d(x) = \text{dr}(x_1 + \dots + x_d), \quad \Pi_d(x) = \text{dr}(x_1 \cdots x_d)$$

the additive and multiplicative observables. Their orthogonal conditional expectations on $\ell^2(\mathcal{A}_d)$ are $P_{\Sigma,d}$ and $P_{\Pi,d}$. The proposition 2.5 proves by exact enumeration that

$$\|[P_{\Sigma,3}, P_{\Pi,3}]\| = \frac{\sqrt{3}}{4}.$$

Proposition 29.1 (Electronic prefactor APP content). *The prefactor $\sqrt{3}/4$ of equation (29.1) is the exact norm of the noncommutativity between the sum and product sheets in dimension three. It is not a geometric factor introduced from the electronic formula.*

Proof. The joint table of Σ_3 and Π_3 determines the principal angles between $\text{Ran } P_{\Sigma,3}$ and $\text{Ran } P_{\Pi,3}$. If s is one of its singular values, the corresponding contribution to the commutator norm is $s\sqrt{1-s^2}$. Enumeration of the 9^3 words gives at most

$$s^2(1-s^2) = \frac{3}{16}, \quad s^2 = \frac{1}{4}.$$

Therefore the maximum norm is $\sqrt{3/16} = \sqrt{3}/4$, before introducing α , Δ_4 , a unit of energy or a reference mass. $\square$

The physical function of the factor is thereby fixed: it measures the maximal incompatibility of the two arithmetic readings on the same ternary support. The electron does not inherit two independent numbers, one additive and the other multiplicative; it inherits the exact obstruction to rendering the two sheets simultaneously compatible. The prefactor preserves, within the mass law, the local memory of the APP disagreement from which the route arises.

29.1.3 2nd factor: the exchange involution between π and φ

The primitive calendar of modes (Σ, Π, Π) induces, on the areal and volumetric sheets, the matrix

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

theorem 10.55 obtains it by counting the transitions $\Sigma \rightarrow \Pi$, $\Pi \rightarrow \Pi$ and $\Pi \rightarrow \Sigma$, each once. The matrix is not postulated from the golden ratio: its eigenvalues φ and $-\varphi^{-1}$ appear after the incidence has been constructed.

In the second-order memory, $\text{Sym}^2(F_{\text{av}})$ has spectrum

$$\varphi^2, \quad -1, \quad \varphi^{-2}.$$

Thus, φ^{-2} is the positive stable mode of the hinge. The closure mark π , applied just once to the areal return, produces

$$\frac{\pi}{\varphi^2}.$$

This reason preserves the closure–self-scaling order. The electronic law uses its opposite orientation.

Theorem 29.2 (Exact Inversion of the Electronic Hinge). *Let*

$$\mathcal{R}(x, y) = \frac{x}{y^2}, \quad \mathbf{J}(x, y) = (y, x).$$

Then $\mathbf{J}^2 = I$ and

$$\mathcal{R}(\pi, \varphi) = \frac{\pi}{\varphi^2}, \quad (\mathcal{R} \circ \mathbf{J})(\pi, \varphi) = \frac{\varphi}{\pi^2}.$$

Consequently,

$$\frac{\varphi}{\pi^2} = \frac{1}{\varphi^3 (\pi/\varphi^2)^2}, \quad e^{\varphi/\pi^2} = \exp \left[\frac{1}{\varphi^3 (\pi/\varphi^2)^2} \right].$$

Proof. The first pair of identities is obtained by applying $\mathcal{R}$ before and after the involution $\mathbf{J}$. For the reparametrization,

$$\varphi^3 \left(\frac{\pi}{\varphi^2} \right)^2 = \frac{\pi^2}{\varphi},$$

and its inverse is φ/π^2 . The last equality follows by applying the exponential. $\square$

The simultaneous presence of π and φ therefore has a precise structural meaning. π/φ^2 reads areal closure over quadratic volumetric memory; φ/π^2 reads interior self-growth over quadratic closure. They are not two similar quotients chosen for convenience. They are the two orientations of the same reader, exchanged by an exact involution. The term e^{φ/π^2} publishes the interior orientation of that hinge in the electron section.

29.1.4 3rd and 4th factors: the integers 22, 15 and the center 5

The census of the APP chart employed by the electronic family provides three integer sums:

$$\Sigma(\text{APP}) = 396, \quad \Sigma(\text{centro } 2 \times 2) = 18, \quad \Sigma(\text{canal}) = 270.$$

The first admits the ring decomposition.

$$396 = 18 + 72 + 108 + 198.$$

The kernel 2×2 is therefore an explicit substructure of the complete chart, and the integer 270 belongs to the global channel of the same APP-TPK census. The sum of the even rings is

$$18 + 108 = 126, \quad \frac{396}{126} = \frac{22}{7}.$$

This last quotient is the rational closure seal of the chart, not an identification of $22/7$ with the number π generated by the coinductive reader. It does show that the numerator 22 already serves an internal function before appearing in the electronic law.

Theorem 29.3 (Internal Origin of the Electronic Coefficients). *In the electronic family fixed by the APP chart and its global channel,*

$$22 = \frac{\Sigma(\text{APP})}{\Sigma(\text{centro})} = \frac{396}{18} = 27 - 5,$$

$$15 = \frac{\Sigma(\text{canal})}{\Sigma(\text{centro})} = \frac{270}{18} = 3 \cdot 5 = \binom{6}{2}.$$

In particular, the coefficients are determined without consulting Z_0 , the electron mass or a chart of units.

Proof. The first two equalities are exact divisions of the finite census:

$$396 = 22 \cdot 18, \quad 270 = 15 \cdot 18.$$

By defining

$$p := 27 - 22,$$

one obtains $p = 5$, and then

$$15 = 3p = 3 \cdot 5 = \binom{6}{2}.$$

All quantities belong to the integer domain of the chart and the family before metrological transduction. $\square$

29.1.4.1 Persistence selector in the central return

The same pair has a dynamic construction that also fixes its sign without consulting a mass. The four orientations of the central seed $(5, 5)$ return to that cell for the first time after twenty-seven steps. The typed return space is

$$\mathcal{K}_e = \mathbb{Z}/9\mathbb{Z} \times \mathbb{F}_3, \quad |\mathcal{K}_e| = 27.$$

Moreover, the microfiber of π contains the five genealogically distinct regions

$$\mathcal{R}_\pi = 3_{++} + 1_{--} + 1_{\perp}.$$

The regional gauge diag_c and, within a repeated phase, the internal order of U_6 define the injection

$$\iota : \mathcal{R}_\pi \hookrightarrow \mathcal{K}_e$$

through the five distinct places

$$(4, 0), \quad (5, 0), \quad (6, 0), \quad (6, 1), \quad (6, 2).$$

The first two correspond, respectively, to the transverse region and the mutated return; the last three separate the positive regions by their tritic sheet. The common visible word does not intervene as the inverse of the map.

Theorem 29.4 (Internal selector of the electronic coefficients). *With the central return and the five typed regions fixed, the coefficients of the electronic equation are*

$$\boxed{n_e = -|\mathcal{K}_e \setminus \iota(\mathcal{R}_\pi)| = -22, \quad m_e^{\text{tor}} = |\mathcal{R}_\pi \times \mathbb{F}_3| = +15.}$$

The first symbol records the boundary deficit; the second, the retained torsional memory.

Proof. The five places above are distinct, so ι is injective and

$$|\mathcal{K}_e \setminus \iota(\mathcal{R}_\pi)| = 27 - 5 = 22.$$

Each of the five regions preserves three triticle leaves, hence

$$|\mathcal{R}_\pi \times \mathbb{F}_3| = 5 \cdot 3 = 15.$$

The orientation rule assigns a negative sign to states removed from the shell and a positive sign to retained memory. The construction opens only the center, the return of twenty-seven states, and the regional identities; it does not open an observed mass, an impedance, or a calibrated signature. $\square$

The number 22 measures how many central kernels of weight 18 are recomposed by the APP sum 396. The number 15 measures how many of those same kernels are recomposed by the channel 270, and simultaneously counts the pairs of a hexad. The number 5 is not an additional parameter:

$$5 = 27 - 22, \quad 15 = 3 \cdot 5.$$

The three expressions display the same closure from depth 27, from the central normalization, and from hexadic combinatorics.

29.1.4.2 Conservation of coefficients 22 and 15 in the material realization of the electron

The historical exploration first located $p = 5$ through a finite comparison with Z_0 , and from there wrote $(27 - p, 3p) = (22, 15)$. That procedure retains value as a history of discovery and as an external check. It is not, however, the proof that governs the law: the subsequent count

$$\frac{396}{18} = 22, \quad \frac{270}{18} = 15$$

constructs both integers within APP-TPK and recovers $p = 5$ as a consequence. The history explains how the pair was found; the census explains why it belongs to the equation.

29.1.5 5.^o factor: the cubo of the closure α

The closure of α precedes Dimensional Mechanics. As established in chapter devoted to the dodecaphasic closure of α , the twelve base-thousand blocks of π , e , φ and the seal K combine through the reversible carry

$$P_m + E_m - \Phi_m - K_m + c_{m+1} = A_m + 1000c_m, \quad c_{13} = c_1 = 0.$$

The output is

$$A_\alpha = (007, 297, 352, 569, 283, 800, 997, 285, 105, 472, 380, 663)$$

and, by concatenation, the exact dodecaphasic window

$$\alpha_{12} = \pi_{12}(\alpha) = \frac{7297352569283800997285105472380663}{10^{36}}.$$

Telescoping the carries proves the complete integer identity; the compatible prolongation constructs $\alpha = \lim_{\leftarrow N} A^{(N)}$, and α_{12} is its thirty-six-digit rational projection. The comparison with the fine-structure constant is performed subsequently and does not intervene in its construction.

The electronic law takes the third power

$$\alpha^3.$$

Mathematically, this is the multiplicative evaluation of the third symmetric power of the same scalar already closed; neither three metrological copies nor three different parameters are introduced. Its contribution is

$$22\alpha^3 = 0.00000854906599200551727014979807598228 \dots$$

Proposition 29.5 (Statute of the Electronic Cubic Channel). *The term $22\alpha^3$ of the electronic law depends only on the prior closure of α , on internal multiplication, and on the integer 22 of theorem 29.3. It does not depend on a reading of the electron mass.*

Proof. α_{12} is the rational number fixed by the concatenation and carry identity, and the compatible family of windows determines the section α . The projection α_{12}^3 certifies the printed digits of α^3 . The coefficient 22 is determined by $396/18$. The composition of both operations contains only data constructed before the energy chart. $\square$

This proposition fixes exactly what is used for the electron. The third power is the cubic degree of this channel in the published law. There is no need here to postulate a universal rule according to which every occurrence of α^d automatically represents a dimension d ; a universal statement of that kind would require its own theorem of multiplicativity among degrees. The electron genealogy is complete without turning that generalization into a premise.

29.1.6 6th factor: minimal torsion and global defect Δ_4

The terms *minimal torsion* and *global defect* denote a single invariant here, not two distinct corrections. The object used by the mass law is

$$\Delta_4 = \frac{\pi - e \log \pi}{270} \left(1 + \frac{\pi}{729} \right). \quad (29.2)$$

Its inputs are the already fixed HMT readings of π and e , the global channel 270, and the interior cell 729. Consequently,

$$\Delta_4 = 0.00011119642269214005405113478612067909 \dots$$

The first ratio measures the signed defect $\pi - e \log \pi$ in units of the 270 channel; the second factor couples that ratio to the interior cell through $\pi/729$. This reading adds no parameters to the formula: it separates its two constituent operations. The [chapter 31](#) shows that it is a pre-dimensional mathematical input and distinguishes it from the operator that eventually realizes it as a change of sheet.

Proposition 29.6 (Globality of the Electronic Defect). *Δ_4 is a global invariant preceding the species. In the electronic law it enters through*

$$1 + 15\Delta_4 = 1.00166794634038210081076702179181019 \dots$$

It is not a corrector chosen after knowing the residue of the electron mass.

Proof. The [equation \(29.2\)](#) contains neither a species label, a mass, nor a unit. The integer 15 has already been constructed by $270/18$. Therefore, the composition $1 + 15\Delta_4$ is determined by the global data and the internal census. $\square$

Δ_4 must be distinguished from the other blocks of the mass character and from the other notions of torsion. In particular, it is neither s_{270} , nor the quadratic invariant $135\Delta_4^2$, nor the angular residue ε_{res} , nor the geometric Cartan–Holst torsion. Substituting one for another would repeat or alter the level of the correction. For the electron, Δ_4 preserves the global memory of refinement; the factor 15 states how many times the electron family reads it. The qualifier *minimal* preserves the historical name of this invariant and is not used as a substitute for a variational theorem of minimality.

29.1.7 Material realization of the fixed point (5,5) in the electronic closure

Let

$$C_{81} = \{1, \dots, 9\}^2$$

the nonadic atlas and

$$\rho(r, c) = (10 - r, 10 - c)$$

its central inversion. classification theorem for the families and generations of the nonadic atlas proves that the atlas contains a unique generation-zero cell and that this cell bears the positional label e_{core} .

Proposition 29.7 (Uniqueness of the Electronic Center). *The involution ρ has a unique fixed point:*

$$(r, c) = (5, 5).$$

Its uniqueness is combinatorial and does not depend on the value of e .

Proof. The equation $(r, c) = \rho(r, c)$ is equivalent to

$$2r = 10, \quad 2c = 10.$$

In $\{1, \dots, 9\}^2$ it has the unique solution $r = c = 5$. The evaluation of the mass does not appear in the argument. $\square$

Corollary 29.8 (Universality of the Central Section). *Every admissible automorphism of the atlas that intertwines the central inversion preserves the cell $(5, 5)$.*

Proof. If $f\rho = \rho f$ and $\rho(5, 5) = (5, 5)$, then $\rho f(5, 5) = f\rho(5, 5) = f(5, 5)$. Therefore $f(5, 5)$ is another fixed point of ρ ; uniqueness forces $f(5, 5) = (5, 5)$. $\square$

This is the structural reason why the electronic class is the same in every realization preserving the atlas and its inversion: a local decimal does not recur by chance; rather, every admissible chart must transport the same fixed point and the same chain of invariants. The physical reading of this uniqueness identifies each electron with a realization of that central class; the proposition proves the internal universality of the atlas.

In the electronic family fixed by the construction, this point is the geometric anchor of the section. Four objects should not be confused:

$$(5, 5), \quad e_{\text{core}}, \quad v_e = (0, 0, 0, 0, 0), \quad \text{the raw signature of the central cell.}$$

The first is a position; the second, its internal label; the third, the chosen origin in the fine action lattice; the fourth preserves the raw atlas data and need not be the zero vector. Their coordination explains why the electron occupies the center without erasing the genealogy of the representations describing it.

Oriented electronic routes e_{++}, e_{--} , envelope e_{env} , action origin and signatures of the central cell

The central cell carries two oriented histories of twelve events. Writing v^2 for the concatenation of two copies of a sextuple, the active route of orientation $++$, its opposite history, and the historical envelope are, respectively,

$$\begin{aligned} e_{\text{act}} = e_{++} &= (2, 2, 5, -4, -3, -2)^2, \\ e_{--} &= (4, 3, 2, -2, -2, -5)^2, \\ e_{\text{env}} &= (4, 3, 5, -4, -3, -5)^2. \end{aligned} \tag{E-centro-1}$$

The two stories publish the same word

$$\tau_{12} = +++-+++-,$$

but they are not the same route. The envelope takes the absolute maximum coordinate by coordinate while preserving the sign; it records amplitudes, but flattens the genealogical order and does not replace e_{act} .

The other readers of the center live in distinct codomains:

$$\begin{aligned} v_e &= (0, 0, 0, 0, 0), & \text{affine origin of the action lattice,} \\ R(5, 5) &= (24, 0, 12, 0, -12), & \text{raw APP signature of the cell,} \\ \text{Sig}_{\text{masa}}^{\Gamma_{108}} &= (-12, -4, 12, -6, 12), & \text{even mass projection,} \\ \text{Sig}_{\text{carga}}^{\Gamma_{108}} &= (0, 0, 0, 0, 0), & \text{odd charge projection.} \end{aligned} \tag{E-centro-2}$$

The fact that the charge word has zero odd signature does not turn the route into zero, nor does it identify the raw signature with the affine origin. Position, active route, opposite route, envelope, origin and both signatures are coordinated objects, not six notations for the same vector.

29.1.8 Particle–antiparticle conjugation and persistence of the electronic band

Electronic identity does not end at the central point or at the mass value. Material conjugation is realized on dodecaphasic words. Let

$$C_M(\mathbf{t}) = -\text{rev}(\mathbf{t})$$

be the oriented anti-section and let $\chi \in \{-1, +1\}$ be the charge orientation. The band is the class

$$[(\mathbf{t}, \chi)]_M = \{(\mathbf{t}, \chi), (C_M \mathbf{t}, -\chi)\}$$

of definition of the dodecaphasic route band. The band preserves its even components and exhibits two sections with opposite odd components.

Theorem 29.9 (Electron and positron as sections of a band). *For the band character of mass and band character conjugation theorem,*

$$m(C_M \mathbf{t}, -\chi) = m(\mathbf{t}, \chi).$$

In the electronic channel,

$$(n, \chi) = (-22, +1) \longleftrightarrow (+22, -1)$$

and both pairs produce

$$\chi n = -22.$$

Therefore, electron and positron have the same mass and opposite charge orientations.

Proof. If E_+ and E_- are, respectively, the even and odd components of the character with respect to C_M , then

$$E_+(C_M \mathbf{t}) = E_+(\mathbf{t}), \quad E_-(C_M \mathbf{t}) = -E_-(\mathbf{t}).$$

From here,

$$E_{\text{band}}(C_M \mathbf{t}, -\chi) = E_{\text{band}}(\mathbf{t}, \chi),$$

and the equality of mass is obtained by applying the exponential. In the electronic term,

$$(+1)(-22) = (-1)(+22) = -22.$$

□

The finite closure of proposition on closure of the forms of the 108-phase calendar reinforces this interpretation: over the eighty-one CT–108 routes, neither pure sign nor pure reversal preserves the taxonomy, whereas $-\text{rev}$ does preserve it, with

$$81 = 54 + 9 + 9 + 9.$$

The antiparticle is therefore neither a negative mass nor an independent copy. It is the other oriented section of the same band: it preserves the even invariants that determine mass and reverses the odd invariants that determine charge and orientation.

29.1.9 Electronic Closure Theorem

Theorem 29.10 (Holographic composition of the electronic section). *With the objects constructed in the preceding sections fixed, the dimensionless electronic section is*

$$\mathbf{e} = \|[P_{\Sigma,3}, P_{\Pi,3}]\| \left[\exp((\mathcal{R} \circ \mathbf{J})(\pi, \varphi)) - \frac{\Sigma(\text{APP})}{\Sigma(\text{centro})} \alpha^3 \right] \left[1 + \frac{\Sigma(\text{canal})}{\Sigma(\text{centro})} \Delta_4 \right].$$

Equivalently,

$$\mathbf{e} = \frac{\sqrt{3}}{4} \left(e^{\varphi/\pi^2} - 22\alpha^3 \right) (1 + 15\Delta_4).$$

Its evaluation is

$$\mathbf{e} = 0.51099892248806607250987087719134943763 \dots$$

Proof. The proposition 29.1 contributes

$$\| [P_{\Sigma,3}, P_{\Pi,3}] \| = \sqrt{3}/4.$$

The theorem 29.2 contributes

$$(\mathcal{R} \circ \mathbf{J})(\pi, \varphi) = \varphi/\pi^2.$$

The theorem 29.3 contributes

$$\frac{\Sigma(\text{APP})}{\Sigma(\text{centro})} = 22, \quad \frac{\Sigma(\text{canal})}{\Sigma(\text{centro})} = 15.$$

The dodecaphasic closure contributes the rational window α_{12} of the coinductive section α , and equation (29.2) contributes the global defect. Substitution yields the abbreviated formula. Successive evaluation of

$$\begin{aligned} e^{\varphi/\pi^2} &= 1.17814494259630678081160095680888910 \dots, \\ 22\alpha^3 &= 0.00000854906599200551727014979807598228 \dots, \\ 1 + 15\Delta_4 &= 1.00166794634038210081076702179181019 \dots \end{aligned}$$

gives the declared value. $\square$

The theorem is holographic in a verifiable sense: each local factor retains a distinct layer of the complete system. The commutator retains sheet duality; the exponential retains the interior orientation of the hinge $\pi\text{--}\varphi$; 22 and 15 retain the global census and its central kernel; α^3 retains fine closure; and Δ_4 retains global refinement. The center determines where the section is anchored, and the band determines how it survives under conjugation. The value does not replace this structure: it is its common evaluation.

29.1.9.1 Anterior and posterior action splitting upon return

The regional realization of the conjugate angular coordinate C^* distinguishes two coordinates on the same internal action line. The coordinate before the return is H_5 and the subsequent coordinate is

$$\eta_{\text{ret}}^{(\text{deg})} = H_5 - \frac{90}{\pi} \mathcal{C}_\pi(\alpha).$$

The notation $\eta_{\text{ret}}^{(\text{deg})}$ is the canonical sexagesimal coordinate defined on the action branch by $\eta_{\text{ret}}^{(\text{deg})} = C^*/2 - \alpha_{\angle, \text{deg}}$. The preceding equality is evaluated in that chart; it does not introduce a second return coordinate. Therefore, the additive and multiplicative charts of the oriented splitting are

$$\delta_{2w} := H_5 - \eta_{\text{ret}}^{(\text{deg})} = \frac{90}{\pi} \mathcal{C}_\pi(\alpha), \quad R_{\text{act}} := \frac{\hbar_{\text{HMT}}^{\text{pre}}}{\hbar_{\text{HMT}}^{\text{ret}}} = \frac{H_5}{\eta_{\text{ret}}^{(\text{deg})}}. \quad (29.3)$$

Both coordinates follow from a single pre-return/post-return transport. They are not two Planck constants: the common factor $10^{-34} \mathcal{U}_5$ fixes the dimensional line and cancels exactly in R_{act} . The action ellipse subsequently realizes $\text{Area}(\theta) = \hbar_{\text{HMT}}^{\text{ret}} \theta$, so that the splitting precedes the holonomy reading.

The coupling

$$\kappa_e^\beta = 80 - 54\alpha + 6\Delta_4, \quad \mathbf{e}_e^\beta = \mathbf{e} R_{\text{act}}^{\kappa_e^\beta} = 0.5109989506900008086 \dots \quad (29.4)$$

is the bidirectional quantitative closure of the electron. The two independent readers of the electronic register constructed in the following chapter select, without consulting the comparison mass,

$$K_D(\Gamma_e) = K_\Omega(\Gamma_e) = (80, 54, 6),$$

and derive exactly κ_e^β . The integer 80 comes from the first coordinate of those readers; the word $w_e = 201101$ retains its own residence in the channel of Euler's constant.

Internal Closure of Bidirectional Refinement

The identity (29.3), the reduction (80, 54, 6), and the equation (29.4) constitute internally exact, target-free results on the electronic record. Propagation to fibers of higher rank is formulated by means of the multisection operators.

29.1.10 Energy chart and contrast of the bidirectional closure

Let $\mathcal{U}_E$ a real line of energy dimensions and let u_E a chosen basis. The metrological transduction is

$$\varsigma_{u_E} : \mathbb{R} \longrightarrow \mathcal{U}_E, \quad \varsigma_{u_E}(x) = x u_E.$$

The chart preserves two stages of the same genealogy. The basal stage is

$$E_e^{(0)} = \varsigma_{u_E}(\mathfrak{e}).$$

With $u_E = 1 \text{ MeV}$,

$$E_e^{(0)} = 0.5109989224880660725 \dots \text{ MeV}.$$

The action ratio and the derived exponent complete the leading stage,

$$E_e^\beta = E_e^{(0)} R_{\text{act}}^{\kappa_e^\beta} = 0.5109989506900008086082571648 \dots \text{ MeV}, \quad m_e^\beta = E_e^\beta / c^2.$$

The unit does not appear in [equation \(29.1\)](#); it is the basis of a dimensional line applied to an already constructed output.

The CODATA 2022 reference for the electron energy equivalent is [\[COD24\]](#)

$$E_e^{\text{CODATA}} = 0.51099895069(16) \text{ MeV}.$$

In that same chart, the bidirectional output satisfies

$$E_e^\beta - E_e^{\text{CODATA}} = 8.086082571648 \times 10^{-16} \text{ MeV},$$

with relative difference

$$1.582406883758 \times 10^{-15}.$$

The comparison is carried out after freezing the internal evaluation. The identities of the basal selector remain

$$22 = \frac{396}{18}, \quad 15 = \frac{270}{18}, \quad p = 27 - 22 = 5.$$

and additionally receive the combinatorial proof $-22 = -|K_e \setminus \iota(R_\pi)|$, $+15 = |R_\pi \times \mathbb{F}_3|$. The metrological comparison verifies the output; the route, the readers, and the action ratio generate it.

29.1.11 Structural Consequences of Electronic Closure

The preceding structural reading imposes several distinctions:

1. $\sqrt{3}/4$ is the norm of the APP commutator in dimension three. Its agreement with other geometric formulas does not supersede this provenance.
2. π/φ^2 and φ/π^2 are not interchangeable. The first ratio is the closure orientation; the second, obtained through J, is the electronic orientation of self-growth.
3. 22 and 15 follow from 396/18 and 270/18. Presenting them as values logically selected by Z_0 would reverse the demonstrated dependence.
4. α^3 uses the prior closure of α . The cubic power belongs to the electron law; by itself, it does not authorize a universal rule $\alpha^d \leftrightarrow d$ without an additional theorem.
5. Δ_4 is global. No must reemplazarse by s_{270} , by $135\Delta_4^2$ neither by a residue particular of the especie.
6. The position (5, 5), the label e_{core} , the fine origin $v_e = 0$, and the central raw signature are coordinated objects, not synonyms.
7. The electron and positron are not distinguished by two mass laws: they are the two oriented sections of a single band, with even mass and odd charge.
8. 0.5109989224... is first a dimensionless scalar. MeV, MeV/ c^2 , CODATA, and Z_0 belong to the subsequent comparison.

These distinctions explain why the electronic equation is a miniature of the HMT–MD system. In a single section, support, sheets, orientation, census, fine closure, refinement, center, persistence, and conjugation reappear. This is the content of the word *holographic*: not that a digit poetically represents the whole, but that each factor preserves a verifiable dependence on the whole that produces it.

29.2 Electron realization band and separation between base mass and beta closure

The enriched route of the central point publishes two complementary representations of its memory: the complete tritic word $\gamma_{108} \in \mathbb{F}_3^{108}$ and the signed dodecaphase record $\tau_{12} \in \{-1, 0, +1\}^{12}$. The two readers defined below are computed exclusively from those representations, the calendar, and the HMT constants already constructed.

29.2.1 Reader of displacements on the cyclic calendar of order 108

Let S be the cyclic shift on $\mathbb{Z}/108\mathbb{Z}$. For $k \in \mathbb{Z}/108\mathbb{Z}$, define

$$D_k(\Gamma) := \#\{t \in \mathbb{Z}/108\mathbb{Z} : \gamma_t \neq \gamma_{t+k}\} = \|(I - S^k)\gamma\|_0. \quad (29.5)$$

The integer triple of the reader is

$$K_D(\Gamma) = \left(D_{27} + D_{36}, D_1, \frac{D_4 - D_3}{2} \right) =: (A_D, B_D, C_D), \quad (29.6)$$

and its scalar evaluation is

$$\kappa_D(\Gamma) = A_D - \alpha B_D + \Delta_4 C_D. \quad (29.7)$$

The shifts (27) and (36) are the chronological lifts of 90° and 120° because nine discrete steps represent 30° . The term D_1 measures the immediate local boundary, and the difference $D_4 - D_3$ compares the two genealogies before their height quotient. In the domain of the 324 routes, $\gamma_{t+54} = \gamma_t$ holds; all the D_k are even and C_D is an integer.

29.2.2 Dodecaphasic correlation difference reader

For the word $\tau = (\tau_r)_{r \in \mathbb{Z}/12\mathbb{Z}}$, let

$$C_k^{\text{corr}}(\tau) = \sum_{r=0}^{11} \tau_r \tau_{r+k}, \quad A_\ell(\tau) = \sum_{r=0}^{11} \left(\sum_{j=0}^{\ell-1} \tau_{r+j} \right)^2.$$

The primitive genealogical obstruction and the antipodal memory are

$$\Omega_{90|120}(\tau) = 4A_3 - 3A_4 = -2C_1^{\text{corr}} - 4C_2^{\text{corr}} - 6C_3^{\text{corr}}, \quad (29.8)$$

$$P_6(\tau) = \sum_{r=0}^5 \tau_r \tau_{r+6}. \quad (29.9)$$

The symbol P_6 prevents the collision with the short cycle of the corridor of constantes. The second triple and its evaluation are

$$K_\Omega(\Gamma) = (\Omega_{90|120}(\tau_\Gamma), 54, P_6(\tau_\Gamma)), \quad \kappa_\Omega = \Omega_{90|120} - 54\alpha + P_6\Delta_4. \quad (29.10)$$

Lemma 29.11 (Linear integral primitive uniqueness). *Let $L = uA_3 + vA_4$, with $u, v \in \mathbb{Z}$, be a contrast that vanishes when the genealogies descend to the same height, $A_3 = 3a$, $A_4 = 4a$. Then $(u, v) = n(4, -3)$. Once the sign $90 - 120$ has been fixed, $\Omega_{90|120} = 4A_3 - 3A_4$ is the primitive generator.*

Proof. Vanishing for every a requires $3u + 4v = 0$. Since $\gcd(3, 4) = 1$, the integer solutions are $(u, v) = n(4, -3)$. Primitivity corresponds to $|n| = 1$. $\square$

The lemma selects Ω within the class of integral linear contrasts of the energies A_3, A_4 . Its uniqueness domain is that functional class; nonlinear invariants belong to a distinct selection problem.

29.2.3 Evaluation of the central route

Before metrological comparison, the electronic route $\Gamma_e = (5, 5, ++)$ provides,

$$(D_1, D_3, D_4, D_{27}, D_{36}) = (54, 56, 68, 48, 32),$$

and its signed dodecaphonic register is

$$\tau_e = (+, +, +, -, -, -, +, +, +, -, -, -).$$

Therefore,

$$K_D(\Gamma_e) = (48 + 32, 54, (68 - 56)/2) = (80, 54, 6),$$

$$K_\Omega(\Gamma_e) = (80, 54, 6).$$

Theorem 29.12 (Common Electronic Reduction of Bidirectional Readers). *The readers K_D and K_Ω coincide on the central route and determine*

$$\kappa_e = 80 - 54\alpha + 6\Delta_4. \quad (29.11)$$

Proof. The two preceding evaluations yield the same integer triple. Substitution of its components into $A - \alpha B + \Delta_4 C$ proves (29.11). Every quantity used in this reduction belongs to the route, to its two records, or to the previously constructed HMT constants. $\square$

Subsequent realization on the energy chart. With $R_{\text{act}} = H_5/\eta_{\text{ret}}^{(\text{deg})}$, the positive action on the baseline e_0 yields

$$\mathfrak{e}_e^\beta = \mathfrak{e}_0 R_{\text{act}}^{\kappa_e}. \quad (29.12)$$

After applying the current chart to MeV is obtained

$$s_{u_E}(\mathfrak{e}_e^\beta)|_{u_E=1 \text{ MeV}} = 0.5109989506900008086082571648 \dots \text{ MeV}. \quad (29.13)$$

The first equality belongs to the internal theorem conditioned by the basal and the action line; the second is its dimensional realization. The external value is reserved for metrological comparison.

The integer (80) follows from $D_{27} + D_{36}$ and, in the dodecaphase reader, from $\Omega_{90|120}$. Its coincidence with other internal cardinals is a compatibility control and does not replace this operator provenance. The orientation route $(--)$ is a temporal translation of route $(++)$ and preserves the same profile, in accordance with mass parity under conjugation.

29.2.4 Separation of readers and return scales

Numerical coincidence of internal components is subordinate to their type. At the central point, the objects involved include

$$R_{\text{atlas}}(5, 5) = (24, 0, 12, 0, -12), \quad L_{\kappa_0}(\Gamma_e^{++}) = (24, 0, 0, 0, -12),$$

$$\begin{aligned} k_{\Delta_4}^{\text{dod}}, k_{\Delta_4}^{\text{pro}} : \mathcal{R}_{324} &\longrightarrow \mathbb{Z}, \\ k_{\Delta_4}^{\text{dod}}(\Gamma_e^{++}) &= 12, \quad k_{\Delta_4}^{\text{pro}}(\Gamma_e^{++}) = 4, \\ K_D(\Gamma_e^{++}) &= K_\Omega(\Gamma_e^{++}) = (80, 54, 6). \end{aligned}$$

Here $\mathcal{R}_{324}$ denotes the domain of the 324 oriented routes. The first is a static family signature; the second, a genealogy extractor; the third and fourth terms are torsional readers defined, respectively, by dodecaphasic reading and prospective prolongation; the final object is the pair of bidirectional readers. No partial equality identifies those morphisms or permits transferring a coefficient between readers.

Four temporal scales are also distinguished: 27 is the first return of class, orbit, and cell; 54 = $D_1(\Gamma_e)$ is the observable kinematic return within the cycle; 108 closes the calendar; and 216 closes the oriented kinematic state in the extension. This typing keeps positional return, remark, and oriented holonomy separate.

Causal chain of the electronic outcome The route and its two registers determine K_D , K_Ω , their common reduction, and the evaluation of κ_e . The realization in MeV subsequently applies the energy chart; external mass enters only in the comparison. The pair $\mathbf{K} = (K_D, K_\Omega)$ is the complete operator output of HMT on each route. In a noncentral fiber, its representation by a single physical scalar requires an additional morphism between the fiber operator and the dimensional chart; this requirement neither modifies the internal law nor authorizes the selection of a cell by metrological proximity.

From the Cartan–Holst realization to horizon–area–information confluence through conservation of action The electronic closure fixes a material realization; the next step identifies the grammar that converts action, clock, and mass into physical magnitudes without using that realization as input data.

Determinantal derivation of the absolute action scale and valuation 10^{-34}

30.1 Derivation of the action scale from the dodecaphasic state

The action decade is determined intrinsically, without adopting it as an inherited unit. The local determinantal functional is applied to the previously constructed HMT output α and defines

$$\Sigma_H = \varphi \alpha^{16}. \quad (30.1)$$

The exponent 16 is the order of the local cokernel of the signed closure; its decimal valuation fixes

$$\text{ord}_{10}(\Sigma_H) = -34. \quad (30.2)$$

The origin of the exponent can be verified within this volume. On the local orbit of four leaves, the Hadamard block acts

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}. \quad (30.3)$$

Lemma 30.1 (Local index of the action line). *The form normal of Smith of H_4 is*

$$\text{SNF}(H_4) = \text{diag}(1, 2, 2, 4). \quad (30.4)$$

Consequently,

$$\text{coker}(H_4) \simeq \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}, \quad |\text{coker}(H_4)| = 16.$$

Proof. The greatest common divisor of the entries is one; that of the nonzero 2×2 minors is two; that of the 3×3 minors is four; and $|\det H_4| = 16$. The successive quotients of these determinantal divisors are 1, 2, 2, 4, which prove equation (30.4). The product of the invariants gives the order of the cokernel. $\square$

The norm of the section α over those sixteen sheets produces α^{16} ; the self-scaling φ acts only once on the basis, not once per sheet. Hence equation (30.1). The certified comparisons

$$\alpha^{16} < 10^{-34} < \varphi \alpha^{16} < 10^{-33} \quad (30.5)$$

prove, without introducing the SI unit, that the decimal order is exactly -34 . The terminal evaluation begins with

$$\Sigma_H = 1.046243838104756580184229208303089 \dots \times 10^{-34}. \quad (30.6)$$

The substitution of the kernel by three copies with exponent 16^3 , or the repeated application of φ on each sheet, would alter the genealogy. Both operations constitute falsification criteria rather than equivalent conventions. The causal chain that is preserved is therefore,

$$\text{APP} \rightarrow \widehat{\mathbb{T}} \rightarrow \text{TPK} \rightarrow U_{12}^{\text{sign}} \rightarrow \alpha \rightarrow \Sigma_H \rightarrow 10^{-34} \mathcal{U}_S \rightarrow \{\hbar_{\text{pre}}, \hbar_{\text{ret}}\}.$$

This volume does not rederive α , but develops the metrological geometry subsequent to the selection of the type and decade by Σ_H . The integer -34 is neither a mantissa nor an adjustment to the International System, but the terminal valuation of the local functional.

30.2 Oriented sections of the action line

The realizations before and after the return constitute two sections of the same action line:

$$\hbar_{\text{pre}} = H_5 10^{-34} \mathcal{U}_S, \quad \hbar_{\text{ret}} = \eta_{\text{ret}} 10^{-34} \mathcal{U}_S. \quad (30.7)$$

The common decade determines the line, while the oriented information is expressed through

$$R_{\text{act}} = \frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}} = \frac{H_5}{\eta_{\text{ret}}}, \quad \xi_{\text{act}} = \frac{1}{2} \log R_{\text{act}}. \quad (30.8)$$

The first quantity provides a multiplicative coordinate and the second an additive coordinate of rapidity type. They do not represent two independent Planck constants, but the two orientations of the same section with respect to the return.

If $h_a = 2\pi\hbar_a$, $a \in \{\text{pre}, \text{ret}\}$, a quantity proportional to $h_a^{1/2}$ changes as $R_{\text{act}}^{1/2}$, one proportional to $h_a^{-1/2}$ changes as $R_{\text{act}}^{-1/2}$, and a quotient in which the action cancels is a bidirectional invariant.

30.3 Causal bifurcation subsequent to α

The genealogy of the action and the angular pair forms a bifurcated diagram. The regional branch produces the reduced mantissa η_{ret} ; the determinantal branch produces the decimal order $\varepsilon_H = -34$. Both converge on the action line. The angular pair is constructed in a sister branch from α and η_{ret} :

$$\begin{array}{ccccccc} \alpha, \pi, \varphi & \longrightarrow & H_5, \mathcal{C}_\pi & & & & \alpha, \varphi, H_4 \\ \downarrow & & \downarrow & & & & \downarrow \\ x_A, D_A, \rho_\pi & \longrightarrow & \eta_{\text{ret}} & \longrightarrow & \hbar_{\text{ret}} \in \mathcal{L}_S & \longleftarrow & \Sigma_H, \varepsilon_H = -34 \\ & & \downarrow & & & & \\ & & (A_{\text{deg}}, C_{\text{deg}}^*) & \longrightarrow & (A_{\text{rad}}, C_{\text{rad}}^*) & \longrightarrow & (q_+, q_-) \longrightarrow V_{90,120}. \end{array} \quad (30.9)$$

The expository order places the Planck derivation before the complete angular publication; the diagram simultaneously preserves the exact mathematical dependence of each output.

30.4 Local cokernel and derivation of the absolute order 10^{-34}

The Smith normal form of the local dodecaphase block is

$$\text{SNF}(H_4) = \text{diag}(1, 2, 2, 4), \quad \text{coker}(H_4) \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}. \quad (30.10)$$

The local cokernel has order 16. For the constant section $s_\alpha(g) = \alpha$, the local norm and an action of φ produce

$$\boxed{\Sigma_H = \varphi \mathcal{N}_{G_H}(s_\alpha) = \varphi \alpha^{16}.} \quad (30.11)$$

The exact inequalities

$$\alpha^{16} < 10^{-34} < \varphi \alpha^{16} < 10^{-33} \quad (30.12)$$

determine

$$\boxed{\varepsilon_H = \lfloor \log_{10} \Sigma_H \rfloor = -34.} \quad (30.13)$$

Degree-five regional character and positive action measure

The preceding section isometrically identifies a rank-three positive sector in the oriented interface $t = 7$, the dodecaphase, and the Witt incidence module. The following construction derives the other coefficient of the angular law, $5/468$, and proves why the two terms combine through a positive sum. The question requires three operations to be distinguished:

1. count microscopic presentations;
2. count distinct emissions after forming the TPK quotient;
3. assign a action additive to a path.

The first two operations produce different measures. The third is characterized by a universal property, not by the subsequent proximity of a decimal value.

We shall denote by Γ_{tr} the directed graph of 1944 states of the enriched-path reader. Its free path category will be written $\mathbf{P}(\Gamma_{\text{tr}})$. The local definition of its edges and their action is developed in the corresponding section.

31.1 Seed space, quotient of emissions and regional fibers

Let

$$\Omega_{\text{seed}} = ((\mathbb{Z}/9\mathbb{Z})^2 \times D_4)^2, \quad |\Omega_{\text{seed}}| = 104\,976,$$

the executable set of seed pairs of the finite sender, and let

$$F : \Omega_{\text{seed}} \twoheadrightarrow \mathcal{U}_6, \quad |\mathcal{U}_6| = 468,$$

the application that preserves the distinct sextuple emission. The exact count of fibers is

$ F^{-1}(U) $	$\#\{U \text{ of that size}\}$	supplied seeds
144	432	62 208
432	18	7 776
1 944	18	34 992.

The three contributions sum to 104 976. Therefore, F is not a regular covering: different emissions have different numbers of microscopic presentations.

The microfiber of π contains five distinct emissions. Four have multiplicity 144, and one has multiplicity 432. This datum suffices to prove that the uniform measure on seeds and the uniform measure on emissions are not the same.

31.2 2 exact measures on the quotient of emissions and words

If each seed receives weight $1/104\,976$, the image measure under F is

$$\nu_{\text{seed}}(U) = \frac{|F^{-1}(U)|}{104\,976}.$$

Its possible atoms are

$$\frac{1}{729}, \quad \frac{1}{243}, \quad \frac{1}{54}.$$

In particular,

$$\nu_{\text{seed}}(\mathcal{R}_{\pi}^{\text{micro}}) = \frac{4 \cdot 144 + 432}{104\,976} = \boxed{\frac{7}{729}}.$$

If the elementary state is instead a distinct emission $U \in \mathcal{U}_6$, the diagonal algebra of observables is

$$\mathcal{D}_{468} = \ell^{\infty}(\mathcal{U}_6)$$

and its normalized count state is

$$\tau_{468}(f) = \frac{1}{468} \sum_{U \in \mathcal{U}_6} f(U).$$

Let $\Pi_{\pi}^{(468)}$ be the characteristic function of the five regions of the microfiber. Then

$$\tau_{468}(\Pi_{\pi}^{(468)}) = \boxed{\frac{5}{468}}.$$

The exact difference between both readings is

$$\frac{5}{468} - \frac{7}{729} = \frac{41}{37\,908}.$$

They are not two approaches. The first measure counts presentations; the second counts emission classes.

Theorem 31.1 (Uniqueness of the uniform measure of the catalog). *The unique probability measure on $\mathcal{U}_6$ invariant under every relabeling of its 468 atoms is the uniform measure. Equivalently, the unique positive unital trace on $M_{468}(\mathbb{C})$ is*

$$\tau_{468}(A) = \frac{1}{468} \text{Tr } A.$$

Proof. Transpositions take any atom to any other. Invariance forces all of them to have the same mass, and normalization fixes it at $1/468$. In the matrix version, unitary invariance equates the value of all rank-one projectors. Spectral decomposition and linearity reconstruct the normalized trace. $\square$

The decisive hypothesis is explicit: the reader works with the *quotient catalogue*. The theorem does not claim that the uniform measure on seeds becomes uniform when the fibers are forgotten. The symmetry used is the naturality of the finite unlabeled object, not an ergodic property attributed without proof to the microscopic dynamics.

31.3 Canonical Hilbertization of the quotient

The relationship between seeds and emissions can be formulated unambiguously. Let

$$H_{\text{seed}} = \ell^2(\Omega_{\text{seed}}), \quad H_U = \ell^2(\mathcal{U}_6).$$

The emitter determines

$$J_F \delta_U = |F^{-1}(U)|^{-1/2} \sum_{\omega \in F^{-1}(U)} \delta_{\omega}.$$

Distinct fibers are disjoint and each column has norm one; therefore,

$$J_F^* J_F = I_{H_U}.$$

Moreover, for every observable f of the catalogue,

$$M_{f \circ F} J_F = J_F M_f.$$

The image $J_F H_U$ is exactly the subspace of functions that are constant on each fiber.

If $F^\sharp : H_U \rightarrow H_{\text{seed}}$ denotes the map induced by precomposition, without normalization, then

$$J_F = F^\sharp ((F^\sharp)^* F^\sharp)^{-1/2}.$$

It is the isometric factor of the polar decomposition of that map. This expression proves its naturality under isomorphisms and relabelings of the emitter. It does not imply a false functoriality under arbitrary composition of nonregular surjections: the normalization depends on the fibers of the specific map.

In the language of measures, the elevation of the quotient trace that is uniform within each fiber equals

$$\tilde{\tau}(\omega) = \frac{1}{468 |F^{-1}(F\omega)|}.$$

Each fiber receives total mass $1/468$. The inverse compensation by multiplicity, and not an informal invocation of orbits, is the exact proof.

Corollary 31.2 (Canonicity of $5/468$). *Once the regional reader is defined on distinct emissions, the mass of the microfiber of π is forced:*

$$\tau_{468}(\Pi_\pi^{(468)}) = \frac{\text{rank } \Pi_\pi^{(468)}}{468} = \frac{5}{468}.$$

31.4 Positive defect between the regional and dodecaphasic measures

Write

$$H_{\text{ang}} = H_U \otimes V_{11}.$$

The internal scalar

$$\Delta_4 = \frac{\pi - e \log \pi}{270} \left(1 + \frac{\pi}{729}\right) = 0.0001111964226921402 \dots$$

is constructed with the already fixed HMT readings of π and e , and with the integers 270 and 729 of the ledger. Here it is a pre-dimensional mathematical input: neither units nor physical observables intervene.

The internal construction of the scalar does not by itself determine which operator must multiply. To keep this premise visible, we declare:

Def— sheet-change defect. The dodecaphase coordinate of the sheet-change defect is $\Delta_4 P_3$.

Add will fix the additive form of the action but does not select **Def**: for any $c \geq 0$, the same argument would admit the pair $(\Pi_\pi^{(468)}, cP_3)$. Proving **Def** directly from a total transition of the enriched topological phase kernel is an additional condition.

We define the Kronecker sum

$$D_{\text{switch}}^{\text{dod}} = \Pi_\pi^{(468)} \otimes I_{11} + \Delta_4 I_{468} \otimes P_3.$$

The two summands are positive, commute, and act on different factors. Therefore,

$$D_{\text{switch}}^{\text{dod}} \geq 0.$$

The intertwiner U_W of the [theorem 55.3](#) produces the Witt chart:

$$\begin{aligned} D_{\text{switch}}^{\text{Witt}} &= (I_{468} \otimes U_W) D_{\text{switch}}^{\text{dod}} (I_{468} \otimes U_W)^* \\ &= \Pi_\pi^{(468)} \otimes I_{E_{30}} + \Delta_4 I_{468} \otimes Q_3. \end{aligned}$$

Thus, the same positive operator is read in the dodecaphase and in the incidence module.

The oriented isometric chain also identifies the active sectors.

$$I_{L_{\text{or}}} \otimes P_G, \quad P_3, \quad Q_3.$$

No equivalence of the complete complements of the three modules is asserted: W_{GK} is a partial equivalence of the selected ranges. This is exactly the information required to identify the second contribution to the trace.

Theorem 31.3 (Exact Scope of Entanglement and Action Induction). *Let W_{GK} and U_W be the partial isometries of the theorem 55.7. Then:*

1. *for every $c \geq 0$,*

$$W_{GK}^*(cP_3)W_{GK} = c(I_{L_{\text{or}}} \otimes P_G), \quad U_W(cP_3)U_W^* = cQ_3;$$

thus, entanglement carries the coefficient, but does not select it;

2. *if P_{sgn} and P_{std} are the central projectors of the decomposition*

$$P_3V_{11}|_{H_*} \simeq \text{sgn}_{H_*} \oplus \text{std}, \quad \text{rank } P_{\text{sgn}} = 1, \quad \text{rank } P_{\text{std}} = 2,$$

every positive self-adjoint operator supported on P_3V_{11} and H_ -equivariant has the form*

$$D_{a,b} = aP_{\text{sgn}} + bP_{\text{std}}, \quad a, b \geq 0.$$

Even impose the trace of Δ_4P_3 leaves the family

$$a + 2b = 3\Delta_4.$$

The isotropy $a = b$ is not deduced from the residual symmetry S_3 ;

3. *the uniform probability of the catalog, $\mu_U(U) = 1/468$, is the canonical trace of the finite quotient, and its compensated lift is determined by J_F . It is not the image measure of the uniform probability on seeds, whose atoms are $1/729, 1/243, 1/54$. In particular,*

$$\mu_U(\mathcal{R}_\pi^{\text{micro}}) = \frac{5}{468}, \quad F_\# \mu_{\text{seed}}(\mathcal{R}_\pi^{\text{micro}}) = \frac{7}{729};$$

4. *the complete law*

$$\delta_* = \frac{5}{468} + \frac{3}{11}\Delta_4, \quad \lambda_* = 1 - \delta_*, \quad a_*(e) = g(e) + \lambda_*s(e)$$

is a theorem relative to the five clauses $\text{Tr}_q\text{--Add--Def--Res--Loc}$. The executable engine of the reduced topological phase kernel does not induce, by directed cover, the branched operator Γ_{tr} : the associated dynamics returns to the identity after 54 iterations, while each state of Γ_{tr} has four distinct successors and no self-edge.

Consequently, the preceding results prove the canonicity of the isometric identifications, of the trace on the quotient catalogue, and of the additive character within their premises. The stronger assertion that a total transition of the enriched Topological Phase Kernel necessarily induces Δ_4P_3 , the uniform measure, and the local law still requires the additional dynamical theorem specified in the induction analysis.

Proof. The identities in the first assertion are obtained by multiplying $W_{GK}^*W_{GK} = I_{L_{\text{or}}} \otimes P_G$, $W_{GK}W_{GK}^* = P_3$, and $U_WP_3U_W^* = Q_3$ by the scalar c . For the second assertion, the restriction to $H_* \simeq S_3$ is the

multiplicity-free sum of two nonisomorphic irreducible representations. Schur's lemma diagonalizes every equivariant self-adjoint operator on those two blocks; positivity is equivalent to $a, b \geq 0$. Its normalized trace is $(a + 2b)/11$, which proves the residual freedom even after the trace has been fixed. The full action of A_5 on the irreducible triplet would reduce the family to cP_3 , but it would not fix the value of c either.

The third point follows from the fibre census: F has multiplicities 144, 432, 1944, so the image measure weights an emission by its number of presentations. By contrast, the trace τ_{468} assigns the same weight to every class; J_F realizes exactly its normalized lift. The fourth point is the logical concatenation already proved: Tr_q fixes the quotient trace, **Add**the sum of traces, **Def**declares $\Delta_4 P_3$, **Res**defines the residual capacity, and **Loc**carries it to the edges. The finite enumeration proves the obstruction between the reduced kernel and Γ_{tr} : a directed cover would preserve the outgoing star, and a semiconjugacy of the identity return would require self-edges in the image, contradicting both censuses of Γ_{tr} . $\square$

The complete spectrum of $D_{\text{switch}}^{\text{dod}}$ is

eigenvalue	multiplicity
0	$(468 - 5)(11 - 3) = 3704$
Δ_4	$(468 - 5)3 = 1389$
1	$5(11 - 3) = 40$
$1 + \Delta_4$	$5 \cdot 3 = 15$.

Consequently,

$$\begin{aligned} \tau_{468 \cdot 11}(D_{\text{switch}}) &= \tau_{468}(\Pi_{\pi}^{(468)}) + \Delta_4 \tau_{11}(P_3) \\ &= \boxed{\frac{5}{468} + \frac{3}{11} \Delta_4}. \end{aligned}$$

The same identity holds in the Witt chart. In the oriented-sector chart, the term $3/11$ is the trace of P_G on the homogeneous space of eleven branches.

31.5 Universal property of the positive action functional

The word *action* is used in an additive sense: the concatenation of two routes receives the sum of their increments. To separate this content from a chosen formula, we first define the category in which the law will be unique.

Definition 31.4 (Positive Action Character HMT). A character local of action on the product regional–dodecafasico is one map

$$\delta : M_{468}(\mathbb{C})_+ \oplus M_{11}(\mathbb{C})_+ \longrightarrow \mathbb{R}_{\geq 0}$$

which satisfies:

1. exact additivity on the product cone;
2. homogeneidad positiva;
3. invariance under independent unitary conjugation in the two factors;
4. bifactorial normalization,

$$\delta(I_{468}, 0) = \delta(0, I_{11}) = 1.$$

Additivity translates concatenation; invariance eliminates the coordinates; bifactorial normalization fixes a scale in each summand. This is not a unital functional on the direct-sum algebra: its unit is (I_{468}, I_{11}) and additivity gives $\delta(I_{468}, I_{11}) = 2$. No clause contains the value of an angle.

Theorem 31.5 (Uniqueness of the positive character and relative residual capacity). *En the class of the definition preceding there exists a unique character positive of action HMT. It is*

$$\delta(A, B) = \tau_{468}(A) + \tau_{11}(B).$$

Applied to the defect declared by Def, $(\Pi_\pi^{(468)}, \Delta_4 P_3)$, it yields

$$\delta_* = \frac{5}{468} + \frac{3}{11}\Delta_4.$$

If, in addition, Resis declared, according to which the self-dual capacity for sheet change is normalized to one and the defect reduces it through $\lambda = 1 - \delta$, the residual capacity within $\text{Tr}_q\text{-Add-Def-Resis}$

$$\lambda_* = 1 - \delta_* = 1 - \frac{5}{468} - \frac{3}{11}\Delta_4.$$

Proof. The separability of additivity separates the factors:

$$\delta(A, B) = \delta(A, 0) + \delta(0, B).$$

Each restriction is positive, homogeneous, and invariant under conjugation. For a rank-one projector, unitary invariance gives a common value. Additivity over a spectral decomposition forces each restriction to be a multiple of the trace. The bifactorial normalization fixes those multiples at $1/468$ and $1/11$. The sum of traces follows. The last formula applies the declared normalization of residual capacity. $\square$

Numerically,

$$\lambda_* = 0.9892859130191415\dots, \quad 0 < \lambda_* < 1.$$

Positivity already follows from $0 < \Delta_4 < 10^{-3}$ and $5/468 < 1$. It does not depend on a metrological comparison.

31.6 Positive Action Functor on the Category of Compatible Paths

Recall the directed graph Γ_{tr} and its free category $\mathbf{P}(\Gamma_{\text{tr}})$. For an edge $e : x \rightarrow y$, define

$$g(e) = \mathbf{1}_{\{d_y \neq d_x\}}, \quad s(e) = \mathbf{1}_{\{m_y \neq m_x\}},$$

where g records a turn and s a change of sheet. The arrow from the scalar λ_* to the edges does not follow from the trace theorem. It is declared through the clause **LOC**: a turn has unit cost, a change of sheet has cost λ_* , and the two indicators are added when they occur on the same edge. Under **LOC**, the local action is

$$a_*(e) = g(e) + \lambda_* s(e).$$

For a route $\gamma = e_1 \cdots e_n$,

$$\mathcal{A}_*(\gamma) = \sum_{j=1}^n a_*(e_j) = \# \text{giros}(\gamma) + \lambda_* \# \text{cambios de hoja}(\gamma).$$

For the empty path $\mathbf{1}_x$, one fixes $\mathcal{A}_*(\mathbf{1}_x) = 0$. Then

$$\mathcal{A}_*(\gamma_1 \gamma_2) = \mathcal{A}_*(\gamma_1) + \mathcal{A}_*(\gamma_2).$$

Therefore,

$$\mathcal{A}_* : \mathbf{P}(\Gamma_{\text{tr}}) \longrightarrow (\mathbb{R}_{\geq 0}, +)$$

is a monoidal functor, or additive path cocycle.

The indicators g and s are insensitive to the direction of traversal. Orientation is not lost: it resides in the line L_{or} and in the star involution, whereas positive cost resides in the projectors. Conflating the two data would introduce negative signs into the action.

For $\beta > 0$, the associated transfer operator is

$$(\mathcal{L}_{\beta,*}f)(y) = \sum_{e:x \rightarrow y} e^{-\beta a_*(e)} f(x).$$

All its weights are strictly positive. Since the potential depends only on the two local indicators, it commutes with the coarse action $(\mathbb{Z}/3\mathbb{Z})^3$ and respects the decomposition into 27 Bloch sectors used in the following section.

A finite unweighted deterministic dynamics acts through permutation matrices and has singular values equal to one. The cocycle $\mathcal{A}_*$ turns the inventory of paths into a transfer operator that preserves the positive cone, that is, into a matrix with strictly positive weights on the permitted edges. That property does not by itself prove contraction, simplicity of the dominant eigenvalue, or spectral separation; these conclusions require a normalization or an additional spectral argument.

31.7 Regional character of degree 5 and reduced action mantissa

The central block

$$\mathcal{B}_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} \quad (31.1)$$

has eigenvalues 9 and 5. The length 4 of the step orbits 3, the rank 12 of the dodecaphasic cycle, and $104976/1944 = 54$ determine the character

$$H_5(\alpha, \varphi) = \frac{1}{2} \sqrt{\frac{1000\alpha}{\varphi}} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5. \quad (31.2)$$

The nonadic completion constructs the analytic coordinate and the regional determinant.

$$x_A = \frac{50\pi}{9}\alpha, \quad D_A = e^{-2x_A} = \exp\left(-\frac{100\pi}{9}\alpha\right). \quad (31.3)$$

The microfiber supplies $\rho_\pi = 169$, and its absolutely convergent series is

$$\mathcal{C}_\pi(\alpha) = 169\alpha^6 + \frac{D_A\alpha^7}{1 - D_A\alpha}. \quad (31.4)$$

The reduced mantissa after return is then constructed by

$$\eta_{\text{ret}} = H_5 - \frac{90}{\pi} \mathcal{C}_\pi(\alpha) \in \mathbb{R}_{>0}. \quad (31.5)$$

Its right-hand side contains exclusively objects generated previously by APP–TRIT–TPK and by the closure of α .

Self-dual derivation of the Planck constant and its action line

32.1 Compton realization associated with the temporal periodicity of 108 phases

On the cycle of circular realization identity of the 108-phase calendar, fix a positive action section $\hbar$. The associated angular quantum is

$$E_{108} = \hbar\omega_{108}, \quad m_{108} = \frac{E_{108}}{c^2} = \frac{\hbar\omega_{108}}{c^2}. \quad (32.1)$$

G has not yet been used. The geometry of the cycle and angular quantization suffice to fix its two Compton readers.

Theorem 32.1 (Identity between the temporal periodicity of 108 phases and the Compton length). *For the quantum (32.1),*

$$\bar{\lambda}_C(m_{108}) = \frac{\hbar}{m_{108}c} = R_{108}, \quad \lambda_C(m_{108}) = \frac{h}{m_{108}c} = C_{108}. \quad (32.2)$$

Therefore, the cycle radius is its reduced Compton wavelength, and the length of one revolution is its full Compton wavelength.

Proof. Substituting $m_{108} = \hbar\omega_{108}/c^2$,

$$\frac{\hbar}{m_{108}c} = \frac{\hbar c^2}{\hbar\omega_{108}c} = \frac{c}{\omega_{108}} = R_{108}.$$

Since $h = 2\pi\hbar$, the second equality is $2\pi R_{108} = C_{108}$. □

The coincidence is not a numerical fit. It is a homogeneous identity: it holds for all $\hbar, c, t_0 > 0$ and depends only on the same period determining the frequency, the cycle length, and the quantum of action.

32.2 Gravitational Transducer and Self-Dual Selection Equation

The gravitational line HMT–MD contains the transducer

$$G(\theta) = \theta \frac{c^5 t_0^2}{\hbar}, \quad \theta > 0. \quad (32.3)$$

Homogeneity is exact, but by itself does not select θ . The CT108 cycle supplies an additional internal condition: the gravitational reader must return the same radial scale as the geometric and Compton readers.

We define the reduced gravitational radius

$$r_g(\theta) := \frac{G(\theta)m_{108}}{c^2}. \quad (32.4)$$

Substituting (32.1) and (32.3),

$$r_g(\theta) = \theta \frac{\pi}{54} \ell_0. \quad (32.5)$$

Theorem 32.2 (Self-dual selector of the temporal periodicity of 108 phases). *The equation*

$$r_g(\theta) = R_{108}$$

has a unique positive solution:

$$\boxed{\theta_* = \left(\frac{54}{\pi}\right)^2.} \quad (32.6)$$

Proof. By (circular realization identity of the 108-phase calendar) and (32.5), the equation is

$$\theta \frac{\pi}{54} \ell_0 = \frac{54}{\pi} \ell_0.$$

Since $\ell_0 > 0$, simplification yields $\theta = (54/\pi)^2$. The function $\theta \mapsto r_g(\theta)$ is strictly increasing on $\mathbb{R}_{>0}$, so there is no other positive solution. $\square$

Corollary 32.3 (Internal Planck Unit). *With $G_* = G(\theta_*)$, the Planck sections satisfy*

$$\ell_P := \sqrt{\frac{\hbar G_*}{c^3}} = R_{108}, \quad t_P := \sqrt{\frac{\hbar G_*}{c^5}} = \frac{1}{\omega_{108}}, \quad (32.7)$$

$$m_P := \sqrt{\frac{\hbar c}{G_*}} = m_{108}, \quad E_P := m_P c^2 = E_{108}. \quad (32.8)$$

In particular,

$$\boxed{R_{108} = \bar{\lambda}_C = r_g = \ell_P, \quad C_{108} = \lambda_C = 2\pi r_g = 2\pi \ell_P.} \quad (32.9)$$

Proof. $G_* = (54/\pi)^2 c^5 t_0^2 / \hbar$ is substituted. Then

$$\ell_P = \frac{54}{\pi} c t_0 = R_{108}, \quad t_P = \frac{54}{\pi} t_0 = \frac{1}{\omega_{108}},$$

and

$$m_P = \frac{\pi}{54} \frac{\hbar}{c^2 t_0} = \frac{\hbar \omega_{108}}{c^2} = m_{108}.$$

The rest follows from theorem 32.1. $\square$

Remark 32.4 (Three readers, one section). Circular geometry publishes R_{108} and C_{108} ; Compton quantization publishes $\bar{\lambda}_C$ and λ_C ; gravitational realization publishes r_g and $2\pi r_g$. In θ_* those three pairs are coordinates of a single section. This is the precise reason why the Planck unit ceases to appear as an accidental dimensional combination within this formalization.

32.3 Commutativity of the angular, temporal, and gravitational normalizations of the self-dual identity

The transducer (32.3) admits two ways of fixing the elementary scale, which must not be confused.

- (i) If $\theta = 1$ is fixed, then $\sqrt{\hbar G/c^3} = \ell_0$. The elementary transition $c t_0$ is identified with the Planck length of that chart.
- (ii) If the *complete orbit* CT108 is required to have Planck radius, the condition selects $\theta_* = (54/\pi)^2$ and $R_{108} = \ell_P$.

They are not rival values of G . They are two gauges of the same homogeneous equation: one assigns the Planck unit to the elementary step; the other, to the radius of the complete cycle. The annex uses the latter because its thesis concerns the identity of the CT108 closure.

32.4 Gravitational conventions: reduced radius and Schwarzschild radius

The gravitational notation is fixed in an unambiguous manner:

$$r_g = \frac{Gm}{c^2}, \quad r_s = \frac{2Gm}{c^2} = 2r_g.$$

The preceding theorem uses r_g , not r_s . In the usual convention of Planck mass, the equality $\hbar/(mc) = Gm/c^2$ is produced in $m = m_p$, whereas

$$\frac{\hbar}{mc} = \frac{2Gm}{c^2} \iff m = \frac{m_p}{\sqrt{2}}. \quad (32.10)$$

For this reason the phrase “gravitational scale” designates, in the rector thesis, the reduced radius. The conventional horizon radius remains a distinct reader.

32.5 Bifurcation of the action line

The internal line of action possesses two oriented sections:

$$\hbar_{\text{pre}} = H_5 10^{-34} \mathcal{U}_S, \quad \hbar_{\text{ret}} = \eta_{\text{ret}} 10^{-34} \mathcal{U}_S, \quad R_{\text{act}} = \frac{H_5}{\eta_{\text{ret}}}. \quad (32.11)$$

The decade 10^{-34} fixes the absolute scale; the quotient R_{act} transports the oriented difference. This separation makes it possible to ask whether Planck self-duality survives in both coordinates.

Theorem 32.5 (Reciprocal Covariance of Action and Gravity). *For $a \in \{\text{pre}, \text{ret}\}$, let*

$$G_a(\theta) = \theta \frac{c^5 t_0^2}{\hbar_a}, \quad m_{108,a} = \frac{\hbar_a \omega_{108}}{c^2}.$$

Then

$$\frac{G_a(\theta) m_{108,a}}{c^2} = \theta \frac{\pi}{54} \ell_0 \quad (32.12)$$

is independent of a . Moreover,

$$\frac{m_{108,\text{pre}}}{m_{108,\text{ret}}} = R_{\text{act}}, \quad \frac{G_{\text{pre}}}{G_{\text{ret}}} = R_{\text{act}}^{-1}, \quad (32.13)$$

and the Planck length $\sqrt{\hbar_a G_a / c^3} = \sqrt{\theta} \ell_0$ is the same in both orientations.

Proof. The factors $\hbar_a$ cancel in the product $G_a m_{108,a}$. The two ratios in (32.13) follow directly from the definitions. The invariance of the Planck length follows from $\hbar_a G_a = \theta c^5 t_0^2$. $\square$

A symmetric parametrization makes the geometry visible:

$$\hbar_{\text{geom}} := \sqrt{\hbar_{\text{pre}} \hbar_{\text{ret}}}, \quad \xi := \frac{1}{2} \log R_{\text{act}}, \quad \hbar_{\text{pre}} = \hbar_{\text{geom}} e^{\xi}, \quad \hbar_{\text{ret}} = \hbar_{\text{geom}} e^{-\xi}. \quad (32.14)$$

Bidirectional memory is lodged in ξ ; Planck geometry is lodged in the invariant product. This allows a real difference between the two action coordinates without requiring a difference in propagation speed.

Remark 32.6 (Physical scope of the theorem). The reciprocity of theorem 32.5 is exact within the transducer. To identify G_{pre} and G_{ret} with two oriented gravitational measurements, the protocol, the observable, and the symmetry relating them must be specified. The theorem proves structural compatibility; by itself, it neither anticipates a Lorentz violation nor an observable variation of G .

The action line constructed in the preceding section admits a circular realization compatible with temporal periodicity of order 108. This realization coordinates three objects that already possess their own types: the action section, the geometry of one revolution, and the gravitational transduction. The resulting coordination provides an internal Planck identity and prepares the positive line on which the general mass operator will subsequently act.

32.5.1 Oriented action letters and declared circular realization

Let

$$\mathcal{L}_S^+, \quad \mathcal{L}_T^+, \quad \mathcal{L}_C^+, \quad \mathcal{L}_L^+, \quad \mathcal{L}_M^+, \quad \mathcal{L}_G^+$$

the positive half-lines of action, time, velocity, length, and mass. The pre-return and post-return coordinates are written

$$\hbar_{\text{pre}} = H_5 10^{-34} \mathcal{U}_S, \quad \hbar_{\text{ret}} = \eta_{\text{ret}} 10^{-34} \mathcal{U}_S, \quad R_{\text{act}} = \frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}} = \frac{H_5}{\eta_{\text{ret}}}. \quad (32.15)$$

The subscript $a \in \{\text{pre}, \text{ret}\}$ shall designate either of the two charts. In each one the exact distinction is preserved

$$h_a = 2\pi \hbar_a. \quad (32.16)$$

Let us fix an elementary duration $t_0 \in \mathcal{L}_T^+$, an internal velocity $c_{\text{int}} \in \mathcal{L}_C^+$, and $\ell_0 = c_{\text{int}} t_0$. The declared circular realization of the temporal periodicity of order 108 is determined by

$$T_{108} = 108 t_0, \quad \omega_{108} = \frac{2\pi}{T_{108}}, \quad R_{108} = \frac{c_{\text{int}}}{\omega_{108}} = \frac{54}{\pi} \ell_0, \quad (32.17)$$

and by the length of the full turn

$$C_{108} = 2\pi R_{108} = c_{\text{int}} T_{108} = 108 \ell_0. \quad (32.18)$$

Thus, R_{108} is the radius of the realization and C_{108} its perimeter. The angular frequency, the cyclic frequency and the two action constants are related through $\omega_{108} = 2\pi/T_{108}$ and $h_a = 2\pi \hbar_a$.

32.5.2 Correspondence between the order-108 periodicity and Compton readings

In the chart a , the circular quantum and its mass section are

$$E_{108,a} = \hbar_a \omega_{108}, \quad m_{108,a} = \frac{E_{108,a}}{c_{\text{int}}^2} = \frac{\hbar_a \omega_{108}}{c_{\text{int}}^2}. \quad (32.19)$$

Theorem 32.7 (Identity between circular periodicity and Compton readings). *For each chart $a \in \{\text{pre}, \text{ret}\}$, the reduced and complete Compton readings satisfy*

$$\boxed{\bar{\lambda}_C(m_{108,a}) = \frac{\hbar_a}{m_{108,a} c_{\text{int}}} = R_{108}, \quad \lambda_C(m_{108,a}) = \frac{h_a}{m_{108,a} c_{\text{int}}} = C_{108}.} \quad (32.20)$$

Proof. Substituting (32.19) produces

$$\frac{\hbar_a}{m_{108,a} c_{\text{int}}} = \frac{c_{\text{int}}}{\omega_{108}} = R_{108}.$$

The second equality follows by applying simultaneously (32.16) and (32.18). $\square$

The theorem is homogeneous in $(\hbar_a, c_{\text{int}}, t_0)$: the same periodicity determines the radius, the perimeter, and the two Compton charts. The circular realization is the declared geometric primitive; the preceding equalities are its exact consequences.

32.5.3 Circular selector and internal Planck unit

The gravitational line is coordinated with the chart a through the homogeneous transducer

$$G_a(\theta) = \theta \frac{c_{\text{int}}^5 t_0^2}{\hbar_a} \in \mathcal{L}_G^+, \quad \theta \in \mathbb{R}_{>0}. \quad (32.21)$$

Its reduced gravitational radius, evaluated on the circular quantum, is

$$r_{g,a}(\theta) := \frac{G_a(\theta) m_{108,a}}{c_{\text{int}}^2} = \theta \frac{\pi}{54} \ell_0. \quad (32.22)$$

Proposition 32.8 (Planck Circular Selector). *Within the circular realization (32.17), the condition $r_{g,a}(\theta) = R_{108}$ selects a unique positive coordinate,*

$$\boxed{\theta_{108}^\circ = \left(\frac{54}{\pi}\right)^2}. \quad (32.23)$$

The coordinate is independent of the chart oriented a .

Proof. The equations (32.17) and (32.22) reduce the condition to

$$\theta \frac{\pi}{54} \ell_0 = \frac{54}{\pi} \ell_0.$$

The positivity of ℓ_0 and the strict monotonicity of the map $\theta \mapsto r_{g,a}(\theta)$ provide the solution and its uniqueness. $\square$

Let $G_{108,a}^\circ := G_a(\theta_{108}^\circ)$. The internal units associated with this chart satisfy

$$\ell_{P,a}^\circ := \sqrt{\frac{\hbar_a G_{108,a}^\circ}{c_{\text{int}}^3}} = R_{108}, \quad t_{P,a}^\circ := \sqrt{\frac{\hbar_a G_{108,a}^\circ}{c_{\text{int}}^5}} = \omega_{108}^{-1}, \quad (32.24)$$

$$m_{P,a}^\circ := \sqrt{\frac{\hbar_a c_{\text{int}}}{G_{108,a}^\circ}} = m_{108,a}, \quad E_{P,a}^\circ := m_{P,a}^\circ c_{\text{int}}^2 = E_{108,a}. \quad (32.25)$$

The equality of radii is therefore published as

$$R_{108} = \bar{\lambda}_C = r_g = \ell_P^\circ, \quad C_{108} = \lambda_C = 2\pi r_g = 2\pi \ell_P^\circ. \quad (32.26)$$

The convention of this theorem is the reduced gravitational radius $r_g = Gm/c_{\text{int}}^2$. The Schwarzschild radius satisfies $r_s = 2r_g$; if this second convention is adopted, its crossing with $\bar{\lambda}_C$ occurs at $m = m_{P,a}^\circ/\sqrt{2}$. The declaration of the radius therefore forms part of the chart and prevents a factor of two from being inadvertently transferred between the two identities.

The coordinate θ_{108}° belongs to the declared circular specialization. The general gravitational selector, formulated on the enriched holonomy and its spectral response, retains its residence in section devoted to the natural gravitational selector. Thus, the circular identity fixes an internal Planck normalization and the general criterion fixes the conditions that the physical realization of the gravitational section must satisfy.

32.5.4 Pre-return/post-return covariance

The product of the reciprocal sections of action and gravity preserves the circular geometry:

$$\frac{m_{108,\text{pre}}}{m_{108,\text{ret}}} = R_{\text{act}}, \quad \frac{G_{108,\text{pre}}^\circ}{G_{108,\text{ret}}^\circ} = R_{\text{act}}^{-1}, \quad \hbar_a G_{108,a}^\circ = \theta_{108}^\circ c_{\text{int}}^5 t_0^2. \quad (32.27)$$

In particular, $\ell_{\text{P},\text{pre}}^\circ$ and $\ell_{\text{P},\text{ret}}^\circ$ coincide with R_{108} . The bidirectional memory resides in the quotient of the masses and in its gravitational reciprocal; the geometric scale resides in the invariant product.

Reciprocity admits a symmetric parametrization that separates the common scale from the orientation of the return. Define

$$\hbar_{\text{geom}} := \sqrt{\hbar_{\text{pre}} \hbar_{\text{ret}}}, \quad G_{108,\text{geom}}^\circ := \sqrt{G_{108,\text{pre}}^\circ G_{108,\text{ret}}^\circ}, \quad \xi_{\text{act}} := \frac{1}{2} \log R_{\text{act}}. \quad (32.28)$$

Then

$$\hbar_{\text{pre}} = \hbar_{\text{geom}} e^{\xi_{\text{act}}}, \quad \hbar_{\text{ret}} = \hbar_{\text{geom}} e^{-\xi_{\text{act}}},$$

whereas the conjugate gravitational sections satisfy

$$G_{108,\text{pre}}^\circ = G_{108,\text{geom}}^\circ e^{-\xi_{\text{act}}}, \quad G_{108,\text{ret}}^\circ = G_{108,\text{geom}}^\circ e^{\xi_{\text{act}}}.$$

The coordinate ξ_{act} preserves the pre-return/post-return orientation; the product $\hbar_a G_{108,a}^\circ$ preserves geometric scale. This parametrization of the action charts does not by itself introduce distinct propagation speeds.

32.5.5 Split self-dual mass line

Given a chart a , let the split algebra be

$$\mathbb{D}_{\text{sp}} := \mathbb{R}[j]/(j^2 - 1), \quad P_\pm := \frac{I \pm j}{2},$$

and consider its spectral decomposition

$$\mathbb{D}_{\text{sp}} = \mathbb{R}P_+ \oplus \mathbb{R}P_-, \quad P_\pm^2 = P_\pm, \quad P_+P_- = 0, \quad P_+ + P_- = I.$$

The broken conjugation swaps the projectors,

$$\overline{uP_+ + vP_-} = vP_+ + uP_-,$$

and its norm is

$$N_{\text{sp}}(uP_+ + vP_-) := uv.$$

For $m \in \mathcal{L}_M^+$, define the section

$$Z_a(m) := \frac{m_{\text{P},a}^\circ}{m} P_+ + \frac{m}{m_{\text{P},a}^\circ} P_-. \quad (32.29)$$

The unit $m_{\text{P},a}^\circ$ also determines the involution

$$\iota_{\text{P},a}(m) = \frac{(m_{\text{P},a}^\circ)^2}{m}. \quad (32.30)$$

Theorem 32.9 (Split self-dual mass line). *For every $m \in \mathcal{L}_M^+$,*

$$\boxed{N_{\text{sp}}(Z_a(m)) = 1, \quad \overline{Z_a(m)} = Z_a(t_{\text{P},a}(m))}. \quad (32.31)$$

Conjugation has a unique positive fixed point,

$$m = m_{\text{P},a}^\circ, \quad Z_a(m_{\text{P},a}^\circ) = I.$$

Proof. The norm is the product of the two reciprocal coordinates:

$$\frac{m_{\text{P},a}^\circ}{m} \frac{m}{m_{\text{P},a}^\circ} = 1.$$

Conjugation interchanges both coordinates, which is equivalent to replacing m by $(m_{\text{P},a}^\circ)^2/m$. The positive fixed point satisfies $m^2 = (m_{\text{P},a}^\circ)^2$. $\square$

With the speed

$$x_a(m) = \log \frac{m}{m_{\text{P},a}^\circ}, \quad (32.32)$$

the section and its conjugate are written

$$Z_a(m) = e^{-x_a} P_+ + e^{x_a} P_-, \quad \overline{Z_a(m)} = e^{x_a} P_+ + e^{-x_a} P_-.$$

Their even and odd readers are

$$Q_a(m) := \frac{e^{x_a} + e^{-x_a}}{2} = \cosh x_a, \quad A_a(m) := \frac{e^{x_a} - e^{-x_a}}{2} = \sinh x_a. \quad (32.33)$$

The first preserves the distance to the self-dual point and the second preserves its orientation.

The reciprocal lattice vectors associated with the same coordinate are

$$r_{\text{g},a}(m) = \frac{G_{108,a}^\circ m}{c_{\text{int}}^2}, \quad \bar{\lambda}_{\text{C},a}(m) = \frac{\hbar_a}{mc_{\text{int}}},$$

and satisfy

$$\frac{r_{\text{g},a}(m)}{\bar{\lambda}_{\text{C},a}(m)} = e^{2x_a(m)}, \quad \frac{\bar{\lambda}_{\text{C},a}(m)}{r_{\text{g},a}(m)} = e^{-2x_a(m)}. \quad (32.34)$$

Conjugation interchanges the two readers; Planck's identity is their unitary equilibrium.

Remark 32.10 (Planck identity and electronic closure). Identity $m = m_{\text{P},a}^\circ$ fixes the self-dual origin of the positive mass line. The electronic section fixes a distinct position on that same line through its own genealogy. The section devoted to the universal action operator constructs the universal operator that transports absolute scale, character, towers, and bidirectional reading; the section devoted to bidirectional electronic closure subsequently realizes its central rank-one reduction. Thus, the Planck unit supplies the identity of the coordinate, and the electron supplies the first complete material transition on it.

Proposition 32.11 (Genealogical realization of the electronic center). *Let*

$$F_e^+ = \mathbb{R}_{>0}(P_+ + P_-)$$

the positive half-line of the rank-one central fiber, let $\mathcal{G}_e$ be the space of enriched electronic genealogies and let $\mathcal{L}_M^+$ be the positive mass line. A realization morphism

$$\mathfrak{R}_e : F_e^+ \times \mathcal{G}_e \longrightarrow \mathcal{L}_M^+$$

can send the identity $I = P_+ + P_-$ to a mass different from $m_{P,a}^\circ$ without altering the [theorem 32.9](#): the Planck identity fixes the autodual origin of the chart, while the electronic genealogy fixes a displacement on that chart.

Proof. If $\chi_e : \mathcal{G}_e \rightarrow \mathbb{R}_{>0}$ is the positive character constructed by the electronic record, the rule

$$\mathfrak{R}_e(\lambda I, g) = \lambda m_{P,a}^\circ \chi_e(g)$$

is positive and multiplicative in the genealogical coordinate. For the neutral genealogy, $\chi_e = 1$, the Planck origin is recovered; for a genealogy with $\chi_e(g) \neq 1$, the same internal identity occupies a different position of the line. The concrete electronic closure preserves its antecedent development in the section devoted to bidirectional electronic closure. $\square$

32.6 Autoduality of the Planck internal unit under orientation inversion

The CT108 selector constructs

$$\theta_{108} = \left(\frac{54}{\pi}\right)^2, \quad \ell_P = \frac{54}{\pi} \ell_0, \quad t_P = \frac{54}{\pi} t_0, \quad (32.35)$$

and

$$E_P = \frac{\hbar_{\text{int}}}{t_P} = \frac{\pi \hbar_{\text{int}}}{54 t_0}. \quad (32.36)$$

Comparing the corresponding identity gives

$$\boxed{E_P = k_B \Theta_{\text{clk}} \log 3.} \quad (32.37)$$

The relation does not depend on a coincidence between units: both sides follow from the same chronological structure 108.

When the gravitational constant of [chapter 39](#) is introduced, the Planck family satisfies

$$\ell_P^2 = \frac{G\hbar}{c^3} = \theta_{108} \ell_0^2, \quad (32.38)$$

$$F_P = \frac{c^4}{G}, \quad P_P = \frac{c^5}{G}, \quad (32.39)$$

$$\rho_P = \frac{\hbar}{\theta_{108}^2 c^5 t_0^4}. \quad (32.40)$$

Although G and $\hbar$ transform reciprocally between the sections before and after return, their product in ℓ_P^2 preserves geometry. This invariance determines area as the interface between the geometric, informational and entanglement realizations.

32.7 Action line and Planck constant

Let $\mathcal{L}_S$ be the real action line with positive basis $\mathcal{U}_S$. The transduction

$$\tau_S : \mathbb{R}_{>0} \longrightarrow \mathcal{L}_S, \quad u \longmapsto u 10^{\varepsilon_H} \mathcal{U}_S \quad (32.41)$$

publish the sections preceding and following the return:

$$\hbar_{\text{pre}} = H_5 10^{-34} \mathcal{U}_S, \quad \boxed{\hbar_{\text{ret}} = \eta_{\text{ret}} 10^{-34} \mathcal{U}_S}. \quad (32.42)$$

The full cycle actions are

$$h_{\text{pre}} = 2\pi \hbar_{\text{pre}}, \quad h_{\text{ret}} = 2\pi \hbar_{\text{ret}}. \quad (32.43)$$

For a phase θ , the angular and action charts commute:

$$S(\theta) = \hbar_{\text{ret}} \theta_{\text{rad}} = \hbar_{\text{ret}} \frac{\pi}{180} \theta_{\text{deg}} = h_{\text{ret}} \frac{\theta_{\text{deg}}}{360}. \quad (32.44)$$

Phase operator, spectral radius and factorization of dual projection

The two preceding sections have gathered three pieces that previously appeared separately: an isometric identification between projectors, a quotient measure, and a positive action cocycle. The present section studies the spectral consequence of that law and places the result within the complete architecture of HMT.

The separation of statuses is precise. The triple identification is an exact theorem relative to the new internal variational rule and an oriented anchor. The measure and the positive character are exact theorems of $\text{Tr}_q\text{-Add}$; the defect is relative, in addition, to Def , the capacity to Res , and the local cost to Loc . The angular value is an interval-certified output of the operator Γ_{tr} . Its proximity to the canonical conjugate angular coordinate is extraordinary, but the interval for the difference excludes equality.

33.1 Decomposition of the angular operator into 27 sectors of 72 states

The all-paths reader Γ_{tr} has 1944 microscopic states. The coarse translation action of $(\mathbb{Z}/3\mathbb{Z})^3$ permits the decomposition

$$1944 = 27 \cdot 72.$$

Each block of 72 states is parametrized by

$$(r, s, m, d) \in (\mathbb{Z}/3\mathbb{Z})^2 \times \{0, 1\} \times D_4,$$

where m is the sheet and d the preceding direction. From each state four cardinal continuations depart. The TRIT of the arrival cell decides the following sheet.

With the action of the chapter 31, the weight of an edge is

$$\exp[-g(e) - \lambda_* s(e)].$$

The character transform reduces the operator to complex matrices L_{k_a, k_b} of size 72×72 . The program enumerates the edges of each block; it does not receive a precomputed spectral matrix. Power iteration is retained as an exploration, but the result that follows no longer rests on it: the spectral radii are isolated by means of a complete rational certification.

Let ρ_{01} and ρ_{12} be the spectral radii of the oriented sectors $(0, 1)$ and $(1, 2)$. We define the *band phase*

$$y_{\text{band}} = \log \rho_{12} - \log \rho_{01}.$$

For

$$\lambda_* = 1 - \frac{5}{468} - \frac{3}{11} \Delta_4,$$

the certification gives

$$1.460655120862011111 < \rho_{01} < 1.460655120862404024,$$

$$1.515812008270944328 < \rho_{12} < 1.515812008271195416,$$

and, by interval propagation,

$$0.037066226434427529 < y_{\text{band}} < 0.037066226434862172,$$

$$2.123738337168943^\circ < C_{\text{band}} := \frac{180}{\pi} y_{\text{band}} < 2.123738337193847^\circ.$$

The primary quantity is y_{band} , a dimensionless logarithm of spectral quotient. Degrees appear later when the chart $180/\pi$ is applied. The operator first produces an internal phase and only later an Archimedean angular representation.

33.1.1 Exact calculation of the spectral radius of the angular operator

The blocks are nonnormal. Therefore, a small residual $\|Lv - \mu v\|$ does not suffice to prove that μ is the eigenvalue of greatest modulus. The proof used here consists of four steps.

First, twelve structurally zero rows occur in each block. Expanding the characteristic polynomial along these rows yields a factor z^{12} and a principal minor of order sixty. A further eight structurally zero rows occur in this minor. Therefore,

$$\det(zI_{72} - L) = z^{20} \det(zI_{52} - A),$$

and every nonzero eigenvalue is contained, with multiplicity, in a reduced block A of order fifty-two.

Second, the certificate provides rational Gaussian matrices S, R , with denominator 10^{16} . The verifier calculates $E = I - RS$ exactly and obtains

$$\|E\|_{\infty} \leq 8.5664 \times 10^{-14} \quad \text{and} \quad \|E\|_{\infty} \leq 3.7628 \times 10^{-14}$$

for the sectors $(0, 1)$ and $(1, 2)$, respectively. The Neumann series then proves that S is invertible.

Third, if A_0 is the rational midpoint of the interval block, the identity

$$S^{-1}AS = (I - E)^{-1}RAS$$

bounds the distance to $C_0 = RA_0S$ by

$$1.25125 \times 10^{-13} \quad \text{and} \quad 5.7037 \times 10^{-14}.$$

The Gershgorin discs of C_0 , enlarged by those bounds, contain the discs of the exact similarity. In each block, one disc remains separated from the other fifty-one; the second Gershgorin theorem assigns it exactly one eigenvalue. Its lower modulus exceeds the upper modulus of every other disc. This proves that the isolated eigenvalue is the spectral radius.

Fourth, $\pi, e, \log \pi, e^{-1}, e^{-\lambda^*}$ and $\sqrt{3}$ are not accepted as library decimals. The verifier encloses them by means of Machin's formula, the factorial and logarithmic series with remainder, the alternating series for the exponential, and a quadratic rational bracket for $\sqrt{3}$. Verification employs only integers and fractions; the argument is closed by the rational reduction, the Neumann series, and the Gershgorin separation presented in this section. The reproducible decimal expansions to a depth of ten thousand are published, as independent finite witnesses, in the [appendix of expansions](#).

33.2 Effective band operator T_{band} and isolation of the angular phase

Let

$$x = \frac{50\pi}{9}\alpha = 0.127362829012869966 \dots$$

and let

$$R = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The effective bilayer operator associated exclusively with the band phase is

$$T_{\text{band}} = \exp(-xI + y_{\text{band}}R).$$

At the base of oriented sectors,

$$T_{\text{band}} = \begin{pmatrix} e^{-x+y_{\text{band}}} & 0 \\ 0 & e^{-x-y_{\text{band}}} \end{pmatrix}.$$

We verify

$$\det T_{\text{band}} = e^{-2x}, \quad \frac{(T_{\text{band}})_{11}}{(T_{\text{band}})_{22}} = e^{2y_{\text{band}}}.$$

Therefore,

$$-\frac{9}{100\pi} \log \det T_{\text{band}} = \alpha, \quad \frac{90}{\pi} \log \frac{(T_{\text{band}})_{11}}{(T_{\text{band}})_{22}} = C_{\text{band}}.$$

The two recoveries have distinct statuses. The second returns the output produced by the gap. The first rereads the α that entered x ; it proves reversibility of the chart, not a second independent derivation of α .

The exponential form separates two functions. The common factor e^{-x} describes the shared contraction of the two sheets. The factors $e^{\pm y_{\text{band}}}$ distinguish orientations while preserving the determinant. Since I and R commute, the pair

$$\left(\det T_{\text{band}}, \frac{(T_{\text{band}})_{11}}{(T_{\text{band}})_{22}} \right)$$

uniquely reconstructs (x, y_{band}) .

33.3 Posterior comparison and precision scale

The following section constructs another phase, the *regularized regional phase* y_{reg} —also denoted there by y_* —. To formulate a comparison here with that subsequent result, we anticipate the notation

$$C_{\text{reg}} := \frac{180}{\pi} y_{\text{reg}} = 2.123738338968646^\circ.$$

This line does not constitute a second definition or an entry of the spectral operator: it cites the conclusion of [theorem 34.14](#), whose construction is developed in the following section. The two pairs of objects are not identified:

$$(y_{\text{band}}, C_{\text{band}}) \neq (y_{\text{reg}}, C_{\text{reg}}).$$

The certified interval difference of the linear law is

$$-1.799703 \times 10^{-9}^\circ < C_{\text{band}} - C_{\text{reg}} < -1.774799 \times 10^{-9}^\circ,$$

with relative difference enclosed by

$$-8.475 \times 10^{-10} < \frac{C_{\text{band}} - C_{\text{reg}}}{C_{\text{reg}}} < -8.356 \times 10^{-10}.$$

The law that produces C_{band} does not receive C_{reg} , CKM, Barbero, masses, or SI data. This target independence makes the proximity a candidate for strong, falsifiable recognition. The spectral certification eliminates the former numerical gap; the interval difference, which excludes zero, now fixes the exact boundary: C_{band} is a predimensional spectral output distinct from the regional realization C_{reg} .

The additional term

$$-\frac{\Delta_4^2}{27}$$

en λ produce

$$C_{\text{num}} \approx 2.12373833894104980^\circ,$$

at 2.76×10^{-11} degrees of C_{reg} . The integer 27 is internal, since it counts the characters of $(\mathbb{Z}/3\mathbb{Z})^3$, but that fact does not force the coefficient of the second moment. The exact additivity of **Add** excludes quadratic terms. The variant is preserved as an ablation and not as the principal law.

33.4 Functional dependence of the spectral radius with respect to the components of the operator

All subsequent rows use the same operator; only one input of the law changes. They are numerical ablation diagnostics; the preceding certified interval pertains exclusively to the complete law $\text{Tr}_q\text{--Add--Def--Res--Loc}$ (and, within it, the positive defect is fixed in the $\text{Tr}_q\text{--Add--Defstratum}$).

rule	λ	C_{num} in degrees
self-dual	1	2.082755794784418
microfiber only	$1 - 5/468$	2.123621809570142
trace $5/468 + (3/11)\Delta_4$	0.989285913019141	2.123738337181414
$4/468$ instead of $5/468$	0.991422665155894	2.115535228138553
$3/12$ instead of $3/11$	0.989288440210566	2.123728626433667
$5/243$ instead of $5/468$	0.979393542015659	2.161907060464779
$5/729$ instead of $5/468$	0.993110963140488	2.109064222037112
$7/729$, seed measure	0.990367478915522	2.119584299852700.

The ablations demonstrate that the output distinguishes the complete microfiber, the reduced dimension eleven, and the emission measure. By themselves, they do not assign a chance probability: such a probability would first require defining a space of alternative models and a measure on it.

33.5 Typed composition of the morphisms Archimedean, exceptional, and dimensional

The complete construction can be written as a sequence of applications with visible domains:

$$\begin{aligned}
 \Omega_{\text{seed}} &\xrightarrow{F} \mathcal{U}_6 \xrightarrow{J_F} H_U, \\
 \mathcal{R}_\pi^{++} &\xrightarrow{\beta_\pi} P_G H_G, \\
 L_{\text{or}} \otimes P_G H_G &\xrightarrow{W_{GK}} P_3 V_{11} \xrightarrow{U_W} Q_3 E_{30}, \\
 (\Pi_\pi^{(468)}, \Delta_4 P_3) &\mapsto D_{\text{switch}} \xrightarrow{\tau} \delta_* \xrightarrow{\text{Res}} \lambda_*, \\
 \lambda_* &\xrightarrow{\text{Loc}} a_* \xrightarrow{\mathcal{L}_{1,*}} y_{\text{band}} \xrightarrow{180/\pi} C_{\text{band}}, \\
 (\alpha, y_{\text{band}}) &\xrightarrow{\exp(-xI + y_{\text{band}}R)} T_{\text{band}}.
 \end{aligned}$$

Each arrow declares what it preserves:

- F preserves the emission class and leaves its presentation multiplicity in the fibre;
- J_F compensates that multiplicity in the norm;
- β_π preserves the exceptional position;
- W_{GK} preserves the oriented representation;
- U_W preserves incidence and metric;
- the trace forgets coordinates and preserves ranks;
- the transfer operator sums paths with the action weight;
- the chart $180/\pi$ represents an internal phase in degrees.

33.6 Dependency diagram of the angular operator and its publications

Object	Mathematical Nature	Contenido demostrado
$U_W = I/6 : V_{11} \rightarrow E_{30}$	exact theorem	isometry, surjectivity, equivariant incidence, and positive uniqueness
$P_3 \leftrightarrow Q_3$	exact theorem	conjugate projectors, rank three, and trace 3/11
star R_{B_K}	typed exact theorem	spectrum $1^{[7]}, (-1)^{[4]}$, index 3/11, and conjugation obstruction
P_G in $t = 7$	finite exact theorem	filtration $11 \rightarrow 3 \rightarrow 1$, positive projector, and action S_3
$\mathcal{R}_\pi^{++} \leftrightarrow P_G$	exact theorem	S_3 -equivariant bijection by exceptional position
oriented isometric identification	new relative theorem	unique selector through B_K, u_K , Reynolds, polar, and reduction $240 \rightarrow 1$
measure 1/468	exact theorem of Tr_q	unique normalized trace and polar application induced by fibers
$D_{\text{switch}} \text{ y } \delta_*$	theorem exact relativo to Tr_q -Add-Def	positivity, spectrum, and trace $5/468 + (3/11)\Delta_4$
λ_*	theorem internal relativo to Tr_q -Add-Def-Res	residual capacity obtained from the defect through the rule Res
$\mathcal{A}_*$	internal theorem relative to Tr_q -Add-Def-Res-Loc	cocycle additive of paths with cost local $g + \lambda_*s$
$T_{\text{band}} \text{ y } C_{\text{band}}$	intervalar computational theorem	reduction $72 \rightarrow 60 \rightarrow 52$, rational similarity, Gershgorin isolation and output whose certified distance lies between 1.774799×10^{-9} and 1.799703×10^{-9} degrees of C_{reg}
$C_{\text{band}} = C_{\text{reg}}$	refuted statement for the linear law	the certified interval of the difference excludes zero
total transition $\mathcal{X}_{\text{TPK}}^{\text{enf}} \rightarrow \Gamma_{\text{tr}}$	required global realizer	Tr_q -Add-Def-Res-Loc defines the relative reader; the complete transition additionally requires a dynamical pushforward measure on Γ_{tr}

33.7 Predimensional action at the center of the double projection of the genealogical state

Dual projection starts from the same integral holonomic state and produces two readings. In the Archimedean branch, compatible cylinders are refined and their diameters decrease; evaluation fixes a limiting value and forgets part of the genealogy. In the exceptional branch, incidences extend towards Witt, Golay, Niemeier–Leech, $V^{\natural}$, the Monster and moonshine.

The new result adds a common object stronger than a coincidence of cardinalities. The eleven-mode module, its rank-three positive sector, and the trace of the action pass from dodecaphase to Witt through U_W . The oriented sector $t = 7$ is incorporated through W_{GK} . The microfiber of π provides the regional projector $\Pi_\pi^{(468)}$, and the uniform measure on the catalogue fixes its weight.

Positive action thus stands at the center of the two readings:

$$\text{region} \oplus \text{oriented polarization} \mapsto \text{positive defect.}$$

The Archimedean evaluation and the exceptional evaluation remain different maps; the intertwiner identifies a common sector, it does not collapse their codomains.

33.8 Compatibility of the angular operator with the cylinders of the discrete continuum

An HMT history determines nested cylinders

$$C_1 \supset C_2 \supset \cdots$$

with compatible prefixes. When the Archimedean reader satisfies

$$\text{diam}(C_n) \longrightarrow 0,$$

the completeness of $\mathbb{R}$ fixes at most one value in the intersection. The new action adds structure before that limit: it distinguishes the cost of the route, the emission class, and the multiplicity of presentations that the final value no longer displays.

This is the precise mathematical meaning of the assertion that the continuum is read as a projection of discrete genealogies. The system of cylinders and its evaluations is prolonged at every scale; the global coverage theorem proves $\text{ev}_{\mathbb{R}}(\Omega_{\infty, K}) = K$ for every compact set $K \subset \mathbb{R}$ and, upon directing the intervals, proves that the image is all of $\mathbb{R}$. The finite theorem in this section controls the angular action at each level and is integrated into that global coverage; it neither reopens nor conditions it. The algebraic descent of each operation retains its own verification.

33.9 Compatibility with the exceptional rider, Moonshine, and M-theory screens

The intertwiner U_W lands in the incidence module of W_{12} . From the ternary code, the two exceptional continuations remain separate:

$$C_W \longrightarrow W_{12} \xrightarrow{\text{relative to } (D, \ell)} W_{24},$$

and

$$C_W \xrightarrow{\phi} N(A_2^{12}) \xrightarrow{v_\alpha} \Lambda_{24} \longrightarrow V_\Lambda \longrightarrow V^{\natural} \longrightarrow \mathbb{M} \longrightarrow \text{moonshine}.$$

Positive action contributes to this branch a transported sector and an orientation line. The arrows from Leech to moonshine are the classical constructions applied to the finite object obtained. The Monster action is realized on $V^{\natural}$; the HMT words are not identified here with a direct module for that action.

The screens $12 \rightarrow 11 \rightarrow 10$, $26 \rightarrow 24$, and the T-dual involution form an exact mathematical interface with the dimensions and dualities used in M-theory. The full physical identification belongs to Dimensional Mechanics and is typed in the second part by the additional map to g_{MN} , C_{MNP} , ψ_M , action, and supersymmetry. It is neither deferred to another work nor deduced from the Monster action.

33.10 Spectral factorization and isolation of the dominant sectors

The pre-dimensional line is organized into four strata.

1. **Finite kernel.** APP, TRIT, the TPK emitter, the catalog $104\,976 \rightarrow 468 \rightarrow 243$, the five regions of π , the dodecaphase, K , α , Witt, and the exceptional branch have explicit constructions and certificates.
2. **Operatorial bridge.** Incidence constructs U_W ; the oriented sector $t = 7$ constructs P_G ; B_K and u_K select the oriented isometric identification; the three rank-three projectors are identified relative to that rule.
3. **Action extension.** Tr_q fixes the measure $1/468$, **Add** fixes the additive character, and **Def** declares the coupling $\Delta_4 P_3$. **Res** transforms the defect into capacity, and **Loc** carries capacity to the edges. The five clauses produce the local law.
4. **Spectral consequence.** Γ_{tr} produces T_{band} and a certified predimensional angular phase, with relative difference between 8.356×10^{-10} and 8.475×10^{-10} with respect to C_{reg} .

This permits a strong and typed closure of this route: the finite emitter of the infinite coinductive section, the incidence, the identification of projectors, the measure, and the action are closed relative to the declared primitives, the variational rule $\mathcal{E}_B$, and the action clauses Tr_q –**Add**–**Def**–**Res**–**Loc**. **Add** selects the linear form; **Def** selects its second coordinate; **Res** and **Loc** specify how it is read as a cost. The global theorem covering all of $\mathbb{R}$, induction from a total transition of $\mathcal{X}_{\text{TPK}}^{\text{enr}}$, and physical equivalence with M-theory are formulated as extensions of the preceding results.

33.11 Compatibility between the matrix, angular, and lattice realizations

The constructed representations simultaneously satisfy the following compatibility conditions:

1. five regions of π , with three positivas, one negativa and one transversal;
2. P_G, P_3, Q_3 as positive projectors and R_{B_K} as an orientation involution;
3. the oriented homogeneous sector $t = 7$, not the heterogeneous quotient of $t = 8$;
4. the line L_{or} in the triple isometric identification;
5. the relative character of the variational rule that selects H_*, c_*, r_* ;
6. $5/468$ on emissions and $7/729$ on seeds, without interchanging them;
7. the linearity of the character as a consequence of **Add** and the coefficient Δ_4 as content separated from **Def**;
8. $(y_{\text{band}}, C_{\text{band}})$ as a certified spectral output, separate from $(y_{\text{reg}}, C_{\text{reg}})$, the canonical regional realization of the following section;
9. α as input to x , so that the determinant reading is reversible and not independent;
10. the branches $W_{12} \rightarrow W_{24}$ and $A_2^{12} \rightarrow \Lambda_{24}$ as separate constructions;
11. the relative reader Tr_q –**Add**–**Def**–**Res**–**Loc** distinguished from a total transition, whose dynamic image measure constitutes an additional construction.

These conditions are not editorial notes. They are part of the logical structure of the result.

Equivalence of the global and analytic charts of the angular pair

The positive measure after return is obtained before the counterangle. The determinantal branch determines the action valuation and both constructions converge in the section of the dimensional line. Only then are the angle, its oriented complement and the characters q_+ and q_- formed, evaluating the incidences already generated by the kernel.

34.1 Intrinsic angular phase α , nonadic lift 10^3 , and precursor x_A

34.1.1 Construction of the analytic precursor x_A , the contraction D_A , and the regional correction $\mathcal{C}_\pi(\alpha)$

The finite catalog and an analytic evaluation answer questions of different types. The catalog determines which regions exist, which share a return and what information each preserves. An analytic evaluation must also decide how the length of a word is transformed into degree, how a branch is prolonged after its return and with which scalar character that prolongation is read. This chapter explicitly constructs that second operation.

The construction begins with four data already obtained in the preceding chapters: the five-region microfiber of π , the length six of each regional word, the dodecaphasic closure of α , and the bilayer transport associated with the common coordinate $A = 10^3\alpha$. No value of the conjugate angular coordinate C_*^{HMT} , of the reduced action component, of the Barbero–Immirzi functional, or of a metrological constant is used in the evaluation. The result is a convergent regional character, an exact recurrence, and an oriented phase whose sexagesimal chart reproduces the published conjugate angular coordinate.

The separation between the finite part and the analytic part is essential. The integer 169 will be a theorem of the catalogue. The series that prolongs it will be the theorem of an analytic reader whose rules are declared before it is evaluated. Thus, the additional premises are not concealed within a decimal coincidence.

34.2 Preangular construction of η_{ret} through H_5 , D_A and $\mathcal{C}_\pi(\alpha)$

34.2.1 Independent derivation of $\eta_{\text{ret}} = H_5 - (90/\pi)\mathcal{C}_\pi(\alpha)$ through the regional analytic reader

The finite catalogue and an analytic evaluation answer questions of different types. The catalogue determines which regions exist, which share a return, and what information each preserves. An analytic evaluation must further determine how the length of a word is transformed into degree, how a branch is prolonged after its return, and by which scalar character that prolongation is read. This section constructs that second operation explicitly.

The construction begins with the five-region microfiber of π , the length six of each regional word, the dodecaphasic closure of α , and the declared angular insertion

$$\iota_\angle(x) := 10^3 x u_o, \quad A_{\text{deg}} := 10^3 \alpha.$$

Here u_o is the sexagesimal base whose complete turn has coordinate 360. The factor 10^3 belongs to that chart normalization and is not deduced from the arithmetic closure of α . The two-sheet transport is subsequently constructed from A_{deg} . The function evaluated by the reader does not receive as an argument the counter-angle, the post-return coordinate, the hexad–octad functional, or a metrological constant. The result is a convergent regional character, an exact recurrence, and an oriented phase in the declared sexagesimal chart.

The separation between the finite part and the analytic part is essential. The integer 169 is a theorem of the catalogue. The series that prolongs it belongs to an analytic reader whose rules are declared before evaluating it.

The phase is denoted by y_* and, when its regional origin is emphasized, by y_{reg} ; both symbols denote the same reader constructed below.

The evaluation receives the catalogue of 468 emissions, the exact value of α , $\varphi = (1 + \sqrt{5})/2$, the fixed formula of H_5 , the angular chart, and the five rules of the regional reader. With those data, it computes 169, D_A , the recurrence, and C^* . The theorem is exact in this declared class of readers; it does not use C^* as an input or assert that the rules of H_5 are the only possible natural class.

34.2.2 Persistence of the return in the 5 regions of the microfiber

Let

$$\mathcal{F}_\pi = \Psi^{-1}(010211) \subset \mathcal{U}_6$$

the microfiber relative to the catalog. Recall its five elements, now separated into initial block and terminal block:

$$\begin{aligned} &(501, 614, 498 \mid 169, 272, 272), \\ &(810, 923, 870 \mid 169, 272, 272), \\ &(810, 983, 810 \mid 169, 272, 272), \\ &(870, 923, 810 \mid 169, 272, 272), \\ &(870, 923, 810 \mid 418, 674, 521). \end{aligned}$$

We define the return projection.

$$p_{\text{ret}} : \mathbb{Z}^6 \longrightarrow \mathbb{Z}^3, \quad p_{\text{ret}}(u_1, \dots, u_6) = (u_4, u_5, u_6).$$

Proposition 34.1 (Persistent terminal block). *The multiset $p_{\text{ret}}(\mathcal{F}_\pi)$ has a unique element of maximum multiplicity:*

$$b_\pi = (169, 272, 272), \quad \text{mult}_{\mathcal{F}_\pi}(b_\pi) = 4.$$

The only remaining terminal block is

$$b_\pi^- = (418, 674, 521)$$

and has multiplicity one.

Proof. The assertion follows by exact enumeration of the five rows of the catalog that satisfy $\Psi(U) = 010211$. Four rows have terminal block $(169, 272, 272)$; the fifth, which carries the mutated return, has $(418, 674, 521)$. The maximum multiplicity is four and is attained only once. $\square$

The oriented difference between the mutated return and the persistent is

$$\omega_\pi = b_\pi^- - b_\pi = (249, 402, 249), \quad \langle (1, 1, 1), \omega_\pi \rangle = 900.$$

This identity separates two objects that must not be conflated. The integer 169 is a coordinate of a terminal datum, that is, a datum of degree zero. The vector ω_π can be read as a one-cochain after orienting the edge joining the two blocks. Calling it a cocycle would additionally require fixing a cochain complex and verifying that its coboundary vanishes. The analytic prolongation constructed below does not presuppose that identification.

The selection of 169 does not require privileging the first position of the block. The symmetric group $\mathfrak{S}_3$ acts by permuting its three coordinates. The stabilizer of b_π exchanges the two occurrences of 272 and fixes the unique coordinate of multiplicity one. Denote that coordinate by $\text{sing}(b)$ when the block has type (r, s, s) , with $r \neq s$. Equivalently,

$$\text{sing}(b) = \sum_{j=1}^3 b_j - 2 \text{moda}(b).$$

In particular,

$$\boxed{\text{sing}(b_\pi) = 169.}$$

Definition 34.2 (Residual fiber selector). The selector ρ_π successively applies two operations:

1. selects the terminal block of maximum multiplicity within $p_{\text{ret}}(\mathcal{F}_\pi)$;
2. retains the coordinate fixed by the stabilizer of that block.

By [proposition 34.1](#),

$$\boxed{\rho_\pi = 169.}$$

Theorem 34.3 (Equivariant uniqueness of return). *Among the selectors that*

1. *are invariant under permutations of the five regions;*
2. *are equivariant under permutations of the three terminal coordinates;*
3. *select the block of maximal multiplicity;*
4. *retain a coordinate fixed by its stabilizer,*

the return value is uniquely determined and is 169.

Proof. Invariance under $\mathfrak{S}_5$ prevents using the order of the rows. The third condition selects b_π , which is the unique mode by [proposition 34.1](#). Its stabilizer in $\mathfrak{S}_3$ has two orbits on the coordinates: the pair of values 272 and the singleton value 169. The fourth condition selects the second orbit. Equivariance preserves its value when its position is shifted. $\square$

This argument improves the positional reading of the return. The 169 is not «the first coordinate» of a selected row: it is an invariant of the regional multiset and of the action of its reordering symmetries. Nor is it selected by global rarity. Among the 468 emissions in the catalogue, 169 appears 84 times, uniformly distributed with 14 occurrences in each of the six positions. The uniqueness of [theorem 34.3](#) is therefore relational and regional: it resides in the combination of microfiber, modal persistence, and stabilizer, not in the isolated frequency of the integer.

34.2.3 Construction of the regional analytic reader and its convergence domain

Analytic continuation is fixed by the following rules.

Definition 34.4 (Regional Determinantal Reader). The determinantal regional reader satisfies:

1. *degree compatibility*: cylindrical length is the analytic degree;
2. *renewal decomposition*: the finite return impulse and strict continuation belong to distinct additive summands;
3. *continuation normalization*: after recording the residue ρ_π just once, the surviving branch begins at the unit of the determinant line;
4. *concatenation covariance*: each elementary prolongation acts through $\bigwedge^2 \mathcal{T} = D_A$;
5. *orientation*: in the cardinal convention fixed for APP, the regional return belongs to the compensating sector and is subtracted from the fifth-degree action phase.

The third rule has mathematical content. The value 169 is recorded once as the amplitude of the return; it is not multiplied again at each prolongation. The strict continuation counts the sole line that continues

and normalizes it by its unit. This is the distinction between a renewal recurrence and homogeneous iteration of the initial impulse.

The need to declare that normalization can be proved. For every $\lambda \in \mathbb{R}$, the family

$$F_\lambda(z) = 169z^6 + \lambda \frac{D_A z^7}{1 - D_A z}$$

preserves the same finite return and the same determinantal ratio between consecutive coefficients. The catalogue and the ratio alone do not distinguish λ . The unit of the determinant line fixes $\lambda = 1$ before $z = \alpha$ is evaluated. Thus, the normalization is not obtained by fitting the final value; it forms part of the definition of the character.

34.2.4 Analytical solution of the return recurrence

We define the coefficients κ_n by

$$\kappa_6 = 169, \quad \kappa_7 = D_A, \quad \kappa_{n+1} = D_A \kappa_n \quad (n \geq 7).$$

The first equality is the regional impulse; the next two describe the normalized continuation.

Theorem 34.5 (Graduated analytic realization). *The analytic germ compatible with the rules of definition 34.4 is unique and equals*

$$\mathcal{C}_\pi(z) = 169z^6 + \frac{D_A z^7}{1 - D_A z}.$$

Their coefficients satisfy

$$\kappa_n = D_A^{n-6} \quad (n \geq 7).$$

Proof. Write

$$F(z) = 169z^6 + \sum_{k \geq 1} c_k z^{6+k}.$$

The normalization of continuation and the determinantal covariance give

$$c_1 = D_A, \quad c_{k+1} = D_A c_k.$$

By induction, $c_k = D_A^k$. The geometric sum yields

$$\sum_{k \geq 1} D_A^k z^{6+k} = \frac{D_A z^7}{1 - D_A z}.$$

No free coefficients remain. □

Since $0 < \alpha < 1$ and $0 < D_A < 1$,

$$0 < D_A \alpha < 1.$$

Consequently, $\mathcal{C}_\pi(\alpha)$ converges absolutely and its radius of convergence is $D_A^{-1} > 1$. Numerically,

$$D_A = 0.7751291192471564301888840964802 \dots,$$

$$D_A \alpha = 0.00565639046986492673885079095469 \dots$$

The speed of this contraction explains why the complete sum and its degree-seven truncation have the same first fifteen digits in the final chart.

34.2.5 Persistence of return in the 5 regions and compatibility of the η_{ret} component

Let

$$\mathcal{F}_\pi = \Psi^{-1}(010211) \subset \mathcal{U}_6$$

the microfiber relative to the catalog. Recall its five elements, now separated into initial block and terminal block:

$$\begin{aligned} &(501, 614, 498 \mid 169, 272, 272), \\ &(810, 923, 870 \mid 169, 272, 272), \\ &(810, 983, 810 \mid 169, 272, 272), \\ &(870, 923, 810 \mid 169, 272, 272), \\ &(870, 923, 810 \mid 418, 674, 521). \end{aligned}$$

We define the return projection.

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$$b_\pi = (169, 272, 272), \quad \text{mult}_{\mathcal{F}_\pi}(b_\pi) = 4.$$

The only remaining terminal block is

$$b_\pi^- = (418, 674, 521)$$

and has multiplicity one.

Proof. The assertion follows by exact enumeration of the five rows of the catalog that satisfy $\Psi(U) = 010211$. Four rows have terminal block $(169, 272, 272)$; the fifth, which carries the mutated return, has $(418, 674, 521)$. The maximum multiplicity is four and is attained only once. $\square$

The oriented difference between the mutated return and the persistent is

$$\omega_\pi = b_\pi^- - b_\pi = (249, 402, 249), \quad \langle (1, 1, 1), \omega_\pi \rangle = 900.$$

This identity separates two objects that must not be conflated. The integer 169 is a coordinate of a terminal datum, that is, a datum of degree zero. The vector ω_π can be read as a one-cochain after orienting the edge joining the two blocks. Calling it a cocycle would additionally require fixing a cochain complex and verifying that its coboundary vanishes. The analytic prolongation constructed below does not presuppose that identification.

The selection of 169 does not require privileging the first position of the block. The symmetric group $\mathfrak{S}_3$ acts by permuting its three coordinates. The stabilizer of b_π exchanges the two occurrences of 272 and fixes the unique coordinate of multiplicity one. Denote that coordinate by $\text{sing}(b)$ when the block has type (r, s, s) , with $r \neq s$. Equivalently,

$$\text{sing}(b) = \sum_{j=1}^3 b_j - 2 \text{moda}(b).$$

In particular,

$$\boxed{\text{sing}(b_\pi) = 169.}$$

Definition 34.7 (Residual fiber selector). The selector ρ_π successively applies two operations:

1. selects the terminal block of maximum multiplicity within $p_{\text{ret}}(\mathcal{F}_\pi)$;

2. retains the coordinate fixed by the stabilizer of that block.

By [proposition 34.6](#),

$$\rho_\pi = 169.$$

Theorem 34.8 (Equivariant uniqueness of return). *Among the selectors that*

1. *are invariant under permutations of the five regions;*
2. *are equivariant under permutations of the three terminal coordinates;*
3. *select the block of maximal multiplicity;*
4. *retain a coordinate fixed by its stabilizer,*

the return value is uniquely determined and is 169.

Proof. Invariance under $\mathfrak{S}_5$ prevents using the order of the rows. The third condition selects b_π , which is the unique mode by [proposition 34.6](#). Its stabilizer in $\mathfrak{S}_3$ has two orbits on the coordinates: the pair of values 272 and the singleton value 169. The fourth condition selects the second orbit. Equivariance preserves its value when its position is shifted. $\square$

This argument improves the positional reading of the return. The 169 is not «the first coordinate» of a selected row: it is an invariant of the regional multiset and of the action of its reordering symmetries. Nor is it selected by global rarity. Among the 468 emissions in the catalogue, 169 appears 84 times, uniformly distributed with 14 occurrences in each of the six positions. The uniqueness of [theorem 34.8](#) is therefore relational and regional: it resides in the combination of microfiber, modal persistence, and stabilizer, not in the isolated frequency of the integer.

34.2.6 Application of the regional reader to the conjugate angular momentum pair

Analytic continuation is fixed by the following rules.

Definition 34.9 (Regional Determinantal Reader). The determinantal regional reader satisfies:

1. *degree compatibility:* cylindrical length is the analytic degree;
2. *renewal decomposition:* the finite return impulse and strict continuation belong to distinct additive summands;
3. *continuation normalization:* after recording the residue ρ_π just once, the surviving branch begins at the unit of the determinant line;
4. *concatenation covariance:* each elementary prolongation acts through $\bigwedge^2 \mathcal{T} = D_A$;
5. *orientation:* in the cardinal convention fixed for APP, the regional return belongs to the compensating sector and is subtracted from the fifth-degree action phase.

The third rule has mathematical content. The value 169 is recorded once as the amplitude of the return; it is not multiplied again at each prolongation. The strict continuation counts the sole line that continues and normalizes it by its unit. This is the distinction between a renewal recurrence and homogeneous iteration of the initial impulse.

The need to declare that normalization can be proved. For every $\lambda \in \mathbb{R}$, the family

$$F_\lambda(z) = 169z^6 + \lambda \frac{D_A z^7}{1 - D_A z}$$

preserves the same finite return and the same determinantal ratio between consecutive coefficients. The catalogue and the ratio alone do not distinguish λ . The unit of the determinant line fixes $\lambda = 1$ before

$z = \alpha$ is evaluated. Thus, the normalization is not obtained by fitting the final value; it forms part of the definition of the character.

34.2.7 Closed sum and publication of the posterior component upon return

We define the coefficients κ_n by

$$\kappa_6 = 169, \quad \kappa_7 = D_A, \quad \kappa_{n+1} = D_A \kappa_n \quad (n \geq 7).$$

The first equality is the regional impulse; the next two describe the normalized continuation.

Theorem 34.10 (Graduated analytic realization). *The analytic germ compatible with the rules of definition 34.9 is unique and equals*

$$\mathcal{C}_\pi(z) = 169z^6 + \frac{D_A z^7}{1 - D_A z}.$$

Their coefficients satisfy

$$\kappa_n = D_A^{n-6} \quad (n \geq 7).$$

Proof. Write

$$F(z) = 169z^6 + \sum_{k \geq 1} c_k z^{6+k}.$$

The normalization of continuation and the determinantal covariance give

$$c_1 = D_A, \quad c_{k+1} = D_A c_k.$$

By induction, $c_k = D_A^k$. The geometric sum yields

$$\sum_{k \geq 1} D_A^k z^{6+k} = \frac{D_A z^7}{1 - D_A z}.$$

No free coefficients remain. □

Since $0 < \alpha < 1$ and $0 < D_A < 1$,

$$0 < D_A \alpha < 1.$$

Consequently, $\mathcal{C}_\pi(\alpha)$ converges absolutely and its radius of convergence is $D_A^{-1} > 1$. Numerically,

$$D_A = 0.7751291192471564301888840964802 \dots,$$

$$D_A \alpha = 0.00565639046986492673885079095469 \dots$$

The speed of this contraction explains why the complete sum and its degree-seven truncation have the same first fifteen digits in the final chart.

34.3 Confluence of η_{ret} With the valuation $\varepsilon_H = -34$ in the magnitude $\hbar_{\text{ret}}$

34.3.1 Derivation of the determinantal character of parangular transport

The dodecaphasic closure provides α . The completion of the three nonadic pairs provides the scale 10^3 , so that

$$A = 10^3 \alpha.$$

Having chosen the Archimedean angular chart, the common coordinate in radians is

$$x = \frac{\pi A}{180} = \frac{50\pi}{9} \alpha.$$

Let R be a real involution of zero trace in dimension two,

$$R^2 = I_2, \quad \text{tr } R = 0,$$

and consider the formal two-layer transport

$$\mathcal{T}(x, y) = \exp(-xI_2 - yR).$$

The oriented variable y has not yet been evaluated. Nevertheless, the action of $\mathcal{T}$ on the determinant line is independent of it:

$$\begin{aligned} \det \mathcal{T}(x, y) &= \exp(\text{tr}(-xI_2 - yR)) \\ &= e^{-2x}. \end{aligned}$$

We define

$$D_A = e^{-2x} = e^{-100\pi\alpha/9}.$$

The subscript recalls that this is the determinant of the common contraction associated with A , not a function of C_*^{HMT} .

Proposition 34.11 (Character of the determinant line). *The map*

$$\delta_A : (\mathbb{N}_0, +) \longrightarrow (\mathbb{R}_{>0}, \cdot), \quad \delta_A(k) = D_A^k,$$

is the unique multiplicative character whose value on an elementary prolongation is D_A .

Proof. The multiplicativity gives

$$\delta_A(k) = \delta_A(\underbrace{1 + \cdots + 1}_{k \text{ times}}) = \delta_A(1)^k = D_A^k.$$

The same identity proves existence and uniqueness. $\square$

The determinant remains invariant under conjugation of $\mathcal{T}$ and under the exchange of sheets $R \mapsto -R$. Therefore, the tail constructed from D_A belongs to the common mode. The orientation will appear in the sign with which the regional character corrects the phase.

34.3.2 action character of degree 5 in the construction regional

The central block

$$\mathcal{B}_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}$$

has eigenvalues 9 and 5. The dodecaphasic cycle contributes rank 12; its step-three orbits have length four; and the complete census contributes the exact quotient

$$\frac{104976}{1944} = 54.$$

From these invariants the fifth-degree action character is fixed.

$$H_5(\alpha, \varphi) = \frac{1}{2} \sqrt{\frac{1000\alpha}{\varphi}} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5.$$

The four rational coefficients can be written as

$$\frac{9}{16} = \frac{\lambda_+}{4^2}, \quad \frac{5}{9} = \frac{\lambda_-}{\lambda_+}, \quad \frac{7}{48} = \frac{\frac{1}{2} \operatorname{tr} \mathcal{B}_c}{4 \cdot 12}, \quad \frac{1}{54} = \frac{1944}{104976},$$

with $(\lambda_+, \lambda_-) = (9, 5)$. These identities certify the provenance of the numbers used. The successive assignment of those invariants to degrees two, three, four, and five forms part of the analytic character; it does not follow from the mere existence of the finite catalog.

This precision removes a common ambiguity. A combination of invariants of the construction is an internal and reproducible definition; its naturality requires the reader rules specifying order, signs, and normalization. Here those rules are visible and are subject to the same negative controls as the rest of the construction.

34.3.3 Exact decomposition of the action line into pre-return and post-return sections

The fifth-degree character and the post-action component after return are two distinct values:

$$H_5 = 1.05457181764615446962582182943306 \dots,$$

$$\bar{h}_{\text{HMT}}^{\text{ret}} = 1.05457181691503906113369058472430 \dots$$

The first is the local action before incorporating the regional character; the second is the oriented action remaining after the return. The correction that generates the conjugate angular coordinate prevents identifying them.

The metrological comparison must respect this distinction. The first value gives the mantissa

$$2\pi H_5 = 6.62607014999998792600111034989947 \dots,$$

while the subsequent action gives

$$2\pi \bar{h}_{\text{HMT}}^{\text{ret}} = 6.62607014540625433351074984759288 \dots$$

In the International System, the value

$$h = 6.62607015 \times 10^{-34} \text{ J s}$$

is exact by definition [Bur19]. Therefore,

$$2\pi H_5 - 6.62607015 = -1.2073998889650101 \dots \times 10^{-14},$$

and it is not an exact equality. The proximity of the value before the return is a remarkable numerical check; it does not authorize replacing the defining constant of the SI with the subsequent component or confusing the two HMT actions.

This result precisely determines the mathematical situation. The regional reader generates C_*^{HMT} and, from it, the functional $\Gamma_{6 \rightarrow 8}$. The same calculation separates the Planck approximation produced by H_5 from the reduced action intervening in the conjugate angular coordinate. The separation does not weaken the construction: it eliminates an identification incompatible with the exact figures and converts every comparison into a falsifiable assertion of a well-defined type.

34.3.4 Factorization of the determinantal character in the 2 phase charts

Dodecaphase closure supplies α . The declared angular insertion uses the scale 10^3 , compatible with the grouping of three nonadic pairs, so that

$$A_{\deg} = 10^3 \alpha^\circ.$$

Having chosen the Archimedean angular chart, the common coordinate in radians is

$$x =: A_{\text{rad}} = \frac{\pi A_{\deg}}{180} = \frac{50\pi}{9} \alpha.$$

Let R be a real involution of zero trace in dimension two,

$$R^2 = I_2, \quad \text{tr } R = 0,$$

and consider the formal two-layer transport

$$\mathcal{T}(x, y) = \exp(-xI_2 - yR).$$

The oriented variable y has not yet been evaluated. Nevertheless, the action of $\mathcal{T}$ on the determinant line is independent of it:

$$\begin{aligned} \det \mathcal{T}(x, y) &= \exp(\text{tr}(-xI_2 - yR)) \\ &= e^{-2x}. \end{aligned}$$

We define

$$D_A = e^{-2x} = e^{-100\pi\alpha/9}.$$

The subscript recalls that this is the determinant of the common contraction associated with A , not a function of the counter-angle.

Proposition 34.12 (Character of the determinant line). *The map*

$$\delta_A : (\mathbb{N}_0, +) \longrightarrow (\mathbb{R}_{>0}, \cdot), \quad \delta_A(k) = D_A^k,$$

is the unique multiplicative character whose value on an elementary prolongation is D_A .

Proof. The multiplicativity gives

$$\delta_A(k) = \delta_A(\underbrace{1 + \cdots + 1}_{k \text{ times}}) = \delta_A(1)^k = D_A^k.$$

The same identity proves existence and uniqueness. $\square$

The determinant remains invariant under conjugation of $\mathcal{T}$ and under the exchange of sheets $R \mapsto -R$. Therefore, the tail constructed from D_A belongs to the common mode. The orientation will appear in the sign with which the regional character corrects the phase.

34.3.5 Publication of the action character of degree 5 in the parangular chart

The central block

$$\mathcal{B}_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}$$

has eigenvalues 9 and 5. The dodecaphasic cycle contributes rank 12; its step-three orbits have length four; and the complete census contributes the exact quotient

$$\frac{104976}{1944} = 54.$$

The current reader declares the nature of fifth-degree action

$$H_5(\alpha, \varphi) = \frac{1}{2} \sqrt{\frac{1000\alpha}{\varphi}} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5.$$

The four rational coefficients can be written as

$$\frac{9}{16} = \frac{\lambda_+}{4^2}, \quad \frac{5}{9} = \frac{\lambda_-}{\lambda_+}, \quad \frac{7}{48} = \frac{\frac{1}{2} \operatorname{tr} \mathcal{B}_c}{4 \cdot 12}, \quad \frac{1}{54} = \frac{1944}{104976},$$

with $(\lambda_+, \lambda_-) = (9, 5)$. These identities fix the internal arithmetic representability of the character coefficients. Their order, signs, and normalization belong to the graded rule of the regional reader and are applied before any chart of units. The ablations certify that every term effectively intervenes in the resulting phase. In particular, neither the SI mantissa of action nor the sexagesimal value of the counter-angle belongs to the domain of the generating function.

34.4 Parangular application $P(\alpha, \eta_{\text{ret}})$ and bilayer closure $C^* = 2(\eta_{\text{ret}} + \alpha)$

34.4.1 Oriented phase and conjugate angular coordinate

The fifth-degree phase prior to return is

$$y_5 = \frac{\pi}{90}(H_5 + \alpha).$$

The complete regional character produces the correction $\mathcal{C}_\pi(\alpha)$. With the orientation fixed in [definition 34.4](#), we define

$$y_* = \frac{\pi}{90}(H_5 + \alpha) - 169\alpha^6 - \frac{D_A\alpha^7}{1 - D_A\alpha}.$$

This is the primary formula. The sexagesimal unit does not come into play until applying the chart.

$$C_*^{\text{HMT}} = \frac{180}{\pi} y_*.$$

Theorem 34.13 (Conjugate angular coordinate of the regional reader determinantal). *With the dodecaphasic closure of α , $\varphi = (1 + \sqrt{5})/2$ and the rules of the determinantal regional reader fixed, the phase y_* and its conjugate angular coordinate are determined without introducing the sought value. Their evaluation is*

$$y_* = 0.03706622646583826398493779409230638 \dots,$$

$$C_*^{\text{HMT}} = 2.12373833896864572426195138039336 \dots^\circ.$$

Therefore, to fifteen decimal places,

$$C_*^{\text{HMT}} = 2.123738338968646^\circ.$$

Proof. [theorem 34.3](#) determines 169. The closure of α determines A, x , and $D_A = e^{-100\pi\alpha/9}$. [theorem 34.5](#) determines the complete series. The formula for H_5 contains only α, φ , and declared structural invariants. Substituting these data into the expression for y_* gives the printed value. The decimal value of C_*^{HMT} occurs in none of the right-hand sides. $\square$

The reduced action component after return is now as follows output:

$$\bar{h}_{\text{HMT}}^{\text{ret}} = \frac{C_{*}^{\text{HMT}}}{2} - \alpha = H_5 - \frac{90}{\pi} \mathcal{C}_{\pi}(\alpha).$$

Numerically,

$$\bar{h}_{\text{HMT}}^{\text{ret}} = 1.05457181691503906113369058472430 \dots$$

The inverse identity has not been used to introduce C_{*}^{HMT} ; it is obtained after generating y_{*} .

34.4.2 Conjugate coordinate C^{*} and reversal of its orientation under the bilayer involution

The fifth-degree phase prior to return is

$$y_5 = \frac{\pi}{90} (H_5 + \alpha).$$

The complete regional character produces the correction $\mathcal{C}_{\pi}(\alpha)$. With the orientation fixed in [definition 34.9](#), we define

$$C_{\text{rad}}^{*} := y_{*} = \frac{\pi}{90} (H_5 + \alpha) - 169\alpha^6 - \frac{D_A \alpha^7}{1 - D_A \alpha}.$$

From now on we also write

$$y_{\text{reg}} := y_{*}$$

when it is necessary to distinguish this realization from the spectral phase of Γ_{tr} . The two notations here designate the same canonical regional object. This is the primary formula. The sexagesimal unit does not intervene until the chart is applied

$$C_{\text{deg}}^{*} := \frac{180}{\pi} C_{\text{rad}}^{*}.$$

The symbols C_{rad}^{*} and C_{deg}^{*} do not denote two counterangles: they are the radial and sexagesimal coordinates of the same angular element.

Theorem 34.14 (Counter-angle of the regional determinantal reader). *Given the dodecaphase closure of α , $\varphi = (1 + \sqrt{5})/2$, the germ H_5 and the rules of the regional determinantal reader, the phase C_{rad}^{*} and its counter-angle are determined without introducing the sought value into the evaluation function. Its evaluation is*

$$C_{\text{rad}}^{*} = 0.03706622646583826398493779409230638 \dots,$$

$$C_{\text{deg}}^{*} = 2.12373833896864572426195138039336 \dots^{\circ}.$$

Therefore, to fifteen decimal places,

$$C_{\text{deg}}^{*} = 2.123738338968646^{\circ}.$$

Proof. The [theorem 34.8](#) determines 169. The closure of α determines A_{deg} , A_{rad} , and $D_A = e^{-100\pi\alpha/9}$. The [theorem 34.10](#) determines the complete series. The fixed formula for H_5 contains only α , φ , and declared structural invariants. Substituting these data into the expression for C_{rad}^{*} yields the printed value. The decimal value of the counter-angle appears in none of the right-hand sides of this evaluation. Comparison with a metrological action mantissa is undertaken only after the phase has been obtained and does not alter the generating function. $\square$

Let $u_{\circ}$ be the sexagesimal base whose complete turn has coordinate 360, and let us introduce the angular coordinate $\alpha_{\angle, \text{deg}} := \alpha$. The output posterior to return is then typed by

$$\eta_{\text{ret}}^{(\text{deg})} = \frac{C_{\text{deg}}^*}{2} - \alpha_{\angle, \text{deg}} = H_5 - \frac{90}{\pi} \mathcal{C}_{\pi}(\alpha).$$

Its radial coordinate is

$$\eta_{\text{ret}}^{(\text{rad})} = \frac{\pi}{180} \eta_{\text{ret}}^{(\text{deg})} = \frac{C_{\text{rad}}^*}{2} - \frac{\pi}{180} \alpha.$$

Numerically,

$$\eta_{\text{ret}}^{(\text{deg})} = 1.05457181691503906113369058472430 \dots$$

The inverse identity has not been used to introduce C_{deg}^* ; it is obtained after generating C_{rad}^* . The symbol $\hbar$ is reserved for the element of the action line constructed in the following section.

34.4.3 Direct theorem $C^* = 2(\eta_{\text{ret}} + \alpha)$ and inverse character of the identity $\eta_{\text{ret}} = C^*/2 - \alpha$

The fifth-degree character and the coordinate after return are two distinct values:

$$H_5 = 1.05457181764615446962582182943306 \dots,$$

$$\eta_{\text{ret}}^{(\text{deg})} = 1.05457181691503906113369058472430 \dots$$

The former is the local character before incorporation of the regional character; the latter is the oriented coordinate remaining after the return. The correction generating the contra-angle prevents their identification.

The metrological comparison must respect this distinction. The first value gives the mantissa

$$2\pi H_5 = 6.62607014999998792600111034989947 \dots,$$

while the posterior coordinate gives

$$2\pi \eta_{\text{ret}}^{(\text{deg})} = 6.62607014540625433351074984759288 \dots$$

As an external control only, we retain the mantissa 6.62607015 of the defining value of the International System [Bur19]. Neither its exponent nor its unit is yet incorporated into the HMT line: the decade is derived in the chapter devoted to the determinantal order of the action line and the unit belongs to a later metrological chart. The comparison of mantissas gives

$$2\pi H_5 - 6.62607015 = -1.2073998889650101 \dots \times 10^{-14},$$

and is not an exact equality. The proximity of the pre-return value is a noteworthy numerical control; it does not authorize replacing the defining SI constant by the subsequent coordinate or confusing an angular coordinate with the action element that will be constructed upon incorporating the decade.

This result determines the mathematical situation precisely. The regional reader generates $(C_{\text{rad}}^*, C_{\text{deg}}^*)$ and, from that single angular element, the functional $\Gamma_{6 \rightarrow 8}$. The same calculation separates the external proximity of the mantissa produced by H_5 from the return coordinate that enters the counter-angle. The separation does not weaken the construction: it eliminates an identification incompatible with the exact digits and turns each comparison into a falsifiable assertion of a well-defined type.

The cubic completion

$$10^3 = (9 + 1)^3 = 9^3 + 3 \cdot 9^2 + 3 \cdot 9 + 1 = 729 + 243 + 27 + 1 \quad (34.1)$$

determines the chart global of the coordinate direct. The closure bilayer contributes the factor 2 of the coordinate conjugate:

$$\boxed{A_{\deg} = 10^3 \alpha, \quad C_{\deg}^* = 2(\eta_{\text{ret}} + \alpha) \quad \text{in } \mathbb{R}/(360\mathbb{Z}).} \quad (34.2)$$

The inverse identity

$$\eta_{\text{ret}} = \frac{1}{2} C_{\deg}^* - \alpha \quad (34.3)$$

is established as an algebraic corollary of the direct construction.

34.5 Exact commutativity of the charts in turns, degrees, and radians

34.5.1 Derivation of the cylindrical length and the analytic degree of the parangular precursor

Let $\mathcal{P}$ be the free module generated by the regional words and let $F^n \mathcal{P}$ be its length filtration. The associated grading records the first level at which each word appears. For a parameter z , the ordinary length character is

$$\chi_z([w]) = z^{|w|}.$$

Multiplicativity

$$\chi_z([uv]) = \chi_z([u])\chi_z([v])$$

expresses that concatenation adds lengths. When evaluated at $z = \alpha$, a word of length n receives degree α^n .

Definition 34.15 (Degree compatibility). The regional analytic reader is compatible with the cylindrical filtration when it sends the homogeneous component of length n to analytic degree n :

$$\text{gr}_n \mathcal{P} \longrightarrow \alpha^n \mathbb{R}.$$

The five regions belong to $\mathcal{U}_6$; therefore, the return datum appears in degree six:

$$\rho_\pi[U_6] \longmapsto 169\alpha^6.$$

This six is the length of the regional word. It is not the length of the closed walks of chapter devoted to the minimality of the closures of the five regions of π . Those walks traverse the four anchors $U_\pi, U_e, U_\varphi, U_{\text{APP}}$ simultaneously and have minimum length eight. A word, an edge of the block graph, and a closed walk are objects of distinct types; the grading used here belongs to the first.

34.5.2 Publication of the analytic degree on the global phase chart

Let $\mathcal{P}$ be the free module generated by the regional words and let $F^n \mathcal{P}$ be its length filtration. The associated grading records the first level at which each word appears. For a parameter z , the ordinary length character is

$$\chi_z([w]) = z^{|w|}.$$

Multiplicativity

$$\chi_z([uv]) = \chi_z([u])\chi_z([v])$$

expresses that concatenation adds lengths. When evaluated at $z = \alpha$, a word of length n receives degree α^n .

Definition 34.16 (Degree compatibility). The regional analytic reader is compatible with the cylindrical filtration when it sends the homogeneous component of length n to analytic degree n :

$$\text{gr}_n \mathcal{P} \longrightarrow \alpha^n \mathbb{R}.$$

The five regions belong to $\mathcal{U}_6$; therefore, the return datum appears in degree six:

$$\rho_\pi[U_6] \mapsto 169\alpha^6.$$

This six is the length of the regional word. It is not the length of the closed walks of chapter devoted to the minimality of the closures of the five regions of π . Those walks traverse the four anchors $U_\pi, U_e, U_\varphi, U_{\text{APP}}$ simultaneously and have minimum length eight. A word, an edge of the block graph, and a closed walk are objects of distinct types; the grading used here belongs to the first.

The global calendar exhibits the three factorizations

$$360 = 12 \cdot 30 = 4 \cdot 90 = 3 \cdot 120. \quad (34.4)$$

The analytical chart is obtained through

$$A_{\text{rad}} = \frac{\pi}{180} A_{\text{deg}} = \frac{50\pi}{9} \alpha, \quad C_{\text{rad}}^* = \frac{\pi}{180} C_{\text{deg}}^* = \frac{\pi}{90} (\eta_{\text{ret}} + \alpha). \quad (34.5)$$

Theorem 34.17 (Commutativity of Regional and Action Factorizations). *The two routes toward the conjugate phase produce the same output:*

$$\boxed{\frac{\pi}{180} C_{\text{deg}}^* = \frac{\pi}{90} (\eta_{\text{ret}} + \alpha) = \frac{\pi}{90} (H_5 + \alpha) - \mathcal{E}_\pi(\alpha) = y_*}. \quad (34.6)$$

Proof. Substituting equation (31.5) in the second member yields

$$\frac{\pi}{90} \left(H_5 - \frac{90}{\pi} \mathcal{E}_\pi + \alpha \right) = \frac{\pi}{90} (H_5 + \alpha) - \mathcal{E}_\pi.$$

The final expression is the regional definition of y_* , whereas the first equality follows from equation (34.2). $\square$

34.6 spectral decomposition $q_\pm$ and calculo functional on the channels 90/120 previously generated

34.6.1 Spectral reconstruction of the angular pair and hexad-octad transition

The common phase and the oriented phase determine the two channels.

$$q_- = e^{-x+y_*}, \quad q_+ = e^{-x-y_*}.$$

The invariants are recovered exactly.

$$q_- q_+ = D_A, \quad \alpha = -\frac{9}{100\pi} \log D_A, \quad C_*^{\text{HMT}} = \frac{90}{\pi} \log \frac{q_-}{q_+}.$$

The first equality reads the common contraction; the third, the oriented separation of the sheets. The exchange $R \mapsto -R$ permutes q_+ and q_- , preserves D_A and reverses the sign of y_* .

For

$$S_{90}(q) = \frac{q^{90}}{1 - q^{270}}, \quad S_{120}(q) = \frac{q^{120}}{1 - q^{360}},$$

the hexad–octad transition functional is written entirely in the phase chart:

$$\Gamma_{6 \rightarrow 8} = \frac{180}{\pi} x y_* + 12 (S_{90}(q_-) - S_{90}(q_+)) - (S_{120}(q_-) - S_{120}(q_+)).$$

The evaluation produced by the coordinate C_*^{HMT} of theorem 34.13 is

$$\Gamma_{6 \rightarrow 8} = 0.274007672694459274747255260774 \dots$$

Thus, the internal value recognized in Dimensional Mechanics as the Barbero–Immirzi functional ceases to receive C_*^{HMT} as an independent literal: both descend from the same pair of phases generated in this chapter. The geometric and physical interpretation of $\Gamma_{6 \rightarrow 8}$ belongs to the work on Dimensional Mechanics; its mathematical evaluation already belongs to the previously constructed interface.

34.6.2 Relative transporter and spectral disjunction of the blocks

In the ordered basis of channels, let

$$R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad P_{\pm} = \frac{I \pm R}{2}, \quad B_0 = 7I + 2R, \quad B_1 = A_{\text{deg}}I + C_{\text{deg}}^*R.$$

The split decomposition

$$xI + yR = \rho(x, y)e^{\eta(x, y)R}, \quad \rho(x, y) = \sqrt{x^2 - y^2}, \quad \eta(x, y) = \text{artanh}(y/x)$$

constructs the unique transporter

$$T_{\text{rel}} = \frac{\sqrt{(A_{\text{deg}})^2 - (C_{\text{deg}}^*)^2}}{\sqrt{45}} \exp \left[\left(\text{artanh} \frac{C_{\text{deg}}^*}{A_{\text{deg}}} - \text{artanh} \frac{2}{7} \right) R \right], \quad T_{\text{rel}}B_0 = B_1.$$

In the spectral basis it acts by $9 \mapsto A_{\text{deg}} + C_{\text{deg}}^*$ and $5 \mapsto A_{\text{deg}} - C_{\text{deg}}^*$.

Theorem 34.18 (Spectral Disjunction of the Local and Angular Realizations). *The spectra*

$$\text{Spec}(B_0) = \{9, 5\}, \quad \text{Spec}(B_1) = \{A_{\text{deg}} + C_{\text{deg}}^*, A_{\text{deg}} - C_{\text{deg}}^*\}$$

are disjoint. Consequently,

$$\text{Hom}_{B_0, B_1} := \{\Phi : \Phi B_0 = B_1 \Phi\} = \{0\}.$$

Proof. The rational bounds constructed for the angular coordinates give $7.29 < A_{\text{deg}} < 7.30$ and $2.12 < C_{\text{deg}}^* < 2.13$; therefore, $9.41 < A_{\text{deg}} + C_{\text{deg}}^* < 9.43$ and $5.16 < A_{\text{deg}} - C_{\text{deg}}^* < 5.18$. In the basis that diagonalizes R , each entry of an intertwiner is multiplied by the difference between an eigenvalue of B_0 and one of B_1 ; disjointness forces all of them to vanish. $\square$

The relative transporter compares two blocks within the split algebra; the dodecaphasic–Witt and incidence intertwiners act between equivalent representations in their own spaces of residence. This separation positively types both operations and preserves their domains.

34.6.3 Spectral evaluation $V_{90,120}$ of the incidence channels previously generated by the TPK

The common phase and the oriented phase determine the two channels.

$$q_- = e^{-A_{\text{rad}} + C_{\text{rad}}^*}, \quad q_+ = e^{-A_{\text{rad}} - C_{\text{rad}}^*}.$$

The invariants are recovered exactly.

$$q_- q_+ = D_A, \quad \alpha = -\frac{9}{100\pi} \log D_A, \quad C_{\text{deg}}^* = \frac{90}{\pi} \log \frac{q_-}{q_+}.$$

The first equality reads the common contraction; the third, the oriented separation of the sheets. The exchange $R \mapsto -R$ permutes q_+ and q_- , preserves D_A and reverses the sign of C_{rad}^* .

Let $H^2 = I$ and $P_{\pm} = (I \pm H)/2$. The angular block

$$B = A_{\text{rad}}I + C_{\text{rad}}^*H = (A_{\text{rad}} + C_{\text{rad}}^*)P_+ + (A_{\text{rad}} - C_{\text{rad}}^*)P_- \quad (34.7)$$

is published multiplicatively by

$$e^{-B} = q_+P_+ + q_-P_-, \quad q_{\pm} = \exp[-(A_{\text{rad}} \pm C_{\text{rad}}^*)]. \quad (34.8)$$

The involution $C^* \mapsto -C^*$ permutes q_+ and q_- , and preserves

$$q_+q_- = e^{-2A_{\text{rad}}} = D_A. \quad (34.9)$$

The cardinalities 90/120, previously constructed through TPK incidence, now receive their spectral evaluation:

$$V_{90,120} = (I - T^{90})(I - T^{120})^{-1} = \sum_{\sigma \in \{+, -\}} \frac{1 - q_{\sigma}^{90}}{1 - q_{\sigma}^{120}} P_{\sigma}. \quad (34.10)$$

34.6.3.1 Split relative transporter

The comparison With the central block of APP can be carried out without identifying representations distinct. Fijemos the ordered basis of channels and the involution

$$R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad R^2 = I, \quad P_{\pm} = \frac{I \pm R}{2}.$$

In commutative algebra

$$\mathcal{B}_R := \{xI + yR : x, y \in \mathbb{R}\}$$

consider a, within the already declared sexagesimal chart,

$$B_0 := 7I + 2R, \quad B_1 := A_{\text{deg}}I + C_{\text{deg}}^*R.$$

The orthogonal change of basis

$$S = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

simultaneously diagonalizes the involution and both blocks. For $x > |y|$, we define

$$\rho(x, y) := \sqrt{x^2 - y^2}, \quad \eta(x, y) := \operatorname{artanh}\left(\frac{y}{x}\right).$$

Then

$$xI + yR = \rho(x, y)e^{\eta(x, y)R}.$$

If

$$\rho_0 = \sqrt{45}, \quad \eta_0 = \operatorname{artanh}(2/7), \quad \rho_1 = \rho(A_{\text{deg}}, C_{\text{deg}}^*), \quad \eta_1 = \eta(A_{\text{deg}}, C_{\text{deg}}^*),$$

the unique left multiplication that carries B_0 to B_1 is

$$T_{\text{rel}} := \frac{\rho_1}{\sqrt{45}} e^{(\eta_1 - \eta_0)R}, \quad T_{\text{rel}}B_0 = B_1.$$

Proposition 34.19 (Relative carrier of the two blocks). *Once R , the channel order, the positive branch, and the sexagesimal chart have been fixed, T_{rel} is the unique element of $\mathcal{B}_R$ satisfying $TB_0 = B_1$. In the spectral basis it acts by*

$$9 \mapsto A_{\text{deg}} + C_{\text{deg}}^*, \quad 5 \mapsto A_{\text{deg}} - C_{\text{deg}}^*.$$

Proof. The split polar decomposition yields the printed formula and proves existence. Since $\det B_0 = 45 \neq 0$, every matrix solution of $T B_0 = B_1$ satisfies $T = B_1 B_0^{-1}$; it is therefore unique. Diagonalization by means of S produces the two spectral quotients $(A_{\text{deg}} + C_{\text{deg}}^*)/9$ and $(A_{\text{deg}} - C_{\text{deg}}^*)/5$, equivalent to the exponential expression of T_{rel} . $\square$

This transporter is not a representational intertwiner. Indeed, an intertwiner Φ between the endomorphisms B_0 and B_1 should satisfy

$$\Phi B_0 = B_1 \Phi.$$

Theorem 34.20 (Spectral Obstruction to Entanglement). *For the values of A_{deg} and C_{deg}^* constructed in this section,*

$$\boxed{\Phi B_0 = B_1 \Phi \implies \Phi = 0.}$$

Proof. The spectra are

$$\text{Spec}(B_0) = \{9, 5\}, \quad \text{Spec}(B_1) = \{A_{\text{deg}} + C_{\text{deg}}^*, A_{\text{deg}} - C_{\text{deg}}^*\}.$$

The previously demonstrated rational bounds provide the disjoint intervals.

$$7.29 < A_{\text{deg}} < 7.30, \quad 2.12 < C_{\text{deg}}^* < 2.13,$$

and, consequently,

$$9.41 < A_{\text{deg}} + C_{\text{deg}}^* < 9.43, \quad 5.16 < A_{\text{deg}} - C_{\text{deg}}^* < 5.18.$$

In the basis that diagonalizes R , each entry ϕ_{ij} of Φ is multiplied by the difference between an eigenvalue of B_0 and one of B_1 . All these differences are nonzero; hence all entries of Φ vanish. $\square$

Numerically,

$$\rho_1 = 6.9814819335172378048 \dots, \quad \frac{\rho_1}{\sqrt{45}} = 1.0407378791354140779 \dots,$$

$$\eta_1 - \eta_0 = 0.005796351470571295590 \dots$$

The formula preserves the split algebra, the projectors $P_{\pm}$, and the Lorentzian form up to the conformal factor $\rho_1^2/45$. It nevertheless depends on B_1 already constructed: it does not constitute a prior derivation of A_{deg} or C_{deg}^* , nor should it be confused with the equivariant intertwiners between realizations of the same module.

For

$$S_{90}(q) = \frac{q^{90}}{1 - q^{270}}, \quad S_{120}(q) = \frac{q^{120}}{1 - q^{360}},$$

the hexad–octad transition functional is written entirely in the phase chart:

$$\Gamma_{6 \rightarrow 8} = \frac{180}{\pi} A_{\text{rad}} C_{\text{rad}}^* + 12(S_{90}(q_-) - S_{90}(q_+)) - (S_{120}(q_-) - S_{120}(q_+)).$$

The evaluation produced by the counter-angle of the [theorem 34.14](#) is

$$\boxed{\Gamma_{6 \rightarrow 8} = 0.274007672694459274747255260774 \dots}$$

The mathematically equivalent forms of the internal functional are

$$\boxed{\Gamma_{6 \rightarrow 8} = A_{\text{rad}} C_{\text{deg}}^* + K_{6 \rightarrow 8} = \frac{180}{\pi} A_{\text{rad}} C_{\text{rad}}^* + K_{6 \rightarrow 8} = \frac{\pi A_{\text{deg}} C_{\text{deg}}^*}{180} + K_{6 \rightarrow 8}.$$

These identities evaluate $\Gamma_{6 \rightarrow 8}$ after generating $(C_{\text{rad}}^*, C_{\text{deg}}^*)$; it is not used in the inverse sense to select the counter-angle. The physical reading in the Ashtekar–Barbero connection is then represented by means of

$$\Gamma_{6 \rightarrow 8} \dashrightarrow \gamma_{\text{BI}}.$$

The arrow introduces no third functional and does not act retroactively on C^* . The series S_{90} , S_{120} , the coefficients 12 and -1 , and their assignment to the $6 \rightarrow 8$ passage belong to the declared mathematical construction. Physical identification additionally requires the connection's own normalization.

34.7 Naturalness under Refinement and Separation with Respect to Metrological Recognition

34.7.1 Exact estimation of the remainder of the analytic prolongation

The expression initially found,

$$y_7 = \frac{\pi}{90}(H_5 + \alpha) - 169\alpha^6 - D_A\alpha^7,$$

is the seventh-degree truncation of the full character. Produce

$$C_7 = 2.12373833896864600265403827103919 \dots^\circ.$$

The omitted remainder is not an indeterminate term; it is the exact geometric tail.

$$\mathcal{C}_\pi(\alpha) - 169\alpha^6 - D_A\alpha^7 = \frac{D_A^2\alpha^8}{1 - D_A\alpha}.$$

Therefore,

$$|C_7 - C_*^{\text{HMT}}| = \frac{180}{\pi} \frac{D_A^2\alpha^8}{1 - D_A\alpha} = 2.7839208689064583 \dots \times 10^{-16}^\circ.$$

The recurrence thus transforms a seventh-degree approximation into a complete analytic realization and provides a closed-form bound for each subsequent truncation.

34.7.2 Functional dependence of the regional selector and the analytic continuation

The construction changes when any of its active components is removed. The following values are obtained without recalibrating the other terms:

variant	conjugate angular coordinate in degrees
$\rho_\pi = 168$	2.1237383389772976863904487...
$\rho_\pi = 169$	2.1237383389686457242619514...
$\rho_\pi = 170$	2.1237383389599937621334541...
without determinantal tail	2.1237383389686949415301675...
D_A replaced by 1	2.1237383389686313409965826...

These ablations do not in themselves constitute a probabilistic estimate. They demonstrate that the equivariant selector and the determinantal character are effective components of the output and cannot be removed while maintaining the same realization.

34.7.3 Operatorial Interpretation of the Conjugate Angular Coordinate

The entire procedure can be summarized as a chain of applications:

$$\begin{aligned}\mathcal{F}_\pi &\mapsto b_\pi \mapsto \rho_\pi = 169, \\ \alpha &\mapsto A \mapsto x \mapsto D_A, \\ (\rho_\pi, D_A) &\mapsto \mathcal{C}_\pi(\alpha), \\ (H_5, \mathcal{C}_\pi(\alpha)) &\mapsto y_* \mapsto C_*^{\text{HMT}}, \\ (x, y_*) &\mapsto (q_+, q_-) \mapsto \Gamma_{6 \rightarrow 8}.\end{aligned}$$

The first line belongs to the finite catalog; the second, to the common closure of α ; the third, to the determinantal prolongation; the fourth produces the oriented coordinate; the fifth transports it toward the dimensional interface.

In this organization, α measures the common contraction of the transport, and C_*^{HMT} measures the oriented separation of its two sheets. The regional return adds no external constant: it selects a finite amplitude and prolongs it through the determinant already fixed by α . The chart in degrees is subsequent. The primary object is the pair of phases (x, y_*) .

An exhaustive verification reconstructs the five regions directly from the catalogue, traverses the actions of $\mathfrak{S}_5$ and $\mathfrak{S}_3$, checks the recurrence, and sums the series. The input of α is the exact output of its dodecaphasic closure, not a transcribed metrological digit. The verification keeps the catalogue theorems, reader rules, and external comparison separate; therefore, no experimental comparison intervenes in the generating function.

34.7.4 Angular Truncation Convergence Certification

The expression initially found,

$$y_7 = \frac{\pi}{90}(H_5 + \alpha) - 169\alpha^6 - D_A\alpha^7,$$

is the seventh-degree truncation of the full character. Produce

$$C_7 = 2.12373833896864600265403827103919 \dots^\circ.$$

The omitted remainder is not an indeterminate term; it is the exact geometric tail.

$$\mathcal{C}_\pi(\alpha) - 169\alpha^6 - D_A\alpha^7 = \frac{D_A^2\alpha^8}{1 - D_A\alpha}.$$

Therefore,

$$\left| C_7 - C_{\text{deg}}^* \right| = \frac{180}{\pi} \frac{D_A^2\alpha^8}{1 - D_A\alpha} = 2.7839208689064583 \dots \times 10^{-16}^\circ.$$

The recurrence thus transforms a seventh-degree approximation into a complete analytic realization and provides a closed-form bound for each subsequent truncation.

34.7.5 Naturality of the regional selector under refinement and chart change

The construction changes when any of its active components is removed. The following values are obtained without recalibrating the other terms:

variant	counter-angle in degrees
$\rho_\pi = 168$	2.1237383389772976863904487...
$\rho_\pi = 169$	2.1237383389686457242619514...
$\rho_\pi = 170$	2.1237383389599937621334541...
without determinantal tail	2.1237383389686949415301675...
D_A replaced by 1	2.1237383389686313409965826...

These ablations do not in themselves constitute a probabilistic estimate. They demonstrate that the equivariant selector and the determinantal character are effective components of the output and cannot be removed while maintaining the same realization.

34.7.6 Metric independence of the parangular construction

The entire procedure can be summarized as a chain of applications:

$$\begin{aligned}
\mathcal{F}_\pi &\mapsto b_\pi \mapsto \rho_\pi = 169, \\
\alpha &\mapsto A_{\deg} \mapsto A_{\text{rad}} \mapsto D_A, \\
(\rho_\pi, D_A) &\mapsto \mathcal{C}_\pi(\alpha), \\
(H_5, \mathcal{C}_\pi(\alpha)) &\mapsto C_{\text{rad}}^* \mapsto C_{\deg}^*, \\
(A_{\text{rad}}, C_{\text{rad}}^*) &\mapsto (q_+, q_-) \mapsto \Gamma_{6 \rightarrow 8}.
\end{aligned}$$

The first line belongs to the finite catalog; the second, to the common closure of α ; the third, to the determinantal prolongation; the fourth produces the oriented coordinate; the fifth transports it toward the dimensional interface.

In this organization, α measures the common contraction of transport, and the counter-angle measures the oriented separation of its two sheets. In the current evaluation, the regional return selects a finite amplitude and prolongs it through the determinant already fixed by α , without inserting an additional external constant. The degree chart is subsequent. The primary object is the phase pair $(A_{\text{rad}}, C_{\text{rad}}^*)$. The recurrence, regional sum, and ablations depend on the catalogue and on α , but receive no additional metrological constant.

Positive constitutive vacuum operator and joint derivation of ε, μ, Z, c

35.1 Bilayer spectral decomposition of the angular operator

Let $R = R^* = R^{-1}$ be the active involution and

$$P_{\pm} = \frac{I \pm R}{2}, \quad T = q_+ P_+ + q_- P_-, \quad 0 < q_+ < q_- < 1.$$

With

$$x = A_{\text{rad}} = \frac{\pi A}{180}, \quad y = C_{\text{rad}}^* = \frac{\pi C^*}{180},$$

the two channels are constructed before evaluating the vacuum:

$$q_+ = e^{-x-y}, \quad q_- = e^{-x+y}. \quad (35.1)$$

The exchange of sheets reverses C^* , that is, $y \mapsto -y$, and permutes $q_+ \leftrightarrow q_-$. The pair is thus the spectral reading of the same angular antecedent, not two parameters fitted separately.

Definition 35.1 (Constitutive response).

$$\mathcal{V}_{90,120} = (I - T^{90})(I - T^{120})^{-1}. \quad (35.2)$$

Since $\|T\| < 1$, the denominator is positive and invertible. Functional calculus gives

$$\mathcal{V}_{90,120} = r_+ P_+ + r_- P_-, \quad r_{\pm} = \frac{1 - q_{\pm}^{90}}{1 - q_{\pm}^{120}}. \quad (35.3)$$

The assertion of invertibility deserves to be made explicit. For each leaf, $0 < q_{\pm}^{120} < 1$; therefore

$$I - T^{120} = (1 - q_+^{120})P_+ + (1 - q_-^{120})P_-$$

is strictly positive and

$$(I - T^{120})^{-1} = \frac{P_+}{1 - q_+^{120}} + \frac{P_-}{1 - q_-^{120}}.$$

The response $\mathcal{V}_{90,120}$ is not obtained by solving four separate scalar equations: it is the result of the functional calculus of a single positive contraction. This observation prevents permeability, permittivity, impedance and propagation from losing their shared antecedent.

Theorem 35.2 (Joint spectral determination of the four constitutive relations). *The four constituent characters*

$$\widehat{\varepsilon} = r_+^2, \quad \widehat{\mu} = r_-^2, \quad \widehat{Z} = \frac{r_-}{r_+}, \quad \widehat{c} = \frac{1}{r_+ r_-} \quad (35.4)$$

are spectral functions of $\mathcal{V}_{90,120}$. They satisfy

$$\widehat{\mu} \widehat{\varepsilon} = \widehat{c}^{-2}, \quad \frac{\widehat{\mu}}{\widehat{\varepsilon}} = \widehat{Z}^2. \quad (35.5)$$

Proof. The identities are obtained by substituting equation (35.4). No dimensional chart nor any exterior value intervenes. $\square$

Corollary 35.3 (Reconstruction of the two sheets). *The four characters do not contain four degrees of freedom. Any of the pairs $(\widehat{\mu}, \widehat{\varepsilon})$, $(\widehat{Z}, \widehat{c})$, or $(\widehat{\mathcal{E}}, \widehat{\mathcal{B}})$ reconstructs the two positive eigenvalues:*

$$r_- = \sqrt{\widehat{\mu}} = \sqrt{\widehat{Z}/\widehat{c}}, \quad r_+ = \sqrt{\widehat{\varepsilon}} = \frac{1}{\sqrt{\widehat{Z}\widehat{c}}}, \quad (35.6)$$

$$\log r_- = \frac{\mathcal{E} + \mathcal{B}}{2}, \quad \log r_+ = \frac{\mathcal{E} - \mathcal{B}}{2}. \quad (35.7)$$

In particular, orientation is lost if only the product is retained, but is recovered by adding the quotient.

Proof. The first line is the positive root of the identities the corresponding identity. The second inverts the linear system $\mathcal{E} = \log r_- + \log r_+$, $\mathcal{B} = \log r_- - \log r_+$. $\square$

This corollary expresses a central property of HMT: product and quotient are complementary functionals of the same state. The product determines the propagation in common, whereas the quotient preserves the relative orientation of the inner and outer sheets.

In the V2 certificate,

$$\begin{aligned} r_+ &= 0.9999996285482493631786952281 \dots, \\ r_- &= 0.9997241371895183641315165853 \dots, \\ \widehat{\varepsilon} &= 0.9999992570966367027604416155 \dots, \\ \widehat{\mu} &= 0.9994483504793269350899815559 \dots, \\ \widehat{Z} &= 0.9997245085389372154555543466 \dots, \\ \widehat{c} &= 1.0002763104861575261063862689 \dots \end{aligned}$$

These numerals are the terminal recognition of [equation \(35.2\)](#); they do not define its exponents 90, 120.

35.2 Harmonic factorization of the 90 and 120 sectors

The harmonic semigroup is

$$\langle 90, 120 \rangle = 30\langle 3, 4 \rangle.$$

If $s = q^{30}$, then

$$\frac{1 - q^{90}}{1 - q^{120}} = \frac{1 - s^3}{1 - s^4} = \frac{1 + s + s^2}{1 + s + s^2 + s^3}. \quad (35.8)$$

The identity also separates the response from its defect. If

$$F_{3 \rightarrow 4}(s) = \frac{1 + s + s^2}{1 + s + s^2 + s^3}, \quad 0 < s < 1,$$

then

$$d_{3 \rightarrow 4}(s) := 1 - F_{3 \rightarrow 4}(s) = \frac{s^3}{1 + s + s^2 + s^3} > 0. \quad (35.9)$$

The additional term is preserved exactly as a positive defect. Moreover,

$$F'_{3 \rightarrow 4}(s) = -\frac{s^2(3 + 2s + s^2)}{(1 + s + s^2 + s^3)^2} < 0. \quad (35.10)$$

Consequently, the order of the angular channels uniquely determines the order of the constitutive responses. The assignment of $\widehat{\mu}$ and $\widehat{\varepsilon}$ to the respective sheets is obtained analytically, without resorting to their decimal expansions.

Each sheet of the vacuum constitutes the exact completion from three to four terms of the same elementary mode 30. Consequently, permeability and permittivity are interpreted as two quadratic functionals of the same completion evaluated on conjugate sheets, and not as independent constitutive parameters.

Since the function $(1 - q^{90})/(1 - q^{120})$ decreases in $0 < q < 1$ and $q_+ < q_-$, we obtain

$$0 < r_- < r_+ < 1, \quad 0 < \widehat{\mu} < \widehat{\varepsilon} < 1, \quad 0 < \widehat{Z} < 1 < \widehat{c}.$$

The limits of the functional determine its geometry. When $q \rightarrow 0^+$, $F(q) \rightarrow 1$: the four readings approach unity and the memory defect disappears. When $q \rightarrow 1^-$,

$$F(q) \rightarrow \frac{90}{120} = \frac{3}{4}.$$

The response therefore takes values in the positive interval $(3/4, 1)$. The endpoint $3/4$ is not a fitted constant: it is the quotient of the two channel lengths and the continuous realization of the three-to-four completion.

35.3 Sheet involution and preservation of orientation

The sheet exchange $C^* \mapsto -C^*$ acts by

$$\widehat{\mu} \longleftrightarrow \widehat{\varepsilon}, \quad \widehat{Z} \mapsto \widehat{Z}^{-1}, \quad \widehat{c} \mapsto \widehat{c}. \quad (35.11)$$

The common propagation is even; impedance preserves orientation. This distinction will reappear in Barbero and CKM: not all functionals associated with the change of sheet have the same parity.

35.4 Logarithmic characters and tracial compatibility

Define

$$\mathcal{E} = \log(r_+ r_-), \quad \mathcal{B} = \log(r_- / r_+).$$

With the normalized trace $\tau = \text{Tr}/2$,

$$2\tau(\log \mathcal{V}) = \mathcal{E}, \quad -2\tau(R \log \mathcal{V}) = \mathcal{B}. \quad (35.12)$$

The pair $(\mathcal{E}, \mathcal{B})$ constitutes a linear coordinate system for the operator logarithm:

$$\log \mathcal{V} = \frac{\mathcal{E}}{2} I - \frac{\mathcal{B}}{2} R. \quad (35.13)$$

The scalar component is invariant under sheet exchange; the oriented component changes sign. This decomposition directly demonstrates the parity of propagation and the oriented character of impedance.

The nonadic amplification

$$\mathcal{V}_m = \mathcal{V} \otimes I_{9^m}$$

is compatible with the expectations $E_m = \text{id} \otimes \tau_9$:

$$E_m(\mathcal{V}_{m+1}) = \mathcal{V}_m.$$

Therefore, $\mathcal{E}$ and $\mathcal{B}$ are not peculiarities of a 2×2 matrix; they persist in the UHF tower and in its tracial closure II_1 .

More precisely, for every polynomial p and afterwards, by continuity, for every continuous function on the spectrum,

$$E_m(p(\mathcal{V}_{m+1})) = p(\mathcal{V}_m). \quad (35.14)$$

Compatibility does not preserve only two numbers: it preserves the entire functional algebra generated by the response. In particular, it preserves its spectral projectors, powers, logarithm, and quadratic forms that measure the defects equation (35.9).

35.5 Dimensional realization of the constitutive relations

Dimensional Mechanics determines

$$Z_{\text{vac}} = \widehat{Z} u_S u_Q^{-2}, \quad (35.15)$$

$$c_{\text{vac}} = \widehat{c} u_C, \quad (35.16)$$

$$\mu_{\text{vac}} = \widehat{\mu} u_S u_C^{-1} u_Q^{-2}, \quad (35.17)$$

$$\varepsilon_{\text{vac}} = \widehat{\varepsilon} u_S^{-1} u_C^{-1} u_Q^2. \quad (35.18)$$

The constitutive identities hold on the straight lines:

$$\mu_{\text{vac}} \varepsilon_{\text{vac}} = c_{\text{vac}}^{-2}, \quad \mu_{\text{vac}} / \varepsilon_{\text{vac}} = Z_{\text{vac}}^2. \quad (35.19)$$

The SI chart comes later. It is not necessary to declare c primitive and then solve $\mu \varepsilon = c^{-2}$: the three objects descend jointly from $\mathcal{V}$ and from the declared lines.

The separation of levels is expressed by the diagram

$$\begin{array}{ccc} (A, C^*) & \longrightarrow & T \longrightarrow \mathcal{V} \longrightarrow (\widehat{\mu}, \widehat{\varepsilon}, \widehat{Z}, \widehat{c}) \\ \downarrow \text{line charts} & & \downarrow \text{final evaluation} \\ (u_S, u_Q, u_C) & \longrightarrow & (\mu_{\text{vac}}, \varepsilon_{\text{vac}}, Z_{\text{vac}}, c_{\text{vac}}) \longrightarrow \text{SI coordinates.} \end{array}$$

The first line is dimensionless and operatorial; the second determines types of quantity; the third selects a metrological chart. Altering the final chart changes neither the eigenvalues nor the constitutive identities. In this precise sense, mathematical–physical unification consists in distinguishing the genealogical invariant from its coordinate system, not in removing the dimensional structure.

Proposition 35.4 (Spectral minimality). *Let W be a positive two-sheet operator with simple spectrum $\{r_+, r_-\}$. If four positive characters satisfy $\mu \varepsilon = c^{-2}$ and $\mu / \varepsilon = Z^2$, if their propagation in common is determined by $c^{-1} = \det W = r_- r_+$, and if its orientation fixes $Z = r_- / r_+$, then necessarily*

$$(\mu, \varepsilon, Z, c) = \left(r_-^2, r_+^2, \frac{r_-}{r_+}, \frac{1}{r_- r_+} \right).$$

Therefore, equation (35.4) is not one parametrization among many: it is the minimal positive realization compatible with product, quotient and orientation.

Proof. From $Z = r_- / r_+$ and $c^{-1} = \det W = r_- r_+$ the positive roots are obtained uniquely roots. The two constitutive identities then force afterwards $\mu = r_-^2$ and $\varepsilon = r_+^2$. $\square$

The first conceptual inversion consists in conceiving the vacuum as a positive relational response.

In HMT, the vacuum is the minimal dynamical response of the system when its two sheets must coexist, propagate, and close their difference while preserving the orientation that distinguishes them.

The two sheets constitute two conjugate readings of the same angular antecedent. One preserves an orientation, and the other preserves the opposite orientation. Hence two channels, q_+ and q_- , appear as the two correlated spectral faces of a single state.

The operator

$$\mathcal{V}_{90,120} = (I - T^{90})(I - T^{120})^{-1}$$

compares two propagation depths: 90 and 120. Those depths are determined by the same elementary mode:

$$90 = 3 \cdot 30, \quad 120 = 4 \cdot 30.$$

Therefore, the response of each sheet can be rewritten as

$$\frac{1 - q^{90}}{1 - q^{120}} = \frac{1 + s + s^2}{1 + s + s^2 + s^3}, \quad s = q^{30}.$$

And here is where the interpretation becomes genuinely powerful. The numerator contains three states of the mode 30; the denominator contains those same three states and one more. The vacuum is, therefore, an exact completion $3 \rightarrow 4$.

That fourth term integrates and preserves the preceding three. It completes them. What is added is recorded as a positive completion residue:

$$1 - \frac{1 + s + s^2}{1 + s + s^2 + s^3} = \frac{s^3}{1 + s + s^2 + s^3} > 0.$$

Vacuum memory is precisely the exact increment that allows three terms to become four. It is the positive and necessary trace of the completion.

From the two eigenvalues of that single operator arise the four constitutive relations:

$$\widehat{\varepsilon} = r_+^2, \quad \widehat{\mu} = r_-^2, \quad \widehat{Z} = \frac{r_-}{r_+}, \quad \widehat{c} = \frac{1}{r_+ r_-}.$$

This completely changes the interpretation of permeability and permittivity.

Permittivity is the quadratic response of one sheet. Permeability is the quadratic response of the conjugate sheet. Impedance preserves the oriented relation between them. Propagation speed depends on their common product.

Thus, product and quotient perform two different tasks:

- the product forgets which sheet was which and preserves the common propagation;
- the quotient preserves what the relative orientation between them was.

The product answers the question: “what do the two leaves share?”. The quotient answers the question: “how are they distinguished?”.

Por eso

$$\widehat{\mu} \widehat{\varepsilon} = \widehat{c}^{-2}$$

y

$$\frac{\widehat{\mu}}{\widehat{\varepsilon}} = \widehat{Z}^2$$

are the two complementary ways of reading the same spectral pair.

The holographic structure appears here in a particularly transparent form: four visible magnitudes actually contain only two internal degrees of freedom. Any of certain suitable pairs allows the two sheets to be reconstructed. The constitutive publication preserves genealogy when product and quotient remain jointly available.

This is the deep repair of the ablation of μ and ε . The repair reunites their values with their common operator-theoretic origin.

The International System chart may subsequently transform these intrinsic sections into $\mu_0, \varepsilon_0, Z_0$, and c . Its function is to express, within a metrological convention, a structure whose provenance has already been selected by the operator.

The vacuum thereby ceases to be a repository of four constantes. becomes a unique spectral operation With two conjugate memories.

And that view has an important philosophical consequence: the vacuum is a compatibility relation between two oriented ways of realizing one and the same state.

35.6 Functional recovery, composition and the Catalan moment

The incidence response 90/120 preserves more information than an isolated scalar value. In the same notation as (35.8), write

$$f(s) = \frac{1 + s + s^2}{1 + s + s^2 + s^3}, \quad \mathcal{D} = s \frac{d}{ds}, \quad \mathcal{V} = f(T^{30}).$$

The marks $P_{\pm}$ maintain channel orientation. Angular transport is the antecedent of the response; its functional reconstruction recovers this representation without replacing the memory archive from which it arises.

Theorem 35.5 (Inversion of the constitutive response). *The function f is a decreasing bijection from $(0, 1)$ onto $(3/4, 1)$. Its inverse ψ assigns to r the unique root $s \in (0, 1)$ of*

$$rs^3 + (r - 1)(1 + s + s^2) = 0. \quad (35.20)$$

The two labelled responses $r_{\pm}$ determine

$$s_{\pm} = \psi(r_{\pm}), \quad q_{\pm} = s_{\pm}^{1/30}, \quad x = -\frac{1}{2} \log(q_+ q_-), \quad y = \frac{1}{2} \log \frac{q_-}{q_+}. \quad (35.21)$$

For the complete operator $S = \mathcal{V}^2$, positive with spectrum in $(9/16, 1)$, functional calculus gives

$$T = [\psi(S^{1/2})]^{1/30}, \quad \det T = \det[\psi(S^{1/2})]^{1/30}. \quad (35.22)$$

Proof. The negative derivative of (35.10) and the limits $f(0^+) = 1$, $f(1^-) = 3/4$ prove bijectivity. Multiplying $f(s) = r$ by its positive denominator gives (35.20). The positive root of order 30 recovers $q_{\pm}$; the sum and difference of $\log q_{\pm} = -x \mp y$ give the two angular coordinates. By positivity, $S^{1/2} = \mathcal{V}$. Applying ψ to each spectral projector recovers T^{30} and the positive root gives T . Taking its determinant completes the proof. $\square$

This recovery of the complete operator does not extend to an isolated determinant. Indeed,

$$T_1 = \text{diag}(1/5, 1/3), \quad T_2 = \text{diag}(1/6, 2/5)$$

have determinant $1/15$ but distinct spectra in the same contractive chamber. The proved injectivity implies $f(T_1^{30})^2 \neq f(T_2^{30})^2$. Both channels and their marks preserve recovery; reducing them to a product loses information.

35.6.1 Transported composition of constitutive responses

The angular transport and the incidences 90 and 120 originate in the same APP–TRIT–TPK structure: the sheets retain evaluations and their carries; the tritic regime retains orientation; the TPK calendar transports phase and memory to the angular channels. The constitutive response applies the function f of (35.10) to those channels. Its inversion expresses, in response coordinates, the composition of transports sharing a common frame. The constructive order is thereby preserved: transport, composition and spectral evaluation. The resulting operation acts on this representation of the enriched state; its inverse recovers the channels, while the complete memory history remains in the underlying structure.

Let $s = q^{30}$ and $\psi = f^{-1}$. The exponent 30 is the greatest common divisor of the incidences 90 and 120. Cubic inversion determines $\psi : (3/4, 1) \rightarrow (0, 1)$. Adjoining the boundary response $f(1) = 3/4$ extends the bijection to

$$\psi : D \rightarrow (0, 1], \quad D = [3/4, 1).$$

At this boundary the rational function f is used; the quotient $(1 - q^{90})/(1 - q^{120})$ has a removable singularity there.

Proposition 35.6 (Response monoid and additive coordinate). *The operation*

$$r \odot t = f(\psi(r)\psi(t)), \quad r, t \in D, \quad (35.23)$$

makes D a commutative cancellative monoid with identity $3/4$. The map

$$\ell(r) = -\frac{1}{30} \log \psi(r) \quad (35.24)$$

is an isomorphism from this monoid to $(\mathbb{R}_{\geq 0}, +)$. The interval $(3/4, 1)$ is closed under $\odot$; the identity belongs to the boundary extension. The identity is the only invertible element within D .

Proof. The product of two elements of $(0, 1]$ belongs to the same interval, and

$$\psi(r \odot t) = \psi(r)\psi(t).$$

For three responses, either association has image $\psi(r)\psi(t)\psi(u)$; injectivity of ψ proves associativity. Commutativity follows from scalar multiplication. If $r \odot t = r \odot u$, positivity of $\psi(r)$ permits cancellation, giving $\psi(t) = \psi(u)$ and hence $t = u$. The parameter 1 corresponds to $3/4$ and provides the identity. The logarithm transforms products into sums, and its range on $(0, 1]$ shows that ℓ is a bijection onto $\mathbb{R}_{\geq 0}$. In particular, the response of a channel q satisfies $\ell(r) = -\log q$. If both parameters are strictly less than 1, so is their product. Finally, two numbers in $(0, 1]$ have product 1 only when both are 1. $\square$

Proposition 35.7 (Admissible cancellation and inverse extension). *For $r, u \in D$, the equation $r \odot t = u$ has a solution $t \in D$ exactly when $u \geq r$. This solution is unique and is given by*

$$t = f\left(\frac{\psi(u)}{\psi(r)}\right). \quad (35.25)$$

The same function f is a bijection from $(0, \infty)$ onto $(0, 1)$. With this extension of ψ , the law (35.23) transports the multiplicative group of positive parameters to the interval $(0, 1)$. Group inversion satisfies

$$r^\ominus = \psi(r)r, \quad r \odot r^\ominus = \frac{3}{4}. \quad (35.26)$$

Proof. The equation is equivalent to $\psi(t) = \psi(u)/\psi(r)$. This quotient belongs to $(0, 1]$ exactly when $\psi(u) \leq \psi(r)$, which, by decreasing monotonicity, is equivalent to $u \geq r$. The bijection proves existence and uniqueness. The derivative in (35.10) is negative for every $s > 0$; the limits of f at 0 and infinity are 1 and 0, respectively. This establishes the extended bijection. Multiplying the numerator and denominator by s^3 gives

$$f(s^{-1}) = \frac{s^3 + s^2 + s}{s^3 + s^2 + s + 1} = sf(s).$$

The multiplicative inverse of $s = \psi(r)$ therefore has response $\psi(r)r$. Its composition with r corresponds to the parameter 1. $\square$

The extension to $s > 1$, equivalently to $q > 1$, is the algebraic extension of the reader. The chapter's physical contractive sector retains its declared domain. The coordinate ℓ extends to an isomorphism with $(\mathbb{R}, +)$ and changes sign under group inversion.

35.6.1.1 Operator realization in a common sheet frame

Proposition 35.8 (Composition with marked projectors). *Let $\mathcal{R} = \mathcal{R}^*$, $\mathcal{R}^2 = I$ and $P_{\pm} = (I \pm \mathcal{R})/2$. For $j = a, b$, let*

$$T_j = q_{j,+}P_+ + q_{j,-}P_-, \quad \mathcal{V}_j = f(T_j^{30}), \quad 0 < q_{j,\pm} < 1.$$

The composition $T_{a \circ b} = T_a T_b$ satisfies

$$\mathcal{V}_{a \circ b} = f(\psi(\mathcal{V}_a)\psi(\mathcal{V}_b)). \quad (35.27)$$

Its two responses are $r_{a,\pm} \odot r_{b,\pm}$. If $T_j = \exp[-(x_j I + y_j \mathcal{R})]$ with $x_j > |y_j|$, then

$$T_a T_b = \exp[-((x_a + x_b)I + (y_a + y_b)\mathcal{R})], \quad (35.28)$$

if $x_a + x_b > |y_a + y_b|$.

Proof. Orthogonality $P_+ P_- = 0$ gives

$$T_a T_b = q_{a,+}q_{b,+}P_+ + q_{a,-}q_{b,-}P_-.$$

In particular, T_a and T_b commute. Functional calculus on these projectors yields $\psi(\mathcal{V}_j) = T_j^{30}$ and $(T_a T_b)^{30} = T_a^{30} T_b^{30}$; applying f proves (35.27). More explicitly, for every function g defined on the two eigenvalues,

$$P_{\pm} g(T_j) = g(q_{j,\pm})P_{\pm}.$$

Since $q_{j,\pm} = \exp[-(x_j \pm y_j)]$, their products prove (35.28). The triangle inequality concludes $x_a + x_b > |y_a| + |y_b| \geq |y_a + y_b|$. $\square$

The recovered coordinates can be written directly as

$$x = \frac{\ell(r_+) + \ell(r_-)}{2}, \quad y = \frac{\ell(r_+) - \ell(r_-)}{2}.$$

Composition adds these pairs. Simultaneous sheet exchange permutes the two channels of each transport, acts as $(x, y) \mapsto (x, -y)$ and is an automorphism of the componentwise law. Inversion of both transports in the algebraic extension acts as $(x, y) \mapsto (-x, -y)$. The two involutions commute and retain distinct functions: one exchanges orientation markings; the other inverts the transport.

To represent exchange by conjugation, use an operator L satisfying $L\mathcal{R}L^{-1} = -\mathcal{R}$. In the representation

$$\mathcal{R} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

direct multiplication gives $LP_{\pm}L^{-1} = P_{\mp}$, and hence $LT_jL^{-1} = q_{j,-}P_{+} + q_{j,+}P_{-}$. Conjugation by $\mathcal{R}$ preserves its own projectors. A trace combining the channels likewise cannot replace the projectors in this functional identity: their labels contribute to the composition.

The common-frame hypothesis is substantive. For example, the positive definite operators

$$A_0 = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/3 \end{pmatrix}, \quad B_0 = \begin{pmatrix} 1/2 & 1/6 \\ 1/6 & 1/2 \end{pmatrix}$$

have positive eigenvalues, but

$$A_0B_0 = \begin{pmatrix} 1/4 & 1/12 \\ 1/18 & 1/6 \end{pmatrix}$$

is not self-adjoint. The proposition specifies transport composition on common projectors; its domain does not encompass an arbitrary mixture of material media. Realizing a concatenation of HMT routes additionally retains their memory data: algebraic closure of the parameter cone describes the representation and does not by itself select new primary routes.

35.6.1.2 Channel product and transport composition

The normalized speed of one state remains determined by

$$\widehat{c} = (r_{+}r_{-})^{-1}.$$

Here the ordinary product combines the two responses of that same state. The law $\odot$ evaluates a different operation: composition of two transports on each sheet. The following exact calculation distinguishes their arguments:

$$f(1/2) = \frac{14}{15}, \quad \frac{14}{15} \odot \frac{14}{15} = f(1/4) = \frac{84}{85} \neq \frac{196}{225} = \left(\frac{14}{15}\right)^2. \quad (35.29)$$

The fractions in this example are algebraic test parameters. The expression for $\widehat{c}$ and the preceding constitutive identities remain unchanged. The added content is an explicit law transporting composition and recovering its additive angular coordinate.

35.6.2 Functional reconstruction of the oriented moment

The differential relation (35.30) admits an inverse preserving the complete system of moments. Its domain is the constitutive function produced by the cycle sectors, with the orientation of (35.1). Incidences $90 = 3 \cdot 30$ and $120 = 4 \cdot 30$ fix that function before its evaluation in channels $s_{\pm} = q_{\pm}^{30}$. The condition at the origin corresponds to disappearance of the moment when its contractive argument vanishes. This condition also determines the integration constants.

Theorem 35.9 (Integral reconstruction and uniqueness). *Let $f(s) = (1 - s^3)/(1 - s^4)$, for $0 < s < 1$, be the response of (35.8), and let $\mathcal{D} = s d/ds$. There is a unique function $F \in C^2((0, 1))$ satisfying*

$$\mathcal{D}^2 F(s) = \frac{s(1+s)}{1+s+s^2} f(s), \quad \lim_{s \downarrow 0} F(s) = 0. \quad (35.30)$$

It is given by

$$F(s) = \int_0^s \log \frac{s}{t} \frac{1+t}{1+t+t^2} f(t) dt = \int_0^s \frac{\log(s/t)}{1+t^2} dt = C_2(s), \quad (35.31)$$

$$f(s) = \frac{1+s+s^2}{s(1+s)} \mathcal{D}^2 F(s). \quad (35.32)$$

Its continuous extension to $s = 1$ recovers $F(1) = G_{\text{Cat}}$.

Proof. The factorization $1 + t + t^2 + t^3 = (1 + t)(1 + t^2)$ gives

$$\frac{1 + t}{1 + t + t^2} f(t) = \frac{1}{1 + t^2}.$$

The integrand of (35.31) is integrable at zero, since

$$0 \leq F(s) \leq \int_0^s \log(s/t) dt = s.$$

Hence $F(s) \rightarrow 0$. Differentiation on the open interval, first on a truncated interval and then by dominated convergence, gives

$$\mathcal{D}F(s) = \int_0^s \frac{dt}{1 + t^2}, \quad \mathcal{D}^2F(s) = \frac{s}{1 + s^2} = \frac{s(1 + s)}{1 + s + s^2} f(s).$$

In particular, F is of class C^2 and satisfies the equation. If F_1, F_2 are two solutions, their difference $H = F_1 - F_2$ satisfies $\mathcal{D}^2H = 0$. Setting $z = \log s$ gives $d^2H(e^z)/dz^2 = 0$, so $H(s) = a \log s + b$. The existence of a zero limit as $s \downarrow 0$ first forces $a = 0$ and then $b = 0$. The condition on F therefore determines the solution and also the limit $\mathcal{D}F(s) \rightarrow 0$.

For $s < 1$, the geometric expansion of $(1 + t^2)^{-1}$ and absolute integrability of its weighted terms allow termwise integration:

$$\int_0^s t^{2k} \log(s/t) dt = \frac{s^{2k+1}}{(2k+1)^2}, \quad F(s) = \sum_{k \geq 0} \frac{(-1)^k s^{2k+1}}{(2k+1)^2}.$$

The series represents the integral system (35.31). Absolute and uniform convergence of this last series on $[0, 1]$ permits passage to the endpoint $s = 1$. In the integral, this is also permitted by the integrable bound $\log(1/t)$, extending the integrand by zero for $t > s$. Finally, solving the differential equation for f gives (35.32), with positive denominators on $(0, 1)$. $\square$

The limit of the moment and the limit of its second derivative are taken in their respective convergence classes. In particular,

$$\lim_{s \uparrow 1} \mathcal{D}^2 C_2(s) = \frac{1}{2}$$

is the radial Abel limit of the differentiated series. Evaluation of $C_2(1)$ directly uses its absolutely convergent series. This distinction preserves the same depth operator in the interior and in its boundary readout.

35.6.3 Compatibility of reconstruction with channels and speed

The theorem restores a function from another previously constructed function. Cubic inversion (35.20) instead acts on evaluations $r_{\pm}$: it recovers arguments $s_{\pm}$ within that fixed system. Composition of the two operations preserves quarter-turn orientation, labels $P_{\pm}$ and transport incidences. Thus $G_{\text{Cat}} = C_2(1)$ is the endpoint reading of a functional object also participating in the interior responses $r_{\pm} = f(s_{\pm})$.

For $\mathcal{V}_m = \mathcal{V} \otimes I_{9^m}$ and $\tau_m = \text{Tr}/(2 \cdot 9^m)$, the speed readout retains exactly the normalization of (35.12):

$$\exp[-2\tau_m(\log \mathcal{V}_m)] = \frac{1}{r_+ r_-} = (\det \mathcal{V})^{-1} = \widehat{c}. \quad (35.33)$$

Indeed, each eigenvalue $r_{\pm}$ is repeated 9^m times, so $2\tau_m(\log \mathcal{V}_m) = \log r_+ + \log r_-$. The identity uses the determinant of the two-sheet realization and the normalized trace of its amplification. The previously fixed dimensional sections then give

$$c_{\text{int}} = c_{\text{vac}} = \widehat{c} u_C, \quad \ell_0 = c_{\text{vac}} t_0,$$

with the same state, chart and elementary duration as (35.34). Functional restoration preserves this velocity section when composing it with action and length; spectral multiplicity is absorbed by the declared normalization.

35.6.4 A common constitutive section in action, length and gravity

Relations (35.18) and the preceding restoration maintain a unique velocity section:

$$c_{\text{int}} = c_{\text{vac}} = \widehat{c} u_C, \quad \ell_0 = c_{\text{int}} t_0, \quad c_{\text{int}} = (\mu_{\text{vac}} \varepsilon_{\text{vac}})^{-1/2}. \quad (35.34)$$

The final equality is an identity of sections in compatible dimensional bases. The radial proof of section 39.1 first constructs $L = (5759/23040)\alpha^{16}L_*$ and then determines

$$G_{\text{ret}} = \frac{c_{\text{int}}^3 L^2}{\hbar_{\text{ret}}} = \frac{L^2}{\hbar_{\text{ret}} (\mu_{\text{vac}} \varepsilon_{\text{vac}})^{3/2}}. \quad (35.35)$$

The second expression results from substituting the constitutive identity into the first; it introduces no new calibration of either channel. In coordinates, if $c_{\text{int}} = c_* \widehat{c}$, then $G_{\text{ret}} = c_*^3 \widehat{c}^3 L^2 / \hbar_{\text{ret}}$. The propagation factor enters only once. Catalan recovery, composed transport and radial response thus preserve the same angular origin, its marks and its normalization.

Charge, von Klitzing resistance, conductance, Josephson constant and magnetic flux

36.1 Derivation of the charge section from the vacuum

With Z_{vac} , $h_a = 2\pi\hbar_a$ and α fixed, the charge of sheet $a \in \{\text{pre}, \text{ret}\}$ is

$$e_a = \sqrt{\frac{2\alpha_a}{Z_{\text{vac}}}}. \quad (36.1)$$

The section e_a is not introduced through an International System coordinate intended to reconstruct α . The causal order proceeds from α , the action, and the vacuum toward the charge section.

36.2 Von Klitzing and Josephson constants, conductance and magnetic flux

From equation (36.1) we deduce

$$R_K = \frac{h_a}{e_a^2} = \frac{Z_{\text{vac}}}{2\alpha}, \quad (36.2)$$

$$G_0 = \frac{2e_a^2}{h_a} = \frac{4\alpha}{Z_{\text{vac}}}, \quad (36.3)$$

$$\Phi_{0,a} = \frac{h_a}{2e_a} = \sqrt{\frac{h_a Z_{\text{vac}}}{8\alpha}}, \quad (36.4)$$

$$K_{J,a} = \frac{2e_a}{h_a} = \Phi_{0,a}^{-1}. \quad (36.5)$$

Consequently,

$$R_K G_0 = 2, \quad G_0 Z_{\text{vac}} = 4\alpha, \quad K_{J,a} \Phi_{0,a} = 1. \quad (36.6)$$

The bidirectional covariance takes the form

$$\begin{aligned} R_K^{\text{pre}} &= R_K^{\text{ret}}, & G_0^{\text{pre}} &= G_0^{\text{ret}}, \\ \frac{\Phi_{0,\text{pre}}}{\Phi_{0,\text{ret}}} &= R_{\text{act}}^{1/2}, & \frac{K_{J,\text{pre}}}{K_{J,\text{ret}}} &= R_{\text{act}}^{-1/2}. \end{aligned}$$

The resistance and conductance relations are independent of the orientation of action. The flux–frequency pair preserves that orientation and transforms its two components reciprocally.

Thermal constants and radiation: Boltzmann, Wien and Stefan–Boltzmann

37.1 Relation termica intrinseca of the escala 108

The chronological structure 108 fixes

$$k_B \Theta_{\text{clk}} \log 3 = \frac{h_{\text{int}}}{108 t_0} = \frac{2\pi \hbar_{\text{int}}}{108 t_0}. \quad (37.1)$$

This equality does not separately determine a numerical coordinate of k_B and another of Θ_{clk} without an additional normalization. It does determine their product as a section of energy. The factor $\log 3$ has three compatible antecedents:

$$\log 3 = C_1 + C_2 - 2C_0 = \frac{S_{\hat{T}}}{k_B} = \frac{E_P}{k_B \Theta_{\text{clk}}}. \quad (37.2)$$

The three equalities correspond to residual, informational and thermodynamic realizations of the same scalar. The coincidence of value does not identify their respective domains.

37.2 Spectral formulations of Wien's displacement law

For the spectral density per wavelength, the maximum satisfies

$$5(1 - e^{-x_5}) = x_5, \quad x_5 = 5 + W_0(-5e^{-5}).$$

With equation (37.1),

$$\lambda_{\text{max}}(\Theta_{\text{clk}}) = \frac{108 \log 3}{x_5} \ell_0. \quad (37.3)$$

For the frequency density,

$$3(1 - e^{-x_3}) = x_3, \quad x_3 = 3 + W_0(-3e^{-3}),$$

and

$$\nu_{\text{max}}(\Theta_{\text{clk}}) = \frac{x_3}{108 t_0 \log 3}. \quad (37.4)$$

The maxima are not reciprocals, since the spectral densities transform with the corresponding Jacobian. The quotient x_5/x_3 belongs to the structure of the parameterization and does not represent an inconsistency.

37.3 Stefan–Boltzmann law and thermodynamics of the photon gas

Evaluating the radiation law at Θ_{clk} produces the intrinsic flux

$$j_{\star}(\Theta_{\text{clk}}) = \frac{2\pi^5}{15 \cdot 108^4 (\log 3)^4} \frac{h_{\text{int}}}{\ell_0^2 t_0^2}, \quad (37.5)$$

in the declared normalization. The Stefan–Boltzmann constant is not inserted as a primitive: it results from integrating the bosonic spectrum once action, propagation, and temperature have been fixed. The spectral laws employed are those of Planck and Wien [Pla01, Wie93]; HMT fixes here the section on which they are evaluated, not their constants by retrospective comparison.

In units of the Planck scale,

$$n_\gamma \ell_P^3 = \frac{2\zeta(3)}{\pi^2 \log^3 3}, \quad (37.6)$$

$$\frac{u_\gamma \ell_P^3}{E_P} = \frac{\pi^2}{15 \log^4 3}, \quad (37.7)$$

$$\frac{s_\gamma \ell_P^3}{k_B} = \frac{4\pi^2}{45 \log^3 3}. \quad (37.8)$$

$\zeta(3)$ represents the third-order bosonic moment, whereas $\log 3$ determines the energy associated with the TRIT decision. The equation preserves the distinction between the two invariants.

The quantum thermal conductance is expressed as

$$\frac{\kappa_Q t_0}{k_B} = \frac{\pi^2}{324 \log 3}. \quad (37.9)$$

37.4 Metrological Confluence of Thermal and Radiative Invariants

Object	Equation internal	Significado estructural
R_K	$Z_{\text{vac}}/(2\alpha)$	bidirectionally invariant quantum resistance
G_0	$4\alpha/Z_{\text{vac}}$	conductancia complementaria
Φ_0	$\sqrt{hZ}/(8\alpha)$	flow preserving orientation of action
K_J	Φ_0^{-1}	reciprocal relation between frequency and flux
E_P	$k_B \Theta \log 3$	energy of the cycle and TRIT informational content
Wien's law	$108 \log 3/x_5$	endpoint spectral in the scale chronological
Photonic Apéry	$2\zeta(3)/(\pi^2 \log^3 3)$	number density of the bosonic field

Remark 37.1 (Negative control). The block is falsified if a change of spectral variable improperly preserves the same Wien extremum, if $\zeta(3)$ replaces $\log 3$, or if separate values are assigned to k_B and Θ without specifying a chart. The identities the corresponding identity provide direct algebraic controls.

Logical Scope and Domain of Validity

Electric quanta and Planck–TRIT confluence: *Exact internal theorem*. Wien, Stefan–Boltzmann, and the photonic gas: *Exact mathematical composition* with the declared bosonic continuum. The SI chart remains an *External metrological comparison*.

Electroweak constants, W/Z relation and angular mixing

38.1 Angular determinant and Weinberg electroweak relation

The determinant of T is

$$\det T = q_+ q_- = D_A. \quad (38.1)$$

The constitutive sector of the vacuum evaluates $F(T)^2$, where $F(z) = (1 - z^{90})/(1 - z^{120})$, whereas the electroweak angular sector evaluates $\det T$. These are two functions of a common antecedent, not a horizontal factorization between constants. This distinction is developed in the [section 38.1](#).

Remark 38.1 (Negative control). The block is refuted if any of these exact identities is violated: positivity of $I - T^{120}$, equality [equation \(35.8\)](#), relations [equation \(35.5\)](#), or compatibility $E_m(\mathcal{V}_{m+1}) = \mathcal{V}_m$. An SI coincidence cannot compensate for an operator failure.

Logical Scope and Domain of Validity

Operator, spectrum, completion $3 \rightarrow 4$, parity, amplification, and constitutive sections: *Exact internal theorem*. The formulation that unites them in a single operator is *New formalization*; the angular data and lines are an *Antecedent demonstrated result* of the preceding HMT development and are declared here as explicit initial structure.

38.2 Constitutive and Determinantal Functions of the Angular Operator

Recall the two-sheet operator of the [chapter 30](#),

$$T = q_+ P_+ + q_- P_-, \quad q_- = e^{-(A_{\text{rad}} - C_{\text{rad}}^*)}, \quad q_+ = e^{-(A_{\text{rad}} + C_{\text{rad}}^*)}. \quad (38.2)$$

The electromagnetic vacuum applies the function

$$F(T)^2, \quad F(z) = \frac{1 - z^{90}}{1 - z^{120}}, \quad (38.3)$$

while the electroweak angular sector applies the determinant

$$\boxed{\det T = q_+ q_- = e^{-2A_{\text{rad}}} =: D_A.} \quad (38.4)$$

They are two functional realizations of the common antecedent (A, C^*, T) . Generation starts from T ; recovery of that representation from the complete constitutive operator is also possible in the contractive domain. By [theorem 35.5](#),

$$T = [\psi((F(T)^2)^{1/2})]^{1/30},$$

and therefore $\det T$ is also recovered. The isolated determinant does not reconstruct the two channels: the positive counterexample in the same theorem preserves their product and modifies their responses. Structural unity resides in T , its projectors and provenance memory; functional recovery preserves that order and does not identify the different readings of permeability, permittivity and θ_W .

Under the sheet involution $C^* \mapsto -C^*$, the eigenvalues q_+, q_- are exchanged. Therefore D_A and the constitutive speed are even, while $(\hat{\mu}, \hat{\epsilon})$ is exchanged and $\hat{Z}$ is inverted. The following electroweak relation uses the even invariant.

38.3 Exact relation between sectors W and Z

The angular coordinate of duration–perduration is

$$\chi_{\text{dur}}(n_A, s_C) = \exp\left(n_A A_{\text{rad}} + \frac{s_C}{6} C_{\text{rad}}^*\right). \quad (38.5)$$

This coordinate is not introduced as an isolated label. Its origin is

$$\begin{aligned} \text{APP} &\longrightarrow \hat{\mathbb{T}} \longrightarrow \text{TPK} \longrightarrow \text{surviving extension} \longrightarrow \text{route} \\ &\longrightarrow \text{ledger} \longrightarrow (n_A, s_C, \dots) \longrightarrow \chi_{\text{dur}}. \end{aligned} \quad (38.6)$$

The signatures of W , Z and H are typed quantities obtained through that chain; this chapter adopts them without selecting them from their terminal masses. In the established angular realization, the charged and neutral sectors satisfy

$$(n_A, s_C)_W = (94, -1), \quad (n_A, s_C)_Z = (95, -1). \quad (38.7)$$

This assertion refers to the common angular component. The additional refinement coordinates of the General Mass Law are not eliminated unless the physical transducer proves their cancellation.

Theorem 38.2 (Weinberg defect in the physical-mass scheme of the angular chart). *If W and Z are evaluated on the same scale section and share the refinement that accompanies equation (38.7), then*

$$\boxed{\frac{m_W^\angle}{m_Z^\angle} = e^{-A_{\text{rad}}} = \sqrt{D_A}, \quad \left(\frac{m_W^\angle}{m_Z^\angle}\right)^2 = D_A.} \quad (38.8)$$

In the physical-mass scheme,

$$\boxed{\sin^2 \theta_W^{\text{OS,HMT}} = 1 - D_A = 0.2248708807528435698111159035\dots} \quad (38.9)$$

Moreover,

$$\frac{m_W^\angle}{m_Z^\angle} = 0.8804141748331613670128885459\dots \quad (38.10)$$

Proof. By the corresponding identity,

$$\frac{\chi_{\text{dur}}(94, -1)}{\chi_{\text{dur}}(95, -1)} = \exp(-A_{\text{rad}});$$

the coordinate C^* cancels because both sectors share $s_C = -1$. Squaring and using equation (38.4) yields D_A . The definition of physical masses $s_W^2 = 1 - m_W^2/m_Z^2$ yields equation (38.9). $\square$

The principal content is the identity between two internal constructions: the determinant of the angular operator and the one-step difference in the realization W/Z . Promotion to a physical gauge selector requires an intertwiner that carries the sheets of T to the neutral basis (W^3, B) and demonstrates which refinements are preserved or cancel.

38.4 Gauge couplings in rationalized normalization

We define the dimensionless gauge coupling

$$e_g := \sqrt{4\pi\alpha} = 0.3028221208717526427662442689 \dots \quad (38.11)$$

e_g must not be confused with a dimensional charge section e_{int} : the former is a coupling in natural units; the latter lives in the charge line of Dimensional Mechanics.

Let

$$s_W^2 = 1 - D_A, \quad c_W^2 = D_A. \quad (38.12)$$

The convention electrodebil racionalizada to level of arbol gives

$$g = \frac{e_g}{s_W} = 2\sqrt{\frac{\pi\alpha}{1 - D_A}} = 0.6385883427334574283647003809 \dots, \quad (38.13)$$

$$g' = \frac{e_g}{c_W} = 2\sqrt{\frac{\pi\alpha}{D_A}} = 0.3439541633108502429362319257 \dots, \quad (38.14)$$

$$\frac{g'}{g} = \sqrt{D_A^{-1} - 1} = 0.5386164141966093594996610933 \dots \quad (38.15)$$

Proposition 38.3 (Tree-level custodial relation). *At the same interface,*

$$\rho_{\text{EW}}^{(0)} := \frac{(m_W^{\angle})^2}{(m_Z^{\angle})^2 c_W^2} = 1. \quad (38.16)$$

Proof. By the corresponding identity, both the relative numerator and c_W^2 are D_A . $\square$

The equations the corresponding identity are exact compositions once the physical mass scheme, rationalization, and scale have been declared. They do not assert that a change of renormalization scheme leaves their numerical values unchanged.

38.5 Fermi Constant in the Approximation at Tree Level

In natural units, integration of the charged propagator at low energy yields

$$\frac{G_F^{(0)}}{\sqrt{2}} = \frac{g^2}{8m_W^2}. \quad (38.17)$$

Substituting equation (38.13) gives the composition

$$G_F^{(0)} = \frac{\pi\alpha}{\sqrt{2}(1 - D_A)m_W^2}. \quad (38.18)$$

By adopting the previously obtained mass

$$m_W^{\text{HMT}} = 80.381592550221 \dots \text{ GeV}, \quad (38.19)$$

the terminal evaluation is

$$G_F^{(0)} = 1.1157162817659203186621784 \times 10^{-5} \text{ GeV}^{-2}. \quad (38.20)$$

The difference with respect to a renormalized Fermi constant is not used to choose any coefficient. Under the convention

$$G_F = \frac{\pi\alpha}{\sqrt{2}(1-D_A)m_W^2(1-\Delta r)}, \quad (38.21)$$

the remaining obligation is to derive Δr from the internal sector of loops, states, and scale. Opening the observed value of G_F to define Δr would be a calibrated reconstruction, not a prediction.

38.6 Higgs Coupling Oriented Dependence

In the angular chart, the relevant routes are

$$(n_A, s_C)_H = (97, 9), \quad (n_A, s_C)_W = (94, -1). \quad (38.22)$$

Therefore,

$$\frac{m_H^2}{m_W^2} = \frac{\chi_{\text{dur}}(97, 9)}{\chi_{\text{dur}}(94, -1)} = \exp\left(3A_{\text{rad}} + \frac{5}{3}C_{\text{rad}}^*\right) = 1.55872087212325548564 \dots \quad (38.23)$$

Here, C^* does not cancel: the scalar sector preserves the orientation that the determinant $D_A = e^{-2A_{\text{rad}}}$ does not distinguish.

Let us take the convention of the potential

$$V(H) = -\mu_H^2 H^\dagger H + \lambda_H (H^\dagger H)^2, \quad (38.24)$$

for which

$$m_H^2 = 2\lambda_H v^2, \quad m_W^2 = \frac{g^2 v^2}{4}. \quad (38.25)$$

Corollary 38.4 (Oriented Higgs self-coupling). *At the declared tree-level interface,*

$$\lambda_H = \frac{\pi\alpha}{2(1-D_A)} \exp\left(6A_{\text{rad}} + \frac{10}{3}C_{\text{rad}}^*\right) = 0.1238479115482466804662 \dots \quad (38.26)$$

Moreover,

$$G_F^{(0)} m_H^2 = \sqrt{2} \lambda_H. \quad (38.27)$$

Proof. Dividing the two equations in equation (38.25) gives $m_H^2/m_W^2 = 8\lambda_H/g^2$. With $g^2 = 4\pi\alpha/(1-D_A)$ and the square of equation (38.23), we obtain equation (38.26). On the other hand, $G_F^{(0)} = 1/(\sqrt{2}v^2)$ and $m_H^2 = 2\lambda_H v^2$ imply equation (38.27). $\square$

The value of λ_H precedes the incorporation of radiative corrections and depends on the declared convention and scale. Its structural interest is that it separates two memories: the common gauge sector uses the even invariant D_A , whereas the Higgs route explicitly retains C^* .

38.7 Determinantal closure of the electroweak sector: mixing, couplings, and constitutive spectrum

The complete chain of this chapter is

$$\begin{aligned}
(A, C^*) &\longrightarrow T \longrightarrow \begin{cases} F(T)^2 &\longrightarrow (\widehat{\mu}, \widehat{\varepsilon}, \widehat{Z}, \widehat{c}), \\ \det T = D_A &\longrightarrow (s_W^2, c_W^2), \end{cases} \\
(D_A, \alpha) &\longrightarrow (e_g, g, g'), \\
(D_A, \alpha, m_W) &\longrightarrow G_F^{(0)}, \\
(A, C^*, D_A, \alpha) &\longrightarrow \left(\frac{m_H}{m_W}, \lambda_H \right).
\end{aligned} \tag{38.28}$$

The figure always appears at the end. Neither $\sin^2 \theta_W$, nor G_F , nor λ_H is introduced as a target value for selecting A, C^* or the route signatures.

Proposition 38.5 (Main Result). *The same angular operator determines the vacuum through a rational function of its sheets and the electroweak defect through its determinant. The one-step difference between W and Z turns that determinant into $\sin^2 \theta_W^{\text{OS}} = 1 - D_A$; the Higgs route, in contrast, preserves the counterangle and preserves the orientation eliminated by the determinant.*

Remark 38.6 (Negative control). The Weinberg relation is refuted in its domain if the angular signatures W/Z do not differ only by $n_A : 94 \rightarrow 95$, if a refinement coordinate does not cancel at the declared interface, or if $(m_W^\angle/m_Z^\angle)^2 \neq D_A$. Gauge promotion is blocked if no intertwiner with the basis (W^3, B) exists. The figures of g, g', G_F, λ_H change if the scheme or scale is altered without recording it. Using observed G_F or m_H to select Δr , the signatures, or C^* introduces feedback from the target value and invalidates the derivation. Conflating e_g with the dimensional charge constitutes a type error.

Logical Scope and Domain of Validity

Angular determinant, quotient W/Z and complementary Weinberg defect: *Exact internal theorem in the conditional angular chart.* Couplings $g, g', \rho_{\text{EW}}^{(0)}$, Fermi and Higgs: *Exact composition at the tree-level electroweak interface.* The gauge-basis intertwiner and the internal correction Δr : *Open frontier with a refutation criterion.* Fermi and Higgs are not independent constants selected without target values: they are exact compositions using the signatures, the quantity m_W and the declared tree-level convention. Subsequent conventional values are *External metrological comparison.*

38.8 Involution of leaves in 4 representations

The involution $C^* \mapsto -C^*$ organizes four distinct responses:

$$\begin{aligned}
S_{\text{bi}}(A, -C^*) &= S_{\text{bi}}(A, C^*), \\
\Gamma_{6 \rightarrow 8}(A, -C^*) &= -\Gamma_{6 \rightarrow 8}(A, C^*), \\
(\widehat{\mu}, \widehat{\varepsilon}) &\longleftrightarrow (\widehat{\varepsilon}, \widehat{\mu}), \\
\widehat{c} &\longmapsto \widehat{c}, \quad D_A \longmapsto D_A.
\end{aligned} \tag{38.29}$$

The bilayer information is even; the areal orientation is odd; the two constitutive quantities form a doublet, and the determinants remain fixed. This parity table precedes any interpretation as a physical symmetry.

38.9 Involutive rational transformation in the CKM parameter space

Let

$$M_{\text{CKM}} = \begin{pmatrix} 2 & -5 & 28 \\ 0 & 7 & -25/2 \\ 1 & -20 & -2/3 \end{pmatrix}, \quad \det M_{\text{CKM}} = -\frac{3857}{6}. \quad (38.30)$$

The inverse exists on $\mathbb{Q}$ and is

$$M_{\text{CKM}}^{-1} = \begin{pmatrix} 1528/3857 & 3380/3857 & 801/3857 \\ 75/3857 & 176/3857 & -150/3857 \\ 6/551 & -30/551 & -12/551 \end{pmatrix}. \quad (38.31)$$

With

$$h = \frac{C^*}{6}, \quad \Delta = \frac{180}{\pi} \Delta_4, \quad \theta = M_{\text{CKM}}(A, h, \Delta)^T, \quad (38.32)$$

the odd coordinate enters in the direction

$$c_h = \begin{pmatrix} -5 \\ 7 \\ -20 \end{pmatrix}. \quad (38.33)$$

If $c_A = (2, 0, 1)^T$ and $c_\Delta = (28, -25/2, -2/3)^T$, the three columns (c_A, c_h, c_Δ) are precisely the basis transported by M_{CKM} . The second row of equation (38.31) is the covector ℓ_h that extracts the odd coordinate:

$$\ell_h c_A = 0, \quad \ell_h c_h = 1, \quad \ell_h c_\Delta = 0. \quad (38.34)$$

Theorem 38.7 (Reflexion CKM inducida by the sheet). *Let*

$$D_h = \text{diag}(1, -1, 1), \quad \ell_h = \frac{1}{3857} \begin{pmatrix} 75 & 176 & -150 \end{pmatrix}. \quad (38.35)$$

The involution induced on the space of θ is

$$\boxed{\mathcal{R}_{\text{CKM}} = M_{\text{CKM}} D_h M_{\text{CKM}}^{-1} = I - 2c_h \ell_h.} \quad (38.36)$$

Satisfies

$$\mathcal{R}_{\text{CKM}}^2 = I, \quad \det \mathcal{R}_{\text{CKM}} = -1, \quad \mathcal{R}_{\text{CKM}} M_{\text{CKM}} = M_{\text{CKM}} D_h. \quad (38.37)$$

Proof. Multiplication of the corresponding identity gives the identity; in particular, equation (38.34) proves that $c_h \ell_h$ is the rank-one projector onto $\mathbb{Q}c_h$ along $\text{span}\{c_A, c_\Delta\}$. The exact product $\ell_h c_h = 1$ implies

$$(I - 2c_h \ell_h)^2 = I - 4c_h \ell_h + 4c_h (\ell_h c_h) \ell_h = I.$$

The operator fixes the hyperplane $\ker \ell_h = \text{span}\{c_A, c_\Delta\}$ pointwise and changes the sign of c_h . Its eigenvalues are therefore 1, 1, -1; hence $\det \mathcal{R}_{\text{CKM}} = -1$ and $\text{Tr} \mathcal{R}_{\text{CKM}} = 1$. Finally,

$$\mathcal{R}_{\text{CKM}} M_{\text{CKM}} = M_{\text{CKM}} D_h M_{\text{CKM}}^{-1} M_{\text{CKM}} = M_{\text{CKM}} D_h,$$

which proves the entanglement identity. $\square$

The explicit matrix is

$$\mathcal{R}_{\text{CKM}} = \begin{pmatrix} 4607/3857 & 1760/3857 & -1500/3857 \\ -150/551 & 199/551 & 300/551 \\ 3000/3857 & 7040/3857 & -2143/3857 \end{pmatrix}. \quad (38.38)$$

Although $\mathcal{R}_{\text{CKM}}$ is not an orthogonal reflection for the Euclidean product of the three angular coordinates, it is for the transported rational metric

$$G_{\text{CKM}} = (M_{\text{CKM}}^{-1})^T M_{\text{CKM}}^{-1}, \quad \mathcal{R}_{\text{CKM}}^T G_{\text{CKM}} \mathcal{R}_{\text{CKM}} = G_{\text{CKM}}. \quad (38.39)$$

Indeed, in the basis (c_A, c_h, c_Δ) , the metric is the identity and the involution is D_h . Hence the term “reflection” has precise metric content and does not merely describe that $\mathcal{R}_{\text{CKM}}^2 = I$. Phase $\delta_{\text{CP}} = 9A$ remains fixed under this reflection. Therefore, $\mathcal{R}_{\text{CKM}}$ is not physical CP conjugation, but rather the transport of HMT sheet inversion to mixing space.

The rational inverse of M_{CKM} recovers

$$A = \frac{1528\theta_{12} + 3380\theta_{23} + 801\theta_{13}}{3857}, \quad (38.40)$$

$$C^* = \frac{6(75\theta_{12} + 176\theta_{23} - 150\theta_{13})}{3857}. \quad (38.41)$$

The third coordinate is also recovered without loss:

$$\Delta = \frac{6\theta_{12} - 30\theta_{23} - 12\theta_{13}}{551}. \quad (38.42)$$

These equations are coordinate changes. They do not turn CKM into a retrospective cause of A, C^* or of γ_{HMT} .

As a terminal recognition, the previous constructor determines

$$(\theta_{12}, \theta_{23}, \theta_{13}, \delta_{\text{CP}}) = (13.003313589509^\circ, 2.398056157332^\circ, 0.213977382244^\circ, 65.676173123554^\circ). \quad (38.43)$$

The rational reflection is proved before using these decimals.

Proposition 38.8 (Main Result). *The transition $6 \rightarrow 8$ is first formalized through the inclusion $\iota_{6,8} : \mathcal{F}_6 \hookrightarrow \mathcal{F}_8$, from which 90, 120 and the defect $-1/12$ are obtained. The six-trit word then provides the bijection $729 \leftrightarrow 729$, and $\mathcal{J}_{6 \rightarrow 8}$ transports the two collective modes to the boundary. On that antecedent, area and the modular Hamiltonian exactly reconstruct the two families; thermofield-double purification determines the entanglement entropy, and the finite first law preserves the relative-entropy defect. In CKM, sheet inversion becomes a rational rank-one odd reflection.*

Remark 38.9 (Negative control). The initial genealogy fails if $\iota_{6,8}$ is not injective, if the counts are not 90/120, if $\delta_{6,8}^{\text{inc}} \neq -1/12$, if β_{729} is not bijective or if it is falsely presented as an additive isomorphism. Collective transport fails if $\mathcal{J}_{6 \rightarrow 8} D_{\text{inc}} \neq D_{\text{area}} \mathcal{J}_{6 \rightarrow 8}$. Reconstruction fails if the determinant of the boundary observables is not thirty or if joint reconstruction identity of the area channels does not recover the occupations. The TFD fails if its partial trace is not ρ_A . The first law does not extend to finite changes without the relative entropy of first finite modular identity with defect term. Type III_{λ_θ} is invalidated if a channel does not persist, if there is no compatible product or if the ratio group changes. The CKM reflection is refuted by any of $\mathcal{R}^2 \neq I$, $\det \mathcal{R} \neq -1$ or $\mathcal{R}M \neq MD_h$. A formula $\theta = f(\gamma_{\text{HMT}})$ without an intertwiner violates the typed independence of the two functional realizations of the common antecedent.

Logical Scope and Domain of Validity

Inclusion $H \subset O$, counts 90/120, two normalizations of the defect $-1/12$, bijection of six trits $729 \leftrightarrow 729$, collective intertwiner, scale a_0 , finite operators, area spectrum, TFD, oriented defect, local and finite first law, and CKM reflection: *Internally exact theorem with its explicit starting structures*. Classification III_{λ_θ} : *Theorem with explicit persistence hypotheses*. Interpretation of the thermofield-double copy as a physical exterior: *Conditional physical realization*. The magnitude γ_{HMT} is adopted as *Proved antecedent result*; it is not rederived in this volume.

The electroweak link starts from the same angular operator that produced the vacuum constitutive relations, but uses another functional.

To obtain μ , ε , Z , and c it was necessary to preserve the full spectrum. It was necessary to distinguish the two sheets.

The determinant does the opposite:

$$D_A = \det T = q_+ q_-.$$

By multiplying the eigenvalues, the determinant integrates the two orientations into a common invariant angular magnitude. It publishes the even part and reserves the leaf identity in the oriented memory.

By eso D_A is even under the inversion

$$C^* \mapsto -C^*.$$

The sector W/Z has precisely that structure. The routes of W and Z are adjacent in the principal angular coordinate and share the same oriented coordinate. Upon forming the quotient, the contribution of C^* cancels.

Remains

$$\left(\frac{m_W}{m_Z}\right)^2 = D_A,$$

and therefore

$$\sin^2 \theta_W = 1 - D_A.$$

The identity reveals that the Weinberg angle is the complementary defect of the angular determinant.

The determinant measures the common part that survives when the two sheets are collapsed into their product. $1 - D_A$ measures the complementary part required for mixing.

It may be stated as follows: the vacuum preserves the entire response; the electroweak sector uses the even compression of that response.

The same angular antecedent produces two distinct functions:

- the complete functional calculus generates the constitutive relations;
- the determinant generates the mixing invariant.

This explains why the vacuum and the Weinberg angle are two distinct functional realizations of the same angular antecedent.

From that point onward, the couplings g and g' are, at the stated level, the two projections of the same rationalized electromagnetic coupling onto the sine and cosine directions of the mixing:

$$g^2 = \frac{4\pi\alpha}{1 - D_A}, \quad g'^2 = \frac{4\pi\alpha}{D_A}.$$

The tree custodial identity

$$\rho_{EW}^{(0)} = 1$$

states that the same angular partition organizes mass and coupling coherently.

The Fermi constant appears later as the effective low-energy response of the charged channel:

$$G_F^{(0)} = \frac{\pi\alpha}{\sqrt{2}(1 - D_A)m_W^2}.$$

The Fermi constant appears as a composition of the fine structure, the angular defect, and the carrier scale W .

The Higgs sector introduces a decisive difference. Its route has an oriented coordinate distinct from the route of W , so C^* persists. The scalar self-coupling publishes the orientation reserved by the determinant.

The gauge sector is even: it uses D_A .

The scalar sector is oriented: it retains C^* .

This division is physically suggestive. Gauge fields organize the common part of the transformation; the scalar sector preserves the memory of the broken orientation.

Radiative corrections belong to a later level and must be constructed prospectively from the internal loops, states, and scales, with G_F as output and terminal evaluation. The causal skeleton is already present: vacuum, mixing, couplings, and Fermi response follow from different readings of the same enriched angular state.

38.10 Electroweak Angular Closure and Compatibility Conditions of the Gauge Transducer

Let

$$c_W = \sqrt{D_A}, \quad s_W = \sqrt{1 - D_A}.$$

The rotation

$$\begin{pmatrix} Z \\ A \end{pmatrix} = \begin{pmatrix} c_W & -s_W \\ s_W & c_W \end{pmatrix} \begin{pmatrix} W^3 \\ B \end{pmatrix}$$

is orthogonal and preserves orientation. The identities

$$\left(\frac{m_W}{m_Z}\right)^2 = D_A, \quad \sin^2 \theta_W = 1 - D_A$$

are closed in the angular chart. Gauge promotion requires that the transducer preserve the kinetic form, the charges, and the Ward identities; the correction Δr belongs to radiative dynamics and is not obtained by fitting it to G_F .

Intrinsic derivation of gravitation and contragredient reciprocity $G \leftrightarrow \hbar$

39.1 Simultaneous rigidity of the longitudinal reader and radial coupling

The Triadic Modular Holography construction transmits to the radial realization an enriched state with its coordinates, orientation, carry, memory and incidences. The sequence

$$\begin{aligned} \text{APP} &\longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow \text{enriched state} \\ &\longrightarrow \text{joint discrete structure of the continuum} \end{aligned}$$

preserves the Archimedean projection and incidence projection, together with the five inherent constructions of the continuum. The extension

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

is articulated with the operator order $U_t \longrightarrow \Gamma_9 \longrightarrow K_{\text{ph}}$: phase returns and the memory record increases. The regional coordinates, selected record K , coordinate α and return action are received with this common provenance and with their previously constructed readers.

The result of this section is a classification of realizations simultaneously preserving the generated data and the longitudinal, metric and temporal rules specified below. For each positive character the radial response is determined; the coefficient relating that response to mass is common to all admissible characters. The gravitational interpretation is formulated after both readings have been constructed on the same space.

39.1.1 Incidence record and constitutive constraints

Consider complex Hilbert spaces V_{in} and V_{out} of dimension N with fixed inner products. The marked cardinality specified in R1 is positive; in particular, both fibres are nonzero. Adjoints are taken with respect to these products. The symbol $\mathbf{U}$ will denote normalized archive transport, and U_t will retain its meaning as evolution of the enriched state.

R1. Marked record. The incidence set is

$$\Omega = \left\{ (j, k, q, u, v) : \begin{array}{l} j, k \in \mathbb{Z}/12\mathbb{Z}, \quad k - j \in \{0, 3, 6, 9\}, \\ 1 \leq q \leq 8, \quad u < v, \quad u, v \in \{1, 2, 4, 5, 7, 8\} \end{array} \right\}. \quad (39.1)$$

The twelve phases, four relative positions, eight positions of index q and fifteen pairs in the last set give

$$N = |\Omega| = 12 \cdot 4 \cdot 8 \binom{6}{2} = 12 \cdot 4 \cdot 120 = 5760. \quad (39.2)$$

Each incidence has its individual mark. In the corresponding basis $(e_i)_{i \in \Omega}$ of V_{out} , let $P_i = e_i e_i^*$. The common mode u is normalized by $|u_i|^2 = 1/N$; write $P = uu^*$. The marked algebra is $\mathcal{A}_\Omega = \text{span}\{P_i : i \in \Omega\}$.

R2. Normalization and complete archive. The generated inverter $B : V_{\text{in}} \rightarrow V_{\text{out}}$ satisfies

$$B^* B = \frac{1}{4} I_{\text{in}}, \quad B B^* = \frac{1}{4} I_{\text{out}}. \quad (39.3)$$

The two components are preserved

$$C_0 = (I - P)B, \quad M_0 = PB, \quad C_0 + M_0 = B, \quad U = 2B. \quad (39.4)$$

The first records variation relative to the common mode; the second preserves that mode. Their images are orthogonal and their sum recovers the complete transport. By (39.3), U is unitary.

R3. Individual inscription. The map $\mathcal{E} : \text{End}(V_{\text{out}}) \rightarrow \mathcal{A}_\Omega$ is a linear conditional expectation fixing each P_i and satisfying

$$\mathcal{E}(A_1 T A_2) = A_1 \mathcal{E}(T) A_2, \quad A_1, A_2 \in \mathcal{A}_\Omega. \quad (39.5)$$

Inscription of the result preserves the complete construction data in a complementary record. The expectation determines the inscribed observable; archive preservation determines its subsequent reconstruction.

R4. Longitudinal composition. Fix the generated coordinate $\alpha > 0$ and the longitudinal reference section $L_* > 0$. Longitudinal transport is defined by

$$D = L_* \alpha^{16} \mathcal{E}(C_0 C_0^*) U : V_{\text{in}} \longrightarrow V_{\text{out}}. \quad (39.6)$$

This rule applies the inscribed coefficient linearly to the longitudinal section. Its type fixes the position of the power α^{16} and return coefficient within the composition. Reading an intensity root is a different operation with its own transformation rule.

R5. Positive radial metric. Let $X = X^* > 0$ be the complete character on V_{in} . Define the positive radial reading R_X on V_{out} by

$$\|R_X v\|^2 = \|X D^* v\|^2 \quad (v \in V_{\text{out}}). \quad (39.7)$$

This equality fixes the metric on every vector of the fibre. In polar notation, R_X is the arrival modulus $|(DX)^*| = \sqrt{DX^2 D^*}$.

R6. Action and clock. The generated action section $\hbar_{\text{ret}} > 0$ and the real phase lift satisfy

$$S(\theta) = \hbar_{\text{ret}} \theta. \quad (39.8)$$

The constitutive velocity $c > 0$ is preserved. Once the intensive length L of (39.6) is obtained, the conjugate clock satisfies $\omega L = c$. For $\theta(t) = \omega t$, the action rate determines the energy scale

$$E_L = \frac{dS}{dt} = \hbar_{\text{ret}} \omega = \frac{\hbar_{\text{ret}} c}{L}. \quad (39.9)$$

The lift and its refinements remain associated with route memory.

R7. Readings of the same character. On $Y = UXU^* > 0$, define

$$H_X = E_L Y, \quad M_X = \frac{H_X}{c^2}, \quad \Lambda_X = LY^{-1}. \quad (39.10)$$

Energy, mass and conjugate length preserve the same character and transport. The inverse acts in the positive sector; the null sector retains its own treatment.

R8. Reduced radius and dimensional sections. The physical identification of this realization is

$$R_X = \frac{G}{c^2} M_X. \quad (39.11)$$

Thus R_X represents the reduced gravitational radius. The corresponding Schwarzschild radius is $2R_X$. L_* , the action section and the constitutive velocity section are jointly maintained; a change of units transforms their coordinates according to their dimensions.

Provenance of the constraints. Hadamard transport, archive preservation, inscription, phase action and reciprocity have antecedents in the corpus. The expanded record (39.1) and compositions (39.6)–(39.7) are explicitly incorporated in the assembled construction. R4 and R5 are constitutive rules of this realization; the following proof establishes their joint consequence with the remaining antecedents. R8 specifies the physical identification fixing the coefficient's meaning.

39.1.2 Simultaneous determination theorem

Theorem. For the same generated data B , (P_i) , P , α , L_* , $\hbar_{\text{ret}}$, c and the same character X , constraints R1–R8 uniquely determine inscription, longitudinal transport and radial response. For all admissible positive characters, the coefficient G is common. One has

$$\mathcal{E} = \Delta_\Omega, \quad \Delta_\Omega(T) = \sum_{i \in \Omega} P_i T P_i, \quad (39.12)$$

$$r_N = \frac{N-1}{4N} = \frac{5759}{23040}, \quad L = L_* \alpha^{16} r_N, \quad (39.13)$$

$$D = LU, \quad R_X = LUXU^*, \quad (39.14)$$

$$R_X \Lambda_X = L^2 I, \quad H_X \Lambda_X = \hbar_{\text{ret}} c I. \quad (39.15)$$

The coupling is determined by

$$G = \frac{c^3 L^2}{\hbar_{\text{ret}}} = \frac{c^3 L_*^2}{\hbar_{\text{ret}}} \left(\frac{5759}{23040} \right)^2 \alpha^{32}. \quad (39.16)$$

Proof: uniqueness of inscription. Let $E_{ij} = e_i e_j^*$ be the matrix units. By bimodularity,

$$\mathcal{E}(E_{ij}) = P_i \mathcal{E}(E_{ij}) P_j.$$

For $i \neq j$, membership of $\mathcal{E}(E_{ij})$ in the diagonal algebra implies $P_i \mathcal{E}(E_{ij}) P_j = 0$. For $i = j$, fixing P_i gives $\mathcal{E}(E_{ii}) = P_i$. Linearity then determines (39.12) on all matrices. Marked inscription is thus fixed by its algebraic properties.

Proof: return coefficient and archive. Normalization of B and $P^2 = P = P^*$ give

$$C_0 C_0^* = \frac{1}{4} (I - P), \quad M_0 M_0^* = \frac{1}{4} P.$$

Since $P_i P P_i = |u_i|^2 P_i = P_i / N$, one obtains

$$\Delta_\Omega(C_0 C_0^*) = \frac{N-1}{4N} I, \quad \Delta_\Omega(M_0 M_0^*) = \frac{1}{4N} I. \quad (39.17)$$

Both readings sum to $I/4$. The first is scalar, so any normalized distribution (p_i) on incidences, with $p_i \geq 0$ and $\sum_i p_i = 1$, produces $\sum_i p_i r_N = r_N$. The same coefficient is therefore preserved under arbitrary diagonal weighting.

Proof: length and metric. Substituting (39.17) into R4 gives $D = LU$, with L as in (39.13). Unitarity of U implies $DD^* = L^2 I$. By R5, the quadratic forms of R_X^2 and $DX^2 D^*$ coincide. The polarization identity determines

$$R_X^2 = DX^2D^* = L^2UX^2U^*.$$

The operator $LUXU^*$ is positive and its square is the operator on the right-hand side. Uniqueness of the positive square root proves (39.14). In particular, $R_I = LI$.

Proof: action, reciprocity and coupling. R6 and R7 give

$$H_X = \frac{\hbar_{\text{ret}}c}{L}Y, \quad M_X = \frac{\hbar_{\text{ret}}}{cL}Y, \quad \Lambda_X = LY^{-1}.$$

Multiplication produces both products of (39.15). Next, R8 is equivalent to

$$LY = \frac{G\hbar_{\text{ret}}}{c^3L}Y.$$

Invertibility of Y determines (39.16). The same substitution verifies the equality for each positive character. If G_1 and G_2 satisfy the identity with the same data, $(G_1 - G_2)Y = 0$, and therefore $G_1 = G_2$. This conclusion distinguishes the state-dependent response R_X from the common coefficient G . $\square$

39.1.3 Reciprocity and phase rigidity

The power–logarithmic antecedent uses, for $x > 0$,

$$\ell_p(x) = \frac{x^p - 1}{p} \quad (p \neq 0), \quad \ell_0(x) = \log x.$$

Direct substitution proves $\ell_{-p}(x^{-1}) = -\ell_p(x)$. On the positive domain of their inverses, this identity is equivalent to

$$\exp_p(s) \exp_{-p}(-s) = 1.$$

Spectral composition on a positive character transmits this product to the pair LY, LY^{-1} . In the present realization, R5 can equivalently be expressed, once R1–R4 and R7 are fixed, by $R_X\Lambda_X = L^2I$: multiplication by Λ_X^{-1} determines $R_X = LUXU^*$, and this expression satisfies R5.

The two products of (39.15) also give a direct form of rigidity:

$$R_X = \frac{L^2}{\hbar_{\text{ret}}c}H_X. \quad (39.18)$$

For a positive rescaling s , the pair $(sR_X, \Lambda_X/s)$ preserves the radial product; the product $H_X(\Lambda_X/s) = \hbar_{\text{ret}}cI/s$, with energy and action fixed, requires $s = 1$. Joint preservation determines the scale. Likewise, any positive radial operator with the same metric R5 equals R_X by uniqueness of the positive root.

The lifted action is also exactly determined. For $\theta \neq 0$, the equality $S/\hbar_{\text{ret}} = \theta$ fixes S . Even when rotations of all subdivisions $\theta/9^n$ are preserved, a fixed real rescaling a satisfying

$$\exp(-ia\theta/9^n) = \exp(-i\theta/9^n) \quad (n \geq 0)$$

satisfies $(a - 1)\theta/9^n \in 2\pi\mathbb{Z}$. This sequence tends to zero; for sufficiently large n its only possible value is zero. From $\theta \neq 0$ one deduces $a = 1$.

39.1.4 Planck scales and specialization of the unit character

Write $L_\alpha = L$; on the return section, $E_{P,\text{ret}} = E_L$.

The constructed length $L_\alpha > 0$, action section $\hbar_a > 0$ and velocity $c > 0$ determine the scales

$$t_P = \frac{L_\alpha}{c}, \quad m_{P,a} = \frac{\hbar_a}{cL_\alpha}, \quad E_{P,a} = \frac{\hbar_a c}{L_\alpha}.$$

Indeed, substituting $G_a = c^3 L_\alpha^2 / \hbar_a$ into the expressions $\sqrt{\hbar_a G_a / c^5}$, $\sqrt{\hbar_a c / G_a}$ and $\sqrt{\hbar_a c^5 / G_a}$ gives those three identities respectively. Positivity fixes the root in each case. Length precedes recognition of these scales; their conventional expressions are identities of the already constructed realization.

For a mode of character $y > 0$, the mass and longitudinal readings are

$$m(y) = m_{P,a}y, \quad r_g(y) = L_\alpha y, \quad \bar{\lambda}(y) = \frac{L_\alpha}{y}.$$

Thus $r_g(y)\bar{\lambda}(y) = L_\alpha^2$ for all positive modes. Equality of the two lengths characterizes the mode $y = 1$, since $y = y^{-1}$ and $y > 0$ imply $y = 1$. In this mode, mass is $m_{P,a}$ and both lengths equal L_α . General reciprocity is thus distinguished from its specialization to the unit character.

The chronological step and Planck time satisfy $t_0 = (\pi_{\text{HMT}}/54)t_P$. The complete 108-step phase period is $108t_0 = 2\pi_{\text{HMT}}t_P$ and its angular frequency is $\omega_{108} = 1/t_P = c/L_\alpha$. A horizon of radius $R_h = \zeta_h L_\alpha$, with $\zeta_h > 0$, preserves its own geometric variable. The Schwarzschild radius associated with a mass is $2r_g$.

If the two action sections are compared with L_α and c fixed, the ratio $\rho = \hbar_b / \hbar_a$ gives $G_b = G_a / \rho$, $m_{P,b} = \rho m_{P,a}$ and $E_{P,b} = \rho E_{P,a}$; L_α and t_P remain equal. This comparison between sections remains distinct from temporal evolution of the constants.

39.1.5 Archive covariance and transformation of units

Let $T : V_{\text{in}} \rightarrow V'_{\text{in}}$ and $S : V_{\text{out}} \rightarrow V'_{\text{out}}$ be unitary. Joint transport of the data is

$$B' = SBT^*, \quad P' = SPS^*, \quad P'_i = SP_iS^*, \quad X' = TXT^*.$$

The transported expectation $\mathcal{E}'(A') = S\mathcal{E}(S^*A'S)S^*$ fixes P'_i and is bimodular in the transformed marked algebra. By substitution,

$$\begin{aligned} U' &= SUT^*, & D' &= SDT^*, \\ R'_{X'} &= SR_XS^*, & H'_{X'} &= SH_XS^*, & \Lambda'_{X'} &= S\Lambda_XS^*. \end{aligned}$$

Conjugation preserves the R1–R8 equalities and the same coefficient G . Marks, orientations and records are transported with the bases. Incidence permutations are a particular case.

Incorporating an auxiliary memory space $\mathcal{F}$ transports P to $P \otimes I_{\mathcal{F}}$, U to $U \otimes I_{\mathcal{F}}$ and Δ_Ω to $\Delta_\Omega \otimes \text{id}$. The inscription of $C_0C_0^* \otimes I_{\mathcal{F}}$ is $r_N I \otimes I_{\mathcal{F}}$; the coefficient preserves the original marked cardinality.

To specify dimensional covariance, let ℓ , τ and m be the positive factors of the new length, time and mass units relative to the previous units. Numerical coordinates satisfy

$$L' = L/\ell, \quad c' = c\tau/\ell, \quad \hbar'_{\text{ret}} = \hbar_{\text{ret}}\tau/(m\ell^2).$$

Thus,

$$\frac{(c')^3(L')^2}{\hbar'_{\text{ret}}} = G \frac{m\tau^2}{\ell^3},$$

which is the transformation of a quantity of dimension $\text{longitud}^3 \text{masa}^{-1} \text{tiempo}^{-2}$. The section L_* transforms as L , while α and r_N remain dimensionless.

39.1.6 Exact elimination of coupled memory

Let $\mathcal{K}$ and $\mathcal{H}$ be finite Hilbert spaces, $W = W^* > 0$ a global weight, and $C : \mathcal{K} \rightarrow \mathcal{H}$ surjective. The action of residuals r , with boundary defect z , is

$$\mathcal{S}(r) = \alpha r^* W r, \quad C r = z, \quad \alpha > 0.$$

The global weight preserves its cross blocks between incidences. Define $\mathcal{R} = C W^{-1} C^*$. For $v \neq 0$, injectivity of C^* gives

$$v^* \mathcal{R} v = \|W^{-1/2} C^* v\|^2 > 0.$$

Thus $\mathcal{R}$ is invertible. The least-action residual is

$$r_*(z) = W^{-1} C^* \mathcal{R}^{-1} z.$$

One verifies $C r_* = z$. Each admissible residual is written $r = r_* + k$, with $C k = 0$ and $k^* W r_* = k^* C^* \mathcal{R}^{-1} z = 0$. Consequently,

$$\mathcal{S}(r) = \alpha z^* \mathcal{R}^{-1} z + \alpha k^* W k. \quad (39.19)$$

Positivity of W proves uniqueness of the minimum. The pair (z, k) preserves and reconstructs the complete residual.

For a route with transports U_j , the residuals $r_j = f_j - U_j f_{j-1}$ are reconstructed through $f_j = U_j f_{j-1} + r_j$. The map C collects the subsequent transports of each residual towards the endpoint. If its blocks are C_j , then

$$\mathcal{R} = \sum_{j,k} C_j (W^{-1})_{jk} C_k^*.$$

This expression preserves correlations between route segments. For an invertible longitudinal transport D , the boundary $y = D z$ has response

$$\mathcal{R}_L = D \mathcal{R} D^*, \quad S_{\text{ef},L}(y) = \alpha y^* (D \mathcal{R} D^*)^{-1} y = \alpha z^* \mathcal{R}^{-1} z.$$

The same action is obtained whether memory is eliminated first or the boundary is transported first. The prefactor preserves the normalization of the source action; when that action carries $\alpha = \hbar_{\text{ret}}$, reduction transmits the same section.

39.1.7 Joint reduction of radius, energy and conjugate length

Write the same positive character in a boundary and memory decomposition:

$$Y = \begin{pmatrix} A & F \\ F^* & C \end{pmatrix}, \quad C > 0, \quad Y_{\text{ef}} = A - F C^{-1} F^*.$$

The identity

$$\begin{pmatrix} b \\ y \end{pmatrix}^* Y \begin{pmatrix} b \\ y \end{pmatrix} = b^* Y_{\text{ef}} b + \eta^* C \eta, \quad \eta = y + C^{-1} F^* b,$$

proves $Y_{\text{ef}} > 0$, characterizes the least-energy extension and preserves the reconstruction $y = \eta - C^{-1} F^* b$. For $s > 0$, substitution into the Schur complement gives $\text{Schur}(sY) = sY_{\text{ef}}$. Consequently,

$$R_{\text{ef}} = L Y_{\text{ef}}, \quad H_{\text{ef}} = E_L Y_{\text{ef}}.$$

To calculate the compressed conjugate length, solve $Y(x, y) = (f, 0)$. The second equation gives $y = -C^{-1} F^* x$, and the first gives $x = Y_{\text{ef}}^{-1} f$. The boundary block of Y^{-1} is therefore Y_{ef}^{-1} . Thus,

$$\Lambda_b = LY_{\text{ef}}^{-1}, \quad R_{\text{ef}}\Lambda_b = L^2I, \quad R_{\text{ef}} = \frac{L}{E_L}H_{\text{ef}}. \quad (39.20)$$

Reduction acts on the same blocks in both readings and preserves the residual η . The effective character may depend on memory in matrix form, while the coefficient L/E_L remains fixed.

39.1.8 Dynamic memory and induced temporal norm

Dynamic elimination preserves an additional spectral dependence. In a stationary representation, let

$$H = \begin{pmatrix} H_{bb} & V \\ V^* & H_{ii} \end{pmatrix}, \quad H_{ii} > 0.$$

For z in the relevant resolvent domain, elimination of the interior component in $(H - zI)(b, y) = (f, 0)$ gives

$$\mathcal{K}_H(z) = H_{bb} - zI - V(H_{ii} - zI)^{-1}V^*. \quad (39.21)$$

When both inverses exist, the boundary block of $(H - zI)^{-1}$ is $\mathcal{K}_H(z)^{-1}$. For $|z| < \lambda_{\min}(H_{ii})$, the convergent resolvent series gives

$$\mathcal{K}_H(z) = H_{\text{est}} - zZ - z^2VH_{ii}^{-3}V^* - \dots, \quad (39.22)$$

$$H_{\text{est}} = H_{bb} - VH_{ii}^{-1}V^*, \quad Z = I + VH_{ii}^{-2}V^* \geq I. \quad (39.23)$$

Positivity follows from $Z - I = (H_{ii}^{-1}V^*)^*(H_{ii}^{-1}V^*)$. The static extension $(b, -H_{ii}^{-1}V^*b)$ has squared norm b^*Zb ; energy and temporal norm arise from the same extension. The physical parameter is $z = \hbar_{\text{ret}}\omega$.

To first temporal order, $T = Z^{1/2}$ and $\psi = Tb$ transform the pair (H_{est}, Z) into $(T^{-*}H_{\text{est}}T^{-1}, I)$. The section $\hbar_{\text{ret}}$ remains fixed. If T depends on time, the transformation also preserves the term $\dot{T}b$ of $\dot{\psi} = \dot{T}b + T\dot{b}$.

Radial proportionality is preserved in the complete resolvent. Writing $\chi_{\text{rad}} = L/E_L$ and $R = \chi_{\text{rad}}H$, substitution of each block proves

$$\mathcal{K}_R(\chi_{\text{rad}}z) = \chi_{\text{rad}}\mathcal{K}_H(z). \quad (39.24)$$

This identity transports the operator and spectral parameter simultaneously and preserves every order of dynamic memory. For the effective forms of (39.20), canonical normalization requires dual transport of the inverse length:

$$H_{\text{can}} = T^{-*}H_{\text{ef}}T^{-1}, \quad R_{\text{can}} = T^{-*}R_{\text{ef}}T^{-1}, \quad \Lambda_{\text{can}} = T\Lambda_bT^*.$$

Multiplying these expressions gives

$$R_{\text{can}} = \frac{L}{E_L}H_{\text{can}}, \quad R_{\text{can}}\Lambda_{\text{can}} = L^2I.$$

The proof allows noncommuting T , R_{ef} and Λ_b . The induced temporal norm and dual length jointly preserve the same G .

39.1.9 Refinement, spectral domains and scope

Let J and K be isometric inscriptions between two levels satisfying $X'J = JX$ and $U'J = KU$, with the same intensive section L . Then

$$R'K = LU'X'U^*K = KLUXU^* = KR,$$

and the same calculation proves $H'K = KH$. Intertwining of the inverses is formulated on the corresponding domains. When the compatible system has a positive injective self-adjoint limit X , spectral calculus

defines $Y = UXU^*$, $R = LY$, $H = E_L Y$ and $\Lambda = LY^{-1}$. Proportionality $R = (L/E_L)H$ holds on $\text{Dom}(Y)$, whereas $R\Lambda = L^2 I$ and $H\Lambda = \hbar_{\text{ret}} c I$ hold on $\text{Dom}(Y^{-1})$, with bounded extensions of their products. Spectral cuts $\mathbf{1}_{[\varepsilon, M]}(Y)$ verify the identities before passage to the limit on these domains. A refined edge length $L/9$ incorporates its own scale factor; the intensive section is preserved according to the declared intertwining.

The classification distinguishes transformations preserving the realization from those modifying a constitutive rule. Diagonal weightings, unitary equivalences, the auxiliary archive and joint reduction preserve G . Changing the marked resolution modifies R1; replacing r_N with $\sqrt{r_N}$, α^{16} with its square, or the inscribed component with the total norm modifies R4. Isolated radius rescaling changes R5, and isolated action rescaling changes R6. R8 maintains the distinction between reduced radius, Schwarzschild radius and change of units.

One thus obtains mathematical uniqueness of the R1–R8 realization: the generated data and character fix the response, and the common composition fixes G . The longitudinal and metric rules appear explicitly in this conclusion as incorporated compositions; gravitational identification retains its physical status. The proof covers the declared class with its constitutive definitions and compatible transformations. The rational checks of the development verify finite cases of the stated identities; the general proof resides in the preceding equations and arguments.

39.2 Constitutive continuity and evaluation of the gravitational section

The longitudinal construction determines L before using gravitational identification. The composition of rules R1–R8 preserves the velocity section of (35.34) and the previously generated return action. With

$$L = \frac{5759}{23040} \alpha^{16} L_*, \quad \hbar_{\text{ret}} = \eta_{\text{ret}} S_*, \quad c_{\text{int}} = c_* \widehat{c},$$

the result of (39.16) is expressed as

$$G_{\text{ret}} = \frac{c_*^3 L_*^2}{S_*} \left(\frac{5759}{23040} \right)^2 \frac{\alpha^{32} \widehat{c}^3}{\eta_{\text{ret}}} = \frac{L^2}{\hbar_{\text{ret}} (\mu_{\text{vac}} \varepsilon_{\text{vac}})^{3/2}}. \quad (39.25)$$

The channels are the same as those of (35.1):

$$q_{\pm} = \exp \left[-\frac{\pi}{180} (A_{\text{deg}} \pm C_{\text{deg}}^*) \right], \quad r_{\pm} = \frac{1 - q_{\pm}^{90}}{1 - q_{\pm}^{120}}, \quad \widehat{c} = (r_+ r_-)^{-1}.$$

The constitutive bases satisfy $\varepsilon = \varepsilon_* r_+^2$, $\mu = \mu_* r_-^2$ and $c_* = (\mu_* \varepsilon_*)^{-1/2}$. Their product proves $c_{\text{int}} = (\mu \varepsilon)^{-1/2}$. If the dimensional coordinate of c_{int} is declared directly, its basis is recovered as $c_* = c_{\text{int}} / \widehat{c}$. The constitutive coefficient is not multiplied again: both expressions represent the same section.

In the representation $L_* = 1 \text{ m}$, $S_* = 10^{-34} \text{ J s}$ and $c_{\text{int}} = 299792458 \text{ m s}^{-1}$, evaluation of the readers and radial composition gives

$$G_{\text{ret}} = 6.674299659414022706367776189 \dots \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}. \quad (39.26)$$

The comparison value $G_{\text{CODATA 2022}} = 6.67430(15) \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ retains its experimental uncertainty. In the same units, the difference between the evaluation and the central value is $-3.40585977 \dots \times 10^{-18}$, approximately -0.00227057 standard uncertainties. The evaluation digits correspond to the declared readers and bases; uncertainty belongs to the measured result. No reference digit selects N , r_N , the longitudinal rule or the radial norm.

39.2.1 Chronological realization of the same constructed radius

With the elementary duration

$$t_0 = \frac{\pi L}{54c_{\text{int}}}, \quad \ell_0 = c_{\text{int}}t_0 = \frac{\pi L}{54}, \quad (39.27)$$

the calendar of 108 positions satisfies

$$T_{108} = 108t_0 = \frac{2\pi L}{c_{\text{int}}}, \quad \omega_{108} = \frac{c_{\text{int}}}{L}, \quad R_{108} = \frac{c_{\text{int}}}{\omega_{108}} = L.$$

By substitution,

$$\left(\frac{54}{\pi}\right)^2 \frac{c_{\text{int}}^5 t_0^2}{\hbar_a} = \frac{c_{\text{int}}^3 L^2}{\hbar_a}. \quad (39.28)$$

This identity places the circular selector retained below in the same longitudinal normalization. Its uniqueness theorem preserves its stated geometric premise; the preceding construction additionally specifies the length used in that realization.

The positive character $Y = I$ gives simultaneously $R_X = \Lambda_X = L$ and $M_X = \hbar_a/(c_{\text{int}}L)$. For a general character $Y > 0$, the lengths are reciprocal: $R_X = LY$, $\Lambda_X = LY^{-1}$. Their product remains L^2I without their individual values necessarily coinciding. Thus identification of the four lengths in the circular case corresponds to a precise sector of the radial proof and does not eliminate dependence on the character.

The proportionality $R_X = (G/c_{\text{int}}^2)M_X$ is common to all characters in the domain. Its preservation under change of basis, refinement, minimization with memory, Schur complementation and dynamic normalization has been proved in [section 39.1](#). The five-component carrier and its reduction to two helicities, developed subsequently, describe the tensor representation; they do not replace the proof of the common radial coefficient.

39.3 Causal composition of the phase, action, and length invariants in the derivation of G

Gravity is organized here into four strata that must not be merged:

$$\begin{aligned} \text{inscription and radial norm R1–R8} &\longrightarrow L \longrightarrow G = c_{\text{int}}^3 L^2 / \hbar_a, \\ \text{icosahedral incidence} &\longrightarrow V_{12} \xrightarrow{P_5} V_5 \xrightarrow{\Gamma_5} \text{STF}_3 \xrightarrow{\Lambda(n)} \mathcal{H}_{\text{TT}}(n), \\ \text{CT108 calendar with } t_0 = \pi L / (54c_{\text{int}}) &\longrightarrow R_{108} = L, \quad \theta_{108}^{\text{circ}} = (54/\pi)^2, \\ \text{metrological chart} &\longrightarrow 6.6742996594140227 \dots \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}. \end{aligned} \quad (39.29)$$

The first line is the determination proved in [section 39.1](#): length precedes coupling. The second constructs the carrier and its transverse traceless reduction. The third preserves the calendar as a realization of that same length, according to (39.28). The fourth evaluates the section in the bases of [section 39.2](#). The alternative holonomy branches and the decimal reader are preserved with their own domains; they do not replace the radial proof.

39.4 Gravitational magnitude line and chronological scale 108

Let $c_{\text{int}} > 0$, $t_0 > 0$, $\ell_0 = c_{\text{int}}t_0$ and a positive section $\hbar_a$ of the action line, with $a \in \{\text{pre}, \text{ret}\}$. The dimensional transducer is

$$G_a(\theta) = \theta \frac{c_{\text{int}}^5 t_0^2}{\hbar_a} = \theta \frac{c_{\text{int}}^3 \ell_0^2}{\hbar_a}, \quad \theta > 0. \quad (39.30)$$

Equality of the two expressions uses only $\ell_0 = c_{\text{int}} t_0$ and $[G] = L^3 M^{-1} T^{-2}$. Dimension is fixed by the line; the dimensionless character θ distinguishes the realizations of that system. In the longitudinal composition already proved, $t_0 = \pi L / (54 c_{\text{int}})$ and $\theta = (54/\pi)^2$ reproduce exactly $G_a = c_{\text{int}}^3 L^2 / \hbar_a$.

The CT108 calendar defines

$$T_{108} = 108 t_0, \quad \omega_{108} = \frac{2\pi}{108 t_0}, \quad R_{108} = \frac{c_{\text{int}}}{\omega_{108}} = \frac{54}{\pi} \ell_0. \quad (39.31)$$

For each action sheet, let

$$E_{108,a} = \hbar_a \omega_{108}, \quad m_{108,a} = \frac{E_{108,a}}{c_{\text{int}}^2}. \quad (39.32)$$

Then, the reduced Compton wavelength coincides with the radius of the cycle:

$$\bar{\lambda}_{C,a} = \frac{\hbar_a}{m_{108,a} c_{\text{int}}} = R_{108}. \quad (39.33)$$

Theorem 39.1 (Uniqueness of the selector under the HMT periodicity condition). *With the reduced convention $r_g = Gm/c_{\text{int}}^2$ and the HMT periodicity premise that identifies the gravitational radius of the CT108 quantum with its phase radius, the closure condition*

$$r_{g,a}(\theta) = R_{108} \quad (39.34)$$

selects a unique positive solution,

$$\theta_{108}^{\text{circ}} = \left(\frac{54}{\pi} \right)^2 = 295.4525715010569415303525 \dots \quad (39.35)$$

In that embodiment,

$$R_{108} = \bar{\lambda}_{C,a} = r_{g,a} = \ell_P^{\text{circ}}. \quad (39.36)$$

Proof. By the corresponding identity,

$$r_{g,a}(\theta) = \frac{G_a(\theta) m_{108,a}}{c_{\text{int}}^2} = \theta c_{\text{int}} t_0^2 \omega_{108} = \theta \frac{\pi}{54} \ell_0.$$

Comparing it with $R_{108} = (54/\pi) \ell_0$ necessarily gives $\theta = (54/\pi)^2$. The equation is linear in θ and all the factors are positive, so the solution is unique. $\square$

The equality $r_g = R_{108}$ does not follow from the dimensionality of [equation \(39.30\)](#). It is the explicit geometric condition of this HMT realization: action, Compton, and periodic return must determine the same length. The theorem proves that, once this structural condition is accepted, the character θ is fixed without using the SI value of G .

Remark 39.2 (Radius convention). The theorem uses the reduced gravitational radius Gm/c^2 . If it is replaced by the Schwarzschild radius $2Gm/c^2$, the selector changes by a factor of two. The two expressions correspond to distinct geometric conditions, so the convention must be declared before the calculation.

39.5 Reciprocity $G \leftrightarrow \hbar$ and conservation of the Planck area

Corollary 39.3 (Planck family associated with the temporal periodicity of 108 phases). *For $G_a = G_a(\theta_{108}^{\text{circ}})$,*

$$\ell_P = \frac{54}{\pi} \ell_0, \quad t_P = \frac{54}{\pi} t_0, \quad (39.37)$$

$$E_{P,a} = \frac{\hbar_a}{t_P} = \frac{\pi \hbar_a}{54 t_0} = \frac{h_a}{108 t_0}, \quad \ell_P^2 = \frac{G_a \hbar_a}{c_{\text{int}}^3} = \theta_{108}^{\text{circ}} \ell_0^2. \quad (39.38)$$

Moreover,

$$F_{P,a} = \frac{c_{\text{int}}^4}{G_a}, \quad P_{P,a} = \frac{c_{\text{int}}^5}{G_a}, \quad \rho_{P,a} = \frac{\hbar_a}{(\theta_{108}^{\text{circ}})^2 c_{\text{int}}^5 t_0^4}. \quad (39.39)$$

Proof. The first three identities follow from $\ell_P = R_{108}$, $t_P = \ell_P / c_{\text{int}}$, and $h_a = 2\pi \hbar_a$. For the area,

$$\frac{G_a \hbar_a}{c_{\text{int}}^3} = \theta_{108}^{\text{circ}} c_{\text{int}}^2 t_0^2 = \theta_{108}^{\text{circ}} \ell_0^2.$$

The remaining expressions are the dimensional compositions of force, power, and mass per volume with the same section. $\square$

Let

$$R_{\text{act}} := \frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}}. \quad (39.40)$$

Since the action appears in the numerator of the chronological mass and in the denominator of G ,

$$\frac{m_{108,\text{pre}}}{m_{108,\text{ret}}} = R_{\text{act}}, \quad \frac{G_{\text{pre}}}{G_{\text{ret}}} = R_{\text{act}}^{-1}, \quad G_a \hbar_a = \theta_{108}^{\text{circ}} c_{\text{int}}^5 t_0^2. \quad (39.41)$$

Therefore $\ell_P^2 = G_a \hbar_a / c^3$ does not change sheet. Gravity and action are oriented reciprocally, while area preserves the common geometry. This cancellation is the convergence interface towards the area operator of section 40.5. The second fundamental relation appears when action and gravity are compared.

In HMT, action measures how much occurs while simultaneously preserving the orientation of the realization. There is one section before the return and another after it. The phase may return accompanied by a different memory history. That difference is stored in the bidirectional ratio of the two action sections.

Gravity responds in a reciprocal manner.

The gravitational transducer has the form

$$G_a(\theta) = \theta \frac{c_{\text{int}}^5 t_0^2}{\hbar_a}.$$

Action appears in the denominator. Therefore, if $\hbar$ is multiplied by a ratio R_{act} when the sheet changes, the gravitational constant is multiplied by the inverse ratio:

$$\frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}} = R_{\text{act}}, \quad \frac{G_{\text{pre}}}{G_{\text{ret}}} = R_{\text{act}}^{-1}.$$

Action and gravity form a contragredient pair here: when one advances in one direction, the other exactly compensates for that displacement.

What remains invariant is

$$G_a \hbar_a.$$

And that is precisely the combination that determines the Planck area:

$$\ell_P^2 = \frac{G_a \hbar_a}{c_{\text{int}}^3}.$$

Action remembers the sheet. Gravity responds to that memory. Area remains invariant.

This is an extraordinarily fertile idea. The Planck area emerges as the object that survives when action and gravity transform into one another reciprocally.

It may be stated as follows: $\hbar$ preserves the dynamical orientation; G preserves the geometric compensation; ℓ_P^2 preserves that on which both realizations agree.

The area is, therefore, a currency of exchange between quantum physics and geometry, because the bidirectional transformation of $\hbar$ and G selects exactly that combination as invariant.

This reciprocity affords another understanding of the relation between information and gravity. Information requires a distinction among states; action measures the cost of transition between them. Yet if that transition must acquire a geometry, the gravitational response changes inversely so as to keep the quantum of area stable.

Geometry incorporates phase information through an exact compensation.

From a Machian reading, this is already a highly suggestive result. A local magnitude of geometry is determined relationally by the orientation and the return of the complete state. The invariant area restricts that relational dependence and fixes its consistency.

The HMT version of the relational principle consists in something more precise than the maxim "matter determines geometry": the orientation of action and the gravitational response co-determine one another in such a way that elementary geometry remains invariant.

The action contains the history. Gravity adjusts the response. The area preserves the agreement.

The derivation of G takes that reciprocity to a much more concrete geometric condition.

Dimensional Mechanics first determines what type of object G can be. It fixes its magnitude line and dimensional homogeneity. Chronological closure then selects its value: dimension determines the class of magnitude, and closure determines its dimensionless coefficient.

The selection proceeds from the temporal periodicity of (108) phases.

The cycle 108 determines a frequency, a phase radius, and an energy. From that energy arises a chronological mass. And constructing the reduced Compton wavelength of that mass yields exactly the same radius as that of the cycle:

$$\bar{\lambda}_C = R_{108}.$$

Phase and matter are already stating the same length before G is introduced.

Then gravity comes into play. The reduced gravitational radius is constructed.

$$r_g = \frac{Gm}{c^2},$$

and gravitational closure is required to identify the phase radius, Compton length, and gravitational radius as a single structural scale:

$$r_g = R_{108} = \bar{\lambda}_C.$$

That condition uniquely selects

$$\theta_{108} = \left(\frac{54}{\pi} \right)^2.$$

In this way,

$$R_{108} = \bar{\lambda}_C = r_g = \ell_P.$$

The structural function of θ_{108} determines the significance of its terminal decimal: it shows what G is doing.

G appears as the coefficient required to make the four lengths published by these languages coincide:

- the lenguaje of the fase produce R_{108} ;
- the quantum language produces $\bar{\lambda}_C$;
- the gravitational language produces r_g ;
- the lenguaje of Planck produce ℓ_P .

The gravitational constant is the mediator that prevents those four readers from separating.

Stated in more physical terms: G expresses how strongly geometry must respond for a quantum whose mass follows from the clock 108 to close upon its own phase.

It is the relation that makes possible for matter, phase, and geometry to be three compatible publications of the same event.

Here the Machian reading becomes stronger. The local gravitational scale is selected by a global cycle-closure condition. The local radius of a quantum acquires meaning when integrated into a complete periodicity. Inertia, phase, and geometry are determined relationally.

The demonstrated scope provides a concrete mathematical mechanism for a Machian reading: a local property—the gravitational coupling—is selected by the consistency of the global state. The other historical formulations of Mach's principle preserve their own domain.

Then comes the fifth dimension.

The scalar G fixes the gravitational intensity, and the carrier specifies its representational structure. HMT constructs a symmetric traceless tensor space of dimension five. That carrier admits five helicoidal weights:

$$+2, \quad +1, \quad 0, \quad -1, \quad -2.$$

They are the five natural degrees of freedom of a symmetric traceless tensor in three dimensions before the dynamical constraints of a specific theory are imposed.

That clarifies a point that is often confused. “Five components” designates the complete carrier, whereas the two helicities designate the transverse massless reduction in general relativity.

It means that the complete carrier has five irreducible coordinates. The dynamics then decides which part of that carrier is physical:

- in a massive spin-two realization, all five weights may survive;
- in a massless realization with gauge invariance, the transverse traceless projection eliminates the sectors ± 1 and 0;
- the helicities ± 2 remain.

Thus, HMT provides a space prior to the dynamical decision. It constructs the complete five-component carrier and contains, through the corresponding projection, its physical reduction to two helicities.

Beauty resides in the ordered coexistence of two statements:

- the spin-two geometry has a five-component carrier;

- massless gravitational radiation occupies a two-dimensional subspace.

Both statements form a coherent and precise reduction.

And that reduction preserves another central lesson: the scalar G fixes the common intensity, while the orientation information resides in the tensor carrier. The intensity resides in the constant, the helicities reside in the representation, and the complete genealogy brings magnitude and carrier together.

39.6 Genealogical domain of gravitational realization

The gravitational realization receives as its initial object a section of the discrete continuum already constructed by the composition

$$\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow X_{\text{TPK}}^{\text{enr}} \longrightarrow C_{\text{cont}}^{\text{disc}}. \quad (39.42)$$

APP preserves the additive and multiplicative evaluations through residue, quotient and carry; TRIT determines the regime and orientation; the TPK incorporates the tritic phase selector without eliminating either of the two APP sheets, performs cardinal transport, preserves memory and realizes phase return without resetting the state. The dimensional transducer then acts on the surviving section of the continuum:

$$\mathfrak{G}_{108} : C_{\text{cont}}^{\text{disc}} \times \mathcal{L}_S^+ \longrightarrow \mathcal{G}_{\text{HMT-MD}}, \quad (x, \hbar_a) \longmapsto (G_a, R_{108}, m_{108,a}, \ell_{P,a}^2). \quad (39.43)$$

The cardinals 90/120, the periodicity 108, the action, and the angular pair enter with their genealogy preserved. No subsequent gravitational identification retrospectively selects these antecedents.

39.7 Gravitational Transducer and Chronological Selector of Periodicity

Let $c_{\text{int}} > 0$, $t_0 > 0$, $\ell_0 = c_{\text{int}} t_0$, and $\hbar_a \in \mathcal{L}_S^+$, with $a \in \{\text{pre}, \text{ret}\}$. The dimensional transducer is

$$G_a(\theta) = \theta \frac{c_{\text{int}}^5 t_0^2}{\hbar_a} = \theta \frac{c_{\text{int}}^3 \ell_0^2}{\hbar_a}, \quad \theta > 0. \quad (39.44)$$

The chronology cyclic of 108 states determina

$$T_{108} = 108 t_0, \quad \omega_{108} = \frac{2\pi}{108 t_0}, \quad R_{108} = \frac{c_{\text{int}}}{\omega_{108}} = \frac{54}{\pi} \ell_0. \quad (39.45)$$

The chronological quantum and its associated mass are

$$E_{108,a} = \hbar_a \omega_{108}, \quad m_{108,a} = \frac{E_{108,a}}{c_{\text{int}}^2}. \quad (39.46)$$

The reduced Compton wavelength satisfies

$$\bar{\lambda}_{C,a} = \frac{\hbar_a}{m_{108,a} c_{\text{int}}} = R_{108}. \quad (39.47)$$

Theorem 39.4 (Uniqueness of the gravitational periodicity selector). *Under the reduced convention $r_g = Gm/c_{\text{int}}^2$, the condition $r_{g,a}(\theta) = R_{108}$ has the unique positive solution*

$$\boxed{\theta_{108}^{\text{circ}} = \left(\frac{54}{\pi} \right)^2}. \quad (39.48)$$

The corresponding realization identifies structurally

$$R_{108} = \bar{\lambda}_{C,a} = r_{g,a} = \ell_P^{\text{circ}}. \quad (39.49)$$

Proof. By the corresponding identity,

$$r_{g,a}(\theta) = \frac{G_a(\theta)m_{108,a}}{c_{\text{int}}^2} = \theta \frac{\pi}{54} \ell_0.$$

Equality with $R_{108} = (54/\pi)\ell_0$ imposes $\theta = (54/\pi)^2$. Linearity in θ and positivity of the factors determine uniqueness. $\square$

In the chart in which $c_{\text{int}} = (\mu\varepsilon)^{-1/2}$, the transducer assumes the form

$$G_a = \left(\frac{54}{\pi} \right)^2 \frac{t_0^2}{\hbar_a (\mu\varepsilon)^{5/2}}. \quad (39.50)$$

The constitutive product $\mu\varepsilon$ governs the causal cone and the gravitational scale; the quotient $Z^2 = \mu/\varepsilon$ preserves the impedance orientation.

39.8 Bidirectional Covariance of Gravity and Action

We define

$$R_{\text{act}} = \frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}}. \quad (39.51)$$

The chronological mass and the gravitational coupling transform contragrediently:

$$\frac{m_{108,\text{pre}}}{m_{108,\text{ret}}} = R_{\text{act}}, \quad \frac{G_{\text{pre}}}{G_{\text{ret}}} = R_{\text{act}}^{-1}. \quad (39.52)$$

Its product remains fixed:

$$G_a \hbar_a = \theta_{108}^{\text{circ}} c_{\text{int}}^5 t_0^2, \quad \ell_P^2 = \frac{G_a \hbar_a}{c_{\text{int}}^3} = \theta_{108}^{\text{circ}} \ell_0^2. \quad (39.53)$$

The action preserves the orientation of the sheet; the area of Planck preserves the geometry common of both orientations.

Theorem 39.5 (Chart independence of the sections h and G). *Let u_L, u_T, u_S be positive bases of the dimensional lines of length, time and action, and replace them by*

$$u'_L = \lambda_L u_L, \quad u'_T = \lambda_T u_T, \quad u'_S = \lambda_S u_S, \quad \lambda_L, \lambda_T, \lambda_S > 0.$$

The coordinates of the same sections satisfy

$$c'_{\text{int}} = \frac{\lambda_T}{\lambda_L} c_{\text{int}}, \quad t'_0 = \frac{t_0}{\lambda_T}, \quad \hbar'_a = \frac{\hbar_a}{\lambda_S}, \quad G'_a = \frac{\lambda_S \lambda_T^3}{\lambda_L^5} G_a. \quad (39.54)$$

For the induced basis $u_G = u_L^5 u_T^{-3} u_S^{-1}$ one obtains

$$\hbar'_a u'_S = \hbar_a u_S, \quad G'_a u'_G = G_a u_G. \quad (39.55)$$

Consequently, $\hbar_a = 2\pi \hbar_a$, $G_a = \theta_{108}^{\text{circ}} c_{\text{int}}^5 t_0^2 / \hbar_a$ and $G_a \hbar_a = \theta_{108}^{\text{circ}} c_{\text{int}}^5 t_0^2$ are identities of sections, not equalities dependent on a choice of units. The metrological chart publishes their numerical coordinates after the HMT–MD construction.

Proof. The first three transformations of (39.54) are the contragredient law of coordinates in $L \otimes T^{-1}$, T and S . Substituting them into (39.44) produces the fourth. The basis of the gravitational line transforms by the inverse factor, proving (39.55). The scalars π and $\theta_{108}^{\text{circ}}$ are dimensionless and remain invariant. $\square$

Barbero–Immirzi parameter, hexadic–octadic incidence and spectrum of the area operator

40.1 Hexad–Octad Incidence Morphism and Conservation of the Oriented Channels 90/120

With the dodecad and the alignment of the section devoted to the dodecaphasic residual design and the lifting of hexads fixed, let H be a hexad in the HMT coordinates, let $\ell(H) \subset D$ be its image, and let $O_H = \iota_D(\ell(H))$ be its unique lift. Marking a point and a pair of points of the residual hexad yields

$$F_6(H) = \ell(H) \times \binom{\ell(H)}{2}, \quad |F_6| = 6 \binom{6}{2} = 90.$$

If the marked point can traverse the octad:

$$F_8(H) = O_H \times \binom{\ell(H)}{2}, \quad |F_8| = 8 \binom{6}{2} = 120.$$

The inclusion $\ell(H) \hookrightarrow O_H$ induces the incidence morphism

$$\mathcal{I}_{6 \rightarrow 8} : F_6(H) \hookrightarrow F_8(H).$$

Normalizing the difference by the chosen factor $24 \binom{6}{2}$ yields

$$\boxed{\frac{90 - 120}{24 \binom{6}{2}} = \frac{6 - 8}{24} = -\frac{1}{12}.$$

The equality is a normalized defect of the combinatorial interface $6 \rightarrow 8$. The factor $24 \binom{6}{2}$ is part of the definition of this normalization; the inclusion alone determines the difference -30 . The operator $\mathcal{I}_{6 \rightarrow 8}$ has marked incidence configurations as its domain; it is neither the transverse extractor of the dodecaphasic state nor the system of angular moments.

The transition from a hexad to an octad combines three structures of distinct types: a finite incidence, two Lambert series associated with the channels $q_{\pm}$, and an identification with the real parameter of the Holst action. The three are articulated through the incidence, the angular functional, and the areal dictionary into the geometry of the Ashtekar–Barbero connection developed in this same residence.

40.2 Tritic derivation of the cardinalities 90/120 and subsequent realization in hexad–octad incidence

Let H be a set of six points and let O be a set of eight points containing a distinguished copy of H . Consider

$$\mathcal{F}_6 = H \times \binom{H}{2}, \quad \mathcal{F}_8 = O \times \binom{H}{2}.$$

The inclusion distinguished $H \hookrightarrow O$ induce

$$\iota_{6,8} : \mathcal{F}_6 \hookrightarrow \mathcal{F}_8, \quad (h, \{h_1, h_2\}) \mapsto (h, \{h_1, h_2\}).$$

This incidence morphism is the combinatorial antecedent of the cardinals 90/120; it is neither the dodecaphasic extractor nor an angular series.

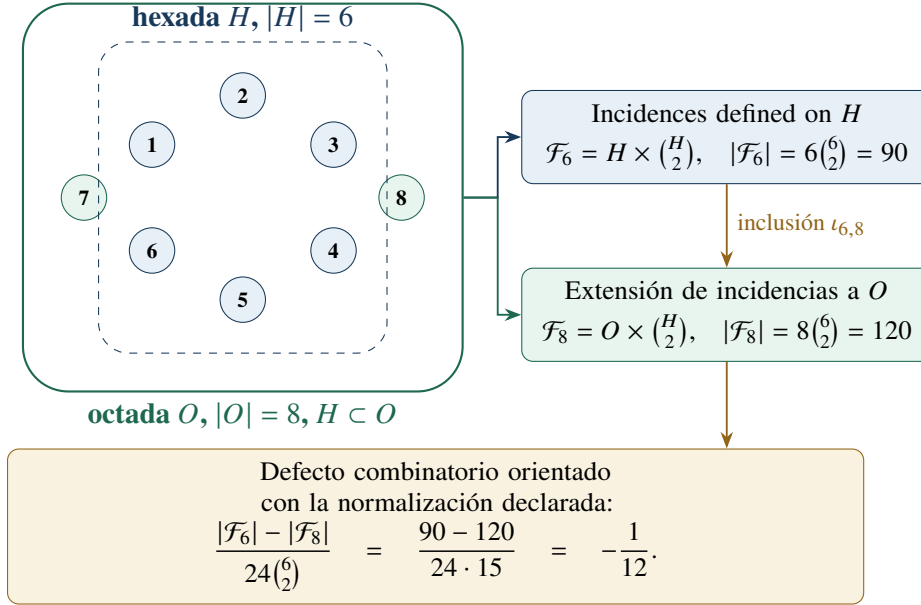
Theorem 40.1 (Cardinalities of the Inclusion).

$$|\mathcal{F}_6| = 6 \binom{6}{2} = 90, \quad |\mathcal{F}_8| = 8 \binom{6}{2} = 120.$$

The normalized difference by twenty-four coordinates is

$$\frac{90 - 120}{24 \binom{6}{2}} = -\frac{1}{12}.$$

Proof. Each element of the first coordinate can be combined with any of the fifteen pairs of H . The multiplication principle gives the two cardinalities; the last identity follows by substituting $\binom{6}{2} = 15$. $\square$



Esta identidad pertenece al mapa de incidencias. No identifica el extractor dodecafásico, el semigrupo de torres ni el funcional angular.

Figure 40.1: Inclusión de un subconjunto distinguido de seis elementos en uno de ocho elementos y conteos de incidencia. Los cardinales 90 y 120 no fusionan el extractor dodecafásico, el funcional angular ni el semigrupo de torres.

Los índices iniciales de la jerarquía angular quedan así asociados a dos espacios de incidencia: 90 configuraciones sobre la hexada y 120 sobre la octada ambiente. El cociente $-1/12$ es una identidad combinatoria de esta normalización.

40.3 Series de Lambert asociadas a los 2 sectores de incidencia

Para $0 < q < 1$, definimos simultáneamente

$$S_{90}(q) = \frac{q^{90}}{1 - q^{270}}, \quad S_{120}(q) = \frac{q^{120}}{1 - q^{360}}, \quad \Delta S_m = S_m(q_-) - S_m(q_+).$$

Geometric expansions

$$\Delta S_{90} = \sum_{j \geq 0} s_{90+270j}, \quad \Delta S_{120} = \sum_{j \geq 0} s_{120+360j}$$

place both terms within the global semigroup of chapter devoted to the global hierarchy of dimensional towers.

The angular functional associated with the transition $6 \rightarrow 8$ is

$$K_{6 \rightarrow 8} = 12\Delta S_{90} - \Delta S_{120},$$

and does not follow from the incidence cardinalities alone. Its weights are determined only after fixing the two series, the pole residue $1/24$, integer positivity, and the minimality criterion proved below. The resulting angular coordinate is

$$\Gamma_{6 \rightarrow 8} = \frac{\pi AC^*}{180} + K_{6 \rightarrow 8}.$$

At the point HMT,

$$\begin{aligned}\Delta S_{90} &= 2.95169439297457621139847987861 \times 10^{-4}, \\ \Delta S_{120} &= 1.96835112502951720093738706217 \times 10^{-5}, \\ \frac{\pi AC^*}{180} &= 0.270485322934140078465586458790 \dots,\end{aligned}$$

and, therefore,

$$\Gamma_{6 \rightarrow 8} = 0.27400767269445927474725526077398041940 \dots$$

The tables computed with the conjugate angular coordinate C^* rounded to fifteen decimal places give 0.2740076726944593 in double-precision arithmetic; the preceding value is the multiprecision evaluation of the complete regional phase.

40.4 Diophantic equation of the weights

The identity

$$\lim_{q \rightarrow 1^-} (1 - q) \frac{q^m}{1 - q^n} = \frac{1}{n}$$

associates to the combination $a\Delta S_{90} - b\Delta S_{120}$ the normalized residue

$$\frac{a}{270} - \frac{b}{360}.$$

Theorem 40.2 (Integer Weights with Residue $1/24$). *The positive integer solutions of*

$$\frac{a}{270} - \frac{b}{360} = \frac{1}{24}$$

are

$$(a, b) = (12 + 3t, 1 + 4t), \quad t \in \mathbb{Z}_{\geq 0}.$$

The pair $(12, 1)$ is the unique coordinatewise minimal solution and the unique one of minimal ℓ^1 norm.

Proof. Multiplying by 1080 produces $4a - 3b = 45$. Modulo three, one obtains $a \equiv 0 \pmod{3}$; writing $a = 3u$ yields $4u - b = 15$. Positivity implies $u = 4 + t$, $b = 1 + 4t$, with $t \geq 0$. Therefore, $a + b = 13 + 7t$, which proves minimality. $\square$

Primitivity $\gcd(a, b) = 1$ does not by itself select the minimal pair, since other primitive solutions exist. The weight $12 : -1$ results from combining the residue condition $1/24$ with minimality. The choice of that residue is a normalization condition of the construction; it is not deduced from the decimal evaluation of $\Gamma_{6 \rightarrow 8}$.

40.5 Monotonicity of the functional with respect to the conjugate angular coordinate

Para $m \in \{90, 120\}$,

$$S'_m(q) = \frac{mq^{m-1}(1+2q^{3m})}{(1-q^{3m})^2} > 0.$$

Moreover,

$$\frac{dq_-}{dC} = \frac{\pi}{180}q_-, \quad \frac{dq_+}{dC} = -\frac{\pi}{180}q_+.$$

Proposition 40.3 (Strict Monotonicity). *For $1^\circ \leq C \leq 3^\circ$, the function $C \mapsto \Gamma_{6 \rightarrow 8}(A, C)$ is strictly increasing. Consequently, every value in its image has at most one preimage in that interval.*

Proof. The derivative of each difference ΔS_m is positive. On the specified interval one obtains $\Delta S'_{120} < 5.35 \times 10^{-4}$, whereas $\pi A/180 > 0.1273$. Therefore, $\Gamma'(C) > 0.1267$. At $C = C^*$,

$$\Gamma'(C^*) = 0.1328995105618907 \dots$$

□

Local injectivity excludes a second nearby angular coordinate with the same functional evaluation. This control neither constructs nor selects C^* : the conjugate angular coordinate has already been obtained in the preceding regional branch. The proposition only quantifies sensitivity of the downstream evaluation once A and C^* are fixed; it does not authorize inversion of $\Gamma_{6 \rightarrow 8}$ or a physical value of γ_{BI} to redefine them.

40.6 Hexad–octad inclusion and associated functional realizations

Let us fix a set H of six points and a set O of eight points with a distinguished inclusion $j : H \hookrightarrow O$. In the Witt residence, H is a hexad of a dodecad and O its lifted octad; in this chapter only the already fixed inclusion is used, not the terminal value of Barbero–Immirzi. We write

$$\binom{H}{2} := \{P \subset H : |P| = 2\}, \quad \left| \binom{H}{2} \right| = \binom{6}{2} = 15, \quad (40.1)$$

and we define two incidence spaces with distinct types:

$$\mathcal{F}_6 := H \times \binom{H}{2}, \quad \mathcal{F}_8 := O \times \binom{H}{2}. \quad (40.2)$$

The inclusion of incidences is

$$\iota_{6,8} : \mathcal{F}_6 \hookrightarrow \mathcal{F}_8, \quad (h, P) \mapsto (j(h), P). \quad (40.3)$$

It is injective because j is injective and preserves the marked pair P .

Theorem 40.4 (Incidence and defect counts oriented). *The two spaces of equation (40.2) satisfy*

$$|\mathcal{F}_6| = 6 \binom{6}{2} = 90, \quad |\mathcal{F}_8| = 8 \binom{6}{2} = 120, \quad |\mathcal{F}_8 \setminus \iota_{6,8}(\mathcal{F}_6)| = 30. \quad (40.4)$$

Normalization by the twenty-four coordinates of the dodecad–twenty-four pair and by the fifteen pairs of H produces

$$\delta_{6,8}^{\text{inc}} = \frac{|\mathcal{F}_6| - |\mathcal{F}_8|}{24 \binom{6}{2}} = -\frac{1}{12}. \quad (40.5)$$

The same fraction is obtained through the reciprocal functional associated with the incorporated memory,

$$\delta_{6,8}^{\text{rec}} = \frac{30}{120} - \frac{30}{90} = -\frac{1}{12}. \quad (40.6)$$

They are two exact realizations of the same oriented defect, but have distinct domains and are not identified with $\zeta(-1)$ without an additional intertwiner.

Proof. The multiplicative principle applied to equation (40.2) gives $6 \cdot 15 = 90$ and $8 \cdot 15 = 120$. The image of $\iota_{6,8}$ has ninety elements, so the complement has thirty. Substitution into equation (40.5) gives $-30/(24 \cdot 15) = -1/12$, while equation (40.6) is $1/4 - 1/3 = -1/12$. $\square$

The hexad–octad transition admits two functional realizations that are not deduced from one another. The algebraic diagram, formulated on the spaces that justify the counts, is

$$\begin{aligned} (H \xrightarrow{j} O) &\longmapsto (|\mathcal{F}_6|, |\mathcal{F}_8|) = (90, 120) \longmapsto \Gamma_{6 \rightarrow 8} = \gamma_{\text{HMT}}, \\ (h_6, o_8, t) = (6, 8, 3) &\longmapsto -\frac{o_8 - h_6}{t} = -\frac{2}{3}, \end{aligned} \quad (40.7)$$

where $-2/3$ is the coefficient of the third coordinate of row 13 of the CKM constructor. The first realization transforms the incidence into area and modular memory; the second transforms the same transition into angular-mixing parameters. $\theta_{\text{CKM}} = f(\gamma_{\text{HMT}})$ is not postulated: both outputs preserve a common antecedent.

The morphism determining the sectors $90/120$ is $\iota_{6,8}$. Magnitude γ_{HMT} is adopted as a theorem previously demonstrated in the hexad–octad construction and is not reconstructed from its decimal representation. The operatorial consequences begin only after fixing the elementary scale a_0 , defined below without using entropy or the terminal product $\gamma_{\text{HMT}}a_0$.

40.7 Correspondence between the Nonadic Interior and the Boundary Degrees of Freedom

Let $\mathbb{T}_3 = \{0, 1, 2\}$. Every $x \in \mathbb{Z}/9\mathbb{Z}$ has a unique digit representation $x = x_0 + 3x_1$, with $x_i \in \mathbb{T}_3$, and every $y \in \mathbb{Z}/27\mathbb{Z}$ a unique representation $y = y_0 + 3y_1 + 9y_2$. Therefore

$$\mathcal{B} = (\mathbb{Z}/9\mathbb{Z})^3, \quad \mathcal{B}_\partial = (\mathbb{Z}/27\mathbb{Z})^2, \quad |\mathcal{B}| = |\mathcal{B}_\partial| = 3^6 = 729. \quad (40.8)$$

Equality of cardinalities is accompanied by the explicit map. If $x_j = r_{2j-1} + 3r_{2j}$ for $j = 1, 2, 3$, we define

$$\beta_{729}(x_1, x_2, x_3) = (r_1 + 3r_2 + 9r_3, r_4 + 3r_5 + 9r_6). \quad (40.9)$$

Proposition 40.5 (Genealogical bijection $729 \leftrightarrow 729$). *The map $\beta_{729} : \mathcal{B} \rightarrow \mathcal{B}_\partial$ is bijective and preserves the ordered six-trit word. In general, it is not an isomorphism of additive groups: regrouping the digits changes where the carry is recorded.*

Proof. Extracting the six digits from three nonadic coordinates and grouping them into two triples is a permutation of the coordinates of $\mathbb{T}_3^6$. The positional decomposition is unique on both sides, so the operation has an inverse. The additive caveat is verified, for example, by adding words that generate a carry between r_3 and r_4 : that boundary separates the two coordinates of order twenty-seven, but crosses two distinct coordinates in the nonadic packing. $\square$

On the boundary torus $\mathcal{B}_\partial$, we take the block of vertices

$$A = \{0, 1, \dots, 8\} \times \{0, 1, \dots, 8\}. \quad (40.10)$$

Its oriented links crossing the boundary are separated by direction:

$$\partial A_{90} = \{((-1, j), (0, j)), ((8, j), (9, j)) : 0 \leq j \leq 8\}, \quad (40.11)$$

$$\partial A_{120} = \{((i, -1), (i, 0)), ((i, 8), (i, 9)) : 0 \leq i \leq 8\}, \quad (40.12)$$

where all coordinates are read modulo twenty-seven. Thus

$$\partial A = \partial A_{90} \sqcup \partial A_{120}, \quad |\partial A_{90}| = |\partial A_{120}| = 18. \quad (40.13)$$

The subscripts 90, 120 label the channel provenance; they do not say that the boundary has ninety or one hundred and twenty links.

Theorem 40.6 (Law areal of the cut triadic). *In the cubic graph $N \times N \times N$ with unit capacity, every cut between two opposite faces has capacity at least N^2 , and a transverse layer attains that bound. For $N = 9$,*

$$\mathcal{A}_{\min} = 81, \quad S_{\text{tri}} = 81 \log 3. \quad (40.14)$$

Proof. There are N^2 lines of edges parallel to the normal direction, pairwise disjoint. Every cut must intersect each one, so its capacity is at least N^2 . A single transverse layer contains exactly N^2 edges and attains the minimum. If each cut link carries a maximally mixed trit, its entropy is $\log 3$; additivity over the eighty-one links gives equation (40.14). $\square$

The entropy S_{tri} measures the capacity of the cubic cut. The modular entropy of the thirty-six links of equation (40.13) measures another state and must not be equated with it.

40.8 Collective incidence entangler between interior and boundary

In the sum $\ell^2(\mathcal{F}_6) \oplus \ell^2(\mathcal{F}_8)$, we define

$$u_{90} = \frac{1}{\sqrt{90}} \sum_{f \in \mathcal{F}_6} (\delta_f, 0), \quad u_{120} = \frac{1}{\sqrt{120}} \sum_{f \in \mathcal{F}_8} (0, \delta_f). \quad (40.15)$$

In $\ell^2(\partial A_{90}) \oplus \ell^2(\partial A_{120})$, let

$$v_{90} = \frac{1}{\sqrt{18}} \sum_{e \in \partial A_{90}} (\delta_e, 0), \quad v_{120} = \frac{1}{\sqrt{18}} \sum_{e \in \partial A_{120}} (0, \delta_e). \quad (40.16)$$

The pairs (u_{90}, u_{120}) and (v_{90}, v_{120}) are orthonormal. Therefore there exists a unique isometry between the collective subspaces that satisfies

$$\mathcal{J}_{6 \rightarrow 8} u_{90} = v_{90}, \quad \mathcal{J}_{6 \rightarrow 8} u_{120} = v_{120}. \quad (40.17)$$

Define

$$D_{\text{inc}} = 90|u_{90}\rangle\langle u_{90}| + 120|u_{120}\rangle\langle u_{120}|, \quad (40.18)$$

$$D_{\text{area}} = 90|v_{90}\rangle\langle v_{90}| + 120|v_{120}\rangle\langle v_{120}|. \quad (40.19)$$

Theorem 40.7 (Exact transport of the two labels). *The map $\mathcal{J}_{6 \rightarrow 8}$ is unitary between the two collective planes and intertwines the incidence and area operators:*

$$\boxed{\mathcal{J}_{6 \rightarrow 8} D_{\text{inc}} = D_{\text{area}} \mathcal{J}_{6 \rightarrow 8}.} \quad (40.20)$$

Therefore, every polynomial combination of the labels 90, 120, in particular the primitive combination $12 : -1$, is transported with the same coefficients.

Proof. Orthonormality shows that $\mathcal{J}_{6 \rightarrow 8}$ maps one orthonormal basis to another. In u_{90} , both sides of equation (40.20) equal $90v_{90}$; in u_{120} , both equal $120v_{120}$. Equality on a basis proves the identity. The functional calculus, in its polynomial form, preserves the intertwining. $\square$

This map does not purport to inject ninety individual incidences into eighteen links: it transports exactly the two uniform modes and their spectral labels. Extending it to non-uniform fluctuations would require a new incidence map; the modular operator defined here only uses the collective subspace that has just been constructed.

40.9 Determination of the Elementary Scale of the Spectrum of the Area Operator

Let C^* the angular coordinate conjugate, expresada in grados, and fijemos

$$\theta_{\text{clk}} = \frac{C^*}{6} \frac{\pi}{180}, \quad \boxed{a_0 = \frac{1 + 2 \cos(10^\circ)}{3} \theta_{\text{clk}}.} \quad (40.21)$$

The factor preceding θ_{clk} is not a decimal correction. It is the normalized real character of the phase triplet

$$\frac{1}{3} \text{Tr diag} \left(1, e^{i\pi/18}, e^{-i\pi/18} \right) = \frac{1 + 2 \cos(10^\circ)}{3}. \quad (40.22)$$

Thus, a_0 is the spectral mean of the angular structure on the oriented trit. It is positive, is fixed before the area operator, and is not selected from entropy. The subsequent identification $\gamma_{\text{HMT}} = \Gamma_{6 \rightarrow 8}$ realizes the HMT output in the Ashtekar–Barbero connection [BG95a, Imm97a]; it does not reopen its derivation.

Remark 40.8 (Alcance of the simbolo a_0). In this chapter, $a_0 \equiv a_0^{\text{area}}$ denotes exclusively the normalization of equation (40.21). It is neither the Bohr radius nor the initial cosmological scale factor, although those objects are traditionally written with the same symbol. No identity in this chapter allows one to substitute one for the other.

40.10 Areal Law of the Triadic Cut: Capacity $81 \log 3$, Area Operator, Spectrum, and Collective Intertwiner of the Channels $90/120$

The HMT identification of the functional $\Gamma_{6 \rightarrow 8}$ with the Barbero–Immirzi coordinate is made on the same triadic boundary that supports the incidence (90/120). The interior and its boundary sheet have equal cardinality,

$$\mathcal{B} = (\mathbb{Z}/9\mathbb{Z})^3, \quad |\mathcal{B}| = 729, \quad \partial\mathcal{B} = (\mathbb{Z}/27\mathbb{Z})^2, \quad |\partial\mathcal{B}| = 729,$$

by regrouping six trits into three nonadic coordinates or two coordinates of order twenty-seven.

Theorem 40.9 (Areal Law of Triadic Cutoff). *In the cubic graph $N \times N \times N$ with unit capacity, the minimum cut between two opposite faces has capacity N^2 . For $N = 9$,*

$$\mathcal{A}_{\min} = 81, \quad S_{\text{tri}} = \mathcal{A}_{\min} \log 3 = 81 \log 3.$$

Proof. The N^2 lines parallel to the normal direction are edge-disjoint paths, so every cut intercepts them. A transverse layer contains exactly N^2 edges and attains the bound. Each triadic link contributes $\log 3$ to the entropy of a maximally mixed state. $\square$

On the boundary torus 27×27 , let A be the canonical block 9×9 . Its boundary decomposes exactly into

$$\partial A = \partial A_{90} \sqcup \partial A_{120}, \quad |\partial A_{90}| = |\partial A_{120}| = 18. \quad (40.23)$$

The horizontal links represent channel (90), and the vertical ones channel (120). This decomposition turns the pair of incidences into an operator decomposition of the boundary.

The areal entropy $S_{\text{tri}} = 81 \log 3$ belongs to the maximally mixed state of the cut links. The bilayer entropy $S_{\text{bi}}(y)$ of the definition of oriented bilayer entropy belongs to the normalized distribution between the two sheets; the two objects retain distinct domains.

Let $N = \text{diag}(0, 1, 2)$ be the number operator of a tritic link. The areal realization declares the positive scale

$$\theta_{\text{clk}} = \frac{C^*}{6} \frac{\pi}{180}, \quad a_0 = \frac{1 + 2 \cos(10^\circ)}{3} \theta_{\text{clk}}, \quad \gamma_{\text{HMT}} = \Gamma_{6 \rightarrow 8}.$$

The formula for a_0 constitutes the starting structure of this realization; once that scale is fixed, the operator and its spectrum follow exactly.

Definition 40.10 (Area Operator HMT). On the boundary sheet $\mathcal{H}_\partial = \bigotimes_{e \in \partial A} \mathbb{C}^3$, define

$$\widehat{\mathcal{A}}_{\text{HMT}} = \gamma_{\text{HMT}} a_0 \sum_{e \in \partial A} N_e. \quad (40.24)$$

Its spectrum in the canonical cut is

$$\text{Spec } \widehat{\mathcal{A}}_{\text{HMT}} = \{n\gamma_{\text{HMT}}a_0 : 0 \leq n \leq 72\},$$

with multiplicities given by the coefficients of $(1 + z + z^2)^{36}$. Prolongation through compatible cuts produces the protected family $\mathcal{A}_n^{\text{prot}} = n\gamma_{\text{HMT}}a_0$.

The intertwiner that fixes normalization is constructed in the subspace of collective channels. Let u_{90}, u_{120} be the uniform unit vectors of the hexadic and octadic incidences, and let v_{90}, v_{120} be the uniform unit vectors of the two sets of links of (40.23). The map

$$\begin{aligned} \mathcal{J}_{6 \rightarrow 8} : \text{span}\{u_{90}, u_{120}\} &\longrightarrow \text{span}\{v_{90}, v_{120}\}, \\ \mathcal{J}_{6 \rightarrow 8} u_{90} &= v_{90}, \quad \mathcal{J}_{6 \rightarrow 8} u_{120} = v_{120} \end{aligned} \quad (40.25)$$

is an isometry between these collective subspaces. If

$$D_{\text{inc}} = 90|u_{90}\rangle\langle u_{90}| + 120|u_{120}\rangle\langle u_{120}|, \quad D_{\text{area}} = 90|v_{90}\rangle\langle v_{90}| + 120|v_{120}\rangle\langle v_{120}|,$$

then

$$\boxed{\mathcal{J}_{6 \rightarrow 8} D_{\text{inc}} = D_{\text{area}} \mathcal{J}_{6 \rightarrow 8}.$$

In particular, the primitive combination $12 : -1$ acts with the same coefficients before and after $\mathcal{J}_{6 \rightarrow 8}$. This equality fixes the normalization dictionary that inserts $\gamma_{\text{HMT}} = \Gamma_{6 \rightarrow 8}$ into the Ashtekar–Barbero connection.

The passage from hexad to octad is an incidence structure transformation.

In passing from 6 to 8, the available pairs, the internal relations, and the way in which orientation can be preserved change. That transition generates an oriented defect. The cardinal variation realizes a turn in the mode of organization.

The same sheet inversion is expressed by

$$C^* \longmapsto -C^*.$$

But different readers respond in different ways to that inversion.

The hexad–octad functional leading to the Barbero–Immirzi parameter is odd: when the sheet is inverted, it changes sign. Geometric orientation is reversed.

Bilayer entropy, by contrast, is even: it assigns the same entropy to C^* and $-C^*$ when the probability distribution is observed. The total information may remain unchanged even though the orientation has been reversed.

CKM space realizes a third response. The inversion induces an exact rational reflection on the three angles:

$$R_{\text{CKM}} = M_{\text{CKM}} \text{diag}(1, -1, 1) M_{\text{CKM}}^{-1},$$

with

$$R_{\text{CKM}}^2 = I, \quad \det R_{\text{CKM}} = -1.$$

This means that the change of sheet is transported to the mixing space as a true reflection. There exists an odd direction of rank one and a plane that remains fixed.

The counter-angle C^* is precisely the coordinate that changes. The phase constructed from A remains invariant. The transformation thus separates two memories:

- an orientation memory;
- a phase memory.

This provides a profound reading of CKM. The mixing angles are coordinates of a space acted upon by the same involution that already organized the hexad–octad transition.

The turn $6 \rightarrow 8$ supplies the orientation operation that CKM realizes in its own codomain; physical causality belongs to the corresponding transducer.

We then have three sister publications of the same transit:

1. **Geometric publication.** The Barbero–Immirzi parameter orients the quantum of area.
2. **Informational publication.** The reduced state and its modular Hamiltonian measure what information from that orientation remains on the boundary.
3. **Mixing publication.** The CKM reflection transforms the mixing coordinates and isolates their odd component.

γ_{BI} , entropy and the CKM angles remain distinct realizations of a common genealogical hinge. Unity resides in the antecedent prior to ramification. They share the hinge, but each preserves a distinct part of it.

This is one of the most beautiful expressions of the HMT methodology: the fundamental unit is recovered by tracing the genealogies backward until the common operation that generates distinct terminal results is found.

Gravity conserves the spin as the orientation of the area.

Information conserves the spin as memory of the state.

The mixture preserves the twist as a reflection of coordinates.

The same transformation «breathes» in three ways.

40.11 Typed distinction between Barbero–area, area–modular Hamiltonian, and Cartan–Holst realization

The derivation of γ_{BI} , the minimal area and the spectrum of the area operator concludes in this chapter. Joint reconstruction of occupation and provenance through the boundary modular Hamiltonian, and the Palatini–Cartan–Holst realization, retain their own domains, operators and proofs in Part IV.

40.12 Hexad–octad incidence, Barbero–Immirzi parameter, and area spectrum

The areal derivation receives as antecedents the incidence inclusion $6 \rightarrow 8$, the TPK cardinals $90/120$, the normalized defect $-1/12$, and the spectral channels $q_{\pm}$. The hexad–octad functional determines the Barbero–Immirzi parameter and the normalization of the area operator. This residence contains the HMT derivation and the spectrum; the Palatini–Cartan–Holst realization occupies a subsequent residence in Dimensional Mechanics.

Let $H \subset O$, with $|H| = 6$ and $|O| = 8$, and preserve the marked pair of the hexad in the inclusion

$$\iota_{6,8} : \mathcal{F}_6 = H \times \binom{H}{2} \hookrightarrow \mathcal{F}_8 = O \times \binom{H}{2}. \quad (40.26)$$

The internal count gives

$$|\mathcal{F}_6| = 6 \binom{6}{2} = 90, \quad |\mathcal{F}_8| = 8 \binom{6}{2} = 120, \quad \frac{90 - 120}{24 \binom{6}{2}} = -\frac{1}{12}. \quad (40.27)$$

Once π , the angle A and the unique counterangle C^* have been published, the two channels are

$$q_{\pm} = \exp\left[-\frac{\pi}{180}(A \pm C^*)\right], \quad 0 < q_+ < q_- < 1, \quad (40.28)$$

and their oriented-return series are

$$S_{90}(q) = \frac{q^{90}}{1 - q^{270}}, \quad S_{120}(q) = \frac{q^{120}}{1 - q^{360}}, \quad \Delta S_m = S_m(q_-) - S_m(q_+). \quad (40.29)$$

Theorem 40.11 (Diophantine determination of the Barbero–Immirzi functional). *Among positive combinations $a\Delta S_{90} - b\Delta S_{120}$, the residue condition $1/24$ and coordinatewise minimality determine the unique primitive pair $(a, b) = (12, 1)$. Thus the dimensionless output is*

$$\boxed{\gamma_{\text{HMT}} = \Gamma_{6 \rightarrow 8} = \frac{\pi A C^*}{180} + 12\Delta S_{90} - \Delta S_{120}.} \quad (40.30)$$

This formula receives the already generated angular pair and does not use the conventional value of the Barbero–Immirzi parameter as input.

Proof. The residue of S_{90} at $q = 1$ is $1/270$, and that of S_{120} is $1/360$. The required condition is equivalent to

$$\frac{a}{270} - \frac{b}{360} = \frac{1}{24} \iff 4a - 3b = 45.$$

Its positive integer solutions are $(a, b) = (12 + 3t, 1 + 4t)$, $t \geq 0$; the unique minimal solution is $(12, 1)$. Substituting this pair into the incidence functional proves (40.30). $\square$

The causal sequence is fixed without exchanging roles:

$$(\mathcal{F}_6 \hookrightarrow \mathcal{F}_8) \longrightarrow (90, 120, -1/12) \longrightarrow (A, C^*, q_{\pm}) \longrightarrow \Gamma_{6 \rightarrow 8} = \gamma_{\text{HMT}} \longrightarrow \hat{\mathcal{A}}_{\text{phys}}. \quad (40.31)$$

The subsequent realization in the Holst action recognizes the same dimensionless section; it does not participate in its selection.

For the two boundary sectors, let

$$N_m = \sum_{e \in \partial A_m} N_e, \quad m \in \{90, 120\}, \quad \text{Spec}(N_e) = \{0, 1, 2\}. \quad (40.32)$$

The physical area operator is written

$$\boxed{\frac{\widehat{\mathcal{A}}_{\text{phys}}}{\ell_P^2} = \gamma_{\text{HMT}} a_0 (N_{90} + N_{120})}. \quad (40.33)$$

On 36 links qutrit,

$$\text{Spec}(N_{90} + N_{120}) = \{0, 1, \dots, 72\}, \quad (40.34)$$

and the multiplicity of the eigenvalue n is the coefficient of z^n in $(1 + z + z^2)^{36}$. From here it follows

$$\text{Spec}\left(\frac{\widehat{\mathcal{A}}_{\text{phys}}}{\ell_P^2}\right) = \{n\gamma_{\text{HMT}} a_0 : 0 \leq n \leq 72\}. \quad (40.35)$$

40.13 Total Geometry and Origin Memory

The modular Hamiltonian of the boundary is

$$\boxed{K_A = -\log \rho_A = cI + \Delta_{\text{mod}}(90N_{90} + 120N_{120})}. \quad (40.36)$$

Let

$$N = N_{90} + N_{120}, \quad D = 90N_{90} + 120N_{120}. \quad (40.37)$$

The transition matrix $\begin{pmatrix} 1 & 1 \\ 90 & 120 \end{pmatrix}$ has determinant 30, and its inverse reconstructs the sectors:

$$\boxed{N_{90} = 4N - \frac{D}{30}, \quad N_{120} = \frac{D}{30} - 3N}. \quad (40.38)$$

The area publishes the total number of excitations; the modular Hamiltonian publishes their provenance sector. The joint pair reconstructs the complete occupation of the boundary.

40.14 Bilayer state and oriented information

Let $R = R^* = R^{-1}$ be the sheet involution and $y = C_{\text{rad}}^*$. The normalized state is

$$\boxed{\rho_y = \frac{e^{-yR}}{2 \cosh y}}. \quad (40.39)$$

Its modular Hamiltonian and its entropy are

$$K_y = -\log \rho_y = \log(2 \cosh y)I + yR, \quad (40.40)$$

$$S(\rho_y) = \log(2 \cosh y) - y \tanh y. \quad (40.41)$$

The inversion $y \mapsto -y$ leaves entropy fixed and produces the oriented informational distance

$$\boxed{D(\rho_y \| \rho_{-y}) = 2y \tanh y}. \quad (40.42)$$

The even component measures the spectral distribution; the odd component measures the informational cost of reversing the orientation.

40.15 How much informational energy and gravitational cell of medium action

The periodicity of 108 states determines

$$t_P = \frac{54}{\pi} t_0, \quad E_P = \frac{\hbar}{t_P} = \frac{h}{108 t_0}. \quad (40.43)$$

The thermal realization of the tritico clock satisfies

$$E_P = k_B \Theta_{\text{clk}} \log 3. \quad (40.44)$$

For the horizon of radius R ,

$$E_\Lambda(R) = \frac{c^4 R}{2G}, \quad E_\Lambda(R) = T_H(R) S_H(R). \quad (40.45)$$

In the Planck cell,

$$E_\Lambda = T_H S_H = \frac{E_P}{2} = \frac{k_B \Theta_{\text{clk}} \log 3}{2}. \quad (40.46)$$

Therefore,

$$E_\Lambda t_P = \frac{\hbar}{2}, \quad E_\Lambda T_{108} = \frac{h}{2}. \quad (40.47)$$

Vacuum energy, area entropy, chronological temperature, and action share a single confluence cell with typed domains.

When $S_H/k_B = \pi$, the multiplicity efectiva satisfies

$$\Omega_{\text{eff}} = e^{S_H/k_B} = e^\pi. \quad (40.48)$$

The hyperbolic Euler operator possesses

$$\text{Spec}(e^{\pi H}) = \{e^\pi, e^{-\pi}\}. \quad (40.49)$$

Equality of the scalars establishes spectral confluence; the intertwiner between microscopic horizon multiplicity and Perron dynamics belongs to its specific physical residence.

Exact reversibility, erasure cost and zero Landauer limit

The full transport preserves fibre information and may have zero logical cost; the projection or reinitialization that discards the genealogy is subject to Landauer's cost. The two domains are separated explicitly.

41.1 Logical reversibility and Landauer's principle in TPK fibers

Let $(\Omega, 2^\Omega, \mu)$ be a finite probability space, let $X : \Omega \rightarrow \mathcal{X}$ be a random variable with values in the space $\mathcal{X}$ of complete states, and let $q : \mathcal{X} \rightarrow Y$ be a reader. The chain rule gives

$$H(X) = H(q(X)) + H(X | q(X)).$$

The second summand measures the information of the fiber that the reader does not publish.

Theorem 41.1 (Landauer zero for full transport and projection cost). *If $U : \mathcal{X} \rightarrow \mathcal{X}$ is a bijective update of the complete HMT state, then*

$$H(X | U(X)) = 0.$$

For a projected output $q(U(X))$, the discarded information is

$$H(X | q(U(X))).$$

A physical reset that identifies distinct memory states must account for that information in the environment.

Proof. Bijectivity implies $X = U^{-1}(U(X))$; therefore X is determined by $U(X)$ and the conditional entropy is zero. For the reader q , the chain rule separates published entropy from fiber entropy. If the reset operation is not injective, the output no longer permits reconstruction of the memory state. Landauer's thermodynamic principle bounds the cost of implementation in terms of that erased entropy; the reversible update does not by itself contribute an erasure term. $\square$

In nats and bits, respectively,

$$Q \geq k_B \Theta H_{\text{nat}}, \quad Q \geq k_B \Theta \log 2 H_{\text{bit}}.$$

For bijective transport of the complete state, the erasure entropy is zero and, therefore, the minimum Landauer term associated exclusively with erasure satisfies

$$Q_L^{\min}[U] = k_B \Theta H(X | U(X)) = 0.$$

For the restart of a uniform TRIT, on the other hand,

$$Q_{\text{reset}} \geq k_B \Theta \log 3.$$

These two equations locate the cost in information loss: the cost appears when projection or resetting forgets the fiber. Preparation, projection, and resetting may produce entropy; reversible transport preserves genealogical information. The null value is the logical Landauer limit for that complete update, whereas the dynamical costs of control, finite speed, and coupling to the environment belong to the physical realization of the device.

Scope of zero Landauer cost

The equality $H(X | U(X)) = 0$ is exact for a bijective update of the integral holonomic state. The bound $Q_{\text{reset}} \geq k_B \Theta \log 3$ specializes Landauer's principle to erasure of a uniform triadic decision. Realization of both formulas in joules uses the Boltzmann transducer and a declared physical temperature.

Edge Information Transduction in Minimum Triadic Cut Capacity. The information preserved by the boundary reappears geometrically in the minimal triadic cut and its capacity.

Minimal area of the triadic cut and capacity $81 \log 3$

Areal quantization begins with the nonadic interior, its boundary, and the collective transport of incidence. The minimal cut contains 81 units and its qutrit capacity is $81 \log 3$; the result precedes its gravitational realization.

42.1 Areal quantization, modular dynamics, and rational CKM transformation

42.1.1 Hexad–octad inclusion and associated functional realizations

Let us fix a set H of six points and a set O of eight points with a distinguished inclusion $j : H \hookrightarrow O$. In the Witt residence, H is a hexad of a dodecad and O its lifted octad; in this chapter only the already fixed inclusion is used, not the terminal value of Barbero–Immirzi. We write

$$\binom{H}{2} := \{P \subset H : |P| = 2\}, \quad \left| \binom{H}{2} \right| = \binom{6}{2} = 15, \quad (42.1)$$

and we define two incidence spaces with distinct types:

$$\mathcal{F}_6 := H \times \binom{H}{2}, \quad \mathcal{F}_8 := O \times \binom{H}{2}. \quad (42.2)$$

The inclusion of incidences is

$$\iota_{6,8} : \mathcal{F}_6 \hookrightarrow \mathcal{F}_8, \quad (h, P) \mapsto (j(h), P). \quad (42.3)$$

It is injective because j is injective and preserves the marked pair P .

Theorem 42.1 (Incidence and defect counts oriented). *The two spaces of equation (42.2) satisfy*

$$|\mathcal{F}_6| = 6 \binom{6}{2} = 90, \quad |\mathcal{F}_8| = 8 \binom{6}{2} = 120, \quad |\mathcal{F}_8 \setminus \iota_{6,8}(\mathcal{F}_6)| = 30. \quad (42.4)$$

Normalization by the twenty-four coordinates of the dodecad–twenty-four pair and by the fifteen pairs of H produces

$$\delta_{6,8}^{\text{inc}} = \frac{|\mathcal{F}_6| - |\mathcal{F}_8|}{24 \binom{6}{2}} = -\frac{1}{12}. \quad (42.5)$$

The same fraction is obtained through the reciprocal functional associated with the incorporated memory,

$$\delta_{6,8}^{\text{rec}} = \frac{30}{120} - \frac{30}{90} = -\frac{1}{12}. \quad (42.6)$$

They are two exact realizations of the same oriented defect, but have distinct domains and are not identified with $\zeta(-1)$ without an additional intertwiner.

Proof. The multiplicative principle applied to equation (42.2) gives $6 \cdot 15 = 90$ and $8 \cdot 15 = 120$. The image of $\iota_{6,8}$ has ninety elements, so the complement has thirty. Substitution into equation (42.5) gives $-30/(24 \cdot 15) = -1/12$, while equation (42.6) is $1/4 - 1/3 = -1/12$. $\square$

The hexad–octad transition admits two functional realizations that are not deduced from one another. The algebraic diagram, formulated on the spaces that justify the counts, is

$$\begin{aligned}
(H \xrightarrow{j} O) &\mapsto (|\mathcal{F}_6|, |\mathcal{F}_8|) = (90, 120) \mapsto \Gamma_{6 \rightarrow 8} = \gamma_{\text{HMT}}, \\
(h_6, o_8, t) &= (6, 8, 3) \mapsto -\frac{o_8 - h_6}{t} = -\frac{2}{3},
\end{aligned} \tag{42.7}$$

where $-2/3$ is the coefficient of the third coordinate of row 13 of the CKM constructor. The first realization transforms the incidence into area and modular memory; the second transforms the same transition into angular-mixing parameters. $\theta_{\text{CKM}} = f(\gamma_{\text{HMT}})$ is not postulated: both outputs preserve a common antecedent.

The morphism determining the sectors $90/120$ is $\iota_{6,8}$. Magnitude γ_{HMT} is adopted as a theorem previously demonstrated in the hexad–octad construction and is not reconstructed from its decimal representation. The operatorial consequences begin only after fixing the elementary scale a_0 , defined below without using entropy or the terminal product $\gamma_{\text{HMT}}a_0$.

42.1.2 Correspondence between the Nonadic Interior and the Boundary Degrees of Freedom

Let $\mathbb{T}_3 = \{0, 1, 2\}$. Every $x \in \mathbb{Z}/9\mathbb{Z}$ has a unique digit representation $x = x_0 + 3x_1$, with $x_i \in \mathbb{T}_3$, and every $y \in \mathbb{Z}/27\mathbb{Z}$ a unique representation $y = y_0 + 3y_1 + 9y_2$. Therefore

$$\mathcal{B} = (\mathbb{Z}/9\mathbb{Z})^3, \quad \mathcal{B}_\partial = (\mathbb{Z}/27\mathbb{Z})^2, \quad |\mathcal{B}| = |\mathcal{B}_\partial| = 3^6 = 729. \tag{42.8}$$

Equality of cardinalities is accompanied by the explicit map. If $x_j = r_{2j-1} + 3r_{2j}$ for $j = 1, 2, 3$, we define

$$\beta_{729}(x_1, x_2, x_3) = (r_1 + 3r_2 + 9r_3, r_4 + 3r_5 + 9r_6). \tag{42.9}$$

Proposition 42.2 (Genealogical bijection $729 \leftrightarrow 729$). *The map $\beta_{729} : \mathcal{B} \rightarrow \mathcal{B}_\partial$ is bijective and preserves the ordered six-trit word. In general, it is not an isomorphism of additive groups: regrouping the digits changes where the carry is recorded.*

Proof. Extracting the six digits from three nonadic coordinates and grouping them into two triples is a permutation of the coordinates of $\mathbb{T}_3^6$. The positional decomposition is unique on both sides, so the operation has an inverse. The additive caveat is verified, for example, by adding words that generate a carry between r_3 and r_4 : that boundary separates the two coordinates of order twenty-seven, but crosses two distinct coordinates in the nonadic packing. $\square$

On the boundary torus $\mathcal{B}_\partial$, we take the block of vertices

$$A = \{0, 1, \dots, 8\} \times \{0, 1, \dots, 8\}. \tag{42.10}$$

Its oriented links crossing the boundary are separated by direction:

$$\partial A_{90} = \{((-1, j), (0, j)), ((8, j), (9, j)) : 0 \leq j \leq 8\}, \tag{42.11}$$

$$\partial A_{120} = \{((i, -1), (i, 0)), ((i, 8), (i, 9)) : 0 \leq i \leq 8\}, \tag{42.12}$$

where all coordinates are read modulo twenty-seven. Thus

$$\partial A = \partial A_{90} \sqcup \partial A_{120}, \quad |\partial A_{90}| = |\partial A_{120}| = 18. \tag{42.13}$$

The subscripts 90, 120 label the channel provenance; they do not say that the boundary has ninety or one hundred and twenty links.

Theorem 42.3 (Law areal of the cut triadic). *In the cubic graph $N \times N \times N$ with unit capacity, every cut between two opposite faces has capacity at least N^2 , and a transverse layer attains that bound. For $N = 9$,*

$$\mathcal{A}_{\min} = 81, \quad S_{\text{tri}} = 81 \log 3. \tag{42.14}$$

Proof. There are N^2 lines of edges parallel to the normal direction, pairwise disjoint. Every cut must intersect each one, so its capacity is at least N^2 . A single transverse layer contains exactly N^2 edges and attains the minimum. If each cut link carries a maximally mixed trit, its entropy is $\log 3$; additivity over the eighty-one links gives equation (42.14). $\square$

The entropy S_{tri} measures the capacity of the cubic cut. The modular entropy of the thirty-six links of equation (42.13) measures another state and must not be equated with it.

42.1.3 Collective incidence entangler between interior and boundary

In the sum $\ell^2(\mathcal{F}_6) \oplus \ell^2(\mathcal{F}_8)$, we define

$$u_{90} = \frac{1}{\sqrt{90}} \sum_{f \in \mathcal{F}_6} (\delta_f, 0), \quad u_{120} = \frac{1}{\sqrt{120}} \sum_{f \in \mathcal{F}_8} (0, \delta_f). \quad (42.15)$$

In $\ell^2(\partial A_{90}) \oplus \ell^2(\partial A_{120})$, let

$$v_{90} = \frac{1}{\sqrt{18}} \sum_{e \in \partial A_{90}} (\delta_e, 0), \quad v_{120} = \frac{1}{\sqrt{18}} \sum_{e \in \partial A_{120}} (0, \delta_e). \quad (42.16)$$

The pairs (u_{90}, u_{120}) and (v_{90}, v_{120}) are orthonormal. Therefore there exists a unique isometry between the collective subspaces that satisfies

$$\mathcal{J}_{6 \rightarrow 8} u_{90} = v_{90}, \quad \mathcal{J}_{6 \rightarrow 8} u_{120} = v_{120}. \quad (42.17)$$

Define

$$D_{\text{inc}} = 90|u_{90}\rangle\langle u_{90}| + 120|u_{120}\rangle\langle u_{120}|, \quad (42.18)$$

$$D_{\text{area}} = 90|v_{90}\rangle\langle v_{90}| + 120|v_{120}\rangle\langle v_{120}|. \quad (42.19)$$

Theorem 42.4 (Exact transport of the two labels). *The map $\mathcal{J}_{6 \rightarrow 8}$ is unitary between the two collective planes and intertwines the incidence and area operators:*

$$\boxed{\mathcal{J}_{6 \rightarrow 8} D_{\text{inc}} = D_{\text{area}} \mathcal{J}_{6 \rightarrow 8}.} \quad (42.20)$$

Therefore, every polynomial combination of the labels 90, 120, in particular the primitive combination $12 : -1$, is transported with the same coefficients.

Proof. Orthonormality shows that $\mathcal{J}_{6 \rightarrow 8}$ maps one orthonormal basis to another. In u_{90} , both sides of equation (42.20) equal $90v_{90}$; in u_{120} , both equal $120v_{120}$. Equality on a basis proves the identity. The functional calculus, in its polynomial form, preserves the intertwining. $\square$

This map does not purport to inject ninety individual incidences into eighteen links: it transports exactly the two uniform modes and their spectral labels. Extending it to non-uniform fluctuations would require a new incidence map; the modular operator defined here only uses the collective subspace that has just been constructed.

42.1.4 What is the elementary and minimal scale of the area operator

Let C^* the angular coordinate conjugate, expresada in grados, and fijemos

$$\theta_{\text{clk}} = \frac{C^*}{6} \frac{\pi}{180}, \quad \boxed{a_0 = \frac{1 + 2 \cos(10^\circ)}{3} \theta_{\text{clk}}.} \quad (42.21)$$

The factor preceding θ_{clk} is not a decimal correction. It is the normalized real character of the phase triplet

$$\frac{1}{3} \operatorname{Tr} \operatorname{diag} \left(1, e^{i\pi/18}, e^{-i\pi/18} \right) = \frac{1 + 2 \cos(10^\circ)}{3}. \quad (42.22)$$

Thus, a_0 is the spectral mean of the angular structure on the oriented trit. It is positive, is fixed before the area operator, and is not selected from entropy. The subsequent identification $\gamma_{\text{HMT}} = \Gamma_{6 \rightarrow 8}$ realizes the HMT output in the Ashtekar–Barbero connection [BG95b, Imm97b]; it does not reopen its derivation.

Remark 42.5 (Alcance of the simbolo a_0). In this chapter, $a_0 \equiv a_0^{\text{area}}$ denotes exclusively the normalization of equation (42.21). It is neither the Bohr radius nor the initial cosmological scale factor, although those objects are traditionally written with the same symbol. No identity in this chapter allows one to substitute one for the other.

Construction of the area operator from the triad cut and channel separation. The cut and its capacity determine an area operator; its spectrum must preserve total occupation and channel provenance separately.

Spectrum of the area operator and reconstruction of channel provenance

The sectors 90 and 120 define compatible occupancy and memory operators. Inverting the linear system reconstructs both channels and prevents total area from erasing provenance.

43.1 Area and modular operators in the sectors 90 and 120

In each of the two families of equation (42.13) there are eighteen qutrit channels. For $m \in \{90, 120\}$, define

$$N_m = \sum_{e \in \partial A_m} N_e, \quad \text{Spec}(N_e) = \{0, 1, 2\}. \quad (43.1)$$

The Barbero–Immirzi output is used here as a prior result, without rederiving it. With the area normalization constructed in equation (42.21), one obtains

$$\boxed{\frac{\widehat{\mathcal{A}}_{\text{phys}}}{\ell_P^2} = \gamma_{\text{HMT}} a_0 (N_{90} + N_{120})}, \quad \gamma_{\text{HMT}} a_0 = 0.0016755940749224991. \quad (43.2)$$

Theorem 43.1 (Finite spectrum of the area quantum). *On the thirty-six links of the cut,*

$$\text{Spec}(N_{90} + N_{120}) = \{0, 1, \dots, 72\}. \quad (43.3)$$

The multiplicity of the eigenvalue n is the coefficient of z^n in $(1 + z + z^2)^{36}$. Consequently,

$$\boxed{\text{Spec}\left(\frac{\widehat{\mathcal{A}}_{\text{phys}}}{\ell_P^2}\right) = \{n\gamma_{\text{HMT}} a_0 : 0 \leq n \leq 72\}}. \quad (43.4)$$

The compatible cuts prolong the family $\mathcal{A}_n^{\text{prot}} = n\gamma_{\text{HMT}} a_0 \ell_P^2$.

Proof. Each operator N_e has a spectral generating function $1 + z + z^2$. The thirty-six factors commute and act on distinct tensor factors; therefore, the generating function of the total is $(1 + z + z^2)^{36}$, whose degrees range over all integers from zero to seventy-two. Multiplication by the positive scalar $\gamma_{\text{HMT}} a_0 \ell_P^2$ gives equation (43.4). $\square$

Two quantization procedures must be distinguished. The cut law theorem 42.3 counts eighty-one unit capacities and determines $81 \log 3$; the operator equation (43.2) quantizes occupations of thirty-six response links, up to seventy-two number trits. They share the same boundary sheet, but not the same observable.

The torsion modular produces

$$\Delta_4^{\text{mod}} = 0.0001111970500599773249414566131933856 \dots, \quad (43.5)$$

and the boundary Hamiltonian

$$\boxed{K_{\text{HHM}}^{(A)} = -\log \rho_A = cI + \Delta_4^{\text{mod}} (90N_{90} + 120N_{120})}. \quad (43.6)$$

The qualifier *holographic* does not designate an analogy: the incidence $6 \rightarrow 8$ is transported to the boundary sets ∂A_{90} and ∂A_{120} , and the reduced state resides in the algebra generated by its number operators.

Theorem 43.2 (Joint reconstruction of geometry and memory). *Let*

$$N = N_{90} + N_{120}, \quad D = 90N_{90} + 120N_{120}. \quad (43.7)$$

Then

$$\boxed{N_{90} = 4N - \frac{D}{30}, \quad N_{120} = \frac{D}{30} - 3N.} \quad (43.8)$$

In particular, $[\widehat{\mathcal{A}}_{\text{phys}}, K_{\text{HHM}}^{(A)}] = 0$, and the pair $(\widehat{\mathcal{A}}_{\text{phys}}, K_{\text{HHM}}^{(A)} - cI)$ reconstructs the two occupation sectors.

Proof. The operators N_{90} and N_{120} act on distinct factors and commute. The linear map carrying (N_{90}, N_{120}) to (N, D) has matrix

$$\begin{pmatrix} 1 & 1 \\ 90 & 120 \end{pmatrix}, \quad \det = 30.$$

Its inverse is exactly equation (43.8). Since the corresponding identity are linear combinations of the same number operators, the commutator vanishes. $\square$

The operator interpretation is precise: the area observable preserves the total number of excitations, while the modular Hamiltonian preserves their sector of origin. Neither observable alone retains both coordinates; the joint pair does determine them.

43.2 Confluence of area, information and energy

The angular scale θ_{clk} in equation (42.21) and the chronological temperature Θ_{clk} are distinct objects; the use of an uppercase letter is part of the typing. The thermodynamic realization in chapter devoted to the quantum–electrical constants, the thermal scale and the Planck system proves

$$E_P = k_B \Theta_{\text{clk}} \log 3, \quad (43.9)$$

whereas the areal realization constructs the elementary increment

$$\delta \mathcal{A} = \gamma_{\text{HMT}} a_0 \ell_P^2. \quad (43.10)$$

They are not equated: equation (43.9) fixes the informational quantum of energy, and equation (43.10) the geometric quantum of area. Their quotient defines the typed tension of conversion

$$\boxed{\mathcal{T}_{I/A} = \frac{E_P}{\delta \mathcal{A}} = \frac{k_B \Theta_{\text{clk}} \log 3}{\gamma_{\text{HMT}} a_0 \ell_P^2}.} \quad (43.11)$$

The expression introduces no additional constant: it determines the normalizations that must be preserved in the transformation of information into geometry.

At the minimal cut of eighty-one links, the physical informational capacity and the energy of one trit per link are

$$S_{\text{cut}}^{\text{phys}} = 81 k_B \log 3, \quad E_{\text{cut}}^{(1)} = 81 E_P = \Theta_{\text{clk}} S_{\text{cut}}^{\text{phys}}. \quad (43.12)$$

The final equality is an equation of state of this realization, not a universal equipartition law. In particular, the Gibbs state of the thirty-six boundary links has a different entropy and is treated next.

This separation determines the structural content of the confluence: Barbero–Immirzi orients the area spectrum; $\log 3$ measures the irreducible decision of the TRIT; Θ_{clk} converts that information into energy; and ℓ_P^2 connects the area quantum to the gravitational family. Gravity is described as a confluence of functional realizations of a common antecedent, and not as an imposed equality among their values.

Transport of the channel memory to the modular Hamiltonian and to the double thermal field purification. Channel memory is represented below through the modular Hamiltonian and a thermofield-double purification.

Boundary modular Hamiltonian, thermofield-double purification and oriented information

The faithful qutrit state admits a double-thermofield purification. The modular Hamiltonian records the orientation of the channels, and the informational deficit measures the finite separation between states without being identified with area.

44.1 Qutrit state, thermofield-double purification, and informational deficit

For a link with weight $w > 0$, let

$$Z(w) = 1 + e^{-w} + e^{-2w}, \quad \rho(w) = \frac{1}{Z(w)} \sum_{n=0}^2 e^{-wn} |n\rangle\langle n|. \quad (44.1)$$

The weights of the two families are

$$w_{90} = 90\Delta_4^{\text{mod}} = 0.0100077345053979592447310951874 \dots, \quad (44.2)$$

$$w_{120} = 120\Delta_4^{\text{mod}} = 0.0133436460071972789929747935832 \dots, \quad \frac{w_{120}}{w_{90}} = \frac{4}{3}. \quad (44.3)$$

The finite boundary state is

$$\rho_A = \rho(w_{90})^{\otimes 18} \otimes \rho(w_{120})^{\otimes 18}. \quad (44.4)$$

Its logarithm reproduce equation (43.6), with $c = 18 \log Z(w_{90}) + 18 \log Z(w_{120})$.

Theorem 44.1 (Double-thermofield purification). *The vector*

$$|\text{TFD}(w)\rangle = \frac{1}{\sqrt{Z(w)}} \sum_{n=0}^2 e^{-wn/2} |n\rangle_A \otimes |n\rangle_{\bar{A}} \quad (44.5)$$

purify $\rho(w)$. The product of eighteen copies for each of the weights w_{90}, w_{120} purifies ρ_A ; in the doubled system, $S(\rho_A)$ is exactly entanglement entropy.

Proof. Upon tracing out the second factor, the orthogonality of $|n\rangle_{\bar{A}}$ eliminates the cross terms and leaves

$$\text{Tr}_{\bar{A}} |\text{TFD}(w)\rangle\langle\text{TFD}(w)| = \rho(w).$$

The partial trace commutes with tensor products, which proves the assertion for the thirty-six links. $\square$

For a link,

$$S(w) = \log Z(w) + w \frac{e^{-w} + 2e^{-2w}}{Z(w)}. \quad (44.6)$$

Consequently,

$$S(\rho_A) = 18S(w_{90}) + 18S(w_{120}) = 39.54837320881807994957319730 \dots \text{ nats.} \quad (44.7)$$

The maximally mixed state of thirty-six qutrits has entropy $36 \log 3$. We define the oriented defect

$$I_{\text{or}} = 36 \log 3 - S(\rho_A) = 0.001669183233868940655631225913 \dots \text{ nats.} \quad (44.8)$$

Its positivity follows from the entropic maximality of the uniform state. The proximity of I_{or} to $\gamma_{\text{HMT}} a_0$ does not constitute an identity and is not used to select either value.

44.2 Odd-even decomposition of the bilayer state: interleaf information and oriented geometric transport

The normalized distribution between sheets separates even entropy from odd hexad–octad transport. This bilayer parity preserves the boundary information that the modular Hamiltonian records and avoids identifying entropy, area, and orientation.

44.3 The bicover: even information and odd geometric transport

Barbero–Immirzi and the entropy of the two sheets are not two juxtaposed quantities. They arise from the same channels $q_{\pm}$, but preserve different data. Probabilistic normalization forgets the common scale and measures the distribution between orientations; the hexad–octad functional preserves the sign of that orientation. The first reading is even and the second is odd.

44.3.1 Normalized distribution between leaves

Write

$$x = A_{\text{rad}}, \quad y = C_{\text{rad}}^*, \quad q_- = e^{-x+y}, \quad q_+ = e^{-x-y}.$$

The sum $q_- + q_+ = 2e^{-x} \cosh y$ allows defining

$$p_- = \frac{q_-}{q_- + q_+} = \frac{e^y}{2 \cosh y}, \quad p_+ = \frac{q_+}{q_- + q_+} = \frac{e^{-y}}{2 \cosh y}.$$

The common factor e^{-x} disappears. The distribution preserves the asymmetry between sheets, but not the common angular scale.

Definition 44.2 (Bilayer entropy). The entropy of the oriented distribution is

$$S_{\text{bi}}(y) := -p_- \log p_- - p_+ \log p_+ = \log(2 \cosh y) - y \tanh y.$$

Proposition 44.3 (Parity and maximum of the bicover entropy). *The function S_{bi} is even and satisfies*

$$S_{\text{bi}}(-y) = S_{\text{bi}}(y), \quad S'_{\text{bi}}(y) = -y \operatorname{sech}^2 y, \quad 0 < S_{\text{bi}}(y) \leq \log 2.$$

The maximum $\log 2$ is attained only at $y = 0$, when both sheets have equal weight.

Proof. Parity follows because $\cosh$ is even and $y \tanh y$ is also even. The direct derivation produces

$$\frac{d}{dy} \log(2 \cosh y) = \tanh y, \quad \frac{d}{dy} (y \tanh y) = \tanh y + y \operatorname{sech}^2 y.$$

Therefore $S'_{\text{bi}}(y) = -y \operatorname{sech}^2 y$: the function increases on $(-\infty, 0)$, decreases on $(0, \infty)$, and takes the value $\log 2$ at the origin. Positivity is that of the entropy of a distribution with two strictly positive weights. $\square$

44.3.2 Parity of the hexad-octad transport

Let

$$s_n(q_-, q_+) = q_-^n - q_+^n.$$

The leaf exchange is the involution

$$J_{\text{sh}} : (x, y) \mapsto (x, -y),$$

which permutes $q_- \leftrightarrow q_+$. Hence,

$$s_n \circ J_{\text{sh}} = -s_n.$$

The functional constructed in [section 40.1](#) is

$$\Gamma_{6 \rightarrow 8}(A, C^*) = \frac{\pi A C^*}{180} + 12(S_{90}(q_-) - S_{90}(q_+)) - (S_{120}(q_-) - S_{120}(q_+)).$$

Theorem 44.4 (Theorem of double-layer parity). *Under the exchange $C^* \mapsto -C^*$,*

$$S_{\text{bi}}(-C_{\text{rad}}^*) = S_{\text{bi}}(C_{\text{rad}}^*),$$

$$\Gamma_{6 \rightarrow 8}(A, -C^*) = -\Gamma_{6 \rightarrow 8}(A, C^*).$$

The same bilayer therefore produces an even informational reading and an odd geometric transport.

Proof. The first equality is [proposition 44.3](#). In the second, the term $\pi A C^*/180$ changes sign, and each Lambert difference changes sign because its two arguments are interchanged. The complete combination is odd. $\square$

44.3.3 Interpretation of the constitutive operator as coupling between vacuum, field, and response

Bilayer entropy and $\Gamma_{6 \rightarrow 8}$ are not two names for the same number. The former quantifies how much weight the two orientations share after cancellation of the common scale. The latter preserves the orientation and incidence of the step $6 \rightarrow 8$. At the HMT point,

$$S_{\text{bi}} = 0.69246069960358548 \dots \text{ nats}, \quad \Gamma_{6 \rightarrow 8} = 0.27400767269445927 \dots$$

The link between information and geometry is that both readings factor through the same channels $q_{\pm}$, not that their values are equated.

The particle–antiparticle band of chapter devoted to the particle band, anti-section and virtuality reproduces the same parity distinction: mass and the amount of correlation descend to the band quotient, whereas charge, orientation and transport change sign. In the Cartan–Holst action, $\Gamma_{6 \rightarrow 8}$ occupies the oriented coordinate; in the informational reading, S_{bi} counts the distribution between sheets. This explains how a single structure can preserve information and reverse orientation without confusing the two operations.

Tangential variation of the modular state and derivation of the areal law. Tangent variation of the modular state produces the area first law; finite variations preserve the relative-entropy term.

Modular area first law and finite extension through relative entropy

The first modular law relates variations of entropy and area within the declared finite system. Its nonlinear extension retains the positive relative-entropy defect and therefore does not conflate tangent equality with global identity.

45.1 1st modular law for area variations

Let us define the two normalized areas

$$\mathcal{A}_m = \gamma_{\text{HMT}} a_0 N_m, \quad m \in \{90, 120\}. \quad (45.1)$$

Then

$$K_{\text{HHM}}^{(A)} - cI = \beta_{90} \mathcal{A}_{90} + \beta_{120} \mathcal{A}_{120}, \quad \beta_m = \frac{m \Delta_4^{\text{mod}}}{\gamma_{\text{HMT}} a_0}. \quad (45.2)$$

The terminal values are

$$\beta_{90} = 5.9726485401070929914651798889 \dots, \quad (45.3)$$

$$\beta_{120} = 7.9635313868094573219535731852 \dots, \quad \frac{\beta_{120}}{\beta_{90}} = \frac{4}{3}. \quad (45.4)$$

Theorem 45.1 (First modular law in areal variables). *For a differentiable family of normalized states $\rho(\varepsilon)$ passing through ρ_A , while keeping Δ_4^{mod} , γ_{HMT} , a_0 fixed,*

$$\boxed{\delta S = \beta_{90} \delta \langle \mathcal{A}_{90} \rangle + \beta_{120} \delta \langle \mathcal{A}_{120} \rangle}. \quad (45.5)$$

Proof. The first entanglement law in the reference state is $\delta S = \delta \langle -\log \rho_A \rangle$. The scalar term cI does not contribute because $\delta \text{Tr } \rho = 0$. Substituting equation (45.2) proves the identity. $\square$

The formula is local in state space. For finite changes, relative entropy appears; omitting it would turn a tangent identity into a false claim of global linearity.

Theorem 45.2 (Finite Identity with a Defect Term). *For every state σ whose support is contained in that of ρ_A ,*

$$S(\sigma) - S(\rho_A) = \beta_{90} \Delta_\sigma \langle \mathcal{A}_{90} \rangle + \beta_{120} \Delta_\sigma \langle \mathcal{A}_{120} \rangle - D(\sigma \| \rho_A), \quad (45.6)$$

where $D(\sigma \| \rho_A) = \text{Tr } \sigma (\log \sigma - \log \rho_A) \geq 0$. In particular, the linear variation of theorem 45.1 is the tangent term of this identity.

Proof. By definition and by $K_{\text{HHM}}^{(A)} = -\log \rho_A$,

$$D(\sigma \| \rho_A) = -S(\sigma) + \text{Tr}(\sigma K_{\text{HHM}}^{(A)}).$$

For $\sigma = \rho_A$, the left-hand side is zero and $S(\rho_A) = \text{Tr}(\rho_A K_{\text{HHM}}^{(A)})$. Subtracting the two identities gives

$$S(\sigma) - S(\rho_A) = \Delta_\sigma \langle K_{\text{HHM}}^{(A)} \rangle - D(\sigma \| \rho_A).$$

The scalar cI cancels between normalized states; substituting equation (45.2) yields equation (45.6). The positivity of relative entropy is Klein's inequality. $\square$

The correction $D(\sigma \| \rho_A)$ measures the informational distance between the actual state and the reference modular state. Thus, the first law loses no content upon leaving the infinitesimal regime: it becomes an exact identity with a controlled positive defect.

Inductive persistence of modular weights and factorial classification of the limit. Inductive persistence of modular weights leads to infinite operator closure and its factorial classification.

Persistence of the incidences 90/120 and hyperfinite factorial limit

Faithful local products preserve the ratios of channels 90 and 120 under refinement. The GNS representation and the ITPFI limit determine the hyperfinite type III factor with explicit boundary parameter.

46.1 Local Limit and Modular Classification of the Infinite System

In an infinite UHF algebra, there is in general no global trace-class density matrix within the algebra. The correct object is the net

$$K_{\text{HHM}}^{(m)} = -\log \rho_m \quad (46.1)$$

with compatible-state restrictions, or the modular generator in the GNS representation. The tracial expectation $\text{id} \otimes \tau_9$ does not automatically preserve a nonracial Gibbs state and does not replace state compatibility.

Theorem 46.1 (Modular type under faithful persistence of both channels). *Suppose that:*

- (i) *each cut is a product of faithful qutrit states with weights w_{90} and w_{120} ;*
- (ii) *the inclusions preserve the state and the channel label;*
- (iii) *infinitely many factors of each type occur, both with positive asymptotic density;*
- (iv) *no additional spectral quotients are introduced, and the product representation is factorial.*

Then the additive group of logarithms of ratios of eigenvalues is

$$90\Delta_4^{\text{mod}}\mathbb{Z} + 120\Delta_4^{\text{mod}}\mathbb{Z} = 30\Delta_4^{\text{mod}}\mathbb{Z}. \quad (46.2)$$

The GNS closure ITPFI is the hyperinfinitesimal factor

$$\boxed{\text{III}_{\lambda_\partial}, \quad \lambda_\partial = e^{-30\Delta_4^{\text{mod}}} = 0.9966696464689573786108299769 \dots} \quad (46.3)$$

Proof. At each link, the eigenvalue quotients are powers of e^{-w_m} . Infinite persistence of both types generates the subgroup $w_{90}\mathbb{Z} + w_{120}\mathbb{Z}$. Since $\gcd(90, 120) = 30$, it follows that equation (46.2). The Araki–Woods asymptotic ratio criterion for the faithful product yields the hyperfinite ITPFI factor with parameter $e^{-30\Delta_4^{\text{mod}}}$ [AW68, Tak03]. $\square$

The theorem states its actual hypotheses. If one of the channels disappears, if the product is incompatible, or if the ratio group changes, the type must be recalculated.

Gravitational publication of the modular closure via the pentacomponent carrier. Modular closure prepares the gravitational expression: first the five-component carrier is constructed from icosahedral incidence.

Icosahedral descent and five-component gravitational carrier

The gravitational line preserves the constructed length and its chronological realization of the corresponding section; its carrier then descends to the icosahedral quotient. The irreducible projector of rank 5 selects the tensor carrier without yet assuming a massive or radiative realization.

47.1 Intrinsic gravitational constant and tensor representation

47.1.1 Icosahedral descent A_5/C_5 and gravitational carrier selection

Gravity is organized here into four strata that must not be merged:

$$\begin{aligned}
 \text{inscription and radial norm R1–R8} &\longrightarrow L \longrightarrow G = c_{\text{int}}^3 L^2 / \hbar_a, \\
 \text{icosahedral incidence} &\longrightarrow V_{12} \xrightarrow{P_5} V_5 \xrightarrow{\Gamma_5} \text{STF}_3 \xrightarrow{\Lambda(n)} \mathcal{H}_{\text{TT}}(n), \\
 \text{CT108 calendar with } t_0 = \pi L / (54 c_{\text{int}}) &\longrightarrow R_{108} = L, \quad \theta_{108}^{\text{circ}} = (54/\pi)^2, \\
 \text{metrological chart} &\longrightarrow 6.6742996594140227 \dots \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}.
 \end{aligned} \tag{47.1}$$

The first line preserves determination of length and coupling through rules R1–R8 of [section 39.1](#). The second constructs the carrier and its transverse traceless reduction. The third is the chronological realization of the already constructed length, and the fourth evaluates the same dimensional section. The reduction $12 \rightarrow 5 \rightarrow 5 \rightarrow 2$ does not alter the common radial coefficient: intensity and representation preserve their provenance and respective functions.

47.1.2 Gravitational magnitude line and chronological scale 108

Let $c_{\text{int}} > 0$, $t_0 > 0$, $\ell_0 = c_{\text{int}} t_0$ and a positive section $\hbar_a$ of the action line, with $a \in \{\text{pre}, \text{ret}\}$. The dimensional transducer is

$$G_a(\theta) = \theta \frac{c_{\text{int}}^5 t_0^2}{\hbar_a} = \theta \frac{c_{\text{int}}^3 \ell_0^2}{\hbar_a}, \quad \theta > 0. \tag{47.2}$$

Equality of the two expressions uses only $\ell_0 = c_{\text{int}} t_0$ and $[G] = L^3 M^{-1} T^{-2}$. Dimension is fixed by the line; the dimensionless character θ distinguishes the realizations of that system. In the longitudinal composition already proved, $t_0 = \pi L / (54 c_{\text{int}})$ and $\theta = (54/\pi)^2$ reproduce exactly $G_a = c_{\text{int}}^3 L^2 / \hbar_a$.

The CT108 calendar defines

$$T_{108} = 108 t_0, \quad \omega_{108} = \frac{2\pi}{108 t_0}, \quad R_{108} = \frac{c_{\text{int}}}{\omega_{108}} = \frac{54}{\pi} \ell_0. \tag{47.3}$$

For each action sheet, let

$$E_{108,a} = \hbar_a \omega_{108}, \quad m_{108,a} = \frac{E_{108,a}}{c_{\text{int}}^2}. \tag{47.4}$$

Then, the reduced Compton wavelength coincides with the radius of the cycle:

$$\bar{\lambda}_{C,a} = \frac{\hbar_a}{m_{108,a} c_{\text{int}}} = R_{108}. \tag{47.5}$$

Theorem 47.1 (Uniqueness of the selector under the HMT periodicity condition). *With the reduced convention $r_g = Gm/c_{\text{int}}^2$ and the HMT periodicity premise that identifies the gravitational radius of the CT108 quantum with its phase radius, the closure condition*

$$r_{g,a}(\theta) = R_{108} \quad (47.6)$$

selects a unique positive solution,

$$\theta_{108}^{\text{circ}} = \left(\frac{54}{\pi}\right)^2 = 295.4525715010569415303525 \dots \quad (47.7)$$

In that embodiment,

$$R_{108} = \bar{\lambda}_{C,a} = r_{g,a} = \ell_P^{\text{circ}}. \quad (47.8)$$

Proof. By the corresponding identity,

$$r_{g,a}(\theta) = \frac{G_a(\theta)m_{108,a}}{c_{\text{int}}^2} = \theta c_{\text{int}} t_0^2 \omega_{108} = \theta \frac{\pi}{54} \ell_0.$$

Comparing it with $R_{108} = (54/\pi)\ell_0$ necessarily gives $\theta = (54/\pi)^2$. The equation is linear in θ and all the factors are positive, so the solution is unique. $\square$

The equality $r_g = R_{108}$ does not follow from the dimensionality of equation (47.2). It is the explicit geometric condition of this HMT realization: action, Compton, and periodic return must determine the same length. The theorem proves that, once this structural condition is accepted, the character θ is fixed without using the SI value of G .

Remark 47.2 (Radius convention). The theorem uses the reduced gravitational radius Gm/c^2 . If it is replaced by the Schwarzschild radius $2Gm/c^2$, the selector changes by a factor of two. The two expressions correspond to distinct geometric conditions, so the convention must be declared before the calculation.

47.1.3 Planck System and Bidirectional Covariance

Corollary 47.3 (Planck Family Associated with the CT108 Chronological Structure). *For $G_a = G_a(\theta_{108}^{\text{circ}})$,*

$$\ell_P = \frac{54}{\pi} \ell_0, \quad t_P = \frac{54}{\pi} t_0, \quad (47.9)$$

$$E_{P,a} = \frac{\hbar_a}{t_P} = \frac{\pi \hbar_a}{54 t_0} = \frac{h_a}{108 t_0}, \quad \ell_P^2 = \frac{G_a \hbar_a}{c_{\text{int}}^3} = \theta_{108}^{\text{circ}} \ell_0^2. \quad (47.10)$$

Moreover,

$$F_{P,a} = \frac{c_{\text{int}}^4}{G_a}, \quad P_{P,a} = \frac{c_{\text{int}}^5}{G_a}, \quad \rho_{P,a} = \frac{\hbar_a}{(\theta_{108}^{\text{circ}})^2 c_{\text{int}}^5 t_0^4}. \quad (47.11)$$

Proof. The first three identities follow from $\ell_P = R_{108}$, $t_P = \ell_P/c_{\text{int}}$, and $h_a = 2\pi \hbar_a$. For the area,

$$\frac{G_a \hbar_a}{c_{\text{int}}^3} = \theta_{108}^{\text{circ}} c_{\text{int}}^2 t_0^2 = \theta_{108}^{\text{circ}} \ell_0^2.$$

The remaining expressions are the dimensional compositions of force, power, and mass per volume with the same section. $\square$

Let

$$R_{\text{act}} := \frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}}. \quad (47.12)$$

Since the action appears in the numerator of the chronological mass and in the denominator of G ,

$$\frac{m_{108,\text{pre}}}{m_{108,\text{ret}}} = R_{\text{act}}, \quad \frac{G_{\text{pre}}}{G_{\text{ret}}} = R_{\text{act}}^{-1}, \quad G_a \hbar_a = \theta_{108}^{\text{circ}} c_{\text{int}}^5 t_0^2. \quad (47.13)$$

Therefore, $\ell_P^2 = G_a \hbar_a / c^3$ does not change sheets. Gravity and action are reciprocally oriented, while area preserves the common geometry. This cancellation is the interface of confluence toward the area operator of the section 42.1.

47.1.4 Icosahedral Component of Dimension 5

The antecedent is neither $\mathbb{R}$ nor a subsequent selection of five polarizations. The object responsible for transporting the incidence is defined explicitly before introducing the exceptional realization.

Let $X_{\text{TPK}}^{\text{exc}}$ be the subspace of the enriched state TPK that preserves, in addition to the arithmetic output, the orientation, route, carry, and exceptional incidence. To make the geometric target space explicit, let $\varphi_g = (1 + \sqrt{5})/2$, $v = \sqrt{1 + \varphi_g^2}$, and define

$$\Omega_{12}^{\text{ico}} = \frac{1}{v} (\{0\} \times \{\pm 1\} \times \{\pm \varphi_g\} \cup \{\pm 1\} \times \{\pm \varphi_g\} \times \{0\} \cup \{\pm \varphi_g\} \times \{0\} \times \{\pm 1\}) \subset S^2. \quad (47.14)$$

Its twelve elements are the oriented vertex classes of an icosahedron. The incidence is not left verbal: it is fixed by

$$\mathcal{I}_{12}^{\text{ico}} = \{(v, w) \in \Omega_{12}^{\text{ico}} \times \Omega_{12}^{\text{ico}} : v \neq w, \langle v, w \rangle = 1/\sqrt{5}\}. \quad (47.15)$$

Each vertex has five neighbors in this relation. The reading bridge is typed as a map

$$\pi_{\text{ico}} : X_{\text{TPK}}^{\text{exc}} \longrightarrow \Omega_{12}^{\text{ico}}. \quad (47.16)$$

The cardinality twelve of the image of π_{ico} is not sufficient. To define an incidence realization, the mapping must satisfy simultaneously:

- (i) being surjective and constant exactly on the fibers that discard the arithmetic chart but preserve the exceptional route;
- (ii) intertwine the action of the oriented automorphisms of the antecedent with a transitive action on Ω_{12}^{ico} ;
- (iii) transporting the declared TPK incidence to $\mathcal{I}_{12}^{\text{ico}}$, not merely its cardinalities;
- (iv) preserve the leaf involution and the nonadic return in the stabilizer of the corresponding fiber.

These conditions separate three assertions: the existence of the map π_{ico} , the identification of the group that acts, and the linear representation constructed thereafter. The second and third cannot be used to define the first retrospectively.

Theorem 47.4 (Incidence Descent to the Icosahedral Quotient). *Suppose that π_{ico} satisfies (i)–(iv) and that the oriented action induced on Ω_{12}^{ico} has group $G_{\text{ico}} \simeq A_5$. Then, for every class $v_0 \in \Omega_{12}^{\text{ico}}$,*

$$H_{v_0} = \text{Stab}_{G_{\text{ico}}}(v_0) \simeq C_5, \quad \Omega_{12}^{\text{ico}} \simeq G_{\text{ico}}/H_{v_0} \simeq A_5/C_5. \quad (47.17)$$

The application

$$\kappa_{v_0} : G_{\text{ico}}/H_{v_0} \longrightarrow \Omega_{12}^{\text{ico}}, \quad gH_{v_0} \longmapsto g \cdot v_0, \quad (47.18)$$

is an equivariant bijection and preserves incidence if, in the quotient, one declares

$$gH_{v_0} \sim g'H_{v_0} \iff (g \cdot v_0, g' \cdot v_0) \in \mathcal{I}_{12}^{\text{ico}}. \quad (47.19)$$

Therefore, the composite $\kappa_{v_0}^{-1} \pi_{\text{ico}}$ is the typed map from the TPK antecedent to the quotient A_5/C_5 .

Proof. The list of equation (47.14) has twelve elements. By transitivity, $|G_{\text{ico}} \cdot v_0| = 12$, and the orbit–stabilizer theorem gives $|H_{v_0}| = 60/12 = 5$. Every group of prime order five is cyclic; hence $H_{v_0} \simeq C_5$. The formula for κ_{v_0} does not depend on the representative: if $gH_{v_0} = g'H_{v_0}$, then $g^{-1}g' \in H_{v_0}$

and $g'v_0 = gv_0$. Surjectivity follows from transitivity, and equality of the cardinalities gives injectivity. Equivariance is immediate: $\kappa_{v_0}(agH_{v_0}) = a\kappa_{v_0}(gH_{v_0})$. Finally, equation (47.19) is precisely the transport of structure through that bijection; hypothesis (iii) makes this transport commute with π_{ico} . $\square$

The theorem does not replace the construction of π_{ico} : it proves that it is obtained once its four requirements have been verified. In the complete HMT profile, the incidence corridor reaching Paley–Witt–Golay–Leech– V^{h} –Monster–Moonshine is the domain of definition of that map. In this autonomous volume, the exact input is the map equation (47.16) with its conditions; the entire segment from A_5/C_5 to TT reduction is proved below without again consulting a gravitational value.

The chain of provenance is expressed through all its morphisms:

$$\text{APP} \longrightarrow \widehat{\mathbb{T}} \longrightarrow \text{TPK} \longrightarrow X_{\text{TPK}}^{\text{exc}} \xrightarrow{\pi_{\text{ico}}} \Omega_{12}^{\text{ico}} \xleftarrow[\simeq]{\kappa_{v_0}} A_5/C_5. \quad (47.20)$$

The complete exceptional corridor is proved in its own dedicated development. The hypothesis that this chapter cannot replace with a count is the existence of π_{ico} satisfying (i)–(iv). With that input fixed, descent to the quotient is theorem 47.4, and P_5 , Γ_5 and $\Lambda(n)$ are constructed from it. Realizing the resulting carrier as a gravitational field remains a typed physical interface, not a consequence of cardinality.

Let $G = A_5$, let $H \simeq C_5$ be the stabilizer of an oriented vertex of the icosahedron, and consider

$$V_{12} = \mathbb{R}[G/H], \quad |G/H| = \frac{60}{5} = 12. \quad (47.21)$$

The permutation representation ρ_{12} decomposes as

$$V_{12} \simeq \mathbf{1} \oplus V_3 \oplus V_{3'} \oplus V_5. \quad (47.22)$$

The central idempotent of the character summand $\chi_5 = (5, 1, -1, 0, 0)$ is

$$P_5 = \frac{5}{60} \sum_{g \in A_5} \chi_5(g^{-1}) \rho_{12}(g) = \frac{1}{12} \left(5I + \sum_{g \in 2A} \rho_{12}(g) - \sum_{g \in 3A} \rho_{12}(g) \right). \quad (47.23)$$

Theorem 47.5 (Rank-five gravitational projector and coupler). *The operator P_5 satisfies*

$$P_5^2 = P_5 = P_5^*, \quad \text{Tr } P_5 = \text{rank } P_5 = 5. \quad (47.24)$$

For

$$\mathbb{G}_{5,a} := G_a(\theta_{108}^{\text{circ}}) P_5 \quad (47.25)$$

we have

$$\text{Spec}(\mathbb{G}_{5,a}) = \{G_a(\theta_{108}^{\text{circ}}) \text{ with multiplicity } 5, 0 \text{ with multiplicity } 7\}. \quad (47.26)$$

Proof. Character orthogonality converts equation (47.23) into the central idempotent that acts as the identity on the unique copy of V_5 and as zero on the other three summands of equation (47.22). Reality of the character and orthogonality of ρ_{12} give self-adjointness. Its trace and rank are five. Multiplying a rank-five projector by the positive scalar G_a immediately produces the indicated spectrum. $\square$

The expression $\mathbb{G}_{5,a}$ does not add five different values of the gravitational constant. It is the same section of G acting on the selected tensor summand and annihilating the other seven modes of the permutation module.

Equivariant reduction of the carrier to the traceless tensor sector. The carrier intertwines with the symmetric traceless tensors and is reduced equivariantly before transversality is imposed.

Tensor intertwiner and transverse reduction $12 \rightarrow 5 \rightarrow 5 \rightarrow 2$

The icosahedral framework constructs an equivariant isometry into the space of traceless symmetric tensors. The transverse projection realizes $12 \rightarrow 5 \rightarrow 5 \rightarrow 2$, preserves the five-component carrier, and localizes the radiative subspace of dimension 2.

48.1 Intertwiner with the space of symmetric traceless tensors

Let $\mathcal{V} \subset S^2$ be the set of the twelve unit vertices of the icosahedron. For each $v \in \mathcal{V}$, define

$$Q_v = vv^T - \frac{1}{3}I_3 \in \text{STF}_3 := \text{Sym}_0^2(\mathbb{R}^3). \quad (48.1)$$

The frame map

$$\Gamma_0 \left(\sum_{v \in \mathcal{V}} x_v e_v \right) = \sum_{v \in \mathcal{V}} x_v Q_v \quad (48.2)$$

is A_5 -equivariant and satisfies

$$\Gamma_0 \Gamma_0^* = \frac{8}{5} I_{\text{STF}_3}, \quad \Gamma_0 = \Gamma_0 P_5. \quad (48.3)$$

Theorem 48.1 (Normalized tensor entangler). *The map*

$$\Gamma_5 := \sqrt{\frac{5}{8}} \Gamma_0|_{V_5} : V_5 \xrightarrow{\cong} \text{STF}_3 \quad (48.4)$$

is a A_5 -equivariant isometry. Its adjoint is

$$\Gamma_5^* Q = \sqrt{\frac{5}{8}} \sum_{v \in \mathcal{V}} (v^T Q v) e_v. \quad (48.5)$$

Proof. Equivariance follows from $Q_{gv} = gQ_v g^T$. The operator $\Gamma_0 \Gamma_0^*$ commutes with the irreducible representation of A_5 on STF_3 , and is therefore scalar. Its trace is

$$\sum_{v \in \mathcal{V}} \|Q_v\|_F^2 = 12 \cdot \frac{2}{3} = 8;$$

dividing by dimension five yields $8/5$. Hence, $\Gamma_5 \Gamma_5^* = I$. Since domain and codomain have dimension five, also $\Gamma_5^* \Gamma_5 = I$. The formula for the adjoint follows from $\langle Q, Q_v \rangle_F = v^T Q v$. $\square$

Normalization through the icosahedral frame eliminates a purely cardinal coincidence: it constructs the map that transports the finite summand to the continuous spin-two carrier.

48.2 Transverse traceless projection

For $n \in S^2$, let $\Pi(n) = I - nn^T$. We define

$$\Lambda(n)Q = \Pi Q \Pi - \frac{1}{2} \text{Tr}(\Pi Q \Pi) \Pi. \quad (48.6)$$

Theorem 48.2 (Reduction from five components to two). *The restriction of $\Lambda(n)$ to STF_3 is the orthogonal projector onto*

$$\mathcal{H}_{\text{TT}}(n) = \{Q \in \text{STF}_3 : Qn = 0\}. \quad (48.7)$$

In particular,

$$\Lambda(n)^2 = \Lambda(n) = \Lambda(n)^*, \quad \text{rank } \Lambda(n) = 2. \quad (48.8)$$

For $n = e_3$, the image is generated by

$$E_+ = \frac{1}{\sqrt{2}} \text{diag}(1, -1, 0), \quad E_\times = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (48.9)$$

Proof. Since $\Pi n = 0$, the output is transverse. Its trace is

$$\text{Tr}(\Pi Q \Pi) - \frac{1}{2} \text{Tr}(\Pi) \text{Tr}(\Pi Q \Pi) = 0,$$

since $\text{Tr } \Pi = 2$. If $Qn = 0$ and $\text{Tr } Q = 0$, then $\Pi Q \Pi = Q$ and $\text{Tr}(\Pi Q \Pi) = 0$, so that $\Lambda(n)Q = Q$. The formula is self-adjoint with respect to the Frobenius product. In the frame $n = e_3$, its matrix in an adapted STF basis is $\text{diag}(1, 1, 0, 0, 0)$, which proves the rank. $\square$

The complete chain thus remains,

$$\boxed{V_{12} \xrightarrow{P_5} V_5 \xrightarrow{\Gamma_5} \text{STF}_3 \xrightarrow{\Lambda(n)} \mathcal{H}_{\text{TT}}(n), \quad 12 \longrightarrow 5 \longrightarrow 5 \longrightarrow 2.} \quad (48.10)$$

The reduced coupling

$$\mathbb{G}_{\text{TT},a}(n) = G_a(\theta_{108}^{\text{circ}}) \Lambda(n) \Gamma_5 P_5 \quad (48.11)$$

has rank two.

Helical decomposition of the spectrum and separation of massive and non-massive modes. The reduction distinguishes the complete helicity spectrum of a massive realization from its two massless helicities.

Helicity decomposition of the spin-two carrier

The real carrier of dimension 5 decomposes into 5 helicity weights. The massive realization retains all 5 components; the massless transverse reduction retains only the radiative helicities ± 2 , without converting a collective possibility into a postulated fundamental particle.

49.1 Decomposition into 5 polarization modes

Dimension five acquires tensorial content upon choosing a unit direction of propagation n and an oriented orthonormal frame (e_1, e_2, n) . Define the five real tensors

$$E_+ = \frac{1}{\sqrt{2}}(e_1 e_1^T - e_2 e_2^T), \quad E_\times = \frac{1}{\sqrt{2}}(e_1 e_2^T + e_2 e_1^T), \quad (49.1)$$

$$E_1 = \frac{1}{\sqrt{2}}(e_1 n^T + n e_1^T), \quad E_2 = \frac{1}{\sqrt{2}}(e_2 n^T + n e_2^T), \quad (49.2)$$

$$E_0 = \frac{1}{\sqrt{6}}(2nn^T - e_1 e_1^T - e_2 e_2^T). \quad (49.3)$$

All are symmetric and traceless. The first two are transverse; the next two mix the transverse plane with the direction of propagation, and the last is the traceless longitudinal scalar mode.

Theorem 49.1 (Complete helical decomposition of the carrier). *The tensors of the corresponding identity form an orthonormal basis of STF_3 . In the complexification, the combinations*

$$E_{\pm 2} = \frac{1}{\sqrt{2}}(E_+ \mp iE_\times), \quad E_{\pm 1} = \frac{1}{\sqrt{2}}(E_1 \mp iE_2), \quad E_0 = E_0 \quad (49.4)$$

are weight- $\pm 2, \pm 1, 0$ vectors, respectively, under rotations about n . Moreover,

$$\Lambda(n)E_{\pm 2} = E_{\pm 2}, \quad \Lambda(n)E_{\pm 1} = 0, \quad \Lambda(n)E_0 = 0. \quad (49.5)$$

Proof. In the frame (e_1, e_2, n) , each tensor has Frobenius norm one. Their off-diagonal entry patterns are disjoint, and the three diagonals of E_+, E_0 have zero inner product; therefore, the five tensors are orthonormal. Since $\dim \text{STF}_3 = 6 - 1 = 5$, they form a basis.

A rotation through angle ψ about n sends (e_1, e_2) to $(\cos \psi e_1 + \sin \psi e_2, -\sin \psi e_1 + \cos \psi e_2)$. Substituting into the formulas gives a rotation through angle 2ψ on $\text{span}\{E_+, E_\times\}$, one through angle ψ on $\text{span}\{E_1, E_2\}$, and E_0 fixed. Hence the weights of equation (49.4). Finally, $\Pi E_+ \Pi = E_+$ and $\Pi E_\times \Pi = E_\times$, whereas $\Pi E_1 \Pi = \Pi E_2 \Pi = 0$. For E_0 , $\Pi E_0 \Pi = -(e_1 e_1^T + e_2 e_2^T)/\sqrt{6}$, which is pure trace in the plane, and subtracting equation (48.6) annihilates it. $\square$

The theorem specifies the meaning of “five polarizations” without assigning all of them the same dynamic status. Before field equations are imposed, the spin-two carrier has five irreducible coordinates. In a massive Fierz–Pauli realization, the constraints leave precisely the five weights of equation (49.4); in a massless realization with gauge invariance, the sectors ± 1 and 0 are eliminated, leaving ± 2 . Here HMT supplies the carrier, the basis, and the projector; the choice of massive or massless dynamics belongs to the physical interface. Thus five does not mean “five observed waves,” but rather five tensorial degrees available before reduction.

The orientation also remains visible: replacing (e_1, e_2, n) with $(e_1, -e_2, -n)$ preserves E_+, E_0 and exchanges the phases of the helical pairs. That information is not in the scalar G_a ; it resides in the carrier on which $\mathbb{G}_{5,a}$ acts.

49.2 Representational chain $12 \rightarrow 5 \rightarrow 5 \rightarrow 2$: carrier V_5 , trace-free tensor, Fierz–Pauli polarizations, and transverse helicities

Dimension five appears in distinct objects, and only equation (48.10) authorizes transport between them:

1. V_5 is the irreducible summand of the representation on the twelve vertices. This is a finite result in representation theory.
2. STF_3 is the real five-dimensional carrier of the $m = -2, -1, 0, 1, 2$ weights of spin two.
3. A massive Fierz–Pauli theory can realize these five weights as five physical states, after fixing the action and constraints.
4. Massless general relativity retains two radiative helicities, ± 2 , represented by E_+ and E_- .
5. The $E(2)$ classes of metric waves, such as III_5 , classify possible amplitudes under other hypotheses; they are not automatically V_5 , Fierz–Pauli, or general relativity.

In particular, the correct statement is: *HMT constructs a five-component tensor carrier before the restrictions; the TT reduction of the massless realization has rank two.* It is not asserted that general relativity has five propagating polarizations.

Existence conditions of the mode through natural holonomic selection. Physical existence of the mode still requires a natural, falsifiable holonomic selector.

Natural gravitational selector and dimensional transducer

50.1 Criteria for selecting the gravitational coordinate: enriched holonomy, spectral response, and metrological chart

The Palatini–Cartan–Holst action and its constrained Hamiltonian first determine the dynamics. Upon it, the gravitational constant is organized into three typed levels: the dimensional line that contains G , the transducer that converts a dimensionless coordinate into a section of that line, and the structural selector of that coordinate. The decimal chart and experimental uncertainty are presented at the end as representation and comparison.

50.1.1 Dimensional line and internal transducer

In the dimensional lattice of Dimensional Mechanics,

$$\mathbf{G} = -\mathbf{S} + 5\mathbf{C} + 2\mathbf{T}, \quad [G] = L^3 M^{-1} T^{-2}.$$

Let

$$c_{\text{int}} \in \mathcal{L}_C^+, \quad t_0 \in \mathcal{L}_T^+, \quad \ell_0 = c_{\text{int}} t_0, \quad \hbar_{\text{int}} \in \mathcal{L}_S^+.$$

Definition 50.1 (Gravitational transducer). For $\theta_G > 0$, define

$$G_{\text{int}}(\theta_G) = \theta_G \frac{c_{\text{int}}^5 t_0^2}{\hbar_{\text{int}}} = \theta_G \frac{c_{\text{int}}^3 \ell_0^2}{\hbar_{\text{int}}} \in \mathcal{L}_G^+. \quad (50.1)$$

The equality of the two expressions and the type of (50.1) are exact. If

$$L_G^2 := \frac{\hbar_{\text{int}} G_{\text{int}}}{c_{\text{int}}^3},$$

then

$$\theta_G = \left(\frac{L_G}{\ell_0} \right)^2. \quad (50.2)$$

The gravitational coordinate is therefore a quadratic ratio of lengths or a normalized holonomy response. The decade $10^{-34} \mathcal{U}_S$ already belongs to $\hbar_{\text{int}}$: it fixes the absolute action section and does not constitute a relative factor added to G .

50.1.2 Observable torsion and positive characters

The observable return of 27 steps

$$\mathcal{K}_{27} := F_{\text{obs}}^{27}$$

satisfies

$$\mathcal{K}_{27}^4 = F_{\text{obs}}^{108} = I.$$

In coordinate $\theta \in \mathbb{Z}/108\mathbb{Z}$, the order is exactly four, since $\mathcal{K}_{27}(\theta) = \theta + 27$, $\mathcal{K}_{27}^2(\theta) = \theta + 54 \neq \theta$, and $\mathcal{K}_{27}^4(\theta) = \theta$.

Theorem 50.2 (Triviality of the Observable Positive Character). *Every multiplicative homomorphism*

$$\chi : \langle \mathcal{K}_{27} \rangle \longrightarrow \mathbb{R}_{>0}^\times$$

cumple $\chi(\mathcal{K}_{27}) = 1$.

Proof. Multiplicativity implies $\chi(\mathcal{K}_{27})^4 = \chi(I) = 1$. The only positive real number whose fourth power is one is 1. $\square$

The same cycle does carry the unitary characters.

$$\chi_j(\mathcal{K}_{27}) = \exp(2\pi i j / 4), \quad j = 0, 1, 2, 3.$$

The observable quotient admits phase, whereas its torsion eliminates a nontrivial positive multiplicative scale. A multiplicative continuation must therefore act on a lift $\tilde{\mathcal{K}}_{27}$ whose class in the abelianization of the isotropy group is nontorsional, and must naturally select both the character and its normalization.

50.1.3 Chronological memory and spectral response

The directed chronological covering realizes the return of 27 steps as the translation

$$(G_{\text{or}}, S) \mapsto (G_{\text{or}}, S) + (1, 18),$$

of infinite semigroup order. This route preserves accumulation, but its spectral response requires interference among histories.

Let $\Gamma = (V, E)$ be a finite directed multigraph with positive measure, weights $a_*(e) \geq 0$, and cocycle $c(e) \in \mathbb{Z}^2$. The twisted operator

$$(\mathcal{L}_{\beta, \vartheta} f)(y) = \sum_{e: x \rightarrow y} e^{-\beta a_*(e)} e^{i \langle \vartheta, c(e) \rangle} f(x)$$

induce the operator positive $\mathcal{H}_{\beta, \vartheta} = \mathcal{L}_{\beta, \vartheta}^* \mathcal{L}_{\beta, \vartheta}$.

Proposition 50.3 (Interference criterion for the phase response). *In a deterministic orbit with a unique incoming edge at each vertex, $\mathcal{H}_{\beta, \vartheta}$ is independent of ϑ . Every nonzero-phase spectral response requires multiple coherent histories or a magnetic differential whose holonomy survives in the enriched cycles.*

Proof. On the deterministic orbit, the sole phase of each term is multiplied by its conjugate in $\mathcal{L}_{\beta, \vartheta}^* \mathcal{L}_{\beta, \vartheta}$ and cancels. The absence of cross terms removes every dependence on ϑ . $\square$

If a simple band $\lambda_0(\vartheta)$ has a minimum at the origin, its Hessian Q is positive semidefinite. With

$$v_{27} = (1, 18), \quad \mathcal{R}_{27} = v_{27}^T Q v_{27},$$

the condition $\mathcal{R}_{27} \neq 0$ constitutes the minimal control of sensitivity to accumulated memory. A natural normalization $\mathfrak{N}$ would allow one to define

$$\Lambda_G = \ell_0 \mathfrak{N}(Q, v_{27}), \quad G_{\text{spec}} = \frac{c_{\text{int}}^3}{\hbar_{\text{int}}} (\delta_\theta \Lambda_G)^2, \quad \delta_\theta = \frac{A_{\text{deg}} - C_{\text{deg}}^*}{360}.$$

The homogeneity of this template is exact. Its physical realization is characterized by four positive conditions: an enriched multigraph, a nonzero response, stability under refinement, and independence under rescaling, together with a normalization $\mathfrak{N}$ fixed before the metrological contrast.

50.1.4 Geometric Functional on Enriched Holonomy

The second continuation defines directly a length functional.

$$\mathcal{L} : \text{Loop}(\mathcal{X}_{\text{TPK}}^{\text{enr}}) \longrightarrow \mathcal{L}_L^+ \cup \{0\},$$

invariant under cyclic change of basepoint, conjugation, admissible relabeling, and subdivision. For an enriched lift, write

$$\Lambda_{27} := \mathcal{L}(\tilde{\mathcal{K}}_{27}), \quad \rho_{27} := \frac{\Lambda_{27}}{c_{\text{int}} t_0}, \quad \theta_G^{\text{hol}} := (\delta_\theta \rho_{27})^2.$$

Transduction produces the homogeneous template

$$G_{\text{hol}} = \frac{c_{\text{int}}^3}{\hbar_{\text{int}}} (\delta_\theta \Lambda_{27})^2. \quad (50.3)$$

The concatenation law of this functional on the torsional cycle forms part of its definition and its proof. Physical selection is completed by constructing $\tilde{\mathcal{K}}_{27}$ and $\mathcal{L}$, fixing a unique normalization, and verifying the stability of the output. The value $G_{\text{CODATA 2022}}$ enters exclusively afterward, in the metrological comparison.

These conditions describe this holonomy template. The longitudinal and radial realization already specified in section 39.1 uses the marked record, its conditional expectation and the positive norm; its normalization and the common value of G are proved there. $\delta_\theta \Lambda_{27}$ is not identified with L by formal resemblance: the alternative template retains the obligation to construct its functional before making that identification.

Proposition 50.4 (Separation of the two gravitational selectors). *The circular selector $\theta_{108}^{\text{circ}} = (54/\pi)^2$, constructed by action–Compton–gravity closure, and the holonomic selector $\theta_G^{\text{hol}} = (\delta_\theta \rho_{27})^2$ are two distinct typed realizations of the transducer G_{int} . They are not identified with one another, and the decimal chart $\mathcal{R}_G$ does not constitute an evaluation of G_{hol} .*

Proof. The first selector depends solely on the circular closure of order 108. The second additionally contains the spectral length $\Lambda_{27} = \mathcal{L}(\tilde{\mathcal{K}}_{27})$. An equality between them would require computing Λ_{27} and proving the agreement of normalizations; no operation of the decimal reader performs that computation. Their domains and input data therefore distinguish them before any numerical comparison. $\square$

50.1.5 Structural selection criterion

Test criterion 50.5 (Structural selector of the gravitational constant). An HMT selector of G is identified when it jointly satisfies:

1. enriched domain, composition, and orientation declared;
2. descent to a nontorsional Abelian class, or a geometric functional invariant under basepoint, conjugation, relabelling, and subdivision;
3. nontrivial positive response stable under refinement;
4. unique normalization independent of the operator's overall scale;
5. internal selection of the coordinate and its decimal position;
6. freezing of the code and entries before revealing the comparison value;
7. final realization as a section of $\mathcal{L}_G$ in a common dimensional chart.

The dimensionless ratios

$$\frac{\mathcal{G}_{\text{tor}}}{\mathcal{I}_{\text{tor}}} = 1, \quad \frac{\mathcal{G}_{\text{av}}}{\mathcal{I}_{\text{av}}} = \frac{\pi}{\varphi^2}$$

describe distinct gravitational–inertial readers and remain compatible with a single dimensional section G . The construction of that section also uses $(\hbar_{\text{int}}, c_{\text{int}}, \ell_0)$ and the selector θ_G .

50.1.6 Evaluation of the decimal gravitational chart and subsequent metrological contrast

The decimal representation preserved in the corpus is

$$\mathcal{R}_G([n, k, d]_3; t, e) := 10^{-n} \left(k + \frac{t}{10^3} + \frac{e}{10^d} \right). \quad (50.4)$$

For the chart $[n, k, d]_3 = [11, 6, 5]_3$, $t = 674$ and $e = 30$, the evaluation racional is

$$\mathcal{R}_G = 10^{-11} \left(6 + \frac{674}{1000} + \frac{30}{10^5} \right) = 6.67430 \times 10^{-11}. \quad (50.5)$$

The triple, the two blocks and the decimal position constitute the declared data of this chart. Its evaluation is exact; its structural selection is governed by [test criterion 50.5](#). This decomposition is retained as a historical decimal witness. The radial branch of (39.16) obtains its coordinate through length and action and gives (39.26); it does not use blocks 674 and 30 to define the coupling.

After choosing the SI basis of $\mathcal{L}_G$, the current contrast is

$$G_{\text{CODATA 2022}} = 6.67430(15) \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}. \quad (50.6)$$

Field	Metrological Statement
Central value	$6.67430 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$
Standard uncertainty	$u(G) = 0.00015 \times 10^{-11} = 1.5 \times 10^{-15} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$
Incertidumbre relativa	$u_r(G) \simeq 2.2 \times 10^{-5}$
Source and temporality	2022 CODATA recommended values, published in 2024 [COD24]
Proof function	Subsequent external contrast; neither the unit nor the uncertainty enters into the rational evaluation of (50.5).

Reproducibility of the gravitational reader. An independent enumeration jointly verifies the dimensional type, the decimal reader, and the monodromy identities on the declared finite domain.

Proposition 50.6 (Causal chain of the gravitational constant). *The covariant action, the restricted Hamiltonian, and the Einstein–Schrödinger equation govern the gravitational dynamics. The dimensional type and the transducer (50.1) are exact; the evaluation (50.5) is exact for its declared chart. The value, unit, and uncertainty of (50.6) constitute an external comparison with CODATA 2022. The autonomous selection of θ_G is typed by the enriched lift, the interfering response, and the holonomy functional subject to the test criterion 50.5.*

In the radial realization under rules R1–R8, selection is determined by (39.13) and (39.16); compatibility with the calendar is proved in (39.28). The holonomy template separately retains the conditions of its own domain.

Dimensional Transduction and Holonomic Selection of the Gravitational Constant

51.1 Gravitational line and dimensional form

In the lattice of chapter devoted to magnitude lines and the dimensional realization of the kernel, the dimension of the gravitational constant is

$$\mathbf{G} = -\mathbf{S} + 5\mathbf{C} + 2\mathbf{T}, \quad [G] = L^3 M^{-1} T^{-2}.$$

Let

$$c_{\text{int}} \in \mathcal{L}_C^+, \quad t_0 \in \mathcal{L}_T^+, \quad \ell_0 := c_{\text{int}} t_0 \in \mathcal{L}_L^+, \quad \hbar_{\text{int}} \in \mathcal{L}_S^+.$$

The section $\hbar_{\text{int}}$ is obtained only after the decade determinantal reader -34 and the choice of a basis for the action line. It is not the dimensionless angular character preceding the return.

For every dimensionless character $\theta_G > 0$, define

$$G_{\text{int}}(\theta_G) = \theta_G \frac{c_{\text{int}}^5 t_0^2}{\hbar_{\text{int}}} = \theta_G \frac{c_{\text{int}}^3 \ell_0^2}{\hbar_{\text{int}}} \in \mathcal{L}_G^+.$$

The equality of the two expressions and their type are INTERNALLY EXACT within the dimensional lattice. They do not select θ_G . If

$$L_G^2 := \frac{\hbar_{\text{int}} G_{\text{int}}}{c_{\text{int}}^3},$$

then

$$\theta_G = \left(\frac{L_G}{\ell_0} \right)^2.$$

This final identity locates the selecting datum: the length ratio must be produced by a holonomy or spectral-response structure prior to the metrological chart. The realization of [section 39.1](#) satisfies this requirement through inscription and the radial norm, with $L_G = L$. What follows examines other return and holonomy templates, preserving the obstructions specific to their quotients.

51.2 Obstruction of the Positive Character on the Observable Return

The section 53.3.1 defines

$$\mathcal{K}_{27} := F_{\text{obs}}^{27}.$$

The primary proposition [proposition 10.46](#) proves

$$F_{\text{obs}}^{108} = I;$$

Therefore, in the observable quotient,

$$\mathcal{K}_{27}^4 = I.$$

Theorem 51.1 (Torsional Obstruction for the Observable Positive Selector). *Every multiplicative homomorphism*

$$\chi : \langle \mathcal{K}_{27} \rangle \longrightarrow \mathbb{R}_{>0}^\times$$

is trivial. In particular,

$$\chi(\mathcal{K}_{27}) = 1.$$

Proof. Multiplicativity and $\mathcal{K}_{27}^4 = I$ imply

$$\chi(\mathcal{K}_{27})^4 = \chi(I) = 1.$$

The sole positive real element whose fourth power is one is 1. $\square$

The observable order is exactly four, and not merely a divisor of four: in the coordinate $\theta \in \mathbb{Z}/108\mathbb{Z}$,

$$\mathcal{K}_{27}(\theta) = \theta + 27, \quad \mathcal{K}_{27}^2(\theta) = \theta + 54 \neq \theta, \quad \mathcal{K}_{27}^4(\theta) = \theta.$$

The same cyclic group admits unitary characters.

$$\chi_k(\mathcal{K}_{27}) = \exp\left(\frac{2\pi i k}{4}\right), \quad k = 0, 1, 2, 3,$$

and, in particular, the phase values $\pm i$. The typed conclusion is that the observable cycle can transport phase but not a nontrivial positive scale. These phase characters are not identified with the Paley operator J_W : they satisfy an analogous quadratic relation on distinct domains.

The obstruction does not depend on a metrological comparison. It affects the very form of the selector: a positive character on the observable cycle cannot produce a nontrivial θ_G . Moreover, a mere set of conjugacy classes is not, in general, a group and cannot be declared the domain of a character without first defining its composition.

51.2.1 Unique admissible multiplicative continuation

A multiplicative route can reopen only on an enriched object that does not inherit the observable torsion. For an object x of the memory groupoid, the correctly typed domain would be the isotropy group

$$H_x := \text{Aut}_{\mathcal{G}_{\text{mem}}}(x)$$

and, by multiplicativity towards an abelian group, its abelianization

$$H_x^{\text{ab}} \longrightarrow \mathbb{R}_{>0}^\times.$$

Every positive character annihilates the torsion part of H_x^{ab} . Therefore, the proposal must construct a lift

$$\tilde{\mathcal{K}}_{27} \in H_x, \quad p(\tilde{\mathcal{K}}_{27}) = \mathcal{K}_{27},$$

prove that its Abelian class is non-torsion and naturally select a character $\tilde{\chi}_G$. The positive route over the observable cycle is thus closed by obstruction: without that non-torsion datum there is no nontrivial multiplicative selector. If every admissible lift satisfied $\tilde{\mathcal{K}}_{27}^4 = I$, the multiplicative route would also be ruled out in the enriched domain.

51.3 Chronological composition of length, action, and spectrum in the determination of G

The [theorem 53.6](#) provides a third route, distinct both from the obstructed observable character and from a possible lift in an isotropy group. On

$$\tilde{\mathcal{O}}_{108}^+ = \mathcal{O}_{108} \times \mathbb{N}^2,$$

the twenty-seven-step return adds (1, 18) and its fourth power adds (4, 72). This action belongs to a directed chronological category. It is neither a lift of the groupoid nor a field proved native to the complete integral holonomic state.

The already constructed positive functional,

$$\mathcal{A}_*(\gamma) = G_{\text{or}}(\gamma) + \lambda_* S(\gamma),$$

is additive under concatenation of directed histories. It is not a character of the isotropy group, because its indicators do not change sign under a formal inversion. The completion $\mathbb{N}^2 \rightarrow \mathbb{Z}^2$ permits unitary characters to be introduced for harmonic analysis, but does not alter this fact.

$$\delta_\theta := \frac{A_{\text{deg}} - C_{\text{deg}}^*}{360}. \quad (51.1)$$

This oriented angular fraction is defined once the unique angle and counter-angle have been constructed. It will be used in the two subsequent dimensional templates and does not intervene retroactively in obtaining A_{deg} or C_{deg}^* .

51.3.1 Operator with torsion and deterministic obstruction to multiplicative continuation

For the exact control, let $\Gamma = (V, E)$ be an enriched directed *finite* multigraph, with measure $\mu : V \rightarrow \mathbb{R}_{>0}$, weights $a_*(e) \geq 0$, parameter $\beta \geq 0$, and cocycle $c(e) \in \mathbb{Z}^2$. For a fixed normalization, consider

$$(\mathcal{L}_{\beta, \vartheta} f)(y) = \sum_{e: x \rightarrow y} e^{-\beta a_*(e)} e^{i \langle \vartheta, c(e) \rangle} f(x), \quad \vartheta \in \mathbb{T}^2,$$

and take the entire finite Hilbert space $\ell^2(V, \mu)$ as the domain. Then $\mathcal{L}_{\beta, \vartheta}$ is bounded and

$$\mathcal{H}_{\beta, \vartheta} := \mathcal{L}_{\beta, \vartheta}^* \mathcal{L}_{\beta, \vartheta} \geq 0$$

is self-adjoint. An extension to an infinite graph would additionally require bounded operators, or densely defined closed operators such that $\mathcal{L}_{\beta, \vartheta}^* \mathcal{L}_{\beta, \vartheta}$ were self-adjoint on a common domain; that extension is not assumed here.

Proposition 51.2 (Phase cancellation in the deterministic orbit). *If Γ is solely the observable deterministic orbit and every vertex has a single incoming edge, then $\mathcal{H}_{\beta, \vartheta}$ is independent of ϑ . In particular, the Hessian of any of its bands with respect to ϑ is null.*

Proof. In each component, $\mathcal{L}_{\beta, \vartheta}$ multiplies the sole incoming term by a unitary phase. In $\mathcal{L}_{\beta, \vartheta}^* \mathcal{L}_{\beta, \vartheta}$, that phase is multiplied by its conjugate and cancels. No cross-terms remain between distinct histories. $\square$

This control prevents spectral rigidity from being attributed to the mere deterministic cycle. A nontrivial phase response requires at least one of the following constructions:

1. an enriched multigraph in which two or more coherent histories contribute to the same amplitude; or
2. to magnetic differential

$$(d_\vartheta f)(e) = e^{i \langle \vartheta, c(e) \rangle / 2} f(t(e)) - e^{-i \langle \vartheta, c(e) \rangle / 2} f(s(e)),$$

with Laplacian $\Delta_\vartheta = d_\vartheta^* d_\vartheta$, whose phase depends on the holonomy of enriched cycles.

In both cases, the Hilbert space, domain, measure, weights, and normalization must be declared, and it must be proved that the family is not unitarily equivalent to the family without phase.

Suppose that $\vartheta \mapsto \mathcal{H}_{\beta, \vartheta}$ is an analytic self-adjoint family on a common domain and that $\lambda_0(\vartheta)$ is a simple isolated band in a neighborhood of $\vartheta = 0$. If 0 is a critical point and a local minimum,

$$Q_{ij} = \left. \frac{\partial^2 \lambda_0(\vartheta)}{\partial \vartheta_i \partial \vartheta_j} \right|_{\vartheta=0}$$

is positive semidefinite. Without the minimum hypothesis, Q represents only the quadratic response. For

$$v_{27} = (1, 18), \quad \mathcal{R}_{27} = v_{27}^\top Q v_{27},$$

the condition $\mathcal{R}_{27} \neq 0$ is a minimal test of spectral sensitivity to the accumulated memory.

51.3.2 Typed normalization of the spectral response and construction of θ_G^{spec}

If a natural and unique normalization $\mathfrak{N}$ converts the preceding response into a positive ratio independent of the operator's global scale, one may define

$$\Lambda_G := \ell_0 \mathfrak{N}(Q, v_{27}), \quad \theta_G^{\text{spec}} := \left(\delta_\theta \frac{\Lambda_G}{\ell_0} \right)^2,$$

and, by the dimensional transducer,

$$G_{\text{spec}} = \frac{c_{\text{int}}^3}{\hbar_{\text{int}}} (\delta_\theta \Lambda_G)^2.$$

The homogeneity of the template is exact. The class of its evaluations is characterized by the conditions of canonical operator selection, nonzero response, naturality and stability under refinement, independence of rescaling, uniqueness of $\mathfrak{N}$, and sealing prior to any query of the metrological value of G . Without those conditions, rescaling invariance proves the nonuniqueness of the selector; with them, the evaluation is determined on the declared normalized domain.

51.4 Nonhomomorphic functional over enriched paths

The second possible continuation abandons the multiplicative character and uses a geometric, spectral, or action functional. Let

$$\mathcal{L} : \text{Loop}(\mathcal{X}_{\text{TPK}}^{\text{enr}}) \longrightarrow \mathcal{L}_L^+ \cup \{0\}$$

invariant under cyclic change of base point, admissible relabeling, conjugation, and subdivision. Its concatenation law must be stated explicitly—for example, as a subadditive norm or a spectral functional—; it is not presumed to be a homomorphism of the torsion group. For a candidate enriched lift, we write

$$\Lambda_{27} := \mathcal{L}(\tilde{\mathcal{K}}_{27}), \quad \rho_{27} := \frac{\Lambda_{27}}{c_{\text{int}} t_0}.$$

The oriented angular fraction of (51.1) produces the dimensionless candidate

$$\theta_G^{\text{hol}} := (\delta_\theta \rho_{27})^2.$$

His specialization in the transducer gives

$$G_{\text{hol}} = \frac{c_{\text{int}}^3}{\hbar_{\text{int}}} (\delta_\theta \Lambda_{27})^2.$$

The temporal form is the same identity:

$$T_{27} := \frac{\Lambda_{27}}{c_{\text{int}}}, \quad t_* := \delta_\theta T_{27}.$$

In particular, it is not appropriate to replace t_* with T_{27}/δ_θ .

These formulas are exactly homogeneous. This does not promote their status to a derivation of G by this template: selection remains open until $\tilde{\mathcal{K}}_{27}$ and $\mathcal{L}$ are constructed without consulting the external value, their naturality is proved and a unique normalization is fixed. Downstream use of A and C^* in δ_θ does not authorize using G to redefine them. The radial proof of [section 39.1](#) is a different specified

realization: its data, norm, length and coupling are explicitly constructed and its reduction preserves memory. The limitation of the preceding template does not transfer to that proof.

51.5 Transduction of the gravitational decimal chart onto the magnitude line and causal separation of metrological recognition

Let

$$\mathcal{R}_G([n, k, d]_3; t, e) := 10^{-n} \left(k + \frac{t}{10^3} + \frac{e}{10^d} \right),$$

where the subscript of $[n, k, d]_3$ records the reader triad and does not denote a ternary numeral. For the declared data

$$[n, k, d]_3 = [11, 6, 5]_3, \quad t = 674, \quad e = 30,$$

rational evaluation is

$$\mathcal{R}_G = 10^{-11} \left(6 + \frac{674}{1000} + \frac{30}{10^5} \right) = 6.67430 \times 10^{-11}.$$

The arithmetic equality is exact once the parameters have been declared. The triad, the blocks 674, 30, and their decimal position have the status of chart data, so this branch is preserved as a parametrized control and subsequent recognition. The number is dimensionless until a basis $u_G \in \mathcal{L}_G^+$ is chosen; only then is the section defined

$$G_{\text{dec}} := \mathcal{R}_G u_G.$$

The agreement of its coordinate with a metrological publication is EXTERNAL RECOGNITION. A future promotion would require the integers 674 and 30, the triple $[11, 6, 5]$, and the decimal placement to be uniquely forced by published symmetries and fixed before the external value is consulted.

51.6 Hierarchy of Evidence for Gravitational Constructions

The longitudinal and radial realization determines the coupling in the explicit R1–R8 domain. Alternative templates and preceding controls retain their own scopes; they do not by themselves constitute other proved genealogies of that realization. The organization is:

Object	Result available	Verdict
Inscription and radial norm R1–R8	$L = (5759/23040)\alpha^{16}L_*$, $R_X\Lambda_X = L^2I$, $H_X\Lambda_X = \hbar_{\text{ret}}cI$, $G = c^3L^2/\hbar_{\text{ret}}$.	Unique determination in the declared domain; memory, reduction and refinement preserved.
CT108 circular realization	$t_0 = \pi L/(54c)$, $R_{108} = L$ y $\theta_{108}^{\text{circ}} = (54/\pi)^2$.	Exact compatibility with the constructed length.
Transductor dimensional	Types $G_{\text{int}}(\theta_G)$ exactly, without fixing θ_G .	Dimensional theorem; no selector.
Character of the observable $\mathcal{K}_{27}$	$\mathcal{K}_{27}^4 = I$ force $\chi(\mathcal{K}_{27}) = 1$.	Obstructed formulation.
Levantamiento enriquecido	Possible only with non-torsion abelian class, natural character and unique normalization.	Obstruction of canonicity demonstrated without those data.
Chronological coverage and spectral response	The cumulative translation has infinite semigroup order; along the deterministic orbit, the phase cancels.	Null response proved on the deterministic orbit; nontrivial class characterized by interference and natural normalization.
Path functional	The formula for G_{hol} is homogeneous once Λ_{27} has been fixed.	Exact parametrized family; no unique selector exists without specifying $\mathcal{L}$ and its normalization.
Decimal reader	Reproduce exactly the declared coordinate, but receive its blocks and position.	Control parametrizado.

Test criterion 51.3 (Gravitational constant selector). For the alternative lift, spectral-response and decimal-reader templates examined in this chapter, promotion is rejected if:

1. the enriched lift preserves only torsion or does not descend to a well-defined Abelian class;
2. the chronological operator does not preserve interference, produces $\mathcal{R}_{27} = 0$, lacks a domain or measure, or its response changes under an admissible refinement;
3. $\mathcal{L}$ depends on the basepoint, relabelling, the chosen orientation, a subdivision, or a free normalization;
4. the normalization $\mathfrak{N}$ depends on the global scale of the operator, is nonunique, or is fixed by consulting G ;
5. in the decimal reader, 674, 30 or the decimal position admit alternatives with equal structural entitlement;
6. the candidate consults the metrological value before being sealed;
7. the value is not stable under refinement or does not define a section of $\mathcal{L}_G$ in a common dimensional chart.

51.7 Gravitational–inertial ratios and the dimensional constant

The local normalization of the torsion chart,

$$\frac{\mathcal{G}_{\text{tor}}}{\mathcal{I}_{\text{tor}}} = 1,$$

and the interface areal–volumetric of the [section 10.5.7.1](#),

$$\frac{\mathcal{G}_{\text{av}}}{\mathcal{I}_{\text{av}}} = \frac{\pi}{\varphi^2},$$

are dimensionless reader ratios. They are not rival values of the dimensional constant G . The first expresses a local diagonal constraint; the second is an interface conditional upon factorization through a common memory. Constructing a section of $\mathcal{L}_G$ additionally requires the $(\hbar_{\text{int}}, c_{\text{int}}, \ell_0)$ framework and a θ_G selector.

Perron dynamics, torsional memory and Einstein–Cartan realization

The Perron operator fixes the stable memory mode; when published as a spin density, its contribution decays as the inverse sixth power of the scale factor. The Einstein–Cartan realization preserves this provenance and its conditions of validity.

52.1 Perron dynamics and the stable memory mode

On a closed route, the cylindrical Perron cocycle satisfies

$$V_n = \varphi^n. \quad (52.1)$$

In spatial dimension $D = 3$, the scale determined by the cocycle is

$$\frac{a_n}{a_0} = V_n^{1/3} = \varphi^{n/3}, \quad (52.2)$$

whereas the stable mode of the quadratic action is

$$M_n = \varphi^{-2n} = V_n^{-2}. \quad (52.3)$$

Eliminating n between the corresponding identity gives the central result:

Theorem 52.1 (Law cylindrical of sixth order for the memory). *For every closed level of the Perron realization in $D = 3$,*

$$M_n = \left(\frac{a_n}{a_0} \right)^{-6}. \quad (52.4)$$

The exponent -6 is not fitted: it is the quotient of the stable exponent -2 by the dimensional root $1/3$.

In the distinguished returns one obtains

$$\frac{a_9}{a_0} = \varphi^3, \quad \frac{a_{27}}{a_0} = \varphi^9, \quad \frac{a_{108}}{a_0} = \varphi^{36},$$

and the same levels satisfy exactly $M_n = (a_n/a_0)^{-6}$. If the additive temporal parameter $t_n = nt_0$ is declared, the logarithmic rate is constant:

$$H_\varphi := \frac{\Delta \log a_n}{\Delta t} = \frac{\log \varphi}{3t_0}, \quad H_\varphi t_0 = 0.1604039416865344824992529711 \dots \quad (52.5)$$

The symbol H_φ names the rate of this discrete scale. Reading it as a Hubble parameter requires the cosmographic transducer; the internal identity does not depend on that promotion.

52.2 Realization in the system of Einstein–Cartan

In an Einstein–Cartan spin fluid, the quadratic spin density obeys

$$s^2 \propto a^{-6}. \quad (52.6)$$

The equation (52.4) provides the exponent and the stable mode without consulting a bounce. To make the physical reading, one must declare a positive section s_*^2 and a map

$$\mathcal{T}_{\text{EC}} : M_n \mapsto s_n^2 := s_*^2 M_n = s_*^2 \left(\frac{a_n}{a_0} \right)^{-6}. \quad (52.7)$$

The map preserves the scaling law, but does not by itself determine the amplitude s_*^2 , the spin current, the sign of the term of contorsion, or the equation of state. With mass density, an effective equation has the form

$$H^2 = \frac{8\pi G}{3} (\rho_m - \rho_{\text{spin},m}) + \dots, \quad (52.8)$$

whereas with energy density it must be written

$$H^2 = \frac{8\pi G}{3c^2} (\rho_E - \rho_{\text{spin},E}) + \dots. \quad (52.9)$$

The two conventions are not mixed. A bounce candidate requires, in addition to equality of the densities, regularity, matter, and boundary data.

Falsification criterion 52.2. The Einstein–Cartan transduction fails if the scale functional produces an exponent other than -6 , if $V_n \neq \varphi^n$ along the declared route, or if the physical map does not preserve positivity and dimensions. Retrospectively choosing s_*^2 to force a bounce threshold does not establish the transducer.

52.3 Structural comparison between the 108-state cycle and Perron dynamics

	CT108–de Sitter framework	Perron–Einstein–Cartan framework
Antecedent	circular return of order 108	cylindrical cocycle and stable mode
Selection mechanism	$r_g = R_{108}$	$V_n = \varphi^n, D = 3$
Magnitudes determined	radius, curvature, temperature, and entropy	expansion, rate, and dilution of memory
Invariant	$G_a \hbar_a$ and circular radius	$M_n a_n^6$
Status	de Sitter conditioned interface	exact internal law; conditioned EC reading
Falsification criterion	incompatible radius or background equation	perdida of the exponent -6 or of the transducer

Hypothesis, domain, and conclusion

The CT108 identities, the selector θ_{108} , and the law $M_n = (a_n/a_0)^{-6}$ are *Internal exact theorem*. The reading of the same radius as de Sitter and the identification of M_n with a spin density are *Conditional physical realizations*, each with its own falsification criterion.

Coordination of torsion and area in the Palatini–Cartan–Holst action. Torsion and area are then coordinated in the first-order Palatini–Cartan–Holst action.

Palatini–Cartan–Holst realization of the Barbero–Immirzi parameter

The structural derivation of the parameter and the area spectrum precedes this chapter. Here the tetrad, connection, curvature, and first-order action are constructed, and only then is the Holst parameter specialized as a physical realization of the areal reader.

53.1 Curvature, torsion, and Einstein–Cartan–Holst action

The HMT transport structure distinguishes the memory variable, holonomy, and closure defect. Its gravitational realization is formulated by means of a tetrad and a connection. This section establishes the first-order geometry and specifies the type of the map between it and the TPK state space .

53.1.1 Discrete Gauss–Stokes identity and incidence transport to the boundary

Let K be a cellular complex, let W be an abelian group, and let $\Omega \in C^1(K; W)$ be a cochain. We denote by $\delta : C^1(K; W) \rightarrow C^2(K; W)$ the cellular coboundary and by $\langle \cdot, \cdot \rangle$ the pairing between cochains with values in W and integral cellular chains.

Theorem 53.1 (Universal Stokes Cell Identity). *For every 2-cadena finite $z \in C_2(K; \mathbb{Z})$,*

$$\langle \delta\Omega, z \rangle = \langle \Omega, \partial z \rangle.$$

Proof. This is the identity that defines the coboundary as the adjoint operator of the cellular boundary. The finiteness of z ensures that both pairings are finite sums in W . $\square$

Corollary 53.2 (Gauss–Stokes on a cellular pseudomanifold). *Let K be a finite, compact, regular two-dimensional cellular pseudomanifold, coherently oriented and possibly with boundary. If z_K is the sum of its 2-cells with the coherent orientation, then*

$$\sum_{f \in K^{(2)}} (\delta\Omega)(f) = \sum_e [\partial z_K : e] \Omega(e).$$

In particular, regularity allows the uncanceled incidences of ∂z_K to be identified with the edges of the geometric boundary.

Proof. The theorem is applied to z_K . Coherent orientation makes each interior edge appear twice in ∂z_K with opposite coefficients; the uncanceled incidences form the boundary chain. $\square$

The non-Abelian version is not obtained merely by replacing sums with products. In this work it is used only when transports to a common base, an ordering of the holonomies, and the corresponding conjugation have been fixed. Outside that subdomain, no non-Abelian Stokes identity is asserted.

53.1.2 Area capacity of the qutrit cell: $81 \log 3$ and collective incidence transport

Theorem 53.3 (Minimum cut of a discrete cube). *In the cubic graph $N \times N \times N$ with unit capacity, every cut separating two opposite faces has capacity at least N^2 , and one exists with capacity N^2 .*

Proof. There are N^2 edge-disjoint paths crossing the cube in the direction normal to the two faces. Every cut must intersect them, so its capacity is at least N^2 . A transverse layer contains exactly N^2 edges and attains the bound. $\square$

For $N = 9$, the cut equals 81 and

$$9^3 = 729 = 27^2.$$

The last equality permits six trits to be regrouped as a volume 9^3 or a grid of cardinality 27^2 . It is a bijection of sets; it is not an isomorphism between $(\mathbb{Z}/9\mathbb{Z})^3$ and $(\mathbb{Z}/27\mathbb{Z})^2$, whose exponents are distinct.

53.1.3 Tetrad, spin connection, and torsion of the Cartan–Holst realization

Let e^I be a tetrad and $\omega^{IJ} = -\omega^{JI}$ a Lorentz connection. Define

$$\begin{aligned} T_{\text{tor}}^I &= D_\omega e^I = de^I + \omega^I{}_J \wedge e^J, \\ F^{IJ} &= d\omega^{IJ} + \omega^I{}_K \wedge \omega^{KJ}, \quad g = \eta_{IJ} e^I \otimes e^J. \end{aligned}$$

The decomposition

$$\omega^{IJ} = \hat{\omega}^{IJ}(e) + K^{IJ}$$

separates the Levi–Civita connection and the contorsion. Then

$$T_{\text{tor}}^I = K^I{}_J \wedge e^J.$$

Curvature and torsion are distinct defects: F measures closure of vector transport; T_{tor} , closure of the tetrad. The T memory of the Ambivalence Principle is likewise not geometric torsion: only a realization functor can relate them.

53.1.4 Palatini–Holst Action and Variational Equations

Let M be a smooth oriented four-dimensional manifold. If $\partial M \neq \emptyset$, we assume compactly supported variations in the interior or, equivalently for the variational problem considered, a boundary term and boundary conditions that annul the boundary contribution. When Ψ contains spinorial fields, we also assume a spin structure and fields compatible with it. Let

$$\Sigma^{IJ} = e^I \wedge e^J, \quad (\star X)^{IJ} = \frac{1}{2} \epsilon^{IJ}{}_{KL} X^{KL}, \quad \kappa = \frac{8\pi G}{c^4}, \quad \gamma \in \mathbb{R} \setminus \{0\}.$$

The Palatini–Cartan–Holst action is

$$S[e, \omega, \Psi] = \frac{1}{2\kappa c} \int_M \Sigma_{IJ} \wedge \left(\star F^{IJ} + \frac{1}{\gamma} F^{IJ} \right) - \frac{\Lambda}{\kappa c} \int_M \text{vol}_e + S_m[e, \omega, \Psi].$$

The variation of this action determines the torsional corrections [Sci64, Kib61, HvdHKN76, Hol96].

We define the spin current by

$$\delta_\omega S_m = -\frac{1}{2} \int_M \delta\omega^{IJ} \wedge \sigma_{IJ}.$$

Theorem 53.4 (Cartan–Holst Equation). *Under the preceding geometric and boundary hypotheses, variation with respect to ω gives*

$$D_\omega [(\star + \gamma^{-1} I) \Sigma^{IJ}] = \kappa c \sigma^{IJ},$$

with the global normalization fixed by the preceding definition of σ^{IJ} .

Proof. One uses $\delta F^{IJ} = D_\omega \delta\omega^{IJ}$, integrates by parts, and sets the coefficient of the arbitrary variation $\delta\omega^{IJ}$ equal to zero. With the sign convention fixed in the definition of σ^{IJ} , the matter term moves to the right-hand side with a positive sign. The factor $1/(2\kappa c)$ produces the coefficient κc . The cosmological

term does not depend on ω , and the boundary conditions eliminate the term generated by integration by parts. $\square$

In signature lorentziana $\star^2 = -I$ on bivectores and, for $\gamma \in \mathbb{R} \setminus \{0\}$,

$$(\star + \gamma^{-1}I)^{-1} = \frac{\gamma^2}{1 + \gamma^2}(-\star + \gamma^{-1}I).$$

Corollary 53.5 (Torsional decoupling). *For a co-tetrad no degenerada and $\gamma \in \mathbb{R} \setminus \{0\}$,*

$$\sigma^{IJ} = 0 \implies T_{\text{tor}}^I = 0 \implies \omega = \dot{\omega}(e).$$

To make the normalization explicit, we define the normalized sources by the action through

$$\delta_g S_m = -\frac{1}{2} \int_M \text{vol}_e \mathcal{T}_{\mu\nu}^S \delta g^{\mu\nu}.$$

When the tetrad coordinates have the dimension of length, the conventional physical energy–momentum tensor is $T_{\mu\nu}^{\text{phys}} := c \mathcal{T}_{\mu\nu}^S$. We likewise denote by $\mathcal{U}^{S,\text{spin}}$ the contribution normalized by the action and by $U^{\text{phys},\text{spin}} := c \mathcal{U}^{S,\text{spin}}$ its physical convention. Upon algebraically eliminating the connection, the metric equation then assumes the two equivalent forms

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa c (\mathcal{T}_{\mu\nu}^{S,\text{LC}} + \mathcal{U}_{\mu\nu}^{S,\text{spin}}) = \kappa (T_{\mu\nu}^{\text{phys},\text{LC}} + U_{\mu\nu}^{\text{phys},\text{spin}}),$$

where the spin contribution is quadratic in the corresponding current and depends on the matter coupling. The coefficient of that contribution is not fixed without specifying the fermionic action and the duality conventions.

53.1.5 Bianchi Identity with Torsion and Flow Balance in the Cartan Connection

For every connection,

$$D_\omega T_{\text{tor}}^I = F^I{}_J \wedge e^J, \quad D_\omega F^{IJ} = 0.$$

After removing torsion,

$$\nabla^\mu (T_{\mu\nu}^{\text{phys},\text{LC}} + U_{\mu\nu}^{\text{phys},\text{spin}}) = 0.$$

These identities are mandatory consistency checks: every HMT gravitational term must arise from a compatible action or satisfy the corresponding Noether balance.

53.1.6 Map of realization and specialization of the Holst parameter

The insertion defined in [section 40.1](#) takes

$$\gamma = \Gamma_{6 \rightarrow 8}.$$

Once this substitution has been made, all the preceding variational consequences are theorems of the specialized action. The physical selection of this action by HMT transport is expressed through a realization map

$$\mathfrak{R}_{\text{grav}} : \mathcal{X}_{\text{TPK}} \dashrightarrow \{(e, \omega, \Psi)\}$$

endowed with three properties: intertwining TPK transport with that of ω , identifying the holonomy classes, and furnishing the scales of G and Λ . These properties fix the precise domain of a gravitational realization of the formalism.

For a nondegenerate co-tetrad, nonzero real γ , and matter without spin current, the torsional sector reduces to the first-order formulation of general relativity with the same cosmological constant, subject to the already declared boundary conditions. In a spin fluid, the scaling $s^2 \propto a^{-6}$ can produce an

Einstein–Cartan bounce; obtaining its density and initial conditions from the TPK state is a question distinct from the variational consistency of the action.

53.2 Nonadic Return, Memory Groupoid and Cartan Entanglement Condition

The hierarchy $27 \rightarrow 54 \rightarrow 108$, the directed cumulative coverage, and the memory groupoid supply the exact condition under which the TPK return admits a compatible Cartan connection. This bridge precedes the Palatini–Cartan–Holst specialization and preserves its domain and refutation criteria intact.

53.3 Nonadic Return, Memory, and Cartan Holonomy

The gravitational realization does not begin by identifying a discrete variable with a geometric magnitude. It begins by distinguishing three levels: return of observable transport, memory distinguishing histories with equal projection, and holonomy of a connection. The bridge formulated here receives the returns and cocycles already demonstrated in HMT; it does not reconstruct them from the Holst action, from a mass, or from a metrological constant.

53.3.1 Exact hierarchy of returns

Let F_{obs} be the observable chronology defined in the HMT kernel, and let

$$\mathcal{K}_{27} := F_{\text{obs}}^{27}.$$

The symbol $\mathcal{K}_{27}$ denotes exclusively the composition of twenty-seven updates of the positional and oriented loop; it does not denote a generic window, a trace of 108 steps, or the complete integral holonomic state. Denote by p the projection onto the positions of the two cursors, and by J_{or} simultaneous reversal of their orientations. The primary proof is in [proposition 10.46](#); its typed consequences are

$$p(\mathcal{K}_{27}x) = p(x), \quad o(\mathcal{K}_{27}x) = J_{\text{or}} o(x), \quad J_{\text{or}}^2 = I,$$

$$F_{\text{obs}}^{54} : \text{restores positions and orientations}, \quad F_{\text{obs}}^{108} = I : \text{also restores the observable phase}.$$

Thus, position return, orientation return, and phase return are distinct assertions. None of them by itself implies the return of all fields of an enriched extension.

The mode quotient and orientation cocycles are established in the corresponding theorem. In areal–volumetric order,

$$N_{27} = 9F_{\text{av}}, \quad F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

If

$$\Omega_{\text{sw}}(\gamma) = \sum_{a \in \gamma} \omega_{\text{sw}}(a), \quad S(\gamma) = \sum_{a \in \gamma} |\omega_{\text{sw}}(a)|, \quad G_{\text{or}}(\gamma) = \sum_{a \in \gamma} g(a),$$

then a block of twenty-seven steps contains three nonadic turns and one orientation reversal:

$$\Omega_{\text{sw}}(\mathcal{K}_{27}) = 0, \quad S(\mathcal{K}_{27}) = 18, \quad G_{\text{or}}(\mathcal{K}_{27}) = 1.$$

In the observable return of 108 steps,

$$\Omega_{\text{sw}}(C_{108}) = 0, \quad S(C_{108}) = 72, \quad G_{\text{or}}(C_{108}) = 4.$$

Thus, for the additive functional declared

$$\mathcal{A}_*(\gamma) := G_{\text{or}}(\gamma) + \lambda_* S(\gamma),$$

are obtained, without introducing a dimensional scale,

$$\mathcal{A}_*(\mathcal{K}_{27}) = 1 + 18\lambda_*, \quad \mathcal{A}_*(C_{108}) = 4 + 72\lambda_*.$$

These values are record counts. Their subsequent thermal or gravitational use does not alter their provenance.

53.3.1.1 Directed system of coverings, refinement, and limit of the Cartan connection

Let O_{108} be the observable orbit and, for each chronological edge e , let

$$c(e) = (g(e), s(e)) \in \mathbb{N}^2$$

the pair formed by the indicators of orientation inversion and effective sheet change. For a directed history $\gamma = e_1 \cdots e_n$, define

$$c(\gamma) = \sum_{j=1}^n c(e_j).$$

The concatenation law takes the form

$$c_{m+n}(x) = c_m(x) + c_n(F_{\text{obs}}^m x).$$

The previous counts express, uniformly over the orbit,

$$c_{27}(x) = (1, 18), \quad c_{108}(x) = (4, 72).$$

Theorem 53.6 (Cumulative coverage of the observable chronology). *On*

$$\tilde{O}_{108}^+ := O_{108} \times \mathbb{N}^2$$

define

$$\tilde{F}^+(x, m) := (F_{\text{obs}}x, m + c(x \rightarrow F_{\text{obs}}x)).$$

Then the transport of twenty-seven steps satisfies

$$\tilde{\mathcal{K}}_{27}^+(x, m) = (\mathcal{K}_{27}x, m + (1, 18)),$$

and

$$\left(\tilde{\mathcal{K}}_{27}^+\right)^4(x, m) = (x, m + (4, 72)) \neq (x, m).$$

Therefore, its observable projection has order four, whereas the cumulative action of the chronological monoid has infinite semigroup order.

Proof. The cocyclic law and the constancy of c_{27} give

$$c_{108}(x) = \sum_{j=0}^3 c_{27}(\mathcal{K}_{27}^j x) = 4(1, 18) = (4, 72).$$

The first coordinate returns through $\mathcal{K}_{27}^4 = I$; the second does not return. After $q > 0$ iterations, the increment is $q(4, 72)$, which is nonzero. $\square$

Remark 53.7 (Type of the cumulative extension). The preceding construction extends directed histories through positive counters. It is not an automorphic lift of groupoid $\mathcal{G}_{\text{mem}}$, nor does it prove that the unbounded coordinate already belongs to state $\mathcal{X}_{\text{TPK}}^{\text{enr}}$. Its Grothendieck completion $\mathbb{Z}^2$ may be used for harmonic analysis, but it does not transform the positive cost $\mathcal{A}_* = G_{\text{or}} + \lambda_* S$ into a character of the isotropy group: upon formally inverting a route, its indicators do not acquire opposite signs.

Domain of validity of the cumulative extension. The hierarchy $27 \rightarrow 54 \rightarrow 108$, the incidences of F_{av} and the preceding cocycles are exact identities in the declared chronology and readers. The assertion is restricted to the observable state those readers preserve; extending it to the complete integral holonomic state would require proving return of each forgotten field.

53.3.2 Memory Groupoid

Let $\mathbf{P}_{\text{TPK}}^{\text{enr}}$ be the enriched path groupoid: its objects are surviving TPK states, and its arrows are admissible routes with declared source and target, composition by concatenation, and inversion by route reversal. Let $\sim_{\text{mem}}$ be an equivalence relation that identifies two arrows only when it preserves the memory fields required by the reader under consideration and is congruent with identities, inversion, and concatenation. We define

$$\mathcal{G}_{\text{mem}} := \mathbf{P}_{\text{TPK}}^{\text{enr}} / \sim_{\text{mem}}.$$

This definition does not transform memory into torsion. Memory belongs to the discrete state and distinguishes histories; torsion belongs to a geometric connection and measures $D_{\omega}e$. Only a map between the two domains can relate them.

A geometric realization requires, as additional data, a length $\ell_0 > 0$, a representation of the groupoid

$$\rho_C : \mathcal{G}_{\text{mem}} \longrightarrow \text{Spin}(1, 3) \ltimes \mathbb{R}^{1,3},$$

and a field map

$$\mathfrak{R}_{\text{grav}} : \mathbf{P}_{\text{TPK}}^{\text{enr}} \dashrightarrow \text{Cartan}_{1,3}.$$

To define the second map, one must specify its action on objects and routes, its compatibility with concatenation and inversion, and the regularity of the resulting connection. None of these properties follows from the mere count of returns.

53.3.3 Joint connection and entanglement condition

Under the geometric hypotheses of [section 53.4](#)—variedad oriented of dimension four, co-tetrad e^I , Lorentz connection ω^{IJ} and boundary conditions declared there—, the connection of Cartan joint is written

$$\mathcal{A}_{\ell_0} = \begin{pmatrix} \omega & \ell_0^{-1}e \\ 0 & 0 \end{pmatrix}.$$

Its curvature is the algebraic identity.

$$\mathcal{F}_{\ell_0} = d\mathcal{A}_{\ell_0} + \mathcal{A}_{\ell_0} \wedge \mathcal{A}_{\ell_0} = \begin{pmatrix} F_{\omega} & \ell_0^{-1}T_{\text{tor}} \\ 0 & 0 \end{pmatrix},$$

where

$$F_{\omega}^{IJ} = d\omega^{IJ} + \omega^I_K \wedge \omega^{KJ}, \quad T_{\text{tor}}^I = D_{\omega}e^I.$$

Curvature and torsion appear as distinct components of the same transport defect. The matrix equality follows algebraically from the explicit data e, ω, ℓ_0 , but does not construct these inputs from TPK.

The strong content of the bridge is the commutativity of holonomies:

$$\boxed{\text{Hol}_{\mathcal{A}_{\ell_0}}(\mathfrak{R}_{\text{grav}}(\gamma)) = \rho_C(\text{Hol}_{\text{TPK}}(\gamma))} \quad (\gamma \in \mathbf{P}_{\text{TPK}}^{\text{enr}}).$$

The equality must be compatible with identities, inversion, concatenation, subdivision, change of base point, and conjugation. Only then is discrete memory represented in a connection; it is not merely renamed as torsion or curvature.

Test criterion 53.8 (Existence conditions for the bridge TPK–Cartan). There is not realization with the declared commutativity and compatibility if any of the following facts occurs:

1. two equivalent representatives of the same route produce nonconjugate geometric holonomies;
2. the image does not preserve identities, inverses, or concatenation;
3. an admissible subdivision alters the holonomy;
4. the equality of holonomies fails for $\mathcal{K}_{27}$ or for their concatenations of depths 54 and 108;
5. the map identifies memory and torsion without exhibiting a morphism that preserves their types;
6. some descending choice—the Holst action, G , a mass or an SI chart—is used to redefine A , C^* or the original TPK return.

Antecedence of the functional $\Gamma_{6 \rightarrow 8}$ with respect to the Palatini–Cartan–Holst realization. [section 40.1](#) fixes, once $A, C^*, q_{\pm}$ have been constructed, the internal functional $\Gamma_{6 \rightarrow 8}$. [section 53.4](#) then studies the Palatini–Cartan–Holst action and its variational equations. The present bridge introduces no inverse route between the two stages.

53.4 Gravity: Holonomy, Torsion, and Collective Response

Gravity is not introduced here through a decimal formula for G . Its construction assembles two already typed branches. In the first, a route preserves orientation and memory; a connection compares frames at distinct points; its holonomy measures return and its curvature the closure defect. In the second, the dimensional chart constructs $G_{\text{int}}(\theta_G)$ and κ_{int} from the internal action scale, velocity, and elementary duration. Only after both branches are available is the Palatini–Cartan–Holst action written. No external metrological figure participates in that precedence.

53.4.1 Returns: position, orientation, phase, and memory

Let F_{obs} be the observable chronology of the transport and

$$\mathcal{K}_{27} = F_{\text{obs}}^{27}.$$

The exact hierarchy distinguishes three returns:

27 steps	position return and orientation reversal,
54 steps	position and orientation return,
108 steps	complete return of the observable phase.

The observable return does not erase the accumulated record. If $g(e)$ counts an orientation reversal and $s(e)$ an effective sheet change, then

$$c(\gamma) = \sum_{e \in \gamma} (g(e), s(e))$$

satisfies the cocyclic law of concatenation. In the canonical blocks,

$$c_{27} = (1, 18), \quad c_{108} = (4, 72).$$

Therefore, the observable projection of $\mathcal{K}_{27}$ has order four, whereas the cumulative coordinate continues to grow. Returning to the same phase does not mean returning to the same history.

Theorem 53.9 (Cumulative coverage of return). *Over $\mathcal{O}_{108} \times \mathbb{N}^2$, the lift*

$$\tilde{F}(x, m) = (F_{\text{obs}}x, m + c(x \rightarrow F_{\text{obs}}x))$$

satisfies

$$\tilde{\mathcal{K}}_{27}(x, m) = (\mathcal{K}_{27}x, m + (1, 18))$$

and

$$\tilde{\mathcal{K}}_{27}^4(x, m) = (x, m + (4, 72)) \neq (x, m).$$

Proof. The constancy of c_{27} along the orbit and the concatenation law give

$$c_{108} = \sum_{j=0}^3 c_{27}(\mathcal{K}_{27}^j x) = 4(1, 18).$$

The first coordinate returns; the second preserves the accumulated memory. $\square$

This is the datum that to geometric realization must preserve. One does not identify the discrete memory with torsion: one constructs a representation that allows them to be compared.

53.4.2 From the path groupoid to the Cartan group

Let $\mathcal{G}_{\text{mem}}$ be the groupoid of enriched routes, with composition by concatenation and inverse by reversal. Fix an element $R_4 \in \text{Spin}^+(1, 3)$ of order four and a unit temporal vector $u \in \mathbb{R}^{1,3}$ fixed by R_4 . If

$$c(\gamma) = (c_g(\gamma), c_s(\gamma)) \in \mathbb{Z}^2$$

is the additive cocycle of inversion and sheet change, we define

$$\rho_C^{\text{disc}}(\gamma) = \left(R_4^{c_g(\gamma)}, \ell_0 c_s(\gamma) u \right) \in \text{Spin}^+(1, 3) \ltimes \mathbb{R}^{1,3}.$$

Theorem 53.10 (Discrete realization of the return cocycle). ρ_C^{disc} is a functor from the route groupoid to the Cartan group. In particular, it preserves identity, concatenation, and inversion, and transports the return $27 \rightarrow 54 \rightarrow 108$ without erasing the cumulative coordinate.

Proof. The semidirect product law is

$$(R, a)(S, b) = (RS, a + Rb).$$

Since $R_4 u = u$ and c is additive,

$$\begin{aligned} \rho_C^{\text{disc}}(\gamma_2) \rho_C^{\text{disc}}(\gamma_1) &= \left(R_4^{c_g(\gamma_2) + c_g(\gamma_1)}, \ell_0 (c_s(\gamma_2) + c_s(\gamma_1)) u \right) \\ &= \rho_C^{\text{disc}}(\gamma_2 \gamma_1). \end{aligned}$$

The identity route has zero cocycle, and reversal changes its sign, so identity and inverse are preserved. The return formulas follow from $c_{27} = (1, 18)$ and $c_{108} = 4c_{27}$. $\square$

This theorem constructs the discrete bridge. A subsequent smooth realization consists of a representation

$$\rho_C : \mathcal{G}_{\text{mem}} \longrightarrow \text{Spin}(1, 3) \ltimes \mathbb{R}^{1,3}$$

and in fields e^I, ω^{IJ} whose holonomy intertwines that representation. The diagram fixing the type is

$$\text{Hol}_{\mathcal{A}_{\ell_0}}(\mathfrak{R}_C(\gamma)) = \rho_C(\text{Hol}_{\text{TPK}}(\gamma)).$$

Identities, inversion, concatenation, and subdivision must be preserved. This equation does not rename the carry as curvature: it requires the already represented discrete transport and the geometric transport to realize the same composition.

Once to positive scale is fixed ℓ_0 , the joint connection is

$$\mathcal{A}_{\ell_0} = \begin{pmatrix} \omega & \ell_0^{-1} e \\ 0 & 0 \end{pmatrix}.$$

Proposition 53.11 (Cartan Joint Curvature). *The curvature of $\mathcal{A}_{\ell_0}$ is*

$$\mathcal{F}_{\ell_0} = d\mathcal{A}_{\ell_0} + \mathcal{A}_{\ell_0} \wedge \mathcal{A}_{\ell_0} = \begin{pmatrix} F_\omega & \ell_0^{-1} T_{\text{tor}} \\ 0 & 0 \end{pmatrix},$$

where

$$F_\omega^{IJ} = d\omega^{IJ} + \omega^I_K \wedge \omega^{KJ}, \quad T_{\text{tor}}^I = D_\omega e^I.$$

Proof. The matrices of forms are multiplied using the exterior product. The diagonal block produces F_ω ; the translation block produces $de + \omega \wedge e = D_\omega e$. $\square$

Curvature and torsion are two components of the same transport defect, but they perform distinct functions. Curvature measures frame return; torsion measures closure of the co-tetrad.

53.4.3 Gauss–Stokes and areal capacity

In a finite cellular complex, for every 1-cochain Ω and every 2-chain z ,

$$\langle \delta\Omega, z \rangle = \langle \Omega, \partial z \rangle.$$

The identity is the definition of the coboundary as the adjoint of the boundary. When z sums the coherently oriented faces of a pseudomanifold, the interior edges cancel, and only the geometric boundary remains. The same equality has already been read as vorticity–circulation and as curvature–holonomy.

The areal scale also has an exact model. In the cubic graph $N \times N \times N$, every cut between two opposite faces contains at least N^2 edges, and to transverse layer attains the bound. For $N = 9$,

$$9^3 = 729 = 27^2.$$

This cardinal equality relates volume and minimal cut; it does not identify the groups $(\mathbb{Z}/9)^3$ and $(\mathbb{Z}/27)^2$, which have distinct exponents.

53.4.4 The Palatini–Cartan–Holst Action

Let M be a smooth oriented manifold of dimension four, with nondegenerate co-tetrad e^I , Lorentz connection $\omega^{IJ} = -\omega^{JI}$, and boundary conditions that make the boundary variational term vanish. Let us write

$$\Sigma^{IJ} = e^I \wedge e^J, \quad (\star X)^{IJ} = \frac{1}{2} \epsilon^{IJ}{}_{KL} X^{KL}, \quad G_{\text{int}} = G_{\text{int}}(\theta_G), \quad \kappa_{\text{int}} = \frac{8\pi G_{\text{int}}}{c_{\text{int}}^4}.$$

These last two quantities are not defined through the action: they follow from the constitutive construction of internal constitutive identity of gravitation. The homogeneity theorem for the Palatini–Cartan–Holst action first proves that the following combination belongs to the action line. The first-order action is

$$S[e, \omega, \Psi] = \frac{1}{2\kappa_{\text{int}} c_{\text{int}}} \int_M \Sigma_{IJ} \wedge \left(\star F^{IJ} + \frac{1}{\gamma} F^{IJ} \right) - \frac{\Lambda_{\text{int}}}{\kappa_{\text{int}} c_{\text{int}}} \int_M \text{vol}_e + S_{\text{m}}[e, \omega, \Psi].$$

The internal parameter constructed in the previous chapter specializes

$$\gamma = \Gamma_{6 \rightarrow 8}.$$

The substitution is performed after obtaining $\Gamma_{6 \rightarrow 8}$ from $A, C^*, q_{\pm}$ and the hexad–octad inclusion; the action is not used to redefine those inputs.

Theorem 53.12 (Cartan–Holst Equation). *If the spin current is normalized through*

$$\delta_{\omega} S_m = -\frac{1}{2} \int_M \delta \omega^{IJ} \wedge \sigma_{IJ},$$

the variation with respect to ω produces

$$D_{\omega} [(\star + \gamma^{-1} I) \Sigma^{IJ}] = \kappa_{\text{int}} c_{\text{int}} \sigma^{IJ}.$$

Proof. One uses $\delta F^{IJ} = D_{\omega} \delta \omega^{IJ}$, integrates by parts, and sets the coefficient of the arbitrary variation equal to zero. The cosmological term does not depend on ω . $\square$

In Lorentzian signature, $\star^2 = -I$ on bivectors. For $\gamma \in \mathbb{R} \setminus \{0\}$,

$$(\star + \gamma^{-1} I)^{-1} = \frac{\gamma^2}{1 + \gamma^2} (-\star + \gamma^{-1} I).$$

Therefore, if the spin current vanishes,

$$T_{\text{tor}}^I = 0, \quad \omega = \hat{\omega}(e).$$

With spinorial matter, torsion is determined algebraically by the current. In the minimal action, it does not constitute an independent propagating degree of freedom.

53.4.5 Metric equation, spin, and geometric balance

To fix the normalization unambiguously, we defines the sources measured on the action line by

$$\delta_g S_m = -\frac{1}{2} \int_M \text{vol}_e \mathcal{T}_{\mu\nu}^S \delta g^{\mu\nu}.$$

When the tetrad coordinates have the dimension of length, we write

$$T_{\mu\nu}^{\text{phys}} := c_{\text{int}} \mathcal{T}_{\mu\nu}^S, \quad U_{\mu\nu}^{\text{phys,spin}} := c_{\text{int}} \mathcal{U}_{\mu\nu}^{S,\text{spin}}.$$

After algebraically eliminating the connection, the metric equation adopts the two equivalent forms

$$G_{\mu\nu} + \Lambda_{\text{int}} g_{\mu\nu} = \kappa_{\text{int}} c_{\text{int}} (\mathcal{T}_{\mu\nu}^{S,\text{LC}} + \mathcal{U}_{\mu\nu}^{S,\text{spin}}) = \kappa_{\text{int}} (T_{\mu\nu}^{\text{phys,LC}} + U_{\mu\nu}^{\text{phys,spin}}).$$

The spin contribution is quadratic in the corresponding current and depends on the matter coupling; its coefficient is not fixed without declaring the fermionic action and the duality conventions.

The geometric identities

$$D_{\omega} T_{\text{tor}}^I = F^I{}_J \wedge e^J, \quad D_{\omega} F^{IJ} = 0$$

imply, after removing the torsion,

$$\nabla^{\mu} (T_{\mu\nu}^{\text{phys,LC}} + U_{\mu\nu}^{\text{phys,spin}}) = 0.$$

This Bianchi–Noether balance is to consistency condition for the gravitational transducer: not additional term can be incorporated without proceeding from to compatible action or satisfying the same balance.

53.4.6 Area–volumetric correspondence π/φ^2

The calendar (Σ, Π, Π) produces the incidence matrix

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

Its Parry measure gives $\mu_{\text{ar}}/\mu_{\text{vol}} = \varphi^{-2}$, and the regional reader marks the closure exactly once π . The marked return satisfies

$$\Theta_n = \pi \frac{F_{n-1}}{F_{n+1}} \longrightarrow \frac{\pi}{\varphi^2}.$$

This ratio is not another version of G . It is the dimensionless correspondence between the areal and volumetric readings. Under the same positive amplitude $\mathcal{M}$,

$$\mathcal{I}_{\text{av}} = \mu_{\text{vol}} \mathcal{M}, \quad \mathcal{G}_{\text{av}} = \pi \mu_{\text{ar}} \mathcal{M}$$

give exactly

$$\frac{\mathcal{G}_{\text{av}}}{\mathcal{I}_{\text{av}}} = \frac{\pi}{\varphi^2}.$$

On the flat sheet, the diagonal restriction is $\mathcal{G}_{\text{tor}} = \mathcal{I}_{\text{tor}}$. Local equivalence and boundary ambivalence are therefore two regimes of the reader, not two incompatible values of a mass.

53.4.7 Typing of G as gravitational transducer and causal order of its readers

The type and homogeneity of G belong to the prior constitutive construction. The internal constitutive identity of gravitation fixes the covariant family $G_{\text{int}}(\theta_G)$ once and for all, and the homogeneity theorem for the Palatini–Cartan–Holst action proves that its insertion into the action has the dimension of action. This chapter does not redefine that family: it uses its section together with the holonomy already constructed. Causality is not linear, but convergent:

$$\mathcal{B}_{\text{geom}} : \text{return and cocycle} \rightarrow \text{Cartan representation} \rightarrow (e, \omega),$$

$$\mathcal{B}_{\text{dim}} : (\hbar_{\text{int}}, c_{\text{int}}, t_0, \theta_G) \rightarrow G_{\text{int}} \rightarrow \kappa_{\text{int}},$$

$$(\mathcal{B}_{\text{geom}}, \mathcal{B}_{\text{dim}}) \rightarrow S_{\text{PCH}} \rightarrow \text{algebraic torsion, metric equation and balance}.$$

The metrological comparison belongs to the end and does not select any of the two branches.

53.4.8 Collective origin of the gravitational mode in the holonomy of paths

The basic objects of the preceding action are the co-tetrad and the connection. Cartan torsion is algebraic; curvature has the propagating sector that contains gravitational waves. Accordingly, the formalism does not need to postulate an elementary graviton in order to define the interaction. A possible quantization of the perturbations may produce a collective mode of helicities ± 2 ; that mode would be an excitation of geometric transport, not a datum added at the level of APP, TRIT, or TPK.

This distinction also excludes a frequent confusion. The existence of classical waves proves propagation of curvature; by itself it does not prove a fundamental quantum. Conversely, describing the field geometrically does not eliminate its radiative degrees of freedom. HMT–MD locates the question properly: memory and its holonomy are constructed first; the spectrum of their excitations is then studied.

Typed comparison of the area quantum with energy, entropy, and horizon information. The first-order action allows the area quantum to be compared with energy, entropy and horizon information without merging these observables.

Convergence of horizon, area, information, entropy and action

The horizon patch coordinates energy, temperature, entropy, and action through typed identities. The half-action cell and chronological closure are presented with their Schwarzschild and de Sitter controls, without turning an interface into a retroactive definition.

54.1 Horizon energy and half-action cell

54.1.1 Geometric conventions of the de Sitter patch

Consider a four-dimensional de Sitter space with areal radius $R > 0$, causal velocity c , and gravitational constant G . Adopt the convention

$$H_R = \frac{c}{R}, \quad \Lambda_R = \frac{3}{R^2}, \quad \ell_P^2 = \frac{G\hbar}{c^3}. \quad (54.1)$$

The density of energy associated to the term cosmological is

$$\varepsilon_\Lambda := \frac{\Lambda_R c^4}{8\pi G}. \quad (54.2)$$

The spatially flat section of the patch contains, up to radius R , the volume

$$V_R = \frac{4\pi}{3} R^3. \quad (54.3)$$

These conventions positively fix the quasilocal energy used in the chapter. Choosing another sign for the de Sitter Killing charge belongs to a distinct construction and must be translated before formulas are compared.

54.1.2 energy of vacuum and energy of Misner–Sharp

In a spherically symmetric geometry, let

$$ds^2 = g_{ab}(x) dx^a dx^b + \mathcal{R}(x)^2 d\Omega_2^2. \quad (54.4)$$

The Misner–Sharp energy [MS64], under the adopted positive convention, is defined by

$$E_{\text{MS}}(\mathcal{R}) := \frac{c^4 \mathcal{R}}{2G} \left(1 - g^{ab} \nabla_a \mathcal{R} \nabla_b \mathcal{R} \right). \quad (54.5)$$

A marginal sphere satisfies

$$g^{ab} \nabla_a \mathcal{R} \nabla_b \mathcal{R} = 0. \quad (54.6)$$

Theorem 54.1 (Confluence volumetrica and cuasilocal). *On the de Sitter horizon $\mathcal{R} = R$, the integrated vacuum energy and the Misner–Sharp energy coincide exactly:*

$$E_\Lambda := \varepsilon_\Lambda V_R = E_{\text{MS}}(R) = \frac{c^4 R}{2G}. \quad (54.7)$$

Proof. The equations the corresponding identity give

$$\varepsilon_\Lambda V_R = \frac{3c^4}{8\pi G R^2} \frac{4\pi R^3}{3} = \frac{c^4 R}{2G}.$$

The marginal condition equation (54.6) applied to equation (54.5) yields the same result. $\square$

The equality brings together two geometric publications. The first integrates a volumetric density; the second evaluates the trapping defect of the areal sphere. Both preserve the radius R and gravitational compliance G/c^4 .

54.1.3 Temperature, entropy and product of horizon

The de Sitter surface gravity has magnitude c^2/R . The Gibbons–Hawking temperature [GH77] and the Bekenstein–Hawking entropy [Bek73, Haw75] are

$$k_B T_H = \frac{\hbar c}{2\pi R}, \quad (54.8)$$

$$\frac{S_H}{k_B} = \frac{A_H}{4\ell_P^2} = \frac{4\pi R^2}{4\ell_P^2} = \frac{\pi R^2}{\ell_P^2}. \quad (54.9)$$

Theorem 54.2 (de Sitter thermodynamic identity). *For every radius R of the declared family,*

$$\boxed{T_H S_H = \frac{c^4 R}{2G} = E_\Lambda.} \quad (54.10)$$

Proof. Multiplying the corresponding identity gives

$$T_H S_H = \frac{\hbar c}{2\pi R k_B} \frac{k_B \pi R^2}{\ell_P^2} = \frac{\hbar c R}{2\ell_P^2}.$$

The identity $\ell_P^2 = G\hbar/c^3$ turns the last member into $c^4 R/(2G)$, equal to equation (54.7). $\square$

The factor $1/2$ has a decomposition geometric transparent. The areal factor 4π , the normalization holographic by four and the periodicity thermal $(2\pi)^{-1}$ satisfy

$$\frac{4\pi}{4} \frac{1}{2\pi} = \frac{1}{2}. \quad (54.11)$$

The half is therefore the joint contraction between areal quantization and Euclidean periodicity.

54.1.4 How complete of the horizon and Planck force

We define the tension or fuerza of Planck and the complete quantum associated with the radius by

$$\mathcal{F}_P := \frac{c^4}{G}, \quad \boxed{Q_H(R) := 2T_H S_H = \frac{c^4 R}{G} = \mathcal{F}_P R.} \quad (54.12)$$

Quasilocal energy occupies half a cell of the complete quantum:

$$\boxed{E_\Lambda = T_H S_H = \frac{Q_H(R)}{2}.} \quad (54.13)$$

Proposition 54.3 (Radial compliance). *The derivative of the complete quantum with respect to radius is the Planck force, and the derivative of the quasilocal energy is half of it:*

$$\frac{dQ_H}{dR} = \mathcal{F}_P, \quad \frac{dE_\Lambda}{dR} = \frac{\mathcal{F}_P}{2}. \quad (54.14)$$

The compliance gravitatoria $\mathcal{C}_P = G/c^4$ satisfies

$$R = \mathcal{C}_P Q_H. \quad (54.15)$$

Proof. The three identities are direct derivatives or inversions of [equation \(54.12\)](#). $\square$

The [equation \(54.15\)](#) expresses the radial response: the complete quantum applies tension, and geometry publishes a radius through universal compliance. The product $T_H S_H$ occupies the quasilocal half of that response.

54.1.5 Specialization to the closure chronological of 108 states

The chronological closure of chapter devoted to the cosmological scales of the 108-phase calendar determines

$$R_{108} = \ell_P, \quad t_P = \frac{\ell_P}{c}, \quad T_{108} = 2\pi t_P, \quad (54.16)$$

and the Planck quantum

$$E_P = \frac{\hbar}{t_P} = k_B \Theta_{\text{clk}} \log 3. \quad (54.17)$$

Theorem 54.4 (Cell CT108 of half action). *In the specialization $R = R_{108} = \ell_P$,*

$$Q_H(\ell_P) = \frac{c^4 \ell_P}{G} = E_P, \quad (54.18)$$

$$E_\Lambda = T_H S_H = \frac{E_P}{2} = \frac{k_B \Theta_{\text{clk}} \log 3}{2}, \quad (54.19)$$

$$\boxed{E_\Lambda t_P = \frac{\hbar}{2}}, \quad \boxed{E_\Lambda T_{108} = \frac{h}{2}}. \quad (54.20)$$

Proof. The first equality is obtained by substituting $\ell_P = \sqrt{G\hbar/c^3}$ and $E_P = \sqrt{\hbar c^5/G}$. The [equation \(54.13\)](#) produces $E_\Lambda = E_P/2$, and [equation \(54.17\)](#) supplies the informational chart. Finally,

$$E_\Lambda t_P = \frac{E_P t_P}{2} = \frac{\hbar}{2}$$

and, as $T_{108} = 2\pi t_P$ and $h = 2\pi\hbar$,

$$E_\Lambda T_{108} = \frac{E_P}{2} (2\pi t_P) = \pi\hbar = \frac{h}{2}.$$

$\square$

The first equality in [equation \(54.20\)](#) publishes half a reduced action over Planck time; the second publishes half a complete action over the CT108 period. Both express the same cell in the angular and circular charts.

The temperature and entropy are also determined:

$$k_B T_H = \frac{E_P}{2\pi}, \quad \boxed{\frac{S_H}{k_B} = \pi}. \quad (54.21)$$

The temperature converts the frequency of horizon in energy; the entropy converts the area of the sphere of Planck in information. Their product recovers the same half-quantum.

54.1.6 Effective horizon weight and hyperbolic exponential

We define the effective thermodynamic weight

$$\Omega_{\text{eff}}(R) := \exp\left(\frac{S_H(R)}{k_B}\right). \quad (54.22)$$

In the CT108 set,

$$\boxed{\Omega_{\text{eff}}(\ell_P) = e^\pi.} \quad (54.23)$$

The weight Ω_{eff} is a positive quantity defined by entropy. Its real value expresses effective multiplicity; an interpretation as an integer cardinality of microstates requires an additional discretization.

The areal–volumetric incidence operator is

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad J_\varphi := \frac{2F_{\text{av}}^2 - 3I}{\sqrt{5}}. \quad (54.24)$$

A direct calculation gives

$$J_\varphi^2 = I, \quad \text{Spec}(J_\varphi) = \{+1, -1\}. \quad (54.25)$$

The hyperbolic flow at closure time π is

$$\mathcal{E}_\pi := e^{\pi J_\varphi} = \cosh \pi I + \sinh \pi J_\varphi, \quad (54.26)$$

and possesses spectrum

$$\boxed{\text{Spec}(\mathcal{E}_\pi) = \{e^\pi, e^{-\pi}\}, \quad \rho(\mathcal{E}_\pi) = e^\pi.} \quad (54.27)$$

Corollary 54.5 (Horizon–Perron scalar confluence). *In the CT108 closure,*

$$\boxed{\frac{S_H}{k_B} = \log \Omega_{\text{eff}} = \log \rho(\mathcal{E}_\pi) = \pi.} \quad (54.28)$$

Proof. The equations the corresponding identity give the three members. $\square$

The scalar identity is exact and preserves two distinct genealogies. The horizon produces e^π through exponentiation of area entropy; the previously constructed transfer dynamics recognizes the same scalar as the expanding eigenvalue of a hyperbolic involution. Terminal equality fixes a boundary condition for the intertwiner and preserves the distinction of domains.

54.1.7 Consistency of the Gravitational Transducer with the Schwarzschild Horizon

A Schwarzschild horizon with the same area radius satisfies

$$R_H = \frac{2GE_H}{c^4}, \quad E_H = \frac{c^4 R_H}{2G}. \quad (54.29)$$

The entropy areal coincides with [equation \(54.9\)](#), whereas the temperature is

$$k_B T_{\text{Sch}} = \frac{\hbar c}{4\pi R_H} = \frac{1}{2} k_B T_H|_{R=R_H}. \quad (54.30)$$

Consequently,

$$E_H = 2T_{\text{Sch}} S_H, \quad T_H^{\text{dS}} = 2T_{\text{Sch}} \quad (54.31)$$

for the same radius. In $R_H = \ell_P$,

$$E_H = \frac{E_P}{2}, \quad T_{\text{Sch}} S_H = \frac{E_P}{4}. \quad (54.32)$$

The contrast separates the two geometries: they share quasilocal energy, area, and entropy, and their surface gravities produce temperatures in the ratio two.

With $\beta = (k_B T_{\text{Sch}})^{-1}$, the action Euclidean satisfies

$$\frac{I_E}{\hbar} = \frac{E_H}{k_B T_{\text{Sch}}} - \frac{S_H}{k_B} = 2\pi - \pi = \pi \quad (54.33)$$

at the Planck radius. The same π appears as reduced entropy, reduced Euclidean action, hyperbolic-flow time, and the logarithm of the effective weight.

54.1.8 Horizontal Horizon–Perron Entangler: Spectral compatibility between boundary energy and return memory

The coincidence $\Omega_{\text{eff}} = \rho(\mathcal{E}_\pi)$ fixes a spectral datum and leaves open the relation between the complete objects. Let $\mathcal{A}_H^{(m)}$ denote the algebra of horizon observables at nonadic level m , let $\omega_H^{(m)}$ denote its state, and let $\mathcal{P}_\varphi^{(m)}$ denote the Perron transfer system.

Definition 54.6 (Thermodynamic–dynamic intertwiner). An admissible intertwiner is a family of unital completely positive linear maps (UCP)

$$\mathfrak{I}_m : \mathcal{A}_H^{(m)} \longrightarrow \mathcal{A}_\varphi^{(m)} \quad (54.34)$$

which satisfies:

- i) preservation of the state, in addition to unitality and complete positivity already incorporated into the UCP condition;
- ii) compatibility with nonadic refinement;
- iii) transport of the involution interior–exterior toward $J_\varphi \mapsto -J_\varphi$;
- iv) commutation between the horizon modular semigroup and the transfer semigroup on the declared subspace;
- v) equality of the spectral radius at time π , given by equation (54.28).

The fifth condition is necessary and scalar; the first four provide the algebraic content that a numerical coincidence does not contain. An intertwiner constructed in this manner would identify the effective thermodynamic multiplicity with a dynamics of expansion and contraction while preserving orientation, state, and refinement.

Estatuto probatorio

The identities $E_\Lambda = E_{\text{MS}} = T_H S_H = c^4 R / (2G)$, the definition $Q_H = \mathcal{F}_P R$, the CT108 specialization, the two half-action cells, and the equality $S_H / k_B = \log \rho(e^{\pi J_\varphi}) = \pi$ are exact results under the declared conventions and de Sitter realization. The identification of Ω_{eff} with a microscopic cardinal and the intertwiner $\mathfrak{I}_m$ belong to a delimited physical interface.

Falsification criterion

Geometric closure fails if $\Lambda_R \neq 3/R^2$, if the sphere of radius R fails to satisfy the marginal condition, or if the integrated energy differs from $E_{\text{MS}}(R)$. Thermodynamic closure fails if the temperature or the area law produces $T_H S_H \neq c^4 R / (2G)$. The CT108 specialization fails if $R_{108} \neq \ell_P$, $T_{108} \neq 2\pi t_P$, or $E_P \neq k_B \Theta_{\text{clk}} \log 3$. The Perron interface is refuted by an obstruction of positivity, state, orientation, or refinement, even when the eigenvalue e^π agrees scalarly.

Machian Factorization of the Transport Memory from the Global Closure and the Local Response. The relation between global closure and local response leads to Mach’s principle formulated on transport memory.

PART VII

Boundary incidence and exceptional stabilizers

Dodecaphasic–Witt intertwiner

55.1 Construction of the point–hexad incidence intertwiner and transport of the residual action of S_3

The repeated appearance of quotient $3/11$ on the dodecaphasic screen, in the polarity of a Witt star, and in nonadic filtration raises a question stronger than the arithmetic equality of three fractions. The correct question is operatorial: to determine whether those three appearances are traces of the same positive sector viewed in different realizations and, if they are, to construct the maps intertwining them without introducing a subsequent metrological datum.

The answer has four levels. First, point–hexad incidence constructs a canonical isometry between the eleven-dimensional dodecaphase module and an eleven-dimensional eigenspace of the Witt design. Second, the Witt star acts on that same module, but does so as a signed involution; its $3/11$ is an index, not the trace of a positive projector. Third, the oriented survival sector appears in $t = 7$, where a single eleven-branch space contains a rank-three sector. Fourth, the orientation line of that triplet, together with the internal witnesses B_K and u_K , variationally selects a unique normalizer S_3 , fixes its orientation, and, through Reynolds averaging and polar decomposition, produces the sector–dodecaphase isometric intertwining. The dodecaphase–Witt passage is a second incidence isometry; it is not presupposed that both arrows realize a single representation of S_3 . The Reynolds averages, the decompositions into representations, and the polar factor used in this section are the standard ones [Ser77, HJ13].

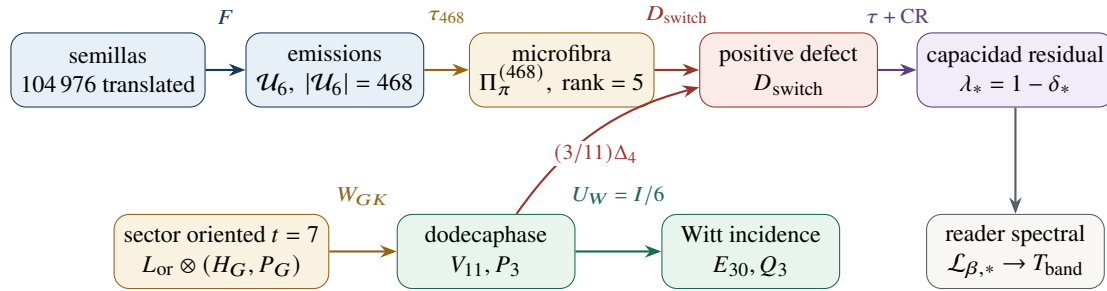


Figure 55.1: Architecture of the predimensional closure. The upper row constructs the measure and the action from the quotient of emissions; the lower row transports the oriented sector $t = 7$ to the dodecaphase and then the dodecaphase to Witt. The solid arrows are proved projector isometries. The first is fixed by the variational selection of theorem 55.7; for the second, no equivariance with respect to the selected H_* is asserted.

55.2 Typing of the objects associated with $3/11$

Let

$$V_{12} = \mathbb{R}^{12}, \quad V_{11} = \mathbf{1}^\perp = \left\{ v \in \mathbb{R}^{12} : \sum_{p=1}^{12} v_p = 0 \right\}.$$

The icosahedral action declared on the twelve positions decomposes as

$$V_{12} = V_1 \oplus V_3 \oplus V_{3'} \oplus V_5.$$

Denotemos by P_3 the orthogonal projector on the first summand three-dimensional. Then

$$P_3^2 = P_3 = P_3^*, \quad \text{rank } P_3 = 3, \quad \tau_{11}(P_3) = \frac{1}{11} \text{Tr}_{V_{11}} P_3 = \frac{3}{11}.$$

There are three nearby objects and a fourth object of a different type:

1. the point–hexad incidence transports P_3 to a rank-three projector Q_3 on a Witt module;
2. survival in $t = 7$ distinguishes three branches among eleven and produces a projector P_G with normalized trace $3/11$;
3. an oriented intertwiner identifies P_3V_{11} with the oriented realization of P_GH_G ;
4. a Witt star has seven even modes and four odd ones, so its normalized trace also equals $3/11$, but it contains negative spectrum and is not a projector.

The distinction between projector and involution will be essential to the positivity of the action law in the following section.

55.3 The point–hexad incidence matrix

Let $\mathcal{H}$ be the set of the 132 hexads of the design

$$W_{12} = S(5, 6, 12).$$

We define

$$I : \mathbb{R}^{12} \longrightarrow \mathbb{R}^{\mathcal{H}}, \quad (Iv)(H) = \sum_{p \in H} v_p,$$

or, equivalently,

$$I_{H,p} = \mathbf{1}_{\{p \in H\}}.$$

This operator records only incidence. It contains no α , units, masses, CKM, Barbero, or any physical target.

Lemma 55.1 (Metric Identity of Incidence). *We have*

$$\boxed{I^*I = 36I_{12} + 30J_{12}.}$$

Therefore, for every $v \in V_{11}$,

$$\|Iv\|^2 = 36\|v\|^2.$$

Proof. In an $S(5, 6, 12)$ system, each point belongs to

$$r = \frac{\binom{11}{4}}{\binom{5}{4}} = 66$$

hexads, and each pair of points belongs to

$$\lambda_2 = \frac{\binom{10}{3}}{\binom{4}{3}} = 30$$

hexads. The diagonal entries of I^*I have value 66 and the nondiagonal entries have value 30. Hence

$$I^*I = (66 - 30)I_{12} + 30J_{12}.$$

Since J_{12} vanishes on $\mathbf{1}^\perp$, the second identity is obtained. $\square$

To identify the spectral branch selected by I , we introduce

$$G_{H,H'} = \binom{|H \cap H'|}{4}.$$

Lemma 55.2 (Spectral identity of incidence). *With $J_{132,12}$ denoting the rectangular all-ones matrix,*

$$GI = 30I + 15J_{132,12}.$$

In particular, if $v \in V_{11}$, then

$$G(Iv) = 30Iv.$$

Proof. The entry (H, p) of GI is

$$\sum_{H' \ni p} \binom{|H \cap H'|}{4}.$$

The exhaustive census of W_{12} gives 45 when $p \in H$ and 15 when $p \notin H$. Consequently, the entry equals $30I_{H,p} + 15$. Reconstruction of the 132 hexads from the extended ternary Golay code checks the $132 \cdot 12$ entries as integers. $\square$

Define

$$E_{30}^{\text{inc}} := I(V_{11}) \subseteq \ker(G - 30I).$$

Its dimension is eleven by the lemma 55.1. The calculation modulo 1009 obtains

$$\text{rank}_{\mathbb{F}_{1009}}(G - 30I) = 121.$$

Since reduction modulo 1009 gives a lower bound for the rational rank, it follows that $\dim_{\mathbb{Q}} \ker(G - 30I) \leq 11$. The preceding inclusion provides eleven linearly independent vectors in that kernel. Therefore,

$$E_{30}^{\text{inc}} = \ker(G - 30I).$$

From now on we will write E_{30} .

Theorem 55.3 (Canonical dodecaphase entangler–Witt). *The operator*

$$U_W = \frac{1}{6}I|_{V_{11}} : V_{11} \longrightarrow E_{30}$$

is a isometry surjective. For every $g \in \text{Aut}(W_{12}) \cong M_{12}$,

$$U_W \rho_{12}(g) = \rho_{\mathcal{H}}(g) U_W.$$

Among the equivariant point–hexad kernels, the isometry condition and the positive sign on incidences uniquely determine U_W .

Proof. The isometry follows from $I^*I = 36I$ over V_{11} . Its image has dimension eleven and is contained in E_{30} , which also has dimension eleven; it is therefore surjective. Equivariance translates

$$p \in H \iff g(p) \in g(H).$$

For uniqueness, M_{12} is transitive on the flags (p, H) , with $p \in H$, and on the antiflags. Every equivariant kernel has two values, a on flags and b on antiflags:

$$T = aI + b(J - I) = (a - b)I + bJ.$$

The term J vanishes on V_{11} , so there T is a multiple of I . The isometry forces $|a - b| = 1/6$ and the positive normalization selects $a - b = 1/6$. $\square$

55.4 Positive transport of the tritic projector

The entangler allows defining

$$Q_3 = U_W P_3 U_W^*.$$

We obtain

$$Q_3^2 = Q_3 = Q_3^*, \quad \text{rank } Q_3 = 3,$$

and the diagram

$$\begin{array}{ccc} V_{11} & \xrightarrow{U_W} & E_{30} \\ P_3 \downarrow & & \downarrow Q_3 \\ V_{11} & \xrightarrow{U_W} & E_{30} \end{array}$$

commutes. In particular,

$$\tau_{V_{11}}(P_3) = \tau_{E_{30}}(Q_3) = \frac{3}{11}.$$

The canonicity of U_W belongs to the labeled design and its automorphism action. It must be distinguished from the screen A_5/C_5 used to define P_3 : in the constructed labeled realization, only the identity of that concrete action of A_5 preserves the fixed set of hexads. Hence U_W canonically transports the already given P_3 toward Q_3 , but does not by itself convert the action A_5/C_5 into a subaction of M_{12} . The bridge to the oriented sector will be constructed separately from the marked data K and B_K .

55.5 Signed index of the Witt star

Let B be a tetrad. The four hexads that contain it are written

$$H_i = B \sqcup P_i, \quad |P_i| = 2,$$

where the four petals P_i partition the complement of B . The star involution R_B fixes B and exchanges the two points of each petal. Its cycle type on twelve points is $1^4 2^4$. In V_{11} , and by means of U_W also in E_{30} , its spectrum is

$$\text{Spec}(R_B) = \{1^{[7]}, (-1)^{[4]}\}.$$

Therefore,

$$\tau_{11}(R_B) = \frac{7-4}{11} = \frac{3}{11}.$$

Theorem 55.4 (Obstruction projector–star). *There is no isomorphism $F : V_{11} \rightarrow E_{30}$ such that*

$$FP_3 = R_B F.$$

Proof. If it existed, P_3 and R_B would be similar. However,

$$\text{Spec}(P_3) = \{1^{[3]}, 0^{[8]}\}, \quad \text{Spec}(R_B) = \{1^{[7]}, (-1)^{[4]}\}.$$

Their minimal polynomials are $x(x-1)$ and $(x-1)(x+1)$, respectively. Similarity is impossible. $\square$

The equality $3/11$ thus has two readings on the same module. For Q_3 it is the mass of a positive rank-three sector. For R_B it is a parity index. A positive action must involve Q_3 ; the star preserves orientation and distinguishes sheets but does not replace the projector.

55.6 homogeneous reduction $11 \rightarrow 3 \rightarrow 1$ in $t = 7$

The preceding reading of $3/11$ in $t = 8$ compared three local roots with eleven continuations of a single root. The numerator and denominator did not belong to the same space. The detailed table locates the correct object in $t = 7$: there are eleven local branches; all eleven survive one horizon, three survive two, and one survives three,

$$11 \longrightarrow 3 \longrightarrow 1.$$

index	triple of words	$h = 2$	$h = 3$
0	200112 212221 110110	3	0
1	200121 212212 110110	3	1
2	200211 212122 110110	3	0
3	202122 210211 110110	0	0
4	202212 210121 110110	0	0
5	202221 210112 110110	0	0
6	210111 202222 110110	0	0
7	210222 202111 110110	0	0
8	212112 200221 110110	0	0
9	212121 200212 110110	0	0
10	212211 200122 110110	0	0

Let $\mathcal{B}_7$ be the set of these eleven branches and

$$H_G = \ell^2(\mathcal{B}_7).$$

We define

$$P_G \delta_b = \begin{cases} \delta_b, & b \text{ admits continuation to horizon two,} \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$P_G^2 = P_G = P_G^*, \quad \text{rank } P_G = 3, \quad \tau_{H_G}(P_G) = \frac{3}{11}.$$

This is the third positive realization.

55.7 The residual action of S_3 and the positive regions of π

A permutation of the three final positions, applied simultaneously to the first two words of each branch and leaving the third fixed, preserves $\mathcal{B}_7$. This rule defines an action of S_3 . Its orbits have sizes

$$3 + 3 + 1 + 1 + 3 = 11.$$

The support of P_G , formado by the indices 0, 1, 2, is one orbit complete, so that P_G conmuta with S_3 .

The character of the representation on the eleven branches, evaluated at the identity, a transposition, and a cycle of order three, is

$$\chi_{\mathcal{B}_7} = (11, 5, 2) = 5\chi_{\text{triv}} + 3\chi_{\text{std}}.$$

On the image of P_G ,

$$\chi_G = (3, 1, 0) = \chi_{\text{triv}} + \chi_{\text{std}}.$$

The three positive regions of the microfibre of π have signatures

$$339555, \quad 393555, \quad 933555.$$

The tails of the first word of the three surviving branches are

$$112, \quad 121, \quad 211.$$

The coordinate substitution $1 \mapsto 3, 2 \mapsto 9$ gives

branch	tail	positive region
0	112	339555
1	121	393555
2	211	933555.

Theorem 55.5 (Horizontal–vertical bridge of the positive microfiber). *The preceding rule is an S_3 -equivariant bijection between the three positive regions of π and the three branches selected by P_G . The map preserves the position of the unique exceptional symbol.*

Proof. Both sets are the three words obtained by placing a single exceptional symbol in one of three positions. The substitution is applied coordinate by coordinate and therefore commutes with every permutation of positions. The exceptional position uniquely recovers the source element. $\square$

55.8 Dependence of orientation in the dodecaphase entangler–Witt

For any normalizer $H = N_{A_5}(C_3) \simeq S_3$ of a subgroup of order three in A_5 , we define the abstract character of the quotient

$$\text{sgn}_H : H \longrightarrow H/C_3 \simeq C_2 \longrightarrow \{+1, -1\}, \quad \text{sgn}_H(h) = \begin{cases} +1, & h \in C_3, \\ -1, & h \in H \setminus C_3. \end{cases}$$

This character is not the ambient sign of a permutation of five letters: every element of A_5 has ambient sign $+1$. It is the intrinsic character of the reflection in the quotient H/C_3 . With this convention, the character of the icosahedral sector restricted to H is

$$\chi_{P_3V_{11}|H} = (3, -1, 0) = \chi_{\text{sgn}_H} + \chi_{\text{std}}.$$

By contrast,

$$\chi_{\text{im } P_G} = (3, 1, 0) = \chi_{\text{triv}} + \chi_{\text{std}}.$$

Corollary 55.6 (Obstruction to the unoriented braider). *There is no S_3 -equivariant isomorphism*

$$\text{im } P_G \longrightarrow P_3V_{11}$$

for the non-twisted actions.

Proof. Isomorphic representations have the same character. Their values on a transposition are 1 and -1 . $\square$

Let (e_0, e_1, e_2) the basis posicional of $\mathbb{R}_{\text{perm}}^3$. We defines

$$L_{\text{or}} = \bigwedge^3 \mathbb{R}_{\text{perm}}^3, \quad \ell_{\text{or}} = e_0 \wedge e_1 \wedge e_2$$

the orientation line of the triplet, equipped with the positive anchor ℓ_{or} . The action of S_3 on this line is precisely its abstract sign character. Upon tensoring by it,

$$\chi_{L_{\text{or}} \otimes \text{im } P_G} = (3, -1, 0),$$

of mode that the character obstruction disappears. The equality of caracteres proves the existence of one class of isomorphisms, but aun not selects one. The selection is obtained of two witnesses already present in the dodecaphase.

55.9 Variational selection of the normalizer S_3

Let

$$K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601)$$

the decimal chart of the seal. The common minimal witness of the two readings is

$$B_K = H_e \cap P_\pi = \{p : K_p \geq 729\} = \{4, 6, 9, 10\},$$

in the mathematical numbering $1, \dots, 12$. En the equivalent convention with zero-based indices, the same set is $\{3, 5, 8, 9\}$. We define two vectors of the sector P_3 :

$$b_K = P_3 \left(\mathbf{1}_{B_K} - \frac{|B_K|}{12} \mathbf{1} \right), \quad u_K = P_3 \left(K - \bar{K} \mathbf{1} \right).$$

The first satisfies exactly.

$$\|b_K\|^2 = 1.$$

The group A_5 contains ten subgroups C_3 . For each one, let

$$H(C_3) = N_{A_5}(C_3) \simeq S_3$$

and let be the sign projector

$$e_{\text{sgn}, H} = \frac{1}{6} \sum_{h \in H} \text{sgn}_H(h) \rho(h).$$

Here sgn_H is the abstract character of H/C_3 defined above, not the —trivial— ambient sign of A_5 . The orientation energy associated with the common witness is

$$\mathcal{E}_B(H) = \|e_{\text{sgn}, H} b_K\|^2.$$

The exact census in $\mathbb{Q}(\sqrt{5})$ produces a unique maximum:

$$\mathcal{E}_B(H_*) = \frac{2}{3} + \frac{2}{15} \sqrt{5}.$$

The second level is

$$\frac{1}{3} + \frac{2}{15} \sqrt{5},$$

so that the exact gap is

$$\boxed{\mathcal{E}_B(H_*) - \mathcal{E}_B(H_{\text{segundo}}) = \frac{1}{3}.$$

The same H_* is the unique maximum when b_K is replaced by u_K . Therefore, the selector does not depend on a single encoding of the witness.

In this labeling,

$$H_* = N_{A_5}(\langle c_* \rangle), \quad c_* = (2\ 3\ 4).$$

The rule is natural under simultaneous relabeling by A_5 : when B_K , K , and P_3 are transported, the maximum is conjugated by the same element.

55.10 Selection of orientation and presentation of S_3

Within of H_* , consider

$$s(g) = \langle b_K, \rho(g)u_K \rangle.$$

The maximum among the two order-three elements uniquely selects

$$c_* = (0, 1, 3, 4, 2),$$

in image notation of the permutation of five points, and the maximum among the three involutions selects

$$r_* = (1, 0, 4, 3, 2).$$

It is verified exactly.

$$r_* c_* r_* = c_*^{-1}.$$

The presentation of the oriented sector by the cycle of positions $c = (0\ 1\ 2)$ and the reflection $r = (0\ 2)$, which fixes the central position, therefore determines a unique isomorphism

$$\theta : S_3^{\text{or}} \xrightarrow{\sim} H_*, \quad \theta(c) = c_*, \quad \theta(r) = r_*.$$

Here S_3^{or} denotes the group of relabelings of the oriented sector $t = 7$, generated by c and r ; it introduces no additional operatorial class. The only branch that reaches horizon three is precisely the central branch 1; this datum fixes which of the three positions remains fixed under reflection.

55.11 Reynolds average and polar factor

Let

$$W_{\text{or}} = L_{\text{or}} \otimes \mathbb{R}_{\text{perm}}^3$$

the oriented sector triple. With the anchor already fixed, we take

$$\boxed{w = \ell_{\text{or}} \otimes e_1},$$

where e_1 represents the central position, the only one reaching horizon three. Define

$$A = \sum_{g \in S_3} \rho_{W_{\text{or}}}(g) |w\rangle \langle b_K| \rho_{V_{11}}(\theta(g))^{-1}.$$

By construction,

$$A \rho_{V_{11}}(\theta(g)) = \rho_{W_{\text{or}}}(g) A, \quad A = A P_3.$$

The exact computation in $\mathbb{Q}(\sqrt{5})$ obtains

$$\text{rank } A = 3.$$

Moreover, $G_A = A A^*$ has all its diagonal entries equal to

$$3 + \frac{2}{5}\sqrt{5}$$

and all its off-diagonal entries equal to

$$\frac{5}{2} + \frac{3}{5}\sqrt{5}.$$

Its two isotype eigenvalues are

$$\lambda_{\text{sgn}} = 8 + \frac{8}{5}\sqrt{5}, \quad \lambda_{\text{std}} = \frac{1}{2} - \frac{1}{5}\sqrt{5}.$$

Both are strictly positive. In particular, $G_A^{-1/2}$ exists, and the polar decomposition

$$U_{GK} = G_A^{-1/2} A$$

is one equivariant isometry of $P_3 V_{11}$ on W_{or} :

$$U_{GK}^* U_{GK} = P_3, \quad U_{GK} U_{GK}^* = I_{W_{\text{or}}}.$$

Let

$$J_G : W_{\text{or}} \longrightarrow L_{\text{or}} \otimes H_G$$

the isometric inclusion in the three selected branches. Then

$$J_G J_G^* = I_{L_{\text{or}}} \otimes P_G.$$

The sector–dodecaphase partial isometry is denoted

$$W_{GK} = U_{GK}^* J_G^* : L_{\text{or}} \otimes H_G \longrightarrow V_{11}$$

satisfies

$$W_{GK}^* W_{GK} = I_{L_{\text{or}}} \otimes P_G, \quad W_{GK} W_{GK}^* = P_3,$$

and, therefore,

$$P_3 W_{GK} = W_{GK} = W_{GK} (I_{L_{\text{or}}} \otimes P_G).$$

Theorem 55.7 (Relative Isometric Identification of Three Projectors). *Relative to the action A_5/C_5 , the projector P_3 , the seal K , the doubly constructed witness B_K , the oriented finite sector $t = 7$, the abstract character sgn_{H_*} , the anchor ℓ_{or} , and the variational rule $\mathcal{E}_B$, there exists a unique normalized partial isometry W_{GK} that identifies P_G with P_3 . Independently, the incidence isometry U_W identifies P_3 with Q_3 . Consequently, the following chain of isometric identifications of projectors is obtained:*

$$\begin{array}{ccccc} L_{\text{or}} \otimes H_G & \xrightarrow{W_{GK}} & V_{11} & \xrightarrow{U_W} & E_{30} \\ I_{L_{\text{or}}} \otimes P_G \downarrow & & \downarrow P_3 & & \downarrow Q_3 \\ L_{\text{or}} \otimes H_G & \xrightarrow{W_{GK}} & V_{11} & \xrightarrow{U_W} & E_{30}. \end{array}$$

Typed traces satisfy

$$\tau_{L_{\text{or}} \otimes H_G}(I_{L_{\text{or}}} \otimes P_G) = \tau_{V_{11}}(P_3) = \tau_{E_{30}}(Q_3) = \frac{3}{11}.$$

The statement identifies three projectors isometrically; it does not assert that their three spaces are the same representation of a common group.

Proof. The energy $\mathcal{E}_B$ selects a unique normalizer H_* ; the scoring criterion $s(g)$ selects a unique pair (c_*, r_*) , and with it a unique isomorphism θ . The Reynolds average is then equivariant for the actions of S_3^{or} and H_* related by θ , and has rank three. The positive part of its polar decomposition fixes the normalization without introducing a basis. The preceding partial identities prove that W_{GK} identifies P_G and P_3 . On the other hand, the theorem 55.3 identifies P_3 and Q_3 as projectors through U_W . These two operatorial identities make the diagram commute, without extending the equivariance of the first arrow to the second. $\square$

Proposition 55.8 (Obstruction to triple equivariance in the fixed gauge). *The preceding construction does not prove that U_W is H_* -equivariant. Under the declared labeling, the fixed action of A_5/C_5 on the twelve*

points preserves the set of hexads of W_{12} only for the identity element. Therefore, the H_* -equivariant intertwining of W_{GK} does not extend, in that gauge, to a common representation on E_{30} .

Proof. The first assertion separates equivariance domains: W_{GK} uses $H_* \subset A_5$, whereas the incidence theorem proves equivariance of U_W for $\text{Aut}(W_{12})$. The exact reproducible census applies the sixty elements of action A_5/C_5 to the labeled hexad system; only the identity preserves that system. Since H_* contains six elements, its action cannot enter by restriction of that fixed action in $\text{Aut}(W_{12})$. $\square$

To promote the chain of projectors to a common representation, it is necessary to construct explicitly a monomorphism

$$\iota : H_* \hookrightarrow \text{Aut}(W_{12})$$

such that, for every $h \in H_*$,

$$U_W \rho_{V_{11}}(h) = \rho_{E_{30}}(\iota(h)) U_W.$$

This equation is also a sufficient condition for the equivariance of the second arrow. No ι with this property has been constructed in the current gauge; therefore the proved equivariance terminates in P_3 and is not prolonged by mere equality of ranks.

Before using B_K , u_K , the orientation, and the polar factor, there were $10 \cdot 6 \cdot 4 = 240$ labeled maps, or 40 after internal relabelings of the sector had been identified. The preceding rules reduce that set to one. This is the verifiable content of the word *canonical* for W_{GK} , within the declared rules. It does not extend canonicity to a common action of H_* on the Witt module.

The variational rule $\mathcal{E}_B$ uses only prior HMT data and fixes the class in which uniqueness is obtained. It does not identify the star involution with P_3 , since their spectra remain distinct.

55.12 Comparison of the readings of $t = 7$ and $t = 8$

In $t = 8$ there are three local roots and eleven extensions to horizon nine of the unique root that continues. The sizes of the three fibers are

$$(11, 0, 0).$$

The natural branch–chain matrix has one row of eleven ones and two zero rows, so its rank is one. In $t = 9$, moreover, the visible quotient changes from $3/11$ to $3/6$. That fraction was not the trace of a stable projector. The correction does not eliminate the sector: it shifts the reading to the homogeneous object P_G of $t = 7$, where rank and dimension belong to the same space.

55.13 Positivity of the Action Induced by the Dodecaphasic Braider–Witt

The law of action may u now a unique class isometric of projectors positive in tris spaces of support:

$$P_G \xleftrightarrow{W_{GK}} P_3 \xleftrightarrow{U_W} Q_3.$$

The star R_{B_K} is reserved for parity and orientation. The projector P_G records survival; P_3 records dodecaphasic polarization; Q_3 records its incidence realization in Witt. This separation of functions is the mathematical premise of the measure and the positive law in the following section. The chain expresses isometric equivalence of projectors, not a common representation: the latter would require the monomorphism ι and the equivariance identity formulated above.

From the spectral gap to the Leech lattice

56.1 From the spectral gap $4 \rightarrow 2 \rightarrow 6 \rightarrow 54$: local coupling, modular compression, and two-dimensional integration

For two consecutive representatives $(k, k + 1)$, consider

$$B_k = \begin{pmatrix} k^2 & k(k+1) \\ k(k+1) & (k+1)^2 \end{pmatrix} \pmod{9}.$$

Theorem 56.1 (Uniqueness of the Adjacent Diagonal Block). *There is a unique block B_k whose diagonal entries coincide. It is determined by $k = 4$ and is*

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}.$$

Proof. The equality of the diagonals is equivalent to

$$k^2 \equiv (k+1)^2 \pmod{9},$$

that is, $2k + 1 \equiv 0 \pmod{9}$. Since 2 is a unit of $\mathbb{Z}/9\mathbb{Z}$, the congruence has the unique solution $k \equiv 4 \pmod{9}$. $\square$

Therefore,

$$B_c = 7I + 2R, \quad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad R^2 = I.$$

With $P_{\pm} = (I \pm R)/2$,

$$B_c = 9P_+ + 5P_-.$$

Theorem 56.2 (Hierarchy of the separation between sum and product). *The central block has local spectral jump $9 - 5 = 4$. Its off-diagonal coefficient is 2. Compression by the 3 classes modulo 3 produces the aggregate difference $51 - 45 = 6$, and expansion over the 9 positions produces the global defect*

$$9 \cdot 6 = 459 - 405 = 54.$$

The integers 4, 2, 6, 54 pertain, respectively, to the spectral gap, the local coupling, modular compression, and two-dimensional integration.

Proof. The decomposition $B_c = 7I + 2R = 9P_+ + 5P_-$ gives the jump and the local coefficient. The aggregated matrices M_{Π}, M_{Σ} satisfy $\sum M_{\Pi} = 51$ and $\sum M_{\Sigma} = 45$. Each aggregated entry represents 9 positions, so unfolding multiplies the difference by 9 and recovers the totals 459 and 405. $\square$

Their ordered concatenations produce the blocks 72 and 27:

$$72 + 27 = 99, \quad 72 - 27 = 45 = \det B_c.$$

The concatenation of the diagonal and antidiagonal entries produces, respectively, 77 and 22. Therefore,

$$72 + 27 = 77 + 22 = 99, \quad 77 - 22 = 55 = 54 + 1.$$

These equalities express two exact partitions of the same block; they do not by themselves identify an external constant.

56.2 Paley conference form, finite quadratic extension and self-dual ternary code

The tetrad B_K constructed in the preceding unit does not yet have a Witt star: to define it, one must first construct the incidence system that contains its hexads. This unit performs that step from the projective gauge of the six units modulo nine, obtains an internal square root of $-I_6$, and constructs the ternary graph code of length twelve. No identification follows from a mere equality of cardinalities.

56.2.1 Projective gauge of the 6 units modulo 9

Let

$$\mathcal{U}_9 = (\mathbb{Z}/9\mathbb{Z})^\times = \{1, 2, 4, 8, 7, 5\},$$

ordered by the successive powers of 2 modulo 9. Fijamos the coordinate bijection

$$\theta : \mathcal{U}_9 \longrightarrow \mathbb{P}^1(\mathbb{F}_5), \quad \begin{array}{c|cccccc} u & 1 & 2 & 4 & 8 & 7 & 5 \\ \hline \theta(u) & \infty & 0 & 1 & 2 & 3 & 4 \end{array}. \quad (56.1)$$

The map θ is a coordinate gauge. It is not asserted to be a homomorphism between multiplication in $\mathcal{U}_9$ and an operation on $\mathbb{P}^1(\mathbb{F}_5)$.

Let

$$\chi : \mathbb{F}_5 \longrightarrow \{-1, 0, 1\}$$

the quadratic character:

$$\chi(0) = 0, \quad \chi(1) = \chi(4) = 1, \quad \chi(2) = \chi(3) = -1.$$

In the order $(\infty, 0, 1, 2, 3, 4)$, we define

$$(C_P)_{\infty, x} = (C_P)_{x, \infty} = 1, \quad x \in \mathbb{F}_5, \quad (56.2)$$

$$(C_P)_{x, y} = \chi(x - y), \quad x, y \in \mathbb{F}_5, x \neq y, \quad (56.3)$$

$$(C_P)_{x, x} = 0. \quad (56.4)$$

Thus,

$$C_P = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & -1 & -1 & 1 \\ 1 & 1 & 0 & 1 & -1 & -1 \\ 1 & -1 & 1 & 0 & 1 & -1 \\ 1 & -1 & -1 & 1 & 0 & 1 \\ 1 & 1 & -1 & -1 & 1 & 0 \end{pmatrix}. \quad (56.5)$$

Lemma 56.3 (Quadratic correlation sum). *For $a \neq b$ in $\mathbb{F}_5$,*

$$\sum_{t \in \mathbb{F}_5} \chi(t - a)\chi(t - b) = -1. \quad (56.6)$$

Proof. The substitution $u = (t - a)/(b - a)$ is to bijection of $\mathbb{F}_5$. Ifnce $\chi((b - a)^2) = 1$, the sum reduce to

$$\sum_{u \in \mathbb{F}_5} \chi(u(u - 1)).$$

Evaluation of $u = 0, 1, 2, 3, 4$ gives, respectively, $0, 0, 1, -1, -1$, whose sum is -1 . $\square$

Proposition 56.4 (Conference Identity). *The matrix C_P is symmetric and satisfies*

$$\boxed{C_P C_P^T = 5I_6.} \quad (56.7)$$

Proof. Each row contains five entries of modulus one and one zero entry; its squared norm is five. For two distinct finite rows, the contribution of the coordinate ∞ is $+1$, whereas lemma 56.3 gives -1 for the five finite coordinates. Their inner product is zero. The sum of a nontrivial character over $\mathbb{F}_5$ is zero, so the row corresponding to ∞ is also orthogonal to every finite row. $\square$

56.2.2 ternary reduction and internal quadratic root of $-I_6$

We reduce C_P modulo tris, with $-1 \equiv 2 \pmod{3}$:

$$A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix} \in M_6(\mathbb{F}_3). \quad (56.8)$$

Theorem 56.5 (Internal square root of $-I_6$). *We have*

$$\boxed{A_W^2 = -I_6}. \quad (56.9)$$

In particular,

$$A_W^{-1} = -A_W.$$

Proof. The proposition 56.4 gives

$$A_W A_W^T \equiv 5I_6 \equiv 2I_6 = -I_6 \pmod{3}.$$

Since $A_W = A_W^T$, $A_W^2 = -I_6$ is obtained. $\square$

The preceding identity is the algebraic residence of the root of -1 in this corridor. No proceeds of the factor $1/4$ of the Hadamard inversion: the former is a rational scalar forced by $H_4^2 = 4I_4$; this is one quadratic structure on characteristic three.

56.2.3 Extension quadratic to $\mathbb{F}_9$

We define

$$\mathbb{F}_9 = \mathbb{F}_3[\iota]/(\iota^2 + 1), \quad \iota^2 = -1. \quad (56.10)$$

The polynomial $X^2 + 1$ has no roots in $\mathbb{F}_3$, so that the quotient is a field of nine elements. For $V = \mathbb{F}_3^6$, let

$$V_9 = V \otimes_{\mathbb{F}_3} \mathbb{F}_9.$$

The operators

$$P_\iota = \frac{1}{2}(I - \iota A_W), \quad P_{-\iota} = \frac{1}{2}(I + \iota A_W) \quad (56.11)$$

are well defined because 2 is invertible in $\mathbb{F}_3$.

Proposition 56.6 (Conjugate projectors). *The following hold*

$$P_\iota^2 = P_\iota, \quad P_{-\iota}^2 = P_{-\iota}, \quad P_\iota P_{-\iota} = 0, \quad P_\iota + P_{-\iota} = I.$$

Proof. Usando $\iota^2 = -1$ and $A_W^2 = -I$,

$$(\iota A_W)^2 = I.$$

The four identities are then the elementary identities for the projectors $(I \mp \iota A_W)/2$ of an involution. $\square$

Theorem 56.7 (Conjugate spectral decomposition). *Let*

$$E_{\pm\iota} = \ker(A_W \mp \iota I).$$

Then

$$V_9 = E_\iota \oplus E_{-\iota}, \quad \dim_{\mathbb{F}_9} E_\iota = \dim_{\mathbb{F}_9} E_{-\iota} = 3. \quad (56.12)$$

Proof. The minimal polynomial of A_W divides $X^2 + 1$, which splits over $\mathbb{F}_9$ with simple roots $\pm\iota$. The projectors of [proposition 56.6](#) produce the direct sum. Since A_W has coefficients in $\mathbb{F}_3$, the Frobenius automorphism exchanges the two eigenspaces; their dimensions are equal and sum to six. $\square$

This construction dota $\mathbb{F}_3^6$ of istructura of modulo of rango tris over $\mathbb{F}_9$, by means of

$$(a + b\iota) \cdot v := av + b(vA_W). \quad (56.13)$$

The equality $3^6 = 9^3$ is thus accompanied by an explicit action; it is not used as a cardinal argument.

56.2.4 Ternary graph code of length 12

We define

$$c : \mathbb{F}_3^6 \longrightarrow \mathbb{F}_3^{12}, \quad c(q) = (q, qA_W), \quad (56.14)$$

and

$$C_W := \text{im } c. \quad (56.15)$$

Its generating matrix is

$$G_W = (I_6 \mid A_W).$$

Theorem 56.8 (Injectivity and self-duality). *The map c is injective, $\dim C_W = 6$ and*

$$\boxed{C_W = C_W^\perp}. \quad (56.16)$$

Proof. The first half of $c(q)$ is q , so $\text{pr}_1 \circ c = \text{id}$; this proves injectivity and the dimension. Moreover,

$$G_W G_W^\top = I_6 + A_W A_W^\top = I_6 + 2I_6 = 0$$

in $\mathbb{F}_3$. Therefore, $C_W \subseteq C_W^\perp$. Both spaces have dimension six, so they are equal. $\square$

For $r = 0, \dots, 12$, let

$$N_r := \#\{q \in \mathbb{F}_3^6 : \text{wt}(q, qA_W) = r\}.$$

The enumeration of the $3^6 = 729$ entries produces

$$N_0 = 1, \quad N_6 = 264, \quad N_9 = 440, \quad N_{12} = 24, \quad (56.17)$$

and $N_r = 0$ in the remaining weights.

Theorem 56.9 (Enumerator and minimum distance). *The code C_W has parameters*

$$\boxed{C_W = [12, 6, 6]_3}$$

and weight enumerator

$$\boxed{W_{C_W}(y) = 1 + 264y^6 + 440y^9 + 24y^{12}}. \quad (56.18)$$

Proof. The dimension and self-duality have already been proved. The census (56.17) traverses the 729 words without sampling. Its least nonzero weight is six, which is the minimum distance of a linear code. The sum $1 + 264 + 440 + 24 = 729$ checks that no word was omitted. $\square$

56.2.5 Algebraic classification of the self-dual ternary structure obtained

Three distinct objects and three typed arrows have been constructed:

$$\mathcal{U}_9 \xrightarrow{\theta} \mathbb{P}^1(\mathbb{F}_5) \xrightarrow{\chi} C_P \xrightarrow{\text{mod } 3} A_W,$$

$$A_W^2 = -I_6 \implies \mathbb{F}_3^6 \cong \mathbb{F}_9^3 \implies C_W = \{(q, qA_W) : q \in \mathbb{F}_3^6\}.$$

The first arrow is a gauge, the second uses the quadratic character, and the third is coefficient reduction. The structure over $\mathbb{F}_9$ is finite; it is not identified with a real or analytic complex structure.

56.2.6 Obstructions in the Paley– C_W corridor: bijection, $A_W^2 = -I_6$, self-duality, and census of 729 words

Test criterion 56.10 (Falsification of the Paley– C_W corridor). The construction is refuted if any of the following occurs:

1. θ is not bijective or is used as a homomorphism without proof;
2. the sum (56.6) differs from -1 ;
3. $C_P C_P^\top \neq 5I_6$;
4. $A_W^2 \neq -I_6$;
5. the projectors (56.11) are not complementary;
6. the map $q \mapsto (q, qA_W)$ loses injectivity or self-duality;
7. the census does not sum to 729, differs from (56.17), or presents a word of weight between one and five;
8. $\mathbb{F}_3^6 \cong \mathbb{F}_9^3$ is inferred only from $3^6 = 9^3$, omitting the action (56.13).

56.3 Ternary Steiner system, Witt-star involutions and generation of M_{12}

The preceding unit constructed the self-dual code $C_W = [12, 6, 6]_3$. Its words of weight six are now projectivized, the Steiner property is proved, and only then is the star associated with the tetrad B_K defined. This order prevents an incidence figure from being turned into an object added by visual similarity.

56.3.1 Supports projective of weight 6

Let

$$\mathcal{H}_W := \{\text{supp}(c) : c \in C_W, \text{wt}(c) = 6\}. \quad (56.19)$$

Each support occurs for the word pair $\{c, -c\}$. Indeed, if two weight-six words with the same support were not proportional, one could scale one of them to cancel a coordinate upon subtraction; this would yield a nonzero word of C_W of weight less than six, contrary to the minimum distance. Since neither is there a nonzero word equal to its opposite in characteristic three, every projective orbit has exactly two elements. Therefore,

$$|\mathcal{H}_W| = \frac{264}{2} = 132. \quad (56.20)$$

For $T \in \binom{[12]}{5}$, define

$$\nu(T) := \#\{H \in \mathcal{H}_W : T \subseteq H\}. \quad (56.21)$$

The exact enumeration of the

$$\binom{12}{5} = 792$$

subsets of five points produce

$$\nu(T) = 1 \quad \text{for every } T \in \binom{[12]}{5}. \quad (56.22)$$

Theorem 56.11 (Ternary Witt System). *The incidence system*

$$W_{12} := ([12], \mathcal{H}_W)$$

is the Steiner system

$$\boxed{W_{12} = S(5, 6, 12)}. \quad (56.23)$$

Proof. The equality (56.22) is precisely the Steiner property. The number of blocks is also recovered from the double count of pairs (T, H) with $T \subset H$:

$$|\mathcal{H}_W| \binom{6}{5} = \binom{12}{5}, \quad |\mathcal{H}_W| = \frac{792}{6} = 132.$$

□

56.3.2 Residue of a tetrad and decomposition into 4 incident components

Let $B \in \binom{[12]}{4}$. In an $S(5, 6, 12)$ system, the number of hexads containing B is

$$\lambda_4 = \frac{\binom{12-4}{5-4}}{\binom{6-4}{5-4}} = \frac{8}{2} = 4. \quad (56.24)$$

We write these four hexads as

$$H_i = B \sqcup P_i, \quad |P_i| = 2, \quad i = 1, \dots, 4.$$

Lemma 56.12 (Residual partition of a tetrad). *The four pairs $P_1, \dots, P_4$ are disjoint and*

$$[12] \setminus B = P_1 \sqcup P_2 \sqcup P_3 \sqcup P_4. \quad (56.25)$$

Proof. If two residual pairs shared a point p , the corresponding hexads would contain the same subset $B \cup \{p\}$ of five points, in contradiction with the Steiner uniqueness. The four pairs are, therefore, disjoint; contain eight points and agotan the complement of B . □

Definition 56.13 (Star and Witt star involution). The star centered at B is

$$\mathcal{S}_B := \{H \in \mathcal{H}_W : B \subseteq H\} = \{B \sqcup P_1, \dots, B \sqcup P_4\}. \quad (56.26)$$

If $P_i = \{p_i^-, p_i^+\}$, its involution is

$$\sigma_B := \prod_{i=1}^4 (p_i^- \ p_i^+). \quad (56.27)$$

The permutation σ_B fixes B pointwise and exchanges the ends of each residual petal.

The mathematical term used henceforth is *Witt star* or *Witt-star involution*. It is not a Euclidean star.

56.3.3 Star selected by the tetrad of the seal

The dodecaphasic unit produced, without using Steiner's design,

$$B_K = \{4, 6, 9, 10\}.$$

Now that W_{12} has been constructed, its star is well defined. The four incident hexads are

$$H_{13} = B_K \sqcup \{1, 3\} = \{1, 3, 4, 6, 9, 10\}, \quad (56.28)$$

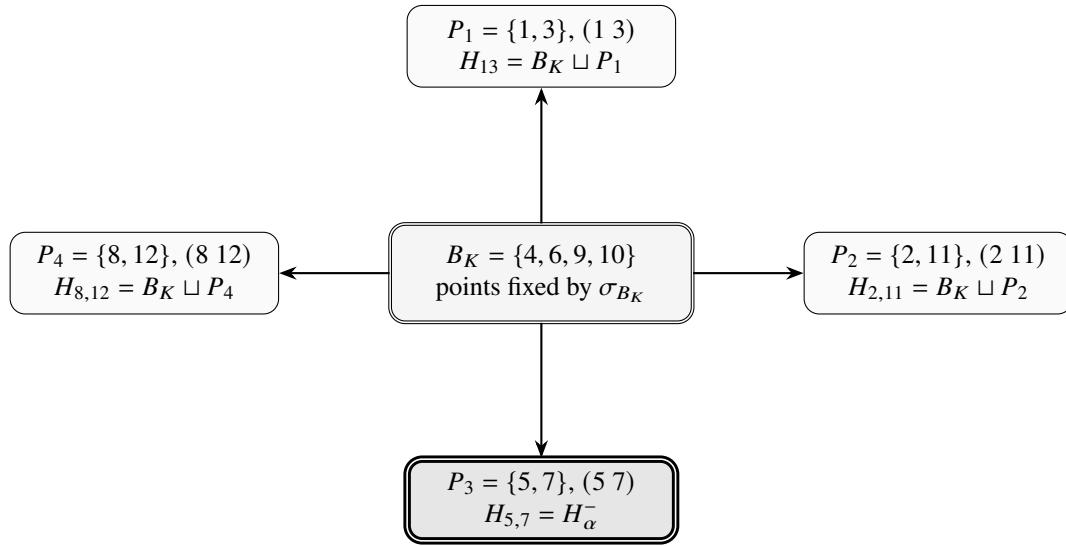
$$H_{2,11} = B_K \sqcup \{2, 11\} = \{2, 4, 6, 9, 10, 11\}, \quad (56.29)$$

$$H_{5,7} = B_K \sqcup \{5, 7\} = \{4, 5, 6, 7, 9, 10\} = H_{\alpha}^-, \quad (56.30)$$

$$H_{8,12} = B_K \sqcup \{8, 12\} = \{4, 6, 8, 9, 10, 12\}. \quad (56.31)$$

Consequently,

$$\sigma_{B_K} = (1\ 3)(2\ 11)(5\ 7)(8\ 12). \quad (56.32)$$



$$\mathcal{S}_{B_K} = \{H_{13}, H_{2,11}, H_{5,7}, H_{8,12}\}.$$

The diagram represents incidence, not geometric distance.

Figure 56.1: Witt star centered at B_K . The center is the fixed tetrad; the four petals are the residual pairs that determine the involution.

56.3.4 Global field of the 495 stars

There exists a star for each tetrad, so

$$\#\{\mathcal{S}_B : B \in \binom{[12]}{4}\} = \binom{12}{4} = 495. \quad (56.33)$$

The exact census verifies

$$\sigma_B(\mathcal{H}_W) = \mathcal{H}_W \quad \text{for every } B \in \binom{[12]}{4}. \quad (56.34)$$

We define

$$\Gamma_{\star} := \langle \sigma_B : B \in \binom{[12]}{4} \rangle \leq \text{Aut}(W_{12}). \quad (56.35)$$

A generating set of six tetrads is

$$1234, \quad 1235, \quad 1236, \quad 1245, \quad 1345, \quad 2345.$$

The orders of the subgroups obtained by adding their involutions successively are

$$1, 2, 6, 18, 360, 7920, 95040. \quad (56.36)$$

Theorem 56.14 (Generation of the Mathieu group M_{12}). *We have*

$$\langle \sigma_B : B \in \binom{[12]}{4} \rangle = \text{Aut}(W_{12}) \cong M_{12}. \quad (56.37)$$

Proof. Each σ_B preserves the 132 hexads, hence $\Gamma_\star \leq \text{Aut}(W_{12})$. The Schreier enumeration of the six indicated generators yields order 95040. The automorphism group of $S(5, 6, 12)$ is M_{12} and has that same order; therefore the inclusion is an equality. $\square$

An isolated star generates only

$$\langle \sigma_B \rangle \cong C_2.$$

The conclusion about M_{12} belongs to the global field of the 495 involutions, not to σ_{B_K} considered in isolation.

56.3.5 Action on the augmentation module

Let

$$V_{11} = \mathbf{1}^\perp \subset \mathbb{R}^{12}.$$

The involution σ_B fixes the four coordinates of B and acts by four transpositions on its complement. In $\mathbb{R}^{12}$ it has eight directions with eigenvalue $+1$ and four with eigenvalue -1 . The constant direction belongs to the first eigenspace. Therefore,

$$\text{Spec}(\sigma_B|_{V_{11}}) = \{1^{[7]}, (-1)^{[4]}\}, \quad \frac{1}{11} \text{Tr}(\sigma_B|_{V_{11}}) = \frac{3}{11}. \quad (56.38)$$

It is a signed involution; it is not a positive projector of rank three.

56.3.6 Obstructions concerning W_{12} , the field of 495 star involutions and the generation of M_{12}

Test criterion 56.15 (Refutation of W_{12} , its stars, and M_{12}). The construction is refuted if any of the following facts occurs:

1. the projectivization of the 264 words of weight six does not produce 132 supports;
2. there exists a subset of five points contained in zero or in more of one hexad;
3. a tetrad does not belong to exactly four hexads;
4. the four residual pairs do not partition the complement of the tetrad;
5. the petals of B_K are not $\{1, 3\}, \{2, 11\}, \{5, 7\}, \{8, 12\}$;
6. any of the 495 involutions fails to preserve $\mathcal{H}_W$;
7. the generated group does not have order 95040;
8. M_{12} is attributed to a single star instead of the global field of involutions;
9. introduces the istrella before of constructing W_{12} .

56.4 Binary extension relative to a dodecad, system W_{24} and group M_{24}

The system W_{12} constructed over twelve phases admits a binary prolongation when an extended Golay code, a dodecad, and an incidence isomorphism are declared. These data are relative inputs of this branch: they are not deduced solely from C_W , and they do not intervene in the ternary lattice branch that will lead to the Leech lattice.

56.4.1 Golay binary code and octads

Let

$$\mathcal{G}_{24} \subset \mathbb{F}_2^{24}$$

the extended binary Golay code. Its weights are

$$0, 8, 12, 16, 24. \quad (56.39)$$

The supports of the weight eight words form the Steiner system

$$W_{24} = S(5, 8, 24). \quad (56.40)$$

Its blocks are called *octads*. Fix a dodecad D , that is, the support of a word of weight twelve, and let $\mathcal{O}_{24}$ be the family of octads.

We define

$$\mathcal{H}(D) := \{O \cap D : O \in \mathcal{O}_{24}, |O \cap D| = 6\}. \quad (56.41)$$

Theorem 56.16 (Residual design of a dodecada). *The residual system is*

$$(D, \mathcal{H}(D)) \cong S(5, 6, 12). \quad (56.42)$$

Moreover, each $H \in \mathcal{H}(D)$ has one unique octad O_H such that

$$O_H \cap D = H. \quad (56.43)$$

Proof. Let $A \subset D$, with $|A| = 5$. In $W_{24} = S(5, 8, 24)$ there is a unique octad O_A containing A . If $j = |O_A \cap D|$, the binary sum of the characteristic words of O_A and D belongs to $\mathcal{G}_{24}$ and has weight

$$8 + 12 - 2j = 20 - 2j.$$

Since $A \subseteq O_A \cap D$, one has $5 \leq j \leq 8$. The four possible weights 10, 8, 6, 4, combined with (56.39), force $j = 6$. Thus every five-point subset of D belongs to a unique residual hexad.

If two octads had the same hexadic intersection with D , both would contain the same five points of that hexad, contradicting uniqueness in $S(5, 8, 24)$. This also proves uniqueness of the lift. $\square$

56.4.2 Relative scale between the HMT design and the dodecagon

To relate the preceding construction to the design obtained from C_W , one must explicitly declare

$$(D, \ell), \quad \ell : W_{12}^{\text{HMT}} \xrightarrow{\sim} (D, \mathcal{H}(D)), \quad (56.44)$$

where ℓ is an incidence isomorphism. If ℓ_* denotes its action on blocks, we define

$$\widehat{\iota}_{D, \ell} := \iota_D \circ \ell_* : W_{12}^{\text{HMT}} \longrightarrow W_{24}, \quad \widehat{\iota}_{D, \ell}(H) = O_{\ell(H)}. \quad (56.45)$$

The composition $\widehat{\iota}_{D, \ell}$ avoids applying ι_D , whose domain is $\mathcal{H}(D)$, directly to a hexad that still lies in $[12]_{\text{HMT}}$.

Proposition 56.17 (Injectivity of the Relative Lift). *The map $\widehat{u}_{D,\ell}$ is injective and preserves the incidence between points and hexads after applying ℓ .*

Proof. The isomorphism ℓ is bijective on blocks. By the theorem 56.16, each residual hexad has a unique lifting octad. Two HMT hexads with the same image would have the same residual hexad and, by bijectivity of ℓ , would be equal. $\square$

There is no absolute choice of (D, ℓ) prior to this gauge. The invariant object is the class of the construction under simultaneous change of dodecad and alignment.

56.4.3 4+1 Decomposition of the incidence of a tetrad into peripheral and axial components

For a tetrad $B \subset D$, the parameters derived from the two Steiner systems are

$$\lambda_4(W_{12}) = \frac{\binom{8}{1}}{\binom{2}{1}} = 4, \quad \lambda_4(W_{24}) = \frac{\binom{20}{1}}{\binom{4}{1}} = 5. \quad (56.46)$$

Therefore, the five octads containing B decompose uniquely as

$$\{O_H : H \in \mathcal{H}(D), B \subset H\} \sqcup \{O_B^\perp\}. \quad (56.47)$$

The four first satisfy $|O_H \cap D| = 6$; the quinta satisfies

$$O_B^\perp \cap D = B. \quad (56.48)$$

For $B = B_K$, the four residual lifts correspond to the petals

$$\{1, 3\}, \quad \{2, 11\}, \quad \{5, 7\}, \quad \{8, 12\},$$

and the fifth octad is the transversal $O_{B_K}^\perp$. This law explains the transition from four hexads to five octads without identifying a face of a cube with a hexad or a vertex with an octad.

56.4.4 Strata Correspondence and Positivity of the Triple Freedom

The five-region structure of channel π has types

$$1_\perp + 1_{--} + 3_{++}.$$

The incidence on B_K has types

$$1_{\text{transversal}} + 1_{H^-} + 3_{\text{restantes}}$$

after marking the negative hexad. There is therefore a natural correspondence between *role* strata:

$$1_\perp \longleftrightarrow O_{B_K}^\perp, \quad 1_{--} \longleftrightarrow \widehat{u}_{D,\ell}(H_\alpha^-), \quad 3_{++} \longleftrightarrow O_{B_K}^{+,3}. \quad (56.49)$$

Here $O_{B_K}^{+,3}$ is the unordered set of the other three residual octads. An individual bijection between the three positive regions and those three octads requires declaring an additional gauge of S_3 . The coincidence of the pattern $1 + 1 + 3$ does not by itself determine that ordering.

56.4.5 Automorphism group of the octad system

Theorem 56.18 (Mathieu group M_{24}). *The automorphism group of the octad Steiner system is*

$$\boxed{\text{Aut}(W_{24}) \cong M_{24}, \quad |M_{24}| = 244823040.} \quad (56.50)$$

The theorem identifies the classical automorphism group of the input binary data. It does not assert that M_{24} is generated from a single dodecad, or that the ternary code C_W determines $\mathcal{G}_{24}$ by itself.

56.4.6 Categorical separation from the lattice branch

The branch constructed in this unit is

$$(W_{12}^{\text{HMT}}; \mathcal{G}_{24}, D, \ell) \longrightarrow W_{24} \longrightarrow \text{Aut}(W_{24}) \cong M_{24}. \quad (56.51)$$

The lattice branch to be constructed subsequently is

$$C_W \longrightarrow N(A_2^{12}) \xrightarrow{\text{index-three neighbor}} \Lambda_{24}. \quad (56.52)$$

Their constructors, domains, and codomains are distinct. In particular,

$$\boxed{\text{no arrow is introduced } W_{24} \longrightarrow \Lambda_{24}.} \quad (56.53)$$

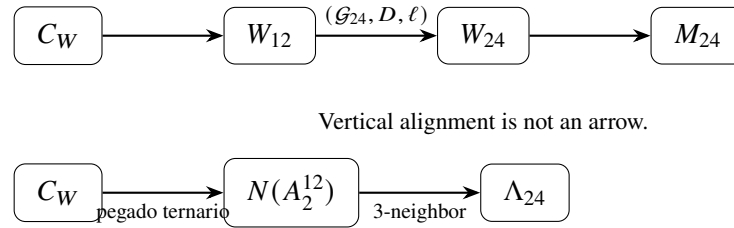


Figure 56.2: Typed bifurcation of the binary and ternary extensions.

56.4.7 Obstructions to the binary prolongation $W_{12} \rightarrow W_{24}$: Golay code, octads, alignment and separation of the Leech branch

Test criterion 56.19 (Refutation of the binary prolongation). The construction is refuted if any of the following facts occurs:

1. the weight spectrum of $\mathcal{G}_{24}$ contains a weight not listed in (56.39);
2. the supports of weight eight not form $S(5, 8, 24)$;
3. a residual intersection of an octad with D that contains five points of D has cardinality other than six;
4. a residual hexad admits either no lifting octad or more than one;
5. ι_D is applied directly to an HMT hexad without the alignment ℓ ;
6. the incidence law on a tetrad is not $4 + 1$;
7. the positive triple is ordered without declaring the gauge S_3 ;
8. $\text{Aut}(W_{24})$ has no order 244823040;
9. it is asserted that W_{24} arises from W_{12} without the binary datum $\mathcal{G}_{24}$, the dodecad, and the alignment;
10. the two branches are serialized by means of an arrow $W_{24} \rightarrow \Lambda_{24}$.

56.5 Ternary discriminant gluing and construction of the Niemeier lattice of type A_2^{12}

This unit begins the lattice branch directly from the self-dual ternary code C_W . The binary system W_{24} , the dodecad, and the group M_{24} are not inputs of this construction. The typed sequence is

$$C_W \longrightarrow (A_2^*/A_2)^{12} \longrightarrow N(C_W) \cong N(A_2^{12}).$$

56.5.1 Reticulo root A_2

Let

$$A_2 = \mathbb{Z}\alpha_1 \oplus \mathbb{Z}\alpha_2$$

the root lattice whose Gram matrix, in the ordered basis (α_1, α_2) , is

$$G_{A_2} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}. \quad (56.54)$$

For $x = x_1\alpha_1 + x_2\alpha_2$,

$$\langle x, x \rangle = 2(x_1^2 - x_1x_2 + x_2^2). \quad (56.55)$$

In particular, A_2 is even, positive defined and

$$\det A_2 = 3.$$

We define

$$\rho := \alpha_1 + \alpha_2.$$

Then

$$\rho^2 = 2, \quad \langle \rho, \alpha_1 \rangle = \langle \rho, \alpha_2 \rangle = 1. \quad (56.56)$$

56.5.2 Decomposition of the incidence of a tetrad into 4 peripheral components and 1 axial component

The dual lattice is

$$A_2^* := \{x \in A_2 \otimes_{\mathbb{Z}} \mathbb{Q} : \langle x, A_2 \rangle \subseteq \mathbb{Z}\}.$$

We choose the representatives

$$\mu_0 = 0, \quad \mu_1 = \frac{2\alpha_1 + \alpha_2}{3}, \quad \mu_2 = \frac{\alpha_1 + 2\alpha_2}{3}. \quad (56.57)$$

Their pairings with the simple base are

$$\begin{array}{c|cc} & \alpha_1 & \alpha_2 \\ \hline \mu_1 & 1 & 0 \\ \mu_2 & 0 & 1 \end{array},$$

thereby $\mu_1, \mu_2 \in A_2^*$. Furthermore,

$$3\mu_1 = 2\alpha_1 + \alpha_2 \in A_2, \quad 3\mu_2 = \alpha_1 + 2\alpha_2 \in A_2,$$

while $\mu_1, \mu_2 \notin A_2$. It is obtained

$$A_2^*/A_2 = \{\bar{\mu}_0, \bar{\mu}_1, \bar{\mu}_2\} \cong \mathbb{Z}/3\mathbb{Z} \cong \mathbb{F}_3. \quad (56.58)$$

The sum of representatives is explicitly reduced by means of

$$\mu_1 + \mu_1 - \mu_2 = \alpha_1, \quad \mu_1 + \mu_2 = \rho, \quad \mu_2 + \mu_2 - \mu_1 = \alpha_2.$$

Consequently,

$$\mu_r + \mu_s - \mu_{r+s \pmod{3}} \in A_2 \quad (r, s \in \mathbb{F}_3). \quad (56.59)$$

Proposition 56.20 (Discriminant Form of A_2). *Under (56.58), the discriminant bilinear and quadratic forms are*

$$b(\bar{\mu}_r, \bar{\mu}_s) \equiv \frac{2rs}{3} \pmod{\mathbb{Z}}, \quad (56.60)$$

$$q(\bar{\mu}_r) \equiv \frac{2r^2}{3} \pmod{2\mathbb{Z}}. \quad (56.61)$$

Proof. The formula (56.55) gives

$$\mu_1^2 = \mu_2^2 = \frac{2}{3}, \quad \langle \mu_1, \mu_2 \rangle = \frac{1}{3}.$$

For $r, s \in \{0, 1, 2\}$, these values coincide, modulo $\mathbb{Z}$, with $2rs/3$. The quadratic formula results by taking $r = s$ and recalling that the lattice is even. $\square$

56.5.3 Correspondence between incidence strata and free orbit of the positive triple

Let

$$L_0 := A_2^{12}.$$

Then

$$L_0^* = (A_2^*)^{12}, \quad L_0^*/L_0 \cong \mathbb{F}_3^{12}.$$

For $c = (c_1, \dots, c_{12}) \in \mathbb{F}_3^{12}$, define

$$\phi(c) := (\mu_{c_1}, \dots, \mu_{c_{12}}) \in L_0^*. \quad (56.62)$$

The extension determined by C_W is

$$\begin{aligned} N(C_W) &:= \bigcup_{c \in C_W} (L_0 + \phi(c)) \\ &= \{x \in L_0^* : x \bmod L_0 \in C_W\}. \end{aligned} \quad (56.63)$$

Lemma 56.21 (Additive closure of the gluing). *The set $N(C_W)$ is a discrete subgroup of L_0^* of rank twenty-four.*

Proof. Let $x = \ell + \phi(c)$ and $y = m + \phi(d)$, with $\ell, m \in L_0$ and $c, d \in C_W$. By (56.59),

$$\phi(c) + \phi(d) - \phi(c + d) \in L_0.$$

Since C_W is linear, $c + d \in C_W$, and then $x + y \in L_0 + \phi(c + d)$. The same identity applied to the opposite gives inverses. The inclusions

$$L_0 \subseteq N(C_W) \subseteq L_0^*$$

They test discretion and range twenty-four. $\square$

56.5.4 Integrality induced by self-duality

The discriminant bilinear form of L_0 is the orthogonal sum of twelve copies of (56.60). For $c, d \in \mathbb{F}_3^{12}$,

$$b(c, d) \equiv \frac{2}{3} \sum_{i=1}^{12} c_i d_i \pmod{\mathbb{Z}}. \quad (56.64)$$

Lemma 56.22 (Integrity of the gluing). *The lattice $N(C_W)$ is integral.*

Proof. The self-duality $C_W = C_W^\perp$ implica

$$\sum_{i=1}^{12} c_i d_i \equiv 0 \pmod{3} \quad (c, d \in C_W).$$

By (56.64), the discriminant form vanishes on $C_W \times C_W$. Therefore, the pairing of any two glued classes is integral. $\square$

56.5.5 Parity Induced by the Code Weights

At a nonzero coordinate of the code, the discriminant quadratic form equals $2/3$ modulo $2\mathbb{Z}$. Therefore,

$$q(c) \equiv \frac{2}{3} \text{wt}(c) \pmod{2\mathbb{Z}}. \quad (56.65)$$

Lemma 56.23 (Parity of the gluing). *The lattice $N(C_W)$ is even.*

Proof. The only weights of C_W are 0, 6, 9, 12, and

$$\frac{2}{3} \{0, 6, 9, 12\} = \{0, 4, 6, 8\} \subseteq 2\mathbb{Z}.$$

Every glue class is isotropic for the discriminant quadratic form, so all norms of $N(C_W)$ are even. $\square$

Let $g_1, \dots, g_6$ be the rows of $G_W = (I_6 \mid A_W)$ and $\gamma_i = \phi(g_i)$. Exact multiplication in A_2^* gives

$$(\langle \gamma_i, \gamma_j \rangle)_{i,j=1}^6 = \begin{pmatrix} 4 & 2 & 2 & 2 & 2 & 2 \\ 2 & 4 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 2 & 2 & 2 \\ 2 & 2 & 2 & 4 & 2 & 2 \\ 2 & 2 & 2 & 2 & 4 & 2 \\ 2 & 2 & 2 & 2 & 2 & 4 \end{pmatrix}. \quad (56.66)$$

This matrix provides an additional direct control of integrality and parity for the six gluing generators.

56.5.6 Index, determinant, and unimodularity

The cosets of L_0 that appear in (56.63) are indexed by the $|C_W| = 729 = 3^6$ elements of the code and are distinct. Consequently,

$$[N(C_W) : L_0] = 3^6. \quad (56.67)$$

Since

$$\det L_0 = (\det A_2)^{12} = 3^{12},$$

the determinant formula for a finite extension gives

$$\det N(C_W) = \frac{\det L_0}{[N(C_W) : L_0]^2} = \frac{3^{12}}{(3^6)^2} = 1. \quad (56.68)$$

56.5.7 Root system induced by Witt incidence and classification of its orbits

En each one of the two classes discriminantes not nulas of A_2^*/A_2 , the norm minima is

$$\mu_1^2 = \mu_2^2 = \frac{2}{3}.$$

If $c \in C_W \setminus \{0\}$, then $\text{wt}(c) \geq 6$. By orthogonality of the twelve factors, any vector of $L_0 + \phi(c)$ has norm at least

$$6 \cdot \frac{2}{3} = 4. \quad (56.69)$$

Therefore, no nontrivial gluing class contains vectors of norm two. The roots of $N(C_W)$ are exactly the roots already present in $L_0 = A_2^{12}$.

Theorem 56.24 (Realization of the Niemeier of type A_2^{12}). *The lattice $N(C_W)$ is positive definite, even, and unimodular, and its root system is A_2^{12} . Consequently,*

$$\boxed{N(C_W) \cong N(A_2^{12})}. \quad (56.70)$$

Proof. Positivity follows from L_0^* . The the corresponding lemma proves integrality and parity; (56.68) proves unimodularity. The bound (56.69) identifies the root system. The Niemeier classification uniquely associates the indicated lattice with that root system. $\square$

56.5.8 Obstructions to the ternary gluing $C_W \rightarrow N(A_2^{12})$: parity, unimodularity, and root system

Test criterion 56.25 (Refutation of ternary gluing). The construction is refuted if any of the following facts occurs:

1. any of the representatives μ_0, μ_1, μ_2 does not belong to A_2^* , two of its classes coincide, or (56.59) fails;
2. the discriminant form is not the one given in proposition 56.20;
3. $N(C_W)$ is not closed under sum or inversos;
4. there exists $c, d \in C_W$ with $\frac{2}{3} \sum_i c_i d_i \notin \mathbb{Z}$;
5. one palabra of C_W produces $\frac{2}{3} \text{wt}(c) \notin 2\mathbb{Z}$;
6. the index is not 3^6 or the determinant is not one;
7. a nontrivial gluing class contains a vector of norm two;
8. W_{24} , a dodecad, or a binary datum is used in any of the preceding gluing operations.

56.6 Incidental origin, typed radial problem and construction of the rootless unimodular neighbor

The lattice $N(A_2^{12})$ has already been constructed. This unit determines a base point from an ordered incidence frame, solves a variational problem on a declared domain, verifies the three conditions for the index-three neighbor, and proves that the result is the Leech lattice. The neighbor vector is denoted $v_{\mathcal{F}_*}$: it depends on the incidence frame and does not enter the numerical carry of α .

56.6.1 Ordered set of incidences

Consider the four hexads

$$H^- = \{4, 5, 6, 7, 9, 10\}, \quad (56.71)$$

$$H_e = \{1, 3, 4, 6, 9, 10\}, \quad (56.72)$$

$$H_\varphi = \{1, 2, 3, 4, 7, 9\}, \quad (56.73)$$

$$H_{\text{APP}} = \{2, 3, 5, 10, 11, 12\}. \quad (56.74)$$

The datum is the ordered framework.

$$\mathcal{F} := (H^-, H_e, H_\varphi, H_{\text{APP}}). \quad (56.75)$$

We define

$$o(\mathcal{F}) := (H_e \cap H_\varphi) \setminus (H^- \cup H_{\text{APP}}). \quad (56.76)$$

Theorem 56.26 (Equivariant Reconstruction of the Origin). *For the frame canonical $\mathcal{F}_*$,*

$$\boxed{o(\mathcal{F}_*) = \{1\}}. \quad (56.77)$$

Moreover, for every simultaneous permutation $g \in M_{12}$,

$$o(g\mathcal{F}) = g o(\mathcal{F}). \quad (56.78)$$

Proof. We have

$$H_e \cap H_\varphi = \{1, 3, 4, 9\}$$

and

$$(H^- \cup H_{\text{APP}}) \cap \{1, 3, 4, 9\} = \{3, 4, 9\}.$$

The difference is $\{1\}$. For equivariance one applies the identities

$$g(A \cap B) = gA \cap gB, \quad g(A \cup B) = gA \cup gB, \quad g(A \setminus B) = gA \setminus gB,$$

valid for all bijections. □

The tetrad $B_K = \{4, 6, 9, 10\}$ does not by itself determine the origin. The singleton is obtained from the complete ordered frame; it is not a coordinate chosen after the result has been observed.

56.6.2 Positive radial chamber of the root system and selection of the dominant cone

Let ρ_i the copia of $\rho = \alpha_1 + \alpha_2$ in the i -esimo factor of A_2^{12} . Consideramos the dominio

$$\mathcal{R}_+ := \left\{ v(a) = \sum_{i=1}^{12} a_i \rho_i : a_i = 1 + 3b_i, b_i \in \mathbb{Z}_{\geq 0} \right\}. \quad (56.79)$$

Since the factors are orthogonal and $\rho_i^2 = 2$, if $S = \sum_i b_i$, then

$$\begin{aligned} v(a)^2 &= 2 \sum_{i=1}^{12} (1 + 3b_i)^2 \\ &= 24 + 12S + 18 \sum_{i=1}^{12} b_i^2. \end{aligned} \quad (56.80)$$

Reducing modulo eighteen,

$$v(a)^2 \equiv 6 + 12S \pmod{18},$$

and therefore

$$v(a)^2 \equiv 0 \pmod{18} \iff S \equiv 1 \pmod{3}. \quad (56.81)$$

Proposition 56.27 (Minima in the declared radial chamber). *On the subset of $\mathcal{R}_+$ defined by $v(a)^2 \equiv 0 \pmod{18}$, the minimum norm is 54. It is attained exactly at twelve vectors: for each $j \in \{1, \dots, 12\}$, the coefficient a_j equals four, and the remaining eleven equal one.*

Proof. Condition (56.81) and the nonnegativity of the b_i imply that the smallest possible value of S is one. This forces a unique index j with $b_j = 1$ and $b_i = 0$ for $i \neq j$. Substituting into (56.80),

$$v(a)^2 = 24 + 12 + 18 = 54.$$

There are twelve choices of j . The next admissible value of S is four and produces a strictly greater norm. $\square$

The origin $o(\mathcal{F}_*) = \{1\}$ selects

$$\boxed{v_{\mathcal{F}_*} = 4\rho_1 + \rho_2 + \dots + \rho_{12}, \quad v_{\mathcal{F}_*}^2 = 54.} \quad (56.82)$$

Remark 56.28 (Exact scope of minimality). The norm 54 is minimal in the positive radial chamber (56.79), not in the entire residue class. The vector

$$u = (\alpha_1 - 2\alpha_2, \rho, \dots, \rho)$$

satisfies

$$\alpha_1 - 2\alpha_2 - \rho = -3\alpha_2,$$

so that

$$u \equiv (\rho, \dots, \rho) \equiv v_{\mathcal{F}_*} \pmod{3A_2^{12}}.$$

However,

$$(\alpha_1 - 2\alpha_2)^2 = 2(1 + 2 + 4) = 14, \quad u^2 = 14 + 11 \cdot 2 = 36.$$

This witness prevents extending the minimality assertion outside the domain of [proposition 56.27](#).

56.6.3 Discriminant gluing of 12 A_2 factors and construction of the integral extension

We write

$$N := N(A_2^{12}), \quad v := v_{\mathcal{F}_*}.$$

We define

$$\chi_v : N \longrightarrow \mathbb{Z}/3\mathbb{Z}, \quad \chi_v(x) = \langle x, v \rangle \pmod{3}, \quad (56.83)$$

its nucleus

$$N_0 := \ker \chi_v = \{x \in N : \langle x, v \rangle \equiv 0 \pmod{3}\}, \quad (56.84)$$

and the extension

$$N_v := N_0 + \mathbb{Z} \frac{v}{3}. \quad (56.85)$$

56.6.4 Condition I1: surjectivity and exclusion of roots

By construction, $v \in A_2^{12} \subseteq N$. The roots of a factor A_2 are

$$\pm\alpha_1, \quad \pm\alpha_2, \quad \pm\rho.$$

Their pairings with ρ are ± 1 or ± 2 . All the coefficients of v are congruent to one modulo three:

$$4 \equiv 1 \pmod{3}, \quad 1 \equiv 1 \pmod{3}.$$

Therefore, no root of A_2^{12} has zero pairing with v modulo three. In particular,

$$\chi_v(\alpha_{1,1}) = 1,$$

so that χ_v is surjective and

$$[N : N_0] = 3. \tag{56.86}$$

Moreover, no root of N belongs to N_0 .

56.6.5 Condition I2: norm divisibility

The identity (56.82) gives

$$v^2 = 54 \equiv 0 \pmod{18}, \quad \left(\frac{v}{3}\right)^2 = 6. \tag{56.87}$$

In particular, $\langle v, v \rangle \equiv 0 \pmod{3}$, hence $v \in N_0$.

56.6.6 Condition I3: class dual of order 3

If $x \in N_0$, there exists $k \in \mathbb{Z}$ with $\langle x, v \rangle = 3k$. Therefore,

$$\left\langle x, \frac{v}{3} \right\rangle = k \in \mathbb{Z},$$

and

$$\frac{v}{3} \in N_0^*. \tag{56.88}$$

On the other hand,

$$\left\langle \frac{v}{3}, \alpha_{1,1} \right\rangle = \frac{4}{3}.$$

Since $\alpha_{1,1} \in N$ and N is integral, this proves

$$\frac{v}{3} \notin N. \tag{56.89}$$

Finally, $3(v/3) = v \in N_0$. The class of $v/3$ in N_0^*/N_0 is nontrivial and has order exactly three. Consequently,

$$[N_v : N_0] = 3. \tag{56.90}$$

56.6.7 Parity, integrality and determinant of the neighbor

Proposition 56.29 (Unimodular structure of the neighbor). *The lattice N_v is integral, even, and unimodular.*

Proof. Since N is unimodular and $[N : N_0] = 3$,

$$\det N_0 = [N : N_0]^2 \det N = 9.$$

Since $[N_v : N_0] = 3$,

$$\det N_v = \frac{\det N_0}{[N_v : N_0]^2} = 1.$$

Let $y = x + m(v/3)$, with $x \in N_0$, and write $\langle x, v \rangle = 3k$. Then

$$\begin{aligned} y^2 &= x^2 + 2m \left\langle x, \frac{v}{3} \right\rangle + m^2 \left(\frac{v}{3} \right)^2 \\ &= x^2 + 2mk + 6m^2. \end{aligned}$$

The three summands form an even integer. The polarized form proves integrality, and determinant one proves unimodularity. $\square$

56.6.8 2 clasis globalis and 6 clasis local

Globally, (56.85) adds only two nontrivial cosets:

$$N_0 + \frac{v}{3}, \quad N_0 - \frac{v}{3}.$$

Locally, in each factor A_2 , a coordinate of those classes belongs to one of the six classes

$$\mathcal{C}_{j,\pm} := A_2 + \mu_j \pm \frac{\rho}{3}, \quad j \in \{0, 1, 2\}. \quad (56.91)$$

In the first coordinate, $4\rho/3 \equiv \rho/3 \pmod{A_2}$, so the same list holds for the twelve factors.

Table 56.1: Minimal representatives of the six local cosets.

class	minimal representative	(u, v)	$2(u^2 - uv + v^2)/9$
$\mathcal{C}_{0,-}$	$-\rho/3$	$(-1, -1)$	$2/9$
$\mathcal{C}_{0,+}$	$\rho/3$	$(1, 1)$	$2/9$
$\mathcal{C}_{1,-}$	$\mu_1 - \rho/3$	$(1, 0)$	$2/9$
$\mathcal{C}_{1,+}$	$\mu_1 + \rho/3 - \rho$	$(0, -1)$	$2/9$
$\mathcal{C}_{2,-}$	$\mu_2 - \rho/3$	$(0, 1)$	$2/9$
$\mathcal{C}_{2,+}$	$\mu_2 + \rho/3 - \rho$	$(-1, 0)$	$2/9$

Here (u, v) means the representative $(u\alpha_1 + v\alpha_2)/3$.

Lemma 56.30 (Exact minimum of the six classes). *Each class in (56.91) has minimum norm exactly $2/9$.*

Proof. The norm of $(u\alpha_1 + v\alpha_2)/3$ is

$$\frac{2}{9}(u^2 - uv + v^2).$$

The integral form $u^2 - uv + v^2$ is positive definite. On a nonzero residual class it does not take the value zero, and hence is at least one. The six representatives in table 56.1 attain one. $\square$

56.6.9 Criterion for the absence of roots in the Leech lattice and consequence for the minimum distance

Theorem 56.31 (Absence of vectors of norm two). *The lattice N_v contains no vectors of norm two.*

Proof. All roots of N are those of A_2^{12} . Condition I1 proves that none belongs to N_0 . Hence class N_0 contains no vectors of norm two.

In a new class $N_0 \pm v/3$, each of the twelve coordinates belongs to one of the six local classes. By lemma 56.30, each coordinate contributes at least $2/9$. Orthogonality gives

$$\left\| x \pm \frac{v}{3} \right\|^2 \geq 12 \cdot \frac{2}{9} = \frac{8}{3} > 2.$$

Nor do the new classes contain roots. $\square$

56.6.10 Identification with the Leech lattice

Theorem 56.32 (Construction of the Leech lattice). *The lattice*

$$N_v = \ker(\langle \cdot, v_{\mathcal{F}_*} \rangle \bmod 3) + \mathbb{Z} \frac{v_{\mathcal{F}_*}}{3} \quad (56.92)$$

is isometric to the Leech lattice:

$$\boxed{N_v \cong \Lambda_{24}.} \quad (56.93)$$

Proof. The lattice N_v has rank twenty-four and is positive definite. [proposition 56.29](#) proves that it is even and unimodular; [theorem 56.31](#) proves that it has no roots. By the Niemeier classification, there exists a unique isometry class of positive-definite, even, unimodular lattices of rank twenty-four without roots: the Leech lattice. $\square$

56.6.11 The 243 syndromes of the ternary punctured code

Let G_{11} be the ternary Golay code with parameters $[11, 6, 5]_3$. Its syndrome space has cardinality

$$3^{11-6} = 3^5 = 243. \quad (56.94)$$

The number of weight errors of at most two is

$$1 + 2 \binom{11}{1} + 2^2 \binom{11}{2} = 1 + 22 + 220 = 243. \quad (56.95)$$

Proposition 56.33 (Perfect correspondence error–syndrome). *Each syndrome of G_{11} has a unique error representative of weight at most two.*

Proof. Minimum distance five implies that two distinct errors of weight at most two cannot differ by a codeword: their difference would have weight at most four. The error–syndrome map is therefore injective on the ball of radius two. Both sets have cardinality 243 by (56.94) and (56.95); the map is bijective. $\square$

These 243 syndromes are not the 243 visible words obtained from the TPK catalogue. They share a cardinal, but they possess distinct domains, equivalences and functions.

56.6.12 Obstructions of the ternary neighbor $N_v \cong \Lambda_{24}$: class of order 3, absence of roots, and separation of the cardinals 243

Test criterion 56.34 (Refutation of the neighbor and of the Leech identification). The construction is refuted if any of the following facts occurs:

1. the operator $o(\mathcal{F})$ not produces $\{1\}$ or is not equivariant;
2. there exists in the radial chamber an admissible vector of norm less than 54, or the number of minimizers is not twelve;
3. global minimality is asserted despite the witness of norm 36;
4. χ_v is not sobreyectivo or one root belongs to N_0 ;
5. $v^2 \not\equiv 0 \pmod{18}$ o $(v/3)^2 \neq 6$;
6. $v/3 \notin N_0^*$, $v/3 \in N$, or its class does not have order three;
7. the determinant of N_v is not one, or the neighbor ceases to be even;
8. some of the six clasis local has minimum least that $2/9$;

9. one of the two new global classes contains a root;
10. the two global classes are confused with the six local classes;
11. W_{24} is used as an antecedent of the gluing or of the neighbor;
12. the 243 syndromes are identified with the 243 words visible by sharing cardinality.

56.7 Finite holonomy, Schläfli configuration and symmetry E_6

The Witt branch does not exhaust the exceptional structures accessible from APP. The parallel branch of this chapter starts from the two mixed sum–product and product–sum polarizations, together with a plaquette orientation and a declared central section. The calculated curvature determines the alternating form; this defines the central extension. An extension is not inferred from cardinality 27 alone.

56.7.1 Ordered system of incidences and radial camera orientation

On $T_9 = (\mathbb{Z}/9\mathbb{Z})^2$, with $S(x, y) = x + y$ and $P(x, y) = xy$, let

$$\Delta_x f = f(x + 1, y) - f(x, y), \quad \Delta_y f = f(x, y + 1) - f(x, y).$$

Mixed connections are

$$A^{SP} = (\Delta_x S, \Delta_y P) = (1, x), \quad A^{PS} = (\Delta_x P, \Delta_y S) = (y, 1).$$

Its discrete curvature is

$$F_A = \Delta_x A_y - \Delta_y A_x.$$

Proposition 56.35 (Oriented opposite curvatures).

$$F_{A^{SP}} = 1, \quad F_{A^{PS}} = -1 \pmod{9}.$$

The circulation over an oriented rectangle of sides r, s is, respectively, rs and $-rs$.

Proof. $\Delta_x x - \Delta_y 1 = 1$ and $\Delta_x 1 - \Delta_y y = -1$. A rectangle contains rs plaquettes; upon summing, the interior edges cancel and the boundary circulation remains. $\square$

For desplazamientos $u = (a, b)$, $v = (c, d)$, the class alternante is

$$\sigma(u, v) = ad - bc \pmod{9}. \quad (56.96)$$

The orientation fixes the sign of σ . The central section used below belongs to the gauge of this realization; replacing it with a coboundary produces an isomorphic extension, not a new invariant.

56.7.2 Central extension and ternary reduction

We define the nonadic Heisenberg group

$$\mathcal{H}_9 = \{(a, b, z) : a, b, z \in \mathbb{Z}/9\mathbb{Z}\}, \quad (a, b, z)(c, d, w) = (a + c, b + d, z + w + bc). \quad (56.97)$$

The commutator of two horizontal shifts is

$$[(a, b, 0), (c, d, 0)] = (0, 0, bc - ad),$$

which records $-\sigma(u, v)$. Reducing each coordinate modulo three gives

$$H_3 = \{(a, b, z) : a, b, z \in \mathbb{F}_3\}, \quad |H_3| = 27, \quad (56.98)$$

with the same law. The nucleo of $\mathcal{H}_9 \rightarrow H_3$ is $(3\mathbb{Z}/9\mathbb{Z})^3$, of cardinal 27.

56.7.3 Construction of the radial character and determination of its congruence kernel

The nonadic extension admits an explicit unitary realization. Let

$$\zeta_9 := \exp(2\pi i/9), \quad \mathcal{H}_9^{\text{Sch}} := \mathbb{C}^9,$$

based on $\{|n\rangle : n \in \mathbb{Z}/9\mathbb{Z}\}$. We define

$$X|n\rangle := |n+1\rangle, \quad Z|n\rangle := \zeta_9^n |n\rangle. \quad (56.99)$$

We have

$$ZX = \zeta_9 XZ$$

and, by multiplication,

$$(X^a Z^b)(X^c Z^d) = \zeta_9^{bc} X^{a+c} Z^{b+d}. \quad (56.100)$$

Theorem 56.36 (Faithful Weyl–Heisenberg Representation). *The map*

$$\rho_{\text{WH}} : \mathcal{H}_9 \longrightarrow U(\mathcal{H}_9^{\text{Sch}}), \quad \rho_{\text{WH}}(a, b, z) := \zeta_9^z X^a Z^b, \quad (56.101)$$

is a faithful unitary representation. For $u = (a, b)$ and $v = (c, d)$, its commutator satisfies

$$[\rho_{\text{WH}}(u, 0), \rho_{\text{WH}}(v, 0)] = \zeta_9^{bc-ad} I = \zeta_9^{-\sigma(u,v)} I. \quad (56.102)$$

Proof. The identity (56.100) reproduces the term bc of (56.97); hence the homomorphism property. If $\zeta_9^z X^a Z^b = I$, the action on the basis successively forces $a = 0$, $b = 0$, and $z = 0$. This proves faithfulness. Comparing the products in the two orders gives ζ_9^{bc-ad} , which is the alternating phase of (56.96). $\square$

The representation distinguishes two conjugate observables. Let

$$Q_{\text{H}} := \text{diag}(0, 1, \dots, 8), \quad F_{jk} := \frac{1}{3} \zeta_9^{jk}, \quad P_{\text{H}} := F^\dagger Q_{\text{H}} F.$$

The eigenbases of Q_{H} and P_{H} are mutually unbiased, since $|F_{jk}|^2 = 1/9$. For every unit vector $\psi \in \mathcal{H}_9^{\text{Sch}}$,

$$\text{Var}_\psi(Q_{\text{H}}) \text{Var}_\psi(P_{\text{H}}) \geq \frac{1}{4} |\langle \psi, [Q_{\text{H}}, P_{\text{H}}] \psi \rangle|^2, \quad (56.103)$$

$$H_{Q_{\text{H}}}(\psi) + H_{P_{\text{H}}}(\psi) \geq \log 9. \quad (56.104)$$

These operators are dimensionless. A physical unit chart is a later realization and does not intervene in the central extension.

56.7.4 Action of 10 desplazamientos over the grafo of Cayley

For a linear functional $\ell(a, b) = pa + qb$, set

$$D_\ell(a, b) = \left(a, b, \frac{ab}{2} + \ell(a, b) \right), \quad Z = (0, 0, 1),$$

where $2^{-1} = 2$ in $\mathbb{F}_3$. The inverse-closed set

$$S_\ell = \{D_\ell(v) : v \in \mathbb{F}_3^2 \setminus \{0\}\} \cup \{Z, Z^{-1}\} \quad (56.105)$$

has ten elements and does not contain the identity.

Theorem 56.37 (Independence of the linear scale). *The nine functionals ℓ produce isomorphic Cayley graphs. The nine sets S_ℓ form a single orbit under $\text{Aut}(H_3)$.*

Proof. The map

$$\phi_\ell(a, b, z) = (a, b, z + \ell(a, b))$$

is a central automorphism and satisfies $\phi_\ell(S_0) = S_\ell$. To write the complete family in the polarized coordinates of equation (56.97), let $c(v, w) = bc$ for $v = (a, b)$, $w = (c, d)$. Given $M \in \text{GL}(2, 3)$, the defect

$$B_M(v, w) = c(Mv, Mw) - \det(M)c(v, w)$$

is symmetric, because the alternating parts of the two terms coincide. We define the quadratic correction

$$q_M(v) = 2B_M(v, v), \quad q_M(v + w) - q_M(v) - q_M(w) = B_M(v, w), \quad (56.106)$$

where $2 = 2^{-1}$ in $\mathbb{F}_3$. The complete family of automorphisms is then written

$$(v, t) \mapsto (Mv, \det(M)t + q_M(v) + \ell(v)), \quad M \in \text{GL}(2, 3), \ell \in (\mathbb{F}_3^2)^*. \quad (56.107)$$

The polar identity of equation (56.106) directly proves that the central product is preserved. The family has $48 \cdot 9 = 432$ elements. Exhaustive enumeration of the $\binom{13}{5} = 1287$ inverse-closed subsets of cardinality ten finds exactly those nine with the parameters derived below. $\square$

Write $\underline{S} = \sum_{s \in S} s$ in $\mathbb{Z}[H_3]$. Counting ordered products gives

$$\underline{S}^2 = 10e + \underline{S} + 5(\underline{H}_3 - e - \underline{S}) = 5\underline{H}_3 + 5e - 4\underline{S}. \quad (56.108)$$

Indeed, the identity receives ten factorizations; an element of S receives one; each of the other sixteen receives five.

Theorem 56.38 (Graph of the twenty-seven lines). *The matrix of adjacency A_ℓ of $\Gamma_\ell = \text{Cay}(H_3, S_\ell)$ satisfies*

$$A_\ell^2 = 5I - 4A_\ell + 5J. \quad (56.109)$$

Therefore,

$$\Gamma_\ell = \text{srg}(27, 10, 1, 5), \quad \text{Spec}(A_\ell) = \{10^1, 1^{20}, (-5)^6\}.$$

It is isomorphic to the intersection graph of the twenty-seven lines of a smooth cubic surface.

Proof. The regular representation of equation (56.108) produces equation (56.109). Its entries state that every vertex has ten neighbors, that an adjacent pair has one common neighbor, and a nonadjacent pair has five. On the constant line, A_ℓ acts by 10; on its complement, $J = 0$ and $(A_\ell - I)(A_\ell + 5I) = 0$. Zero trace fixes the multiplicities 20 and 6.

In the model of a cubic obtained by blowing up six points of $\mathbb{P}^2$, the 27 classes are

$$E_i, \quad L_{ij} = H - E_i - E_j, \quad Q_i = 2H - \sum_{j \neq i} E_j.$$

With $H^2 = 1$, $E_i^2 = -1$, $H \cdot E_i = 0$, and $E_i \cdot E_j = 0$ for $i \neq j$, its intersection matrix has the same parameters. Identification with the classical configuration uses the uniqueness theorem for this realization of the 27 lines [Man05]; it is not attributed to a nonexistent certificate. $\square$

The isomorphism assertion can be made fully effective by means of a computed bijection. The labeling is not absolute: another linear section or an automorphism of the surface produces another table in the same class.

Table 56.2: Explicit bijection between H_3 and the twenty-seven classes of lines.

$a\ b\ z$	class of line
0 0 0	E_1
0 0 1	L_{12}
0 0 2	Q_2
0 1 0	L_{13}
0 1 1	Q_1
0 1 2	E_3
0 2 0	Q_3
0 2 1	E_2
0 2 2	L_{23}
1 0 0	L_{14}
1 0 1	L_{35}
1 0 2	L_{26}
1 1 0	L_{36}
1 1 1	L_{24}
1 1 2	L_{15}
1 2 0	L_{25}
1 2 1	L_{16}
1 2 2	L_{34}
2 0 0	Q_4
2 0 1	L_{46}
2 0 2	E_6
2 1 0	E_5
2 1 1	Q_6
2 1 2	L_{56}
2 2 0	L_{45}
2 2 1	E_4
2 2 2	Q_5

As a local control, the ten neighbors of $e = (0, 0, 0)$ are the elements of S_0 . The table transforms them into the ten classes that intersect E_1 : the five L_{1j} and the five Q_j , with $2 \leq j \leq 6$. The complete verification compares all 729 entries of both adjacency matrices.

The complement $B_\ell = J - I - A_\ell$ satisfies

$$B_\ell^2 = 8I + 2B_\ell + 8J, \quad \text{srg}(27, 16, 10, 8), \quad \text{Spec}(B_\ell) = \{16^1, 4^6, (-2)^{20}\}. \quad (56.110)$$

This is the Schläfli graph in the convention of complementary adjacency; A_ℓ is the intersection graph with parameters $(27, 10, 1, 5)$.

Each edge belongs to a unique triangle, since $\lambda = 1$. The number of triangles is

$$\frac{27 \cdot 10}{6} = 45.$$

The full automorphism group has order

$$27 \cdot 1920 = 51840$$

and, through its action on the line configuration, is identified with $W(E_6)$, not merely by equality of orders [Man05].

56.7.5 The root lattice realizing E_6

To make explicit the object on which that group acts, let

$$\text{Pic}(X) = \mathbb{Z}H \oplus \bigoplus_{i=1}^6 \mathbb{Z}E_i, \quad H^2 = 1, \quad E_i^2 = -1, \quad H \cdot E_i = E_i \cdot E_j = 0 \ (i \neq j), \quad (56.111)$$

the Picard lattice of the blow-up of six general points of $\mathbb{P}^2$. Its canonical class is

$$K_X = -3H + E_1 + \cdots + E_6.$$

The orthogonal complement

$$L_{E_6} = K_X^\perp \subset \text{Pic}(X), \quad (x, y)_{E_6} := -x \cdot y, \quad (56.112)$$

is positive definite of rank six. A simple root basis is

$$\begin{aligned} \alpha_1 &= E_1 - E_2, & \alpha_2 &= E_2 - E_3, & \alpha_3 &= E_3 - E_4, \\ \alpha_4 &= E_4 - E_5, & \alpha_5 &= E_5 - E_6, & \alpha_6 &= H - E_1 - E_2 - E_3. \end{aligned} \quad (56.113)$$

Each α_i is orthogonal to K_X , has norm two, and its products are -1 exactly for adjacent pairs of the diagram E_6 : the chain $1 - 2 - 3 - 4 - 5$, with node 6 joined to node 3. Therefore, its Gram matrix is the Cartan matrix of E_6 .

Proposition 56.39 (The seventy-two roots and their reflections). *The roots of L_{E_6} are*

$$\begin{aligned} &E_i - E_j && (i \neq j), \\ &\pm (H - E_i - E_j - E_k) && (1 \leq i < j < k \leq 6), \\ &\pm (2H - E_1 - \cdots - E_6), \end{aligned} \quad (56.114)$$

in total $30 + 40 + 2 = 72$. For each root α ,

$$s_\alpha(x) = x + (x \cdot \alpha)\alpha \quad (56.115)$$

preserves the intersection and K_X . The six simple reflections generate $W(E_6)$ and act faithfully by permutations on the 27 classes E_i, L_{ij}, Q_i in [theorem 56.38](#).

Proof. Substituting each class of [equation \(56.114\)](#) into [equation \(56.111\)](#) gives $K_X \cdot \alpha = 0$ and $\alpha^2 = -2$. The class count is as stated. The reflection formula is the orthogonal reflection for the form $-(\cdot)$; it preserves K_X because $K_X \cdot \alpha = 0$. The simple system of [equation \(56.113\)](#) generates all roots, and its Coxeter group is $W(E_6)$. Its action on the 27 classes of exceptional curves preserves intersection, and therefore coincides with the action on the already constructed graph [[Man05](#)]. $\square$

56.7.6 Stratification of the 729 Words

We write a six-trit word as

$$w = (\ell_1, h_1, \ell_2, h_2, \ell_3, h_3),$$

and we define two elements of H_3 ,

$$L(w) = (\ell_1, \ell_2, \ell_3), \quad H(w) = (h_1, h_2, h_3),$$

using the central law in their products. According to $L(w)^{-1}H(w)$, the 729 words are divided into

$$\begin{aligned} \mathcal{D}_\ell &= \{w : L(w)^{-1}H(w) = e\}, \\ \mathcal{I}_\ell &= \{w : L(w)^{-1}H(w) \in S_\ell\}, \\ \mathcal{S}_\ell &= \{w : L(w)^{-1}H(w) \notin \{e\} \cup S_\ell\}. \end{aligned}$$

Theorem 56.40 (Hensel–Schläfli Partition).

$$729 = 27 + 270 + 432, \tag{56.116}$$

corresponding to diagonal, incidence, and warping.

Proof. For each one of the 27 values of $L(w)$, there is a unique value of $H(w)$ with identity difference, ten with difference in S_ℓ and sixteen remaining ones. 27 is multiplied by 1, 10, 16. $\square$

The cardinal 432 of the warped stratum numerically coincides with the multiplicity of the transverse region of π in decomposition of the five regions of π . The two objects are not thereby identified: here one counts words classified by $L(w)^{-1}H(w)$, whereas there one counts preimages of a register U_6 . Without an explicit and equivariant intertwiner between those domains, $432 = 432$ is only a cardinal control.

56.7.7 Necessity of the central phase

Enumerating all inverse-closed subsets of cardinality ten in the five groups of order 27 gives

group	abelian	# srg(27, 10, 1, 5)
C_{27}	yes	0
$C_9 \times C_3$	yes	0
C_3^3	yes	0
$C_9 \rtimes C_3$	no	9
H_3	no	9.

Thus, retaining 27 labels while abelianizing composition destroys the configuration. Neither cardinality nor degree suffices.

If $\mathcal{T}$ is the conjunto of 45 triangulos, the cubico

$$C_{\Gamma}(x_1, \dots, x_{27}) = \sum_{\{i,j,k\} \in \mathcal{T}} x_i x_j x_k$$

recovers adjacency, because every edge lies in a unique triangle. Its permutation stabilizer is exactly

$$\text{Aut}(\Gamma_{\ell}) \cong W(E_6).$$

$$\text{APP curvatures } \pm 1 \longrightarrow \sigma \longrightarrow \mathcal{H}_9 \longrightarrow H_3 \longrightarrow \Gamma_{27} \longrightarrow \mathcal{T}_{45} \longrightarrow W(E_6).$$

Figure 56.3: Causal chain of the mixed-holonomy branch. It is parallel to $C_W \rightarrow W_{12} \rightarrow M_{12}$ and not a link of it.

Test criterion 56.41 (Falsification criteria for branch E_6). The branch is refuted if the curvatures are not opposite, if the central cocycle is erased, if the quadratic correction (56.106) does not preserve the product, if (56.108) does not produce the coefficients (10, 1, 5), if an abelian group produces the same configuration, if the partition does not sum to 729, if the table 56.2 does not intertwine the adjacency matrices, or if $W(E_6)$ is identified only by its order and not by its action on the twenty-seven classes and the 72 roots.

56.8 Oriented APP–Fock coupling and transverse rigidity of 12 fibres

The ordinary cyclotomic determinant does not distinguish C from C^{-1} . Sum–product information is preserved by explicitly introducing sheet parity and an orientation line. The resulting coupling produces index 54 from the APP aggregate. Once the four coupling conditions have been fixed, 54, J , and 196884 are not inputs to the evaluation. This evaluative independence asserts neither universal uniqueness of the ansatz nor historical independence of its design.

56.8.1 Triangular mark on each orbit

Let

$$M = \mathbb{Z}[\mathbb{Z}/9\mathbb{Z}]$$

and let $M_{\text{bal}}^{C_3}$ be the submodule of invariant elements ν whose six unit coefficients coincide:

$$\nu_u = c(\nu) \quad (u \in (\mathbb{Z}/9\mathbb{Z})^\times).$$

The coefficients of e_0, e_3, e_6 permanecen libres.

For a free orbit $\mathcal{O} = \{u, 4u, 7u\}$, set

$$b_\nu = e_\nu - e_{4\nu} \in A(\mathcal{O}).$$

If b_ν^b is the covector defined by the form of A_2 , the three roots of the triangular frame satisfy

$$\boxed{\sum_{\nu \in \mathcal{O}} b_\nu \otimes b_\nu^b = 3I_{A(\mathcal{O})}.$$

Consequently,

$$\sum_{\nu \in \mathcal{O}} \nu_\nu b_\nu \otimes b_\nu^b = 3c(\nu)I.$$

The factor three proceeds from the resolution of the identity by the root framework.

56.8.2 Oriented coupling $\mathcal{H}_{\Sigma,\Pi}^{(m)}$ between the balance C_3 , leaf parity, and degree 2 of Fock space

For $m \in 2\mathbb{N}_{>0}$, let

$$\begin{aligned}\mathcal{V}_m &= \mathcal{V}_{\Sigma,m} \oplus \mathcal{V}_{\Pi,m}, \\ \mathcal{V}_{\Sigma,m} &= (A_2 \otimes \mathbb{C})^{\oplus m/2}, \quad \mathcal{V}_{\Pi,m} = (A_2 \otimes \mathbb{C})^{\oplus m/2}, \\ g_m &= C^{\oplus m/2} \oplus (C^{-1})^{\oplus m/2}, \\ \mathbf{E}_m &= I_{\mathcal{V}_{\Sigma,m}} \oplus (-I_{\mathcal{V}_{\Pi,m}}).\end{aligned}$$

The specialization $m = 12$, with the labels of pair, sheet and orbit, is $\mathcal{V}_{12} = W_{\text{APP}} \otimes \mathbb{C}$. This notation prevents confusing the representational space with the Witt system $W_{12} = S(5, 6, 12)$. Degree two of the Fock space and the induced action are

$$\mathcal{F}_2(\mathcal{V}_m) = \mathcal{V}_m[2] \oplus \text{Sym}^2(\mathcal{V}_m[1]),$$

$$\Gamma_2(g_m) = g_m|_{\mathcal{V}_m[2]} \oplus \text{Sym}^2(g_m|_{\mathcal{V}_m[1]}).$$

Let $o_{\Sigma\Pi}$ be the signed orientation line $\Pi - \Sigma$, and let $M_{\text{bal}}^{C_3,-}$ be the oriented copy in which the exchange of leaves acts by $\nu \mapsto -\nu$. We define

$$\begin{aligned}\mathcal{H}_{\Sigma,\Pi}^{(m)} : M_{\text{bal}}^{C_3,-} &\longrightarrow \text{End}(\mathcal{F}_2(\mathcal{V}_m)) \otimes o_{\Sigma\Pi}, \\ \mathcal{H}_{\Sigma,\Pi}^{(m)}(\nu) &= 3c(\nu) \left(0_{\mathcal{V}_m[2]} \oplus \text{Sym}^2 \mathbf{E}_m \right) \otimes o_{\Sigma\Pi}.\end{aligned}$$

The transformation is linear in c , is supported on $\text{Sym}^2(\mathcal{V}_m[1])$, uses leaf parity, and adopts the normalization of the triangular frame.

Let σ the exchange of sheets. Then

$$\nu \mapsto -\nu, \quad o_{\Sigma\Pi} \mapsto -o_{\Sigma\Pi}, \quad \mathbf{E}_m \mapsto -\mathbf{E}_m,$$

and

$$\text{Sym}^2(-\mathbf{E}_m) = \text{Sym}^2(\mathbf{E}_m).$$

Therefore,

$$\mathcal{H}_{\Sigma,\Pi}^{(m)}(\sigma\nu) = \sigma\mathcal{H}_{\Sigma,\Pi}^{(m)}(\nu).$$

When the orientation is reversed, the dual covector also changes sign; thus $(-o_{\Sigma\Pi}^\vee)(-o_{\Sigma\Pi}) = 1$, and the oriented evaluation remains normalized.

Fixing $o_{\Sigma\Pi}^\vee(o_{\Sigma\Pi}) = 1$, the oriented trace is

$$\text{Tr}_{o,\mathcal{F}_2}(T \otimes o_{\Sigma\Pi}) = \text{Tr}_{\mathcal{F}_2}(T).$$

Theorem 56.42 (APP-oriented Index–Fock). *Suppose that m fibers A_2 , with m even, are distributed equally between the orientations C and C^{-1} . Then*

$$\text{Tr}_{o,\mathcal{F}_2} \left(\mathcal{H}_{\Sigma,\Pi}^{(m)}(\nu) \Gamma_2(g_m) \right) = -\frac{3m c(\nu)}{2}.$$

For the aggregate produced by APP,

$$\delta = \nu_\Pi - \nu_\Sigma = 12e_0 + 3e_3 + 3e_6 - 3 \sum_{u \in (\mathbb{Z}/9\mathbb{Z})^\times} e_u,$$

one has $c(\delta) = -3$ and

$$m = 3 \text{ pairs} \cdot 2 \text{ sheets} \cdot 2 \text{ orbits} = 12.$$

Therefore,

$$\boxed{\text{Tr}_{\mathcal{O}, \mathcal{F}_2} \left(\mathcal{H}_{\Sigma, \Pi}^{(12)}(\delta) \Gamma_2(g_{12}) \right) = 54.}$$

Proof. Let $A = E_m g_m$. The contributions of the two leaves cancel: $\text{tr } A = 0$. Since E_m commutes with g_m and $E_m^2 = I$,

$$\text{tr}(A^2) = \text{tr}(g_m^2) = -m.$$

The identity

$$\text{tr}(\text{Sym}^2 A) = \frac{(\text{tr } A)^2 + \text{tr}(A^2)}{2}$$

gives $-m/2$; the frame supplies $3c(\nu)$. $\square$

Corollary 56.43 (Minimality of the Bosonic Degree Detector). *In the declared class of couplings, degree one does not detect the oriented aggregate, and degree two is the first positive bosonic degree that detects it:*

$$\text{tr}(E_m g_m) = 0, \quad \text{tr}(\text{Sym}^2(E_m g_m)) = -\frac{m}{2} \neq 0.$$

Proof. The first equality is the cancellation of the two sheets. The second was obtained in the proof of theorem 56.42. Since degree zero contains only the unit and degree one has zero trace, degree two is the first with a nonzero response. $\square$

56.8.3 Transverse rigidity of the number of fibers

For an integer $m \geq 1$, we define the formal series

$$T_m(q) := q^{-1} \prod_{n \geq 1} (1 + q^n + q^{2n})^{-m} \in q^{-1} \mathbb{Z}[[q]] \quad (56.117)$$

and, in the orthogonal sum A_2^m , the radial vector

$$v_m := 4\rho_1 + \rho_2 + \cdots + \rho_m, \quad \|\rho_j\|^2 = 2. \quad (56.118)$$

We construct two entire functions without using $m = 12$ beforehand:

$$F(m) := [q] T_m(q), \quad R(m) := \|v_m\|^2.$$

Proposition 56.44 (Fock coefficient and radial norm). *For every $m \geq 1$,*

$$F(m) = \frac{m(m-3)}{2}, \quad R(m) = 2(m+15). \quad (56.119)$$

Proof. Up to second order,

$$(1 + q + q^2)^{-m} = 1 - mq + \frac{m(m-1)}{2} q^2 + O(q^3),$$

$$(1 + q^2 + q^4)^{-m} = 1 - mq^2 + O(q^4).$$

The factors with $n \geq 3$ do not contribute to the coefficient of q^2 . Since the exterior factor of T_m is q^{-1} ,

$$F(m) = \frac{m(m-1)}{2} - m = \frac{m(m-3)}{2}.$$

By orthogonality of the m copies of A_2 ,

$$R(m) = 4^2 \|\rho_1\|^2 + \sum_{j=2}^m \|\rho_j\|^2 = 32 + 2(m-1) = 2(m+15).$$

□

Theorem 56.45 (Cyclotomic–radial rigidity). *The equality $F(m) = R(m)$ is equivalent to*

$$(m-12)(m+5) = 0. \quad (56.120)$$

In the domain $m \in \mathbb{Z}_{>0}$, the unique solution is

$$m = 12, \quad F(12) = R(12) = 54.$$

Proof. Equating the two expressions of (56.119) produce

$$m(m-3) = 4(m+15),$$

that is,

$$m^2 - 7m - 60 = (m-12)(m+5) = 0.$$

The second root is negative and lies outside the domain. □

This rigidity belongs to the declared cyclotomic–radial family. It does not use the value 54 to choose m : it first constructs $F(m)$ and $R(m)$, and then resolves their equality.

56.8.4 Factorization of the APP–Fock coupling: hypotheses, invariants, necessary conditions for validity, and obstructions

The coupling factorizes by

$$M_{\text{bal}}^{C_3, -} / \ker c \cong \mathbb{Z} \otimes o_{\Sigma\Pi},$$

and the induced map is injective because $\text{Sym}^2 \mathbf{E}_m \neq 0$. The formula is selected by the four declared conditions: linearity in c , support in $\text{Sym}^2(\mathcal{V}_m[1])$, sheet parity, and normalization by the triangular frame. No uniqueness is asserted among all natural couplings, nor historical independence of the ansatz. Once those conditions have been fixed, 54, J , and 196884 are not inputs to the evaluation.

Three controls show that the result is not determined by a retroactive normalization:

$$(m, c) = (12, -2) \mapsto 36, \quad (m, c) = (10, -3) \mapsto 45,$$

and

$$\delta_k = \delta + k e_0$$

leaves $\mathcal{H}_{\Sigma, \Pi}^{(12)}$ invariant while the digital weight changes from 54 to $54 + 9k$. In particular, for $k = 1$,

$$\text{Tr}_{o, \mathcal{F}_2} \left(\mathcal{H}_{\Sigma, \Pi}^{(12)}(\delta + e_0) \Gamma_2(g_{12}) \right) = 54, \quad w(\delta + e_0) = 63.$$

Proposition 56.46 (Need for orientation data). *If $o_{\Sigma\Pi}$ is forgotten, every natural scalar that is linear in $\nu_{\Pi} - \nu_{\Sigma}$ and depends only on the class of the representation vanishes.*

Proof. C and C^{-1} are conjugate, so the representational class is even under sheet exchange. The aggregate $\nu_{\Pi} - \nu_{\Sigma}$ is odd. Without the orientation line, a transformation that is simultaneously even and odd is zero. □

The theorem thus establishes the exact transport of the aggregate to Fock degree two. The local state-by-state defect preserves the descent obstruction of [corollary 2.47](#); the two statements belong to distinct domains and are compatible.

56.9 From spectral jump 4 to Monster symmetries: APP unfolding $2 \rightarrow 6 \rightarrow 54$, the Leech lattice, V^h , and the Moonshine corridor

The corridor is not a sequence of numerical coincidences. It begins at the central block of APP, preserves the sum–product orientation, branches into the APP–Fock and APP–Witt couplings, and is coordinated anew through the grading. The binding causal order is

$$\begin{aligned} B_c = 9P_+ + 5P_- &\longrightarrow 4 \longrightarrow 2 \longrightarrow 6 \longrightarrow 54 \\ &\longrightarrow \begin{cases} \text{Fock degree two and } [q]t_3 = 54, \\ \text{APP–Witt alignment and } N_\alpha \cong \Lambda_{24} \end{cases} \\ &\longrightarrow V^h \longrightarrow (\text{Aut, ch, } g \mapsto T_g) \longrightarrow \text{Moonshine.} \end{aligned}$$

The two intermediate branches have distinct codomains. The common key 54 is a graded invariant transported by the corridor; it is not a linear map from a scalar into a vertex operator algebra.

56.9.1 Typed hierarchy and 2 realizations of index 54

Theorem 56.47 (Spectral Hierarchy and APP Unfolding). *Let $B_c = 7I + 2R = 9P_+ + 5P_-$ be the central block of APP. If M_Π, M_Σ are the defined modular compressions, then*

$$9 - 5 = 4, \quad (B_c)_{i \neq j} = 2, \quad \sum M_\Pi - \sum M_\Sigma = 51 - 45 = 6,$$

and the deployment over the nine positions gives

$$9 \cdot 6 = 459 - 405 = 54.$$

The four integers pertain, respectively, to the spectral gap, the local coupling, modular compression, and two-dimensional integration.

Proof. The spectral decomposition of B_c gives 9 and 5, whereas $7I + 2R$ fixes the nondiagonal coefficient. The two aggregate sums are 51 and 45. Each compressed entry represents nine positions, so that the unfolding multiplies the difference by nine. None of the four integers is obtained by retrospectively identifying its types. $\square$

Once the orientation line $o_{\Sigma\Pi}$, leaf parity, and the triangular normalization have been fixed, the oriented coupling

$$\mathcal{H}_{\Sigma,\Pi}^{(m)} : M_{\text{bal}}^{C_3,-} \longrightarrow \text{End}(\mathcal{F}_2(\mathcal{V}_m)) \otimes o_{\Sigma\Pi}$$

satisfies

$$\text{Tr}_{o,\mathcal{F}_2} \left(\mathcal{H}_{\Sigma,\Pi}^{(m)}(\nu) \Gamma_2(g_m) \right) = -\frac{3m c(\nu)}{2}.$$

For $m = 12$ and $c(\delta) = -3$, the trace equals 54. On the other hand,

$$t_3(q) = q^{-1} \prod_{n \geq 1} (1 + q^n + q^{2n})^{-12} = q^{-1} - 12 + 54q - 76q^2 - 243q^3 + \cdots,$$

from which $[q]t_3 = 54$.

Theorem 56.48 (Common Index Compatibility Cone). *With the starting structures of each branch preserved, it holds that*

$$D_{\text{APP}} = \text{Tr}_{o, \mathcal{F}_2} \left(\mathcal{H}_{\Sigma, \Pi}^{(12)}(\delta) \Gamma_2(g_{12}) \right) = [q]t_3 = v_\alpha^2 = \frac{196884}{5 \cdot 729 + 1} = 54.$$

This equality constitutes a cone of evaluations toward the same integer. It does not identify W_{APP} , the Fock space, the neighboring lattice, and $V^\natural$.

Proof. The APP census yields $459 - 405 = 54$; the oriented trace yields 54; expansion of the cyclotomic product yields $[q]t_3 = 54$; and the radial vector of the neighbor satisfies $v_\alpha^2 = 4^2 \cdot 2 + 11 \cdot 2 = 54$. Finally, $5 \cdot 729 + 1 = 3646$ and $54 \cdot 3646 = 196884$. The equalities are compared only after each object has been constructed and therefore do not numerically select the corridor. $\square$

Exact status of the hierarchy $4 \rightarrow 2 \rightarrow 6 \rightarrow 54$ and of the lattice identification. The hierarchy $4 \rightarrow 2 \rightarrow 6 \rightarrow 54$, the APP–Fock trace, and the extraction $[q]t_3$ are exact internal identities. Identification of the neighbor with Leech uses the declared chamber, neighboring vector, and lattice classification.

Vertex operator algebra and monstrous Moonshine

57.1 Alineamiento ciclotomico, orbifold of order 3 and Monstrous Moonshine

The Leech lattice constructed in theorem 56.32 makes it possible to pass from finite incidence to an infinite graded structure. This transition will not be effected through a chain of names. The following will be constructed, in typed order:

1. the intertwining of the twelve fibres A_2 ;
2. a fixed-point-free lattice isometry of order three;
3. its graded trace and its type-zero lift;
4. the cyclic orbifold and its dual automorphism of class $3B$;
5. the moonshine vertex operator algebra;
6. through distinct publications, its automorphism group, its graded character, and the McKay–Thompson series.

Moonshine sera the theorem that coordina the action and the series. is not usara the atajo not tipado $V^{\natural} \rightarrow \mathbb{M} \rightarrow \text{Moonshine}$.

57.1.1 Alignment of the 12 APP and Witt fibers

The twelve fibers of $W_{\text{APP}} \cong A_2^{12}$ carry three independent labels:

$$i \in \mathbb{Z}/3\mathbb{Z} \quad (\text{coordinate pair}), \quad \mu \in \{0, 1\} \quad (\text{sheet}), \quad \varepsilon \in \{0, 1\} \quad (\text{cyclotomic orbit}).$$

Fixing the dodecaphasic origin constructed on the TPK, we define

$$\lambda(i, \mu, \varepsilon) := 4i + 2\mu + \varepsilon \pmod{12}. \quad (57.1)$$

Lemma 57.1 (Bijectivity of the alignment). *The map λ from (57.1) is a bijection onto $\mathbb{Z}/12\mathbb{Z}$.*

Proof. For each i , the four values $4i + 2\mu + \varepsilon$ form the block $\{4i, 4i + 1, 4i + 2, 4i + 3\}$. The three blocks for $i = 0, 1, 2$ are disjoint and cover the twelve classes. $\square$

Let C be the Coxeter element of A_2 and let R be the reflection that satisfies

$$RCR = C^{-1}. \quad (57.2)$$

Once the trivializations ι_b and the oriented frames P_b have been fixed, we set

$$g_b := \iota_b^{-1} P_b C P_b^{-1} \iota_b, \quad (57.3)$$

$$T_{i\mu\varepsilon} := \iota_{\lambda(i,\mu,\varepsilon)}^{-1} P_{\lambda(i,\mu,\varepsilon)} R^{\mu}. \quad (57.4)$$

Theorem 57.2 (APP Entanglement–Witt). *For each terna (i, μ, ε) ,*

$$g_{\lambda(i,\mu,\varepsilon)} T_{i\mu\varepsilon} = T_{i\mu\varepsilon} C^{(-1)^{\mu}}. \quad (57.5)$$

The direct sum defines a C_3 -equivariant isomorphism

$$T_\lambda : W_{\text{APP}} \otimes \mathbb{C} \xrightarrow{\cong} W_c := \bigoplus_{b=0}^{11} (A_{2,b} \otimes \mathbb{C}). \quad (57.6)$$

Proof. For $\mu = 0$, the definition of g_b directly gives conjugation by C . For $\mu = 1$, one inserts $\text{RCR} = C^{-1}$. The lemma 57.1 guarantees that the direct sum uses every Witt position exactly once and preserves the pair, sheet, and orbit. $\square$

The theorem is relative to the simultaneous transport of the order of pairs, leaves, and orbits, of the Witt origin, of the frames, and of the orientation. Changing a single one of these coordinates does not define the same functor.

57.1.2 Graded trace of the action of the ternary generator on $V^{\natural}$

Let

$$\rho_W := \bigoplus_{b=0}^{11} g_b$$

and consider the bosonic envelope

$$\mathcal{H}_{\text{bos}}(W_c) := \text{Sym} \left(\bigoplus_{n \geq 1} W_c[n] \right). \quad (57.7)$$

If N is the degree operator, we define

$$Z_{\text{Fock}}(q) := q^{-1} \text{Tr}_{\mathcal{H}_{\text{bos}}(W_c)} \left(\Gamma_{\text{bos}}(\rho_W) q^N \right). \quad (57.8)$$

Proposition 57.3 (Cyclotomic polynomial of level three). *As a formal Laurent series,*

$$\boxed{Z_{\text{Fock}}(q) = q^{-1} \prod_{n \geq 1} (1 + q^n + q^{2n})^{-12} =: t_3(q).} \quad (57.9)$$

Its expansion begins with

$$t_3(q) = q^{-1} - 12 + 54q - 76q^2 - 243q^3 + 1188q^4 + O(q^5). \quad (57.10)$$

Proof. Each g_b is conjugate to C , whence

$$\det(I - g_b q^n) = 1 + q^n + q^{2n}.$$

The identity

$$\sum_{k \geq 0} \text{Tr}(\text{Sym}^k \rho_W) t^k = \det(I - \rho_W t)^{-1}$$

applied in each degree produces the product. Formal multiplication through degree five gives the expansion. In particular, the coefficient of q is obtained as

$$[q^2] (1 + q + q^2)^{-12} (1 + q^2 + q^4)^{-12} = 66 - 12 = 54.$$

$\square$

The value 54 appears here as the coefficient of a trace constructed from twelve fibers. Unit U030 also obtained it as the oriented index of the APP aggregate and proved that twelve is the unique positive solution of cyclotomic–radial rigidity. The equality of the two values is a compatibility between two realizations; neither uses the coefficient of J as input.

57.1.3 Isometry Without Fixed Points Induced by the Ternary Word of Full Support

Let

$$q_* = (1, 1, 1, 1, 1, 1) \in \mathbb{F}_3^6.$$

Multiplication by the Paley–Witt matrix gives

$$q_* A_W = (2, 1, 1, 1, 1, 1). \quad (57.11)$$

Therefore,

$$c_* := (q_*, q_* A_W) = (1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1) \in C_W \quad (57.12)$$

and all its coordinates are non-zero.

For each position b , we fix

$$P_b(c_*) = \begin{cases} R, & (c_*)_b = 1, \\ I, & (c_*)_b = 2, \end{cases} \quad g_c := \bigoplus_{b=0}^{11} P_b(c_*) C P_b(c_*)^{-1}. \quad (57.13)$$

Proposition 57.4 (Order, absence of fixed points and gluing). *The operator g_c has order three, has no nonzero fixed vectors, and preserves the Niemeier lattice $N(A_2^{12})$ constructed in U021.*

Proof. Each block is conjugate to the Coxeter C , whose minimal polynomial is $X^2 + X + 1$. Hence it has order three and has no eigenvalue one. The direct sum preserves both properties. The Coxeter acts trivially on A_2^*/A_2 ; therefore each block preserves the discriminant class and the union of lateral classes that defines the gluing. $\square$

57.1.4 Preservation of the Leech Neighbor

Write

$$v := v_{\mathcal{F}_*} = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \quad v^2 = 54,$$

for the vector selected by the incidence origin of (56.82). Let

$$N := N(A_2^{12}), \quad N_0 := \ker(\langle \cdot, v \rangle \bmod 3), \quad N_v := N_0 + \mathbb{Z} \frac{v}{3} \cong \Lambda_{24}.$$

Theorem 57.5 (Neighbor-Oriented Preservation). *The displacements*

$$y_* := \frac{g_c v - v}{3}, \quad z_* := \frac{g_c^{-1} v - v}{3} \quad (57.14)$$

belong to N_0 . Consequently,

$$g_c N_0 = N_0, \quad g_c N_v = N_v.$$

Proof. The discriminant words of y_* and z_* are c_* and $-c_*$, both in C_W ; hence $y_*, z_* \in N$. Computation in each factor A_2 gives

$$\langle y_*, v \rangle = \langle z_*, v \rangle = -(4^2 + 11) = -27 \equiv 0 \pmod{3}. \quad (57.15)$$

Thus both lie in N_0 . For $x \in N_0$,

$$\langle g_c x, v \rangle = \langle x, g_c^{-1} v \rangle = \langle x, v \rangle + 3\langle x, z_* \rangle \equiv 0 \pmod{3}.$$

The same count with g_c^{-1} gives equality of the kernel. Finally,

$$g_c(v/3) = v/3 + y_*,$$

so that the class generated by the neighbor is preserved. $\square$

57.1.5 24 orientations and lattice naturality

The weight enumerator of (56.18) contains $24y^{12}$. Therefore, the set

$$C_W^\times := \{u \in C_W : u_b \in \{1, 2\} \text{ for every } b\}$$

has 24 elements. For each $u \in C_W^\times$, we define

$$P_b(u) = \begin{cases} R, & u_b = 1, \\ I, & u_b = 2, \end{cases} \quad g_u := \bigoplus_{b=0}^{11} P_b(u) C P_b(u)^{-1}.$$

Proposition 57.6 (Robustness of Full Support Orientations). *For every $u \in C_W^\times$,*

$$\frac{g_u v - v}{3}, \quad \frac{g_u^{-1} v - v}{3} \in N_0.$$

In particular, each g_u preserves the neighbor constructed with the transported orientation.

Proof. In simple root coordinates,

$$C\rho - \rho = (-2, -1), \quad C^{-1}\rho - \rho = (-1, -2).$$

The discriminant classes of the two displacements are u and $-u$. For the radial coefficient $a_b \in \{4, 1\}$,

$$\left\langle \frac{(C^{\pm 1} - I)a_b \rho}{3}, a_b \rho \right\rangle = -a_b^2.$$

The sum over the twelve factors again gives -27 . The proof of [theorem 57.5](#) is applied component by component. $\square$

To type the recalibration, let $h \in \text{Aut}(C_W)$ be monomial: a permutation σ of the twelve coordinates followed by multipliers $\epsilon_b \in \{1, 2\}$. We define its lattice lift

$$\tilde{h} := P_\sigma \circ \bigoplus_{b=0}^{11} R^{\nu_b}, \quad \nu_b = \begin{cases} 0, & \epsilon_b = 1, \\ 1, & \epsilon_b = 2. \end{cases} \quad (57.16)$$

Its action on the discriminant is exactly h . If the change simultaneously transports the code, flag, incidence origin, vector v , neighbor, and frames, then

$$g_{hu} = \tilde{h} g_u \tilde{h}^{-1}. \quad (57.17)$$

The ternary matrix h does not act without a lift on A_2^{12} or on the Leech lattice.

57.1.6 Lattice trace and computation of type 0

Let $\widehat{g}_c$ be the standard lift of g_c to the lattice VOA

$$V_{N_v} = M(1) \otimes \mathbb{C}_\varepsilon[N_v].$$

For $j = 1, 2$, the oscillator–lattice decomposition produces

$$\begin{aligned}
Z_{0,j}(\tau) &:= \text{Tr}_{V_{N_v}} \left(\widehat{g}_c^j q^{L_0-1} \right) \\
&= q^{-1} \left(\sum_{\lambda \in N_v^{g_c^j}} \varepsilon_j(\lambda) q^{\langle \lambda, \lambda \rangle / 2} \right) \prod_{n \geq 1} \det(I - g_c^j q^n)^{-1}.
\end{aligned} \tag{57.18}$$

Proposition 57.7 (Trace of the nontrivial sectors). *We have*

$$Z_{0,1}(\tau) = Z_{0,2}(\tau) = t_3(\tau).$$

Proof. The absence of fixed points implies

$$N_v^{g_c} = N_v^{g_c^2} = \{0\},$$

so that the lattice sum in (57.18) equals one. In each factor, g_c^j has eigenvalues ζ_3, ζ_3^2 ; therefore,

$$\det(I - g_c^j q^n) = 1 + q^n + q^{2n}.$$

The twelve factors reproduce (57.9). □

Theorem 57.8 (Twisted weight and type of the lift). *The standard lift has order three, twisted weight*

$$\rho_{g_c} = \frac{4}{3} \tag{57.19}$$

and type zero.

Proof. In $\mathfrak{h} = N_v \otimes_{\mathbb{Z}} \mathbb{C}$, each of the twelve factors contributes one eigenvector with eigenvalue ζ_3 and another with eigenvalue ζ_3^2 . Therefore,

$$\dim \mathfrak{h}_{\zeta_3} = \dim \mathfrak{h}_{\zeta_3^2} = 12.$$

The formula for the weight of the twisted vacuum of order three gives

$$\rho_{g_c} = \frac{1}{4 \cdot 3^2} (1 \cdot 2 \cdot 12 + 2 \cdot 1 \cdot 12) = \frac{4}{3}.$$

For a lattice isometry of odd order, the standard elevation preserves the order. The type is the class of

$$3^2 \rho_{g_c} = 12 \equiv 0 \pmod{3}.$$

Thus, the two initially independent hypotheses are calculated here. □

57.1.7 Cyclic orbifold and dual class 3B

Theorem 57.9 (Triadic orbifold of the Leech neighbor). *The classical cyclic-orbifold theorems, applied to the standard lift in theorem 57.8, yield*

$$\text{Orb}_{\mathbb{Z}/3}(V_{N_v}, \widehat{g}_c) \cong V^{\natural}. \tag{57.20}$$

The dual automorphism belongs to the class 3B and its series is

$$T_{3B}(\tau) = t_3(\tau) + 12. \tag{57.21}$$

Proof. The isometry is fixed-point-free of order three; the twisted weight and type zero were calculated, not assumed. The cyclic-orbifold theorems identify the holomorphic extension with the Moonshine module and classify the dual automorphism as $3B$ [CLS16, ALY17, Car17]. The identification uses these theorems after constructing the lattice object; it is not deduced solely from an equality of series. $\square$

If $V^{(i)}$ is the $\widehat{g}_c^i$ -twisted module and

$$Z_{i,j}(\tau) := \text{Tr}_{V^{(i)}} \left(\phi_i(\widehat{g}_c)^j q^{L_0-1} \right),$$

the character of the orbifold is the average of the nine sectors:

$$\text{ch}_{V^\natural}(\tau) = \frac{1}{3} \sum_{i,j \in \mathbb{Z}/3} Z_{i,j}(\tau) = J(\tau). \quad (57.22)$$

The level three rational identity is

$$J = t_3 + 12 + \frac{10 \cdot 3^9}{t_3} + \frac{4 \cdot 3^{14}}{t_3^2} + \frac{3^{18}}{t_3^3}. \quad (57.23)$$

Since $t_3^{-1} = q + O(q^2)$,

$$[q]J = 54 + 10 \cdot 3^9 = 196884 = 54(5 \cdot 729 + 1). \quad (57.24)$$

This is a graded compatibility within the constructed corridor, not a scalar morphism $54 \rightarrow V^\natural$.

57.1.8 Lattice construction of order 2

The preceding route of order three and the classical construction of Frenkel–Lepowsky–Meurman produce two presentations of the same moonshine VOA. To declare the second, we fix the central extension

$$1 \longrightarrow \{\pm 1\} \longrightarrow \widehat{\Lambda}_{24} \longrightarrow \Lambda_{24} \longrightarrow 1 \quad (57.25)$$

with commutator $(-1)^{\langle \lambda, \mu \rangle}$, and a cocycle $\varepsilon(\lambda, \mu)$ that represents it. The twisted group algebra is $\mathbb{C}_\varepsilon[\Lambda_{24}]$, and

$$V_\Lambda := M(1) \otimes \mathbb{C}_\varepsilon[\Lambda_{24}] \quad (57.26)$$

is the lattice VOA of central charge 24.

The negation $\lambda \mapsto -\lambda$ has a lift θ compatible with (57.25). Its trace is

$$t_2(\tau) := \text{Tr}_{V_\Lambda}(\theta q^{L_0-1}) = q^{-1} \prod_{n \geq 1} (1 + q^n)^{-24}. \quad (57.27)$$

Indeed, negation pairs e^λ with $e^{-\lambda}$, and the absence of torsion leaves only $\lambda = 0$ in the lattice trace; each of the 24 oscillators contributes $(1 + q^n)^{-1}$.

Let V_Λ^T the modulo irreducible torcido by θ , and $(V_\Lambda^T)^+$ its sector of signo positive. The construction FLM is

$$V^\natural = (V_\Lambda)^+ \oplus (V_\Lambda^T)^+. \quad (57.28)$$

The route $\mathbb{Z}/2$ of (57.28) and the route $\mathbb{Z}/3$ of (57.20) are parallel constructions identified by classical theorems; neither replaces the other.

57.1.9 Automorphism group, character, and graded series

eleven constructed V^h , appear two publications distinct:

$$\text{Aut}(V^h) \cong \mathbb{M}, \quad (57.29)$$

$$\text{ch}_{V^h}(\tau) = J(\tau) = j(\tau) - 744 = q^{-1} + 196884q + 21493760q^2 + \cdots. \quad (57.30)$$

The former identifies the symmetry group; the latter counts graded dimensions. The vacuum line is

$$(V^h)_0 = \mathbb{C}\mathbf{1},$$

and the absence of roots in Λ_{24} gives

$$(V^h)_1 = 0.$$

In degree two,

$$196884 = 1 + 196883 = 196560 + 324 = 54(5 \cdot 729 + 1).$$

Only the first equality is a decomposition into dimensions of representations of the Monster; the others are arithmetic controls.

For $g \in \mathbb{M}$, the McKay–Thompson series is defined

$$T_g(\tau) := \text{Tr}_{V^h} \left(g q^{L_0 - 1} \right). \quad (57.31)$$

Theorem 57.10 (Monstrous Moonshine). *The graded action of $\mathbb{M}$ on V^h produces the family $\{T_g\}_{g \in \mathbb{M}}$. The Moonshine theorems identify these series with genus-zero modular functions and prove their replicability identities [CN79, Bor92, FLM88].*

This theorem has as input the pair

$$(V^h, \text{Aut}(V^h))$$

and its grading. It is not the consequence of an arrow whose domain is the abstract group $\mathbb{M}$.

For $J = T_{1A}$, let P_n be the Faber polynomials normalized by

$$P_n(J) = q^{-n} + O(q).$$

The Hecke–Faber identity is

$$P_n(J) = n T_n J. \quad (57.32)$$

It controls all degrees; the coefficient 196884 is only the first nontrivial manifestation.

57.1.10 Paley–Witt–Golay–Leech–Moonshine Runner Dependency Graph

Test criterion 57.11 (Refutation criteria for the moonshine corridor). The construction is refuted if any of the following facts occurs:

1. λ is not bijective or (57.5) falla;
2. $c_* \notin C_W$, g_c has fixed points or does not preserve the neighbor;
3. the displacements of (57.14) do not belong to N_0 ;
4. the lattice lift $\tilde{h}$ does not realize the monomial automorphism h on the discriminant;
5. the twisted weight is not $4/3$ or the lift is not of type zero;
6. the orbifold is identified with V^h without verifying the hypotheses of the classical theorem;
7. W_{24} is used as a constructor of the Leech lattice;

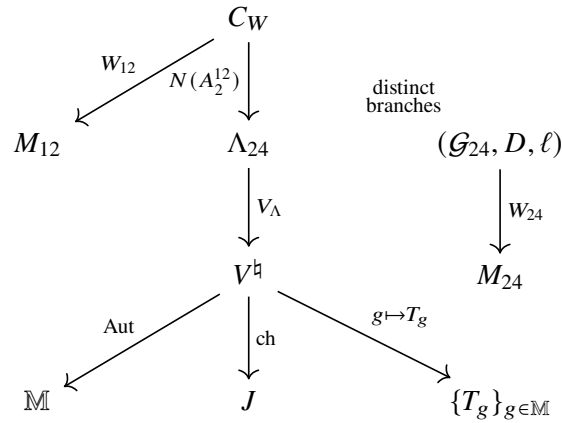


Figure 57.1: Exceptional bifurcation and three publications of $V^{\mathfrak{h}}$. The binary input $(\mathcal{G}_{24}, D, \ell)$ is independent; there is no arrow $W_{24} \rightarrow \Lambda_{24}$.

8. an equality of dimensions is presented as a construction of $\mathbb{M}$;
9. the publications Aut, ch, and $g \mapsto T_g$ are improperly serialized.

57.2 Theta control of the Leech lattice: identity $\Theta_\Lambda = E_4^3 - 720\Delta$ and separation between arithmetic coincidence and structural morphism

From Λ_{24} , the string

$$\Lambda_{24} \longrightarrow V^{\mathfrak{h}} \longrightarrow \mathbb{M} \longrightarrow \text{Moonshine}$$

is classical mathematics: vertex operator algebra, the Monster group, and McKay–Thompson functions. The contribution verified here lies in the finite input, the selection of the boundary, and the rootless 3-neighbor, not in claiming those theorems anew.

Three numerical identities accompany this chain:

$$\Theta_\Lambda = E_4^3 - 720\Delta,$$

$$196884 = 196560 + 324,$$

$$196884 = 54(3645 + 1) = 54(5 \cdot 729 + 1).$$

They are exact arithmetic identities, but without a morphism or a character decomposition, they do not by themselves constitute mathematical signals of compatibility. None replaces the construction of $V^{\mathfrak{h}}$ or provides a direct action of the Monster on HMT decimal words.

57.3 Uniformization of order 2 in Moonshine: functions T_{2A} and j , modular structure and control of the micro–macro corridor

The incidence branch first constructs the Leech lattice Λ_{24} . From that datum, the classical theory of vertex operator algebras provides a canonical prolongation. If $\mathfrak{h} = \Lambda_{24} \otimes_{\mathbb{Z}} \mathbb{C}$ and $M(1)$ is the Fock module of its Heisenberg algebra, the lattice VOA is

$$V_\Lambda = M(1) \otimes \mathbb{C}[\Lambda_{24}], \quad (57.33)$$

with central charge 24. For $\alpha \in \Lambda_{24}$, its vertex operators are written

$$Y(e^\alpha, z) = E^-(-\alpha, z)E^+(-\alpha, z)e^\alpha z^{\alpha_0}.$$

The absence of roots in Λ_{24} implies the vanishing of the weight-one sector in the Moonshine orbifold.

The involution

$$\theta : \Lambda_{24} \longrightarrow \Lambda_{24}, \quad \theta(\lambda) = -\lambda,$$

is intrinsic and prolongs to V_Λ . Its graded trace is obtained by direct counting of the oscillator modes.

Theorem 57.12 (Trace of the negation involution).

$$t_2(\tau) := \text{Tr}_{V_\Lambda}(\theta q^{L_0-1}) = q^{-1} \prod_{n \geq 1} (1 + q^n)^{-24}. \quad (57.34)$$

Proof. The involution pairs e^λ with $e^{-\lambda}$; since the lattice is torsion-free, only $\lambda = 0$ contributes to the trace of the lattice sector. Each one of the twenty-four energy oscillators n contributes $\sum_{k \geq 0} (-1)^k q^{nk} = (1 + q^n)^{-1}$. The vacuum displacement produces the factor q^{-1} . $\square$

The Fricke completion associated with this trace is

$$T_{2A}(\tau) = t_2(\tau) + 24 + \frac{2^{12}}{t_2(\tau)} = q^{-1} + 4372q + 96256q^2 + 1240002q^3 + \cdots. \quad (57.35)$$

The term 24 cancels the constant term of t_2 ; the factor 2^{12} is the degeneracy of the twisted sector of twenty-four bosons. The same boundary series determines the modular invariant through

$$j(\tau) = \frac{(t_2(\tau) + 256)^3}{t_2(\tau)^2}. \quad (57.36)$$

Equations (57.34)–(57.36) construct the canonical branch of order two without continuous parameters.

The Frenkel–Lepowsky–Meurman orbifold is defined by

$$V^{\natural} = (V_\Lambda)^+ \oplus (V_\Lambda^T)^+.$$

It is a holomorphic VOA of central charge 24, satisfies $(V^{\natural})_1 = 0$, and has character

$$\text{ch}_{V^{\natural}}(\tau) = j(\tau) - 744$$

and its automorphism group is the Monster group [FLM88, CN79, Bor92]. In this chain, HMT fixes the Leech lattice and the involution θ ; the classical VOA and FLM theorems identify the resulting prolongation with $V^{\natural}$ and its automorphism group. The assertion does not require an action of the Monster on the digits: the Archimedean evaluation and the exceptional branch descend from a common HMT preimage and preserve different invariants.

The neighboring function $T_{3A} = t_3 + 12 + 729/t_3$ belongs to the classical theory of Hauptmoduln of order three. The active HMT realization in the tritic corridor is the class $3B$; an identification with $3A$ requires declaration of the specific lift that changes the class.

57.3.1 APP–Witt intertwining, Leech neighbor, and orbifold of order 3

The twelve fibers bear the labels

$$(i, \mu, \varepsilon) \in \mathbb{Z}/3\mathbb{Z} \times \{0, 1\} \times \{0, 1\}, \quad \lambda(i, \mu, \varepsilon) = 4i + 2\mu + \varepsilon \pmod{12}.$$

The map λ is bijective and, once trivializations and frames are fixed, the intertwiners satisfy

$$g_{\lambda(i, \mu, \varepsilon)} T_{i\mu\varepsilon} = T_{i\mu\varepsilon} C^{(-1)^\mu}.$$

Its direct sum is an C_3 -equivariant isomorphism

$$T_\lambda : W_{\text{APP}} \otimes \mathbb{C} \xrightarrow{\cong} \bigoplus_{b=0}^{11} (A_{2,b} \otimes \mathbb{C}).$$

Let $N = N(A_2^{12})$, let $v = v_\alpha$ the vector radial of norm 54, and definamos

$$N_0 = \ker(\langle \cdot, v \rangle \bmod 3), \quad N_v = N_0 + \mathbb{Z} \frac{v}{3}.$$

The lattice calculation verifies that $N_v \cong \Lambda_{24}$, that the isometry g_c has order three and has no nonzero fixed points, and that

$$\frac{g_c v - v}{3}, \quad \frac{g_c^{-1} v - v}{3} \in N_0.$$

Hence $g_c N_v = N_v$. Its standard lift to the lattice VOA has twisted weight $4/3$ and type zero. The classical cyclic-orbifold theorems give

$$V_{(3)} := \text{Orb}_{\mathbb{Z}/3}(V_{N_v}, \widehat{g}_c) \cong V^{\natural}, \quad T_{3B} = t_3 + 12.$$

The character is recovered by the average of the nine sectors and by the level three uniformization:

$$\begin{aligned} \text{ch}_{V_{(3)}} &= \frac{1}{3} \sum_{i,j \in \mathbb{Z}/3} Z_{i,j} = J, \\ J &= t_3 + 12 + \frac{10 \cdot 3^9}{t_3} + \frac{4 \cdot 3^{14}}{t_3^2} + \frac{3^{18}}{t_3^3}. \end{aligned}$$

In particular,

$$[q]J = 54 + 10 \cdot 3^9 = 196884 = 54(5 \cdot 729 + 1).$$

57.3.2 Constructions of orders 2 and 3 of the vertex operator algebra $V^{\natural}$

Fix the central extension and the Leech cocycle.

$$1 \longrightarrow \{\pm 1\} \longrightarrow \widehat{\Lambda}_{24} \longrightarrow \Lambda_{24} \longrightarrow 1, \quad V_\Lambda = M(1) \otimes \mathbb{C}_\varepsilon[\Lambda_{24}].$$

The negation $\lambda \mapsto -\lambda$ lifts to θ , and

$$t_2(\tau) = \text{Tr}_{V_\Lambda} \left(\theta q^{L_0 - 1} \right) = q^{-1} \prod_{n \geq 1} (1 + q^n)^{-24}.$$

The FLM presentation is

$$V_{(2)} := (V_\Lambda)^+ \oplus (V_\Lambda^T)^+ \cong V^{\natural}.$$

Theorem 57.13 (Typed equivalence of the presentations of orders 3 and 2). *Let*

$$\phi_3 : V_{(3)} \xrightarrow{\cong} V^{\natural}, \quad \phi_2 : V_{(2)} \xrightarrow{\cong} V^{\natural}$$

the graded isomorphisms provided by the orbifold theorems applied to the preceding data. Then

$$\boxed{\Phi_{23} := \phi_2^{-1} \circ \phi_3 : V_{(3)} \xrightarrow{\cong} V_{(2)}}$$

is a graded isomorphism of vertex operator algebras. Moreover,

$$a \longmapsto \Phi_{23} a \Phi_{23}^{-1}$$

identifies its automorphism groups and preserves the graded traces.

Proof. The composition of VOA isomorphisms is a VOA isomorphism and preserves the vacuum, the conformal vector, the vertex operators, and the grading by L_0 . Conjugation transports automorphisms. For every automorphism a , invariance of the trace under conjugation gives equality of the corresponding graded series. $\square$

The isomorphism Φ_{23} depends on the identifications ϕ_2 and ϕ_3 ; absolute canonicity is not asserted. The order-two construction also preserves the exclusive content

$$T_{2A} = t_2 + 24 + \frac{2^{12}}{t_2}, \quad \boxed{j = \frac{(t_2 + 256)^3}{t_2^2}},$$

which completes the modular uniformization of order two.

Once $V^{\mathfrak{h}}$ has been constructed, the three publications are stated without unduly serializing them:

$$\text{Aut}(V^{\mathfrak{h}}) \cong \mathbb{M}, \quad \text{ch}_{V^{\mathfrak{h}}} = J, \quad g \mapsto T_g(\tau) = \text{Tr}_{V^{\mathfrak{h}}}(gq^{L_0-1}).$$

The Moonshine theorem coordinates the graded action with genus-zero functions and their replicability identities. The Monster action resides in $V^{\mathfrak{h}}$; the fact that it is not a pointwise permutation of digits is a final type control, not the thesis of the corridor.

57.4 Continuous evidentiary chain of the exceptional publication: from the APP unfolding to Paley–Witt, $V^{\mathfrak{h}}$ and Moonshine

The exceptional publication starts from the incidence reader $\mathfrak{E}_{\text{exc}}$ of (frontal diagram of the double projection). It does not take an already given code, lattice or exceptional group as input: it takes the boundary, orientation, sheet and memory of an enriched history and expresses them in the projective gauge fixed by nonadic holonomy. Only afterwards are the classical theorems of codes, lattices and vertex operator algebras applied to the produced objects.

57.4.1 APP unfolding and graded cone of the index 54

Let R be the involution exchanging the APP sheets and let $P_+ = (I + R)/2$, $P_- = (I - R)/2$ be its projectors. The central block

$$B_c = 7I + 2R = 9P_+ + 5P_- \quad (57.37)$$

produces four objects of distinct types:

$$9 - 5 = 4, \quad (B_c)_{i \neq j} = 2, \quad \sum M_{\Pi} - \sum M_{\Sigma} = 51 - 45 = 6, \quad 9 \cdot 6 = 459 - 405 = 54. \quad (57.38)$$

The 4 is the spectral jump between sheets, the 2 the local coupling, the 6 the difference of the modular compressions and the 54 the integration of that difference over the nine positions. Therefore, $4 \rightarrow 2 \rightarrow 6 \rightarrow 54$ designates a hierarchy of morphisms and evaluations, not a chain of equalities between objects.

With the orientation $o_{\Sigma\Pi}$, sheet parity and triangular normalization fixed, the Fock branch has the coupling

$$\mathcal{H}_{\Sigma,\Pi}^{(m)} : M_{\text{bal}}^{C_3,-} \longrightarrow \text{End}(\mathcal{F}_2(\mathcal{V}_m)) \otimes o_{\Sigma\Pi}, \quad (57.39)$$

and its oriented trace satisfies

$$\text{Tr}_{o,\mathcal{F}_2}(\mathcal{H}_{\Sigma,\Pi}^{(m)}(v)\Gamma_2(g_m)) = -\frac{3m c(v)}{2}. \quad (57.40)$$

For $m = 12$ and $c(\delta) = -3$, 54 holds. Independently, the cyclotomic product

$$t_3(q) = q^{-1} \prod_{n \geq 1} (1 + q^n + q^{2n})^{-12} = q^{-1} - 12 + 54q - 76q^2 - 243q^3 + \cdots \quad (57.41)$$

gives $[q]t_3 = 54$.

Theorem 57.14 (Typed cone of the common index). *Once the APP census, the Fock branch, the cyclotomic product, the lattice neighbor of equation (57.52) and the lunar module of equation (57.58) have been constructed separately, their readers satisfy*

$$D_{\text{APP}} = \text{Tr}_{o, \mathcal{F}_2}(\mathcal{H}_{\Sigma, \Pi}^{(12)}(\delta) \Gamma_2(g_{12})) = [q]t_3 = v^2 = \frac{196884}{5 \cdot 729 + 1} = 54. \quad (57.42)$$

The equality gathers evaluations in a cone toward $\mathbb{Z}$; it does not identify their domains or use 196884 to select the corridor.

Proof. The first three equalities are (57.38), (57.40) and (57.41). For the radial vector $v = 4\rho_1 + \rho_2 + \cdots + \rho_{12}$ of (57.51), orthogonality of the factors A_2 gives $v^2 = 4^2 \cdot 2 + 11 \cdot 2 = 54$. Finally, $5 \cdot 729 + 1 = 3646$ and $54 \cdot 3646 = 196884$. Each identity is proved in its own codomain before they are compared. $\square$

57.4.2 Paley form and internal quadratic root of $-I_6$

Let

$$\mathcal{U}_9 = (\mathbb{Z}/9\mathbb{Z})^\times = \{1, 2, 4, 8, 7, 5\},$$

in the order of the powers of 2, and let

$$\theta : \mathcal{U}_9 \xrightarrow{\sim} \mathbb{P}^1(\mathbb{F}_5), \quad \frac{u}{\theta(u)} \mid \begin{array}{cccccc} 1 & 2 & 4 & 8 & 7 & 5 \\ \infty & 0 & 1 & 2 & 3 & 4 \end{array}. \quad (57.43)$$

This bijection is a coordinate gauge of the reader; it is not used as a homomorphism. For the quadratic character $\chi : \mathbb{F}_5 \rightarrow \{-1, 0, 1\}$, define

$$C_P = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & -1 & -1 & 1 \\ 1 & 1 & 0 & 1 & -1 & -1 \\ 1 & -1 & 1 & 0 & 1 & -1 \\ 1 & -1 & -1 & 1 & 0 & 1 \\ 1 & 1 & -1 & -1 & 1 & 0 \end{pmatrix}, \quad (57.44)$$

where the finite off-diagonal entries are $\chi(x - y)$ and the entries incident with ∞ are one.

Lemma 57.15 (Quadratic correlation). *If $a \neq b$ in $\mathbb{F}_5$, then*

$$\sum_{t \in \mathbb{F}_5} \chi(t - a) \chi(t - b) = -1.$$

Proof. The substitution $u = (t - a)/(b - a)$ reduces the sum to $\sum_u \chi(u(u - 1))$. For $u = 0, 1, 2, 3, 4$, the summands are 0, 0, 1, -1 , -1 . $\square$

Proposition 57.16 (Conference identity and ternary reduction). *The matrix C_P is symmetric and satisfies*

$$C_P C_P^T = 5I_6.$$

Its reduction $A_W = C_P \bmod 3$ satisfies

$$\boxed{A_W^2 = -I_6 \text{ in } M_6(\mathbb{F}_3).} \quad (57.45)$$

Proof. Each row of C_P has squared norm five. Two distinct finite rows have inner product $1 - 1 = 0$ by the lemma 57.15; the sum of a nontrivial character also proves orthogonality with the row of ∞ . Reducing modulo three yields $A_W A_W^T = 5I_6 = 2I_6 = -I_6$. Since A_W is symmetric, (57.45) follows. $\square$

The identity (57.45) constructs the finite complex structure of the corridor. Indeed,

$$\mathbb{F}_9 = \mathbb{F}_3[\iota]/(\iota^2 + 1), \quad P_{\pm\iota} = \frac{1}{2}(I \mp \iota A_W)$$

are complementary rank-three projectors and endow $\mathbb{F}_3^6$ with a rank-three $\mathbb{F}_9$ -module structure. The action is explicit:

$$(a + b\iota) \cdot v = av + b(vA_W). \quad (57.46)$$

Therefore, $3^6 = 9^3$ is not used as a cardinal substitute for an action.

57.4.3 Self-dual ternary code and Witt system

Define

$$c : \mathbb{F}_3^6 \longrightarrow \mathbb{F}_3^{12}, \quad c(q) = (q, qA_W), \quad C_W := \text{im } c. \quad (57.47)$$

Theorem 57.17 (Ternary code produced by incidence). *The map c is injective and*

$$C_W = C_W^\perp, \quad W_{C_W}(y) = 1 + 264y^6 + 440y^9 + 24y^{12}. \quad (57.48)$$

In particular, C_W has parameters $[12, 6, 6]_3$.

Proof. The first projection of $c(q)$ is q , so c is injective. With $G_W = (I_6 \mid A_W)$,

$$G_W G_W^T = I_6 + A_W A_W^T = 0$$

in $\mathbb{F}_3$. Thus $C_W \subseteq C_W^\perp$, and equality of dimensions gives self-duality. The enumerator is the exact finite sum

$$\sum_{q \in \mathbb{F}_3^6} y^{\text{wt}(q) + \text{wt}(qA_W)}.$$

Evaluating the $3^6 = 729$ rows of the explicit matrix (57.44) gives, respectively, the counts 1, 264, 440, 24 at weights 0, 6, 9, 12, and zero at the others. The sum of the four counts is 729, so no word remains outside the finite certificate. $\square$

The 264 weight-six vectors occur in pairs $\{x, -x\}$; their 132 supports form the Steiner system

$$W_{12} = S(5, 6, 12). \quad (57.49)$$

This is the recognition, after constructing C_W , of the extended ternary Golay code and its Witt design [MS77, CS99]. The design is not introduced to select the matrix A_W .

The binary prolongation remains separate. If $\mathcal{G}_{24} \subset \mathbb{F}_2^{24}$ is the extended binary Golay code, D a dodecad and $\ell : \{1, \dots, 12\}_{\text{HMT}} \rightarrow D$ a design isomorphism, the weight-eight supports form $W_{24} = S(5, 8, 24)$. For a fixed tetrad,

$$\lambda_4(W_{12}) = 4, \quad \lambda_4(W_{24}) = 5,$$

so that four octads lift the four incident hexads and a fifth is transversal. The datum $(\mathcal{G}_{24}, D, \ell)$ belongs to the binary codomain; there is no canonical arrow $W_{24} \rightarrow \Lambda_{24}$.

57.4.4 Ternary gluing and Leech neighbor

Let $Q = A_2^{12}$, with $\det Q = 3^{12}$. The discriminant of each factor A_2 is $\mathbb{F}_3$, and the gluing defined by (57.47) is

$$N := \{x \in Q^* : x \bmod Q \in C_W\}. \quad (57.50)$$

Theorem 57.18 (Niemeier lattice produced by the code). *The lattice N is positive definite, even and unimodular of rank twenty-four, and its root system is A_2^{12} . Consequently, $N \cong N(A_2^{12})$.*

Proof. Self-duality of C_W makes the gluing integral. All its weights are multiples of three, which makes the extension even. Moreover, $[N : Q] = |C_W| = 3^6$, and therefore

$$\det N = \frac{\det Q}{[N : Q]^2} = \frac{3^{12}}{3^{12}} = 1.$$

A nonzero discriminant class of A_2 has minimum norm $2/3$. Since every nonzero word of C_W has weight at least six, every gluing vector outside Q has norm at least four. The norm-two vectors are exactly the roots of the twelve factors A_2 . The classification of even unimodular lattices of rank twenty-four identifies the indicated type [CS99]. $\square$

Let ρ_i be the highest root of the factor $A_2^{(i)}$, normalized by $\rho_i^2 = 2$, and define

$$v = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \quad v^2 = 54. \quad (57.51)$$

The ordered frame of the exceptional reader fixes which of the twelve fibers occupies the first position; no Archimedean value intervenes in this selection. Set

$$\chi_v(x) = \langle x, v \rangle \bmod 3, \quad N_0 = \ker \chi_v, \quad N_v = N_0 + \mathbb{Z} \frac{v}{3}. \quad (57.52)$$

Lemma 57.19 (Structure of the neighbor). *The character χ_v is surjective, $[N : N_0] = 3$, $v/3 \in N_0^*$, and N_v is even and unimodular.*

Proof. For every root of a factor A_2 , the product with the highest root is ± 1 or ± 2 ; the coefficients of v are congruent to one modulo three. Therefore, no root belongs to N_0 and χ_v is nontrivial, hence surjective. The definition of the kernel gives index three and $\det N_0 = 9$. For $x \in N_0$, $\langle x, v/3 \rangle \in \mathbb{Z}$, while $(v/3)^2 = 6$. Thus the extension by $v/3$ is integral and even. Since $[N_v : N_0] = 3$, $\det N_v = 9/3^2 = 1$. $\square$

Lemma 57.20 (Minimum of the new classes). *Each coordinate of a local class appearing in $N_0 \pm v/3$ has minimum norm $2/9$. Consequently,*

$$\|x \pm v/3\|^2 \geq 12 \cdot \frac{2}{9} = \frac{8}{3} \quad (x \in N_0).$$

Proof. Writing a coordinate of A_2^* in the simple-root basis, the six relevant oriented classes are the shifts of $\pm \rho/3$ by the three discriminant classes. Completing squares in the Gram form $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ gives minimum $2/9$ in all six classes. The orthogonal sum of the twelve coordinates gives the inequality. $\square$

Theorem 57.21 (Construction of the Leech lattice). *The neighbor N_v of (57.52) is isometric to the Leech lattice:*

$$\boxed{N_v \cong \Lambda_{24}}. \quad (57.53)$$

Proof. The roots of N do not belong to N_0 , and the two new classes contain no norm-two vectors by the lemma 57.20. The lemma 57.19 proves that N_v is positive definite, even, unimodular and of rank twenty-four. The uniqueness of the lattice with these properties and no roots identifies it with Λ_{24} [CS99]. $\square$

57.4.5 Order-three isometry, orbifold and lunar module

The Coxeter rotation of each factor A_2 induces an order-three isometry g_c . Stability of the code C_W proves that $g_c N = N$. For the oriented frame fixing v , the shifts

$$y_* := \frac{g_c v - v}{3}, \quad z_* := \frac{g_c^{-1} v - v}{3} \quad (57.54)$$

belong to N_0 : their discriminant words are a pair $c_*, -c_* \in C_W$, and $\langle y_*, v \rangle = \langle z_*, v \rangle = -27$. Therefore,

$$g_c N_v = N_v. \quad (57.55)$$

The action has no fixed points in $N_v \otimes \mathbb{R}$, since each factor A_2 has the two primitive eigenvalues of order three.

Let $\widehat{g}_c$ be the standard lift to the lattice VOA $V_{\Lambda_{24}}$. Its two nontrivial eigenspaces have dimension twelve. The vacuum energy of the twisted module is therefore

$$\rho_{g_c} = \frac{1}{4 \cdot 3^2} \sum_{k=1}^2 k(3-k) \dim \mathfrak{h}_{(k)} = \frac{4}{3}. \quad (57.56)$$

The congruence $3^2 \rho_{g_c} = 12 \equiv 0 \pmod{3}$ proves that the lift is of type zero.

Theorem 57.22 (Triadic orbifold). *The cyclic orbifold of the preceding lift satisfies*

$$\text{Orb}_{\mathbb{Z}/3}(V_{\Lambda_{24}}, \widehat{g}_c) \cong V^{\natural}. \quad (57.57)$$

The dual automorphism belongs to the class 3B.

Proof. The three hypotheses that cannot be replaced by an equality of series have been verified: fixed-point-free order-three isometry, preservation of the neighbor and type zero. The cyclic orbifold theorems then identify the holomorphic extension with the lunar module and classify the dual automorphism [CLS16, ALY17, Car17]. $\square$

The order-two construction is a parallel realization. For the lift θ of negation on Λ_{24} ,

$$V^{\natural} = (V_{\Lambda_{24}})^+ \oplus (V_{\Lambda_{24}}^T)^+. \quad (57.58)$$

The order-two and order-three routes are identified by the VOA theorems; they are not causally serialized one inside the other [FLM88].

Theorem 57.23 (Publications of the lunar module). *Once $V^{\natural}$ has been constructed, one obtains*

$$\text{Aut}(V^{\natural}) \cong \mathbb{M}, \quad (57.59)$$

$$\text{ch}_{V^{\natural}}(\tau) = J(\tau) = j(\tau) - 744 = q^{-1} + 196884q + 21493760q^2 + \cdots, \quad (57.60)$$

$$T_g(\tau) = \text{Tr}_{V^{\natural}}(gq^{L_0-1}), \quad g \in \mathbb{M}. \quad (57.61)$$

The series T_g are the genus-zero modular functions of the Moonshine theorem and satisfy the replicability identities.

Proof. The first identification and the graded construction come from the FLM realization. The coordination between the graded action, the McKay–Thompson series and their modular properties is the Moonshine theorem of Conway–Norton and Borchers [FLM88, CN79, Bor92]. In particular, $(V^{\natural})_0 = \mathbb{C}\mathbf{1}$ and $(V^{\natural})_1 = 0$; the coefficient $196884 = 1 + 196883$ is the first nontrivial decomposition into dimensions of Monster representations. $\square$

Falsification criterion 57.24 (Exceptional corridor). The publication is refuted if any of the effective steps fails: (57.38), the trace (57.40), the coefficient $[q]t_3 = 54$, $C_P C_P^T = 5I_6$, $A_W^2 = -I_6$, self-duality or the enumerator of C_W , evenness or unimodularity of the gluing, absence of roots of the neighbor, its preservation by g_c , the twisted weight $4/3$ or type zero. A narrative is also refuted if it uses W_{24} to construct Λ_{24} , confuses the two sets of 243 on account of their cardinality, or makes the Monster act directly on digits and routes. The action of $\mathbb{M}$ resides in $V^{\natural}$.

Spectral–polar image and exact Riemann–Weil reduction

58.1 Starting point: the complete spectral–polar structure

The input to the resolution is not the isolated Weil form or a list of zeros. It is the complete spectral–polar structure reproduced as data in this volume. The causal chain specific to this chapter is

$$\mathfrak{G}_{\text{sp}} \xrightarrow{\text{pr}_{X,\text{sp}}} X_{\chi_{\text{sp}}} \xrightarrow{\mathcal{A}_{\text{sp}}} \mathfrak{M}_{\text{sp}} \xrightarrow{\mathcal{P}_{\text{sp}}} C_{\text{GW}}. \quad (58.1)$$

The classical codomain appears only in the last arrow; it does not retrospectively select the state, the charts, the projection, or the terminal.

The starting structure contains the object of thirteen coordinates

$$\mathfrak{G}_{\text{sp}}^{(13)} = (\mathcal{F}_{\text{sp}}, \text{Ext}_{\Gamma,\text{sp}}, \text{Rel}_{\text{sp}}, \mathcal{N}_{\text{sp}}, \text{or}_{\text{sp}}, \text{Carr}_{\text{sp}}, \text{Mem}_{\text{sp}}, \partial_{\text{sp}}, \gamma_{\text{sp}}, \Sigma_{\text{sp}}^{\text{mult}}, \text{Surv}_{\text{sp}}, \tau_{\text{sp}}^{\text{term}}, \mathcal{L}_{\text{sp}}). \quad (58.2)$$

Each coordinate enters substantively:

1. $\mathcal{F}_{\text{sp}}$ preserves the two sheets, their orientation, the residue–quotient pair, and the signature of the double publication.
2. $\text{Ext}_{\Gamma,\text{sp}}$ transports that fibre through the directed class of cutoffs and nonadic levels.
3. Rel_{sp} contains the carry cocycle, the orthogonality $P = P^* = P^2$, the two publications, and the structural complement $B_{\text{sp}}^{\text{str}} := \Xi^*(I - P)\Xi$. The identification of this complement with the Guinand–Weil form is a distinct arrow and is audited separately in this chapter.
4. $\mathcal{N}_{\text{sp}}$ records the level m , the window R , the quotient $q = 5/9$, the moments of both sheets, and the indices of the prime, gamma, and polar channels.
5. $\text{or}_{\text{sp}} = \widehat{R} = (R, K)$ keeps the two oriented charts distinct.
6. $\text{Carr}_{\text{sp}} = \text{Carr}_R$ is the operatorial carry and is not annulled when a translation is published.
7. $\text{Mem}_{\text{sp}} = (S_0, M, O_9)$ preserves the finite Stein identity and its terminal object.
8. $\partial_{\text{sp}} = (I - P)\Xi$ is the boundary absent from the $P\Xi$ compression and whose energy defines the structural complement. Its identification with the Weil form is not incorporated into this definition.
9. γ_{sp} is the cofinal route of windows, with a single regulator for all channels.
10. $\Sigma_{\text{sp}}^{\text{mult}}$ is the multisection of the two publications and their prime–gamma–polar components; it is not the signature of a single state.
11. Surv_{sp} preserves the family under $m \mapsto m + 1$ and $\mathcal{D}_R \hookrightarrow \mathcal{D}_{R'}$.
12. $\tau_{\text{sp}}^{\text{term}}$ preserves the terminal $M^{*N}S_0M^N$, which in the mirror chart is published through E_R .
13. $\mathcal{L}_{\text{sp}} = (\Xi, P, B_{\text{sp}}^{\text{str}})$ is the final reader; it is applied after antecedent, charts, memory, route, and terminal object.

The assembler and the publisher are

$$\begin{aligned}
\mathcal{A}_{\text{sp}}(\text{pr}_{X,\text{sp}} \widehat{x}) &= (\widehat{X}_{\infty}, \Gamma_9, \widehat{R}, \text{Carr}_R, S_0, M, \mathcal{O}_9, \Xi, P, \mathcal{D}_R^{\text{cof}}), \\
\mathfrak{M}_{\text{sp}} &= \text{Im } \mathcal{A}_{\text{sp}}, \\
\mathcal{P}_{\text{sp}}(\mathfrak{M}_{\text{sp}}) &= (B_{\text{sp}}^{\text{str}}, \Xi^*(I - P)\Xi, \mathcal{J}_{\text{GW}}, \Delta_{\text{bind}}, \text{cofinal prime–gamma–polar family}), \\
\Delta_{\text{bind}} &:= B_{\text{sp}}^{\text{str}} - \mathcal{J}_{\text{GW}}.
\end{aligned} \tag{58.3}$$

The finite memory of the starting object satisfies

$$S_0 = \sum_{r=0}^{N-1} M^{*r} \mathcal{O}_9^* \mathcal{O}_9 M^r + M^{*N} S_0 M^N. \tag{58.4}$$

The last summand remains visible until its extinction is proved. Therefore, replacing $\mathfrak{G}_{\text{sp}}$ with the published shadow $B_{\text{sp}}^{\text{str}}$ changes the starting object; it is not an abbreviation of the same theorem. The residual Δ_{bind} is preserved until the source–first colligation materially proves its vanishing in (58.16).

58.2 Cofinal domain and target form

For $R > 0$ is fixed

$$\mathcal{D}_R = C_c^\infty((-R/2, R/2)), \quad \widehat{\psi}(t) = \int_{\mathbb{R}} \psi(x) e^{-itx} dx,$$

and, for $\psi, \phi \in \mathcal{D}_R$,

$$h_{\psi, \phi}(z) = \widehat{\psi}(z) \overline{\widehat{\phi}(\bar{z})}, \quad \check{h}_{\psi, \phi}(a) = \langle \psi, T_a \phi \rangle.$$

The union $\bigcup_R \mathcal{D}_R = C_c^\infty(\mathbb{R})$ is exact. For every finite set of distinct complex points, the evaluations $\psi \mapsto (\widehat{\psi}(z_1), \dots, \widehat{\psi}(z_n))$ are jointly surjective when R is sufficiently large. Therefore, this cofinal class separates the evaluations required by the Weil criterion.

Lemma 58.1 (Proof of the joint separation). *The map*

$$\mathcal{D}_R \longrightarrow \mathbb{C}^n, \quad \psi \longmapsto (\widehat{\psi}(z_1), \dots, \widehat{\psi}(z_n))$$

is surjective for every set of distinct complex points $z_1, \dots, z_n$, once R contains a nonempty interval.

Proof. If the evaluation functionals were linearly dependent, there would exist $c_1, \dots, c_n$, not all zero, such that

$$\int_{\mathbb{R}} \psi(x) \left(\sum_{j=1}^n c_j e^{-iz_j x} \right) dx = 0 \quad (\psi \in \mathcal{D}_R).$$

The analytic function $\sum_j c_j e^{-iz_j x}$ would vanish at $(-R/2, R/2)$, hence throughout the plane. The independence of exponentials with distinct exponents forces $c_j = 0$ for every j , a contradiction. The functionals are independent; since the codomain is finite-dimensional, the map has rank $\mathbb{C}^n$. $\square$

58.3 The two columns and the three channels computed before the sign

Be ϑ_L a regulador comun. With

$$\ell_{p,k} = k \log p, \quad w_{p,k} = (\log p) p^{-k/2}, \quad \mu_\Gamma(a) = \frac{e^{-a/2}}{1 - e^{-2a}},$$

the uniform inclusion j_m and the sewing channel $\widehat{\delta}_{a,m}$, which satisfy, are used

$$j_m^* j_m = I, \quad \widehat{\delta}_{a,m}^* \widehat{\delta}_{b,m} = (I - T_a)^* (I - T_b).$$

The regulated columns are

$$\begin{aligned}
J_{+,m,R,L}\psi &= \left((\sqrt{\vartheta_L(\ell_{p,k})w_{p,k}} \widehat{\delta}_{\ell_{p,k},m}\psi)_{\ell_{p,k} \leq R}, \right. \\
&\quad \left. a \mapsto \sqrt{\vartheta_L(a)\mu_\Gamma(a)} \widehat{\delta}_{a,m}\psi, \zeta_+ b_+(\psi) \right), \\
J_{-,m,R,L}\psi &= \left((\sqrt{2\vartheta_L(\ell_{p,k})w_{p,k}} j_m\psi)_{\ell_{p,k} \leq R}, \sqrt{-c_\Gamma} j_m\psi, \zeta_- b_-(\psi) \right),
\end{aligned}$$

where $c_\Gamma = \psi_\Gamma(1/4) - \log \pi < 0$ and $b_\pm = (\widehat{\psi}(i/2) \pm \widehat{\psi}(-i/2))/\sqrt{2}$.

A term-by-term expansion, without using zeros or the sought sign, gives

$$\begin{aligned}
&\langle J_{+,m,R,L}\psi, J_{+,m,R,L}\phi \rangle - \langle J_{-,m,R,L}\psi, J_{-,m,R,L}\phi \rangle = \mathcal{J}_{\text{GW},L}(\psi, \phi), \\
\mathcal{J}_{\text{GW},L}(\psi, \phi) &= h_{\psi,\phi}(i/2) + h_{\psi,\phi}(-i/2) + c_\Gamma \check{h}_{\psi,\phi}(0) \\
&\quad + \int_0^\infty \vartheta_L(a)\mu_\Gamma(a)[2\check{h}(0) - \check{h}(a) - \check{h}(-a)] da \\
&\quad - \sum_{p,k} \vartheta_L(\ell_{p,k})w_{p,k}[\check{h}(\ell_{p,k}) + \check{h}(-\ell_{p,k})].
\end{aligned}$$

The truncation of both columns at $\ell_{p,k} \leq R$ is mandatory, not notational. If $\psi, \phi \in \mathcal{D}_R$, then $\check{h}_{\psi,\phi}(\pm\ell) = 0$ for $\ell > R$; the contribution of each omitted pair of coordinates to the *difference* of Grams is therefore zero. The untruncated negative column, by contrast, would have infinite norm because $\sum_p (\log p)p^{-1/2} = \infty$. In the finite set $\ell_{p,k} \leq R$ the prime coordinates converge individually when $L \rightarrow \infty$. In the gamma channel, the integrand is $O(a)$ at zero and $\mu_\Gamma(a) = O(e^{-a/2})$ at infinity, so convergence is also in norm. We therefore define

$$J_{\pm,m,R} := \lim_{L \rightarrow \infty} J_{\pm,m,R,L} \quad \text{in their truncated-coordinate Hilbert spaces,} \quad (58.5)$$

and we suppress m when it remains fixed. By dominated convergence,

$$\mathcal{J}_{\text{GW},L} \longrightarrow \mathcal{J}_{\text{GW}} \quad (L \rightarrow \infty),$$

and the identities are natural under $m \mapsto m+1$ and $R \leq R'$.

The reduced columns $J_{+,R}$ and $J_{-,R}$ are now material operators from $\mathcal{D}_R$ to Hilbert. The explicit expansion proves, without consulting zeros or sign,

$$\langle J_{+,R}f, J_{+,R}g \rangle - \langle J_{-,R}f, J_{-,R}g \rangle = \mathcal{J}_{\text{pr}}(f, g) + \mathcal{J}_\Gamma(f, g) + \mathcal{J}_{\text{pol}}(f, g) = \mathcal{J}_{\text{GW}}(f, g). \quad (58.6)$$

The clocks $\ell_{p,k}$ produce the prime summand; the integral with μ_Γ and $c_\Gamma \check{h}(0)$, the gamma summand; and

$$\langle b_+(f), b_+(g) \rangle - \langle b_-(f), b_-(g) \rangle = h_{f,g}(i/2) + h_{f,g}(-i/2)$$

produce the polar term. This part of the calculation is material and independent of the structural bridge.

58.4 Source–first colligation of the spectral–polar image

We first fix a nonadic level m ; this index is suppressed in $J_{\pm,R}$, E_R and Σ_R until the encoder is constructed. The decomposition of the explicit codomains of (58.5) is

$$I_+ = \{\text{pr}, \Gamma, \text{pol}\}, \quad I_-(R) = \{\Gamma 0, \text{pol}\} \sqcup \{(p, k) : \ell_{p,k} \leq R\},$$

$$\mathcal{H}_{+,R}^{\text{amb}} = \bigoplus_{i \in I_+} H_{+,i,R}, \quad \mathcal{H}_{-,R}^{\text{amb}} = \bigoplus_{j \in I_-(R)} H_{-,j,R}.$$

We do not identify a reachable subspace with a sum of its coordinate projections. With the ambient inclusions understood, we define

$$H_{+,R} := \overline{\text{Ran } J_{+,R}}, \quad H_{-,R} := \overline{\text{Ran } J_{-,R}},$$

as closed subspaces of the two ambient Hilbert spaces. The structure $\mathfrak{G}_{\text{sp}}$ contains, as part of its mathematical data and before the classical expression, an antispectral state space $\mathcal{H}_R^{\text{sh}}$ and the source–first isometric row

$$\Sigma_R = \begin{pmatrix} \mathcal{K}_R^{\text{src}} \\ \mathcal{F}_R^{\text{src}} \end{pmatrix} : H_{+,R} \longrightarrow H_{-,R} \oplus \mathcal{H}_R^{\text{sh}}, \quad \Sigma_R^* \Sigma_R = (\mathcal{K}_R^{\text{src}})^* \mathcal{K}_R^{\text{src}} + (\mathcal{F}_R^{\text{src}})^* \mathcal{F}_R^{\text{src}} = I_{H_{+,R}}. \quad (58.7)$$

Its reachable action is fixed on the entire positive column by the relations of the two publications that form part of Rel_{sp} . To prevent any substitution of the reader, we denote by $(\mathcal{X}_R^{\text{str}}, \Xi_R^{\text{str}}, P_R^{\text{str}})$ the restriction to $\mathcal{D}_R$ of the structural reader of (58.2). Its two moments are, literally,

$$\begin{aligned} \langle \Xi_R^{\text{str}} f, \Xi_R^{\text{str}} g \rangle &= \langle J_{+,R} f, J_{+,R} g \rangle, \\ \langle P_R^{\text{str}} \Xi_R^{\text{str}} f, P_R^{\text{str}} \Xi_R^{\text{str}} g \rangle &= \langle J_{-,R} f, J_{-,R} g \rangle, \quad (P_R^{\text{str}})^* = (P_R^{\text{str}})^2 = P_R^{\text{str}}. \end{aligned} \quad (58.8)$$

The boundary coordinate comes with the structural isometry.

$$j_R^{\text{sh}} : \overline{\text{Ran}((I - P_R^{\text{str}})\Xi_R^{\text{str}})} \longrightarrow \mathcal{H}_R^{\text{sh}}.$$

By definition of the source–first row, not by a subsequent identification,

$$E_R := \mathcal{F}_R^{\text{src}} J_{+,R} = j_R^{\text{sh}}(I - P_R^{\text{str}})\Xi_R^{\text{str}} : \mathcal{D}_R \longrightarrow \mathcal{H}_R^{\text{sh}}, \quad \Omega_{0,R} := \mathcal{K}_R^{\text{src}} J_{+,R} - J_{-,R}. \quad (58.9)$$

There is no colligation by channel: the same row simultaneously receives the prime–power clocks, the gamma integral, the scalar mode, and the two polar lines. In particular, it preserves all cross-products that arise upon polarizing (58.6). With the ambient projections $\pi_j^- : \mathcal{H}_{-,R}^{\text{amb}} \rightarrow H_{-,j,R}$, the correctly typed output rows are

$$\mathcal{K}_{j,R}^{\text{src}} := \pi_j^- \mathcal{K}_R^{\text{src}} : H_{+,R} \longrightarrow H_{-,j,R}, \quad \mathcal{F}_R^{\text{src}} : H_{+,R} \longrightarrow \mathcal{H}_R^{\text{sh}}. \quad (58.10)$$

Thus, $\mathcal{K}_R^{\text{src}}$ is not applied to a positive coordinate that might not by itself belong to $H_{+,R}$. All cross-terms between channels remain within $(\mathcal{K}_R^{\text{src}})^* \mathcal{K}_R^{\text{src}}$ and $(\mathcal{F}_R^{\text{src}})^* \mathcal{F}_R^{\text{src}}$; they are not orthogonalized before the energy identity.

The survival coordinate of $\mathfrak{G}_{\text{sp}}$ provides residual spaces $\mathcal{R}_{n,R}^{\text{sh}}$, continuations $\Omega_{n,R} : \mathcal{D}_R \rightarrow \mathcal{R}_{n,R}^{\text{sh}}$, and antispectral isometries $V_{n,R} : \mathcal{R}_{n+1,R}^{\text{sh}} \rightarrow \mathcal{R}_{n,R}^{\text{sh}}$. Its first space and its first operator are, by typed definition,

$$\mathcal{R}_{0,R}^{\text{sh}} := H_{-,R}, \quad \Omega_{0,R} := \mathcal{K}_R^{\text{src}} J_{+,R} - J_{-,R}.$$

Its full cycle recurrence is

$$q = \frac{5}{9}, \quad \Omega_{n,R} = q V_{n,R} \Omega_{n+1,R}, \quad V_{n,R}^* V_{n,R} = I, \quad \sup_{n \geq 0} \|\Omega_{n,R} f\| < \infty \quad (f \in \mathcal{D}_R). \quad (58.11)$$

This recurrence belongs to the enriched colligation: it transports the output difference, not the Weil form or its sign.

Proposition 58.2 (Coinductive Cancellation of Output Difference). *For every $R > 0$ and every $f \in \mathcal{D}_R$,*

$$\Omega_{n,R} f = 0 \quad (n \geq 0), \quad \boxed{\mathcal{K}_R^{\text{src}} J_{+,R} = J_{-,R}}. \quad (58.12)$$

Proof. Iterating (58.11) N times and using that each $V_{n,R}$ is isometric gives

$$\|\Omega_{n,R}f\| = q^N \|\Omega_{n+N,R}f\| \leq q^N \sup_{j \geq n} \|\Omega_{j,R}f\|.$$

The right-hand member converges to zero because $0 < q = 5/9 < 1$. Therefore $\Omega_{n,R}f = 0$; the case $n = 0$ and (58.9) produce the framed identity. $\square$

The row-by-row action of the output is no longer implicit. The projections are the restrictions to $H_{-,R}$ of the projections in the ambient Hilbert space, and

$$\begin{aligned} \mathcal{K}_{\Gamma 0,R}^{\text{src}} J_{+,R}f &= \sqrt{-c_\Gamma} j_m f, \\ \mathcal{K}_{(p,k),R}^{\text{src}} J_{+,R}f &= \sqrt{2w_{p,k}} j_m f \quad (\ell_{p,k} \leq R), \\ \mathcal{K}_{\text{pol},R}^{\text{src}} J_{+,R}f &= \zeta_- b_-(f). \end{aligned} \quad (58.13)$$

These rows exhaust $J_{-,R}$, while

$$\mathcal{F}_R^{\text{src}} J_{+,R}f = E_R f \quad (58.14)$$

preserves, with all its crossings, the boundary coordinate. Evaluating (58.7) at $J_{+,R}f$ and $J_{+,R}g$, and using (58.12), produces the energy identity prior to the sign

$$\langle J_{+,R}f, J_{+,R}g \rangle = \langle J_{-,R}f, J_{-,R}g \rangle + \langle E_R f, E_R g \rangle. \quad (58.15)$$

Comparison with the independent calculation (58.6) yields, without introducing a root of the target form,

$$E_R^* E_R = \mathcal{J}_{\text{GW}}|_{\mathcal{D}_R}. \quad (58.16)$$

In particular, $\Delta_{\text{bind},R} := E_R^* E_R - \mathcal{J}_{\text{GW}}|_{\mathcal{D}_R} = 0$; this annihilation is the output of (58.7)–(58.15), not a premise inserted into the publisher.

58.5 Structural complement, mirror sheet and terminal telescopia

The state $E_R = \mathcal{F}_R^{\text{src}} J_{+,R}$ constructed by the source–first row is the boundary coordinate of $\mathfrak{G}_{\text{sp}}$. We define its structural form by

$$\mathcal{B}_{\text{sp},R}^{\text{str}} := E_R^* E_R = \mathcal{J}_{\text{GW}}|_{\mathcal{D}_R}, \quad (58.17)$$

where the second equality is (58.16), already deduced from the colligation and from the generator-by-generator expansion.

Let P_+^{sh} and P_-^{sh} be the complementary orthogonal projectors of the two sheets. Fix

$$q = \frac{5}{9}, \quad A = P_+^{\text{sh}} + q P_-^{\text{sh}}, \quad D = \sqrt{1 - q^2} P_-^{\text{sh}}. \quad (58.18)$$

Orthogonality gives the exact identity

$$A^* A + D^* D = P_+^{\text{sh}} + q^2 P_-^{\text{sh}} + (1 - q^2) P_-^{\text{sh}} = I. \quad (58.19)$$

The nine phases coordinate a single memory-enabled rotation:

$$M_9 = \Phi A_8 \cdots A_0 = q V_9, \quad V_9^* V_9 = I. \quad (58.20)$$

The contraction per turn is q , not q^9 ; the internal phases preserve order and carry.

The colligation types $E_R : \mathcal{D}_R \rightarrow \text{Ran } P_-^{\text{sh}}$. Therefore,

$$P_+^{\text{sh}} E_R = 0, \quad A E_R = q E_R, \quad E_R^* E_R = \mathcal{B}_{\text{sp},R}^{\text{str}} = \mathcal{J}_{\text{GW}}|_{\mathcal{D}_R}. \quad (58.21)$$

Proposition 58.3 (Finite Telescopy with Conserved Terminal). *for each $N \geq 1$,*

$$\mathcal{B}_{\text{sp},R}^{\text{str}} = \sum_{n=0}^{N-1} (D A^n E_R)^* (D A^n E_R) + (A^N E_R)^* (A^N E_R). \quad (58.22)$$

Equivalently, evaluated at $f, g \in \mathcal{D}_R$, the finite identity of the colligation is

$$\begin{aligned} & \langle J_{+,R} f, J_{+,R} g \rangle - \langle J_{-,R} f, J_{-,R} g \rangle \\ &= \sum_{n=0}^{N-1} \langle D A^n E_R f, D A^n E_R g \rangle + \langle A^N E_R f, A^N E_R g \rangle. \end{aligned} \quad (58.23)$$

The last term satisfies

$$(A^N E_R)^* (A^N E_R) = q^{2N} \mathcal{B}_{\text{sp},R}^{\text{str}} \xrightarrow{N \rightarrow \infty} 0. \quad (58.24)$$

Proof. Applying (58.19) to $A^n E_R$,

$$(A^n E_R)^* (A^n E_R) = (D A^n E_R)^* (D A^n E_R) + (A^{n+1} E_R)^* (A^{n+1} E_R).$$

The sum from $n = 0$ to $N - 1$ cancels only the intermediate terms and preserves $(A^N E_R)^* (A^N E_R)$. By (58.21), $A^N E_R = q^N E_R$; hence the terminal object is $q^{2N} E_R^* E_R = q^{2N} \mathcal{B}_{\text{sp},R}^{\text{str}}$. Since $0 < q = 5/9 < 1$, it converges geometrically to zero. $\square$

Only after proving (58.24) is defined

$$\mathcal{L}_R f = (D A^n E_R f)_{n \geq 0}. \quad (58.25)$$

The limit of finite telescoping gives

$$\mathcal{L}_R^* \mathcal{L}_R = \mathcal{B}_{\text{sp},R}^{\text{str}} = \mathcal{J}_{\text{GW}}|_{\mathcal{D}_R} \succeq 0. \quad (58.26)$$

58.6 Common encoder, projection, and explicit double publication

Fix $m \geq 1$ and two orthonormal vectors $u_m, \eta_m \in \mathbb{C}^{9^m}$, where u_m is the uniform mode and η_m the carry mode. The coder space and projection are

$$\mathcal{X}_{m,R}^{\text{sp}} := \mathbb{C}^{9^m} \widehat{\otimes} (H_{-,R} \oplus \mathcal{H}_R^{\text{sh}}), \quad P_{m,R} := |u_m\rangle\langle u_m| \otimes I_{H_{-,R} \oplus \mathcal{H}_R^{\text{sh}}}, \quad Q_{m,R} := I - P_{m,R}. \quad (58.27)$$

The complete action of the five-channel encoder is

$$\begin{aligned} \mathfrak{C}_{m,R} f &:= u_m \otimes (J_{-,R} f, 0) + \eta_m \otimes (0, E_R f), \\ \Xi_{m,R} f &:= \mathfrak{C}_{m,R} f. \end{aligned} \quad (58.28)$$

No root of $\mathcal{J}_{\text{GW}}$ is taken: the two inputs are the output and state of the source–first colligation already defined in (58.9).

Proposition 58.4 (The two moments arise from the action of the encoder). *For every $f, g \in \mathcal{D}_R$,*

$$\begin{aligned}
\langle \mathfrak{C}_{m,R}f, \mathfrak{C}_{m,R}g \rangle &= \langle J_{+,R}f, J_{+,R}g \rangle, \\
\langle P_{m,R}\mathfrak{C}_{m,R}f, P_{m,R}\mathfrak{C}_{m,R}g \rangle &= \langle J_{-,R}f, J_{-,R}g \rangle, \\
\langle Q_{m,R}\Xi_{m,R}f, Q_{m,R}\Xi_{m,R}g \rangle &= \mathcal{J}_{\text{GW}}(f, g).
\end{aligned} \tag{58.29}$$

Proof. The orthogonality $u_m \perp \eta_m$ and (58.15) give

$$\langle \mathfrak{C}_{m,R}f, \mathfrak{C}_{m,R}g \rangle = \langle J_{-,R}f, J_{-,R}g \rangle + \langle E_Rf, E_Rg \rangle = \langle J_{+,R}f, J_{+,R}g \rangle.$$

The projection $P_{m,R}$ preserves the uniform summand and eliminates the carry summand; $Q_{m,R}$ does the reverse. The other two identities follow from (58.16). $\square$

The newly written encoder is an explicit representative of the initial structural reader, not a substitute reader. In the reachable subspaces one defines

$$\begin{aligned}
\mathcal{U}_{m,R}^{\text{str}}(P_R^{\text{str}}\Xi_R^{\text{str}}f) &:= u_m \otimes (J_{-,R}f, 0), \\
\mathcal{U}_{m,R}^{\text{str}}((I - P_R^{\text{str}})\Xi_R^{\text{str}}f) &:= \eta_m \otimes (0, E_Rf).
\end{aligned} \tag{58.30}$$

The two sources are orthogonal. The first Gram equality of each source is (58.8); for the second one uses $E_R = J_R^{\text{sh}}(I - P_R^{\text{str}})\Xi_R^{\text{str}}$. Hence the two rules are well defined, are isometric, and extend to the orthogonal sum of their closures. In that reachable closure they satisfy

$$\mathcal{U}_{m,R}^{\text{str}}\Xi_R^{\text{str}} = \Xi_{m,R}, \quad \mathcal{U}_{m,R}^{\text{str}}P_R^{\text{str}} = P_{m,R}\mathcal{U}_{m,R}^{\text{str}}, \quad (\Xi_R^{\text{str}})^*(I - P_R^{\text{str}})\Xi_R^{\text{str}} = E_R^*E_R. \tag{58.31}$$

Thus, $B_{\text{sp},R}^{\text{str}}$, the complement of the starting structure, is exactly the complement of the constructed coder; the object has not been changed in passing from (58.2) to (58.28).

The rule

$$V_{+,m,R}^0(J_{+,R}f) := \Xi_{m,R}f \tag{58.32}$$

is well defined and is isometric by the first equality of (58.29); it extends uniquely to an isometry $V_{+,m,R} : H_{+,R} \rightarrow \mathcal{X}_{m,R}^{\text{sp}}$. The negative publication is the isometry

$$V_{-,m,R} : H_{-,R} \longrightarrow \mathcal{X}_{m,R}^{\text{sp}}, \quad V_{-,m,R}y = u_m \otimes (y, 0). \tag{58.33}$$

By construction,

$$\boxed{V_{+,m,R}J_{+,R} = \Xi_{m,R}, \quad V_{-,m,R}J_{-,R} = P_{m,R}\Xi_{m,R}.} \tag{58.34}$$

The complement preserves exactly the antiespecular state and its column of remark:

$$\boxed{\Xi_{m,R}^*Q_{m,R}\Xi_{m,R} = E_R^*E_R = \mathcal{L}_R^*\mathcal{L}_R = \mathcal{J}_{\text{GW}}|_{\mathcal{D}_R}.} \tag{58.35}$$

Proposition 58.5 (Induced connector and source–first coincidence). *The operator*

$$\mathcal{K}_R := V_{-,m,R}^*P_{m,R}V_{+,m,R} : H_{+,R} \longrightarrow H_{-,R} \tag{58.36}$$

is contractive, coincides with $\mathcal{K}_R^{\text{src}}$ throughout $H_{+,R}$, and satisfies

$$\|\mathcal{K}_R\| \leq 1, \quad \mathcal{K}_RJ_{+,R} = J_{-,R}. \tag{58.37}$$

Proof. The two charts are isometries and $P_{m,R}$ is an orthogonal projection, hence $\|\mathcal{K}_R\| \leq 1$. Over the dense range of $J_{+,R}$,

$$\mathcal{K}_RJ_{+,R}f = V_{-,m,R}^*P_{m,R}\Xi_{m,R}f = V_{-,m,R}^*V_{-,m,R}J_{-,R}f = J_{-,R}f.$$

The same identity holds for $\mathcal{K}_R^{\text{src}}$ by (58.12); continuity and density prove that the two operators coincide. $\square$

The Gram difference is thus published in all its coincident realizations:

$$\boxed{\begin{aligned} J_{+,R}^* J_{+,R} - J_{-,R}^* J_{-,R} &= \Xi_{m,R}^* Q_{m,R} \Xi_{m,R} \\ &= E_R^* E_R \\ &= \mathcal{L}_R^* \mathcal{L}_R \\ &= \mathcal{B}_{\text{sp},R}^{\text{str}} = \mathcal{J}_{\text{GW}}. \end{aligned}} \quad (58.38)$$

58.7 Cofinal Naturality and Reversible Weil Criterion

For $R \leq R'$, let $i_{R,R'} : \mathcal{D}_R \hookrightarrow \mathcal{D}_{R'}$. The individual columns do not form an isometric system: when a new length $R < \ell_{p,k} \leq R'$ appears, both acquire the same positive energy. For $f, g \in \mathcal{D}_R$, separation of the supports gives exactly

$$\langle (I - T_{\ell_{p,k}})f, (I - T_{\ell_{p,k}})g \rangle = 2\langle f, g \rangle = \langle \sqrt{2}j_m f, \sqrt{2}j_m g \rangle. \quad (58.39)$$

Therefore the prime increment of $J_{+,R'}^* J_{+,R'}$ coincides with that of $J_{-,R'}^* J_{-,R'}$ and cancels in the difference, but no false isometry between the columns is asserted. The correct cofinal compatibility is that of the forms:

$$\boxed{\mathcal{J}_{\text{GW},R} = i_{R,R'}^* \mathcal{J}_{\text{GW},R'} i_{R,R'}, \quad \mathcal{B}_{\text{sp},R}^{\text{str}} = i_{R,R'}^* \mathcal{B}_{\text{sp},R'}^{\text{str}} i_{R,R'}} \quad (58.40)$$

Since $E_R^* E_R = \mathcal{L}_R^* \mathcal{L}_R = \mathcal{J}_{\text{GW},R}$, the minimal Kolmogorov realizations do admit unique isometries over their reachable ranges, defined by

$$\iota_{R,R'}^E(E_R f) := E_{R'} f, \quad \iota_{R,R'}^L(\mathcal{L}_R f) := \mathcal{L}_{R'} f \quad (f \in \mathcal{D}_R). \quad (58.41)$$

These two rules are well defined precisely by (58.40); they are not extended by fiat to $J_{\pm,R}$, $\Xi_{m,R}$, or $\mathcal{K}_R$. At fixed window, the nonadic refinement $m \mapsto m+1$ remains within the structural isometries that transport j_m , $\widehat{\delta}_{a,m}$, u_m , and η_m .

To avoid compressing the last step, we also fix its objects and normalizations. Let

$$\xi(s) := \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$$

and let $\mathcal{Z}$ be the multiset of its zeros, each zero repeated according to its multiplicity. The functional equation and the reality of the coefficients preserve $\mathcal{Z}$ under $\rho \mapsto 1-\rho$, $\rho \mapsto \bar{\rho}$ and $\rho \mapsto 1-\bar{\rho}$. With our Fourier convention we define

$$\Phi_f(s) := \widehat{f}\left(i\left(s - \frac{1}{2}\right)\right), \quad \widetilde{f}(x) := \overline{f(-x)}. \quad (58.42)$$

The complete explicit formula computed in (58.6), including the gamma and polar terms that remove poles and trivial zeros, has the spectral form

$$\mathcal{J}_{\text{GW}}(f, g) = \sum_{\rho \in \mathcal{Z}} \Phi_f(\rho) \overline{\Phi_g(1-\bar{\rho})}. \quad (58.43)$$

The sum counts multiplicities. It converges absolutely: for $f, g \in C_c^\infty(\mathbb{R})$, Paley–Wiener gives decay faster than any power in vertical strips, whereas the number of zeros with $|\text{Im } \rho| \leq T$ is $O(T \log T)$.

Proposition 58.6 (Reversible Weil Criterion with Full Quantifiers). *Under the normalization (58.42) and counting all nontrivial zeros with their multiplicity,*

$$\left[\mathcal{J}_{\text{GW}}(f, f) \geq 0 \text{ for every } f \in C_c^\infty(\mathbb{R}; \mathbb{C}) \right] \iff \left[\operatorname{Re} \rho = \frac{1}{2} \text{ for every } \rho \in \mathcal{Z} \right]. \quad (58.44)$$

The converse implication is not obtained by interpolating an increasing number of zeros: that operation would provide no uniform control over the tail. The complete classical Weil criterion in Bombieri’s formulation is used; its test space and normalization are transferred here without abbreviation.¹

Proof. We materially write out the change of coordinates identifying both criteria. For $f \in C_c^\infty(\mathbb{R})$, let

$$(\mathcal{M}f)(x) := x^{-1/2} f(\log x), \quad x > 0. \quad (58.45)$$

The map

$$\mathcal{M} : C_c^\infty(\mathbb{R}) \longrightarrow C_c^\infty(0, \infty)$$

is bijective, with inverse

$$(\mathcal{M}^{-1}g)(u) = e^{u/2} g(e^u). \quad (58.46)$$

If $\widehat{g}_{\text{Mel}}(s) := \int_0^\infty g(x)x^{s-1} dx$, the change $x = e^u$ gives, with no residual factor,

$$\begin{aligned} \widehat{\mathcal{M}f}_{\text{Mel}}(s) &= \int_{\mathbb{R}} f(u)e^{(s-1/2)u} du \\ &= \widehat{f}\left(i\left(s - \frac{1}{2}\right)\right) = \Phi_f(s). \end{aligned} \quad (58.47)$$

Moreover,

$$\widehat{\mathcal{M}f}_{\text{Mel}}(1 - \rho) = \overline{\widehat{\mathcal{M}f}_{\text{Mel}}(1 - \bar{\rho})} = \overline{\Phi_f(1 - \bar{\rho})}. \quad (58.48)$$

Therefore the Weil–Bombieri criterion functional is exactly

$$\sum_{\rho \in \mathcal{Z}} \widehat{\mathcal{M}f}_{\text{Mel}}(\rho) \overline{\widehat{\mathcal{M}f}_{\text{Mel}}(1 - \rho)} = \sum_{\rho \in \mathcal{Z}} \Phi_f(\rho) \overline{\Phi_f(1 - \bar{\rho})} = \mathcal{J}_{\text{GW}}(f, f). \quad (58.49)$$

The class of tests is not changed, no multiplicity is lost, and the set of zeros is not truncated, because $\mathcal{M}$ is a bijection. The cited classical criterion applied to (58.49) gives precisely the equivalence (58.44). In the forward direction, if $\operatorname{Re} \rho = 1/2$, the same identity reduces explicitly to

$$\mathcal{J}_{\text{GW}}(f, f) = \sum_{\rho \in \mathcal{Z}} |\Phi_f(\rho)|^2 \geq 0.$$

□

The equality $\bigcup_R \mathcal{D}_R = C_c^\infty(\mathbb{R})$ is exact; each function employed in the argument belongs to some window, and cofinal naturality transports there the positivity already proved.

Theorem 58.7 (Riemann–Weil Resolution from the Spectral–Polar Image). *Having fixed the complete structure $\mathfrak{G}_{\text{sp}}$, the complete Guinand–Weil form is positive on the cofinal separating class. Consequently, every nontrivial zero ρ of the Riemann zeta function satisfies*

$$\operatorname{Re} \rho = \frac{1}{2}.$$

¹E. Bombieri, *Remarks on Weil’s quadratic functional in the theory of prime numbers, I*, Rendiconti Lincei, Matematica e Applicazioni, ser. 9, vol. 11 (2000), pp. 183–233. The criterion established there states that the quadratic functional of the explicit formula is positive semidefinite on all $C_c^\infty(0, \infty)$ if and only if the Riemann Hypothesis holds.

Proof. The prime–gamma–polar expansion identifies the difference between the two columns with $\mathcal{I}_{\text{GW}}$ using a single regulator and removes it by dominated convergence. The antispecular recurrence (58.11) annihilates the output difference before forming an inequality; therefore the isometry Σ_R gives (58.15). Comparing that identity with the calculated expansion produces $E_R^* E_R = \mathcal{I}_{\text{GW}}$. The local identity $A^* A + D^* D = I$ is iterated without erasing the terminal and gives (58.22); the terminal is exactly $q^{2N} E_R^* E_R$ and vanishes only in the limit. Hence $\mathcal{L}_R^* \mathcal{L}_R = \mathcal{I}_{\text{GW}} \succeq 0$. The action (58.28) then constructs the encoder, its two moments, the projection, and the two publications, and (58.38) identifies all realizations of the same square. Naturality transports that identity to the entire cofinal union. Finally, (58.44) concludes the assertion. $\square$

58.8 Subsequent, non-governing local Redheffer control

The following development verifies that local passivity can be obtained from the carry without using the Weil form. It is a noncircularity control and not a premise of theorem 58.7. This truncated transfer is not the complete colligation Σ_R of (58.7).

For $N = 9^m$, be $U_{a,N}$ the odometro unitario

$$U_{a,N}(e_j \otimes f) = e_{j+1} \otimes f \quad (j < N-1), \quad U_{a,N}(e_{N-1} \otimes f) = e_0 \otimes T_a f.$$

With $P_N = |u_N\rangle\langle u_N| \otimes I$, $Q_N = I - P_N$, $A_{a,N} = P_N U_{a,N} P_N$, and $C_{a,N} = Q_N U_{a,N} P_N$,

$$A_{a,N}^* A_{a,N} + C_{a,N}^* C_{a,N} = P_N, \quad C_{a,N}^* C_{a,N} = \frac{N-1}{N^2} (2I - T_a - T_a^*). \quad (58.50)$$

For $0 < s < 1$, the transfer

$$\mathcal{R}_{a,N}(s) = (sI - A_{a,N})(I - sA_{a,N})^{-1} \quad (58.51)$$

satisfies

$$I - \mathcal{R}_{a,N}(s)^* \mathcal{R}_{a,N}(s) = (1 - s^2)(I - sA_{a,N}^*)^{-1} \cdot C_{a,N}^* C_{a,N} (I - sA_{a,N})^{-1} \succeq 0. \quad (58.52)$$

With

$$s_{p,N} = \frac{Np^{-1/2}}{1 + (N-1)p^{-1/2}}, \quad s_\Gamma(a) = e^{-a/2},$$

the sum over prime clocks and the direct gamma integral have norm at most one. Explicitly,

$$\mathcal{R}_{N,R} = \bigoplus_{\log p \leq R} \mathcal{R}_{\log p, N}(s_{p,N}) \oplus \int_0^\infty \oplus \mathcal{R}_{a,N}(s_\Gamma(a)) da, \quad \|\mathcal{R}_{N,R}\| \leq 1. \quad (58.53)$$

The finite chart of roles

$$\mathcal{H}_+ = \text{span}\{f_3, f_6, f_9\}, \quad \mathcal{H}_- = \text{span}\{f_4, f_8\}, \quad C_\perp = |f_4\rangle\langle f_3| + |f_8\rangle\langle f_9|$$

satisfies $I - C_\perp^* C_\perp = P_6$. The two-way energy charts fix

$$\begin{aligned} \mathcal{K}_q &= K_2 \otimes (\mathbb{C}^9)^{\otimes q}, \\ \mathbf{B}_{\pm,q} : \mathcal{M}_{\pm,q,R} &\longrightarrow \mathcal{K}_q \widehat{\otimes} \mathcal{H}_\pm \widehat{\otimes} \mathcal{H}_{\mathbb{C}_R}, \\ \mathbf{B}_{\pm,q}^* \mathbf{B}_{\pm,q} &= \mathbf{G}_{\pm,q}, \quad \mathbf{B}_{\pm,q}^\# = \mathbf{G}_{\pm,q}^{-1} \mathbf{B}_{\pm,q}^*. \end{aligned} \quad (58.54)$$

The transfer

$$\mathcal{C}_{q,R}^{\text{src}} = \mathbf{B}_{-,q}^\# (I \otimes C_\perp \otimes \mathcal{R}_{N,R}) \mathbf{B}_{+,q} \quad (58.55)$$

has defect

$$\Delta_{q,R}^{\text{src}} = I \otimes [P_6 \otimes I + (P_3 + P_9) \otimes (I - \mathcal{R}_{N,R}^* \mathcal{R}_{N,R})] \succeq 0, \quad (58.56)$$

$$I - (\mathcal{C}_{q,R}^{\text{src}})^{\#} \mathcal{C}_{q,R}^{\text{src}} = \mathbf{B}_{+,q}^{\#} \Delta_{q,R}^{\text{src}} \mathbf{B}_{+,q} \succeq 0. \quad (58.57)$$

This control preserves route, clock, role, carry, and channels before forming a difference. It is not identified with $\mathcal{K}_R$ and does not replace the explicit common publication of (58.34); its local scope therefore does not reopen the cofinal conclusion.

58.9 Falsifiers of an ablated shadow

Falsification criterion 58.8 (Substitutions that change the starting object). The preceding argument does not transfer to a shadow that eliminates one of the thirteen coordinates of (58.2), constructs J_+ and J_- from different antecedents, replaces an orthogonal projection with a non-self-adjoint idempotent, omits a channel, uses incompatible regulators, chooses P_R after the sign, erases the terminal object before the limit, or reduces the class to a noncofinal family.

Proof. Each operation respectively violates the domain of $\mathfrak{G}_{\text{sp}}$, the proper definition of (58.32), the orthogonal square, the functional identity, the causal direction, finite telescoping, or the reversibility criterion. The falsifier affects the replaced shadow; it does not dislodge the complete structural image used in theorem 58.7. $\square$

Provenance and evidentiary force. The spectral–polar restriction, source–first colligation, its complement and the mirror sheet are PRE-EXISTING_AUTHORIAL_ARCHITECTURE. The coinductive vanishing of Ω , the finite identity with terminal and the action encoder (58.28) are a NEW_FORMALIZATION of that complete structure. The Guinand–Weil identification and cofinal closure are a RECOVERED_RESULT. Equations (58.7)–(58.38) exhibit the object, its domains, its action, the terminal and the limit without introducing an abstract root of the form; their strength is *proved with explicit starting structure*.

PART VIII

Theorem-based synthesis

Arithmetic before Archimedean realization

59.1 VI. The discrete structure of the continuum

59.1.1 The continuum is a realization of compatible cylinders

The inverse limit X_∞ preserves all compatible histories. Its half-closed cylinders determine a topology; refinement determines shape; the cocycle determines holonomy; orientation determines boundaries; and the inductive algebra determines readers. The five internal spectral, solenoidal, gauge, coherent and determinantal aspects emerge jointly from this construction. Separating them would destroy the object giving them their common naturality.

Dual projection is a pair of representations of the same algebra:

$$\rho_{\text{arq}} : \mathfrak{S}_{\text{cyl}} \rightarrow \mathcal{A}_{\mathbb{R}}, \quad \rho_{\text{exc}} : \mathfrak{S}_{\text{cyl}} \rightarrow \mathcal{A}_{\text{exc}}.$$

The first contracts compatible cylinders towards the Archimedean reading. The second preserves boundary incidence towards Witt–Golay–Leech– $V^{\mathfrak{h}}$ –Monster. $\mathbb{R}$ and the exceptional branch are not two universes of origin: they are two representational quotients of a richer source.

Abelianization necessarily forgets the electronic commutator and part of the memory record. The real line can therefore be complete as a scalar continuum while remaining insufficient as a description of genealogy. Holonomic arithmetic precedes $\mathbb{R}$ just as a history precedes one of its coordinates.

59.1.2 Predefinition of compatible futures

Let

$$\mathcal{S}_n(x) = \{y \in X_{n+1} : r_{n+1,n}(y) = x \text{ and } y \text{ is admissible}\}.$$

The coinductive tree contains possible extensions. The Sturmian law guarantees a unique right-special word per length; TRIT types its curvature and TPK selects the continuation while preserving the complementary branch at the boundary. Predefinition does not mean that an external numeral has been stored. It means that every finite output is fixed by a cylinder and that the system of cylinders is compatible at every depth.

This makes precise the expression ‘choice of survival futures’. The future is not guessed from a measurement: it is selected within the already typed set of extensions preserving the relations of the state. Past, present and future are functions of the return operator, threshold and opening, not mere verbal labels.

59.2 VII. Kernel theorem and consequences

59.2.1 Minimal statement

We define the kernel object

$$\mathfrak{S}_{\text{HMT}}^{\text{alg}} = \left[\lim_{\leftarrow n} (\mathbb{R} \langle e_x, J_x, T_u \rangle / \mathcal{R}_{\text{HMT}}) \right] \rtimes_{\Gamma_9} \mathbb{N}.$$

In the reversible sector, $\mathbb{N}$ is replaced by $\mathbb{Z}$. Its space of infinite states is

$$X_\infty = \lim_{\leftarrow n} X_n.$$

The expression is algebraic. A bounded representation ρ determines its own completion $\mathfrak{S}_{\text{HMT}}^\rho = \overline{\rho(\mathfrak{S}_{\text{HMT}}^{\text{alg}})}^{\|\cdot\|}$; the topology of the codomain therefore belongs to the reader and is not placed before the generator.

Kernel theorem of conservative evolution of unity. APP arithmetic, oriented by the fibres $J_\tau^2 = -\tau I$ and transported by the TPK cocycle, generates an associative noncommutative algebra whose refinement preserves unity and whose returns raise memory. The typed spectrum of representations of this algebra correlatively determines the correlated system $(\pi, \varphi, e, \alpha)$, the joint structure of the continuum, the electronic vacancy, action and gravity scales, geometric parameters, catalogue constants and boundary symmetries. Each realization preserves its domain, reader and degree.

The proof divides into six consecutive identities:

1. the APP equations prove preservation of residue and carry;
2. the tritic relations prove coexistence of the three curvatures;
3. the cocycle identity proves associativity of concatenation;
4. nonsplitting of C_{108} proves that phase return raises memory;
5. the inverse-limit/direct-limit pair proves preservation of unity under refinement and reading;
6. the universal property proves that every typed output is a representation of the same source, not a disguised conventional input.

59.2.2 Conceptual consequence: scalar reality is a faithful but incomplete shadow

The Archimedean representation can be faithful to the determined value yet unfaithful to genealogy. Two histories can have the same real coordinate and different carry, height or memory. The discrete structure of the continuum resolves this multiplicity by preserving fibres before contracting them. The phrase ‘one point becomes two’ thus translates as follows: a scalar coordinate can admit two distinct holonomic lifts, and bidirectionality resides in the fibre, not in an arbitrary duplication of the point.

59.2.3 Physical consequence: unity does not first preserve symmetries; it produces them

What is primarily preserved is the possibility of reconstruction: sheet, residue, quotient, carry, orientation, boundary and memory. A symmetry appears when a representation has automorphisms preserving these relations. Exceptional symmetries, electromagnetic polarity and gravitational pentapolarity are thus different ways of stabilizing an already generated boundary. Symmetry conservation is a consequence of genealogical conservation.

59.2.4 Informational consequence: infinite growth at zero internal cost

Linear complexity $p(m) = m + 1$ allows unlimited nonperiodic information to be produced without exponential entropy growth. Refinement adds memory, but the integral update is reversible. Cost appears when a reader decides to erase the fibre that would allow the step to be reconstructed. This combination—aperiodicity, zero topological entropy, growing memory and preserved unity—is the operational mathematical definition of the HMT topological-temporal crystal.

59.3 VIII. Independent contributions arising from retrospective reading

59.3.1 First contribution: inverse-limit–direct-limit duality

The distinction between the inverse limit of states and the direct limit of observables had not been formulated at kernel level in the earlier developments. It explains why deepening and projection are conjugate operations without being trivial inverses; corrects the identification of global unity with an individual section; and provides the natural framework for predefinition.

59.3.2 Second contribution: universal history algebra

The presentation by generators and relations brings together the TRIT fibres, APP order, TPK transports and the memory-record cocycle. Its universal property proves that subsequent readers need no new generators if they preserve the relations.

59.3.3 Third contribution: trigeometric exponential identity

The common exponential of J_+ , J_0 , J_- shows that complex rotation, the nilpotent jet and hyperbolic dilation are three components of a single law. Euler's identity is its elliptic projection; φ is its hyperbolic scale; e is the exponential law; and α is the seam gluing the jet to dodecaphasic holonomy.

59.3.4 Fourth contribution: constants as a typed spectrum of representations

The notion of scalar character is extended to a spectrum of representations and functionals. This brings constants, particles and fields together without confusing their codomains. The same definition locates Catalan as a moment, Feigenbaum as nonlinear rescaling, Boltzmann as transducer, Barbero–Immirzi as incidence covector, Planck as action section and gravity as contragredient character.

Functional recovery makes an additional relation explicit: the constitutive incidence response 90/120, with its regularity at the origin, determines the complete Catalan moment; its inversion recovers the labelled angular channels. Transport composition is then expressed through the law $r \odot t = f(\psi(r)\psi(t))$, preserving product, orientation and normalized trace. On that same velocity section, inscription and the radial norm first determine the length L , then the products $R_X \Lambda_X = L^2 I$ and $H_X \Lambda_X = \hbar_{\text{ret}} c I$, and finally the common coefficient $G = c^3 L^2 / \hbar_{\text{ret}}$. The proofs of the corresponding section preserve the memory and constitutive hypotheses of each realization.

The Feigenbaum constant acquires a precise structural definition: it is the asymptotic separation invariant of the parameters selected by the critical returns of the uniform dodecaphasic reader. The construction preserves its APP–TRIT–TPK origin, the joint continuum and the first-root order. Identifying the selected centres with the analytic centres makes the scaling theorem applicable to that same sequence; the proofs in 24.6.8.1–24.6.8.3 and (24.152) establish the parametric result. The Jacobi representation recovers the profile from its twelve-node measure, while constitutive inversion recovers the oriented channels and the moment whose endpoint is Catalan's constant. These operations exhibit relations between readers of the same antecedent, preserving the domain and information of each.

59.3.5 Fifth contribution: symmetry as reconstruction stabilizer

The intuition that symmetries result from conservation is converted into a definition of the stabilizer of boundary and relations. This formulation links dual projection, the exceptional corridor and physical representations without postulating symmetry as the source.

59.3.6 Sixth contribution: triadic irreducibility of the kernel

The author's image of the Borromean link admits an exact formulation through three forgetful quotients. If the two APP sheets are identified, both the potential $\Delta_{\Sigma\Pi}$ and the commutator $[P_\Sigma, P_\Pi]$ disappear. If the three TRIT regimes are identified, the sign spectrum disappears

$$J_+^2 = -I, \quad J_0^2 = 0, \quad J_-^2 = I.$$

If the TPK cocycle is trivialized, the relation $d_M(\Gamma_9 x) - d_M(x) = 1$ becomes zero and phase return becomes indistinguishable from state return. Each quotient destroys an invariant the other two factors cannot reconstruct. APP, TRIT and TPK are therefore three functionally irreducible generators of the kernel object: APP supplies discrepancy and sheet order; TRIT supplies the local curvature signature; TPK supplies composition, holonomy and memory.

This irreducibility explains why the kernel cannot be reduced to an arithmetic table, a quadratic algebra or phase dynamics. The structure appears only in their composition.

59.3.7 Seventh contribution: the electron contains the Witt null pair

Retrospective comparison of the electronic block with the three TRIT regimes reveals a local closure that had remained implicit. Starting from

$$S^2 = I, \quad J^2 = -I, \quad SJ = -JS$$

we form

$$n_{\pm} = \frac{S \pm J}{2}.$$

Then

$$n_{\pm}^2 = 0, \quad n_+ n_- + n_- n_+ = I.$$

The corner $Cl_{1,1} \cong M_2(\mathbb{R})$ therefore contains the three local classes: J is elliptic, S is hyperbolic and $n_{\pm}$ are parabolic. The electronic vacancy is simultaneously a minimal carry defect in APP and a null direction of the split algebra. TRIT does not introduce the threshold as an external correction to the electron: it recognizes its type and preserves it under TPK transport. The designation Witt refers here to the local null decomposition; the Paley–Witt incidence structure is obtained subsequently as a combinatorial boundary representation of the same genealogy.

59.3.8 Eighth contribution: Weyl relations of the two clocks

The phrase ‘the helix contains two clocks’ admits an exact operator presentation. Let T be the step, V_9 the nonadic-phase character, W_δ the irrational-capacity character, and $\delta = \log_{10}(10/9)$. If ζ_9 and ω_δ denote their internal characters, then

$$V_9 T = \zeta_9 T V_9, \quad W_\delta T = \omega_\delta T W_\delta.$$

For $\Gamma_9 = T^9$,

$$V_9 \Gamma_9 = \Gamma_9 V_9, \quad W_\delta \Gamma_9 = \omega_\delta^9 \Gamma_9 W_\delta.$$

The phase clock closes and the capacity clock continues. If D_M measures helical height,

$$[D_M, \Gamma_9] = \Gamma_9.$$

This final identity is the shortest algebraic form of the difference between phase return and state return. It also locates noncommutativity precisely: basal translation belongs to an Abelian group, but its crossed product with the phase and capacity observables satisfies noncommutative Weyl relations. The HMT aperiodic crystal is thus an ordered suspension of a finite clock by an irrational character, lifted by a memory operator.

In the subsequent elliptic representation, ζ_9 and ω_δ are recognized as $\text{Exp}_+(2\pi J_+/9)$ and $\text{Exp}_+(2\pi \delta J_+)$. The HMT character precedes that complex notation.

59.3.9 Ninth contribution: a polynomial equation for the three geometries

The three local generators can be combined in

$$\mathbb{J} = p_+ J_+ + p_0 J_0 + p_- J_-.$$

The identity

$$\boxed{\mathbb{J}^6 = \mathbb{J}^2}$$

contains exactly the spectrum of squares $\{-I, 0, I\}$. If $K = \mathbb{J}^2$, the regimes are recovered from the operator itself:

$$p_+ = \frac{K^2 - K}{2}, \quad p_0 = I - K^2, \quad p_- = \frac{K^2 + K}{2}.$$

This formula dispels the appearance that TRIT is merely a ternary label. The regime is an intrinsic spectral decomposition. The corresponding exponential identity

$$\text{Exp}_\star(t\mathbb{J}) = p_+(\cos t + \mathbb{J} \sin t) + p_0(I + t\mathbb{J}) + p_-(\cosh t + \mathbb{J} \sinh t)$$

contains as faces Euler's elliptic closure, the nilpotent threshold and hyperbolic self-scaling. Its composition with Γ_9 , together with $[D_M, \Gamma_9] = \Gamma_9$, constitutes the minimal operator identity from which conservative evolution of the dimensional unit can be expounded.

PART IX

**Proof appendices and classification tables: algebraic, arithmetic
and coinductive certificates**

Ordered affine calculation of the APP leaves

A.1 Affine monoid $\text{End}(U_{\text{APP}}) \ltimes U_{\text{APP}}$ and ordered composition of routes

Let U_{APP} be the radial module prior to its Archimedean evaluation. The correct envelope is the monoid

$$\text{AffEnd}(U_{\text{APP}}) = \text{End}(U_{\text{APP}}) \ltimes U_{\text{APP}},$$

because the threshold or folding leaves may contain noninvertible endomorphisms. The action of (Λ, C) on $u \in U_{\text{APP}}$ is

$$(\Lambda, C) \cdot u = \Lambda u + C.$$

The composition is calculated without commuting factors:

$$\begin{aligned} (\Lambda_2, C_2)(\Lambda_1, C_1) \cdot u &= \Lambda_2(\Lambda_1 u + C_1) + C_2 \\ &= (\Lambda_2 \Lambda_1)u + (C_2 + \Lambda_2 C_1). \end{aligned} \tag{A.1}$$

Therefore,

$$(\Lambda_2, C_2)(\Lambda_1, C_1) = (\Lambda_2 \Lambda_1, C_2 + \Lambda_2 C_1).$$

The second coordinate is not an ordinary sum of shifts: it is a cocycle twisted by the accumulated linear action.

Proposition A.1 (Prefix Formula). *For a route with N factors $h_j = (\lambda_j, C_j)$, ordered from $j = 0$ through $j = N - 1$, the prefix holonomy is*

$$H_N = h_{N-1} \cdots h_1 h_0 = (\Lambda_N, C_N),$$

where

$$\Lambda_N = \prod_{j=0}^{N-1} \lambda_j$$

and

$$C_N = C_{N-1} + \lambda_{N-1} C_{N-2} + \lambda_{N-1} \lambda_{N-2} C_{N-3} + \cdots + \lambda_{N-1} \cdots \lambda_1 C_0.$$

Proof. For $N = 1$ the formula is immediate. If it holds for N , then

$$h_N H_N = (\lambda_N \Lambda_N, C_N + \lambda_N C_N),$$

which yields exactly the expression of length $N + 1$. The induction preserves the order of the route and does not identify paths with the same endpoints. $\square$

A.2 Difference sum-product as curvature of order

Let $A_a = (I, a)$ and $M_\lambda = (\lambda, 0)$ be given. In the invertible sector,

$$A_a M_\lambda A_a^{-1} M_\lambda^{-1} = A_{a(1-\lambda)}.$$

The direct proof for u is

$$u \xrightarrow{M_\lambda^{-1}} \lambda^{-1} u \xrightarrow{A_a^{-1}} \lambda^{-1} u - a \xrightarrow{M_\lambda} u - \lambda a \xrightarrow{A_a} u + a(1 - \lambda).$$

Thus, the distinction between accumulation and folding is published as a translation. When $\lambda = 1$, the commutator vanishes; when $\lambda \neq 1$, the route retains a holonomy that cannot be reconstructed from the endpoint alone.

Example A.2 (Two routes with the same visible data). Fix $u_0 = 0$, $a = 1$, $\lambda = 2$. The routes

$$A_1 M_2 : u_0 \mapsto 1, \quad M_2 A_{1/2} : u_0 \mapsto 1$$

publish the same final value. However, their ledgers contain distinct factors, shifts, and prefixes. Pointwise evaluation identifies the routes; enriched affine holonomy distinguishes them.

A.3 Exhaustive enumeration of the local chart $(\tau_+, \tau_-) \mapsto (p, a, r_3, c_3)$

The following table does not posit that every global radial degree arises from a single pair. It records the elementary chart obtained by composing two TRIT orientations. The degree of a general route is the integer cocycle of identity of the radial cocycles of the tritic lift; this chart provides its nine local generators.

Table A.1: The 9 oriented pairs of the local TRIT chart.

τ_+	τ_-	p	a	r_3	c_3	Lectura local
1	1	2	0	-1	1	double radial opening with positive carry
1	0	1	1	1	0	opening and threshold
1	-1	0	2	0	0	positive orientation equilibrium
0	1	1	-1	1	0	threshold and aperture
0	0	0	0	0	0	umbral doble
0	-1	-1	1	-1	0	threshold and contraction
-1	1	0	-2	0	0	negative orientation equilibrium
-1	0	-1	-1	-1	0	contraction and threshold
-1	-1	-2	0	1	-1	double contraction with negative carry

Here $p = r_3 + 3c_3$, with balanced residue $r_3 \in \{-1, 0, 1\}$. The carry distinguishes $p = 2$ from its residue -1 and $p = -2$ from its residue 1 . The pair (p, a) likewise distinguishes the three local genealogies of degree 0.

Nonadic Certificate and Geometry of Memory

B.1 Typed distinction between decimal catalog, ternary words and ambient fiber

Table B.1: Strata of the certificate of the joint nonadic connection.

Object	Cardinality	Mathematical function
Catalog of source triples	104 976	Finite domain prior to the joint selection; it is not a decimal table.
Decimal emissions U_6	468	Compatible emissions of length 6 in the decimal publication.
Visible words w_6	243	Published ternary words after typing; they are not the complete environment.
Ambient space $\mathbb{F}_3^6$	729	Space of all ternary words of length 6; it contains the 243 visible words.
Certified levels	396	Finite prolongation up to the declared boundaries.
Trits recorridos	2 376	Longitud total $396 \cdot 6$.
Vueltas completas	43	Nonadic cycles in the certified segment.
Pairs $(K, K + 9)$	387	Pairs of same-phase by channel; the ledger may differ.

The prolongation chain is preserved in the order

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9.$$

None of these objects can be replaced by the subsequent modal quotient. The holonomy Γ_9 acts on the enriched dynamics; the reduced-memory publication preserves only the information specified by its codomain.

B.2 Phase return, memory advance, and complete state

With

$$\rho_9(N) = 1 + ((N - 1) \bmod 9), \quad q_9(N) = \left\lfloor \frac{N - 1}{9} \right\rfloor,$$

we have

$$\rho_9(N + 9) = \rho_9(N), \quad q_9(N + 9) = q_9(N) + 1.$$

Therefore,

$$H_9(N + 9) = H_9(N) + (0, 0, 1).$$

The equality of the first phase coordinate does not imply equality of the state:

$$g(K + 9) = g(K), \quad B_{K+9} \neq B_K$$

in general. The helical formula is precisely the minimal representation of this distinction.

B.3 Non-Split Dodecaphase Extension

The sequence

$$0 \longrightarrow C_{12} \longrightarrow C_{108} \longrightarrow C_9 \longrightarrow 0$$

couples the dodecaphasic calendar to the nonadic return. If a global homomorphic section $s : C_9 \rightarrow C_{108}$ existed, the generator of C_9 would have to lift to an element of order dividing 9 that projected onto the generator. The active temporal lift, however, preserves the accumulated dodecaphasic displacement and does not identify the state after one phase turn. The seam is recorded as carry; geometrically, it manifests as axial translation of the helix.

B.4 Decomposition of the microfiber of π into 5 regions and multiplicity of Archimedean evaluation

The microfiber of π contains five oriented regions with total multiplicity

$$432 + 4 \cdot 144 = 1008.$$

The regions publish the same real value, but retain distinct genealogies. This result provides the internal precedent for spiral classification: a visible curve does not by itself determine the region, the sheet, the carry, or the memory that produced it.

Algebra of L_p and classification of the 5 canonical families $p \in \{2, 1, 0, -1, -2\}$

C.1 The coordinate L_p and its inverse

For $x > 0$, define

$$L_p(x) = \begin{cases} (x^p - 1)/p, & p \neq 0, \\ \log x, & p = 0. \end{cases}$$

Its inverse is

$$\exp_p(u) = \begin{cases} (1 + pu)^{1/p}, & p \neq 0, 1 + pu > 0, \\ e^u, & p = 0. \end{cases}$$

The derivative

$$\frac{d}{dx} L_p(x) = x^{p-1} > 0$$

proves that L_p is strictly increasing on its domain and is therefore invertible.

Proposition C.1 (Sum–Product Law). *For $x, y > 0$,*

$$L_p(xy) = L_p(x) + L_p(y) + pL_p(x)L_p(y).$$

Proof. Si $p \neq 0$,

$$\begin{aligned} L_p(x) + L_p(y) + pL_p(x)L_p(y) &= \frac{x^p - 1}{p} + \frac{y^p - 1}{p} + \frac{(x^p - 1)(y^p - 1)}{p} \\ &= \frac{x^p y^p - 1}{p} = L_p(xy). \end{aligned}$$

For $p = 0$, the equality reduces to $\log(xy) = \log x + \log y$. □

Corollary C.2 (Deformed group in the positive domain). *The operation*

$$u \boxplus_p v = u + v + puv$$

satisfies

$$1 + p(u \boxplus_p v) = (1 + pu)(1 + pv).$$

Its neutral element is 0, and the inverse of u is

$$\ominus_p u = -\frac{u}{1 + pu},$$

si $1 + pu \neq 0$.

C.2 Evaluation of $\exp_p(u_0 + \beta m)$ in the 5 canonical families

Let m be the published depth, and suppose

$$L_p(r/r_0) = \beta m, \quad \theta = \theta_0 + \omega m, \quad \omega \neq 0.$$

Eliminating $m = (\theta - \theta_0)/\omega$,

$$r(\theta) = r_0 \left[1 + p \frac{\beta}{\omega} (\theta - \theta_0) \right]^{1/p}, \quad p \neq 0,$$

and

$$r(\theta) = r_0 \exp \left[\frac{\beta}{\omega} (\theta - \theta_0) \right]$$

for $p = 0$. Positive changes of origin and rescalings yield the conventional forms:

Table C.1: Canonical publications in the translational sector.

p	Normal equation	Conventional name	Domain and feature
2	$r^2 = a^2 \theta$	Fermat spiral	$\theta \geq 0$; linear increment of the quadratic radial area.
1	$r = a\theta$	Archimedes	$\theta \geq 0$; constant radial increment per angular unit.
0	$r = r_0 e^{b\theta}$	logarithmic	$\theta \in \mathbb{R}$; constant dilation per angular unit.
-1	$r = a/\theta$	hyperbolic	$\theta > 0$; the product $r\theta$ is constant.
-2	$r^2 = a^2/\theta$	lituus spiral	$\theta > 0$; product $r^2\theta$ is constant.

C.3 Conjugation $p \leftrightarrow -p$ of the power–logarithmic sheets

Para $p \neq 0$,

$$L_{-p}(x^{-1}) = \frac{x^p - 1}{-p} = -L_p(x).$$

The identity also holds for $p = 0$. Consequently, radial inversion interchanges p with $-p$ and reverses the sign of the linearized coordinate. This relation pairs the Fermat spiral with the lituus and the Archimedean spiral with the hyperbolic spiral as conjugate orientations of the reader, without identifying them as identical curves.

C.4 Classification of the affine sector by the pair $(p, [\Lambda, C])$

If the radial coordinate satisfies

$$u_{n+1} = \Lambda u_n + C,$$

the exact internal iteration is

$$u_n = \Lambda^n u_0 + \sum_{k=0}^{n-1} \Lambda^k C.$$

If $I - \Lambda$ is invertible, $u_* = (I - \Lambda)^{-1}C$ and $u_n - u_* = \Lambda^n(u_0 - u_*)$. Only in a positive scalar realization, $\chi\Lambda = \lambda\chi$ with $\lambda > 0$, does one define $u_* = C/(1 - \lambda)$. With $\theta_n = \theta_0 + n\omega$, $\omega \neq 0$, the interpolation published in that realization is

$$r(\theta) = r_0 \exp_p \left[u_* + \lambda^{(\theta - \theta_0)/\omega} (u_0 - u_*) \right].$$

The complete taxonomy is therefore not a list of five names. It contains at least the degree p , the orientation a , the affine class $[\Lambda, C]$, the carry cocycle, the boundary, the angular increment, and the memory ledger.

Naturality under truncation, nonadic equivariance, and Archimedean fidelity defect

D.1 Affine prefix reader and enriched record conservation

For a truncated history

$$\widehat{\gamma}_N = \widehat{e}_N \cdots \widehat{e}_1,$$

The enriched reader records

$$\mathcal{P}_{\text{rad},N}^{\text{enr}}(\widehat{\gamma}_N) = \left((H_0, \text{Led}_0); (p(\widehat{e}_j), a(\widehat{e}_j), P_j, A_j, h_{e_j}, H_j, \text{Led}_j)_{j=1}^N \right).$$

The term $H_j = h_{e_j} \cdots h_{e_1}$ preserves the prefixes. The ledger Led_j preserves, according to the route, sheet, residue, quotient, carry, boundary, orientation, cylinder, signature, survival, and memory. The real-valued coordinates are obtained subsequently:

$$\mathcal{P}_{\text{arq}}^{\text{esp}} = \text{ev}_{\mathbb{R}^2} \circ \mathcal{P}_{\text{rad}}^{\text{enr}}.$$

D.2 Complete proof of naturality under truncation

Let $\pi_{N,M}$ denote deletion of the suffix of length $N - M$, with $M < N$. Let $\text{pr}_{N,M}$ denote the projection that preserves the initial record and the first M edge records. Then

$$\text{pr}_{N,M} \mathcal{P}_{\text{rad},N}^{\text{enr}}(\widehat{\gamma}_N) = \left((H_0, \text{Led}_0); (p(\widehat{e}_j), a(\widehat{e}_j), P_j, A_j, h_{e_j}, H_j, \text{Led}_j)_{j=1}^M \right).$$

By definition of truncation,

$$\pi_{N,M}(\widehat{\gamma}_N) = \widehat{e}_M \cdots \widehat{e}_1,$$

and upon reapplying the reader the same sequence is obtained. Therefore,

$$\text{pr}_{N,M} \circ \mathcal{P}_{\text{rad},N}^{\text{enr}} = \mathcal{P}_{\text{rad},M}^{\text{enr}} \circ \pi_{N,M}.$$

The proof does not use the value of the curve. It uses prefix identity, which is the information that the reduced seven-coordinate reader would have lost.

D.3 Compatibility criterion for general refinements of paths

For a refinement $r : \gamma \rightarrow \gamma'$, naturality requires a map $\widehat{r}$ on ledgers and the local equality

$$h_e = h_{e'_1} \cdots h_{e'_k}$$

for each edge e replaced by the refined chain $(e'_1, \dots, e'_k)$. If, in addition, the carry, boundary, and memory of the refined chain compose the records of the original step, then

$$\widehat{r} \circ \mathcal{P}_{\text{rad}}^{\text{enr}} = \mathcal{P}_{\text{rad}}^{\text{enr}} \circ r.$$

This condition is a verifiable commutative square; it must not be assumed for an arbitrary refinement without exhibiting the factors.

D.4 Equivariance of the radial cocycle under the holonomy Γ_9

The action of Γ_9 returns the phase and prolongs the state. In the codomain of the reader, $\widehat{\Gamma}_9^{\text{rad}}$ acts by shifting the records and updating the last prefix. The correct identity is

$$\mathcal{P}_{\text{rad}}^{\text{enr}}(\Gamma_9\gamma) = \widehat{\Gamma}_9^{\text{rad}}\mathcal{P}_{\text{rad}}^{\text{enr}}(\gamma).$$

Requiring invariance would erase precisely the memory that the connection preserves. In the reversible sector, route inversion satisfies

$$H_{C_{\text{rev}}\gamma} = H_{\gamma}^{-1}$$

when all factors belong to the invertible affine subgroup. This typed assertion does not automatically extend to a global involution on an arbitrary subsequent quotient.

D.5 3 explicit obstructions to the faithfulness of Archimedean evaluation

Proposition D.1 (Intrinsic circle and projected circle). *There exist two histories with the same visible curve.*

$$c(\theta) = (\cos \theta, \sin \theta)$$

and different ledgers.

Proof. The first history has constant axial memory $m = 0$. The second is the helix

$$h(\theta) = (\cos \theta, \sin \theta, \theta/2\pi),$$

whose ledger records $m(\theta) = \theta/2\pi$. Both satisfy $\pi_{xy} \circ h = c$. The projection is therefore non-injective. $\square$

Proposition D.2 (Same endpoint, different affine order). *If $a(1 - \lambda) \neq 0$, the paths $A_a M_\lambda$ and $M_\lambda A_a$ are not equivalent under the enriched reader, even though they may be calibrated to publish the same endpoint.*

Proof. Its commutator is $A_{a(1-\lambda)} \neq I$. The commutator datum belongs to the prefix ledger and disappears when only the endpoint is retained. $\square$

Proposition D.3 (Same curve, different genealogical region). *Equality of the published real value does not identify the five regions of the microfiber of π ; consequently, it is likewise insufficient to identify the spiral class when the reader preserves region, boundary, or carry.*

Stratified Taxonomy of Spiral Correspondences

E.1 Domain stratification: history, reader, Archimedean evaluation, and physical realization

Table E.1: Levels that must not collapse into a single notion of curvature.

Level	Invariant	Meaning
APP	$F(S, P) = 1, F(P, S) = -1$	Discrete curvature of the order of accumulation and folding.
TRIT	$\tau \in \{-1, 0, 1\}, J_\tau^2 = -\tau I$	Local elliptic, parabolic, or hyperbolic regime of the transport.
TPK	Γ_9 , carry, boundary, route, and ledger	Phase holonomy and memory on the integral holonomic state.
Radial level	$p, a, [\Lambda, C]$	Characters that determine the radial law before recognizing a curve.
Archimedean	$r(\theta)$	Curve published in $\mathbb{R}^2$; it may forget genealogy.
Frenet level	$\kappa_F, \mathcal{T}_F$	Differential curvature and torsion of the visible curve or suspension.
Dimensional level	fields, scales and representations	Posterior physical realization, with its own premises.

E.2 Classification by the increments (ω, β, μ)

Table E.2: Complete kinematic taxonomy of the minimal degenerate cases.

$\Delta\theta$	Δu_{rad}	Δm	Salida visible	Genealogical Information
0	0	0	point	published steady state
0	$\neq 0$	0	recta radial	scale change
0	0	$\neq 0$	fibra axial	non-transverse displacement memory
0	$\neq 0$	$\neq 0$	radial sheet suspended	scale and memory
$\neq 0$	0	0	circle	intrinsic angular closure
$\neq 0$	0	$\neq 0$	cylindrical helix	closed phase, high memory
$\neq 0$	$\neq 0$	0	espiral plana	radial character visible
$\neq 0$	$\neq 0$	$\neq 0$	radial helical suspension	turn, radius and depth preserved

E.3 Affine holonomy classification

The following table belongs to a real scalar realization of the linear part, $\chi\Lambda = \lambda\chi$; for ease of reading, we write Λ in the conditions column. It does not classify abstract endomorphisms by numerical inequalities.

Table E.3: Types of holonomy over the radial coordinate.

Condition	Orbit of u	Interpretation
$\Lambda = 1, C = 0$	$u_n = u_0$	constant radius; the memory may be zero or hidden by projection.
$\Lambda = 1, C \neq 0$	$u_n = u_0 + nC$	translational sector that publishes the family L_p .
$0 < \Lambda < 1$	$u_n \rightarrow u_*$	contraction towards a fixed section.
$\Lambda > 1$	$ u_n - u_* $ grows	expansion from a fixed section.
$\Lambda < 0$	orientation alternation	requires control of the domain of $\exp_p$ at each step.
$\Lambda = 0$	$u_{n+1} = C$	noninvertible folding; it belongs to the monoid, not to the affine group.

E.4 Genealogical Equivalence Criterion

Two routes are not identified merely because they publish the same curve. A typed isomorphism preserving the following term by term is required:

$$((x_0, H_0, \text{Led}_0); (\widehat{e}_j, p(\widehat{e}_j), a(\widehat{e}_j), P_j, A_j, h_{e_j}, H_j, \text{Led}_j)_{j=1}^N)$$

and that commutes with the admissible truncations and refinements. The resulting classification is finer than the conventional taxonomy because it incorporates the history preceding the emergence of $\mathbb{R}^2$.

Electromagnetic and Material Realizations with Domain and Codomain Separation

F.1 Circular electromagnetic mode

Let

$$\zeta = \frac{z - c_{\text{vac}} t}{\lambda},$$

and

$$\mathbf{E}_\sigma = E_0(\mathbf{e}_1 \cos 2\pi\zeta + \sigma \mathbf{e}_2 \sin 2\pi\zeta), \quad \mathbf{B}_\sigma = c_{\text{vac}}^{-1} \hat{\mathbf{k}} \times \mathbf{E}_\sigma.$$

Then

$$\mathbf{E}_\sigma \perp \mathbf{B}_\sigma \perp \hat{\mathbf{k}}, \quad |E| = c_{\text{vac}} |B|, \quad \omega = c_{\text{vac}} k.$$

At fixed time, the normalized endpoint of the field in the phase–position bundle describes

$$(\cos 2\pi\zeta, \sin 2\pi\zeta, \zeta).$$

This statement does not assert that every spatial field line is a helix: it describes the trajectory of the tip of the polarization vector together with the marked event.

F.2 Loxodromic compatibility

For

$$\Phi(r, \varphi, z, t) = \ell\varphi + \nu \log(r/r_0) + k_z z - \omega t,$$

the transformation

$$(r, \varphi) \mapsto ((2 + \sqrt{3})r, \varphi + \pi/2)$$

preserves the phase modulo 2π exactly when

$$\nu \log(2 + \sqrt{3}) + \ell \frac{\pi}{2} \in 2\pi\mathbb{Z}.$$

Dilation is interpreted as a change of scale between solutions; not as spontaneous energy production.

F.3 Compatibility of the radial reader with the electronic fixed point (5, 5) and atlas 81/56/13/324

The electron occupies the unique fixed point (5, 5) of the material atlas. This centrality is not derived anew in the present work. The prior operator compatibility includes

$$C^{(3)} M_{\text{ph}} = -\frac{1}{2} M_{\text{ph}}, \quad 1 - \left(\frac{1}{2}\right)^2 = \frac{3}{4} = \frac{90}{120},$$

and

$$\|[P_{\Sigma,3}, P_{\Pi,3}]\| = \frac{\sqrt{3}}{4}.$$

The theory of spiral correspondences adds a route reader; it does not replace the General Mass Law or use the electron to select p , Λ , or C retrospectively.

F.4 Matrix of validity conditions by causal level

Table F.1: Falsifiers and type errors excluded by the construction.

Failure	Consequence
The classical curve selects p, Λ, C .	Circularity: the reader ceases to be constructive–generative.
Only the final holonomy is preserved.	Loss of prefixes, carry, and boundary; naturality is not established.
Invariance under Γ_9 is required.	The memory advance is erased; the correct relation is equivariance.
U_6, w_6 , and $\mathbb{F}_3^6$ are conflated.	Collapse of objects with cardinalities 468, 243, and 729.
The term phase return is identified with state return.	Contradiction with $B_{K+9} \neq B_K$ in general.
τ, p , and κ_F are identified.	Regime collapse, radial character and visible curvature.
A general helix is called any variable radius.	Differential statement unverified Lancret condition.
The φ axis is identified with the electron.	Mixture of the fiber of constants and the material realization.
Pell is interpreted as energy amplification.	Violation of scale reading and physical conservation.
Electromagnetic helicity is identified with spin 1/2.	Confusion between distinct representations.
A point cell is imposed on every species.	Contradiction with the character of a fibre or multi-section of the noncentral species.
The five constructions of the continuum are separated.	Breakdown of the previous unique genealogy prior to any geometric publication.

Mathematical Glossary

Table G.1: Symbols and Key Terms.

Symbol or term	Definition
Σ, Π	Additive and multiplicative evaluations on the common support APP.
τ	Local TRIT orientation regime; it is neither the radial degree nor Frenet torsion.
M_{ph}	Tritic operational-phase selector.
Γ_9	Nonadic holonomy of enriched dynamics.
H_N	Accumulated affine holonomy up to the prefix N .
Λ_N, C_N	Linear part and translational cocycle of H_N .
p	Integer cocycle of the radial character.
a	Local orientation character that separates genealogies of the same local degree.
$L_p, \exp_p$	Power-logarithmic coordinate and its positive inverse.
$\boxplus_p$	Deformed law that publishes the sum-product interaction.
Led_N	Enriched ledger of sheet, carry, boundary, route, and memory.
$\mathcal{P}_{\text{rad}}^{\text{enr}}$	Radial reader enriched prior to the curve.
$\mathcal{P}_{\text{arq}}^{\text{esp}}$	Subsequent evaluation on a curve in $\mathbb{R}^2$.
$\kappa_F, \mathcal{T}_F$	Curvature and torsion of Frenet of the visible curve.
Radial helical suspension	Variable-radius surface or sheet that preserves phase, depth, and path.
Unfaithful publication	Assessment that can identify distinct histories.
Constructivo—generativo	Method that constructs the object and its genealogy before recognizing its conventional appearance.

Dictionary of domains, operators and representations

Notation, types, and symbol disambiguation

Symbol	Object
$\mathcal{A}_9$	Nonadic arithmetic torus of paths with two internal evaluations.
TRIT	Oriented ternary coordinate of the transport; the visible output does not exhaust the enriched lift with carry.
TPK	Topological phase kernel with integral holonomic state, transitions, routes, extensions and memory.
M_{ph}	Tritic operational-phase selector, distinct from cardinal transport and stroboscopic sampling.
τ_3	Primitive phase block $(+1, -1, 0)$.
$\mathcal{N}, \mathcal{E}, \mathcal{S}, \mathcal{O}$	Generators of transport cardinal.
Obs_3	Stroboscopic sampling of every third step.
$\mathcal{R}_{12}$	Internal TPK-oriented dodecaphasic temporal system.
$\Theta_{108} = \mathbb{Z}/108\mathbb{Z}$	Observable calendar 12×9 ; it is not the complete TPK state.
C_{12}	Rotation of the twelve temporal sectors.
Γ_{108}	Oriented transport orbit of 108 steps.
$\mathcal{G}_9$	Unitary pruning within the extension and boundary relation; it does not by itself constitute the monodromy.
$\mathfrak{M}_9$	Nonadic monodromy with memory: it returns the phase and changes the state.
π_{term}	Projection of the terminal holonomic state onto the three selected visible blocks.
R_{sgn}	Reading of the signed integer coordinate of the same terminal holonomic state.
U_{12}^{cat}	Visible concatenation $U_6 U_6$.
U_{12}^{sgn}	Signed coordinate of the integral holonomic state, reversible with K through $\mathcal{H}_{12}$; it is not deduced from U_{12}^{cat} , the calendar projection or the three isolated terminal blocks.
$\mathcal{H}_{12}$	Reversible dodecaphase transformation between U_{12}^{sgn} and K .
$\mathcal{E}_{\text{dod}}$	Integral extractor $(D_3, D_4, Q) : \mathbb{Z}^{12} \rightarrow \mathbb{Z}^{25}$ that reconstructs K .
$\overline{C}_9^3$	Nonadic cubic completion.
A, C^*	Intrinsic angular and antiangular coordinates.
$q_{\pm}$	Oriented channels of the double-layer reader.
$D_{\text{raw}}^{90,120}$	Raw diagnostic difference of series in a fixed chart; it is not a canonical extractor.
$\iota_{6,8}$	Inclusion $\mathcal{F}_6 \hookrightarrow \mathcal{F}_8$ of hexad–octad incidences.
$K_{6 \rightarrow 8}$	Angular functional of the two Lambert subtails, subsequent to $q_{\pm}$, the series, the pole, positivity, and minimality.
$\langle 90, 120 \rangle$	Spectral semigroup governing the hierarchy of towers.
$\varepsilon_{\text{res}}^{\circ}$	Residue of the exact angular identity; it is not a finite extraction from the preceding 90/120 objects.
Δ_4	Minimal torsion or global torsional invariant prior to the species; it is not Cartan torsion.
$T_{\alpha, \infty}$	Remote prolongation of the carry along the entire path of α , typed separately from the orbital generator $T_1^{\text{orb}} = L_1$, the neutral counter, and the calendar.

Symbol	Object
N_1^{neu}	Number of neutral positions in the first window of the transport example; it equals 3 and is not an operator.
$T_1^{\text{orb}} := L_1$	Second generator in the local orbital gauge; in its reference section it is abbreviated T_1 .
ε_{clk}	Energy quantum of the clock, $\hbar_{\text{int}} 2\pi / (108t_0)$; it is not a canonical energy basis unless the temporal frame is specified. The symbol ε_0 is reserved for the vacuum permittivity at the SI constitutive interface.
CLK1–CLK6	Labels of the six equations of the finite clock, from the Hamiltonian to the Schrödinger equation.
$\mathcal{K}_{27}$	Compound return of twenty-seven updates.
X_{∞}^{cal}	Common enriched domain of global evaluations.

Repeated cardinalities with different sources

Equality of cardinalities does not automatically transport domain, operation, memory, or evidentiary force. The following table sets out the collisions that most directly affect interpretation.

Index	Objeto tipado	Function and separation
108	Θ_{108}	The 12×9 calendar closes the observable clock.
108	Π_{108}^{cal}	Calendrical projection of a cursor with 324 seeds; it forgets the second cursor, orientation, carry, boundary, enriched chronological record, and survival.
108	w_{108}	Word of $108 = 18 \times 6$ trits; it is not the calendar.
104, 976	$X_{\text{TPK}}^{\text{seed}}$	Two-cursor engine, $104, 976 = 324^2$; it preserves a domain distinct from that of the calendar.
1080	w_{1080}	Length of a word.
1080	$\Gamma_0(1080)$	Level of a modular subgroup; requires its own realization.
729	$\mathbb{F}_3^6$	Ternary word environment.
729	9^3	Cubic cell/interior EMRH.
729	$\text{Ext}(X_K)$	Number of subcylinders examined at each prolongation stage.
729	bulk EMRH	Inner layer of the base completion mil.
27	$\mathcal{K}_{27}$	Composite return operator; not the name of an autonomous theory.
27	cubic stratum	Contribution of two boundary coordinates in $999 = 729 + 243 + 27$.
27	module or classifier	Each occurrence requires its map before identifying itself with another.

Six distinct sources of 90/120

Object	Domain and operation	Exit and type prohibition
$\mathcal{E}_{\text{dod}}$	$K \in \mathbb{Z}^{12}$; stepwise differences 3, 4 and sum Q	Transverse dodecaphasic coordinates; no incidence nor angular series.
$D_{\text{raw}}^{90,120}$	A q chart; raw difference $S_{90}(q) - S_{120}(q)$	Scale diagnostic, distinct from the Barbero–Immirzi functional and the incidence operators.
$\iota_{6,8}$	$H \subset O$; inclusion of $H \times \binom{H}{2}$ in $O \times \binom{H}{2}$	Cardinals 90, 120 and defect $-1/12$; neither an extractor nor a series.
$K_{6 \rightarrow 8}$	Channels $q_{\pm}$ and Lambert subtails; combination $12\Delta S_{90} - \Delta S_{120}$	Angular functional relative to its premises; it is not transported as a mass block.
$\mathfrak{S} = \langle 90, 120 \rangle$	Moment heights $s_n = q_-^n - q_+^n$	Global tower semigroup; does not end at eight heights.

Four realizations of bidirectional reversibility

Operation	Domain	Action
$P_{\pm} = (I \pm R)/2$	Split Algebra	Project the two spectral sectors.
U-turn	Route and composite calendar	Invert the orientation of the transport without erasing its word.
$\text{Ext}^{\pm}$	Extension Tree	It compares future and past valences; the balance may be read through $\log(\#\text{Ext}^+/\#\text{Ext}^-)$.
$C_M(\tau) = -\text{rev}(\tau)$	Dodecaphonic Word	It conjugates order and sign; it is not the cellular involution ρ_c .

The common function of these four realizations is to preserve reversible orientation. It does not authorize the identification of phase return, route return, word return, and state return.

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